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Practice Questions

CALCULUS

JEE Main & Advanced

Specification

- Contains more than 1500+ fully Solved.
- Contains Famous Questions of All Institutes of Kota.

PRACTICE QUESTIONS

CALCULUS

FOR
JEE MAIN & ADVANCED

1. **Function**
2. **Inverse Trigonometric Function**
3. **Limit**
4. **Continuity & Differentiability**
5. **Method of Differentiation**
6. **Application of Derivative**
7. **Definite Integration**
8. **Indefinite Integration**
9. **Area Under Curve**
10. **Differential Equation**

About this Book

The content of this book is developed by **GB SIR** by selecting most famous Kota's institutes with extensive research on various competitive exam papers and their pattern. The whole content is focused towards conceptual learning for the preparation of engineering entrance exam.

CONTENT

FUNCTION

ITF

LIMIT

CONTINUITY & DIFF

AOD

MOD

DEFINITE INTEGRATION

INDEFINITE INTEGRATION

AREA UNDER CURVE

DIFFERENTIAL EQUATION

Questions

1. Range of $f(x) = [\sin x + 1] + \left[\dfrac{\sin x}{2} + 2\right] + \left[\sin\dfrac{x}{3} + 3\right] + \ldots + \left[\sin\dfrac{x}{100} + 100\right]$ $\forall\, x \in [0, \pi]$ is

2. $f: R \to R,\ f(x) = \dfrac{3x^2 + mx + n}{x^2 + 1}$. If the range of this function is $[-4, 3)$, then find the value of $m^2 + n^2$.

3. Find the range of $f(x) = \log_e\left(\dfrac{1}{\sqrt{[\cos x] - [\sin x]}}\right)$ where [x] denotes the integral part of x.

4. Set A consists of 6 different elements and set B consists of 4 different elements. Number of mappings which can be defined from the set $A \to B$ which are surjective, is
(A) 256 (B) 432
(C) 840 (D) 1560

5. If $f(x) = \pi\left(\dfrac{\sqrt{x+7} - 4}{x - 9}\right)$, then the range of function $y = \sin(2f(x))$ is
(A) [0, 1] (B) $\left(0, \dfrac{1}{\sqrt{2}}\right]$
(C) $\left(0, \dfrac{1}{\sqrt{2}}\right) \cup \left(\dfrac{1}{\sqrt{2}}, 1\right]$ (D) (0, 1]

6. Which of the following pair of functions have the same graph?
[**Note :** [k], {k} and sgn k denote the largest integer less than or equal to k, fractional part of k and signum function of k respectively.]
(A) $f(x) = ln[1 + \{x\}]$ and $g(x) = ln(1 + [\{x\}])$
(B) $f(x) = \dfrac{1}{1 + \tan^2 x} + \dfrac{1}{1 + \cot^2 x}$ and $g(x) = 2\sin^2 x + \cos 2x$
(C) $f(x) = 2^{\text{sgn}(x^2 + 3x + 3)}$ and $g(x) = 2\,\text{sgn}(x^2 + 3x + 3)$
(D) $f(x) = \dfrac{\text{sgn}\, x}{\text{sgn}\, x}$ and $g(x) = \dfrac{|x|}{|x|}$

7. Let $f: R \to R$ be defined as $f(x) = 3^{-|x|} - 3^x + \text{sgn}(e^{-x}) + 2$, then which one of the following is correct?
(A) f is injective but not surjective
(B) f is surjective but not injective
(C) f is injective as well as surjective
(D) f is neither injective nor surjective
[**Note:** sgn x denotes signum function of x.]

8. The set of all real values of a so that the range of the function $y = \dfrac{x^2 + a}{x + 1}$ is R, is
(A) $[1, \infty)$ (B) $(-\infty, -1)$
(C) $(1, \infty)$ (D) $(-\infty, -1]$

9. If a polynomial function 'f' satisfies the relation $\log_2(f(x)) = \log_2\left(2 + \dfrac{2}{3} + \dfrac{2}{9} + \ldots..\infty\right) \cdot \log_3\left(1 + \dfrac{f(x)}{f\left(\frac{1}{x}\right)}\right)$ and $f(10) = 1001$, then the value of $f(20)$ is
(A) 2002 (B) 7999
(C) 8001 (D) 16001

10. In the square ABCD with side $AB = 2$, two points M and N are on the adjacent sides of the square such that MN is parallel to the diagonal BD. If x is the distance of MN from the vertex A and $f(x) = \text{Area}(\Delta\, AMN)$,

then range of $f(x)$ is :

(A) $\left(0, \sqrt{2}\right]$ (B) $(0, 2]$

(C) $\left(0, 2\sqrt{2}\right]$ (D) $\left(0, 2\sqrt{3}\right]$

11. If the functions
$f(x) = (k^2 - 3k + 2)x^2 + (k^2 - 1)\ \forall\ x \in R$ and
$g(x) = (k^2 - 6k + 5)x^3 + (k^2 - 2k + 1)x + (k^2 - k)\ \forall\ x \in R$ have the same graph then the number of real values of k, is

(A) 0 (B) 1

(C) 2 (D) 3

12. Let $f : X \to Y$ be a function such that $f(x) = \sqrt{x-2} + \sqrt{4-x}$, then the set of X and Y for which f(x) is both injective as well as surjective, is

(A) [2, 4] and $\left[\sqrt{2}, 2\right]$

(B) [3, 4] and $\left[\sqrt{2}, 2\right]$

(C) [2, 4] and [1, 2]

(D) [2, 3] and [1, 2]

13. The function $f(x) = \dfrac{x+1}{x^3+1}$ $(x \neq \pm 1)$ can be written as the sum of an even function $g(x)$ and an odd function $h(x)$. The even function $g(x)$, is

(A) $\dfrac{x^4-1}{2(x^6+1)}$ (B) $\dfrac{x^4-1}{x^6+1}$

(C) $\dfrac{x^4-1}{x^6-1}$ (D) $\dfrac{x^4-1}{2(x^6-1)}$

14. Let $f(x) = \sin^{23}x - \cos^{22}x$ and $g(x) = 1 + \dfrac{1}{2}\tan^{-1}|x|$, then the number of values of x in interval $[-10\pi, 20\pi]$ satisfying the equation $f(x) = \text{sgn}(g(x))$, is

(A) 6 (B) 10

(C) 15 (D) 20

Paragraph for question nos. 15 to 17

Let $f(x) = \begin{cases} \beta x^2 + 3, & -\infty < x < -1 \\ 2x + \alpha, & -1 \le x < \infty \end{cases}$ and

$g(x) = \begin{cases} x + 4, & 0 \le x \le 8 \\ -3x - 2, & -\infty < x < 0 \end{cases}$.

15. The function $g(f(x))$ is not defined if

(A) $\alpha \in (10, \infty), \beta \in (5, \infty)$

(B) $\alpha \in (4, 10), \beta \in (5, \infty)$

(C) $\alpha \in (10, \infty), \beta \in (0, 1)$

(D) $\alpha \in (4, 10), \beta \in (1, 5)$

16. If $\alpha = 2$ and $\beta = 3$, then range of $g(f(x))$ is equal to

(A) (– 2, 12] (B) (0, 12]

(C) [4, 12] (D) [– 1, 12]

17. If $\alpha = 3$ then the value of x in interval [1, 3] for which $f(x) + g(x) = 12$, is

(A) 1 (B) $\dfrac{3}{2}$

(C) $\dfrac{5}{3}$ (D) 3

18. Which of the following statement(s) is(are) correct?

(A) If f is a one-one mapping from set A to A, then f is onto.

(B) If f is an onto mapping from set A to A, then f is one-one.

(C) Let f and g be two functions defined from $R \to R$ such that gof is injective, then f must be injective.

(D) If set A contains 3 elements while set B contains 2 elements, then total number of functions from A to B is 8.

19. The sum of all positive integral values of 'a', $a \in [1, 500]$ for which the equation $[x]^3 + x - a = 0$ has solution is (where [ ] denote the greatest integer function)

(A) 462 (B) 512

(C) 784 (D) 812

20. Let $f: R \to R$ be defined as
$f(x) = \sin \pi [x] + e^{-|x|} - e^{x}$, then $f(x)$ is
[**Note:** [x] denotes the largest integer less than or equal to x.]
(A) injective but not surjective.
(B) surjective but not injective.
(C) injective as well as surjective.
(D) neither injective nor surjective

21. Set A consists of 3 distinct elements and set B consists of 6 distinct elements. Number of many one functions which can be defined from $A \to B$, is
(A) 120 (B) 96
(C) 180 (D) 60

22. Which of the following are identical functions?
(A) $f(x) = \text{sgn}\left(|x| + 1\right)$
(B) $g(x) = \sin^2(\ln x) + \cos^2(\ln x)$
(C) $h(x) = \dfrac{2}{\pi}\left(\sin^{-1}\{x\} + \cos^{-1}\{x\}\right)$
(D) $k(x) = \sec^2[\{x\}] - \tan^2\{[x]\}$
(where [x] denotes greatest integer less than or equal x, {x} denotes fractional part of x and sgn x denotes signum function of x respectively.)

23. The polynomial R(x) is the remainder upon dividing x^{2007} by $x^2 - 5x + 6$. If R(0) can be expressed as $ab(a^c - b^c)$, find the value of $(a + b + c)$.

24. Find the domain and range of function.
$f(x) = 4 \tan x \cos x$;

25. Suppose the domain of the function $y = f(x)$ is $-1 \le x \le 4$ and the range is $1 \le y \le 10$. Let $g(x) = 4 - 3f(x-2)$. If the domain of g (x) is $a \le x \le b$ and the range of g (x) is $c \le y \le d$ then which of the following relations hold good?
(A) $2a + 4b + c + d = 0$
(B) $a + b + d = 8$
(C) $5b + c + d = 4$
(D) $a + b + c + d + 18 = 0$

26. Consider the function $f(x) = \sqrt{\dfrac{\log_3 x - 1}{\log_3 \dfrac{x}{9}}}$.
Find the sum of all the integral values which lie in [0, 100] in the range of the function.

27. Find the domains of definitions of the following functions:
(Read the symbols [] and {} as greatest integers and fractional part functions respectively.)

(i) $f(x) = \sqrt{\dfrac{1 - 5^x}{7^{-x} - 7}}$

(ii) $y = \log_{10} \sin(x-3) + \sqrt{16 - x^2}$

(iii) $f(x) = \dfrac{1}{\sqrt{4x^2 - 1}} + \ln x(x^2 - 1)$

(iv) $f(x) = \sqrt{\log_{\frac{1}{2}} \dfrac{x}{x^2 - 1}}$

28. Find the domains of definitions of the following functions:
(Read the symbols [] and {} as greatest integers and fractional part functions respectively.)

(i) $f(x) = \dfrac{\sqrt{\cos x - (1/2)}}{\sqrt{6 + 35x - 6x^2}}$

(ii) $f(x) = \dfrac{[x]}{2x - [x]}$

(iii) $f(x) = \log_x \sin x$

(iv) $f(x) = \log_2\left(-\log_{1/2}\left(1 + \dfrac{1}{\sin\left(\dfrac{x^\circ}{100}\right)}\right)\right)$

$\sqrt{\log_{10}(\log_{10} x) - \log_{10}(4 - \log_{10} x) - \log_{10} 3}$

(v) $f(x) = \sqrt{(5x-6-x^2)\,[\{ln\{x\}\}]} + \sqrt{(7x-5-2x^2)} + \left(ln\left(\frac{7}{2}-x\right)\right)^{-1}$

29. Find the number of integers in the domain of the function $f(x)=\sqrt{x^2-|x|}+\frac{1}{\sqrt{9-x^2}}$.

30. Prove that the function defined as, $f(x) = \begin{cases} e^{-\sqrt{|\ln\{x\}|}} - \{x\}^{\sqrt{\frac{1}{|\ln\{x\}|}}} & \text{where ever it exists} \\ \{x\} & \text{otherwise} \end{cases}$, then $f(x)$ is odd as well as even. (where $\{x\}$ denotes the fractional part function)

31. Find the range of k for which the equation $k\,ln\,x - x = 0$ posses two distinct solution
(A) (e, ∞)
(B) $(0, e)$
(C) $[e, \infty)$
(D) cannot be determined

32. The sum of all positive integral values of 'a', $a \in [1, 500]$ for which the equation $[x]^3 + x - a = 0$ has solution is (where [ ] denote the greatest integer function)
(A) 462 (B) 512
(C) 784 (D) 812

33. If $f(x)=\begin{cases}2+x, & x\ge 0\\ 2-x, & x<0\end{cases}$ then $f(f(x))$ is given by

(A) $f(f(x)) = \begin{cases}4+x, & x\ge 0\\ 4-x, & x<0\end{cases}$

(B) $f(f(x)) = \begin{cases}2+x, & x\ge 0\\ 2-x, & x<0\end{cases}$

(C) $f(f(x)) = \begin{cases}4+x, & x< 0\\ x, & x\ge 0\end{cases}$

(D) $f(f(x)) = \begin{cases}4-x, & x\ge 0\\ x, & x<0\end{cases}$

34. Column I gives functions and column 2 the nature of the functions.

Column-I

(A) $f:[0,\infty)\to[0,\infty),\ f(x)=\frac{x}{1+x}$

(B) $f:R-\{0\}\to R,\ f(x)=x-\frac{1}{x}$

(C) $f:R-\{0\}\to R,\ f(x)=x+\frac{1}{x}$

(D) $f:R\to R,\ f(x)=2x+\sin x$

Column-II

(P) one-one onto
(Q) one-one but not onto
(R) onto but not one-one
(S) neither one-one nor onto

35. **Column-I**

(A) $f:R\to R,\ f(x)=\left[x+\frac{1}{2}\right]+\left[x-\frac{1}{2}\right]+2[-x]$
where [ ] denotes greatest integer function

(B) $f:R\to R,\ f(x)=x^3+x^2+3x+\sin x$

(C) $f:R\to R,\ f(x)=\frac{e^{|x|}-e^{-x}}{e^x+e^{-x}}$

(D) $f:R\to R, f(x)=e^{\sin\{x\}}+\sin\left(\frac{\pi}{2}[x]\right)$
where { } and [ ] denotes fractional part function and greatest integer respectively

Column-II

(P) periodic
(Q) one one
(R) many one
(S) onto
(T) into

\

36. Let $f(x) = (3x+2)^2 - 1, \quad -\infty < x \le \frac{-2}{3}$.
If $g(x)$ is the function whose graph is the reflection of the graph of $f(x)$ with respect to line $y = x$, then $g(x)$ equals
(A) $\frac{1}{3}\left(-2-\sqrt{x+1}\right)$, $x \ge -1$
(B) $\frac{1}{3}\left(-2+\sqrt{x+1}\right)$, $x \ge -1$
(C) $\frac{1}{3}\left(-1-\sqrt{x+2}\right)$, $x \ge -2$
(D) $\frac{1}{3}\left(-1+\sqrt{x+2}\right)$, $x \ge -2$

37. Let $f_1(x) = \begin{cases} x & \text{for } 0 \le x \le 1 \\ 1 & \text{for } x > 1 \\ 0 & \text{otherwise} \end{cases}$
and $f_2(x) = f_1(-x)$ for all x
$f_3(x) = -f_2(x)$ for all x
$f_4(x) = f_3(-x)$ for all x
Which of the following is necessarily true?
(A) $f_4(x) = f_1(x)$ for all x
(B) $f_1(x) = -f_3(-x)$ for all x
(C) $f_2(-x) = f_4(x)$ for all x
(D) $f_1(x) + f_3(x) = 0$ for all x

38. Number of integer in the range of the function,
$f(x) = \sqrt{\sin\frac{\pi x}{2}} + \sqrt{16-x^2} + \sqrt{x} + \log_2\left(x(x-2)\right)$.

39. Let R be the region in the first quadrant bounded by the x and y axis and the graphs of $f(x) = \frac{9}{25}x + b$ and $y = f^{-1}(x)$. If the area of R is 49, then the value of b, is
(A) $\frac{18}{5}$ (B) $\frac{22}{5}$ (C) $\frac{28}{5}$
(D) none

40. If f(x) is defined on (0, 1), then the domain of definition of $f(e^x) + f(\ln|x|)$ is
(A) $(-e, -1)$
(B) $(-e, -1) \cup (1, e)$
(C) $(-\infty, -1) \cup (1, \infty)$
(D) $(-e, e)$

41. If $f: R \to R$ and $g: R \to R$ are defined by $f(x) = 2x + 3$ and $g(x) = x^2 + 7$, then the sum of values of x for which $g(f(x)) = 8$, is
(A) 0 (B) – 1
(C) – 2 (D) – 3

42. Let $f: D \to R$ be defined as
$f(x) = \frac{x^2+2x+a}{x^2+4x+3a}$ where D and R denote the domain of f and the set of all real numbers respectively. If f is surjective mapping then the range of a is
(A) $0 \le a \le 1$ (B) $0 < a \le 1$
(C) $0 \le a < 1$ (D) $0 < a < 1$

43. Let $f: R \to [1, \infty)$ be a quadratic surjective function such that $f(2+x) = f(2-x)$ and $f(1) = 2$. If $g: (-\infty, \ln 2] \to [1, 5)$ is given by $g(\ln x) = f(x)$ then which of the following is(are) **correct**?
(A) The value of $f(3)$ is equal to 2.
(B) $g^{-1}(x) = \ln\left(2-\sqrt{x-1}\right)$
(C) $g^{-1}(x) = \ln\left(2+\sqrt{x-1}\right)$
(D) The sum of values of x satisfying the equation $f(x) = 5$ is 4.

44. If $f(x) = [x]$ and $g(x) = |x|$ then which of the following is(are) **correct**?
(A) $fog\left(\frac{5}{2}\right) - gof\left(\frac{-5}{2}\right) = -1$
(B) $(f+2g)(-1) = 0$
(C) If $\text{sgn}\left(fog(x)\right) = 0$ then $x \in (-1, 1)$.
(D) If $\text{sgn}\left(gof(x)\right) = 0$ then $x \in [0, 1)$.
[**Note :** [k] and sgn (k) denote greatest integer less than or equal to k and signum function of k respectively.]

45. Let $A = \{1, 2, 3, 4\}$. Number of function $f : A \to A$ are there satisfying $fof(x) = x$ $\forall\, x \in A$, is
(A) 10 (B) 11
(C) 12 (D) 13

46. Let f(x) =

$$\begin{cases} x^2 - 3, & x \le -5 \\ x + \lambda & -5 < x < -1 \\ (\mu - 7)\left(|1 - x| + |1 + x|\right) & -1 \le x \le 1 \\ x + 6 & 1 < x < 5 \\ 3 - x^2 & x \ge 5 \end{cases}$$

If f(x) is an odd function then the value of $(\lambda + \mu)$, is
(A) 1 (B) 5
(C) 10 (D) 13

47. Let $f(x) = \begin{cases} x^2 - 4, & \text{if } |x| \le 3 \\ 5\,\text{sgn}|x - 3|, & \text{if } |x| > 3 \end{cases}$

and $g(x) = 2\tan^{-1}(e^x) - \frac{\pi}{2}$ for all $x \in R$,

then which of the following is(are) **correct**?
[**Note:** sgn(k) denotes the signum function of k.]
(A) fog(x) is an even function.
(B) gof(x) is an even function.
(C) gog(x) is an odd function .
(D) fof(x) is an odd function.

48. Find the Number of functions that can be defined from the set $A = \{1, 2, 3\}$ to the set $B = \{1, 2, 3, 4, 5\}$, such that $f(i) \le f(j)$ for $i < j$.

49. Let $f(x) = \log_2(\log_3(\log_4(\log_5(\sin x + b^2))))$ find real values of b^2 so that f(x) is defined $\forall\, x \in R$.

50. Find the number of real values of x satisfying the equation $x^2 + [x] + 3 = 4x$ where [x] denotes largest integer less than or equal to x.

51. If the range of the function

$f(x) = \log_2\left(4^{x^2} + 4^{(x-1)^2}\right)$ is $\left[\frac{p}{q}, b\right)$

(where $p, q \in N$), find the least value of $(p + q)$.

52. Suppose that f is a function such that $f(\cos x) = \cos 17x$. Which one of the following functions g has the property that $g(\sin x) = \sin 17x$.

(A) $g(x) = f\left(\sqrt{1 - x^2}\right)$

(B) $g(x) = f\left(x - \frac{\pi}{4}\right)$

(C) $g(x) = \sqrt{1 - f(x)^2}$

(D) $g(x) = f(x)$

53. Let $f : R \to R$ defined by $f(x) = \text{Min.}(|x|, 1 - |x|)$
Then which of the following hold(s) good?
(A) Range of f is $(-\infty, 1]$
(B) f is aperiodic.
(C) f is neither even nor odd.
(D) f is neither injective nor surjective.

54. Let $f(x) = \left|\, |x^2 - 9| - |x - a| \,\right|$. Find the number of integers in the range of a so that $f(x) = 0$ has 4 distinct real root.

55. **Column I**

(A) Let $f : R^+ \to \{-1, 0, 1\}$ defined by $f(x) = \text{sgn}(x - x^4 + x^7 - x^8 - 1)$ where sgn denotes signum function, then f(x) is
(B) Let $f : R \to R$ and satisfies $f(x) + x\,f(-x) = x + 1$, then f(x) is
(C) Let $f : [0, 4] \to [0, 9]$ defined by $f(x) = \sqrt{6x - x^2}$, then f(x) is
(D) Let $f : [0, 3] \to [2, 8]$ defined by $f(x) = 2^{|x-1| + |x-2|}$, then f(x) is

Column II
(P) Into
(Q) One-one
(R) Many-one
(S) Even
(T) Onto

56. **Column-I**
(A) Number of integers in the range of the function $f(x) = \sqrt{{}^{x^2+4x}C_{2x^2+3}}$ is
(B) Number of integers in the range of the function
$$f(x) = \frac{\sin x}{\sqrt{1+\tan^2 x}} - \frac{\cos x}{\sqrt{1+\cot^2 x}} \text{ is}$$
(C) Fundamental period of the function
$f(x) = \sin(\sin(\pi x)) + e^{x-[x]}$
where [x] denotes the greatest integer less than or equal to x is

Column-II
(P) 1
(Q) 2
(R) 3
(S) 4

57. Let f(x) and g(x) are two functions such that $f(x) = \frac{2}{\pi}\tan^{-1}x$ and $g(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ then which of the following statement(s) is/are correct for the composite functions
$fog : R \to \left[-\frac{1}{2}, \frac{1}{2}\right]$ and
$gof : R \to \left[-\frac{1}{2}, \frac{1}{2}\right]$?
(A) fog is a bijective function
(B) fog is surjective
(C) gof is bijective
(D) gof is into function

58. The function $f : [0, 2] \to [1, 4]$, defined by $f(x) = x^3 - 5x^2 + 7x + 1$, is
(A) one-one and onto
(B) onto but not one-one
(C) one-one but not onto
(D) neither one-one nor onto

59. The function
$f : R \to \{x \in R : -1 < x < 1\}$
defined by $f(x) = \frac{x}{1+|x|}$ is
(A) one-one and into
(B) one-one and onto
(C) many-one and into
(D) many-one and onto

60. The function f(x) =
$$\begin{cases} 5x-4; & \text{for } 0 \le x \le 1 \\ 4x^2-3x; & \text{for } 1 < x \le 2 \\ -3x+16; & \text{for } 2 < x \le 3 \end{cases}$$. Then the correct statements.
(A) The no. of integers in the range to f(x) is 15.
(B) The no. of integers in the range of f(x) is 14.
(C) f(x) is one-one in the given interval [0, 3]
(D) f(x) is many-one in the given interval [0, 3]

61. Let $f : R \to [0, \infty)$ be defined as $f(x) = \log\left(\sqrt{9x^2 - 12x + \lambda} + 1\right)$ be an onto function where λ is a real parameter belong to (0, 10). Find the greatest possible value of λ.

62. If the function $g : [-1, 0] \to [1, 2]$ is defined as $g(x) = 2 - 2x^2 + x^4$, then $g^{-1}(x)$ is equal to
(A) $\sqrt{1+\sqrt{x-1}}$

(B) $-\sqrt{1+\sqrt{x-1}}$

(C) $-\sqrt{1-\sqrt{x-1}}$

(D) $\sqrt{1-\sqrt{x-1}}$

63. A function $f(x) = \sqrt{1-2x} + x$ is defined from $D_1 \to D_2$ and is onto. If the set D_1 is its complete domain then the set D_2 is

(A) $\left(-\infty, \frac{1}{2}\right]$ (B) $(-\infty, 2)$

(C) $(-\infty, 1)$ (D) $(-\infty, 1]$

64. Find the value of expression $\left[\frac{18}{35}\right]+\left[\frac{18(2)}{35}\right]+\left[\frac{18(3)}{35}\right]+ \ldots\ldots + \left[\frac{18(33)}{35}\right]+\left[\frac{18(34)}{35}\right]$

[Note: [y] denotes the greatest integer function less than or equal to y.]

65. $f: R \to R$, be a polynomial function such that $f(x^2+x+3)+2f(x^2-3x+5) = 6x^2-10x+17 \ \forall \ x \in R$ then find $f(x)$.

66. Column I gives functions and column 2 the nature of the functions.

Column-I

(A) $f:[0,\infty) \to [0,\infty),\ f(x) = \frac{x}{1+x}$

(B) $f: R-\{0\} \to R,\ f(x) = x - \frac{1}{x}$

(C) $f: R-\{0\} \to R,\ f(x) = x + \frac{1}{x}$

(D) $f: R \to R,\ f(x) = 2x + \sin x$

Column-II

(P) one-one onto

(Q) one-one but not onto

(R) onto but not one-one

(S) neither one-one nor onto

67. If $f(x) = \frac{4x(x^2+1)}{x^2+(x^2+1)^2}$, $x \geq 0$ then range of $f(x)$ is

(A) $\left[\frac{8}{5}, \infty\right) \cup \{0\}$ (B) $\left[0, \frac{8}{5}\right]$

(C) $\left(0, \frac{8}{5}\right]$ (D) $\left[2, \frac{8}{5}\right]$

68. In which of the following intervals, the minimum value of $f(x) = |\sin x| + |\cos x + \sec x| + \frac{x^2+2x+2}{|x+1|}$ is equal to 4?

(A) $\left(-\frac{\pi}{2}, 0\right] \sim \{-1\}$ (B) $\left[0, \frac{\pi}{2}\right)$

(C) $\left(\frac{3\pi}{2}, 2\pi\right]$ (D) $\left[2\pi, \frac{5\pi}{2}\right)$

69. **Column-I**

(A) $f: R \to R,\ f(x) = \left[x+\frac{1}{2}\right]+\left[x-\frac{1}{2}\right]+2[-x]$

where [ ] denotes greatest integer function

(B) $f: R \to R,\ f(x) = x^3+x^2+3x+\sin x$

(C) $f: R \to R,\ f(x) = \frac{e^{|x|}-e^{-x}}{e^x+e^{-x}}$

(D) $f: R \to R,\ f(x) = e^{\sin\{x\}} + \sin\left(\frac{\pi}{2}[x]\right)$

where { } and [ ] denotes fractional part function and greatest integer respectively

Column-II

(P) one one

(Q) many one

(R) onto

(S) into

70. If $f(x) = \{x\} + \left\{x + \left[\frac{x}{1+x^2}\right]\right\} + \left\{x + \left[\frac{x}{1+2x^2}\right]\right\} + \ldots\ldots + \left\{x + \left[\frac{x}{1+99x^2}\right]\right\}$ then value of $\left[f\left(\sqrt{3}\right)\right]$ is

(A) 5050 (B) 4950
(C) 17 (D) 73

Note :- [k] and {k} denote greatest integer and fractional part functions of k respectively.

71. Let f be a bijective function and $a \neq 0$, then the function $g(x) = a\, f\left(\frac{x+a}{a}\right)$ has an inverse function which is

(A) $\frac{1}{a} f^{-1}(x-1)$ (B) $a\left(f^{-1}\left(\frac{x}{a}\right) - 1\right)$

(C) $a f^{-1}\left(\frac{x}{a}\right) - 1$ (D) $\frac{1}{a} f^{-1}(ax-1)$

72. Number of elements in the domain of function $f(x) = 3^{\sin^{-1} x^2} + \frac{(7x+1)!}{\sqrt{x+1}}$, is

(A) 5 (B) 6
(C) 8 (D) 9

Paragraph for question nos. 73 & 74

Consider the rational function $f(x) = \frac{x^2 - 5}{3 - x}$, $x \in R - \{3\}$.

73. The range of function f(x) is

(A) $(-\infty, -10] \cup [-2, \infty)$
(B) $(-\infty, -1] \cup [2, \infty)$
(C) $(-\infty, -8] \cup [-1, \infty)$
(D) $(-\infty, -5] \cup [4, \infty)$

74. If $f : (-\infty, 1] \to [-2, \infty)$ is a bijective mapping then $f^{-1}(x)$ is

(A) $\frac{x + \sqrt{x^2 + 12x + 20}}{2}$

(B) $\frac{-x + \sqrt{x^2 + 12x + 20}}{2}$

(C) $\frac{x - \sqrt{x^2 + 12x + 20}}{2}$

(D) $\frac{-x - \sqrt{x^2 + 12x + 20}}{2}$

75. Let $f(x) = \begin{cases} |2x-1| & -2 \leq x \leq 1 \\ x^2 - 4 & 1 < x \leq 8 \end{cases}$, then possible integers in the domain of $y = f(f(x))$ is/are

(A) – 2 (B) 0
(C) 1 (D) 5

76. Which of the following is(are) incorrect?

(A) If $f(x) = \sin x$ and $g(x) = ln\, x$ then range of $g(f(x))$ is $[-1, 1]$.

(B) If $x^2 + ax + 9 > x \ \forall x \in R$ then $-5 < a < 7$.

(C) If $f(x) = \left(2011 - x^{2012}\right)^{\frac{1}{2012}}$ then $f(f(2)) = \frac{1}{2}$.

(D) The function $f : R \to R$ defined as $f(x) = \frac{x^2 + 4x + 30}{x^2 - 8x + 18}$ is not surjective.

77. Let $f : I \to R$, defined as $f(x) = 5 \cos 4\pi x - 13 \sin 7\pi x + 2$. Then which of the following alternative(s) is/are **TRUE**?

(A) Range of f is a singleton set.
(B) f is an even function.
(C) $f(f(x)) = f(x) \ \forall x \in I$
(D) Inverse function of f is non existent.

78. Given $f(x) = (x+1)(x+2)(x+3)(x+4)$ defined as $f : A \to B$ where A and B are domain and codomain. Which of the following

are correct
(A) $A = (-\infty, -1]$ then the function is one-one
(B) $A = (-\infty, -3.7]$ then the function is one-one
(C) $B = [-1, \infty)$ then the function is onto.
(D) None are correct

Paragraph for Question no. 79 to 81

A line PQ parallel to the diagonal BD of a square ABCD with side length 'a' unit is drawn at a distance x from the vertex A, where $x \in \left[0, \sqrt{2}\,a\right]$ cuts the adjacent sides at P and Q. Let f(x) be the area of the segment of a square cut off by PQ, with A as one of the vertex.

79. Let $g(x) = f^{-1}(x)$, then the domain of g(x) is
(A) $x \in \left[0, \sqrt{2}\,a\right]$ (B) $x \in [0, 2a^2]$
(C) $x \in \left[\sqrt{2}\,a, a^2\right]$ (D) $x \in [0, a^2]$

80. For $a = 2$, the domain of the function
$\phi(x) = \sqrt{f^{-1}(x) - f(x)}$ is/are
(A) $x \in [0, 1]$ (B) $x \in [0, 2]$
(C) $x \in [1, 2]$ (D) $x \in [2, \infty)$

81. If the equation $f(x) = f^{-1}(x)$ has exactly three solutions $\forall\, x \in \left[0, \sqrt{2}\,a\right]$, then the value of a is
(A) 1 (B) $\sqrt{2}$
(C) 2 (D) $2\sqrt{2}$

Paragraph for questions nos. 82 to 84

Consider, f(x) =

$$f(x) = \begin{cases} \sqrt{1-x^2} & -1 \le x < 0 \\ x^2 + 1 & 0 \le x < 1 \\ \dfrac{(x-1)^2}{4} + 2 & x \ge 1 \end{cases}$$

One more function g is defined such that $g(f(x)) = x \;\forall\; x \ge -1$ and $f(g(x)) = x \;\forall\; x \ge 0$.

82. The range of the function $y = f(f(f(g(x))))$ is
(A) $[-1, \infty)$ (B) $[0, \infty)$
(C) $[1, \infty)$ (D) $[2, \infty)$

83. The domain of $y = g(g(g(f(x))))$ is
(A) $[-1, \infty)$ (B) $[0, \infty)$
(C) $[1, \infty)$ (D) $[2, \infty)$

84. The number of solution(s) of the equation $f(x) = g(x)$ is(are) –
(A) 0 (B) 1
(C) 2 (D) 3

85. Find the domain and range of

$$f(x) = \sin \log_e \left(\frac{\sqrt{4-x^2}}{1-x} \right)$$

86. Which of the following pair(s) of function have same graphs?
(A) $f(x) = \dfrac{\sec x}{\cos x} - \dfrac{\tan x}{\cot x}$,
$g(x) = \dfrac{\cos x}{\sec x} + \dfrac{\sin x}{\operatorname{cosec} x}$
(B) $f(x) = \operatorname{sgn}(x^2 - 4x + 5)$,
$g(x) = \operatorname{sgn}\left(\cos^2 x + \sin^2\left(x + \dfrac{\pi}{3}\right)\right)$ where sgn denotes signum function.

(C) $f(x) = e^{\ln(x^2+3x+3)}$, $g(x) = x^2 + 3x + 3$

(D) $f(x) = \dfrac{\sin x}{\sec x} + \dfrac{\cos x}{\operatorname{cosec} x}$,

$g(x) = \dfrac{2\cos^2 x}{\cot x}$

87. Let $f : R \to [1, \infty)$ be a function defined by $f(x) = x^2 - 10ax + 5 - a + 25a^2$. If f(x) is surjective on R, then the value of a is

(A) 0 (B) 1
(C) 2 (D) 4

88. Let [x] = the greatest integer less than or equal to x. If all the values of x such that the product $\left[x - \frac{1}{2}\right]\left[x + \frac{1}{2}\right]$ is prime, belongs to the set $[x_1, x_2) \cup [x_3, x_4)$, find the value of $x_1^2 + x_2^2 + x_3^2 + x_4^2$.

89. Possible integral values of x which can lie in the domain of the function $f(x) = \log\left(ax^3 + (a+b)x^2 + (b+c)x + c\right)$ if $b^2 - 4ac < 0$ and $a > 0$, is

(A) – 1 (B) 0
(C) 1 (D) 2

90. The function $f: R \to R$ defined as

$f(x) = \ln\left(\sqrt{\sqrt{x^2+1} + x} + \sqrt{\sqrt{x^2+1} - x}\right)$ is

(A) One-one and onto both
(B) One-one but not onto
(C) Onto but not one-one
(D) Neither one-one nor onto

91. If $a_1, a_2, a_3, \ldots a_n$ be the integers which are not included in the range of

$f(x) = \dfrac{40}{(x-5)(x+1)}$

where $a_i > a_{i+1}\ \forall$ i and value of

$\sum_{i=1}^{n} {}^{5-a_i}C_{i-1} = {}^{x}C_{y}$ then x + y can be

(A) 14 (B) 21
(C) 16 (D) 15

92. Let $f(x) = ax^2 + bx + c$ $(a < b)$ and $f(x) \geq 0$ $\forall\ x \in R$. Find the minimum value of $\dfrac{a+b+c}{b-a}$.

93. Let function f(x) be defined as $f(x) = x^2 + bx + c$, where b, c are real numbers and $f(1) - 2f(5) + f(9) = 32$.
Number of ordered pairs (b, c) such that $|f(x)| \leq 8$ for all x in the interval [1, 9].

94. The maximum value of the function

$f(x) = \dfrac{x^4 - x^2}{x^6 + 2x^3 - 1}$ where $x > 1$ is equal to

(A) $\frac{1}{3}$ (B) $\frac{1}{6}$
(C) $\frac{1}{18}$ (D) $\frac{1}{2}$

95. Find the largest integral value of 'a' for which every solution of the equation $x([x] - 5) + 2\{x\} + 6 = 0$ satisfies the inequality $(a-3)x^2 + 2(a+3)x - 8a \leq 0$. where [y] and {y} denote greatest integer and fractional part functions respectively.

96. Let $f(x) = \left||x^2 - 4x + 3| - 2\right|$. Which of the following is/are correct?
(A) f(x) = m has exactly two real solutions of different sign $\forall\ m > 2$.
(B) f(x) = m has exactly two real solutions $\forall\ m \in (2, \infty) \cup \{0\}$.
(C) f(x) = m has no solutions $\forall\ m < 0$.
(D) f(x) = m has four distinct real solutions $\forall\ m \in (0, 1)$.

97. If graph of curve
$f(x) = \{\sin B - \sin(B+C)\}x^2 + \{\sin A - \sin(A+B)\}x + \{\sin C - \sin(C+A)\}$
touches x-axis and triangle ABC is a scalene triangle then a, b, c are in
(A) A.P. (B) G.P.
(C) H.P. (D) none of these

98. If $f(x) = 3[2x] - 2[3x]$, where [x] represent the greatest integer function less than or equal to x, then which one of the following alternative is true?
(A) f is aperiodic.
(B) f is periodic with period $\frac{1}{6}$.
(C) f is periodic with period 1.
(D) f is periodic with no fundamental period.

99. Let $g(x) = ax + b$, where $a < 0$ and g is defined from [1, 3], onto [0, 2] then the value of
$$\cot\begin{pmatrix}\cos^{-1}(|\sin x| + |\cos x|) \\ + \sin^{-1}(-|\cos x| - |\sin x|)\end{pmatrix}$$
is equal to
(A) g (1) (B) g (2)
(C) g (3) (D) g (1) + g (3)

100. Let $f(x) + 2f(1-x) = x^2 + 1,\ \forall x \in R$. Then the number of integer in the range of 'x' for which $\sqrt{f(x)}$ is not a real number, is
(A) one (B) two
(C) three (D) four

101. If $e^x + e^{g(x)} = e$, then
(A) Domain of g is $(-\infty, 1)$
(B) Range of g is $(-\infty, 1)$
(C) Domain of g is $(-\infty, 0]$
(D) Range of g is $(-\infty, 1]$

102. Let $f(x) = |x - 2|$ and
$g(x) = \underbrace{f(f(f(f.....(f(x)).....)}_{n \text{ times}}$. If the equation $g(x) = k,\ k \in (0, 2)$ has 8 distinct solutions then the value n is equal to
(A) 3 (B) 4 (C) 5 (D) 8

103. Let a function $f: R \to R$ such that $f(1) = 2$ and $f(x+y) = 2^x f(y) + 4^y f(x)\ \forall x, y \in R$. If $f'(2) = k\, ln\, 2$ then find the value of k.

104. Let $f(x) = 4x(1-x),\ 0 \le x \le 1$. The number of solution of $f(f(f(x))) = \frac{x}{3}$ is
(A) 2 (B) 4
(C) 8 (D) 16

105. Find the number of function $f: \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4, 5\}$ which assumes exactly 3 distinct values

106. Let $f(x) = a \sin x + c$, where $a, c \in R$ and $a < 0$. Then $f(x) < 0\ \forall x \in R$ if
(A) $c < -a$ (B) $c > -a$
(C) $-a < c < a$ (D) $c < a$

107. The period of the function
$f(x) = \frac{|\sin x| + |\cos x|}{|\sin x - \cos x|}$ is
(A) $\pi/2$ (B) $\pi/4$
(C) π (D) 2π

108. $f(x) = \frac{x}{lnx}$ and $g(x) = \frac{lnx}{x}$. Then identify the CORRECT statement
(A) $\frac{1}{g(x)}$ and $f(x)$ are identical functions
(B) $\frac{1}{f(x)}$ and g(x) are identical functions
(C) $f(x) \cdot g(x) = 1 \quad \forall x > 0$
(D) $\frac{1}{f(x) \cdot g(x)} = 1 \quad \forall x > 0$

109. Let $f(x) = \sin^2 x + \cos^4 x + 2$ and $g(x) = \cos(\cos x) + \cos(\sin x)$. Also let period of f (x) and g (x) be T_1 and T_2 respectively then
(A) $T_1 = 2T_2$ (B) $2T_1 = T_2$
(C) $T_1 = T_2$ (D) $T_1 = 4T_2$

110. Which of the following function is surjective but not injective
(A) $f: R \to R \quad f(x) = x^4 + 2x^3 - x^2 + 1$
(B) $f: R \to R \quad f(x) = x^3 + x + 1$
(C) $f: R \to R^+ \quad f(x) = \sqrt{1+x^2}$
(D) $f: R \to R \quad f(x) = x^3 + 2x^2 - x + 1$

111. Let $f: A \to B$ and $g: B \to C$ be two functions and $gof: A \to C$ is defined. Then which of the following statement(s) is true?
(A) If gof is onto then f must be onto.
(B) If f is into and g is onto then gof must be onto function.
(C) If gof is one-one then g is not necessarily one-one.
(D) If f is injective and g is surjective then gof must be bijective mapping.

112. Let $f: R \to R$ be defined by $f(x) = (a^2-1)(a^2-4)x^3 + (a^2-1)(a+2)x^2 + (a+1)(a+2)x + a + 5$.
If f(x) is into then number of possible values of 'a' are
(A) 1 (B) 2
(C) 3 (D) more than 3.

113. $f: (0, \infty) \to (0, \infty)$ be a real valued function given by
$\alpha x^2 f\left(\frac{1}{x}\right) + f(x) = \frac{x}{x+1} \quad \forall\ x > 0$. The set of values of α will be
(A) $(-1, 1)$ (B) $(-1, 0]$
(C) $(-\infty, -2) \cup (1, \infty)$ (D) $(1, \infty)$

114. Let $f: R \to R$ and $g: R \to R$ be functions such that $f[g(x)]$ is one-one function
Column-I
(A) Then $g(x)$
(B) then $f(x)$
(C) If $g(x)$ is onto then $f(x)$ is
(D) If $g(x)$ is into then $f(x)$ is
Column-II
(P) may be one-one
(Q) must be one-one
(R) may be many-one
(S) must be many-one

115. Let $f(x) = (\sec x + \tan x)^{\left(\{x\} + \frac{\sin 2\pi x}{x^2+1}\right)}$ and $g(x) = (\sec x - \tan x)^{\left(\{x\} + \frac{\sin 2\pi x}{x^2+1}\right)}$.
If $h(x) = f(x)\,g(x)$ then fundamental period of $h(x)$ is
(A) 1 (B) $\frac{\pi}{2}$
(C) π (D) Not defined

116. If $f(x) = \frac{4x(x^2+1)}{x^2 + (x^2+1)^2}$, $x \ge 0$ then range of $f(x)$ is
(A) $\left[\frac{8}{5}, \infty\right) \cup \{0\}$ (B) $\left[0, \frac{8}{5}\right]$
(C) $\left(0, \frac{8}{5}\right]$ (D) $\left[2, \frac{8}{5}\right]$

Paragraph for question nos. 117 to 119
Given, $f(x) = (x + 2a)(x + a - 4)$,
$g(x) = k(x^2 + x) + 3k + x$,
$h(x) = (1 - \sin\theta)x^2 + 2(1 - \sin\theta)x - 3\sin\theta$

117. If $f(x) < 0\ \forall\ x \in [-1, 1]$, then 'the range of a is
(A) $\frac{1}{2} < a < 3$ (B) $\frac{-1}{2} < a < \frac{1}{2}$
(C) $-3 < a < \frac{-1}{2}$ (D) $3 < a < 5$

118. If $g(x) + 3 > 0\ \forall\ x \in R$, then the range of k, is
(A) $-1 < k < \frac{1}{11}$ (B) $-1 < k < 0$
(C) $0 < k < \frac{1}{11}$ (D) $k > \frac{1}{11}$

119. If the quadratic equation h(x) = 0 has both roots complex then θ belongs to

(A) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (B) $\left(0, \frac{3\pi}{2}\right)$

(C) $\left(\frac{\pi}{6}, \frac{7\pi}{6}\right)$ (D) $\left(\frac{7\pi}{6}, \frac{11\pi}{6}\right)$

120. Let $f(x) = \sqrt{\sin^4 x + 4\cos^2 x} - \sqrt{\cos^4 x + 4\sin^2 x}$ and $g(\sin 2t) = \sin t + \cos t \ \forall\, t \in \left[\frac{-\pi}{4}, \frac{\pi}{4}\right]$.

If the range of $g(f(x))$. is [a, b] then find the value of $a^2 + b^2$.

121. Find the number of integers in the range of the function $f(x) = \cos x\left(\sin x + \sqrt{\sin^2 x + \frac{1}{2}}\right)$.

122. Find the domains of definition of the given functions.

$f(x) = \sqrt{1 - \sqrt{2 - \sqrt{3 - x}}}$

123. The range of function $f(x) = \sin x + \cos x - [\sin x] - [\cos x] \ \forall\, x \in R$, is

(A) [0, 1) (B) [0, 2)

(C) [1, 2) (D) $[0, \sqrt{2}]$

[**Note:** [k] denotes greatest integer less than or equal to k.]

Paragraph for question nos. 124 to 126

Consider, $f(x) = [x]^2 - [x + 6]$ and $g(x) = 3kx^2 + 2x + 4(1 - 3k)$

where [α] denotes the largest integer less than or equal to α.

Let $A = \{x \mid f(x) = 0\}$ and $k \in [a, b]$ for which every element of set A satisfies the inequality $g(x) \ge 0$.

124. The set A is equal to

(A) {– 2, 3}

(B) (– 3, – 2] ∪ [3, 4)

(C) [– 2, – 1) ∪ [3, 4)

(D) [– 2, 4]

125. The value of $(6b - 3a)$ is equal to

(A) 1 (B) – 1

(C) – 2 (D) 2

126. If $k = a$ and $g : A \to B$ is onto then set B is equal to

(A) [0, 5] (B) (– 5, 5)

(C) (– 5, 0] (D) [–5, 5]

127. If Max. $\left\{\frac{ax + b}{\sqrt{1 + x^2}}; x \in R\right\} = 3$ then the value of $a^2 + b^2$ is

(A) 3 (B) 6

(C) 9 (D) 27

128. Let $f(x) = \begin{cases} x + 3 & \text{if } x \in [-4, -2) \\ 1 & \text{if } x \in [-2, 2) \\ 3 - x & \text{if } x \in [2, 4] \end{cases}$

$g(x) = \begin{cases} x + 6 & x < 0 \\ 2x + 6 & x \ge 0 \end{cases}$ then ;

(A) gof(x) = k will have one atleast solution if $k \in [5, 8]$

(B) Range of fog(x) is [–1, 1]

(C) $\lim_{x \to -2} fog(x) = -1$

(D) gof(x) is an even function.

129. Let $f(x) = \begin{cases} 2x + 3\,; & 0 < x < 1 \\ 2 + 3x\,; & 1 \le x < 2 \end{cases}$;

$g(x) = \begin{cases} x + 1 & 2 < x < 4 \\ 4[x + 1] & x \ge 4 \end{cases}$

then find the number of integers in the range of gof(x).

[**Note:** [ ] denotes greater integer function.]

130. Let $f(x) = \begin{cases} x+3, & -3 < x < 0 \\ 3, & 0 \le x < 1 \\ 4-x, & 1 \le x \le 4 \end{cases}$

and $g(x) = x^2 + 4, x > 3$. Then,
(A) gof(x) is not defined and fog(x) is defined
(B) fog(x) is not defined and gof(x) is defined
(C) Both fog (x) and gof (x) are defined
(D) Both gof(x) and fog(x) are not defined.

131. If number of ordered pairs (p, q) from the set $S = \{1, 2, 3, 4, 5\}$ such that the function $f(x) = \frac{x^3}{3} + \frac{p}{2}x^2 + qx + 10$, defined from R to R is injective, is n then find number of divisors of n.

132. Let $f(x) = mx$ and $g(x) = |x - \pi|$ where $m \in R$. If there exist exactly two real numbers x such that $f(x) = g(x)$ then the range of m is (a, b). Find the value of (a + b).

133. If $\alpha^3 - 3\alpha^2 + 5\alpha - 4 = 0$ and $\beta^3 + 3\beta^2 + 5\beta + 5 = 0$; $\alpha, \beta \in R$, then the value of $[|\alpha + \beta|]$ is
(where $[\cdot]$ denotes greatest integer function)
(A) 0 (B) 1
(C) 2 (D) 3

134. The equation $x^2 - 2 = [\sin x]$, where [.] denotes the greatest integer function, has
(A) infinitely many roots
(B) only one root which is an integer
(C) only one root which is irrational
(D) exactly two roots

135. Let f be a odd periodic function from R to R. If the period of f is 2 and it is given that

$$f\left(3.65 + \log_{\frac{9}{4}}\left(\frac{1}{2\sqrt{3}}\sqrt{6 - \frac{1}{2\sqrt{3}}\sqrt{6 - \frac{1}{2\sqrt{3}}\ldots\ldots\infty}}\right)\right) = 1$$

then find the absolute value of f(0.85).

136. Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6, 7\}$ are two sets. Let m is the number of one-one functions $f: A \to B$ such that $f(i) \ne i$ $\forall i \in A$ and n is the number of one-one functions $f: A \to B$ such that $|f(i) - i| \le 3$ $\forall i \in A$, then find the value of $(m + n)$.

137. Find the sum of all integral values of a where $a \in [-10, 10]$ such that the graph of the function $f(x) = ||x - 2| - a| - 3$ has exactly three x-intercepts.

138. Find the domains of definition of the given functions.
(i) $f(x) = \sqrt{x^{14} - x^{11} + x^6 - x^3 + x^2 + 1}$
(ii) $f(x) = \sqrt{1 - \sqrt{2 - \sqrt{3 - x}}}$
(where [ ] greatest integer function.)

139. Find the Range of the function
$f(x) = \sqrt{-x^2 + 4x - 3} + \sqrt{\sin\left(\frac{\pi}{2}\left(\sin\frac{\pi}{2}(x-1)\right)\right)}$

ANSWER KEY

1. $\{5050, 5051\}$ 2. 16 where $m = 0$; $n = -4$ 3. domain $= \left[-\frac{\pi}{2}, 0\right]$, range f $= \{0\}$

4.	D	5.	C	6.	ACD	7.	D	8.	B	9.	C	10.	B
11.	B	12.	B	13.	C	14.	C	15.	A	16.	C	17.	C
18.	CD	19.	D	20.	D	21.	B	22.	ACD	23.	2011		
24.	– 4, 4	25.	BD	26.	5049								

27. (i) $(-\infty, -1) \cup [0, \infty)$] (ii) $(3 - 2\pi < x < 3 - \pi) \cup (3 < x \le 4)$

(iii) $(-1 < x < -1/2) \cup (x > 1)$ (iv) $\left[\frac{1-\sqrt{5}}{2}, 0\right) \cup \left[\frac{1+\sqrt{5}}{2}, \infty\right)$

28. (i) $\left(-\frac{1}{6}, \frac{\pi}{3}\right] \cup \left[\frac{5\pi}{3}, 6\right)$ (ii) $R - \left\{-\frac{1}{2}, 0\right\}$

(iii) $2K\pi < x < (2K+1)\pi$ but $x \ne 1$ where K is non–negative integer

(iv) $\{x \mid 1000 \le x < 10000\}$

(v) (xviii) $(1, 2) \cup (2, 5/2)$

29. 5 31. A 32. D 33. A

34. (A) Q ; (B) R ; (C) S ; (D) P 35. (A) P, R, T ; (B) Q, S; (C) R, T ; (D) P, R, T

36.	A	37.	B	38.	1	39.	C	40.	A	41.	D	42.	D
43.	ABD	44.	ACD	45.	D	46.	A	47.	ABC	48.	0035		

50. no value 51. 5 52. D 53. BD

54. 17 55. (A) P, R; (B) P, R, S; (C) P, R; (D) R, T 56. (A) P; (B) R; (C) Q

57. D 58. B 59. B 60. AD 61. 0004 62. C 63. D

64. 289

65. $f(x) = 0$ has a root in $(0, 2)$; $f(x)$ is invertible

66. (A) Q ; (B) R ; (C) S ; (D) P 67. B 68. AB

69. (A) Q, S ; (B) P, R; (C) Q, S ; (D) Q, S 70. D 71. B 72. D

73. A 74. D 75. ABC 76. AC 77. ABCD 78. BC

79. D 80. A 81. B 82. D 83. C 84. B

85. $[-2, 1)$, $[-1, 1]$ 86. ABCD 87. D 88. 11 89. BCD 90. D

91.	AC	92.	3	93.	1	94.	B	95.	1	96.	ABCD		
97.	A	98.	C	99.	C	100.	A	101.	A	102.	B	103.	28
104.	C	105.	1500			106.	D	107.	C	108.	A	109.	C

110. D 111. C 112. B 113. B 114. (A) Q ; (B) P, R ; (C) Q ; (D) PR

115. C 116. B 117. A 118. D 119. D 120. 2

121. 3 122. $x \in [-1, 2]$ 123. B 124 C 125. D 126. A

127. C 128. ABCD 129. 4 130. D 131. 4 132. 1

133. A 134. D 135. 1 136. 94 137. 0003

138. (i) $(-\infty, \infty)$ (ii) $[-1, 2]$ 139. 0, 2

1. $f(x) = [\sin x] + \left[\sin \frac{x}{2}\right] + 5050$

$$f(x) \begin{cases} 5050, & x \in \left[0, \frac{\pi}{2}\right) \\ 5051, & x \in \frac{\pi}{2}, \pi \\ 5050, & x \in \left(\frac{\pi}{2}, \pi\right) \end{cases}$$

2. $f(x) = \frac{3(x^2+1)+mx+n-3}{1+x^2}$; $f(x) = 3 + \frac{mx+n-3}{1+x^2}$; $y = 3 + \frac{mx+n-3}{1+x^2}$

for y to lie in $[-4, 3)$

$mx + n - 3 < 0 \quad \forall \ x \in R$

this is possible only if $m = 0$

when, $m = 0$ then $y = 3 + \frac{n-3}{1+x^2}$

note that $n - 3 < 0$ (think !)

$n < 3$

if $x \to \infty$, $y_{max} \to 3^-$

now y_{min} occurs at $x = 0$ (as $1 + \pi^2$ is maximum)

$y_{min} = 3 + n - 3 = n \quad \Rightarrow \quad n = -4$

Alternative : $y = \frac{12n - m^2}{-12}$.

$x^2 (y-3) - mx + y - n = 0$

$x \in R$

$D \geq 0$

$m^2 - 4(y-3)(y-x) \geq 0$

$m^2 - 4(y^2 - ny - 3y + 3x) \geq 0$

$4y^2 - 4y(n+3) + 12n - m^2 \leq 0$ .... (1)

Also given $(y+4)(y-3) \leq 0$

$y^2 + y - 12 \leq 0$ ... (2)

compare (1) & (2) we get $\frac{4}{1} = -\frac{4(n+3)}{1} = \frac{12n-m^2}{-12}$

$\Rightarrow \quad m = 0$ & $n = -4$

3. Given, $f(x) = \log_e\left(\frac{1}{\sqrt{[\cos x] - [\sin x]}}\right)$ ....(1)

Domain of f :

For f (x) to be defined

$[\cos x] - [\sin x] > 0$

$\Rightarrow \quad [\cos x] > [\sin x]$

$\Rightarrow \quad -\frac{\pi}{2} \le x \le 0$

$[\cos x] = -1$ $[\sin x] = 0$ $[\cos x] = -1, [\sin x] = 0$	$\rightarrow [\cos x] = 0, [\sin x] = 1$ $[\cos x] = 0$ $[\sin x] = 0$ $[\cos x] = 1, [\sin x] = 0$
$[\cos x] = -1$ $[\sin x] = -1$	$[\cos x] = 0$ $[\sin x] = -1$ $\rightarrow [\cos x] = 1, [\sin x] = -1$

$\therefore \quad \text{domain} = \left[-\frac{\pi}{2}, 0\right]$ **Ans.**

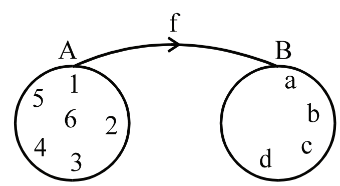
Range of f : In $\left[-\frac{\pi}{2}, 0\right]$, $[\cos x] - [\sin x] = 1$

$\therefore \quad$ from (1), $\quad f(x) = \log_e 1 = 0$

hence range f = $\{0\}$ **Ans.**

Note : Here f is a many one function.

4. Groups 1,1,1,3 or 1,1,2,2



Number of mappings $= \frac{6!4!}{3!3!} + \frac{6!4!}{2!2!2!2!}$

$= 480 + 1080 = 1560$ **Ans.**

5. Given, $f(x) = \pi\left(\frac{\sqrt{x+7} - 4}{x - 9}\right)$

Clearly, domain of $f(x) = [-7, \infty) - \{9\}$.

Now, $f(x) = \frac{\pi\,(x + 7 - 16)}{(x-9)\left(\sqrt{x+7} + 4\right)}$ (Rationalise) $= \frac{\pi}{\sqrt{x+7} + 4}$.

So, range of $f(x)$ is $\left(0, \frac{\pi}{4}\right] - \left\{\frac{\pi}{8}\right\}$.

Hence, range of $y = \sin\left(2f(x)\right)$ is $(0, 1] - \left\{\frac{1}{\sqrt{2}}\right\}$. **Ans.**

6.

(A) Clearly, domain of $f =$ domain of $g = R$.

and range of $f =$ range of $g = \{0\}$. Also $f(x) = g(x)\ \forall\ x \in R$, so f(x) and g(x) are identical functions.

(B) Domain of $f = R - \left\{x \,\middle|\, x = (2n+1)\frac{\pi}{2}, n\pi \text{ where } n \in I\right\}$ and domain of $g = R$.

(C) Clearly, domain of $f =$ domain of $g = R$.
and range of $f =$ range of $g = \{2\}$. Also $f(x) = g(x)\ \forall\ x \in R$, so $f(x)$ and $g(x)$ are identical functions.

(D) Clearly, domain of $f =$ domain of $g = R - \{0\}$.
and range of $f =$ range of $g = \{1\}$. Also $f(x) = g(x)\ \forall\ x \in R$, so $f(x)$ and $g(x)$ are identical functions.]

7. We have $f(x) = \begin{cases} 3, & x \le 0 \\ 3^{-x} - 3^{x} + 3, & x > 0 \end{cases}$ (As $\text{sgn}(e^{-x}) = 1\ \ \forall\ x \in R$)

Clearly $f(x)$ is many one.
For $x > 0$, $f(x)$ is decreasing, hence range of $f(x)$ for $x > 0$ is $(-\infty, 3)$
$\therefore$ Range of the function $f(x)$ is $(-\infty, 3]$, which is subset of R.
Hence f is neither injective nor surjective.

8. We have $y = \dfrac{x^2 + a}{x + 1}$

$\Rightarrow\ x^2 - yx + (a - y) = 0$
As $x \in R$, so discriminant ≥ 0
$\Rightarrow\ y^2 - 4(a - y) \ge 0,\ \forall\ y \in R$
$\Rightarrow\ y^2 + 4y + 4 \ge 4a + 4,\ \forall\ y \in R$
$\Rightarrow\ (y + 2)^2 \ge 4(a + 1),\ \forall\ y \in R$
$\Rightarrow\ 4a + 4 \le 0$
$\Rightarrow\ a \le -1$ (But $a \ne -1$, think ?)
Hence $a \in (-\infty, -1)$. Ans.

9. $$\log_2(f(x)) = \log_2\left(\frac{2}{1 - \frac{1}{3}}\right) \cdot \log\left(1 + \frac{f(x)}{f\left(\frac{1}{x}\right)}\right)$$

$$\Rightarrow \log_2(f(x)) = \log_2 3 \cdot \log\left(1 + \frac{f(x)}{f\left(\frac{1}{x}\right)}\right)$$

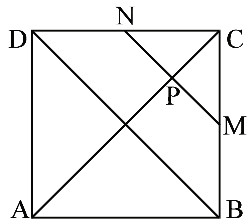
$$f(x) = 1 + \frac{f(x)}{f\left(\frac{1}{x}\right)}$$

$$\Rightarrow f(x) \cdot f\left(\frac{1}{x}\right) = f\left(\frac{1}{x}\right) + f(x) \qquad \Rightarrow f(x) = 1 \pm x^n$$

$f(x) = 1 \pm 10^n = 1001$
$\Rightarrow\ \pm 10^n = 1000\ \Rightarrow$ positive sign must be taken and $n = 3$
$\therefore\ f(x) = 1 + x^3$
$f(x) = 1 + (20)^3 = 8001$ **Ans.**

10. *$AP = x$; $MN = y$; $BD = 2\sqrt{2}$*

hence, $\dfrac{y}{2\sqrt{2}} = \dfrac{2\sqrt{2}-x}{\sqrt{2}}$



$\Rightarrow$ Δ's CNM and CDB are similar $\quad y = 2\left(2\sqrt{2} - x\right)$

$$f(x) = \frac{xy}{2} = x\left(2\sqrt{2} - x\right) \qquad = 2 - \left(x - \sqrt{2}\right)^2$$

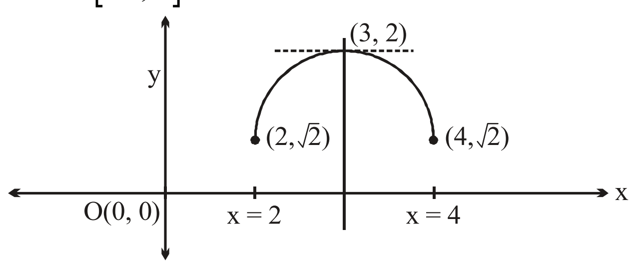
$$\left.\begin{array}{l} f(x)]_{max} = 2 \quad \text{when} \quad x = \sqrt{2} \\ f(x)]_{min} = 0 \quad \text{when} \quad x = 2\sqrt{2} \end{array}\right]$$

$\Rightarrow$ range is $(0, 2]$

11. $k^2 - 3k + 2 = 0$, $k^2 - 1 = 0$, $k^2 - 6k + 5 = 0$, $k^2 - 2k + 1 = 0$ and $k^2 - k = 0$
must be satisfied simultaneously.
So, $k = 1$.
Hence, number of real values of k is one (i.e., $k = 1$). **Ans.**

12. Clearly domain is $x \in [2, 4]$
Now, $f'(x) = 0 \Rightarrow x = 3$ and $f(3) = 2$

$\therefore$ Range $\left[\sqrt{2}, 2\right]$



Graph of $f(x) = \sqrt{x-2} + \sqrt{4-x}$ in $[2, 4]$

But, domain has to be restricted in either $[2, 3]$ or $[3, 4]$ for function to be invertible. Ans.

13. Even function $= \dfrac{f(x)+f(-x)}{2} = \dfrac{1}{2}\left[\dfrac{x+1}{x^3+1} + \dfrac{1-x}{1-x^3}\right] = \dfrac{1}{2}\left[\dfrac{1}{x^2-x+1} + \dfrac{1}{1+x+x^2}\right]$

$$= \frac{1}{2}\left[\frac{2(x^2+1)}{(x^2+1)^2 - x^2}\right] = \frac{x^2+1}{x^4+x^2+1} = \frac{x^4-1}{x^6-1} \quad \textbf{Ans.}$$

14. $g(x) = \frac{1}{2}\tan^{-1}|x| + 1 \Rightarrow \operatorname{sgn}(g(x)) = 1$

$\sin^{23}x - \cos^{22}x = 1$

$\sin^{23}x = 1 + \cos^{22}x$ which is possible if $\sin x = 1$ and $\cos x = 0$

$\sin x = 1,\ x = 2n\pi + \frac{\pi}{2}$

hence $-10\pi \le 2n\pi + \frac{\pi}{2} \le 20\pi$

$\Rightarrow -\frac{21}{2} \le 2n \le \frac{39}{2} \Rightarrow -\frac{21}{4} \le n \le \frac{39}{4}$

$\Rightarrow -5 \le n \le 9$

hence number of values of $x = 15$ **Ans.**

Paragraph for question nos. 15 to 17

Sol. (15-17)

15. Clearly, $g(f(x))$ is not defined

if $-2 + \alpha > 8$ and $\beta + 3 > 8$

$\Rightarrow \alpha > 10$ and $\beta > 5$

$\Rightarrow \alpha \in (10, \infty)$ and $\beta \in (5, \infty)$ **Ans.**.

16. If $\alpha = 2, \beta = 3$ then

$$f(x) = \begin{cases} 2x+2, & x \ge -1 \\ 3x^2+3, & x < -1 \end{cases}$$

So, range of $f(x) \in [0, \infty)$

$\Rightarrow$range of $g(f(x)) = [4, 12]$ **Ans.**

17. For $x \in [1, 3]$ and $\alpha = 3$

Now, $f(x) + g(x) = 12$

$\Rightarrow 3x = 5 \Rightarrow x = \frac{5}{3}$. **Ans.**

18. (A) Let $f : N \to N$ such that $f(x) = 2x$. Clearly f is one-one but not onto.

Note: If f is a one-one mapping from set A to A, then f is onto only if A is finite set.

(B) $f : R \to R$ such that $f(x) = x^3 - x^2 - 4x + 4$. Clearly $f(-2) = f(2) = f(1) = 0$.

Hence f is many one but since it is an odd degree polynomial therefore its range is R hence it is onto.

Note: If f is a onto mapping from set A to A then f is one-one only if A is finite set.

(C) Suppose f is not one-one then there are atleast two real numbers $x_1, x_2 \in R,\ x_1 \ne x_2$ such that $f(x_1) = f(x_2)$

$\therefore g(f(x_1)) = g(f(x_2))$

i.e. gof is not one-one which is a contradiction to the given hypothesis that gof is one-one.

Hence f must be one-one.

(D) Clearly, total number of functions from A to B = $2 \times 2 \times 2 = 8$.

19. $x = a - [x]^3 \quad \Rightarrow \quad x \in I$

$\therefore \quad a = x^3 + x$

$$\sum a = \sum_{r=1}^{7} r^3 + \sum_{r=1}^{7} r = \left(\frac{7\times 8}{2}\right)^2 + \frac{7\times 8}{2}$$

$= 784 + 28 = 812$. **Ans.**

20. We have $f(x) = e^{-|x|} - e^{x}$ $\quad$ (As $\sin \pi [x] = 0 \ \forall \ x \in R$)

$$= \begin{cases} 0 & ; \ x < 0 \\ e^{-x} - e^{x} & ; \ x \ge 0 \end{cases}$$

Clearly, f(x) is many-one and into function. So, f(x) is neither injective nor surjective.

21. $6^3 - {}^6C_3 \cdot 3! = 216 - 120 = 96$ **Ans**

22. (A) $\operatorname{sgn}\left(|x| + 1\right) = 1 \ \forall \ x \in R$.

(B) $\sin^2(\ln x) + \cos^2(\ln x) = 1 \ \forall \ x \in R^+$

(C) $\frac{2}{\pi}\left(\sin^{-1}\{x\} + \cos^{-1}\{x\}\right) = 1 \ \forall \ x \in R$.

(D) $\sec^2[\{x\}] - \tan^2\{[x]\} = \sec^2 0° - \tan^2 0° = 1 \ \forall \ x \in R$.

23. $$x^2-5x+6\overline{\big)\, x^{2007}\,}\big(\, Q(x)$$
$$\overline{R(x)=ax+b}$$

hence $x^{2007} = Q(x) \cdot (x^2 - 5x + 6) + ax + b$

$x^{2007} = Q(x) \cdot (x-2)(x-3) + \underbrace{ax + b}_{R(x)}$ ....(1)

now $R(0) = b = ?$

Put $x = 2$ in (1)

$2a + b = 2^{2007}$ ....(2)

put $x = 3$ in (1)

$3a + b = 3^{2007}$ ....(3)

(3) – (2) gives

$a = 3^{2007} - 2^{2007}$

now $b = 2^{2007} - 2a =$

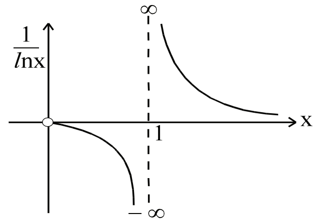
$2^{2007} - 2(3^{2007} - 2^{2007}) = 2^{2007} + 2 \cdot 2^{2007} - 2 \cdot 3^{2007})$

$= 3 \cdot 2^{2007} - 2 \cdot 3^{2007} = 6[2^{2006} - 3^{2006}]$

$= 2 \cdot 3[2^{2006} - 3^{2006}] \equiv ab(a^c - b^c)$

hence $a = 2; \quad b = 3; \quad c = 2006$

$\therefore \quad a + b + c = 2 + 3 + 2006 = 2011$ Ans.

24. D : $x \in R$; R : [1, 2]  .

25. for domain of g (x) is the set of x for which

$-1 \le x-2 \le 4 \quad \Rightarrow \quad 1 \le x \le 6$

hence a = 1 and b = 6

for range, $1 \le y \le 10$

$\Rightarrow \quad 1 \le f(x) \le 10$

$\Rightarrow \quad 1 \le f(x-2) \le 10$

$\Rightarrow \quad 3 \le 3f(x-2) \le 30$

hence $-30 \le -3f(x-2) \le -3$

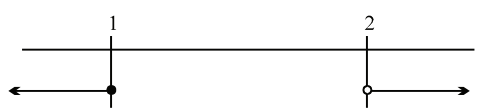
$\therefore \quad -26 \le 4-3f(x-2) \le 1$

hence c = – 26 and d = 1

now verify

26. $f(x) = \sqrt{\dfrac{\log_3 x - 1}{\log_3 \dfrac{x}{9}}}$

For domain $\dfrac{\log_3 x - 1}{\log_3 x - 2} \ge 0$



$\therefore \quad \log_3 x > 2 \quad \Rightarrow \quad x > 9$ or $\quad \log_3 x \le 1 \Rightarrow 0 < x \le 3$

domain is $(0, 3] \cup (9, \infty)$

Now $y = \sqrt{\dfrac{t-1}{t-2}}$ when $t = \log_3 x$

$= \sqrt{1 + \dfrac{1}{t-2}}$ if $t = 1,\ y = 0$

$y \in [0, 1) \cup (1, \infty)$ $\quad$ ($y \to \infty$ when $t \to 2^+$)

Hence sum of all integral values = 2 + 3 + 4 + ... + 100 = 5049 **Ans.**

27.

(i) $f(x) = \sqrt{\dfrac{1-5^x}{7^{-x}-7}}$

Sol. $f(x) = \sqrt{\dfrac{1-5^x}{7^{-x}-7}}$

$N^r \geq 0$ and $D^r > 0$ $\cup$ $N^r \leq 0$ and $D^r < 0$

$f(x) \sqrt{\dfrac{1-5^x}{7^{-x}-7}} = \sqrt{\dfrac{7^x(1-5^x)}{1-7^{x+1}}}$

$$\underset{>0}{\overset{7^x}{\Downarrow}} \left(\frac{1-5^x}{1-7^{x+1}}\right) \geq 0$$

Case I: $1 - 5^x \geq 0$ & $1 - 7^{x+1} > 0$

$5^x \leq 1$; $7^{x+1} < 1$

$x \leq 0$ ... (1) ; $x + 1 < 0$

$x < -1$ ... (2)

$(1) \cap (2)$

$x < -1$

Case II $1 - 5^x \leq 0$ & $1 - 7^{x+1} < 0$

$5^x \geq 1$; $7^{x+1} \geq 1$

$x \geq 0$ ... (3) & $x + 1 > 0$ $\Rightarrow$ $x > -1$ ... (4)

$(3) \cap (4)$

$x \geq 0$

$\therefore$ Case I $\cup$ Case II

$x \in (-\infty, -1) \cup [0, \infty)$]

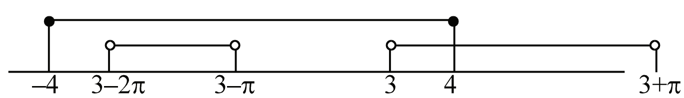
(ii) $y = \log_{10} \sin(x-3) + \sqrt{16-x^2}$

Sol. $f(x) = \log_{10} \sin(x-3) + \sqrt{16-x^2}$

D_1: $\sin(x-3) > 0 \Rightarrow 0 < x - 3 < \pi \Rightarrow 2k\pi < x - 3 < (2k+1)\pi$

$\Rightarrow$ $3 + 2k\pi < x < 3 + (2k+1)\pi$

D_2: $x \in [-4, 4]$



$(3 - 2\pi, 3 - \pi) \cup (3, 4]$ Ans.]

(iii) $f(x) = \dfrac{1}{\sqrt{4x^2-1}} + \ln x(x^2-1)$

Sol. $f(x) = \dfrac{1}{\sqrt{4x^2-1}} + \ln x(x^2-1)$

$4x^2 - 1 > 0$ and $x(x^2 - 1) > 0$

$\Rightarrow$ $\left(-1, -\dfrac{1}{2}\right) \cup (1, \infty)$ **Ans.**

(iv) $f(x) = \sqrt{\log_{\frac{1}{2}} \dfrac{x}{x^2-1}}$

Sol. $f(x) = \sqrt{\log_{\frac{1}{2}} \dfrac{x}{x^2 - 1}}$

$f(x) = \sqrt{-\log_2 \dfrac{x}{x^2 - 1}}$

$0 < \dfrac{x}{x^2 - 1} \le 1$

$\Rightarrow \quad \left[\dfrac{1-\sqrt{5}}{2}, 0\right) \cup \left[\dfrac{1+\sqrt{5}}{2}, \infty\right)$ **Ans.**

28.

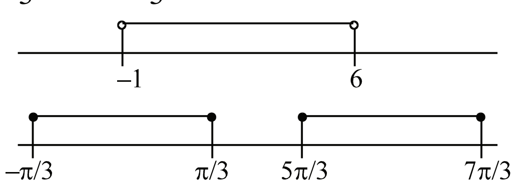
(i) $f(x) = \dfrac{\sqrt{\cos x - (1/2)}}{\sqrt{6 + 35x - 6x^2}}$

Sol. $f(x) = \dfrac{\sqrt{\cos x - (1/2)}}{\sqrt{6 + 35x - 6x^2}}$

$\cos x \ge \dfrac{1}{2}$ i.e. $6x^2 - 35x - 6 < 0$

$-\dfrac{\pi}{3} \le x \le \dfrac{\pi}{3}$ $6x^2 - 36x + x - 6 < 0$

$\dfrac{5\pi}{3} \le x \le \dfrac{7\pi}{3}$ $(6x + 1)(x - 6) < 0$



$\therefore \quad x \in \left(-\dfrac{1}{6}, \dfrac{\pi}{3}\right] \cup \left[\dfrac{5\pi}{3}, 6\right)$ **Ans.**

(ii) $f(x) = \dfrac{[x]}{2x - [x]}$

Sol. $f(x) = \dfrac{[x]}{x + x - [x]} = \dfrac{[x]}{x + \{x\}}$

for $x > 0$ function is obvious defined

function is not defined at $x = 0$

for $x < 0$, function is defined everywhere

except at $x = -1/2$

hence Domain : $R - \left\{-\frac{1}{2}, 0\right\}$ **Ans.**

(iii) $f(x) = \log_x \sin x$

Sol. $f(x) = \log_x \sin x$

$\sin x > 0$ and $x > 0, \neq 1$

$0 < x < \pi$

$2K\pi < x < (2K + 1)\pi$ but $x \neq 1$ where K is non–negative integer

(iv) $f(x) = \log_2\left(-\log_{1/2}\left(1 + \frac{1}{\sin\left(\frac{x°}{100}\right)}\right)\right) + \sqrt{\log_{10}(\log_{10} x) - \log_{10}(4 - \log_{10} x) - \log_{10} 3}$

Sol. $f(x) = \log_2\left(-\log_{1/2}\left(1 + \frac{1}{\sin\left(\frac{x°}{100}\right)}\right)\right) + \sqrt{\log_{10}(\log_{10} x) - \log_{10}(4 - \log_{10} x) - \log_{10} 3}$

D_1 D_2

D_1: $\log_2\left(\log_2\left(1 + \operatorname{cosec}\frac{\pi x}{18000}\right)\right)$

$0 < \frac{\pi x}{18 \times 10^3} < \pi$

$2k\pi < \frac{\pi x}{18 \times 10^3} < (2k + 1)\pi$

$(18 \times 10^3)k < x < (2k + 1)$ $k \in I$

D_2 $\log_{10}\left(\frac{\log_{10} x}{(4 - \log_{10} x) \times 3}\right) \geq 0$ $[x > 1$ and $x < 10^4]$

put $\log_{10} x = y$

$\frac{y}{3(4 - y)} - 1 \geq 0;$ $\frac{y - 3(4 - y)}{3(4 - y)} \geq 0;$ $\frac{4(y - 3)}{3(4 - y)} \geq 0$

$\frac{(y - 3)}{(y - 4)} \leq 0;$ $3 \leq y < 4$

i.e. $3 \leq \log_{10} x < 4 \Rightarrow 10^3 \leq x < 10^4$.]

(v) $f(x) = \sqrt{(5x - 6 - x^2)\,[\{ln\{x\}\}]} + \sqrt{(7x - 5 - 2x^2)} + \left(ln\left(\frac{7}{2} - x\right)\right)^{-1}$

Sol. $f(x) = \sqrt{(5x - 6 - x^2)\,[\{ln\{x\}\}]} + \sqrt{(7x - 5 - 2x^2)} + \left(ln\left(\frac{7}{2} - x\right)\right)^{-1}$

D_1 D_2 D_3

D_1 $\quad x \notin I \quad \forall\, x \in R - I$ function is zero

D_2 $\quad 2x^2 - 7x + 5 \le 0$

$2x^2 - 2x - 5x + 5 \le 0$

$(2x - 5)(x - 1) \le 0$

$x \in [1, 5/2]$

$D_3 \quad \dfrac{7}{2} - x > 0 \qquad\qquad \dfrac{7}{2} - x \ne 1$

$x < \dfrac{7}{2} \qquad\qquad x \ne \dfrac{5}{2}$

$\therefore \quad D_1 \cap D_2 \cap D_3$

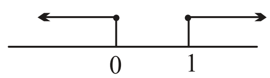
$x \in (1, 2) \cup (2, 5/2)$ Ans.

29. $f(x) = \sqrt{x^2 - |x|} + \dfrac{1}{\sqrt{9 - x^2}}$

Let $\quad |x| = t$

$f(x) = \sqrt{t^2 - t} + \dfrac{1}{\sqrt{9 - t^2}}$

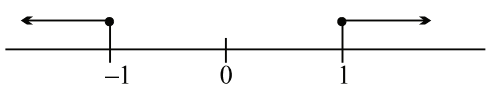
$t(t - 1) \ge 0$ $\quad$ and $\quad 9 - t^2 > 0$

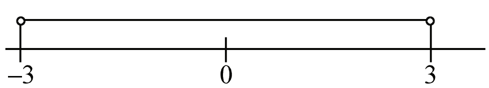


and $\quad t \in (-3, 3)$

$|x| \ge 1 \qquad x \ge 1$ or $x \le -1$

or $\quad |x| \le 0 \;\Rightarrow\; x = 0$





$(-3, -1] \cup [1, 3) \cup \{0\}$

Hence number of integer in the domain of function is 5 **Ans.**

30. $f(x) = \begin{cases} e^{-\sqrt{|\ln\{x\}|}} - \{x\}^{\sqrt{\frac{1}{|\ln\{x\}|}}} & x \in R - I \\ \{x\} & x \in I \end{cases}$

$f(x) = \begin{cases} e^{-\left(\sqrt{-\ln\{x\}}\right)} - \{x\}^{\left(\sqrt{\frac{-1}{\ln\{x\}}}\right)} & x \in R - I \\ \{x\} & x \in I \end{cases}$

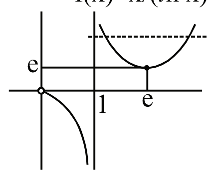
Let $\quad -\ln\{x\} = t \quad \Rightarrow \quad \{x\} = e^{-t}$

$$f(x) = \begin{cases} e^{-\sqrt{t}} - e^{\frac{-t}{\sqrt{t}}} = 0 & x \in R - I \\ 0 & x \in I \end{cases}$$

$\therefore \quad f(x) = 0$

$\Rightarrow \quad f(x)$ is odd is well as even.]

31. for $\dfrac{x}{\ln x} = k$



32. $x = a - [x]^3 \quad \Rightarrow \quad x \in I$

$\therefore \quad a = x^3 + x$

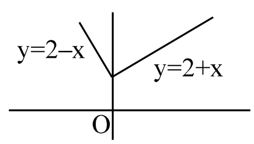
$$\sum a = \sum_{r=1}^{7} r^3 + \sum_{r=1}^{7} r = \left(\frac{7\times 8}{2}\right)^2 + \frac{7\times 8}{2}$$

$= 784 + 28 = 812$ **Ans.**

33. $f(f(x)) = \begin{cases} 2 + f(x), & f(x) \ge 0 \\ 2 - f(x), & f(x) < 0 \end{cases}$

since $f(x) \ge 2 \quad \forall\, x \in R$

$\therefore \quad f(f(x)) = \begin{cases} 4 + x, & x \ge 0 \\ 4 - x, & x < 0 \end{cases}$



34. (A) $f'(x) = \dfrac{1}{(1+x)^2} > 0 \quad \therefore$ one-one

$f'(x) \to 1$ as $x \to \infty \quad \therefore$ not onto

(B) $f'(x) = 1 + \dfrac{1}{x^2}$, f(x) increases both for $x > 0$ and $x < 0$

$f(1) = f(-1) = 0$

$\therefore$ f(x) is onto but not one-one

(C) $f'(x) = 1 - \dfrac{1}{x^2}$ changes sign

$\therefore$ not one-one

f(x) is not onto since

$f(x) \ge 2$ and $f(x) \le -2$

(D) $f'(x) = 2 + \cos x > 0 \quad \therefore$ one-one

$f(x) \to \pm\infty$ as $x \to \pm\infty \quad \therefore$ onto.

35. **(A)** $f(x) = \left[x+\frac{1}{2}\right] + \left[x-\frac{1}{2}\right] + 2[-x]$

clearly $f(x)$ is into as it will not take non integral values

now write $[x] = x - \{x\}$

$$f(x) = \left(x+\frac{1}{2}\right) - \left\{x+\frac{1}{2}\right\} + \left(x-\frac{1}{2}\right) - \left(x-\frac{1}{2}\right) + 2\left(-x-\{-x\}\right)$$

$$= -\left\{x+\frac{1}{2}\right\} - \left\{x-\frac{1}{2}\right\} - 2\{-x\}$$

this is periodic with period 1.

so function is periodic, many one and into $\Rightarrow$ **P, R, T**

(B) $f(x) = x^3 + x^2 + 3x + \sin x$

clearly onto and not periodic

now $f'(x) = 3x^2 + 2x + 3\cos x$

$$= \underbrace{3x^2 + 2x + 2}_{\substack{D<0 \\ \text{so always +ve}}} + \underbrace{1 + \cos x}_{\text{always +ve}}$$

so $f'(x)$ is always + ve so one-one, onto $\Rightarrow$ **Q, S**

(C) $f(x) = \dfrac{e^{|x|} - e^{-x}}{e^{x} + e^{-x}}$

now if x is < 0 then $f(x) = 0$

so many one

$$f(x) = \begin{cases} 0 & x \le 0 \\ \dfrac{e^x - e^{-x}}{e^x + e^{-x}} & x > 0 \end{cases}$$

so for $x > 0$, $f(x) = \dfrac{e^{2x} - 1}{e^{2x} + 1}$

now in $x > 0$

$$y = \frac{e^{2x} - 1}{e^{2x} + 1} = \frac{e^{2x} + 1 - 2}{e^{2x} + 1} = 1 - \frac{2}{e^{2x} + 1}$$

so $1 < e^{2x} < \infty$

$2 < e^{2x} + 1 < \infty$

$$\frac{1}{2} > \frac{1}{e^{2x} + 1} > 0$$

$1 > \dfrac{2}{e^{2x} + 1} > 0$. So range is $[0, 1)$ hence into

so many one, into $\Rightarrow$ **R, T**

(D) $f(x) = e^{\sin\{x\}} + \sin\left(\frac{\pi}{2}[x]\right)$

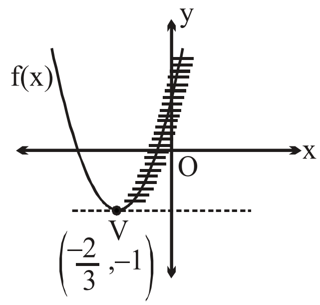
clearly into as $e^{\sin\{x\}}$ and $\sin\frac{\pi}{2}[x]$ both bounded

and periodic with period 4. So many one

i.e. Many one, periodic, into$\Rightarrow$ **P, R, T**]

36. We have $f(x) = y = (3x + 2)^2 - 1$

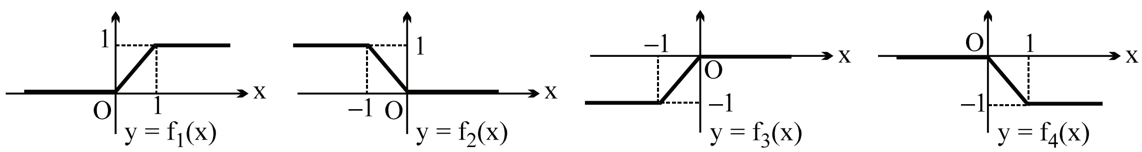
$\Rightarrow (3x + 2)^2 = y + 1 \quad \Rightarrow 3x + 2 = -\sqrt{y+1} \quad \left(\text{As } x \le \frac{-2}{3}\right)$



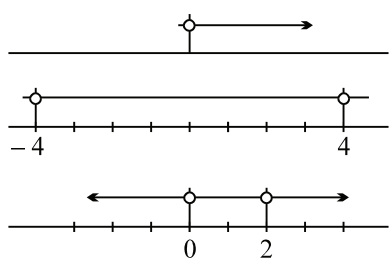
$\Rightarrow \quad x = \left(\frac{-2-\sqrt{y+1}}{3}\right) = f^{-1}(y) = g(y)$

Hence $g(x) = \left(\frac{-2-\sqrt{x+1}}{3}\right)$ **Ans.**

37.

Hint: 

38. Domain of $\sqrt{x}$ is $x \ge 0$ ....(i)



domain of $\sqrt{16 - x^2}$ is $x \in [-4, 4]$ ....(ii)

domain of $\log_2 x(x-2)$ is $(-\infty, 0) \cup (2, \infty)$ ....(iii)

intersect of (i), (ii) and (iii) is $(2, 4]$ ....(iv)

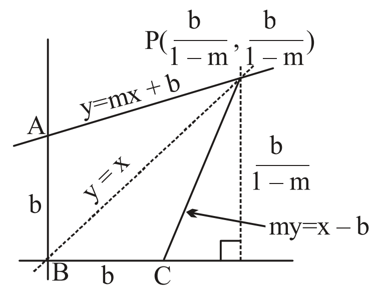
domain of $\sqrt{\sin\frac{\pi x}{2}}$ is such that $\frac{\pi x}{2} \in [2n\pi, (2n+1)\pi]$, $n \in I$

$\Rightarrow \quad x \in [4n, 4n + 2]$, $n \in I$ ....(v)

intersection of (iv) and (v) is $x = 4$

$\Rightarrow$ hence domain of $f(x)$ is $x \in \{4\}$. Hence range of f is $\{5\}$

39. If $f(x) = mx + b$, then $f^{-1}(x) = \dfrac{x-b}{m}$ and their point of intersection can be found by setting $x = my + b$ since they intersect on $y = x$.



Thus $x = \dfrac{b}{1-m}$ and the point of intersection is $\left(\dfrac{b}{1-m}, \dfrac{b}{1-m}\right)$.

Region R can be broken up into congruent triangles PAB and PCB which both have a base of b and a height of $\dfrac{b}{1-m}$.

The area of R is $2\left(\dfrac{b}{2}\right)\left(\dfrac{b}{1-m}\right) = \dfrac{b^2}{1-m} = 49.$

For $m = \dfrac{9}{25}$, $b^2 = \dfrac{16}{25} \cdot 49 \quad \Rightarrow \quad b = \dfrac{28}{5}$ Ans.

40. $0 < e^x < 1 \Rightarrow x < 0$

$0 < \ln|x| < 1$

$\Rightarrow 1 < |x| < e$

$\Rightarrow \ x \in (-e, -1) \cup (1, e)$

$\therefore \qquad x \in (-e, -1)$. **Ans.**

41. Given, $g(f(x)) = 8 \Rightarrow (f(x))^2 + 7 = 8 \Rightarrow f(x) = \pm 1 \Rightarrow 2x + 3 = \pm 1$

$\Rightarrow \qquad 2x = -3 \pm 1 \Rightarrow x = -1, -2$

Hence, sum of values of $x = (-1) + (-2) = -3$. **Ans.**

42. Let $\quad y = \dfrac{x^2 + 2x + a}{x^2 + 4x + 3a}$ ....(i)

$\Rightarrow \qquad (y-1)x^2 + (4y-2)x + (3ay - a) = 0$ ....(ii)

If range of eq. (i) is R, then roots of eq. (ii) must be real for all $y \in R$.

$\Rightarrow \qquad (4y-2)^2 - 4(y-1)(3ay-a) \ge 0 \quad \forall\, y \in R$

$\Rightarrow \qquad (4-3a)y^2 - 4(1-a)y + (1-a) \ge 0\ \forall\, y \in R$ ....(iii)

Now, eq. (iii) is true $\forall\, y \in R$, if $4 - 3a > 0 \Rightarrow a < \dfrac{4}{3}$

$$\left[\begin{array}{l}\text{If } x^2 + 2x + a = 0 \text{ and } x^2 + 4x + 3a = 0 \\ \text{have a common root then } a = 0, 1. \\ \text{So range of } f \text{ is not equal to R.}\end{array}\right]$$

and $\qquad 16(1-a)^2 - 4(4-3a)(1-a) \le 0$

$\Rightarrow \qquad a(a-1) \le 0$

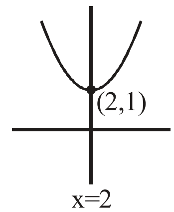
Hence, we get $0 < a < 1$. **Ans.**

43. $f(x) = a(x-2)^2 + 1$

$\because f(1) = 2 \qquad \therefore a = 1$

$f(x) = x^2 - 4x + 5$

$g(\ln x) = x^2 - 4x + 5 = (x-2)^2 + 1$



$g(x) = (e^x - 2)^2 + 1$

$x \in (-\infty, \ln 2] \Rightarrow g(x) \in [1, 5)$

$\Rightarrow g(x)$ is invertible function.

$y = (e^x - 2)^2 + 1$

$e^x = 2 + \sqrt{y-1}, 2 - \sqrt{y-1}$

$x = \ln\left(2 - \sqrt{y-1}\right)$ as $x \in (-\infty, \ln 2]$

$\therefore \qquad g^{-1}(x) = \ln\left(2 - \sqrt{x-1}\right)$. **Ans.**

44. $fog(x) = [\,|x|\,]$

$gof(x) = |\,[x]\,|$

(A) $fog\left(\frac{5}{2}\right) - gof\left(\frac{-5}{2}\right) = \left[\left|\frac{5}{2}\right|\right] - \left|\left[\frac{5}{2}\right]\right| = 2 - 3 = -1$

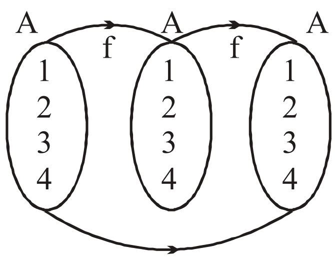
(B) $(f + 2g)(x) = [x] + 2|x|$

$(f + 2g)(-1) = -1 + 2 = 1$

(C) $\text{sgn}(fog(x)) = 0 \Rightarrow fog(x) = 0 \Rightarrow [|x|] = 0 \Rightarrow 0 \le |x| < 1 \Rightarrow x \in (-1, 1)$

(D) $\text{sgn}(gof(x)) = 0 \Rightarrow gof(x) = 0 \Rightarrow |[x]| = 0 \Rightarrow 0 \le [x] < 1 \Rightarrow x \in [0, 1)$. **Ans.**

45.



$fof : A \to A$

46. $$f(x) = \begin{cases} x^2 - 3, & x \le -5 \\ x + \lambda & -5 < x < -1 \\ (\mu - 7)\left(|1-x| + |1+x|\right) & -1 \le x \le 1 \\ x + 6 & 1 < x < 5 \\ 3 - x^2 & x \ge 5 \end{cases}$$

$$f(-x) = \begin{cases} x^2 - 3 & -x \le -5 \Rightarrow x \ge 5 \\ -x + \lambda & -5 < -x < -1 \Rightarrow 1 < x < 5 \\ (\mu - 7)\left(|1+x| + |1-x|\right) & -1 \le x \le 1 \\ -x + 6 & 1 < -x < -1 \\ 3 - x^2 & x \le -5 \end{cases}$$

$$f(x) + f(-x) = \begin{cases} 0 & x \le -5 \\ \lambda + 6 & -5 < x < -1 \\ -2\,(\mu - 7)\left(|1-x| + |1+x|\right) & -1 \le x \le 1 \\ \lambda + 6 & 1 < x < 5 \\ 0 & x \ge 5 \end{cases}$$

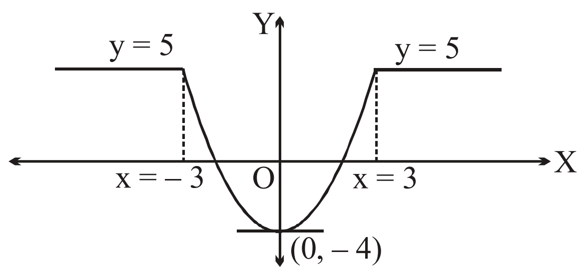
for $f(x)$ to be an odd function
$f(x) + f(-x) = 0\ \forall\ x \in R$
$\therefore\ \lambda = -6$ and $\mu = 7 \Rightarrow (\lambda + \mu) = -6 + 7 = 1$. **Ans.**

47. $f(x)$ is an even function.

Also, $g(x) = 2\tan^{-1}(e^x) - \dfrac{\pi}{2}$

$g(-x) = 2\tan^{-1}(e^{-x}) - \dfrac{\pi}{2}$

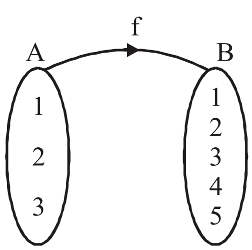
$= 2\left(\dfrac{\pi}{2} - \tan^{-1} e^x\right) - \dfrac{\pi}{2}$



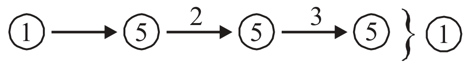
Graph of y = f(x) (even function)

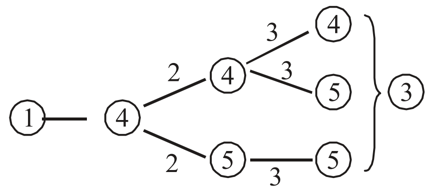
$= \dfrac{\pi}{2} - 2\tan^{-1}(e^x) = -g(x) \Rightarrow g(x)$ is an odd function.

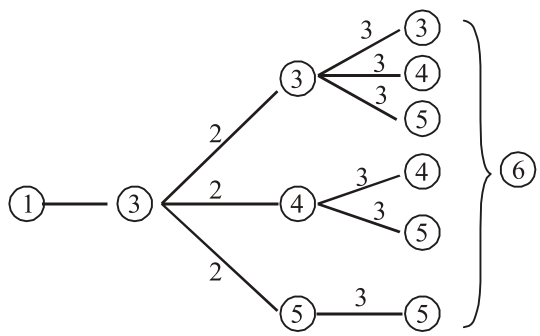
Hence, fog(x), gof, fof(x) is an even function. But gog(x) is an odd function. **Ans.**]

48. 

$n(S) = 5^3$
For n(A) :



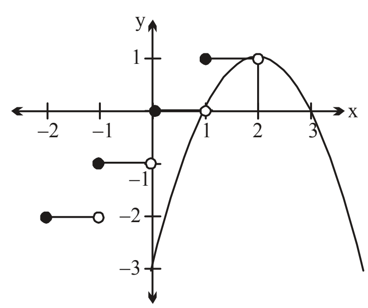




Similarly when 1 is associated with (2) then number of such functions are 10 and when 1 is associated when (1) then number of such functions are 15.
Total functions = 1 + 3 + 6 + 10 + 15 = 35. **Ans.**

49. Now $\log_3 \log_4 \log_5 (\sin x + b^2) > 0$ or $\log_4 \log_5 (\sin x + b^2) > 1$
or $\log_5 (\sin x + b^2) > 4$ or $\sin x + b^2 > 5^4$
Now for all x we want $\sin x > 5^4 - b^2$
$\Rightarrow 5^4 - b^2 < \sin x$
$\Rightarrow 625 - b^2 < -1 \Rightarrow b^2 > 626$

50. Given $x^2 + [x] + 3 = 4x$
$[x] = -x^2 + 4x - 3$
$[x] = -(x-1)(x-3)$



$\therefore$ This equation has no solution
$\Rightarrow$ Answer is 0.

Alternate: We are given that, $x^2 - 4x + x - \{x\} + 3 = 0$ ....(1)
$\Rightarrow x^2 - 3x + 3 = \{x\} \Rightarrow 0 \le x^2 - 3x + 3 < 1$

Now, $x^2 - 3x + 3 = \left(x - \frac{3}{2}\right)^2 + \frac{3}{4} > 0$

$\therefore x^2 - 3x + 3 > 0 \ \forall x \in R$
Also, $x^2 - 3x + 3 < 1$
$\Rightarrow x^2 - 3x + 2 < 0$
$\Rightarrow (x-1)(x-2) < 0 \Rightarrow 1 < x < 2$
so, $[x] = 1$
$\therefore$ From the equation (1) we get, $x^2 - 4x + 4 = 0$
i.e., $(x-2)^2 = 0$
$\Rightarrow x = 2$, which does not satisfy $1 < x < 2$.
Hence, no real value of x satisfy the given equation.

51. Consider $4^{x^2} + 4^{(x-1)^2}$

$AM \geq GM$ for two positive numbers 4^{x^2} and $4^{(x-1)^2}$

$$\frac{4^{x^2}+4^{(x-1)^2}}{2} \geq \left[4^{x^2}\cdot 4^{(x-1)^2}\right]^{1/2} = 2^{x^2}\cdot 2^{(x-1)^2} = 2^{x^2+(x-1)^2};$$

$$4^{x^2}+4^{(x-1)^2} \geq 2^{x^2+(x-1)^2+1}$$

now $z = x^2 + (x-1)^2 + 1$

$$\Rightarrow 2x^2 - 2x + 2 = 2[x^2 - x + 1] = 2\left[\left(x-\frac{1}{2}\right)^2+\frac{3}{4}\right]$$

$\therefore$ $z_{max} \to \infty$ hence $z_{min} = \frac{3}{2}$

$\therefore$ $4^{x^2}+4^{(x-1)^2}$ has the minimum value $= 2^{3/2}$

hence $f(x) \geq \log_2 (2)^{3/2} = \frac{3}{2}$

$\therefore$ $y \geq \frac{3}{2}$ $\Rightarrow$ range is $\left[\frac{3}{2}, \infty\right)$ **Ans.**

$\Rightarrow$ $p + q = 3 + 2 = 5$ **Ans.**

Range: where derivative of f is to be considered]

52. replacing x by $\frac{\pi}{2} - x$; $f\left(\cos\left(\frac{\pi}{2}-x\right)\right) = \cos 17\left(\frac{\pi}{2}-x\right)$

$f(\sin x) = \sin 17x = g(\sin x)$

hence $f = g$.

53. $f(x) = \text{Min.}(\,|x|\,,\ 1-|x|\,)$

$D_f = R$

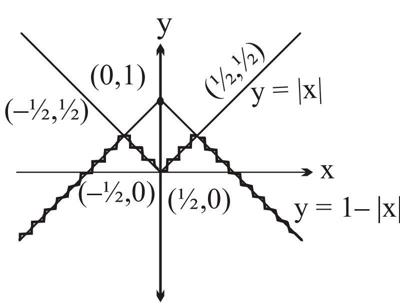
$f(-x) = f(x)$

$\Rightarrow$ f is even function.

Put

$|x| = 1 - |x|$

$\therefore$ $x = \pm\frac{1}{2}$



Clearly $R_f = \left(-\infty, \frac{1}{2}\right]$

Also f is aperiodic.

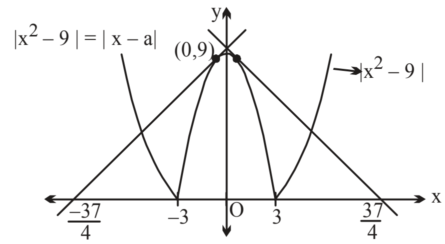
54. For tangency, $x^2 - 9 = x - a$

$\Rightarrow\ x^2 - x + a - 9 = 0$

Put $D = 0$

$\Rightarrow 1 - 4a + 36 = 0 \ \Rightarrow\ a = \frac{37}{4}$

|||ly $a = \frac{-37}{4}$



$\therefore$ For 4 distinct solution, $a \in \left(\frac{-37}{4}, -3\right) \cup (-3, 3) \cup \left(3, \frac{37}{4}\right)$

Hence, number of integers are 17 **Ans.**

55.

(A) Clearly $x - x^4 + x^7 - x^8 - 1 < 0 \ \forall\ x \in R^+$.

Hence $\text{sgn}\,(x - x^4 + x^7 - x^8 - 1) = -1\ \forall\ x \in R^+$.

(B) We have $f(x) + x\,f(-x) = x + 1$ .... (1)

Replace x by $-x$ in (1), we get

$-x\,f(x) + f(-x) = 1 - x$ .... (2)

$\therefore$ From (1) and (2), we get

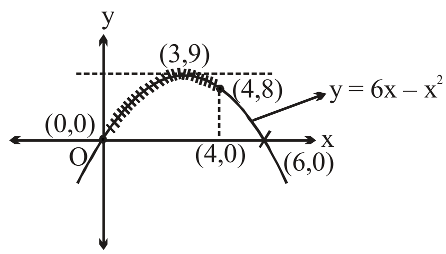
$x^2 f(x) - x\,f(-x) = x^2 - x$ .... (3)

$\Rightarrow (1 + x^2)\,f(x) = 1 + x^2$

Hence $f(x) = 1$

(C) We have $f(x) = \sqrt{6x - x^2}$

Clearly range of $f = [0, 3]\ \forall\ 0 \le x \le 4$.



(D) We have $f(x) = 2^{|x-1| + |x-2|}$

$$= \begin{cases} 2^{-2x+3} & ;\quad 0 \le x \le 1 \\ 2 & ;\quad 1 < x < 2 \\ 2^{2x-3} & ;\quad 2 \le x \le 3 \end{cases}$$

Clearly range of $f = [2, 8]\ \forall\ 0 \le x \le 3$.]

56.

(A) For domain $x^2 + 4x \ge 2x^2 + 3$

$\therefore\quad x^2 - 4x + 3 \le 0$

$(x - 3)(x - 1) \le 0$

$1 \le x \le 3$

Domain is {1, 2, 3}

$f(1) = 1$; $f(2) = \sqrt{12}$; $f(3) = \sqrt{{}^{21}C_{21}} = 1$

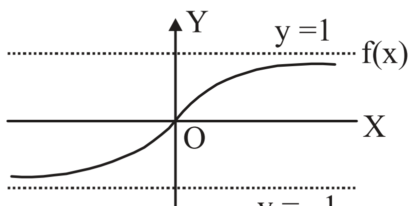
$\Rightarrow$ Hence only 1 lie in the range which is an integer]

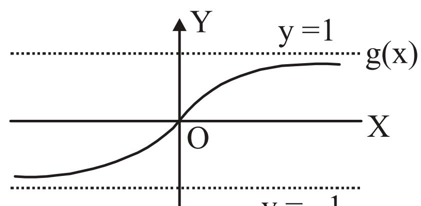
(B) $f(x) = \sin x\,|\cos x| - \cos x\,|\sin x|$

$$f(x) = \begin{cases} 0 & \text{if } x \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right) \\ -\sin 2x & \text{if } x \in \left(\frac{\pi}{2}, \pi\right) \\ \sin 2x & \text{if } x \in \left(\frac{3\pi}{2}, 2\pi\right) \end{cases}$$

Hence range is $[-1, 1]$
range consists of 3 elements i.e. $\{-1, 0, 1\}$

57.





from graph
$f(x)$ is one - one , $-1 < f(x) < 1$
$g(x)$ is one - one, $-1 < g(x) < 1$
Thus $f(g(x))$ and $g(f(x))$ both will be one-one

$$f(g(x)) = \frac{2}{\pi} \tan^{-1}(g(x))$$

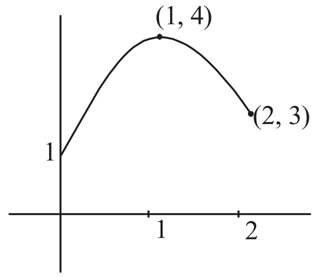
$$-1 < g(x) < 1 \Rightarrow \frac{-\pi}{4} < \tan^{-1}(g(x)) < \frac{\pi}{4}$$

$$-\frac{1}{2} < \frac{2}{\pi}\tan^{-1}(g(x)) < \frac{1}{2}$$ thus fog is into.

$$g(f(x)) = \frac{e^{f(x)} - e^{-f(x)}}{e^{f(x)} + e^{-f(x)}}$$ as $-1 < f(x) < 1$

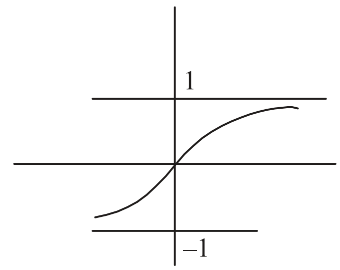
$$g(f(x)) \in \left(\frac{e^{-1} - e^{1}}{e^{1} - e^{-1}}, \frac{e^{1} + e^{-1}}{e + e^{-1}}\right)$$ thus gof is alos into.

58. $f'(x) = 3x^2 - 10x + 7 = (3x - 7)(x - 1)$



so many one but onto.

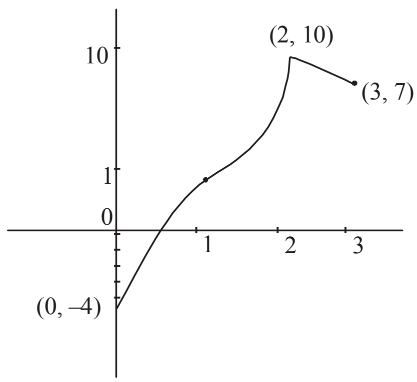
59. f(x) is odd function



$$\frac{dy}{dx} = - \frac{(1-x)\cdot 1 - x\cdot 1}{(1+x)^2} \quad \frac{1}{(1+x)^2}$$

$\frac{dy}{dx} > 0$, so one-one. Range is (–1, 1).]

60.



So range is [–4, 10]. No. of integers = 15 Many one.

61. Given $f(x) = \log\left(\sqrt{9x^2 - 12 + \lambda} + 1\right)$

$D_f = R$ & $R_f = [0, \infty)$

So, it is possible when

$9x^2 - 12x + \lambda = 0$

has equal roots

$\Rightarrow$ Disc. = 0

So, $144 = 36\lambda$

$\Rightarrow$ $\lambda = 4$ **Ans.**

62. We have $g(x) = x^4 - 2x^2 + 2 = y$

$\Rightarrow$ $x^2 = 1 \pm \sqrt{y-1}$

$\Rightarrow$ $x = -\sqrt{1-\sqrt{y-1}}$

Hence $g^{-1}(x) = -\sqrt{1-\sqrt{y-1}}$.]

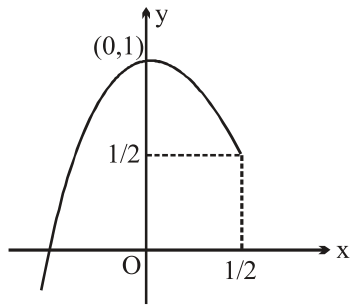
63. $f(x) = \frac{1}{2}\left(2\sqrt{1-2x} + 2x\right)$

$= -\frac{1}{2}\left(-2\sqrt{1-2x} - 2x\right)$

$= -\frac{1}{2}\left(\left(\sqrt{1-2x} - 1\right)^2 - 2\right)$

$y = 1 - \frac{1}{2}\left[\left(\sqrt{1-2x} - 1\right)^2\right]$

$y_{max} = 1$ when $x = 0$



x=0 is the inflection point

Alternatively: $f'(x) = 1 - \frac{1}{\sqrt{1-2x}}$; $f'(x) = 0$

$\Rightarrow \quad 1 - 2x = 1 \quad \Rightarrow \quad x = 0$

also $\quad f(-\infty) = \rightarrow -\infty$

Note: $x = 0$ is the inflection point.

64. $N = \left[\frac{18(1)}{35}\right] + \left[\frac{18(2)}{35}\right] + \left[\frac{18(3)}{35}\right] + \ldots\ldots\ldots + \left[\frac{18(33)}{35}\right] + \left[\frac{18(34)}{35}\right] \quad \ldots (1)$

$N = \left[18 - \frac{18}{35}\right] + \left[18 - \frac{18(2)}{35}\right] + \left[18 - \frac{18(3)}{35}\right] + \ldots\ldots + \left[18 - \frac{18(33)}{35}\right] + \left[18 - \frac{18(34)}{35}\right] \ldots (2)$

$(1) + (2) \Rightarrow 2N = \underbrace{(18-1) + (18-1) + \ldots\ldots\ldots + (18-1)}_{34 \text{ times}} = 17\ (34)$

$\Rightarrow N = 17\ (17) = 289.$

65. obviously f is a linear polynomial

let $\quad f(x) = ax + b$ hence $f(x^2 + x + 3) + 2f(x^2 - 3x + 5) \equiv 6x^2 - 10x + 17$

or $\quad [a(x^2 + x + 3) + b] + 2[a(x^2 - 3x + 5) + b] \equiv 6x^2 - 10x + 17$

$\left.\begin{array}{ll} a + 2a = 6 & \ldots.(1) \\ a - 6a = -10 & \ldots.(2) \end{array}\right] \Rightarrow a = 2$ *(comparing coeff. of x^2 and coeff. of x both sides)*

again, $\quad 3a + b + 10a + 2b = 17$

$\Rightarrow \quad 6 + b + 20 + 2b = 17$

$26 + 3b = 17 \quad \Rightarrow \quad b = -3$

$\therefore \quad f(x) = 2x - 3$

66.

(A) $f'(x) = \dfrac{1}{(1+x)^2} > 0 \quad \therefore$ one-one

$f'(x) \to 1$ as $x \to \infty \quad \therefore$ not onto

(B) $f'(x) = 1 + \dfrac{1}{x^2}$, f(x) increases both for $x > 0$ and $x < 0$

$f(1) = f(-1) = 0$

$\therefore$ f(x) is onto but not one-one

(C) $f'(x) = 1 - \dfrac{1}{x^2}$ changes sign

$\therefore$ not one-one

f(x) is not onto since

$f(x) \geq 2$ and $f(x) \leq -2$

(D) $f'(x) = 2 + \cos x > 0 \quad \therefore$ one-one

$f(x) \to \pm\infty$ as $x \to \pm\infty \quad \therefore$ onto.

67. $$y = \frac{4x\,(x^2+1)}{x^2+(x^2+1)^2} = \frac{4\left(x+\frac{1}{x}\right)}{1+\left(x+\frac{1}{x}\right)^2}$$

Let $x + \frac{1}{x} = t$

for $x > 0,\ t \geq 2$

$$y = \frac{4t}{1+t^2} = \frac{4}{\frac{1}{t}+t}$$

for $t \geq 2,\ t + \frac{1}{t} \geq \frac{5}{2}$

$0 < y \leq \frac{8}{5}$ & $y = 0$ for $x = 0$

$\therefore$ Range of f(x) is $\left[0, \frac{8}{5}\right]$ Ans.

68. Range of $|\sin x|$ is $[0, 1]$ $\forall x \in R$

Range of $|\cos x + \sec x|$ is $[2, \infty)$ $\forall$ x such that $\cos x \neq 0$

Range of $\dfrac{x^2+2x+2}{|x+1|} = |x+1| + \dfrac{1}{|x+1|}$ is $[2, \infty)$ $\forall\ x + 1 \neq 0$ i.e. $x \neq -1$

Given that $f(x)_{min} = \left\{|\sin x|+|\cos x+\sec x|+\frac{x^2+2x+2}{|x+1|}\right\}_{min} = 4$

which is possible only if every term has its minimum value which can be obtained at $x = 0$ only
$\therefore$ the domain will be the set containing $x = 0$ and not containing $x = -1$.]

69.

(A) $f(x) = \left[x+\frac{1}{2}\right]+\left[x-\frac{1}{2}\right]+2[-x]$

clearly $f(x)$ is into as it will not take non integral values
now write $[x] = x - \{x\}$

$$f(x) = \left(x+\frac{1}{2}\right) - \left\{x+\frac{1}{2}\right\} + \left(x-\frac{1}{2}\right) - \left(x-\frac{1}{2}\right) + 2\left(-x-\{-x\}\right)$$

$$= -\left\{x+\frac{1}{2}\right\} - \left\{x-\frac{1}{2}\right\} - 2\{-x\}$$

this is periodic with period 1.
so function is periodic, many one and into $\Rightarrow$ **P, R, T**

(B) $f(x) = x^3 + x^2 + 3x + \sin x$
clearly onto and not periodic
now $f'(x) = 3x^2 + 2x + 3\cos x$

$$= \underbrace{3x^2+2x+2}_{\substack{D<0 \\ \text{so always +ve}}} + \underbrace{1+\cos x}_{\text{always +ve}}$$

so $f'(x)$ is always + ve so one-one, onto $\Rightarrow$ **Q, S**

(C) $f(x) = \frac{e^{|x|}-e^{-x}}{e^x+e^{-x}}$

now if x is < 0 then $f(x) = 0$
so many one

$$f(x) = \begin{cases} 0 & x \le 0 \\ \frac{e^x-e^{-x}}{e^x+e^{-x}} & x > 0 \end{cases}$$

so for $x > 0$, $f(x) = \frac{e^{2x}-1}{e^{2x}+1}$

now in $x > 0$

$$y = \frac{e^{2x}-1}{e^{2x}+1} = \frac{e^{2x}+1-2}{e^{2x}+1} = 1 - \frac{2}{e^{2x}+1}$$

so $1 < e^{2x} < \infty$
$2 < e^{2x} + 1 < \infty$

$$\frac{1}{2} > \frac{1}{e^{2x}+1} > 0$$

$1 > \frac{2}{e^{2x}+1} > 0$. So range is [0, 1) hence into

so many one, into $\Rightarrow$ **R, T**

(D) $f(x) = e^{\sin\{x\}} + \sin\left(\frac{\pi}{2}[x]\right)$

clearly into as $e^{\sin\{x\}}$ and $\sin\frac{\pi}{2}[x]$ both bounded

and periodic with period 4. So many one

i.e. Many one, periodic, into $\Rightarrow$ **P, R, T**

70. $f(x) = \{x\} + \left\{x + \left[\frac{x}{1+x^2}\right]\right\} + \left\{x + \left[\frac{x}{1+2x^2}\right]\right\} + \ldots\ldots + \left\{x + \left[\frac{x}{1+99x^2}\right]\right\}$

$= 100\{x\}$ $(\because \{x+m\} = \{x\}, m \in I)$

$f(\sqrt{3}) = 100\ (0.732) = 73.2$

$\left[f\left(\sqrt{3}\right)\right] = 73$ **Ans.**

71. Let $y = g(x) \Rightarrow x = g^{-1}(y)$

$y = a\ f\left(\frac{x}{a}+1\right)$

$\therefore\ f\left(\frac{x}{a}+1\right) = \frac{y}{a} \Rightarrow \frac{x}{a} + 1 = f^{-1}\left(\frac{y}{a}\right)$

$\Rightarrow x = a\left[f^{-1}\left(\frac{y}{a}\right) - 1\right]$

$\Rightarrow g^{-1}(x) = a\left(f^{-1}\left(\frac{x}{a}\right) - 1\right)$. **Ans.**

72. Also, $\sin^{-1}x^2$ is defined for $-1 \le x \le 1$ $\Rightarrow$ $3^{\sin^{-1}x^2}$ is defined for $x \in [-1, 1]$.

Now, $(7x + 1)$ is defined for $7x + 1 \ge 0$ with $7x + 1 \in N \cup \{0\} \Rightarrow x \in \left\{\frac{-1}{7}, 0, \frac{1}{7}, \frac{2}{7}, \ldots\ldots\right\}$

Also, $\frac{1}{\sqrt{x+1}}$ is defined for $x \in (-1, \infty)$.

So, domain of $f(x) = \left\{\frac{-1}{7}, 0, \frac{1}{7}, \frac{2}{7}, \frac{3}{7}, \frac{4}{7}, \frac{5}{7}, \frac{6}{7}, 1\right\}$.

Paragraph for question nos. 73 & 74

73 Let $y = \dfrac{x^2 - 5}{3 - x} \Rightarrow x^2 + yx - (3y + 5) = 0$

As, $x \in R$, so $D \geq 0 \Rightarrow y^2 + 4\,(3y + 5) \geq 0 \Rightarrow (y + 10)\,(y + 2) \geq 0$

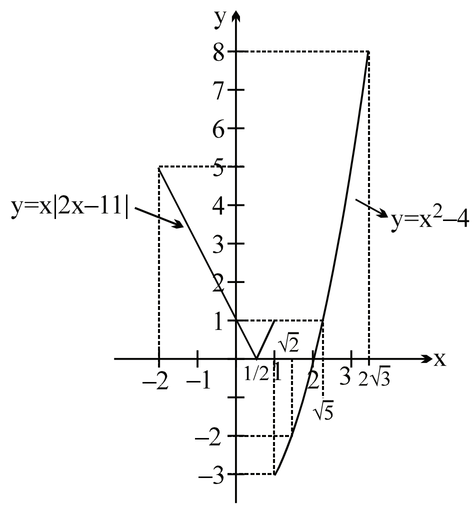
$\Rightarrow y \in (-\infty, -10] \cup [-2, \infty) =$ range of f(x).

74 Given, $y = \dfrac{x^2 - 5}{3 - x} \Rightarrow x^2 + yx - (3y + 5) = 0$

$\therefore\ x = \dfrac{-y \pm \sqrt{y^2 + 12y + 20}}{2} \Rightarrow f^{-1}(x) = \dfrac{-x - \sqrt{x^2 + 12x + 20}}{2}$

[**Note:** $f^{-1}(x) = \dfrac{-x + \sqrt{x^2 + 12x + 20}}{2}$ is rejected. Think!]

75. $D_{fof\,(x)} = [-2, 1] \cup \left[\sqrt{2}, 2\sqrt{3}\,\right]$



76.

(A) Clearly, $g\big(f(x)\big) = g(\sin x) = ln\,(\sin x)$

For domain, $0 < \sin x \leq 1$

Clearly, Range of $g\big(f(x)\big) = (-\infty, 0]$.

(B) As, $x^2 + (a - 1)\,x + 9 > 0\ \forall\ x \in R$, so

$D < 0 \Rightarrow (a - 1)^2 - 36 < 0 \Rightarrow (a - 1)^2 - (6)^2 < 0 \Rightarrow (a - 7)\,(a + 5) < 0 \Rightarrow -5 < a < 7.$

(C) Given, $f(x) = \left(2011 - x^{2012}\right)^{\frac{1}{2012}}$

$\Rightarrow\quad f\big(f(x)\big) = \left(2011 - \big(f(x)\big)^{2012}\right)^{\frac{1}{2012}} = \left(2011 - \left(2011 - x^{2012}\right)\right)^{\frac{1}{2012}} = x$

So, $f\big(f(x)\big) = x \Rightarrow f\big(f(2)\big) = 2$

(D) Given, $f : R \to R$ be $y = \dfrac{x^2 + 4x + 30}{x^2 - 8x + 18} = \dfrac{5}{3}$ (say)

$\Rightarrow\quad 3x^2 + 12x + 90 = 5x^2 - 40x + 90 \Rightarrow 0 = 2x^2 - 52x \Rightarrow 0 = 2x\,(x - 26)$

So, $x = 0, 26 \Rightarrow f(0) = f(26) = \frac{5}{3}$.

Hence f(x) is many-one function.

Also, $x^2 + 4x + 30 > 0 \ \forall \ x \in R$ and $x^2 - 8x + 18 > 0 \ \forall \ x \in R$

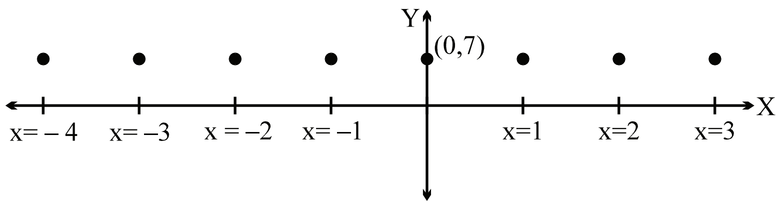
Hence, y can't be zero or negative for any $x \in R$. But co-domain is R.

Hence, f(x) is into function. **Ans.**

77. Clearly $f(x) = 7, \ \forall \ x \in I$

$\therefore \quad R_f = \{7\}$

The graph of f (x) is shown below.



Note : f is periodic function with period 1.

78. $f(x) = (x^2 + 5x + 4)(x^2 + 5x + 6)$

Put $x^2 + 5x = t$

$(t + 4)(t + 6) = t^2 + 10t + 24 = (t + 5)^2 - 1$

$\because$ minimum value of $f(x) = -1$, $R_f = [-1, \infty)$

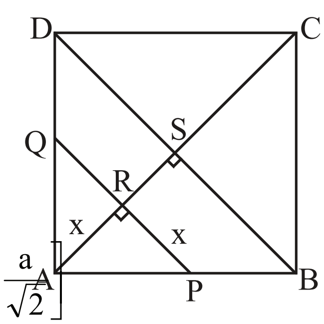
at $x^2 + 5x = -5$

$$\Rightarrow x = \frac{-5 \pm \sqrt{25-20}}{2} = \frac{-5 \pm \sqrt{5}}{2}$$

$$= \frac{-5 \pm 2.25}{2} = -3.625 \text{ or } -1.375 \text{ approx.}$$

Paragraph for Question no. 79 to 81

Sol. Two cases are possible for PQ w.r.t. diagonal CA

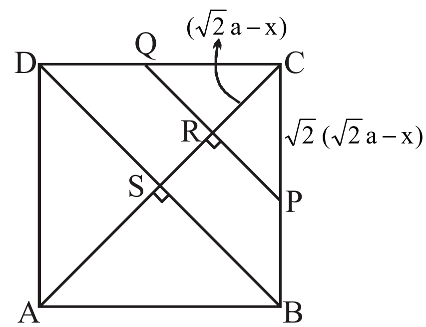


Case-I: When $x = AR \le AS$, i.e. $x \in \left[0, \frac{a}{\sqrt{2}}\right]$

Triangle ARP is an isosceles right angle triangle

hence $AP = \sqrt{2}\,x = AQ$

$$\Rightarrow \text{area of segment i.e. } f(x) = \frac{\left(\sqrt{2}x\right)^2}{2} = x^2,\ x \in \left[0, \frac{a}{\sqrt{2}}\right]$$



Case-II: When $x = AR \geq AS$ i.e. $x \in \left[\frac{a}{\sqrt{2}}, \sqrt{2}a\right]$

$\Rightarrow$ $CP = CQ = \sqrt{2}\,(\sqrt{2}\,a - x)$

area of segment $f(x) = a^2 - (\sqrt{2}\,a - x)^2,\ x \in \left[\frac{a}{\sqrt{2}}, \sqrt{2}a\right]$

i.e. $f : [0, \sqrt{2}\,a] \to [0, a^2]$

$$f(x) = \begin{cases} x^2, & 0 \leq x \leq \frac{a}{\sqrt{2}} \\ a^2 - \left(\sqrt{2}a - x\right)^2, & \frac{a}{\sqrt{2}} \leq x \leq a \end{cases}$$

when $0 \leq x \leq \frac{a}{\sqrt{2}}$, $f(x) = x^2,\ 0 \leq f(x) \leq \frac{a^2}{2}$

Put $x = f^{-1}(x) \Rightarrow \left(f^{-1}(x)\right)^2 = x,\ 0 \leq x \leq \frac{a^2}{2},\ 0 \leq f^{-1}(x) \leq \frac{a}{\sqrt{2}}$

$\Rightarrow f^{-1}(x) = \sqrt{x},\ 0 \leq x \leq \frac{a^2}{2}$

when $\frac{a}{\sqrt{2}} \leq x \leq \sqrt{2}\,a$, $f(x) = a^2 - (\sqrt{2}\,a - x)^2,\ \frac{a^2}{2} \leq f(x) \leq a^2$

Put $x = f^{-1}(x),\ x = a^2 - \left(\sqrt{2}a - f^{-1}(x)\right)^2,\ \frac{a^2}{2} \leq x \leq a^2,\ \frac{a}{\sqrt{2}} \leq f^{-1}(x) \leq \sqrt{2}\,a$

$\Rightarrow \left(\sqrt{2}\,a - f^{-1}(x)\right)^2 = a^2 - x$

$\Rightarrow \sqrt{2}\,a - f^{-1}(x) = \pm\sqrt{a^2 - x}$

$\Rightarrow f^{-1}(x) = \sqrt{2}\,a \mp \sqrt{a^2 - x}$

Neglecting positive sign since $\frac{a}{\sqrt{2}} \leq f^{-1}(x) \leq \sqrt{2}\,a$

$$\Rightarrow f(x) = \sqrt{2}\,a - \sqrt{a^2 - x} \Rightarrow f^{-1}(x) = \begin{cases} \sqrt{x}, & 0 \leq x \leq \frac{a^2}{2} \\ \sqrt{2}a - \sqrt{a^2 - x}, & \frac{a^2}{2} \leq x \leq a^2 \end{cases}$$

Hence domain of $f^{-1}(x)$ is $x \in [0, a^2]$

For $a = 2$

$$f(x) = \begin{cases} x^2, & 0 \leq x \leq \sqrt{2} \\ 4 - \left(2\sqrt{2} - x\right)^2, & \sqrt{2} \leq x \leq 2 \end{cases}$$

and $f^{-1}(x) = \begin{cases} \sqrt{x}, & 0 \le x \le 2 \\ 2\sqrt{2} - \sqrt{4-x}, & \sqrt{2} \le x \le 2 \end{cases}$

Domain of $\sqrt{f^{-1}(x) - f(x)}$ is when $f^{-1}(x) - f(x) \ge 0$

Case-I: $0 \le x \le \sqrt{2}$

$\sqrt{x} - x^2 \ge 0 \Rightarrow \sqrt{x}\ (1 - x^{3/2}) \ge 0 \Rightarrow 0 \le x \le 1$

Case-II: $\sqrt{2} \le x \le 2$

$f^{-1}(x) = \sqrt{x}$ i.e. $f^{-1}(x) \in \left[2^{1/4}, \sqrt{2}\right]$ and $f(x) = 4 - (2\sqrt{2} - x)^2$, $f(x) \in [2, 8(\sqrt{2} - 1)]$

Hence $f^{-1}(x) - f(x) \ge 0\ \forall\ x \in \phi$.

Hence domain of $\phi(x) = \sqrt{f^{-1}(x) - f(x)}$ is $x \in [0, 1]$ for $a = 2$.

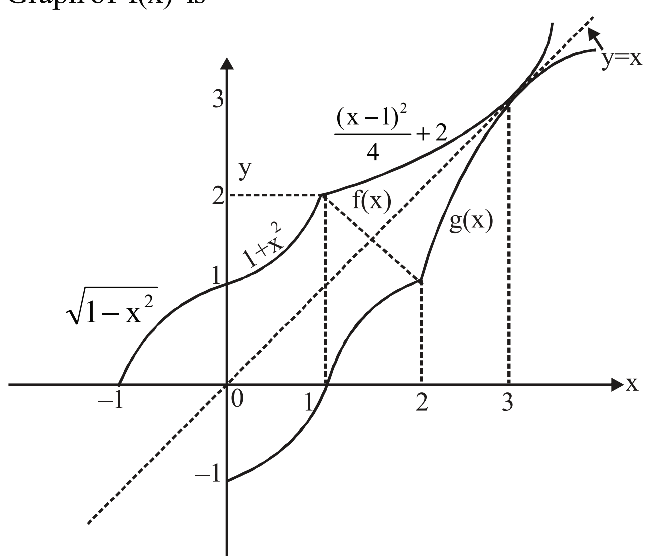
The equation $f(x) - f'(x) = 0$ has exactly three solutions $f(x) = \begin{cases} x^2, & 0 \le x \le \dfrac{a}{\sqrt{2}} \\ a^2 - \left(\sqrt{2}a - x\right)^2, & \dfrac{a}{\sqrt{2}} \le x \le a \end{cases}$

$$f^{-1}(x) = \begin{cases} \sqrt{x}, & 0 \le x \le \dfrac{a^2}{2} \\ \sqrt{2}a - \sqrt{a^2 - x}, & \dfrac{a^2}{2} \le x \le a^2 \end{cases}$$

Exactly three solutions exists if $\dfrac{a^2}{2} = \dfrac{a}{\sqrt{2}}$ i.e. $a = \sqrt{2}$. **Ans.**]

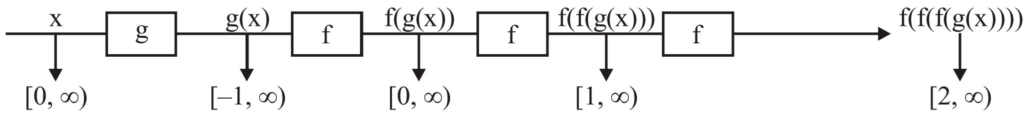
Paragraph for questions nos. 82 to 84

Graph of $f(x)$ is



82 $\quad y = f\left(f\left(f\left(g(x)\right)\right)\right)$

From the graph it is clear that domain of $g(x)$ is $[0, \infty)$ and range of $g(x)$ is $[-1, \infty)$



Hence, range of $f(f(f(g(x))))$ is $[2, \infty)$

83. For the domain of $y = g(g(g(f(x))))$

$g(g(f(x))) \in [0, \infty)$

$\Rightarrow g(f(x)) \in [1, \infty)$

$\Rightarrow f(x) \in [2, \infty)$

$\Rightarrow x \in [1, \infty)$

84. The solution of equation $f(x) = g(x)$ is same as solution of the equation $f(x) = x$.

$\dfrac{(x-1)^2}{4} + 2 = x \Rightarrow x^2 - 2x + 1 = 4(x-2)$

$\Rightarrow x^2 - 6x + 9 = 0 \Rightarrow (x-3)^2 = 0 \Rightarrow x = 3.$

85. Given, $f(x) = \sin \log_e \left(\dfrac{\sqrt{4-x^2}}{1-x}\right)$ ....(1)

Domain of f: For f (x) to be defined

(i) $4 - x^2 \ge 0$

$\Rightarrow$ $-2 \le x \le 2$ ....(2)

(ii) $\dfrac{\sqrt{4-x^2}}{1-x} > 0$

$\Rightarrow$ $1 - x > 0$

$\Rightarrow$ $-\infty < x < 1$ ....(3)

from (2) and (3), $-2 \le x < 1$

$\therefore$ domain of $f = [-2, 1)$ **Ans**

Range of f:

$\lim\limits_{x \to -2+0} \log_e \left(\dfrac{\sqrt{4-x^2}}{1-x}\right) = -\infty$ and $\lim\limits_{x \to 1-0} \log_e \left(\dfrac{\sqrt{4-x^2}}{1-x}\right) = \infty$

also $\log_e \left(\dfrac{\sqrt{4-x^2}}{1-x}\right)$ is a continuous function in domain [–2, 1),

therefore it attains all values between $-\infty$ and ∞.

$\therefore$ $-\infty < \log_e \left(\dfrac{\sqrt{4-x^2}}{1-x}\right) < \infty$

$\Rightarrow$ $-1 \le \log_e \left(\dfrac{\sqrt{4-x^2}}{1-x}\right) \le 1$s $\therefore$ range of $f = [-1, 1]$ **Ans.**

86.

(A) We have $f(x) = \dfrac{\sec x}{\cos x} - \dfrac{\tan x}{\cot x}$, $g(x) = \dfrac{\cos x}{\sec x} + \dfrac{\sin x}{\operatorname{cosec} x}$

Clearly both $f(x)$ and $g(x)$ are identical functions as $x \neq \dfrac{k\pi}{2}$ $\forall$ $k \in I$.

(B) As $x^2 - 4x + 5 = (x-2)^2 + 1 > 0$
Hence $f(x) = 1$ $\forall$ $x \in R$.

Also $\cos^2 x + \sin^2\left(x + \dfrac{\pi}{3}\right) > 0$

Hence $g(x) = 1$ $\forall$ $x \in R$.
$\Rightarrow$ f(x) and g(x) are identical.

(C) $f(x) = e^{\ln(x^2+3x+3)}$
As $x^2 + 3x + 3 = \left(x + \dfrac{3}{2}\right)^2 + \dfrac{3}{4} > 0$ $\forall$ $x \in R$.

Hence $f(x) = x^2 + 3x + 3$ $\forall$ $x \in R$.
$\Rightarrow$ f(x) is identical to g(x).

(D) We have $f(x) = \dfrac{\sin x}{\sec x} + \dfrac{\cos x}{\operatorname{cosec} x}$, $g(x) = \dfrac{2\cos^2 x}{\cot x}$

Clearly both f(x) and g(x) are identical functions as $x \neq \dfrac{k\pi}{2}$ $\forall$ $k \in I$.

87. $f(x) = (x - 5a)^2 + 5 - a$. $\Rightarrow$ $5 - a = 1$ $\Rightarrow$ $a = 4$.

88. Product of two integers is prime if one of them is 1.

now $\left[x - \dfrac{1}{2}\right]\left[x + \dfrac{1}{2}\right]$ is to be prime

Case-I: Let $\left[x - \dfrac{1}{2}\right] = 1$ and $\left[x + \dfrac{1}{2}\right] = 2$

$\therefore$ $1 \le x - \dfrac{1}{2} < 2$ and $2 \le x + \dfrac{1}{2} < 3$; $\dfrac{3}{2} \le x < \dfrac{5}{2}$ and $\dfrac{3}{2} \le x < \dfrac{5}{2}$

Hence $x \in \left[\dfrac{3}{2}, \dfrac{5}{2}\right)$ ....(1)

Case-II: Let $\left[x - \dfrac{1}{2}\right] = -1$ and $\left[x + \dfrac{1}{2}\right] = -2$ (we will find no solution)

Case-III: Let $\left[x + \dfrac{1}{2}\right] = 1$ and $\left[x - \dfrac{1}{2}\right] = 2$ (we will find no solution)

Case-IV: Let $\left[x+\frac{1}{2}\right] = -1$ and $\left[x-\frac{1}{2}\right] = -2$

$\therefore \quad -1 \le x + \frac{1}{2} < 0 \quad$ and $\quad -2 \le x - \frac{1}{2} < -1$

$-\frac{3}{2} \le x < -\frac{1}{2} \quad$ and $\quad -\frac{3}{2} \le x < -\frac{1}{2}$

Hence $x \in \left[-\frac{3}{2}, -\frac{1}{2}\right)$ ....(2)

from (1) and (2) $\quad x \in \left[-\frac{3}{2}, -\frac{1}{2}\right) \cup \left[\frac{3}{2}, \frac{5}{2}\right)$

$\therefore \quad x_1^2 + x_2^2 + x_3^2 + x_4^2 = \frac{9}{4} + \frac{1}{4} + \frac{9}{4} + \frac{25}{4} = \frac{44}{4} = 11.$ **Ans.**

89. Given, $f(x) = \log\left(ax^3 + (a+b)x^2 + (b+c)x + c\right)$

For $f(x)$ to be defined

$ax^3 + (a + b)x^2 + (b + c)x + c > 0$

$\Rightarrow \quad (ax^3 + bx^2 + cx) + (ax^2 + bx + c) > 0$

$\Rightarrow \quad x(ax^2 + bx + c) + ax^2 + bx + c > 0$

$\Rightarrow \quad (x + 1)(ax^2 + bx + c) > 0$

$\Rightarrow \quad x + 1 > 0$

[$\because$ $b^2 - 4ac < 0$ and $a > 0$, $\therefore$ $ax^2 + bx + c > 0$ for all real x]

$\Rightarrow \quad x > -1$

hence domain of $f = (-1, \infty)$ **Ans.**

90. $f(x) \quad = \frac{1}{2} \ln\left(\sqrt{\sqrt{x^2+1}+x} \pm \sqrt{\sqrt{x^2+1}-x}\right)^2$

$= \frac{1}{2} \ln\left\{\sqrt{x^2+1} + x + \sqrt{x^2+1} - x \pm 2\sqrt{\left(\sqrt{x^2+1}\right)^2 - x^2}\right\}$

$= \frac{1}{2} \ln\left\{2\sqrt{x^2+1} \pm 2\right\}$

An even function hence neither one-one nor onto.

91. $y = \dfrac{40}{x^2 - 4x - 5}$

$\Rightarrow$ $x^2 - 4xy - 5y - 40 = 0$

$\because$ x is real $\therefore$ $D \geq 0$

$16y^2 + 4y(5y + 40) \geq 0$

$9y^2 + 40y \geq 0 \Rightarrow y(9y + 40) \geq 0$

$\therefore$ $y \in \left(-\infty, -\dfrac{40}{9}\right] \cup [0, \infty)$ but $y \neq 0$

$\therefore$ $y \in \left(-\infty, -\dfrac{40}{9}\right] \cup (0, \infty)$

Integers which are not included in the range are – 4, – 3, – 2, – 1, 0

Now, $\sum_{i=1}^{5} {}^{5-a_i}C_{i-1} = {}^5C_0 + {}^6C_1 + {}^7C_2 + {}^8C_3 + {}^9C_4$

$= {}^{10}C_4 = {}^{10}C_6$

$\therefore$ $24y = 10 + 4$ or $10 + 6$

$= 14$ or 16

92. $f(1) = a + b + c$
$f(-2) = 4a - 2b + c$
hence $f(1) - f(-2) = 3(b - a)$

$$E = \frac{a+b+c}{b-a} = \frac{3f(1)}{f(1)-f(-2)} = \frac{3}{1-\frac{f(-2)}{f(1)}}$$

Hence $E_{min.}$ occurs when $f(-2) = 0$
Hence $E_{min.} = 3$ **Ans.**

Aliter: $f(x) \geq 0 \ \forall \ x$
$\Rightarrow \ a > 0$ and $b^2 - 4ac \leq 0$

$\therefore \quad c \geq \frac{b^2}{4a}$

$\therefore \quad a + b + c \geq a + b + \frac{b^2}{4a}$

$$a + b + c \geq \frac{4a^2 + 4ab + b^2}{4a}$$

since $b - a > 0$

$$\therefore \quad \frac{a+b+c}{b-a} \geq \frac{4a^2+4ab+b^2}{4a(b-a)}$$

$$= \frac{(2a+b)^2}{4a(b-a)} = \frac{\left(\frac{3a+(b-a)}{2}\right)^2}{a(b-a)}$$

Using A.M. $\geq$ G.M. on $3a$ and $b - a$

$$\geq \frac{3a(b-a)}{a(b-a)}$$

$$= 3$$

Equality holds when $3a = b - a$
$\therefore \quad 4a = b$
$\Rightarrow \quad b = c = 4a$. **Ans.**

93. $f(1) = 1 + b + c$
$f(5) = 25 + 5b + c$
$f(9) = 81 + 9b + c$
Now $f(1) - 2f(5) + f(9) = 32$ ......(1)
Since $|f(x)| \leq 8$ for all x in the interval $[1, 9]$, we have
$|f(1)| \leq 8; \quad |f(5)| \leq 8; \quad |f(9)| \leq 8$
Now from (1),
$32 = |f(1) - 2f(5) + f(9)| \leq |f(1)| + 2|f(5)| + |f(9)| \leq 32$
$\Rightarrow \quad f(1) = f(9) = 8; \ f(5) = -8$

i.e. $b + c + 1 = 8;\ 9b + c + 81 = 8;\ 5b + c + 25 = -8$

The only pair (b, c) that satisfies the condition when $b = -10$ and $c = 17$

$\Rightarrow$ 1 ordered pair. **Ans.**

94. $$\frac{x^4 - x^2}{x^6 + 2x^3 - 1} = \frac{x - \frac{1}{x}}{x^3 + 2 - \frac{1}{x^3}}$$

$$= \frac{x - \frac{1}{x}}{\left(x - \frac{1}{x}\right)^3 + 1 + 1 + 3\left(x - \frac{1}{x}\right)} \le \frac{x - \frac{1}{x}}{3\left(x - \frac{1}{x}\right) + 3\left(x - \frac{1}{x}\right)}$$

$$= \frac{1}{6}$$

95. $x[x] - 5x + 2x - 2[x] + 6 = 0$

$[x]\,(x - 2) - 3\,(x - 2) = 0$

$([x] - 3)(x - 2) = 0 \Rightarrow x = 2$ or $[x] = 3 \Rightarrow x \in [3, 4) \cup \{2\}$

$(a - 3)x^2 + 2\,(a + 3)x - 8a \le 0$

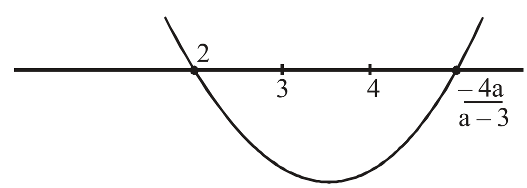
$ax^2 - 3x^2 + 2ax + 6x - 8a \le 0$

$ax^2 - 2ax - 3x^2 + 6x + 4ax - 8a \le 0$

$ax(x - 2) - 3x(x - 2) + 4a(x - 2) \le 0$

$(x - 2)\ (x(a - 3) + 4a) \le 0$

Case-I $a - 3 > 0 \Rightarrow a > 3$

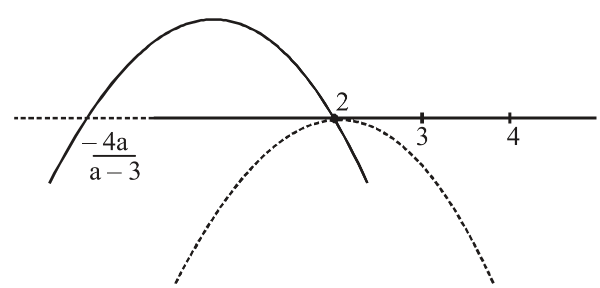


$$\frac{-4a}{a-3} \ge 4 \Rightarrow -4a \ge 4a - 12$$

$$8a \le 12 \Rightarrow a \le \frac{3}{2}$$

No solution.

Case-II $a - 3 < 0$

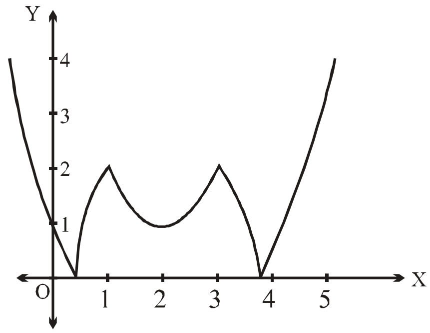


$$\frac{-4a}{a-3} \le 2 \Rightarrow -4a \ge 2a - 6$$

$$6 \ge 6a \Rightarrow a \le 1$$

$\therefore$ greatest integral value of a is 1. **Ans.**

96. See graph $y = f(x) = \left|\,|x^2 - 4x + 3| - 2\,\right|$, $y = m$ is a horizontal line with intersection points, from which the x-values have different signs, only if $m > 2$.



97. $\because$ $A + B + C = \pi$

$\therefore$ $\sin(B + C) = \sin(\pi - A) = \sin C$

$\sin(C + A) = \sin B$ and $\sin(A + B) = \sin C$

$\therefore$ $f(x) = (\sin B - \sin A)x^2 + (\sin A - \sin C)x + (\sin C - \sin B)$

$\because$ Graph of $y = f(x)$ touches the x-axis.

$\therefore$ $f(x) = 0$ will have real and equal roots.

$\because$ $x = 1$ is a root

$\therefore$ other root is also $x = 1$

$\therefore$ product of roots = 1

$\Rightarrow$ $\dfrac{\sin C - \sin B}{\sin B - \sin A} = 1$

$\Rightarrow$ $2 \sin B = \sin A + \sin C$

$\Rightarrow$ $2b = a + c$

$\therefore$ a, b, c are in A.P. **Ans.**

98. We have $f(x) = 3[2x] - 2[3x] = 3(2x - \{2x\}) - 2(3x - \{3x\}) = 2\{3x\} - 3\{2x\}$
Clearly $f(x)$ is periodic with period 1.

99. Consider $F(x) = \cot\left(\cos^{-1}(|\sin x| + |\cos x|\right) + \sin^{-1}\left(-|\cos x| - |\sin x|\right)$

But $|\sin x| + |\cos x| \in [1, \sqrt{2}]$

$\therefore F(x) = \cot(\cos^{-1}(1) + \sin^{-1}(-1)) = \cot\left(0 - \dfrac{\pi}{2}\right) = 0$

$= g(3)$ (As $F(x) = 0, \forall x \in D_F$)

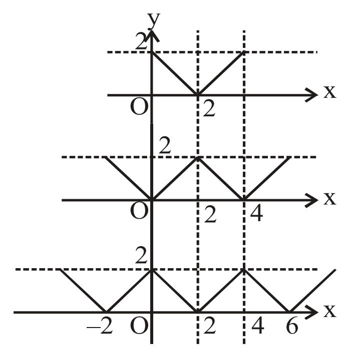
100. We have $f(x) + 2f(1 - x) = x^2 + 1, \ \forall x \in R$ ....(1)
Replacing x by $1 - x$ in (1), we get
$f(1 - x) + 2f(x) = (1 - x)^2 + 1$ ....(2)

Now, $2 \times 2 - (1) \Rightarrow f(x) = \dfrac{x^2 - 4x + 3}{3}$

Clearly $\sqrt{f(x)}$ is not defined when $f(x) < 0$

$\Rightarrow x \in (1, 3)$

101. For domain
$e^{g(x)} = e - e^x$
$g(x) = ln(e - e^x)$
$\therefore \ e - e^x > 0$
$e > e^x \Rightarrow \ x < 1$
$\therefore$ domain is $(-\infty, 1)$
For range
argument of ln varies from $(0, e)$
range $(ln\, e, ln\, 0)$
i.e. $(-\infty, 1)$ **Ans.**

102.


Graph of $f(x) = |x - 2|$; shift the graph on x-axis by 2 units.

Graph of $f\big(f(x)\big) = \big||x-2|-2\big|$

Graph of $f\Big(f\big(f(x)\big)\Big) = \Big|\big||x-2|-2\big|-2\Big|$

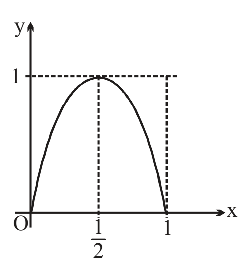
Obviously, if the equation $g(x) = k$, $k \in (0, 2)$ has 8 distinct solutions, then $n = 4$. **Ans.**

103. $f(x + y) = 2^x f(y) + 4^y f(x)$ ....(1)
Also, $f(y + x) = 2^y f(x) + 4^x f(y)$ ....(2)
Now, $2^x f(y) + 4^y f(x) = 2^y f(x) + 4^x f(y)$

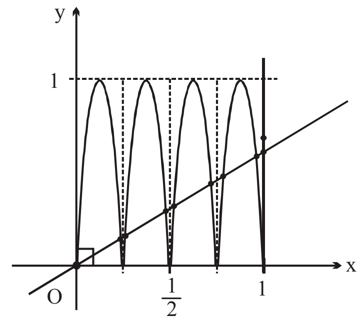
$$\Rightarrow \quad \frac{f(x)}{4^x - 2^x} = \frac{f(y)}{4^y - 2^y} = k$$

$\Rightarrow \quad f(x) = k \cdot (4^x - 2^x)$
$\Rightarrow \quad f(1) = 2 = k \cdot 2 \Rightarrow k = 1$
$f(x) = 4^x - 2^x$
$f'(x) = 4^x\, ln\, 4 - 2^x\, ln\, 2$
$f'(2) = 16\, ln\, 4 - 4\, ln\, 2 = 28\, ln\, 2$
$\Rightarrow \quad k = 28.$ **Ans.**

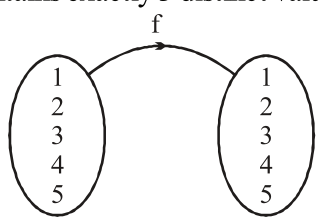
104. $f(x) = 4x(1 - x), 0 \le x \le 1$



$y = f(f(f(x))),\ 0 \le x \le 1$



105. Since range contains exactly 3 distinct values



$\therefore \quad {}^5C_3\left[\frac{5!}{1!\,1!\,3!}\frac{1}{2!}+\frac{5!}{1!\,2!\,2!}\frac{1}{2!}\right]\times 3! = 1500.$ **Ans.**

106. $f(x) = a\sin x + c < 0 \quad \forall x \in R$

$\therefore \quad a\sin x < -c$

$\Rightarrow \sin x > \frac{-c}{a} \quad \forall x \in R$

$\therefore \frac{-c}{a} < -1 \Rightarrow \frac{c}{a} > 1$

$\Rightarrow \quad c < a.$ **Ans.**

107. when $p = \pi/2$ then $D^r \to \cos x + \sin x \Rightarrow \pi/2$ can not be the period

108. (A) $\frac{1}{g(x)} = \frac{1}{\ln x / x}$; $f(x) = \frac{x}{\ln x}$ $\qquad x > 0,\ x \ne 1$ for both

(B) $\frac{1}{f(x)} = \frac{1}{x/\ln x}$; $g(x) = \frac{\ln x}{x}$

$\Rightarrow \quad \frac{1}{f(x)}$ is not defined at $x = 1$ but $g(1) = 0$

(C) $f(x) \cdot g(x) = \frac{x}{\ln x} \cdot \frac{\ln x}{x} = 1$ $\qquad$ if $x > 0,\ x \ne 1 \Rightarrow$ N. I.

(D) $\frac{1}{f(x).g(x)} = \frac{1}{\frac{x}{\ln x} \cdot \frac{\ln x}{x}} = 1$ $\qquad$ only for $x > 0$ and $x \ne 1$]

109. $f(x) = \sin^2 x + (1 - \sin^2 x)^2 + 2$

$= 3 - \sin^2 x + \sin^4 x$

$= 3 - \sin^2 x \cos^2 x$

$= 3 - \dfrac{\sin^2 2x}{4}$

$\Rightarrow \quad T_1 = \dfrac{\pi}{2}$, and $T_2 = \dfrac{\pi}{2}$

110. (A) $f(x) = x^4 + 2x^3 - x^2 + 1 \rightarrow$ A polynomial of degree even will always be into

say $\quad f(x) = a_0 x^{2n} + a_1 x^{2n-1} + a_2 x^{2n-2} + ... + a_{2n}$

$$\underset{x \to \pm\infty}{\text{Limit}}\ f(x) = \underset{x \to \pm\infty}{\text{Limit}}\ [x^{2n}\left(a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + ... + \frac{a_{2n}}{x^{2n}}\right) = \begin{bmatrix} \infty & \text{if } a_0 > 0 \\ -\infty & \text{if } a_0 < 0 \end{bmatrix}$$

Hence it will never approach $\infty / -\infty$

(B) $f(x) = x^3 + x + 1$

$\Rightarrow \quad f'(x) = 3x^2 + 1$

$\Rightarrow$ injective as well as surjective

(C) $f(x) = \sqrt{1 + x^2}$ – neither injective nor surjective (minimum value = 1)

$f(x) = x^3 + 2x^2 - x + 1$

$\Rightarrow \quad f'(x) = 3x^2 + 4x - 1$

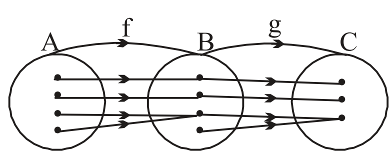
$\Rightarrow\ D > 0$

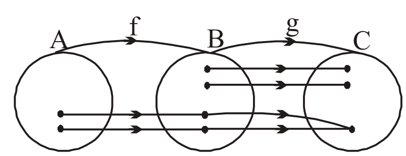
Hence $f(x)$ is surjective but not injective.

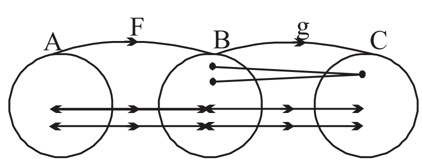
111. (A) We have $\quad f : A \rightarrow B$,

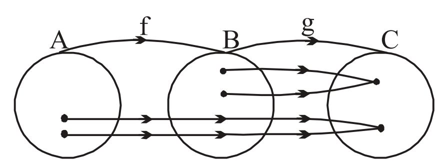
$g : B \rightarrow C$

and $\quad gof : A \rightarrow C$

(A) 

(B) 

(C) 

(D) 

]

clearly gof is one-one but g is many one function

112. for into range $\subset R$

we know every odd degree polynomial is onto and hence a polynomial degree even

coefficient of x^3 is zero if $a = 2, -2, 1$ or -1

if $a = 2$, $f(x)$ is a quadratic polynomial

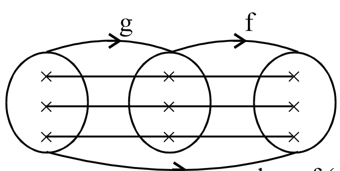
$\Rightarrow$ into

if $a = -2$, $f(x)$ is constant which is into
if $a = 1$, $f(x)$ is linear hence onto
if $a = -1$, $f(x)$ is constant hence into
hence $a \in \{2, -2, -1\}$

113. Replace x by $\frac{1}{x}$ and by solving, we have

$$f(x) = \frac{x(1-\alpha x)}{(x+1)(1-\alpha^2)}$$

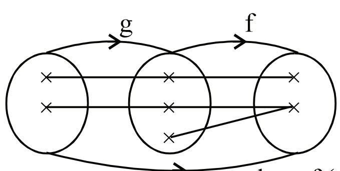
$\because$ x and $f(x) > 0$
$\therefore$ $\alpha < 0$ and $1-\alpha^2 > 0$
$\therefore$ $\alpha \in (-1, 0)$ Ans

114. 

here f (x) is one–one

figure-1

If fog is one-one then g (x) must be one-one and f (x) can be one-one and can be many one



here f (x) is Many–one

figure-2

If g (x) is onto then f (x) must be one-one (in figure-2, g (x) is not onto) and
If g (x) is into then f (x) may be one-one; may be many-one

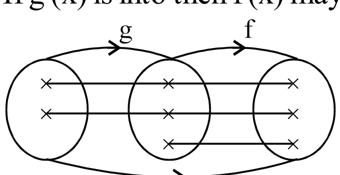
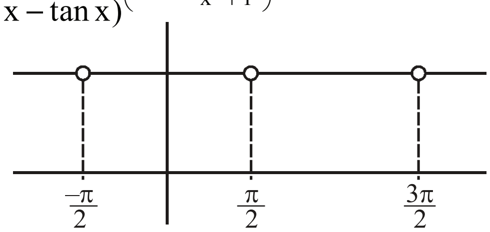


figure-2

g (x) is into and f (x) is one-one and in figure-2 also g (x) is into but f (x) is Many-one.

115. $$f(x) \,.\, g(x) = (\sec x + \tan x)^{\left(\{x\} + \frac{\sin 2\pi x}{x^2+1}\right)} \cdot (\sec x - \tan x)^{\left(\{x\} + \frac{\sin 2\pi x}{x^2+1}\right)}$$

$$= (\sec^2 x - \tan^2 x)^{\left(\{x\} + \frac{\sin 2\pi x}{x^2+1}\right)}$$



$= 1 \qquad x \in R - \{(2n+1)\frac{\pi}{2}\}, n \in I$

$\therefore$ h (x) is periodic with period π. Ans.

116. $y = \dfrac{4x\,(x^2+1)}{x^2+(x^2+1)^2} = \dfrac{4\left(x+\frac{1}{x}\right)}{1+\left(x+\frac{1}{x}\right)^2}$

Let $x + \frac{1}{x} = t$

for $x > 0,\ t \ge 2$

$y = \dfrac{4t}{1+t^2} = \dfrac{4}{\frac{1}{t}+t}$

for $t \ge 2,\ t + \frac{1}{t} \ge \frac{5}{2}$

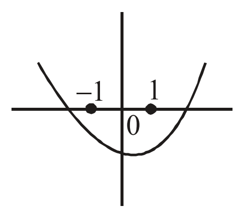
$0 < y \le \frac{8}{5}$ & $y = 0$ for $x = 0$

$\therefore$ Range of f(x) is $\left[0, \frac{8}{5}\right]$ Ans.

Paragraph for question nos. 117 to 119

Sol.

117. Given $f(x) = (x + 2a)(x + a - 4) = x^2 + (3a - 4)x + 2a(a - 4)$.



$\left.\begin{matrix} f(-1) < 0 \\ f(1) < 0 \end{matrix}\right\}$

on solving, we get $\boxed{\frac{1}{2} < a < 3}$

118. $g(x) = k(x^2 + x) + 3k + x > -3 \ \forall\ x.$

$\Rightarrow \quad k(x^2 + x) + 3k + x + 3 > 0 \ \forall\ x.$

or $\quad k x^2 + (k + 1)x + (3k + 3) > 0 \ \forall\ x.$

$\left.\begin{matrix} k > 0 \\ D < 0 \end{matrix}\right\} n$

here $D = (k + 1)^2 - 4k \cdot 3(k + 1) < 0$

$\Rightarrow \quad (k + 1)(k + 1 - 12k) < 0$

$(k + 1)(11k - 1) > 0$

$\Rightarrow \quad k < -1 \text{ or } k > \frac{1}{11}$

$\Rightarrow \quad \boxed{k > \frac{1}{11}} \quad (\because k > 0)$

119. Given, $(1-\sin\theta)x^2 + 2(1-\sin\theta)x - 3\sin\theta = 0$ has both roots complex, then $D < 0$

$4(1-\sin\theta)^2 + 4(1-\sin\theta)3\sin\theta < 0$

$(1-\sin\theta)(1+2\sin\theta) < 0$

$\underbrace{(\sin\theta - 1)}_{(-)\text{ve number}}(2\sin\theta + 1) > 0$

$\Rightarrow \quad 2\sin\theta + 1 < 0$

$\boxed{\sin\theta < -\frac{1}{2}} \quad \Rightarrow \theta \in \left(\frac{7\pi}{6}, \frac{11\pi}{6}\right)$

120. $f(x) = \sqrt{\sin^4 x + 4\cos^2 x} - \sqrt{\cos^4 x + 4\sin^2 x}$

$= \sqrt{\sin^4 x + 4 - 4\sin^2 x} - \sqrt{\cos^4 x + 4 - 4\cos^2 x}$

$= 2 - \sin^2 x - (2 - \cos^2 x)$

$= \cos^2 x - \sin^2 x = \cos 2x$

$g(\sin 2t) = \sin t + \cos t$

$(g(\sin 2t))^2 = 1 + \sin 2t$

$\Rightarrow g(\sin 2t) = \sqrt{1 + \sin 2t}$

$\therefore \ g(x) = \sqrt{1+x} \quad -1 \le x \le 1$

Now, $g(f(x)) = g(\cos 2x) = \sqrt{1 + \cos 2x}$

$= \sqrt{2}\,|\cos x|$

$\therefore$ Range of $g(f(x))$ is $[0, \sqrt{2}]$

$\Rightarrow a^2 + b^2 = 2$ Ans.

121. $y = \cos x\left(\sin x + \sqrt{\sin^2 x + \frac{1}{2}}\right)$

$(y\sec x - \sin x)^2 = \sin^2 x + \frac{1}{2}$

$y^2(1 + \tan^2 x) - 2y\tan x = \frac{1}{2}$

$y^2\tan^2 x - 2y\tan x + y^2 - \frac{1}{2} = 0$

$\tan x \in R \qquad \therefore \ D \ge 0$

$4y^2 - 4y^2\left(y^2 - \frac{1}{2}\right) \geq 0$

$1 - \left(y^2 - \frac{1}{2}\right) \geq 0$

$y^2 - \frac{3}{2} \leq 0 \Rightarrow \left(y - \frac{\sqrt{3}}{2}\right)\left(y + \frac{\sqrt{3}}{2}\right) \leq 0$

$y \in \left[-\sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}}\right]$

Hence number of integers are 3 i.e. $\{-1, 0, 1\}$.

122. $3 - x \geq 0$

$\Rightarrow \quad 3 \geq x$

$\Rightarrow \quad x \leq 3$

$2 - \sqrt{3-x} \geq 0$

$\Rightarrow \quad 4 \geq 3 - x$

$\Rightarrow \quad x \geq -1$

and $\quad 1 \geq \sqrt{2-\sqrt{3-x}}$

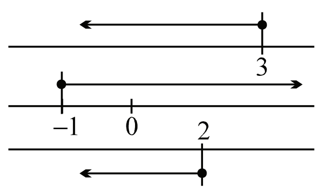
$\Rightarrow \quad \sqrt{3-x} \geq 1$

$\Rightarrow \quad 3 - x \geq 1$

$2 \geq x$

$\Rightarrow \quad x \leq 2$

hence $\quad x \in [-1, 2]$



123. We have, $f(x) = \{\sin x\} + \{\cos x\}$

So, $\quad R_f = [0, 2)$. **Ans.**

Paragraph for question nos. 124 to 126

Sol.

124. Given, $f(x) = 0 \Rightarrow [x]^2 - [x] - 6 = 0$

$\Rightarrow ([x]-3)([x]+2) = 0$

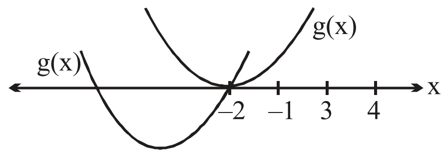
$\Rightarrow [x] = -2$ or 3.

So, $\quad x \in [-2, -1) \cup [3, 4) \equiv A$.

125. Every solution of set A satisfies the inequality $g(x) \geq 0$

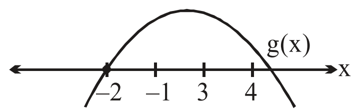
$3kx^2 + 2x + 4(1 - 3k) \geq 0 \quad \forall\ x \in A$

Case-1: If $k > 0$ here, $g(-2) = 0$



$$\frac{b}{2a} \leq -2 \Rightarrow \frac{-2}{2 \cdot 3k} \leq -2 \Rightarrow \frac{1}{3k} \geq 2 \Rightarrow k \leq \frac{1}{6}$$

$$k \in \left(0, \frac{1}{6}\right]$$



Case-2: If $k < 0$

$f(4) \geq 0$

$$48k + 8 + 4(1 - 3k) \geq 0 \Rightarrow 12 + 36k \geq 0 \Rightarrow k \geq \frac{-1}{3}$$

$$\therefore \quad k \in \left[\frac{-1}{3}, 0\right)$$

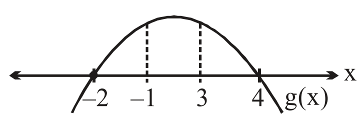
Case-3 : For $k = 0$, $2x + 4 \geq 0 \Rightarrow x \geq -2$

$\therefore$ $k = 0$ is also the solution.

$$\therefore \quad k \in \left[\frac{-1}{3}, \frac{1}{6}\right] = [a, b]$$

$$\therefore \quad 6b - 3a = 1 - (-1) = 2.$$

126. $k = \dfrac{-1}{3}$,



$g(x) = -x^2 + 2x + 8$

Range of $g(x)$ is

when $x \in A$

$A \equiv [-2, -1) \cup [3, 4)$

$[0, 5)$. **Ans.**

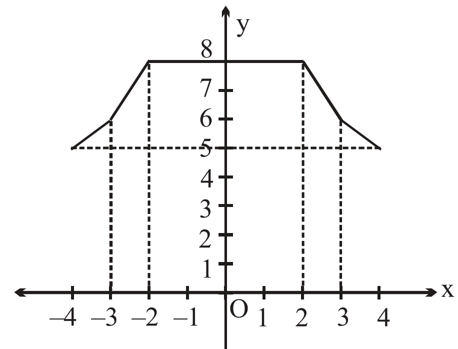
127. Let $x = \tan\theta$

$$f(q) = \frac{a\tan\theta + b}{\sqrt{1 + \tan^2\theta}}, \ \theta \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$$

$$= \frac{a\tan\theta + b}{\sec\theta}, \ \theta \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$$

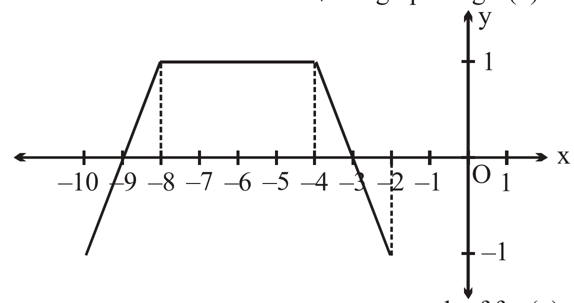
$$= a\sin\theta + b\cos\theta, \ \theta \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$$

Maximum $f(\theta) = \sqrt{a^2 + b^2} = 3 \Rightarrow a^2 + b^2 = 9$. **Ans.**



graph of gof(x)

128. $\text{gof}(x) = \begin{cases} x+9 & [-4, -3) \\ 2x+12 & [-3, -2) \\ 8 & [-2, 2) \\ 12-2x & [2, 3) \\ 9-x & [3, 4] \end{cases}$



graph of fog(x)

$\text{fog}(x) = \begin{cases} x+9 & [-10, -8) \\ 1 & [-8, -4) \\ -3-x & [-4, -2] \end{cases}$

129. $g\big(f(x)\big) = \begin{cases} 2x+4 & \left(0, \dfrac{1}{2}\right) \\ 20 & \left[\dfrac{1}{2}, 1\right) \\ 24 & \left[1, \dfrac{4}{3}\right) \\ 28 & \left[\dfrac{4}{3}, \dfrac{5}{3}\right) \\ 32 & \left[\dfrac{5}{3}, 2\right) \end{cases}$

130. Domain of f(x) is (–3, 4] and range of f(x) is [0, 3]
Domain of g(x) is $(3, \infty)$ and range of g(x) is $(13, \infty)$
Now, $\text{gof}(x) = g\big(f(x)\big)$
Range of f(x) $\neq$ is not an element of domain of g(x)
Hence, gof(x) is not defined.
Simililarly $\text{fog}(x) = f[g(x)]$
Range of g(x) $\neq$ is not an element of domain of f(x)
Hence, fog(x) is not defined. **Ans.**

131. $\because \quad f(x) = \dfrac{x^3}{3} + \dfrac{p}{2}x^2 + qx + 10$

$\Rightarrow \quad f'(x) = x^2 + px + q$

for f(x) to be one one

$f'(x) \geq 0, \ \forall x \in R$

$\therefore \quad p^2 - 4q \leq 0$

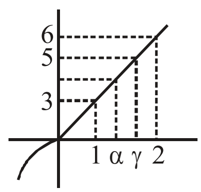
p	q
1	1, 2, 3, 4, 5
2	1, 2, 3, 4, 5
3	3, 4, 5
4	4, 5

$\therefore$ number of ordered pairs, n = 15

$\therefore$ number of divisors of n = 4. **Ans.**

132. Range : (0, 1)

133. $\alpha^3 - 3\alpha^2 + 5\alpha = 4$



$\beta^3 + 3\beta^2 + 5\beta = -5$

$(\beta)^3 - 3(-\beta)^2 + 5(-\beta) = 5$

$-\beta = \gamma$

$\gamma^3 - 3\gamma^2 + 5\gamma = 5$

$f(x) = x^3 - 3x^2 + 5x$

$f'(x) = 3x^2 - 6x + 5$

$|\alpha + \beta| = |\alpha - \gamma|$

$\Rightarrow [|\alpha + \beta|] = 0$

134. We have $[\sin x] = -1, 0, 1$

So, we have the following cases

Case – I :- when $[\sin x] = -1$, In this case, we have $x^2 - 2 = -1$

$\Rightarrow \quad x = \pm 1$

$\therefore \ x = -1$ is the solution in this case

Case – II :- when $[\sin x] = 0$, In this case, we have $x^2 - 2 = 0 \Rightarrow x = \pm\sqrt{2}$

But, $\left[\sin\sqrt{2}\right] = 0$ and $\left[\sin\left(-\sqrt{2}\right)\right] = -1$,

$\therefore \ x = \sqrt{2}$ is the solution in this case

Case–III :- when $[\sin x] = 1$, In this case, we have $x^2 - 2 = 1 \Rightarrow x = \pm\sqrt{3}$

But, $\left[\sin\sqrt{3}\right] = 0$ and $\left[\sin\left(-\sqrt{3}\right)\right] = -1$, Therefore, there is no solution in this case.

Hence, the given equation has two solutions only, namely, $x = -1$ and $x = \sqrt{2}$.

135. Let $\quad x = \dfrac{1}{2\sqrt{3}}\sqrt{6 - \dfrac{1}{2\sqrt{3}}\sqrt{6 - \dfrac{1}{2\sqrt{3}}\sqrt{6 - \dfrac{1}{2\sqrt{3}}}}} \ldots\ldots\ldots\infty$

$$x = \frac{1}{2\sqrt{3}}\sqrt{6 - x} \Rightarrow 12x^2 + x - 6 = 0$$

$$x = \frac{-1 \pm \sqrt{1+288}}{24} = \frac{-1 \pm 17}{24} = \frac{16}{24}$$

$$x = \frac{2}{3}.$$

$\therefore \quad f\left(3.65 + \log_{9/4}\frac{2}{3}\right) = 1 \Rightarrow f\left(3.65 - \frac{1}{2}\right) = 1 \Rightarrow f(3.15) = 1$

$f(3.15) = f(3.15 - 2) = f(1.15) = f(1.15 - 2) = f(-0.85)$

But f is odd function.

$\Rightarrow \quad f(3.15) = -f(0.85)$

$\Rightarrow \quad f(0.85) = -f(3.15) = -1.$

136. $m = {}^5C_4 \cdot 4! = 5! = \text{Total}$

When exactly 2 elements of A maps to itself i.e. $f(3) = 3$, $f(4) = 4$

$\therefore$ from 5, 6, 7 select any 2 in ${}^3C_2 \times 2! = 6$

When exactly one element of A maps to itself say $f(3) = 3$

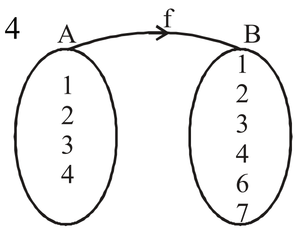
Now 4 can be map in 3 ways and remaining elements $3 \times 2 = 6$

$\therefore \quad {}^2C_1 \times 3 \times 6 = 36$

Total $= 36 + 6 = 42$

$\Rightarrow \quad 5! - 42 = 78 = m$

$n = {}^2C_1 \cdot {}^2C_1 \cdot {}^2C_1 \cdot {}^2C_1 = 16.$ **Ans.**



137. For x-intercept y = 0

$\therefore \quad \left| |x - 2| - a \right| = 3$

$\Rightarrow |x - 2| - a = 3$ or -3

$\Rightarrow |x - 2| = a + 3$ or $a - 3$

For 3 x-intercepts

$a + 3 > 0$ and $a - 3 = 0$ ......(1)

or

$a + 3 = 0$ and $a - 3 > 0$ ......(2)

From (1) a = 3 and (2) is rejected.

Hence, sum = 3. **Ans.**

138.

(i) Given, $f(x) = \sqrt{x^{14} - x^{11} + x^6 - x^3 + x^2 + 1}$ ....(1)

for $f(x)$ to be defined,

$x^{14} - x^{11} + x^6 - x^3 + x^2 + 1 \geq 0$ ....(2)

Case-I: If $x \geq 1$,

$x^{14} - x^{11} + x^6 - x^3 + x^2 + 1$

$= (x^{14} - x^{11}) + (x^6 - x^3) + (x^2 + 1) > 0$

Case-II: If $0 \leq x \leq 1$,

$x^{14} - x^{11} + x^6 - x^3 + x^2 + 1$

$= x^{14} + [(x^6 - x^{11}) + (x^2 - x^3)] + 1 > 0 \; [\because \; x^6 - x^{11} \geq 0, \; x^2 - x^3 \geq 0]$

Case-III: If $x < 0$

$$x^{14} - x^{11} + x^6 - x^3 + x^2 + 1 > 0 \qquad [\because \; x^{11} < 0, x^3 < 0, x^{14}, x^6, x^2 > 0]$$

Thus for all real x

$$x^{14} - x^{11} + x^6 - x^3 + x^2 + 1 > 0$$

hence f(x) is defined for all real x.

$\therefore$ domain f = $(-\infty, \infty)$ **Ans.**

(ii) $3 - x \geq 0$

$\Rightarrow \quad 3 \geq x$

$\Rightarrow \quad x \leq 3$

$2 - \sqrt{3-x} \geq 0 \quad \Rightarrow \quad 4 \geq 3 - x$

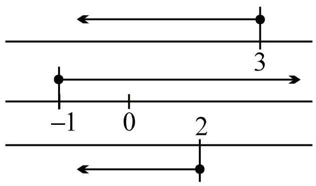
$\Rightarrow \quad x \geq -1$

and $\quad 1 \geq \sqrt{2-\sqrt{3-x}} \quad \Rightarrow \quad \sqrt{3-x} \geq 1$

$\Rightarrow \quad 3 - x \geq 1$

$2 \geq x \quad \Rightarrow \quad x \leq 2$

hence $\quad x \in [-1, 2]$ **Ans.**



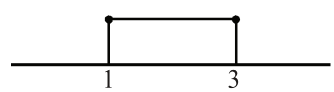
139. First find domain

$-x^2 + 4x - 3 \leq 0$

i.e. $\quad x^2 - 4x + 3 \leq 0$

i.e. $\quad (x - 3 (x - 1) \leq 0$

Now $\quad 1 \leq x \leq 3$



$0 \leq x - 1 \leq 2$

$$0 \leq \frac{\pi}{2}(x-1) \leq \pi$$

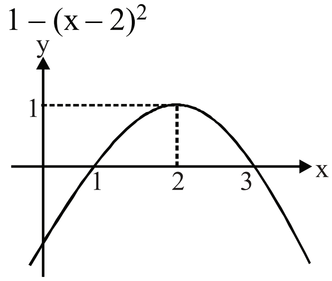
So $\quad \sin\frac{\pi}{2}(x-1)$ is always positive and varies from 0 to 1.

Now $\quad \frac{\pi}{2}\sin\left(\frac{\pi}{2}(x-1)\right)$ varies from 0 to $\frac{\pi}{2}$

so its sin is also positive so the domain is $x \in [1, 3]$.

Now $\quad y = -x^2 + 4x - 3 = -[x^2 - 4x + 3\backslash = -[(x-2)^2 - 1] = 1 - (x-2)^2$

Now its range is [0, 1] also max at x = 2 and min. at x = 3.



also $\sin\left(\frac{\pi}{2}\left(\sin\frac{\pi}{2}(x-1)\right)\right)$ will be maximum at x = 2 and

its value is 1 and minimum at x = 3 and its value is 0.

Now both are maximum at x = 2 and

both are minimum at x = 3.

so $y_{max} = 2$ and $y_{min} = 0$

so range $\in [0, 2]$

Questions

1(mcq) Let x_1, x_2, x_3, x_4 be four non zero numbers satisfying the equation

$$\tan^{-1}\frac{a}{x} + \tan^{-1}\frac{b}{x} + \tan^{-1}\frac{c}{x} + \tan^{-1}\frac{d}{x} = \frac{\pi}{2}$$

then which of the following relation(s) hold good?

(A) $\sum_{i=1}^{4} x_i = a + b + c + d$

(B) $\sum_{i=1}^{4} \frac{1}{x_i} = 0$

(C) $\prod_{i=1}^{4} x_i = abcd$

(D) $(x_1 + x_2 + x_3)(x_2 + x_3 + x_4)(x_3 + x_4 + x_1)(x_4 + x_1 + x_2) = abcd$

2(mcq) Let function f(x) be defined as $f(x) = |\sin^{-1}x| + \cos^{-1}\left(\frac{1}{x}\right)$. Then which of the following is/are **TRUE?**

(A) f(x) is injective in its domain.
(B) f(x) is many-one in its domain.
(C) Range of f is a singleton set.
(D) $\text{sgn}(f(x)) = 1$ where sgn x denotes signum function of x.

3(mcq) Which of the following statement(s) is/are correct?

(A) The graph of the function $f(x) = \text{sgn}\left(x^3 + \sqrt{x^6 + 1}\right)$ is symmetric with respect to origin, where sgn x denotes signum function of x.

(B) The range of the function $f(x) = \cos^{-1}\left(\frac{1}{e^x + e^{-x}}\right)$ is $\left[\frac{\pi}{3}, \frac{\pi}{2}\right)$.

(C) The domain of the definition of function $f(x) = \sqrt{1-x} + \sqrt{x-2}$ is [1, 2].

(D) The value of $2\cos^{-1}(\cos 7) - \sin^{-1}(\sin 11)$ is equal to 3.

4.

	Column I	Column II
(A)	Let $f: R \to R$ and $f_n(x) = f(f_{n-1}(x))\ \forall\ n \geq 2, n \in N$, the roots of equation $f_3(x)\,f_2(x)\,f(x) - 25\,f_2(x)\,f(x) + 175\,f(x) = 375$. Which also satisfy equation $f(x) = x$ will be	(P) 1
(B)	Let $f: [5, 10]$ onto $[4, 17]$, the integers in the range of $y = f(f(f(x)))$ is/are	(Q) 5
(C)	Let $f(x) = 8\cot^{-1}(\cot x) + 5\sin^{-1}(\sin x) + 4\tan^{-1}(\tan x) - \sin(\sin^{-1}x)$ then possible integral values which f(x) can take	(R) 10

(D) Let 'a' denote the roots of equation (S) 15

$$\cos(\cos^{-1}x) + \sin^{-1}\sin\left(\frac{1+x^2}{2}\right) = 2\sec^{-1}(\sec x)$$

then possible values of [| 10a |]
where [·] denotes the greatest integer function will be

5. Find the sum of all possible values of x satisfying $\text{arc}\cos\left(\frac{2}{\pi}\text{arc}\cos x\right) = \text{arc}\sin\left(\frac{2}{\pi}\text{arc}\sin x\right)$.

6. Let $f(x) = \begin{cases} \cos^{-1}\left(\dfrac{1+x}{\sqrt{2(1+x^2)}}\right), & x \le 0 \\ \tan^{-1}x, & x > 0 \end{cases}$

If the range of values of k for which the equation f(x) = k has exactly two solutions is [a, b) then find the value of $\left(\frac{1}{a}+\frac{1}{b}\right)\pi$.

7. Let g(x) = ax + b, where a < 0 and g is defined from [1,3] onto [0,2] then the value of $\cot\left(\cos^{-1}(|\sin x| + |\cos x|) + \sin^{-1}(-|\cos x| - |\sin x|)\right)$ is equal to

(A) g(1) (B) g (2) (C) g(3) (D) g(1) + g(3)

8. If $(\sin^{-1}a)^2 + (\cos^{-1}b)^2 + (\sec^{-1}c)^2 + (\text{cosec}^{-1}d)^2 = \frac{5\pi^2}{2}$,

then the value of $(\sin^{-1}a)^2 - (\cos^{-1}b)^2 + (\sec^{-1}c)^2 - (\text{cosec}^{-1}d)^2$ is equal to

(A) $-\pi^2$ (B) $-\frac{\pi^2}{2}$ (C) 0 (D) $\frac{\pi^2}{2}$

9. Find the range of the following function
$(\sec^{-1}x)^2 + (\text{cosec}^{-1}x)^2$

10. Domain of $f(x) = \frac{1}{\sqrt{\ln \cot^{-1} x}}$

11. If x_1 and x_2 are two real values of x satisfying the equation, $\sin^{-1}\frac{14}{|x|} + \sin^{-1}\frac{2\sqrt{15}}{|x|} = \frac{\pi}{2}$, then find the value of $\left(x_1^2 + x_2^2\right)$.

12. If range of the function $f(x) = \left(\cos^{-1}\frac{x}{2}\right)^2 + \pi\sin^{-1}\frac{x}{2} - \left(\sin^{-1}\frac{x}{2}\right)^2 + \frac{\pi^2}{12}\left(x^2+6x+8\right)$ is $[a\pi^2$, $b\pi^2]$, then find the value of 2(a + b).

13(mcq) Which of the following is/are correct?

(A) $\sin^{-1}\left(\sin\frac{6}{\pi}\right)+\frac{6}{\pi}=\pi$ (B) $\cos^{-1}\left(\cos\frac{12}{\pi}\right)+\frac{12}{\pi}=2\pi$

(C) $-\tan^{-1}\left(\tan\frac{30}{\pi}\right)+\frac{30}{\pi}=3\pi$ (D) $-\cot^{-1}\left(\cot\frac{36}{\pi}\right)+\frac{36}{\pi}=4\pi$

14. $2\cos^{-1}\frac{3}{\sqrt{13}}+\cot^{-1}\frac{16}{63}+\frac{1}{2}\cos^{-1}\frac{7}{25}$ is equal to

(A) π (B) 2π (C) $\frac{\pi}{2}$ (D) none

15. Solve the inequality $(\text{arc sec } x)^2-6(\text{arc sec } x)+8>0$

16.

	Column - I **Equation**		**Column - II** **Relation between x and y**
(A)	$\sin^{-1}x+\sin^{-1}y=\frac{\pi}{2}$	(P)	$x^2+y^2=1$
(B)	$\tan^{-1}x+\tan^{-1}y=\frac{\pi}{2}$	(Q)	$xy=1$
(C)	$\cos^{-1}x+\cos^{-1}y=\frac{\pi}{2}$	(R)	$x^2+y^2=x^2y^2$
(D)	$\sec^{-1}x+\sec^{-1}y=\frac{\pi}{2}$	(S)	$x^2-y^2=1$
		(T)	$x^2-y^2=x^2y^2$

17. Let $f: R\to[-1, 1]$ and $g: R\to B$, where R be the set of all real numbers and

$g(x)=\sin^{-1}\left(\frac{f(x)}{2}\sqrt{4-f^2(x)}\right)+\frac{\pi}{3}$. If $y=f(x)$ and $y=g(x)$ are both surjective, then set B is given by

(A) $\left[-\frac{\pi}{3},\frac{\pi}{3}\right]$ (B) $\left[0,\frac{2\pi}{3}\right]$ (C) $\left[0,\frac{\pi}{3}\right]$ (D) $[0,\pi]$

18(mcq) If $\sum_{n=1}^{\infty}\tan^{-1}\left(\frac{4n}{n^4+5}\right)=\frac{\pi}{A}+\tan^{-1}(B), A>0, B>0$, then

(A) $AB=1$ (B) $AB=2$ (C) $\frac{A}{B}=4$ (D) $\frac{A}{B}=8$

19(mcq) If $f(x)=\sin^{-1}\left[x^2+\frac{1}{2}\right]+\cos^{-1}\left[x^2-\frac{1}{2}\right]$ where $[\cdot]$ denotes the greatest integer function, then which of the following is/are true?

(A) $y = f(x)$ represents a line segment (B) domain of $f(x)$ is $\left(-\sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}}\right)$

(C) $y = \frac{\pi}{2}$ if $x \in \left(-\sqrt{\frac{3}{2}}, 0\right)$ (D) $y = \pi$ if $x \in \left(0, \sqrt{\frac{3}{2}}\right)$

20. If $x = \alpha, y = \beta$ is a solution of the equation $12\sin x + 5\cos x = 2y^2 - 8y + 21$, then the value of $(2\beta + 1)\tan\alpha$ is
(A) 8 (B) 9 (C) 12 (D) 15

21. The number of solutions of the equation $\tan^{-1}\left(\frac{1}{2x+1}\right) + \tan^{-1}\left(\frac{1}{4x+1}\right) = \tan^{-1}\left(\frac{2}{x^2}\right)$ is
(A) 0 (B) 1 (C) 2 (D) 3

22. If $f(x) = \text{cosec}^{-1}(\text{cosec}x)$ and $\text{cosec}(\text{cosec}^{-1}x)$ are equal functions then maximum range of values of x is

(A) $\left[-\frac{\pi}{2}, -1\right] \cup \left[1, \frac{\pi}{2}\right]$ (B) $\left[-\frac{\pi}{2}, 0\right] \cup \left[0, \frac{\pi}{2}\right]$

(C) $(-\infty, -1] \cup [1, \infty)$ (D) $[-1, 0) \cup [0, 1)$

23. Let $\cos^{-1}(x) + \cos^{-1}(2x) + \cos^{-1}(3x) = \pi$. If x satisfies the cubic $ax^3 + bx^2 + cx - 1 = 0$, then $(a + b + c)$ has the value equal to
(A) 24 (B) 25 (C) 26 (D) 27

24. Number of integral solutions of the equation $\text{sgn}\left(\sin^{-1}\left[\frac{\pi x}{6}\right]\right) = 1$,

where [x] denotes the greatest integer less than or equal to x and sgn x denotes signum function of x.
(A) 2 (B) 3 (C) 5 (D) 7

25. Range of the function $f(x) = \cos^{-1}\left(\frac{1}{e^x + e^{-x}}\right)$ is

(A) $(0, \pi)$ (B) $\left[\frac{\pi}{6}, \frac{\pi}{2}\right)$ (C) $\left[\frac{\pi}{3}, \frac{\pi}{2}\right)$ (D) $\left[\frac{\pi}{2}, \frac{2\pi}{3}\right)$

26. The true set of values of k for which $\sin^{-1}\left(\frac{1}{1+\sin^2 x}\right) = \frac{k\pi}{6}$ may have a solution, is

(A) $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ (B) $[1, 3]$ (C) $\left[\frac{1}{6}, \frac{1}{2}\right]$ (D) $[2, 4]$

27. The value of x satisfying the equation $(\sin^{-1}x)^3 - (\cos^{-1}x)^3 + (\sin^{-1}x)(\cos^{-1}x)(\sin^{-1}x - \cos^{-1}x) = \frac{\pi^3}{16}$, is

(A) $\cos\frac{\pi}{5}$ (B) $\cos\frac{\pi}{4}$ (C) $\cos\frac{\pi}{8}$ (D) $\cos\frac{\pi}{12}$

28. Let $g : R \to \left(0, \frac{\pi}{3}\right]$ is defined by $g(x) = \cos^{-1}\left(\frac{x^2 - k}{1 + x^2}\right)$.

Then the possible values of 'k' for which g is surjective function, is

(A) $\left\{\frac{1}{2}\right\}$ (B) $\left(-1, -\frac{1}{2}\right]$ (C) $\left\{-\frac{1}{2}\right\}$ (D) $\left[-\frac{1}{2}, 1\right)$

29. For $n \in N$, if $\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{4} + \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{n} = \frac{\pi}{4}$ then n is equal to

(A) 43 (B) 47 (C) 49 (D) 51

30. The domain of definition of function $f(x) = \sqrt{\cos^{-1}x - 2\sin^{-1}x}$ is equal to

(A) $\left[\frac{-1}{2}, \frac{1}{2}\right]$ (B) $\left[\frac{1}{2}, 1\right]$ (C) $\left[-1, \frac{1}{2}\right]$ (D) $\left[-1, \frac{\sqrt{3}}{2}\right]$

31. If $f(x) = \tan^{-1}x - \frac{2}{\pi}(\tan^{-1}x)^2 + \frac{4}{\pi^2}(\tan^{-1}x)^3 - \ldots\ldots\ldots\ldots\infty$,

then the sum of integral values of a for which the equation $f^2(x) + (\sin^{-1}x)^2 = a$, possess solution, is

(A) 6 (B) 7 (C) 9 (D) 10

32. If $0 < x < 1$, then the solution of equation

$$3\sin^{-1}\left(\frac{2x}{1+x^2}\right) + 2\tan^{-1}\left(\frac{2x}{1-x^2}\right) - 4\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \frac{\pi}{4}, \text{ is}$$

(A) $\sqrt{2} - 1$ (B) $2 - \sqrt{3}$ (C) $2 - \sqrt{2}$ (D) $\sqrt{3} - \sqrt{2}$

33. Number of integral values in the domain of function $f(x) = \sin^{-1}\left|\frac{x^2 - 4}{3x}\right|$, is

(A) 5 (B) 6 (C) 7 (D) 8

34. Let $f(x) = \sin^{-1}x - x$ and $g(x) = x^2 + 5x + 6 + \cos^{-1}x$. If $|f(x)| + |g(x)| = |f(x) + g(x)|$, then the true set of real values of x, is

(A) $\{x \mid x \in R, -1 \le x \le 1\}$ (B) $\{x \mid x \in R, -1 \le x \le 0\}$

(C) $\{x \mid x \in R, 0 \le x \le 1\}$ (D) $\left\{x \mid x \in R, \frac{-1}{2} \le x \le \frac{1}{2}\right\}$

35. Let $S_n = \cot^{-1}\left(3x + \frac{2}{x}\right) + \cot^{-1}\left(6x + \frac{2}{x}\right) + \cot^{-1}\left(10x + \frac{2}{x}\right) + \ldots\ldots\ldots\ldots + n$ terms, where $x > 0$. If $\lim\limits_{n\to\infty} S_n = 1$, then x equals

(A) $\frac{\pi}{4}$ (B) 1 (C) tan 1 (D) cot 1

36(**mcq**) If $A = \tan^{-1}\left(\frac{1}{7}\right)$ and $B = \tan^{-1}\left(\frac{1}{3}\right)$, then

(A) $\cos 2A = \frac{24}{25}$ (B) $\cos 2B = \frac{4}{5}$ (C) $\cos 2A = \sin 4B$ (D) $\tan 2B = \frac{3}{4}$

37. If $\sin^{-1}(1-x) + \cos^{-1}(mx) = \frac{\pi}{2}$ has some solution for x. Then the range of m, is

(A) $(-\infty, -1)$ (B) $(-1, \infty)$ (C) $(\frac{1}{2}, \infty)$ (D) $[-\frac{1}{2}, \infty)$

Paragraph for question nos. 38 & 39

Consider a real-valued function $f(x) = \sqrt{\sin^{-1}x + 2} + \sqrt{1 - \sin^{-1}x}$

38. The domain of definition of $f(x)$ is

(A) $[-1, 1]$ (B) $[\sin 1, 1]$ (C) $[-1, \sin 1]$ (D) $[-1, 0]$

39. The range of $f(x)$ is

(A) $\left[0, \sqrt{3}\right]$ (B) $\left[1, \sqrt{3}\right]$ (C) $\left[1, \sqrt{6}\right]$ (D) $\left[\sqrt{3}, \sqrt{6}\right]$

40. $\cos\left(\cos^{-1}\cos\left(\frac{8\pi}{7}\right) + \tan^{-1}\tan\left(\frac{8\pi}{7}\right)\right)$ has the value equal to

(A) 1 (B) -1 (C) $\cos\frac{\pi}{7}$ (D) 0

41. Let two functions are $f(x) = \sqrt{\sin^{-1}x - \cos^{-1}x}$ and $g(x) = \sqrt{\cos^{-1}x - |\sin^{-1}x|}$.

	Column-I		**Column-II**
(A)	Set of values of x for which $f(x)$ is defined is	(P)	$\left[-1, \frac{1}{\sqrt{2}}\right]$
(B)	Set of values of x for which $g(x)$ is defined is	(Q)	$\left[0, \frac{\pi}{\sqrt{2}}\right]$
(C)	The images of mapping $y = f(x)$ lies in the interval	(R)	$\left[0, \frac{\pi}{\sqrt{2}}\right]$
(D)	The images of mapping $y = g(x)$ lies in the interval	(S)	$\left[\frac{1}{\sqrt{2}}, 1\right]$
		(T)	$[-1, 1]$

42. The range of the function $f(x)=\sqrt{\cos^{-1}\left(\sqrt{\log_4 x}\right)-\frac{\pi}{2}+\sin^{-1}\left(\frac{1+x^2}{4x}\right)}$

(A) $\left(0, \frac{\pi}{2}+\sqrt{\frac{\pi}{2}}\right]$ (B) $\left[\frac{\pi}{2}, \frac{\pi}{2}+\sqrt{\frac{\pi}{2}}\right]$ (C) $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$ (D) None

43. MULTIPLE CORRECT (43 a , b , c)

(a) If the equation $\sin^{-1}\sqrt{x}+\cos^{-1}\sqrt{x^2-1}+\tan^{-1}(\tan y)=a$ has at least one solution then the possible integral value which 'a' can take is

(A) 1 (B) 2 (C) 3 (D) 4

(b) If the equation $\sqrt{\sin^{-1}|\sin x|}+\sqrt{\tan^{-1}|\tan x|}=2\sqrt{a}$ has atleast one solution then the possible integral value which 'a' can take is

(A) 0 (B) 1 (C) 2 (D) 3

(c) If the equation $\sin^{-1}|\cos x|-\cos^{-1}|\sin x|=a$ has atleast one solution then the number of possible integral values of a is

(A) 0 (B) 1 (C) 2 (D) 3

44. Let $S_n=\sum_{n=1}^{n}\sin^{-1}\left(\frac{(2n+1)}{n(n+1)\left(\sqrt{n(n+2}+\sqrt{(n+1)(n-1)}\right)}\right)$. Find the value of 100 cos (S_{99}).

45. Let $f(x)=x^2-2ax+a-2$ and $g(x)=\left[2+\sin^{-1}\frac{2x}{1+x^2}\right]$. If the set of real values of 'a' for which $f(g(x))<0\ \forall x\in R$ is (k_1, k_2) then find the value of $(10k_1+3k_2)$.

[**Note :** [k] denotes greatest integer less than or equal to k.]

46. The number of value(s) of $a\in R$ for which the equation $5\tan^{-1}(x^2+x+a)+3\cot^{-1}(x^2+x+a)=2\pi$, has unique solution is

(A) 0 (B) 1 (D) 2 (D) more than two

47. If $f(x)=\tan^{-1}\left(\sqrt{\left(\frac{1-\cos 2}{1+\cos 2}\right)}+x^2\right)$ and $g(x)=\begin{cases}|x| & x\neq 0\\ 1 & x=0\end{cases}$ then number of solution(s) of the equation $f(x)=g(x)$ is(are)

(A) 0 (B) 1 (C) 2 (D) 3

48$_{(\mathbf{mcq})}$ Which of the following is(are) correct?

(A) Domain of $f(x)=\sin^{-1}(\cos^{-1}x+\tan^{-1}x+\cot^{-1}x)$ is null set.

(B) Domain of $f(x)=\cos^{-1}(\tan^{-1}x+\cot^{-1}x+\sin^{-1}x)$ is $[-1,-\cos 1]$.

(C) Domain of $f(x)=\sin^{-1}(\cos^{-1}x)$ is $[\cos 1, 1]$

(D) Domain of $f(x)=\cos^{-1}(\sin^{-1}x)$ is $[-\sin 1, \sin 1]$,

49. If $\sin(30^\circ + \text{arc}\tan x) = \dfrac{13}{14}$ and $0 < x < 1$, the value of x is $\dfrac{a\sqrt{3}}{b}$, where a and b are positive integers with no common factors. Find the value of $\left(\dfrac{a+b}{2}\right)$.

50. The value of $a \in R$ for which the equation $x^2 + 2ax + 8\cos^{-1}(x^2 - 4x + 5) = 8\sin^{-1}(4x - 5 - x^2)$ has atleast one solution is
(A) $\pi + 1$ (B) $-(1 + \pi)$ (C) $2 - \pi$ (D) $\pi - 2$

51.

Column-I	Column-II
(A) $f(x) = \sin^{-1}\left(\dfrac{2}{\lvert\sin x - 1\rvert + \lvert\sin x + 1\rvert}\right)$	(P) $f(x)$ is many one
(B) $f(x) = \cos^{-1}(\lvert x-1\rvert - \lvert x-2\rvert)$	(Q) Domain of $f(x)$ is R
(C) $f(x) = \sin^{-1}\left(\dfrac{\pi}{\lvert\sin^{-1}x - (\pi/2)\rvert + \lvert\sin^{-1}x + (\pi/2)\rvert}\right)$	(R) Range contain only irrational number
(D) $f(x) = \cos(\cos^{-1}\lvert x\rvert) + \sin^{-1}(\sin x) - \text{cosec}^{-1}(\text{cosec}\, x) + \text{cosec}^{-1}\lvert x\rvert$	(S) $f(x)$ is even.

52. Let $f(x) = \tan^{-1}(\cot x - 2\cot 2x)$ and $\sum\limits_{r=1}^{5} f(r) = a - b\pi$, where $a, b \in N$. Find the value of $(a + b)$.

53. Let $f : D_1 \to R_1$ defined by $f(x) = \sin^{-1}(x - 2) + \cos^{-1}(4 - x)$
and $g : D_2 \to R_2$ defind by $g(x) = \sin^{-1}\sqrt{4 - x^2} + \cos^{-1}\sqrt{x - 1}$.
If p and q are number of integers in the domain of f and g respectively, then $(p + q)$ is equal to
(A) 1 (B) 2 (C) 3 (D) 4

54. Let $f : R \to \left(\dfrac{-\pi}{2}, 0\right]$ be a function defined by $f(x) = \tan^{-1}(2x - x^2 + \lambda)$. If f is onto, then the number of integers in the range of λ, is
(A) 0 (B) 1 (C) 2 (D) 3

55$_{(mcq)}$ Let a function $f : A \to B$ be defined as $f(x) = \sin^{-1}(\tan x) - \text{cosec}^{-1}(\cot x)$.
Which of the following statement(s) is/are **TRUE** for the function $f(x)$?
(A) $f(x)$ is periodic with fundamental period π.
(B) The function $f(x)$ is non-invertible.

(C) The composite function $f(f(x))$ is not defined.
(D) The function $f(x)$ is an even function.

56$_{(\mathbf{mcq})}$ Let $f(x) = \sin^{-1}|\sin x| + \cos^{-1}(\cos x)$. Which of the following statement(s) is/are **TRUE** ?
(A) $f(f(3)) = \pi$
(B) $f(x)$ is periodic with fundamental period 2π.
(C) $f(x)$ is neither even nor odd.
(D) Range of $f(x)$ is $[0, 2\pi]$

57. For $x, y, z, t \in R$, if $\sin^{-1}x + \cos^{-1}y + \sec^{-1}z \geq t^2 - \sqrt{2\pi}\ t + 3\pi$, then find the value of

$$\sec\left(\tan^{-1}x + \tan^{-1}y + \tan^{-1}z + \tan^{-1}\left(\sqrt{\frac{2}{\pi}}\ t\right)\right).$$

58$_{(\mathbf{mcq})}$ If $f(x) = \sin^{-1}x \cdot \cos^{-1}x \cdot \tan^{-1}x \cdot \cot^{-1}x \cdot \sec^{-1}x \cdot \operatorname{cosec}^{-1}x$, then which of the following statement(s) **hold(s) good** ?
(A) The graph of $y = f(x)$ does not lie above x axis.
(B) The non-negative difference between maximum and minimum value of the function $y = f(x)$ is $\frac{3\pi^6}{64}$.
(C) The function $y = f(x)$ is not injective.
(D) Number of non-negative integers in the domain of $f(x)$ is two.

59. Let $f(x) = \cos^{-1}x + \sin^{-1}x$ and $g(x) = \cot^{-1}x + \tan^{-1}x$. Then which one of the following alternative is true?
(A) Both f and g are periodic.
(B) Both f and g are aperiodic.
(C) f is periodic but g is periodic.
(D) g is periodic but f is aperiodic.

60. Let, $f(x) = \begin{cases} \frac{8}{\pi}\tan^{-1}(-|x|+3), & |x| > 2 \\ \left[\frac{3x^2 - |x| + 3}{x^2+1}\right], & |x| \leq 2 \end{cases}$

Number of integers in the range of $f(x)$ is
Where [ ] denotes greatest integer function.
(A) 5
(B) 6
(C) 7
(D) more than 7

61. If $x^2 + ax + b = 0$ has two distinct negative integral roots

and $\left(\log_{1/2}\left(\frac{2}{\pi}\cot^{-1}x + 1\right)\right)^2 + a\log_{1/2}\left(1 - \frac{\tan^{-1}x}{\pi}\right) = a - b$ has no real solution.

Then find the minimum value of a.

62. Let $f(x) = \sqrt{\frac{\pi^2}{2} + \frac{\pi}{2}x - x^2}$, $g(x) = \sqrt{\sin^{-1}x}$ and $h(x) = \sqrt{\cos^{-1}x}$ be three real valued function.

If range of f(x) is [aπ, bπ], provided $f(g(x))$ and $f(h(x))$ both exist then find the value of $16(a^2 + b^2)$.

63. If $a_0, a_1, a_2, \ldots\ldots\ldots\ldots, a_n$ are poitive real number satisfying $a_i \cdot a_{n-i} = 1$ for all $i = 0, 1, 2, \ldots\ldots, n$ and k is any real number such that $S_n = \sum_{i=0}^{n} \frac{2}{1+a_i^k}$. If $[\sin^{-1} S_n] = 1$.

64. If $2^{\left(\cos^{-1}x\right)^2}\sqrt{\left(\sin^{-1}y\right)^4 - 2\left(\sin^{-1}y\right)^2 + 2} = 1$, then the value of $(x - y)$ can be

(A) $1 - \frac{\pi}{2}$ (B) $1 + \frac{\pi}{2}$ (C) $1 - \sin 1$ (D) $2 - \sin 1$

Paragraph for question nos. 65 to 66

Consider a function f such that $f(x, y) = x^3 + y^3 + 3xy \; \forall \; x, y \in R$.

65. If $f(3\sin^{-1}x, 2\cos^{-1}x) = 1$ then the value of x given by

(A) $-\sin 1$ (B) $-\cos 2$ (C) $\cos 2$ (D) none

66. If $h(x) = f\left(\left(\sin^{-1}x\right)^{\frac{2}{3}}, 0\right) + f\left(\left(\cos^{-1}x\right)^{\frac{2}{3}}, 0\right)$ then the range of $h(x)$ is given as

(A) $\left[\frac{\pi^2}{8}, \frac{5\pi^2}{4}\right]$ (B) $\left[\frac{\pi^2}{4}, \frac{5\pi^2}{4}\right]$ (C) $\left[\frac{\pi^2}{16}, \frac{\pi^2}{4}\right]$ (D) $\left[\frac{\pi^2}{4}, \frac{3\pi^2}{4}\right]$

67. If $a\sin^{-1}x - b\cos^{-1}x = c$, then the value of $a\sin^{-1}x + b\cos^{-1}x$ (whenever exists) is equal to

(A) 0 (B) $\frac{\pi ab + c(b-a)}{a+b}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi ab + c(a-b)}{a+b}$

68. The value of $\tan^{-1}\frac{4}{7} + \tan^{-1}\frac{4}{19} + \tan^{-1}\frac{4}{39} + \tan^{-1}\frac{4}{67} + \ldots\ldots\ldots \infty$ equals

(A) $\tan^{-1}1 + \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}$ (B) $\tan^{-1}1 + \cot^{-1}3$

(C) $\cot^{-1}1 + \cot^{-1}\frac{1}{2} + \cot^{-1}\frac{1}{3}$ (D) $\cot^{-1}1 + \tan^{-1}3$

69. The solution set of equation $\cos^{-1}\left(\frac{x^2}{2} + \sqrt{1-x^2}\sqrt{1-\frac{x^2}{4}}\right) = \cos^{-1}\left(\frac{x}{2}\right) - \cos^{-1}x$ is

(A) [–1, 1] (B) [–1, 0] (C) [0, 1] (D) (–1, 1)

70. If 'α' is the only real root of the equation $x^3 + bx^2 + cx + 1 = 0$ $(b < c)$, then the value of $\tan^{-1}\alpha + \tan^{-1}(\alpha^{-1})$ is equal to

(A) $\frac{-\pi}{2}$ (B) $\frac{\pi}{2}$ (C) 0 (D) can't be determined

71. If the roots of the equation $x^3 - 10x + 11 = 0$ are α, β and γ then the value of $\tan(\tan^{-1}\alpha + \tan^{-1}\beta + \tan^{-1}\gamma)$ equals

(A) $\frac{9}{11}$ (B) $\frac{-11}{9}$ (C) $\frac{11}{9}$ (D) 1

72. $\tan^{-1}2 + \tan^{-1}5 + \tan^{-1}8$ has the value equal to

(A) $\cot^{-1}1 + 2\cot^{-1}\left(\frac{1}{2}\right) + 3\cot^{-1}\left(\frac{1}{3}\right)$ (B) $\tan^{-1}(1) + \tan^{-1}(2) + \tan^{-1}(3)$

(C) $2\tan^{-1}(1) + \tan^{-1}(2) + \tan^{-1}(3)$ (D) $\cot^{-1}(1) + 2\cot^{-1}(2) + 3\cot^{-1}(3)$

73. Let $f(x) = (\text{arc}\tan x)^3 + (\text{arc}\cot x)^3$. If the range of $f(x)$ is $[a, b)$ then find the value of $\frac{b}{a}$.

74. If $\cos^{-1}\left(\frac{x}{2}\right) + \cos^{-1}\left(\frac{y}{3}\right) = \theta$, (where $x, y > 0$) then find the maximum value of $9x^2 - 12xy\cos\theta + 4y^2$.

75. Number of positive integers x satisfying the equation $\tan\left(\tan^{-1}\frac{x}{10} + \tan^{-1}\frac{1}{x+1}\right) = \tan\frac{\pi}{4}$, is

(A) 0 (B) 1 (C) 2 (D) 3

76. The number of integral values of k for which the equation $\sin^{-1}x + \tan^{-1}x = k$ has a solution, is

(A) 5 (B) 7 (C) 11 (D) 13

77. Let $f(x) = \sin^{-1}(\sin x)$, $g(x) = \cos^{-1}(\cos x)$, then which one of the following **incorrect**?

(A) $f(x) = g(x)\ \forall\ x \in \left(0, \frac{\pi}{4}\right)$ (B) $f(x) < g(x)\ \forall\ x \in \left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$

(C) $f(x) < g(x)\ \forall\ x \in \left(\pi, \frac{5\pi}{4}\right)$ (D) $f(x) > g(x)\ \forall\ x \in \left(\frac{3\pi}{2}, 2\pi\right)$

78. Number of solutions of the equation $\left|\sin^{-1}(\sin x)\right| = \cos x$, for $x \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$ is equal to

(A) 0 (B) 2 (C) 4 (D) 6

79. The value of $\sum_{r=2}^{\infty}\tan^{-1}\left(\frac{1}{r^2 - 5r + 7}\right)$, is

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) $\frac{5\pi}{4}$

80. Match the entries of Column-I with one or more than one entries of column-II. Note that [x], {x} and sgn x denote largest integer less than or equal to x, fractional part of x and signum function of x respectively.

	Column I	Column II
(A)	Let $f:[-1,1] \to R$ be defined by $f(x) = \sqrt[5]{x} + \sin^{-1}x$ then f(x) is	(P) Odd
(B)	Let $f: R \to \{-1, 0, 1\}$ be defined by $f(x) = \text{sgn}\left(\frac{1-\|x\|}{1+\|x\|}\right)$ then f(x) is	(Q) Even
(C)	Let $f:[-4, 2] \to [0, 3]$ be defined by $f(x) = \sqrt{8-2x-x^2}$ then f(x) is	(R) Onto
(D)	Let f: $(-\infty, 0] \to [0, \infty)$ be defined by $f(x) = \frac{2^{-[x]}}{2^{\{x\}}} - 2^{\|x\|}$ then f(x) is	(S) One-One
		(T) Many-One

81.

	Column I	Column II	
(A)	$2\cot\left(\cot^{-1}(3)+\cot^{-1}(7)+\cot^{-1}(13)+\cot^{-1}(21)\right)$ has the value equal to	(P)	3
(B)	If $\tan\left(\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)+\ldots\ldots+\tan^{-1}\left(\frac{1}{381}\right)\right) = \frac{m}{n}$. where $m, n \in N$, then the least value of $(m+n)$ is divisible by	(Q)	4
(C)	Number of integral ordered pairs (x, y) satisfying the equation $\arctan\frac{1}{x} + \arctan\frac{1}{y} = \arctan\frac{1}{10}$, is	(R)	5
		(S)	8
(D)	The smallest positive integral value of n for which $(n-2)x^2 + 8x + n + 4 > \sin^{-1}(\sin 12) + \cos^{-1}(\cos 12)\ \forall\ x \in R$, is	(T)	10

82. Let α and β be the roots of the equation

$$3x^2 - \left(2a + 4\sin^{-1}\left(\sqrt{[a]+[-a]}\right)\right)x + \sqrt{3-|a|} = 0, a \in R^+ \text{ (the set of all positive real numbers).}$$

Find the sum of all values of a for which $\alpha, \beta \in R$ (the set of all real numbers).
[Note : [k] denotes the largest integer less than or equal to k.]

83. If $0 < \cos^{-1}x < 1$ and $1 + \sin(\cos^{-1}x) + \sin^2(\cos^{-1}x) + \sin^3(\cos^{-1}x) + \ldots\ldots \infty = 2$, then x equals

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2\sqrt{3}}$

84. Number of solutions of the equation $\left|\sin^{-1}(\sin x)\right| = \cos x$, for $x \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$ is equal to

(A) 0 (B) 2 (C) 4 (D) 6

85. If $S_n = \sum_{r=1}^{n} r!$ then for $n > 6$ $\left(\text{given } \sum_{r=1}^{6} r! = 873\right)$

Column-I		Column-II	
(A)	$\sin^{-1}\left(\sin\left(S_n - 7\left[\frac{S_n}{7}\right]\right)\right)$	(P)	$5 - 2\pi$
(B)	$\cos^{-1}\left(\cos\left(S_n - 7\left[\frac{S_n}{7}\right]\right)\right)$	(Q)	$2\pi - 5$
(C)	$\tan^{-1}\left(\tan\left(S_n - 7\left[\frac{S_n}{7}\right]\right)\right)$	(R)	$6 - 2\pi$
(D)	$\cot^{-1}\left(\cot\left(S_n - 7\left[\frac{S_n}{7}\right]\right)\right)$	(S)	$5 - \pi$
		(T)	$\pi - 4$

(where [ ] denotes greatest integer function)

86. If the area enclosed by the curves $f(x) = \cos^{-1}(\cos x)$ and $g(x) = \sin^{-1}(\cos x)$ in $x \in \left[\frac{9\pi}{4}, \frac{15\pi}{4}\right]$ is $\frac{a\pi^2}{b}$ (where a and b are coprime), then find (a + b).

87. Prove that: $\tan^{-1}\left(\frac{3\sin 2\alpha}{5+3\cos 2\alpha}\right) + \tan^{-1}\left(\frac{\tan\alpha}{4}\right) = \alpha$ (where $0 < \alpha < \frac{\pi}{2}$)

88. The complete solution set of the equation

$$\sin^{-1}\sqrt{\frac{1+x}{2}} - \sqrt{2-x} = \cot^{-1}\left(\tan\sqrt{2-x}\right) - \sin^{-1}\sqrt{\frac{1-x}{2}}, \text{ is}$$

(A) $[-1, 1]$ (B) $\left(2-\frac{\pi^2}{4}, 1\right]$ (C) $\left[-1, 2-\frac{\pi^2}{4}\right)$ (D) $[0, 1]$

89. Let $f(x) = \frac{1}{\pi}(\sin^{-1}x + \cos^{-1}x + \tan^{-1}x) + \frac{x+1}{x^2+2x+10}$. If the absolute maximum value of f(x) is M. find 52 M.

90. If the equation $x^3 + bx^2 + cx + 1 = 0$, $(b < c)$, has only one root α. Then the value of $2\tan^{-1}(\operatorname{cosec}\alpha) + \tan^{-1}(2\sin\alpha\sec^2\alpha)$, is

(A) $-\pi$ (B) $-\frac{\pi}{2}$ (C) $\frac{\pi}{2}$ (D) π

91. If the sum $\sum_{n=1}^{10}\sum_{m=1}^{10}\tan^{-1}\left(\frac{m}{n}\right) = k\pi$, find the value of k.

92. Let m be the number of solutions of $\sin 2x + \cos 2x + \cos x + 1 = 0$ in $x \in \left(0, \frac{\pi}{2}\right)$ and

$n = \sin\left(\tan^{-1}\left(\tan\frac{7\pi}{6}\right) + \cos^{-1}\left(\cos\frac{7\pi}{3}\right)\right)$, then find the value of (m + n).

93. The value of $a \in R$ for which the equation $x^2 + 2ax + 8\cos^{-1}(x^2 - 4x + 5) = 8\sin^{-1}(4x - 5 - x^2)$

has atleast one solution is
(A) $\pi + 1$ (B) $-(1+\pi)$ (C) $2-\pi$ (D) $\pi - 2$

94. Let m be the number of elements in the domain of $f(x)$ and n be the number of elements in the range of $f(x)$ where $f(x) = \sin^{-1}\left(\sec\left(3\tan^{-1}x\right)\right) + \cos^{-1}\left(\operatorname{cosec}\left(3\cot^{-1}x\right)\right)$, then the value of $(m+n)$ is
(A) 3 (B) 4 (C) 5 (D) 6

95$_{(mcq)}$ Consider a function $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) + \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right) - a\tan^{-1}x$.
Where a is any real constant. Find the value of 'a' if $f(x) = 0$ for all x.
(A) 6 (B) -6 (C) 2 (D) -2

96. Let $f(x) = e^{\cos(\sin^{-1}x)}$ and $g(x)$ be inverse function of $f(x)$. The area of the triangle formed by joining the points $(1, g(1))$, $(e, g(e)$ and $(0, 0)$ is
(A) $\frac{e^2-1}{2e}$ (B) $\frac{e}{2}$ (C) $\frac{e-1}{2}$ (D) $\frac{e^2-1}{2}$

97. If $\theta \in \left(-\frac{\pi}{2}, 0\right)$, then the value of $\sec\left(\tan^{-1}(\cot\theta) - \cot^{-1}(\tan\theta)\right)$ equals
(A) 2 (B) $\sqrt{2}$ (C) -1 (D) 1

98. Let $f : (-\infty, -1] \to \left(\frac{\pi}{2}, \pi\right]$ be defined as $f(x) = \sec^{-1}(-x^2 + x + a)$. If $f(x)$ is surjective, then the range of a is
(A) $\{1\}$ (B) $\left\{\frac{-5}{4}\right\}$ (C) $\left(-\infty, \frac{-5}{4}\right]$ (D) $(-\infty, 1]$

99. If the sum of the series $\sum_{n=1}^{\infty}\left(\frac{\pi - \sec^{-1}\sqrt{|x|+1} - \operatorname{cosec}^{-1}\sqrt{|x|+1}}{\pi a}\right)^n$ is finite, where $x \in R$ and $a > 0$ then a lies in the interval
(A) $(1, \infty)$ (B) $\left(0, \frac{1}{2}\right)$ (C) $\left(\frac{1}{2}, \infty\right)$ (D) $\left[\frac{1}{2}, \infty\right)$

100. If k times the sum of first n natural numbers is equal to the sum of squares of first n natural numbers, then $\sin^{-1}\left(\frac{9k^2 - 4n^2}{6k + 4n}\right)$ is equal to
(A) π (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

101. If $\frac{x}{a}+\frac{y}{b}=1$ is the equation of tangent drawn from a point P (α, β) on the curve $y=\sin^{-1}\left(1-\sqrt{x-2}\right)+\cos^{-1}\left(\sqrt{2-[x]}\right)$ to the curve $y=-x+2\cos(x-2)+\pi$ where α is a prime number and $[y]$ denotes largest integer less than or equal to y,then find the value of $|a+b-\alpha-2\beta|$.

Paragraph for question nos. 102 to 104

If α is the minimum value of the expression

$$y=\frac{4x^2-4x+17}{7+4x-4x^2} \quad \text{and} \quad \beta=\left[\frac{1002\cdot1003\cdot1004\cdot1006\cdot1007\cdot1008}{\frac{1}{6}(1005)^6}\right]$$

where $[y]$ denotes largest integer less than or equal to y.

102. $\alpha+\beta$ is equal to

(A) 8 (B) 7 (C) 5 (D) 2

103. If $k\in[\alpha,\beta]$ then least integral value of k for which $f(x)=\tan^{-1}\left(\sin(k\cos x)\right)$ has its range $\left[\frac{-\pi}{4},\frac{\pi}{4}\right]$ is

(A) – 1 (B) 0 (C) 2 (D) 4

104. If $g(x)=\log_{\frac{19}{9}}\left(x+e^{\ln\left(\tan\left(\cot^{-1}\left(5+4x-x^2\right)\right)\right)}\right)$ and $x\in[\alpha,\beta]$ then range of $g(x)$ is

(A) $(-\infty,\infty)$ (B) $[1,\infty)$ (C) $[-1,\infty)$ (D) $\left[\log_{\frac{19}{9}}\left(2+\frac{\pi}{2}\right),\ \infty\right)$

Paragraph for question nos.105 to 107

Let $f(x)=\sin^{-1}\left(\frac{3x^4+6x^2-1}{\left(x^2+1\right)^3}\right)$ and $g(x)$ is defined as $g(x)=$

$$\begin{cases} f(x)+3\,\text{cosec}^{-1}\left(x^2+1\right) & |x|\le 1 \\ f(x)+3\sec^{-1}\left(x^2+1\right) & |x|>1 \end{cases}$$

105. The value of $g\left(\sqrt{\tan 22\frac{1^\circ}{2}}\right)+g\left(\sqrt[3]{\cot 15^\circ}\right)+g\left(\sqrt[5]{\sin 18^\circ}\right)$ is equal to

(A) 2π (B) 4π (C) $\frac{7\pi}{2}$ (D) $\frac{9\pi}{2}$

106. If $h(x)=\frac{1}{5}(3\sin x+4\cos x)$ then domain and range of g(h(x)) are respectively,

(A) $[-1, 1]$; $[\pi, 3\pi]$ (B) R ; $\{\pi\}$ (C) R ; $\left\{\frac{3\pi}{2}\right\}$ (D) $[-1, 1]$; $\left[\frac{\pi}{2},\frac{3\pi}{2}\right]$

107. Number of solutions of the equation $g(x)=\tan\left(\cot^{-1}\left(\left|\frac{1}{4x}\right|\right)\right)$ is –

(A) 0 (B) 1 (C) 2 (D) 4

108. If $\cos^{-1}\sqrt{x^3-x+1}+\cos^{-1}\sqrt{x-x^3}+\cos^{-1}\sqrt{1-|y|}=\frac{2\pi}{3}$ then $|y|$ is equal to

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{1}{8}$

109. If $f(x)=\sin^{-1}\left(\frac{4x}{4+x^2}\right)$ then the value of $f\left(\sqrt{7}-2\right)-f\left(\frac{4}{\sqrt{7}-2}\right)$ is

(A) 0 (B) $\frac{\pi}{2}$ (C) π (D) $\frac{3\pi}{4}$

110. The maximum value of the function $f(x)=\left(\sin^{-1}(\sin x)\right)^2-\sin^{-1}(\sin x)$, is

(A) $\frac{\pi}{4}(\pi+2)$ (B) $\frac{\pi}{4}(\pi-2)$ (C) $\frac{\pi}{2}(\pi+2)$ (D) $\frac{\pi}{2}(\pi-2)$

111. If $f(x)=\sin^{-1}\left(\frac{x^2(2x^2+1)-1}{1+x^2}\right)=\begin{cases}a\pi+b\cos^{-1}x, & 0\le x\le 1\\ p\pi+q\sin^{-1}x, & -1\le x<0\end{cases}$, then find the absolute of (a+b+p+q).

112. If $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{2}{3x}+\tan^{-1}\frac{3}{4}=\frac{\pi}{2}$, then the value of $\tan(\pi-2\tan^{-1}x)$, is

(A) $\frac{24}{7}$ (B) $\frac{7}{24}$ (C) $\frac{25}{7}$ (D) $\frac{7}{25}$

113. In the interval $\left[0,\frac{\pi}{2}\right]$, the equation $\cos^2 x-\cos x-x=0$ has

(A) no solution (B) exactly one solution
(C) exactly two solution (D) more than two solutions

114$_{(mcq)}$ The value of x satisfying the equation $\sin^3 x+\sin^2 x\cos x+\sin x\cos^2 x+\cos^3 x=\frac{\sqrt{6}}{2}$ can be

(A) $\tan^{-1}(2-\sqrt{3})$ (B) $\tan^{-1}(\sqrt{2}-1)$ (C) $\tan^{-1}(\sqrt{2}+1)$ (D) $\sin^{-1}\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)$

115. The number of real solution of the equation $\cos^{-1}x + 2\cos^{-1}2x + 3\cos^{-1}3x = 6\pi$ is
(A) 0 (B) 1 (C) 2 (D) infinitely many

116. If range of the function $f(x) = \cot^{-1}\left(\frac{x^2}{x^2+1}\right)$ is (a, b] then find the value of $\frac{b}{a}$.

117$_{(mcq)}$ Let $f(x) = \tan^{-1}(1 + x + x^2 + ...\ \infty)$, $g(x) = \cot^{-1}\left(\frac{1}{1-x+x^2-x^3+....\infty}\right)$ then find the correct statement $\forall\ x \in (0, 1)$

(A) Range of $f(x) \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (B) Range of $f(x) \in \left(0, \frac{\pi}{4}\right)$

(C) Range of $g(x) \in \left(\cot^{-1}2, \frac{\pi}{4}\right)$ (D) Range of $g(x) \in \left(\frac{\pi}{4}, \pi\right)$

118. Find the value of x satisfying the equation,

$$\log_{10}\sqrt{5\cot^{-1}x - 1} + \frac{1}{2}\log_{10}(2\cot^{-1}x + 3) + \log_{10}\sqrt{5} = 1.$$

119. The exhaustive set of values of 'a' for which the function $f(x) = \tan^{-1}(x^2 - 18x + a) > 0\ \forall\ x \in R$ is
(A) $(81, \infty)$ (B) $[81, \infty)$ (C) $(-\infty, 81]$ (D) $(-\infty, 81)$

120. The range of $f(x) = \cot^{-1}(-x) - \tan^{-1}x + \sec^{-1}x$ is

(A) $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$ (B) $\left[\frac{\pi}{2}, \pi\right) \cup \left(\pi, \frac{3\pi}{2}\right]$

(C) $\left(\frac{-\pi}{2}, \frac{3\pi}{2}\right)$ (D) $\left(\frac{\pi}{2}, \pi\right) \cup \left(\pi, \frac{3\pi}{2}\right)$

121. The value of $\sin^{-1}(\cos 2) - \cos^{-1}(\sin 2) + \tan^{-1}(\cot 4) - \cot^{-1}(\tan 4) + \sec^{-1}(\text{cosec}\,6) - \text{cosec}^{-1}(\sec 6)$ is
(A) 0 (B) 3π (C) $8 - 3\pi$ (D) $5\pi - 16$

122. Let $S_n = \sum_{n=1}^{n}\sin^{-1}\left(\frac{(2n+1)}{n(n+1)\left(\sqrt{n(n+2} + \sqrt{(n+1)(n-1)}\right)}\right)$. Find the value of 100 cos (S_{99}).

123. Let $f(x) = x^2 - 2ax + a - 2$ and $g(x) = \left[2 + \sin^{-1}\frac{2x}{1+x^2}\right]$. If the set of real values of 'a' for which $f(g(x)) < 0\ \forall\ x \in R$ is (k_1, k_2) then find the value of $(10k_1 + 3k_2)$.
[**Note :** [k] denotes greatest integer less than or equal to k.]

124. The number of value(s) of a $\in$ R for which the equation $5\tan^{-1}(x^2+x+a)+3\cot^{-1}(x^2+x+a)=2\pi$, has unique solution is

(A) 0 (B) 1 (D) 2 (D) more than two

125. If $f(x)=\tan^{-1}\left(\sqrt{\left(\frac{1-\cos 2}{1+\cos 2}\right)}+x^2\right)$ and $g(x)=\begin{cases}|x| & x\neq 0\\ 1 & x=0\end{cases}$ then number of solution(s) of the equation f(x) = g(x) is(are)

(A) 0 (B) 1 (C) 2 (D) 3

126$_{\textbf{(mcq)}}$ Which of the following is(are) correct?

(A) Domain of $f(x)=\sin^{-1}(\cos^{-1}x+\tan^{-1}x+\cot^{-1}x)$ is null set.

(B) Domain of $f(x)=\cos^{-1}(\tan^{-1}x+\cot^{-1}x+\sin^{-1}x)$ is $[-1,-\cos 1]$.

(C) Domain of $f(x)=\sin^{-1}(\cos^{-1}x)$ is $[\cos 1, 1]$

(D) Domain of $f(x)=\cos^{-1}(\sin^{-1}x)$ is $[-\sin 1, \sin 1]$,

127. If $\sin(30°+\text{arc}\tan x)=\frac{13}{14}$ and $0<x<1$, the value of x is $\frac{a\sqrt{3}}{b}$, where a and b are positive integers with no common factors. Find the value of $\left(\frac{a+b}{2}\right)$.

128. The value of a $\in$ R for which the equation $x^2+2ax+8\cos^{-1}(x^2-4x+5)=8\sin^{-1}(4x-5-x^2)$ has atleast one solution is

(A) $\pi+1$ (B) $-(1+\pi)$ (C) $2-\pi$ (D) $\pi-2$

129.

Column-I	Column-II
(A) $f(x)=\sin^{-1}\left(\frac{2}{\lvert\sin x-1\rvert+\lvert\sin x+1\rvert}\right)$	(P) f(x) is many one
(B) $f(x)=\cos^{-1}(\lvert x-1\rvert-\lvert x-2\rvert)$	(Q) Domain of f(x) is R
(C) $f(x)=\sin^{-1}\left(\frac{\pi}{\lvert\sin^{-1}x-(\pi/2)\rvert+\lvert\sin^{-1}x+(\pi/2)\rvert}\right)$	(R) Range contain only irrational number
(D) $f(x)=\cos(\cos^{-1}\lvert x\rvert)+\sin^{-1}(\sin x)-\text{cosec}^{-1}(\text{cosec } x)+\text{cosec}^{-1}\lvert x\rvert$	(S) f(x) is even.

130. Let $f(x)=\tan^{-1}(\cot x-2\cot 2x)$ and $\sum_{r=1}^{5} f(r) = a-b\pi$, where a, b $\in$ N. Find the value of (a + b).

131. Let $\alpha=\cot^{-1}\left(\frac{\pi}{3}\right)$, $\beta=\sin^{-1}\left(\frac{\pi}{4}\right)$ and $\gamma=\sec^{-1}\left(\frac{2\pi}{3}\right)$, then the correct order sequence is

(A) $\alpha<\gamma<\beta$ (B) $\beta<\alpha<\gamma$ (C) $\gamma<\beta<\alpha$ (D) $\alpha<\beta<\gamma$

132. Let $f: R\to\left(0,\frac{\pi}{6}\right]$ be defined as $f(x)=\sin^{-1}\left(\frac{4}{4x^2-12x+17}\right)$ then f(x) is

(A) injective as well as surjective. (B) surjective but not injective.

(C) injective but not surjective. (D) neither injective nor surjective.

133. Find the value of
$[\cos 1 - \cos^{-1} 1] - [\sin 1 - \sin^{-1} 1] + [\tan 1 - \tan^{-1} 1] - [\cot 1 - \cot^{-1} 1] + [\sec 1 - \sec^{-1} 1] - [\operatorname{cosec} 1 - \operatorname{cosec}^{-1} 1]$ where $[x]$ denotes greatest integer less than or equal to x.

134. Match the entries of Column-I with one or more than one entries of column-II. Note that $[x]$, $\{x\}$ and sgn x denote largest integer less than or equal to x, fractional part of x and signum function of x respectively.

Column I	Column II
(A) Let $f : [-1, 1] \to R$ be defined by $f(x) = \sqrt[5]{x} + \sin^{-1}x$ then $f(x)$ is	(P) Odd
(B) Let $f : R \to \{-1, 0, 1\}$ be defined by $f(x) = \operatorname{sgn}\left(\frac{1-\lvert x\rvert}{1+\lvert x\rvert}\right)$ then $f(x)$ is	(Q) Even
(C) Let $f : [-4, 2] \to [0, 3]$ be defined by $f(x) = \sqrt{8-2x-x^2}$ then $f(x)$ is	(R) Onto
(D) Let $f: (-\infty, 0] \to [0, \infty)$ be defined by $f(x) = \frac{2^{-[x]}}{2^{\{x\}}} - 2^{\lvert x\rvert}$ then $f(x)$ is	(S) One-One
	(T) Many-One

135. The value of sum $\sum_{n=1}^{\infty} \tan^{-1}\left(\frac{8n}{n^4 - 2n^2 + 5}\right)$ equals

(A) $\tan^{-1}2$ (B) $\cot^{-1}2$ (C) $\cot^{-1}2 + \cot^{-1}3$ (D) $\tan^{-1}2 + \cot^{-1}2$

136. The number of solutions of the equation $|y| = \cos x$ and $y = \cot^{-1}(\cot x)$ in $\left(-\frac{3\pi}{2}, \frac{5\pi}{2}\right)$ is

(A) 2 (B) 4 (C) 6 (D) none of these

137. The number of non zero roots of the equation $\sqrt{\sin x} = \cos^{-1}(\cos x)$ in $(0, 2\pi)$

(A) 0 (B) 1 (C) 2 (D) infinite

138$_{(mcq)}$ If $f(x) = \tan^{-1}\left(\frac{\sqrt{3}x - 3x}{3\sqrt{3} + x^2}\right) + \tan^{-1}\left(\frac{x}{\sqrt{3}}\right)$, $0 \le x \le 3$, then

(A) the least value f(x) is $-\frac{\pi}{3}$ (B) the greatest value of f(x) is $\frac{\pi}{4}$

(C) the least value of f(x) is 0 (D) the greatest value of f(x) is $\frac{\pi}{3}$

139. Let $f: R \to [\alpha, \infty), f(x) = x^2 + 3ax + b, g(x) = \sin^{-1}\frac{x}{4}$ $(\alpha \in R)$.

	Column - I		**Column - II**
(A)	The possible integral values of 'a' for which f(x) is many one in interval [– 3, 5] is/are	(P)	– 2
(B)	Let a = – 1 and gof(x) is defined for x ∈ [– 1, 1] then possible integral values of b can be	(Q)	– 1
(C)	Let a = 2, α = – 8 the value(s) of b for which f(x) is surjective is/are	(R)	0
(D)	If a = 1, b = 2, then integers in the range of fog(x) is/are	(S)	1

140. If the minimum value of function $f(x) = 8^{\sin^{-1}x} + 8^{\cos^{-1}x}$ is m, then the value of $\log_2 m$ is equal to

(A) $1 + \frac{\pi}{4}$ (B) $-1 + \frac{3\pi}{4}$ (C) $1 + \frac{3\pi}{4}$ (D) $-1 + \frac{\pi}{2}$

141(mcq) The value of $\cos^{-1}\{\sin(\cos^{-1}(\cos x) - \sin^{-1}(\sin x)\}$ where $x \in \left(\pi, \frac{3\pi}{2}\right)$ is equal to

(A) $-\frac{\pi}{2}$ (B) $\frac{\pi}{2}$ (C) $-\pi$ (D) π

142.

	Column - I		**Column - II**
(A)	$\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{13}\right) + ... + \infty$	(P)	$\frac{\pi}{2}$
(B)	$\sin^{-1}\left(\frac{12}{13}\right) + \cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(\frac{63}{16}\right)$	(Q)	$\frac{\pi}{4}$
(C)	$\sin^{-1}\left(\frac{4}{5}\right) + 2\tan^{-1}\left(\frac{1}{3}\right)$	(R)	π
(D)	$\cot^{-1} 9 + \operatorname{cosec}^{-1}\left(\frac{\sqrt{41}}{4}\right)$	(S)	$\frac{\pi}{3}$

]

143. Column-I contain four functions and column-II contain their properties. Match every entry of column-I with one or more entries of column-II.

	Column-I		**Column-II**
(A)	$f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$	(P)	range is $[0, \pi]$
(B)	$g(x) = \sin^{-1}\lvert x\rvert + 2\tan^{-1}\lvert x\rvert$	(Q)	is increasing $\forall\ x \in (0, 1)$
(C)	$h(x) = 2\sin^{-1}\left(\frac{2x}{1+x^2}\right),\ x \in [0, 1]$	(R)	period is 2π
(D)	$k(x) = \cot(\cot^{-1}x)$	(S)	is decreasing $\forall\ x \in (0, 1)$

144. If $\sin^{-1}(x^2 + y^2) + \tan^{-1}\sqrt{4y^2 + 1} + \sec^{-1}x = a$ then find the sum of the possible values of a.

145. The sum of the infinite terms of the series

$$\cot^{-1}\left(1^2+\frac{3}{4}\right)+\cot^{-1}\left(2^2+\frac{3}{4}\right)+\cot^{-1}\left(3^2+\frac{3}{4}\right)+\ldots\ldots \text{ is equal to :}$$

(A) $\tan^{-1}(1)$ (B) $\tan^{-1}(2)$ (C) $\tan^{-1}(3)$ (D) $\tan^{-1}(4)$

146. Find the set of values of 'a' for which the equation $2\cos^{-1}x = a + a^2(\cos^{-1}x)^{-1}$ posses a solution.

147. Find the integral values of K for which the system of equations;

$$\begin{cases} \arccos x + (\arcsin y)^2 = \dfrac{K\pi^2}{4} \\ (\arcsin y)^2 \cdot (\arccos x) = \dfrac{\pi^4}{16} \end{cases}$$ possesses solutions & find those solutions.

148. Find the domain of definition the following functions.
(Read the symbols [] and {} as greatest integers and fractional part functions respectively.)

$$f(x) = \sqrt{\sin(\cos x)} + \ln(-2\cos^2 x + 3\cos x + 1) + e^{\cos^{-1}\left(\frac{2\sin x+1}{2\sqrt{2\sin x}}\right)}$$

149. Solve the equation:

$$\cos^{-1}\left(\sqrt{6}x\right)+\cos^{-1}\left(3\sqrt{3}x^2\right)=\frac{\pi}{2} \qquad \left[\text{Ans. } x=\frac{1}{3}\right]$$

150. If $\sin^{-1}\left(x-\frac{x^2}{2}+\frac{x^3}{4}-\ldots\ldots\right)+\cos^{-1}\left(x^2-\frac{x^4}{2}+\frac{x^6}{4}-\ldots\ldots\right)=\frac{\pi}{2}$ for $0<|x|<\sqrt{2}$ then x equals to

(A) 1/2 (B) 1 (C) – 1/2 (D) – 1

151. Equation of the image of the line $x + y = \sin^{-1}(a^6+1) + \cos^{-1}(a^4+1) - \tan^{-1}(a^2+1)$, $a \in R$ about x axis is given by

(A) $x - y = 0$ (B) $x - y = \frac{\pi}{2}$ (C) $x - y = \pi$ (D) $x - y = \frac{\pi}{4}$

152.

	Column I		Column II
(A)	$\cot^{-1}\left(\tan(-37^\circ)\right)$	(P)	143°
(B)	$\cos^{-1}\left(\cos(-233^\circ)\right)$	(Q)	127°
(C)	$\sin\left(\frac{1}{2}\cos^{-1}\left(\frac{1}{9}\right)\right)$	(R)	$\frac{3}{4}$
(D)	$\cos\left(\frac{1}{2}\text{arc}\cos\left(\frac{1}{8}\right)\right)$	(S)	$\frac{2}{3}$

153. The exhaustive set of values of 'x' for which $\sin^{-1}\left(\frac{4x}{x^2+4}\right)+2\tan^{-1}\left(\frac{-x}{2}\right)$ is independent of 'x', is

(A) $(-\infty, -2]$ (B) $[2, \infty)$ (C) $(-\infty, -2] \cup [2, \infty)$ (D) $[-2, 2]$

154. If $\alpha = \sum_{n=1}^{\infty} \tan^{-1}\left\{\frac{16(2n+1)}{(2n+1)^4 - 4(2n+1)^2 + 16}\right\}$ and $\sin\alpha = \frac{a}{b}$ where a, b $\in$ N, then find the least value of $a^2 + b^2$.

155. Sum of the squares of all the solution(s) of the equation,
$2\sin^{-1}(x+2) = \cos^{-1}(x+3)$ is
(A) 4 (B) 6.25 (C) 10.25 (D) none

156. $\alpha = \sin^{-1}\left(\cos\left(\sin^{-1} x\right)\right)$ and $\beta = \cos^{-1}\left(\sin\left(\cos^{-1} x\right)\right)$, then :
(A) $\tan\alpha = \cot\beta$ (B) $\tan\alpha = -\cot\beta$ (C) $\tan\alpha = \tan\beta$ (D) $\tan\alpha = -\tan\beta$

157. Let $\alpha = \sqrt{\cot^{-1} x}$; $\beta = \cot^{-1} x$ and $\gamma = (\cot^{-1} x)^2$. If $\alpha > \beta > \gamma$ then 'x' can be
(A) 0 (B) 1 (C) 2 (D) 3

158. If $x = \alpha$ satisfies the equation $\frac{\sin^{-1} x^2 + \cos^{-1} x}{\cos^{-1} x^2 + \sin^{-1} x} = -3$,
then find the value of $(\alpha^2 + 2\alpha + 3)$.

159. Let $S = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \ldots\ldots\ldots\infty$

	Column-I		**Column-II**
(A)	Sum of real root(s) of the equation $\cos^{-1}(x^2 - 5) + 2\cos^{-1}(x^3 - x^2 - 4x + 3) = 3\pi S$, is	(P)	3
(B)	The value of $2\tan\left(\cot^{-1}\left(\log_{10}\left(\sqrt{10}\right)^{4S}\right)\right)$ equals	(Q)	0
(C)	Number of integral point(s) in the domain of function $f(x) = \sqrt{\log_3(\cos^{-1} Sx)}$ is	(R)	2
(D)	The value of $\left[5\tan^{-1}\left(\cos\left(\tan^{-1}\{S\}\right)\right)\right]$ equals	(S)	1

where {x} denotes fractional part function of x and [x] denotes greatest integer less than or equal to x respectively.

160. Let $f(n) = \sum_{k=-n}^{n}\left(\cot^{-1}\frac{1}{k} - \tan^{-1} k\right)$, $k \neq 0$. If the value of $\sum_{n=2}^{10}\left(f(n) + f(n-1)\right)$ is $m\pi$ then find the value of m.

161. If $\theta = \frac{1}{2}\sin^{-1}\frac{1}{4}$ then $64\sin\theta + 64\cos\theta - 8\sec\theta - 8\operatorname{cosec}\theta + \tan\theta + \cot\theta$ equals
(A) 7 (B) 8 (C) 9 (D) 10

162. The range of the function $f(x) = \sin^{-1} x\,(1 + \cos^{-1} x + (\cos^{-1} x)^2 + \ldots\ldots \infty)$ where $x \in (\cos 1, 1)$ is

(A) $\left(\frac{\pi}{2}, \infty\right)$ (B) $\left(1, \frac{\pi}{2}\right)$ (C) $(-\infty, 1)$ (D) None

163. The domain of $f(x) = \sqrt{\log_{\tan^{-1} x}(x^2 + x + 1)}$, is

(A) $(\tan 1, \infty)$ (B) $(-1, \tan 1)$ (C) $(-1, \tan 1)$ (D) $(-\tan 1, \tan 1)$

164$_{\text{(mcq)}}$ If $\sin^{-1} x + \sin^{-1} y = A$ and $\cos^{-1} x - \cos^{-1} y = B$ then find the correct statements.

(A) $x + y = 2\sin\left(\frac{A}{2}\right) \cos\left(\frac{B}{2}\right)$ (B) $x + y = 2\cos\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right)$

(C) $x - y = 2\cos\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right)$ (D) $x - y = -2\cos\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right)$

165. If $\sin^{-1}(e^x) + \cos^{-1}(x^2) = \frac{\pi}{2}$, then find the number of solutions of this equation

(A) 1 (B) 2 (C) 3 (D) 0

166$_{\text{(mcq)}}$ If $A = \tan^{-1}\left(\frac{1}{7}\right)$ and $B = \tan^{-1}\left(\frac{1}{3}\right)$, then

(A) $\cos 2A = \frac{24}{25}$ (B) $\cos 2B = \frac{4}{5}$ (C) $\cos 2A = \sin 4B$ (D) $\tan 2B = \frac{3}{4}$

167. If $\sin^{-1}(1 - x) + \cos^{-1}(mx) = \frac{\pi}{2}$ has some solution for x. Then the range of m, is

(A) $(-\infty, -1)$ (B) $(-1, \infty)$ (C) $(\frac{1}{2}, \infty)$ (D) $[-\frac{1}{2}, \infty)$

ANSWER KEY

1.	BCD	**2.**	AD	**3.**	BD	**4.**	(A) Q, S; (B) Q, R, S; (C) P, Q, R, S; (D) P, R						
5.	1	**6.**	6	**7.**	C	**8.**	C	**9.**	$t = \pi$	**10.**	$(-\infty, \cot 1)$		
11.	0512	**12.**	5	**13.**	ABC	**14.**	A	**15.**	$[1, \infty)$				
16.	(A) P, (B) Q, (C) P, (D) R					**17.**	B	**18.**	BD	**19.**	ABD	**20.**	C
21.	B	**22.**	A	**23.**	C	**24.**	A	**25.**	C	**26.**	B	**27.**	C
28.	C	**29.**	B	**30.**	C	**31.**	D	**32.**	A	**33.**	D	**34.**	C
35.	D	**36.**	ABCD			**37.**	D	**38.**	C	**39.**	D	**40.**	B
41.	(A) S ; (B) P ; (C) R; (D) R					**42.**	D						
43.	(a) BCD (b) AB (c) B					**44.**	1	**45.**	20	**46.**	B	**47.**	D
48.	ABCD	**49.**	8			**50.**	B						
51.	(A) P, Q, R, S; (B) **P,** Q; (C) P, R, S; (D) **P,** R, S							**52.**	20	**53.**	B	**54.**	B
55.	ABCD	**56.**	AB			**57.**	1	**58.**	AB	**59.**	D	**60.**	C
61.	5	**62.**	17.			**63.**	1	**64.**	C	**65.**	D	**66.**	A
67.	D	**68.**	B	**69.**	C	**70.**	A	**71.**	D	**72.**	C	**73.**	28
74.	36	**75.**	B	**76.**	A	**77.**	D	**78.**	B	**79.**	C		
80.	(A) P, S (B) Q, R, T (C) R, T (D) T							**81.**	(A) P; (B) Q, R, S, T; (C) Q; (D) R				
82.	5	**83.**	C	**84.**	B	**85.**	(A) P; (B) Q ; (C) P ; (D) S						
86.	17	**88.**	B	**89.**	47	**90.**	A	**91.**	0001	**92.**	1	**93.**	B
94.	C	**95.**	ACD	**96.**	B	**97.**	C	**98.**	A				
99.	C	**100.**	D	**101.**	2	**102.**	B	**103.**	C	**104.**	B	**105.**	C
106.	B	**107.**	D	**108.**	A	**109.**	A	**110.**	A	**111.**	4	**112.**	A
113.	B	**114.**	AD	**115.**	A	**116.**	2	**117.**	AC	**118.**	cot 1	**119.**	A
120.	B	**121.**	D	**122.**	1	**123.**	20	**124.**	B	**125.**	D		
126.	ABCD			**127.**	8	**128.**	B						
129.	(A) P, Q, R, S; (B) P, Q; (C) P, R, S; (D) P, R, S									**130.**	20		
131.	D	**132.**	B	**133.**	4	**134.**	(A) P, S (B) Q, R, T (C) R, T (D) T						
135.	D	**136.**	A	**137.**	B	**138.**	BC						
139.	(A) P,Q,R,S ; (B) P, Q,R ; (C) S ; (D) R,S							**140.**	C	**141.**	BC		
142.	(A) Q; (B) R; (C) P ; (D) Q					**143.**	(A) P, Q, R; (B) P, Q; (C) P, Q; (D) Q						
144.	$a = \frac{5\pi}{2}$			**145.**	B	**146.**	$a \in [-2\pi, \pi] - \{0\}$						
147.	$K = 2$; $\cos\frac{\pi^2}{4}, 1$ & $\cos\frac{\pi^2}{4}, -1$.					**148.**	$n \in I$	**149.**	$x = \frac{1}{3}$			**150.**	B
151.	D	**152.**	(A) Q; (B) Q; (C) S; (D) R					**153.**	D	**154.**	0041		
155.	A	**156.**	A	**157.**	$(\cot 1, \infty)$			**158.**	2				
159.	(A) Q ; (B) S ;(C) R ;(D) P			**160.**	99	**161.**	B	**162.**	A	**163.**	A		
164.	AD	**165.**	A	**166.**	ABCD			**167.**	D				

1. Let $\tan^{-1}\frac{a}{x} = \alpha \quad \Rightarrow \quad \tan\alpha = \frac{a}{x}$ etc.

$\alpha + \beta + \gamma + \delta = \frac{\pi}{2}$

$\tan(\alpha + \beta + \gamma + \delta) = \tan\frac{\pi}{2}$

$\frac{S_1 - S_3}{1 - S_2 + S_4} = \infty \quad \Rightarrow \quad 1 - S_2 + S_4 = 0 \Rightarrow \quad S_4 - S_2 + 1 = 0$

Now, $S_4 = \tan\alpha \cdot \tan\beta \cdot \tan\gamma \cdot \tan\delta = \frac{abcd}{x^4}$

$S_2 = \sum \tan\alpha \cdot \tan\beta = \frac{\sum ab}{x^2}$

$\therefore \quad \frac{abcd}{x^4} - \frac{\sum ab}{x^2} + 1 = 0$

$x^4 - \sum ab\, x^2 + abcd = 0$ ⟶ x_1, x_2, x_3, x_4

$\therefore \quad x_1 + x_2 + x_3 + x_4 = 0 \quad ....(1)$

$\sum x_1x_2x_3 = 0 \quad ....(2)$

$\sum x_1x_2x_3 = \underbrace{x_1x_2x_3x_4}_{\text{non zero}}\left[\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{1}{x_4}\right] = 0$

$x_1x_2x_3x_4 = abcd$

2. $f(x) = |\sin^{-1}x| + \cos^{-1}\left(\frac{1}{x}\right)$

Domain of $f(x)$ is $\{-1, 1\}$

$f(1) = \frac{\pi}{2}, f(-1) = \frac{3\pi}{2}$

So function $f(x)$ is injective

$\text{sgn}(f(x)) = 1 \quad (f(x) > 0)$

Range of $f(x) = \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\}$]

3.

(A) We have $f(x) = \text{sgn}\left(x^3 + \sqrt{x^6 + 1}\right) = 1 \ \forall x \in R \qquad \left(x^2 + \sqrt{x^2 + 1}\right) > 0 \ \forall x \in R$

$\Rightarrow$ $f(x)$ is an even function.

Hence graph of $f(x)$ is symmetric about y-axis i.e. $x = 0$.

(B) $\frac{1}{e^x + e^{-x}}$ is even

domain $(-\infty, \infty) \rightarrow \left[\cos^{-1} f(0), \cos^{-1} f(\infty)\right]$ hence range $\left[\frac{\pi}{3}, \frac{\pi}{2}\right)$

(C) Clearly, domain of $f(x) = \phi$.

(D) As $\cos^{-1}(\cos 7) = 7 - 2\pi$ and $\sin^{-1}(\sin 11) = 11 - 4\pi$
So, $2\cos^{-1}(\cos 7) - \sin^{-1}(\sin 11) = 2(7 - 2\pi) - (11 - 4\pi) = 14 - 4\pi - 11 + 4\pi = 3$ **Ans.**

4.
(A) $f_2(x) = f(f(x)) = f(x) = x$
$f_3(x) = f(f_2(x) = f(x) = x$
$\Rightarrow$ $x^3 - 25x^2 + 175x - 375 = 0$
$(x-5)(x^2 - 20x + 75)$
$(x-5)^2(x-15) = 0$
$\Rightarrow$ $x = 5, 15$ $\Rightarrow$ **Q, S**

(B) Range of $f(f(f(x)))$ is $[4, 17]$ $\Rightarrow$ **Q, R, S**
Domain of $f(x)$ is $[-1, 1]$

(C) If $x \in [-1, 0)$, $f(x) = 8(x + \pi) + 5x + 4x - x = 16x + 8\pi$
$f(x) \in [8\pi - 16, 8\pi]$
If $x \in (0, 1]$, $f(x) = 8x + 5x + 4x - x = 16x$
$f(x) \in (0, 16]$ $\Rightarrow$ **P, Q, R, S**

(D) $x \in [-1, 0]$

$$x + \frac{1+x^2}{2} = -2x$$

$$x^2 + 6x + 1 = 0 \quad \Rightarrow \quad x = \frac{-6 \pm \sqrt{36-4}}{2} = -3 \pm 2\sqrt{2}$$

$$x = 2\sqrt{2} - 3 \quad \Rightarrow |10a| = \left[\left|20\sqrt{2} - 30\right|\right] = 30 - 20\sqrt{2}\,|$$

$x \in [0, 1]$

$$x + \frac{1+x^2}{2} = 2x$$

$1 + x^2 = 2x \;\Rightarrow\; x = 1 \;\Rightarrow$ $|10a| = 10$
$|10a| = 10,\ |20\sqrt{2} - 30|$
$\Rightarrow$ $[\,|10a|\,] = 1, 10$ $\Rightarrow$ **P, R**]

5. $$\text{arc}\cos\left(\frac{2}{\pi}\text{arc}\cos x\right) = \text{arc}\sin\left(\frac{2}{\pi}\text{arc}\sin x\right)$$

$$\cos^{-1}\left(\frac{2}{\pi}\left(\frac{\pi}{2} - \sin^{-1} x\right)\right) = \sin^{-1}\left(\frac{2}{\pi}\sin^{-1} x\right)$$

$$\cos^{-1}\left(1-\frac{2}{\pi}\sin^{-1}x\right) = \sin^{-1}\left(\frac{2}{\pi}\sin^{-1}x\right)$$

Let $\frac{2}{\pi}\sin^{-1}x = \alpha$ where $\alpha \in [0, 1]$ think!

$\Rightarrow \quad \cos^{-1}(1-\alpha) = \sin^{-1}\alpha$

$\Rightarrow \quad \sin^{-1}\sqrt{2\alpha-\alpha^2} = \sin^{-1}\alpha$

$\Rightarrow \quad \sqrt{2\alpha-\alpha^2} = \alpha \qquad \Rightarrow \quad 2\alpha - \alpha^2 = \alpha^2$

$\Rightarrow \quad 2\alpha = 2\alpha^2$

Hence α is either 0 or 1.

If $\alpha = 0$ then $x = 0$

if $\alpha = 1$ then $x = 1$

hence sum of all possible value of x is 1 **Ans.**

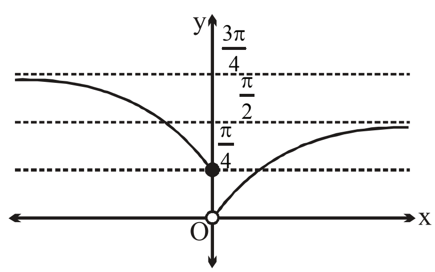
6. For $x \le 0$,

$$f(x) = \cos^{-1}\left(\frac{1+x}{\sqrt{2(1+x^2)}}\right) \qquad \text{put } x = \tan\theta \Rightarrow \theta = \tan^{-1}x \; ; \theta \in \left(\frac{-\pi}{2}, 0\right]$$

$$= \cos^{-1}\left(\frac{1+\tan\theta}{\sqrt{2\sec^2\theta}}\right) = \cos^{-1}\left(\frac{1}{\sqrt{2}}(\cos\theta+\sin\theta)\right) = \cos^{-1}\left(\cos\left(\theta-\frac{\pi}{4}\right)\right) \; ; \theta-\frac{\pi}{4} \in \left(\frac{-3\pi}{4}, \frac{-\pi}{4}\right]$$

$$= -\left(\theta-\frac{\pi}{4}\right) = \frac{\pi}{4} - \tan^{-1}x$$

$$f(x) = \begin{cases} \frac{\pi}{4} - \tan^{-1}x; & x \le 0 \\ \tan^{-1}x; & x > 0 \end{cases}$$



From the graph it is clear that equation $f(x) = k$ has exactly two roots then $k \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right) \equiv [a, b)$

Hence, $\left(\frac{1}{a}+\frac{1}{b}\right)\pi = \left(\frac{4}{\pi}+\frac{2}{\pi}\right) \times \pi = 6.$ **Ans.**

7. Consider $F(x) = \cot\left(\cos^{-1}(|\sin x| + |\cos x|) + \sin^{-1}(-|\cos x| - |\sin x|)\right)$

But $|\sin x| + |\cos x| \in [1, \sqrt{2}] \; \forall x \in R$

$\therefore F(x) = \cot(\cos^{-1}(1) + \sin^{-1}(-1)) = \cot\left(0-\frac{\pi}{2}\right) = 0 = g(3)$

(As $F(x) = 0, \forall x \in D_F$)]

8. As $0 \le (\sin^{-1}a)^2 \le \dfrac{\pi^2}{4}$, $0 \le (\cos^{-1}b)^2 \le \pi^2$, $0 \le (\sec^{-1}c)^2 \le \pi^2 \left(\text{except } \dfrac{\pi^2}{4}\right)$ and

$0 < (\text{cosec}^{-1}d)^2 \le \dfrac{\pi^2}{4}$

So $0 < (\sin^{-1}a)^2 + (\cos^{-1}b)^2 + (\sec^{-1}c)^2 + (\text{cosec}^{-1}d)^2 \le \dfrac{5\pi^2}{2}$

$\therefore \quad (\sin^{-1}a)^2 + (\cos^{-1}b)^2 + (\sec^{-1}c)^2 + (\text{cosec}^{-1}d)^2 = \dfrac{5\pi^2}{2}$ (Given)

$\Rightarrow (\sin^{-1}a)^2 = \dfrac{\pi^2}{4}$, $(\cos^{-1}b)^2 = \pi^2$, $(\sec^{-1}c)^2 = \pi^2$ and $(\text{cosec}^{-1}d)^2 = \dfrac{\pi^2}{4}$

Hence $(\sin^{-1}a)^2 - (\cos^{-1}b)^2 + (\sec^{-1}b)^2 - (\text{cosec}^{-1}d)^2 = 0$]

9. $y = (\sec^{-1}x)^2 + (\text{cosec}^{-1}x)^2 = (\sec^{-1}x + \text{cosec}^{-1}x)^2 - 2\sec^{-1}x \cdot \text{cosec}^{-1}x$

put $t = \sec^{-1}x$; $\sec^{-1}x + \text{cosec}^{-1}x = \dfrac{\pi}{2}$

$y = \dfrac{\pi^2}{4} - 2t\left(\dfrac{\pi}{2} - t\right) = 2t^2 - \pi t + \dfrac{\pi^2}{4}$

$= 2\left[t^2 - \dfrac{\pi}{2}t + \dfrac{\pi^2}{8}\right] = 2\left[\left(t - \dfrac{\pi}{4}\right)^2 + \dfrac{\pi^2}{16}\right] = \dfrac{\pi^2}{8} + 2\left(t - \dfrac{\pi}{4}\right)^2$

$\therefore \quad y_{min} = \dfrac{\pi^2}{8}$; $\quad y_{max} = \dfrac{5\pi^2}{4}$ at $t = \pi$ **Ans.**

10. $ln(\cot^{-1}x) > 0 \Rightarrow \cot^{-1}x > 1 \Rightarrow x < \cot 1$

11. $\sin^{-1}\dfrac{14}{|x|} = \dfrac{\pi}{2} - \sin^{-1}\dfrac{2\sqrt{15}}{|x|} = \cos^{-1}\dfrac{2\sqrt{15}}{|x|}$

$= \sin^{-1}\sqrt{1 - \left(\dfrac{2\sqrt{15}}{|x|}\right)^2}$, for or $|x| \ge 2\sqrt{15}$

$\Rightarrow \left(\dfrac{14}{|x|}\right)^2 = 1 - \left(\dfrac{2\sqrt{15}}{|x|}\right)^2$

$\Rightarrow \quad |x| = 16 \Rightarrow x = \pm 16$, which satisfy $|x| \ge 2\sqrt{15}$

Clearly, $\left(x_1^2 + x_2^2\right) = 256 + 256 = 512$. **Ans.**

12. $f(x) = \left(\cos^{-1}\frac{x}{2}\right)^2 + \pi \sin^{-1}\frac{x}{2} - \left(\sin^{-1}\frac{x}{2}\right)^2 + \frac{\pi^2}{12}\left(x^2+6x+8\right)$

$= \left(\cos^{-1}\frac{x}{2} + \sin^{-1}\frac{x}{2}\right)\left(\cos^{-1}\frac{x}{2} - \sin^{-1}\frac{x}{2}\right) + \pi \sin^{-1}\frac{x}{2} + \frac{\pi^2}{12}\left(x^2+6x+8\right)$

$= \frac{\pi}{2}\cos^{-1}\frac{x}{2} - \frac{\pi}{2}\sin^{-1}\frac{x}{2} + \pi\sin^{-1}\frac{x}{2} + \frac{\pi^2}{12}\left(x^2+6x+8\right)$

$= \frac{\pi}{2}\left(\cos^{-1}\frac{x}{2} + \sin^{-1}\frac{x}{2}\right) + \frac{\pi^2}{12}\left(x^2+6x+8\right) = \frac{\pi^2}{4} + \frac{\pi^2}{12}\left(x^2+6x+8\right)$

Domain of f(x) is [– 2, 2]
f(x) is increasing in [– 2, 2]

$\therefore$ Range of f(x) is $\left[\frac{\pi^2}{4}, \frac{9\pi^2}{4}\right] = [a\pi^2, b\pi^2] \Rightarrow 2(a + b) = 2\left(\frac{1}{4}+\frac{9}{4}\right) = 5$. **Ans.**

13. (A) $\sin^{-1}\left(\sin\frac{6}{\pi}\right) + \frac{6}{\pi} = \pi - \frac{6}{\pi} + \frac{6}{\pi} = \pi$

(B) $\cos^{-1}\left(\cos\frac{12}{\pi}\right) + \frac{12}{\pi} = 2\pi - \frac{12}{\pi} + \frac{12}{\pi} = 2\pi$

(C) $-\tan^{-1}\left(\tan\frac{30}{\pi}\right) + \frac{30}{\pi} = 3\pi - \frac{30}{\pi} + \frac{30}{\pi} = 3\pi$

(D) **is incorrect.**

14. $2\cos^{-1}\frac{3}{\sqrt{13}} + \cot^{-1}\frac{16}{63} + \frac{1}{2}\cos^{-1}\frac{7}{25} = \pi$

LHS $= 2\tan^{-1}\frac{2}{3} + \tan^{-1}\frac{63}{16} + \frac{1}{2}\tan^{-1}\frac{24}{7}$

let $\tan^{-1}\frac{24}{7} = \alpha, \quad \alpha \in (0, \pi/2)$

$\therefore$ $\tan\alpha = \frac{24}{7}$

$\frac{2\tan(\alpha/2)}{1-\tan^2(\alpha/2)} = \frac{24}{7} \Rightarrow \frac{\tan(\alpha/2)}{1-\tan^2(\alpha/2)} = \frac{12}{7}$

$12\tan^2(\alpha/2) + 7\tan(\alpha/2) - 12 = 0$

$\therefore$ $\tan(\alpha/2) = \frac{3}{4}$ $\quad \tan(\alpha/2) = -\frac{4}{3}$ (rejected)

$$\tan^{-1}\frac{(4/3)}{1-(4/9)} + \tan^{-1}\frac{63}{16} + \frac{\alpha}{2}$$

$$\tan^{-1}\frac{12}{5} + \tan^{-1}\frac{63}{16} + \tan^{-1}\frac{3}{4}$$

$$\pi + \tan^{-1}\left(\frac{\frac{12}{5}+\frac{3}{4}}{1-\frac{36}{20}}\right) + \tan^{-1}\left(\frac{63}{16}\right)$$

$$\pi - \tan^{-1}\left(\frac{63}{16}\right) + \tan^{-1}\left(\frac{63}{16}\right) = \pi.$$

15. let $\sec^{-1}x = t$

$t^2 - 6t + 8 > 0$

$(t-4)(t-2) > 0 \Rightarrow t > 4$ or $t < 2$

but $\quad \sec^{-1}x \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$

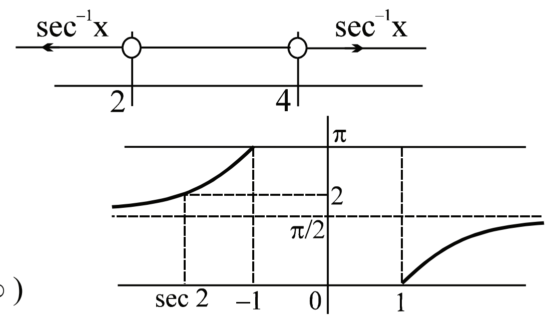
Hence $\sec^{-1}x > 4$ no solution

hence $\sec^{-1}x < 2$

$\Rightarrow \quad \frac{\pi}{2} < \sec^{-1}x < 2$ or $\frac{\pi}{2} < \sec^{-1}x \le 0$

Hence $x \in (-\infty, \sec 2)$ or $x \in [1, \infty)$

Hence $x \in (-\infty, \sec 2) \cup [1, \infty)$ **Ans.**



16. $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$ $\quad$ so $\quad \cos^{-1}x = \sin^{-1}y = \cos^{-1}\sqrt{(1-y^2)}$

$x^2 + y^2 = 1$

$\tan^{-1}x + \tan^{-1}y = \frac{\pi}{2}$ $\quad$ as $\quad \tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$

$\cot^{-1}x = \tan^{-1}y \Rightarrow xy = 1$

$\cos^{-1}x + \cos^{-1}y = \frac{\pi}{2} \Rightarrow x^2 + y^2 = 1$

$\sec^{-1}x + \operatorname{cosec}^{-1}x = \frac{\pi}{2}$

$\sec^{-1}x + \sec^{-1}y = \pi/2$

$\operatorname{cosec}^{-1}x = \sec^{-1}y = \operatorname{cosec}^{-1}\frac{y}{\sqrt{(y^2-1)}}$

$x^2(y^2-1) = y^2 \Rightarrow x^2 + y^2 = x^2y^2$. **Ans.**

17. Let $\sin^{-1}\dfrac{f(x)}{2} = \theta \Rightarrow \theta \in \left[-\dfrac{\pi}{6}, \dfrac{\pi}{6}\right], 2\theta \in \left[-\dfrac{\pi}{3}, \dfrac{\pi}{3}\right]$ (as $f(x) \in [-1, 1]$)

$$g(x) = \sin^{-1}\left(\cancel{2}\frac{f(x)}{\cancel{2}}\sqrt{1-\left(\frac{f(x)}{2}\right)^2}\right) + \frac{\pi}{3} = \sin^{-1}(\sin 2\theta) + \frac{\pi}{3} = 2\theta + \frac{\pi}{3} = 2\sin^{-1}\left(\frac{f(x)}{2}\right) + \frac{\pi}{3}$$

$$g(x) = 2\sin^{-1}\frac{f(x)}{2} + \frac{\pi}{3} \Rightarrow g(x) \in \left[2\left(-\frac{\pi}{6}\right)+\frac{\pi}{3}, 2\left(\frac{\pi}{6}\right)+\frac{\pi}{3}\right]$$

$g(x) = \left[0, \dfrac{2\pi}{3}\right] \Rightarrow$ (B)

as f is onto hence

Alternatively: when $f(x) = -1 \Rightarrow \quad g(x) = \sin^{-1}\left(-\dfrac{\sqrt{3}}{2}\right) = -\dfrac{\pi}{3}$

when $f(x) = 1 \Rightarrow \quad g(x) = \sin^{-1}\left(\dfrac{\sqrt{3}}{2}\right) = \dfrac{\pi}{3}$

hence $g(x) \in \left[\left(-\dfrac{\pi}{3}+\dfrac{\pi}{3}\right), \left(\dfrac{\pi}{3}+\dfrac{\pi}{3}\right)\right]$ i.e. $g(x) \in \left[0, \dfrac{2\pi}{3}\right]$

18. $n^4 + 4 = (n^2+2)^2 - 4n^2 = \left((n-1)^2+1\right)\left((n+1)^2+1\right)$

$$\tan^{-1}\frac{\left((n+1)^2+1\right) - \left((n-1)^2+1\right)}{1+\left((n+1)^2+1\right)\left((n-1)^2+1\right)} = \tan^{-1}\left((n+1)^2+1\right) - \tan^{-1}\left((n-1)^2+1\right)$$

$$\therefore S_\infty = \frac{3\pi}{4} - \tan^{-1}2 = \frac{\pi}{4} + \frac{\pi}{2} - \tan^{-1}2 = \frac{\pi}{4} + \cot^{-1}2 = \frac{\pi}{4} + \tan^{-1}\frac{1}{2} = \frac{\pi}{A} + \tan^{-1}B$$

$\therefore$ A = 4 and B = $\dfrac{1}{2}$.

19. $Sin^{-1}\left[x^2+\dfrac{1}{2}\right]$ is defined if $\left[x^2+\dfrac{1}{2}\right] = -1 \text{ or } 0 \text{ or } 1$

and correspondingly $\left[x^2-\dfrac{1}{2}\right] = -2 \text{ or } -1 \text{ or } 0$

but $\cos^{-1}\left[x^2-\dfrac{1}{2}\right]$ is not defined if $\left[x^2-\dfrac{1}{2}\right] = -2$

$$\therefore \left[x^2+\frac{1}{2}\right] = 0 \text{ or } 1 \Rightarrow 0 \le \left(x^2+\frac{1}{2}\right) < 2 \Rightarrow 0 \le x^2 < \frac{3}{2}$$

$\therefore 0 \le |x| < \sqrt{\frac{3}{2}}$

Then $f(x) = \left(\sin^{-1}(0) + \cos^{-1}(-1)\right)$ or $\left(\sin^{-1}(1) + \cos^{-1}(0)\right)$

$\Rightarrow f(x) = \pi \ \forall\ x \in domain \left(-\sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}}\right)$.

20. $LHS \le \sqrt{a^2+b^2} = 13$ and $RHS = 2(y-2)^2 + 13 \ge 13$

solution exists if LHS = RHS = 13

then $\sin(\alpha+\theta) = 1 \Rightarrow \alpha = \frac{\pi}{2} - \theta = \frac{\pi}{2} - \left(\tan^{-1}\frac{5}{12}\right) = \tan^{-1}\frac{12}{5}$

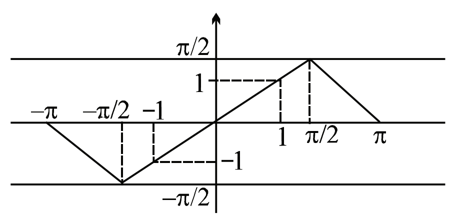
Also RHS = 13 if $y = \frac{-b}{2a} \Rightarrow \beta = 2$

$\therefore (2\beta+1)\tan\alpha = 12$.

21. x = 3 is the only solution, x = 0 and x = – 2/3 is rejected]

22. $y = \text{cosec}^{-1}(\text{cosec}x)$; $x \in R - \{n\pi, n \in I\}$; $y \in [-\pi/2, 0) \cup (0, \pi/2]$

and $\quad y = \text{cosec}(\text{cosec}^{-1}x)$; $|x| \ge 1$; $|y| \ge 1$



$\therefore$ range of value of x $\quad y \in \left[-\frac{\pi}{2}, -1\right] \cup \left[1, \frac{\pi}{2}\right]$

23. $\cos^{-1}(2x) + \cos^{-1}(3x) = \pi - \cos^{-1}(x) = \cos^{-1}(-x)$

$\cos^{-1}[(2x)(3x) - \sqrt{1-4x^2}\sqrt{1-9x^2}\,] = \cos^{-1}(-x)$

$6x^2 - \sqrt{1-4x^2}\cdot\sqrt{1-9x^2} = -x$

$(6x^2 + x)^2 = (1 - 4x^2)(1 - 9x^2)$

$\Rightarrow \quad x^2 + 12x^3 = 1 - 13x^2 \quad \Rightarrow \quad 12x^3 + 14x^2 - 1 = 0$

$\therefore \quad a = 12;\ \ b = 14;\ c = 0 \ \Rightarrow \quad a + b + c = 26$ **Ans.**

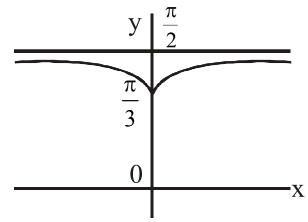
24. We have $\sin^{-1}\left[\frac{\pi x}{6}\right] > 0 \Rightarrow \left[\frac{\pi x}{6}\right] = 1$

$\Rightarrow \quad 1 \le \frac{\pi x}{6} < 2 \Rightarrow \frac{6}{\pi} \le x < \frac{12}{\pi}$

$\therefore \quad x = 2, 3$ only.

Hence two integral solution will satisfy above equation.

25. $\dfrac{1}{e^{x}+e^{-x}}$ is even



domain $(-\infty, \infty) \to \left(0, \dfrac{1}{2}\right] \Rightarrow$ range is (B).

26. As, $\quad 1 \le 1 + \sin^2 x \le 2 \ \Rightarrow\ \dfrac{1}{2} \le \dfrac{1}{1+\sin^2 x} \le 1$

$\Rightarrow \quad \dfrac{\pi}{6} \le \sin^{-1}\left(\dfrac{1}{1+\sin^2 x}\right) \le \dfrac{\pi}{2}$

$\Rightarrow \quad \dfrac{\pi}{6} \le \dfrac{k\pi}{6} \le \dfrac{\pi}{2} \Rightarrow 1 \le k \le 3.$ **Ans.**

27. L.H.S. $(\sin^{-1}x - \cos^{-1}x)^3 + 3\sin^{-1}x\cos^{-1}x\,(\sin^{-1}x - \cos^{-1}x) + \sin^{-1}x\cos^{-1}x\,(\sin^{-1}x - \cos^{-1}x)$

$= (\sin^{-1}x - \cos^{-1}x)\left[\left(\sin^{-1}x - \cos^{-1}x\right)^2 + 4\sin^{-1}x\cos^{-1}x\right]$

$= (\sin^{-1}x - \cos^{-1}x)\left[\left(\sin^{-1}x + \cos^{-1}x\right)^2\right] = \dfrac{\pi^3}{16}$

$= \left(\sin^{-1}x - \cos^{-1}x\right)\dfrac{\pi^2}{4} = \dfrac{\pi^3}{16}$

$\sin^{-1}x - \cos^{-1}x = \dfrac{\pi}{4}$

$\sin^{-1}x + \cos^{-1}x = \dfrac{\pi}{2}$

$2\cos^{-1}x = \dfrac{\pi}{4}$

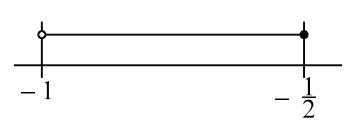
$x = \cos\dfrac{\pi}{8}.$ **Ans.**

28. We have $\dfrac{1}{2} \le \dfrac{x^2-k}{1+x^2} < 1 \ \Rightarrow\ \dfrac{1}{2} \le 1 - \dfrac{(k+1)}{x^2+1} < 1 \quad \forall x \in R \ \Rightarrow\ k+1 > 0$

So $\quad k > -1$ and $\dfrac{k+1}{x^2+1} \le \dfrac{1}{2}$

$\Rightarrow \quad x^2 + 1 \ge 2k + 2$

So $\quad x^2 - (2k+1) \ge 0 \quad \forall x \in R \ \Rightarrow\ 4(2k+1) \le 0$



$\therefore \quad k \le -\dfrac{1}{2}.$

29. We have, $\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{4} = \tan^{-1}\left(\frac{\frac{1}{3}+\frac{1}{4}}{1-\frac{1}{12}}\right) = \tan^{-1}\left(\frac{7}{11}\right)$

Again, $\tan^{-1}\frac{7}{11}+\tan^{-1}\frac{1}{5} = \tan^{-1}\left(\frac{\frac{7}{11}+\frac{1}{5}}{1-\frac{7}{55}}\right) = \tan^{-1}\left(\frac{46}{48}\right) = \tan^{-1}\frac{23}{24}$

$\therefore\ \tan^{-1}\frac{1}{n} = \tan^{-1}1 - \tan^{-1}\frac{23}{24} = \tan^{-1}\left(\frac{1-\frac{23}{24}}{1+\frac{23}{24}}\right) = \tan^{-1}\left(\frac{1}{47}\right) \Rightarrow n = 47$ **Ans.**

30. We have $f(x) = \sqrt{\cos^{-1}x - 2\sin^{-1}x}$

Clearly, for domain of $f(x)$, $\cos^{-1}x - 2\sin^{-1}x \geq 0$

$\Rightarrow \frac{\pi}{2} \geq 3\sin^{-1}x \Rightarrow \sin^{-1}x \leq \frac{\pi}{6} \Rightarrow x \leq \frac{1}{2}$

So, $D_f = \left[-1, \frac{1}{2}\right]$ **Ans.**

31. Clearly, $f(x) = \dfrac{\tan^{-1}x}{1+\frac{2}{\pi}\tan^{-1}x}$

Now domain of equation $f^2(x) + (\sin^{-1}x)^2 = a$, is $x \in [-1, 1]$.

Now, $f(x) = \frac{\pi}{2}\left(\dfrac{\frac{2}{\pi}\tan^{-1}x + 1 - 1}{1+\frac{2}{\pi}\tan^{-1}x}\right) = \frac{\pi}{2}\left(1 - \dfrac{1}{1+\frac{2}{\pi}\tan^{-1}x}\right)$

So, $f(x)|_{min} = \frac{-\pi}{2}$ at $x = -1$.

$f(x)|_{max} = \frac{\pi}{6}$ at $x = 1$.

So, $(f(x))^2 \in \left[0, \frac{\pi^2}{4}\right]$ and minimum and maximum is attained at $x = 0$ and $x = -1$.

Also $(\sin^{-1}x)^2 \in \left[0, \frac{\pi^2}{4}\right]$

$\therefore\ a_{max} = \dfrac{\pi^2}{2}$ at x = – 1 and $a_{min} = 0$ at x = 0.

So, $a \in \left[0, \dfrac{\pi^2}{2}\right]$

$\therefore\ a_{integer} = 0, 1, 2, 3, 4$
Hence, sum of integral values of a = 10. **Ans.**

32. Given, $3\sin^{-1}\left(\dfrac{2x}{1+x^2}\right) + 2\tan^{-1}\left(\dfrac{2x}{1-x^2}\right) - 4\cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right) = \dfrac{\pi}{4}$

As, $x \in (0, 1)$ so the given equation becomes $3(2\tan^{-1}x) + 2(2\tan^{-1}x) - 4(2\tan^{-1}x) = \dfrac{\pi}{4}$

$\Rightarrow \tan^{-1}x = \dfrac{\pi}{8} \Rightarrow x = \tan\dfrac{\pi}{8} = \left(\sqrt{2}-1\right)$. **Ans.**]

33. We must have, $-1 \le \left|\dfrac{x^2-4}{3x}\right| \le 1 \Rightarrow -1 \le \dfrac{x^2-4}{3x} \le 1$

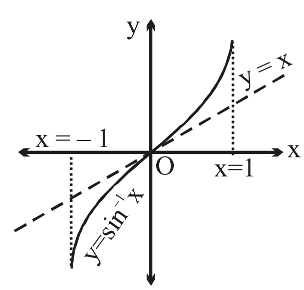
$\Rightarrow \dfrac{x^2-4}{3x} + 1 \ge 0$ and $\dfrac{x^2-4}{3x} - 1 \le 0$

$\Rightarrow \dfrac{(x+4)(x-1)}{3x} \ge 0$ and $\dfrac{(x-4)(x+1)}{3x} \le 0$

$\Rightarrow x = -4, -3, -2, -1, 1, 2, 3, 4$. Hence, number of integral values is 8. **Ans.**

34. As, $|x| + |y| = |x+y|$ is true if $xy \ge 0$.

So, $(\sin^{-1}x - x)\left(\underbrace{x^2+5x+6+\cos^{-1}x}_{+\text{ve for all } x\in[-1,1]}\right) \ge 0$



$\Rightarrow \sin^{-1}x - x \ge 0 \Rightarrow \sin^{-1}x \ge x$
Hence, $x \in [0, 1]$. **Ans.**

35. As, $T_n = \cot^{-1}\left[\dfrac{(n+1)(n+2)}{2}x + \dfrac{2}{x}\right] \Rightarrow T_n = \tan^{-1}\left(\dfrac{2x}{(n+2)(n+1)x^2+4}\right)$

$\therefore\ T_n = \tan^{-1}\left(\left(\dfrac{n+2}{2}\right)x\right) - \tan^{-1}\left(\left(\dfrac{n+1}{2}\right)x\right)$

So, $S_n = \tan^{-1}\left(\left(\dfrac{n+2}{2}\right)x\right) - \tan^{-1}x$

$\Rightarrow \quad \lim\limits_{n \to \infty} S_n = \frac{\pi}{2} - \tan^{-1} x = \cot^{-1} x = 1 \text{ (Given)} \Rightarrow x = \cot 1.$ **Ans.**

36. $\cos 2A = \frac{1-\tan^2 A}{1+\tan^2 A} = \frac{1-\frac{1}{49}}{1+\frac{1}{49}} = \frac{48}{50} = \frac{24}{25}$

$\cos 2B = \frac{1-\tan^2 B}{1+\tan^2 B} = \frac{1-\frac{1}{9}}{1+\frac{1}{9}} = \frac{8}{10} = \frac{4}{5}$

$\therefore \quad \sin 2A = \frac{7}{25}, \ \sin 2B = \frac{3}{5}$

$\sin 4B = 2\cdot \frac{3}{5}\cdot\frac{4}{5} = \frac{24}{25}$

37. $-1 \le 1-x \le 1 \quad \Rightarrow \quad 0 \le x \le 2$

and $-1 \le mx \le 1$ with $1-x = mx$ so $x = \frac{1}{m+1}$

$-1 \le \frac{m}{m+1} \le 1$ so $-1 \le \frac{m}{m+1}, \ \frac{(2m+1)}{m+1} \ge 0, \ m \in (-\infty, -1) \cup -[\frac{1}{2}, \infty)$

$\frac{m}{m+1} - 1 \le 0, -\frac{1}{m+1} \le 0, \ \frac{1}{m+1} \ge 0 \quad \Rightarrow \quad m > -1$

$m \in [-\frac{1}{2}, \infty)$.

Paragraph for question nos. 38 & 39

Sol.

38. Given, $f(x) = \sqrt{\sin^{-1} x + 2} + \sqrt{1-\sin^{-1} x}$

Clearly, for domain of $f(x)$, $1 - \sin^{-1} x \ge 0$ $\left(\text{As, } \sin^{-1} x + 2 > 0 \ \forall \ x \in [-1, 1]\right)$

$\Rightarrow \quad \sin^{-1} x \le 1 \Rightarrow x \le \sin 1$

$\therefore \quad D_f = [-1, \sin 1]$.

39. Given, $f(x) = \sqrt{\sin^{-1} x + 2} + \sqrt{1-\sin^{-1} x}$, where $x \in [-1, \sin 1] = y$ (say)

Clearly, $y > 0 \ \forall \ x \in [-1, \sin 1]$

Now, $y^2 = (\sin^{-1} x + 2) + (1 - \sin^{-1} x) + 2\sqrt{\left(\sin^{-1} x + 2\right)\left(1-\sin^{-1} x\right)}$

$\Rightarrow \quad y^2 = 3 + 2\sqrt{\frac{9}{4} - \left(\sin^{-1} x + \frac{1}{2}\right)^2}$

Clearly, $y^2_{max.}\left(\sin^{-1} x = \frac{-1}{2}\right) = 3 + 2\sqrt{\frac{9}{4} - 0} = 3 + 3 = 6$

$\Rightarrow y_{max.}\left(x = -\sin\frac{1}{2}\right) = \sqrt{6}$

Also, $y^2_{min.}(\sin^{-1}x = 1) = 3 + 2\sqrt{\frac{9}{4} - \frac{9}{4}} = 3$

$\Rightarrow y_{min.}(x = \sin 1) = \sqrt{3}$.

Hence, rane of $f = \left[\sqrt{3}, \sqrt{6}\right]$. **Ans.**

40. $\cos^{-1}\left(\cos\frac{8\pi}{7}\right) = \cos^{-1}\left(\cos\left(2\pi - \frac{8\pi}{7}\right)\right) = \cos^{-1}\left(\cos\frac{6\pi}{7}\right) = \frac{6\pi}{7}$

$\tan^{-1}\left(\tan\frac{8\pi}{7}\right) = \tan^{-1}\left(\tan\frac{\pi}{7}\right) = \frac{\pi}{7}$

41. For domain of $y = f(x)$

$\sin^{-1}x - \cos^{-1}x \geq 0 \Rightarrow \sin^{-1}x \geq \cos^{-1}x \Rightarrow \frac{\pi}{2} \geq \sin^{-1}x \geq \frac{\pi}{4} \Rightarrow \frac{1}{\sqrt{2}} \leq x \leq 1$

and range of f(x) is $\forall\, x \in \left[\frac{1}{\sqrt{2}}, 1\right]$

$\sin^{-1}x - \cos^{-1}x = \frac{\pi}{2} - 2\cos^{-1}x$

$\Rightarrow f(x) = \sqrt{\frac{\pi}{2} - 2\cos^{-1}x} \Rightarrow f(x) \in \left[0, \sqrt{\frac{\pi}{2}}\right]$

Domain of $y = g(x)$, for g(x) to be defined $\cos^{-1}x - |\sin^{-1}x| \geq 0 \Rightarrow \cos^{-1}x \geq |\sin^{-1}x|$

For $x \in [-1, 0]$, $\cos^{-1}x \in \left[\frac{\pi}{2}, \pi\right]$

Hence, $\cos^{-1}x \geq |\sin^{-1}x|$

For $x \in (0, 1] \Rightarrow 2\cos^{-1}x \geq \frac{\pi}{2} \Rightarrow \cos^{-1}x \geq \frac{\pi}{4} \Rightarrow x \in \left(0, \frac{1}{\sqrt{2}}\right)$

Hence domain is $x \in \left[-1, \frac{1}{\sqrt{2}}\right]$.

Range of $g(x)\ \forall\ x \in \left[-1, \frac{1}{\sqrt{2}}\right]$.

For $x \in [-1, 0]$, $\cos^{-1}x - |\sin^{-1}x| \Rightarrow \cos^{-1}x + \sin^{-1}x = \frac{\pi}{2}$

$\Rightarrow\ g(x) = \sqrt{\frac{\pi}{2}}$

For $x \in \left(0, \frac{1}{\sqrt{2}}\right)$

$g(x) = \sqrt{\cos^{-1}x - \sin^{-1}x} = \sqrt{\frac{\pi}{2} - 2\sin^{-1}x} \Rightarrow g(x) \in \left[0, \sqrt{\frac{\pi}{2}}\right)$.

Hence, range of $g(x) \in \left[0, \sqrt{\frac{\pi}{2}}\right)$.

42. Domain is $x = 1$ only

Hence range is $\sqrt{0} + \sin^{-1}\left(\frac{2}{4}\right) = \frac{\pi}{6}$ **Ans.**

43.

(a) $0 \le x \le 1$; $x \ge 1$ or $x \le -1$ $\Rightarrow$ $x = 1$

hence $\pi + \tan^{-1}(\tan y) = a$

$\tan^{-1}(\tan y) = a - \pi$

$\Rightarrow\quad -\frac{\pi}{2} < a - \pi < \frac{\pi}{2}$

$\Rightarrow\quad \frac{\pi}{2} < a < \frac{3\pi}{2}$. Hence $a \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

$\therefore$ 'a' can be 2, 3, 4

(b) $2\sqrt{\sin^{-1}|\sin x|} = 2\sqrt{a}$ $\quad \{x \ne (2n+1)\frac{\pi}{2}\}$

$\Rightarrow\quad \sin^{-1}|\sin x| = a$

$\Rightarrow\quad a \in [0, \pi/2)$]

(c) $a = \sin^{-1}|\cos x| - \cos^{-1}|\sin x| = \left[\begin{array}{ll} 0 & x \in \left[0, \frac{\pi}{2}\right] \\ \pi - x - \left(\pi - \cos^{-1}(\cos x)\right) = 0 & x \in \left[\frac{\pi}{2}, \pi\right] \end{array}\right.$

and this function has period $= \pi$.

$\therefore\quad a = 0$]

44. After rationalization.

$$S_n = \sum \sin^{-1}\left(\frac{\sqrt{n^2+2n}-\sqrt{n^2-1}}{n(n+1)}\right) = \sum\left(\sin^{-1}\frac{1}{n} - \sin^{-1}\frac{1}{n+1}\right)$$

$t_1 = \sin^{-1} 1 - \sin^{-1}\frac{1}{2}$, $t_2 = \sin^{-1}\frac{1}{2} - \sin^{-1}\frac{1}{3}$ and so on

$$S_n = \sin^{-1} 1 - \sin^{-1}\left(\frac{1}{n+1}\right) = \frac{\pi}{2} - \sin^{-1}\left(\frac{1}{n+1}\right)$$

$\cos S_n = \frac{1}{n+1}$, $\cos S_{99} = \frac{1}{100}$

$100 \cos S_{99} = 1$. **Ans.**

45. We have $g(x) = \left[2 + \sin^{-1}\frac{2x}{1+x^2}\right] = 2 + \left[\sin^{-1}\frac{2x}{1+x^2}\right]$

As, $\sin^{-1}\frac{2x}{1+x^2} \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$\therefore$ $\left[\sin^{-1}\frac{2x}{1+x^2}\right] = -2, -1, 0, 1.$

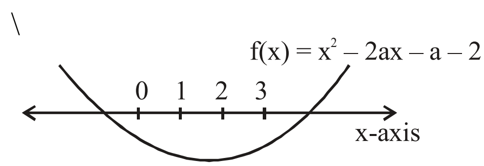
Range of $g(x) = \{0, 1, 2, 3\}$ for $f(g(x)) < 0 \ \forall \ x \in R$

$\Rightarrow$ $f(0) < 0$ and $f(3) < 0$

Now, $f(0) < 0 \Rightarrow a - 2 < 0 \Rightarrow a < 2$

and $f(3) < 0 \Rightarrow 9 - 6a + a - 2 < 0$

$a > \frac{7}{5}$



$\therefore$ $a \in \left(\frac{7}{5}, 2\right)$.

Hence, $k_1 = \frac{7}{5}$, $k_2 = 2$

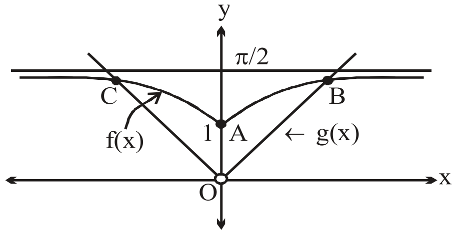
$\therefore$ $(10k_1 + 3k_2) = 14 + 6 = 20$. **Ans.**

46. We have $2\pi = \frac{3\pi}{2} + 2\tan^{-1}(x^2 + x + a)$ $\left(\text{As } \tan^{-1}\alpha + \cot^{-1}\alpha = \frac{\pi}{2} \ \forall \ \alpha \in R\right)$

$\Rightarrow$ $\tan^{-1}(x^2 + x + a) = \frac{\pi}{4}$ $\Rightarrow x^2 + x + a = 1 \Rightarrow x^2 + x + (a - 1) = 0$

$\therefore$ For required condition, put $D = 0 \Rightarrow 1 - 4(a - 1) = 0 \Rightarrow a = \frac{5}{4}$. **Ans.**

47. $f(x) = \tan^{-1}\left(\sqrt{\left(\dfrac{1-\cos 2}{1+\cos 2}\right)} + x^2\right) = \tan^{-1}\left(\tan 1 + x^2\right)$



From the graph it is clear that f(x) and g(x) intersect at three distinct points i.e. at A, B and C.

48. $\sin^{-1}\ (\cos^{-1} x + \tan^{-1} x + \cot^{-1} x) = \sin^{-1}\left(\dfrac{\pi}{2} + \cos^{-1} x\right)$

For domain $\quad -1 \le \dfrac{\pi}{2} + \cos^{-1} x \le 1 \Rightarrow -1 - \dfrac{\pi}{2} \le \cos^{-1} x \le 1 - \dfrac{\pi}{2}$

But $\quad 0 \le \cos^{-1} x \le \pi$ so No solution

similarly for $\quad \cos^{-1}\ (\tan^{-1} x + \cot^{-1} x + \sin^{-1} x) = \cos^{-1}\left(\dfrac{\pi}{2} + \sin^{-1} x\right)$

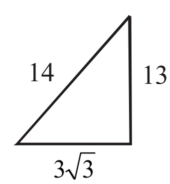
$-1 \le \dfrac{\pi}{2} + \sin^{-1} x \le 1 \Rightarrow -\dfrac{\pi}{2} - 1 \le \sin^{-1} x \le 1 - \dfrac{\pi}{2}$

$-\dfrac{\pi}{2} \le \sin^{-1} x \le 1 - \dfrac{\pi}{2} \Rightarrow -1 \le x \le -\cos 1$

$\sin^{-1}(\cos^{-1} x) \Rightarrow -1 \le \cos^{-1} x \le 1 \Rightarrow 0 \le \cos^{-1} x \le 1 \Rightarrow \cos 1 \le x \le 1$

$\cos^{-1}(\sin^{-1} x) \Rightarrow -1 \le \sin^{-1} x \le 1 \Rightarrow -\sin 1 \le x \le \sin 1$. **Ans.**

49. $\sin\left(\dfrac{\pi}{6} + \tan^{-1} x\right) = \dfrac{13}{14}$



$\dfrac{\pi}{6} + \tan^{-1} x = \sin^{-1}\left(\dfrac{13}{14}\right) = \tan^{-1}\left(\dfrac{13}{3\sqrt{3}}\right)$

$$\tan^{-1} x = \tan^{-1}\left(\frac{13}{3\sqrt{3}}\right) - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \tan^{-1}\left[\frac{\frac{13}{3\sqrt{3}} - \frac{1}{\sqrt{3}}}{1 + \frac{13}{9}}\right] = \tan^{-1}\left[\frac{10}{3\sqrt{3}}\left(\frac{9}{22}\right)\right]$$

$$\tan^{-1} x = \tan^{-1}\left[\frac{5\sqrt{3}}{11}\right]$$

$$\therefore\ x = \frac{5\sqrt{3}}{11} \Rightarrow \frac{a\sqrt{3}}{b} \Rightarrow a + b = 16 \Rightarrow \left(\frac{a+b}{2}\right) = 8.\ \textbf{Ans.}$$

Aliter: $\sin(30° + \tan^{-1}x) = \dfrac{13}{14}$

$\Rightarrow \dfrac{1}{2}\cos(\tan^{-1}x) + \dfrac{\sqrt{3}}{2}\sin(\tan^{-1}x) = \dfrac{13}{14} \Rightarrow 7(1 + \sqrt{3}x) = 13\sqrt{1+x^2}$

$\Rightarrow 11x^2 - 49\sqrt{3}x + 60 = 0$ (On squaring)

$\therefore\ x = \dfrac{49\sqrt{3} \pm 39\sqrt{3}}{22} = \dfrac{5\sqrt{3}}{11}, 4\sqrt{3}$ (Reject)

$\therefore\ x = \dfrac{5\sqrt{3}}{11}$. **Ans.**

50. As, $x^2 - 4x + 5 = (x-2)^2 + 1$

So, $\sin^{-1}(x^2 - 4x + 5)$ and $\cos^{-1}(x^2 - 4x + 5)$ will be defined only for $x = 2$.

So, $4 + 4a + 8\left(\dfrac{\pi}{2}\right) = 0 \Rightarrow a = -(1+\pi)$. **Ans.**

51. (A) Let $|\sin x - 1| + |\sin x + 1| = y$

if $\sin x = t \Rightarrow |t-1| + |t+1| = y$

if $t \in [-1, 1] \Rightarrow y = 2$

if $t \in (-\infty, -1] \cup [1, \infty) \Rightarrow y > 2$

$t \in [-1, 1] \Rightarrow \sin x = t \Rightarrow x \in R \Rightarrow y = 2$

$t \in (-\infty, -1) \cup (1, \infty) \Rightarrow$ not possible because $\sin x \in [-1, 1]$

$\therefore\ f(x) = \sin^{-1}(2/2) = \sin^{-1}(1) = \pi/2 =$ constant (periodic)

$\Rightarrow$ **P, Q, R, S**

(B) Let $|x-1| - |x-2| = y$

if $x \in (-\infty, 1] \Rightarrow y = -1$

if $x \in [1, 2] \Rightarrow y = [-1, 1]$

if $x \in [2, \infty) \Rightarrow y = 1$

$\therefore\ f(x) = \cos^{-1}(|x-1| - |x-2|)$

domain is R, range is $[0, \pi]$; $\therefore$ **P, Q** is correct

(C) Let $|\sin^{-1}x - (\pi/2)| + |\sin^{-1}x + (\pi/2)| = y$

let $\sin^{-1}x = t \Rightarrow t \in \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$

$\left[t - \dfrac{\pi}{2}\right] + \left[t + \dfrac{\pi}{2}\right] = y$

$t \in \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right] \Rightarrow y = \pi$

$\therefore\ f(x) = \sin^{-1}\left(\dfrac{\pi}{\pi}\right) = \dfrac{\pi}{2}$

$\therefore$ f (x) is periodic and constant function also range contain only irrational number.

$\Rightarrow$ **P, R, S**

(D) Domain is $\{\pm 1\}$

$f(x) = \frac{\pi}{2} + 1$ for $x = \pm 1$

range contain only irrational value and also constant function.

$\Rightarrow$ **P, R, S**

52. Given, $f(x) = \tan^{-1}(\cot x - 2\cot 2x)$

$$= \tan^{-1}\left(\frac{1}{\tan x} - \frac{1-\tan^2 x}{\tan x}\right) = \tan^{-1}\left(\frac{\tan^2 x}{\tan x}\right) = \tan^{-1}(\tan x).$$

Now, $\sum_{r=1}^{5} f(r) = f(1) + f(2) + f(3) + f(4) + f(5)$

As, $f(1) = 1,\ f(2) = 2 - \pi,\ f(3) = 3 - \pi,\ f(4) = 4 - \pi,\ f(5) = 5 - 2\pi$

$\Rightarrow \sum_{r=1}^{5} f(r) = 15 - 5\pi = a - b\pi$

$\Rightarrow a = 15, b = 5$

Hence, the value of $(a + b) = 20$.**Ans.**

53. Domain of $f(x) = -1 \le (x-2) \le 1$ and $-1 \le (4-x) \le 1$

$\Rightarrow$ $x = 3$ hence $p = 1$

Domain of $g(x) = 0 \le \sqrt{4-x^2} \le 1$ and $0 \le (x-1) \le 1 \Rightarrow x \in [\sqrt{3}, 2]$

$\therefore$ $q = 1$

$\therefore$ $p + q = 2$ **Ans.**

54. $\because$ f is onto

$\therefore$ Range of $\tan^{-1}(2x - x^2 + \lambda)$ should be $\left(\frac{-\pi}{2}, 0\right]$

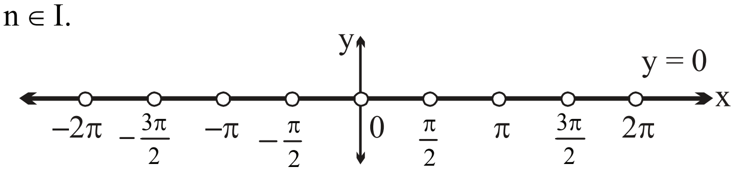
$\Rightarrow$ Range of $2x - x^2 + \lambda$ should be $(-\infty, 0]$; hence $D = 0$

$\Rightarrow 4 + 4\lambda = 0$, hence $\lambda = -1$

Hence number of integer in the range of λ is one i.e. -1.

55. We have $f(x) = \sin^{-1}(\tan x) - \operatorname{cosec}^{-1}(\cot x) = \sin^{-1}(\tan x) - \sin^{-1}\left(\frac{1}{\cot x}\right) = 0 \ \forall x \ne \frac{n\pi}{2}$

where $n \in I$.



Graph of $y = f(x)$

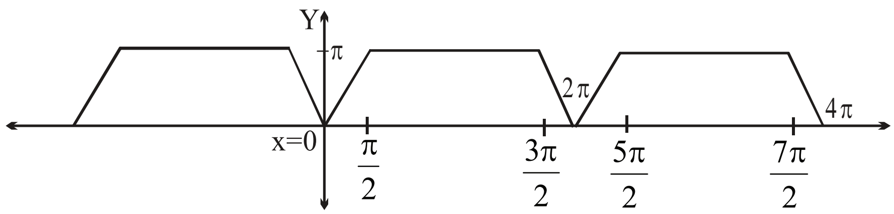
Clearly $y = f(x)$ is periodic function with fundamental period $\frac{\pi}{2}$.

The graph of $y = f(x)$ is union of line segments on x axis and not a straight line.

The function $y = f(x)$ is many-one, so non-invertible.
Also $y = f(x)$ is even and odd both simultaneously.
As $f(0)$ is not defined, so $f(f(x))$ is not defined.]

56. $f(x) = \begin{cases} 2x & ; 0 \le x \le \dfrac{\pi}{2} \\ \pi & ; \dfrac{\pi}{2} < x \le \dfrac{3\pi}{2} \\ 4\pi - 2x & ; \dfrac{3\pi}{2} < x \le 2\pi \end{cases}$

Clearly f(x) is periodic function with period 2π. The graph of $f(x)$ is shown below.



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57. As, we know that

$\sin^{-1} x \in \left[\dfrac{-\pi}{2}, \dfrac{\pi}{2}\right]$, $\cos^{-1} y \in [0, \pi]$ and $\sec^{-1} z \in \left[0, \dfrac{\pi}{2}\right) \cup \left(\dfrac{\pi}{2}, \pi\right]$

So, $\sin^{-1}x + \cos^{-1}y + \sec^{-1}z \le \dfrac{\pi}{2} + \pi + \pi = \dfrac{5\pi}{2}$.

Also, $t^2 - \sqrt{2\pi}\ t + 3\pi = t^2 - 2\sqrt{\dfrac{\pi}{2}}\ t + \dfrac{\pi}{2} - \dfrac{\pi}{2} + 3\pi$

$= \left(t - \sqrt{\dfrac{\pi}{2}}\right)^2 + \dfrac{5\pi}{2} \ge \dfrac{5\pi}{2}$

Hence, the given in equation exists if equality holds, i.e.,

L.H.S. = R.H.S. = $\dfrac{5\pi}{2} \Rightarrow x = 1, y = -1, z = -1, t = \sqrt{\dfrac{\pi}{2}}$.

Now, $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z + \tan^{-1}\left(\dfrac{\sqrt{2}}{\sqrt{\pi}}\, t\right) = \dfrac{\pi}{4} - \dfrac{\pi}{4} - \dfrac{\pi}{4} + \dfrac{\pi}{4} = 0$

So, $\sec(0) = 1$. **Ans.**

58. Domain of $\sin^{-1}x$ and $\cos^{-1}x$,each is $[-1, 1]$ and that of $\sec^{-1} x$ and $\text{cosec}^{-1} x$, each is $(-\infty, -1] \cup [1, \infty)$
$\therefore$ Domain of f(x) must be $\{-1, 1\}$
$\therefore$ Range of f(x) will be $\{f(-1), f(1)\}$
where $f(-1) = \sin^{-1}(-1) \cdot \cos^{-1}(-1) \cdot \tan^{-1}(-1) \cdot \cot^{-1}(-1) \cdot \sec^{-1}(-1) \cdot \text{cosec}^{-1}(-1)$

$$= \left(\frac{-\pi}{2}\right)\cdot(\pi)\cdot\left(\frac{-\pi}{4}\right)\cdot\left(\frac{3\pi}{4}\right)\cdot(\pi)\cdot\left(\frac{-\pi}{2}\right) = \frac{-3\pi^6}{64} \text{ and } f(1) = 0 \ \{\text{as } \cos^{-1} 1 = 0\}$$

(i) Thus, the graph of f(x) is a two point graph which doesn't lie above x - axis.

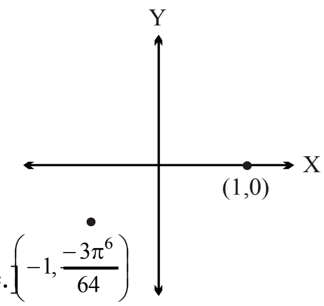
(ii) $f(x)_{max} = 0$ and $f(x)_{min} = \dfrac{-3\pi^6}{64}$

Hence $\left|f(x)_{max} - f(x)_{min}\right| = \dfrac{3\pi^6}{64}$

(iii) f(x) is one-one hence injective.

(iv) Domain is {–1, 1}

∴ Number of non-negative integers in the domain of f(x) is one.



59. $D_f = [-1, 1]$

As f(x + T) = f(x) (T > 0) is not verified for x = 1

⇒ f is aperiodic function

Similarly, $D_g = (-\infty, -\infty)$

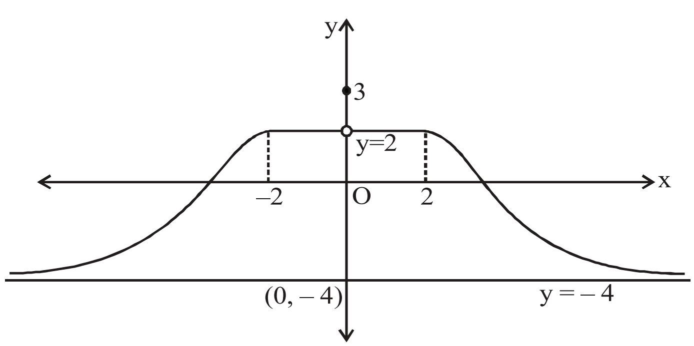
As $g(x) = \dfrac{\pi}{2}, \forall\, x \in R$

⇒ g is periodic function with no fundamental period.

60. We have $H(x) = \left[\dfrac{3(x^2+1) - |x|}{x^2+1}\right] = \left[3 - \dfrac{|x|}{x^2+1}\right]$ $\left(\text{As } 0 < \dfrac{|x|}{x^2+1} = \dfrac{1}{|x| + \dfrac{1}{|x|}} \le \dfrac{1}{2}\right)$

$$\text{So, } H(x) = \begin{cases} \dfrac{8}{\pi}\tan^{-1}(-|x|+3), & x \in (-\infty, -2) \cup (2, \infty) \\ 3, & x = 0 \\ 2, & x \in [-2, 2] - \{0\} \end{cases}$$

Also, H(x) is an even function on R, so graph of H(x) is symmetrical about y-axis.



Graph of H(x)

Note : $H(x) = \frac{8}{\pi}\tan^{-1}(3 - |x|), |x| > 2$

Also for $|x| > 2,\ 3 - |x| \in (-\infty, 1)$, so

$$H(x) \in \frac{8}{\pi}\left(\frac{-\pi}{2}, \frac{\pi}{4}\right) = (-4,\ 2)$$

From the above graph, it is clear that H(x) is discontinuous at only one point i.e. $x = 0$.
Range of H(x) is $(-4, 2] \cup \{3\}$.
Hence number of integers $\{-3, -2, -1, 0, 1, 2, 3) \Rightarrow$ 7 integers **Ans.**]

61. $$\left(\log_{1/2}\left(\frac{2}{\pi}\cot^{-1}x + 1\right)\right)^2 + a\log_{1/2}\left(1 - \frac{\tan^{-1}x}{\pi}\right) - a + b = 0$$

$$= \left(\log_{1/2}\left(\frac{2}{\pi}\cot^{-1}x + 1\right)\right)^2 + a\left(\log_{1/2}\left(1 - \frac{\tan^{-1}x}{\pi}\right) - 1\right) + b = 0 \qquad (\log_{1/2}2 = -1)$$

$$= \left(\log_{1/2}\left(\frac{2}{\pi}\cot^{-1}x + 1\right)\right)^2 + a\left(\log_{1/2}\left(2 - \frac{2}{\pi}\tan^{-1}x\right)\right) + b = 0$$

$$= \left(\log_{1/2}\left(\frac{2}{\pi}\cot^{-1}x + 1\right)\right)^2 + a\left(\log_{1/2}\left(2 - \frac{2}{\pi}\left(\frac{\pi}{2} - \cot^{-1}x\right)\right)\right) + b = 0$$

$$= \left(\log_{1/2}\left(\frac{2}{\pi}\cot^{-1}x + 1\right)\right)^2 + a\left(\log_{1/2}\left(1 + \frac{2}{\pi}\cot^{-1}x\right)\right) + b = 0$$

$$\Rightarrow \qquad y^2 + ay + b = 0 \text{ where } y = \left(\log_{1/2}\left(\frac{2}{\pi}\cot^{-1}x + 1\right)\right)$$

now it is given that equation $y^2 + ay + b = 0$ has no real solutions so the roots of $x^2 + ax + b = 0$ should not lie in range of y.

so if x_1, x_2 are roots of $x^2 + ax + b = 0$ then these should not lie in the range of y i.e.

$$\log_{1/2}\left(\frac{2}{\pi}\cos^{-1}x + 1\right)$$

$$x_1, x_2 \notin \text{range of } \log_{1/2}\left(\frac{2}{\pi}\cot^{-1}x + 1\right)$$

$$\notin (\log_{1/2}3, \log_{1/2}1)$$

$$\notin (\log_{1/2}3, 0)$$

As x_1, x_2 both are (– ve) integral roots and $\notin (\log_{1/2}3, 0)$

so $x_1, x_2 \in (-\infty, -2]$
now $a = -(x_1 + x_2)$

$$a_{min.} = -\left(\max.\text{ of } (x_1 + x_2)\right)$$

maximum of $x_1 + x_2 = -2 - 3 = -5$
$a_{minimum} = -(-5) = 5$ Ans.]

62. $f(x) = \sqrt{\frac{\pi^2}{2} + \frac{\pi}{2}x - x^2}$

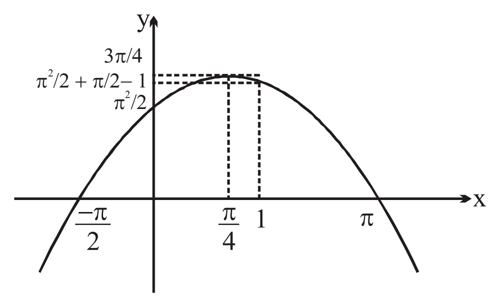
$$= \sqrt{\left(x + \frac{\pi}{2}\right)(\pi - x)}$$

$\therefore$ domain of $f(x)$ is $\left[\frac{-\pi}{2}, \pi\right]$

but $f(g(x)) = f\left(\sqrt{\sin^{-1}x}\right)$ and $f(h(x)) = f\left(\sqrt{\cos^{-1}x}\right)$ both exit for $x \in [0, 1]$

$\therefore$ Modified domain of f(x) is [0, 1]

Graph of $y = \frac{\pi^2}{2} + \frac{\pi}{2}x - x^2 = \left(x + \frac{\pi}{2}\right)(\pi - x)$ is



$$f(0) = \sqrt{\frac{\pi^2}{2}}$$

$$f(1) = \sqrt{\frac{\pi^2}{2} + \frac{\pi}{2} - 1} > f(0)$$

$$f\left(\frac{\pi}{4}\right) = \sqrt{\frac{\pi^2}{2} + \frac{\pi^2}{8} - \frac{\pi^2}{16}} = \sqrt{\frac{9\pi^2}{16}} = \frac{3\pi}{4}$$

$\therefore$ Range of $f(x)$ is $\left[\frac{\pi}{\sqrt{2}}, \frac{3\pi}{4}\right]$

$\therefore$ $16(a^2 + b^2) = 16\left(\frac{1}{2} + \frac{9}{16}\right) = 17$. **Ans.**

63. $S_n = \frac{2}{1 + a_0{}^k} + \frac{2}{1 + a_1{}^k} + \frac{2}{1 + a_2{}^k} + \ldots\ldots\ldots + \frac{2}{1 + a_n{}^k}$

$$S_n = \frac{2}{1 + a_n{}^k} + \frac{2}{1 + a_{n-1}{}^k} + \frac{2}{1 + a_{n-2}{}^k} + \ldots\ldots\ldots + \frac{2}{1 + a_0{}^k} \qquad \left(\because a_i = \frac{1}{a_{n-1}}\right)$$

$\therefore$ $2S_n = 2 \times (n + 1)$
$S_n = (n + 1)$

Now, $[\sin^{-1}(n+1)] = 1$

$$1 \le \sin^{-1}(n+1) \le \frac{\pi}{2}$$

$\sin 1 \le n + 1 \le 1$

$\sin 1 - 1 \le n \le 0$

$\therefore$ $n = 0$ (only integral value). **Ans.**

64. $$2^{(\cos^{-1}x)^2}\sqrt{(\sin^{-1}y)^4 - 2(\sin^{-1}y)^2 + 2} = 1$$

$$2^{(\cos^{-1}x)^2}\sqrt{\left((\sin^{-1}y)^2 - 1\right)^2 + 1} = 1$$

Now $2^{(\cos^{-1}x)^2} \ge 1$ and $\sqrt{\left((\sin^{-1}y)^2 - 1\right)^2 + 1} \ge 1$

$\Rightarrow$ $2^{(\cos^{-1}x)^2}\sqrt{(\sin^{-1}y - 1)^2 + 1} \ge 1$

$\Rightarrow$ $2^{(\cos^{-1}x)^2} = 1$ and $\left((\sin^{-1}y)^2 - 1\right)^2 + 1 = 1$

$\Rightarrow$ $(\cos^{-1}x)^2 = 0$ and $(\sin^{-1}y)^2 = 1$

$\Rightarrow$ $x = 1$ and $\sin^{-1}y = \pm 1$

$\therefore$ $y = \sin 1$ or $-\sin 1$

$\therefore$ $x - y = 1 + \sin 1$ or $1 - \sin 1$ **Ans.**

Paragraph for question nos. 65 to 66

65 $f(3\sin^{-1}x, 2\cos^{-1}x) = 1$

$\Rightarrow$ $(3\sin^{-1}x)^3 + (2\cos^{-1}x)^3 + 3 \cdot 3\sin^{-1}x \cdot 2\cos^{-1}x - 1 = 0$

$\Rightarrow$ $(3\sin^{-1}x)^3 + (2\cos^{-1}x)^3 + (-1)^3 - 3 \cdot 3\sin^{-1}x \cdot 2\cos^{-1}x(-1) = 0$

$\Rightarrow$ $a^3 + b^3 + c^3 - 3abc = 0$

$\Rightarrow$ $a + b + c = 0$ $(\because a \ne b \ne c)$

$\Rightarrow$ $3\sin^{-1}x + 2\cos^{-1}x - 1 = 0$

$\Rightarrow$ $\sin^{-1}x + 2(\sin^{-1}x + \cos^{-1}x) - 1 = 0$

$\Rightarrow$ $\sin^{-1}x + 2(\sin^{-1}x + \cos^{-1}x) = 1 \Rightarrow \sin^{-1}x = 1 - \pi$

$\Rightarrow$ no solution

66 $h(x) = (\sin^{-1}x)^2 + (\cos^{-1}x)^2$

$= (\sin^{-1}x + \cos^{-1}x)^2 - 2\sin^{-1}x\cos^{-1}x)$

$$= \frac{\pi^2}{4} - 2t\left(\frac{\pi}{2} - t\right) = \frac{\pi^2}{4} - 2\left(t^2 - \frac{\pi}{2}t + \frac{\pi^2}{16} - \frac{\pi^2}{16}\right)$$ where $t = \sin^{-1}x$

$$= \frac{\pi^2}{4} - 2\left(\left(t - \frac{\pi}{4}\right)^2 - \frac{\pi^2}{16}\right) = \frac{\pi^2}{8} + 2\left(t - \frac{\pi}{4}\right)^2$$ $t = \sin^{-1}x \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$\therefore$ least value $= \dfrac{\pi^2}{8}$

greatest value $= \dfrac{\pi^2}{8} + 2\left(-\dfrac{3\pi}{4}\right)^2 = \dfrac{\pi^2}{8} + \dfrac{9\pi^2}{8} = \dfrac{5\pi^2}{4}$

$\therefore$ Range of $h(x) = \left[\dfrac{\pi^2}{8}, \dfrac{5\pi^2}{4}\right]$. **Ans.**

67. We have $b\sin^{-1}x + b\cos^{-1}x = \dfrac{b\pi}{2}$ ...... (1)

and $a\sin^{-1}x - b\cos^{-1}x = c$ ...... (2) (given)

$\therefore$ On adding (1) and (2), we get $(a+b)\sin^{-1}x = \dfrac{b\pi}{2} + c$

$\Rightarrow \sin^{-1}x = \dfrac{\frac{b\pi}{2}+c}{a+b}$. Similarly $\cos^{-1}x = \dfrac{\frac{a\pi}{2}-c}{a+b}$

Hence $(a\sin^{-1}x + b\cos^{-1}x) = \dfrac{\pi ab + c(a-b)}{a+b}$

68. Let $S = 7 + 19 + 39 + 67 + \ldots\ldots + T_n$

$S = 0 + 7 + 19 + 39 + \ldots\ldots\ldots. + T_{n-1} + T_n$

(Subtracting) $-$ $-$ $-$ $-$ $-$ $-$ $-$

$T_n = 7 + 12 + 20 + 28 + \ldots\ldots + (T_n - T_{n-1})$

$= 7 + \dfrac{(n-1)}{2}\,[\,24 + 8\,(n-2)\,] = 4n^2 + 3$

$\therefore T_n' = \tan^{-1}\dfrac{4}{4n^2+3} = \tan^{-1}\dfrac{1}{n^2+\frac{3}{4}} = \tan^{-1}\dfrac{1}{1+\left(n^2-\frac{1}{4}\right)}$

$= \tan^{-1}\left[\dfrac{\left(n+\frac{1}{2}\right)-\left(n-\frac{1}{2}\right)}{1+\left(n+\frac{1}{2}\right)\left(n-\frac{1}{2}\right)}\right] = \tan^{-1}\left(n+\dfrac{1}{2}\right) - \tan^{-1}\left(n-\dfrac{1}{2}\right)$

Hence $S_\infty = \sum\limits_{n=1}^{\infty} T_n' = \dfrac{\pi}{2} - \tan^{-1}\dfrac{1}{2}$

$= \tan^{-1}1 + \tan^{-1}\dfrac{1}{2} + \tan^{-1}\dfrac{1}{3} - \tan^{-1}\dfrac{1}{2} = \tan^{-1}1 + \cot^{-1}3$

69. Let $x = \cos\alpha$ and $\frac{x}{2} = \cos\beta$

Then $\cos^{-1}\left(\frac{x^2}{2} + \sqrt{1-x^2}\sqrt{1-\frac{x^2}{4}}\right) = \cos^{-1}\left(\frac{x}{2}\right) - \cos^{-1} x$

$\Rightarrow \cos^{-1}(\cos\alpha\cos\beta + \sin\alpha\sin\beta) = \beta - \alpha$

$\Rightarrow \cos^{-1}(\cos(\beta - \alpha)) = \beta - \alpha$

which is possible if $0 \le \beta - \alpha \le \pi$ and $0 \le \cos\alpha, \cos\beta \le 1$

$\therefore\ 0 \le x \le 1$ and $0 \le \frac{x}{2} \le 1$

$\therefore$ common set of values of x will be [0, 1]

70. Let $f(x) = x^3 + bx^2 + cx + 1$

so, $f(0) = 1 > 0$, $f(-1) = b - c < 0$

so, $-1 < \alpha < 0$, therefore $\tan^{-1}\alpha + \tan^{-1}\left(\frac{1}{\alpha}\right)$

$= \tan^{-1}\alpha + (\cot^{-1}\alpha - \pi)$

$= \frac{\pi}{2} - \pi = \frac{-\pi}{2}$ **Ans.**

71. $\tan(\tan^{-1}\alpha + \tan^{-1}\beta + \tan^{-1}\gamma) = \frac{\alpha+\beta+\gamma-\alpha\beta\gamma}{1-\alpha\beta-\beta\gamma-\gamma\alpha}$

From given equation

$\alpha + \beta + \gamma = 0$, $\alpha\beta + \beta\gamma + \gamma\alpha = -10$

and $\alpha\beta\gamma = -11$

$\therefore\ \tan(\tan^{-1}\alpha + \tan^{-1}\beta + \tan^{-1}\gamma) = \frac{0-(-11)}{1-(-10)} = 1.$

72. (A) $\left(\cot^{-1}1 + \cot^{-1}\frac{1}{2} + \cot^{-1}\frac{1}{3}\right) + \cot^{-1}\frac{1}{2} + \cot^{-1}\frac{1}{3} + \cot^{-1}\frac{1}{3}$

$= \pi + \frac{3\pi}{4} + \tan^{-1}(3) > 2\pi$

(C) $\tan^{-1}2 + \tan^{-1}\left(\frac{5+8}{1-40}\right) = \pi + \tan^{-1}(2) - \tan^{-1}\left(\frac{1}{3}\right)$

$= \pi + \tan^{-1}\left(\frac{2-\frac{1}{3}}{1+\frac{2}{3}}\right) = \pi + \tan^{-1}\left(\frac{5}{5}\right)$

$$= \pi + \tan^{-1}(1) = \frac{5\pi}{4}$$ **Ans.**

(D) $\left(\cot^{-1}1 + \cot^{-1}2 + \cot^{-1}2\right) + \cot^{-1}2 + 2\cot^{-1}3$

$$\frac{\pi}{2} + \frac{\pi}{4} + \cot^{-1}3 = \frac{3\pi}{4} + \tan^{-1}\frac{1}{3} < \pi.$$

73. We have

$$f(x) = (\tan^{-1}x)^3 + (\cot^{-1}x)^3 = (\tan^{-1}x + \cot^{-1}x)\left((\tan^{-1}x)^2 - (\tan^{-1}x)(\cot^{-1}x) + (\cot^{-1}x)^2\right)$$

$$= \frac{\pi}{2}\left((\tan^{-1}x)^2 - (\tan^{-1}x)\left(\frac{\pi}{2} - \tan^{-1}x\right) + \left(\frac{\pi}{2} - \tan^{-1}x\right)^2\right) \qquad \left(\text{Using } \cot^{-1}x = \frac{\pi}{2} - \tan^{-1}x\right)$$

$$= \frac{3\pi}{2}\left(\left(\tan^{-1}x - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{48}\right)$$

Clearly, $f(x)$ will be minimum when $\left(\tan^{-1}x - \frac{\pi}{4}\right)^2 = 0$

and $f(x)$ will be maximum when $\left(\tan^{-1}x - \frac{\pi}{4}\right)^2 = \left(-\frac{\pi}{2} - \frac{\pi}{4}\right)^2$

$$\therefore \quad a = f(x)_{\min} = \frac{3\pi}{2}\left(0 + \frac{\pi^2}{48}\right) = \frac{\pi^3}{32} \text{ and } b = f(x)_{\max}$$

$$= \frac{3\pi}{2}\left(\left(\frac{-3\pi}{4}\right)^2 + \frac{\pi^2}{48}\right) = \frac{7\pi^3}{8}$$

Hence $\frac{b}{a} = \frac{\frac{7\pi^3}{8}}{\frac{\pi^3}{32}} = 28$ **Ans.**

74. Here $\cos^{-1}\left(\frac{x}{2}\right) + \cos^{-1}\left(\frac{y}{3}\right) = \theta$

so that $\cos^{-1}\left[\frac{x}{2}\cdot\frac{y}{3} - \sqrt{1-\frac{x^2}{4}}\sqrt{1-\frac{y^2}{9}}\right] = \theta$ or $\frac{xy}{6} - \sqrt{1 - \frac{x^2}{4} - \frac{y^2}{9} + \frac{x^2y^2}{36}} = \cos\theta$

or $\frac{x^2y^2}{36} + \cos^2\theta - \frac{xy\cos\theta}{3} = 1 - \frac{x^2}{4} - \frac{y^2}{9} + \frac{x^2y^2}{36}$

or $9x^2 - 12xy\cos\theta + 4y^2 = 36(1 - \cos^2\theta) = 36\sin^2\theta$

75. $\tan\left(\tan^{-1}\frac{x}{10}+\tan^{-1}\frac{1}{x+1}\right) = \tan\frac{\pi}{4}$

$$\Rightarrow \frac{\frac{x}{10}+\frac{1}{x+1}}{1-\frac{x}{10}\left(\frac{1}{x+1}\right)} = 1$$

$$\Rightarrow \frac{x}{10}+\frac{1}{x+1} = 1-\frac{x}{10}\left(\frac{1}{x+1}\right)$$

$\Rightarrow$ x (x + 1) + 10 = 10 (x + 1) – x

$\Rightarrow$ $x^2 + x + 10 = 10x + 10 - x$

$\Rightarrow$ $x^2 - 8x = 0 \Rightarrow x = 0, 8$. **Ans.**

76. $\sin^{-1}x+\tan^{-1}x = k, x \in [-1, 1]$ $\left[\text{As, } \sin^{-1}x + \tan^{-1}x \text{ is an increasing function in } [-1, 1]\right]$

$\therefore$ $k \in \left[\frac{-3\pi}{4}, \frac{3\pi}{4}\right]$

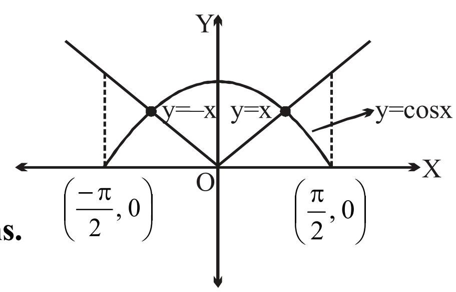
But $k \in I$; $k = -2, -1, 0, 1, 2$.

77. $$f(x) = \sin^{-1}(\sin x) = \begin{cases} x, & \frac{-\pi}{2} \le x \le \frac{\pi}{2} \\ \pi - x, & \frac{\pi}{2} < x \le \frac{3\pi}{2} \\ x - 2\pi, & \frac{3\pi}{2} < x \le \frac{5\pi}{2} \end{cases}$$

$$g(x) = \cos^{-1}(\cos x) = \begin{cases} x, & 0 \le x \le \pi \\ 2\pi - x, & \pi < x \le 2\pi \end{cases}$$

Now, verify alternatives.

78. As, $\left|\sin^{-1}(\sin x)\right| = |x|$, for $x \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$



$\therefore$ From above graph, the equation

$\left|\sin^{-1}(\sin x)\right| = \cos x$ has two solutions, in $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$. **Ans.**

79. $\sum_{r=2}^{\infty} \tan^{-1}\left(\frac{(r-2)-(r-3)}{1+(r-3)(r-2)}\right) = \sum_{r=2}^{\infty}\left(\tan^{-1}(r-2)-\tan^{-1}(r-3)\right)$

$= \tan^{-1} 0 - \tan^{-1}(-1)$
$\tan^{-1}1 - \tan^{-1}0$
$\tan^{-1}2 - \tan^{-1}1$
$\vdots$
$\vdots$
$\tan^{-1}(n-2) - \tan^{-1}(n-3)$

$S_n = \tan^{-1}(n-2) + \frac{\pi}{4}$

$\therefore S_{\infty} = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$. **Ans.**

80.

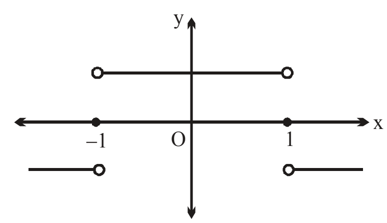
(A) We have

$f(x) = \sqrt[5]{x} + \sin^{-1}x$

Cleary, domain of f(x) = [–1, 1].

Also, f(x) is increasing so f(x) is one-one function.

(B) $f(x) = \text{sgn}\left(\frac{1-|x|}{1+|x|}\right)$



$D_f = R$

$R_f = \{-1, 0, 1\}$ even function

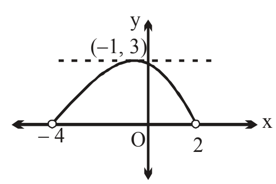
(C) For domain of f(x), we must have $8-2x-x^2 \ge 0$

$\Rightarrow x^2+2x-8 \le 0$

$\Rightarrow (x+4)(x-2) \le 0$

$\Rightarrow x \in [-4, 2]$

$R_f = [0, 3]$



(D) $f(x) = \frac{2^{-[x]}}{2^{\{x\}}} - 2^{|x|} = 2^{-x} - 2^{|x|} = 0 \ \forall\ x \le 0$

81.

(A) Consider $\sum \cot^{-1}(n^2+n+1)$ where n = 1, 2, 3, 4

$\sum \tan^{-1} \frac{1}{1+n(n+1)}$

$\sum \tan^{-1}\left(\frac{(n+1)-n}{1+n(n+1)}\right) = \tan^{-1}(n+1) - \tan^{-1}(n)$

$\therefore \quad S = T_1 + T_2 + T_3 + T_4$

$= \left(\tan^{-1}(2)-\tan^{-1}(1)\right) + \left(\tan^{-1}(3)-\tan^{-1}(2)\right) + \left(\tan^{-1}(4)-\tan^{-1}(3)\right) +$

$\left(\tan^{-1}(5) - \tan^{-1}(4)\right)$

$$= \tan^{-1}(5) - \tan^{-1}(1) = \tan^{-1}\frac{5-1}{1+5} = \tan^{-1}\frac{2}{3}$$

$\therefore \quad 2\cot\left(\cot^{-1}\frac{3}{2}\right) = 2\cdot\frac{3}{2} = 3$ **Ans.**

(B) We have $\tan\left(\tan^{-1}\left(\frac{1}{1+1\cdot 2}\right) + \tan^{-1}\left(\frac{1}{1+2\cdot 3}\right) + ... + \tan^{-1}\left(\frac{1}{1+19\cdot 20}\right)\right)$

$= \tan\left(\tan^{-1}\left(\frac{2-1}{1+2\cdot 1}\right) + \tan^{-1}\left(\frac{3-2}{1+2\cdot 3}\right) + ... + \tan^{-1}\left(\frac{20-19}{1+19\cdot 20}\right)\right)$

$= \tan[\tan^{-1} 2 - \tan 1 + \tan^{-1} 3 - \tan^{-1} 2 + ... + \tan^{-1} 20 - \tan^{-1} 19] = \tan(\tan^{-1} 20 - \tan^{-1} 1)$

$= \tan\left(\tan^{-1}\left(\frac{20-1}{1+20\cdot 1}\right)\right) = \frac{20-1}{1+20} = \frac{19}{21} = \frac{m}{n}$ (Given)

$\therefore$ m = 19, n = 21

Hence $(m+n)_{Least} = 40$. **Ans.**

(C) Using, $\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$ and $\tan(\text{arc}\tan a) = a \ \forall a \in R$, we have

$$\tan\left(\text{arc}\tan\frac{1}{x} + \text{arc}\tan\frac{1}{y}\right) = \tan\left(\text{arc}\tan\frac{1}{10}\right)$$

$$\Rightarrow \quad \frac{\frac{1}{x}+\frac{1}{y}}{1-\frac{1}{xy}} = \frac{1}{10} \Rightarrow (x-10)(y-10) = 101$$

The following four ordered pair of integer numbers are solutions of this equation:
(11, 111); (111, 11), (9, – 91), (– 91, 9) $\Rightarrow$ 4 ordered pairs **Ans.**

(D) We have
$\sin^{-1}(\sin 12) + \cos^{-1}(\cos 12) = -(4\pi - 12) + (4\pi - 12) = 0$
$\therefore \quad (n-2)x^2 + 8x + n + 4 > 0 \ \forall x \in R$
$\Rightarrow \quad (n-2) > 0 \Rightarrow n \geq 3$ and $(8)^2 - 4(n-2)(n+4) < 0$ or $n^2 + 2n - 24 > 0$
$\Rightarrow \quad n > 4 \Rightarrow n \geq 5$
So, $n_{smallest} = 5$. **Ans.**

82. As, $[a] + [-a] = \begin{cases} 0, & \text{if } a \in I \\ -1, & \text{if } a \notin I \end{cases}$

$\Rightarrow \quad \sqrt{[a]+[-a]}$ is defined only for integral values of a.

$\therefore \qquad \sin^{-1}\left(\sqrt{[a]+[-a]}\right)=0$

Also, $\quad 3-|a| \geq 0 \Rightarrow |a| \leq 3 \Rightarrow -3 \leq a \leq 3.$

So, $\quad a = 1, 2, 3 \qquad$ (Given, $a \in R^+$)

Now, the given quadratic equation becomes

$$3x^2 - 2ax + \sqrt{3-|a|} = 0$$

Clearly, discriminant $= D = 4a^2 - 4 \cdot 3\sqrt{3-|a|} = 4\left(a^2 - 3\sqrt{3-|a|}\right)$

$\therefore \qquad$ for $\alpha, \beta \in R, \quad D \geq 0 \Rightarrow a = 2$ and 3.

Hence, the sum of all values of $a = 2 + 3 = 5$ **Ans.**

83. We have $\quad 1 + \sin(\cos^{-1} x) + \sin^2(\cos^{-1} x) + \ldots\ldots\ldots \infty = 2$

$$\Rightarrow \frac{1}{1-\sin\left(\cos^{-1} x\right)} = 2 \Rightarrow \frac{1}{2} = 1 - \sin\left(\cos^{-1} x\right)$$

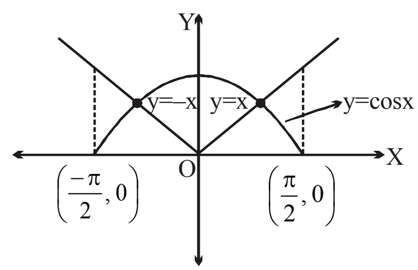
$$\Rightarrow \sin(\cos^{-1} x) = \frac{1}{2}$$

$$\Rightarrow \cos^{-1} x = \frac{\pi}{6} \qquad \Rightarrow x = \frac{\sqrt{3}}{2} \text{ Ans.}$$

84. As, $\quad \left|\sin^{-1}(\sin x)\right| = |x|$, for $x \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$\therefore$ From above graph, the equation

$\left|\sin^{-1}(\sin x)\right| = \cos x$ has two solutions, in $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$. **Ans.**



85. $\qquad S_n = \underline{1! + 2! + 3! + 4! + 5! + 6!} + 7I$ where I be an integer

$S_n = 873 + 7I$

$$\frac{S_n}{7} = 124.71 + \mathbf{I} \qquad \Rightarrow \qquad \left[\frac{S_n}{7}\right] = 124 + I$$

$$\therefore \qquad 7\left[\frac{S_n}{7}\right] = 868 + 7I \qquad \Rightarrow \qquad S_n - 7\left[\frac{S_n}{7}\right] = (873 + 7I) - (868 + 7I) = 5$$

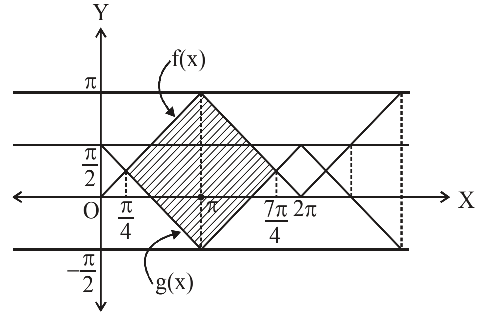
now $\quad \sin^{-1}(\sin 5) = \sin^{-1}\left(\sin(5-2\pi)\right) = 5 - 2\pi \qquad \Rightarrow \qquad$ **(P)**

$\quad \cos^{-1}(\cos 5) = \cos^{-1}\left(\cos(2\pi-5)\right) = 2\pi - 5 \qquad \Rightarrow \qquad$ **(Q)**

$\quad \tan^{-1}(\tan 5) = \tan^{-1}\left(\tan(5-2\pi)\right) = 5 - 2\pi \qquad \Rightarrow \qquad$ **(P)**

and $\quad \cot^{-1}(\cot 5) = \cot^{-1}\left(\cot(5-\pi)\right) = 5 - \pi \qquad \Rightarrow \qquad$ **(R)**

86.



We have $g(x) = \sin^{-1}(\cos x) = \dfrac{\pi}{2} - \cos^{-1}(\cos x)$

Both the curves bound the regions of same area

in $\left[\dfrac{\pi}{4}, \dfrac{7\pi}{4}\right], \left[\dfrac{9\pi}{4}, \dfrac{15\pi}{4}\right]$ and so on

$\therefore$ Required area = area of shaded square $= \dfrac{9\pi^2}{8} = \dfrac{a\pi^2}{b}$

$\therefore$ a = 9 and b = 8

Hence a + b = 17 **Ans.**

87. $\tan^{-1}\left(\dfrac{3\sin 2\alpha}{5+3\cos 2\alpha}\right) + \tan^{-1}\left(\dfrac{\tan\alpha}{4}\right)$ where $-\dfrac{\pi}{2} < \alpha < \dfrac{\pi}{2}$

$$\tan^{-1}\left(\frac{\left(3\dfrac{2\tan\alpha}{1+\tan^2\alpha}\right)}{5+3\dfrac{1-\tan^2\alpha}{1+\tan^2\alpha}}\right) = \tan^{-1}\left(\frac{6\tan\alpha}{8+2\tan^2\alpha}\right) = \tan^{-1}\left(\frac{3\tan\alpha}{4+\tan^2\alpha}\right) = \tan^{-1}\left(\frac{\frac{3}{4}\tan\alpha}{1+\frac{1}{4}\tan^2\alpha}\right)$$

$$\tan^{-1}\left(\frac{\tan\alpha - \dfrac{\tan\alpha}{4}}{1+\tan\alpha\dfrac{\tan\alpha}{4}}\right) = \tan^{-1}\left(\tan\left(\alpha-\frac{\alpha}{4}\right)\right) = \alpha - \frac{\alpha}{4}$$

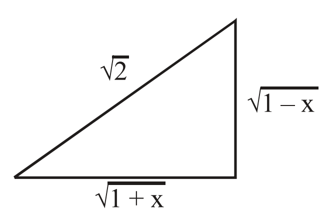
$$-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$$

$\therefore \quad a - \dfrac{\alpha}{4} + \dfrac{\alpha}{4} = \alpha$ Hence proved.

88. $\sin^{-1}\sqrt{\dfrac{1+x}{2}} - \sqrt{2-x} = \cot^{-1}\left(\tan\left(\sqrt{2-x}\right)\right) - \sin^{-1}\sqrt{\dfrac{1-x}{2}}$

$$\text{for domain} \quad \left.\begin{array}{lll} 0 \le \dfrac{1+x}{2} \le 1 & \Rightarrow & -1 \le x \le 1 \\ 2-x \ge 0 & \Rightarrow & x \le 2 \\ 0 \le \dfrac{1-x}{2} \le 1 & \Rightarrow & -1 \le x \le 1 \end{array}\right\} -1 \le x \le 1 \qquad ....(1)$$

$$\sin^{-1}\sqrt{\frac{1+x}{2}} - \sqrt{2-x} = \frac{\pi}{2} - \tan^{-1}\left(\tan\left(\sqrt{2-x}\right)\right) - \cos^{-1}\sqrt{\frac{1+x}{2}}$$



$$\sin^{-1}\sqrt{\frac{1+x}{2}}+\cos^{-1}\sqrt{\frac{1+x}{2}}-\sqrt{2-x}=\frac{\pi}{2}-\tan^{-1}\left(\tan\left(\sqrt{2-x}\right)\right)$$

$\tan^{-1}\left(\tan\left(\sqrt{2-x}\right)\right)=\sqrt{2-x}$ which is true. Now $\tan^{-1}(\tan x) = x$ when $x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

$$0 \le \sqrt{2-x} < \frac{\pi}{2} \quad \Rightarrow \quad 0 \le 2 - x < \frac{\pi^2}{4}$$

$$\Rightarrow \quad -2 \le -x < \frac{\pi^2}{4} - 2$$

$$\Rightarrow \quad 2-\frac{\pi^2}{4} < x \le 2 \qquad \text{..........(2)}$$

From (1) and (2), the complete solution set is $\left(2-\frac{\pi^2}{4}, 1\right]$ **Ans.**

89. Domain = [– 1, 1]

$$f(x) = \frac{1}{\pi}(\sin^{-1}x + \cos^{-1}x + \tan^{-1}x) + \frac{(x+1)}{(x+1)^2+9}$$

$$f_{max.} = f(1) = \frac{3}{4}+\frac{2}{13} = \frac{47}{52} = M$$

$\Rightarrow$ $52M = 47$. **Ans.**

90. Let $f(x) = x^3 + bx^2 + cx + 1$.
$f(0) = 1 > 0$, $f(-1) = b - c < 0$ so, $\alpha \in (-1, 0)$.
So, $2\tan^{-1}(\text{cosec}\,\alpha) + \tan^{-1}(2\sin\alpha\sec^2\alpha)$

$$= 2\tan^{-1}\left(\frac{1}{\sin\alpha}\right) + \tan^{-1}\left(\frac{2\sin\alpha}{1-\sin^2\alpha}\right) = 2\left[\tan^{-1}\left(\frac{1}{\sin\alpha}\right)+\tan^{-1}(\sin\alpha)\right]$$

$$= 2\left(-\frac{\pi}{2}\right) = -\pi \qquad (\text{as } \sin\alpha < 0) \quad \textbf{Ans.}$$

91. $S = \sum_{n=1}^{10}\left(\tan^{-1}\frac{1}{n}+\tan^{-1}\frac{2}{n}+\tan^{-1}\frac{3}{n}+........+\tan^{-1}\frac{10}{n}\right)$

now consider

$$\sum_{n=1}^{10}\tan^{-1}\frac{1}{n} = \underbrace{\tan^{-1}1}+\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{3}+......................+\tan^{-1}\frac{1}{9}+\tan^{-1}\frac{1}{10}$$

$$\sum_{n=1}^{10}\tan^{-1}\frac{2}{n} = \tan^{-1}\frac{2}{1}+\underbrace{\tan^{-1}\frac{2}{2}}+\tan^{-1}\frac{2}{3}+\tan^{-1}\frac{2}{4}+\tan^{-1}\frac{2}{5}+..........+\tan^{-1}\frac{2}{10}$$

$$\sum_{n=1}^{10} \tan^{-1}\frac{3}{n} = \tan^{-1}\frac{3}{1} + \tan^{-1}\frac{3}{2} + \underbrace{\tan^{-1}\frac{3}{3}} + \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{3}{5} + \ldots\ldots\ldots + \tan^{-1}\frac{3}{10}$$

$$\vdots \qquad \vdots \qquad \vdots$$

$$\sum_{n=1}^{10} \tan^{-1}\frac{10}{n} = \tan^{-1}\frac{10}{1} + \tan^{-1}\frac{10}{2} + \tan^{-1}\frac{10}{3} + \tan^{-1}\frac{10}{4} + \ldots\ldots\ldots\ldots + \underbrace{\tan^{-1}\frac{10}{10}}$$

$$S = \underbrace{\left(10\cdot\frac{\pi}{4}\right)} + \underbrace{\left(\tan^{-1}\frac{1}{2} + \tan^{-1}2\right)} + \left(\tan^{-1}\frac{1}{3} + \tan^{-1}3\right) + \left(\tan^{-1}\frac{1}{4} + \tan^{-1}4\right) + \ldots\ldots\ldots\ldots$$

[45 such pair each pair have value equal to $\pi/2$]

$$S = \frac{5\pi}{2} + \frac{45\pi}{2} = \frac{50\pi}{2} = 25\pi \Rightarrow k = 25 \text{ Ans.}$$

92. $\underbrace{\sin 2x}_{>0} + \underbrace{(1+\cos 2x)}_{>0} + \underbrace{\cos x}_{>0} > 0$ for $x \in \left(0, \frac{\pi}{2}\right)$. Hence no solution

$m = 0$

$$n = \sin\left[\tan^{-1}\left(\tan\frac{7\pi}{6}\right) + \cos^{-1}\left(\cos\frac{7\pi}{3}\right)\right]$$

$$n = \sin\left[\frac{\pi}{6} + \frac{\pi}{3}\right] = \sin\frac{\pi}{2} = 1$$

$m + n = 0 + 1 = 1$. **Ans.**

93. As, $x^2 - 4x + 5 = (x-2)^2 + 1$

So, $\sin^{-1}(x^2 - 4x + 5)$ and $\cos^{-1}(x^2 - 4x + 5)$ will be defined only for $x = 2$.

So, $4 + 4a + 8\left(\frac{\pi}{2}\right) = 0$

$\Rightarrow a = -(1 + \pi)$. **Ans.**

94. $\sec(3\tan^{-1}x) = 1$ or -1 and $-\frac{3\pi}{2} < 3\tan^{-1}x < \frac{3\pi}{2}$

$\Rightarrow 3\tan^{-1}x = 0, \pi, -\pi \Rightarrow \tan^{-1}x = 0, \frac{\pi}{3}, \frac{-\pi}{3} \Rightarrow x = 0, \sqrt{3}, -\sqrt{3} \Rightarrow$ Domain $\in \{0, \sqrt{3}, -\sqrt{3}\}$

Now, $f(x) = \sin^{-1}(\sec(3\tan^{-1}x)) + \cos^{-1}(\text{cosec}(3\cot^{-1}x))$

$$= \sin^{-1}(\sec(3\tan^{-1}x)) + \cos^{-1}\left(\text{cosec}\left(\frac{3\pi}{2} - 3\tan^{-1}x\right)\right) = \frac{\pi}{2} + 2\sin^{-1}(\sec(3\tan^{-1}x))$$

Range $\in \left\{\frac{3\pi}{2}, \frac{-\pi}{2}\right\}$. **Ans.**

95. When $x < -1$,

$$\sin^{-1}\left(\frac{2x}{1+x^2}\right) + \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

$= -\pi - 2\tan^{-1}x - 2\tan^{-1}x + \pi + 2\tan^{-1}x = -2\tan^{-1}x$

when $-1 < x < 0$, then sum of the above three terms $= 2\tan^{-1}x - 2\tan^{-1}x + 2\tan^{-1}x = 2\tan^{-1}x$

when $0 < x < 1$, the sum of the three terms $= 2\tan^{-1}x + 2\tan^{-1}x + 2\tan^{-1}x = 6\tan^{-1}x$

when $x > 1$, the sum of threeterms $= \pi - 2\tan^{-1}x + 2\tan^{-1}x + 2\tan^{-1} - \pi = 2\tan^{-1}x$

So $a = 6, 2, -2$

If we check at terminal points at $x = \pm 1, 0$ then we get $a = 6, 2$. **Ans.**

96. $f(x) = e^{\cos(\sin^{-1}x)} = y$

$\Rightarrow \cos(\sin^{-1}x) = \ln y$

$\Rightarrow \sin^{-1}x = \cos^{-1}(\ln y)$

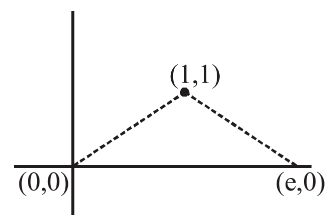
$\Rightarrow x = \sin(\cos^{-1}(\ln y)) = f^{-1}(y)$

$\therefore g(x) = \sin(\cos^{-1}(\ln x))$

$g(1) = \sin(\cos^{-1}0) = 1$

$g(e) = \sin(\cos^{-1}1) = 0$

$\therefore$ The points are (1, 1), (e, 0) and (0, 0)



Area of the triangle $= \frac{1}{2} \times e \times 1 = \frac{e}{2}$. **Ans.**

97. We have

$$\tan^{-1}(\cot\theta) - \cot^{-1}(\tan\theta) = \left(\frac{\pi}{2} - \cot^{-1}(\cot\theta)\right) - \left(\frac{\pi}{2} - \tan^{-1}(\tan\theta)\right)$$

$= \tan^{-1}(\tan\theta) - \cot^{-1}(\cot\theta)$

$= \theta - (\pi + \theta) = -\pi \quad \left(\text{as } \theta \in \left(\frac{-\pi}{2}, 0\right)\right)$

Hence $\sec\left(\tan^{-1}(\cot\theta) - \cot^{-1}(\tan\theta)\right) = \sec(-\pi) = \sec\pi = -1$ **Ans.**

98. For f(x) to be surjective.

Range of f(x) must be $\left(\frac{\pi}{2}, \pi\right]$ and

hence range of $y = -x^2 + x + a$ must be $(-\infty, -1]\ \forall\ x \in (-\infty, -1]$

$\therefore\ y = -(x^2 - x - a) = -\left[\left(x - \frac{1}{2}\right)^2 - \left(a + \frac{1}{4}\right)\right]$

$\Rightarrow\ y = a + \frac{1}{4} - \left(x - \frac{1}{2}\right)^2 \ \Rightarrow\ y \in (-\infty, a - 2]$

[Maximum value occurs at x = – 1]

$\therefore\ a - 2 = -1 \Rightarrow a = 1$. **Ans.**

99. Given expression = $\sum_{n=1}^{\infty}\left(\frac{\pi-\pi/2}{\pi a}\right)^n = \sum_{n=1}^{\infty}\frac{1}{2^n a^n} = \left(\frac{1}{2a}\right)+\left(\frac{1}{2a}\right)^2+\left(\frac{1}{2a}\right)^3+\ldots\ldots\ldots\infty$ terms.

sum is finite if $\frac{1}{2a} < 1 \quad \Rightarrow a > \frac{1}{2}$

$\therefore\ a \in \left(\frac{1}{2}, \infty\right)$

100. We have

$$k \cdot \frac{n(n+1)}{2} = \frac{n(n+1)(2n+1)}{6}$$

$\Rightarrow \qquad 2n + 1 = 3k$

$$\therefore \qquad \sin^{-1}\left(\frac{9k^2-4n^2}{6k+4n}\right) = \sin^{-1}\left(\frac{(3k+2n)(3k-2n)}{2(3k+2n)}\right) = \sin^{-1}\frac{1}{2} = \frac{\pi}{6}.\ \textbf{Ans.}$$

101. $y = \sin^{-1}\left(1-\sqrt{x-2}\right) + \cos^{-1}\left(\sqrt{2-[x]}\right)$

$$\underbrace{\begin{array}{ll} -1 \le 1 - \sqrt{x-2} \le 1 & 0 \le \sqrt{2-[x]} \le 1 \\ -2 \le -\sqrt{x-2} \le 0 & 0 \le 2-[x] \le 1 \\ 0 \le \sqrt{x-2} \le 2 & -2 \le -[x] \le -1 \\ 0 \le x-2 \le 4 & 1 \le [x] \le 2 \\ 2 \le x \le 6 & 1 \le x < 3 \end{array}}_{x\in[2,3)}$$

$\therefore\ \alpha = 2$

& $f(2) = \frac{\pi}{2} + \frac{\pi}{2} = \pi$

$\therefore$ P is $(2, \pi)$ and this point also lies on the second curve

$\therefore$ equation of tangent to the curve $y = -x + 2\cos(x-2) + \pi$ at point P is

$$\frac{dy}{dx} = -1 - 2\sin(x-2)$$

$$\left.\frac{dy}{dx}\right|_{x=2} = -1$$

$\therefore\ y - \pi = -1\,(x - 2)$

$$x + y = \pi + 2 \Rightarrow \frac{x}{\pi+2} + \frac{y}{\pi+2} = 1 \Rightarrow a = \pi + 2 = b$$

$\Rightarrow$ $a + b - \alpha - 2\beta = 2(\pi + 2) - 2 - 2\pi = 2$ **Ans.**

Paragraph for question nos. 102 to 104

102. $y = \dfrac{4x^2 - 4x + 17}{-(4x^2 - 4x) + 7} = \dfrac{4\left(x - \frac{1}{2}\right)^2 + 16}{-4\left(x - \frac{1}{2}\right)^2 + 8} = \dfrac{16 + 4\left(x - \frac{1}{2}\right)^2}{8 - 4\left(x - \frac{1}{2}\right)^2}$

$y|_{min.} = \dfrac{16}{8} = 2 = \alpha \qquad \left\{\because \text{ at } x = \dfrac{1}{2} \text{ numerator is min. \& denominator is max.}\right\}$

$\beta = \left[\dfrac{(1005-3)\,(1005-2)\,(1005-1)\,(1005+1)\,(1005+2)\,(1005+3)}{\frac{1}{6}\,(1005)^6}\right]$

$= \left[\dfrac{6.\left((1005)^2-1\right)\left((1005)^2-4\right)\left((1005)^2-9\right)}{(1005)^6}\right]$

$= \left[6\left(1-\dfrac{1}{(1005)^2}\right)\left(1-\dfrac{4}{(1005)^2}\right)\left(1-\dfrac{9}{(1005)^2}\right)\right] = 5$

$\therefore \quad \alpha + \beta = 7$

103. $f(x) = \tan^{-1}\left(\sin(k\cos x)\right)$

$-1 \le \sin(k\cos x) \le 1 \qquad \left\{\because \text{range of } f(x) \text{ is } \left[-\dfrac{\pi}{4}, \dfrac{\pi}{4}\right]\right\}$

$\because \quad \cos x \in [-1, 1]$

$\therefore \quad k\cos x \in [-k, k]$

but $k\cos x \in \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right] \Rightarrow k \ge \dfrac{\pi}{2}$

$\therefore$ Least integral value of k is 2.

104. $g(x) = \log_{\frac{19}{9}}\left(x + e^{ln\left(\tan\left(\cot^{-1}\left(5+4x-x^2\right)\right)\right)}\right)$

$= \log_{\frac{19}{9}}\left(x + \tan\left(\cot^{-1}\left(5+4x-x^2\right)\right)\right) = \log_{\frac{19}{9}}\left(x + \tan\left(\cot^{-1}\left(9-(x-2)^2\right)\right)\right)$

$$g(x)|_{min.} = \log_{\frac{19}{9}}\left(2+\tan\left(\cot^{-1}9\right)\right) = \log_{\frac{19}{9}}\left(2+\frac{1}{9}\right) = 1 \qquad \text{at } x=2$$

$$g(x)|_{max.} = \log_{\frac{19}{9}}\left(5+\tan\left(\frac{\pi}{2}\right)\right) = \infty \quad \text{at } x = 5$$

$\therefore$ Range is $[1, \infty)$ Ans.]

Paragraph for question nos.105 to 107

Sol: $$f(x) = \sin^{-1}\left(\frac{3\left(x^4+2x^2+1\right)-4}{\left(x^2+1\right)^3}\right)$$

$$= \sin^{-1}\left(\frac{3}{x^2+1} - \frac{4}{\left(x^2+1\right)^3}\right) \quad \text{let } \frac{1}{x^2+1} = \sin\theta \Rightarrow \theta = \operatorname{cosec}^{-1}(1+x^2)$$

$$\Rightarrow \theta \in \left(0, \frac{\pi}{2}\right] \ \forall\ x \in R$$

$$= \sin^{-1}(3\sin\theta - 4\sin^3\theta) = \sin^{-1}(\sin 3\theta)$$

$$f(x) = \sin^{-1}(\sin 3\theta) = \begin{cases} 3\theta = 3\operatorname{cosec}^{-1}\left(1+x^2\right) & 3\theta \in \left(0, \frac{\pi}{2}\right) \Rightarrow \theta \in \left(0, \frac{\pi}{6}\right) \Rightarrow x \in (-\infty, -1) \cup (1, \infty) \\ \pi - 3\theta = \pi - 3\operatorname{cosec}^{-1}\left(1+x^2\right) & 3\theta \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right] \Rightarrow \theta \in \left[\frac{\pi}{6}, \frac{\pi}{2}\right] x \in [-1, 1] \end{cases}$$

$$f(x) = \begin{cases} 3\operatorname{cosec}^{-1}\left(1+x^2\right), & |x| > 1 \\ \pi - 3\operatorname{cosec}^{-1}\left(1+x^2\right), & |x| \le 1 \end{cases}$$

$$g(x) = \begin{cases} 3\operatorname{cosec}^{-1}\left(1+x^2\right) + 3\sec^{-1}\left(1+x^2\right) = \frac{3\pi}{2} & |x| > 1 \\ \pi - 3\operatorname{cosec}^{-1}\left(1+x^2\right) + 3\operatorname{cosec}^{-1}\left(1+x^2\right) = \pi & |x| \le 1 \end{cases}$$

105 $$\sqrt{\tan 22\frac{1^\circ}{2}} = \sqrt{\sqrt{2}-1} < 1 \Rightarrow g\left(\sqrt{\tan 22\frac{1^\circ}{2}}\right) = \pi$$

$$\sqrt[3]{\cot 15^\circ} = \left(2+\sqrt{3}\right)^{\frac{1}{3}} > 1 \Rightarrow g\left(\sqrt[3]{\cot 15^\circ}\right) = \frac{3\pi}{2}$$

$$\sqrt[5]{\sin 18^\circ} = \left(\frac{\sqrt{5}-1}{4}\right)^{\frac{1}{5}} < 1 \Rightarrow g\left(\sqrt[5]{\sin 18^\circ}\right) = \pi$$

$$\therefore\ g\left(\sqrt{\tan 22\frac{1^\circ}{2}}\right) + g\left(\sqrt[3]{\cot 15^\circ}\right) + g\left(\sqrt[5]{\sin 18^\circ}\right) = \frac{7\pi}{2}$$

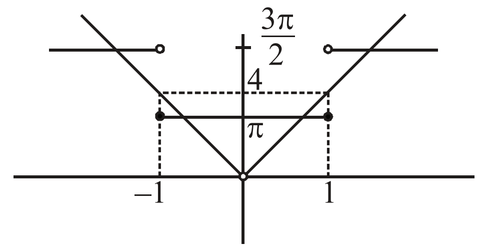
106. $h(x) = \frac{1}{5}(3\sin x + 4\sin x) \Rightarrow h(x) \in [-1, 1] \quad \forall\, x \in R$

$\therefore \quad g(h(x)) = \pi \quad \forall\, x \in R$

domain is R and range is $\{\pi\}$

107. $\tan\left(\cot^{-1}\left(\left|\frac{1}{4x}\right|\right)\right) = \tan(\tan^{-1}|4x|) = |4x|, \quad x \neq 0$

$\Rightarrow g(x) = |4x|$



Both curves intersect at 4 points

$\therefore$ Number of solutions is 4.

108. $\cos^{-1}\sqrt{x^3 - x + 1} + \cos^{-1}\sqrt{x - x^3} + \cos^{-1}\sqrt{1 - |y|} = \frac{2\pi}{3}$

$\sin^{-1}\sqrt{x - x^3} + \cos^{-1}\sqrt{x - x^3} + \cos^{-1}\sqrt{1 - |y|} = \frac{2\pi}{3}$

$\frac{\pi}{2} + \cos^{-1}\sqrt{1 - |y|} = \frac{2\pi}{3}$

$\cos^{-1}\sqrt{1 - |y|} = \frac{\pi}{6}$

$1 - |y| = \frac{3}{4}$

$|y| = \frac{1}{4}$ **Ans.**

109. $f(x) = \sin^{-1}\left(\frac{4x}{4 + x^2}\right) = \sin^{-1}\left(\frac{x}{1 + \frac{x^2}{4}}\right) = \sin^{-1}\left(\frac{2 \cdot \frac{x}{2}}{1 + \frac{x^2}{4}}\right)$

$-\left(\pi + 2\tan^{-1}\frac{x}{2}\right)$	$2\tan^{-1}\frac{x}{2}$	$\pi - 2\tan^{-1}\frac{x}{2}$
$\frac{x}{2} < -1$	$-1 \le \frac{x}{2} \le 1$	$\frac{x}{2} > 1$
$x < -2$	$-2 \le x \le 2$	$x > 2$

$f\left(\sqrt{7}-2\right) = 2\tan^{-1}\left(\frac{\sqrt{7}-2}{2}\right) \qquad \left\{\because \ \sqrt{7}-2 \simeq 0.6...\right\}$

$f\left(\frac{4}{\sqrt{7}-2}\right) = \pi - 2\tan^{-1}\left(\frac{4}{2\left(\sqrt{7}-2\right)}\right) \qquad \left\{\because \frac{4}{\sqrt{7}-2} > 2\right\}$

$\therefore \quad f\left(\sqrt{7}-2\right) - f\left(\frac{4}{\sqrt{7}-2}\right) = 2\tan^{-1}\left(\frac{\sqrt{7}-2}{2}\right) - \pi + 2\tan^{-1}\left(\frac{2}{\sqrt{7}-2}\right)$

$= 2\cdot\frac{\pi}{2} - \pi = 0$. **Ans.**

110. $y = \left(\sin^{-1}(\sin x)\right)^2 - \sin^{-1}(\sin x)$

$= \left(\sin^{-1}(\sin x) - \frac{1}{2}\right)^2 - \frac{1}{4}$

$\therefore$ For maximum value of y, $\sin^{-1}(\sin x) = \frac{-\pi}{2}$.

Hence $y_{maximum} = \left(\frac{\pi}{2}+\frac{1}{2}\right)^2 - \frac{1}{4} = \frac{\pi}{4}(\pi + 2)$. **Ans.**

111. $f(x) = \sin^{-1}\left(\frac{2x^4+x^2-1}{1+x^2}\right) = \sin^{-1}\left(\frac{\left(2x^2-1\right)\left(x^2+1\right)}{x^2+1}\right) = \frac{\pi}{2} - \cos^{-1}(2x^2-1)$

$x = \cos\theta \Rightarrow$ if $x \in [0, 1] \Rightarrow \theta \in \left[0, \frac{\pi}{2}\right]$

if $x \in [-1, 0) \Rightarrow \theta \in \left(\frac{\pi}{2}, \pi\right]$

$\therefore f(x) = \frac{\pi}{2} - \cos^{-1}(\cos 2\theta) =$

$$\begin{cases} \frac{\pi}{2} - 2\theta, & 2\theta \in [0, \pi] \Rightarrow \theta \in \left[0, \frac{\pi}{2}\right] \Rightarrow x \in [0, 1] \\ \frac{\pi}{2} - (2\pi - 2\theta), & 2\pi - 2\theta \in [\pi, 2\pi] \Rightarrow \theta \in \left[\frac{\pi}{2}, \pi\right] \text{but} \theta \in \left(\frac{\pi}{2}, \pi\right] \Rightarrow x \in [-1, 0) \end{cases}$$

$$f(x) = \begin{cases} \dfrac{\pi}{2} - 2\cos^{-1} x & 0 \le x \le 1 \\ -\dfrac{\pi}{2} - 2\sin^{-1} x & -1 \le x < 0 \end{cases}$$

$\therefore \quad a + b + p + q = \dfrac{1}{2} \ \ 2 - \dfrac{1}{2} - 2 = -4.$

$\therefore \quad |a + b + p + q| = 4$ Ans.]

112. $\tan^{-1}\dfrac{2}{3x} + \tan^{-1}\dfrac{3}{4} = \dfrac{\pi}{2} - \tan^{-1}\dfrac{1}{2}$. Taking tangent on both sides.

$$\frac{\dfrac{2}{3x} + \dfrac{3}{4}}{1 - \dfrac{2}{3x}\cdot\dfrac{3}{4}} = \cot\left(\tan^{-1}\frac{1}{2}\right) = 2 \Rightarrow \frac{8+9x}{12-6x} = 2 \Rightarrow x = \frac{4}{3}$$

$$2\tan^{-1}x = \pi + \tan^{-1}\left(\frac{2x}{1-x^2}\right) \quad \text{as } x > 1$$

$$= \pi + \tan^{-1}\left(\frac{-24}{7}\right)$$

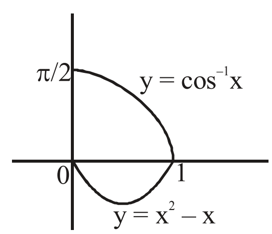
$$\tan\left(-\tan^{-1}\left(\frac{-24}{7}\right)\right) = \frac{24}{7}$$

$\therefore$ 24 + 7 = 31. **Ans.**

113. $\cos^2 x - \cos x - x = 0$

$\cos^2 - \cos x = \cos^{-1}(\cos x)$

Let $\cos x = t$

number of solutions of $t^2 - t = \cos^{-1} t$ in [0, 1] = 1 

. **Ans.**

114. $(\sin x + \cos x)(\sin^2 x + \cos^2 x) = \dfrac{\sqrt{6}}{2} \quad \Rightarrow \quad \sin x + \cos x = \dfrac{\sqrt{6}}{2}$

$\therefore \quad \sin\left(\dfrac{\pi}{4} + x\right) = \dfrac{\sqrt{3}}{2}$

Now verify $\Rightarrow$ (A) and (D)

115. $\cos^{-1}x + 2\cos^{-1}2x + 3\cos^{-1}3x = \pi + 2\pi + 3\pi$

$\Rightarrow x = -1,\ 2x = -1,\ 3x = -1$

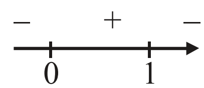
i.e., not possible simultaneously.

116. Let $\dfrac{x^2}{x^2+1} = y$

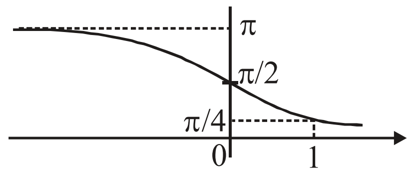
$x^2 = yx^2 + y$

$x^2 - yx^2 = y$

$x^2 = \dfrac{y}{1-y}$



$y \in [0, 1)$



$\Rightarrow \quad \dfrac{\pi}{4} < f(x) \le \dfrac{\pi}{2}$

$\Rightarrow \quad \dfrac{b}{a} = \dfrac{\frac{\pi}{2}}{\frac{\pi}{4}} = 2$]

117. $f(x) = \tan^{-1}\left(\dfrac{1}{1-x}\right)$, $0 < x < 1$, $-1 < -x < 0$, $0 < 1 - x < 1$

Range of $f(x) = \left(\dfrac{\pi}{4}, \dfrac{\pi}{2}\right)$, $1 < \dfrac{1}{1-x} < \infty$

$g(x) = \cot^{-1}\left(\dfrac{1}{\frac{1}{1+x}}\right) = \cot^{-1}(1+x), \quad 0 < x < 1$

Range of $g(x) = (\cot^{-1} 2, \cot^{-1} 1) = (\cot^{-1} 2, \pi/4)$.

118. Let $\quad \cot^{-1}x = t$

$\Rightarrow \quad x = \cot t,\ t \in (0, \pi)$

$\log_{10}(5t - 1) + \log_{10}(2t + 3) + \log_{10}5 = 2$

$\log_{10}(5t - 1)(10t + 15) = 2$

$(5t - 1)(10t + 15) = 100$

$50t^2 + 65t - 115 = 0$

$10t^2 + 13t - 23 = 0 \quad \Rightarrow \quad (10t + 23)(t - 1) = 0$

$t = -\dfrac{23}{10}$ (rejected) ; $t = 1$

Hence possible value of $t = 1$

$\Rightarrow \quad \cot^{-1}x = 1 \quad \Rightarrow \quad x = \cot 1$ Ans.

119. $x^2 - 18x + a > 0\ \forall\ x \in R$

$\Rightarrow a > 81\ \forall\ x \in R$

120. We have $f(x) = \pi - \cot^{-1} x - \tan^{-1} x + \sec^{-1} x = \pi - \frac{\pi}{2} + \sec^{-1} x = \frac{\pi}{2} + \sec^{-1} x$

As domain of $f(x)$ is $(-\infty, -1] \cup [1, \infty)$ $\left(\text{As } \cot^{-1}(-x) = \pi - \cot^{-1} x\right)$

$\therefore$ Range of $f(x)$ is $\left[\frac{\pi}{2}, \pi\right) \cup \left(\pi, \frac{3\pi}{2}\right]$.

121. Given expression $= \left\{\frac{\pi}{2} - \cos^{-1}(\cos 2)\right\} - \left\{\frac{\pi}{2} - \sin^{-1}(\sin 2)\right\}$

$+ \left\{\frac{\pi}{2} - \cot^{-1}(\cot 4)\right\} - \left\{\frac{\pi}{2} - \tan^{-1}(\tan 4)\right\} + \left\{\frac{\pi}{2} - \operatorname{cosec}^{-1}(\operatorname{cosec} 6)\right\} - \left\{\frac{\pi}{2} - \sec^{-1}(\sec 6)\right\}$

$= \sin^{-1}(\sin 2) - \cos^{-1}(\cos 2) + \tan^{-1}(\tan 4) - \cot^{-1}(\cot 4) + \sec^{-1}(\sec 6) - \operatorname{cosec}^{-1}(\operatorname{cosec} 6)$

$= (\pi - 2) - 2 + (4 - \pi) - (4 - \pi) + (2\pi - 6) - (6 - 2\pi) = \pi - 4 + 4\pi - 12 = 5\pi - 16.$]

122. After rationalization.

$$S_n = \sum \sin^{-1}\left(\frac{\sqrt{n^2+2n} - \sqrt{n^2-1}}{n(n+1)}\right) = \sum \left(\sin^{-1}\frac{1}{n} - \sin^{-1}\frac{1}{n+1}\right)$$

$t_1 = \sin^{-1} 1 - \sin^{-1}\frac{1}{2}$, $t_2 = \sin^{-1}\frac{1}{2} - \sin^{-1}\frac{1}{3}$ and so on

$$S_n = \sin^{-1} 1 - \sin^{-1}\left(\frac{1}{n+1}\right) = \frac{\pi}{2} - \sin^{-1}\left(\frac{1}{n+1}\right)$$

$\cos S_n = \frac{1}{n+1}$, $\cos S_{99} = \frac{1}{100}$

$100 \cos S_{99} = 1$. **Ans.**

123. We have $g(x) = \left[2 + \sin^{-1}\frac{2x}{1+x^2}\right] = 2 + \left[\sin^{-1}\frac{2x}{1+x^2}\right]$

As, $\sin^{-1}\frac{2x}{1+x^2} \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$\therefore$ $\left[\sin^{-1}\frac{2x}{1+x^2}\right] = -2, -1, 0, 1.$

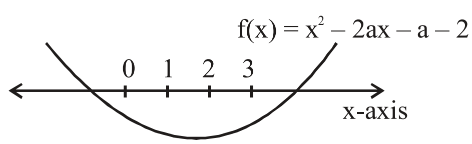
Range of $g(x) = \{0, 1, 2, 3\}$ for $f(g(x)) < 0 \ \forall \ x \in R$

$\Rightarrow$ $f(0) < 0$ and $f(3) < 0$

Now, $f(0) < 0 \Rightarrow a - 2 < 0 \Rightarrow a < 2$

and $f(3) < 0 \Rightarrow 9 - 6a + a - 2 < 0$

$a > \frac{7}{5}$



$\therefore \quad a \in \left(\frac{7}{5}, 2\right)$.

Hence, $k_1 = \frac{7}{5}, k_2 = 2$

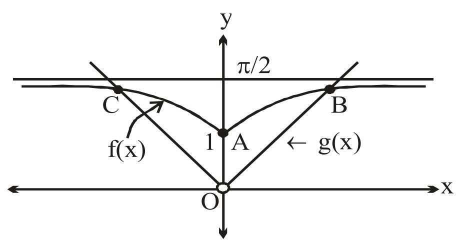
$\therefore \quad (10k_1 + 3k_2) = 14 + 6 = 20.$ **Ans.**

124. We have $2\pi = \frac{3\pi}{2} + 2\tan^{-1}\left(x^2 + x + a\right)$ $\quad \left(\text{As } \tan^{-1}\alpha + \cot^{-1}\alpha = \frac{\pi}{2}\ \forall\ \alpha \in R\right)$

$\Rightarrow \quad \tan^{-1}(x^2 + x + a) = \frac{\pi}{4} \Rightarrow x^2 + x + a = 1 \Rightarrow x^2 + x + (a-1) = 0$

$\therefore$ For required condition, put $D = 0 \Rightarrow 1 - 4(a-1) = 0 \Rightarrow a = \frac{5}{4}$. **Ans.**

125. $f(x) = \tan^{-1}\left(\sqrt{\left(\frac{1-\cos 2}{1+\cos 2}\right)} + x^2\right) = \tan^{-1}\left(\tan 1 + x^2\right)$



From the graph it is clear that f(x) and g(x) intersect at three distinct points i.e. at A, B and C. **Ans.**

126. $\sin^{-1}(\cos^{-1}x + \tan^{-1}x + \cot^{-1}x) = \sin^{-1}\left(\frac{\pi}{2} + \cos^{-1}x\right)$

For domain $\quad -1 \le \frac{\pi}{2} + \cos^{-1}x \le 1 \Rightarrow -1 - \frac{\pi}{2} \le \cos^{-1}x \le 1 - \frac{\pi}{2}$

But $\quad 0 \le \cos^{-1}x \le \pi$ so No solution

similarly for $\quad \cos^{-1}(\tan^{-1}x + \cot^{-1}x + \sin^{-1}x) = \cos^{-1}\left(\frac{\pi}{2} + \sin^{-1}x\right)$

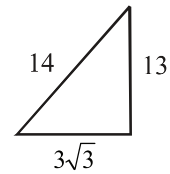
$$-1 \le \frac{\pi}{2} + \sin^{-1}x \le 1 \Rightarrow -\frac{\pi}{2} - 1 \le \sin^{-1}x \le 1 - \frac{\pi}{2}$$

$$-\frac{\pi}{2} \le \sin^{-1}x \le 1 - \frac{\pi}{2} \Rightarrow -1 \le x \le -\cos 1$$

$\sin^{-1}(\cos^{-1}x) \Rightarrow -1 \le \cos^{-1}x \le 1 \Rightarrow 0 \le \cos^{-1}x \le 1 \Rightarrow \cos 1 \le x \le 1$

$\cos^{-1}(\sin^{-1}x) \Rightarrow -1 \le \sin^{-1}x \le 1 \Rightarrow -\sin 1 \le x \le \sin 1.$ **Ans.**

127. $\sin\left(\frac{\pi}{6}+\tan^{-1}x\right)=\frac{13}{14}$



$$\frac{\pi}{6}+\tan^{-1}x=\sin^{-1}\left(\frac{13}{14}\right)=\tan^{-1}\left(\frac{13}{3\sqrt{3}}\right)$$

$$\tan^{-1}x=\tan^{-1}\left(\frac{13}{3\sqrt{3}}\right)-\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)=\tan^{-1}\left[\frac{\frac{13}{3\sqrt{3}}-\frac{1}{\sqrt{3}}}{1+\frac{13}{9}}\right]=\tan^{-1}\left[\frac{10}{3\sqrt{3}}\left(\frac{9}{22}\right)\right]$$

$$\tan^{-1}x=\tan^{-1}\left[\frac{5\sqrt{3}}{11}\right]$$

$$\therefore\ x=\frac{5\sqrt{3}}{11}\ \Rightarrow\ \frac{a\sqrt{3}}{b}\ \Rightarrow\ a+b=16\ \Rightarrow\ \left(\frac{a+b}{2}\right)=8.\ \textbf{Ans.}$$

Aliter: $\sin(30^\circ+\tan^{-1}x)=\frac{13}{14}$

$$\Rightarrow\ \frac{1}{2}\cos(\tan^{-1}x)+\frac{\sqrt{3}}{2}\sin(\tan^{-1}x)=\frac{13}{14}\ \Rightarrow\ 7(1+\sqrt{3}\,x)=13\sqrt{1+x^2}$$

$\Rightarrow 11x^2-49\sqrt{3}\,x+60=0$ (On squaring)

$$\therefore\ x=\frac{49\sqrt{3}\pm 39\sqrt{3}}{22}=\frac{5\sqrt{3}}{11},\ 4\sqrt{3}\ \text{(Reject)}$$

$$\therefore\ x=\frac{5\sqrt{3}}{11}.\ \textbf{Ans.}]$$

128. As, $x^2-4x+5=(x-2)^2+1$

So, $\sin^{-1}(x^2-4x+5)$ and $\cos^{-1}(x^2-4x+5)$ will be defined only for $x=2$.

So, $4+4a+8\left(\frac{\pi}{2}\right)=0\Rightarrow a=-(1+\pi)$. **Ans.**

129. (A) Let $|\sin x-1|+|\sin x+1|=y$

if $\sin x=t \Rightarrow |t-1|+|t+1|=y$

if $t\in[-1,1] \Rightarrow y=2$

if $t\in(-\infty,-1]\cup[1,\infty) \Rightarrow y>2$

$t\in[-1,1] \Rightarrow \sin x=t$

$\Rightarrow x\in R \Rightarrow y=2$

$t\in(-\infty,-1)\cup(1,\infty)$

$\Rightarrow$ not possible because $\sin x\in[-1,1]$

$\therefore$ $f(x)=\sin^{-1}(2/2)=\sin^{-1}(1)=\pi/2=$ constant (periodic)

$\Rightarrow$ **P, Q, R, S**

(B) Let $|x-1|-|x-2|=y$

if $x \in (-\infty, 1] \Rightarrow \quad y = -1$

if $x \in [1, 2] \quad \Rightarrow \quad y = [-1, 1]$

if $x \in [2, \infty) \quad \Rightarrow \quad y = 1$

$\therefore \quad f(x) = \cos^{-1}(|x-1|-|x-2|)$

domain is R, range is $[0, \pi]$; $\quad \therefore \quad$ **P, Q** is correct

(C) Let $|\sin^{-1} x - (\pi/2)| + |\sin^{-1} x + (\pi/2)| = y$

let $\sin^{-1}x = t \Rightarrow \quad t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\left[t-\frac{\pi}{2}\right] + \left[t+\frac{\pi}{2}\right] = y$

$t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \Rightarrow \quad y = \pi$

$\therefore \quad f(x) = \sin^{-1}\left(\frac{\pi}{\pi}\right) = \frac{\pi}{2}$

$\therefore$ f(x) is periodic and constant function also range contain only irrational number.

$\Rightarrow$ **P, R, S**

(D) Domain is $\{\pm 1\}$

$f(x) = \frac{\pi}{2} + 1 \quad$ for $x = \pm 1$

range contain only irrational value and also constant function.

$\Rightarrow$ **P, R, S**]

130. Given, $f(x) = \tan^{-1}(\cot x - 2\cot 2x)$

$$= \tan^{-1}\left(\frac{1}{\tan x} - \frac{1-\tan^2 x}{\tan x}\right) = \tan^{-1}\left(\frac{\tan^2 x}{\tan x}\right) = \tan^{-1}(\tan x).$$

Now, $\sum_{r=1}^{5} f(r) = f(1) + f(2) + f(3) + f(4) + f(5)$

As, $f(1) = 1,\ f(2) = 2-\pi,\ f(3) = 3-\pi,\ f(4) = 4-\pi,\ f(5) = 5-2\pi$

$\Rightarrow \quad \sum_{r=1}^{5} f(r) = 15 - 5\pi = a - b\pi \Rightarrow a = 15, b = 5$

Hence, the value of $(a+b) = 20$.**Ans.**

131. As $\frac{\pi}{3} > 1 \Rightarrow \cot^{-1}\frac{\pi}{3} < \cot^{-1} 1 = \frac{\pi}{4} \Rightarrow \cot^{-1}\frac{\pi}{3} < \frac{\pi}{4} = 0.7856$ ....(i)

As $\frac{\pi}{4} > \frac{1}{\sqrt{2}} \Rightarrow \sin^{-1}\frac{\pi}{4} > \sin^{-1}\frac{1}{\sqrt{2}} = \frac{\pi}{4} \Rightarrow \sin^{-1}\frac{\pi}{4} > \frac{\pi}{4} = 0.7856$ ....(ii)

As $\frac{2\pi}{3} > 2 \Rightarrow \sec^{-1}\frac{2\pi}{3} > \sec^{-1} 2 = \frac{\pi}{3} \Rightarrow \sec^{-1}\frac{2\pi}{3} > \frac{\pi}{3} \simeq 1.0476$ ....(iii)

Clearly $\alpha < \beta < \gamma$.

132. We have $f(x) = \sin^{-1}\left(\dfrac{4}{4x^2 - 12x + 17}\right)$

$$= \sin^{-1}\left(\frac{1}{x^2 - 3x + \frac{17}{4}}\right) = \sin^{-1}\left(\frac{1}{\left(x - \frac{3}{2}\right)^2 + \frac{17}{4} - \frac{9}{4}}\right) = \sin^{-1}\left(\frac{1}{\left(x - \frac{3}{2}\right)^2 + 2}\right)$$

Hence $f(x) \in \left(0, \frac{\pi}{6}\right]$, so co-domain = Range

$\left(\text{As } x^2 - 3x + \frac{17}{4} = \left(x - \frac{3}{2}\right)^2 + 2 \in [2, \infty)\right)$

Also $y = 4x^2 - 12x + 17$ is many one function.
Hence $f(x)$ is surjective but not injective.

133. As $\cos 1 - \cos^{-1} 1 = \cos 1 - 0 \in (0, 1)$

$\Rightarrow$ $[\cos 1 - \cos^{-1} 1] = 0$

$\sin 1 - \sin^{-1} 1 = \sin 1 - \frac{\pi}{2} \in (-1, 0)$

$\Rightarrow$ $[\sin 1 - \sin^{-1} 1] = -1$

$\tan 1 - \tan^{-1} 1 = \tan 1 - \frac{\pi}{4} \approx 1.73 - 0.78 > 0$ and $\in (0, 1)$

$\Rightarrow [\tan 1 - \tan^{-1} 1] = 0$ $\left\{\tan 1 \approx \tan\frac{\pi}{3}\right\}$

$\cot 1 - \cot^{-1} 1 = \cot 1 - \frac{\pi}{4} \approx 0.57 - 0.78 < 0$ and $\in (-1, 0)$

$\Rightarrow$ $[\cot 1 - \cot^{-1} 1] = -1$

$\sec 1 - \sec^{-1} 1 = \sec 1 - 0 < \sec\frac{\pi}{3} = 2.$ As $\sec 1 \in (1, 2)$

$\Rightarrow$ $[\sec 1 - \sec^{-1} 1] = 1$

and $\operatorname{cosec} 1 - \operatorname{cosec}^{-1} 1 = \operatorname{cosec} 1 - \frac{\pi}{2}$

As $\operatorname{cosec}\frac{\pi}{3} < \operatorname{cosec} 1 < \operatorname{cosec}\frac{\pi}{4} < \frac{\pi}{2}$

$\Rightarrow \operatorname{cosec} 1 \in \left(\frac{2}{\sqrt{3}}, \sqrt{2}\right)$

$\therefore$ $\operatorname{cosec} 1 - \frac{\pi}{2} \in (-1, 0) \Rightarrow [\operatorname{cosec} 1 - \operatorname{cosec}^{-1} 1] = -1$

Hence the value of given exprersion $= 0 - (-1) + 0 - (-1) + 1 - (-1)$
$= 1 + 1 + 1 + 1 = 4$ **Ans.**

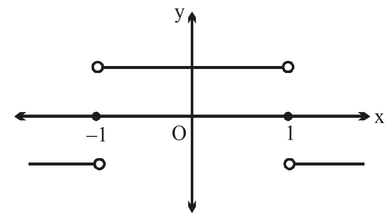
134.

(A) We have

$f(x) = \sqrt[5]{x} + \sin^{-1}x$

Cleary, domain of $f(x) = [-1, 1]$.

Also, f(x) is increasing so f(x) is one-one function.

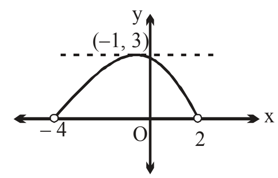


(B) $f(x) = \text{sgn}\left(\dfrac{1-|x|}{1+|x|}\right)$

$D_f = R$

$R_f = \{-1, 0, 1\}$ even function

(C) For domain of f(x), we must have $8 - 2x - x^2 \geq 0$



$\Rightarrow x^2 + 2x - 8 \leq 0$

$\Rightarrow (x+4)(x-2) \leq 0$

$\Rightarrow x \in [-4, 2]$

$R_f = [0, 3]$

(D) $f(x) = \dfrac{2^{-[x]}}{2^{\{x\}}} - 2^{|x|} = 2^{-x} - 2^{|x|} = 0 \ \forall\ x \leq 0]$

135. $\tan^{-1}\left(\dfrac{8n}{n^4 - 2n^2 + 5}\right) = \tan^{-1}\dfrac{2\{(n+1)^2 - (n-1)^2\}}{(n^2-1)^2 + 4}$

$$= \tan^{-1}\dfrac{\left(\dfrac{n+1}{2}\right)^2 - \left(\dfrac{n-1}{2}\right)^2}{1 + \left(\dfrac{n+1}{2}\right)^2 \cdot \left(\dfrac{n-1}{2}\right)^2}$$

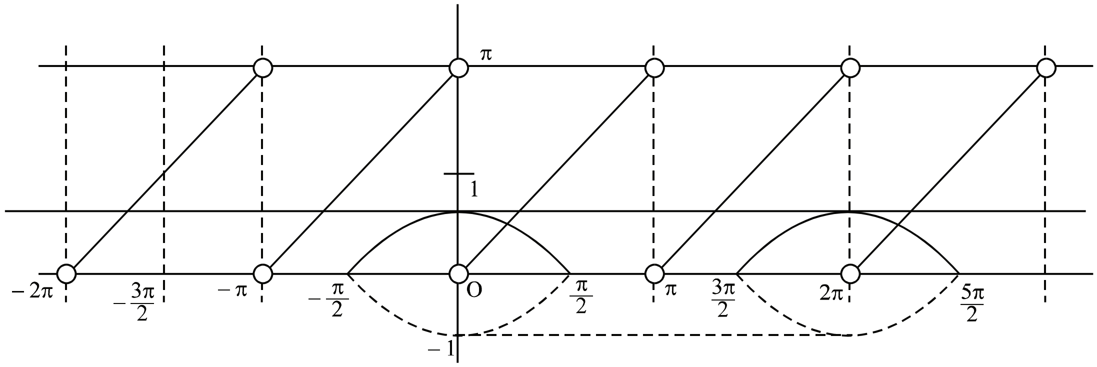
$$= \tan^{-1}\left(\frac{n+1}{2}\right)^2 - \tan^{-1}\left(\frac{n-1}{2}\right)^2$$

$$\therefore \sum_{n=1}^{\infty} \tan^{-1}\left(\frac{8n}{n^4 - 2n^2 + 5}\right) = \lim_{n\to\infty}\left[\left(\tan^{-1}1^2 - 0\right) + \left\{\tan^{-1}\left(\frac{3}{2}\right)^2 - \tan^{-1}1^2\right\} + \right.$$

$$\left.\left\{\tan^{-1}2^2 - \tan^{-1}\left(\frac{3}{2}\right)^2\right\} + \ldots\ldots + \left\{\tan^{-1}\left(\frac{n+1}{2}\right)^2 - \tan^{-1}\left(\frac{n-1}{2}\right)^2\right\}\right]$$

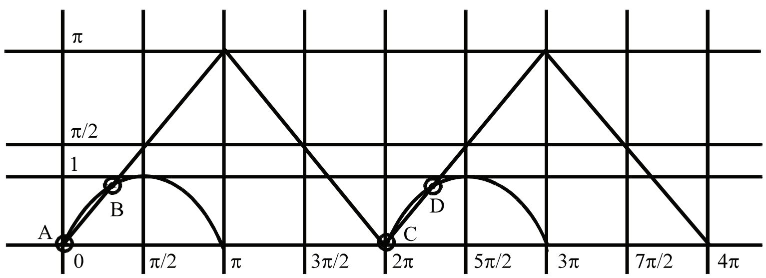
$$= \lim_{n\to\infty}\left\{\tan^{-1}\left(\frac{n+1}{2}\right)^2 - 0\right\} = \frac{\pi}{2} = \tan^{-1}2 + \cot^{-1}3$$

136.



Hence two solutions]

137.



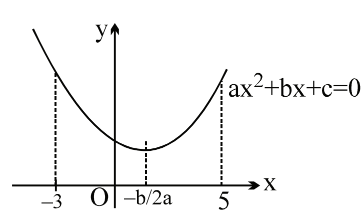
138. $f(x) = \tan^{-1}\left(\dfrac{\dfrac{x}{3} - \dfrac{x}{\sqrt{3}}}{1 + \dfrac{x^2}{3\sqrt{3}}}\right)$

$= \tan^{-1}\dfrac{x}{3} - \tan^{-1}\dfrac{x}{\sqrt{3}}$; $0 \le x \le 3$

Hence $f(x) = \tan^{-1}\left(\dfrac{x}{3}\right)$

$\Rightarrow$ range of $f(x)$ is $\left[0, \dfrac{\pi}{4}\right]$

139.

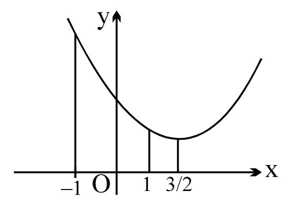


(A) $-3 < -\dfrac{3a}{2} < 5$ $\Rightarrow -\dfrac{10}{3} < a < 2$

$\therefore$ $a \in \{-3, -2, -1, 0, 1\}$ $\Rightarrow$ **P, Q, R, S**

(B) $f(x) = x^2 - 3x + b$

$gof(x) = \sin^{-1}\left(\dfrac{f(x)}{4}\right)$ is defined $\forall$ $x \in [-1, 1]$



$\Rightarrow \quad -1 \le \dfrac{f(x)}{4} \le 1 \qquad \forall\ x \in [-1, 1]$

$-4 \le x^2 - 3x + b \le 4 \quad \forall\ x \in [-1, 1]$

$f_{min} = f(+1) = b - 2$

$f_{max} = f(-1) = b + 4$

$\Rightarrow \quad b - 2 \ge -4$ and $b + 4 \le 4$

$b \ge -2$ and $b \le 0$

$\Rightarrow \quad b \in [-2, 0]$

$\therefore \quad b = -2, -1, 0 \Rightarrow$ **P, Q, R**

(C) $f : R \to [-8, \infty),\ f(x) = x^2 + 6x + b$

$f_{mim} = f(-3) = b - 9$

$b - 9 = -8 \quad \Rightarrow \quad b = 1 \quad \Rightarrow$ **S**

(D) $h(x) = fog(x) = \left(\sin^{-1}\dfrac{x}{4}\right)^2 + 3\sin^{-1}\dfrac{x}{4} + 2$

$h(t) = t^2 + 3t + 2,\ t = \sin^{-1}\dfrac{x}{4},\quad t \in \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$

$= \left(t + \dfrac{3}{2}\right)^2 - \dfrac{1}{4}$

$h(t)|_{max}$ occurs when $t = \dfrac{\pi}{2}$ and $h(t)|_{min}$ occurs when $t = -\dfrac{3}{2}$

$\therefore \quad h(t)|_{min} = -\dfrac{1}{4}$ and $h(t)|_{max} = \dfrac{\pi^2}{4} + \dfrac{3\pi}{2} + 2$

$\therefore \quad$ Range of $fog(x) = \left[-\dfrac{1}{4}, \dfrac{\pi^2}{4} + \dfrac{3\pi}{2} + 2\right]$

$\therefore \quad$ Possible integers are 0, 1 $\quad \Rightarrow \quad$ **R, S**

140. By using A.M. – G.M. inequality, we have

$$\frac{8^{\sin^{-1}x} + 8^{\cos^{-1}x}}{2} \ge \sqrt{8^{\sin^{-1}x} \cdot 8^{\cos^{-1}x}} \Rightarrow f(x) \ge 2\sqrt{8^{\sin^{-1}x + \cos^{-1}x}}$$

$$\therefore f(x)_{min.} = m = 2\sqrt{8^{\pi/2}} = 2\left(2^{3\pi/4}\right) = 2^{1+\frac{3\pi}{4}}$$

Hence $\log_2 m = 1 + \dfrac{3\pi}{4}$

141. $f x \in \left(\pi, \dfrac{3\pi}{2}\right)$ then $\sin^{-1}(\sin x) = \pi - x$ and $\cos^{-1}(\cos x) = 2\pi - x$

$\therefore \quad \cos^{-1}\{\sin(\cos^{-1}(\cos x) - \sin^{-1}(\sin x)\} = \cos^{-1}\{\sin\{(2\pi - x) - (\pi - x)\}\}$

$= \cos^{-1}\sin\pi = \frac{\pi}{2}$

142.

(A) $\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{13}\right) + ... + \infty$

$T_n = \tan^{-1}\frac{1}{1+n+n^2} = \tan^{-1}\frac{(n+1)-n}{1+(n+1)n} = \tan^{-1}(n+1) - \tan^{-1}n$

putting n = 1, 2, 3, ..., and adding, we get

$S_n = \tan^{-1}(n+1) - \tan^{-1} 1$

$S_\infty = \tan^{-1}(\infty)\ \frac{-\pi}{4} = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}.$

(B) Since $\sin^{-1}\left(\frac{12}{13}\right) = \tan^{-1}\left(\frac{12}{5}\right) = \cos^{-1}\frac{4}{5} = \tan^{-1}\frac{3}{4}$

$\therefore$ LHS $= \tan^{-1}\frac{12}{5} + \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{63}{16}$

since $\frac{12}{5} \times \frac{3}{4} > 1$, we have

$$= \tan^{-1}\frac{12}{5} + \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{\frac{12}{5}+\frac{3}{4}}{1-\frac{12}{5}.\frac{3}{4}} = \pi - \tan^{-1}\frac{63}{16}.$$

(C) $2\tan^{-1}\frac{1}{3} = \tan^{-1}\frac{2/3}{1-\frac{1}{9}} = \tan^{-1}\frac{3}{4}$ and $\sin^{-1}\frac{4}{5} = \tan^{-1}\frac{4}{3}$

$\therefore\ \sin^{-1}\frac{4}{5} + 2\tan^{-1}\frac{1}{3} = \tan^{-1}\frac{4}{3} + \tan^{-1}\frac{3}{4} = \tan^{-1}\frac{4}{3} + \cot^{-1}\frac{4}{3} = \frac{\pi}{2}.$

(D) $\operatorname{cosec}^{-1} x = \cot^{-1}\sqrt{x^2-1}$

$\therefore\ \operatorname{cosec}^{-1}\frac{\sqrt{41}}{4} = \cot^{-1}\sqrt{\frac{41}{16}-1} = \cot^{-1}\left(\frac{5}{4}\right);$

$\therefore\ \cot^{-1} 9 + \operatorname{cosec}^{-1}\frac{\sqrt{41}}{4} = \cot^{-1} 9 + \cot^{-1}\frac{5}{4}$

$$= \tan^{-1}\frac{1}{9} + \tan^{-1}\frac{4}{5} = \tan^{-1}\frac{\frac{1}{9}+\frac{4}{5}}{1-\frac{1}{9}.\frac{4}{5}} = \tan^{-1} 1 = \frac{\pi}{4}$$

143.

(A) $f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$

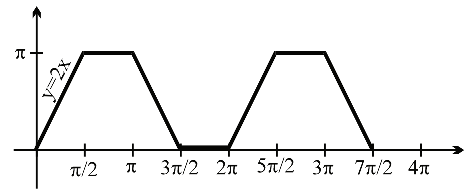
$= x + x = 2x, \qquad x \in [0, \pi/2]$

$= \pi - x + x = \pi, \qquad x \in [\pi/2, \pi]$

$= \pi - x + 2\pi - x = 3\pi - 2x, \qquad x \in [\pi, 3\pi/2]$

$= x - 2\pi + 2\pi - x = 0, \qquad x \in [3\pi/2, 2\pi]$

$R_f = [0, \pi]$



(B) $g(x) = \sin^{-1}|x| + 2\tan^{-1}|x|$

g is even, $\quad D_g \in [-1, 1]$

$|x| \in [0, 1]$

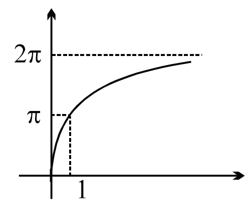
if $x \geq 0$, $g(x) = \sin^{-1}x + 2\tan^{-1}x$

$$R_g = \left[0+0, \frac{\pi}{2}+2\left(\frac{\pi}{4}\right)\right] = [0, \pi]$$

g(x) is increasing $\forall\ x \in (0, 1)$

g(x) is decreasing $\forall\ x \in (-1, 0)$

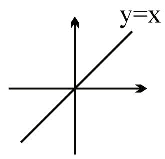
(C) $h(x) = 2\sin^{-1}\left(\dfrac{2x}{1+x^2}\right),\ x \in [0, 1]$



$\tan^{-1}x = \theta, \qquad \theta \in [0, \pi/4], \quad 2\theta \in [0, \pi/2]$

$h(x) = 2\sin^{-1}(\sin 2\theta) = 4\theta = 4\tan^{-1}(x)$

(D) $\cot(\cot^{-1}x) = x\ \forall\ x \in R$



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144. $1 \geq x^2 + y^2 \geq 0$ ...... (1) $\qquad$ & $\qquad |x| \geq 1 \Rightarrow x^2 \geq 1$ .... (2)

equation (1) & (2) $x = \pm 1$ & $y = 0$

$$\Rightarrow \text{ If } x = 1 \text{ then } a = \frac{\pi}{2} + \frac{\pi}{4} + 0 = \frac{3\pi}{4} \ \& \ \text{ if } x = -1 \text{ then } a = \frac{\pi}{2} + \frac{\pi}{4} + \pi = \frac{7\pi}{4}$$

$$\Rightarrow \text{ sum of values of } a = \frac{5\pi}{2}$$

145. $$T_n = \cot^{-1}\left(n^2 + \frac{3}{4}\right) = \tan^{-1}\left(\frac{1}{n^2 + (3/4)}\right) = \tan^{-1}\left(\frac{1}{1 + n^2 - (1/4)}\right)$$

$$= \tan^{-1}\left(\frac{1}{1+\left(n-\frac{1}{2}\right)\left(n+\frac{1}{2}\right)}\right)$$

$$= \tan^{-1}\left(\frac{\left(n+\frac{1}{2}\right)-\left(n-\frac{1}{2}\right)}{1+\left(n+\frac{1}{2}\right)\left(n-\frac{1}{2}\right)}\right)$$

146. Let $\cos^{-1}x = t,\ t \in (0, \pi]$ as $\cos^{-1}x \neq 0$, being in D^r

$$2t = a + \frac{a^2}{t}$$

$$2t^2 - at - a^2 = 0$$

$$t = \frac{a \pm \sqrt{a^2 + 8a^2}}{4} = \frac{a \pm 3a}{4}$$

with +ve sign

$$t = a \in (0, \pi] \quad ....(1)$$

with – ve sign

$$t = -\frac{a}{2}$$

$$0 < -\frac{a}{2} \leq \pi$$

$$0 < -a \leq 2\pi$$

$$-2\pi \leq a < 0 \quad ....(2)$$

from (1) and (2) $\quad a \in [-2\pi, \pi] - \{0\}$ **Ans.**

147. $\cos^{-1} x + (\sin^{-1} y)^2 = \dfrac{K\pi^2}{4}$

$$(\sin^{-1} y)^2 \cdot (\cos^{-1} x) = \frac{\pi^4}{16}$$

$$-\frac{\pi}{2} \leq \sin^{-1} y \leq \frac{\pi}{2}$$

$$0 \leq (\sin^{-1} y)^2 \leq \frac{\pi}{4}$$

$$0 \leq \cos^{-1} x \leq \pi$$

$$0 \leq \cos^{-1} x + (\sin^{-1} y)^2 \leq \pi + \frac{\pi^2}{4}$$

$$0 \leq \frac{K\pi^2}{4} \leq \pi + \frac{\pi^2}{4} \qquad 0 \leq K\ \frac{\pi x}{\pi} + 1$$

$$0 \leq \frac{K\pi}{4} \leq 1 + \frac{\pi}{4} \Rightarrow 0 \leq k \leq \frac{\pi}{4} + 1 \Rightarrow k \in \{0, 1, 2\} \qquad ....(1)$$

$$\cos^{-1} x + \frac{\pi^4}{16\cos^{-1} x} = \frac{K\pi^2}{4} \text{ from given equation}$$

Put $\cos^{-1} x = t$

$$t + \frac{\pi^4}{16t} = \frac{K\pi^2}{4}$$

Also AM ≥ GM

$$\frac{K\pi^2}{8} \geq \left(\frac{\pi^4}{16}\right)^{1/2}$$

$\dfrac{16t^2+\pi^4}{416t} = \dfrac{K\pi^2}{4}$

$16\,t^2 + \pi^4 = 4K\pi^2 t$

for (1) and (2) k = 2

$16\,t^2 - 4t\,K\pi^2 + \pi^4 = 0$

$t = \dfrac{8\pi^2 \pm \sqrt{64\pi^4 - 64\pi^4}}{32}$

$t = \dfrac{\pi^2}{4}$

$\Rightarrow \cos^{-1} x = \dfrac{\pi^2}{4} \Rightarrow x = \cos\left(\dfrac{\pi^2}{4}\right) \Rightarrow (\sin^{-1} y)^2\ \dfrac{\pi^2}{4} = \dfrac{\pi^4}{16} \Rightarrow \sin^{-1} y = \pm\dfrac{\pi}{2}$

$y = \pm 1$]

$\dfrac{K\pi^2}{8} \geq \dfrac{\pi^2}{4}$

$K \geq 2$ ....(2)

148. $f(x) = \sqrt{\sin(\cos x)} + \ln(-2\cos^2 x + 3\cos x + 1) + e^{\cos^{-1}\left(\frac{2\sin x + 1}{2\sqrt{2\sin x}}\right)}$

Let $\sqrt{2\sin x} = t \quad t > 0$

$-1 \leq \dfrac{t^2+1}{2t} \leq 1$

$-2t \leq t^2 + 1 \leq 2t$

$(t+1)^2 \geq 0$ and $(t-1)^2 \leq 0$

$t \in R$ ....(1) $\quad t = 1$ ....(2)

(1) $\cap$ (2)

$t = 1;\quad \sin x = \dfrac{1}{2};\qquad \cos x = \dfrac{\sqrt{3}}{2}$ or $-\dfrac{\sqrt{3}}{2}$

for $\cos x = -\dfrac{\sqrt{3}}{2}$

$\sqrt{\sin(\cos x)}$ is not defined

and for $\cos x = \dfrac{\sqrt{3}}{2}$ both $\sqrt{\sin(\cos x)}$ and log function are defined

$\therefore \quad \sin x = \dfrac{1}{2}$ and $\cos x = \dfrac{\sqrt{3}}{2}$

$\therefore \quad x = 2n\pi + \dfrac{\pi}{6},\ n \in I$ **Ans.**

149. $\cos^{-1}\left(\sqrt{6}x\right) + \cos^{-1}\left(3\sqrt{3}x^2\right) = \dfrac{\pi}{2}$

$$\cos^{-1}\left[\left(\sqrt{6}x\right)\left(3\sqrt{3}x^2\right)-\sqrt{1-6x^2}\sqrt{1-27\times x^4}\right]=\frac{\pi}{2}$$

$$9\sqrt{2}\,x^3=\sqrt{\left(1-6x^2\right)\left(1-27x^4\right)}$$

Squaring

$162\,x^6 = 1 - 27\,x^4 - 6x^2 + 162\,x^6$

$27\,x^4 + 6\,x^2 - 1 = 0$

$27\,x^4 + 9\,x^2 - 3\,x^2 - 1 = 0$

$9x^2\,(3x^2 + 1) - (3x^2 + 1) = 0$

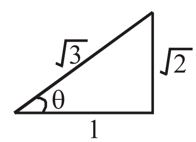
$9x^2 = 1 \qquad \Rightarrow \qquad x = \frac{1}{3} \text{ or } \frac{-1}{3}$

If $x = \frac{1}{3}$

$$\cos^{-1}\frac{\sqrt{6}}{3} + \cos^{-1}\left(3\sqrt{3}.\frac{1}{9}\right)$$

$$\cos^{-1}\sqrt{\frac{2}{3}}+\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\underbrace{\tan^{-1}\frac{1}{\sqrt{2}}+\tan^{-1}\sqrt{2}}_{\frac{\pi}{2}}=\frac{\pi}{2}$$



$$\cos^{-1}\left(-\sqrt{\frac{2}{3}}\right)+\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\pi-\cos^{-1}\sqrt{\frac{2}{3}}+\cos^{-1}\frac{1}{\sqrt{3}}$$

$$\pi-\tan^{-1}\frac{1}{\sqrt{2}}+\tan^{-1}\sqrt{2}=\frac{\pi}{2}+\cot^{-1}\frac{1}{\sqrt{2}}+\tan^{-1}\sqrt{2}=\frac{\pi}{2}+\tan^{-1}\sqrt{2}\ .]$$

150. $\sin^{-1}\left(x-\frac{x^2}{2}+\frac{x^3}{4}-........\right)+\cos^{-1}\left(x^2-\frac{x^4}{2}+\frac{x^6}{4}-........\right)=\frac{\pi}{2}$ for $0<|x|<\sqrt{2}$

$$S_1=\frac{x}{1+\frac{x}{2}}$$

$$=\frac{2x}{x+2} \qquad 0<|x|<\sqrt{2}$$

$$S_2 = \frac{\dfrac{x^2}{1+\dfrac{x^2}{2}} = \dfrac{2x^2}{2+x^2}}{\dfrac{2x}{x+2} = \dfrac{2x^2x}{2+x^2}}$$

$x^2 + 2x = 2 + x^2 \qquad \Rightarrow \qquad x = 1$

151. $\because$ $\sin^{-1}$ is defined for $[-1, 1]$

$\therefore$ $a = 0$

$\therefore$ $x + y = \sin^{-1} 1 + \cos^{-1} 1 - \tan^{-1} 1 = \dfrac{\pi}{4}$

Clearly image about x axis will be $x - y = \dfrac{\pi}{4}$.]

152. (A) $\cot^{-1}(\tan(-37°)) = \cot^{-1}(\cot(90+37°)) = \cot^{-1}(\cot(127°)) = 127° \quad \Rightarrow \quad$ (Q)

(B) $\cos^{-1}(\cos(233°)) = \cos^{-1}(\cos(360° - 127°)) = \cos^{-1}(\cos(127°)) = 127° \Rightarrow$ (Q)

(C) $\sin\dfrac{\theta}{2}$ where $\cos\theta = \dfrac{1}{9}$; $1 - 2\sin^2\dfrac{\theta}{2} = \dfrac{1}{9}$

$\therefore \qquad 2\sin^2\dfrac{\theta}{2} = 1 - \dfrac{1}{9} = \dfrac{8}{9} \Rightarrow \qquad \sin^2\dfrac{\theta}{2} = \dfrac{4}{9}$

$\Rightarrow \qquad \sin\dfrac{\theta}{2} = \dfrac{2}{3} \qquad \Rightarrow \qquad$ (S)

(D) $\cos\dfrac{\theta}{2}$ where $\cos\theta = \dfrac{1}{8}$

$2\cos^2\dfrac{\theta}{2} - 1 = \dfrac{1}{8} \qquad \Rightarrow \qquad 2\cos^2\dfrac{\theta}{2} = 1 + \dfrac{1}{8} = \dfrac{9}{8}$

$\Rightarrow \qquad \cos^2\dfrac{\theta}{2} = \dfrac{9}{16}$

$\cos\dfrac{\theta}{2} = \dfrac{3}{4} \qquad \Rightarrow \qquad$ (R)]

153. $\sin^{-1}\left(\dfrac{4x}{x^2+4}\right) = \sin^{-1}\dfrac{2\left(\dfrac{x}{2}\right)}{1+\left(\dfrac{x}{2}\right)^2}$

$$= \begin{cases} 2\tan^{-1}\left(\frac{x}{2}\right) & \text{if } \left|\frac{x}{2}\right| \le 1 \\ \pi - 2\tan^{-1}\left(\frac{x}{2}\right) & \text{if } \frac{x}{2} > 1 \\ -\left(\pi + 2\tan^{-1}\left(\frac{x}{2}\right)\right) & \text{if } \frac{x}{2} < -1 \end{cases} = \begin{cases} -2\tan^{-1}\left(\frac{-x}{2}\right) & \text{if } |x| \le 2 \\ \pi + 2\tan^{-1}\left(\frac{-x}{2}\right) & \text{if } x > 2 \\ -\pi + 2\tan^{-1}\left(\frac{-x}{2}\right) & \text{if } x < -2 \end{cases} \quad \{\text{Using } \tan^{-1}x = -\tan^{-1}(-x)\}$$

$$\therefore \sin^{-1}\left(\frac{4x}{x^2+4}\right) + 2\tan^{-1}\left(\frac{-x}{2}\right) = \begin{cases} 0 & \text{if } x \in [-2, 2] \\ \pi + 4\tan^{-1}\left(\frac{-x}{2}\right) & \text{if } x > 2 \\ -\pi + 4\tan^{-1}\left(\frac{-x}{2}\right) & \text{if } x < -2 \end{cases}$$

$\therefore$ Given expression is independent of x if $x \in [-2, 2]$ **Ans.**

154. $$\alpha = \lim_{r\to\infty} \sum_{n=1}^{r} \tan^{-1}\left\{\frac{16(2n+1)}{(2n+1)^4 - 4(2n+1)^2 + 16}\right\}$$

$$= \lim_{r\to\infty} \sum_{n=1}^{r} \tan^{-1}\left\{\frac{2\left(n+\frac{1}{2}\right)}{1+\left(n+\frac{1}{2}\right)^4 - \left(n+\frac{1}{2}\right)^2}\right\} \quad \{\text{Dividing numerator and denominator by } 16\}$$

$$= \lim_{r\to\infty} \sum_{n=1}^{r} \tan^{-1}\left\{\frac{\left\{\left(n+\frac{1}{2}\right)^2 + \left(n+\frac{1}{2}\right)\right\} - \left\{\left(n+\frac{1}{2}\right)^2 - \left(n+\frac{1}{2}\right)\right\}}{1+\left\{\left(n+\frac{1}{2}\right)^2 - \left(n+\frac{1}{2}\right)\right\}\left\{\left(n+\frac{1}{2}\right)^2 + \left(n+\frac{1}{2}\right)\right\}}\right\} \quad \{\text{Using } : n^4 - n^2 = (n^2 - n)(n^2 + n)\}$$

$$= \lim_{r\to\infty} \sum_{n=1}^{r} \left[\tan^{-1}\left\{\left(n+\frac{1}{2}\right)^2 + \left(n+\frac{1}{2}\right)\right\} - \tan^{-1}\left\{\left(n+\frac{1}{2}\right)^2 - \left(n+\frac{1}{2}\right)\right\}\right]$$

$$\{\text{Using} : \tan^{-1}\left(\frac{x-y}{1+xy}\right) = \tan^{-1}x - \tan^{-1}y \ \forall\ x, y > 0\}$$

$$= \lim_{r\to\infty} \sum_{n=1}^{r} \left[\left\{\tan^{-1}\left(\frac{9}{4}+\frac{3}{2}\right) - \tan^{-1}\left(\frac{9}{4}-\frac{3}{2}\right)\right\} + \left\{\tan^{-1}\left(\frac{25}{4}+\frac{5}{2}\right) - \tan^{-1}\left(\frac{25}{4}-\frac{5}{2}\right)\right\} + \left\{\tan^{-1}\left(\frac{49}{4}+\frac{7}{2}\right)\right.\right.$$

$$-\tan^{-1}\left(\frac{49}{4}-\frac{7}{2}\right)\Big\} \quad + ... + \left\{\tan^{-1}\left(r+\frac{1}{2}\right)^2+\left(r+\frac{1}{2}\right)\right\}-\tan^{-1}\left\{\left(r+\frac{1}{2}\right)^2-\left(r+\frac{1}{2}\right)\right\}\Bigg]$$

$$=\lim_{r\to\infty}\Bigg[\left\{\tan^{-1}\left(\frac{15}{4}\right)-\tan^{-1}\left(\frac{3}{4}\right)\right\}+\left\{\tan^{-1}\left(\frac{35}{4}\right)-\tan^{-1}\left(\frac{15}{4}\right)\right\}+\left\{\tan^{-1}\left(\frac{63}{4}\right)-\tan^{-1}\left(\frac{35}{4}\right)\right\}$$

$$+ ....+ \left\{\tan^{-1}\left(r+\frac{1}{2}\right)^2-\tan^{-1}\left(r+\frac{1}{2}\right)\right\}+\left\{\tan^{-1}\left(r+\frac{1}{2}\right)^2-\left(r+\frac{1}{2}\right)\right\}\Bigg]$$

$$=\lim_{r\to\infty}\left[\left\{\tan^{-1}\left(r+\frac{1}{2}\right)^2+\left(r+\frac{1}{2}\right)\right\}-\tan^{-1}\left(\frac{3}{4}\right)\right]=\frac{\pi}{2}-\tan^{-1}\left(\frac{3}{4}\right)=\cot^{-1}\left(\frac{3}{4}\right)$$

$\therefore \cot\alpha = \dfrac{3}{4}$ $\quad\therefore \operatorname{cosec}\alpha = \sqrt{1+\cot^2\alpha} = \dfrac{5}{4}$ $\quad\therefore \sin\alpha = \dfrac{4}{5} = \dfrac{a}{b}$

$\therefore (a^2 + b^2)_{least} = 4^2 + 5^2 = 16 + 25 = 41$ **Ans.**

155. Let $\sin^{-1}(x + 2) = \alpha \quad \Rightarrow \quad x + 2 = \sin\alpha$

$\therefore \quad 2\alpha = \cos^{-1}(x + 3)$

$\cos 2\alpha = x + 3 = (x + 2) + 1 = 1 + \sin\alpha$

$1 - 2\sin^2\alpha = 1 + \sin\alpha$

$\sin\alpha\,(1 + 2\sin\alpha) = 0$

$\Rightarrow \quad \sin\alpha = 0 \quad$ or $\quad \sin\alpha = -1/2$

$\therefore \quad x = -2 \quad$ or $\quad x = -2.5$ (rejected) as it does not satisfy the original equation

$\therefore \quad x^2 = 6.25$

156. $\beta = \cos^{-1}\left(\cos\left(\dfrac{\pi}{2}-\cos^{-1}x\right)\right)$; $\beta = \cos^{-1}\{\cos(\sin^{-1}x)\}$

also $\quad \alpha = \sin^{-1}[\cos(\sin^{-1}x)]$

$\alpha + \beta = \pi/2 \quad \Rightarrow \quad \tan\alpha = \cot\beta$ Ans.

157. for $\alpha > \beta > \gamma$, $\cot^{-1}x < 1 \quad \Rightarrow \quad$ x is greater than cot 1 ; **Ans.** $(\cot 1, \infty)$]

158. Given, $\dfrac{\sin^{-1}x^2+\cos^{-1}x}{\cos^{-1}x^2+\sin^{-1}x} = -3$

$\Rightarrow \quad \sin^{-1}x^2 + \cos^{-1}x = -3\cos^{-1}x^2 - 3\sin^{-1}x$

$\Rightarrow \quad \sin^{-1}x^2 + 3\cos^{-1}x^2 + \cos^{-1}x + 3\sin^{-1}x = 0$

$\Rightarrow \quad \pi + 2\cos^{-1}x^2 + 2\sin^{-1}x = 0$

$\Rightarrow \quad \cos^{-1}x^2 + \sin^{-1}x = \dfrac{-\pi}{2}$

Since, $\quad 0 \le \cos^{-1}x^2 \le \dfrac{\pi}{2}$ and $\dfrac{-\pi}{2} \le \sin^{-1}x \le \dfrac{\pi}{2}$

So, $\cos^{-1} x^2 = 0 \Rightarrow x = \pm 1$ and $\sin^{-1} x = \frac{-\pi}{2} \Rightarrow x = -1$.

$\therefore$ Only solution is $x = \alpha = -1$.

Hence, $(-1)^2 + 2(-1) + 3 = 2$. **Ans.**

159. We have $S = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \ldots\ldots$

$\Rightarrow S = 1$

(A) Given $\cos^{-1}(x^2 - 5) + 2\cos^{-1}(x^3 - x^2 - 4x + 3) = 3\pi$

$\Rightarrow \cos^{-1}(x^2 - 5) = \pi$ and $\cos^{-1}(x^3 - x^2 - 4x + 3) = \pi$

$\Rightarrow x^2 - 5 = -1$ and $x^3 - x^2 - 4x + 3 = -1$

$\Rightarrow x^2 = 4$ and $x^3 - x^2 - 4x + 4 = 0$

$\Rightarrow x = 2, -2$ and $(x - 1)(x - 2)(x + 2) = 0$

$\Rightarrow x = 1, 2, -2$

Finally $x = 2, -2$

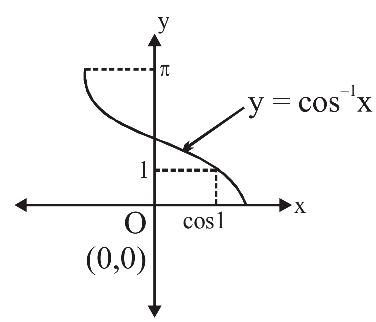
Hence sum $= 2 - 2 = 0$

(B) We have $2\tan\left(\cot^{-1}\left(\log_{10}\left(\sqrt{10}\right)^4\right)\right) = 2\tan(\cot^{-1}2)$

Let $\cot^{-1}2 = \theta \Rightarrow \cot\theta = 2 \Rightarrow \tan\theta = \frac{1}{2}$

Hence $2\tan(\cot^{-1}2) = 2\tan\theta = 2\left(\frac{1}{2}\right) = 1$

(C) We have $\log_3(\cos^{-1}Sx) \geq 0$



$\Rightarrow \log_3(\cos^{-1}x) \geq 0$

$\Rightarrow \cos^{-1}x \geq 1$

$\Rightarrow \cos^{-1}x \in [1, \pi]$

$\Rightarrow x \in [-1, \cos 1\ ]$

Number of integers in domain = 2 (–1 and 0)

(D) We have $\left[5\tan^{-1}\left(\cos\left(\tan^{-1}\{1\}\right)\right)\right] = \left[5\tan^{-1}\left(\cos\left(\tan^{-1} 0\right)\right)\right] = \left[5\left(\tan^{-1}\cos(0)\right)\right] = \left[5\left(\tan^{-1} 1\right)\right]$

$= \left[\frac{5\pi}{4}\right] = [3 \cdot 92] = 3\ ]$

160. $f(n) = \sum_{k=-n}^{n}\left(\cot^{-1}\frac{1}{k} - \tan^{-1}k\right)$

$$= \sum_{k=-n}^{-1}\left(\cot^{-1}\frac{1}{k} - \tan^{-1}k\right) + \underbrace{\sum_{k=1}^{n}\left(\cot^{-1}\frac{1}{k} - \tan^{-1}k\right)}_{\substack{\text{zero} \\ \text{As } \cot^{-1}\frac{1}{k} = \tan^{-1}k}} = \sum_{k=-n}^{-1}\tan^{-1}k + \pi - \tan^{-1}k = n\pi$$

$$\Rightarrow \sum_{n=2}^{10}\left(n\pi + (n-1)\pi\right) = \sum_{n=2}^{10}(2n-1)\pi = 99\pi. \textbf{ Ans.}$$

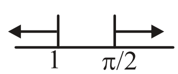
161. $\sin 2\theta = \frac{1}{4} \Rightarrow \sin\theta\cos\theta = \frac{1}{8}$

$$64(\sin\theta + \cos\theta) - 8\left(\frac{1}{\cos\theta} + \frac{1}{\sin\theta}\right) + \left(\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}\right)$$

$$= 64(\sin\theta + \cos\theta) - \frac{8(\sin\theta + \cos\theta)}{1/8} + \left(\frac{\sin^2\theta + \cos^2\theta}{1/8}\right) = 8. \textbf{ Ans.}$$

162. $y = \dfrac{\frac{\pi}{2} - \cos^{-1}x}{1 - \cos^{-1}x}$, $\cos^{-1}x = \dfrac{\pi - 2y}{2 - 2y}$

$0 < \cos^{-1}x < 1 \Rightarrow \quad 0 < \dfrac{\pi - 2y}{2 - 2y} < 1$

$\dfrac{\pi - 2y}{2 - 2y} > 0$  $\dfrac{\pi - 2y}{2 - 2y} - 1 < 0$

$2 - 2y < 0,\ 2y > 2$ $\quad \dfrac{\pi - 2}{2 - 2y} < 0$

$2 - 2y < 0,\ y > 1$

Taking interaction, $y \in \left(\frac{\pi}{2}, \infty\right)$.

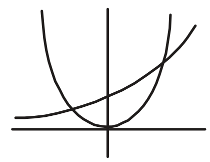
163. Here $x^2 + x + 1 \geq 1$ for $\tan^{-1}x > 1$
so $x^2 + x \geq 0 \Rightarrow x(x+1) \geq 0 \Rightarrow x \geq 0$ or $x \leq -1$
$\tan^{-1}x > 1 \Rightarrow x > \tan 1$ taking interaction.
$x \in (\tan 1, \infty)$
Now ... $x^2 + x + 1 \leq 1$ & $\tan^{-1}x < 1$ but $\tan^{-1}x > 0$
so $x^2 + x \leq 0 \Rightarrow x(x+1) \leq 0 \Rightarrow -1 \leq x \leq 0$
and $0 < x < \tan 1$ taking interaction Null set
so $x \in (\tan 1, \infty)$. **Ans.**

164. $\sin^{-1}x + \sin^{-1} y = A$
$-\sin^{-1}x + \sin^{-1} y = B$

$\sin^{-1} y = \dfrac{A+B}{2}$, $\quad \sin^{-1} x = \dfrac{A-B}{2}$

$x = \sin\left(\dfrac{A-B}{2}\right)$, $\quad y = \sin\left(\dfrac{A+B}{2}\right)$.

165.



$-1 \le e^x \le 1 \quad \Rightarrow \quad \infty < x \le 0$

$-1 \le x^2 \le 1 \quad \Rightarrow \quad -1 \le x \le 1 \quad$ so $-1 \le x \le 0$

$e^x = x^2$ one solution is positive (not acceptable) and one is negative.

166. $\cos 2A = \dfrac{1-\tan^2 A}{1+\tan^2 A} = \dfrac{1-\frac{1}{49}}{1+\frac{1}{49}} = \dfrac{48}{50} = \dfrac{24}{25}$

$\cos 2B = \dfrac{1-\tan^2 B}{1+\tan^2 B} = \dfrac{1-\frac{1}{9}}{1+\frac{1}{9}} = \dfrac{8}{10} = \dfrac{4}{5}$

$\therefore \quad \sin 2A = \dfrac{7}{25}$, $\sin 2B = \dfrac{3}{5}$

$\sin 4B = 2\cdot\dfrac{3}{5}\cdot\dfrac{4}{5} = \dfrac{24}{25}$]

167. $-1 \le 1-x \le 1 \quad \Rightarrow \quad 0 \le x \le 2$

and $-1 \le mx \le 1$ with $1-x = mx$ $\quad$ so $\quad x = \dfrac{1}{m+1}$

$-1 \le \dfrac{m}{m+1} \le 1 \quad$ so $\quad -1 \le \dfrac{m}{m+1}$, $\quad \dfrac{(2m+1)}{m+1} \ge 0$, $m \in (-\infty, -1) \cup -[\dfrac{1}{2}, \infty)$

$\dfrac{m}{m+1} - 1 \le 0, -\dfrac{1}{m+1} \le 0$, $\dfrac{1}{m+1} \ge 0 \quad \Rightarrow \quad m > -1$

$m \in [-\dfrac{1}{2}, \infty)$.

Questions

1. Let $\tan(2\pi\,|\sin\theta|) = \cot(2\pi\,|\cos\theta|)$, where $\theta \in R$ and $f(x) = (\,|\sin\theta| + |\cos\theta|\,)^x$.

Find The value of $\lim\limits_{x\to\infty}\left[\dfrac{2}{f(x)}\right]$. [Note: [ ] represents greatest integer function.]

2. If $\alpha = \sum\limits_{n=1}^{\infty}\tan^{-1}\left\{\dfrac{16(2n+1)}{(2n+1)^4 - 4(2n+1)^2 + 16}\right\}$ and $\sin\alpha = \dfrac{a}{b}$ where $a, b \in N$, then find the least value of $a^2 + b^2$.

3. If $\lim\limits_{n\to\infty}\dfrac{n\cdot 3^n}{n(x-2)^n + n\cdot 3^{n+1} - 3^n} = \dfrac{1}{3}$ then the range of x is $(n \in N)$

(A) $[2, 5)$ (B) $(1, 5)$ (C) $(-1, 5)$ (D) $(-\infty, 5)$

4. Through a point A on a circle, a chord AP is drawn & on the tangent at A a point T is taken such that AT = AP. If TP produced meet the diameter through A at Q, prove that the limiting value of AQ when P moves upto A is double the diameter of the circle.

5. Evaluate: $\lim\limits_{n\to\infty} n^2\left(n^2 \ln\left(\dfrac{n^2}{n^2+1}\right) + 1\right)$

6. At the end points A, B of the fixed segment of length L, lines are drawn meeting in C and making angles θ and 2θ respectively with the given segment. Let D be the foot of the altitude CD and let x represents the length of AD. Find the value of x as θ tends to zero i.e. $\lim\limits_{\theta\to 0} x$.

7. If $f : (0, \infty) \to N$ and

$$f(x) = \left[\frac{x^2+x+1}{x^2+1}\right] + \left[\frac{4x^2+x+2}{2x^2+1}\right] + \left[\frac{9x^2+x+3}{3x^2+1}\right] + \ldots\ldots + \left[\frac{n^2x^2+x+n}{nx^2+1}\right],\ n \in N$$

then find the value of $\lim\limits_{n\to\infty}\left(\dfrac{f(x) - n}{(f(x))^2 - \dfrac{n^3(n+2)}{4}}\right)$.

[**Note:** [y] denotes the greatest integer less than or equal to y.]

8. For $p, n \in N$, let $f(x) = 1 - x^p$ and $g_n(x) = \dfrac{n}{\dfrac{1}{f(x)} + \dfrac{1}{f(2x)} + \ldots\ldots + \dfrac{1}{f(nx)}}$.

Find the value of $\lim\limits_{x\to 0}\dfrac{1 - g_n(x)}{x^p}$ at $n = 5$ and $p = 3$.

9. Let $f(\theta) = \dfrac{1}{\tan^9\theta}\left((1+\tan\theta)^{10} + (2+\tan\theta)^{10} + \ldots + (20+\tan\theta)^{10}\right) - 20\tan\theta$. The left hand limit of $f(\theta)$ as $\theta \to \dfrac{\pi}{2}$ is

(A) 1900 (B) 2000 (C) 2100 (D) 2200

10. Find the value of $\displaystyle\lim_{n\to\infty}\sum_{k=0}^{n}\frac{(k+1)(k+2)}{2^k}$.

11. Let $S=\frac{1}{2}\tan\frac{\pi}{4}+\frac{1}{2^2}\tan\frac{\pi}{8}+\frac{1}{2^3}\tan\frac{\pi}{16}+.......+\frac{1}{2^n}\tan\left(\frac{\pi}{2^{n+1}}\right)$. If $\lim\limits_{n\to\infty}S=L$; then find the value of $(100\pi)L$.
(Use may use the fact $\cot x-\tan x=2\cot 2x$.)

12. If $f(x)=\lim\limits_{n\to\infty}\dfrac{x^n\left(a+\sin(x^n)\right)+\left(b-\sin(x^n)\right)}{(1+x^n)\sec\left(\tan^{-1}(x^n+x^{-n})\right)}$ is continuous at $x=1$, then

(A) $a-b=2\sqrt{5}$ (B) $a+b=2\sqrt{5}$ (C) $a-b=0$ (D) $a+b=0$

13. If $\lim\limits_{x\to0}\dfrac{(1-\cos^n x)(\tan x-\sin x)^m}{x^n}=1$, $(m, n\in N)$ then find the minimum value of $(m+n)$.

14. Let $f: R\to R$ be $f(x)=\begin{cases}x^3-2x, & \text{if } x\in Q\\ x^2-2, & \text{if } x\in R-Q\end{cases}$

Find the sum of all possible values of $\alpha\in R$ for which $\lim\limits_{x\to\alpha}f(x)$ exists.

15. If $\lim\limits_{x\to0}\dfrac{1-\cos(1-\cos x)}{x^4}$ is equal to $\dfrac{m}{n}$, where m and n are relatively prime positive integers, then the sum of the digits of (m^2+n^2) equals

(A) 10 (B) 11 (C) 12 (D) 13

16. Let L(m) be the x-coordinate of the left end point of the intersection of the graphs of $y=x^2-6$ and $y=m$, where $-6<m<6$. The value of $\lim\limits_{m\to0}\left(\dfrac{L(-m)-L(m)}{m}\right)$ equals

(A) zero (B) $\dfrac{1}{\sqrt{6}}$ (C) $\dfrac{2}{\sqrt{6}}$ (D) 1

17. Let α be the fundamental period of the function $f(x)=|\cos(2\pi\{2x\})|+|\sin(2\pi\{2x\}|$ (where $\{\cdot\}$ denotes fractional part of x) and $g(x)=\lim\limits_{n\to\infty}\dfrac{\cos\pi x-x^{2n}\sin(x-2)}{1+x^{2n+1}}$. If $\lim\limits_{x\to2\alpha}g(x)=\dfrac{m}{\sqrt{n}}$ where $m, n\in N$ then find the minimum value of $(m+n)$.

18. Let the isosceles triangle T_1 be inscribed a circle C. Now again an isosceles triangle T_2 is inscribed in the circle with base as one of the equal sides of T_1. This process is continued indefinitely. If θ_i is vertical angle of triangle T_i then $\left[\lim\limits_{i\to\infty}\theta_i\right]$, is

[**Note:** [y] denotes the largest integer less than or equal to y.]

19. Let a function $f(x)$ be defined as $f(x) = \begin{cases} \dfrac{\ln(2-\cos 2x)}{\ln^2(1+\sin 3x)} & \text{for } x<0 \\ \\ \dfrac{e^{\sin 2x}-1}{\ln(1+\tan 9x)} & \text{for } x>0 \end{cases}$.

Find the value of $9\,[f(0^+)+f(0^-)]$.

20. Let θ_k (k = 1, 2, 3, ... , n), $\theta_1 < \theta_2 < ..... < \theta_n$ be the solution of θ such that $\sqrt{3}\sin n\theta + \cos n\theta = 0$. If $\lim\limits_{n\to\infty} \dfrac{1}{n}\sum\limits_{k=1}^{n} \cos\dfrac{\theta_k}{2} = \dfrac{k}{\pi}$, then find the value of k ($k \in N$).

Paragraph for Question no. 21 to 23

Let $P(x) = x^5 - 9x^4 + px^3 - 27x^2 + qx + r$ ($p, q, r \in R$) be divisible by x^2 and α, β and γ are the positive roots of the equation $\dfrac{P(x)}{x^2} = 0$.

21. The value of $(p+q+r)$ is equal to

(A) 9 (B) 27 (C) 81 (D) 108

22. If $\alpha - 1$, $\beta + 3$ and $\gamma + 7$ are the first three terms of a sequence whose sum of first n terms is given by S_n then $\sum\limits_{n=2}^{\infty} \dfrac{1}{\sqrt{S_n \cdot S_{n-1}}}$ is equal to

(A) 1 (B) $\dfrac{1}{4}$ (C) $\dfrac{1}{2}$ (D) 2

23. The value of $\lim\limits_{n\to\infty}\left(\dfrac{1}{p+q+r} + \dfrac{1}{p^2+q^2+r^2} + \dfrac{1}{p^3+q^3+r^3} + ... + \dfrac{1}{p^n+q^n+r^n}\right)$ is equal to

(A) $\dfrac{1}{26}$ (B) $\dfrac{1}{27}$ (C) $\dfrac{25}{26}$ (D) $\dfrac{26}{27}$

24. The value of $\lim\limits_{x\to\pi} \dfrac{2^{\cot x} + 3^{\cot x} - 5^{1+\cot x} + 2}{(2^{\cot x})^2 + (9^{\cot x})^{1/2} - 5^{\cot x} + 1}$ is

(A) 5 (B) 2 (C) non existent (D) – 2

25. The value of $\lim\limits_{x\to 0}\left(\dfrac{\ln(1+x^3\tan^2 x)}{e^{x^3} - e^{\sin^3 x}}\right)$ is equal to

(A) $\dfrac{1}{2}$ (B) $\dfrac{1}{6}$ (C) 2 (D) 6

26. $\lim\limits_{x\to 0}\left(\dfrac{e}{4x}-\dfrac{e}{2x\left(e^{ex}+1\right)}\right)$ equals

(A) $\dfrac{e^2}{2}$ (B) $\dfrac{e^2}{4}$ (C) $\dfrac{e^2}{8}$ (D) none of these

27. $\lim\limits_{x\to 1}\left(\dfrac{1}{\ln x}-\dfrac{1}{x-1}\right)$ equals

(A) 0 (B) $\dfrac{1}{3}$ (C) $\dfrac{1}{2}$ (D) 1

28. Let $f: R^+ \to A$ (where A is co-domain of a non empty set) be a function defined as $f(x) = \lim\limits_{a\to\infty}\dfrac{x^{2a}-3x^a+2}{x^{2a}+x^a+1}$. Find the number of elements in A for which f is surjective.

29. The value of $\lim\limits_{x\to 0^+}\left(\dfrac{\cot^{-1}(\ln\{x\})}{\pi}\right)^{-\ln\{x\}}$ is equal to

[**Note:** where { } denotes fractional part function.]

(A) $e^{\frac{1}{\pi}}$ (B) $e^{\frac{-1}{\pi}}$ (C) $e^{\frac{-2}{\pi}}$ (D) $e^{\frac{2}{\pi}}$

30. $\lim\limits_{x\to 0}\dfrac{e^{\sin^{-1}\frac{x}{\sqrt{x^2+1}}}-e^{\sin^{-1}x}}{e^{\tan x}-e^{\sin x}}$ is equal to

(A) 0 (B) 1 (C) – 1 (D) none

31. $\lim\limits_{x\to 0}\dfrac{e^{x^2}-\sqrt{\cos 2x}}{\ln(\sec x)}$ equals

(A) 2 (B) 3 (C) 4 (D) 5

32. Let $k \in N$ and $a \in R^+$ $(a \neq 1)$ then $\lim\limits_{n\to\infty} n^k\left(a^{\frac{1}{n}}-1\right)\left(\sqrt{\dfrac{n-1}{n}}-\sqrt{\dfrac{n+1}{n+2}}\right)$ is

(A) 0 if $k \in \{1, 2\}$
(B) $-\ln a$ if $k = 3$
(C) non-existent if $k \geq 4$ and $a \in (0, 1)$
(D) non-existent if $k \geq 4$ and $a > 1$.

33. If the value of $\lim\limits_{x\to 0}\dfrac{(1+3x+2x^2)^{\frac{1}{x}}-(1+3x-2x^2)^{\frac{1}{x}}}{x} = k\,e^3$, find the value of 12k

34. $\underset{x\to 0}{\text{Limit}}\ \dfrac{(1+x)^{\frac{1}{x}}+e(x-1)}{\sin^{-1}x}$ is equal to

(A) $\dfrac{-e}{2}$ (B) $\dfrac{e}{2}$ (C) e (D) – e

Paragraph for question nos. 35 to 37

Consider $P(x)=ax^2+bx+c$, where $a, b, c \in R$ and $P(2)=9$. Let α and β be the roots of the equation $P(x)=0$.

35. If $\alpha\to\infty$ and $P'(3)=5$ then $\underset{x\to\infty}{\text{Lim}}\left(\dfrac{P(x)}{5(x-1)}\right)^x$ is equal to

(A) 1 (B) $e^{\frac{1}{5}}$ (C) $e^{\frac{4}{5}}$ (D) $e^{\frac{2}{5}}$

36. If α and β both tends to infinity then $\underset{x\to 3}{\text{Lim}}\ \dfrac{\sqrt{P(x)}-3}{\sin(x-3)}$ is equal to

(A) 0 (B) 1 (C) 9 (D) non-existent

37. If $\alpha=\beta$, then the value of $\underset{\alpha\to 0}{\text{Lim}}\left(\underset{x\to\alpha}{\text{Lim}}\ \dfrac{P(x)}{\left[1-\tan^2\left(\dfrac{\pi}{4}-x+\alpha\right)\right]\left(e^{x-\alpha}-1\right)}\right)$ is equal to

(A) $\dfrac{3}{16}$ (B) $\dfrac{9}{16}$ (C) $\dfrac{5}{4e^2}$ (D) $\dfrac{e^2}{4}$

38. Let $\displaystyle\sum_{r=1}^{n}\frac{r^4}{(2r-1)(2r+1)}=\frac{n^3}{A}+\frac{n^2}{B}+\frac{5n}{C}+\frac{f(n)}{D}$ (A, B, C, D,$\in$ N)

where $f(n)$ is the ratio of two linear polynomials such that $\underset{n\to\infty}{\text{Lim}}\ f(n)=\dfrac{1}{2}$

Find the value of (A+B+C+D).

39. Let $f(x)=\begin{cases}\underset{n\to\infty}{\text{Lim}}\displaystyle\prod_{k=1}^{n}\left(1-\tan^4\frac{x}{2^k}\right), & x\neq 0\\ 1 & x=0\end{cases}$.

The value of $f'(0)$ is

(A) 1 (B) – 1 (C) 0 (D) does not exist

40. At the end-points and the midpoint of a circular arc AB tangent lines are drawn, and the points A and B are joined with a chord. Prove that the ratio of the areas of the two triangles thus formed tends to 4 as the arc AB decreases indefinitely.

41. Given a right triangle ABC which is right angled at A with b < c. If h_a, w_a and m_a are its altitude bisector and median from the vertex A respectively, then find the value of $\lim\limits_{b\to c}\dfrac{m_a - h_a}{w_a - h_a}$.

42. Let $f:[0,\infty)\to R$ be a function satisfying $\sqrt{f(x)}\ e^{f(x)} = x^2\ \forall x \in [0,\infty)$. The value of $\lim\limits_{x\to\infty}\dfrac{\ln x}{f(x)}$ is

(A) 1 (B) $\dfrac{1}{2}$ (C) 2 (D) non-existent

43. The value of $\lim\limits_{n\to\infty}\left(\dfrac{n^2}{n-1}\right)^{\tan\frac{1}{\sqrt{n}}}$ is

(A) 0 (B) 1 (C) e (D) $\dfrac{1}{e}$

44. Evaluate: $\lim\limits_{x\to 1}\dfrac{1-x+\ln x}{1+\cos\pi x}$

45. Evaluate: $\lim\limits_{x\to\frac{\pi}{2}}\dfrac{2^{-\cos x}-1}{x\left(x-\frac{\pi}{2}\right)}$

46. Evaluate: $\lim\limits_{x\to 3}\dfrac{(x^3+27)\ln(x-2)}{x^2-9}$

47. Evaluate: $\lim\limits_{x\to 1}\dfrac{x^2-x\cdot\ln x+\ln x-1}{x-1}$

48. Evaluate: $\lim\limits_{x\to 0}\dfrac{\sin(\pi\cos^2 x)}{x^2}$

49. Evaluate $\lim\limits_{n\to\infty} n^2\left(\sqrt[n]{a}-\sqrt[n+1]{a}\right)\left(a>0,\ n\in N\right)$.

50. If $a_n = 1+\dfrac{1}{2^2}+\dfrac{1}{3^2}+\dfrac{1}{4^2}+.......\infty=\dfrac{\pi^2}{6}$ then $b_n = 1+\dfrac{1}{3^2}+\dfrac{1}{5^2}+\dfrac{1}{7^2}+.......\infty$ has the value equal to

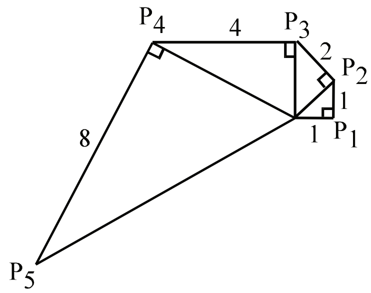
(A) $\dfrac{\pi^2}{8}$ (B) $\dfrac{\pi^2}{12}$ (C) $\dfrac{\pi^2}{16}$ (D) $\dfrac{\pi^2}{4}$

51. If $\lim\limits_{x\to 1}\left(\dfrac{\sin(n\cos^{-1}x)}{\sqrt{1-x^2}}+\dfrac{1-\cos(n\cos^{-1}x)}{1-x^2}\right)$ (where $n \in N$) exist and is equal to $\dfrac{3}{2}$, then find the sum of all possible values of n.

52. Let the equations $x^3+2x^2+px+q=0$ and $x^3+x^2+px+r=0$ have two roots in common and the third root of each equation are represented by α and β respectively.

If $f(x)=\begin{cases} e^{x\log_{1+x}|\alpha+\beta|}, & -1<x<0 \\ a, & x=0 \\ b\dfrac{ln\left(e^{x^2}+\alpha\beta\sqrt{x}\right)}{\tan\sqrt{x}}, & 0<x<1 \end{cases}$ is continuous at $x=0$, then find the value of $2(a+b)$.

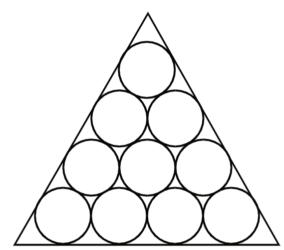
53. Let $\{P_n\}$ be a sequence of points determined as in the figure.



Thus $|AP_1| = 1$, $|P_nP_{n+1}| = 2^{n-1}$, and angle AP_nP_{n+1} is a right angle. $\lim\limits_{n\to\infty} \angle P_nAP_{n+1}$ equals

(A) $\dfrac{\pi}{6}$ (B) $\dfrac{\pi}{4}$ (C) $\dfrac{\pi}{3}$ (D) $\dfrac{5\pi}{12}$

54. Suppose that circle of equal diameter are packed tightly in *n* rows inside an equilateral triangle. (The figure illustrates the case n=4.) If A is the area of the triangle and A_n is the total area occupied by the circles in *n* rows then $\lim\limits_{n\to\infty}\dfrac{A_n}{A}$ equals



(A) $\dfrac{\pi}{\sqrt{3}}$ (B) $\dfrac{2\pi}{\sqrt{3}}$ (C) $\dfrac{\pi}{2\sqrt{3}}$ (D) $\dfrac{\sqrt{3}\pi}{2}$

55. Let $L=\lim\limits_{n\to\infty} n\left(\sqrt[3]{n^3+3n^2+2n+1}+\sqrt{n^2-2n+3}-2n\right)$. If L can be expressed in the form of $\dfrac{p}{q}$ $(p, q \in N)$ in the lowest form then find the value of $(p+q)$.

56. Two circles have centres which are 'd' units apart and each has diameter $\sqrt{d}$. For any 'd' let A(d) be the area of the smallest circle that just contains both these circles. $\lim\limits_{d\to\infty}\dfrac{A(d)}{d^2}$ equals

(A) $\dfrac{\pi}{8}$ (B) $\dfrac{\sqrt{2}\pi}{4}$ (C) $\dfrac{\pi}{4}$ (D) $\dfrac{\pi}{2}$

57. Let $f(x)=x^2+ax+3$ and $g(x)=x+b$, where $F(x)=\lim\limits_{n\to\infty}\dfrac{f(x)+x^{2n}g(x)}{1+x^{2n}}$.

If F(x) is continuous at x = 1 and x = – 1 then find the value of (a^2+b^2).

58. $\lim\limits_{n\to\infty}\left(1+\frac{1}{n^2}\right)^{\frac{8n^3}{\pi}\sin\left(\frac{\pi}{2n}\right)}$ is equal to

(A) 4 (B) e^4 (C) 1 (D) ∞

59. P(x) is a polynomial such that $P(x) + P(2x) = 5x^2 - 18$. Then $\lim\limits_{x\to3}\left(\frac{P(x)}{x-3}\right) =$

(A) 6 (B) 9 (C) 18 (D) 0

60. Let $f(x)$, $g(x)$ are 2 function defined from $[0, 1] \to [0, \infty)$ be continuous functions such that maximum value of $f(x)$ is equal to maximum value of $g(x)$ in $[0, 1]$. Prove that $\exists$ some $c \in [0, 1]$ such $f^2(c) + 3f(c) = g^2(c) + 3g(c)$.

61. Find the value of $\lim\limits_{x\to0}\frac{\left(\left(\sin^{-1}x\right)^{2010} - x^{2010}\right)\cos^{2010}x}{ln\left(1+x^{2010}\right)\left(1+\sin^{2010}x\right)\ \tan^2 x}$.

62. If $\lim\limits_{n\to\infty} 2^n\left(\sum_{k=1}^{n}\frac{1}{k(k+2)} - \frac{1}{4}\right)^n = e^{-\lambda}$ $(\lambda \in N)$, then find the value λ.

63. Find the value of $\lim\limits_{x\to\sin^{-1}\sqrt{\frac{7}{8}}}\left(\frac{\tan^2 x - 7}{8\sin^2 x - 7}\right)$.

64. If $f(x) = \lim\limits_{n\to\infty} x\left(\frac{3}{2}+[\cos x]\left(\sqrt{n^2+1}-\sqrt{n^2-3n+1}\right)\right)$ where [y] denotes largest integer $\le$ y, then find (a) $\lim\limits_{x\to0} f(x)$, (b) $\lim\limits_{x\to\frac{\pi}{2}} f(x)$.

65. Evaluate : $\lim\limits_{x\to0}\left[\frac{\ell n(1+x)^{1+x}}{x^2} - \frac{1}{x}\right]$

66. Evaluate : $\lim\limits_{x\to0}\frac{2(\tan x - \sin x) - x^3}{x^5}$

67. If $\alpha = \lim\limits_{x\to0}\cos^{-1}\left(\frac{\{-x^2\}}{x^2+2x+2}\right)$ and $\beta = \lim\limits_{x\to\infty}\tan^{-1}\left(\frac{e^{-x^2}-e^x}{2e^{-x^2}+e^x}\right)$, then $(\cos\alpha + \tan\beta)$, equals

(A) $\frac{-1}{2}$ (B) $\frac{1}{2}$ (C) 1 (D) does not exist

[Note : {y} denotes the fractional part function of y.]

68. Let $a_r = \frac{n^2+n+r}{r}$, where $n \in N$ and r = 1, 2, 3, ..., n. If H_n is the harmonic mean of a_1, a_2, ...,a_n then find the value of $\lim\limits_{n \to \infty} \frac{H_n}{n}$.

69. If the value limit, $\lim\limits_{x\to 0} \frac{(1+x)^{1/x} - e + \frac{ex}{2}}{x^2} = \frac{Ae}{B}$ where A and B are coprime, then find the value of (A + B).

70. Let $< a_m >$ be the m^{th} term of the sequence, $\left(\frac{2^2}{1^2} - \frac{2}{1}\right)^{-1}, \left(\frac{3^3}{2^3} - \frac{3}{2}\right)^{-2}, \left(\frac{4^4}{3^4} - \frac{4}{3}\right)^{-3}$,.......

The value of $\lim\limits_{m\to\infty} (a_m)^{1/m}$ is equals to

(A) e + 1 (B) e – 1 (C) $\frac{1}{e-1}$ (D) $\frac{1}{e+1}$

71. If $\lim\limits_{n\to\infty} \left(\left((2009)^{2010}\right)^n + \left((2010)^{2009}\right)^n\right)^{1/n}$ is equal to a^b where $a, b \in N$ then a – b is equal to

(A) 2009 (B) 1 (C) – 1 (D) 0

72. If $\lim\limits_{x\to 0} \frac{p\cos x + xe^{\frac{1}{x}}}{1 + \sin x + q\cos x \cdot e^{\frac{1}{x}}} = 0$, then which of the following is/are incorrect about p, q?

(A) p = 0 $q \in R$ (B) p = 4 q = 2 (C) p = 2 $q \in R$ (D) p = 0 q = 2

73. Let $L = \lim\limits_{x\to 0} \frac{x^4 + 3\left(a^2 - \sqrt{a^4 + x^4}\right)}{x^8}$, $a > 0$. If L is finite, then

(A) $a = \frac{3}{2}$ (B) $a = \sqrt{\frac{3}{2}}$ (C) $L = \frac{1}{3}$ (D) $L = \frac{1}{9}$

74. Let x_n is defined by $\left(1+\frac{1}{n}\right)^{n+x_n} = e$, then the value of $\lim\limits_{n\to\infty} x_n$, is

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) 2

75. The sequence $\{a_n\}_{n=1}^{+\infty}$ is defined by $a_1 = 0$ and $a_{n+1} = a_n + 4n + 3$, $n \geq 1$.

Find the value of $\lim\limits_{n\to\infty} \frac{\sqrt{a_n} + \sqrt{a_{4n}} + \sqrt{a_{4^2 n}} + \sqrt{a_{4^3 n}} + ...+ \sqrt{a_{4^{10} n}}}{\sqrt{a_n} + \sqrt{a_{2n}} + \sqrt{a_{2^2 n}} + \sqrt{a_{2^3 n}} + ...+ + \sqrt{a_{2^{10} n}}}$.

76. $L = \lim\limits_{x\to 0} \dfrac{\sqrt{\dfrac{\cos 2x + (1+3x)^{1/3}}{2}} - \sqrt[3]{\dfrac{4\cos^3 x - ln(1+x)^4}{4}}}{x}$

If $L = a/b$ where 'a' and 'b' are relatively primes find $(a+b)$.

77. Let $f(\theta) = 3\sin\theta + 2\cos\theta - 4\left((\sin\theta+\cos\theta)(1-\sin\theta\cos\theta) - \sin^2\theta\cos\theta\right)\ \forall\ \theta \in R$.
Also a and b are respectively the minimum and maximum values of $f(\theta)$.

If $l_1 = \lim\limits_{x\to[a]^-} \dfrac{x^2 - x[x]}{2x-[x]}$ and $l_2 = \lim\limits_{x\to[b]^+} \dfrac{x^2 - x[x]}{2x-[x]}$ where [y] denotes largest integer $\le y$, then find the value of $(6l_1 - l_2)$.

78. If $\lim\limits_{n\to\infty} n a_n = 0$ and $L = \lim\limits_{n\to\infty}\left(1+\dfrac{1}{n}+a_n\right)^n$ then $[L]$ has the value equal to

(A) 0 (B) 1 (C) 2 (D) None

Where [y] denotes greatest integer less than or equal to y.

79. $\lim\limits_{n\to\infty} \dfrac{e^n}{\left(1+\dfrac{1}{n}\right)^{n^2}}$ equals

(A) 1 (B) $\dfrac{1}{2}$ (C) e (D) $\sqrt{e}$

80. If $\lim\limits_{n\to\infty}\left(\dfrac{\left(1+\dfrac{1}{n}\right)^n}{\left(1-\dfrac{1}{n}\right)^n} - e^2\right)n^2 = \dfrac{ae^2}{b}$. Find the minimum value of $(a+b)$.

81. If x is measured in radians and $\lim\limits_{x\to\infty}\left(\sqrt{Ax^2+Bx} - Cx\right) = 2$, then the value of $\dfrac{BC}{A}$ equals $(A,B,C\in R)$

(A) 4 (B) 2 (C) $\dfrac{1}{2}$ (D) none

82. $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\left(\dfrac{4r^3}{n^4+4n^3+2n^2+n+r}\right)$ equals

(A) 1 (B) $\dfrac{1}{2}$ (C) $\dfrac{1}{4}$ (D) $\dfrac{1}{3}$

83. Evaluate: $\lim\limits_{x\to 0}\dfrac{x\sin(\sin x) - \sin^2 x}{x^6}$

84. $\lim\limits_{x\to 0}\left(\frac{2}{x^3}\left(\sin^{-1}x-\tan^{-1}x\right)\right)^{\frac{2}{x^2}}=$

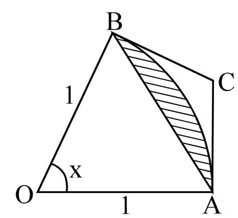
(A) e (B) $\sqrt{e}$ (C) $\frac{1}{e}$ (D) $\frac{1}{\sqrt{e}}$

85. $\lim\limits_{n\to\infty} n^2\left(n^2 \ln\left(\frac{n^2}{n^2+1}\right)+1\right)$ is equal to

(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) ∞

86. Let $f(x)=ax+b$ where a, b are real numbers. Define $f^1(x)=f(x)$ and $f^{n+1}(x)=f\left(f^n(x)\right)$ for every positive integers n. If $f^7(x)=128x+381$, find the value of $(a+b)$.

87. A circular arc of radius 1 subtends an angle of x radians, $0<x<\frac{\pi}{2}$ as shown in the figure. The point C is the intersection of the two tangent lines at A & B. Let T(x) be the area of triangle ABC & let S(x) be the area of the shaded region. Compute:



(a) T(x)(b) S(x) & (c) the limit of $\frac{T(x)}{S(x)}$ as $x\to 0$.

88. If $p<0<q$ and $p, q\in I$ such that

$$\lim_{n\to\infty}\lim_{x\to 0}\frac{(1+x)^{\frac{1}{x}}+(1+2x)^{\frac{1}{2x}}+(1+3x)^{\frac{1}{3x}}+\dots\dots+(1+nx)^{\frac{1}{nx}}-ne}{n^2x}$$ has the value equal to $\left(\frac{p}{q}\right)e$

then find the least value of $(p+q)$.

89. Let a, b be constants such that $\lim\limits_{x\to 1}\frac{x^2+ax+b}{(\ln(2-x))^2}$ exist and have the value equal to l. Find the value of $(a+b+l)$.

90. Consider the sequence $a_n=\left(1+\frac{1}{2}\sum\limits_{k=1}^{n}\frac{1}{(2k-1)(2k+1)}\right)^n$

(a) Find $\lim\limits_{n\to\infty} a_n$. (b) Find $\lim\limits_{n\to\infty}\frac{a_{n+1}}{a_n}$.

91. Find $\lim\limits_{n\to\infty}\left(\frac{1}{\sqrt{n^2+1}}+\frac{1}{\sqrt{n^2+2}}+\dots\dots+\frac{1}{\sqrt{n^2+n}}\right)^n$.

92. If $\lim\limits_{x\to 0}(1+ax+bx^2)^{2/x}=e^3$, then find a and b.

93. Let $\lim_{x\to0}\left(1+\frac{P(x)}{x^4}\right)^{\frac{1}{x-\sin(x)}}$ exists and is equal to e^8. Find P(x), where P(x) is least degree polynomial.

94. If $\lim_{x\to0}\frac{a\sin x-\sin 2x}{\tan^3 x}$ is finite then find the value of 'a' & the limit.

Paragraph for question nos. 95 to 97

Let $f(x) = \lim_{n\to\infty}\left(\cos\sqrt{\frac{x}{n}}\right)^n$, $g(x) = \lim_{n\to\infty}\left(1-x+x\sqrt[n]{e}\right)^n$. Now, consider the function $y=h(x)$, where $h(x)=\tan^{-1}(g^{-1}f^{-1}(x))$.

95. $\lim_{x\to0}\frac{\ln(f(x))}{\ln(g(x))}$ is equal to

(A) 1/2 (B) – 1/2 (C) 0 (D) 1

96. Domain of the function $y=h(x)$ is

(A) $(0,\infty)$ (B) R (C) (0, 1) (D) [0, 1]

97. Range of the function $y=h(x)$ is :

(A) $\left(0,\frac{\pi}{2}\right)$ (B) $\left(-\frac{\pi}{2},0\right)$ (C) R (D) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

98. If $\lim_{x\to\infty}\left(\frac{x+A}{x-2A}\right)^x=5$, then A equals

(A) $\ln 5$ (B) $\frac{1}{3}\ln 5$ (C) 5 (D) $\frac{1}{3}e^5$

99. If $\lim_{x\to0}\frac{\sin(ax)+bx}{x^3}=36$, then

(A) a = 6, b = – 6 (B) a = – 6, b = 6 (C) a = 6, b = 6 (D) a = – 6, b = – 6

100. If the value of $\lim_{n\to\infty}\left(\frac{n^2+n+3}{n^2+3n+5}\right)^n$ is equal to $\frac{p}{e^2}$, then the value of p is

(A) 4 (B) 3 (C) 2 (D) 1

101. $\lim\limits_{x\to-\infty}\dfrac{x^2\tan\left(\dfrac{1}{x}\right)-x}{1-|x|}$ equals

(A) 0 (B) – 1 (C) 1 (D) does not exist

102. $\lim\limits_{n\to\infty}(n)\cdot \ln\left(\tan\left(\dfrac{\pi}{4}+\dfrac{\pi}{n}\right)\right)$ equals

(A) 4π (B) 2 (C) 2π (D) 4

103. If $\lim\limits_{x\to0}\dfrac{x+\sin x-x\cos x-\tan x}{x^n}$ exists and is non-zero finite value, then the value of n is

(A) 3 (B) 4 (C) 5 (D) 6

104. $\lim\limits_{x\to0}\dfrac{5(1-\cos x+4\sin^{-1}x-\sin^2 x)}{(10\tan^{-1}x-x^2+x^4)}$ equals

(A) 0 (B) 1 (C) 2 (D) 3

105. If $\alpha=\lim\limits_{n\to\infty}\left(\dfrac{1}{n^3+1}+\dfrac{4}{n^3+1}+\dfrac{9}{n^3+1}+......+\dfrac{n^2}{n^3+1}\right)$ and $\beta=\lim\limits_{x\to0}\dfrac{\sin 2x}{\sin 8x}$, then the quadratic equation whose roots are α,β is

(A) $12x^2-7x+1=0$ (B) $x^2+19x-120=0$

(C) $x^2-17x+66=0$ (D) $x^2-7x+12=0$

106. $\lim\limits_{x\to I^-}\dfrac{e^{\{x\}}-\{x\}-1}{\{x\}^2}$ equals, where { } denotes the fractional part function and I is an integer.

(A) 1/2 (B) e – 2 (C) 2 – e (D) 1

Paragraph for question nos. 107 & 108

Consider the function $f(x)=\min\left(1,x^{2n},x^{2n+1}\right), n\in N$.

107. The natural number n, for which $\lim\limits_{x\to0}\dfrac{\left(27^x-9^x-3^x+1\right)\left(\cos x-e^x\right)}{f(x)}$ is a finite non zero number, is equal to

(A) 1 (B) 2 (C) 3 (D) 4

108. $\lim\limits_{x\to0}\dfrac{e^{\tan(f(x))}-e^{\sin(f(x))}}{\tan(f(x))-\sin(f(x))}$, is equal to

(A) 0 (B) 1 (C) 2 (D) does not exist.

109. Let $l_1=\lim\limits_{x\to0}\dfrac{\ln(\cos 3x)}{2x^2}$; $l_2=\lim\limits_{x\to0}\dfrac{\sin^2 3x}{x(1-e^x)}$; $l_3=\lim\limits_{x\to1}\dfrac{\sqrt{x}-x}{\ln x}$, then find the value of $(2l_1-l_2+l_3)$.

110. Let $a_1 = 1$ and $a_n = \sin(a_{n-1})$ $\quad n > 1,\ n \in N$

If $\lim\limits_{n\to\infty} \dfrac{2^{2a_n} - 2^{1+a_n}\cdot 3^{a_n} + 3^{2a_n}}{\cos a_n + 1 - e^{a_n} - e^{-a_n}} = -a\, ln^2 a$ then, find the value of 3a.

111. If $f(x) = \begin{cases} ln\,\text{cosec}(x\pi) & 0 < x < 1 \\ ln\sin(2x\pi) & 1 < x < \dfrac{3}{2} \end{cases}$ and $g(x) = \dfrac{2^{f(x)}+1}{3^{f(x)}+1}$ then find $\tan^{-1}(g(1^-))$ and $\sec^{-1}(g(1^+))$.

112. Let $f(x) = x - [x]$, then the value of $\lim\limits_{n\to\infty} f\left(\sqrt{n^2+n+1}\right)$ $n \in I$, equals $\dfrac{p}{q}$ (where p & q are coprime).
The value of (p + q) is equal to
[**Note:** [k] denotes greatest integer function less than or equal to k.]

(A) 1 (B) 2 (C) 3 (D) 4

113. Let $L = \lim\limits_{x\to 0} \dfrac{e^{x\sin(2007)x} - 1}{x\, ln(1+x)}$ and $M = \lim\limits_{x\to\infty} x\left(2x - \left(\sqrt[3]{x^3+x^2+1} + \sqrt[3]{x^3-x^2+1}\right)\right)$
then find the value of LM.

114. If *x* is measured in radians and $\lim\limits_{x\to\infty}\left(\sqrt{Ax^2+Bx} - Cx\right) = 2$, the the value of $\dfrac{BC}{A}$ equals (A,B,C$\in$R)

(A) 4 (B) 2 (C) $\dfrac{1}{2}$ (D) none

115. Let a and b be two positive real numbers such that $\lim\limits_{x\to 0}\dfrac{\sin^2 x}{e^{ax} - bx - 1}$ exists and is equal to $\dfrac{1}{2}$. If the circles $x^2 + y^2 + ax + by + 1 = 0$ and $x^2 + y^2 + 4x + 6y + k = 0$ intersect orthogonally, then find the value of k.

Paragraph for question nos. 116 to 118

Let $f(x) = \dfrac{\sin^{-1}(1-\{x\})\times\cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}\times(1-\{x\})}$ where $\{x\}$ denotes fractional part of x.

116. $L = \lim\limits_{x\to 0^+} f(x)$ is equal to

(A) $\dfrac{\pi}{2}$ (B) $\dfrac{\pi}{2\sqrt{2}}$ (C) $\dfrac{\pi}{\sqrt{2}}$ (D) $\sqrt{2}\pi$

117. $R = \lim\limits_{x\to 0^-} f(x)$ is equal to

(A) $\dfrac{\pi}{2}$ (B) $\dfrac{\pi}{2\sqrt{2}}$ (C) $\dfrac{\pi}{\sqrt{2}}$ (D) $\sqrt{2}\pi$

118. Which of the following is not true ?

(A) cos L < cos R (B) tan (2L) > tan R (C) sin L > sin R (D) tan (2L) < tan R

119. If $\displaystyle\lim_{n\to\infty} \frac{4^n}{\left(4-\sqrt{3}+2\sin\theta\right)^{n+2}}$ exists and is equal to p ($p \neq 0$) where $\theta \in \left(0, \frac{\pi}{2}\right)$, then find the value of $\left(\frac{p+\cos\theta}{p}\right)$.

120. Let $f(x) = \displaystyle\lim_{n\to\infty} \frac{\cos(\pi x) - e^{-nx}\cdot\sin(x-1)}{1+e^{-nx}\cdot(x-1)}$, then which one of the following is **correct**?

(A) The value of $f(1)$ is equal to zero.
(B) The value of $f(0^+)$ is equal to $-\sin 1$.
(C) The value of $f(0^-)$ is equal to -1.
(D) The value of $\left(f(0^+)-f(0^-)\right)$ is equal to $(1+\sin 1)$.

121.

	Column-I	Column-II
(A)	If $\displaystyle\lim_{x\to\infty}\left(\sqrt{x^2-x-1}-ax-b\right)=0$, where $a>0$, then there exists at least one a and b for which point (a, 2b) lies on the line.	(P) $y = -3$
(B)	If $\displaystyle\lim_{x\to\infty}\frac{(1+a^3)+8e^x}{1+(1-b^3)e^x}=2$, then there exists at least one a and b for which point (a, b^3) lies on the line.	(Q) $3x-2y-5=0$
(C)	If $\displaystyle\lim_{x\to\infty}(\sqrt{x^4-x^2+1}-ax^2-b)=0$, when there exists at least one a and b for which point (a, 2b) lies on the line.	(R) $15x-2y-11=0$
(D)	If $\displaystyle\lim_{x\to -a}\frac{x^7+a^7}{x+a}=7$, where $a<0$, then there exists at least one a for which point (a, 2) lies on the line.	(S) $y = 2$

122. The value of the $\displaystyle\text{Lt}_{x\to 0}\frac{\sin\left(5x^5+4x^4+3x^3+2x^2\right)}{\ln\left(\cos\left(x^3+x\right)\right)}$, is equal to

(A) – 1 (B) – 2 (C) – 3 (D) – 4

123. The sequence $\langle a_{n-1}\rangle$, $n \in N$ is an arithmetical progression and d is its common difference.

If $\displaystyle\lim_{n\to\infty}\left(1-\frac{d^2}{a_1^2}\right)\left(1-\frac{d^2}{a_2^2}\right)......\left(1-\frac{d^2}{a_n^2}\right)$ converges to $\frac{1}{4}$ and $a_1 = 8$, then find the value of d.

124. A triangle ABC whose sides are represented by three straight lines $L_1 = 0$, $L_2 = 0$ and $L_3 = 0$ which are given as,

$$L_1 : y - \lim_{n\to\infty}\frac{4}{n^2}\sum_{r=1}^{n}([r x]+3) = 0$$

L_2 : Obtained from L_1 by rotating it through an angle of $\sum_{i=2}^{3} \cot^{-1}(i)$ in anti clockwise sense about the point whose abscissa is 2.

$L_3 : y = \left(1 + \lim_{\theta \to 0^-} \left[\theta^2 + \theta + \sin\theta\right]\right) x$

where [x] denote the largest integer less than or equal to x.
Find the sum of tangents of interior angles of the triangle.

125. Let $f : \left(\frac{-\pi}{2}, \frac{\pi}{2}\right) \to R$, be defined as $f(x) = \begin{cases} \lim_{n \to \infty} \left(\dfrac{(\tan x)^{2n} + x^2}{\sin^2 x + (\tan x)^{2n}}\right), & x \neq 0 \\ 1, & x = 0 \end{cases}$.

Which of the following **holds good**?

(A) $f\left(\frac{\pi^-}{4}\right) = f\left(\frac{\pi^+}{4}\right)$ (B) $f\left(\frac{-\pi^-}{4}\right) = f\left(\frac{\pi^+}{4}\right)$

(C) $f\left(\frac{-\pi^+}{4}\right) = f\left(\frac{\pi^-}{4}\right)$ (D) $f(0^+) = f(0^-) = f(0)$

126. If $\lim_{x \to 0} \left(1 + x + \frac{f(x)}{x}\right)^{1/x} = e^3$, then find the function f(x) of least degree. Also find the coefficient of x^2.

127. If $a^2 = \lim_{\theta \to \pi} \left[\dfrac{\sec^2\theta + 12\sec\theta + 11}{\sec\theta + 1}\right]$

where [ ] denotes the greatest integer function, then the possible value(s) of a is(are)

(A) $-\sqrt{10}$ (B) $\sqrt{10}$ (C) 3 (D) – 3

128. If $L = \lim_{x \to 0} \left(\dfrac{1}{\ln(1+x)} - \dfrac{1}{\ln(x + \sqrt{1+x^2})}\right)$ then find the value of $\dfrac{L + 153}{L}$.

129. Let $f(n) = \lim_{x \to 0} \dfrac{x - n\tan\left(\dfrac{\tan^{-1} x}{n}\right)}{n\sin\left(\dfrac{\tan^{-1} x}{n}\right) - x}$ $(n \in N)$. Find the value of $\sum_{n=1}^{10} \dfrac{2 - f(n)}{1 + f(n)}$. [Ans. 770]

130. If $m \in N$, then the value of $\lim_{x \to \infty} \left(\dfrac{\sum_{k=1}^{100} (x + 2k)^m}{x^m + 5^{1000}}\right)$ is

(A) 0 (B) 1 (C) 100 (D) 1000

131. The value of $\lim\limits_{x\to 1}\dfrac{\sin\left(\sqrt{1+\ln x}\right)-\sin 1}{\ln x}$ is equal to

(A) cos 1 (B) $\dfrac{1}{2}\cos 1$ (C) $\dfrac{1}{4}\cos 1$ (D) 2 cos 1

132. Let $f(x)=\dfrac{2x+2}{2x-1}$ and $g(x)=x+\sqrt{x}$. The value of $\lim\limits_{x\to\infty}\left(f(x)\right)^{g(x)}$ equals

(A) $e^{\frac{1}{2}}$ (B) e (C) $e^{\frac{3}{2}}$ (D) 1

133. If $l=\lim\limits_{x\to\infty} x\left(\tan^{-1}(x+5)+\tan^{-1}(x+1)-e^{\ln\left(\pi\left[1+\cos\frac{1}{x}\right]\right)}\right)$, then l is

(A) – 2 (B) 2 (C) 3 (D) 0

[**Note:** [y] denotes greatest integer less than or equal to y.]

134. The value of $\lim\limits_{x\to 0^+}\left(\dfrac{\cot^{-1}(\ln\{x\})}{\pi}\right)^{-\ln\{x\}}$ is equal to

[**Note:** where { } denotes fractional part function.]

(A) $e^{\frac{1}{\pi}}$ (B) $e^{\frac{-1}{\pi}}$ (C) $e^{\frac{-2}{\pi}}$ (D) $e^{\frac{2}{\pi}}$

Paragraph for question nos. 135 & 136

Let $f(x)=\lim\limits_{n\to\infty}\dfrac{x^2+2(x+1)^{2n}}{(x+1)^{2n+1}+x^2+1}$, $n\in N$ and $g(x)=\tan\left(\dfrac{1}{2}\sin^{-1}\left(\dfrac{2f(x)}{1+f^2(x)}\right)\right)$.

135. If $x\in(-2, 0)$ then range of g(x) is

(A) $\left(0,\dfrac{4}{5}\right)$ (B) $\left(0,\dfrac{40}{9}\right)$ (C) $\left(0,\dfrac{5}{4}\right)$ (D) $\left(0,\dfrac{25}{16}\right)$

136. $\lim\limits_{x\to -3}\dfrac{\sin(x+3)\cdot g(x)}{x^2+4x+3}$ is equal to

(A) 0 (B) – 2 (C) $\dfrac{1}{2}$ (D) non-existent

137. Let $f(x)=\lim\limits_{n\to\infty}\dfrac{3^n\underbrace{\sin(\sin.......\sin(x))}_{n\text{ times}}+\left(\sqrt{2}\cos x+2\right)^n+2^n\cos x}{3^n+\sin x\left(\sqrt{2}\cos x+2\right)^n}$,

if $l=\lim\limits_{x\to\frac{\pi}{4}^+}f(x)$ and $m=\lim\limits_{x\to\frac{\pi}{4}^-}f(x)$ then find the value of l^2+m^2.

138. If a point P(x, y) lies on the curve $y = f(x)$ such that

$$\lim_{(x,y)\to(1,2)} \left(\frac{\tan^{-1}x + \tan^{-1}\frac{1}{y} - \tan^{-1}3}{(x-1)(y-2)}\right) \sin^{-1}(y-2) \text{ exists,}$$

then find $10 \lim_{x\to\frac{1}{3}} \frac{f^{-1}(x)}{(3x-1)}$.

139. If $L = \prod_{n=0}^{\infty}\left(1+\frac{1}{2^{2^n}}\right)$ and $M = \sum_{k=1}^{\infty}\frac{15^k}{\left(5^k-3^k\right)\left(5^{k+1}-3^{k+1}\right)}$. Then value of $L + 4M$.

140. Let $f(x) = x - [x]$, where $[x]$ denotes gratest integer function,

then the value of $\lim_{n\to\infty} f\left(\sqrt{n^2+n+1}\right)$ $n \in I$, equals $\frac{p}{q}$, (where p & q are coprime).

The value of p + q is equal to

(A) 1 (B) 2 (C) 3 (D) 4

141. If $\lim_{n\to\infty} n\,a_n = 0$ and $L = \lim_{n\to\infty}\left(1+\frac{1}{n}+a_n\right)^n$ then [L] has the value equal to

(A) 0 (B) 1 (C) 2 (D) None

Where [y] denotes greatest integer less than or equal to y.

142. $\lim_{x\to1} \frac{\sin\left(\pi\operatorname{cosec}^2\left(\frac{\pi x}{2}\right)\right)\sec\left(\pi\operatorname{cosec}^2\left(\frac{\pi x}{2}\right)\right)}{\ln^2 x}$ is equal to

(A) $\frac{\pi}{2}$ (B) $\frac{\pi^2}{4}$ (C) $\frac{\pi^3}{4}$ (D) $\frac{\pi^3}{8}$

Paragraph for question nos. 143 & 144

Consider a circle $x^2 + y^2 = a^2$ and a point P on it in 1st quadrant. Another circle is drawn concentric with the given circle and radius is 'b' more than the x-co-ordinate of point P. This circle intersects positive x-axis at Q and line OP at R (where O is the origin).

143. If angle subtended by arc QR at origin is 'θ' and $\lim_{\theta\to0}\frac{\text{Length of arc QR}}{\theta^n} = l$ (where l is a non zero finite quantity, $n \neq 0$) then the value of $(a + n + l)$ for $b = 1$ is equal to

(A) $\frac{5}{2}$ (B) $\frac{7}{2}$ (C) $\frac{9}{2}$ (D) $\frac{11}{2}$

144. If from a point M on the x-axis perpendicular is dropped on the line OP with foot of perpendicular as N and if $\lim_{\theta\to0}\frac{MN + \text{Length of arc QR}}{\theta^3} = K$ (where k is a non zero finite quantity) then coordinates of M for b = 2 is equal to

(A) (–1 – a, 0) (B) (– 2 – a, 0) (C) (– a, 0) (D) (1 + a, 0)

Paragraph for question nos. 145 & 146

Let a function $y = f(x)$ is represented parametrically by the equations $x = e^{2t} - 2e^{t} + 3,\ y = e^{t} - 1$ where $t \in (-\infty, 0]$. One more function $g(x)$ is defined such that $(gof)^{-1}(x) = f^{-1}o\, g^{-1}(x) = x$ $\forall\ x \in$ domain of $f(x)$.

145. Identify the incorrect statement(s) for the function $y = g(x)$.

(A) $g(x) = 2 + x^2\ \ \forall\ x \in (-1, 0]$ (B) $g(x)$ is an even function

(C) $g(x)$ is an odd function (D) odd extension of $g(x)$ in (0, 1) is, $g(x) = -(2 + x^2)$

146. The value of $\displaystyle\lim_{x \to -\frac{1}{2}} \frac{9 - 4g(x)}{\sin^{-1}(2x+1)\left(\pi - \cos^{-1}(2x+1)\right)}$ is equal to

(A) $\dfrac{2}{\pi}$ (B) $\dfrac{-4}{\pi}$ (C) $\dfrac{-2}{\pi}$ (D) $\dfrac{4}{\pi}$

147. $f(x) = \dfrac{3x^2 + ax + a + 1}{x^2 + x - 2}$ then which of the following can be correct?

(A) $\displaystyle\lim_{x\to 1} f(x)$ exists $\Rightarrow$ a = – 2 (B) $\displaystyle\lim_{x\to -2} f(x)$ exists $\Rightarrow$ a = 13

(C) $\displaystyle\lim_{x\to 1} f(x) = 4/3$ (D) $\displaystyle\lim_{x\to -2} f(x) = -1/3$

148. Let $f(x) = x + \sqrt{x^2 + 2x}$ and $g(x) = \sqrt{x^2 + 2x} - x$, then

(A) $\displaystyle\lim_{x\to\infty} g(x) = 1$ (B) $\displaystyle\lim_{x\to\infty} f(x) = 1$ (C) $\displaystyle\lim_{x\to-\infty} f(x) = -1$ (D) $\displaystyle\lim_{x\to-\infty} g(x) = -1$

149. Let $f : R^+ \to A$ (where A is co-domain of a non empty set) be a function defined as

$f(x) = \displaystyle\lim_{a\to\infty} \frac{x^{2a} - 3x^a + 2}{x^{2a} + x^a + 1}$. Find the number of elements in A for which f is surjective.

150. $\displaystyle\lim_{n\to\infty} n\, ln\left(\tan\left(\frac{\pi}{4} + \frac{\pi}{n}\right)\right)$ equals

(A) $\dfrac{\pi}{4}$ (B) $\dfrac{\pi}{2}$ (C) π (D) 2π

151. If α, β and γ be three distinct real values such that $\dfrac{\sin\alpha + \sin\beta + \sin\gamma}{\sin(\alpha+\beta+\gamma)} = \dfrac{\cos\alpha + \cos\beta + \cos\gamma}{\cos(\alpha+\beta+\gamma)} = 2$

and $\cos(\alpha + \beta) + \cos(\beta + \gamma) + \cos(\gamma + \alpha) = a$, then find the value of $\displaystyle\lim_{x\to a} \frac{\sqrt{x^2 - a^2}}{\sqrt{x-a} + \sqrt{x} - \sqrt{a}}$.

152. If $\lim\limits_{x\to 0} \sin\left(\dfrac{\pi(1-\cos^m x)}{x^n}\right)$ exists and is non-zero where m, n $\in$ N then

(A) m = 1 ; n = 1 (B) m = 1 ; n = 2 (C) m = 2 ; n = 2 (D) m = 3 ; n = 2

153. If $\lim\limits_{n\to\infty} \dfrac{e\left(1-\frac{1}{n}\right)^n - 1}{n^\alpha}$ exist and is equal to non-zero constant c, then find the value of $24(c-\alpha)$.

154. The value of $\lim\limits_{x\to-\infty} \dfrac{x^5\cos\left(\frac{1}{\pi x^2}\right)+x^6\sin\left(\frac{1}{\pi x}\right)+7}{|x|^5+6|x|+7}$ is

(A) $(\pi + 1)$ (B) $-(1+\pi)$ (C) $-\left(1+\dfrac{1}{\pi}\right)$ (D) $-\left(1+\dfrac{2}{\pi}\right)$

155. The value of $\lim\limits_{k\to\infty}\left(\sum\limits_{n=1}^{k}\dfrac{4n}{1+4n^4}\right)^{1+8k+k^2}$ is equal to

(A) $\sqrt{e}$ (B) $\dfrac{1}{\sqrt{e}}$ (C) $\dfrac{1}{2e}$ (D) $\dfrac{2}{e}$

156. Consider, $f(x) = x^2 - 4(p-1)\left(\dfrac{a^3+2a-3}{a^2+2a-3}\right)x + \left(\dfrac{4a^3+5a^2-6a-3}{a^2+2a-3}\right)$ and $g(x) = \lim\limits_{a\to 1} f(x)$. If the set of values of p is $[\alpha, \beta]$ for which $\lim\limits_{x\to k} \text{sgn}(g(x))$ exists for all k $\in$ R, then find the value of $(\alpha+\beta)$.

157. If $0<\alpha,\beta<\pi$, $\lim\limits_{x\to 0}\left(\dfrac{(\sin\alpha)^x+(\sin\beta)^x}{2}\right)^{\frac{1}{x}}=1$ then the value of $\tan\left(\dfrac{\alpha+\beta}{3}\right)$ is

(A) $\dfrac{1}{\sqrt{3}}$ (B) 1 (C) $\dfrac{1}{2}$ (D) $\sqrt{3}$

158. $\lim\limits_{x\to\infty} \dfrac{\frac{\pi}{2}-\tan^{-1}x}{\ln\left(1+\frac{1}{x}\right)}$ is equal to

(A) 1 (B) 2 (C) 3 (D) 4

159. $\lim\limits_{n\to\infty}\prod\limits_{r=2}^{n}\left(1-\dfrac{1}{r^3}\right)\left(1-\dfrac{1}{r^2+r+1}\right)$ is equal to

(A) 2 (B) $\dfrac{2}{3}$ (C) $\dfrac{1}{2}$ (D) $\dfrac{3}{2}$

160. $\lim\limits_{x\to 0}(\sin^{-1}[\sin x] + \cos^{-1}[\cos x] - 2\tan^{-1}[\tan x])$ is equal to

[**Note:** [k] denotes greatest integer function less than or equal to k.]

(A) π (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{2}$ (D) does not exist

ANSWER KEY

1. 0000 **2.** 0041 **3.** C **4.** 4r **5.** $\frac{1}{2}$ **6.** $\lim\limits_{\theta\to 0} x = 2L$

7. 2 **8.** 45 **9.** C **10.** 16 **11.** 200 **12.** D

13. 10 **14.** 0001 **15.** B **16.** B **17.** 3 **18.** 1

19. 4 **20.** 2 **21.** B **22.** C **23.** A **24.** C **25.** C

26. C **27.** C **28.** 3 **29.** B **30.** C **31.** C

32. ABCD **33.** 48 **34.** B **35.** C **36.** A **37.** B

38. 84 **39.** C **40.** 4 **41.** 4 **42.** B **43.** B **44.** $\frac{-1}{\pi^2}$

45. $\frac{21n2}{\pi}$ **46.** 9 **47.** 2 **48.** π **49.** $\ln a$ **50.** A

51. 0001 **52.** 9 **53.** C **54.** C **55.** 5 **56.** C

57. 17 **58.** B **59.** A **61.** 0335 **62.** 0002 **63.** 0008

64. $\lim\limits_{x\to\frac{\pi}{2}^-} = \frac{3\pi}{2}$ and $\lim\limits_{x\to\frac{\pi}{2}^+} = 0$. **65.** 1/2 **66.** $\frac{1}{4}$ **67.** A

68. 2 **69.** 35 **70.** C **71.** C **72.** ABC **73.** BD

74. B **75.** 683 **76.** 19 **77.** 0009 **78.** C **79.** D

80. 5 **81.** A **82.** A **83.** 1/18 **84.** D **85.** B

86. 5 **87.** $T(x) = \frac{1}{2}\tan^2\frac{x}{2}.\sin x$ or $\tan\frac{x}{2} - \frac{\sin x}{2}$, $S(x) = \frac{1}{2}x - \frac{1}{2}\sin x$, limit $= \frac{3}{2}$

88. 0003 **89.** 0 **90.** (a) DNE; (b) $\frac{5}{4}$ **91.** $e^{\frac{-1}{4}}$ **92.** $a = 3/2$ and $b \in R$

93. $P(x) = \frac{4}{3}x^7$ **94.** a = 2 ; limit = 1 **95.** B **96.** C **97.** D

98. B **99.** B **100.** D **101.** A **102.** C **103.** C **104.** C

105. A **106.** B **107.** A **108.** B **109.** 4 **110.** 2 **112.** C

113. 446 **114.** A **115.** 9 **116.** A **117.** B **118.** B **119.** 9

120. D **121.** (A) Q ; (B) P, Q, R ; (C) Q ; (D) S **122.** D **123.** 6 **124.** 6

125. BCD **126.** $2, 2x^2$ **127.** CD **128.** 307 **129.** 770 **130.** C **131.** B

132. C **133.** A **134.** B **135.** A **136.** C **137.** 2 **138.** 3

139. 5 **140.** C **141.** C **142.** C **143.** A **144.** B

145. BCD **146.** D **147.** ABCD **148.** AC **149.** 3 **150.** D **151.** 2

152. BD **153.** 12 **154.** C **155.** B **156.** 2 **157.** D **158.** A

159. C **160.** B

1. $\tan(2\pi\,|\sin\theta|) = \tan\left(\frac{\pi}{2} - |\cos\theta|\,2\pi\right)$

$2\pi\,|\sin\theta| = n\pi + \frac{\pi}{2} - |\cos\theta|\,2\pi$

$2\pi\,(|\sin\theta| + |\cos\theta|) = n\pi + \frac{\pi}{2}$

$|\sin\theta| + |\cos\theta| = \frac{n}{2} + \frac{1}{4}$ ....(1)

since $1 \le |\sin\theta| + |\cos\theta| \le \sqrt{2}$; $1 \le \frac{n}{2} + \frac{1}{4} \le \sqrt{2}$

$4 \le 2n + 1 \le 4\sqrt{2}$

$\frac{3}{2} \le n \le \frac{4\sqrt{2}-1}{2}$; Thus n = 2 is only possible value.

putting in (i) $|\sin\theta| + |\cos\theta| = \frac{5}{4}$

$g(x) = \lim\limits_{x\to\infty}\left[2\left(\frac{4}{5}\right)^x\right] = 0$ **Ans.**

2. $\alpha = \lim\limits_{r\to\infty}\sum\limits_{n=1}^{r}\tan^{-1}\left\{\frac{16(2n+1)}{(2n+1)^4 - 4(2n+1)^2 + 16}\right\}$

$= \lim\limits_{r\to\infty}\sum\limits_{n=1}^{r}\tan^{-1}\left\{\frac{2\left(n+\frac{1}{2}\right)}{1+\left(n+\frac{1}{2}\right)^4 - \left(n+\frac{1}{2}\right)^2}\right\}$ {Dividing numerator and denominator by 16}

$= \lim\limits_{r\to\infty}\sum\limits_{n=1}^{r}\tan^{-1}\left\{\frac{\left\{\left(n+\frac{1}{2}\right)^2 + \left(n+\frac{1}{2}\right)\right\} - \left\{\left(n+\frac{1}{2}\right)^2 - \left(n+\frac{1}{2}\right)\right\}}{1+\left\{\left(n+\frac{1}{2}\right)^2 - \left(n+\frac{1}{2}\right)\right\}\left\{\left(n+\frac{1}{2}\right)^2 + \left(n+\frac{1}{2}\right)\right\}}\right\}$ {Using : $n^4 - n^2 = (n^2 - n)(n^2 + n)$}

$= \lim\limits_{r\to\infty}\sum\limits_{n=1}^{r}\left[\tan^{-1}\left\{\left(n+\frac{1}{2}\right)^2 + \left(n+\frac{1}{2}\right)\right\} - \tan^{-1}\left\{\left(n+\frac{1}{2}\right)^2 - \left(n+\frac{1}{2}\right)\right\}\right]$

{Using : $\tan^{-1}\left(\frac{x-y}{1+xy}\right) = \tan^{-1}x - \tan^{-1}y \ \forall\ x, y > 0$}

$$= \lim_{r\to\infty}\sum_{n=1}^{r}\left[\left\{\tan^{-1}\left(\frac{9}{4}+\frac{3}{2}\right)-\tan^{-1}\left(\frac{9}{4}-\frac{3}{2}\right)\right\}+\left\{\tan^{-1}\left(\frac{25}{4}+\frac{5}{2}\right)-\tan^{-1}\left(\frac{25}{4}-\frac{5}{2}\right)\right\}+\left\{\tan^{-1}\left(\frac{49}{4}+\frac{7}{2}\right)\right.\right.$$

$$\left.\left.-\tan^{-1}\left(\frac{49}{4}-\frac{7}{2}\right)\right\}+....+\left\{\tan^{-1}\left(r+\frac{1}{2}\right)^2+\left(r+\frac{1}{2}\right)\right\}-\tan^{-1}\left\{\left(r+\frac{1}{2}\right)^2-\left(r+\frac{1}{2}\right)\right\}\right]$$

$$= \lim_{r\to\infty}\left[\left\{\tan^{-1}\left(\frac{15}{4}\right)-\tan^{-1}\left(\frac{3}{4}\right)\right\}+\left\{\tan^{-1}\left(\frac{35}{4}\right)-\tan^{-1}\left(\frac{15}{4}\right)\right\}+\left\{\tan^{-1}\left(\frac{63}{4}\right)-\tan^{-1}\left(\frac{35}{4}\right)\right\}\right.$$

$$\left.+....+\left\{\tan^{-1}\left(r+\frac{1}{2}\right)^2-\tan^{-1}\left(r+\frac{1}{2}\right)\right\}+\left\{\tan^{-1}\left(r+\frac{1}{2}\right)^2-\left(r+\frac{1}{2}\right)\right\}\right]$$

$$= \lim_{r\to\infty}\left[\left\{\tan^{-1}\left(r+\frac{1}{2}\right)^2+\left(r+\frac{1}{2}\right)\right\}-\tan^{-1}\left(\frac{3}{4}\right)\right] = \frac{\pi}{2}-\tan^{-1}\left(\frac{3}{4}\right) = \cot^{-1}\left(\frac{3}{4}\right)$$

$\therefore\ \cot\alpha = \frac{3}{4}\quad \therefore\ \operatorname{cosec}\alpha = \sqrt{1+\cot^2\alpha} = \frac{5}{4}\quad \therefore\ \sin\alpha = \frac{4}{5} = \frac{a}{b}$

$\therefore\ (a^2 + b^2)_{least} = 4^2 + 5^2 = 16 + 25 = 41$ **Ans.**]

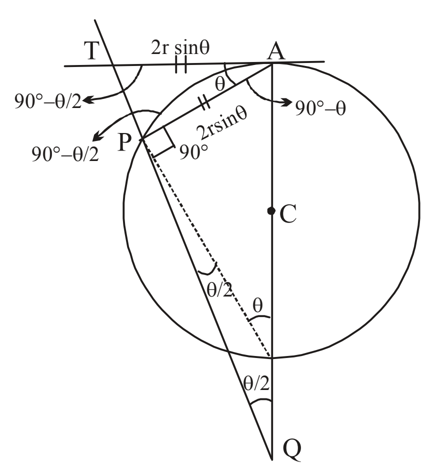
3. $\lim\limits_{n\to\infty}\dfrac{1}{\dfrac{(x-2)^n}{3^n}+3-\dfrac{1}{n}}$ (dividing N^r and D^r by $n\cdot 3^n$)

for $\lim\limits_{n\to\infty}$ to be equal to $\frac{1}{3}$

$\lim\limits_{n\to\infty}\frac{1}{n}\to 0$ (which is True) and $\lim\limits_{n\to\infty}\left(\frac{x-2}{3}\right)^n \to 0.$

$-1 < \frac{x-2}{3} < 1 \Rightarrow -3 < (x-2) < 3 \Rightarrow -1 < x < 5 \Rightarrow (-1, 5)$. **Ans.**

4. $\cos(90°-\theta) = \frac{AP}{2r}$



$AT = AP = 2r\sin\theta \quad ....(1) = AT$

now in ΔATQ $\quad \tan\left(90-\frac{\theta}{2}\right) = \frac{AQ}{AT} = \frac{AQ}{2r\sin\theta}$

$(AQ) = 2r\sin\theta\cdot\cot\frac{\theta}{2}$

$$\lim_{P\to A} AQ = \lim_{\theta\to 0} 2r\cdot 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\frac{\cos(\theta/2)}{\sin(\theta/2)} = \lim_{\theta\to 0} 4r\cos^2\frac{\theta}{2} = 4r \text{ **Ans.**}$$

5. $\lim\limits_{n\to\infty} n^2\left(n^2 \ln\left(\frac{n^2}{n^2+1}\right)+1\right)$.

Putting $n = \frac{1}{x}$, we get

$$\text{given limit} = \lim_{x\to 0}\frac{1}{x^2}\left[\frac{2}{x^2}\ln\left(\frac{1}{x^2+1}\right)+1\right] = \lim_{x\to 0}\frac{\{x^2 - \ln(1+x^2)\}}{x^4}$$

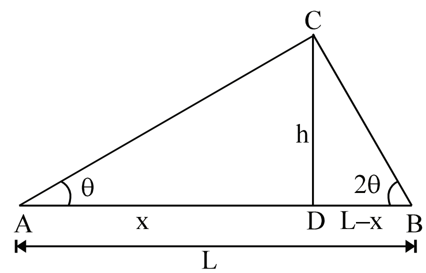
$$= \lim_{x\to 0}\frac{x^2 - \left(x^2 - \frac{x^4}{2} + \ldots\ldots\right)}{x^4} = \frac{1}{2} \text{ **Ans.**}$$

6. $CD = h;\quad \tan\theta = \frac{h}{x};\quad \tan 2\theta = \frac{h}{L-x}$

now $\quad x\tan\theta = (L-x)\tan 2\theta$

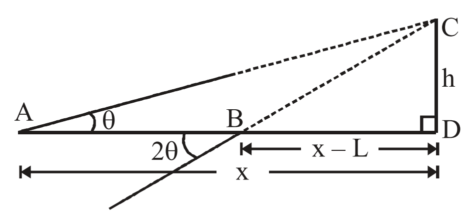
$x(\tan\theta + \tan 2\theta) = L\tan 2\theta$

$$x = \left(\frac{\tan 2\theta}{\tan\theta + \tan 2\theta}\right)L$$



$$x = \left(\frac{2\theta\frac{\tan 2\theta}{2\theta}}{\theta\frac{\tan\theta}{\theta} + 2\theta\frac{\tan 2\theta}{2\theta}}\right)L$$

$$\lim_{\theta\to 0} x = \frac{2L}{3} \text{ **Ans.**}$$



Now, $\quad \tan\theta = \frac{h}{x}$ ......(1)

$\tan 2\theta = \frac{h}{x-L}$ ......(2)

on equate h, we get

$x\tan\theta = (x-L)\tan 2q$

$L\tan 2\theta = x(\tan 2\theta - \tan\theta)$

$$x = \frac{L\sin 2\theta}{\tan 2\theta - \tan\theta}$$

$\therefore \quad \lim\limits_{\theta\to 0} x = 2L$ **Ans.**

7. $$f(x)=\left[\frac{x^2+x+1}{x^2+1}\right]+\left[\frac{4x^2+x+2}{2x^2+1}\right]+\left[\frac{9x^2+x+3}{3x^2+1}\right]+\ldots\ldots+\left[\frac{n^2x^2+x+n}{nx^2+1}\right]$$

$$=\left[1+\frac{x}{x^2+1}\right]+\left[2+\frac{x}{2x^2+1}\right]+\left[3+\frac{x}{3x^2+1}\right]+\ldots\ldots+\left[n+\frac{x}{nx^2+1}\right]$$

$$=1+2+\ldots\ldots+n+\left[\frac{x}{x^2+1}\right]+\left[\frac{x}{2x^2+1}\right]+\left[\frac{x}{3x^2+1}\right]+\ldots\ldots+\left[\frac{x}{nx^2+1}\right]$$

for $x>0$, $\dfrac{x}{x^2+1}=\dfrac{1}{x+\frac{1}{x}}<\dfrac{1}{2}$

$$\frac{x}{2x^2+1}=\frac{1}{2x+\frac{1}{x}}<\frac{1}{2\sqrt{2}}$$

$$\vdots \qquad \vdots \qquad \vdots$$

$$\frac{x}{nx^2+1}=\frac{1}{nx+\frac{1}{x}}<\frac{1}{2\sqrt{n}}$$

$$\therefore \quad \left[\frac{x}{x^2+1}\right]=\left[\frac{x}{2x^2+1}\right]=\ldots\ldots\ldots\ldots=\left[\frac{x}{nx^2+1}\right]=0$$

$$f(x)=\frac{n(n+1)}{2}$$

Now, $$\lim_{n\to\infty}\left(\frac{f(x)-n}{(f(x))^2-\frac{n^3(n+2)}{4}}\right)=\lim_{n\to\infty}\left(\frac{\frac{n(n+1)}{2}-n}{\frac{n^2(n+1)^2}{4}-\frac{n^3(n+2)}{4}}\right)$$

$$=\lim_{n\to\infty}\left(\frac{\frac{n^2-n+1}{2}}{\frac{n^2(n^2+2n+1)}{4}-\frac{n^3(n+2)}{4}}\right)=\lim_{n\to\infty}\left(\frac{\frac{n^2-n+1}{2}}{\frac{n^2}{4}}\right)=2 \textbf{ Ans.}$$

8. We have $f(x)=1-x^3$ and $g_5(x)=\dfrac{5}{\frac{1}{f(x)}+\frac{1}{f(2x)}+\ldots\ldots+\frac{1}{f(5x)}}$

Now, $$\lim_{x\to0}\frac{1-g_5(x)}{x^3}=\lim_{x\to0}\frac{1-\dfrac{5}{\frac{1}{f(x)}+\ldots\ldots+\frac{1}{f(5x)}}}{x^3}$$

$$= \lim_{x \to 0} \frac{\left(\frac{1}{f(x)} - 1\right) + \left(\frac{1}{f(2x)} - 1\right) + \ldots\ldots + \left(\frac{1}{f(5x)} - 1\right)}{x^3\left(\frac{1}{f(x)} + \frac{1}{f(2x)} + \frac{1}{f(3x)} + \frac{1}{f(4x)} + \frac{1}{f(5x)}\right)}$$

$$= \lim_{x \to 0} \frac{\frac{(1 - f(x))}{f(x)} + \frac{(1 - f(2x))}{f(2x)} + \ldots\ldots. + \frac{(1 - f(5x))}{f(5x)}}{5x^3} = \frac{1^3 + 2^3 + 3^3 + 4^3 + 5^3}{5}$$

$$= \frac{\left(\frac{5(5+1)}{2}\right)^2}{5} = 45. \textbf{ Ans.}$$

9. $f(\theta) = \dfrac{1}{\tan^9 \theta} \left((1 + \tan\theta)^{10} + (2 + \tan\theta)^{10} + \ldots. + (20 + \tan\theta)^{10}\right) - 20\tan\theta \cdot$

$$\lim_{\theta \to \frac{\pi}{2}^-} f(\theta) = 10\ (1 + 2 + 3 + \ldots\ldots.. + 20)$$

$$= 10 \times \frac{20 \times 21}{2} = 2100. \textbf{ Ans.}$$

10. $S = \displaystyle\sum_{k=0}^{\infty} \frac{(k+1)^2 + k + 1}{2^k}$

$$S = \sum_{k=0}^{\infty} \frac{(k+1)^2}{2^k} + \sum_{k=0}^{\infty} \frac{k}{2^k} + \sum_{k=0}^{\infty} \frac{1}{2^k}$$

$$S_1 = \frac{1^2}{1} + \frac{2^2}{2} + \frac{3^2}{2^2} + \frac{4^2}{2^3} + \ldots\ldots \infty$$

$$S_2 = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \frac{4}{2^4} + \ldots\ldots. \infty$$

$$S_3 = \frac{1}{1 - \frac{1}{2}} = 2$$

Now, $S_1 = \dfrac{1^2}{1} + \dfrac{2^2}{2} + \dfrac{3^2}{2^2} + \dfrac{4^2}{2^3} + \ldots\ldots$

$$\frac{S_1}{2} = \quad + \frac{1^2}{2} + \frac{2^2}{2^2} + \frac{3^2}{2^3} + \ldots\ldots$$

$$\frac{S_1}{2} = 1+\frac{3}{2}+\frac{5}{2^2}+\frac{7}{2^3}+.....$$

$$\frac{S_1}{4} = \quad +\frac{1}{2}+\frac{3}{2^2}+\frac{5}{2^3}+.....$$

$$\frac{S_1}{4} = 1+\frac{2}{2}+\frac{2}{2^2}+\frac{2}{2^3}+.....$$

$$\frac{S_1}{4} = 1+\frac{1}{1-\frac{1}{2}} = 3$$

$\therefore \quad S_1 = 12$

$$S_2 = \frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+\frac{4}{2^4}+.....\infty$$

$$\frac{S_2}{2} = \quad +\frac{1}{2^2}+\frac{2}{2^3}+\frac{3}{2^4}+.....\infty$$

$$\frac{S_2}{2} = \frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+.....\infty$$

$S_2 = 2$

$S = S_1 + S_2 + S_3 = 12 + 2 + 2 = 16$ **Ans.**

11. $S = \sum_{n=1}^{n} \frac{1}{2^n}\tan\frac{x}{2^{n-1}}$; $x = \frac{\pi}{4}$

Since $\tan x = \cot x - 2\cot 2x$ ....(1)

$\therefore \quad \frac{1}{2}\tan x = \frac{1}{2}\cot x - \cot 2x$

$$\frac{1}{2^2}\tan\frac{x}{2} = \frac{1}{2^2}\cot\frac{x}{2} - \frac{1}{2^2}\cot x$$

$$\vdots \qquad \qquad \vdots$$

$$\frac{1}{2^n}\tan\frac{x}{2^{n-1}} = \frac{1}{2^n}\cot\left(\frac{x}{2^{n-1}}\right) - \frac{1}{2^{n-1}}\cot\left(\frac{x}{2^{n-2}}\right)$$

$$\therefore \ S = \left(\frac{\cot\left(\frac{x}{2^{n-1}}\right)}{2^n} - \cot 2x\right)$$

$$S = \left(\frac{1}{2^n}\tan\frac{\pi}{2^{n-1}} - \cot 2x\right)$$

$$\lim_{n\to\infty} S = \left(\frac{1}{2}\cdot\frac{x}{2^{n-1}}\cdot\frac{1}{x}\cot\left(\frac{x}{2^{n-1}}\right) - \cot 2x\right)$$

$$S = \frac{1}{2x} - \cot 2x \qquad \text{put } x = \frac{\pi}{4}$$

$$\therefore\ S = \frac{2}{\pi}$$

$$\therefore\ (100\pi)L = 100\pi\cdot\frac{2}{\pi} = 200.\ \textbf{Ans.}$$

12. $f(x) = \lim_{n\to\infty}\dfrac{x^n\left(a+\sin(x^n)\right)+\left(b-\sin(x^n)\right)}{(1+x^n)\sec\left(\tan^{-1}(x^n+x^{-n})\right)}$

for continuity at $x = 1$

$\lim_{x\to 1} f(x)$ must exist and equals $f(1)$

$$f(1) = \lim_{n\to\infty}\frac{1^n(a+\sin 1^n)+b-\sin(1^n)}{(1+1^n)\cdot\sec\left(\tan^{-1}(1^n+1^{-n})\right)} = \frac{a+\sin 1+b-\sin 1}{2\sec(\tan^{-1}2)} = \frac{a+b}{2\sqrt{5}}$$

Now for $x > 1$ in the immediate neighbourhood

$$f(x) = \lim_{n\to\infty}\frac{a+\sin(x^n)+\dfrac{b-\sin x^n}{x^n}}{\left(1+\dfrac{1}{x^n}\right)\sec\left(\tan^{-1}(x^n+x^{-n})\right)} = \frac{a+(\text{some quantity between 1 and }-1)+0}{1\cdot\sec(\tan^{-1}\infty)} = 0$$

|||ly for $x < 0$ in the immediate neighbourhood $= 0$

$$f(x) = \frac{b}{1\cdot\sec(\tan^{-1}\infty)} = 0$$

Hence $f(x) = 0$ for $x \neq 1$ $\quad\therefore\ \lim_{x\to 1} f(x) = 0 = a+b$

13. $\lim_{x\to 0}\dfrac{(1-\cos^n x)\,(\tan x-\sin x)^m}{x^n} = 1 \Rightarrow \lim_{x\to 0}\left(\dfrac{1-\cos^n x}{x^2}\right)\cdot\left(\dfrac{\tan x-\sin x}{x^3}\right)^m\cdot\left(\dfrac{x^{3m+2}}{x^n}\right) = 1$

$\Rightarrow\ 3m+2 = n \qquad$ ....(1)

and, $\dfrac{n}{2^{m+1}} = 1 \qquad$ ....(2)

$\Rightarrow\ 3m+2 = 2^{m+1}$

So, $m = 2$ and $n = 8$ (minimum values).

14. For existence of limit,

$$\lim_{\substack{x \to \alpha, \\ x \in Q}} f(x) = \lim_{\substack{x \to \alpha, \\ x \in R-Q}} f(x)$$

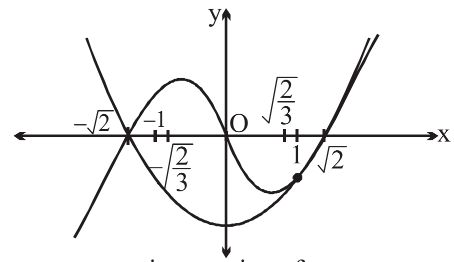
Thus, α must be root of the equation

$x^3 - 2x = x^2 - 2$

$x^2 (x - 1) - 2(x - 1) = 0$

$\Rightarrow \quad x = 1$ or $x = \sqrt{2}$ or $x = -\sqrt{2}$

Sum of all the values of $\alpha = 1$. **Ans.**



15. $$\lim_{x\to 0} \frac{(1-\cos x)}{x} = \frac{1}{2}.$$

$$\lim_{x\to 0} \frac{1-\cos(1-\cos x)}{x^4} = \lim_{x\to 0} \frac{1-\cos(1-\cos x)}{(1-\cos x)^2} \frac{(1-\cos x)^2}{x^4} = \frac{1}{8} \equiv \frac{m}{n}$$

$\Rightarrow \quad m^2 + n^2 = 1^2 + 8^2 = 65 \qquad \Rightarrow \qquad$ sum of the digits = 11. **Ans.**

16. $L(m) = -\sqrt{m+6}$ and $\lim_{m\to 0} \dfrac{-\sqrt{-m+6}+\sqrt{m+6}}{m}$.

$$\lim_{m\to 0} \frac{-\sqrt{-m+6}+\sqrt{m+6}}{m} = \lim_{m\to 0} \frac{1}{2\sqrt{-m+6}} + \frac{1}{2\sqrt{m+6}} = \frac{1}{\sqrt{6}} \textbf{ Ans.}$$

17. $f(x) = |\cos(2\pi(2x-[2x]))| + |\sin(2\pi(2x-[2x])| = |\cos(4\pi x)| + |\sin(4\pi x)|$

$\therefore$ period of $f(x) = \dfrac{1}{8}$

$$g(x) = \lim_{n\to\infty} \frac{\cos \pi x n - x^{2n} \sin(x-2)}{1+x^{2n+1}} = \begin{cases} \cos \pi x, & 0 < x < 1 \\ \dfrac{-\sin(x-2)}{x}, & 1 < x < \infty \end{cases}$$

$$\lim_{x\to 2\alpha} g(x) = \lim_{x\to 1/4} \cos \pi x = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}} = \frac{m}{\sqrt{n}}$$

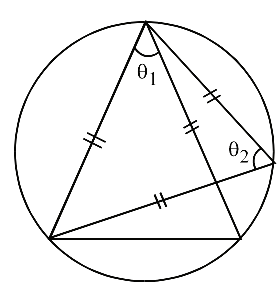
$\Rightarrow m = 1, n = 2$

$\therefore \ (m + n) = 3$. **Ans.**

18. $\theta_2 = 90 - \dfrac{\theta_1}{2}$

$\theta_3 = 90 - \dfrac{\theta_2}{2}$

$\vdots \qquad \vdots \qquad \vdots$



$$\theta_n = 90 - \frac{\theta_{n-1}}{2}$$

Now let $\lim\limits_{i\to\infty} \theta_i = \theta \Rightarrow \quad \lim\limits_{n\to\infty} \theta_n = 90 - \lim\limits_{n\to\infty} \frac{\theta_{n-1}}{2}$

$$\theta = 90 - \frac{\theta}{2} \Rightarrow \frac{3\theta}{2} = 90° \Rightarrow \theta = 60° = \frac{\pi}{3} \Rightarrow \left[\frac{\pi}{3}\right] = 1 \ \textbf{Ans.}$$

19. $\text{Limit}_{h\to 0} f(0+h) = \text{Limit}_{h\to 0} \dfrac{e^{\sin 2h}-1}{ln(1+\tan 9h)} \dfrac{\sin 2h.(2h)}{\sin 2h.(2h)}$

$$= \text{Limit}_{h\to 0} \frac{e^{\sin 2h}-1}{\sin 2h} . \text{Limit}_{h\to 0} \frac{\sin 2h}{2h} \ \text{Limit}_{h\to 0} \frac{2h}{ln(1+\tan 9h)} = (1).(1)$$

$$\text{Limit}_{h\to 0} \frac{2h}{\tan 9h \ ln(1+\tan 9h)\cot 9h} = \text{Limit}_{h\to 0} \frac{2h}{\tan 9h} = \frac{2}{9}$$

$$\text{Limit}_{h\to 0} f(0-h) = \lim_{h\to 0} \frac{ln(1+2\sin^2 h)}{2\sin^2 h} . \lim_{h\to 0} \frac{2\sin^2 h}{ln^2(1-\sin 3h)}$$

$$= \lim_{h\to 0} \frac{2\sin^2 h}{\sin^3 3h\left[\dfrac{ln(1-\sin 3h)}{-\sin 3h}\right]^2} = \frac{2}{9}$$

since $\quad f(0^+) = f(0^-) = \dfrac{2}{9}$ hence limit exist.

Therefore it is possible to define f (0) such that $\quad \text{Limit}_{h\to 0} f(x) = \dfrac{2}{9} = f(0)$

$$\Rightarrow 9\,[f(0^+) + f(0^-)] = 9\left[\frac{2}{9}+\frac{2}{9}\right] = 9 \times \frac{4}{9} = 4. \ \textbf{Ans.}$$

20. $\sqrt{3}\sin n\theta + \cos n\theta = 0$ or $\qquad \tan n\theta = \dfrac{-\sqrt{3}}{3}$

$$n\theta = \frac{-\pi}{6} + k\pi \qquad \text{or} \qquad \theta_k = \frac{(6k-1)\pi}{6n}$$

So, $\dfrac{\theta_k}{2} = \dfrac{(6k-1)\pi}{12n}$

Let $A = \sum\limits_{k=1}^{n} \cos\dfrac{\theta_k}{2} = \sum\limits_{k=1}^{n} \cos\dfrac{(6k-1)\pi}{12n}$

multiply with $2\sin\dfrac{\pi}{4n}$

$$\sum_{k=1}^{n} 2\sin\frac{\pi}{4n}\cos\frac{(6k-1)\pi}{12n} = -\sin\frac{\pi}{6n} + \sin\frac{(3n+1)\pi}{6n} = \cos\frac{\pi}{6n} - \sin\frac{\pi}{6n}$$

$$\text{Then } A = \frac{\cos\frac{\pi}{6n} - \sin\frac{\pi}{6n}}{3\sin\frac{\pi}{4n}}$$

$$\lim_{h\to 0}\left[h\cdot\frac{\cos\frac{\pi h}{6} - \sin\frac{\pi h}{6}}{3\sin\frac{\pi h}{4}}\right] = \lim_{h\to 0}\left[h\cdot\frac{\cos\frac{\pi h}{6}}{2\sin\frac{\pi h}{4}}\right] - 0$$

$$= \lim_{h\to 0}\frac{h}{2\sin\frac{\pi h}{4}} = \lim_{h\to 0}\frac{1}{2\cos\frac{\pi h}{4}\cdot\frac{\pi}{4}} = \frac{2}{\pi}.$$

Paragraph for Question no. 21 to 23

Sol 21 to 23

$P(x) = x^5 - 9x^4 + px^3 - 27x^2 + qx + r$

$\because$ $P(x)$ is divisible by x^2

$\therefore$ $q = r = 0$

$P(x) = x^5 - 9x^4 + px^3 - 27x^2$

$$\frac{P(x)}{x^2} = x^3 - 9x^2 + px - 27 = 0 \quad \begin{matrix}\alpha\\ \beta\\ \gamma\end{matrix}$$

$\alpha, \beta, \gamma \in R^+$

$\alpha + \beta + \gamma = 9,\ \alpha\beta\gamma = 27$

A.M. of $\alpha, \beta, \gamma = \dfrac{\alpha+\beta+\gamma}{3} = 3$

G.M. of $\alpha, \beta, \gamma = (\alpha\beta\gamma)^{1/3} = (27)^{1/3} = 3$

$\because$ A.M. = G.M. $\Rightarrow$ $\alpha = \beta = \gamma$

$\therefore$ $3\alpha = 9 \Rightarrow \alpha = 3 = \beta = \gamma$

$p = \alpha\beta + \beta\gamma + \gamma\alpha = 27$

21. $p + q + r = 27 + 0 + 0 = 27$

22. $\alpha - 1 = 2,\ \beta + 3 = 6,\ \gamma + 7 = 10$

$\therefore$ $S_n = 2 + 6 + 10 + \ldots\ldots.$ to n terms $= 2\,(1 + 3 + 5 + \ldots\ldots$ to n terms$) = 2n^2$

$$\text{Now, } \sum_{n=2}^{\infty}\frac{1}{\sqrt{S_n\cdot S_{n-1}}} = \sum_{n=2}^{\infty}\frac{1}{\sqrt{2n^2\cdot 2\,(n-1)^2}} = \frac{1}{2}\sum_{n=2}^{\infty}\frac{1}{n\,(n-1)} = \frac{1}{2}\sum_{n=2}^{\infty}\left(\frac{1}{n-1} - \frac{1}{n}\right)$$

$$= \frac{1}{2}\left(\frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \ldots\ldots.\infty\right) = \frac{1}{2}. \textbf{ Ans.}$$

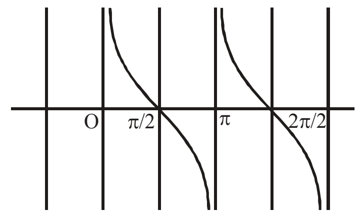
23. $\lim\limits_{n\to\infty}\left(\dfrac{1}{27}+\dfrac{1}{(27)^2}+\dfrac{1}{(27)^3}+........+\dfrac{1}{(27)^n}\right)$ $\{\because q = r = 0\}$

$$= \frac{1}{27}+\frac{1}{(27)^2}+\frac{1}{(27)^3}+.......\infty = \frac{\frac{1}{27}}{1-\frac{1}{27}} = \frac{1}{26}\text{. } \textbf{Ans.}$$

24. In right viccinity of $x = \pi$ $\qquad \cot x \to +\infty \quad \Rightarrow 2^{\cot x}, 3^{\cot x}, 5^{\cot x} \to \infty$

In left viccinity of $x = \pi$ $\qquad \cot x \to -\infty \quad \Rightarrow 2^{\cot x}, 3^{\cot x}, 5^{\cot x} \to 0$

$\therefore$ R.H.L. $\lim\limits_{x\to\pi^+}\dfrac{\left(\dfrac{2}{5}\right)^{\cot x}+\left(\dfrac{3}{5}\right)^{\cot x}-5+\dfrac{2}{5\cot x}}{\left(\dfrac{4}{5}\right)^{\cot x}+\left(\dfrac{3}{5}\right)^{\cot x}-1+\dfrac{1}{5\cot x}} = \dfrac{-5}{-1} = 5.$



L.H.L $\lim\limits_{x\to\pi^-}\dfrac{2^{\cot x}+3^{\cot x}-5\cdot 5^{\cot x}+2}{4^{\cot x}+2^{\cot x}-5^{\cot x}+1} = \dfrac{2}{1} = 2 \Rightarrow l = \text{D.N.E.}$ **Ans.**

25. $f(0) = \lim\limits_{x\to 0} f(x) = \lim\limits_{x\to 0}\dfrac{x^3\tan^2 x}{e^{\sin^3 x}\left(e^{x^3-\sin^3 x}-1\right)} = \lim\limits_{x\to 0}\dfrac{x^3\tan^2 x}{(x-\sin x)(x^2+\sin^2 x+x\sin x)}$

$$= \frac{\dfrac{\tan^2 x}{x^2}}{\dfrac{x-\sin x}{x^3}\cdot\dfrac{x^2+\sin^3 x+x\sin x}{x^2}} = \frac{1}{\frac{1}{6}\times 3} = 2 \text{ Ans.}$$

26. $\dfrac{e}{4x}-\dfrac{e}{2x\left(e^{ex}+1\right)} = \dfrac{e}{4x}\left[1-\dfrac{2}{e^{ex}+1}\right] = \dfrac{e}{4x}\left[\dfrac{e^{ex}+1-2}{\left(e^{ex}+1\right)}\right]$ $\quad(\infty-\infty \text{ form})$

$$= \frac{e^2}{8}\frac{(e^{ex}-1)}{ex} = \frac{e^2}{8}\text{. } \textbf{Ans.}$$

27. $\lim\limits_{x\to 1}\left(\dfrac{1}{ln\,x}-\dfrac{1}{x-1}\right) = \lim\limits_{h\to 0}\left(\dfrac{1}{ln\,(1+h)}-\dfrac{1}{h}\right)$

$$= \lim_{h\to 0}\frac{h-ln(1+h)}{h^2}\lim_{h\to 0}\frac{h}{ln\,(1+h)}$$

Put $\quad ln\,(1+h) = t$

$\qquad 1 + h = e^t$

$$l = \lim_{t\to 0} \frac{e^t - 1 - t}{\left(\frac{e^t - 1}{t}\right)^2 \cdot t^2} = \lim_{t\to 0} \frac{e^t - 1 - t}{t^2} \qquad ....(1)$$

Also $\quad l = \lim_{t\to 0} \frac{e^{-t} - 1 + t}{t^2} \qquad ....(2)$

add (1) and (2)

$$2l = \lim_{t\to 0} \frac{e^t + e^{-t} - 2}{t^2} = \lim_{t\to 0} \frac{e^{2t} - 2e^t + 1}{t^2}$$

$$= \lim_{h\to 0} \left(\frac{e^t - 1}{t}\right)^2 = 2l \;\Rightarrow\; l = \frac{1}{2}.$$

28. $f(x) = 2 \quad 0 < x < 1$

$\quad 0 \qquad x = 1$

$\quad 1 \qquad x > 1$

$\therefore$ Range of $f = \{0, 1, 2\} = A$. (for surjective)

$\Rightarrow$ Number of elements in $A = 3$. **Ans.**

29. $L = \lim_{x\to 0^+} \left(\frac{\cot^{-1}(\ln(x))}{\pi}\right)^{-\ln x} \qquad (1^\infty)$

$$= e^{\lim_{x\to 0^+} \left(\frac{\cot^{-1}\ln(x) - \pi}{\pi}\right)(-\ln x)} = e^{\lim_{x\to 0^+} \frac{\cot^{-1}(-\ln x)}{\pi}(\ln x)} = e^{-\lim_{x\to 0^+} \frac{\tan^{-1}\left(-\frac{1}{\ln x}\right)}{\left(-\frac{1}{\ln x}\right)\pi}} = e^{\frac{-1}{\pi}}$$

$\therefore$ L.H.L. $= f(0^-) = e^{\left(-\frac{x}{\pi}\cdot\frac{1}{x}\right)} = e^{-\frac{1}{\pi}}$

$$f(0^+) = f(0^-) = e^{-\frac{1}{\pi}} = k \text{ **Ans.**}$$

30. $\lim_{x\to 0} \frac{e^{\tan^{-1} x} - e^{\sin^{-1} x}}{e^{\tan x} - e^{\sin x}} = \lim_{x\to 0} \frac{\tan^{-1} x - \sin^{-1} x}{\tan x - \sin x} = \lim_{x\to 0} \left(\frac{\tan^{-1} x - \sin^{-1} x}{x^3}\right)\left(\frac{x^3}{\tan x - \sin x}\right)$

$$= \lim_{x\to 0} \frac{\left(x - \frac{x^3}{3} - \left(x + \frac{x^3}{3!}\right)\right)}{x^3} \cdot \frac{x^3}{\left(x + \frac{x^3}{3}\right) - \left(x - \frac{x^3}{3!}\right)} = \frac{-1}{2} \times 2 = -1 \text{ **Ans.**]}$$

31. $\lim\limits_{x \to 0} \dfrac{\left(e^{x^2}-1\right)+\left(1-\sqrt{\cos 2x}\right)}{x^2\ ln\,(\sec x)^{\frac{1}{x^2}}}$

limit of N^r = $\lim\limits_{x \to 0} \dfrac{e^{x^2}-1}{x^2} + \lim\limits_{x \to 0} \dfrac{1-\cos x}{x^2} \cdot \dfrac{1}{1+\sqrt{\cos 2x}} = 1 + 1 = 2$

Limit of D^r = $\lim\limits_{x \to 0} ln\,(\sec x)^{\frac{1}{x^2}} = \lim\limits_{x \to 0} \dfrac{1}{x^2}(\sec x - 1) = \dfrac{1}{2}$

$\therefore \qquad l = \dfrac{2 \cdot 2}{1} = 4.$ **Ans.**

32. $\lim\limits_{n\to\infty} n^k \left(\dfrac{a^{1/n}-1}{1/n}\right) \cdot \dfrac{1}{n} \left(\left(1-\dfrac{1}{n}\right)^{1/2} - \left(1-\dfrac{1}{n+2}\right)^{1/2}\right)$

$ln\, a \lim\limits_{n\to\infty} \dfrac{n^{k-1}\left(\dfrac{1}{n+2}-\dfrac{1}{n}\right)}{\left(1-\dfrac{1}{n}\right)^{1/2}+\left(1-\dfrac{1}{n+2}\right)^{1/2}} = \dfrac{1}{2} ln\ a \lim\limits_{n\to\infty} \dfrac{n^{k-1}(-2)}{n(n+2)}$

Now analyse.

33. $L = \lim\limits_{x \to 0} \dfrac{(1+3x+2x^2)^{\frac{1}{x}} - (1+3x-2x^2)^{\frac{1}{x}}}{x} \quad \left(\dfrac{0}{0}\right)$

$= \lim\limits_{x \to 0} \dfrac{\left[(1+3x+2x^2)^{\frac{1}{x}} - e^3\right] - \left[(1+3x-2x^2)^{\frac{1}{x}} - e^3\right]}{x}$

$= \lim\limits_{x \to 0} \dfrac{(1+3x+2x^2)^{\frac{1}{x}} - e^3}{x} - \lim\limits_{x \to 0} \dfrac{(1+3x-2x^2)^{\frac{1}{x}} - e^3}{x} = L_1 - L_2$

$L_1 = \lim\limits_{x \to 0} \dfrac{e^{\frac{ln\,(1+3x+2x^2)}{x}} - e^3}{x} = \lim\limits_{x \to 0} \dfrac{e^{3+2x-\frac{x(3+2x)^2}{2}} - e^3}{x}$

$= \lim\limits_{x \to 0} e^3 \left[\dfrac{e^{2x - \frac{x}{2}(3+2x)^2 + ...} - 1}{x}\right] = e^3 \left[2 - \dfrac{9}{2}\right] = \dfrac{-5}{2} e^3$

And $\quad L_2 = \frac{-13}{2} e^3$, $L = 4e^3 \quad \Rightarrow k = 4$. Hence $12k = 48$. **Ans.**

34. $l = \underset{x\to 0}{\text{Limit}} \dfrac{(1+x)^{\frac{1}{x}} + ex - e}{\dfrac{\sin^{-1}x}{x} . x} = \underset{x\to 0}{\text{Limit}} \dfrac{e^{\frac{1}{x}\ell n(1+x)} - e + ex}{x}$

$= \underset{x\to 0}{\text{Limit}} \dfrac{e\left[e^{\overbrace{\ell n(1+x)^{1/x}-1}^{s}} - 1\right] + ex}{x} = e . \underset{s\to 0}{\text{Limit}} \dfrac{e^s - 1}{s} . \underset{x\to 0}{\text{Limit}} \dfrac{\ell n(1+x)^{1/x} - 1}{x} + e$

Let $1 + x = e^z \Rightarrow$ as $x \to 0$; $z \to 0$

$= e . \underset{z\to 0}{\text{Limit}} \dfrac{\frac{z}{e^z - 1} - 1}{e^z - 1} + e \quad = e . \underset{z\to 0}{\text{Limit}} \dfrac{z - e^z + 1}{\left(\frac{e^z - 1}{z}\right)^2 . z^2} + e \quad = e . \underset{z\to 0}{\text{Limit}} \dfrac{z - e^z + 1}{z^2} + e$

$l = \underset{z\to 0}{\text{Limit}} \dfrac{-z - e^{-z} + 1}{z^2} \Rightarrow 2l = \dfrac{2 - e^z - e^{-z}}{z^2} = -\left[\dfrac{e^z + e^{-z} - 2}{z^2}\right]$

$\Rightarrow l = -\dfrac{1}{2} \quad = -\left[\left(\dfrac{e^2 - 1}{z}\right)^2\right] = 1$

Hence Limit $= -\dfrac{e}{2} + e = \dfrac{e}{2}$

Sol. **35 to 37**

35. If $\alpha \to \infty$, then $a \to 0$

$\therefore P(x) = bx + c$

Now, $P(2) = 2b + c = 9$ and $P'(3) = b = 5 \Rightarrow c = -1 \Rightarrow P(x) = 5x - 1$

Now, $\underset{x\to\infty}{\text{Lim}} \left(\dfrac{5x-1}{5x-5}\right)^x$ (1^∞ form) $= e^{\underset{x\to\infty}{\text{Lim}}\left(\frac{5x-1-5x+5}{5x-5}\right)x} = e^{\frac{4}{5}}$

36. When $\alpha, \beta \to \infty$ then $a, b \to 0$

$\therefore P(x) = c$, but $P(2) = 9 \Rightarrow P(x) = 9$

Now, $\underset{x\to 3}{\text{Lim}} \dfrac{\sqrt{P(x)} - 3}{\sin(x-3)} = \underset{x\to 3}{\lim} \dfrac{3-3}{\sin(x-3)} = 0 \quad \left[\text{Since } \dfrac{\text{Exact zero}}{\text{tending towards zero}} = 0\right]$

37. For $\alpha = \beta$, $P(x) = a(x-\alpha)^2$

As $P(2) = 9 \Rightarrow P(2) = a(2-\alpha)^2 = 9$

$\Rightarrow a = \dfrac{9}{(2-\alpha)^2}$

Now, $\displaystyle \lim_{x\to\alpha} \frac{\frac{9}{(2-\alpha)^2}\cdot (x-\alpha)^2}{\left[1-\tan^2\left(\frac{\pi}{4}-(x-\alpha)\right)\right]\left(e^{x-\alpha}-1\right)}$

$$= \lim_{x\to\alpha} \frac{9}{(2-\alpha)^2}\cdot\frac{(x-\alpha)^2}{\left[1+\tan\left(\frac{\pi}{4}-(x-\alpha)\right)\right]\left[1-\tan\left(\frac{\pi}{4}-(x-\alpha)\right)\right]\left(e^{x-\alpha}-1\right)}$$

Putting $x=\alpha+h$, we get

$$= \lim_{h\to 0} \frac{9}{(2-\alpha)^2}\cdot\frac{h^2}{\left[1+\tan\left(\frac{\pi}{4}-h\right)\right]\left[1-\tan\left(\frac{\pi}{4}-h\right)\right]\left(e^{h}-1\right)}$$

$$\lim_{h\to 0} \frac{9}{(2-\alpha)^2}\cdot\frac{h^2}{2\cdot\left(1-\frac{1-\tan h}{1+\tan h}\right)(e^h-1)}$$

$$\lim_{h\to 0} \frac{9}{(2-\alpha)^2\cdot 2}\cdot\frac{h^2(1+\tan h)}{\frac{2\tan h}{h}\cdot\frac{(e^h-1)}{h}\cdot h^2} = \frac{9}{4(2-\alpha)^2} = \lim_{\alpha\to 0}\frac{9}{4(2-\alpha)^2} = \frac{9}{16}.\ \textbf{Ans.}$$

38. $\displaystyle T_r = \frac{r^4}{(2r-1)(2r+1)} = \frac{1}{16}\left[\frac{16r^4-1+1}{(2r-1)(2r+1)}\right]$

$$T_r = \frac{1}{16}\left[\frac{(4r^2-1)(4r^2+1)+1}{4r^2-1}\right] = \frac{1}{16}\left[(4r^2+1)+\frac{1}{(2r-1)(2r+1)}\right]$$

$$= \frac{1}{16}\left[(4r^2+1)+\frac{1}{2}\left(\frac{(2r+1)-(2r-1)}{(2r-1)(2r+1)}\right)\right] = \frac{1}{16}\left[4r^2+1+\frac{1}{2}\left(\frac{1}{(2r-1)}-\frac{1}{(2r+1)}\right)\right]$$

$$\therefore \quad S_n = \frac{1}{4}\sum r^2+\frac{1}{16}\sum 1+\frac{1}{32}\sum\left(\frac{1}{2r-1}-\frac{1}{2r+1}\right)$$

$$= \frac{n(n+1)(2n+1)}{4\cdot 6}+\frac{n}{16}+\frac{1}{32}\left[\left(1-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{5}\right)+.....+\left(\frac{1}{2n-1}-\frac{1}{2n+1}\right)\right]$$

$$= \frac{2n^3+3n^2+n}{24}+\frac{n}{16}+\frac{1}{32}\left(1-\frac{1}{2n+1}\right)$$

$$= \frac{n^3}{12}+\frac{n^2}{8}+\frac{5n}{48}+\frac{1}{16}\left(\frac{n}{2n+1}\right) = \frac{n^3}{A}+\frac{n^2}{B}+\frac{5n}{C}+\frac{f(n)}{D} \quad \text{(Given)}$$

So $A = 12, B = 8, C = 48 ; D = 16$, Hence $A + B + C + D = 84$ **Ans.**

39. $\prod_{k=1}^{n}\left(1-\tan^4\frac{x}{2^k}\right) = \prod_{k=1}^{n}\left(1-\tan^2\frac{x}{2^k}\right)\left(1+\tan^2\frac{x}{2^k}\right)$

$$f(x) = \prod_{k=1}^{n}\frac{\cos\frac{x}{2^{k-1}}}{\cos^4\left(\frac{x}{2^k}\right)} = \frac{\cos x\cdot\cos\frac{x}{2}\cdot\ldots\ldots\cdot\cos\frac{x}{2^{n-1}}}{\left(\cos\frac{x}{2}\cdot\cos\frac{x}{2^2}\cdot\ldots\ldots\cdot\cos\frac{x}{2^n}\right)^4}$$

$$\therefore \quad f(x) = \frac{\sin 2x}{2^n\sin\frac{x}{2^{n-1}}}\cdot\frac{1}{\left(\frac{\sin x}{2^n\sin\frac{x}{2^n}}\right)^4} = \frac{2\sin x\cos x 2^n\cdot 2^n\cdot 2^n\cdot 2^n\sin^x\frac{x}{2^n}}{2^n\sin\frac{x}{2^{n-1}}\sin^n x\sin^3 x} = \frac{x^3\cos x}{\sin^3 x}$$

$$\lim_{n\to\infty}\prod_{k=1}^{n}\left(1-\tan^4\frac{x}{2^k}\right) = \frac{x^3\cos x}{\sin^3 x}$$

$$f'(0) = \lim_{h\to 0}\frac{f(h)-f(0)}{h} = \lim_{h\to 0}\frac{h^3\cos h-\sin^3 h}{h\sin^3 h} = 0.$$

Alternative

Product of n even function is an even function.

Hence $f'(0)$ is 0

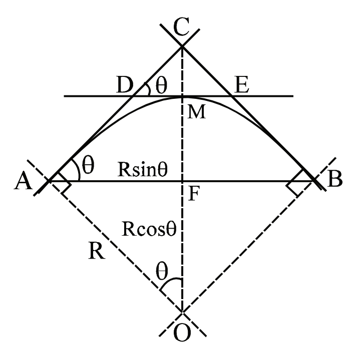
Note that if $f(x)$ is even then $f'(0) = 0$.

40. Δ_1 = Area of $\Delta ABC = R^2\sin\theta(\sec\theta-\cos\theta)$

$$\Delta_1 = R^2\tan\theta(1-\cos^2\theta)$$

$$\text{Area of }\Delta CDE = \frac{R^2(1-\cos\theta)^2}{\cos^2\theta.\tan\theta} = \Delta_2$$

CM = R secθ R
DM = CM cotθ



$$\therefore \frac{\Delta_1}{\Delta_2} = \frac{\tan\theta(1-\cos^2\theta).\cos^2\theta.\tan\theta}{(1-\cos\theta)^2} = l\text{(say)}$$

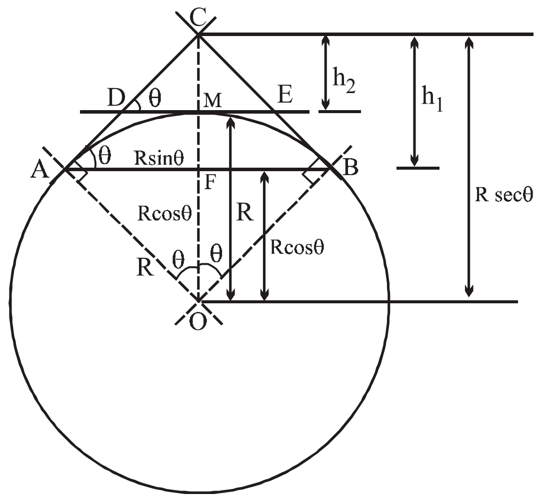
as $\theta \to 0$ $\quad l = \underset{\theta\to 0}{\text{Limit}} \dfrac{(\tan^2\theta)\cos\theta.(1-\cos\theta).(1+\cos\theta)}{(1-\cos\theta)^2}$

$$= 1.\,2\ \underset{\theta\to 0}{\text{Limit}} \frac{\tan^2\theta}{\theta^2}\cdot\frac{\theta^2}{1-\cos\theta} = 4 \text{ Ans}$$

Alternatively: $\Delta_1 = \Delta ABC$

$\Delta_2 = \Delta CDE$

$\dfrac{\Delta_1}{\Delta_2} \to 4$ as $AB \to 0$



$$\frac{\Delta_1}{\Delta_2} = \frac{h_1^2}{h_2^2} = \frac{R^2(\sec\theta-\cos\theta)^2}{R^2(\sec\theta-1)^2}$$

$$= \frac{(1-\cos^2\theta)^2\cdot\cos^2\theta}{\cos^2\theta\,(1-\cos\theta)^2}$$

$$= \underset{\theta\to 0}{\text{Lim}}\ (1+\cos\theta)^2 = 4$$

arc $AB \to 0 \quad \Rightarrow \quad \theta \to 0$

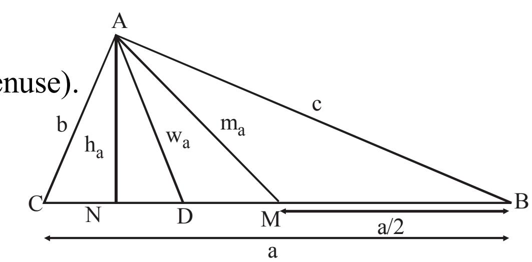
$= 4$ Ans.

41. $b^2 + c^2 = a^2$ ....(1)

also $2m_a = a$

(median to the hypotenuse is half as along as hypotenuse).

$a \cdot h_a = bc = \Delta$; $A = 90°$ ....(2)



and $\quad w_a = \dfrac{2bc\cos\dfrac{A}{2}}{b+c} = \dfrac{\sqrt{2}\,bc}{b+c}$ ....(3)

Now $\quad \dfrac{2m_a - 2h_a}{2w_a - 2h_a} = \dfrac{2a\cdot m_a - 2a\cdot h_a}{2a\cdot w_a - 2a\cdot h_a} = \dfrac{a^2 - 2bc}{\dfrac{2\sqrt{2}\,abc}{b+c} - 2bc}$

$$= \frac{(b^2+c^2-2bc)\,(b+c)}{2\sqrt{2}\ abc - 2bc\,(b+c)} = \frac{(b-c)^2(b+c)}{2bc\left[\sqrt{2(b^2+c^2)}-(b+c)\right]}$$

$$= \frac{(b-c)^2(b+c)\left[\sqrt{2(b^2+c^2)}+(b+c)\right]}{2bc\,\underbrace{\left[2(b^2+c^2)-(b+c)^2\right]}_{=(b-c)^2}} = \frac{(b+c)\left[\sqrt{2(b^2+c^2)}+(b+c)\right]}{2bc}$$

$$\therefore\ \underset{b\to c}{\text{Lim}}\frac{(b+c)\left[\sqrt{2(b^2+c^2)}+(b+c)\right]}{2bc}$$

$b \to c \Rightarrow \dfrac{b}{c} = t \to 1$

$$\therefore \ \lim_{t\to 1}\frac{\left(1+\frac{1}{t}\right)\left[\sqrt{2(1+t^2)}+(t+1)\right]}{2} = \frac{2[2+2]}{2} = 4. \ \textbf{Ans.}$$

42. Since, $f(x) > 0 \ \forall \ x \in (0, \infty)$ and $\lim\limits_{x\to\infty} f(x) = \infty$

$$\Rightarrow \quad \frac{1}{2} \ ln\, f(x) + f(x) = 2 \ ln\, x \quad \Rightarrow \quad \frac{1}{2} \ \frac{ln\, f(x)}{f(x)} + 1 = \frac{2 ln\, x}{f(x)}$$

$$\lim_{x\to\infty}\frac{ln\, x}{f(x)} = \frac{1}{2}. \ \textbf{Ans.}$$

43. Let $\quad L = \lim\limits_{n\to\infty}\left(\frac{n^2}{n-1}\right)^{\tan\frac{1}{\sqrt{n}}} \ (\infty^{\circ})$

$$ln\, L = \lim_{n\to\infty} \tan\frac{1}{\sqrt{n}} \ ln\left(\frac{n^2}{n-1}\right) = \lim_{n\to\infty}\frac{2\, ln\, n - ln(n-1)}{\cot\left(\frac{1}{\sqrt{n}}\right)} \ \left(\frac{\infty}{\infty}\right)$$

Put $\quad n = t^2 \ = \lim\limits_{t\to\infty}\dfrac{4\, ln\, t - ln\,(t^2-1)}{\cot\left(\frac{1}{t}\right)} = \lim\limits_{t\to\infty}\dfrac{2\,(t^2-2)}{t\,(t^2-1)\,\text{cosec}^2\left(\frac{1}{t}\right)\left(\frac{1}{t^2}\right)}$

$$= \lim_{t\to\infty}\frac{2\,(t^2-2)}{t\,(t^2-1)}\left(\frac{\sin\frac{1}{t}}{\frac{1}{t}}\right)^2 = 0$$

$L = 1.$ **Ans.**

44. $\lim\limits_{x\to 1}\dfrac{1-x+ln\, x}{(1+\cos\pi x)} \qquad x = 1 + h$

$$\lim_{h\to 0}\frac{1-(1+h)+ln\,(1+h)}{1-\cos\pi h}\cdot\frac{\pi^2h^2}{\pi^2h^2}$$

$$= \frac{2}{\pi^2}\cdot\lim_{h\to 0}\frac{-h+ln\,(1+h)}{h^2} \qquad \left(\text{as } \lim_{h\to 0}\frac{\pi^2h^2}{1-\cos\pi h} = 2\right)$$

put $\quad \ln(1+h) = t$

$$= \frac{2}{\pi^2}\cdot\lim_{t\to 0}\frac{t-(e^t-1)}{\left(\frac{e^t-1}{t}\right)^2 - t^2} = -\frac{2}{\pi^2}\cdot\lim_{t\to 0}\frac{e^t-1-t}{t^2} = -\frac{2}{\pi^2}\cdot\frac{1}{2} = -\frac{1}{\pi^2} \ \text{Ans.}$$

45. $x \to \frac{\pi}{2} - h; \quad h \to 0$

$$\lim_{h\to 0} \frac{2^{-\sin h} - 1}{\left(\frac{\pi}{2} - h\right)(-h)} = \lim_{h\to 0} \frac{1 - 2^{-\sin h}}{h\left(\frac{\pi}{2} - h\right)} = \lim_{h\to 0} \frac{1 - \frac{1}{2^{\sin h}}}{h\left(\frac{\pi}{2} - h\right)} = \lim_{h\to 0} \frac{2^{\sin h} - 1}{2^{\sin h} \cdot h\left(\frac{\pi}{2} - h\right)}$$

$$= \lim_{h\to 0} \frac{\left(\frac{2^{\sin h} - 1}{\sin h}\right)\sin h}{2^0 \cdot h\left(\frac{\pi}{2} - h\right)} = \frac{\ln 2 \cdot 1}{\pi/2} = \frac{2(\ln 2)}{\pi} \quad \left(\text{as } \lim_{x\to 0} \frac{a^x - 1}{x} = \ln a\right) \textbf{ Ans.}$$

46. $l = \frac{x^3 + 27}{x + 3} \cdot \lim_{x\to 3} \frac{\ln(x-2)}{x-3}; \quad x = 3 + h \qquad ; = 9 \cdot \lim_{h\to 0} \frac{\ln(1+h)}{h} = 9$ **Ans.**

47. $\lim_{x\to 1} \frac{x^2 - x \cdot \ln x + \ln x - 1}{x - 1} = \lim_{x\to 1} \frac{(x^2 - 1) - \ln x\,(x-1)}{x-1} = \lim_{x\to 1} \frac{(x + 1 - \ln x)(x - 1)}{x - 1}$

$\lim_{x\to 1} (x + 1 - \ln x) = 2$ **Ans.**

48. $\lim_{x\to 0} \frac{\sin(\pi\cos^2 x)}{x^2} = \lim_{x\to 0} \frac{\sin(\pi - \pi\cos^2 x)}{\pi\left(1 - \cos^2 x\right)} \cdot \frac{\pi \sin^2 x}{x^2} = 1 \cdot \pi$ **Ans.**

49. $\lim_{n\to\infty} n^2\left(a^{1/n} - a^{1/(n+1)}\right) \quad a > 0,\ n \in N$

put $n = 1/h$

$$\lim_{h\to 0} \frac{1}{h}\left(a^h - a^{h/(1+h)}\right) = \lim_{h\to 0} \frac{1}{h^2}\left(e^{h\ln a} - e^{h(\ln a)/(1+h)}\right)$$

$$= \lim_{h\to 0} \frac{1}{h^2} e^{\frac{h}{1+h}\ln a} \frac{\left(e^{h\ln a - \frac{h}{1+h}\ln a} - 1\right)}{h\ln a - \frac{h}{1+h}\ln a} \cdot h\ln a - \frac{h}{1+h}\ln a$$

$$= \lim_{h\to 0} \frac{1}{h^2} \ln a\left(h - \frac{h}{1+h}\right) = \lim_{h\to 0} \ln a \frac{(h + h^2 - h)}{h^2(1+h)} = \ln a$$

50. $a_n = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \ldots\ldots\infty$

$$\therefore \qquad a_n = \left(1+\frac{1}{3^2}+\frac{1}{5^2}+\frac{1}{7^2}+.......\infty\right) + \frac{1}{4}\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+.......\infty\right)$$

$$\frac{\pi^2}{6} = b_n + \frac{1}{4}\cdot\frac{\pi^2}{6}$$

$$\Rightarrow \qquad b_n = \frac{\pi^2}{6} - \frac{\pi^2}{24} = \frac{\pi^2}{8} \quad \textbf{Ans.}$$

51. $L_1 = \lim\limits_{x\to 1} \dfrac{\sin(n\cos^{-1}x)}{\sqrt{1-x}\,\sqrt{1+x}} = \lim\limits_{x\to 1} \dfrac{n}{\sqrt{1+x}}\left(\dfrac{\cos^{-1}x}{\sqrt{1-x}}\right) = n$

$L_2 = \lim\limits_{x\to 1} \dfrac{1-\cos(n\cos^{-1}x)}{(1-x)(1+x)} = \lim\limits_{x\to 1} \dfrac{n^2}{2}\dfrac{(\cos^{-1}x)^2}{(1-x)}\left(\dfrac{1}{2}\right)$

$\therefore\ L_2 = \dfrac{n^2}{2}$

Given, $\dfrac{n^2}{2} + n = \dfrac{3}{2}$

$\therefore\ n^2 + 2n = 3 \Rightarrow n^2 + 2n - 3 = 0 \Rightarrow (n+3)(n-1) = 0$

$\therefore\ n = 1$ or $n = -3$

Since, $n \in N$

$\therefore\ n = 1 \Rightarrow$ Sum = 1. **Ans.**

52. Subtracting equation (2) from (1), common roots are the roots of $x^2 + q - r = 0$ which has two roots say x_1 & x_2 such that $x_1 + x_2 = 0$

$\Rightarrow \qquad x_1 + x_2 + \alpha = -2 \qquad$ and $\qquad x_1 + x_2 + \beta = -1$

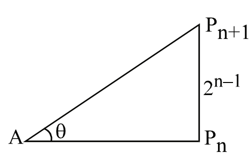
$\qquad\quad \alpha = -2, \qquad\qquad\qquad\qquad \beta = -1$

L.H.L. $= \lim\limits_{h\to 0} e^{-h\frac{\ln 3}{\ln(1-h)}} = e^{\ln 3} = 3 \Rightarrow f(0^-) = f(0) = a = 3$

R.H.L. $= b\lim\limits_{h\to 0} \dfrac{\ln\left(1+e^{h^2}-1+2\sqrt{h}\right)}{\left(e^{h^2}-1+2\sqrt{h}\right)} \cdot \dfrac{\left(e^{h^2}-1+2\sqrt{h}\right)}{\sqrt{h}} \cdot \dfrac{\sqrt{h}}{\tan\sqrt{h}} = 2b \Rightarrow b = \dfrac{3}{2}$

Hence $2(a+b) = 9$ **Ans.**

53. Consider a general triangle



$(AP_{n+1})^2 - (AP_n)^2 = (P_n P_{n+1})^2 = (2^{n-1})^2$

put $n = 1, 2, 3, ..., n$

$(AP_2)^2 - (AP_1)^2 = (P_1P_2)^2 = 1$

$(AP_3)^2 - (AP_2)^2 = (P_2P_3)^2 = 2^2$
$(AP_4)^2 - (AP_3)^2 = (P_3P_4)^2 = (2^2)^2$
$\vdots \qquad \vdots \qquad \vdots \qquad \vdots$
$(AP_{n+1})^2 - (AP_n)^2 = (P_n P_{n+1})^2 = (2^{n-1})^2$

$$(AP_{n+1})^2 - \underbrace{(AP_1)^2}_{=1} = \underbrace{1 + 2^2 + (2^2)^2 + (2^3)^2 + ...+ + (2^{n-1})^2}_{\text{G.P. with common ratio } 2^2} = \frac{2^{2n}-1}{2^2-1} = \frac{4^n-1}{3}$$

$$\therefore \qquad (AP_{n+1})^2 = \frac{4^n-1}{3} + 1 = \frac{4^n+2}{3} \qquad \Rightarrow \qquad (AP_{n+1}) = \sqrt{\frac{4^n+2}{3}}$$

Now $\quad \sin\theta = \dfrac{|P_n P_{n+1}|}{|AP_{n+1}|} = \dfrac{2^{n-1}}{\sqrt{\dfrac{4^n+2}{3}}} = \dfrac{2^n}{\sqrt{4^n+2}}\dfrac{\sqrt{3}}{2}$

as $n \to \infty$, $\dfrac{2^n}{\sqrt{4^n+2}} = 1$

$\Rightarrow \qquad \sin\theta = \dfrac{\sqrt{3}}{2}$ as $n \to \infty \Rightarrow \qquad \theta = \dfrac{\pi}{3}$

54. If L is the length of a side of the equilateral triangle, then the area is

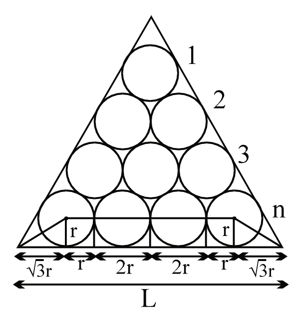
$$A = \frac{\sqrt{3}}{4}L^2 \text{ and so } L^2 = \frac{4}{\sqrt{3}}A \quad ....(1)$$

Let r be the radius of one of the circle. When there are n rows of circles, the figure shows that

$$L = \sqrt{3}\,r + r + (n-2)(2r) + r + \sqrt{3}\,r = r(2n - 2 + 2\sqrt{3})$$

$$\therefore \qquad L = 2r(n - 1 + \sqrt{3}) \qquad \Rightarrow \qquad r = \frac{L}{2(n-1+\sqrt{3})} \qquad ....(2)$$

The number of circles is $1 + 2 + ... + n = \dfrac{n(n+1)}{2}$, and so the total area of the circle is



$$A_n = \frac{n(n+1)}{2}\pi r^2 = \left(\frac{n(n+1)\pi}{2}\right)\left(\frac{L^2}{4\left(n+\sqrt{3}-1\right)^2}\right)$$

$$= \frac{n(n+1)\pi}{2}\cdot\frac{4A/\sqrt{3}}{4\left(n+\sqrt{3}-1\right)^2} = \frac{n(n+1)}{\left(n+\sqrt{3}-1\right)^2}\cdot\frac{\pi A}{2\sqrt{3}}$$

$$\Rightarrow \qquad \lim_{n\to\infty}\frac{A_n}{A} = \lim_{n\to\infty}\frac{n(n+1)}{\left(n+\sqrt{3}-1\right)^2}\cdot\frac{\pi}{2\sqrt{3}} = \lim_{n\to\infty}\frac{n^2\left(1+\dfrac{1}{n}\right)}{n^2\left(1+\dfrac{\sqrt{3}-1}{n}\right)^2}\cdot\frac{\pi}{2\sqrt{3}} = \frac{\pi}{2\sqrt{3}} \textbf{ Ans.}$$

55. $= \lim\limits_{n\to\infty} n\left\{\sqrt[3]{n^3+3n^2+2n+1}-(n+1)\right\} + n\left\{\sqrt{n^2-2n+3}-(n-1)\right\}$

$$= \lim_{n\to\infty} \frac{n\left\{\left(n^3+3n^2+2n+1\right)-(n+1)^3\right\}}{(n^3+3n^2+2n+1)^{2/3}+(n+1)\left(n^3+3n^2+2n+1\right)^{1/3}+(n+1)^2} + \lim_{n\to\infty}\frac{n\left\{n^2-2n+3-(n-1)^2\right\}}{\sqrt{n^2-2n+3}+(n-1)}$$

$$= \lim_{n\to\infty} \frac{n(-n)}{(n^3+3n^2+2n+1)^{2/3}+(n+1)(n^3+3n^2+2n+1)^{1/3}+(n+1)^2} + \lim_{n\to\infty}\frac{n(2)}{\sqrt{n^2-2n+3}+(n-1)}$$

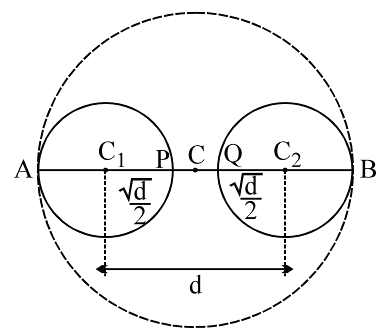
$$= \frac{-1}{1+1+1} + \frac{2}{1+1} = -\frac{1}{3} + 1 = \frac{2}{3} \;]$$

56. $\therefore$ $C_1C_2 = d$

and $r_1 + r_2 = \frac{\sqrt{d}}{2}+\frac{\sqrt{d}}{2} = \frac{2\sqrt{d}}{2} = \sqrt{d}$

hence $r_1 + r_2 < C_1C_2$

hence circle are separated and of course they are congurent



$$PQ = d - \frac{2\sqrt{d}}{2} = d - \sqrt{d}$$

$$CB = CQ + \sqrt{d} = \frac{d-\sqrt{d}}{2} + \sqrt{d} = \frac{d+\sqrt{d}}{2}$$

$$A(d) = \frac{\pi(d+\sqrt{d})^2}{4}$$

$$\therefore \quad \lim_{d\to\infty}\frac{\pi(d+\sqrt{d})^2}{4} = \lim_{d\to\infty}\frac{\pi d^2\left(1+\frac{1}{\sqrt{d}}\right)^2}{4d^2} = \frac{\pi}{4} \text{ Ans.}$$

57. $$F(x) = \begin{cases} f(x), & -1<x<1 \\ g(x), & x\in(-\infty,-1)\cup(1,\infty) \\ \frac{f(1)+g(1)}{2}, & x=1 \\ \frac{f(-1)+g(-1)}{2}, & x=-1 \end{cases}$$

continuous at $x = 1$ $\Rightarrow$ $F(1^-) = F(1^+) = F(1)$

$\Rightarrow$ $f(1^-) = g(1^+) = \frac{f(1)+g(1)}{2}$

$\Rightarrow$ $4 + a = 1 + b = \frac{(4+a)+(1+b)}{2} = \frac{5+a+b}{2}$.

$\therefore$ $b - a = 3$. ....(1)

continuous at $x = -1$ $\Rightarrow$ $F(-1^-) = F(-1^+) = F(-1)$

$\Rightarrow \quad b - 1 = 4 - a = \dfrac{(4-a)+(b-1)}{2}$

$\Rightarrow \quad b - 1 = 4 - a = \dfrac{3+b-a}{2}$

$\therefore \quad a + b = 5 \qquad ....(2)$

(1) and (2)

$\Rightarrow \quad a = 1, b = 4 \Rightarrow a^2 + b^2 = 17.$ **Ans.**

58. We have

$$\text{Given limit} = \lim_{n\to\infty}\left[\left(1+\frac{1}{n^2}\right)^{n^2}\right]^{\frac{4\left(\sin\frac{\pi}{2n}\right)}{\frac{\pi}{2n}}} = e^{4\times 1} = e^4 \quad \textbf{Ans.}$$

59. Since $5x^2 - 18$ is a quadratic polynomial and $P(x) + P(2x) = 5x^2 - 18$ it follows that $P(x)$ must be a quadratic polynomial. Suppose

$P(x) = ax^2 + bx + c$

By hypothesis

$(ax^2 + bx + c) + (4ax^2 + 2bx + c) = 5x^2 - 18$

or $\quad 5x^2 + 3bx + 2c = 5x^2 - 18$

This gives

$a = 1, b = 0, c = -9$

So $\quad P(x) = x^2 - 9$

Therefore $\quad \displaystyle\lim_{x\to 3}\frac{P(x)}{x-3} = \lim_{x\to 3}(x+3) = 6$ **Ans.**

60. Since $f(x)$ and $g(x)$ are 2 continuous function in $[0, 1]$

$\therefore \quad \exists\ a, b \in [0, 1]$

such that $f(\alpha) = M$ and $g(\beta) = M$ (where M is the maximum value)

Now consider a function $h(x) = f(x) - g(x)$ in $[\alpha, \beta]$

$h(\alpha) = M - g(\alpha) \ge 0$

$h(\beta) = f(\beta) - M \le 0$

$\therefore \quad h(\alpha) \cdot h(\beta) \le 0$

$\Rightarrow \quad$ Using I.V.T. $\exists\ c \in [0, 1]$ such that $h(c) = 0$

hence $f(c) = g(c) \quad \Rightarrow \quad f^2(c) + 3f(c) = g^2(c) + 3g(c)$.

61. $$\text{Given limit} = \lim_{x\to 0}\frac{\left(\left(\sin^{-1}x\right)^{2010} - x^{2010}\right)\cos^{2010}x}{x^{2010}\left(\dfrac{\ln\left(1+x^{2010}\right)}{x^{2010}}\right)\left(1+\sin^{2010}x\right)\left(\dfrac{\tan^2 x}{x^2}\right)x^2}$$

$$= \lim_{x\to 0}\frac{\left(\sin^{-1}x\right)^{2010} - x^{2010}}{x^{2012}} \quad (\text{Let } x = \sin\theta)$$

$$\therefore \text{given limit} = \lim_{\theta \to 0} \frac{\theta^{2010} - \sin^{2010}\theta}{\theta^{2012}}$$

$$= \lim_{\theta \to 0} \frac{\theta^{2010} - \left(\theta - \dfrac{\theta^3}{3!} + \ldots\right)^{2010}}{\theta^{2012}} = \lim_{\theta \to 0} \frac{1 - \left(1 - \dfrac{\theta^2}{3!} + \ldots.\right)^{2010}}{\theta^2}$$

$$= \lim_{\theta \to 0} \frac{1 - \left(1 - \dfrac{2010}{6}\theta^2 + \ldots\ldots\right)}{\theta^2} = \frac{2010}{6} = 335 \textbf{ Ans.}$$

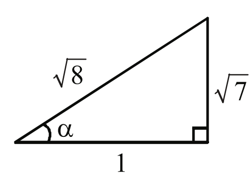
62. $\displaystyle \frac{1}{2}\sum_{k=1}^{n}\left(\frac{1}{k} - \frac{1}{k+1}\right) = \frac{1}{2}\left(\frac{3}{2} - \left(\frac{1}{n+1} + \frac{1}{n+2}\right)\right)$ working

$$\frac{1}{2}\left(\begin{array}{ccc} \Big\{1 & - & \dfrac{1}{3} \\ \dfrac{1}{2} & - & \dfrac{1}{4} \\ \dfrac{1}{3} & - & \dfrac{1}{5} \\ \vdots & & \vdots \\ \dfrac{1}{n-1} & - & \dfrac{1}{n+1} \\ \dfrac{1}{n} & - & \dfrac{1}{n+2}\Big\} \end{array}\right)$$

$$L = \lim_{n \to \infty}\left(2\left(\frac{3}{4} - \frac{1}{2}\left(\frac{1}{n+1} + \frac{1}{n+2}\right) - \frac{1}{n}\right)\right)^n = \lim_{n \to \infty}\left(1 - \left(\frac{2n+3}{(n+1)(n+2)}\right)\right)^n \quad (1^\infty)$$

$$e^{\lim\limits_{n \to \infty} \frac{n(2n+3)}{(n+1)(n+2)}} = e^{-2} \textbf{ Ans.}$$

63. Let $\alpha = \sin^{-1}\sqrt{\dfrac{7}{8}}$



$$\lim_{x \to \alpha} \frac{\tan^2 x - 7}{8\sin^2 x - 7(\sin^2 x + \cos^2 x)}$$

$$= \lim_{x \to \alpha} \frac{\tan^2 x - 7}{\sin^2 x - 7\cos^2 x} = \lim_{x \to \alpha} \frac{1}{\cos^2 \alpha} = 8 \quad \textbf{Ans.}$$

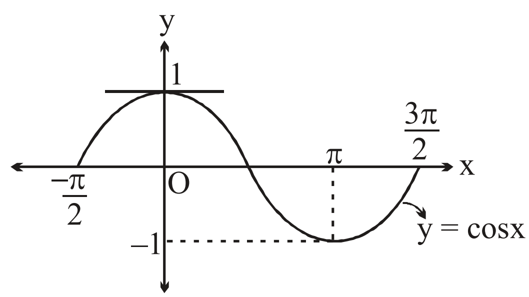
64. $f(x) = \lim_{n\to\infty} x\left\{\frac{3}{2}+[\cos x]\left(\sqrt{n^2+1}-\sqrt{n^2-3n+1}\right)\right\}$

$= \frac{3x}{2} + x\cdot[\cos x]\cdot \lim_{n\to\infty}\left(\sqrt{n^2+1}-\sqrt{n^2-3n+1}\right)$

$= \frac{3x}{2} + x[\cos x]\cdot \lim_{n\to\infty}\frac{n^2+1-(n^2-3n+1)}{\sqrt{n^2+1}+\sqrt{n^2-3n+1}}$

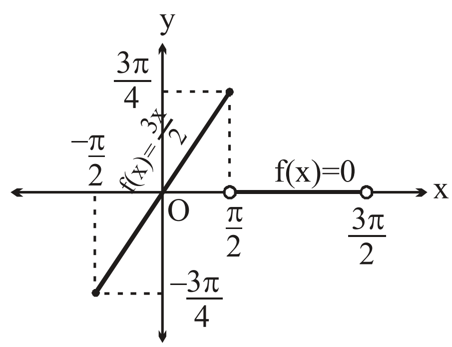
$f(x) = \frac{3x}{2} + x[\cos x]\cdot\frac{3}{2} = \frac{3x}{2}\left(1+[\cos x]\right)$. Now interpret

$$[\cos x] = \begin{cases} 0 & x\in\left[\frac{-\pi}{2},\frac{\pi}{2}\right]-\{0\} \\ 1 & x=0 \\ -1 & x\in\left(\frac{\pi}{2},\frac{3\pi}{2}\right)\end{cases}$$



$$f(x) = \begin{cases} \frac{3x}{2} & -\frac{\pi}{2}\le x\le\frac{\pi}{2} \\ 0 & \frac{\pi}{2}<x<\frac{3\pi}{2}\end{cases}$$

$\therefore$ Graph of $f(x)$ in $\left(-\frac{\pi}{2},\frac{3\pi}{2}\right)$



$f(x) = \frac{3x}{2}\left(1+[\cos x]\right)$

$\lim_{x\to 0^+} = 0$ and $\lim_{x\to 0^-} = 0$

$\lim_{x\to\frac{\pi}{2}^-} = \frac{3\pi}{2}$ and $\lim_{x\to\frac{\pi}{2}^+} = 0$.

65. $\displaystyle \lim_{x\to 0} \frac{(1+x)\ln(1+x)-x}{x^2}$

$\ln(1+x) = t \quad \Rightarrow \quad 1 + x = e^t$

$$= \lim_{t\to 0} \frac{e^t\cdot t-(e^t-1)}{\left(\frac{e^t-1}{t}\right)^2\cdot t^2} = \lim_{t\to 0}\frac{te^t-e^t+1+t-t}{t^2}$$

$$= \lim_{t\to 0}\left[\frac{t(e^t-1)}{t^2}-\frac{e^t-t-1}{t^2}\right] = 1-\frac{1}{2} = \frac{1}{2} \textbf{ Ans.}$$

66. $\displaystyle \lim_{x\to 0}\frac{2(\tan x-\sin x)-x^3}{x^5}$

$$=\lim_{x\to 0}\frac{2\tan x(1-\cos x)-x^3}{x^5} = \lim_{x\to 0}\frac{2\sin x(1-\cos x)-x^3\cos x}{x^5}$$

$$= \lim_{x\to 0}\frac{4\sin^2(x/2)\cdot 2\sin(x/2)\cos(x/2)-x^3\cos x}{x^5} = \lim_{x\to 0}\frac{8\sin^3(x/2)\cdot\left(1-2\sin^2(x/4)\right)-x^3\cos x}{x^5}$$

$$= \lim_{x\to 0}\frac{8\sin^3(x/2)-16\sin^3(x/2)\cdot\sin^2(x/4)-x^3\left(1-2\sin^2(x/2)\right)}{x^5}$$

$$= \lim_{x\to 0}\frac{8\sin^3(x/2)-x^3}{x^5}-\frac{16\sin^3(x/2)\cdot\sin^2(x/4)+2x^3\sin^2(x/2)}{x^5}$$

$x/2 = t$

$$\lim_{x\to 0}\frac{8\sin^3 t-8t^3}{32t^5}-16\times\frac{\sin^3(x/2)}{8(x/2)^3}\times\frac{\sin^3(x/4)}{16(x/4)^2}+\frac{2\sin^2(x/2)}{4(x/2)^2}$$

$$= \lim_{x\to 0}\frac{\sin^3 t-t^3}{4t^5}-\frac{1}{8}+\frac{1}{2} = \lim_{x\to 0}\frac{1}{2}-\frac{1}{8}+\frac{(\sin t-t)}{4t^3}\times\frac{(\sin^2 t+t^2+t\sin t)}{t^2}$$

$$= \frac{3}{8}-\frac{3}{4}\times\frac{1}{6} = \frac{2}{8} = \frac{1}{4} \text{ Ans.}$$

Alternatively: $\displaystyle \lim_{x\to 0}\frac{2(\tan x-\sin x)-x^3}{x^5}$

$x = 2t$

$$\lim_{t\to 0}\frac{2(\tan 2t-\sin 2t)-8t^3}{32t^5}$$

$$= \lim_{t\to 0}\frac{2\left(\frac{2\tan t}{1-\tan^2 t}-\frac{2\tan t}{1+\tan^2 t}\right)-8t^3}{32t^5}$$

$$= \lim_{t\to 0} \frac{4\tan t(2\tan^2 t) - 8t^3(1-\tan^4 t)}{32t^5(1-\tan^4 t)} = \lim_{t\to 0} \frac{\tan^3 t - t^3 + t^3\tan^4 t}{4t^5}$$

$$= \lim_{t\to 0} \frac{\tan^3 t - t^3}{4t^5} + 0 = \lim_{t\to 0} \left[\frac{(\tan t - t)}{t^3} \times \frac{\tan^2 t + t^2 + t\tan t}{t^2}\right]$$

67. for $-1 < x < 1$, $\{-x^2\} = -x^2 + 1$

$$\alpha = \lim_{x\to 0} \cos^{-1}\left(\frac{-x^2+1}{x^2+2x+2}\right) = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

$\therefore \quad \cos\alpha = \cos\frac{\pi}{3} = \frac{1}{2}$

$$\beta = \lim_{x\to\infty} \tan^{-1}\left(\frac{e^{-x^2} - e^x}{2e^{-x^2} + e^x}\right) = \lim_{x\to\infty} \tan^{-1}\left(\frac{\frac{e^{-x^2}}{e^x} - 1}{\frac{2e^{-x^2}}{e^x} + 1}\right) = \tan^{-1}(-1)$$

$\therefore \quad \tan\beta = -1$

$\cos\alpha + \tan\beta = \frac{1}{2} - 1 = \frac{-1}{2}.$

68. $$H_n = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + ... + \frac{1}{a_n}} = \frac{n}{\sum \frac{1}{a_r}}$$

$$\therefore \quad \frac{n}{1+n} = \sum_{r=1}^{n} \frac{r}{n^2+n+r}$$

$$\lim_{n\to\infty} \frac{n}{H_n} = \lim_{n\to\infty}\left(\frac{1}{n^2+n+1} + \frac{2}{n^2+n+2} + ... + \frac{n}{n^2+n+n}\right) = \frac{1}{2}$$

$\therefore \quad \lim_{n\to\infty} \frac{H_n}{n} = 2.$ **Ans.**

69. We expand the numerator in series

$$(1+x)^{1/x} - e + \frac{ex}{2} = e^{\frac{1}{x}\ln(1+x)} - e + \frac{ex}{2}$$

$$= e^{\left(1 - \frac{x}{2} + \frac{x^2}{3} + ...\right)} - e + \frac{ex}{2} = e \cdot e^{\left(-\frac{x}{2} + \frac{x^2}{3} + ...\right)} - e + \frac{ex}{2} =$$

$$= e\left[1+\left(-\frac{x}{2}+\frac{x^2}{3}-.....\right)+\frac{\left(-\frac{x}{2}+\frac{x^2}{3}-.........\right)^2}{2!}\right]-e+\frac{ex}{2} = e\left(\frac{1}{3}+\frac{1}{8}\right)x^2+.......$$

$\therefore$ The limit $= e\left(\frac{1}{3}+\frac{1}{8}\right)=\frac{11e}{24}$ **Ans.**

70. We have, $a_m = \left[\left(\frac{m+1}{m}\right)^{m+1} - \left(\frac{m+1}{m}\right)\right]^{-m}$

$$\therefore (a_m)^{1/m} = \left[\left(\frac{m+1}{m}\right)^{m+1} - \frac{m+1}{m}\right]^{-1} = \left(\frac{m+1}{m}\right)^{-1}\left[\left(1+\frac{1}{m}\right)^m - 1\right]^{-1}$$

$$\therefore \lim_{m\to\infty}(a_m)^{1/m} = \lim_{m\to\infty}\left(1+\frac{1}{m}\right)^{-1}\left[\left(1+\frac{1}{m}\right)^m - 1\right]^{-1} = 1\cdot(e-1)^{-1} = \frac{1}{e-1}$$ **Ans.**

71. Let $f(x) = x^{\frac{1}{x}}$

$\Rightarrow \quad f'(x) = x^{\frac{1}{x}}\left(\frac{\ln x - 1}{x^2}\right)$

If $x > e$ then $f(x)$ is decreasing

$\because \quad 2010 > 2009 \quad \Rightarrow \quad f(2010) < f(2009)$

$\Rightarrow \quad (2010)^{\frac{1}{2010}} < (2009)^{\frac{1}{2009}}$

$\Rightarrow \quad (2010)^{2009} < (2009)^{2010}$

$$\because \quad \lim_{n\to\infty}\left\{\left((2009)^{2010}\right)^n + \left((2010)^{2009}\right)^n\right\}^{\frac{1}{n}}$$

$$= \lim_{n\to\infty}(2009)^{2010}\left\{1+\left(\frac{(2010)^{2009}}{(2009)^{2010}}\right)^n\right\}^{\frac{1}{n}} = (2009)^{2010}$$

72. $$\lim_{x\to 0^+}\frac{\frac{p\cos x}{e^{\frac{1}{x}}}+x}{\frac{1+\sin x}{e^{\frac{1}{x}}}+q\cos x} = \frac{0}{0+q} = 0 \Rightarrow q \in R-\{0\}$$

$$\lim_{x\to 0^-}\frac{p\cos x + xe^{\frac{+1}{x}}}{1+\sin x + q\cos x e^{\frac{1}{x}}} = p = 0$$

$\Rightarrow$ $p = 0,\ q \in R - \{0\}$

73. $L = \lim_{x\to 0}\dfrac{x^4 + 3\left(a^2 - \sqrt{a^4+x^4}\right)}{x^8}\left(\dfrac{0}{0}\right) = \lim_{x\to 0}\dfrac{(x^4+3a^2) - 3\sqrt{a^4+x^4}}{x^8}\left(\dfrac{0}{0}\right)$

$$= \lim_{x\to 0}\frac{\left(x^4+3a^2\right)^2 - 9\left(a^4+x^4\right)}{x^8\left(\left(x^4+3a^2\right)+3\sqrt{a^4+x^4}\right)} \text{ (on rationalising) } = \frac{1}{6a^2}\lim_{x\to 0}\left(\frac{x^8+6a^2x^4-9x^4}{x^8}\right)$$

$$L = \frac{1}{6a^2}\lim_{x\to 0}\left(1+\frac{\left(6a^2-9\right)}{x^4}\right)$$

Clearly $6a^2 - 9 = 0 \Rightarrow a^2 = \frac{9}{6} = \frac{3}{2}$

$\therefore\ a = \sqrt{\frac{3}{2}}$ (As $a > 0$)

Also $L = \dfrac{1}{6a^2} = \dfrac{1}{6\left(\frac{9}{6}\right)} = \dfrac{1}{9}$

74. $\left(n + x_n\right)\ln\left(\dfrac{n+1}{n}\right) = 1$

$$x_n = \frac{1}{\ln\frac{n+1}{n}} - n = \frac{1 - n\ln\frac{n+1}{n}}{\ln\frac{n+1}{n}}$$

Let $1 + \frac{1}{n} = t,\ \ n = \frac{1}{t-1}$ as $n \to \infty,\ t \to 1$

$$\therefore \quad \lim_{t\to 1}\frac{1-\frac{\ln t}{t-1}}{\ln t} = \frac{(t-1)-\ln t}{(t-1)\ln t}$$

Put $t = 1 + h$

$$\lim_{h\to 0} \frac{h - \ln(1+h)}{h^2 \frac{\ln(1+h)}{h}} = \frac{1}{2} \textbf{ Ans.}$$

75. Given $a_{n+1} - a_n = 4n + 3$

Now $a_k - a_1 = \sum_{n=1}^{k-1}(a_{n+1} - a_n) = \sum_{n-1}^{k-1}(4n+3) = 4(k-1)\frac{(k)}{2} + 3(k-1) = 2k^2 + k - 3$

$\Rightarrow \quad a_k = 2k^2 + k - 3.$

So $\lim_{k\to\infty} \frac{\sqrt{2k^2+k-3} + \sqrt{2(4k)^2+(4k)-3} + \ldots\ldots + \sqrt{2(4^{10}k)^2+(4^{10}k)-3}}{\sqrt{2k^2+k-3} + \sqrt{2(2k)^2+(2k)-3} + \ldots\ldots + \sqrt{2(2^{10}k)^2+(2^{10}k)-3}}$

$$= \lim_{k\to\infty} \frac{\sqrt{2+\frac{1}{k}-\frac{3}{k^2}} + \sqrt{2(4^2)+\frac{4}{k}-\frac{3}{k^2}} + \ldots\ldots + \sqrt{2(4^{10})^2+\frac{4^{10}}{k}-\frac{3}{k^2}}}{\sqrt{2+\frac{1}{k}-\frac{3}{k^2}} + \sqrt{2(2)^2+\frac{2}{k}-\frac{3}{k^2}} + \ldots\ldots + \sqrt{2(2^{10})^2+\frac{2^{10}}{k}-\frac{3}{k^2}}}$$

$$= \frac{\sqrt{2} + \sqrt{2(4^2)} + \ldots\ldots + \sqrt{2(4^{10})^2}}{\sqrt{2} + \sqrt{2(2)^2} + \ldots\ldots + \sqrt{2(2^{10})^2}} = \frac{\sqrt{2}\left(1+4+4^2+\ldots\ldots+4^{10}\right)}{\sqrt{2}\left(1+2+2^2+\ldots\ldots+2^{10}\right)} = \frac{\frac{1}{3}(4^{11}-1)}{2^{11}-1} = 683 \textbf{ Ans.}$$

76. Denote $L_1 = \lim_{x\to 0} \frac{\sqrt{\frac{\cos 2x + (1+3x)^{1/3}}{2}} - 1}{x}$ and $L_2 = \lim_{x\to 0} \frac{\sqrt[3]{\frac{4\cos^3 x - \ln(1+x)^4}{4}} - 1}{x}$

we have $\quad L = L_1 - L_2$

Now for L_1 on rationalisation

$$L_1 = \lim_{x\to 0} \frac{\frac{\cos 2x + (1+3x)^{1/3}}{2} - 1}{x\cdot\left(\sqrt{\frac{\cos 2x + (1+3x)^{1/3}}{2}} + 1\right)}$$

$$L_1 = \frac{1}{2}\cdot \lim_{x\to 0} \frac{[(1+3x)^{1/3} - 1] - [1 - \cos 2x]}{2x} = \frac{1}{4}\cdot\left[\lim_{x\to 0}\frac{(1+3x)^{1/3}-1}{x} - \lim_{x\to 0}\underbrace{\frac{1-\cos 2x}{x}}_{\text{zero}}\right]$$

$$= \frac{1}{4}\cdot\left[\lim_{x\to 0}\frac{(1+x+\ldots\ldots-1)}{x} - 0\right] = \frac{1}{4}$$

for $\quad L_2 = \lim_{x\to 0} \frac{\left[\frac{4\cos^3 x - 4\ln(1+x)}{4}\right]^{1/3} - 1}{x} = \lim_{x\to 0}\frac{[\cos^3 x - \ln(1+x)]^{1/3} - 1}{x}$

$$= \frac{1}{3} \cdot \lim_{x \to 0} \frac{\cos^3 x - \ln(1+x) - 1}{x} \qquad \left(\text{Using } a - b = \frac{a^3 - b^3}{a^2 + ab + b^2}\right)$$

now $\quad L_2 = \frac{1}{3} \cdot \left[\lim_{x \to 0} \frac{(\cos x - 1)}{x} \cdot \lim_{x \to 0} (\cos^2 x + 1 + \cos x) - \lim_{x \to 0} \frac{\ln(1+x)}{x} \right]$

$$= \frac{1}{3}\ [0 - 1] = -\frac{1}{3}$$

$\therefore \quad L = \frac{1}{4} + \frac{1}{3} = \frac{7}{12} \quad \Rightarrow \quad \boxed{a + b = 19}$ Ans.

77. $f(\theta) = 3\sin\theta + 2\cos\theta - 4(\sin^3\theta + \cos^3\theta) + 4(1 - \cos^2\theta)\cos\theta$
$= 3\sin\theta - 4\sin^3\theta + 6\cos\theta - 8\cos^3\theta = \sin 3\theta - 2\cos 3\theta$

$\therefore \quad -\sqrt{5} \le f(\theta) \le \sqrt{5}$

$\Rightarrow \quad a = -\sqrt{5},\ b = \sqrt{5} \quad \Rightarrow \quad [a] = -3$ and $[b] = 2$

Now, $\quad l_1 = \lim_{x \to -3^-} \frac{x^2 - x[x]}{2x - [x]} = \lim_{x \to -3^-} \frac{x^2 + 4x}{2x + 4} = \frac{9 - 12}{-6 + 4} = \frac{3}{2}$

$$l_2 = \lim_{x \to 2^+} \frac{x^2 - x[x]}{2x - [x]} = \lim_{x \to 2^+} \frac{x^2 - 2x}{2x - 2} = \frac{4 - 4}{4 - 2} = 0$$

Hence $(6l_1 - l_2) = 6\left(\frac{3}{2}\right) - 0 = 9$ Ans.

78. $l = \lim_{n \to \infty} \left(1 + \frac{na_n + 1}{n}\right)^n \to 1^\infty$

$$= e^{\lim_{n \to \infty} n \cdot \left(\frac{na_n + 1}{n}\right)} = e \quad \Rightarrow \quad [e] = 2 \textbf{ Ans.}$$

79. $L = \lim_{n \to \infty} \frac{e^n}{\left(1 + \frac{1}{n}\right)^{n^2}}$

$$\ln L = \lim_{n \to \infty} \left(n - n^2 \ln\left(1 + \frac{1}{n}\right) \right)$$

$$n = \frac{1}{y}$$

$$\ln L = \lim_{y \to 0} \frac{y - \ln(1+y)}{y^2}$$

$$\ln L = \lim_{y\to 0} \frac{y - \left(y - \frac{y^2}{2} + \ldots\ldots\right)}{y^2} = \frac{1}{2}$$

$$\therefore\ L = e^{\frac{1}{2}} = \sqrt{e}\ \textbf{Ans.}$$

80. Put $n = \frac{1}{y}$

$$\lim_{y\to 0} \frac{\left(\frac{1+y}{1-y}\right)^{\frac{1}{y}} - e^2}{y^2} = \lim_{y\to 0} \frac{e^{\frac{1}{y}\ln\left(\frac{1+y}{1-y}\right)} - e^2}{y^2}$$

$$= e^2 \lim_{y\to 0} \frac{e^{\frac{\ln\left(\frac{1+y}{1-y}\right)-2}{y}} - 1}{y^2} = e^2 \lim_{y\to 0} \frac{\frac{\ln\left(\frac{1+y}{1-y}\right) - 2}{y} - 1}{y^2}$$

$$= e^2 \lim_{y\to 0} \frac{\ln(1+y) - \ln(1-y) - 2y}{y^3} = e^2\left(\frac{2}{3}\right) = \frac{2e^2}{3} \Rightarrow a = 2\ ;\ b = 3$$

$\therefore\ a + b = 5$ **Ans.**

81. $$\lim_{x\to\infty} \frac{(Ax^2 + Bx) - C^2x^2}{\sqrt{Ax^2 + Bx} + Cx} = 2; \qquad \lim_{x\to\infty} \frac{(A - C^2)x^2 + Bx}{x\left[\sqrt{A + (B/x)} + C\right]} = 2$$

for existence of limit $A = C^2$ hence $\frac{BC}{A} = \frac{B}{C}$ (using $A = C^2$)

$$\therefore\quad L = \frac{B}{\sqrt{A} + C} = 2$$

if $C = -\sqrt{A}$ then limit does not exist hence $C = \sqrt{A}$

$$\therefore\quad \frac{B}{2C} = 2 \qquad \Rightarrow \qquad \frac{B}{C} = 4\ \textbf{Ans.}$$

82. $$\sum_{r=1}^{n} \frac{4r^3}{(n^4 + 4n^3 + 2n^2 + n + r)}$$

$$= 4\left\{\frac{1^3}{n^4 + 4n^3 + 2n^2 + n + 1} + \frac{2^3}{n^4 + 4n^3 + 2n^2 + n + 2} + \ldots\ldots + \frac{n^3}{n^4 + 4n^3 + 2n^2 + n + n}\right\}$$

$\because \quad \dfrac{4\{1^3+2^3+.....n^3\}}{n^4+4n^3+2n^2+n+n} < \sum_{r=1}^{n} \dfrac{4r^3}{(n^4+4n^3+2n^2+n+r)} < \dfrac{4\{1^3+2^3+........n^3\}}{n^4+4n^3+2n^2+n+1}$

$\Rightarrow \quad \dfrac{n^2(n+1)^2}{n^4+4n^3+2n^2+n+n} < \sum_{r=1}^{n} \dfrac{4r^3}{(n^4+4n^3+2n^2+n+r)} < \dfrac{n^2(n+1)^2}{n^4+4n^3+2n^2+n+n}$

$\because \quad \lim_{n\to\infty} \dfrac{n^2(n+1)^2}{n^4+4n^3+2n^2+n+n} = 1 \quad \& \quad \lim_{n\to\infty} \dfrac{n^2(n+1)^2}{n^4+4n^3+2n^2+n+1} = 1$

$\therefore$ According to sandwich theorem.

$$\lim_{n\to\infty} \sum_{r=1}^{n} \frac{4n^3}{(n^4+4n^3+2n^2+n+r)} = 1.$$

83. $\lim_{x\to 0} \dfrac{x\sin(\sin x)-\sin^2 x}{x^6}$

Put $\sin x = t$

$$\Rightarrow \quad \lim_{x\to 0} \frac{(\sin^{-1} t)(\sin t)-t^2}{t^6}$$

Now $\sin^{-1}x = x+\dfrac{x^3}{6}+\dfrac{3}{40}x^5+\dfrac{5}{112}x^7$ ....

and $\tan^{-1}x = x-\dfrac{x^3}{3}+\dfrac{x^5}{5}-\dfrac{x^7}{7}$.........

Use this $\Rightarrow$ get the result. $= \dfrac{1}{18}$ **Ans.**

84. $\sin^{-1}x = x+\dfrac{x^3}{6}+\dfrac{3}{40}x^5+\dfrac{5}{112}x^7$ .... and $\tan^{-1}x = x-\dfrac{x^3}{3}+\dfrac{x^5}{5}-\dfrac{x^7}{7}$.........

$$\lim_{x\to 0}\left(\frac{2}{x^3}\left(\sin^{-1}x-\tan^{-1}x\right)\right)^{\frac{2}{x^2}} \quad (1^\infty)$$

$e^{\overbrace{\lim_{x\to 0}\frac{2}{x^2}\left(\frac{2}{x^3}\left(\sin^{-1}x-\tan^{-1}x\right)-1\right)}^{L}}$ So ; $L = 2\lim_{x\to 0}\dfrac{2\left(\sin^{-1}x-\tan^{-1}x\right)-x^3}{x^5}$

$$= 2\lim_{x\to 0}\frac{2\left(x+\frac{x^3}{6}+\frac{3}{40}x^5+\frac{5}{112}x^7-x+\frac{x^3}{3}-\frac{x^5}{5}+\frac{x^7}{7}.........\right)-x^3}{x^5} = -\frac{1}{2}$$

$e^{-1/2} \quad \Rightarrow$ **(D) Ans.**

85. Putting $n = \frac{1}{x}$, we get

given limit $= \lim\limits_{x \to 0} \frac{1}{x^2}\left[\frac{1}{x^2} ln\left(\frac{1}{x^2+1}\right)+1\right] = \lim\limits_{x \to 0} \frac{\left\{x^2 - ln\,(1+x^2)\right\}}{x^4}$

$$= \lim_{x \to 0} \frac{x^2 - \left(x^2 - \frac{x^4}{2} + ...\right)}{x^4} = \frac{1}{2} \textbf{ Ans.}$$

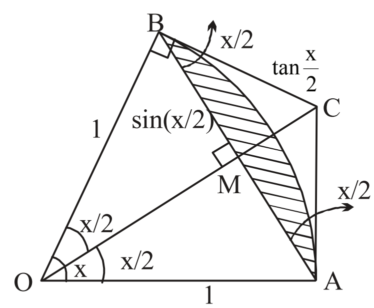
86. a = 2 and b = 3

87. S(x) = area of shaded region
= area of circle sector – area of ΔOAB

$= \frac{1}{2}x - \frac{1}{2}\sin x = \frac{x - \sin x}{2}$

T(x) = Area of ΔABC = area of quadrilateral OACB – Area of ΔOAB

$$= \tan\frac{x}{2} - \frac{1}{2}x = \frac{2\tan\frac{x}{2} - x}{2}$$



$$\therefore \quad \lim_{x\to 0} \frac{T(x)}{S(x)} = \lim_{x\to 0} \frac{2\tan\frac{x}{2} - x}{\frac{x - \sin x}{x^3}\cdot x^3} = \frac{3}{2} \textbf{ Ans.}$$

T (x) = area of quadrilateral OACB – area of triangle OAB

$$= \tan\left(\frac{x}{2}\right) - \frac{1}{2}\sin x = \tan\frac{x}{2} - \frac{\sin x}{2} = \tan\frac{x}{2} - \sin\frac{x}{2}\cos\frac{x}{2}$$

$$= \frac{\cos\frac{x}{2} - \sin\frac{x}{2}\cos\frac{x}{2}}{\cos\frac{x}{2}} = \frac{\sin^3\frac{x}{2}}{\cos\frac{x}{2}}$$

88. $$L = \lim_{\substack{x\to 0 \\ n\to\infty}} \frac{1}{n^2}\left[\frac{(1+x)^{\frac{1}{x}} - e}{x} + \frac{(1+2x)^{\frac{1}{2x}} - e}{x} + ... + \frac{(1+nx)^{\frac{1}{nx}} - e}{x}\right]$$

Consider $l_n = \lim\limits_{x\to 0} \frac{(1+nx)^{\frac{1}{nx}} - e}{x} = \lim\limits_{x\to 0} \frac{e^{\frac{ln(1+nx)}{nx}} - e}{x} = e\lim\limits_{x\to 0} \frac{e^{\frac{ln(1+nx)}{nx}-1} - 1}{x}$

Let $\dfrac{\ln(1+nx)}{nx} - 1 = y$

so that as $x \to 0, y \to 0$

$$\therefore \qquad l_n = e \lim_{y\to 0} \frac{e^y - 1}{y} \cdot \frac{y}{x} = e \cdot \lim_{x\to 0} \frac{\ln(1+nx) - nx}{n x^2}$$

$$l_n = \lim_{x\to 0} \frac{nx - \dfrac{n^2x^2}{2} + ... - nx}{n x^2} \cdot e = \frac{-ne}{2}$$

$$L = \lim_{n\to\infty} \frac{-1}{n^2}\frac{e}{2}\sum n = \frac{-1}{n^2}\frac{e}{2}\,[1+2+3+......+n] = \frac{-e}{2n^2}\frac{n(n+1)}{2}$$

$$= \frac{-e}{4} = \left(\frac{-1}{4}\right)e = \frac{p}{q}\cdot e$$

$\therefore \qquad p = -1;\ q = 4 \quad \Rightarrow \quad p + q = 3$ **Ans.**

89. as $x \to 1$; $\quad D^r \to 0$ and $N^r \to 1 + a + b$

hence for existence of limit $a + b + 1 = 0$ $\quad$ ....(1)

now $\quad l = \lim\limits_{x\to 1} \dfrac{x^2 + ax - a - 1}{(\ln(2-x))^2} = \lim\limits_{x\to 1} \dfrac{(x-1)(x+1+a)}{\ln^2(2-x)}$ $\qquad$ (put $x = 1 + h$)

$$l = \lim_{h\to 0} \frac{h(2+a+h)}{\ln(1-h)\ \ln(1-h)} = \lim_{h\to 0} \frac{(2+a+h)}{h}$$

for existence $2 + a = 0$; $a = -2$ and $l = 1$

from (1), $\qquad b = 1.$]

90. Consider $\dfrac{1}{2}\displaystyle\sum_{k=1}^{n} \frac{1}{(2k-1)(2k+1)} = \frac{1}{2}\cdot\frac{1}{2}\sum_{k=1}^{n} \frac{(2k+1)-(2k-1)}{(2k-1)(2k+1)}$

$$= \frac{1}{4}\sum_{k=1}^{n}\left(\frac{1}{(2k-1)} - \frac{1}{(2k+1)}\right)$$

$$= \frac{1}{4}\left[\frac{1}{1} - \frac{1}{3}\right]$$

$$+ \frac{1}{4}\left[\frac{1}{5} - \frac{1}{7}\right]$$

$$\vdots \qquad \vdots \qquad \vdots$$

$$+ \frac{1}{4}\left[\frac{1}{2n-1} - \frac{1}{2n+1}\right]$$

$$\frac{1}{4}\left[\frac{1}{1}-\frac{1}{2n+1}\right]$$

$$a_n=\left[1+\frac{1}{4}\left(1-\frac{1}{2n+1}\right)\right]^n=\left[\frac{5}{4}-\frac{1}{4(2n+1)}\right]^n \qquad ......(1)$$

(i) $\lim\limits_{n\to\infty} a_n \to \infty$ i.e. does not exist. **Ans.**

(ii) Now $a_{n+1}=\left[\frac{5}{4}-\frac{1}{4(2n+3)}\right]^{n+1}=\left[\frac{5}{4}-\frac{1}{4(2n+3)}\right]^n\cdot\left[\frac{5}{4}-\frac{1}{4(2n+3)}\right]$

$$\therefore \quad \frac{a_{n+1}}{a_n}=\lim_{n\to\infty}\left[\frac{5}{4}-\frac{1}{4(2n+3)}\right]\cdot\left[\frac{\frac{5}{4}-\frac{1}{4(2n+3)}}{\frac{5}{4}-\frac{1}{4(2n+1)}}\right]^n \; ; \quad \frac{a_{n+1}}{a_n}=\frac{5}{4}\lim_{n\to\infty}\left[\frac{\frac{5}{4}-\frac{1}{4(2n+3)}}{\frac{5}{4}-\frac{1}{4(2n+1)}}\right]^n$$

Consider $\lim\limits_{n\to\infty}\left[\dfrac{\frac{5}{4}-\frac{1}{4(2n+3)}}{\frac{5}{4}-\frac{1}{4(2n+1)}}\right]^n \qquad (1^\infty)$

On dividing N^r and D^r by $\frac{5}{4}$, we get

$$\left[\frac{1-\frac{1}{5(2n+3)}}{1-\frac{1}{5(2n+1)}}\right]^n=e^l \text{ (say)}$$

$$\text{where } l=\lim_{n\to\infty}\left[\frac{\frac{1}{5(2n+1)}-\frac{1}{5(2n+3)}}{1-\frac{1}{5(2n+1)}}\right]=\lim_{n\to\infty} n\left[\frac{\frac{2}{5}\cdot\frac{1}{(2n+1)(2n+3)}\cdot 5(2n+1)}{5(2n+1)-1}\right]$$

$$=\lim_{n\to\infty} n\left[\frac{2}{(2n+3)}\cdot\frac{1}{10n+4}\right]=0 \quad\Rightarrow\quad e^l=1$$

Hence $\lim\limits_{n\to\infty}\dfrac{a_{n+1}}{a_n}=\dfrac{5}{4}$ **Ans.**

91. $\lim\limits_{n\to\infty}\left(\dfrac{1}{\sqrt{n^2+1}}+\dfrac{1}{\sqrt{n^2+2}}+.......+\dfrac{1}{\sqrt{n^2+n}}\right)\to 1$

this limit is 1^∞.

Required limit = $\lim\limits_{n\to\infty} e^{n\left[\sum\limits_{k=1}^{n}\frac{1}{\sqrt{n^2+k}}-1\right]} = e^{L}$.

Now, $L = \lim\limits_{n\to\infty} n\sum\limits_{k=1}^{n}\left(\frac{1}{\sqrt{n^2+k}}-\frac{1}{n}\right) = \lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\frac{n-\sqrt{n^2+k}}{\sqrt{n^2+k}}$

$$= -\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\frac{k}{\sqrt{n^2+k}\left(n+\sqrt{n^2+k}\right)}$$

$$= -\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\frac{k}{n\sqrt{n^2+k}+n^2+k}$$

Now, $\frac{k}{n\sqrt{n^2+n}+n^2+n} \le \frac{k}{n\sqrt{n^2+k}+n^2+k} \le \frac{k}{n\sqrt{n^2+1}+n^2+1}$

Apply Sandwich theorem, we get $L = \frac{-1}{4}$. **Ans.**

92. $\lim\limits_{x\to 0}(1+ax+bx^2)^{2/x} = e^3$

now this is 1^∞

$e^{\lim\limits_{x\to 0}\frac{2}{x}(ax+bx^2)} = e^{\lim\limits_{x\to 0} 2a+2bx} = e^{2x}$ so $e^{2a} = e^3$; $a = 3/2$ and $b \in R$

93. $\lim\limits_{x\to 0}\left(1+\frac{P(x)}{x^4}\right)^{\frac{1}{x-\sin(x)}} = e^8$

now this mut be 1^∞ so P(x) must be of degree greater than ≥ 4
i.e. $P(x) = ax^4 + bx^5 + cx^6$ ....
now apply the formula

$e^{\lim\limits_{x\to 0}\frac{1}{x-\sin x}\cdot\frac{P(x)}{x^4}} = e^8$ i.e. $\lim\limits_{x\to 0}\frac{1}{x-\sin x}\cdot\frac{P(x)}{x^4} = 8$ i.e. $\lim\limits_{x\to 0}\frac{x^3}{x-\sin x}\cdot\frac{P(x)}{x^4\cdot x^3} = 8$

i.e. $\lim\limits_{x\to 0}\frac{P(x)}{x^7} = \frac{8}{6}$

now P(x) must be of least degree so its degree must be 7
so $P(x) = ax^4 + bx^5 + cx^6 + dx^7$

when $a = b = c = 0$ and $d = \frac{4}{3}$

so $P(x) = \frac{4}{3}x^7$ Ans.

94. $\displaystyle \lim_{x\to 0}\frac{a\sin x-\sin 2x}{\frac{\tan^3 x}{x^3}\cdot x^3}=\lim_{x\to 0}\frac{\sin x[a-2\cos x]}{x\cdot x^2}=\frac{a-2\cos x}{x^2}$

$\therefore$ $x\to 0$, $a-2\cos x$ should be zero for limit to exist $\Rightarrow \displaystyle\lim_{x\to 0}(a-2\cos x)=0$

$\therefore$ $a-2=0 \quad\Rightarrow\quad a=2$ **Ans.**

$\therefore$ $\displaystyle l=\lim_{x\to 0}\frac{2[1-\cos x]}{x^2}=1$ **Ans.**

Sol. **95. to 97**

now $f(x)=e^{\lim\limits_{n\to\infty} n\left(\cos\sqrt{\frac{x}{n}}-1\right)}=e^{\lim\limits_{n\to\infty}\frac{-\sin^2\left(\sqrt{\frac{x}{n}}\cdot\frac{1}{2}\right)}{\left(\sqrt{\frac{x}{n}}\cdot\frac{1}{2}\right)^2}\cdot\left(\sqrt{\frac{x}{n}}\cdot\frac{1}{2}\right)^2\cdot 2n}$

$\displaystyle =e^{-2x/4}=e^{-x/2}\;\Rightarrow\; f^{-1}(x)=\ln\frac{1}{x^2}$

$g(x)=e^{\lim\limits_{n\to-\infty} n\left(x e^{1/n}-x\right)}=e^{x\lim\limits_{n\to\infty}\frac{e^{1/n}-1}{1/n}}=e^x \quad\Rightarrow\quad g^{-1}(x)=\ln x$

now $\displaystyle\lim_{x\to 0}\frac{\ln(f(x))}{\ln(g(x))}=\frac{-x/2}{x}=-\frac{1}{2}$ **Ans(i).**

$h(x)=\tan^{-1}(g^{-1}f^{-1}(x))$; $\qquad h(x)=\tan^{-1}\left(\ln\left(\ln\frac{1}{x^2}\right)\right)$

and hence domain of $h(x)$ is $(0, 1)$ and range is $(-\pi/2, \pi/2)$

98. $\because$ $\displaystyle\lim_{x\to\infty}\left(\frac{x+A}{x-2A}\right)^x=5\;\Rightarrow\; e^{\lim\limits_{x\to\infty} x\left(\frac{x+A}{x-2A}-1\right)}=5$

$\Rightarrow \displaystyle\lim_{x\to\infty}\frac{x\,(3A)}{x-2A}=\ln 5\;\Rightarrow 3A=\ln 5\;\Rightarrow\; A=\frac{1}{3}\ln 5.$ **Ans.**

99. Given, $\displaystyle\lim_{x\to 0}\frac{\sin(ax)+bx}{x^3}=36$

$$\Rightarrow \lim_{x\to 0}\frac{\left(ax-\frac{a^3\cdot x^3}{3!}+\frac{a^5\cdot x^5}{5!}-........\infty\right)+bx}{x^3}=36$$

$$\Rightarrow \lim_{x \to 0} \frac{\left((a+b)x - \frac{a^3}{6} \cdot x^3 + \frac{a^5}{120} x^5 ....\infty\right)}{x^3} = 36.$$

So, $a + b = 0$.

Also, $\frac{-a^3}{6} = 36 \Rightarrow a = -6$ and $b = 6$. **Ans.**

100. $\lim_{n \to \infty} \left(\frac{n^2+n+3}{n^2+3n+5}\right)^n$ (1^∞)

$$= e^{\lim_{n\to\infty} n\left(\frac{n^2+n+3}{n^2+3n+5}-1\right)} = e^{\lim_{n\to\infty} n\left(\frac{-2n-2}{n^2+3n+5}\right)} = e^{-2} = \frac{1}{e^2} \equiv \frac{p}{e^2}$$ hence $p = 1$ **Ans.**

101. $\lim_{x \to -\infty} \frac{x^2 \tan\left(\frac{1}{x}\right) - x}{1-|x|}$

Put $x = \frac{1}{y}$, so $\lim_{y \to 0} \frac{\frac{1}{y^2} \cdot \tan y - \frac{1}{y}}{1+\frac{1}{y}} = \lim_{y \to 0} \frac{\left(\frac{\tan y}{y} - 1\right)}{y+1} = \frac{1-1}{0+1} = 0$. **Ans.**

102. $\lim_{n\to\infty} ln \left[\tan\left(\frac{\pi}{4}+\frac{\pi}{n}\right)\right]^n$ (1^∞ form)

$$= \lim_{n\to\infty} ln\, e^{\lim_{n\to\infty} n \cdot \left[\tan\left(\frac{\pi}{4}+\frac{\pi}{n}\right)-1\right]} = \lim_{n\to\infty} n \left(\frac{1+\tan\frac{\pi}{n}}{1-\tan\frac{\pi}{n}} - 1\right) = \lim_{n\to\infty} \frac{n \cdot 2\tan\frac{\pi}{n}}{\left(1-\tan\frac{\pi}{n}\right)}; \quad \frac{\pi}{n} = x$$

$$= 2\pi \lim_{x\to 0} \left(\frac{\tan x}{x(1-\tan x)}\right) = 2\pi$$ **Ans.**

103. As, $(x + \sin x - x\cos x - \tan x) = x(1 - \cos x) + \sin x \left(1 - \frac{1}{\cos x}\right)$

$= x(1 - \cos x) - \tan x (1 - \cos x) = (x - \tan x) \cdot (1 - \cos x)$

So, $\lim_{x \to 0} \frac{\left(\frac{x-\tan x}{x^3}\right)\left(\frac{1-\cos x}{x^2}\right)}{x^{n-5}}$ = exist and non-zero,

so, $n = 5$. **Ans.**

104. $\displaystyle \lim_{x\to 0}\frac{5\left(\frac{1-\cos x}{x}+\frac{4\sin^{-1}x}{x}-\frac{\sin^2 x}{x}\right)}{\left(\frac{10\tan^{-1}x}{x}-x+x^3\right)}$ (Divide numerator and denominator by x)

$$=\frac{5(0+4-0)}{(10-0+0)}=\frac{20}{10}=2.\ \textbf{Ans.}$$

105. We have

$$\lim_{n\to\infty}\left(\frac{1^2+2^2+3^2+......+n^2}{n^3+1}\right)=\lim_{n\to\infty}\left(\frac{n\cdot(n+1)\cdot(2n+1)}{6\cdot(n^3+1)}\right)=\frac{2}{6}=\frac{1}{3}.$$

Also, $\displaystyle \beta=\lim_{x\to 0}\frac{\sin 2x}{\sin 8x}=\lim_{x\to 0}\frac{2x\cdot\left(\frac{\sin 2x}{2x}\right)}{3x\cdot\left(\frac{\sin 8x}{8x}\right)}=\frac{2}{8}=\frac{1}{4}.$

Now, $\displaystyle \alpha+\beta=\frac{1}{3}+\frac{1}{4}=\frac{4+3}{12}=\frac{7}{12}.$

Also, $\displaystyle \alpha\beta=\left(\frac{1}{3}\right)\cdot\left(\frac{1}{4}\right)=\frac{1}{12}.$

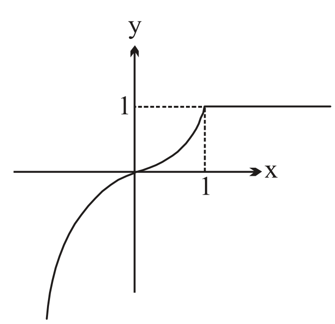
So, required quadratic equation is $\displaystyle x^2-\frac{7}{12}\cdot x+\frac{1}{12}=0\Rightarrow 12x^2-7x+1=0.$ **Ans.**

106. $\displaystyle \lim_{h\to 0}\frac{e^{1-h}-1+h-1}{(1-h)^2}=e-2$ **Ans.**

Sol 107. to 108.

$$f(x)=\min\left(1,x^{2n},x^{2n+1}\right)=\begin{cases}1, & x\ge 1\\ x^{2n+1}, & x<1\end{cases}$$

Graph of f(x) will be as shown in the figure.



107. $\displaystyle \lim_{x\to 0}\frac{\left(27^x-9^x-3^x+1\right)\left(\cos x-e^x\right)}{f(x)}$

$$=\lim_{x\to 0}\frac{\left(3^x-1\right)}{x}\frac{\left(9^x-1\right)}{x}\frac{\left((\cos x-1)-(e^x-1)\right)}{x^{2n-1}}$$

$$=\ln 3\ \ln 9\ \lim_{x\to 0}\frac{\left((\cos x-1)-(e^x-1)\right)}{x^{2n-1}}$$

$\because$ Limit is non zero finite $\Rightarrow$ $2n-1=1$ $\Rightarrow$ $n=1$

108. $\displaystyle \lim_{x\to 0}\frac{e^{\tan f(x)}-e^{\sin f(x)}}{\tan f(x)-\sin f(x)} = \lim_{x\to 0}\frac{e^{\tan x^{2n+1}}-e^{\sin x^{2n+1}}}{\tan x^{2n+1}-\sin x^{2n+1}}$

$$=\lim_{x\to 0}\frac{e^{\sin x^{2n+1}}\left\{e^{\left(\tan x^{2n+1}-\sin x^{2n+1}\right)}-1\right\}}{\left(\tan x^{2n+1}-\sin x^{2n+1}\right)}=1\times 1=1 \quad]$$

109. $\displaystyle l_1 = \lim_{x\to 0} ln(\cos 3x)^{\frac{1}{2x^2}} = \lim_{x\to 0}\frac{\cos 3x-1}{2x^2} = -\lim_{x\to 0}\frac{1-\cos 3x}{(3x)^2}\cdot\frac{9x^2}{2x^2} = -\frac{9}{4}$

$$l_2 = -\lim_{x\to 0}\frac{\sin^2 3x}{(3x)^2}\cdot\frac{9x^2}{x(e^x-1)} = -9$$

$$l_3 = \lim_{x\to 1}\frac{x-x^2}{ln\,x}\cdot\frac{1}{\sqrt{x}+x} = \frac{1}{2}\lim_{h\to 0}\frac{(1+h)(-h)}{ln(1+h)} = -\frac{1}{2}$$

Hence $l_2 < l_1 < l_3$

110. Given $a_1 = 1$ $\Rightarrow$ $a_2 = \sin a_1 = \sin 1$

$a_3 = \sin a_2 = \sin(\sin 1)$

$a_4 = \sin a_3 = \sin(\sin(\sin 1))$

$a_5 = \sin a_4 = \sin(\sin(\sin(\sin 1)))$

$\Rightarrow$ $a_1 = 1, a_2, a_3, ..., a_n$ is a decreasing sequence

and hence as $n\to\infty$, $\displaystyle\lim_{n\to\infty} a_n = 0$

Let $a_n = y$

$\therefore$ as $n\to\infty$, $y\to 0$

$$\therefore \quad \lim_{y\to 0}\frac{2^{2y}-2(2^y)(3^y)+3^{2y}}{\cos y+1-e^y-e^{-y}} = -\lim_{y\to 0}\frac{(2^y-3^y)^2}{(e^y-e^{-y}-2)+(1-\cos y)}$$

$$= -\lim_{y\to 0}\frac{\left(\dfrac{2^y-3^y}{2}\right)^2}{\left(\dfrac{e^y-e^{-y}-2}{y^2}\right)+\left(\dfrac{1-\cos y}{y^2}\right)} = -\frac{ln^2\frac{2}{3}}{1+\frac{1}{2}} = \frac{-2}{3}ln^2\frac{2}{3} \Rightarrow a = \frac{2}{3}$$

$\therefore$ $3a = 2$ **Ans.**

111. $f(x) = ln\,\text{cosec}(x\pi)$ $0<x<1$ and $g(x)=\dfrac{2^{f(x)}+1}{3^{f(x)}+1}$

$\quad = ln\sin(2x\pi)$ $1<x<3/2$

now see $\tan^{-1}\left(g(1^-)\right)$

$$f(1^-) = \lim_{x\to 1^-} \ln \operatorname{cosec}(\pi x) = +\infty$$

so $\displaystyle \lim_{x\to\infty} \frac{2^x+1}{3^x+1}$ divide N^r and D^r by 3^x

$$= \lim_{x\to\infty} \frac{(2/3)^x + (1/3)^x}{1+(1/3)^x} = 0$$

$\tan^{-1}(0) = 0$

now see $\sec^{-1}\left(g\left(1^+\right)\right)$

$$\lim_{x\to 1^+} \ln \sin(2\pi x) = -\infty\,; \qquad \lim_{x\to-\infty} \frac{2^x+1}{3^X+1} = 1$$

So $\sec^{-1}(1) = \dfrac{\pi}{2}$

112. $n < \sqrt{n^2+n+1} < n+1$

So, $\displaystyle \lim_{n\to\infty} f\left(\sqrt{n^2+n+1}\right) = \lim_{n\to\infty} \sqrt{n^2+n+1} - n = \lim_{n\to\infty} \frac{n+1}{\sqrt{n^2+n+1}+n} = \frac{1}{2}$

113. $\displaystyle L = \lim_{x\to 0} \frac{e^{x\sin(2007)x}-1}{x\sin(2007x)} \cdot \frac{x\sin(2007x)}{x\ln(1+x)} = \lim_{x\to 0} \frac{\sin(2007x)}{(2007x)} \cdot \frac{(2007x)}{\ln(1+x)} = 2007$

$$M = \lim_{x\to\infty} -x[(x^3+x^2+1)^{1/3} + (x^3-x^2+1)^{1/3} - 2x]$$

$$= \lim_{x\to\infty} -x\left[x\left(1+\frac{1}{x}+\frac{1}{x^3}\right)^{1/3} + x\left(1-\frac{1}{x}+\frac{1}{x^3}\right)^{1/3} - 2x\right]$$

$$= \lim_{x\to\infty} -x^2[(1+X)^{1/3} + (1-Y)^{1/3} - 2] \quad \text{where } X = \frac{1}{x}+\frac{1}{x^3} \text{ and } Y = \frac{1}{x}-\frac{1}{x^3}$$

$$M = \lim_{x\to\infty} -x^2\left[\left(1+\frac{X}{3}+\frac{1}{3}\left(\frac{1}{3}-1\right)\frac{1}{2!}X^2+\ldots\ldots\right)+\left(1-\frac{Y}{3}+\frac{1}{3}\left(\frac{1}{3}-1\right)\frac{1}{2!}Y^2\right)-2\right]$$

$$= \lim_{x\to\infty} -x^2\left[\left(\frac{X-Y}{3}\right)+\frac{1}{3}\left(\frac{1}{3}-1\right)\frac{1}{2!}(X^2+Y^2)+\ldots\ldots\right]$$

but $\displaystyle X - Y = \frac{2}{x^3}$ and $\displaystyle X^2+Y^2 = \left(\frac{1}{x}+\frac{1}{x^3}\right)^2 + \left(\frac{1}{x}-\frac{1}{x^3}\right)^2 = \left(\frac{2}{x^2}+\frac{2}{x^6}\right)$

$$M = \lim_{x\to\infty} -x^2\left[\frac{2}{3x^3} - \frac{1}{9}\left(\frac{2}{x^2}+\frac{2}{x^6}\right)+\ldots\right] = \frac{2}{9}$$

$\therefore \quad LM = \frac{2}{9} \times 2007 = 2 \times 223 = 446$ Ans.]

114. $\lim\limits_{x\to\infty} \frac{(Ax^2+Bx)-C^2x^2}{\sqrt{Ax^2+Bx}+Cx} = 2;\ \lim\limits_{x\to\infty} \frac{(A-C^2)x^2+Bx}{x\left[\sqrt{A+(B/x)}+C\right]} = 2$

for existence of limit $A = C^2$ hence $\frac{BC}{A} = \frac{B}{C}$ (using $A = C^2$)

$\therefore \quad L = \frac{B}{\sqrt{A}+C} = 2$

if $C = -\sqrt{A}$ then limit does not exist hence $C = \sqrt{A}$

$\therefore \quad \frac{B}{2C} = 2 \quad \Rightarrow \quad \frac{B}{C} = 4$ **Ans.**

115. $\lim\limits_{x\to 0} \frac{\sin^2 x}{e^{ax}-bx-1} = \frac{1}{2}$

$\lim\limits_{x\to 0} \frac{x^2}{e^{ax}-bx-1} = \frac{1}{2} \qquad \left(\frac{0}{0}\right)$

$\lim\limits_{x\to 0} \frac{2x}{ae^{ax}-b} = \frac{1}{2} \qquad \therefore \quad a - b = 0$

$\lim\limits_{x\to 0} \frac{2}{a^2 e^{ax}} = \frac{1}{2}$

$\frac{2}{a^2} = \frac{1}{2} \qquad \therefore \quad a = 2 \quad \Rightarrow \quad b = 2$

$S_1 : x^2 + y^2 + 2x + 2y + 1 = 0$

$S_2 : x^2 + y^2 + 4x + 6y + k = 0$

Now, $2(1)(2) + 2(1)(3) = 1 + k \Rightarrow k = 9$. **Ans.**

Sol. 116.to 118.

We have $f(x) \frac{\sin^{-1}(1-\{x\})\cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}(1-\{x\})}$

$\therefore \lim\limits_{x\to 0^+} f(x) = \lim\limits_{h\to 0} f(0+h) = \lim\limits_{h\to 0} \frac{\sin^{-1}(1-\{0+h\})\cos^{-1}(1-\{0+h\})}{\sqrt{2\{0+h\}}(1-\{0+h\})}$

$= \lim\limits_{h\to 0} \frac{\sin^{-1}(1-h)\cos^{-1}(1-h)}{\sqrt{2h}(1-h)} = \lim\limits_{h\to 0} \frac{\sin^{-1}(1-h)}{(1-h)} \lim\limits_{h\to 0} \frac{\cos^{-1}(1-h)}{\sqrt{2h}}$

= In second limit put $1 - h = \cos\theta$

$$= \lim_{h\to 0}\frac{\sin^{-1}(1-h)}{(1-h)}\lim_{\theta\to 0}\frac{\cos^{-1}(\cos\theta)}{\sqrt{2(1-\cos\theta)}} = \lim_{h\to 0}\frac{\sin^{-1}(1-h)}{(1-h)}\lim_{\theta\to 0}\frac{\theta}{2\sin\left(\frac{\theta}{2}\right)}$$

$$= \sin^{-1} 1 \times 1 = \frac{\pi}{2} \text{ and } \lim_{x\to 0^-} f(x) = \lim_{h\to 0} f(0-h)$$

$$= \lim_{h\to 0}\frac{\sin^{-1}(1-\{0-h\})\cos^{-1}(1-\{0-h\})}{\sqrt{2\{0-h\}}\,(1-\{0-h\})}$$

$$= \lim_{h\to 0}\frac{\sin^{-1}(1+h-1)\cos^{-1}(1+h-1)}{\sqrt{2(-h+1)}\,(1+h-1)}$$

$$= \lim_{h\to 0}\frac{\sin^{-1}h}{h}\lim_{h\to 0}\frac{\cos^{-1}h}{\sqrt{2(1-h)}} = 1\,\frac{\frac{\pi}{2}}{\sqrt{2}} = \frac{\pi}{2\sqrt{2}}$$

119. $p = \lim_{n\to\infty}\left(\frac{4}{\left(4-\sqrt{3}+2\sin\theta\right)}\right)^n\left(\frac{1}{\left(4-\sqrt{3}+2\sin\theta\right)^2}\right)$

$$p = \lim_{n\to\infty}\left(\frac{2}{2+\sin\theta-\frac{\sqrt{3}}{2}}\right)^n\left(\frac{1}{\left(4-\sqrt{3}+2\sin\theta\right)^2}\right)$$

If $\theta \in \left(0, \frac{\pi}{3}\right) \Rightarrow \sin\theta \in \left(0, \frac{\sqrt{3}}{2}\right) \Rightarrow$ p does not exist.

If $\theta \in \left(\frac{\pi}{3}, \frac{\pi}{2}\right) \Rightarrow \sin\theta \in \left(\frac{\sqrt{3}}{2}, 1\right) \Rightarrow p = 0$

$\therefore$ For existence of limit $\theta = \frac{\pi}{3} \Rightarrow p = \frac{1}{4^2} = \frac{1}{16}$

Hence, $\frac{p+\cos\theta}{p} = 1 + \frac{\cos\theta}{p} = 1 + \frac{16}{2} = 9$. **Ans.**

120. We have, $f(1) = \cos\pi = -1$

Also, $f(x) = \begin{cases} \frac{-\sin(x-1)}{(x-1)}; & x \in (-\delta, 0) \\ \cos(\pi x); & x \in (0, \delta) \end{cases}$

So, $f(0^+) = 1$ and $f(0^-) = -\sin 1$.

$\therefore$ $\left(f(0^+) - f(0^-)\right) = 1 - (-\sin 1) = (1 + \sin 1)$. **Ans.**

121.

(A) Here, a > 0, if $a \le 0$, then limit = ∞

$$\therefore \lim_{x\to\infty} \frac{(\sqrt{x^2-x+1}-ax-b)(\sqrt{x^2-x+1}+ax+b)}{(\sqrt{x^2-x+1}+ax+b)}$$

$$\Rightarrow \lim_{x\to\infty} \frac{(x^2-x+1)-(ax+b)^2}{\sqrt{x^2-x+1}+ax+b} \qquad \Rightarrow \lim_{x\to\infty} \frac{(1-a^2)x^2-(1+2ab)x+(1-b^2)}{\sqrt{x^2-x+1}+ax+b}$$

This is possible only when $1 - a^2 = 0$ and $1 + 2ab = 0$

$\therefore\ a \neq 1$

$\Rightarrow a = 1 \qquad (\because\ a > 0)$

$\Rightarrow b = -\frac{1}{2} \ \Rightarrow (a, 2b) = (1, -1)$

(B) Divide numerator and denominator by e^x, then

$$\lim_{x\to\infty} \frac{(1+a^3)e^{-x}+8}{e^{-x}+(1-b^3)} \Rightarrow \frac{0+8}{0+1-b^3} = 2 \qquad \Rightarrow 1 - b^3 = 4$$

$\therefore\ b^3 = -3 \ \Rightarrow b = -3^{1/3}$

Then, $a \in R$

$\Rightarrow (a, b^3) = (a, -3)$

(C) $$\lim_{x\to\infty} \left(\left(\sqrt{x^4-x^2+1}\right) - ax^2 - b\right) = 0$$

Put $x = \frac{1}{t}$ $\qquad \therefore \lim_{t\to 0} \left(\sqrt{\left(\frac{1}{t^4}-\frac{1}{t^2}+1\right)} - \frac{a}{t^2} - b\right) = 0 \ \Rightarrow \lim_{t\to 0} \frac{\sqrt{1-t^2+t^4}-a-bt^2}{t^2} = 0$ ....

(1)

Since R. H. S. is finite, numerator must be equal to 0 at $t \to 0$.

$\therefore\ 1 - a = 0 \quad \therefore\ a = 1$

From equation (1). $\lim_{t\to 0} \frac{\sqrt{1-t^2+t^4}-1-bt^2}{t^2} = 0$

$$\lim_{x\to\infty} (-1+t^2) \left(\frac{(1-t^2+t^4)^{1/2}-(1)^{1/2}}{(1-t^2+t^4)-1}\right) = b$$

$$\Rightarrow (-1)\left(\frac{1}{2}\right) = b \Rightarrow a = 1,\ b = -\frac{1}{2} \Rightarrow (a, -2b) = (1, 2)$$

(D) $$\lim_{x\to -a} \frac{a^7-(-x)^7}{a-(-x)} = 7 \ \Rightarrow 7a^6 = 7 \ \Rightarrow a^6 = 1 \ \Rightarrow a = -1$$

122. $$\mathop{Lt}_{x\to 0} \frac{\sin\left(5x^5+4x^4+3x^3+2x^2\right)}{\left(5x^5+4x^4+3x^3+2x^2\right)} \cdot \frac{\left(5x^5+4x^4+3x^3+2x^2\right)}{\ln\left(1-2\sin^2\left(\frac{x^3+x}{2}\right)\right)}$$

$$= \mathop{Lt}_{x\to 0} \frac{\left(5x^5+4x^4+3x^3+2x^2\right)}{\dfrac{\ln\left(1-2\sin^2\left(\frac{x^3+x}{2}\right)\right)}{-2\sin^2\left(\frac{x^3+x}{2}\right)}\left(-2\sin^2\left(\frac{x^3+x}{2}\right)\right)}$$

$$= \mathop{Lt}_{x\to 0} \frac{\left(5x^5+4x^4+3x^3+2x^2\right)}{2\dfrac{\sin^2\left(\frac{x^3+x}{2}\right)}{\left(\frac{x^3+x}{2}\right)^2}\left(\frac{x^3+x}{2}\right)^2} = -\frac{2}{2\left(\frac{1}{4}\right)} = -4$$

123. $$\prod_{k=1}^{n}\left(1-\frac{d^2}{a_k^2}\right) = \prod_{k=1}^{n}\frac{a_k^2-d^2}{a_k^2} = \prod_{k=1}^{n}\frac{(a_k-d)}{a_k}\cdot\frac{(a_k+d)}{a_k} = \prod_{k=1}^{n}\frac{a_{k-1}}{a_k}\cdot\frac{a_{k+1}}{a_k} \qquad \begin{bmatrix} a_k-d=a_{k-1} \\ a_k+d=a_{k+1}\end{bmatrix}$$

$$= \left(\frac{a_0}{a_1}\cdot\frac{a_1}{a_2}\,......\,\frac{a_{n-1}}{a_n}\right)\left(\frac{a_2}{a_1}\cdot\frac{a_3}{a_2}\,......\,\frac{a_{n+1}}{a_n}\right) = \left(\frac{a_0}{a_n}\right)\left(\frac{a_{n+1}}{a_1}\right) = \left(\frac{a_0}{a_1}\right)\left(\frac{a_{n+1}}{a_n}\right) \quad [\text{but } a_0 = a_1 - d]$$

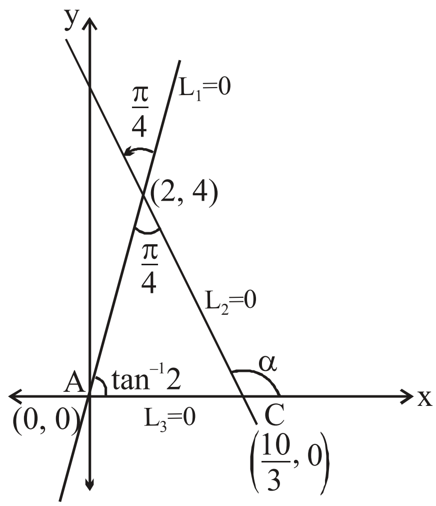
$$= \left(\frac{a_1-d}{a_1}\right)\left(\frac{a_n+d}{a_n}\right) = \left(1-\frac{d}{a_1}\right)\left(1+\underbrace{\frac{d}{a_1+(n-1)\,d}}_{\text{zero as } n\to\infty}\right) = 1-\frac{d}{a_1}$$

but $1-\dfrac{d}{a_1} = \dfrac{1}{4} \Rightarrow \dfrac{3}{4} = \dfrac{d}{a_1} \Rightarrow \dfrac{8\cdot 3}{4} = d$

Hence $d = 6$. **Ans.**

124. $$L_1 : y - \lim_{n\to\infty}\frac{4}{n^2}\sum_{r=1}^{n}\left([r\,x]+3\right) = 0$$

$x + 3 - 1 < [\,x] + 3 \le x + 3$
$2x + 3 - 1 < [2x] + 3 \le 2x + 3$
$3x + 3 - 1 < [3x] + 3 \le 3x + 3$



$\vdots$

$nx + 3 - 1 < [nx] + 3 \le nx + 3$

$$\frac{n(n+1)}{2}x + 2n < \sum_{r=1}^{n}\left([rx]+3\right) \le \frac{n(n+1)}{2}\cdot x + 3n$$

$$\underbrace{\lim_{n\to\infty}\frac{4}{n^2}\left(\frac{n(n+1)}{2}\cdot x + 2n\right)}_{2x} < \lim_{n\to\infty}\frac{4}{n^2}\sum_{r=1}^{n}\left([rx]+3\right) \le \underbrace{\lim_{n\to\infty}\frac{4}{n^2}\left(\frac{n(n+1)}{2}x + 3n\right)}_{2x}$$

$$\therefore \quad \lim_{n\to\infty}\frac{4}{n^2}\sum_{r=1}^{n}\left([rx]+3\right) = 2x$$

$L_1 : y - 2x = 0$

$$\sum_{i=2}^{3}\cot^{-1}(i) = \cot^{-1}(2) + \cot^{-1}(3) = \frac{\pi}{4}$$

From the figure slope of the line $L_2 = 0$ is

$$\tan\alpha = \tan\left(\frac{\pi}{4} + \tan^{-1}2\right) = \frac{1+2}{1-2} = -3$$

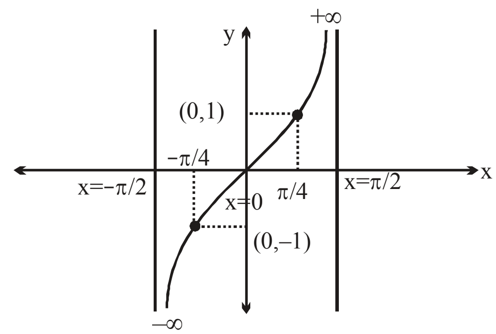
$\therefore \; L_2 : y - 4 = -3(x-2) \Rightarrow 3x + y = 10$

$$L_3 : y = \left(1 + \underbrace{\lim_{\theta\to 0^-}\left[\theta^2 + \theta + \sin\theta\right]}_{-1}\right)x \Rightarrow y = 0 \quad \text{i.e. x-axis}$$

Hence $\tan A + \tan B + \tan C = \tan(\tan^{-1}2) + \tan\frac{\pi}{4} + \tan(\tan^{-1}3) = 2 + 1 + 3 = 6$ **Ans.**

125. Given

$$f(x)=\begin{cases}\lim\limits_{n\to\infty}\left(\dfrac{(\tan x)^{2n}+x^2}{\sin^2 x+(\tan x)^{2n}}\right), & x\neq 0\\ 1, & x=0\end{cases}$$



y = tan x

Note : f(x) is an even function on $\left(\dfrac{-\pi}{2},\dfrac{\pi}{2}\right)$.

Now, $f\left(\dfrac{\pi^+}{4}\right)=1=f\left(\dfrac{-\pi^-}{4}\right)$

Also, $f\left(\dfrac{\pi^-}{4}\right)=\dfrac{\frac{\pi^2}{16}}{\frac{1}{2}}=\dfrac{\pi^2}{8}=f\left(\dfrac{-\pi^+}{4}\right)$

Also, $f(x)=\begin{cases}\dfrac{x^2}{\sin^2 x}, & x\in(0,\delta)\\[2ex] \dfrac{x^2}{\sin^2 x}, & x\in(-\delta,0)\end{cases}$

So, $f(0^+)=f(0^-)=1=f(0)$

Hence, options (B), (C), (D) are correct.

126. $\lim\limits_{x\to 0}\left(1+x+\dfrac{f(x)}{x}\right)^{1/x}=e^3$

now this is of form 1^∞

so $\dfrac{f(x)}{x}=0$ i.e. $f(x)$ has atleast degree 2

$e^{\lim\limits_{x\to 0}\frac{1}{x}\left(x+\frac{f(x)}{x}\right)}=e^3$; $\lim\limits_{x\to 0}\dfrac{1}{x}\left(x+\dfrac{f(x)}{x}\right)=3$;

$\lim\limits_{x\to 0}1+\dfrac{f(x)}{x^2}=3$; $\lim\limits_{x\to 0}\dfrac{f(x)}{x^2}=2$

so $f(x)=2x^2$ (as before)]

127. $a^2=\lim\limits_{\theta\to\pi}\left[\dfrac{\sec^2\theta+12\sec\theta+11}{\sec\theta+1}\right]=\lim\limits_{\theta\to\pi}\left[\dfrac{(\sec\theta+11)(\sec\theta+1)}{(\sec\theta+1)}\right]=\lim\limits_{\theta\to\pi}[\sec\theta+11]$

$=9$ $\{\because$ as $\theta\to\pi$, $\sec\theta$ is slightly less than $-1\}$

$\therefore$ $a=\pm 3$

128. $$L = \lim_{x\to 0}\left(\frac{1}{\ln(1+x)} - \frac{1}{\ln(x+\sqrt{1+x^2})}\right)$$

$$L = \lim_{x\to 0}\frac{\ln(x+\sqrt{1+x^2}) - \ln(x+1)}{x\cdot\frac{\ln(x+1)}{x}\cdot\ln(x+\sqrt{1+x^2})} = \lim_{x\to 0}\frac{\ln\left(\frac{x+\sqrt{1+x^2}}{1+x}\right)}{\frac{x\cdot\ln\left((x+\sqrt{1+x^2}-1)+1\right)(x+\sqrt{1+x^2}-1)}{(x+\sqrt{1+x^2}-1)}}$$

note that $$\lim_{x\to 0}\frac{\ln\left(\overbrace{(x+\sqrt{1+x^2}-1)}^{y}+1\right)}{\underbrace{\left(x+\sqrt{1+x^2}-1\right)}_{y}} \to 1;\quad \lim_{y\to 0}\frac{\ln(1+y)}{y} = 1$$

hence $$L = \lim_{x\to 0}\frac{\ln\left(\underbrace{\left(\frac{x+\sqrt{1+x^2}}{1+x}-1\right)}_{y}+1\right)\cdot\left(\underbrace{\left(\frac{x+\sqrt{1+x^2}}{1+x}-1\right)}_{y}\right)}{x\cdot\left(\underbrace{\left(\frac{x+\sqrt{1+x^2}}{1+x}-1\right)}_{y}\right)(x+\sqrt{1+x^2}-1)}$$

Note that $$\lim_{x\to 0}\frac{\ln\left(\left(\frac{x+\sqrt{1+x^2}}{1+x}-1\right)+1\right)}{\left(\frac{x+\sqrt{1+x^2}}{1+x}-1\right)} = 1$$

$\therefore$ $$L = \lim_{x\to 0}\frac{x+\sqrt{1+x^2}-1-x}{x(1+x)(x+\sqrt{1+x^2}-1)} = \lim_{x\to 0}\frac{\sqrt{1+x^2}-1}{x(\sqrt{1+x^2}+x-1)} \qquad (\text{as } \lim_{x\to 0}(1+x) = 1)$$

$$L = \lim_{x\to 0}\frac{(\sqrt{x^2+1}-1)(\sqrt{x^2+1}+1)}{(\sqrt{x^2+1}+1)\cdot x(\sqrt{x^2+1}+x-1)} = \frac{1}{2}\lim_{x\to 0}\frac{[(x^2+1)-1][\sqrt{x^2+1}-(x-1)]}{x\cdot(\sqrt{x^2+1}+x-1)\cdot[\sqrt{x^2+1}-(x-1)]}$$

$$= \lim_{x\to 0}\frac{1}{2}\frac{x\cdot 2}{(x^2+1)-(x-1)^2} = \lim_{x\to 0}\frac{x}{2x} = \frac{1}{2};\qquad \therefore\quad L = \frac{1}{2}$$

hence $$\frac{L+153}{L} = \frac{(1/2)+153}{(1/2)} = 1 + 2\cdot 153 = 1 + 306 = 307 \text{ Ans.}$$

129. $L = \lim_{x\to 0}\left(\dfrac{x - n\tan\left(\dfrac{\tan^{-1}x}{n}\right)}{n\sin\left(\dfrac{\tan^{-1}x}{n}\right) - x}\right)$

Let $\tan^{-1}x = t$

$$\therefore\ L = \lim_{t\to 0}\frac{\tan t - n\tan\left(\dfrac{t}{n}\right)}{n\sin\left(\dfrac{t}{n}\right) - \tan t}$$

$$= \lim_{t\to 0}\frac{\left(t + \dfrac{t^3}{3} + \dfrac{2}{15} + 5 + \ldots.\right) - n\left(\dfrac{t}{n} + \dfrac{t^3}{n^3} - \dfrac{1}{3} + \ldots.\right)}{n\left(\dfrac{t}{n} + \dfrac{t^3}{n^3} - \dfrac{1}{3!} + \ldots.\right) - \left(t + \dfrac{t^3}{3} + \ldots.\right)}$$

$$= \lim_{t\to 0}\frac{t^3\left(\dfrac{1}{3} + \dfrac{1}{3n^2}\right) + \ldots}{t^3\left(-\dfrac{1}{3} - \dfrac{1}{6n^2}\right) + \ldots} = \frac{\dfrac{1}{3}\left(1 - \dfrac{1}{n^2}\right)}{\dfrac{1}{3}\left(1 - \dfrac{1}{n^2}\right)} = -2\left(\frac{n^2 - 1}{2n^2 + 1}\right) = 1 - \frac{3}{2n^2 + 1}$$

$$\therefore\ f(n) = 1 - \frac{3}{2n^2 + 1}$$

$$\therefore\ \frac{3}{2n^2 + 1} = \frac{1}{1 + f(x)}$$

$$2n^2 = \frac{3}{1 + f(x)} - 1 = \frac{2f(x)}{1 + f(x)}$$

$$\therefore\ \sum_{n=1}^{10}\frac{2 - f(n)}{1 + f(n)} = 2\sum_{n=1}^{10} n^2 = \frac{2(10)\,(11)\,(21)}{6} = 10 \times 11 \times 7 = 770.\ \textbf{Ans.}$$

130. $\lim_{x\to\infty}\dfrac{\sum_{k=1}^{100}(x + 2k)^m}{x^m + 5^{1000}} = \lim_{x\to\infty}\dfrac{\sum_{k=1}^{100}\left(1 + \dfrac{2k}{x}\right)^m}{1 + \left(\dfrac{5^{1000}}{x^m}\right)}$

$$= \frac{\sum_{k=1}^{100} 1}{1 + 0} = 100\quad \text{Ans.}$$

131. $L = \lim\limits_{x\to 1} \dfrac{2\cos\dfrac{\sqrt{1+\ln x}+1}{2}\sin\dfrac{\sqrt{1+\ln x}-1}{2}}{\ln x} = 2\cos 1 \lim\limits_{x\to 1}\dfrac{\sin(M)}{M}\cdot\dfrac{M}{x-1}.$

$$= \frac{2\cos 1}{2}\lim_{x\to 1}\frac{(1+\ln x)^{1/2}-1}{x-1} = \frac{\cos 1}{2}\lim_{x\to 1}\frac{\ln x}{x-1} = \frac{1}{2}\cos 1 \text{ **Ans.**}$$

132. $\lim\limits_{x\to\infty}\left(\dfrac{2x+2}{2x-1}\right)^{x+\sqrt{x}}$ $\quad (1^\infty)$

$$= e^{\lim\limits_{x\to\infty}\left(x+\sqrt{x}\right)\left(\frac{2x+2}{2x-1}-1\right)} = e^{\lim\limits_{x\to\infty}\left(x+\sqrt{x}\right)\left(\frac{3}{2x-1}\right)} = e^{\frac{3}{2}} \text{ **Ans.**}$$

133. $\lim\limits_{x\to\infty} x\left(\pi - \tan\dfrac{2x+6}{x^2+6x+4} - e^{\ln\left(\pi\left[1+\cos\frac{1}{x}\right]\right)}\right)$

$$\lim_{x\to\infty} x\left(\pi - \frac{\tan^{-1}\dfrac{2x+6}{x^2+6x+4}}{\dfrac{2x+6}{x^2+6x+4}}\cdot\frac{2x+6}{x^2+6x+4} - \pi\right) = -2. \text{ **Ans.**]}$$

134. $L = \lim\limits_{x\to 0^+}\left(\dfrac{\cot^{-1}(\ln(x))}{\pi}\right)^{-\ln x}$ $\quad (1^\infty)$

$$= e^{\lim\limits_{x\to 0^+}\left(\frac{\cot^{-1}\ln(x)-\pi}{\pi}\right)(-\ln x)} = e^{\lim\limits_{x\to 0^+}\frac{\cot^{-1}(-\ln x)}{\pi}(\ln x)} = e^{-\lim\limits_{x\to 0^+}\frac{\tan^{-1}\left(-\frac{1}{\ln x}\right)}{\left(-\frac{1}{\ln x}\right)\pi}} = e^{\frac{-1}{\pi}}$$

$\therefore \quad$ L.H.L. $= f(0^-) = e^{\left(-\frac{x}{\pi}\cdot\frac{1}{x}\right)} = e^{-\frac{1}{\pi}}$

$f(0^+) = f(0^-) = e^{-\frac{1}{\pi}} = k$ **Ans.**

Sol. **135. to 136.**

135. If $x \in (-2, 0)$ then $x + 1 \in (-1, 1) \Rightarrow \quad (x + 1)^2 \in [0, 1)$

$\therefore \quad (x + 1)^{2n} \to 0$ as $n \to \infty$

$\therefore \quad f(x) = \lim\limits_{n\to\infty}\dfrac{x^2+2(x+1)^{2n}}{(x+1)^{2n+1}+x^2+1} = \dfrac{x^2}{x^2+1} \quad \Rightarrow \quad f(x) = 1 - \dfrac{1}{x^2+1}$

$f(x) \in \left(0, \dfrac{4}{5}\right) \subset \left[\dfrac{-\pi}{2}, \dfrac{\pi}{2}\right]$ $\qquad$ As. $x \in (-2, 0)$

$$f(x) = \begin{cases} \dfrac{x^2}{x^2+1}; & x \in (-2,0) \\ \dfrac{2}{x+1}; & (-\infty,-2)\cup(0,\infty); \\ 1 & x = 0 \\ 3 & x = -2 \end{cases} \qquad f(x) = \tan\theta \quad \therefore \quad \theta = \tan^{-1}(f(x))$$

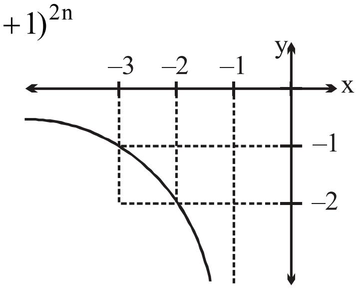
$$\Rightarrow \quad g(x) = \tan\left(\frac{1}{2}\sin^{-1}\left(\frac{2f(x)}{1+f^2(x)}\right)\right) \quad \Rightarrow g(x) = \tan\left(\frac{1}{2}\ 2\tan^{-1} f(x)\right)$$

As $f(x) \in \left(0, \frac{4}{5}\right) \qquad = f(x)$

so, range of $g(x)$ is $\left(0, \frac{4}{5}\right)$.

136. If $x < -2$ then $x + 1 < -1$ and $(x + 1)^{2n} \to \infty$ as $n \to \infty$

$$\therefore \quad f(x) = \lim_{n\to\infty} \frac{x^2 + 2(x+1)^{2n}}{(x+1)^{2n+1} + x^2 + 1} = \lim_{n\to\infty} \frac{\dfrac{x^2}{(x+1)^{2n}} + 2}{x+1+\dfrac{x^2+1}{(x+1)^{2n}}} = \frac{2}{x+1}$$



$f(x) \in (-2, 0)$ as $x \in (-\infty, -2)$
If $x \in (-\infty, -3]$, then $f(x) \in [-1, 0)$

so, $g(x) = \tan\left(\frac{1}{2}\cdot 2\tan^{-1} f(x)\right) = f(x) = \frac{2}{x+1}$

If $x \in (-3, -2)$ then $f(x) \in (-2, -1)$

$$\therefore \quad g(x) = \tan\left(\frac{1}{2}\left(-\pi - 2\tan^{-1} f(x)\right)\right) = -\tan\left(\frac{\pi}{2} + \tan^{-1} f(x)\right) = \cot\left(\tan^{-1} f(x)\right)$$

$$= \cot\left(-\pi + \cot^{-1}\frac{1}{f(x)}\right) = \frac{1}{f(x)} = \frac{x+1}{2}.$$

Now, $\displaystyle\lim_{x\to -3} \frac{\sin(x+3)\, g(x)}{x^2+4x+3}$

$$\Rightarrow \quad \lim_{x\to -3^-} \frac{\sin(x+3)}{(x+3)(x+1)} \cdot \frac{2}{(x+1)} = \frac{2}{4} = \frac{1}{2}$$

and $\displaystyle\lim_{x\to -3^+} \frac{\sin(x+3)}{(x+3)(x+1)} \cdot \frac{x+1}{2} = \frac{1}{2}$. **Ans.**

137. $\sin x < x\,,\ \ x \in \left(0, \frac{\pi}{2}\right) \Rightarrow \sin(\sin x) < \sin x \Rightarrow \underbrace{\sin(\sin \ldots\ldots\sin(x))}_{\text{n times}} \to 0$ as $n \to \infty$

Also as $x \in \frac{\pi}{4}^{+}$; $\left(\sqrt{2}\cos x + 2\right) < 3$

$$\therefore f(x) = \begin{cases} \lim\limits_{n\to\infty} \dfrac{\sin\ldots\ldots\sin(\sin x) + \left(\dfrac{\sqrt{2}\cos x + 2}{3}\right)^n + \left(\dfrac{2}{3}\right)^n \cos x}{1 + \sin x\left(\dfrac{2+\sqrt{2}\cos x}{3}\right)^n}, & x \in \left(\dfrac{\pi}{4}, \dfrac{\pi}{4}+\delta\right)\ (\div \text{by } 3^n) \\[2ex] \lim\limits_{n\to\infty} \dfrac{\left(\dfrac{3}{2+\sqrt{2}\cos x}\right)^n \sin\sin\ldots\sin(x) + 1 + \left(\dfrac{2}{2+\sqrt{2}\cos x}\right)^n \cos x}{\left(\dfrac{3}{2+\sqrt{2}\cos x}\right)^n + \sin x}, & x \in \left(\dfrac{\pi}{4}-\delta, \dfrac{\pi}{4}\right)\ \left(\div \text{by}\left(\sqrt{2}\cos x + 2\right)^n\right) \end{cases}$$

$$f(x) = \begin{cases} 0, & x \in \left(\dfrac{\pi}{4}, \dfrac{\pi}{4}+\delta\right) \\[1ex] \dfrac{1}{\sin x}, & x \in \left(\dfrac{\pi}{4}-\delta, \dfrac{\pi}{4}\right) \end{cases}$$

$$\Rightarrow \lim_{x\to\frac{\pi}{4}^{+}} f(x) = 0 = l \Rightarrow \lim_{x\to\frac{\pi}{4}^{-}} f(x) = \sqrt{2} = m$$

$\Rightarrow \quad l^2 + m^2 = 2$. **Ans.**

138. $$L = \lim_{(x,y)\to(1,2)} \frac{\tan^{-1}x + \tan^{-1}\frac{1}{y} - \tan^{-1}3}{(x-1)} \cdot \frac{\sin^{-1}(y-2)}{(y-2)}$$

$$= \lim_{(x,y)\to(1,2)} \frac{\tan^{-1}\dfrac{x+\frac{1}{y}}{1-\frac{1}{y}} - \tan^{-1}3}{(x-1)} \cdot \frac{\sin^{-1}(y-2)}{(y-2)}$$

$$= \lim_{(x,y)\to(1,2)} \left(\frac{\tan^{-1}\dfrac{(xy+1)}{y-x} - \tan^{-1}3}{(x-1)}\right) \cdot \frac{\sin^{-1}(y-2)}{(y-2)}$$

L to exist $\tan^{-1}\dfrac{xy+1}{y-x} = \tan^{-1} 3$ (Important step)

$$\frac{xy+1}{y-x} = 3$$

$$xy + 1 = 3y - 3x$$

$$x = \frac{3y-1}{3+y}$$

$\therefore \quad f^{-1}(x) = \dfrac{3x-1}{3+x}$

$\therefore \quad \lim\limits_{x \to \frac{1}{3}} \dfrac{f^{-1}(x)}{3x-1} = \dfrac{1}{3+\frac{1}{3}} = \dfrac{3}{10}$. **Ans.**

139. $L_n = \left(1+\dfrac{1}{2^1}\right)\left(1+\dfrac{1}{2^2}\right)\left(1+\dfrac{1}{2^{2^3}}\right).......\left(1+\dfrac{1}{2^{2^n}}\right)$

multiply and divided by $1-\dfrac{1}{2}$

$$L_n = 2\left(1-\frac{1}{2}\right)\left(1+\frac{1}{2}\right)\left(1+\frac{1}{2^2}\right)\left(1+\frac{1}{2^{2^3}}\right).......\left(1+\frac{1}{2^{2^n}}\right)$$

$$L_n = 2\left(1-\frac{1}{2^{2^{n+1}}}\right) = 2.$$

For M

$$\frac{5^k \cdot 3^k}{\left(5^k - 3^k\right)\left(5^{k+1} - 3^{k+1}\right)} = \frac{5^k}{2}\left(\frac{1}{5^k - 3^k} - \frac{5}{5^{k+1} - 3^{k+1}}\right)$$

$$= \frac{1}{2}\left(\frac{5^k}{5^k - 3^k} - \frac{5^{k+1}}{5^{k+1} - 3^{k+1}}\right)$$

$\therefore \quad M_n = \dfrac{1}{2}\sum\limits_{k=1}^{n}\left(\dfrac{5^k}{5^k - 3^k} - \dfrac{5^{k+1}}{5^{k+1} - 3^{k+1}}\right) = \dfrac{1}{2}\left(\dfrac{5}{5-3} - \dfrac{5^{n+1}}{5^{n+1} - 3^{n+1}}\right)$

$$M = \lim_{n - \infty} \frac{1}{2}\left(\frac{5}{2} - \frac{5^{n+1}}{5^{n+1} - 3^{n+1}}\right) = \frac{1}{2}\left(\frac{5}{2} - 1\right) = \frac{3}{4}$$

$\Rightarrow \quad L + 4M = 5.$ **Ans.**

140. $n < \sqrt{n^2+n+1} < n+1$

so, $\lim\limits_{n\to\infty} f\left(\sqrt{n^2+n+1}\right) = \lim\limits_{n\to\infty} \sqrt{n^2+n+1} - n = \lim\limits_{n\to\infty} \dfrac{n+1}{\sqrt{n^2+n+1}+n} = \dfrac{1}{2}$

141. $l = \lim\limits_{n\to\infty}\left(1+\dfrac{na_n+1}{n}\right)^n \to 1^\infty$

$= e^{\lim\limits_{n\to\infty} n\cdot\left(\frac{na_n+1}{n}\right)} = e \Rightarrow [e] = 2$ **Ans.**

142. Given limit $= \lim\limits_{x\to1} \dfrac{\tan\left(\pi \operatorname{cosec}^2\left(\frac{\pi x}{2}\right)\right)}{\ln^2 x} = \lim\limits_{h\to0} \dfrac{\tan\left(\pi\operatorname{cosec}^2\frac{\pi}{2}(1+h)\right)}{\ln^2(1+h)}$ (where $x = 1+h$)

$= \lim\limits_{h\to0} \dfrac{\tan\left(\pi \sec^2\frac{\pi h}{2}\right)}{h^2}$ $\left(\because \lim\limits_{h\to0}\dfrac{\ln(1+h)}{h} = 1\right)$

$= \lim\limits_{h\to0} \dfrac{\tan\left(\pi\sec^2\frac{\pi h}{2} - \pi\right)}{h^2} \times \dfrac{\left(\pi\sec^2\frac{\pi h}{2}-\pi\right)}{\left(\pi\sec^2\frac{\pi h}{2}-\pi\right)} = \lim\limits_{h\to0}\dfrac{\pi\tan^2\frac{\pi h}{2}}{h^2\times\left(\frac{\pi}{2}\right)^2}\times\left(\dfrac{\pi}{2}\right)^2 = \dfrac{\pi^3}{4}$. **Ans.**

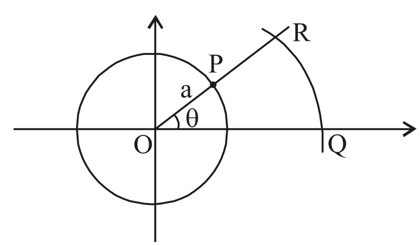
Sol. **143. to 144**

143. Let $P(a\cos\theta, a\sin\theta)$ then $OQ = b + a\cos\theta$

$= 1 + a\cos\theta$ (as b is 1)

so $QR = \theta(1+a\cos\theta)$

so $\lim\limits_{\theta\to0}\dfrac{\theta(1+a\cos\theta)}{\theta^n}$ i.e. $\lim\limits_{\theta\to0}\dfrac{1+a\cos\theta}{\theta^{n-1}}$



Clearly $a = -1$

$n-1 = 2$ i.e. $n = 3$ and $l = 1/2$

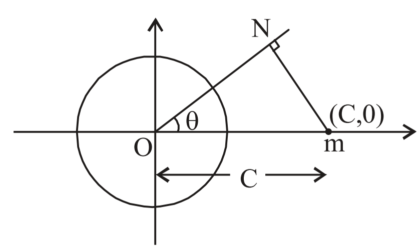
so $a + n + l = -1 + 3 + \dfrac{1}{2} = 2 + \dfrac{1}{2} = \dfrac{5}{2}$

144. Let m be (C, 0)

So $MN = C\sin\theta$

Also $QR = \theta(b + a\cos\theta)$

So $\lim\limits_{\theta\to0}\dfrac{\theta(b+a\cos\theta)+C\sin\theta}{\theta^3}$



$$= \lim_{\theta\to 0} \frac{\theta\left(b+a\left(1-\frac{\theta^2}{2!}+\frac{\theta^4}{4!}-\right)\right)+C\left(\theta-\frac{\theta^3}{3!}+\frac{\theta^5}{5!}\right).......}{\theta^3}$$

$$= \lim_{\theta\to 0} \frac{\theta(a+b+c)+\theta^3\left(\frac{-a}{2!}-\frac{C}{3!}\right)+\theta^5\left(\frac{a}{4!}+\frac{C}{5!}\right)......}{\theta^3}$$

Now for limit to exist $a+b+c=0$

& Limit is $-\left(\frac{a}{2}+\frac{C}{6}\right)$

So $b = 2$ so $C = (-2-a)$ i.e. point is $(-2-a, 0)$

Sol. **145. to 146**

$x = e^{2t} - 2e^t + 3$
$x = (e^t - 1)^2 + 2$
$x = y^2 + 2$
$y^2 = x - 2$
As $t \in (-\infty, 0]$
$\Rightarrow$ $e^t \in (0, 1]$
$\Rightarrow$ $e^t - 1 \in (-1, 0]$
$\Rightarrow$ $(e^t - 1)^2 \in [0, 1)$
$x = (e^t - 1)^2 + 2 \in [2, 3)$
$y = e^t - 1 \in (-1, 0]$
$\therefore$ $y = -\sqrt{x-2} = f(x)$
$(gof)^{-1}(x) = f^{-1}o\, g^{-1}(x) = x\ \forall\ x \in$ domain of f.
$\therefore$ f and g are inverse of each other.
$y^2 + 2 = x \Rightarrow g(x) = x^2 + 2$

145. (A) $g(x) = x^2 + 2\ \forall\ x \in (-1, 0] \to$ Range of 'f' = Domain of 'g'.
(B) and (C) Domain of $g(x)$ is $(-1, 0]$, therefore function can not be even or odd.
(D) Since, $g(0) = 2$ therefore its odd extension cannot exist

146. $$\lim_{x\to -\frac{1}{2}} \frac{9-4g(x)}{\sin^{-1}(2x+1)\left(\pi-\cos^{-1}(2x+1)\right)} = \lim_{x\to -\frac{1}{2}} \frac{9-4(x^2+2)}{\sin^{-1}(2x+1)\left(\pi-\cos^{-1}(2x+1)\right)}$$

$$= \lim_{x\to -\frac{1}{2}} \frac{1-4x^2}{\sin^{-1}(2x+1)\left(\pi-\cos^{-1}(2x+1)\right)} = \lim_{x\to -\frac{1}{2}} \frac{(1+2x)}{\sin^{-1}(2x+1)}\ \frac{(1-2x)}{(\pi-\cos^{-1}(2x+1))}$$

$$= \frac{1-2\left(\frac{-1}{2}\right)}{\pi-\frac{\pi}{2}} = \frac{2}{\frac{\pi}{2}} = \frac{4}{\pi}$$

147. $f(x)=\dfrac{3x^2+ax+a+1}{(x+2)(x-1)}$

as $x\to 1,\ D^r\to 0$ hence as $x\to 1,\ N^r\to 0$

$\therefore$ $3+2a+1=0 \Rightarrow$ $a=-2\Rightarrow$ **(A)**

as $x\to -2,\ D^r\to 0$ hence as $x\to -2,\ N^r\to 0$

$\therefore$ $12-2a+a+1=0$ $\Rightarrow$ $a=13\Rightarrow$ **(B)**

now $\displaystyle\lim_{x\to 0} f(x)=\lim_{x\to 1}\frac{3x^2-2x-1}{(x+2)(x-1)}=\lim_{x\to 1}\frac{(3x+1)(x-1)}{(x+2)(x-1)}=4/3\Rightarrow$ **(C)**

now $\displaystyle\lim_{x\to -2}\frac{3x^2+13x+14}{(x+2)(x-1)}=\lim_{x\to -2}\frac{(3x+7)(x+2)}{(x+2)(x-1)}=-\frac{1}{3}\Rightarrow$ **(D)**]

148. $\displaystyle\lim_{x\to-\infty} f(x)=\lim_{x\to-\infty}\frac{\left(\sqrt{x^2+2x}-x\right)\left(\sqrt{x^2+2x}+x\right)}{\left(\sqrt{x^2+2x}-x\right)}=\lim_{x\to-\infty}\frac{2x}{-x\left(\sqrt{1+\frac{2}{x}}+1\right)}=\frac{2}{-2}=-1$

And $\displaystyle\lim_{x\to\infty} g(x)=\lim_{x\to\infty}\frac{\left(\sqrt{x^2+2x}-x\right)\left(\sqrt{x^2+2x}+x\right)}{\sqrt{x^2+2x}+x}=\lim_{x\to\infty}\frac{2x}{x\left(\sqrt{1+\frac{2}{x}}+1\right)}=\frac{2}{2}=1$ **Ans.**

149. $f(x)=2 \quad 0<x<1$

$0 \quad x=1$

$1 \quad x>1$

$\therefore$ Range of $f=\{0, 1, 2\}=A$. (for surjective)

$\Rightarrow$ Number of elements in $A=3$. **Ans.**

150. $\displaystyle L=\lim_{n\to\infty}\left(\tan\left(\frac{\pi}{4}+\frac{\pi}{n}\right)\right)^n$ (1^∞)

$\displaystyle = e^{\lim_{n\to\infty} n\left(\frac{1+\tan(\pi/n)}{1-\tan(\pi/n)}-1\right)} = e^{\lim_{n\to\infty}\frac{2\tan(n)}{\pi/n}\cdot\pi} = e^{2\pi}$

$\therefore$ $l=2\pi$ **Ans.**]

151. Let $\alpha+\beta+\gamma=\theta$

Hence $\cos(\alpha+\beta)+\cos(\beta+\gamma)+\cos(\gamma+\alpha)=\cos(\theta-\alpha)+\cos(\theta-\beta)+\cos(\theta-\gamma)$

$=\cos\theta\sum\cos\alpha+\sin\theta\sum\sin\alpha$

$=2\cos^2\theta+2\sin^2\theta=2$

Hence $a=2$

$\therefore$ $\displaystyle\lim_{x\to 2}\frac{\sqrt{x^2-4}}{\sqrt{x-2}+\sqrt{x}-\sqrt{2}}=2$ **Ans.**

152. When m = 1 , n = 1

$$\lim_{x\to 0} \sin\left(\frac{\pi(1-\cos x)}{x}\right) = \sin 0 = 0$$

When m = 1; n = 2

$$\lim_{x\to 0} \sin\left(\frac{\pi(1-\cos x)}{x^2}\right) = \sin(\pi/2) = 1$$

When m = n = 2

$$\lim_{x\to 0} \sin\left(\frac{\pi(1-\cos^2 x)}{x^2}\right) = \lim_{x\to 0} \sin\left(\frac{\pi(1-\cos x)(1+\cos x)}{x^2}\right) = \sin(\pi) = 0$$

When m = 3; n = 2

$$\lim_{x\to 0} \sin\left(\frac{\pi(1-\cos^3 x)}{x^2}\right) = \lim_{x\to 0} \sin\left(\frac{\pi(1-\cos x)(1+\cos^2 x+\cos x)}{x^2}\right) = \sin\left(\frac{3\pi}{2}\right) = -1 \text{ **Ans.**}$$

153. $$\lim_{n\to\infty} \frac{e\, e^{n\, \ln\left(1-\frac{1}{n}\right)} - 1}{n^{\alpha}}$$

$$\lim_{n\to\infty} \frac{\left(e^{\overbrace{n\,\ln\left(1-\frac{1}{n}\right)+1}^{M}} - 1\right)}{n^{\alpha}} = \lim_{n\to\infty} \frac{(e^M - 1)}{M} \cdot \frac{M}{n^{\alpha}}$$

$$\lim_{n\to\infty} \frac{n\ \ln\left(1-\frac{1}{n}\right)+1}{n^{\alpha}} = c$$

$$\lim_{n\to\infty} \frac{n\left[\frac{-1}{n} - \frac{1}{n^2}\frac{1}{2} + \ldots\ldots\right]+1}{n^{\alpha}}$$

for limit to exist $\alpha = -1$.

$\therefore$ Limt is $\frac{-1}{2} = c$

$\Rightarrow$ $c - \alpha = \frac{-1}{2} - (1) = \frac{1}{2}$

$\therefore$ $24(c - \alpha) = 12$. **Ans.**

154. $\displaystyle \lim_{x\to-\infty} \frac{x^5\cos\left(\frac{1}{\pi x^2}\right)+x^6\sin\left(\frac{1}{\pi x}\right)+7}{|x|^5+6|x|+7} = \lim_{x\to-\infty} \frac{\left(\cos\left(\frac{1}{\pi x^2}\right)+\frac{\sin\left(\frac{1}{\pi x}\right)}{\frac{1}{\pi x}\times\pi}+\frac{7}{x^5}\right)}{\left(-1-\frac{6}{x^4}+\frac{7}{x^5}\right)}$

$$=\frac{\left(1+\frac{1}{\pi}+0\right)}{-1-0+0}=-\left(1+\frac{1}{\pi}\right)\textbf{ Ans.}$$

155. As, $\quad T_n=\dfrac{4n}{1+4n^4}$

$$\Rightarrow \quad T_n=\frac{n}{\frac{1}{4}+n^4}=\frac{n}{\left(n^2+n+\frac{1}{2}\right)\left(n^2-n+\frac{1}{2}\right)}=\frac{1}{2}\left(\frac{\left(n^2+n+\frac{1}{2}\right)-\left(n^2-n+\frac{1}{2}\right)}{\left(n^2-n+\frac{1}{2}\right)\left(n^2+n+\frac{1}{2}\right)}\right)$$

$$\Rightarrow \quad T_n=\frac{1}{2}\left[\frac{1}{n^2-n+\frac{1}{2}}-\frac{1}{n^2+n+\frac{1}{2}}\right]$$

Putting $n=1, 2, 3, ..., n-1, k$, we get

$$T_1=\frac{1}{2}\left(\frac{1}{\frac{1}{2}}-\frac{1}{2+\frac{1}{2}}\right)$$

$$T_2=\frac{1}{2}\left(\frac{1}{2+1/2}-\frac{1}{6+1/2}\right)$$

$$T_3=\frac{1}{2}\left(\frac{1}{6+\frac{1}{2}}-\frac{1}{12+\frac{1}{2}}\right)$$

$$T_k=\frac{1}{2}\left(\frac{1}{k^2-k+1/2}-\frac{1}{k^2+k+1/2}\right)$$

(Adding)

$$= \frac{1}{2}\left(2 - \frac{1}{k^2 + k + \frac{1}{2}}\right)$$

So, $$\lim_{k \to \infty}\left(\sum_{n=1}^{k} \frac{4n}{1+4n^4}\right)^{1+8k+k^2} = \left(1 - \frac{1}{2k^2+2k+1}\right)^{1+8k+k^2} \quad (1^\infty)$$

$$= e^{\lim_{k\to\infty} \frac{k^2+8k+1}{-(2k^2+2k+1)}} = e^{\frac{-1}{2}} = \frac{1}{\sqrt{e}} \text{ Ans.}$$

156. $$g(x) = \lim_{a\to 1}\left(x^2 - 4(p-1)\left(\frac{(a-1)(a^2+a+3)}{(a-1)(a+3)}\right)x + \frac{(a-1)(4a^2+9a+3)}{(a-1)(a+3)}\right)$$

$g(x) = x^2 - 5(p-1)x + 4$

For the existence of limit for all $k \in R$ as $x \to k$

$D \le 0$

$25(p-1)^2 - 16 \le 0 \Rightarrow (5(p-1)-4)(5(p-1)+4) \le 0$

$$\Rightarrow p \in \left[\frac{1}{5}, \frac{9}{5}\right].$$

157. Given limit $= \sqrt{\sin\alpha \sin\beta} = 1$ this is possible only when $\alpha = \beta = \dfrac{\pi}{2}$

159. $$T_r = \frac{(r-1)r(r+1)}{r^3} = \frac{r-1}{r}\cdot\frac{r+1}{r}$$

$$\prod_{r=2}^{n} T_r = \left(\frac{1}{2}\cdot\frac{2}{3}\cdots\cdots\frac{n-2}{n-1}\cdot\frac{n-1}{n}\right)\left(\frac{3}{2}\cdot\frac{4}{3}\cdots\cdots\frac{n}{n-1}\,\frac{n+1}{n}\right) = \frac{n+1}{2n} = \frac{1}{2}. \qquad]$$

160. $$\text{LHL} = \lim_{x\to 0^-}(\sin^{-1}[\sin x] + \cos^{-1}[\cos x] - 2\tan^{-1}[\tan x])$$

$$= \sin^{-1}(-1) + \cos^{-1}(0) - 2\tan^{-1}(-1)$$

$$= -\frac{\pi}{2} + \frac{\pi}{2} - 2\left(-\frac{\pi}{4}\right) = \frac{\pi}{2}$$

$$\text{RHL} = \lim_{x\to 0^+}(\sin^{-1}[\sin x] + \cos^{-1}[\cos x] - 2\tan^{-1}[\tan x]) = \sin^{-1}0 + \cos^{-1}0 - 2\tan^{-1}0$$

$$= 0 + \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

$\therefore$ Given limit $= \dfrac{\pi}{2}$

Questions

01 If $f(x) = \begin{cases} \dfrac{a^x - 1}{x^n \sin x}\left(\dfrac{b\sin x - \sin bx}{\cos x - \cos bx}\right)^n & x > 0 \\ \\ \dfrac{a^x \sin bx - b^x \sin ax}{\tan bx - \tan ax} & x < 0 \end{cases}$

is continuous at $x = 0$ $(a, b > 0, b \neq 1, a \neq b)$. Find $f(O^+)$ and $f(O^-)$ and obtain relation between a, b and n.

02 Let $f(x) = \text{cosec } 2x + \text{cosec } 2^2 x + \text{cosec } 2^3 x + \ldots\ldots\ldots\ \text{cosec } 2^n x,\ x \in (0, \pi/2)$
and $g(x) = f(x) + \cot 2^n x$

If $H(x) = \begin{cases} (\cos x)^{g(x)} + (\sec x)^{\text{cosec } x} & \text{if } x > 0 \\ p & \text{if } x = 0 \\ \dfrac{e^x + e^{-x} - 2\cos x}{x \sin x} & \text{if } x < 0 \end{cases}$

Find the value of p, if possible to make the function H (x) continuous at x = 0.

03$_{\text{(mcq)}}$ Let $f(x) = \cos x$ and $H(x) = \begin{cases} \text{Min}[f(t)/0 \le t \le x] & \text{for } 0 \le x \le \frac{\pi}{2} \\ \frac{\pi}{2} - x & \text{for } \frac{\pi}{2} < x \le 3 \end{cases}$, then

(A) H (x) is continuous & derivable in [0, 3]
(B) H(x) is continuous but not derivable at $x = \pi/2$
(C) H(x) is neither cont. nor deri. at $x = \pi/2$
(D) Maximum value of H(x) in [0,3] is 1

04 $f(x) = \begin{cases} x^2 + x + 1, \text{if } x < 0 \\ (x-3)^2 + 2b, \text{if } x \ge 0 \end{cases}$ and $g(x) = \begin{cases} 2x + a, \text{ if } x < 0 \\ |x - 2|, \text{ if } x \ge 0 \end{cases}$ where a is non negative real number and b is any real number. If fog(x) is continuous for all real x, then find the sum of all possible distinct values of a & b.

05 Let f be a real valued differentiable function on R such that $f'(1) = 6$ and $f'(2) = 2$.

Then $\displaystyle\lim_{h \to 0} \frac{f(3\cos h + 4\sin h - 2) - f(1)}{f(3e^h - 5\sec h + 4) - f(2)}$ is equal to

(A) 24 (B) 8 (C) 4 (D) 2

06 Let $g(x) = e^{f(x)}$ where g(x) is a differentiable function on $(0, \infty)$ such that $g(x+1) = (x+1)g(x)$.
Then for n = 1, 2, 3........
$f'(n+1) - f'(1) =$

(A) $-4\left(\dfrac{1}{1^2} + \dfrac{1}{2^2} + \ldots\ldots + \dfrac{1}{n^2}\right)$
(B) $4\left\{\dfrac{1}{1^2} + \dfrac{1}{2^2} + \ldots\ldots + \dfrac{1}{n^2}\right\}$

(C) $1 + \dfrac{1}{2} + \dfrac{1}{3} + \ldots\ldots + \dfrac{1}{n}$
(D) $\dfrac{1}{2} + \dfrac{1}{3} + \ldots\ldots + \dfrac{1}{n+1}$

07. If $f(x)=\begin{cases}[x], \text{ if } -3\le x<0\\ 2x+1, \text{ if } 0\le x\le 2\end{cases}$ and $g(x)=f(|x|)+|f(x)|$, then evaluate $\lim\limits_{x\to 0} g(x)$

08 Given $f(x)=\sum\limits_{k=1}^{n}\tan\left(\frac{x}{2^k}\right)\sec\left(\frac{x}{2^{k-1}}\right)$; $k, n \in N$

& $g(x)=\lim\limits_{n\to\infty}\prod\limits_{r=1}^{n}\left(1-\tan^2\frac{x}{2^r}\right)$; $r, n\in N$

If $h(x)=\begin{cases}\dfrac{\ln\left(1+f(x)+\tan\dfrac{x}{2^n}\right)}{x} & ; \quad x<0\\ g(x) & ; \quad x>0\\ k & ; \quad x=0\end{cases}$ then

(A) if $k=0$, f(x) is continuous at $x=0$ (B) if $k=1$, h(x) is continuous at $x=0$
(C) if $k\in R$, h(x) is cotinuous at $x=0$ (D) h(x) can not be made continuous at $x=0$ for any k

09 Let α be the number of possible integral values of a for which $ax-x^2<1\ \forall\ x\in R$.
Number of points of non-derivability of the function $g(x)=[3+4\alpha\sin x]$ in $(0,\pi)$ is
[**Note:** [k] denotes largest integer less than or equal to k.]

10. If $f(x)=\dfrac{\sqrt{a^2-ax+x^2}-\sqrt{a^2+ax+x^2}}{\sqrt{a+x}-\sqrt{a-x}}$, $x\ne 0$ then the value of f(0) such that f(x) is continuous at $x=0$ (where $a>0$), is

(A) $\frac{1}{\sqrt{a}}$ (B) $\frac{-1}{\sqrt{a}}$ (C) $\sqrt{a}$ (D) $-\sqrt{a}$

11. Let f(x), $f:R\to R$ be a non-constant continuous function such that $(e^x-1)f(2x)=(e^{2x}-1)f(x)$

If $f'(0)=1$, then $\lim\limits_{x\to 0}\left(\frac{f(x)}{x}\right)^{\frac{1}{x}}$ equal to

(A) e (B) $e^{1/2}$ (C) e^2 (D) e^{-2}

12. Let $f(x)=\begin{cases}|[x]|, & 0\le\{x\}<\frac{1}{2}\\ |x|, & \frac{1}{2}\le\{x\}<1\end{cases}$, $\forall\ x\in\left[\frac{-7}{2},\frac{7}{2}\right]$.

If L is number of point of discontinuity and M is the number of point on non-differentiability of the function f(x), then find the value of (L + M).

13. Let f be an injective function with domain $[a, b]$ and range $[c, d]$. If α is a point in (a, b) such that f has left hand derivative l and right hand derivative r at $x = \alpha$ with both l and r non-zero different and negative, then left hand derivative and right hand derivative of f^{-1} at $x = f(\alpha)$ respectively, is

(A) $\frac{1}{r}, \frac{1}{l}$ (B) r, l (C) $\frac{1}{l}, \frac{1}{r}$ (D) l, r

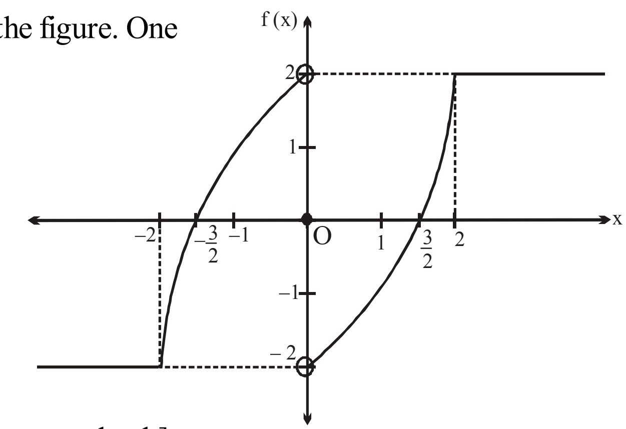
14. **Statement-1:** The number of points where function $f(x) = \left| x(e^x - 1)(x-1)\tan^{-1}(x^2 + 3x - 4) \right|$ is non-derivable is 1.

Statement-2: For a differentiable function $g(x)$, $h(x) = |g(x)|$ is differentiable at $x = x_0$ where x_0 is either a repeated root of the equation $g(x) = 0$ or $g(x_0) \neq 0$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

15.$_{\text{(mcq)}}$ The graph of a function $y = f(x)$ is shown in the figure. One more function $y = g(x)$ is defined as

$$g(x) = \begin{cases} f(x) - 2, & x < 0 \\ f(x), & x = 0 \\ \dfrac{-1}{f(x)} - \dfrac{1}{2}, & 0 < x < \dfrac{3}{2} \\ -f(x), & x \geq \dfrac{3}{2} \end{cases}$$



Identify the correct statement(s)?
[**Note :** [k] denotes greatest integer less than or equal to k]

(A) Range of $g(x)$ is $[-4, \infty)$.
(B) $\lim_{x \to 0^-} [g(x)] = -1$.
(C) $g(x)$ is continuous at $x = 0$.
(D) $g(x)$ is discontinuous at $x = \pm\frac{3}{2}$.

16.$_{\text{(mcq)}}$ Which of the following statement(s) is(are) **correct**?

(A) If $f(x)$ is discontinuous and $g(x)$ is continuous at $x = a$ $(g(a) \neq 0)$, then $\frac{f(x)}{g(x)}$ must be discontinuous at $x = a$.
(B) If $f(x)$ is defined in $[a, b]$ and $f(a)\,f(b) < 0$, then the equation $f(x) = 0$ has atleast one real root in (a, b).
(C) If $f(x)$ is continuous at $x = a$ $(f(a) \neq 0)$ and $g(x)$ is discontinuous at $x = a$, then the functionh $(x) = f(x)\,g(x)$ must be discontinuous at $x = a$.
(D) If $fog(x)$ is continuous at $x = a$ and $g(x)$ is also continuous at $x = a$, then $f(x)$ must be continuous at $x = a$.

Paragraph for Question no. 17 to 19

Consider a function $f(x) = \sin^{-1}[x]\cos^{-1}\{x\}$ where [ ] denotes G.I.F. and { } denotes fractional part of x then answer following questions?

17$_{(mcq)}$ Find the correct statements above f(x)?

(A) Domain is $[-1, 1]$

(B) Domain is $[-1, 2)$

(C) Range is $\left[\dfrac{-\pi^2}{4}, \dfrac{\pi}{4}\right]$.

(D) Range is $\left[\dfrac{-\pi^2}{4}, \dfrac{\pi}{4}\right)$.

18 Find the correct statements above continuity and differentiability of f(x)?

(A) continuous at $x = 0$.

(B) continuous at $x = 1$.

(C) differentiale at $x = 0$.

(D) differentiable at $x = 1$.

19$_{(mcq)}$ Find the correct statements about nature of f(x).

(A) $\lim\limits_{x\to 0} f(x)$ exists and equal to zero.

(B) $\lim\limits_{x\to 1} f(x)$ exists and equal to zero.

(C) f(x) is injective $\forall\ x \in (-1, 0)$.

(D) f(x) is injective $\forall\ x \in (1, 2)$.

20$_{(mcq)}$ Let $f : R^+ \to R$ be a continuous function satisfying $f\left(\dfrac{x}{y}\right) = f(x) - f(y)\ \ \forall\ x, y \in R^+$. If $f'(1) = 1$, then

(A) f is unbounded

(B) $f(x) = \lim\limits_{n\to\infty} n\left(x^{1/n} - 1\right),\ \forall\ x > 0$

(C) $\lim\limits_{x\to 0}\dfrac{f(1+x)}{x} = 1$

(D) $\lim\limits_{x\to 0} x\,f(x) = 0$

21$_{(mcq)}$ Let f(x) be a differentiable function satisfying the relation $\left|f(x) - \sum\limits_{k=1}^{n} x^k - f(y) + \sum\limits_{k=1}^{n} y^k\right| \le (x-y)^2$, for all $x, y \in R$ and $f(0) = 1$

(A) $\lim\limits_{n\to\infty} f\left(\dfrac{1}{2}\right) = 2$

(B) For $n = 100$, $\lim\limits_{x\to 1}\dfrac{f(x) - f(1)}{x - 1} = 5050$

(C) For $n = 3$, sgn (f(x)) is discontinuous at exactly are point

(D) For $n = 4$, sgn (f '(x)) is discontinuous at exactly are point.

22. If f(x) is defined $\forall\ x \in R$ and is discontinuous only at $x = 0$ such that $f^3(x) - 6f^2(x) + 11f(x) - 3 = 3\ \forall\ x \in R$, then the number of such functions is equal to

(A) 4 (B) 8 (C) 16 (D) 24

23. Given $f(x) = \sum_{r=1}^{n} \tan\left(\frac{x}{2^r}\right) \sec\left(\frac{x}{2^{r-1}}\right)$; r, n ∈ N

$$g(x) = \lim_{n\to\infty} \frac{\ln\left(f(x)+\tan\frac{x}{2^n}\right) - \left(f(x)+\tan\frac{x}{2^n}\right)^n \cdot \left[\sin\left(\tan\frac{x}{2}\right)\right]}{1+\left(f(x)+\tan\frac{x}{2^n}\right)^n}$$

$= k$ for $x = \frac{\pi}{4}$ and the domain of g (x) is (0 , π/2).

where [ ] denotes the greatest integer function.

Find the value of k, if possible, so that g (x) is continuous at x = π/4. Also state the points of discontinuity of g (x) in (0 , π/4) , if any.

24. **(mcq)** A function f is defined on an interval [a, b]. Which of the following statement(s) is/are INCORRECT?

(A) If f(a) and f(b), have opposite sign, then there must be a point c ∈ (a, b) such that f(c) = 0.

(B) If f is continuous on [a, b], f(a) < 0 and f(b) > 0, then there must be a point c ∈ (a, b) such that f(c) = 0

(C) If f is continuous on [a, b] and there is a point c in (a, b) such that f(c) = 0, then f(a) and f(b) have opposite sign.

(D) If f has no zeroes on [a, b], then f(a) and f(b) have the same sign.

25. The function $f(x) = \left(\frac{2+\cos x}{x^3 \sin x} - \frac{3}{x^4}\right)$ is not defined at x = 0. How should the function be defined at x = 0 to make it continuous at x = 0.

26. Let f be an odd continuous function defined on set of real number such that $f(p) = (-1)^p\, p$, where p is any prime natural number then find the minimum number of real roots of f(x) = 0.

27 If for a differentiable function y = f(x), $\left(f'(x) > 0\right)$ $\lim_{x\to 1} \frac{e^{2f(x)} - 2e^{f(x)+1} + e^2\cos(x-1)}{\sec^2(x-1)-1} = \frac{7}{2}e^2$ and area of the triangle formed by the tangent drawn to the curve y = f(x) at $\left(1, f(1)\right)$ and co-ordinate axes is Δ, then find the value of $\frac{1}{\Delta}$.

28 **(mcq)** Let α is the number of solution of $3^{\{x\}} + [x^2 - 2x + 3] = 0$ and $f(x) = \frac{xa^x + bx\cos x + c\sin x}{x^5}$, $x \neq \alpha (a \geq 1)$ is continuous at x = α, then

(A) $f(\alpha) = \frac{1}{120}$ (B) a + b + c = 0 (C) 4 abc = – 3 (D) a – c = 5b

[**Note :** { y } and [y] denotes fraction part of function and greatest integer function of y.]

29. Let $f(x) = \lim_{n\to\infty} \sum_{r=0}^{n-1} \frac{x}{(rx+1)\ \{(r+1)x+1\}}$, then

(A) $f(x)$ is continuous but not derivable at $x = 0$
(B) $f(x)$ is both continuous and derivable at $x = 0$
(C) $f(x)$ is neither continuous nor derivable at $x = 0$
(D) $f(x)$ has non removable finite type of discontinuity at $x = 0$.

30$_{\text{(mcq)}}$ Let $f(x) = [x]$ and $g(x) = \begin{cases} 0, & x \in I \\ x^2, & x \notin I \end{cases}$, then (Here [ ] denotes G.I.F.)

(A) $\lim\limits_{x\to1} g(x)$ exists, but $g(x)$ is not continuous at $x = 1$
(B) $g(x)$ is discontinuous $\forall\, x \in I$
(C) *gof* is continuous for all x
(D) *fog* is continuous for all x

31$_{\text{(mcq)}}$ Consider the function $f : R \to A$ given by $f(x) = x + [x]$ where $[x]$ denotes greatest integer less than or equal to x and A is the range of the function. Then which of the following is/are correct?

(A) $f^{-1}(x)$ exists and is given by $f^{-1}(x) = x - \dfrac{[x]}{2}\ \forall\, x \in A$.
(B) $f^{-1}(x)$ exists and is given by $f^{-1}(x) = x - [x]\ \forall\, x \in A$.
(C) $f(x)$ is a continuous function.
(D) $f^{-1}(x)$ is a continuous function.

32. For $x \in (0, \infty)$, let $f(x) = \ln x + x + 1$.

If $g(x) = \begin{cases} \left(1+\dfrac{15(x-2)}{x^2+1}\right)^{\frac{\ln(3e)}{\tan(x-2)}} & ; \quad 1 \le x < 2 \\ 27e^3 & ; \quad x = 2 \\ \lambda\left(\dfrac{e^{f(x)}-2e^3}{\sin x\cos 2-\cos x\sin 2}\right) & ; \quad 2 < x \le 3 \end{cases}$

is continuous in $[1,3]$ then find the value of λ.

33. A derivable function $f : R^+ \to R$ satisfies the condition $f(x) - f(y) \ge \ln(x/y) + x - y$ for every $x, y \in R^+$. If g denotes the derivative of f then compute the value of the sum $\sum\limits_{n=1}^{100} g\left(\dfrac{1}{n}\right)$.

34. Let f(x) be a real valued function not identically zero satisfies the equation,
$f(x + y^n) = f(x) + (f(y))^n$ for all real x & y and $f'(0) \ge 0$ where $n\ (> 1)$ is an odd natural number. Find f(10).

Paragraph for questions nos. 35 to 37

Let f be a differentiable function satisfying the relation

$$f(x+y) = f(x) + f(y) - 2xy + (e^x - 1)(e^y - 1) \quad \forall\ x, y \in R \text{ and } f'(0) = 1.$$

35. $\{f(2)\}$ is equal to

(A) $e^2 - 5$ (B) $e^2 - 6$ (C) $e^2 - 7$ (D) $e^2 - 8$

Note: $\{y\}$ denotes the fraction part function of y.

36. The value of $\lim\limits_{x\to 0} \dfrac{f(2x) + 4x^2 - 2x}{x^2}$ is

(A) 2 (B) 4 (C) 8 (D) 16

37. Let $g(x) = f(x) + x^2 - 2$. If the equation $|g(|x|)| = k$ has four distinct solutions then the set of values of k is

(A) (0, 1) (B) (– 2, 2) (C) (0, 3) (D) (0, 2)

38. Let $f(x) = \begin{cases} \{x^2\}, & -1 \le x < 1 \\ |1-2x|, & 1 \le x < 2 \\ (1-x^2)\,\text{sgn}(x^2-3x-4), & 2 \le x \le 4 \end{cases}$

If m denotes the number of points of discontinuity of $f(x)$ in $[-1, 4]$ and n denotes the number of points of non-derivability of $f(x)$ in $(-1, 4)$ then $(m+n)$ equals

(A) 2 (B) 4 (C) 5 (D) 6

[**Note:** {k} and sgn(k) denote fractional part function and signum function of k respectively)]

39$_{(mcq)}$ Let g(x) and h(x) be two quadratic polynomials with leading coefficient as unity. If $g(0) = c$, $h(0) = b$, $g(1) = 0$, $h(1) = 0$ and $b + c = -1$, also $f(x) = \dfrac{g([x])}{h([x])}$

([ ] represents greatest integer function). Which of the following holds good?

(A) domain of $f(x)$ is $R - [1, 2)$ if $b \notin I$

(B) domain of $f(x)$ is $R - [1, 2) \cup [b, b+1)$ if $b \in I$

(C) $\lim\limits_{x\to 1^-} f(x) = \dfrac{c}{b}$, if $b \ne 0$

(D) $f(x)$ is continuous if $b = c = -\dfrac{1}{2}$

40. Let $f(x) = \begin{cases} \lim\limits_{t\to 0}\left[\dfrac{q\tan t}{t}\right]\dfrac{(1-\sin x)}{(\pi-2x)^2}, & x > \dfrac{\pi}{2} \\ \dfrac{1}{2}\lim\limits_{\alpha\to 0}\left(\left[\dfrac{p\sin\alpha}{\alpha}\right]\right), & x = \dfrac{\pi}{2} \\ \dfrac{1-\sin^3 x}{3\cos^2 x}, & x < \dfrac{\pi}{2} \end{cases}$

where p, q are natural numbers and [ ] denotes greatest integer function. If $f(x)$ is continuous at

$x = \frac{\pi}{2}$ then find the value of $(p + q)$.

41$_{\text{(mcq)}}$ If $f(x) = \text{sgn}\left((\sin^2 x - \sin x - 1)(\sin^2 x + \sin x + 1)\right) = 0$ has exactly 4 points of discontinuity for $x \in (0, n\pi)$, $n \in N$ then the value of n may be equal to

(A) 2 (B) 3 (C) 4 (D) 5

42$_{\text{(mcq)}}$ Let $f(x) = \sum_{r=1}^{n} (x^r + x^{-r})^2,\ x \neq \pm 1$ and $g(x) = \begin{cases} \lim\limits_{n\to\infty} \left(f(x) - 2n\right) x^{-2n-2}(1-x^2),\ x \neq 0, \pm 1 \\ -1,\quad x = 0, \pm 1 \end{cases}$

then g (x)

(A) is discontinuous at $x = -1$

(B) is continuous at $x = 2$

(C) has a removable discontinuity at $x = 1$

(D) has an irremovable discontinuity at $x = 1$

43$_{\text{(mcq)}}$ Which of the following function(s) is/are continuous at $x = 0$ (where Q is the set of rational numbers.)?

(A) $f(x) = \begin{cases} |x| & x \in Q \\ -|x| & x \notin Q \end{cases}$

(B) $f(x) = \begin{cases} x^2 - 2 & x \in Q \\ -x^2 - 2 & x \notin Q \end{cases}$

(C) $f(x) = \begin{cases} \{x\} & x \in Q \\ [x] & x \notin Q \end{cases}$ where $\{x\}$ and $[x]$ denote the fractional part function and greatest integer function of x, respectively.

(D) $f(x) = \begin{cases} x & x \in Q \\ 1 - x & x \notin Q \end{cases}$

44. Let $f(x) = \text{cosec } 2x + \text{cosec } 2^2 x + \text{cosec } 2^3 x + \ldots\ldots\ldots\ \text{cosec } 2^n x,\ x \in (0, \pi/2)$

and $g(x) = f(x) + \cot 2^n x$

If $H(x) = \begin{cases} (\cos x)^{g(x)} + (\sec x)^{\text{cosec}\, x} & \text{if } x > 0 \\ p & \text{if } x = 0 \\ \dfrac{e^x + e^{-x} - 2\cos x}{x \sin x} & \text{if } x < 0 \end{cases}$

Find the value of p, if possible to make the function H (x) continuous at $x = 0$.

45 Let $X = \{1, 2, 3, 4, 5, 6\}$ and a, c are natural numbers selected from set X with replacement.

Let $f(x) = \begin{cases} x^2 + x, & x \leq 1 \\ x^2 - ax, & x > 1 \end{cases}$, $g(x) = \begin{cases} x^2 - cx, & x \leq 2 \\ x^2 - 4x, & x > 2 \end{cases}$

and $h(x) = f(x) \cdot g(x),\ x \in R$.

N_1 = Number of function h(x) such that f (x) and g(x) are both discontinuous.

N_2 = Number of function h (x) such that h (x) is continuous at x = 1 and discontinuous at x = 2 but f (x) and g(x) are both discontinuous.
N_3 = Number of function h (x) such that h (x) is discontinuous at x = 1 and continuous at x = 2.
Find the value of $(N_1 + N_2 + N_3)$.

46 Let f(x) be a continuous function satisfying $f^3(x) - 5f^2(x) + 10f(x) - 12 \le 0$, $f^2(x) - 4f(x) + 3 \ge 0$ and $f^2(x) - 6f(x) + 8 \le 0$ and A is the area bounded by the line $y = x$, $y = f(x)$ and $x = 0$. Find the value of 10A.

47. Let $y = f(x)$ be a differentiable function which satisfies the functional rule $f(xy) = f(x) + f(y) - x - y + xy$, $x \in R^+$, $y \in R$ and $f'(1) = 2$. If g is the inverse of f, then find the value of $\left[g\left(f'\left(\frac{1}{e}\right)\right)\right]$.

[**Note :** [y] denotes greatest integer function $\le$ y.]

48. Let $f(x) = x^2 + ax + 3$ and $g(x) = x + b$, where $F(x) = \lim\limits_{n \to \infty} \dfrac{f(x) + x^{2n} g(x)}{1 + x^{2n}}$.
If F(x) is continuous at x = 1 and x = – 1 then find the value of $(a^2 + b^2)$.

49. Let f(x) be a function defined in [0, 5] such that $f^2(x) = 1 \ \forall\ x \in [0, 5]$ and f(x) is discontinuous only at all integers in [0, 5]. Find total number of possible functions.

50. Consider a function,

$$f(x) = \begin{cases} \dfrac{\ln\left(1 - ax\left[\dfrac{4x^2}{x^4+1}\right] + a\right)}{e^{bx} - e^{b\left[\frac{4x^2}{x^4+1}\right]}}, & -1 < x < 1;\ x \ne 0 \\[2ex] \dfrac{(\tan bx - \tan b)(1 + \cos 2bx)}{2(x-1)}, & 1 < x < 2 \end{cases}$$

where [k] denote greatest integer less than or equal to k.
If f(x) is continuous in $(-1,0) \cup (0, 2)$ and passing through the point $(1, \ln 2)$ then find the value of $(1 + a + 2b^2)$.

51. Let $f : R \to R$ be a function defined as

$$f(x) = \begin{cases} 1 - |x|, & |x| \le 1 \\ 0, & |x| > 1 \end{cases}$$

and $g(x) = f(x - 1) + f(x + 1)\ \forall\ x \in R$.
Find number of points of non-differentiability of g(x) on R.

52$_{\textbf{(mcq)}}$ Let f be a function defined by $f(x)=\begin{cases}\tan^{-1}\left|\dfrac{x-1}{x+1}\right|, & x\neq -1\\ 0, & x=-1\end{cases}$

then which of the following is(are) incorrect.

(A) $\lim\limits_{x\to -1^+} f(x)$ does not exist.

(B) $\lim\limits_{x\to -1^-} f(x)$ does not exist.

(C) f is discontinuous at $x=-1$ because $\lim\limits_{x\to -1^-} f(x)\neq f(-1)$.

(D) f is discontinuous because $\lim\limits_{x\to -1^+} f(x)\neq \lim\limits_{x\to -1^-} f(x)$.

53$_{\textbf{(mcq)}}$ Consider the function $f(x)=\begin{cases}1+e^x, & x<0\\ x+2, & 0\le x\le 3\\ 6-\dfrac{3}{x}, & x>3\end{cases}$

Which of the following is(are) correct?

(A) f(x) is continuous $\forall\ x\in R$

(B) f(x) is non-derivable $\forall\ x\in R$

(C) f(x) is bounded

(D) f(x) has exactly one horizontal asymptote

54. Let $f: R^+\to R^+$ be a differentiable function such that $f(x^n)=2f(x^{n-1})$, $n\in N$, $n\ge 2$. If $k\cdot f'(2^n)=f'(2)$, then the value of k is

(A) $n!$

(B) $\dfrac{1}{n!}$

(C) n

(D) $\dfrac{1}{n}$

55. Let f(x) be continuous function $\forall\ x\in R$ except at $x=1$ such that $f'(x)>0\ \forall\ x\in(1,\infty)$ and $f'(x)<0\ \forall\ x\in(-\infty,1)$ such that $f(1^+)=\dfrac{1}{2}$ and $f(1^-)=2$, $f(1)=1$ then $\lim\limits_{x\to 0} f(\cos^3 x+\sin^2 x)$ is equal to

(A) $\dfrac{1}{2}$

(B) 1

(C) 2

(D) 0

56(mcq) Let $f(x)=\begin{cases}\dfrac{x\,e^{[x]+|x|}-\pi}{[x]+|x|}, & x\neq 0\\ \dfrac{22}{7}, & x=0\end{cases}$

where [y] denotes largest integer $\le$ y. Then which of the following hold(s) good?

(A) $f(x)$ is continuous at $x=0$.
(B) $f(x)$ is discontinuous at $x=0$.
(C) $f(x)$ is derivable at $x=0$.
(D) $f(x)$ is non-derivable at $x=0$.

57. Let f be a differentiable function on R and f '(3) = 2, then $\displaystyle\lim_{h\to 0}\left(\frac{f(3+\sin^3 h)-f(3-\sin^2 h)}{h^2}\right)$ is equal to

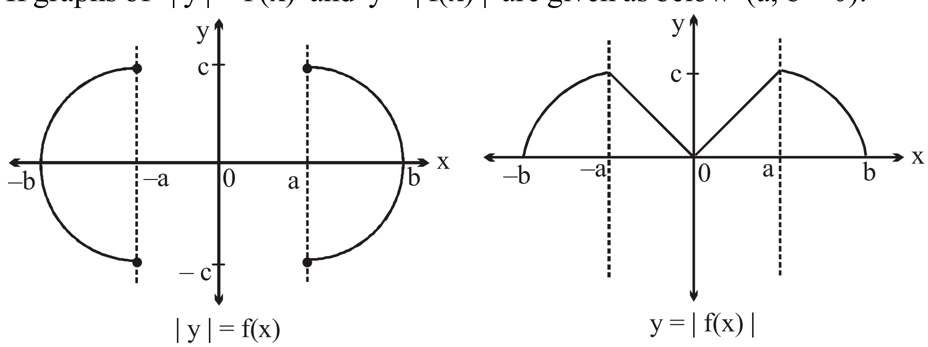
(A) 1 (B) 2 (C) 3 (D) 4

58. Let f(x) be a differentiable function satisfies $f(x+y)=f(x)+f(y)$ for all $x, y\in R$ and $f'(0)=1$,

then $\displaystyle\lim_{x\to 0}\frac{2^{f(\tan x)}-2^{f(\sin x)}}{x^2 f(\sin x)}$ is equal to

(A) ln 2 (B) $\frac{1}{2}\ln 2$ (C) $\frac{1}{4}\ln 2$ (D) $\frac{1}{8}\ln 2$

59. If graphs of $|y|=f(x)$ and $y=|f(x)|$ are given as below $(a, b>0)$.



Then identify the correct statement.

(A) f(x) is discontinuous at 2 points in [–b, b] and non-differentiable at 2 points in (–b, b).
(B) f(x) is discontinuous at 2 points in [–b, b] and non-differentiable at 3 points in (–b, b).
(C) f(x) is discontinuous at 3 points in [–b, b] and non-differentiable at 3 points in (–b, b).
(D) f(x) is discontinuous at 3 points in [–b, b] and non-differentiable at 4 points in (–b, b).

60. If A, B, C are angles of an acute angled triangle with A < B < C and

$f(x)=\begin{cases}\left(1+(\sin A)^{\frac{1}{x}}\right)^{(\operatorname{cosec} B\sin C)^{\frac{1}{x}}}, & x>0\\ k, & x=0\end{cases}$ is right continuous at $x=0$ then the value of k is equal to

(A) $e^{\frac{\sin A\sin C}{\sin B}}$ (B) e^{ABC} (C) e (D) 1

61. Let, $f(x)=\begin{cases}\dfrac{8}{\pi}\tan^{-1}(-|x|+3), & |x|>2\\ \left[\dfrac{3x^2-|x|+3}{x^2+1}\right], & |x|\le 2\end{cases}$

where [x] denotes largest integer less than or equal to x.

Number of points where the function $f(x)$ is discontinuous is equal to

(A) 0 (B) 1 (C) 2 (D) 4

62. Let $f(x)=\begin{cases}\dfrac{\pi^2}{e}(1-x)\dfrac{\left(x^{\frac{1}{x-1}}-e\right)}{(1+\cos\pi x)}, & x\ne 1\\ k, & x=1\end{cases}$

If $f(x)$ is continuous at $x=1$, then the value of k, is

(A) 1 (B) 2 (C) 3 (D) 4

63 Let $f(t) = |t|+|t-1|\ \forall\ t\in R$ and $g(x)=\begin{cases}\max.(f(t)), & x-1\le t\le x, \quad 0\le x\le 1\\ 3-x, & 1<x\le 2\end{cases}$

Number of points where g(x) is non-derivable in [0, 2], is

(A) 0 (B) 1 (C) 2 (D) 3

64. Let $f(x)=\lim\limits_{n\to\infty}\sin^{2n}x$, then number of points where $f(x)$ is discontinuous is

(A) 0 (B) 1 (C) 2 (D) infinitely many

65. If $f(x)=\begin{cases} f(x)=ln\left(1+|x|+(b-2)|x+1|\right)+a\tan^{-1}x+2, & -2<x<0\\ b, & x=0\\ \dfrac{(c+2)\sin^{-1}\sqrt{\{x\}}+e^{2x}-e^{ln\left(4\left\{\frac{x}{2}\right\}\right)}-1}{x^2}, & 0<x<2\end{cases}$

is derivable in $(-2, 2)$ then find the value of $(9a^2+b^2+c^2)$.

[**Note:** {y} denotes fractional part of y.]

Paragraph for question nos. 66 to 68

Let $f(x)=\begin{cases}\max.\ \{t^3-t^2+t+1, 0\le t\le x\}, & 0\le x\le 1\\ \min.\ \{3-t, 1<t\le x\} & 1<x\le 2\end{cases}$

and $g(x)=\begin{cases}\max.\ \{\frac{3}{8}t^4+\frac{1}{2}t^3-\frac{3}{2}t^2+1, 0\le t\le x\}, & 0\le x<1\\ \min.\ \{\frac{3}{8}t+\frac{1}{32}\sin^2\pi t+\frac{5}{8}, 1\le t\le x\} & 1\le x\le 2\end{cases}$

66 The function f(x), $\forall\ x\in[0,2]$ is

(A) continuous and differentiable
(B) continuous but not differentiable
(C) discontinuous and not differentiable
(D) none

67 Which of the following is true?

(A) $\lim\limits_{x\to1^-} fog(x) > \lim\limits_{x\to1^+} gof(x)$
(B) $\lim\limits_{x\to1^-} fog(x) < \lim\limits_{x\to1^+} gof(x)$
(C) $\lim\limits_{x\to1^-} fog(x) = \lim\limits_{x\to1^+} gof(x)$
(D) none of these

68. Let $Z(x)=\frac{d}{dx}\left(f(x)^{g(x)}\right)$ and $Y(x)=\frac{d}{dx}\left(g(x)^{f(x)}\right)$ then Z(x) and Y(x) vanish simultaneously at

(A) $x=-\frac{1}{3}$
(B) $x=0$
(C) $x=1$
(D) No real value of x

69. Consider the function f defined as

$$f(x)=\begin{cases}\dfrac{e^{[3\cot x-1]}+\sin 2x-a+[\cos x-\sin x]}{\left(x-\frac{\pi}{4}\right)^2}, & x>\frac{\pi}{4}\\ P([P]-2)-2, & x=\frac{\pi}{4}.\\ b\left[\dfrac{2}{3}-\dfrac{1+\sin 2x}{1+\sin 2x+\sin^2 2x}\right], & x<\frac{\pi}{4}\end{cases}$$

If f(x) is continuous at $x=\frac{\pi}{4}$ then the value of $[a+b+n]$ where n is number of integral values of P.

[**Note :** [y] denotes greatest integer less than or equals to y

70 If $f(x)=\begin{cases}\dfrac{\tan[x^2]\pi}{ax^2}+ax^3+b, & 0<x\le1\\ 2\cos\pi x+\tan^{-1}x, & 1<x\le2\end{cases}$ is differentiable in $x\in(0,2]$. Then $a=\frac{1}{k}$ and $b=\frac{\pi}{4}-\frac{26}{k_2}$. Then find the value of (k_2-k_1).

Paragraph for question nos 71 to 73

A function $f(x)$ is said to be differentiable at $x = c$, if $f'(c^+) = f'(c^-)$ = a finite quantity. Also $f(x)$ is said to be bounded if $|f(x)| < n \ \forall\ x$ for which $f(x)$ is defined (where n is finite)

71. Let $f : R \to R$ and $c \in R$ such that $\lim\limits_{n\to\infty} n\left(f\left(c+\frac{1}{n}\right) - f(c)\right) = a\ (a \in R)$ then

(A) $f'(c)$ exists and is equal to a
(B) $f'(c)$ does not exists
(C) $f'(c)$ exists but is not equal to a
(D) $f'(c)$ may or may not exist.

72. If $\lim\limits_{x\to\infty} f(x) = 0$ and $\lim\limits_{x\to-\infty} f(x) = 0$ and f(x) be continuous on R (f(x) is not a constant function). Then

(A) $f(x)$ is bounded on R and attains both maximum and minimum on R.
(B) $f(x)$ is unbounded on R but attains minimum on R.
(C) $f(x)$ is bounded on R and attains either maximum or minimum or both on R.
(D) f is unbounded.

73. $f(x)$ is defined on [0, 1] and $f(x) = \begin{cases} x^a \sin(x^{-c}) & x \neq 0 \\ 0 & x = 0 \end{cases}$ where $a, c \in R$ and $c > 0$

(A) $f'(0)$ exists if $0 < a < 1$
(B) $f(x)$ is continuous if $a < 0$
(C) $f'(x)$ is bounded and continuous if $a > c$
(D) $f(x)$ and $f'(x)$ are continuous if $a > 1 + c$.

74. Let $f(x)$, $g(x)$ and $h(x)$ be continuous and differentiable function such that

$$f\left(\frac{1}{n}\operatorname{cosec}\frac{1}{n}\right) = g\left(n\tan\frac{1}{n}\right) = h\left(2n^2\left(1-\cos\frac{1}{n}\right)\right) = 0\ \forall\ n \in N.$$

If $F(x) = f(x) + g(x) + h(x)$ then

(A) $F(x) = 0\ \forall\ x \in (0, 1]$
(B) $F(1) = 0$ and $F'(1) = 0$
(C) $F'(x) = 0 = F''(x)\ \ \forall\ x \in (0, 1]$
(D) $F(1) = 0$ but $F'(1)$ need not to be zero.

75. Let $f(x) = \max.\left\{\left|x^2 - 2|x|\right|, |x|\right\}$ and $g(x) = \min.\left\{\left|x^2 - 2|x|\right|, |x|\right\}$. If L denotes number of points where f(x) is non-derivable and M denotes the number of points where g(x) is non-derivable, then find the value of $(L + M)$.

76. Let $f : \left[0, \frac{\pi}{2}\right] \to [-1, 1]$ and $g : \left[0, \frac{\pi}{2}\right] \to [-1, 1]$ be two functions defined as $f(x) = \sin\left(\frac{\pi}{2}[x^2] - x^2\right)$ and $g(x) = \cos\left(\frac{\pi}{2}[x^3] - x^3\right)$ respectively, where [x] denotes greatest integer function of x and p denotes number of points of discontinuity of $f(x)$ and q denotes number of points of non-differentiability of g(x) then find the value of $(p^2 + q^2)$

77. Let f(x) be a differentiable function satisfying $f(xy) = f(x) + f(y) + xy - x - y - 1 \ \forall \ x, y > 0$

and $\lim\limits_{h\to 0} \dfrac{2-f(1-h)}{h} = 2$. If $L = \lim\limits_{x\to\infty} \dfrac{2+\dfrac{2}{x}-f\left(\dfrac{x+1}{x}\right)}{1-\sqrt{\cos\dfrac{1}{3x}}}$ then find the absolute value of L.

78. Let R be the set of real numbers and $f: R \to R$, be a differentiable function such that $|f(x) - f(y)| \le |x - y|^3 \ \forall \ x, y \in R$. If $f(10) = 100$, the value of $f(20)$ is equal to

79. Let $f(x) \begin{cases} \sqrt{(2n+1)x - x^2 - (n^2+n)} & n \le x < n + \dfrac{1}{2} \\ n + 1 - x & n + \dfrac{1}{2} \le x < n+1 \end{cases}$ $(n \in I)$.

Find the number of values of x where f(x) is non derivable in $(-5, 5)$.

80. Let $f(x) = \dfrac{\sqrt{x^2 + kx + 1}}{x^2 - k}$. The interval(s) of all possible values of k for which f is continuous for every $x \in R$, is

(A) $(-\infty, -2]$ (B) $[-2, 0)$ (C) $R - (-2, 2)$ (D) $(-2, 2)$

81$_{\text{(mcq)}}$ The $f(x) = [\cos x]$ in $[0, 2\pi]$ is discontinuous at

(A) $x = 0$ (B) $x = \pi/2$ (C) $x = \pi$ (D) $x = 3\pi/2$

where [ ] denotes greatest integer function.

82. If f(x) is a continuous function from $R \to R$ and attains only irrational values then $\sum\limits_{r=1}^{100} f(r)$ equals

(A) $\sum\limits_{r=1}^{99} f(r)$ (B) $\sum\limits_{r=1}^{100} f(2r)$ (C) $\sum\limits_{r=1}^{100} \left(f(2r)+1\right)$ (D) $\sum\limits_{r=1}^{101} f(2r)$

83. Let $f(x) = \begin{bmatrix} \sqrt{4x^2 - 12x + 9}.\{x\} & \text{for } 1 \le x \le 2 \\ \cos\left(\dfrac{\pi}{2}(|x| - \{x\}\right) & -1 \le x < 1 \end{bmatrix}$

Number of points where f(x) is non-derivable in $[-1, 2]$

84. Find the number of points where the function $f(x) = \sin^{-1}\left(\left||x|-1\right|\right)$ is non derivable.

85. Let f be differentiable at x = a and f ' (a) = 4 let $f(a) \neq 0$. Evaluate $\lim\limits_{n\to\infty}\left\{\frac{f(a+1/n)}{f(a)}\right\}^n$.

86 Let $f'(x^2) = \frac{1}{x}$ for $x > 0$, $f(1) = 1$ and $g'(\sin^2 x - 1) = \cos^2 x + p\ \forall\ x \in R$, $g(-1) = 0$.

If $h(x) = \begin{cases} f(x), & x > 0 \\ g(x), & -1 \le x \le 0 \end{cases}$

is a continuous function then find the absolute value of 2p.

87. Let $f(x) = \begin{cases} \dfrac{(e^{2x}+1)-(x+1)(e^x+e^{-x})}{x(e^x-1)} & \text{if} \quad x \neq 0 \\ k & \text{if} \quad x = 0 \end{cases}$

if $f(x)$ is continuous at x = 0 then k is equal to

(A) 1/2 (B) 1 (C) 3/2 (D) 2

88. Let $f(x) = \begin{cases} \dfrac{\tan\left(\frac{\pi}{4}+2x\right)-2\tan\left(\frac{\pi}{4}+x\right)+\tan\frac{\pi}{4}}{\sin\left(\frac{\pi}{4}+2x\right)-2\sin\left(\frac{\pi}{4}+x\right)+\sin\frac{\pi}{4}} & ;\ x \neq 0 \\ k & ;\ x = 0 \end{cases}$

If f(x) is continuous at x = 0 then k equals

(A) $4\sqrt{2}$ (B) $2\sqrt{2}$ (C) $-4\sqrt{2}$ (D) $-2\sqrt{2}$

89 Let $f(x) = \lim\limits_{n\to\infty}\left(\left(a^n+b^n\right)^{\frac{1}{n}}\sin x + \left\{e^x\right\}^n\right)\left(\left[\frac{1}{n\cot^{-1}n}\right]+1\right)$, $\forall\ x \in R$ where $a > b > 0$.

If $H(x) = \operatorname{sgn}\left(f(x)-3\right)$ has exactly one point of discontinuity $\forall\ x \in [0, 2\pi]$, then find the value of 24a.

[**Note:** {k} and [k] denotes fractional part of k and greatest integer less than or equal to k respectively. and sgn α denote signum function of α.]

90$_{(mcq)}$ If $f(x) = \sqrt{x+2\sqrt{2x-4}} + \sqrt{x-2\sqrt{2x-4}}$ then which of the following is/are **correct**?

(A) f(x) is continuous at all points in its domain

(B) f(x) is differentiable at all points in its domain

(C) f '(x) = 0 for all $x \in [2, 4)$

(D) Range of function is $\left[2\sqrt{2}, \infty\right)$

Paragraph for question nos. 91 and 92

Let $f : \left(\frac{-\pi}{2}, \frac{\pi}{2}\right) \to R$, such that f(x) is differentiable at x = 0 and f(x) = $\begin{cases} \frac{1}{2}(1+\sec x)\, f\left(\frac{x}{2}\right), & x \neq 0 \\ 2, & x = 0 \end{cases}$

91 The value of $f\left(\frac{\pi}{4}\right) + f'\left(\frac{\pi}{4}\right)$ is

(A) $\frac{24\pi-16}{\pi}$ (B) $\frac{24\pi+16}{\pi^2}$ (C) $\frac{8(3\pi-4)}{\pi^2}$ (D) $\frac{8(3\pi+4)}{\pi}$

92 The value $\lim\limits_{x \to 0} \left(\left\{ \frac{f(x)-2}{x^2} \right\} + \left[\frac{f(x)-2\cos x}{x^2} \right] \right)$ is

[**Note :** where [y] denotes greatest integer function less than or equal to y and {y} denotes fractional part function of y.]

(A) $\frac{3}{5}$ (B) $\frac{5}{3}$ (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

93. Let $f(x) = \max.\left\{ \left| x^2 - 2|x| \right|, |x| \right\}$ and $g(x) = \min.\left\{ \left| x^2 - 2|x| \right|, |x| \right\}$. If L denotes number of points where f(x) is non-derivable and M denotes the number of points where g(x) is non-derivable, then find the value of $(L+M)$.

94. Let $f(x) = \begin{cases} \lim\limits_{n\to\infty} \dfrac{\left|\log_e |x|\right| - e^{nx}\{\sin x\}}{2x + e^{(n+1)x}}, & x \in (-\pi, \pi) - \{0\} \\ 0 & x = 0 \end{cases}$

If l and m are number of points where f(x) is discontinuous and non-differentiable in $(-\pi, \pi)$ respectively then find the value of $(l + m)$

(Note: {y} denotes fractional part of y.)

95$_{\textbf{(mcq)}}$ Let $f(x) = \begin{cases} \lim\limits_{t\to\infty} \dfrac{(1+\sin \pi x)^t - 1}{(1+\sin \pi x)^t + 1}; & x \neq I \\ 0, & x \in I \end{cases}$

which of the following is/are correct?

(A) f(x) is continuous at all integers.

(B) f(x) has non-removable finite type of discontinuity at all odd integers.

(C) Left hand limit of f(x) at all even integers is equal to right hand limit of f(x) at all odd integers.

(D) Right hand limit of f(x) at all odd integers is equal to right hand limit of f(x) at all even integers.

Paragraph for question nos. 96 to 98

Let f(x) be a polynomial satisfying $\lim\limits_{x\to\infty} \dfrac{x^4 f(x)}{x^8+1} = 3$,

$f(2) = 5$, $f(3) = 10$, $f(-1) = 2$, and $f(-6) = 37$.

96 The value of $f(0)$ is equal to

(A) 1 (B) 0 (C*) 109 (D) 119

97 The value of $\lim\limits_{x\to-6} \dfrac{f(x)-x^2-1}{3(x+6)}$, is equal to

(A) – 6! (B) 6! (C) $\dfrac{6!}{2}$ (D) $\dfrac{-6!}{2}$

98. The number of points of discontinuity of $g(x) = \dfrac{1}{x^2+1-f(x)}$ in $\left[\dfrac{-15}{2}, \dfrac{5}{2}\right]$, is equal to

(A) 4 (B) 3 (C) 1 (D) 0

99 Let $f(x) = |\cos x| + [\cos x]$ $x \in [0, 2\pi]$ if x is the number of points of discontinuity of f (x) and p be the number of solutions to equation $(f(x))^2 = 1$ then the value of n + p is

(A) 3 (B) 5 (C) 4 (D) 2

100 Let n_1, n_2 and n_3 denotes the number of points of non-derivability of functions $f(x) = \cot^2 x - \text{cosec}^2 x$ in $(0, 2\pi)$, $g(x) = \cot^{-1}x + \sin^{-1}x + \tan^{-1}x$ in $(-1, 1)$ and $h(x) = [5 + \cos x]$ in $[0, 2\pi)$ respectively. Find the value of $(n_1 + n_2 + n_3)$.

Note: [y] denotes largest integer $\le$ y.

Paragraph for questions nos. 101 & 102

$$\text{Let } g(x) = \begin{cases} \dfrac{ax^2+bx+c(\cot^n x)}{4+\cot^n x}, & x \in \left(0, \dfrac{\pi}{4}\right) \\ 1, & x = \dfrac{\pi}{4} \\ \dfrac{\sin x + \cos x + \tan^n x}{1+c(\tan^n x)}, & x \in \left(\dfrac{\pi}{4}, \dfrac{\pi}{2}\right) \end{cases}$$

where a, b, c are real constants, and $f(x) = \lim\limits_{n\to\infty} g(x)$

101. If $\lim\limits_{x\to\frac{\pi}{4}} f(x)$ exists then c may be equal to

(A) 2 (B) $\frac{1}{2}$ (C) 3 (D) – 1

102. If $f(x)$ is continuous at $x = \frac{\pi}{4}$ then c is equal to

(A) – 1 (B) 1 (C) 2 (D) – 2

103 Let $f : [0, 2] \to R$ be continuous and $f(0) = f(2)$. Prove that there exists x_1 and x_2 in $(0, 2)$ such that $x_2 - x_1 = 1$ and $f(x_2) = f(x_1)$

104$_{(mcq)}$ Let two lines be $f_1(t)x + f_2(t)y = 5$ and $g_1(t)x + g_2(t)y = 4$, $t \in [0, 1]$, where f_1, f_2, g_1, g_2 are continuous functions. $f_1(0) = g_2(0) = f_2(1) = g_1(1) = 2$, $f_2(0) = g_1(0) = 3$ and $f_1(1) = g_2(1) = 4$ then both the lines

(A) are parallel for some $t \in [0, 1]$

(B) are perpendicular for some $t \in [0, 1]$

(C) make complimentary angles with + ve x-axis for some $t \in [0, 1]$

(D) none of these

105. Find the number of points in [–2, 2] where the function $f(x) = x + \{-x\} + [x]$ is discontinuous.
[**Note :** [y] and {y} denote greatest integer less than or equal to y and fractional part functions of y respectively.]

106. If $f(x) = \text{sgn}(3x\cos^{-1}x - 6\cos^{-1}x - \pi x + 2\pi)$, then the number of points of discontinuity of $f(x)$ is/are
[**Note :** sgn k denotes signum function of k.]

(A) 2 (B) 0 (C) 1 (D) 3

107$_{(mcq)}$ Let $f: [0, 1] \to R$ be defined as $f(x) = x$ for $x \in$ rational and $f(x) = 1 - x$ for $x \in$ irrational, then which of the following is/are correct?

(A) $f(x)$ is injective in [0, 1] (B) $f[f(x)] = x \ \forall x \in [0, 1]$

(C) $f(x)$ is continuous only at $x = \frac{1}{2}$ (D) $f(x)$ has its own inverse

108. If $f(x) = \lim\limits_{n \to \infty} \frac{(1+\sin x)^n + e^x}{1+(1+\sin x)^n}$ where $x \in [-2\pi, 2\pi]$,

then find the total number of points of discontinuity of f(x).

109. Prove that the function $f(x) = a\sqrt{x-1} + b\sqrt{2x-1} - \sqrt{2x^2-3x+1}$ where $a + 2b = 2$ and $a, b \in R$ always has a root in $(1, 5)$ $\forall\, b \geq 0$ and a root in $(1, 2)$.

110$_{(mcq)}$Let $f(x) = -x^3 + x^2 - x + 1$ and $g(x) = \begin{cases} \min.(f(t);\ 0 \le t \le x), & 0 \le x \le 1 \\ x - 1, & 1 < x \le 2 \end{cases}$.

Which of the following statement(s) is(are) correct ?

(A) $\lim\limits_{x \to 1} g(g(x))$ exist and is equals to 1 (B) g(x) is continuous at $x = 1$

(C) g(x) is non-differentiable at $x = 1$ (D) $[g(x)] = 0 \ \forall \ x \in (0, 2)$

[**Note :** [y] denotes greatest integer less than or equals to y]

111$_{\text{(mcq)}}$ Let $f(x) = \begin{cases} x^3 + 1, & \text{if } x \in (0,3), \text{ where } x \text{ is rational} \\ 3x - 1, & \text{if } x \in (0,3), \text{ where } x \text{ is irrational} \end{cases}$,

then which of the following statement(s) is(are) correct?

(A) f(x) is continuous at $x = 1$.

(B) f(x) is continuous for all $x \in (0, 3)$.

(C) f(x) is differentiable at $x = 1$ and $f'(1) = 1$.

(D) f(x) is differentiable at $x = 1$ and $f'(1) = 3$.

112$_{\text{(mcq)}}$ If $f(x) = \begin{cases} \dfrac{e^{\cos^{2n} x} + \sin^{2m} x - 2}{(2x - \pi)^2}, & x \neq \dfrac{\pi}{2} \\ 1, & x = \dfrac{\pi}{2} \end{cases}$

is continuous at $x = \dfrac{\pi}{2}$ $(m \in I, n \in N, m \neq 0)$, then

(A) $n = 1$ & $m = -3$ (B) $n = 2$ & $m = -4$

(C) $n = 3$ & $m = -4$ (D) $n = 4$ & $m = -3$

113$_{\text{(mcq)}}$ Let $y = f(x) = \text{Limit}_{n \to \infty} \dfrac{x}{1 + \left(4 \sin^2 x\right)^n}$ in $[0, \pi]$. Which of the following statement(s) is/are correct?

(A) f(x) is discontinuous at one value of x in $[0, \pi]$.

(B) f(x) is discontinuous at two values of x in $[0, \pi]$.

(C) Derivative of $\tan^{-1} x$ w.r.t. $f(x)$ at $x = \dfrac{1}{2}$ is $\dfrac{4}{5}$.

(D) $y'\left(\dfrac{\pi}{6}\right) = \dfrac{1}{2}$

114. **Column - I** **Column-II**

(A) If $\lim\limits_{x \to 0} \left(\dfrac{\sin x}{x}\right)^{\frac{1}{1 - \cos x}}$ is L then $\ln$ (L) equals k. (P) 0

The value of $\dfrac{1}{k^2}$ is equal to (Q) 1

(B) The value of $\lim\limits_{n \to \infty} \left(\frac{n^2}{n-1}\right)^{\tan\frac{1}{\sqrt{n}}}$ is (R) 3

(C) Number of solution of the equation $8\sin^4 x + 8\cos^4 x = 5$ in the interval $0 < x < 2\pi$ is

(D) Let $f(x)$ be a non-constant polynomial function and (S) 8

$g(x) = \left| x(x-1)(x-2) f(x) \right|$. If $g(x)$ is differentiable $\forall\, x \in R$,

then minimum number of distinct roots of $f(x) = 0$ is (T) 9

115. If $f(x) = 23 - 2^{-x^2+2x+3}\ \forall\, x \in R$ and $g(x) = \left[\frac{f(x)}{\mu}\right]$, where $[k]$ denotes greatest integer function less than or equal to k and $\mu \in N$, then find the sum of all values of μ for which $g(x)$ is discontinuous for at least one real value of x.

116. The sum of all values of x for which the function $f(x) = \sqrt{|x^2 - 5x + 6|}$ is continuous, but not differentiable is

(A) – 5 (B) 0 (C) 5 (D) 6

117. If $f(x) = \frac{6}{\pi}\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ then number of points where $y = [f(x)]$ is discontinuous in its domain, is

(A) 3 (B) 5 (C) 7 (D) 9

[**Note :** [k] denotes greatest integer less than or equal to k.]

118 Consider two functions $f(x) = \sin x$ and $g(x) = |f(x)|$.

Statement-1: The function $h(x) = f(x)\, g(x)$ is not differentiable in $[0, 2\pi]$

Statement-2: $f(x)$ is differentiable and $g(x)$ is not differentiable in $[0, 2\pi]$

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

119$_{\text{(mcq)}}$ Let $f(x) = \lim\limits_{\alpha \to 0} \left\{\frac{a \sin\alpha}{\alpha}\right\} x + \left\{\frac{b \tan\alpha}{\alpha}\right\}$ (where $a \in I^-$ and $b \in I^+$, $\{y\}$ denotes fractional part of y), then

(A) f(x) is continuous and differentiable for all values of x

(B) f(x) is an even function

(C) f(x) is an odd function

(D) the area bounded by $y = f(x)$ and $y = \sqrt{4 - x^2}$ is equal to $\frac{3\pi}{2}$.

120. Let $f(x)$ be a differentiable function. Evaluate: $\displaystyle\lim_{n\to\infty}\binom{n}{k}\left\{f\left(\frac{x}{n}\right)-f(0)\right\}^k$

121. If $\displaystyle\sum_{r=1}^{n}\sin^{-1}\alpha_r=\frac{n\pi}{2}$ for any $n\in N$ and $p=\displaystyle\prod_{r=1}^{n}(\alpha_r)^r$.

If $f(x)=\begin{cases}\dfrac{x^{1/3}-(3-2x)^{1/4}}{x^2-x}, & x\neq p\\ k, & x=p\end{cases}$ is continuous at $x=p$, then find the value of $6(k+p)$.

122$_{\textbf{(mcq)}}$ Let $P(x)$ be a polynomial of n degree and $f(x)=\begin{cases}P\left(\dfrac{1}{x^3}\right)e^{-\frac{1}{x^4}}, & x\neq 0\\ 0, & x=0\end{cases}$, then

(A) $f(x)$ is discontinuous at $x=0$
(B) $f(x)$ is continuous at $x=0$
(C) $f(x)$ is non differentiable at $x=0$
(D) $f'(0)=\displaystyle\lim_{x\to 0}f(x)$.

123. Let $f(x)=ax^9+b\sin x+cx^2\,\text{sgn}(x)+\dfrac{(e^x-e^{-x})}{(e^x+e^{-x})}$ be defined on set of real numbers, $(a>0, b, c, \in R)$. If $f(-5)=5$, $f(-2)=-3$, then find the minimum number of zeros of the equation $f(x)=0$.

124. If $f(x)$ and $g(x)$ are two differentiable function and graph of $y=g(x)$ is reflection of graph of $y=f(x)$ with respect to line $y=x$. If point (a, b) lies on curve $y=f(x)$ and $f'(a)=3$, then find the value of $\dfrac{670}{\gamma}$ (where g is ordinate of the point on the curve $y=g'(x)$ whose abscissa is b.)

125. A continuous, even periodic function f with period 8 is such that $f(0)=0$, $f(1)=-2$, $f(2)=1$, $f(3)=2$, $f(4)=3$, then the value of $\tan^{-1}\left(\tan\left(f(-5)+f(20)+\cos^{-1}(f(-10))+f(17)\right)\right)$ is equal to

(A) $2\pi-3$
(B) $3-2\pi$
(C) $2\pi+3$
(D) $3-\pi$

Paragraph for Question Nos. 126 to 128

Let f be a differentiable function satisfying $f(x+y)=f(x)\cdot f(y)\ \forall\, x,y\in R$; $f(x)>0$ and if $f(1)=\dfrac{1}{3}$.

126 $\displaystyle\lim_{n\to\infty}(f(x)+2f(x+1)+3f(x+2)+\ldots\ldots..(n+1)f(x+n))$ is equal to

(A) $f(x)$
(B) $\dfrac{4}{3}f(x)$
(C) $\dfrac{9f(x)}{4}$
(D) $9f(x)$

127. $\lim\limits_{x\to\infty}\left(f(x)+f\left(\frac{1}{x}\right)\right)^{\frac{1}{x}}$ is equal to

(A) f(0) (B) f(1) (C) f(–1) (D) f(1) + f(–1)

128. Let $f_n(x) = \dfrac{f_{n-1}(x)}{1+a_{n-1}\cdot f_{n-1}(x)}$ where $n \in N$ and $f_0(x) = f(x)$. If $f_n(x) = \dfrac{f(x)}{1+\lambda f(x)}$ then λ is equal to

(A) 0 (B) 1 (C) $\sum\limits_{i=0}^{n-1} a_i$ (D) a_{n-1}

129.

	Column-I		Column-II
(A)	If $f:[0, 1]\to[0, 1]$ is continuous, then the number of roots of the equation $f(x) = x^3$ is	(P)	one
(B)	If f is derivable, then between the consecutive roots of $f'(x) = 0$, the number of roots of $f(x) = 0$ is	(Q)	at least one
(C)	The number of values of x at which $f(x)= \|1 - \|x-1\|\|$ is not differentiable is	(R)	at most one
(D)	Let f(x) = {x, if x is rational = 2 – x, if x is irrational. The number of values of x at which f(x) is continuous is	(S)	more than one

Paragraph for Quesrtion no. 130 to 133

Let $f(x) = \min.\left\{x^{2/3}, \sqrt{2-x^2}\right\}$ and $g(x) = \max\left\{x^{2/3}, \sqrt{2-x^2}\right\}$, $|x| \le \sqrt{2}$

130 The number of points at which f (x) is not differentiable is

(A) 0 (B) 1 (C) 2 (D) > 2

131. The range of f (x) is

(A) [0, 1] (B) $[0, 2^{1/3}]$ (C) $[1, 2^{1/3}]$ (D) $[2^{1/3}, \sqrt{2}]$

132. The number of points at which g (x) is not differentiable is

(A) 0 (B) 1 (C) 2 (D) > 2

133. The range of g (x) is

(A) $[1, 2^{1/3}]$ (B) $[1, \sqrt{2}]$ (C) $[2^{1/3}, \sqrt{2}]$ (D) $[\sqrt{2}, 2^{1/3}]$

134. Let f be a differentiable function on $(0, \infty)$ and suppose that $\lim\limits_{x\to\infty}\left(f(x)+f'(x)\right) = L$ where L is a finite quantity, then which of the following must be true?

(A) $\lim\limits_{x\to\infty} f(x) = 0$ and $\lim\limits_{x\to\infty} f'(x) = L$

(B) $\lim\limits_{x\to\infty} f(x) = \frac{L}{2}$ and $\lim\limits_{x\to\infty} f'(x) = \frac{L}{2}$

(C) $\lim\limits_{x\to\infty} f(x) = L$ and $\lim\limits_{x\to\infty} f'(x) = 0$

(D) nothing definite can be said

135. Consider, $f(x) = \min\left\{x^3 - 1, -\frac{1}{4}(|x-2| + |x+2|), 7 - x^3\right\}$

p and q denote number of points where f(x) is discontinuous and non-derivable in [–2, 3] respectively then p+q is

(A) 0 (B) 1 (C) 3 (D) 4

Paragraph for question no. 136 to 138

For $0 < x < \frac{\pi}{2}$, let $f_1(x) = \sum_{r=1}^{n} \sec\left(x + \frac{r\pi}{6}\right)\sec\left(x + (r-1)\frac{\pi}{6}\right)$ and $f_1(x) - 2f_2(x) = 2\tan\left(x + \frac{n\pi}{6}\right)$

Also $f_2(x) + f_3(x) = 0$ and $f_4(x) = \begin{cases} (\exp.)\dfrac{e^{\left(e^{x + f_2(x) + \tan x} - 1\right)} - 1}{2\cdot(e^x - 1)}; & x < 0 \\ k_1 \quad ; & x = 0 \\ \left(1 + |f_2(x)|\right)^{\frac{k_2}{f_3(x)}}; & x > 0 \end{cases}$

136. The value of k_1 and k_2, if $f_4(x)$ is continuous at $x = 0$ is

(A) $\frac{1}{\sqrt{e}}, 2$ (B) e, 1 (C) $\frac{e}{2}, 2$ (D) $\sqrt{e}, \frac{1}{2}$

137. $y = f_3(x)$ is

(A) discontinuous and non derivable at $x = \frac{\pi}{4}$ and $x = \frac{\pi}{3}$

(B) neither continuous nor derivable at $x = \frac{2\pi}{5}$

(C) continuous and derivable in $\left(0, \frac{\pi}{2}\right)$

(D) continuous but not derivable at $x = \frac{2\pi}{5}$

138. For n = 3, the solution of equation $f_1(x) + 4 = 0$, in $\left(0, \frac{\pi}{2}\right)$ is

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) non existent

139. Let $f(x) = \begin{cases} \tan\left(\frac{1}{2}\sin^{-1}\left(\frac{2x}{1+x^2}\right)\right), & x \ge 1 \\ \lambda\cos^{-1}(2x-3) + \mu, & 0 \le x < 1 \end{cases}$

If $f(x)$ is continuous in $[0, \infty)$ then the value of $(\lambda + \mu)$, is equal to

(A) 1 (B) $2 - \pi$ (C) $3 - 2\pi$ (D) any real number

140. Let f be a derivable function satisfying $f(x+y)=f(x)+f(y),\ \forall\ x,\ y\in R$, then
(A) $|f(x)|$ must be non-derivable. (B) $|f(x)|$ may be non-derivable.
(C) $\lim\limits_{x\to 0}(1+f(x))^{\frac{1}{x}}$ must be equal to e. (D) $\left|\dfrac{f(x)}{x}\right|$ must be continuous.

141. Let $f: R\to R$ be a derivable function satisfying
$f(y)\,f(x-y)=f(x),\ \forall\ x, y\in R$. If $f'(0)=p$; $f'(5)=q$ $(p, q\neq 0)$ then $f(5)$ equals
(A) $\dfrac{p}{q}$ (B) $\dfrac{q}{p}$ (C) $\dfrac{p^2}{q}$ (D) $\dfrac{q^2}{p}$

142 Let $f(x)=|x-2|\,e^x g(x)+\sin x$, where $g(x)$ is a continuous function.
If $f(x)$ is differentiable at $x=2$, then $g(2)$ is equal to
(A) 0 (B) 1 (C) 2 (D) 3

143. Let $\dfrac{\pi}{3}<A<\dfrac{\pi}{2}$ and a function f(x) is defined as

$$f(x)=\begin{cases}\lim\limits_{n\to\infty}\dfrac{ax^2(\sin A-\sin^3 A)-(5x-b)\left|\sin A-\sin^3 A\right|^n}{(\sin A-\sin^3 A)-\left|\sin A-\sin^3 A\right|^n}, & x\in Q\\[2ex] \lim\limits_{n\to\infty}\dfrac{ax^2(\sin A+\sin^3 A)+(5x-b)\left|\sin A+\sin^3 A\right|^n}{(\sin A+\sin^3 A)+\left|\sin A+\sin^3 A\right|^n}, & x\notin Q\end{cases}$$

If f(x) is continuous at $x=2$ and $x=3$ then the value of $b-a$, is
(A) 3 (B) 4 (C) 5 (D) 6

144. Compare each value of Column-I with (ae^b) and then match with corresponding true statement(s) of Column-II :

Column-I	Column-II
(A) If $f(x)=\left(\dfrac{1-\tan x}{1+\sin x}\right)^{\operatorname{cosec} x}$ is to be made continuous at $x=0$, then f(0) should be equal to	(P) $ab\leq 0$
(B) $\lim\limits_{x\to\infty}\left(\dfrac{e^{\frac{1}{x^2}}-1}{2\tan^{-1}(x^2)-\pi}\right)$ is equal to	(Q) $b-a=\dfrac{1}{2}$

(C) $\lim_{x\to 0}\left(\frac{e-(1+x)^{\frac{1}{x}}}{\tan x}\right)$ is equal to (R) $a \le b$

(D) $\lim_{x\to \infty}\left(\sin\frac{1}{x}+\cos\frac{1}{x}\right)^{x}$ is equal to (S) (a, b) lies on straight line $x = 1$

145$_{(mcq)}$ Let $f(x) = \min.\,(|e-x|, \pi-|x|)$. Then which of the following statement(s) is (are) correct?
(A) f(x) is many one but not even function.
(B) Range of f(x) is $\left(-\infty, \frac{\pi-e}{2}\right]$.
(C) f(x) is continuous and derivable at all integral points.
(D) f(x) is continuous everywhere but non-derivable at exactly two points.

146$_{(mcq)}$ Let $f(x) = \begin{cases} |x|^p \sin\frac{1}{x} + x\,|\tan x|^q, & x \ne 0 \quad p, q \in N \\ 0, & \text{if} \quad x = 0 \end{cases}$

If f(x) is derivable at x = 0 then the ordered pair (p, q) is
(A) (1, 2) (B) (2, 1) (C) (3, 2) (D) (2, 2)

147. Let $f(x)$ be a differentiable function such that $\lim_{h\to 0}\left(\frac{f(x+h)}{f(x)}\right)^{\frac{1}{h}} = e^{(\tan x)\,f(-x)}$ and $f(0) = 1$, then the value of $f'(0)$ is equal to
(A) 0 (B) 1 (C) e (D) – e

148. Let g(x) be the inverse of an invertible function f(x), which is differentiable for all real x, then $g''(f(x))$ equals

(A) $\frac{-f''(x)}{(f'(x))^3}$

(B) $\frac{f'(x)\,f''(x)-(f'(x))^2}{(f'(x))^2}$

(C) $\frac{f'(x)\,f''(x)+(f'(x))^2}{(f'(x))^2}$

(D) $\frac{-(f'(x))^3}{f''(x)}$

149. Let $g(x)=\begin{cases}\left(\dfrac{1}{1+[\sin x]}\right)^{\frac{2}{2x-\pi}}; & x\neq\dfrac{\pi}{2}\\ p^2-1 \quad ; & x=\dfrac{\pi}{2}\end{cases}$

then the value of p for which f(x) is continuous at $x=\dfrac{\pi}{2}$ is

[Note: [k] denotes the largest integer less than or equal to k.]

(A) ±1 (B) $\pm\dfrac{1}{2}$ (C) $\pm\sqrt{2}$ (D) 0

150. Let $g(x)=\begin{cases}2x^2-3x+\dfrac{3}{2}, & \forall\ x<1\\ \dfrac{x^2}{2}, & \forall\ x\geq1\end{cases}$

If L and M denotes the point of discontinuity and non-derivability of $g(x)$, then the value of $(L+M)$ is

(A) 0 (B) 1 (C) 2 (D) 3

151. $\lim\limits_{x\to\infty} x\left(\dfrac{1}{e}-\left(\dfrac{x}{x+1}\right)^x\right)$ equals

(A) $\dfrac{1}{e}$ (B) $\dfrac{-1}{2e}$ (C) $\dfrac{1}{2e}$ (D) $\dfrac{-1}{e}$

152. If $\lim\limits_{x\to1}\sec^{-1}\left(\dfrac{k}{\ln x}-\dfrac{k}{x-1}\right)=\dfrac{\pi}{3}$, then the value of k is equal to

(A) 2 (B) 3 (C) 4 (D) 6

153. Which one of the following function is discontinuous for atleast one real value of x?

(A) $f(x)=\sqrt{3+2\sin x}$ (B) $g(x)=\dfrac{e^x+1}{e^x+3}$

(C) $h(x)=\left(\dfrac{2^{2x}+1}{2^{3x}+5}\right)^{\frac{5}{7}}$ (D) $k(x)=\sqrt{1+\operatorname{sgn} x}$

[**Note :** sgn x denotes signum function of x.]

154. Let $P(x)$ be a polynomial satisfying $P(x) - 2P'(x) = 3x^3 - 27x^2 + 38x + 1$.

If the function $f(x) = \begin{cases} \dfrac{P''(x)+18}{6}, & x \neq \dfrac{\pi}{2} \\ \sin^{-1}(ab) + \cos^{-1}(a+b-3ab), & x = \dfrac{\pi}{2} \end{cases}$

is continuous at $x = \dfrac{\pi}{2}$, then find $(a+b)$.

155. If $f(x) = \begin{cases} x^{\alpha} \cdot \sin\dfrac{1}{x}, & x > 0 \\ 0, & x = 0 \end{cases}$,

then $f'(x)$ is continuous for $\alpha \in$

(1) (0, 1] (2) (0, 2) (3) (1, 2] (4*) $(2, \infty)$

Paragraph for Question no.156 to 158

One of the most famous functions in calculus as the Dirichlet's function $D(x) = \begin{cases} 1, & x \in Q \\ 0, & x \notin Q \end{cases}$.

This function is one of the rare functions whose graph can not be drawn.

Let us define another such function $f(x) = \begin{cases} x^3 + 2x^2, & x \in Q \\ -x^3 + 2x^2 + ax, & x \notin Q \end{cases}$.

156 The value of a so that this function is differentiable at x = 0 is

(A) 1 (B) –1 (C) 0 (D) None of these

157 For the value of a obtained in above question. f(x) is

(A) one - one and onto (B) many one and onto

(C) one-one and into (D) many one and into

158 $\lim\limits_{x \to 0} |f'(x)| =$

(A) 1 (B) 2 (C) 3 (D) Does not exist

159$_{\text{(mcq)}}$ The number of points of discontinuity of function $f(x) = \lim\limits_{n \to \infty} \dfrac{x^{2n} - 2\sin x^{2n}}{x^{2n} + 2\sin x^{2n}}$ is equal to number of point of non differentiability of which of the following function(s).

(A) $g(x) = \text{sgn}(x^2 - 1)$ (B) $h(x) = \sin^{-1}\left(\dfrac{2x}{1+x^2}\right)$

(C) $k(x) = \cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)$ (D) $\ell(x) = \text{sgn}(|x| - 1)$

Paragraph for question nos. 160 to 162

Let f be a real valued function defined on R (the set of all real numbers) where $y = f(x)$ is defined parametrically as $x = 5t - |2 - t|$, $y = -3t + 2|t - 1|$ $\forall$ $t \in R$.

160. The function $f(x)$ is
(A) continuous and derivable everwhere on R except at $x = 4$ and $x = 10$.
(B) continuous and derivable everywhere on R.
(C) continuous and derivable everywhere on R except at $x = 10$.
(D) continuous everywhere on R but not derivable at $x = 4$ and $x = 10$.

161. The value of $f'(\pi)$ is equal to
(A) $\frac{-7}{3}$ (B) $\frac{-1}{4}$ (C) $\frac{-5}{6}$ (D) $\frac{-2}{3}$

162. Number of solution of the equation $f(x) = 2$ is equal to
(A) 1 (B) 0 (C) 2 (D) 3

163. The function $f(x) = \begin{cases} e^{-1/x^2} \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$ at $x = 0$ is
(A) continuous but not differentiable
(B) continuous and differentiable
(C) discontinuous with oscillatory type discontinuity
(D) none

01 $f(0) = 1$; $f(0^+) = \ln a \cdot \left(\frac{b}{3}\right)^n$; $f(0^-) = 1$ **02** p = 2 **03** **A, D**

04	**2**	**05**	(C)	**06**	D	**07.**	LDNE	**08**	B
09	23	**10.**	D	**11.**	B	**12.**	21	**13.**	A
14.	**A**	**15.**	ABC	**16.**	AC	**17.**	BC	**18**	A
19	ACD	**20.**	ABCD	**21.**	ABCD	**22.**	D		

23. $k = 0$; $g(x) = \begin{cases} \ell n(\tan x) & \text{if } 0 < x < \frac{\pi}{4} \\ 0 & \text{if } \frac{\pi}{4} \le x < \frac{\pi}{2} \end{cases}$. Hence g (x) is continuous everywhere.

24.	A CD	**25.**	$\frac{1}{60}$	**26.**	3	**27.**	4		
28	ABCD	**29.**	C	**30.**	AC	**31.**	AD	**32.**	9
33.	5150	**34.**	10	**35.**	C	**36.**	A	**37.**	D
38.	A	**39.**	ABCD	**40.**	6	**41**	CD	**42.**	**ABD**
43.	AB	**44.**	2	**45**	45, where $N_1 = 30$; $N_2 = 5$; $N_3 = 10$				
46	45	**47.**	2	**48.**	17	**49.**	162	**50.**	1
51.	5	**52.**	ABD	**53**	ABC	**54.**	C	**55.**	C
56.	BD	**57.**	B	**58.**	B	**59.**	B	**60.**	D
61.	B	**62.**	A	**63**	B	**64.**	D	65.	57
66	B	**67**	A	68.	D	**69.**	6	70.	6
71.	D	**72**	C	**73.**	D	**74**	B	**75.**	12
76	12	**77.**	18	**78.**	100	**79**	19	**80.**	B
81.	ABD	**82.**	B						
83.	not differentiable at 0, 1, $\frac{3}{2}$ and 2			**84.**	5	**85.**	$e^{\frac{f'(a)}{f(a)}}$	**86**	3
87.	B	**88.**	C	**89**	36	**90.**	ACD	**91**	C
92	B	**93.**	12	**94**	5	**95**	BC	**96**	C
97	D	**98.**	B	**99.**	C	**100**	4	**101.**	D
102.	B	**104.**	AC	**105.**	5	**106.**	C	**107.**	ABCD
108.	4	**110.**	ABCD	**111.**	AD	**112.**	ABC	**113.**	**BC**
114.	(A) T ; (B) Q ; (C) S ; (D) R			**115.**	253	**116.**	C	**117**	D
118.	D	**119.**	ABC	**120.**	$f'(0)^k \cdot \frac{x^k}{k!}$	**121.**	11		**122.**	BD
123.	5	**124.**	2010	**125.**	D	**126**	C	**127.**	C
128.	C	**129.**	(A) Q ; (B) R ; (C) S ; (D) P			**130**	D	**131.**	A
132	C	**133**	B	**134**	C	**135.**	B	**136.**	D
137.	C	**138.**	B	**139.**	A	**140.**	B	**141.**	B
142.	A	**143.**	C						
144.	(A) PS; (B) PQR; (C) QR; (D) RS			**145.**	AC	**146.**	BCD	**147.**	A
148.	A	**149.**	C	**150.**	A	**151.**	B	**152.**	C
153.	D	**154.**	2	**155.**	D	**156.**	C	**157.**	B
158.	D	**159.**	ABD	**160.**	D	**161.**	C	**162.**	A
163.	B								

SOLUTIONS

01 $\displaystyle \lim_{h\to0} f(0+h) = \lim_{h\to0} \frac{a^h-1}{\sinh.h^n}\left[\frac{b\sinh - \sin bh}{\cosh - \cos bh}\right]^n$

$$=\lim_{h\to0}\frac{a^h-1}{h}\cdot\frac{h}{\sinh}\left[\left(\frac{b\sin(h)-\sin(bh)}{h.h^2}\right)\cdot\frac{h^2}{\cosh-\cos bh}\right]^n = \ln a\cdot l^n$$

Use $\exp^n$ here to get $\ln(a)\cdot\left(\frac{b}{3}\right)^n = 1$

where $\displaystyle l = \lim_{h\to0}\frac{b\sinh-\sin bh}{h^3}\cdot\frac{h^2}{\cosh-\cos bh} = \frac{l_1}{l_2}$

where $\displaystyle l_1 = \frac{b\sinh-\sin bh}{h^3}$ and $\displaystyle l_2 = \frac{\cosh-\cos bh}{h^2}$

$$=\lim_{h\to0}\left[\frac{b(\sinh-h)-(\sin bh-bh)}{h^3}\right]\cdot\left[b^2\left(\frac{1-\cos bh}{b^2h^2}\right)-\left(\frac{1-\cosh}{h^2}\right)\right]^{-1}$$

$$=\lim_{h\to0}\left[b\left(\frac{\sinh-h}{h^3}\right)-\left(\frac{\sin bh-bh)}{b^3h^3}\right)b^3\right]\cdot\left[\left(\frac{b^2}{2}-\frac{1}{2}\right)^{-1}\right]$$

$$=\lim_{h\to0}\left[b\left(-\frac{1}{6}\right)-b^3\left(-\frac{1}{6}\right)\right]\cdot\frac{2}{(b^2-1)}\quad\left[\lim_{h\to0}\frac{\sin x-x}{x^3}=-\frac{1}{6}\right]$$

$$=\lim_{h\to0}\frac{b(b^2-1)}{6}\cdot\frac{2}{(b^2-1)}=\left(\frac{b}{3}\right)\ ;\text{ hence }\ f(0^+)=\ln a\cdot\left(\frac{b}{3}\right)^n$$

$$f(0^-)=\lim_{h\to0} f(0-h)=\lim_{h\to0}\frac{a^{-h}\sin(-bh)-b^{-h}\sin(-ah)}{\tan(-bh)-\tan(-ah)}\quad(\text{multiply } D^r \,\&\, N^r \text{ by } a^h\cdot b^h)$$

$$=\lim_{h\to0}\frac{a^h\sin ah-b^h\sin bh}{a^h.b^h[\tan ah-\tan bh]}$$

$$=\lim_{h\to0}\frac{ah\cdot\dfrac{a^h\sin(ah)}{ah}-\dfrac{b^h\sin(bh)}{bh}\cdot bh}{ah\cdot\dfrac{\tan(ah)}{ah}-bh\cdot\dfrac{\tan(bh)}{bh}}$$

$$\lim_{h\to0}\frac{(a-b)}{(a-b)}=1$$

02 $f(x) = \text{cosec } 2x + \text{cosec } 2^2x + \ldots\ldots + \text{cosec } 2^n x$

now $\displaystyle \text{cosec } 2x = \frac{1}{\sin 2x} = \frac{\sin(2x-x)}{\sin x\sin 2x} = \cot x - \cot 2x$

|||ly $\quad \text{cosec } 2^2 x = \cot 2x - \cot 2^2 x$

$\text{cosec } 2^3 x = \cot 2^2 x - \cot 2^3 x$

$\vdots$

$\operatorname{cosec} 2^n x = \cot 2^{n-1} x - \cot 2^n x$

$\therefore \quad f(x) = \cot x - \cot 2^n x$

$\therefore \quad g(x) = f(x) + \cot 2^n x = \cot x$

now $\quad H(0+h) = \lim_{h\to 0}\left((\cos h)^{\cot h} + (\sec h)^{\operatorname{cosec} h}\right)$

$= e^{\lim_{h\to 0} \coth\,(\cos h - 1)} + e^{\lim_{h\to 0} \operatorname{cosec} h\,(\sec h - 1)}$

$= 1 + 1 = 2 \quad ....(1)$

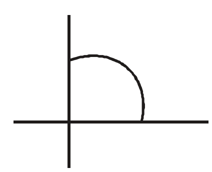
$$H(0-h) = \lim_{h\to 0} \frac{e^{-h} + e^{h} - 2\cos h}{h \sin h}$$

$$= \lim_{h\to 0}\left\{\frac{e^h + e^{-h} - 2}{h^2} + \frac{2(1-\cos h)}{h^2}\right\} = 2 \quad ....(2)$$

From (1) and (2) H (x) will be cont. if p = 2

03 $\quad f(x) = \cos x$ and $H(x) = \begin{cases} \operatorname{Min}[f(t)/0 \le t \le x] \\ \dfrac{\pi}{2} - x \end{cases}$

i.e. $\quad H(x) = \begin{cases} \operatorname{Min} f(t)/0 \le t \le \dfrac{\pi}{2} \\ \end{cases}$



Now Let $x = \pi/6$ at certain instant

then $\quad 0 \le t \le \pi/6$

and $\quad H(x) = \text{Min of } f(t) \quad$ i.e. $\quad$ Min of $\cos 0 - \cos \pi/6$

$\Rightarrow \quad \cos \pi/6$

$$H(x) = \begin{cases} \cos x & 0 \le x \le \pi/2 \\ \dfrac{\pi}{2} - x & \dfrac{\pi}{2} < x \le 3 \end{cases}$$

$$f'\left(\frac{\pi}{2}^-\right) = \lim_{h\to 0} \frac{\cos\left(\frac{\pi}{2} - h\right) - 0}{-h} = -1$$

$$f'\left(\frac{\pi}{2}^+\right) = \lim_{h\to 0} \frac{\frac{\pi}{2} - \left(\frac{\pi}{2} + h\right) - 0}{h} = \frac{-h}{h} = -1$$

Hence this function is derivable & continuous at $x = \pi/2$

continuous at all other points.

Max. value of H(x) is [0, 3) = 1

04 $f(g(x)) = \begin{cases} (g(x))^2 + (g(x)) + 1, & \text{if } g(x) < 0 \\ (g(x) - 3)^2 + 2b, & \text{if } g(x) \geq 0 \end{cases}$

$$\Rightarrow \quad f(g(x)) = \begin{cases} (2x+a)^2 + (2x+a) + 1, & \text{if } 2x + a < 0 \ \& \ x < 0 \\ |x-2|^2 + |x-2| + 1, & \text{if } |x-2| < 0 \ \& \ x \geq 0 \\ (2x+a-3)^2 + 2b, & \text{if } 2x + a \geq 0 \ \& \ x < 0 \\ (|x-2|-3)^2 + 2b, & \text{if } |x-2| \geq 0 \ \& \ x \geq 0 \end{cases}$$

$$\Rightarrow \quad f(g(x)) = \begin{cases} (2x+a)^2 + (2x+a) + 1, & \text{if } x < -\frac{a}{2} \\ (2x+a-3)^2 + 2b, & \text{if } -\frac{a}{2} \leq x < 0 \\ (|x-2|-3)^2 + 2b, & \text{if } x \geq 0 \end{cases} \quad (\because \text{ a is non negative real number})$$

$\because$ f(g(x)) is continuous function

$\therefore$ at $x = -\dfrac{a}{2}$

$\text{LHL} = f\left(\dfrac{-a}{2}\right) = \text{RHL}$

$\Rightarrow 0 + 0 + 1 = (-3)^2 + 2b$

$\Rightarrow b = -4$

at $x = 0$

LHL = f(0) = RHL

$\Rightarrow$ $(a-3)^2 + 2b = (-1)^2 + 2b$

$\Rightarrow$ $(a-3)^2 = 1 \Rightarrow a - 3 = \pm 1$

$\Rightarrow$ $a = 2$ or 4

$\therefore$ possible values of a are 2 & 4

$\therefore$ sum of all possible values = 2 + 4 + (– 4) = 2. **Ans.**

05 $\displaystyle \lim_{h \to 0} \frac{f(3\cos h + 4\sin h - 2) - f(1)}{f(3e^h - 5\sec h + 4) - f(2)} \left(\frac{0}{0} \text{ form}\right)$

$$= \lim_{h \to 0} \frac{(-3\sin h + 4\cos h)\ f'(3\cos h + 4\sin h - 2)}{\left(3e^h - 5\sec h\ \tan h\right)\ f'\left(3e^h - 5\sec h + 4\right)}$$

$$= \frac{4}{3} \times \frac{f'(1)}{f'(2)} = \frac{4 \times 6}{3 \times 2}$$

$$= \frac{24}{6} = 4 \textbf{ Ans.}$$

06. $\because\ f(x) = \ln g(x)$

$\therefore\ f(x+1) = \ln g(x+1)$

$\therefore\ f(x+1) - f(x) = \ln g(x+1) - \ln g(x)$

$$= \ln\frac{g(x+1)}{g(x)} = \ln(x+1)$$

$$\Rightarrow\ f'(x+1) - f'(x) = \frac{1}{x+1}$$

Putting $x = 1, 2, 3.....n$ and adding, we get

$$(f'(2) - f'(1)) + (f'(3) - f'(2)) + ... + (f'(n+1) - f'(n)) = \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{n+1}$$

$$\Rightarrow\ f'(n+1) - f'(1) = \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{n+1}.$$

07. $\lim\limits_{x\to 0^+} g(x) = \lim\limits_{x\to 0^+} f(|x|) + |f(x)| = \lim\limits_{x\to 0^+} f(x) + |f(x)| = \lim\limits_{x\to 0^+} 2x+1+|2x+1| = 2$

$\lim\limits_{x\to 0^-} g(x) = \lim\limits_{x\to 0^-} \underbrace{f(|x|)}_{0^+} + |f(x)| = 1 + |-1| = 2$

, g (x) is discontinuous at $x = -2, -1$

08 $f(x) = \sum\limits_{k=1}^{n} \tan\left(\frac{x}{2^k}\right)\sec\left(\frac{x}{2^{k-1}}\right) = \sum\limits_{k=1}^{n} \frac{\sin\left(\frac{x}{2^k}\right)}{\cos\left(\frac{x}{2^k}\right)\cos\left(\frac{x}{2^{k-1}}\right)}$

$$= \sum_{k=1}^{n}\left(\tan\frac{x}{2^{k-1}} - \tan\frac{x}{2^k}\right) = \tan x - \tan\frac{x}{2^n}$$

$$\therefore \text{LHL at } x = 0 = \lim_{x\to 0^-} \frac{\ln(1+\tan x)}{x} = 1$$

$$g(x) = \lim_{x\to\infty} \prod_{r=1}^{n} \left(\frac{\cos^2\frac{x}{2^r} - \sin^2\frac{x}{2^r}}{\cos^2\frac{x}{2^r}}\right)$$

$$\lim_{n\to\infty} \prod_{r=1}^{n} \frac{2\tan\frac{x}{2^r}}{\tan\frac{x}{2^{r-1}}} = \frac{x}{\tan x}$$

$$\text{RHL at } x = 0 = \lim_{x\to 0^+} g(x) = 1$$

$\therefore\quad k = \text{LHL} = \text{RHL} = 1$

09 As $-1 + ax - x^2 < 0 \ \forall x \in R \Rightarrow x^2 - ax + 1 > 0 \ \forall x \in R$

$\Rightarrow$ $a^2 - 4 < 0 \Rightarrow -2 < a < 2$

So, number of integral values of $\alpha = 3$.

$\therefore$ $g(x) = [3 + 4\alpha \sin x]$

$= [3 + 12 \sin x] = 3 + [12 \sin x]$

So, In $(0, \pi)$, number of points of discontinuity $= (2 \times 11) + 1 = 23$.

Hence, number of points of non-derivability $= 23$. **Ans.**

10. $f(0) = \lim\limits_{x \to 0} f(x)$

Applying the method of multiplication and division by conjugate, we get

$$\lim_{x \to 0} \frac{-2ax}{2x} \frac{\sqrt{a+x} + \sqrt{a-x}}{\sqrt{a^2 - ax + x^2} + \sqrt{a^2 + ax + x^2}} = \frac{-a \cdot 2\sqrt{a}}{2a} = -\sqrt{a}$$

11. $\dfrac{f(2x)}{e^{2x} - 1} = \dfrac{f(x)}{e^x - 1}$ (Given)

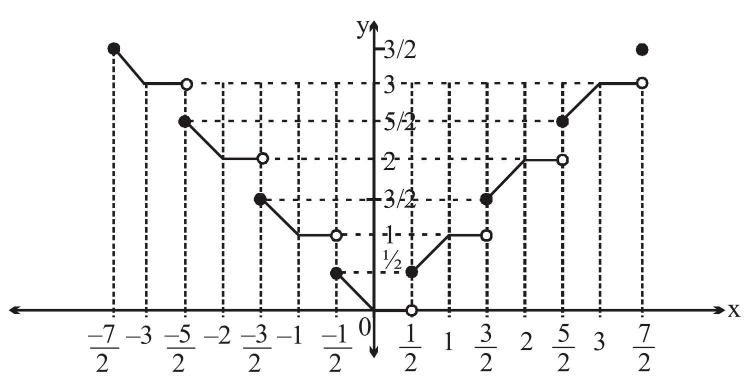
$$\Rightarrow \quad \frac{f(x)}{e^x - 1} = \frac{f\left(\frac{x}{2}\right)}{e^{\frac{x}{2}} - 1} = \frac{f\left(\frac{x}{4}\right)}{e^{\frac{x}{2^2}} - 1} = \ldots\ldots = \frac{f\left(\frac{x}{2^n}\right)}{e^{\frac{x}{2^n}} - 1}$$

$$\Rightarrow \quad \frac{f(x)}{e^x - 1} = \lim_{n \to \infty} \frac{f\left(\frac{x}{2^n}\right)}{e^{\frac{x}{2^n}} - 1} = \lim_{h \to 0} \frac{f(h)}{e^h - 1} = \lim_{h \to 0} \frac{f(h)}{h} = \lim_{h \to 0} \frac{f(h) - f(0)}{h} = f'(0)$$

$\therefore$ $f(x) = e^x - 1$

$$\lim_{x \to 0} \left(\frac{f(x)}{x}\right)^{\frac{1}{x}} = \lim_{x \to 0} \left(\frac{e^x - 1}{x}\right)^{\frac{1}{x}} = e^{\lim\limits_{x \to 0} \frac{e^x - 1 - x}{x^2}} = e^{\frac{1}{2}}. \ \textbf{Ans.}]$$

12.



L = Number of point of discontinuous = 7

M = Number of point non-differentiability = 14

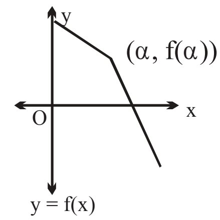
Hence (L + M) = 21. **Ans.**

13. Let $f(\alpha) = b$

$\Rightarrow \quad f^{-1}(b) = \alpha$

$f(\alpha + h) = b - k \quad \Rightarrow \quad f^{-1}(b - k) = \alpha + h$

$f(\alpha - h) = b + k \quad \Rightarrow \quad f^{-1}(b + k) = \alpha - h$



Now L.H.D. of $f^{-1}(x)$ at $x = f(\alpha) = b$

$$(f^{-1})'(b^-) = \lim_{k\to 0} \frac{f^{-1}(b-k) - f^{-1}(b)}{(b-k)-b}$$

$$= \lim_{h\to 0} \frac{\alpha + h - \alpha}{f(\alpha+h) - f(\alpha)} = \lim_{h\to 0} \frac{1}{\frac{f(\alpha+h)-f(\alpha)}{h}}$$

$$= \frac{1}{f'(\alpha^+)} = \frac{1}{r}$$

||ly Now R.H.D. of $f^{-1}(x)$ at $x = f(\alpha) = b$

$$(f^{-1})'(b^+) = \lim_{k\to 0} \frac{f^{-1}(b+k) - f^{-1}(b)}{(b-k)-b} = \lim_{h\to 0} \frac{(\alpha-h)-\alpha}{f(\alpha-h)-f(\alpha)}$$

$$= \lim_{h\to 0} \frac{1}{\frac{f(\alpha-h)-f(\alpha)}{-h}} = \frac{1}{f'(\alpha^-)} = \frac{1}{l}$$

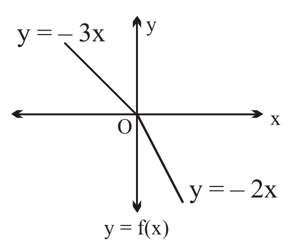
Aliter: Verification by taking an example

$$y = f(x) = \begin{cases} -3x & \text{for } x \le 0 \\ -2x & \text{for } x > 0 \end{cases}$$

$f'(0^-) = -3 = l$

$f'(0^+) = -2 = r$

$$g(y) = x = f^{-1}(y) = \begin{cases} \dfrac{-y}{3} & \text{for } y \ge 0 \\ \dfrac{-y}{2} & \text{for } y < 0 \end{cases}$$



$$g'(0^+) = f'(0^+) = \frac{-1}{3} = \frac{1}{l}$$

$$g'(0^-) = f'(0^+) = \frac{-1}{2} = \frac{1}{r}$$

Note: If l and r are positive, then L.H.D and R.H.D of f^{-1} are $\frac{1}{l}$ & $\frac{1}{r}$ and if l and r are negative then L.H.D. and R.H.D are $\frac{1}{r}$ & $\frac{1}{l}$. (Think) **Ans.**

14. Given $f(x) = \left| x(e^x - 1)(x-1)\tan^{-1}(x^2 + 3x - 4) \right|$

$= \left| x(e^x - 1)(x-1)\tan^{-1}\left((x+4)(x-1)\right) \right|$

$x(e^x - 1)(x-1)\tan^{-1}\left((x+4)(x-1)\right) = 0$

$\Rightarrow \quad x = 0, 1, -4$

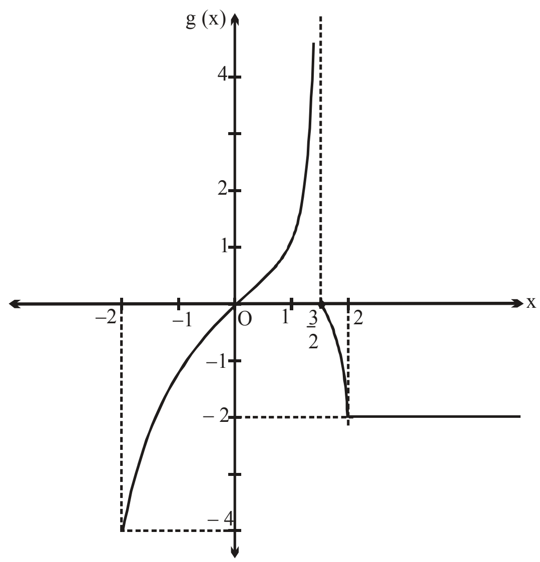
(**Note that** $x = 0$ and 1 are repeated roots)

So, $f(x)$ is non-derivable at $x = -4$ only

Hence, statement-2 is obviously true and

statement-2 is the correct explanation of statement-1 also. **Ans.**

15. Graph of $y = g(x)$



From the graph it is clear that

(A) range of $g(x)$ is $[-4, \infty)$

(B) $\lim\limits_{x \to 0^-} [g(x)] = -1$

(C) $g(x)$ is continuous at $x = 0$

(D) $g(x)$ is continuous at $x = \dfrac{-3}{2}$

but discontinuous at $x = \dfrac{3}{2}$

$\therefore$ options (A), (B), (C) are correct. **Ans.**

16.

(A) Let $h(x) = \dfrac{f(x)}{g(x)}$ be continuous at x = a. Also given g(x) is continuous at x = a.

Hence h(x) g(x) = f(x) must be continuous at x = a but given f(x) is discontinuous at x = a.
Hence our assumption is false $\Rightarrow$ h(x) must be discontinuous.

(B) False because f(x) can be discontinuous in [a, b].

(C) Obviously true.

(D) D is false. Example sgn $(\cot^{-1}x)$ at x = 0 is continuous and $(\cot^{-1} x)$ is also continuous at x = 0. But sgn (x) is discontinuous at x = 0. **Ans.**

Paragraph for Question no. 17 to 19

$f(x) = \sin^{-1}[x] \cos^{-1}\{x\}$

so $-1 \le [x] \le 1$ so $-1 \le x < 2$

$-1 \le \{x\} \le 1 \ \forall x$ Domain $[-1, 2)$

For $-1 \le \{x\} \le 1$ $[x] = -1$,

$$f(x) = \sin^{-1}(-1)\cos^{-1}(x-1) = -\frac{\pi}{2}\cos^{-1}(x-1)$$

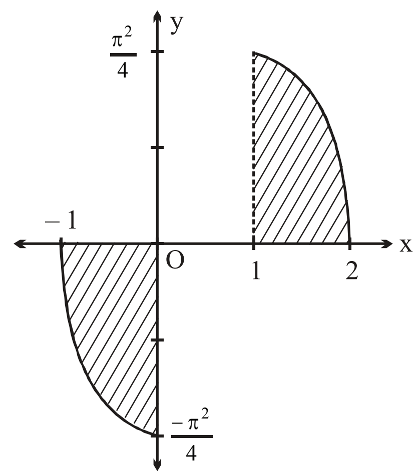
$$f(-1) = -\frac{\pi}{2}\cos^{-1}(0) = -\frac{\pi}{2}\cdot\frac{\pi}{2} = \frac{-\pi^2}{4},$$

$$f(0) = -\frac{\pi}{2}\cos^{-1}1 = 0$$

For $0 \le x < 1$ $[x] = 0$, $f(x) = \sin^{-1}(0)\cos^{-1}x = 0$

For $1 \le x < 2$ $[x] = 1$, $f(x) = \sin^{-1}(1)\cos^{-1}(x-1) = \dfrac{\pi}{2}\cos^{-1}(x-1)$

$$f(1) = \frac{\pi}{2}\cdot\frac{\pi}{2} = \frac{\pi^2}{4}$$. **Ans.**



20. $f\left(\dfrac{x}{y}\right) = f(x) - f(y)$

$$\therefore\ f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{f\left(1+\frac{h}{x}\right)}{\left(\frac{h}{x}\right)\cdot x}$$

Also $f(1) = 0$

$$\therefore\ f'(x) = \lim_{h\to 0}\frac{f\left(1+\frac{h}{x}\right)-f(1)}{\left(\frac{h}{x}\right)\cdot x} = \frac{f'(1)}{x} = \frac{1}{x}$$

$f(x) = \ln x + c$ but $f(1) = 0$ $\therefore$ $f(x) = \ln x$

(B) $\lim\limits_{n\to\infty} \dfrac{x^{1/n}-1}{(1/n)} = \ln x$

(C) $\lim\limits_{x\to0} \dfrac{f(1+x)}{x} = \lim\limits_{x\to0} \dfrac{\ln(1+x)}{x} = 1$

(D) $\lim\limits_{x\to0} x\,f(x) = \lim\limits_{x\to0} x \ln x = \lim\limits_{x\to0} \dfrac{\ln x}{\left(\dfrac{1}{x}\right)} = \lim\limits_{x\to0} \dfrac{1/x}{-1/x^2} = 0$. **Ans.**

21. $\left| f(x) - f(y) - \sum\limits_{k=1}^{n} (x^k - y^k) \right| \le |x-y|^2$

$$\left| \frac{f(x)-f(y)}{x-y} - \sum_{k=1}^{n} \frac{x^k - y^k}{x-y} \right| \le |x-y|$$

$$\lim_{y\to x} \left| \frac{f(x)-f(y)}{x-y} - \sum_{k=1}^{n} \frac{x^k - y^k}{x-y} \right| \le \lim_{y\to x} |x-y|$$

$$\left| f'(x) - \sum_{k=1}^{n} kx^{k-1} \right| \le 0$$

$$f'(x) = \sum_{k=1}^{n} k\,x^{k-1} = 1 + 2x + 3x^2 + \ldots\ldots + nx^{n-1}$$

$f(x) = x + x^2 + x^3 + \ldots\ldots + x^n + c$

$f(0) = 1$

$\therefore\ f(x) = 1 + x + x^2 + \ldots\ldots + x^n$

$$\lim_{n\to\infty} f\left(\frac{1}{2}\right) = 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \ldots\ldots + \infty = \frac{1}{1-\frac{1}{2}} = 2$$

For $n = 100$, $\lim\limits_{x\to1} \dfrac{(1+x+x^2+\ldots\ldots+x^{100})-101}{(x-1)}$

$$= \lim_{x\to1} \frac{(x-1)}{(x-1)} + \frac{(x^2-1)}{(x-1)} + \ldots\ldots + \frac{(x^{100}-1)}{(x-1)}$$

$= 1 + 2 + 3 + \ldots\ldots + 100 = 5050$

For $n = 3$, $\operatorname{sgn}(f(x)) = \operatorname{sgn}(1 + x + x^2 + x^3) = \operatorname{sgn}\big((1+x)(1+x^2)\big)$

$(1 + x)(1 + x^2) = 0$ when $x = -1$

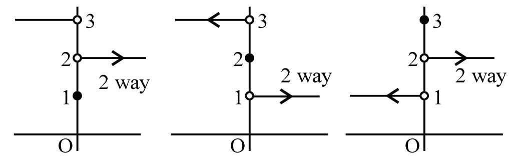
For $n = 4$, $\operatorname{sgn}(f'(x)) = \operatorname{sgn}(1 + 2x + 3x^2 + 4x^3)$

Let $F(x) = 1 + 2x + 3x^2 + 4x^3$

$F'(x) = 2 + 6x + 4x^2 > 0$

$\Rightarrow$ F(x) has exactly one real zero.

22. $\because$ $f^3(x) - 6f^2(x) + 11\, f(x) - 6 = 0$



$\Rightarrow$ $\left(f^2(x) - 3f(x) + 2\right)\left(f(x) - 3\right) = 0$

$\Rightarrow$ $\left(f(x) - 1\right)\left(f(x) - 2\right)\left(f(x) - 3\right) = 0$

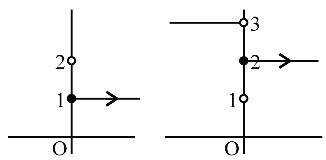
$\Rightarrow$ $f(x) = 1$ or 2 or 3

(i) If L.H.L. $\neq$ R.H.L. $\neq f(0)$
then number of such functions $= 3! = 6$. (See diagram)

(ii) If any two of L.H.L. $f(0^-)$, R.H.L. $f(0^+)$ and $f(0)$ are equal and third one is not equal then number of such functions $= {}^3C_2 \times {}^3C_2 \times {}^2C_1 = 18$

$\therefore$ Number of such functions $= 24$. Ans.

(ii) Out of 1, 2, 3 we can select any two of ${}^3C_2$ say 1 and 2 and out of $f(0^+)$, $f(0^-)$ and $f(0)$ we can select any two which are equal in ${}^3C_2$ ways.
Now total ways ${}^3C_2 \cdot {}^3C_2 \cdot 2 = 18$



Alternative: Total number of function are $= 3^3 = 27$
Total number of continuous function $= 3$
Hence total number of discontinuous function $27 - 3 = 24$ **Ans.**

23. $f(x) = \left(\tan\frac{x}{2}\right)(\sec x) + \left(\tan\frac{x}{2^2}\right)\left(\sec\frac{x}{2}\right) + \ldots\ldots.$

$$T_1 = \frac{\sin(x/2)}{\cos(x/2)} \cdot \frac{1}{\cos x} = \frac{\sin(x - (x/2))}{\cos(x/2)\cos x} = \tan x - \tan\frac{x}{2}$$

$$T_2 = \frac{\sin(x/4)}{\cos(x/4)} \cdot \frac{1}{\cos(x/2)} = \frac{\sin((x/2) - (x/4))}{\cos(x/4)\cos(x/2)} = \tan\frac{x}{2} - \tan\frac{x}{2^2}$$

$\vdots$

|||ly $\quad T_n = \tan\frac{x}{2^{n-1}} - \tan\frac{x}{2^n}$

$\therefore$ $\quad f(x) = \tan x - \tan\frac{x}{2^n}$

now $\quad g(x) = \lim\limits_{n\to\infty} \dfrac{ln(\tan x) - (\tan x)^n \cdot \left[\sin\left(\tan\frac{x}{2}\right)\right]}{1 + (\tan x)^n}$ if $x \neq \frac{\pi}{4}$ and $g\left(\frac{\pi}{4}\right) = k$

$$\therefore \quad g(x) = \begin{cases} \ell n(\tan x) & \text{if } 0 < x < \frac{\pi}{4} \\ 0 & \text{if } \frac{\pi}{4} < x < \frac{\pi}{2} \\ k & x = \frac{\pi}{4} \end{cases}$$

now $\quad g\left(\frac{\pi}{4}^{-}\right) = 0; \quad g\left(\frac{\pi}{4}^{+}\right) = 0$

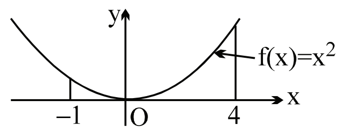
$\Rightarrow \quad k = 0$

obvious $g(x)$ is continuous every where in $(0, \pi/4)$

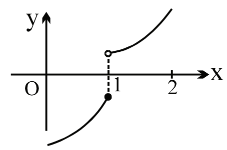
24. (A) is wrong as the clause of continuity is a must for $f(x)$

(B) is correct from I.V.T.

(C) as per the graph (C) is incorrect



(D) is wrong if f is discontinuous



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25. $$l = \lim_{x\to 0} \frac{2x + x\cos x - 3\sin x}{x^4 \sin x} = \lim_{x\to 0} \frac{2x + x\cos x - 3\sin x}{x^4}$$

$$= \lim_{t\to 0} \frac{6t + 3t\cos 3t + 3\sin 3t}{243\,t^5} \qquad \text{(put } x = 3t)$$

$$= \frac{2t + t(4\cos^3 t - 3\cos t) - (3\sin t - 4\sin^3 t)}{81\cdot t^5} \qquad \text{(cancelling by 3)}$$

$$= \frac{2t + t\cos t(1 - 4\sin^2 t) - (3\sin t - 4\sin^3 t)}{81\cdot t^5}$$

$$= \frac{(2t + t\cos t - 3\sin t)}{81t^5} + \frac{4\sin^2 t(\sin t - t\cos t)}{81t^5}$$

$$l = \frac{l}{81} + \frac{4}{81}\lim_{t\to 0}\frac{\sin^2 t}{t^2}\lim_{t\to 0}\frac{\cos t(\tan t - t)}{t^3}$$

$$l = \frac{l}{81} + \frac{4}{81}\lim_{t\to 0}\frac{\tan t - t}{t^3}$$

$$l = \frac{l}{81} + \frac{4}{81}\cdot\frac{1}{3}$$ [Note : l can be checked using expansion of $\sin x$ and $\cos x$ as

$$\frac{80l}{81} = \frac{4}{81\cdot 3} \Rightarrow l = \frac{1}{60} \text{ Ans.}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} \ldots\ldots \text{ and } \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} \ldots\ldots$$

Alternatively: $\lim\limits_{x\to 0} \dfrac{x(2+\cos x)-3\sin x}{x^5}$

$$= \frac{x\left[2+1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+.....\right]-3\left[x-\frac{x^3}{3!}+\frac{x^5}{5!}.......\right]}{x^5}$$

$$= \frac{1}{4!}-\frac{3}{5!} = \frac{1}{24}-\frac{3}{120} = \frac{5-3}{120} = \frac{1}{60} \text{ Ans.}$$

26. $f(0) = 0$ as f is odd.

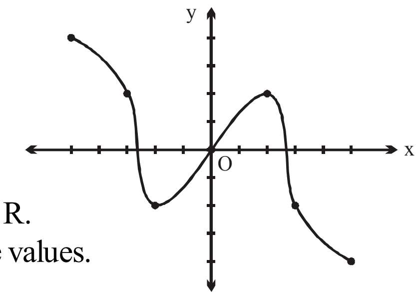
Also, $f(2) = 2 \Rightarrow f(-2) = -2$

$f(3) = -3 \Rightarrow f(-3) = 3$

$f(5) = -5 \Rightarrow f(-5) = 5$

So, by I.V.T, $f(x) = 0$ has atleast 3 roots in R.

As real prime numbers will yeald only negative values.



27 $\lim\limits_{x\to 1} \dfrac{e^{2f(x)}-2e^{f(x)+1}+e^2\cos(x-1)}{\sec^2(x-1)-1} = \dfrac{7}{2}e^2$

Here limit exists. Therefore as $x \to 1$, $N^r \to 0$ ($\because D^r$. also tends to zero)

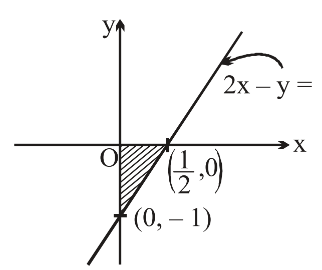
$e^{2f(1)} - 2 \cdot e \cdot e^{f(1)} + e^2 = 0$

$\left(e^{f(1)}-e\right)^2 = 0 \Rightarrow e^{f(1)} = e \Rightarrow f(1) = 1$

$$\lim_{x\to 1} \frac{e^{2f(x)}-2\cdot e\cdot e^{f(x)}+e^2-e^2\left(1-\cos(x-1)\right)}{\tan^2(x-1)} = \frac{7}{2}e^2$$

$$\lim_{x\to 1} \frac{\left(e^{f(x)}-e\right)^2-e^2\left(1-\cos(x-1)\right)}{\frac{\tan^2(x-1)}{(x-1)^2}(x-1)^2} = \frac{7}{2}e^2$$

$$\lim_{x\to 1}\left(\frac{e^2\left(e^{f(x)-1}-1\right)^2}{\left(f(x)-1\right)^2}\cdot\left(\frac{f(x)-1}{x-1}\right)^2-e^2\frac{\left(1-\cos(x-1)\right)}{(x-1)^2}\right) = \frac{7}{2}e^2$$



$$e^2 \cdot 1 \cdot \left(f'(1)\right)^2 - e^2\cdot\frac{1}{2} = \frac{7}{2}e^2$$

$\left(f'(1)\right)^2 = 4 \Rightarrow f'(1) = \pm 2 \Rightarrow f'(1) = 2 \;\{\because f'(x) > 0\}$

Now, equation of tangent at $\left(1, f(1)\right)$ is

$y - 1 = 2(x - 1)$

$2x - y = 1$

$\therefore \quad \Delta = \dfrac{1}{2}\times\dfrac{1}{2}\times 1 = \dfrac{1}{4} \Rightarrow \dfrac{1}{\Delta} = 4.$ **Ans.**

28 $\alpha = 0$

$$f(0) = \lim_{x \to 0} \frac{xa^x + bx\cos x + c\sin x}{x^5}$$

$$= \lim_{x \to 0} \frac{x\left(1 + x\, ln\, a + \frac{x^2 (ln\, a)^2}{2!} + ...\right) + bx\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!}\right) + c\left(x - \frac{x^3}{3!} + \frac{x^5}{5!}\right)}{x^5}$$

$$= \lim_{x \to 0} \frac{x(1 + b + c) + x^2\, ln\, a + x^3\left[\frac{(ln\, a)^2}{2} - \frac{b}{2} - \frac{c}{6}\right] + x^5\left[\frac{b}{24} + \frac{c}{120}\right]}{x^5}$$

To exist, $1 + b + c = 0$, $ln\, a = 0 \Rightarrow a = 1$.

$$-\frac{b}{2} - \frac{c}{6} = 0 \Rightarrow 3b + c = 0$$

$$\Rightarrow \quad b = \frac{1}{2},\ c = \frac{-3}{2}$$

$f(0) = \dfrac{1}{120}$ and $f'(0) = 0$. **Ans.**

29. Consider $\displaystyle \lim_{n \to \infty} \sum_{r=0}^{n-1} \frac{x}{(rx+1)\{(r+1)x+1\}}$

$$\sum_{r=0}^{n-1} \frac{((r+1)x+1) - (rx+1)}{(rx+1)\{(r+1)x+1\}} = \sum_{r=0}^{n-1} \frac{1}{(rx+1)} - \frac{1}{rx+x+1}$$

$$= \left(\frac{1}{1} - \frac{1}{x+1}\right)$$

$$\left(\frac{1}{x+1} - \frac{1}{2x+1}\right)$$

$$\left(\frac{1}{2x+1} - \frac{1}{3x+1}\right)$$

$$\vdots \qquad \vdots$$

$$\left(\frac{1}{(n-1)x+1} - \frac{1}{nx+1}\right)$$

$\therefore \quad f(x) = \displaystyle \lim_{n \to \infty} \left(\frac{1}{1} - \frac{1}{nx+1}\right) = \begin{bmatrix} 1 & x \neq 0 \\ 0 & x = 0 \end{bmatrix}$ limit does not exist at $x = 0$ and hence discontinuous

$\Rightarrow$ non derivable

30. $gof(x) = g([x]) = 0$
$\Rightarrow$ continuous for every x

31. Let $x \in [m, m+1)$ where $m \in I$ then
$f(x) = x + [x] = x + m$
$\because f'(x) = 1 > 0 \ \forall \ x \in (m, m+1)$
$\therefore$ f(x) is strictly increasing
$\therefore f^{-1}(x)$ exists.
$\therefore \quad y = f(x) = x + m \Rightarrow x = y - m$
but $\quad m \le x < m + 1$
$\therefore 2m \le x + m < 2m + 1$
$\therefore 2m \le y < 2m + 1$
$\therefore [y] = 2m$

$\therefore \ m = \dfrac{[y]}{2} \qquad \therefore \ x = y - m = y - \dfrac{[y]}{2} = f^{-1}(y)$

$\therefore \ f^{-1}(x) = x - \dfrac{[x]}{2} \forall x \in [2m, 2m+1) \qquad$ where $m \in I$

Now $\quad f(x) = x + [x] = \begin{cases} x & \forall x \in [0,1) \\ x+1 & \forall x \in [1,2) \\ x+2 & \forall x \in [2,3) \end{cases}$

which is discontinuous at every integer point

Consider, $f^{-1}(x) = x - \dfrac{[x]}{2}$ where $x \in [2m, 2m+1)$

$$= \begin{cases} x & \forall x \in [0,1) \\ x-1 & \forall x \in [2,3) \\ x-2 & \forall x \in [4,5) \end{cases}$$

Clearly, $f^{-1}(x)$ is continuous $\forall \ x \in [2m, 2m+1)$ where $m \in I$

Considering $\lim\limits_{x \to 2m} f^{-1}(x)$, only R.H.L. exists and is equal to $f^{-1}(2m)$

$\therefore \ f^{-1}(x)$ is continuous $\forall \ x \in [2m, 2m+1)$ $(m \in I)$ i.e. $f^{-1}(x)$ is continuous in its domain.

32. $f(x) = \ln x + x + 1$

$g(2^-) = e^l$ where $l = \lim\limits_{x \to 2^-} \dfrac{\ln(3e)}{\tan(x-2)} \left(1 + \dfrac{15(x-2)}{x^2+1} - 1\right)$

$= \ln(3e) \lim\limits_{x \to 2} \dfrac{15(x-2)}{\tan(x-2)} \left(\dfrac{1}{1+x^2}\right)$

$l = 3 \ln(3e) = \ln(27e^3)$

$g(2^-) = e^{\ln(27e^3)} = 27e^3 = g(2)$

Again $g(2^+) = \lim\limits_{x\to 2^+} \dfrac{\lambda\left(e^{f(2^+)} - 2e^3\right)}{\left(\dfrac{\sin(x-2)}{x-2}\right)(x-2)}$

$$= \lim_{h\to 0} \lambda\left[\frac{e^{\ln(1+h)+3+h} - 2e^3}{h}\right]$$

$$= \lambda\left[e^3\left\{\frac{e^{\ln(2+h)} - 2}{h}\right\}\right] = \lambda e^3\left(\frac{(2+h)e^h - 2}{h}\right)$$

$$= \lambda e^3\left[\lim_{h\to 0}\frac{2(e^h-1)}{h} + e^h\right]$$

For continuous $3e^3\lambda = 27e^3 \Rightarrow 3\lambda = 27 \Rightarrow \lambda = 9$. **Ans.**

Aliter: $\lim\limits_{x\to 2^-}\left(1+\dfrac{15(x-2)}{1+x^2}\right)^{\frac{\ln(3e)}{\tan(x-2)}} = e^{\lim\limits_{x\to 2^-}\left(1+\frac{15(x-2)}{1+x^2}-1\right)\frac{\ln(3e)}{\tan(x-2)}} = e^{3\ln(3e)} = 27e^3.$

Also, $\lim\limits_{x\to 2^+}\lambda\left(\dfrac{e^{f(x)} - 2e^3}{\sin(x-2)}\right) = \lim\limits_{x\to 2^+}\lambda\left(\dfrac{e^{f(x)} - e^{\ln 2e^3}}{\sin(x-2)}\right)$

$$= \lim_{x\to 2^+}\lambda\, e^{\ln 2e^3}\,\frac{\left(e^{f(x)-\ln(2e^3)} - 1\right)}{\sin(x-2)}$$

$$= \lim_{x\to 2^+}\lambda \,.\, 2e^3 \,.\left(\frac{e^{\ln x + x + 1 - \ln 2 - 3} - 1}{\sin(x-2)}\right)$$

$$= \lim_{x\to 2^+}\lambda \,.\, 2e^3 \,.\,\frac{\ln x + x - \ln 2 - 2}{\sin(x-2)}$$

$$= \lim_{x\to 2^+}\lambda \,.\, 2e^3 \,.\left(\frac{\ln\left(1+\frac{x}{2}-1\right)+(x-2)}{\sin(x-2)}\right) = \lim_{x\to 2^+} 1 \,.\, 2e^3\left(\frac{\dfrac{\ln\left(1+\frac{x-2}{2}\right)}{2\left(\frac{x-2}{2}\right)} + \dfrac{(x-2)}{x-2}}{\dfrac{\sin(x-2)}{x-2}}\right)$$

$$= 1 \, . \, 2e^3\left(\frac{1}{2}+1\right) = \lambda \cdot 3e^3$$

$\therefore \quad \lambda \cdot 3e^3 = 27e^3$

$\Rightarrow \quad \lambda = 9$ **Ans.**

33. $f(x) - f(y) \ge \ln x - \ln y + x - y$ ....(1)

$x \to y$ and $y \to x$,

$f(y) - f(x) \ge \ln y - \ln x + y - x$

i.e. $f(x) - f(y) \le \ln x - \ln y + x - y$ ....(2)

from (1) and (2)

$f(x) - f(y) = \ln x - \ln y + x - y$ ....(3)

$\therefore \quad f(x) - f(y) = (\ln x + x) - (\ln y + y)$

hence $f(x) = \ln x + x$

Alternatively: $f'(x) = \lim\limits_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$ $\quad (x \to x+h,\ y \to x \text{ in } (3))$

$$= \lim_{h\to 0} \frac{\ln(x+h) - \ln x + h}{h} = \lim_{h\to 0} \frac{\ln\left(\frac{x+h}{x}\right)+h}{h}$$

$$= \lim_{h\to 0}\left[1 + \frac{\ln(1+(h/x))}{x\,(h/x)}\right]$$

$$= 1 + \frac{1}{x}\lim_{t\to 0}\frac{\ln(1+t)}{t}$$

$$f'(x) = 1 + \frac{1}{x}$$

$$\therefore \quad g(x) = 1 + \frac{1}{x}$$

$$\sum_{n=1}^{100} g\left(\frac{1}{n}\right) = \sum_{n=1}^{100}(1+n) = 100 + (1+2+3+4+\ldots\ldots+100)$$

$= 100 + 5050 = 5150$ Ans.

34. $f(x + y^n) = f(x) + (f(y))^n \qquad f'(0) \ge 0$

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{f\left(x+(h^{1/n})^n\right)-f(x)}{h}$$

$$f'(0) = \lim_{t\to 0}\frac{(0+t)-f(0)}{t}$$

$$= \lim_{h\to 0}\frac{f(x)+\left(f(h^{1/n})\right)^n - f(x)}{(h^{1/n})^n}$$

$$\lim_{t\to 0}\frac{f(t)}{t} = f'(0) = l \text{ (say)}$$

also $f(0) = 0$

$$f'(x) = \lim_{t\to 0}\left(\frac{f(t)}{t}\right)^n \quad \text{where } h^{1/n} = t$$

$f'(x) = [f'(0)]^n \Rightarrow f'(0) = [f'(0)]^n$

$l = l^n$ (where $f'(0) = 1 \Rightarrow$ $l = 0$ or -1 or 1

If $l = 0 \Rightarrow f'(x) = 0 \Rightarrow f(x) = c$, but $f(0) = 0$

$\Rightarrow$ $c = 0$, $\therefore$ $f(x) = 0$ (not possible)

If $l = 1 \Rightarrow f'(x) = -1$ (not possible);

$\therefore$ $f'(0) = 1$

$\therefore$ $f'(x) = 1$ $\Rightarrow f(x) = x + c$

$\Rightarrow$ $f(x) = x$ $\Rightarrow$ $f(0) = 10$ **Ans.**

Paragraph for questions nos. 35 to 37

Sol. $$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$= \lim_{h\to 0}\frac{f(x)+f(h)-2xh+(e^x-1)(e^h-1)-f(x)}{h}$$

$$= \lim_{h\to 0}\left(\frac{f(h)}{h}+(e^x-1)\frac{(e^h-1)}{h}\right)-2x$$

$= f'(0) + e^x - 1 - 2x$

$= 1 + e^x - 1 - 2x = e^x - 2x$

$f(x) = e^x - x^2 + \lambda$

$f(0) = 1 - 0 + 1 \Rightarrow \lambda = -1$

$\therefore f(x) = e^x - x^2 - 1$

35. $f(2) = e^2 - 4 - 1 = e^2 - 5$

$\{f(2)\} = e^2 - 5 - 2 = e^2 - 7]$ $\{\because 7 < e^2 < 8\}$

36. $$\lim_{x\to 0}\frac{f(2x)+4x^2-2x}{x^2}$$

$$= \lim_{x\to 0}\frac{e^{2x}-4x^2-1+4x^2-2x}{x^2}$$

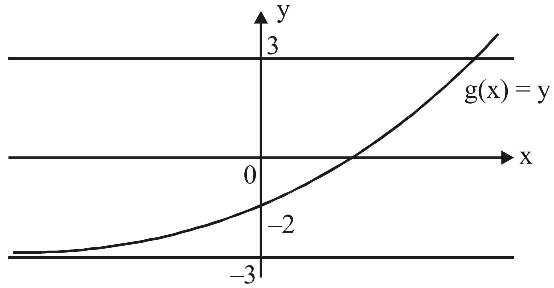
$$= \lim_{x\to 0}\frac{e^{2x}-2x-1}{x^2}$$

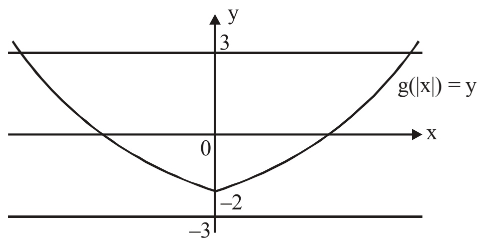
$$= \lim_{x\to 0}\frac{1+2x+\frac{4x^2}{2!}+\ldots\ldots-2x-1}{x^2} = 2$$

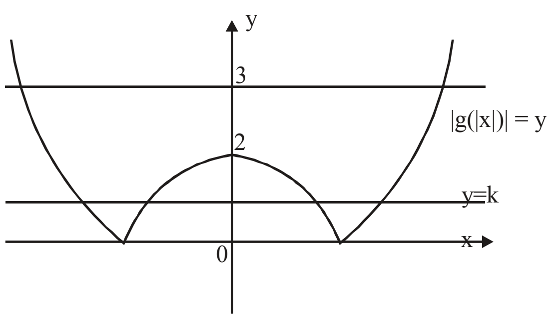
37 $g(x) = f(x) + x^2 - 2 = e^x - x^2 - 1 + x^2 - 2$

$g(x) = e^x - 3$

graph $= |\,g(|\,x\,|)\,|$ is –







Now the $|\,g(|\,x\,|)\,| = k$ has four roots

$\therefore\ k \in (0, 2)$

38. $f(x) = \begin{cases} 0, & x = -1 \\ x^2, & -1 < x < 1 \\ 2x - 1, & 1 \le x < 2 \\ x^2 - 1, & 2 \le x < 4 \\ 0, & x = 4 \end{cases}$

Clearly, $f(x)$ is discontinuous at $x = -1$ and $x = 4$ in $[-1, 4]$ but $f(x)$ is derivable everywhere in $(-1, 4)$

$\therefore\quad m = 2$ and $n = 0$

Hence $m + n = 2$. **Ans.**

39. $\because\ g(0) = c\ \therefore\ g(x) = x^2 + k_1x + c$

$\Rightarrow\ g(1) = 1 + k_1 + c$

$\because\ h(0) = b\ \therefore\ h(x) = x^2 + k_2x + b$

$\Rightarrow\quad h(1) = 1 + k_2 + c$

$f(x) = 0$ & $h(x) = 0$ have one common root.

$g(1) = h(1) = 0$, if $k_1 = b$ & $k_2 = c$ $\quad (\because\ 1 + b + c = 0)\quad$ ....(1)

$\therefore$ $g(x) = x^2 + bx + c$

$h(x) = x^2 + cx + b$

$$f(x) = \frac{[x]^2 + b[x] + c}{[x]^2 + c[x] + b} = \frac{([x]-1)([x]-c)}{([x]-1)([x]-b)}$$

$$f(x) = \frac{([x]-c)}{([x]-b)} \quad ([x] \neq 1)$$

$\Rightarrow \quad x \neq [1, 2)$

now if 'b' is not an integer then D^r can not be zero and

hence domain of $f(x)$ is $R - [1, 2)$, if $b \notin I$.

if 'b is an integer then domain of $f(x)$ is $R - [1, 2) \cup [b, b+1)$, if $b \in I$.

$f(x)$ is continuous only iff $b = c = -1/2$ i.e. when $f(x)$ becomes constant

because if $b \neq c$ then limit of $f(x) = \dfrac{([x]-c)}{([x]-b)}$ will not exist at any x in the domain of $f(x)$

40. $$f\left(\frac{\pi}{2}\right) = \frac{1}{2}\lim_{\alpha \to 0}\left[\frac{p\sin\alpha}{\alpha}\right] = \left(\frac{p-1}{2}\right) \left(\begin{array}{l} \because \dfrac{\sin\alpha}{\alpha} \text{ is little less than 1 as } \alpha \to 0 \\ \therefore\ p-1 < \dfrac{p\sin\alpha}{\alpha} < p \quad \text{where } p \in I^+ \end{array}\right)$$

$$\text{L.H.L. at } x = \frac{\pi}{2} = \lim_{x \to \frac{\pi}{2}^-} \frac{1-\sin^3 x}{3\cos^2 x} = \lim_{x \to \frac{\pi}{2}^-} \frac{(1-\sin x)(1+\sin^2 x + \sin x)}{3(1-\sin x)(1+\sin x)} = \frac{1}{2}$$

$$\text{R.H.L. at } x = \frac{\pi}{2} = \lim_{x \to \frac{\pi}{2}^+} \left\{\lim_{t \to 0}\left[\frac{q\tan t}{t}\right]\frac{1-\sin x}{(\pi-2x)^2}\right\} \left(\begin{array}{l} \because \dfrac{\tan t}{t} \text{ is little greater than 1 as } t \to 0 \\ \therefore\ q < \dfrac{q\tan t}{t} < q+1 \quad \text{for } q \in I^+ \end{array}\right)$$

$$= \lim_{x \to \frac{\pi}{2}^+} \frac{q(1-\sin x)}{(\pi-2x)^2} = \lim_{h \to 0} \frac{q(1-\cos h)}{4h^2} = \frac{q}{8} \qquad \left(x = \frac{\pi}{2} + h\right)$$

$\because$ function is continuous

$$\therefore \text{ L.H.L.} = \text{R.H.L.} = f\left(\frac{\pi}{2}\right)$$

$$\Rightarrow \frac{q}{8} = \frac{1}{2} \Rightarrow q = 4\ ;$$

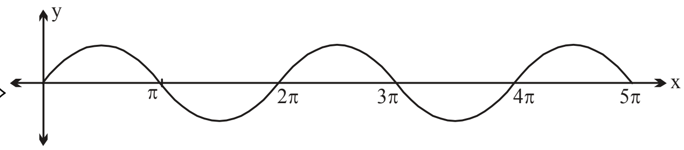
$$\frac{p-1}{2} = \frac{1}{2} \Rightarrow p = 2\ ; \qquad \therefore\ p + q = 6$$

41 f(x) is discontinuous where

$(\sin^2 x - \sin x - 1)(\sin^2 x + \sin x + 1) = 0$

$\Rightarrow \sin^2 x - \sin x - 1 = 0 \qquad (\because\ \sin^2 x + \sin x + 1 = 0$ has no real roots.)

$$\Rightarrow \sin x = \frac{1 \pm \sqrt{5}}{2} \quad \text{or} \quad \sin x = \frac{1-\sqrt{5}}{2} \quad (\because\ -1 \le \sin x \le 1)$$

graph of sin x $\Rightarrow$ 

There will be two values of x between π and 2π for which $\sin x = \dfrac{1-\sqrt{5}}{2}$.

$\therefore$ For 4 points of discontinuity, n can take the value 4 or 5. **Ans.**

42. We have $f(x) = \sum_{r=1}^{n}\left(x^{2r} + \frac{1}{x^{2r}} + 2\right) = \sum_{r=1}^{n} x^{2r} + \sum_{r=1}^{n}\frac{1}{x^{2r}} + 2n$

$$= (x^2 + x^4 + ... + x^{2n}) + \left(\frac{1}{x^2} + \frac{1}{x^4} + ... + \frac{1}{x^{2n}}\right) + 2n = \frac{x^2(1-x^{2n})}{(1-x^2)} + \frac{1}{x^2}\frac{\left(1-\frac{1}{x^{2n}}\right)}{1-\frac{1}{x^2}} + 2n$$

$$f(x) = \frac{x^2(1-x^{2n})}{(1-x^2)} + \frac{\left(1-x^{2n}\right)}{(1-x^2)x^{2n}} + 2n$$

$$\boxed{f(x) = \frac{(1-x^{2n})}{(1-x^2)}\left(x^2 + \frac{1}{x^{2n}}\right) + 2n} \quad x \neq \pm 1$$

$$\therefore \quad (f(x) - 2n)(1-x^2) = (1-x^{2n})\left(x^2 + \frac{1}{x^{2n}}\right)$$

Now consider $g(x) = \lim\limits_{n\to\infty}\left((f(x)-2n)x^{-2n-2}(1-x^2)\right)$ for $x \neq \pm 1, 0$

$$= \lim_{n\to\infty} (1-x^{2n})\left(x^2 + \frac{1}{x^{2n}}\right)\cdot\frac{1}{x^{2n+2}};\ x \neq \pm 1, 0$$

$$= \lim_{n\to\infty} \frac{(1-x^{2n})(x^{2n+2}+1)}{(x^{2n})(x^{2n+2})} = \lim_{n\to\infty}\left(-1+\frac{1}{x^{2n}}\right)\left(1+\frac{1}{x^{2n+2}}\right)$$

Now $\quad g(x) = \begin{cases} -1 & \text{if } |x| > 1 \\ +\infty & \text{if } |x| < 1 - \{0\} \\ -1 & \text{if } x = \pm 1, 0 \end{cases}$

Now clearly g has non removable infinite type of discontinuity at x = 1 and – 1
Also g is continuous at x = 2

43. (A) f(x) is continuous at $|x| = -|x|$ i.e. at $x = 0$
(B) f(x) is continuous at $x^2 - 2 = -x^2 - 2$ i.e. at $x = 0$
(C) f(x) is only right continuous at $x = 0$.
(D) f(x) is continuous at $x = 1 - x$ i.e. at $x = \dfrac{1}{2}$

44. Let $f(x) = \text{cosec } 2x + \text{cosec } 2^2 x + \text{cosec } 2^3 x + \ldots\ldots\ldots\ \text{cosec } 2^n x,\ x \in (0, \pi/2)$
and $g(x) = f(x) + \cot 2^n x$

$$\text{If } H(x) = \begin{cases} (\cos x)^{g(x)} + (\sec x)^{\text{cosec } x} & \text{if } x > 0 \\ p & \text{if } x = 0 \\ \dfrac{e^x + e^{-x} - 2\cos x}{x \sin x} & \text{if } x < 0 \end{cases}$$

Find the value of p, if possible to make the function H (x) continuous at x = 0.

Method 2. $f(x) = \text{cosec } 2x + \text{cosec } 2^2 x + \ldots\ldots + \text{cosec } 2^n x$

now $\text{cosec } 2x = \dfrac{1}{\sin 2x} = \dfrac{\sin(2x - x)}{\sin x \sin 2x} = \cot x - \cot 2x$

|||ly $\text{cosec } 2^2 x = \cot 2x - \cot 2^2 x$
$\text{cosec } 2^3 x = \cot 2^2 x - \cot 2^3 x$
$\vdots$
$\text{cosec } 2^n x = \cot 2^{n-1} x - \cot 2^n x$

$\therefore\ f(x) = \cot x - \cot 2^n x$
$\therefore\ g(x) = f(x) + \cot 2^n x = \cot x$

now $H(0 + h) = \lim\limits_{h \to 0} \left((\cos h)^{\cot h} + (\sec h)^{\text{cosec } h} \right)$

$= e^{\lim\limits_{h \to 0} \coth (\cos h - 1)} + e^{\lim\limits_{h \to 0} \text{cosec} h (\sec h - 1)}$

$= 1 + 1 = 2$ ....(1)

$$H(0 - h) = \lim_{h \to 0} \frac{e^{-h} + e^{h} - 2\cos h}{h \sin h} = \lim_{h \to 0} \left\{ \frac{e^h + e^{-h} - 2}{h^2} + \frac{2(1 - \cos h)}{h^2} \right\} = 2 \quad \text{....(2)}$$

From (1) and (2) H (x) will be cont. if p = 2

45 Given $f(x) = \begin{cases} x^2 + x, & x \le 1 \\ x^2 - ax, & x > 1 \end{cases}$, $g(x) = \begin{cases} x^2 - cx, & x \le 2 \\ x^2 - 4x, & x > 2 \end{cases}$ and $h(x) = f(x) . g(x),\ x \in R$

Now, define h(x) in the n.b.d. of x = 1 and x = 2.

$h(x) = f(x) \cdot g(x) = \begin{cases} (x^2 + x)(x^2 - cx), & x \le 1 \\ (x^2 - ax)(x^2 - cx), & x > 1 \end{cases}$ and

$h(x) = \begin{cases} (x^2 - ax)(x^2 - cx), & x \le 2 \\ (x^2 - ax)(x^2 - 4x), & x > 2 \end{cases}$

Note: f(x) is discontinuous $\forall\ a \in \{1, 2, 3, 4, 5, 6\}$
g(x) is discontinuous if $c \ne 4$
N_1: Given both f(x) and g(x) are discontinuous
Hence, $a \to 6$ ways $\quad \therefore \quad N_1 = 6 \times 5 = 30$
$c \to 5$ ways

N_2 since g(x) is discontinuous $\therefore$ $c \neq 4$

Now, h(x) is continuous at x = 1

$h(1^-) = h(1^+)$

$\Rightarrow$ $2(1-c) = (1-a)(1-c) \Rightarrow (1-c)(1+a) = 0$

$\therefore$ $1 - c = 0$ $(a \neq -1)$ (Given)

$\Rightarrow$ $c = 1$

Also, given h(x) is discontinuous at x = 2

$(4-2a)(4-2c) \neq (4-2a)(-4)$

$(4-2a)(8-2c) \neq 0$

$(2-a)(4-c) \neq 0$

Since c = 1

$\therefore$ $a \neq 2 \Rightarrow$ $a \to 5$ ways

$c \to 1$ way

$\therefore$ $N_2 = 5.$

Now, N_3 given h(x) is discontinuous at x = 1

$\Rightarrow$ $(1-c)(a+1) \neq 0$ $\therefore$ $c \neq 1$

and given h(x) is continuous at x = 2.

$\Rightarrow$ $(2-a)(4-c) = 0$

if c = 4 then $a \to 6$ ways

if a = 2 then $c \to 4$ ways

Total 10 ways.

$\therefore$ $N_1 = 30$

$N_2 = 5$

$N_3 = 10$

Total $N_1 + N_2 + N_3 = 45$. **Ans.**

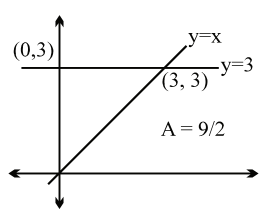
46 Let f(x) = y

$\therefore$ $y^2 - 4y + 3 \geq 0$ $(y-1)(y-3) \geq 0$ ....(1)

$y^2 - 6y + 8 \leq 0$ $(y-4)(y-2) \leq 0$ ....(2)

$y^3 - 5y^2 + 10y - 12 \leq 0$ $(y-3)(y^2 - 2y + 4) \leq 0$ ....(3)

combining (1), (2) and (3), we get f(x) = 3 i.e. constant function



Hence 10A = 45 sq. units **Ans.**

47. $$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{f\left(x\left(1+\frac{h}{x}\right)\right) - f(x)}{h}$$

$$= \lim_{h\to 0} \frac{f(x)+f\left(1+\frac{h}{x}\right)-x-\left(1+\frac{h}{x}\right)+x\left(1+\frac{h}{x}\right)-f(x)}{h}$$

$$= \lim_{h\to 0} \frac{f\left(1+\frac{h}{x}\right)-1}{h}+1-\frac{1}{x} = \lim_{h\to 0} \frac{f\left(1+\frac{h}{x}\right)-f(1)}{x\cdot\frac{h}{x}}+1-\frac{1}{x}$$

(On puting $x=y=1$ in the given functional rule, we get $f(1)=1$.)

$\therefore \quad f'(x) = \frac{1}{x}f'(1)+1-\frac{1}{x} = \frac{2}{x}+1-\frac{1}{x}$

$f'(x) = \frac{1}{x}+1 \Rightarrow f(x) = \ln x + x + c$

put $\quad x = 1, c = 0$

$\therefore \quad f(x) = \ln x + x$

at $\quad x = e,\ f(e) = 1 + e \Rightarrow f^{-1}(1+e) = e \Rightarrow g(1+e) = e$

$f'\left(\frac{1}{e}\right) = e+1 \quad \therefore \quad g\left(f'\left(\frac{1}{e}\right)\right) = g(e+1) = e$

$\left[g\left(f'\left(\frac{1}{e}\right)\right)\right] = [e] = 2$. **Ans.**

48. $F(x) = \begin{cases} f(x), & -1 < x < 1 \\ g(x), & x \in (-\infty,-1)\cup(1,\infty) \\ \frac{f(1)+g(1)}{2}, & x = 1 \\ \frac{f(-1)+g(-1)}{2}, & x = -1 \end{cases}$

continuous at $x = 1 \Rightarrow F(1^-) = F(1^+) = F(1)$

$\Rightarrow f(1^-) = g(1^+) = \frac{f(1)+g(1)}{2}$

$\Rightarrow 4 + a = 1 + b = \frac{(4+a)+(1+b)}{2} = \frac{5+a+b}{2}$.

$\therefore \quad b - a = 3. \quad$ ....(1)

continuous at $x = -1 \Rightarrow F(-1^-) = F(-1^+) = F(-1)$

$\Rightarrow b - 1 = 4 - a = \frac{(4-a)+(b-1)}{2}$

$\Rightarrow b - 1 = 4 - a = \frac{3+b-a}{2}$

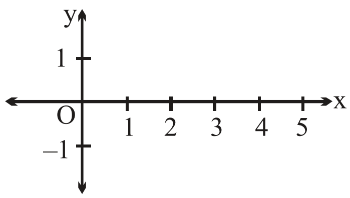
$\therefore \quad a + b = 5 \quad$ ....(2)

(1) and (2)

$\Rightarrow \quad a = 1, b = 4 \Rightarrow a^2 + b^2 = 17$. **Ans.**

49. f(0) can be taken in 2 ways $\left\langle \begin{array}{l} f(0) = 1 \\ f(0) = -1 \end{array}\right.$

$|||^{ly}$ at x = 1, 2, 3, 4 3 possibilities $\Big]$ think!



and at x = 5 only one possibilites

$\therefore$ Total number of functions = $2 \times 3^4 \times 1 = 162$. **Ans.**

50. For $-1 < x < 1$,

$$\frac{4x^2}{x^4+1} = \left\langle \begin{array}{ll} 0, \text{ when } x = 0 & \\ \dfrac{4}{x^2+\dfrac{1}{x^2}} & \text{when } x \neq 0 \end{array}\right.$$

$$\therefore \quad \frac{4x^2}{x^4+1} \leq 2^- \quad \left(\because\ x^2+\frac{1}{x^2} \geq 2\right)$$

Sign of equality holds at $x = \pm 1$, but $x \in (-1, 1) - \{0\}$

$$\therefore \quad \frac{4x^2}{x^4+1} < 2$$

$$\Rightarrow \quad \left[\frac{4x^2}{x^4+1}\right] = 1 \quad \text{as} \quad x \to 1 \text{ from LHS.}$$

Now, $\displaystyle \lim_{x\to 1^-} \frac{\ln(1-ax+a)}{e^{bx}-e^b} = \lim_{x\to 1^-} \frac{\frac{\ln\left(1+a(1-x)\right)}{a(1-x)}\, a(1-x)}{e^b\,\frac{\left(e^{b(x-1)}-1\right)}{b(x-1)}\, b(x-1)} = -\frac{a}{be^b}$

Again, $\displaystyle \lim_{x\to 1^+} \frac{(\tan bx - \tan b)(1+\cos 2bx)}{2(x-1)}$

$$= \lim_{x\to 1^+} \frac{\tan\left(b(x-1)\right)\left(1+\tan bx\ \tan b\right)\cdot 2\cos^2 bx}{2b\,(x-1)} \cdot b$$

$= (1 + \tan^2 b)(\cos^2 b)\cdot b = b$

$\because$ f(x) is continuous at x = 1

$$\therefore \quad \frac{-a}{be^b} = b = \ln 2$$

$\Rightarrow \quad b = \ln 2 \quad$ and $\quad \dfrac{-a}{be^b} = \ln 2$

$\Rightarrow \quad -a = (\ln 2)^2 \cdot 2$

$\Rightarrow \; a = -2(\ln 2)^2$

$a + 2b^2 = -2(\ln 2)^2 + 2(\ln 2)^2 = 0.$

$\therefore \quad 1 + a + 2b^2 = 1.$ **Ans.**

51. We have

$$f(x) = \begin{cases} 0, & x \in (-\infty, -1) \\ 1+x, & x \in [-1, 0] \\ 1-x, & x \in (0, 1] \\ 0, & x \in (1, \infty) \end{cases}$$

$$\Rightarrow \; f(x-1) = \begin{cases} 0, & x-1 \in (-\infty, -1) \\ 1+(x-1), & x-1 \in [-1, 0] \\ 1-(x-1), & x-1 \in (0, 1] \\ 0, & x-1 \in (1, \infty) \end{cases}$$

or $$f(x-1) = \begin{cases} 0, & x < 0 \\ x, & 0 \le x \le 1 \\ 2-x, & 1 < x \le 2 \\ 0, & x > 2 \end{cases}$$

Also, $$f(x+1) = \begin{cases} 0, & x+1 \in (-\infty, -1) \\ 1+(x+1), & x+1 \in [-1, 0] \\ 1-(x+1), & x+1 \in (0, 1] \\ 0, & x+1 \in (1, \infty) \end{cases}$$

or $$f(x+1) = \begin{cases} 0, & x < -2 \\ 2+x, & -2 \le x \le -1 \\ -x, & -1 < x \le 0 \\ 0, & 0 < x < \infty \end{cases}$$

Now, $$g(x) = f(x-1) + f(x+1) = \begin{cases} 0, & x < -2 \\ 2+x, & -2 \le x \le -1 \\ -x, & -1 < x \le 0 \\ x, & 0 < x \le 1 \\ 2-x, & 1 < x \le 2 \\ 0, & x > 2 \end{cases}$$

Clearly,
It is easy to check that $g(x)$ is continuous $\forall\, x \in R$ and non-differentiable at $x = -2, -1, 0, 1, 2$
and differentiable elsewhere.
Hence, number of points of non-differentiability of $g(x)$ are 5. **Ans.**

52. L.H.L. at $x=-1$

$$\lim_{h\to 0}\tan^{-1}\left|\frac{-1-h-1}{-1-h+1}\right| = \lim_{h\to 0}\tan^{-1}\left|\frac{-2-h}{-h}\right| = \lim_{h\to 0}\tan^{-1}\left(1+\frac{2}{h}\right) = \frac{\pi}{2}$$

R.H.L. at $x=-1$

$$\lim_{h\to 0}\tan^{-1}\left|\frac{-1+h-1}{-1+h+1}\right| = \lim_{h\to 0}\tan^{-1}\left|\frac{-2+h}{h}\right| = \lim_{h\to 0}\tan^{-1}\left(\frac{2}{h}-1\right) = \frac{\pi}{2}$$

$\therefore$ L.H.L = R.H.L. $\neq f(-1)$

i.e. $\lim_{x\to -1} f(x) \neq f(-1)$ **Ans.**

53 Differentiability at $x=0$

$$f'(0^+) = \lim_{h\to 0}\frac{(h+2)-2}{h} = 1$$

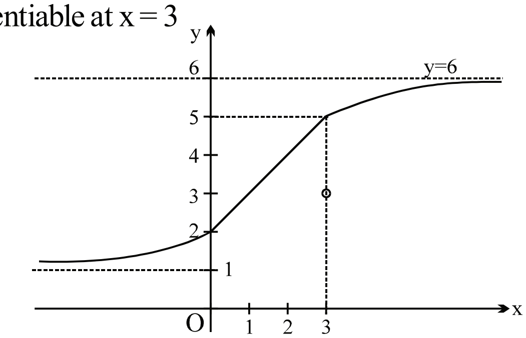
$$f'(0^-) = \lim_{h\to 0}\frac{1+e^{-h}-2}{-h} = \lim_{h\to 0}\frac{e^{-h}-1}{-h} = 1$$

Hence f is differentiable hence continuous at $x=0$.
Differentiability at $x=3$

$$f'(3^+) = \lim_{h\to 0}\frac{\frac{6(3+h)-3}{3+h}-5}{h} = \lim_{h\to 0}\frac{6(3+h)-3-15-5h}{h(3+h)} = \frac{1}{3}$$

$$f'(3^-) = \lim_{h\to 0}\frac{(3-h)+2-5}{-h} = 1$$

$\Rightarrow$ f is continuous but not differentiable at $x=3$
Graph shown below :



54. $f(x^n) = 2f(x^{n-1})$

$f'(x^n)\cdot n\,x^{n-1} = 2\,f'(x^{n-1})\cdot(n-1)\,x^{n-2}$

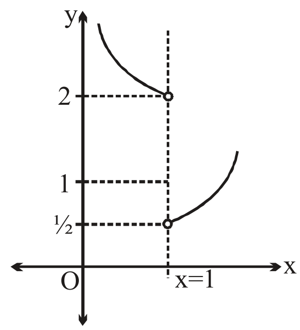
Put $x=2$.

$f'(2^n)\cdot n\cdot 2^{n-1} = 2\cdot f'(2^{n-1})\cdot(n-1)\,2^{n-2}$

$\Rightarrow \quad f'(2^n) = \frac{(n-1)}{(n)}\,f'(2^{n-1}) \qquad \Rightarrow f'(2^n) = \frac{(n-1)}{n}\times\frac{(n-2)}{(n-1)}\,\ldots\ldots\ldots\,\frac{1}{2}\cdot f'(2)$

$\Rightarrow \quad f'(2^n) = \frac{f'(2)}{n}$. **Ans.**

55. $\underset{x \to 0}{\text{Lim}}\ f(\cos^3 x + \sin^2 x)$



$x \to 0\ \ \cos^3 x - \cos^2 x < 0$

$\therefore \qquad 1 + \cos^3 x - \cos^2 x < 1$

$\therefore \qquad \underset{x \to 0}{\text{Lim}}\ f(1 + \cos^3 x - \sin^2 x) = 2$. **Ans.**]

56. $\underset{h\to 0}{\text{Lim}}\ f(0-h) = \underset{h\to 0}{\text{Lim}}\ \dfrac{(-h)e^{[-h]+|-h|} - \pi}{[-h]+|-h|} = \underset{h\to 0}{\text{Lim}}\ \dfrac{(-h)e^{-1+h} - \pi}{-1+h} = \pi$

But we are given $f(0) = \dfrac{22}{7} \Rightarrow f(x)$ is discontinuous at $x = 0 \Rightarrow f(x)$ is non–differentiable at $x = 0$.]

57. $\underset{h\to 0}{\text{Lim}}\left(\dfrac{f(3+\sin^3 h) - f(3-\sin^2 h)}{h^2}\right) = \underset{h\to 0}{\text{Lim}}\left(\dfrac{f(3+\sin^3 h) - f(3-\sin^2 h)}{\sin^3 h + \sin^2 h}\right)\cdot\left(\dfrac{\sin^3 h + \sin^2 h}{h^2}\right)$

$= f'(3)\cdot 1 = 2$. **Ans.**

58. $f(x + y) = f(x) + f(y)$ ....(1)

$f'(0) = 1$

put $x = 0$, $y = 0$ in (1)

$f(0) = 0$

consider $f'(x) = \underset{h\to 0}{\text{Lim}}\ \dfrac{f(x+h) - f(x)}{h}$

or $\qquad f'(x) = \underset{h\to 0}{\text{Lim}}\ \dfrac{f(x) + f(h) - f(x)}{h} \quad = f'(0) = 1$

$\therefore \qquad f(x) + x + c \qquad\qquad \because f(0) = 0 \qquad\qquad \therefore c = 0$

$\therefore \qquad f(x) = x$

$\therefore\ l = \underset{x\to 0}{\text{Lim}}\ \dfrac{2^{\tan x} - 2^{\sin x}}{x^2\ \dfrac{\sin x}{x}\cdot x}$

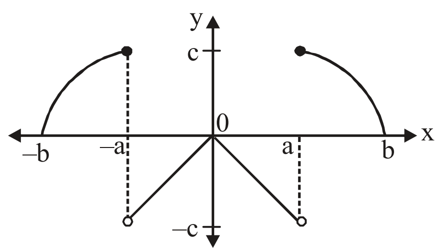
$\underset{x\to 0}{\text{Lim}}\ \dfrac{2^{\tan x} - 2^{\sin x}}{x^3}$

$\underset{x\to 0}{\text{Lim}}\ 2^{\sin x} \Rightarrow \underset{x\to 0}{\text{Lim}}\ \dfrac{2^{\tan x - \sin x} - 1}{x^3}$

$$\lim_{x\to 0}\frac{2^{\tan x-\sin x}-1}{\tan x-\sin x}\times\lim_{x\to 0}\frac{\tan x-\sin x}{x^3}=\ln 2\cdot\frac{\tan x}{x}\left[\frac{1-\cos x}{x^2}\right]$$

$$\ln 2\times\frac{1}{2}\qquad\Rightarrow\qquad\frac{1}{2}\ln 2.\ \textbf{Ans.}$$

59. From the 1st graph if $x\in(-a,a)$ then no value of y and hence $f(x)$ must be negative in $(-a,a)$.
Also from the above two graph if $x\in[-b,-a]\cup[a,b]$ then $f(x)$ is positive
and hence from the given graphs, graph of f(x) is as follows.



Clearly, f(x) is discontinuous at 2 points and non-differentiable at 3 points

60. $$k=\lim_{x\to 0^+}\left(1+(\sin A)^{\frac{1}{x}}\right)^{(\operatorname{cosec} B\sin C)^{\frac{1}{x}}}\qquad(1^\infty)$$

$(\because C>B\Rightarrow \sin C>\sin B\Rightarrow \sin C\operatorname{cosec} B>1)$

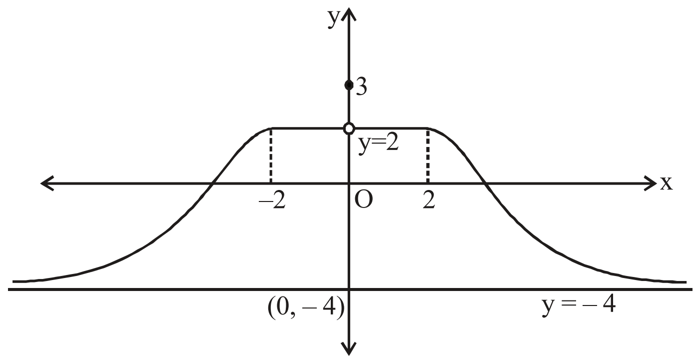
$$=e^{\lim_{x\to 0^+}(\operatorname{cosec}B\sin C)^{\frac{1}{x}}(\sin A)^{\frac{1}{x}}}\qquad=e^{\lim_{x\to 0^+}\left(\frac{\sin C\sin A}{\sin B}\right)^{\frac{1}{x}}}\qquad=e^0=1.$$

$$\boxed{\begin{array}{l}\because A<B<C<\dfrac{\pi}{2}\\ \sin A<\sin B<\sin C<1\\ \therefore 0<\dfrac{\sin A}{\sin B}<1\ \text{ and }\ 0<\sin C<1\\ \Rightarrow\ 0<\dfrac{\sin A\sin C}{\sin B}<1\end{array}}\ .\ \textbf{Ans.}$$

61. We have $H(x)=\left[\dfrac{3(x^2+1)-|x|}{x^2+1}\right]=\left[3-\dfrac{|x|}{x^2+1}\right]\left(\text{As}\quad 0<\dfrac{|x|}{x^2+1}=\dfrac{1}{|x|+\dfrac{1}{|x|}}\le\dfrac{1}{2}\right)$

$$\text{So,}\ H(x)=\begin{cases}\dfrac{8}{\pi}\tan^{-1}(-|x|+3), & x\in(-\infty,-2)\cup(2,\infty)\\ 3, & x=0\\ 2, & x\in[-2,2]-\{0\}\end{cases}$$

Also, H(x) is an even function on R, so graph of H(x) is symmetrical about y-axis.



Graph of H(x)

Note : $H(x) = \dfrac{8}{\pi}\tan^{-1}(3 - |x|), |x| > 2$

Also for $|x| > 2,\ 3 - |x| \in (-\infty, 1)$, so

$$H(x) \in \frac{8}{\pi}\left(\frac{-\pi}{2}, \frac{\pi}{4}\right) = (-4,\ 2)$$

From the above graph, it is clear that H(x) is discontinuous at only one point i.e. $x = 0$.]\

62. $x = 1 + h$

$$\underset{h\to 0}{\mathrm{Lt}}\ \frac{\pi^2}{e}(-h)\frac{\left((1+h)^{\frac{1}{h}} - e\right)}{(1+\cos(\pi+\pi h))}$$

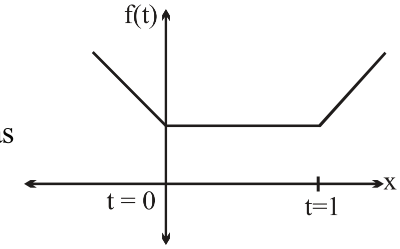
$$= \underset{h\to 0}{\mathrm{Lt}} -\frac{\pi^2}{e}h\frac{\left((1+h)^{\frac{1}{h}} - e\right)}{(1-\cos\pi h)} \qquad = \underset{h\to 0}{\mathrm{Lt}} -\frac{\pi^2}{e}h\frac{\left(e^{\frac{1}{h}\ln(1+h)} - e\right)}{\left(2\sin^2\left(\frac{\pi h}{2}\right)\right)}$$

$$= \underset{h\to 0}{\mathrm{Lt}} -\frac{\pi^2}{e}h\frac{\left(e^{\frac{1}{h}\left(h-\frac{h^2}{2!}+\frac{h^2}{3!}+\ldots\right)} - e\right)}{\dfrac{2\sin^2\left(\frac{\pi h}{2}\right)}{\left(\frac{\pi h}{2}\right)^2}\dfrac{\pi^2h^2}{4}} \quad \Rightarrow \quad = \underset{h\to 0}{\mathrm{Lt}} -\frac{2}{e}\frac{e}{h}\left(e^{-\frac{h}{2!}+\frac{h^2}{3!}+\ldots} - 1\right)$$

= 1

63 Given $f(t) = \begin{cases} 1-2t, & t<0 \\ 1, & 0 \le t \le 1 \\ 2t-1, & t>1 \end{cases} \Rightarrow g(x) = \begin{cases} f(x-1), & 0 \le x \le 1 \\ 3-x, & 1 < x \le 2 \end{cases}$

$\Rightarrow \quad g(x) = \begin{cases} 3-2x, & 0 \le x \le 1 \\ 3-x, & 1 < x \le 2 \end{cases}$

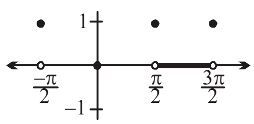


$\therefore$ The function $g(x)$ is discontinuous at $x = 1$, as $\lim\limits_{x \to 1} g(x)$ does not exist.

So, g(x) is non-derivable at x = 1. **Ans.**

64. Discontinuous at all odd integral multiple of $\frac{\pi}{2}$

(Isolated point removable)



65. Since, f(x) is derivable in (– 2, 2)
Derivability of f(x) at x = – 1 is possible only when $b - 2 = 0 \Rightarrow b = 2$
$\{\because\ |x+1|$ is non-derivable at x = – 1.$\}$
Similarly derivability of f(x) at x = 1 is possible only when $c + 2 = 0 \Rightarrow c = -2$.
Now, f(x) becomes

$$f(x) = \begin{cases} \ln(1-x) + a\tan^{-1}x + 2, & -2 < x < 0 \\ 2, & x = 0 \\ \dfrac{e^{2x} - e^{\ln\left(4\left\{\frac{x}{2}\right\}\right)} - 1}{x^2}, & 0 < x < 2 \end{cases}$$

$$f(x) = \begin{cases} \ln(1-x) + a\tan^{-1}x + 2, & -2 < x < 0 \\ 2, & x = 0 \\ \dfrac{e^{2x} - 2x - 1}{x^2}, & 0 < x < 2 \quad \left(\because 0 < \dfrac{x}{2} < 1 \Rightarrow \left\{\dfrac{x}{2}\right\} = \dfrac{x}{2}\right) \end{cases}$$

$$= \lim_{x \to 0^+} \frac{e^{2x} - 2x - 1}{4x^2} \cdot 4 = \frac{1}{2} \times 4 = 2$$

$$\therefore \quad \lim_{x \to 0^-} \ln(1-x) + a\tan^{-1}x + 2 = 2$$

$$f'(0^-) = \lim_{h \to 0} \frac{ln(1+h) + a\tan^{-1}(-h) + 2 - 2}{-h}$$

$$\lim_{h \to 0} \frac{ln(1+h)}{-h} + \frac{a\tan^{-1}(-h)}{-h} = -1 + a$$

$$f'(0^+) = \lim_{h \to 0} \frac{\frac{e^{2h} - 2h - 1}{h^2} - 2}{h} = \lim_{h \to 0} \frac{e^{2h} - 2h - 1 - 2h^2}{h^3} = \frac{4}{3}$$

$\therefore$ $a - 1 = \frac{4}{3} \Rightarrow a = \frac{7}{3}$

$\Rightarrow$ $9a^2 + b^2 + c^2 = 9 \cdot \frac{49}{9} + 4 + 4 = 57$ **Ans.**

Paragraph for question nos. 66 to 68

Sol. Consider $f(x)$ in $[0, 1]$

$f'(t) = 3t^2 - 2t + 1 > 0 \ \forall\ t \in (0, 1)$ $\Rightarrow$ f is $\uparrow$ in $(0, 1)$

maximum occurs at $t = x$.

and hence $f(x)$ is $\begin{cases} x^3 - x^2 + x + 1 & 0 \le x \le 1 \\ 3 - x & 1 < x \le 2 \end{cases}$

again consider $g(x) =$ in $[0, 1)$

$$g(t) = \frac{3}{8}t^4 + \frac{1}{2}t^3 - \frac{3}{2}t^2 + 1$$

$$g'(t) = \frac{3}{2}t^3 + \frac{3}{2}t^2 - 3t = \frac{3}{2}t(t^2 + t - 2) = \frac{3}{2}t(t-1))(t+2)$$

$g(t)$ decreases in $[0, 1)$

$\Rightarrow$ maximum occurs when $t = 0$ and $g(0) = 1$

again consider $g(x)$ function in $[1, 2]$

$$g(t) = \frac{3}{8}t + \frac{1}{32}\sin^2 \pi t + \frac{5}{8}$$

$$g'(t) = \frac{3}{8} + \frac{\pi}{32}\sin(2\pi t) > 0 \ \forall\ t \in R$$

$\therefore$ g is an increasing function in $[1, 2]$

$\Rightarrow$ minimum occurs when $t = 1$ and $g(1) = 1$

hence $g(x) = \begin{cases} 1 & 0 \le x < 1 \\ 1 & 1 < x \le 2 \end{cases} = 1 \ \forall\ x \in [0, 2]$

66. $f(x)$ is continuous but not differentiable at $x = 1$ Ans.

67. $\lim_{x \to 1^-} fog(x) = f(1)$ and $\lim_{x \to 1^+} gof(x) = 1$ also $f(1) > 1$ Ans.

68. $Z(x) = \dfrac{d}{dx} f(x)^{g(x)} = \dfrac{d}{dx} f(x)^1 \ \forall \ x \in [0, 1) \cup (1, 2]$

$= f'(x) \ \forall \ x \in [0, 1) \cup (1, 2]$

& $Y(x) = \dfrac{d}{dx} g(x)^{f(x)} = \dfrac{d}{dx} 1^{f(x)} = 0 \quad \forall \ x \in [0, 1) \cup (1, 2]$

Hence the functions Y(x) and Z(x) can vanish simultaneously at $f'(x) = 0$ which is not possible for any real x. Ans.

69. R.H.L. $\lim\limits_{x \to \frac{\pi}{4}^+} \dfrac{e^{[3\cot x - 1]} + \cos\left(2x - \frac{\pi}{2}\right) - a + [\cos x - \sin x]}{\left(x - \frac{\pi}{4}\right)^2}$

$\cot x < 1$ in $x \in \dfrac{\pi}{4}^+ \Rightarrow e^{[3\cot x - 1]} = e$ in $x \in \dfrac{\pi}{4}^+$

Also $\cos x < \sin x$ in $x \in \dfrac{\pi}{4}^+ \Rightarrow [\cos x - \sin x] = -1$ in $x \in \dfrac{\pi}{4}^+$.

Also $\cos\left(2x - \dfrac{\pi}{2}\right) = 2\cos^2\left(x - \dfrac{\pi}{4}\right) - 1$

$= 2\left(\cos\left(x - \dfrac{\pi}{4}\right) - 1\right)\left(\cos\left(x - \dfrac{\pi}{4}\right) + 1\right) + 1$

$\therefore \quad \lim\limits_{x \to \frac{\pi}{4}^+} \dfrac{-a + e - 2\left(1 - \cos\left(x - \frac{\pi}{4}\right)\right)\left(1 + \cos\left(x - \frac{\pi}{4}\right)\right)}{\left(x - \frac{\pi}{4}\right)^2}$

If limit exists a = e & R.H.L. is –2. 1/2.2 = – 2

for L.H.L. Let $y = \dfrac{1 + \sin 2x + \sin^2 2x}{1 + \sin 2x}$

$y = 1 + \dfrac{\sin^2 2x}{1 + \sin 2x} = 1 + \dfrac{1}{\frac{1}{\sin 2x} + \frac{1}{\sin 2x}} < 1 + \dfrac{1}{2} = \dfrac{3}{2}$

$\dfrac{1}{y} > \dfrac{2}{3}$.

$\lim\limits_{x \to -\frac{\pi}{4}^-} \left[\dfrac{2}{3} - \dfrac{1 + \sin 2x}{1 + \sin x + \sin^2 2x}\right] = -1$

$\therefore \quad -b = -2 \Rightarrow b = 2$.

Also, $f\left(\frac{\pi}{4}\right) = P\,([p]-2)-2 = -2.$

$$P([P]-2) = 0$$

$\Rightarrow \quad p = 0, 2 \Rightarrow n = 2$

$\therefore \quad [a+b+n] = [e+2+2] = [e+4] = 6.$ **Ans.**

70 $f(x) = \begin{cases} ax^3 + b, & 0 < x \le 1 \\ 2\cos\pi + \tan^{-1} x, & 1 < x \le 2 \end{cases} \Rightarrow a + b = -2 + \frac{\pi}{4}$

$$f'(x) = \begin{cases} 3ax^2, & 0 < x \le 1 \\ -2\pi \sin \pi x + \dfrac{1}{1+x^2}, & 1 < x \le 2 \end{cases}$$

$3a = \frac{1}{2} \quad \Rightarrow a = \frac{1}{6}$

$b = \frac{\pi}{4} - 2 - \frac{1}{6} = \frac{\pi}{4} - \frac{13}{6} = \frac{\pi}{4} - \frac{26}{12}$

$\Rightarrow \; k_1 = 6, \;\; k_2 = 12 \quad \therefore \; k_2 - k_1 = 6.$ **Ans.**

Paragraph for question nos 71 to 73

71. $\lim\limits_{n\to\infty} n\left(f\left(c+\frac{1}{n}\right) - f(c)\right) = a = \lim\limits_{n\to\infty} \dfrac{\left(f\left(c+\frac{1}{n}\right)\right) - f(c)}{\frac{1}{n}}$

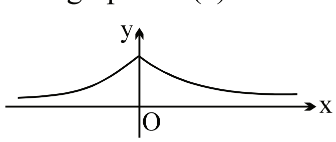
$= \lim\limits_{h\to 0} \dfrac{f(c+h) - f(c)}{h} = a \quad (\text{where } \frac{1}{n} = h)$

$h \to 0$ from RHS, hence it is just RHD. Hence nothing can be said about LHD

$\therefore \quad f'(c)$ may or may not exist. **Ans.**

72. As f is continuous on R and $\lim\limits_{x\to\infty} f(x) = \lim\limits_{x\to-\infty} f(x) = 0$ it clearly implies f is bounded because if its tends to ∞ or $-\infty$ at any point $x = a$ then it will become discontinuous at $x = a$.

Also graph of f (x) can be of the form



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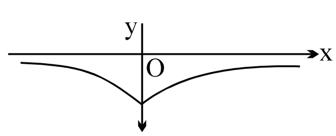

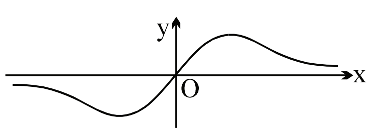

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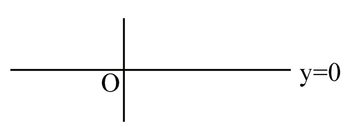


in all above cases f attains either a maximum or minimum or both.

73. $\lim_{x\to 0} x^{+a}\sin(x^{-c}) = 0$ if $a > 0$

hence if $a > 0$, $\lim_{x\to 0} x^{+a}\sin(x^{-c}) = f(0)$ hence f is continuous at $x = 0$ and clearly on remaining real line f is continuous. Hence f is continuous if $a > 0$ ....(1)

now $f'(0^+) = \lim_{h\to 0}\frac{h^a \sin(h^{-c})}{h} = \lim_{h\to 0} h^{a-1}\sin(h^{-c})$ which exist if $a - 1 > 0$ i.e. $a > 1$ ....(2)

now $f'(x) = \begin{cases} 0 & x = 0 \\ a\,x^{a-1}\sin(x^{-c}) + x^a\cos(x^{-c})(cx^{-c-1}) & x \neq 0 \end{cases}$

clearly f' is continuous in (0, 1] but at x = 0

$\lim_{h\to 0} f'(0+h) = \lim_{h\to 0} ah^{a-1}\sin(h^{-c}) + h^a\cos(h^{-c})(-ch^{-c-1})$

$= \lim_{h\to 0} ah^{a-1}\sin(h^{-c}) + h^{a-c-1}(-c)\cos(h^{-c})$

which exist if $a - c - 1 > 0 \quad\Rightarrow\quad a > 1 + c$ ....(3)
by intersection of conditions (1), (2) and (3).

74 f(x), g(x) and h(x) are continuous at x = 1

$\therefore \quad \lim_{n\to\infty} f\left(\frac{1}{n}\operatorname{cosec}\frac{1}{n}\right) = f(1) = 0$

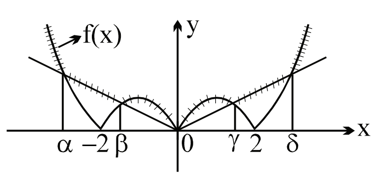
$\lim_{n\to\infty} g\left(n\tan\frac{1}{n}\right) = g(1) = 0$

and $\quad \lim_{n\to\infty} h\left(2n^2\left(1-\cos\frac{1}{n}\right)\right) = h(1) = 0$

$\therefore \quad \lim_{n\to\infty} F(x) = f(1) + g(1) + h(1) = 0$

Also as $n \to \infty$, the graph of y = F (x) approaches to be a straight line
$\therefore$ for y = F (x) to be differentiable at x = 1.
F ' (1) must be zero.

75. $f(x)$ is non differentiable at $x = \alpha, \beta, 0, \gamma, \delta$ i.e. 5 points



and $\quad g(x)$ is non differentialbe at $x = \alpha, \beta, 0, -2, 2$ i.e. 7 points
$\Rightarrow \quad L + M = 5 + 7 = 12$

76 In $\left[0, \frac{\pi}{2}\right]$, $0 \le x^2 \le \frac{\pi^2}{4} \approx 2.4674$

and $0 \le x^3 \le \frac{\pi^3}{8} \approx 3.8758$

$\therefore$ $[x^2]$ will be discontinuous at x = 1 and 2 and $[x^3]$ will be discontinuous hence not differentiable at x = 1, 2 and 3.

$\therefore$ p = 2 and q = 3 $\therefore$ p + q = 5 and $p^2 + q^2 = 13$ Ans.

77. $(xy) = f(x) + f(y) + xy - x - y - 1$

$x = y = 1$

$f(1) = 2f(1) + 1 - 1 - 1 - 1 \quad \Rightarrow \quad f(1) = 2$

$$\lim_{h \to 0} \frac{2 - f(1-h)}{h} = 2$$

$$\Rightarrow \quad \lim_{h \to 0} \frac{f(1-h) - f(1)}{-h} = 2$$

$$\Rightarrow \quad f'(1) = 2$$

Now, $f'(x) = \lim\limits_{h \to 0} \dfrac{f(x+h) - f(x)}{h} = \lim\limits_{h \to 0} \dfrac{f\left(x\left(1+\frac{h}{x}\right)\right) - f(x)}{h}$

$$= \lim_{h \to 0} \frac{f(x) + f\left(1+\frac{h}{x}\right) + x\left(1+\frac{h}{x}\right) - x - \left(1+\frac{h}{x}\right) - 1 - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{f\left(1+\frac{h}{x}\right) - 2}{x\frac{h}{x}} + 1 - \frac{1}{x} = \frac{1}{x} \,.\, f'(1) + 1 - \frac{1}{x}$$

$$= \frac{2}{x} + 1 - \frac{1}{x} = \frac{1}{x} + 1$$

$f(x) = \ln x + x + c$

putting x = 1

$2 = 0 + 1 + c \Rightarrow c = 1$

$\therefore \quad f(x) = \ln x + x + 1$

$$L = \lim_{x \to \infty} \frac{2 + \frac{2}{x} - f\left(1+\frac{1}{x}\right)}{1 - \sqrt{\cos\frac{1}{3x}}} = \lim_{t \to 0} \frac{2 + 2t - f(1+t)}{1 - \sqrt{\cos\frac{t}{3}}} \quad \left(\text{where } x = \frac{1}{t}\right)$$

$$= \lim_{t \to 0} \frac{2+2t-\{ln(1+t)+1+t+1\}}{1-\cos\frac{t}{3}} \left(1+\sqrt{\cos\frac{t}{3}}\right)$$

$$= \lim_{t \to 0} \frac{t-\left(t-\frac{t^2}{2}+\frac{t^3}{3}........\right)}{\frac{1-\cos\frac{t}{3}}{\frac{t^2}{9}}\cdot\frac{t^2}{9}} \cdot \left(1+\sqrt{\cos\frac{t}{3}}\right)$$

$$= \frac{1}{2} \times \frac{9\times 2}{1/2} = 18$$

78. Put $x = x + h$

$y = x$

$\therefore$ $|f(x+h) - f(x)| \le |h|^3$

$$\lim_{h \to 0}\left|\frac{f(x+h)-f(x)}{h}\right| \le \lim_{h \to 0} h^2$$

$\therefore$ $|f'(x)| \le 0$

$\Rightarrow$ $|f'(x)| = 0 \Rightarrow f(x) = \text{constant}$

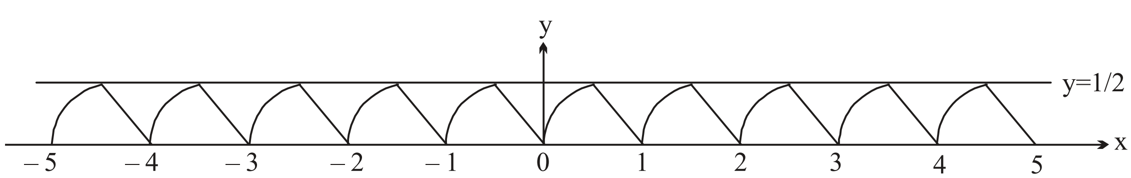
$\therefore$ $f(x) = 100$

Hence $f(20) = 100$. **Ans**

79

$$f(x) = \begin{cases} \sqrt{\frac{1}{4}-\left\{x-\left(n+\frac{1}{2}\right)\right\}^2} & n \le x < n+\frac{1}{2} \\ n+1-x & n+\frac{1}{2} \le x < n+1 \end{cases}$$

$\therefore$ Graph of f(x) will be



Clearly f(x) is continuous in R but not differentiable at $x = n + \frac{1}{2}$ and n $\forall$ $n \in I$

$\therefore$ not differentiable at 19 points in (– 5, 5).

Also f(x) is periodic with fundamental period 1.

$\Rightarrow$ 19 points **Ans.**

80. $f(x) = \dfrac{\sqrt{x^2 + kx + 1}}{x^2 - k}$

for f to be continuous $\forall\ x \in R$

$x^2 + kx + 1 \geq 0$ and $x^2 - k$ must not have any root i.e. $k < 0$

$\therefore \quad k^2 - 4 \leq 0 \quad$ and $\quad k < 0 \quad ....(1)$

$\Rightarrow \quad k \in [-2, 2] \quad ....(2)$

from (1) and (2)

$k \in [-2, 0)$ Ans.

81. $f(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x \in (0, \pi/2] \\ -1 & \text{if } x \in (\pi/2, 3\pi/2) \\ 0 & \text{if } x \in (3\pi/2, 2\pi) \\ 1 & \text{if } x = 2\pi \end{cases}$

clearly function is discontinuous at $\quad x = 0,\ \pi/2,\ 3\pi/2,\ 2\pi$

82. continuity

$\Rightarrow \quad f(1) = f(2) = ...f(100) =$ a fixed irrational value

if (A) is correct then

$f(1) + f(2) + ... + f(100) = f(1) + f(2) + ... + f(99) \Rightarrow f(100) = 0$

which is not possible as $f(100)$ must be irrational

if (B) is correct then

$f(1) + f(2) + ... + f(100) = f(2) + f(4) + ... + f(100) + f(102) + ... + f(200)$

$\therefore \quad \underbrace{f(1) + f(3) + ... + f(99)}_{50 \text{ terms}} = \underbrace{f(102) + f(104) + ... + f(200)}_{50 \text{ terms}}$ equality holds

83. f is defined as follows in $[-1, 2]$

$$f(x) = \begin{cases} 0 & \text{if } x = 2 \text{ or } x = -1 \\ (2x-3)(x-1) & \text{if } \frac{3}{2} \leq x < 2 \\ (3-2x)(x-1) & \text{if } 1 \leq x < \frac{3}{2} \\ \cos\left(\frac{\pi}{2}(x - x + 0)\right) = 1 & \text{if } 0 \leq x < 1 \\ \cos(2x+1)\frac{\pi}{2} & \text{if } -1 < x < 0 \end{cases}$$

84. Plot the graph of $y = |x| - 1$.

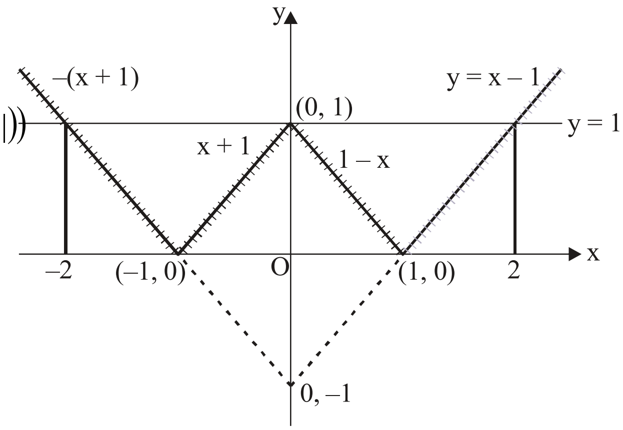
Now graph $y = \big|\,|x| - 1\,\big|$

Now for domain of the function $\sin^{-1}(f(|x|))$

$0 \le f(|x|) \le 1$

$\therefore\ x \in [-2, 2]$

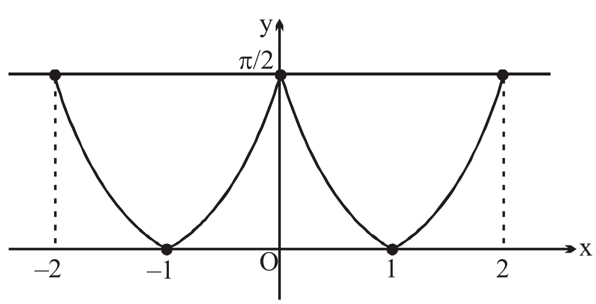
Now graph of $\sin^{-1}(g(x))$



Graph of y = g(x)

$$y = \sin^{-1}(g(x)) = \begin{cases} \sin^{-1}(x-1) \text{ in } [1, 2] \\ \sin^{-1}(1-x) \text{ in } [0, 1] \\ \sin^{-1}(x+1) \text{ in } [-1, 0] \\ \sin^{-1}(x+1) \text{ in } [-2, -1] \end{cases}$$

Graph of $y = \sin^{-1}(g(x))$



Now derivable at $\{-2, -1, 0, 1, 2\}$ vertical tangent at -2 and 2

85. $$l = \lim_{n\to\infty}\left\{\frac{f(a+1/n)}{f(a)}\right\}^n \qquad (1^\infty \text{ form})$$

$$l = (\exp.)\left(\lim_{n\to\infty} n\left\{\frac{f(a+1/n)-f(a)}{f(a)}\right\}\right); \qquad \text{put } n = \frac{1}{h}$$

$$= (\exp.)\left(\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}\cdot\frac{1}{f(a)}\right) = (\exp.)\left(\frac{f'(a)}{f(a)}\right) = e^{\frac{f'(a)}{f(a)}} \text{ Ans.}$$

86 $$f'(x^2) = \frac{1}{x} \quad\Rightarrow\quad f'(x) = \frac{1}{\sqrt{x}},\quad x > 0$$

$\Rightarrow\quad f(x) = 2\sqrt{x} + c \qquad$ (c = integration constant)

$\because\quad f(1) = 1 \quad\Rightarrow\quad c = -1$

$\therefore\quad f(x) = 2\sqrt{x} - 1,\ x > 0$

and $\quad g'(\sin^2 x - 1) = \cos^2 x + p\ \forall\ x \in R$

$\Rightarrow\quad g'(-\cos^2 x) = \cos^2 x + p$

$\Rightarrow\quad g'(x) = p - x,\ \forall\ x \in [-1, 0]$

$\Rightarrow \quad g(x) = px - \frac{x^2}{2} + k \quad$ (where k = integration constant)

$\because \quad g(-1) = 0 \Rightarrow 0 = -p - \frac{1}{2} + k \Rightarrow k = \frac{1}{2} + p$

$\therefore \quad g(x) = px - \frac{x^2}{2} + \frac{1}{2} + p$

$$\therefore \quad h(x) = \begin{cases} 2\sqrt{x} - 1 & , \ x > 0 \\ px - \frac{x^2}{2} + \frac{1}{2} + p & , -1 \le x \le 0 \end{cases}$$

$\therefore$ At $x = 0$

L.H.L = R.H.L = f(0)

$\Rightarrow -1 = \frac{1}{2} + p \Rightarrow p = \frac{-3}{2}$

Hence $2p = -3$

$\therefore$ Absolute value of $2p$ is 3

87. $$l = \lim_{x \to 0} \frac{(e^{2x}+1) - (e^x + e^{-x}) - x(e^x + e^{-x})}{x^2\left(\frac{e^x - 1}{x}\right)}$$

multiply N^r and D^r by e^x

$$\lim_{x \to 0} \frac{e^x(e^{2x}+1) - (e^{2x}+1) - x(e^{2x}+1)}{x^2 \cdot e^x}$$

$$= \lim_{x \to 0} (e^{2x}+1)\left[\frac{(e^x - 1) - x}{x^2}\right] = 2 \cdot \frac{1}{2} = 1 \text{ ans.}$$

88. $f(0) = k = \lim_{x \to 0} f(x)$

$$= \lim_{x \to 0} \frac{\tan\left(\frac{\pi}{4} + 2x\right) - 2\tan\left(\frac{\pi}{4} + x\right) + \tan\frac{\pi}{4}}{\sin\left(\frac{\pi}{4} + 2x\right) - 2\sin\left(\frac{\pi}{4} + x\right) + \sin\frac{\pi}{4}}$$

$$= \lim_{x \to 0} \frac{\left(\tan\left(\frac{\pi}{4} + 2x\right) - \tan\left(\frac{\pi}{4} + x\right)\right) + \left(\tan\frac{\pi}{4} - \tan\left(\frac{\pi}{4} + x\right)\right)}{\left(\sin\left(\frac{\pi}{4} + 2x\right) + \sin\frac{\pi}{4}\right) - 2\sin\left(\frac{\pi}{4} + x\right)}$$

$$= \lim_{x\to 0} \frac{\tan x\left(1+\tan\left(\frac{\pi}{4}+2x\right)\tan\left(\frac{\pi}{4}+x\right)\right)-\tan x\left(1+\tan\left(\frac{\pi}{4}+x\right)\tan\frac{\pi}{4}\right)}{2\sin\left(\frac{\pi}{4}+x\right)\cos x-2\sin\left(\frac{\pi}{4}+x\right)}$$

$$= \lim_{x\to 0} \frac{\tan x\ \tan\left(\frac{\pi}{4}+x\right)\left(\tan\left(\frac{\pi}{4}+2x\right)-\tan\frac{\pi}{4}\right)}{-2\sin\left(\frac{\pi}{4}+x\right)\left(\frac{1-\cos x}{x^2}\right)x^2}$$

$$= \lim_{x\to 0} \frac{\tan x\tan\left(\frac{\pi}{4}+x\right)\tan 2x\left(1+\tan\left(\frac{\pi}{4}+2x\right)\tan\frac{\pi}{4}\right)}{-2\sin\left(\frac{\pi}{4}+x\right)\left(\frac{1-\cos x}{x^2}\right)x^2} = -4\sqrt{2}$$

Hence $k = -4\sqrt{2}$ **Ans.**

89 $$f(x) = \lim_{n\to\infty}\left(\left(a\left(1+\left(\frac{b}{a}\right)^n\right)^{\frac{1}{n}}\sin x + \{e^x\}^n\right)\left(\left[\frac{\frac{1}{n}}{\tan^{-1}\frac{1}{n}}\right]+1\right)\right)$$

$\therefore$ $f(x) = (a \sin x + 0)(1 + 1)$

$\therefore$ $f(x) = 2a \sin x$

$H(x) = \text{sgn}(2a \sin x - 3)$ has exactly one point of discontinuity in $[0, 2\pi]$, then $2a \sin x - 3 = 0$ must have one real root in $[0, 2\pi]$, $\sin x = \frac{3}{2a}$

$\therefore$ $a = \frac{3}{2}$ only. $\Rightarrow$ $24a = 36$. **Ans.**

90. $f(x) = \sqrt{x+2\sqrt{2x-4}} + \sqrt{x-2\sqrt{2x-4}}$

Its domain is $[2, \infty)$

Let $t = \sqrt{2x-4}$

$$f(x) = \sqrt{\frac{t^2}{2}+2+2t} + \sqrt{\frac{t^2}{2}+2-2t} = \frac{1}{\sqrt{2}}(t+2+|t-2|)$$

$$= \begin{cases} 2\sqrt{2} & \text{if} \quad t < 2 \\ \sqrt{2}t & \text{if} \quad t \ge 2 \end{cases} = \begin{cases} 2\sqrt{2} & \text{if} \quad x \in [2, 4) \\ 2\sqrt{x-2} & \text{if} \quad x \in [4, \infty) \end{cases}$$

$$f'(x) = \begin{cases} 0 & \text{if} \quad x \in [2, 4) \\ \frac{1}{\sqrt{x-2}} & \text{if} \quad x \in [4, \infty) \end{cases},$$ $f(x)$ is not derivable at $x = 4$. **Ans.**

Paragraph for question nos. 91 and 92

Let $f : \left(\frac{-\pi}{2}, \frac{\pi}{2}\right) \to R$, such that $f(x)$ is differentiable at $x = 0$ and

$$f(x) = \begin{cases} \frac{1}{2}(1+\sec x)\, f\left(\frac{x}{2}\right), & x \neq 0 \\ 2, & x = 0 \end{cases}$$

Sol. $f(x) = \dfrac{(1+\cos x)}{2\cos x}\, f\left(\dfrac{x}{2}\right) = \dfrac{\cos^2 \frac{x}{2}}{\cos x}\, f\left(\dfrac{x}{2}\right)$

$$= \frac{\cos^2\left(\frac{x}{2}\right)\cdot\cos^2\left(\frac{x}{2^2}\right)\cdot\cos^2\left(\frac{x}{2^3}\right)\ldots\ldots\cos^2\left(\frac{x}{2^n}\right)}{\cos x\cdot\cos\left(\frac{x}{2}\right)\cdot\cos\left(\frac{x}{2^2}\right)\ldots\ldots\ldots\cos\left(\frac{x}{2^{n-1}}\right)}\cdot f\left(\frac{x}{2^n}\right).$$

$$= \frac{\tan x}{x}\cdot f(0)$$

$$f(x) = \begin{cases} \frac{2\tan x}{x}, & x \neq 0 \\ 2, & x = 0 \end{cases}$$

$\therefore \quad f\left(\frac{\pi}{4}\right) = \dfrac{2\times 1}{\frac{\pi}{4}} = \dfrac{8}{\pi}$

91. $f'(x) = \dfrac{2\,[x\sec^2 x - \tan x]}{x^2}, x \neq 0$

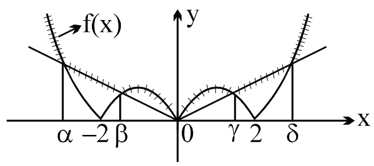
$\Rightarrow \quad f'\left(\frac{\pi}{4}\right) = \dfrac{2\left(\frac{\pi}{2}-1\right)}{\frac{\pi}{4}\cdot\frac{\pi}{4}} = \dfrac{8}{\pi}\left(\dfrac{\pi}{2}-1\right)$

$$= \left(4 - \frac{8}{\pi}\right)\frac{4}{\pi} = \left(\frac{16}{\pi} - \frac{32}{\pi}\right)$$

$\Rightarrow \quad f\left(\frac{\pi}{4}\right) + f'\left(\frac{\pi}{4}\right) = \dfrac{8\,(3\pi - 4)}{\pi^2}$

92. Limit $= \dfrac{5}{3}$. **Ans.**

93. $f(x)$ is non differentiable at $x = \alpha, \beta, 0, \gamma, \delta$ i.e. 5 points



and $g(x)$ is non differentialbe at $x = \alpha, \beta, 0, -2, 2$ i.e. 7 points $\Rightarrow L + M = 5 + 7 = 12$]

94. $$f(x) = \begin{cases} \dfrac{\left|\log_e |x|\right|}{2x} & -\pi < x < 0 \\ 0 & x = 0 \\ \dfrac{-\{\sin x\}}{e^x} & 0 < x < \pi \end{cases}$$

f(x) is discontinuous at $x = 0, \dfrac{\pi}{2}$ and non-differentiable at $x = 0, \dfrac{\pi}{2}$ and -1.

$(l + m) = 5$. **Ans.**

95. $$f(x) = \begin{cases} 1 & x \in (2I)^+ \\ -1 & x \in (2I)^- \\ -1 & x \in (2I+1)^+ \\ 1 & x \in (2I+1)^- \\ 0 & x \in I \end{cases}$$

Paragraph for question nos. 96. to 98.

According to given information, we must have f(x) a polynomial of degree 4 with leading coefficient 3.
So, $f(x) = 3(x-2)(x-3)(x+1)(x+6) + (x^2+1)$

96. $f(0) = 109$. **Ans.**

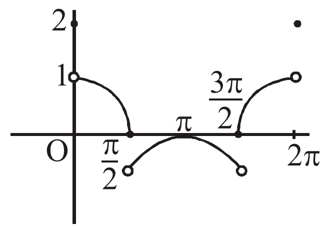
97. $$\lim_{x\to -6} \frac{f(x) - x^2 - 1}{3(x+6)} = \lim_{x\to -6} \frac{3(x-2)(x-3)(x+1)(x+6)}{3(x+6)} = (-8)(-9)(-5) = -360. \text{ Ans.}$$

98. We have $$g(x) = \frac{1}{x^2+1-f(x)} = \frac{1}{(x^2+1) - 3(x-2)(x-3)(x+1)(x+6) - (x^2+1)}$$

$$= \frac{-1}{3(x-2)(x-3)(x+1)(x+6)},\ x \ne -6, -1, 2, 3$$

But $x \in \left[\dfrac{-15}{2}, \dfrac{5}{2}\right]$, so points of discontinuous are three i.e., $x = -6, -1, 2$. **Ans.**

99. $\begin{cases} 2 & x = 0, 2\pi \\ \cos x & 0 < x \le \frac{\pi}{2} \\ -\cos x - 1 & \frac{\pi}{2} < x < \frac{3\pi}{2} \\ 0 & x = \frac{3\pi}{2} \\ \cos x & \frac{3\pi}{2} < x < 2\pi \end{cases}$



$x = 4 \quad p = 0$ become $f(x)$ is not equal to 1 or -1 in $[0, 2\pi]$

$\therefore \quad n + p = 4$

100. $f(x) = \cot^2 x - \operatorname{cosec}^2 x$

Clearly $f(x)$ is discontinuous at $x =$

π. $\quad n_1 = 1$

$g(x) = \frac{\pi}{2} + \sin^{-1} x$

$\therefore$ $g(x)$ is differentiable in $(-1,$

$\therefore$ $1)$

$n_2 = 0$

$h(x) = 5 + [\cos x] = \begin{cases} 6\ ; & x = 0 \\ 5; & 0 < x \le \frac{\pi}{2} \\ +4\ ; & \frac{\pi}{2} < x < \frac{3\pi}{2} \\ 5\ ; & \frac{3\pi}{2} \le x < 2\pi \end{cases}$

$\Rightarrow$ h(x) is non-derivable at $x = 0, \frac{\pi}{2}, \frac{3\pi}{2}$

$\therefore n_3 = 3$

Hence $(n_1 + n_2 + n_3) = 1 + 0 + 3 = 4$ **Ans.**

Paragraph for questions nos. 101 & 102

Sol. Clearly $f\left(\frac{\pi^-}{4}\right) = c;\ f\left(\frac{\pi^+}{4}\right) = \frac{1}{c}$

101 As $\lim\limits_{x \to \frac{\pi}{4}} f(x)$ exists, so $c = \frac{1}{c} \Rightarrow c = \pm 1$ **Ans.**

102. As $f(x)$ is continuous at $x = \frac{\pi}{4}$, so $\lim\limits_{x \to \frac{\pi}{4}} f(x) = f\left(\frac{\pi}{4}\right)$

$\Rightarrow \quad c = \frac{1}{c} = 1 \quad \Rightarrow \quad c = 1$ **Ans.**

103. Consider continuous function g as

$g(x) = f(x+1) - f(x) \qquad (x_2 = x_1 + 1)$

now, $g(0) = f(1) - f(0) = f(1) - f(2) \qquad ....(1)$

$g(1) = f(2) - f(1) = f(2) - f(1) \qquad ....(2)$

hence g (0) and g (1) are of opposite signs, hence $\exists$ some $c \in (0, 1)$ where $g(c) = 0$

i.e. $\quad f(c+1) = f(c) \qquad [c + 1 \in (1, 2)$ as $c \in (0, 1)$]

put $\quad c = x_1$; $c + 1 = x_2$

$\therefore \quad f(x_2) = f(x_1)$ where $x_2 - x_1 = 1$

obviously $x_1, x_2 \in (0, 2)$

104.

(A) If lines are parallel for some $t \in (0, 1)$ then we have to prove,

$$\frac{-f_1(t)}{f_2(t)} = \frac{-g_1(t)}{g_2(t)}$$

$\Rightarrow f_1(t)\, g_2(t) - f_2(t)\, g_1(t) = 0$

Let $\phi(t) = f_1(t)\, g_2(t) - f_2(t)\, g_1(t)$

$\because f_1, f_2, g_1, g_2$ are continuous functions

$\therefore \phi(t)$ is continuous function

$\phi(0) = f_1(0)\, g_2(0) - f_2(0)\, g_1(0)$

$= 2 \times 2 - 3 \times 3 = -5$

and $\phi(1) = f_1(1)\, g_2(1) - f_2(1)\, g_1(1)$

$= 4 \times 4 - 2 \times 2 = 12$

$\therefore$ According to I.V.T. $\phi(t) = 0$ for some $t \in (0, 1)$

$\therefore$ Lines are parallel for some $t \in (0, 1)$

(B) If lines are $\perp^r$ then $\left(\frac{-f_1(t)}{f_2(t)}\right)\left(\frac{-g_1(t)}{g_2(t)}\right) = -1$

$\Rightarrow f_1(t)\, g_1(t) + f_2(t)\, g_2(t) = 0$

Let $h(t) = f_1(t)\, g_1(t) + f_2(t)\, g_2(t)$

$\Rightarrow h(0) = 12$ and $h(1) = 16$

(C) If lines make complimentary angles then

$$\left(\frac{-f_1(t)}{f_2(t)}\right)\left(\frac{-g_1(t)}{g_2(t)}\right) = 1$$

$\Rightarrow f_1(t)\, g_1(t) - f_2(t)\, g_2(t) = 0$

let $k(t) = f_1(t)\, g_1(t) - f_2(t)\, g_2(t)$

here $k(0) = 0$ and $k(1) = 0$

105. $f(x) = x + \{-x\} + [x]$

$= x + (-x - [-x]) + [x]$ as $\{x\} = x - [x]$

$= x - x - [-x] + [x]$

$f(x) = [x] - [-x]$

i.e. $f(x) = \begin{bmatrix} -4 & x = -2 \\ -3 & -2 < x < -1 \\ -2 & x = -1 \\ -1 & -1 < x < 0 \\ 0, & x = 0 \\ 1 & 0 < x < 1 \\ 2 & x = 1 \\ 3 & 1 < x < 2 \\ 4 & x = 2 \end{bmatrix}$ $\Rightarrow$ discontinuous at all integral value of x

i.e. – 2, – 1, 0, 1, 2. i.e. five points. **Ans.**

106. $f(x) = \text{sgn}\left((3\cos^{-1}x - \pi)(x-2)\right)$

$= \text{sgn}(x)$ is discontinuous at $x = 0$

$\Rightarrow$ $f(x)$ is discontinuous at $x = \cos\dfrac{\pi}{3}$ only

Since $x = 2 \notin D_f$ & no graph of f exist in viccinity of $x = 2$.

$\therefore$ Only one point of discontinuity i.e. 1 **Ans.**

108. We have, $f(x) = \lim\limits_{n \to \infty} \dfrac{(1+\sin x)^n + e^x}{1 + (1+\sin x)^n}$

$$= \begin{cases} e^x, & x \in ((2n-1)\pi,\ 2n\pi) \\ \dfrac{1+e^x}{2}, & x = n\pi \\ 1, & x \in (2n\pi,\ (2n+1)\pi) \end{cases} \qquad \text{where } n \in I.$$

$\because$ when $x \in (2n\pi, (2n+1)\pi) \Rightarrow 0 < \sin x \le 1$ and $x \in ((2x-1)\pi, 2n\pi)$

$\Rightarrow -1 \le \sin x < 0$.

So, $f(x) = \begin{cases} \dfrac{1+e^{-2\pi}}{2}, & x = -2\pi \\ 1, & -2\pi < x < -\pi \\ \dfrac{1+e^{-\pi}}{2}, & x = -\pi \\ e^x, & -\pi < x < 0 \\ 1, & x = 0 \\ 1, & 0 < x < \pi \\ \dfrac{1+e^x}{2}, & x = \pi \\ e^x, & \pi < x < 2\pi \\ \dfrac{1+e^x}{2}, & x = 2\pi \end{cases}$

Clearly, number of points of discontinuity of $f(x)$ in $[-2\pi, 2\pi] = 4$
i.e., $x = -2\pi, -\pi, \pi, 2\pi$.
Note : The function $f(x)$ is continuous at $x = 0$. **Ans.**

109. Let $b > 0$, then $f(1) = b > 0$
and $f(5) = 2a + 3b - 6 = 2(a + 2b) - b - 6 = 4 - b - 6 = -(2 + b) < 0$
Hence by IVT, $\exists$ some $c \in (1, 5)$ s.t. $\Rightarrow$ $f(c) = 0$
If $b = 0$ then $a = 2$

$f(x) = 2\sqrt{x-1} - \sqrt{2x^2 - 3x + 1} = 0$ (finding roots of $f(x) = 0$)

$\Rightarrow$ $4(x - 1) = 2x^2 - 3x + 1 = (2x - 1)(x - 1)$

$(x - 1)(2x - 5) = 0 \Rightarrow x = \dfrac{5}{2}$

Hence $f(x) = 0$ if $x = \dfrac{5}{2}$ which lies in $(1, 5)$

If $b < 0$, $f(1) = b < 0$ and

$f(2) = a + b\sqrt{3} - \sqrt{3}$

$= (a + 2b) + \left(\sqrt{3} - 2\right)b - \sqrt{3}$

$= (2 - \sqrt{3}) - (2 - \sqrt{3})\,b$

$= (2 - \sqrt{3})(1 - b) > 0$ (as $b < 0$)

Hence $f(1)$ as $f(2)$ have opposite signs
$\exists$ some $c \in (1, 2) \subset (1, 5)$ for which $f(c) = 0$

110. $f'(x) = -3x^2 + 2x - 1 < 0 \ \forall\ x$
$f(x)$ decreasing function

$\therefore$ $g(x) = \begin{cases} f(x); & 0 \le x \le 1 \\ x - 1; & 1 < x \le 2 \end{cases} = \begin{cases} -x^3 + x^2 - x + 1; & 0 \le x \le 1 \\ x - 1; & 1 < x \le 2 \end{cases}$

$$g\left(g(1^+)\right) = g(0^+) = 1$$

$$g\left(g(1^-)\right) = g(0^+) = 1.$$ **Ans.**

111. $f(x)$ is continuous and differentiable at only $x = 1$ and $f'(1) = 3$.

112. f(x) is continuous at $x = \frac{\pi}{2}$

$$\Rightarrow \quad \lim_{x \to \frac{\pi}{2}} \frac{e^{\cos^{2n} x} + \sin^{2m} x - 2}{(2x - \pi)^2} = 1 \Rightarrow \lim_{h \to 0} \frac{\left(e^{\sin^{2n} h} - 1\right) - \left(1 - \cos^{2m} h\right)}{4h^2} = 1$$

$$\Rightarrow \quad \lim_{h \to 0} \left(\frac{\left(e^{\sin^{2n} h} - 1\right)}{h^2} - \frac{1 - \cos^{2m} h}{h^2} \right) = 4 \Rightarrow \lim_{h \to 0} \frac{\left(e^{\sin^{2n} h} - 1\right)}{(\sin^{2n} h)} \cdot \frac{\sin^{2n} h}{h^2} - m = 4$$

If $n = 1$, then $m = -3$ and if $n > 1$ then $m = -4$.
In both cases : $f'(0) = 0$. **Ans.**

113 for $x = \frac{\pi}{6}$ or $\frac{5\pi}{6}$, $4\sin^2 x = 1$

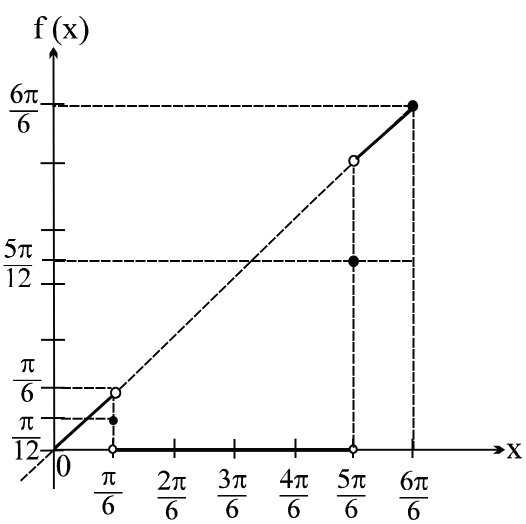
$$\therefore \quad \lim_{n\to\infty} f(x) = \lim_{n\to\infty} \frac{x}{1 + 1^n} = \frac{x}{2}$$

for $\frac{\pi}{6} < x < \frac{5\pi}{6}$, $4\sin^2 x > 1$

$$\therefore \quad \lim_{n\to\infty} f(x) = \lim_{n\to\infty} \frac{x}{1 + (\text{greater than } 1)^n} = 0$$

for $0 \le x < \frac{\pi}{6}$ or $\frac{5\pi}{6} < x \le \pi$, $4\sin^2 x < 1$

$$\therefore \quad f(x) = \operatorname*{Limit}_{n\to\infty} \frac{x}{1 + (\text{less than } 1)^n} = x$$



$$\text{Hence } f(x) = \begin{cases} x & \text{for } 0 \le x < \frac{\pi}{6} \text{ or } \frac{5\pi}{6} < x \le \pi \\ \frac{x}{2} & \text{for } x = \frac{\pi}{6} \text{ or } \frac{5\pi}{6} \\ 0 & \text{for } \frac{\pi}{6} < x < \frac{5\pi}{6} \end{cases}$$

and graph is as shown above.

From the graph it is clear that $f(x)$ is continuous everywhere in $[0,\pi]$ except at $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ and has non removable discontinuity of finite type. Jump at $x = \frac{\pi}{6}$ it is $\frac{\pi}{6}$ units and at $x = \frac{5\pi}{6}$ it is $\frac{5\pi}{6}$ units.

114. (B) Let $L = \lim_{n \to \infty} \left(\frac{n^2}{n-1}\right)^{\tan\frac{1}{\sqrt{n}}} \quad (\infty^0)$

$$\ln L = \lim_{n \to \infty} \tan\frac{1}{\sqrt{n}} \ln\left(\frac{n^2}{n-1}\right) = \lim_{n \to \infty} \frac{2\ln n - \ln(n-1)}{\cot\left(\frac{1}{\sqrt{n}}\right)} \quad \left(\frac{\infty}{\infty}\right)$$

Put $n = t^2$ $= \lim_{t \to \infty} \frac{4\ln t - \ln(t^2-1)}{\cot\left(\frac{1}{t}\right)} = \lim_{t \to \infty} \frac{2(t^2-2)}{t(t^2-1)\operatorname{cosec}^2\left(\frac{1}{t}\right)\left(\frac{1}{t^2}\right)}$

$$= \lim_{t \to \infty} \frac{2(t^2-2)}{t(t^2-1)}\left(\frac{\sin\frac{1}{t}}{\frac{1}{t}}\right)^2 = 0$$

$L = 1$. **Ans.**

(C) $\sin 2x = \pm\frac{\sqrt{3}}{2}$

$2x \in (0, 4\pi)$

Hence, 8 solutions. **Ans.**

(D) Since $g(x)$ is differentiable $\forall x \in R$

$\therefore$ $f(x)$ must have the factor $x(x-1)(x-2)$ atleast once.

$\therefore$ minimum 3 roots of $f(x) = 0$. **Ans.**

115. $f(x) = 23 - 2^{-x^2+2x-1+4} = 23 - \frac{16}{2^{(x-1)^2}}$

Range of $f(x)$ is $= [7, 23)$

Now, $g(x) = \left[\frac{f(x)}{\mu}\right]$ will be discontinuous at all points where $\frac{f(x)}{\mu}$ is an integer.

When $\mu \ge 23$ then $0 < \frac{f(x)}{\mu} < 1 \Rightarrow \left[\frac{f(x)}{\mu}\right] = 0\ \forall x \in R$ and $g(x)$ becomes continuous in R.

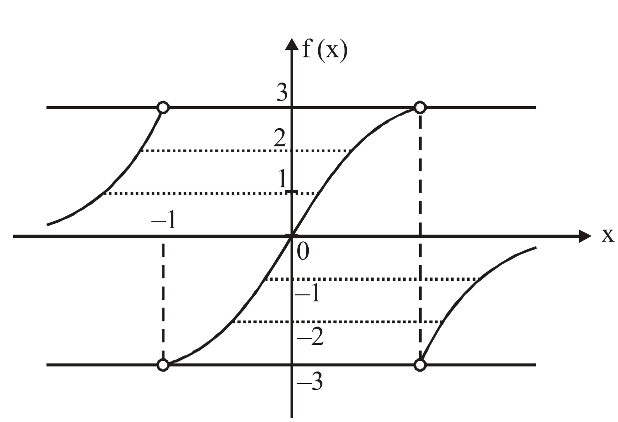
When $1 \le \mu < 23$ then $\frac{f(x)}{\mu}$ becomes integer at some values of x and $g(x)$ will be discontinuous at these value of x.

$\therefore$ Sum of all values of μ is

$$1 + 2 + 3 + \dots\dots\dots + 22 = \frac{22 \cdot 23}{2} = 253. \textbf{ Ans.}$$

116. The function is only non-differentiable where the quadratic in the absolute value is equal to zero - which occurs at $x = 2$ and $x = 3$.

117. [Sol.$_2$ Graph of $f(x) = \frac{6}{\pi}\tan^{-1}\left(\frac{2x}{1-x^2}\right)$



Range of f(x) is (–3, 3) integral values of $f(x)$ are –2, –1, 0, 1, 2.

For each integral value of f(x) (except zero) there are exactly two values of x and for f(x) = 0, there is exaclty are value of x, x = 0.

$\therefore$ Total number of points where $f(x)$ is an integer and where $[f(x)]$ is discontinuous is 9.

118. $f(x) = \sin x$ is differentiable in $[0, 2\pi]$
$g(x) = |\sin x|$ is not differentiable at $x = \pi$.
Let $h(x) = f(x)\, g(x) = |\sin x| \sin x$

$$h'(\pi) = \lim_{h\to 0}\frac{|\sin(\pi - h)|\sin(\pi - h) - 0}{h} = \lim_{h\to 0}\frac{|\sin h|\sin h}{h} = 0$$

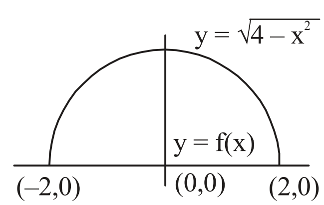
$\Rightarrow$ $h(x)$ is differentiable at $x = \pi$
but $g(x)$ is not differentiable at $x = \pi$]

119. $$f(x) = \lim_{\alpha\to 0}\left\{\frac{a\sin\alpha}{\alpha}\right\}x + \left\{\frac{b\tan\alpha}{\alpha}\right\}$$

$$= \lim_{\alpha\to 0}\left(\frac{a\sin\alpha}{\alpha} - \left[\frac{a\sin\alpha}{\alpha}\right]\right)x + \left(\frac{b\tan\alpha}{\alpha} - \left[\frac{b\tan\alpha}{\alpha}\right]\right)$$

$$= \lim_{\alpha\to 0}\left(a - \left[\frac{a\sin\alpha}{\alpha}\right]\right)x + \left(b - \left[\frac{b\tan\alpha}{\alpha}\right]\right)$$

$$\left\{\begin{array}{l} \because \dfrac{\sin\alpha}{\alpha} \text{ is little less than 1 as } \alpha \to 0 \\ \because \dfrac{a\sin\alpha}{\alpha} \text{ is little greater than a as } \alpha \to 0 \ (\because a \in I^-) \\ \because \dfrac{\tan\alpha}{\alpha} \text{ is little greater than 1 and } \alpha \to 0 \\ \therefore \dfrac{b\tan\alpha}{\alpha} \text{ is little greater than b as } \alpha \to 0 \ (\because b \in I^+) \end{array}\right\}$$



$f(x) = (a - a)x + (b - b) = 0$

$\therefore$ f(x) is continuous for all values of x and f(x) is both even and odd function.

Area bounded by y = f(x) and $y = \sqrt{4 - x^2}$ is 2π.

120. Let $L = \lim_{n\to\infty} \binom{n}{k}\left\{f\left(\frac{x}{n}\right)-f(0)\right\}^k$. Now $L = \lim_{n\to\infty}\left\{\dfrac{f\left(\frac{x}{n}\right)-f(0)}{\frac{x}{n}-0}\right\}^k\left(\frac{x}{n}\right)^k \cdot {}^nC_k$

As f(x) is differentiable, hence $\lim_{n\to\infty}\dfrac{f\left(\frac{x}{n}\right)-f(0)}{\frac{x}{n}-0}=f'(0)$

$\therefore\ L = \lim_{n\to\infty}\{f'(0)\}^k x^k \cdot \frac{n}{n}\frac{(n-1)}{n}\frac{(n-2)}{n}\ ....\ \frac{(n-k+1)}{n}\cdot\frac{(n-k)!}{k!(n-k)!}$

$= f'(0)^k \cdot \frac{x^k}{k!} \cdot \lim_{n\to\infty} 1\cdot\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\left(1-\frac{3}{n}\right)....\left(1-\frac{k-1}{n}\right) = f'(0)^k \cdot \frac{x^k}{k!}$ **Ans.**

121. As $\sin^{-1}\alpha_1 + \sin^{-1}\alpha_2 \ .... + \sin^{-1}\alpha_n = \frac{n\pi}{2}$

$\Rightarrow \quad \sin^{-1}\alpha_1 = \sin^{-1}\alpha_2 = ... = \sin^{-1}\alpha_n = \frac{\pi}{2}$ $\quad\left(\because -\frac{\pi}{2}\le \sin^{-1}\alpha \le \frac{\pi}{2}\right)$

$\Rightarrow \quad \alpha_1 = \alpha_2 = ...\alpha_n = 1$

So, $p = \prod_{r=1}^{n}(1)^r = 1$

Hence, $f(x) = \begin{cases}\dfrac{x^{1/3}-(3-2x)^{1/4}}{x^2-x}, & x\ne 1\\ k, & x=1\end{cases}$

Clearly, f(x) is continuous at $x = 1$.

$\therefore \quad k = \dfrac{x^{1/3}-(3-2x)^{1/4}}{x^2-x}$

$= \lim_{h\to 0}\dfrac{(1+h)^{1/3}-(1-2h)^{1/4}}{(1+h)^2-(1+h)}$ $\quad$ (putting $x = 1+h$)

$= \frac{1}{3}+\frac{1}{2}=\frac{5}{6}$

Hence, $6(k+p) = 6\left(\frac{5}{6}+1\right) = 11$]

122. $\lim_{x\to 0} f(x) = \lim_{x\to 0}\dfrac{P\left(\frac{1}{x^3}\right)}{e^{\frac{1}{x^4}}} = \lim_{t\to\infty}\dfrac{P(t^3)}{e^{t^4}} = 0$

(As we know, $\lim\limits_{x \to \infty} \frac{x^n}{e^x} = 0, n \in N$) $\Rightarrow$ $\lim\limits_{x \to 0} f(x) = f(0) = 0$

$$f'(0) = \lim_{h \to 0} \frac{P\left(\frac{1}{h^3}\right)}{he^{\frac{1}{h^4}}} = \lim_{t \to \infty} \frac{tP(t^3)}{e^{t^4}} = 0.$$ **Ans.**

123. $f_1(x) = ax^9$ is an odd continuous function

$f_2(x) = b \sin x$ is an odd continuous function

$$f_3(x) = cx^2 \operatorname{sgn}(x) = \begin{cases} cx^2, & x > 0 \\ 0, & x = 0 \\ -cx^2, & x < 0 \end{cases}$$ is an odd continuous function

$$f_4(x) = \frac{d(e^x - e^{-x})}{(e^x + e^{-x})}$$

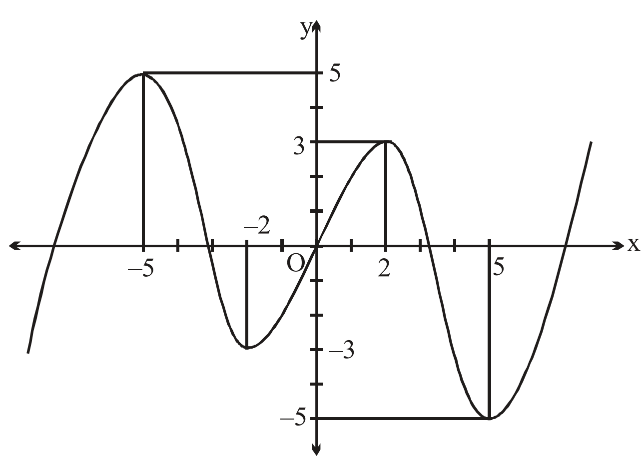
$$f_4(-x) = \frac{(e^{-x} - e^{-x})}{(e^{-x} + e^x)} = -f_4(x)$$

$\therefore$ $f_4(x)$ is odd function and continuous $\forall x \in R$ ($\because$ denominator is never zero).

$\therefore$ f(x) will be an odd continuous $\Rightarrow$ $f(-x) = -f(x)$

$\therefore$ $f(5) = -5$ and $f(2) = +3$, $f(0) = 0$

If $a > 0$, then when $x \to -\infty$, $f(x) \to -\infty$ and when $x \to +\infty$ $f(x) \to +\infty$



Clearly, $f(x) = 0$ will have minimum five roots. **Ans.**

124. $\because$ Graph of $y = g(x)$ is reflection ofgraph of $y = f(x)$ with respect to line $y = x$.

g (x) is inverse function of $f(x)$

$\because$ (a, b) lies on $y = f(x)$

$f(a) = b$

$$g(f(x)) = x \quad \Rightarrow \quad g'(f(x))\, f'(x) = 1$$

$$\Rightarrow \quad g'(f(x)) = \frac{1}{f'(x)} \quad \Rightarrow \quad g'(f(\alpha)) = \frac{1}{f'(\alpha)}$$

$$\Rightarrow \quad g'(b) = \frac{1}{3}$$

$$g = \frac{1}{3} \quad \Rightarrow \quad \frac{670}{\gamma} = 2010 \ \textbf{Ans.}$$

125. $f(-5) = 2,\ f(20) = f(4) = 3,\ f(-10) = 1,\ f(17) = -2$

$\therefore$ Given expression $= \tan^{-1}\tan\{2 + 3 + 0 - 2\} = 3 - \pi$. **Ans.**

Paragraph for Question Nos. 126 to 128

126. $\because \quad f'(x) = \underset{h \to 0}{\text{Lim}} \dfrac{f(x+h) - f(x)}{h}$

$$= f(x) \underset{h \to 0}{\text{Lim}} \frac{f(h) - 1}{h} \qquad \ldots (1)$$

putting $x = 0$, $y = 1$ in given functional relation.

$$f(1) = f(0) \cdot f(1) \quad \Rightarrow \quad f(0) = 1$$

$$\therefore \quad f'(x) = f(x) \underset{h \to 0}{\text{Lim}} \frac{f(h) - f(0)}{h} = f'(x) \cdot f(x)$$

$$\Rightarrow \quad \frac{f'(x)}{f(x)} = f'(0) \quad \Rightarrow \quad \ln f(x) = f'(0)\, x + c$$

$\because \quad x = 0, \quad f(0) = 1 \quad \therefore \quad c = 0$

$\therefore \quad \ln f(x) = f'(0)x \quad \Rightarrow \quad f(x) = e^{f'(0) \cdot x}$

putting $x = 1$

$$\frac{1}{3} = e^{f'(0)}$$

$$\therefore \quad f(x) = \left(\frac{1}{3}\right)^x$$

Let $\quad S = \underset{n \to \infty}{\text{Lim}} \{f(x) + 2f(x+1) + 3f(x+2) \ldots\ldots.. (n+1)\, f(x+n)\}$

$$= \left(\frac{1}{3}\right)^x + 2\left(\frac{1}{3}\right)^{x+1} + 3\left(\frac{1}{3}\right)^{x+2} \ldots\ldots \infty \text{ terms} \quad \ldots (2)$$

$$\frac{S}{3} = \left(\frac{1}{3}\right)^{x+1} + 2\left(\frac{1}{3}\right)^{x+2} \text{ .... (3)}$$

(2) – (3),

$$\frac{2S}{3} = \left(\frac{1}{3}\right)^{x} + \left(\frac{1}{3}\right)^{x+1} + \left(\frac{1}{3}\right)^{x+2} \text{ ....} = \frac{\left(\frac{1}{3}\right)^{x}}{1-\frac{1}{3}} = \frac{3f(x)}{2}$$

$$\Rightarrow \qquad S = \frac{9f(x)}{4}$$

127. $\lim\limits_{x\to\infty}\left(\left(\frac{1}{3}\right)^{x} + 3^{x}\right)^{\frac{1}{x}} = \lim\limits_{x\to\infty} 3\left(\left(\frac{1}{9}\right)^{x} + 1\right)^{\frac{1}{x}} = 3 = f(-1)$

128. $f_1(x) = \frac{f_0(x)}{1+a_0 f_0(x)} = \frac{\left(\frac{1}{3}\right)^{x}}{1+a_0\left(\frac{1}{3}\right)^{x}}$; $f_2(x) = \frac{f_1(x)}{1+a_1 f_1(x)} = \frac{\frac{(1/3)^x}{1+a_0(1/3)^x}}{1+a_1\left(\frac{(1/3)^x}{1+a_0(1/3)^x}\right)} = \frac{3^{-x}}{1+(a_0+a_1)3^{-x}}$

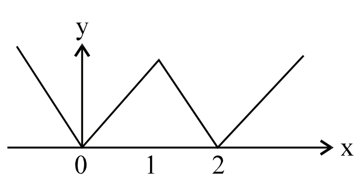
$$\therefore \qquad f_n(x) = \frac{3^{-x}}{1+(a_0+a_1 .....a_{n-1})^{3-x}}$$

$$\therefore \qquad \lambda = a_0 + a_1 + ... + a_{n-1} = \sum_{r=0}^{n\cdot 1} a_r$$

129.

(A) If f(0) = 0 or f(1) = 1, the root is 0 or 1
If f(0) > 1 and f(1) < 1, then $g(x) = f(x) - x^3$ satifies, g(0) > 0, g(1) < 1
Hence by intermediate value theorem
g(x) = 0 has at least one root.

(B) If f '(a) = f ' (b) = 0 and f(α) = f(β) = 0 where a < α < β < b, then by Rolle's theorem f '(x) vanishes between α and β contradicting the fact that a and b are two consecutive roots of f '(x) = 0.

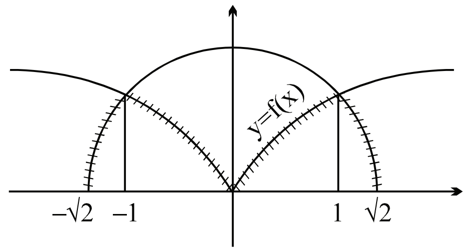
(C) 

f(x) is not differentiable at x = 0, 1, 2

(D) x = 2 – x → x = 1.
f(x) is continuous only at x = 1.

Paragraph for Quesrtion no. 130 to 133

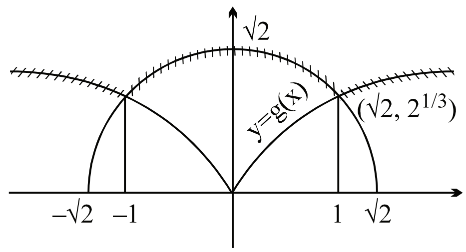
130. $f(x) = \begin{cases} \sqrt{2-x^2} & \text{in } [-\sqrt{2},-1] \\ x^{2/3} & \text{in } [-1,1] \\ \sqrt{2-x^2} & \text{in } [1, \sqrt{2}] \end{cases}$



$f(x)$ is not differentiable at $x = 0, \pm 1$

131. The range of $f(x)$ is [0, 1]

132. $g(x) = \begin{cases} x^{2/3} & \text{in } [-\sqrt{2},-1] \\ \sqrt{2-x^2} & \text{in } [-1,1] \\ x^{2/3} & \text{in } [1, \sqrt{2}] \end{cases}$



$y = f(x)$ is the continuous curve
$y = g(x)$ is the curve
$g(x)$ is not differentiable at $x = \pm 1$

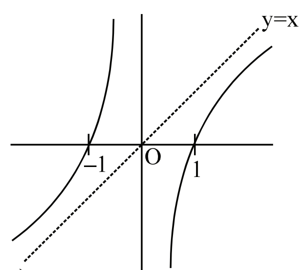
133. the range is $[1, \sqrt{2}]$

134. $\therefore$ $f(x)$ is differentiable in $(0, \infty)$

hence $\lim\limits_{x\to\infty} f(x)$ must exist and is finite

$\therefore$ $y = f(x)$ must have a horizontal asymptote

as $x \to \infty$ then only $\lim\limits_{x\to\infty} f(x)$ will exist



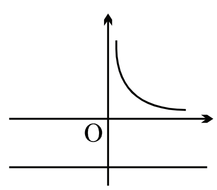
If $f(x)$ has an inclined asymptotes as $y = x - \frac{1}{x}$ then $\lim\limits_{x\to\infty} f(x) \to \infty$

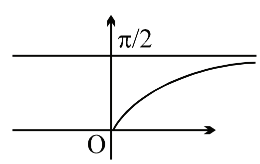
$\therefore$ $f(x)$ has a horizontal asymptote

hence $\lim\limits_{x\to\infty} f'(x) \to 0$

$\Rightarrow$ **(C) (also see figure for $f(x) = \tan^{-1}x$)**

e.g. Take the example given





$f(x)=\tan^{-1}x$

(i) Let $f(x) = x \sin \frac{1}{x}$ which is differentiable in $(0, \infty)$

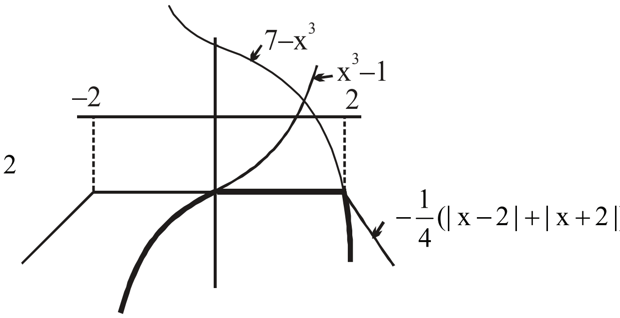
$$f'(x) = \sin\frac{1}{x} - \frac{1}{x}\cos\frac{1}{x}$$

$$f(x) + f'(x) = \underbrace{\left(x\sin\frac{1}{x}\right)}_{\lim\limits_{x\to\infty} \to 1} + \underbrace{\left(\sin\frac{1}{x} - \frac{1}{x}\cos\frac{1}{x}\right)}_{\lim\limits_{x\to\infty} \to 0}$$

hence $\lim\limits_{x\to\infty} f(x) = L$ and $\lim\limits_{x\to\infty} f'(x) = 0$ **Ans.**

(ii) $f(x) = \tan^{-1}x$ in $(0, \infty)$]

135. [Sol. $f(x) = \begin{cases} x^3-1 & \text{for } x<0 \\ -\frac{1}{4}(|x-2|+|x+2|) \text{ for } 0\le x\le 2 \\ \quad = -1 \\ 7-x^3 & \text{for } x>2 \end{cases}$



f(x) is cont. every where but non-derivable at x= 2 only.
$\therefore$ p = 0 and q =1

Paragraph for question no. 136 to 138

SOl : We have $f_1(x) = 2\sum\limits_{r=1}^{n} \frac{\sin\left(\left(x+\frac{r\pi}{6}\right)-\left(x+(r-1)\frac{\pi}{6}\right)\right)}{\cos\left(x+(r-1)\frac{\pi}{6}\right)\cdot\cos\left(x+\frac{r\pi}{6}\right)}$

$$= 2\cdot\left[\left(\tan\left(x+\frac{\pi}{6}\right)-\tan x\right)+\left(\tan\left(x+\frac{2\pi}{6}\right)-\tan\left(x+\frac{\pi}{6}\right)\right)+\left(\tan\left(x+\frac{3\pi}{6}\right)-\tan\left(x+\frac{2\pi}{6}\right)\right)\right.$$

$$\left.+..............+\left(\tan\left(x+\frac{n\pi}{6}\right)-\tan\left(x+(n-1)\frac{\pi}{6}\right)\right)\right]$$

$\Rightarrow \quad f_1(x) = 2\cdot\left(\tan\left(x+\frac{n\pi}{6}\right)-\tan x\right)$

for n = 3, $f_1(x) = 2\left(\tan\left(\frac{\pi}{2}+x\right)-\tan x\right) = 2(-\cot x-\tan x) = -2\frac{1}{\sin x\cos x}$

Now $\quad 2f_2(x) = f_1(x) - 2\cdot\tan\left(x+\frac{n\pi}{6}\right)$

$\quad = 2\tan\left(x+\frac{n\pi}{6}\right) - 2\tan x - 2\tan\left(x+\frac{n\pi}{6}\right)$

$\therefore \quad f_2(x) = -\tan x \Rightarrow \quad f_3(x) = -f_2(x)$, so

$f_3(x) = \tan x$

Now $f_4(x) = \begin{cases} e^{\frac{e^{(e^x-1)}-1}{2\cdot(e^x-1)}} & ; \; x<0 \\ k_1 & ; \; x=0 \\ (1+|\tan x|)^{\frac{k_2}{\tan x}} & ; \; x>0 \end{cases}$

Clearly $f(0^-) = e^{\frac{1}{2}\ln e} = \sqrt{e}$

$f(0^+) = e^{k_2}$ & $f(0) = k_1$

136. As $f_4(x)$ is continuous at $x = 0$, so by definition of continuity, we get

$f(0^-) = f(0^+) = f(0)$

$\Rightarrow \quad \sqrt{e} = e^{k_2} = k_1$

$\therefore \quad k_1 = \sqrt{e}$ & $k_2 = \dfrac{1}{2}$

137. As $y = f_3(x) = \tan x$

clearly $f_3(x)$ is continuous as well as derivable everywhere in $(0, \pi/2)$.

138. for $n = 3$,

$f_1(x) = -4$

$\Rightarrow \quad -2(\tan x + \cot x) = -4$

$\Rightarrow \quad \dfrac{\sin x}{\cos x} + \dfrac{\cos x}{\sin x} = 2$

$\Rightarrow \quad 1 = 2\sin x \cdot \cos x$

$\therefore \quad x = \dfrac{\pi}{4} \in \left(0, \dfrac{\pi}{2}\right)$

139. $f(1) = \tan\left(\dfrac{1}{2}\sin^{-1}\left(\dfrac{2x}{1+x^2}\right)\right) = \tan\dfrac{\pi}{4} = 1$

$$f(1^+) = \lim_{x\to 1^+} \tan\left(\frac{1}{2}\sin^{-1}\left(\frac{2x}{1+x^2}\right)\right) = \lim_{x\to 1^+} \tan\left(\frac{1}{2}\left(\pi - 2\tan^{-1}x\right)\right) = \tan\left(\frac{1}{2}\left(\pi - 2\cdot\frac{\pi}{4}\right)\right)$$

$= \tan\dfrac{\pi}{4} = 1$

$f(1^-) = \lim_{x\to 1^-} \left(\lambda\cos^{-1}(2x-3) + \mu\right)$

Here, $-1 \le 2x - 3 \le 1 \;\Rightarrow\; 2 \le 2x \le 4$

$\Rightarrow \quad 1 \le x \le 2$

It is clear that $\cos^{-1}(2x-3)$ is not defined.

for $0 \le x < 1$, therefore $\lambda = 0$.

$f(1^-) = \lim\limits_{x \to 1^-} (0 + \mu) = \mu$

since, f is continuous in $[0, \infty)$

$\therefore \quad \mu = 1$

Hence, $\lambda + \mu = 0 + 1 = 1$ **Ans.**

140. $f(x + y) = f(x) + f(y)$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{f(x) + f(h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{f(0+h) - f(0)}{h} \quad (\text{as } f(0) = 0)$$

$f'(x) = f'(0)$

$\therefore \quad f(x) = f'(0)x + C$

if $\quad x = 0$; $f(0) = 0 \quad \Rightarrow \quad C = 0$

hence $\quad f(x) = f'(0)x$

$\quad f(kx) = f'(0)\,kx = kf(x) \qquad \{\text{as } f'(0)\,x = f(x)\}$

Hence $f(kx) = kf(x)$.]

141. put $x \to x + h$ and $y \to h$

$f(x) \cdot f(h) = f(x + h)$

$$\therefore \; f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{f(x)[f(h) - 1]}{h} \quad (x{=}0,\ y{=}0;\ f(0){=}0 \text{ not possible, } f(0){=}1)$$

$f'(x) = f(x) \cdot f'(0)$

$\Rightarrow \quad f'(5) = f(5) \cdot f'(0)$

$q = f(5) - p \quad \Rightarrow f(5) = \dfrac{q}{p}$ **Ans.**

142. As $g(x)$ is a continuous function, so

$$\lim_{h \to 0} g(2+h) = \lim_{h \to 0} g(2-h) = g(2) \qquad \text{........(1)}$$

Also $\quad f(x)$ is differentiable at $x = 2$,

so $f'(2^+) = f'(2^-)$.

$$\text{Now,} \quad f'(2^-) = \lim_{h \to 0} \frac{e^{2-h}\left|(2-h) - 2\right| g(2-h) + \sin(2-h) - 0 - \sin 2}{-h}$$

$$= \lim_{h \to 0} \frac{e^{2-h}\, h\, g(2)}{-h} + \frac{2\cos\left(\frac{4-h}{2}\right)\sin\left(\frac{-h}{2}\right)}{-h} = -\,e^2 g(2) + \cos 2 \qquad \text{.......(2)}$$

(Using equation (1))

$$\text{And } f'(2^+) = \lim_{h \to 0} \frac{e^{2+h}\left|(2+h) - 2\right| g(2+h) + \sin(2+h) - 0 - \sin 2}{h}$$

$$= \lim_{h\to 0} \frac{e^{2+h}\, h\, g(2)}{h} + \frac{2\cos\left(\frac{4+h}{2}\right)\sin\left(\frac{h}{2}\right)}{h}$$

$= e^2 g(2) + \cos 2$ ......(3)

(Using equation (1))

As $f'(2^+) = f'(2^-)$

$\Rightarrow -e^2 g(2) + \cos 2 = e^2 g(2) + \cos 2$

Hence $g(2) = 0$

143. $\because$ $\frac{\pi}{3} < A < \frac{\pi}{2} \Rightarrow \frac{\sqrt{3}}{2} < \sin A < 1$ and $\frac{3\sqrt{3}}{8} < \sin^3 A < 1$

$0 < \sin A - \sin^3 A < 1$ and $\sin A + \sin^3 A > 1$

$\therefore$ $f(x) = \begin{cases} ax^2, & x \in Q \\ 5x - b, & x \notin Q \end{cases}$

$\because$ f(x) is continuous at x = 2 and 3.

$\therefore$ $ax^2 = 5x - b$ should have roots 2 and 3 $\Rightarrow ax^2 - 5x + b = 0$

$\therefore$ $\frac{5}{a} = 5 \Rightarrow a = 1$ and $\frac{b}{a} = 6 \Rightarrow b = 6$

$\therefore$ $b - a = 5$. **Ans.**

144.

(A) $L = \lim_{x\to a} \left(f(x)\right)^{g(x)} = (1)^{\infty}$

$\Rightarrow L = e^k \text{ where } k = \lim_{x\to a}\left[\left(f(x) - 1\right).g(x)\right]$

$\therefore L = e^k \text{ where } k = \lim_{x\to 0}\left[\frac{-(\tan x + \sin x)}{(1+\sin x)\sin x}\right]$

$$= \lim_{x\to 0}\left[\frac{-\left(\frac{1}{\cos x} + 1\right)}{1 + \sin x}\right] = -2$$

$L = e^{-2} = ae^b \Rightarrow a = 1,\ b = -2$

(B) $$\lim_{\frac{1}{x}\to 0}\left[\frac{\left(\frac{e^{1/x^2} - 1}{1/x^2}\right)}{\frac{-2\tan^{-1}(1/x^2)}{(1/x^2)}}\right] = -\frac{1}{2} = ae^b \Rightarrow a = \frac{-1}{2},\ b = 0$$

Here, $\dfrac{\pi}{2}-\tan^{-1}\left(x^2\right)=\cot^{-1}\left(x^2\right)=\tan^{-1}\left(\dfrac{1}{x^2}\right)$

(C) $\quad L=\underset{x\to 0}{\text{Lim}}-\left[\dfrac{e^{\frac{1}{x}\log(1+x)}-e}{x}\right]$

where $\log(1+x)=x-\dfrac{x^2}{2}+\dfrac{x^3}{3}-\dfrac{x^4}{4}+....\infty$

$\therefore\ L=\underset{x\to 0}{\text{Lim}}\,(-e)\left[\dfrac{\left(e^{-x/2+x^2/3-x^3/4+.....\infty}\right)-1}{\left(-x/2+x^2/3-x^3/4+...\infty\right)}\times\left(-\dfrac{1}{2}+\dfrac{x}{3}-\dfrac{x^2}{4}+....\infty\right)\right]$

$\therefore L=(-e)(-1/2)=e/2=ae^b\Rightarrow a=\dfrac{1}{2},\ b=1$

(D) $\quad$ Required limit $=\underset{\theta\to 0}{\text{Lim}}\left(\sin\theta+\cos\theta\right)^{1/\theta}=(1)^{\infty}$ form

$=e^k\ \ where\ k=\underset{\theta\to 0}{\text{Lim}}\left[\dfrac{\sin\theta+\cos\theta-1}{\theta}\right]=\dfrac{0}{0}\ form$

$=\underset{\theta\to 0}{\text{Lim}}\left[\dfrac{\cos\theta-\sin\theta}{1}\right]=1$

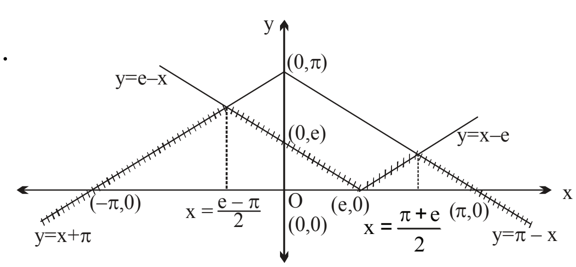
$\therefore L=e=ae^b\ \ where\ a=1,\ b=1$

145. Clearly f(x) is continuous everywhere

but non-derivable at 3 points viz. $x=\dfrac{e-\pi}{2},e,\dfrac{\pi+e}{2}$.

Now verify alternatives from graph.

Also, Range of f(x) is $\left(-\infty,\dfrac{\pi+e}{2}\right]$



Graph of y = f (x)

146. $\quad f'(0^+)=\underset{h\to 0}{\text{Lim}}\dfrac{h^p\sin\dfrac{1}{h}+h(\tan h)^q}{h}$

$=\underset{h\to 0}{\text{Lim}}\,h^{p-1}\sin\dfrac{1}{h}+h(\tan h)^q$

For this limit to exist $p-1>0$ and $q\in N$

$\therefore\quad p\geq 2$ and $q\in N$

again

$$f'(0^-) = \lim_{h \to 0} \frac{-h^p \sin\frac{1}{h} - h(\tan h)^q}{-h}$$

$$\lim_{h \to 0} h^{p-1} \sin\frac{1}{h} + (\tan h)^q$$

Which is same hence B, C, D. **Ans.**

147. $\lim_{h\to 0}\left(\frac{f(x+h)}{f(x)}\right)^{\frac{1}{h}} = e^{(\tan x)\, f(-x)}$ (Given)

$$e^{\lim_{h\to 0}\frac{1}{h}\left(\frac{f(x+h)-f(x)}{f(x)}\right)} = e^{(\tan x)\, f(-x)}$$

$$e^{\frac{f'(x)}{f(x)}} = e^{(\tan x)\, f(-x)}$$

$\therefore \quad \frac{f'(x)}{f(x)} = (\tan x)\, f(-x)$

Put $x = 0 \quad \Rightarrow \quad \frac{f'(0)}{f(0)} = 0$

$\therefore \quad f'(0) = 0$

$\therefore \quad f'(0) = 0$ **Ans.**

148. Given that $g^{-1}(x) = f(x) \Rightarrow x = g(f(x)) \Rightarrow 1 = g'(f(x))f'(x)$

$$\Rightarrow g'(f(x)) = \frac{1}{f'(x)}$$

$$\Rightarrow g''(f(x))\, f'(x) = \frac{-f''(x)}{(f'(x))^2}$$

$$\Rightarrow g''(f(x)) = \frac{-f''(x)}{(f'(x))^3}$$

149. As $g(x)$ is continuous at $x = \frac{\pi}{2}$, so

$$g\left(\frac{\pi}{2}\right) = \lim_{x\to\frac{\pi}{2}}\left(\frac{1}{1+[\sin x]}\right)^{\frac{2}{2x-\pi}} = (\text{exact } 1)^{\infty} = 1$$

$p^2 - 1 = 1 \quad \Rightarrow \quad p^2 = 2 \Rightarrow p = \pm\sqrt{2}$. **Ans.**

150. We have, $g(x) = \begin{cases} \dfrac{x^2}{2}, & \forall\ x \ge 1 \\ 2x^2 - 3x + \dfrac{3}{2}, & \forall\ x < 1 \end{cases}$

$\therefore$ g(x) is continuous $\forall\ x \in R$.

Now, $g'(1^-) = \lim\limits_{h \to 0}\left(\dfrac{g(1-h) - g(1)}{-h}\right) = \lim\limits_{h \to 0}\left(\dfrac{2(1-h)^2 - 3(1-h) + \dfrac{3}{2} - \dfrac{1}{2}}{-h}\right)$

$$= \lim_{h \to 0} \frac{2(1+h^2-2h) - 3 + h + 1}{-h} = \lim_{h \to 0} \frac{2h^2 - h}{-h} = \lim_{h \to 0}(1 - 2h) = 1$$

and $g'(1^+) = \lim\limits_{h \to 0} \dfrac{g(1+h) - g(1)}{h}$

$$= \lim_{h \to 0} \frac{\dfrac{(1+h)^2}{2} - \dfrac{1}{2}}{h} = \lim_{h \to 0} \frac{1 + h^2 + 2h - 1}{2h} = \lim_{h \to 0}\left(1 + \frac{h}{2}\right) = 1$$

Hence, $g'(1) = 1 \Rightarrow$ g(x) is continuous and differentiable $\forall\ x \in R$.

So, $L = 0, M = 0$

Hence, $L + M = 0 + 0 = 0$. **Ans.**

151. $\lim\limits_{x \to \infty} x\left(\dfrac{1}{e} - \left(\dfrac{x}{x+1}\right)^x\right)$

Let $x = \dfrac{1}{t}$

$$\therefore \quad \lim_{t \to 0} \frac{1}{t}\left(\frac{1}{e} - \left(\frac{1}{1+t}\right)^{1/t}\right) = \frac{1}{e}\lim_{t \to 0} \frac{(1+t)^{1/t} - e}{t \cdot (1+t)^{1/t}} = \frac{1}{e^2}\lim_{t \to 0} \frac{e^{\frac{\ln(1+t)}{t}} - e}{t}$$

$$= \frac{1}{e}\lim_{t \to 0} \frac{e^{\frac{\ln(1+t)}{t} - 1} - 1}{\left(\dfrac{\ln(1+t)}{t} - 1\right)} \cdot \left(\frac{\ln(1+t)}{t} - 1\right)$$

$$= \frac{1}{e}\lim_{t \to 0} \frac{\ln(1+t) - t}{t^2} = \frac{-1}{2e}. \textbf{ Ans.}$$

152. $\lim\limits_{x \to 1} \left(\dfrac{k}{\ln x} - \dfrac{k}{x-1} \right) = 2$

Put $x = 1 + h$

$$\lim_{h\to 0} \frac{k\,(h - \ln(1+h))}{h \ln(1+h)} = 2$$

$$\lim_{h\to 0} \frac{k\left[h - \left(h - \dfrac{h^2}{2} + ...\right)\right]}{h^2} = 2$$

$\Rightarrow \dfrac{k}{2} = 2. \Rightarrow k = 4$

153. $k(x) = \sqrt{1 + \operatorname{sgn} x} = \begin{cases} 0\,; & x < 0 \\ 1\,; & x = 0 \\ 2\,; & x > 0 \end{cases}$

So, $k(x)$ is discontinuous at $x = 0$. **Ans.**

154. Let $P(x) = Ax^3 + Bx^2 + Cx + D$

So, $P'(x) = 3Ax^2 + 2Bx + C$

Now, putting above result in $P(x) - 2P'(x) = 3x^3 - 27x^2 + 38x + 1$,

We have

$A = 3, B = -9, C = 2, D = 5.$

$\therefore$ $P(x) = (3x^3 - 9x^2 + 2x + 5)$

$P'(x) = 9x^2 - 18x + 2$

$P''(x) = 18x - 18$

So, $f(x) = \begin{cases} 3x; & x \neq \dfrac{\pi}{2} \\ \sin^{-1}(ab) + \cos^{-1}(a + b - 3ab); & x = \dfrac{\pi}{2} \end{cases}$

As, $f(x)$ is continuous at $x = \dfrac{\pi}{2}$, so

$$\frac{3\pi}{2} = \left(\sin^{-1}(ab) + \cos^{-1}(a + b - 3ab)\right)$$

$\therefore$ $ab = 1$ and $a + b - 3ab = -1$.

$\therefore$ $a = 1, b = 1$

Hence $(a + b) = 2$. **Ans.**

155. $f'(x)$ is continuous

$\Rightarrow$ $f(x)$ must be differentiable.

$\Rightarrow$ $f(x)$ must be continuous at $x = 0$.

$\therefore \quad f(0) = \lim_{x \to 0^+} f(x)$

$= \lim_{x \to 0^+} x^{\alpha} \cdot \sin\left(\frac{1}{x}\right) = 0$, if $\alpha > 0$ ....(1)

Also,

$f'(x) = \alpha\, x^{\alpha - 1} \sin\left(\frac{1}{x}\right) - x^{\alpha - 2} \cos \frac{1}{x}$

So, $\quad f'(x)$ exist at $x = 0$,

if $\quad \alpha > 2$ ....(2)

$\therefore \quad (1) \cap (2)$

$\Rightarrow \quad \alpha \in (2, \infty)$. **Ans.**

Paragraph for Question no.156 to 158

156. $x^3 + 2x^2 = x^2 (x + 2)$ therefore $f(x)$ is differentiable at $x = 0$ if x^2 is a factor of $(-x^3 + 2x^2 + ax)$

$\Rightarrow \quad a = 0$

157. $f(0) = f(-2) \quad \Rightarrow \quad$ many one function

$$\left.\begin{matrix} x^3 + 2x^2 = p \\ -x^3 + 2x^2 = p \end{matrix}\right\}$$ atleast one of the equation have solution $\forall\, p \in R$

$\Rightarrow$ onto function.

158. Domain of $f'(x)$ is a singlton set $\{0\}$.

$\therefore$ limit does not exist.]

159. $$f(x) = \lim_{n\to\infty} \frac{x^{2n} - 2\sin x^{2n}}{x^{2n} + 2\sin x^{2n}} = \begin{cases} 1, & |x|>1 \\ -\frac{1}{3}, & 0 \le |x| < 1 \\ \frac{1-2\sin 1}{2+2\sin 1}, & |x|=1 \end{cases}$$

$\therefore \quad$ number of points of discontinuity = 2.

(a) $$g(x) = \text{sgn}\,(x^2 - 1) = \begin{cases} 1, & |x| > 1 \\ -1, & |x| < 1 \\ 0, & |x| = 1 \end{cases}$$

$\therefore \quad$ number of points of non-differentiability = 2

(b) $$h(x) = \sin^{-1} \frac{2x}{1 + x^2} = \begin{cases} 2\tan^{-1} x, & -1 \le x \le 1 \\ \pi - 2\tan^{-1} x, & x > 1 \\ -\pi - 2\tan^{-1} x, & x < -1 \end{cases}$$

clearly h(x) is non-differentiable at x = 1, – 1.

$\therefore$ h(x) is non differentiable at two points.

(c) $k(x) = \cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x, & x \ge 0 \\ -2\tan^{-1}x, & x < 0 \end{cases}$

$\therefore$ k(x) is non differentiable at only one point.

(d) $\ell(x) = \text{sgn}(|x|-1) = \begin{cases} 1, & |x| > 1 \\ -1, & |x| < 1 \\ 0, & |x| = 1 \end{cases}$

$\therefore$ function $\ell(x)$ is non differentiable at x = 1, – 1.

$\therefore$ $\ell(x)$ is non differentiable at two points.]

Paragraph for question nos. 160 to 162

We have, $x = 5t - |t-2|,\ y = -3t + 2|t-1|\ \forall\ t \in R$

Case-I: $-\infty < t < 1$

Now, $x = 5t + (t-2) = 6t - 2 \Rightarrow t = \dfrac{x+2}{6}$

and $y = -3t - 2(t-1) = -5t + 2 \Rightarrow y = \dfrac{-5}{6}(x+2) + 2 = \dfrac{-5x+2}{6}$

As $t < 1 \Rightarrow \dfrac{x+2}{6} < 1 \Rightarrow x < 4$

Case-II: $1 \le t < 2$

Now, $x = 5t + (t-2) = 6t - 2 \Rightarrow t = \dfrac{x+2}{6}$ and $y = -3t + 2(t-1) = -t - 2$

$\Rightarrow y = -\dfrac{(x+2)}{6} - 2 = \dfrac{-x-14}{6}$

As $1 \le t < 2 \Rightarrow 1 \le \dfrac{x+2}{6} < 2 \Rightarrow 4 \le x < 10$

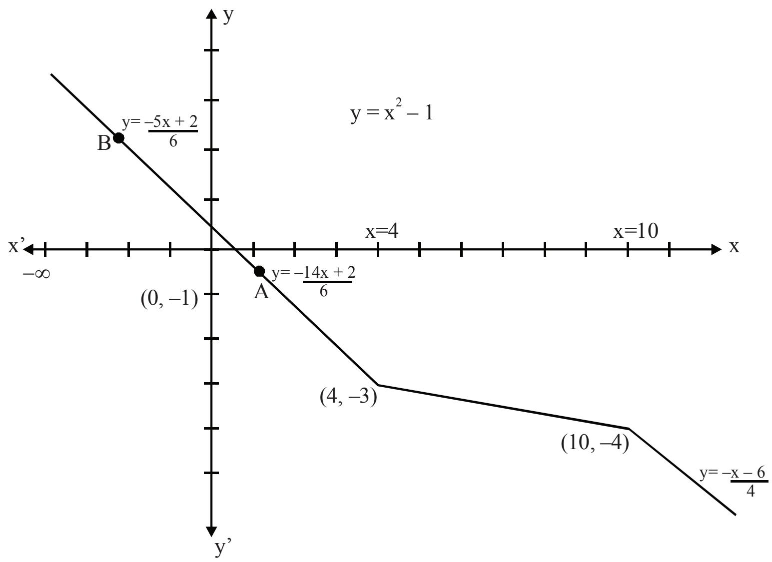
Case-III: $t \ge 2$

Now, $x = 5t - (t-2) = 4t + 2 \Rightarrow t = \left(\dfrac{x-2}{4}\right)$

and $y = -3t + 2(t-1) = -t - 2 \Rightarrow y = -\left(\dfrac{x-2}{4}\right) - 2 = \dfrac{-x-6}{4}$

As $t \ge 2 \Rightarrow \dfrac{x-2}{4} \ge 2 \Rightarrow x \ge 10$

$$\therefore\quad y = f(x) = \begin{cases} \dfrac{-5x+2}{6}; & -\infty < x < 4 \\ \dfrac{-14-x}{6}; & 4 \le x < 10 \\ \dfrac{-x-6}{4}; & 10 \le x < \infty \end{cases}$$



Graph of function y = f(x)

160. Clearly, from above graph,
f(x) is continuous everywhere on R but not derivable at x = 4 and x = 10.

161. Clearly, $f'(\pi) = \frac{-5}{6}$.

162. Clearly, from above graph, function f(x) intersects line y = 2 at exactly one point.
So, f(x) = 2 has exactly one solution. **Ans.**

163. [Sol. $f'(0) = \lim\limits_{n\to 0} \frac{e^{-1/h^2}\sin(1/h)}{h}$

$\lim\limits_{h\to 0} \frac{e^{-1/h^2}}{h}$ put h = 1/t

$= \lim\limits_{t\to\infty} \frac{e^{-t^2}}{1/t} = \lim\limits_{t\to\infty} \frac{t}{e^{t^2}}\left(\frac{\infty}{\infty}\right)$

$= \lim\limits_{t\to\infty} \frac{t}{2tet^2} = 0 \Rightarrow \lim\limits_{t-0} \frac{e^{-1/h^2}.\sin 1/h}{h} = 0 \times$ (a quantity oscillating between 1 & –1) = 0

hence function f(x) is differentiable at x = 0

Questions

1. Let m and n be number of integers in the domain and range of $f(x) = \sqrt{16 - x^2} + \cos^{-1}\left(\frac{x^2}{16}\right)$, then find the value of m + n.

2. Let P(x) be a polynomial of degree 3 such that P(x) + 2 is divisible by $(x - 2)^2$ and P(x) – 2 is divisible by $(x + 2)^2$. If the value of P(–3) is equal to $\frac{a}{b}$. (where a, b ∈ N) then find the minimum value of (a – b).

Paragraph for Question Nos. 3 to 5

If $f(x) = x^2 - 2|x|$

$$g(x) = \begin{cases} \text{minimum } \{ f(t) : -2 \le t \le x, & -2 \le x < 0\} \\ \text{maximum } \{ f(t) : 0 \le t \le x, & 0 \le x \le 3\} \end{cases}$$ then

3. The function y = |f(x)| is differentiable for

(A) $x \in R$ (B) $x \in R - \{0\}$ (C) $x \in R - \{0,2\}$ (D) None of these

4. Number of values of x in [–2, 3] where g(x) is non-derivable is

(A) 0 (B) 1 (C) 2 (D) 3

5. $\int_0^1 fog(x)\,dx$ is equal to

(A) 0 (B) $\frac{33}{2}$ (C) 21 (D) None of these

6. Let $f(x) = c_1 e^{2x} + c_2 e^{5x}$ satisfies the relation $f''(x) + a f'(x) + b f(x) = 0$ for any real number c_1, c_2. Find the value of $(a^2 + b^2)$.

7. If $f(x) = ln(lnx)$ where x > e

Prove that $\frac{1}{(m+1)ln(m+1)} < f(m+1) - f(m) < \frac{1}{m \cdot ln(m)}$ for m > e.

8. If minimum value of $f(x) = x^2\, ln\, x$ is p, then the value of $ln\,|2p|$ is equal to

(A) 1 (B) – 1 (C) $\frac{-1}{2e}$ (D) $\frac{1}{2}$

9. If the function $f(x) = 2\tan x + (2a + 1)\ln|\sec x| + (a - 2)x$ is increasing in $\left(0, \frac{\pi}{2}\right)$ then range of 'a' is equal to

(A) $(-\infty, 0]$ (B) [0, 1] (C) [0, 3] (D) $[0, \infty)$

10. Suppose that f(x) is defined and positive for all real numbers x. If $f(x)$ increases for $x<0$ and decreases for all $x>0$, and if $g(x)=\dfrac{1}{f\left(\dfrac{1}{x}\right)}$, then find the intervals of mononocity.

11. Find all possible values of 'a' for which $f(x)=\log_a(4ax-x^2)$ is monotonically increasing for every $x\in\left[\dfrac{3}{2},2\right]$.

12. If equation of tangent drawn to the curve y = f(x) at its point P(3, 5) is 5x – 4y + 5 = 0 and
$$\lim_{x\to3}\frac{\left(3^{4f(x)}-2(1+3+3^2+.......+3^{19})-1\right)^2}{1-\cos\left(ln(4-x)\right)} = 2\,(a\,3^b\,ln\,c)^2$$
where a and c are prime numbers and $b\in N$, then find the value of $(a+b+c)$.

13. The normal at the point $P\left(2,\dfrac{1}{2}\right)$ on the curve $xy=1$, meets the curve again at Q. If m is the slope of the curve at Q, then find $|m|$.

14. Find the number of solutions satisfying the equation $\begin{vmatrix} x^2+\sin x & \cos x & x(1+\sin x) \\ x+\cos x & & x+1 \end{vmatrix}=0$.

15. Let a, b, c be 3 real numbers $a<b<c$ and f(x) is continuous in [a, c] and differenitable in (a, b). Also f '' (x) > 0 in (a, c) prove that $(c-b)\,.\,f(a)+(b-a)\,f(c)>(c-a)\,f(b)$

16. If the exhaustive set of all possible values of c such that $f(x)=e^{2x}-(c+1)\,e^x+2x+\cos 2+\sin 1$, is monotonically increasing for all $x\in R$, is $(-\infty,\lambda]$, then find the value of λ.

17. A function f(x) which satisfies the relation $f(x)=e^x+\int_0^1 e^x f(t)\,dt$, then

(A) $f(0)<0$

(B) $f(x)$ is decreasing function

(C) $f(x)$ is an increasing function

(D) $\int_0^1 f(x)\,dx>0$

18. If the curves $y=\dfrac{(x-a)^2}{4}$ and $y=e^x$ touches each other then find the sum of the squares of all possible values of a.

19. Let $a_n\ (n\ge1)$ be the value of x for which $\int_x^{2x} e^{-t^n}\,dt\ \ (x>0)$ is maximum.

If $L = \lim\limits_{n\to\infty} ln(a_n)$ then find the value of e^{-L}.

20. If $f''(x) > 0$, $\forall x \in R$, $f'(3) = 0$

and $g(x) = f(\tan^2 x - 2\tan x + 4)$ $0 < x < \frac{\pi}{2}$ then g(x) is increasing in

(A) $\left(0, \frac{\pi}{3}\right)$ (B) $\left(0, \frac{\pi}{6}\right)$ (C) $\left(0, \frac{\pi}{4}\right)$ (D) None of these

21. If the point P(a, b) lies on the curve $9y^2 = x^3$ such that the normal to the curve at P makes equal intercepts with the axes. Find the value of (a + 3b).

[Hint : Normal makes equal intercept at P $\Rightarrow \left.\frac{dy}{dx}\right|_P = 1$]

22. Let $f: R \to R$ be a continuous function and let $g(x) = f(x)\int_0^x f(t)\,dt$ $\forall x \in R$. If $g(x) \le 0$, $\forall x$ and

$g(0) = 0$, then the value of $f\{g(\ln 5)\} + g\{f(\ln 5)\}$ is equal to

(A) 0 (B) 1 (C) 5 (D) 10

23. Let *f* be a continuous function on [a, b]. If $F(x) = \left(\int_a^x f(t)\,dt - \int_x^b f(t)\,dt\right)\big(2x - (a+b)\big)$ then there exist

some $c \in (a, b)$ such that

(A) $\int_a^c f(t)\,dt = \int_c^b f(t)\,dt$

(B) $\int_a^c f(t)\,dt - \int_c^b f(t)\,dt = f(c)(a + b - 2c)$

(C) $\int_a^c f(t)\,dt - \int_c^b f(t)\,dt = f(c)\big(2c - (a+b)\big)$

(D) $\int_a^c f(t)\,dt + \int_c^b f(t)\,dt = f(c)\big(2c - (a+b)\big)$

Paragraph for question nos. 24 to 26

Let

$$f(x) = \begin{cases} \max.\{t^3 - t^2 + t + 1, 0 \le t \le x\}, & 0 \le x \le 1 \\ \min.\ \{3 - t, 1 < t \le x\} & 1 < x \le 2 \end{cases}$$

$$\text{and } g(x) = \begin{cases} \max.\{\frac{3}{8}t^4 + \frac{1}{2}t^3 - \frac{3}{2}t^2 + 1, 0 \le t \le x\}, & 0 \le x < 1 \\ \min.\ \{\frac{3}{8}t + \frac{1}{32}\sin^2 \pi t + \frac{5}{8}, 1 \le t \le x\} & 1 \le x \le 2 \end{cases}$$

24. The function f(x), $\forall x \in [0, 2]$ is
(A) continuous and differentiable
(B) continuous but not differentiable
(C) discontinuous and not differentiable
(D) none

25. Which of the following is true?
(A) $\lim_{x\to 1^-} fog(x) > \lim_{x\to 1^+} gof(x)$
(B) $\lim_{x\to 1^-} fog(x) < \lim_{x\to 1^+} gof(x)$
(C) $\lim_{x\to 1^-} fog(x) = \lim_{x\to 1^+} gof(x)$
(D) none of these

26. Let $Z(x) = \frac{d}{dx}\left(f(x)^{g(x)}\right)$ and $Y(x) = \frac{d}{dx}\left(g(x)^{f(x)}\right)$ then Z(x) and Y(x) vanish simultaneously at
(A) $x = -\frac{1}{3}$
(B) $x = 0$
(C) $x = 1$
(D) No real value of x

27.

	Column-I		Column-II
(A)	If $f : [0, 1] \to [0, 1]$ is continuous, then the number of roots of the equation $f(x) = x^3$ is	(P)	one
(B)	If f is derivable, then between the consecutive roots of $f'(x) = 0$, the number of roots of $f(x) = 0$ is	(Q)	at least one
(C)	The number of values of x at which $f(x)= \|1 - \|x-1\|\|$ is not differentiable is	(R)	at most one
		(S)	more than one

28. Let $f(x) = (1 + x^2)\, e^x$. If n_1 and n_2 are the number of roots of the equation $(1 + x^2)\, e^x = k$ according as $k > 0$ or $k \le 0$ then
(A) $n_1 + n_2 = 2$
(B) $n_1 - n_2 = 1$
(C) $n_1 > n_2$
(D) $n_1 = n_2$

29. If f(x) is a thrice differentiable function and given that $f(1) = 1, f(2) = 8, f(3) = 27, f(4) = 64$, then
(A) $f''(x) = f'''(x) = 6\ \forall\ x \in (1, 4)$
(B) $f'(x) = f''(x) = f'''(x) = 6$ from some $x \in (1, 4)$
(C) $f'''(x) = 6\ \forall\ x \in (1, 4)$
(D) $f'''(x) = 6$, for some $x \in (1, 4)$

30. The cost of fuel in running an engine is proportional to the square of the speed and is Rs. 48 per hour for a speed of 16 km/hr. Other costs amounts to Rs. 300 per hour. The most economical speed (in km/hr), is
(A) 30
(B) 32
(C) 40
(D) 24

31. Let $f(x) = ax^2 + bx + c$ where $|f(x)| \le 1\ \forall\ x \in [0, 1]$. The maximum value of $|a| + |b| + |c|$, is
(A) 1
(B) 10
(C) 17
(D) None

32. Find the sum of all integral values of a such that $a(x^2 + x - 1) \le (x^2 + x + 1)^2\ \forall\ x \in R$.

33. If f is a continuous function defined for all real numbers x and if the maximum value of f(x) is 5 and the minimum value of f(x) is – 7, then which of the following must be true?

(i) The maximum value of $f(|x|)$ is 5.

(ii) The maximum value of $|f(x)|$ is 7.

(iii) The minimum value of $f(|x|)$ is 0.

(A) (i) only (B) (ii) only (C) (i) & (ii) only (D) (ii) & (iii) only

Paragraph for question nos. 34 to 36

Let $f(x) = \int_0^1 ((12x + 20y) f(y) + 8) xy \, dy$

34. The maximum value of $f(x)$ is

(A) 8 (B) $\frac{1}{8}$ (C) 4 (D) $\frac{1}{4}$

35. The number of solution of the equation $|f(|x|)| = 2$ is

(A) 0 (B) 2 (C) 4 (D) 8

36. The range of $f(-2^x)$ is

(A) $(-\infty, 0)$ (B) $(0, \infty)$ (C) $\left(-\infty, \frac{1}{8}\right]$ (D) $\left[\frac{1}{4}, \infty\right)$

37. A line is tangent to the curve $f(x) = \frac{41x^3}{3}$ at the point P in the first quadrant, and has a slope of 2009. This line intersects the y-axis at (0, b). Find the value of 'b'.

38. A curve is given by the equations $x = at^2$ & $y = at^3$. A variable pair of perpendicular lines through the origin 'O' meet the curve at P & Q. Show that the locus of the point of intersection of the tangents at P & Q is $4y^2 = 3ax - a^2$.

39. A straight line is drawn through the origin and parallel to the tangent to a curve $\frac{x + \sqrt{a^2 - y^2}}{a} = ln\left(\frac{a + \sqrt{a^2 - y^2}}{y}\right)$ at an arbitary point M. Show that the locus of the point P of intersection of the straight line through the origin & the straight line parallel to the x-axis & passing through the point M is $x^2 + y^2 = a^2$.

40. If the tangent at the point (x_1, y_1) to the curve $x^3 + y^3 = a^3$ $(a \neq 0)$ meets the curve again in (x_2, y_2) then show that $\frac{x_2}{x_1} + \frac{y_2}{y_1} = -1$.

41. For all $x \in (0, 1)$
(A) $e^x < 1 + x$ (B) $\log_e(1+x) < x$ (C) $\sin x > x$ (D) $\log_e x > x$

42. Let $f(x) = \int e^x (x-1)(x-2)\, dx$ then f decreases in the interval
(A) $(-\infty, 2)$ (B) $(-2, -1)$ (C) $(1, 2)$ (D) $(2, +\infty)$

Paragraph for question nos. 43 to 45

Let f be a differentiable function on R such that $f'(x) = f(x) + \int_0^2 f(x)\, dx$, $f(0) = 1$

43. Number of points of intersection of the graph of $y = f(x)$ with x-axis is equal to
(A) 0 (B) 1 (C) 2 (D) 3

44. Let $g : R \to R$ be a function defined as $g(x) = \begin{cases} (4-e^2)f(x) + e^2 - 1, & x \ge 0 \\ x, & x < 0 \end{cases}$ then $g(x)$ is
(A) one one into function (B) one one onto function
(C) many one onto function (D) many one into function

45. The number of points of intersection of graph $y = g(x)$ and $y = x^3$, is
(where $g(x)$ is function as in above question)
(A) 4 (B) 3 (C) 2 (D) 1

46. Let $f''(x) > 0\ \forall x \in R$ and $g(x) = 2f\left(\frac{x^2}{2}\right) + f(6 - x^2)$. Then
(A) g attains maximum at $x = \pm 1$. (B) g attains minimum at $x = \pm 1$.
(C) g attains maximum at $x = 0$. (D) g attains minimum at $x = \pm 2$.

47. The number of integral values of 'a' for which the function $f(x) = \begin{bmatrix} x^3 - x^2 + 10x - 5, & x \le 1 \\ -2x + \log_2(a^2 - 2), & x > 1 \end{bmatrix}$

has the greatest value at $x = 1$, is
(A) 8 (B) 10 (C) 15 (D) 20

48. Let $f : [0, \infty) \to R$ be a continuous strictly increasing function such that $f^4(x) = \int_0^x t^2 f^3(t)\, dt$.

If $f(0) = \frac{-1}{4}$, then the value of $f(3)$ is

(A) 1 (B) 2 (C) 3 (D) 4

49. Find all the lines that pass through the point (1, 1) and are tangent to the curve represented parametrically as $x = 2t - t^2$ and $y = t + t^2$.

50. Let $x(t) = \cos^3 t$ and $y(t) = \sin^2 t$ defines a curve where t is a parameter. This curve passes through the point $\left(\frac{1}{8}, \frac{3}{4}\right)$ at some $t = t_0 \in \left(0, \frac{\pi}{2}\right)$. If the absolute value of the slope of the curve at that point is $\frac{p}{q}$ (where p and q are in their lowest form). Find (p + q)

51. Find sum of the intercepts of the tangent at any point to the curve $x^{\frac{1}{2}} + y^{\frac{1}{2}} = 20$, on the coordinate axes.

52. For constant number 'a', consider the function $f(x) = ax + \cos 2x + \sin x + \cos x$ on R (the set of real numbers) such that $f(u) < f(v)$ for $u < v$. If the range of 'a' for any real numbers u, v is $\left[\frac{m}{n}, \infty\right)$, then find the minimum value of $(m + n)$.

Paragraph for question nos. 53 to 56

Consider the function $f(x) = |\sin x| + [\sin x]$, $x \in [-\pi, \pi]$, where [ ] denotes greatest integer function.

53. The number of discontinuities of $f(x)$ is
(A) 1 (B) 2 (C) 3 (D) 4

54. The number of solutions of the equation $(f(x))^2 = 1$ is
(A) 0 (B) 1 (C) 2 (D) 3

55. The number of points of maxima of $f(x)$ is
(A) 0 (B) 1 (C) 2 (D) 3

56. $\int_{-\pi/2}^{\pi/2} f(x)\,dx$ equals
(A) 1 (B) 2 (C) $1 + \frac{\pi}{2}$ (D) $2 - \frac{\pi}{2}$

Paragraph for question nos. 57 to 59

If a continuous function f defined on the real line R, assumes positive and negative values in R then the equation $f(x) = 0$ has a root in R. For example, if it is known that a continuous function f on R is positive at some point and its minimum value is negative then the equation $f(x) = 0$ has a root in R.
Consider $f(x) = |\ln x| - px$, where p is a real constant.

57. The line $y = px$ meets the curve $y = |\ln x|$ for $p \geq 1$ at
(A) no point (B) one point (C) two point (D) three point

58. The number of values of p for which $f(x) = 0$ has exactly one real root is

(A) 1 (B) 2 (C) 3 (D) Infinite

59. The value of p for which $|\ln x| - px = 0$ has two distinct real roots is

(A) $\frac{1}{e}$ (B) 1 (C) e (D) 0

60. If $f(x) = \left(\frac{a^2 - 1}{3}\right)x^3 + (a - 1)x^2 + 2x + 1$ is monotonic increasing for every $x \in R$ then find the range of values of 'a'.

61. Construct the graph of the function $f(x) = -\left|\frac{x^2 - 9}{x + 3} - x + \frac{2}{x - 1}\right|$ and comment upon the following

(a) Range of the function,
(b) Intervals of monotonocity,
(c) Point(s) where f is continuous but not diffrentiable,
(d) Point(s) where f fails to be continuous and nature of discontinuity.
(e) Gradient of the curve where f crosses the axis of y.

62. Find the difference between the global maximum and global minimum values of the function $f(x) = 1 + 12\,|x| - 3x^2$ defined on $[-2, 5]$.

63. If $9 + f''(x) + f'(x) = x^2 + f^2(x)$, where f(x) is twice differentiable function such that f "(x)¹ 0"x Î R and let P be the point of maxima of f(x) then find the number of tangents which can be drawn from P to the circle $x^2 + y^2 = 9$.

Paragraph for question nos.64 to 66

If $y = f(x)$ differentiable $\forall\, x \in R - \{0\}$ and $y = g(x)$ differentiable $\forall\, x \in R - \{0\}$ are two curves such that they pass through (1, 1) and (2, 3) respectively. Also tangents to the two curves where their abscissa are same intersect on y-axis and normals to the curves at the point where their abscissa are equal intersect on x-axis.

64. The curve f(x) is

(A) $\frac{2}{x} - x$ (B) $2x^2 - \frac{1}{x}$ (C) $\frac{2}{x^2} - x$ (D) none of these

65. The curve g(x) is

(A) $x - \frac{1}{x}$ (B) $x + \frac{2}{x}$ (C) $x^2 - \frac{1}{x^2}$ (D) none of these

66. The number of positive integral solutions of f(x) = g(x) are

(A) 4 (B) 5 (C) 6 (D) none of these

Paragraph for question nos. 67 to 69

Consider the function $y = ax^4 - bx^3$

67. If out of the points (0,0) and (3,–27) one is point of inflection and one the point of local minima then the other point of inflection will have the x-co-ordinate as 'c' then a + b + c has the value.
(A) 5 (B) 6 (C) 7 (D) 8

68. If the function $y = cx^3 - (b-a)^2 Kx^2 + b(c+a)K^2 x + a$ has local minimum at $x = x_1$ and local maximum at $x = x_2$ such that $x_1 = x_2^2$ (x_1 , x_2 are distinct) then K equals.
(A) a (B) b (C) c (D) none

69. $\displaystyle\int \frac{dx}{(x+a)^{\frac{bc}{a+b+c}}(x-b)^{\frac{b+c}{a+b+c}}}$

(A) $\frac{7}{5}\left(\frac{x-4}{x+1}\right)^{1/7} + C$ (B) $\frac{7}{5}\left(\frac{x+1}{x-4}\right)^{1/7} + C$

(C) $\frac{5}{7}\left(\frac{x+1}{x-4}\right)^{1/7} + C$ (D) $\frac{5}{7}\left(\frac{x-4}{x+1}\right)^{1/7} + C$

Paragraph for question nos. 70 to 72

The function $f(x) = \sqrt{ax^3 + bx^2 + cx + d}$ has its non-zero local minimum and local maximum values at -2 and 2 respectively. Given 'a' is root of the equation $x^2 - x - 6 = 0$.

70. The value of $(a + b + c)$ is equal to
(A) 16 (B) 18 (C) 20 (D) 22

71. The roots of the equation $ax^2 + bx + c = 0$
(A) are opposite in sign (B) are imaginary
(C) are both positive (D) are both negative.

72. The smallest positive integral value of d is equal to
(A) 1 (B) 32 (C) 33 (D) 34

73. Let $y = 2x + c$ is tangent to the curve $y = \sin^{-1} x$, then the possible values of c is

(A) $\frac{\pi}{3} - \sqrt{3}$ (B) $\frac{\pi}{3} + \sqrt{3}$ (C) $-\frac{\pi}{3} + \sqrt{3}$ (D) $-\frac{\pi}{3} - \sqrt{3}$

74. If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, where $a > 0$, attains its maximum and minimum value at p and q respectively such that $p^2 = q$, then find the value of a.

75. Let C_1 be the graph of the curve represented by the equation $x^2 - 13x + 4y = 1$. If C_2 is the new curve obtained when C_1 is reflected in the origin, then the equation of C_2 is

(A) $x^2 - 13x - 4y = 1$ (B) $x^2 + 13x + 4y = 1$
(C) $x^2 - 13x - 4y = -1$ (D) $x^2 + 13x - 4y = 1$

76. Prove that the equation $\dfrac{x^3+1}{x^2+1} = 5$ has no root in [0, 2]

77. Let $f(x) = 2x^3 + 3x^2 - 12x - 4$. If domain and range of f(x) is R– {a} and R respectively then the true set of value of a is [m, n]. Find the value of $4(n - m)$, (where $a, m, n \in R$).

78. Let P(x) be a polynomial function of degree n satisfying the equation

$(P(x))^2 \cdot P'''(x) = (P''(x))^3\ P'(x)\ \forall\ x \in R$ and f(x) is a polynomial of degree same as that of P(x). Also $f'(0) = f'(1) = f'(-1) = 0$, $f(0) = 4$, $f(1) = 3 = f(-1)$. If $f(x) = k$ has four distinct real solution then find the number of integral values of k.

79. Let $f(x) = \begin{cases} x^3 - x^2 + 10x - 5; & x \le 1 \\ -2x + \log_2(b^2 - 1); & x > 1 \end{cases}$.

If f(x) has greatest value at x = 1, then the largest integral value of b is

(A) 9 (B) 10 (C) 11 (D) 12

80. Let $f(x) = \displaystyle\int_2^x \frac{dt}{\sqrt{x^2+x+3}}$. If g(x) is the inverse of f(x) then g'(0) has value equal to

(A) 1 (B) 2 (C) 3 (D) 4

Paragraph for question nos. 81 to 83

Let f be a cubic polynomial function such that $f(x) + f(-x) = 0\ \forall\ x \in R$ and $\displaystyle\lim_{h \to \frac{1}{\sqrt{3}}} \frac{f'(h) - 2}{h - \frac{1}{\sqrt{3}}} = 4\sqrt{3}$.

81. The value of $f'\left(\dfrac{1}{\sqrt{6}}\right)$ equals

(A) 1 (B) 2 (C) 3 (D) 6

82. If $\phi(x) = \displaystyle\int_1^x e^t\ f(x+1-t)\,dt + \frac{5}{4}$ then $\phi'(1)$ equals

(A) $\dfrac{2}{e}$ (B) 2e (C) $\dfrac{1}{e}$ (D) e

83. If $g\circ f(x) = x\ \forall\ x \in R$ and $g(2) = a$, $g(16) = b$ then the value of $\int_a^b f(x)\,dx + \int_2^{16} g(x)\,dx$ equals

(A) 14 (B) 28 (C) 30 (D) 32

84. Let x, y, k be real number such that $8k^2\left(\frac{x^2}{y^2}+\frac{y^2}{x^2}\right)+4k\left(\frac{x}{y}+\frac{y}{x}\right)=15$. If the maximum value of k is M, find 100M.

85. The maximum value of $f(x)=\frac{1+\cos x}{1+\cos x+\cos^2 x}$ is

(A) $\frac{1}{2}$ (B) 1 (C) 2 (D) $\frac{3}{4}$

86. Which of the following are correct?

(A) $x^4 + 2x^2 - 6x + 2 = 0$ has exactly two solutions

(B) $x^5 + 5x + 1 = 0$ has exactly one solutions.

(C) $x^n + ax + b = 0$ where n is an even natural number has atmost two solution $a, b \in R$.

(D) $x^3 - 3x + c = 0$, $c > 0$ does not have two solutions for $x \in (0, 1)$.

87. For a certain curve $\frac{d^2y}{dx^2} = 6x - 4$ and y has a local minimum value 5 when x = 1. If the global maximum and minimum values of y in [0, 2] is M and m respectively, then find the value of M + m.

88. Let $f(x) = \sin^{-1}x + \cos^{-1}x^2 + \sin^{-1}x^3 + \cos^{-1}x^4 + ... + \sin^{-1}x^{2n-1} + \cos^{-1}x^{2n}$
and $g(x) = \cos^{-1}x + \sin^{-1}x^2 + \cos^{-1}x^3 + \sin^{-1}x^4 + ... + \cos^{-1} x^{2n-1} + \sin^{-1}x^{2n}$.
where $n \in N$
If sum of least value of f(x) and greatest value of g(x) be 2010π then the value of n will be

89. The complete set of non-zero values of k such that the equation $|x^2 - 10x + 9| = kx$ is satisfied by atleast one and atmost three values of x, is

(A) $(-\infty, -16] \cup [4, \infty)$ (B) $(-\infty, -16] \cup [16, \infty)$

(C) $(-\infty, -4] \cup [4, \infty)$ (D) $(-\infty, 4] \cup [16, \infty)$

90. The coordinates of the two points on the curve $y = x^4 - 2x^2 - x$ that have a common tangent line, can be

(A) (1, –2) (B) (0, 0) (C) (–1, 0) (D) (2, 1)

91. The range of real constant 't' such that $(1 - t)\sin\theta + t\tan\theta > \theta$ always holds $\forall\ \theta \in \left(0, \frac{\pi}{2}\right)$ is

(A) $[1, \infty)$ (B) $\left[\frac{1}{2}, \infty\right)$ (C) $\left[\frac{1}{3}, \infty\right)$ (D) None

92. Let $y = f(x) = \int_0^x \left(\sqrt[3]{8t^3 + mt^2} - nt\right)dt$. If the line y = x is tangent at infinity to the curve y = f(x) then which of the following is / are true?
(A) $m = 6n$ (B) $m^2 + n^2 = 148$ (C) $m + n = 14$ (D) $|m - n| = 10$

93. If the tangent at a point P_1 (other than (0,0)) on the curve $ax^3 - y + b = 0$ meets the curve again at P_2. The tangent at P_2 meets the curve at P_3 and so on . If the abscissae of P_1, P_2, P_3 .....P_n form a G.P. then (a, b) may be
(A) (1, 0) (B) (2 , 7) (C) (3, 5) (D) (4, 9)

94. If $f(x) = |x - e|^{\pi} + |x - \pi|^{e}$, then the number of values of x at which f''' (x) does not exist is
(A) 0 (B) 1 (C) 2 (D) 3

95. If the value of

$\int_0^{2\pi} \text{Min.}\left(|x - \pi|, \cos^{-1}(\cos x)\right)dx$ is $k\pi^2$, then k is equal to

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{8}$ (D) 1

96. Let f be a continuous function satisfying the equation $\int_0^x f(t)dt + \int_0^x t\,f(x - t)dt = e^{-x} - 1$.

Find the numbers of integers in the range of f(x) in the interval [–10, 10].

97. Let f be a real valued function defined on R (the set of real numbers) such that
$f'(x) = 100\,(x - 1)\,(x - 2)^2\,(x - 3)^3$ .... $(x - 100)^{100}$, for all $x \in R$.

If g is a function defined on R such that $\int_a^x e^{f(t)}dt = \int_0^x g(x - t)dt + 2x + 3$ and the sum of all the values of x for which g (x) has a local extremum is 500λ, then find the value of λ.

98. Let f(x) & g(x) be differentiable for $0 \le x \le 1$, such that $f(0) = 2$, $g(0) = 0$, $f(1) = 6$. If there exist a real number c in [0, 1] such that $f'(c) = 2\,g'(c)$, then the value of $g(1)$ must be :
(A) 1 (B) 2 (C) –2 (D) –1

99. Let $f(x) = ||x^2 - 4x + 3| - 2|$. Which of the following is/are correct?
(A) f(x) = m has exactly two solutions of different sign $\forall\ m > 2$.
(B) f(x) = m has exactly two solutions $\forall\ m \in (2, \infty) \cup 0$.
(C) f(x) = m has no solutions $\forall\ m < 0$.
(D) f(x) = m has four distinct solutions $\forall\ m \in (0, 1)$.

100. If $f(x)=\left(\sqrt{4-x^2}-3\right)^2+\left(\sqrt{4-x^2}+1\right)^3$, then the maximum value of $f(x)$, is

(A) 25 (B) 28 (C) 36 (D) 40

101. If $c \in [0, 1]$ then the minimum value of $\int_0^{\frac{4\pi}{3}} |\sin x - c|\,dx$ occurs when c is equal to

(A) $\frac{1}{4}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$

102. Let $f(x)=\int_0^x 3^t(3^t-4)(x-t)\,dt$ $(x \ge 0)$. If $x = a$ is the point where $f(x)$ attains its local minimum value then find the value of 3^a.

103. $f(x)=\frac{x^2-1}{x^2+1}$, for every real number x, then the minimum value of f

(A) does not exist because f is unbounded
(B) is not attained even though f is bounded
(C) is equal to 1
(D) is equal to – 1.

Paragraph for Question no. 104 to 106

Let f and g be two functions defined as

$$f(x)=\begin{cases} |2x-3|-|x+1|, & -1 \le x < 0 \\ \ln(\{2x\}+1), & 0 \le x \le 2 \end{cases}$$

and $g(x) = 3 - 2f(3 - 2x)$.
[**Note:** $\{y\}$ denotes the fractional part function of y.]

104. The number of integers in the range of f(x) is

(A) 3 (B) 4 (C) 5 (D) 6

105. The number of solution(s) of the equation $4x^2 - 4x + 1 - f(x) = 0$ is(are)

(A) 2 (B) 3 (C) 4 (D) 5

106. The range of g(x) is

(A) $[0, \ln 2) \cup (2, 5]$
(B) $[-10, -4) \cup (-2\ln 2, 0]$
(C) $[-7, -1) \cup (3 - 2\ln 2, 3]$
(D) $[-5, -2) \cup (-\ln 2, 0]$

107. If the range of all real values of b for which the function $f(x) = (b^2 - 3b + 2)(\cos^2 x - \sin^2 x) + (b-1)x + \sin 2$ does not possess any critical points on R is (b_1, b_2), then find the value of $(b_1 + b_2)$.

Paragraph for question nos. 108 to 110

Let $f(x) = 2x^3 - 6x^2 - 18x + a$, $(a \in R)$ is cubic polynomial for all real value of x.

108. The interval of x in which f (x) is injective is

(A) $\left(\frac{-9}{7}, 0\right)$ (B) $\left(0, \frac{14}{3}\right)$ (C) (– 1, 3) (D) [– 1, 4)

109. The maximum value of 'a' for which f (x) has minimum value which is less than or equal to – 40 in (0, 4) is

(A) 12 (B) 14 (C) 15 (D) 17

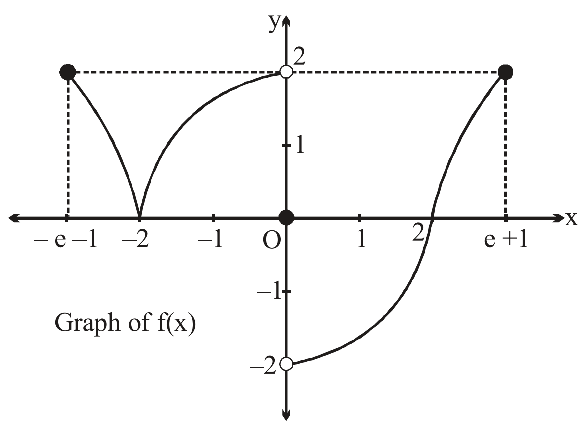
110. If $g(x) = f'(x)$ and $h(x) = |g(|x|)|$, then the number of points where h'(x) does not exist, is/are.

(A) 0 (B) 2 (C) 3 (D) 4

Paragraph for questions nos. 111 to 113

$$\text{Let } f(x) = \begin{cases} 2\ln(-x-1), & -e-1 \le x \le -2 \\ \frac{1}{2}(4-x^2), & -2 < x < 0 \\ 0, & x = 0 \\ \frac{1}{2}(x^2-4), & 0 < x < 2 \\ 2\ln(x-1), & 2 \le x \le e+1 \end{cases}$$

and graph of f(x) is as shown.



$$\text{Also, } g(x) = \begin{cases} \min\{f(t) : -e-1 \le t \le x\}, & -e-1 \le x < 0 \\ \max\{f(t) : 0 \le t \le x\}, & 0 \le x \le e+1 \end{cases}$$

111. Which one of the following statement **does not** hold good?

(A) Range of g(x) is [0, 2]. (B) g(x) is non-monotonic in [–2, 2].
(C) g(x) is a continuous function. (D) g(x) is an odd function.

112. If $x = \alpha$ is the point of non-differentiability of g(x) in $(-e-1, e+1)$ then the values of α is

(A) – 2, 1 (B) – 2, – 1 (C) – 2, 2 (D) – 2, 0

113. If the equation $g(x) = k$ has exactly two distinct solutions in $[-e-1, e+1]$ then the sum of all possible integral values of k is

(A) 0 (B) 1 (C) 2 (D) 3

114. The function $f(x) = \int_{-1}^{x} t(e^t - 1)(t-1)(t-2)^3 (t-3)^5 dt$ has a local minimum at x =

(A) 0 (B) 1 (C) 2 (D) 3

115. Let

$$f(x) = \int_{-2}^{x} (t^2 - t + 2)(t^2 - t - 2)(t^2 - t - 6)(t^2 - t - 12)\, dt$$

for all $x \in R$, then which of the following statements is (are) correct?

(A) The equation of normal to f(x) at x = –2 is x = –2
(B) f(x) increases in $(-3, -2) \cup (-1, 2) \cup (3, 4)$
(C) f(x) decreases in $(-\infty, -3) \cup (-2, -1) \cup (2, 3) \cup (4, \infty)$
(D) The sum of values of x at which f(x) has local maximum equals –1

116. Let $f(x) = \cot x - \tan x - 2\tan 2x - 4\tan 4x - 8\cot 8x$, $x \neq \frac{n\pi}{8}$, $n \in I$ and $g(x) = x^3 + 6x - 1$.

One more function h (x) is defined as $h : R - \left\{\frac{n\pi}{8}, n \in I\right\} \to R$, $h(x) = f(x) + g(x)$ then identify the correct statement(s).

(A) $h''\left(\frac{\pi}{24}\right) = \frac{\pi}{4}$
(B) h (x) is odd function
(C) h (x) is increasing in the domain.
(D) If the equation $h(x) = \lambda$ has a solution in (0, 3) then number of integral values of λ is 7.

117. Suppose $f(x) = x^3 + ax^2 + bx + c$ satisfies $f(-2) = -10$ and takes the extreme value $\frac{50}{27}$ at $x = \frac{2}{3}$. Find (a + b + c).

118. For $a>0$, find the minimum value of the integral $\int_0^{1/a}(a^3+4x-a^5x^2)e^{ax}\,dx$.

119. If the maximum area of the region enclosed by the curves $y=|x|e^{|x|}$ and the line $y=a$ $(0\le a\le e)$ in $x\in[-1,1]$ is A, then find the value of [A].
[**Note:** [k] denotes greatest integer function less than or equal to k.]

120. If the acute angle between the tangents drawn from (0, 1) to the graph of $f(x) = x^3 - 3x^2 + 3x$ is $\tan^{-1}\left(\frac{m}{n}\right)$, where m and n are co-prime natural number. Find the value of $(m+n)$.

Paragraph for question nos. 121 to 123

Let $f(x)=e^{(p+1)x}-e^x$ for real number $p>0$.

121. The value of $x=s_p$ for which $f(x)$ is minimum, is

(A) $-\frac{\ln(p+1)}{p}$ (B) $-\ln(p+1)$ (C) $-\ln p$ (D) $\ln\left(\frac{p+1}{p}\right)$

122. Let $g(t)=\int_t^{t+1}f(x)e^{t-x}\,dx$. The value of $t=t_p$, for which g (t) is minimum is

(A) $-\ln\left(\frac{e^p-1}{p}\right)$
(B) $-\frac{1}{p}\ln\left(\frac{e^p-1}{p}\right)$
(C) $-\frac{1}{p}\ln\left(\frac{(p+1)(e^p-1)}{p}\right)$
(D) $-\ln\left((p+1)(e^p-1)\right)$

123. Use the fact that $1+\frac{p}{2}\le\frac{e^p-1}{p}\le1+\frac{p}{2}+p^2$ $(0<p\le1)$, the value of $\lim_{p\to0^+}(s_p-t_p)$ is

(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) non existent

124. If the line $y=ax$ and the curve $y=x^3-3x^2+2x$ intersects at the point other than origin then the range of a is

(A) $\left(-\infty,\frac{1}{4}\right]$ (B) $\left[\frac{1}{4},\infty\right)$ (C) $\left[\frac{-1}{4},\infty\right)$ (D) $\left(-\infty,\frac{-1}{4}\right]$

125. Consider a function $f(x)=\int_{-4\pi^2}^{x^2}\frac{\sin\sqrt{\theta}}{1+\cos^2\sqrt{\theta}}\,d\theta$ then which of the following is(are) **correct**?

(A) f(x) has a local minima at $x=2\pi$.
(B) f(x) has a local maxima at $x=\pi$
(C) f(x) has a local minima at $x=0$
(D) f(x) has a local maxima at $x=-\pi$

126. For the function $f(x)=\dfrac{\sqrt{x^2+1}}{x-1}$ which of the following hold(s) good?

(A) f has a local maxima but no local minima.
(B) f has 2 asymptotes
(C) There are exactly two integral values which are not in the range of the function.
(D) f is continuous and differentiable everywhere in its domain.

127. Consider the curve given by the equation $y^2 = x^2 + 33$ (where $y > 0$). The y-intercept of the normal line drawn to the curve at $x = 4$ is

(A) 7 (B) 12 (C) 14 (D) 21

128. Let $P(x)$ be a polynomal of degree 4 such that $P(1) = 7$ and attains its minimum value 3 at both $x = 2$ and $x = 3$, then the vale of $P(5)$ is

(A) 25 (B) 39 (C) 47 (D) 55

129. Let $f(x) = \cos^3 x - \lambda \cos^2 x,\ \ 0 < x < \pi,$ then number of integral values of λ so that $f(x)$ has exactly one minimum and exactly one maximum, is

(A) 0 (B) 1 (C) 2 (D) 3

130. Let $h(x)=\dfrac{g^2(x)}{2}+3x^3-5$, where $g(x)$ is a continuous and differentiable function. It is given that $h(x)$ is a monotonically increasing function and $g(0) = 4$, then

(A) $g^2(1) > 10$ (B) $h(5) > 3$ (C) $h\left(\dfrac{5}{2}\right) < 2$ (D) $g^2(-1) > 22$

131. Let a function $f:\left[\dfrac{\pi}{2},\pi\right]\to R$ be defined as $f(x)=\displaystyle\int_{\frac{\pi}{2}}^{x}(\cos t-2\sin t)\,dt$. The greatest value of f(x) is

(A) 0 (B) $\dfrac{\pi}{4}$ (C) $\dfrac{\pi}{3}$ (D) $\dfrac{\pi}{2}$

132. If the function $f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$ has a positive point of maxima then find the largest natural value of 'a'.

133. Let f(x) be a polynomial one-to-one function such that $f(x) f(y) + 2 = f(x) + f(y) + f(xy)\ \forall\ x, y \in R - \{0\}$.

Also $f(1) \neq 1$, $f'(1) = 3$. If $g(x)=\dfrac{x}{4}\left(f(x)+3\right)-\displaystyle\int_0^x f(t)\,dt$. Then which one of the following is **correct**?

(A) $g(x) = 0$ has exactly one root in (0, 1) (B) $g(x) = 0$ has exactly two root in (0, 1)
(C) $g(x) \neq 0\ \forall\ x \in R - \{0\}$ (D) $g(x) = 0\ \forall\ x \in R - \{0\}$

Paragraph for Question Nos. 134 to 136

Let $f(x) = \begin{cases} 1-x & \text{if } 0 \le x \le 1 \\ 0 & \text{if } 1 < x \le 2 \\ (2-x)^2 & \text{if } 2 < x \le 3 \end{cases}$ and $F(x) = \int_0^x f(t)\,dt$ then

134. The function F (x) is
(A) increasing in (0, 1) and decreasing in (2, 3)
(B) decreasing in (0, 1) and increasing in (2, 3)
(C) decreasing in $(0, 1) \cup (2, 3)$
(D) increasing in $(0, 1) \cup (2, 3)$

135. Range of F (x) is
(A) $\left[0, \frac{1}{2}\right]$
(B) $\left[0, \frac{1}{3}\right]$
(C) $\left[0, \frac{5}{6}\right]$
(D) [0, 1]

136. Area enclosed by the curve y = F (x) and x-axis as *x* varies from 0 to 3, is
(A) $\frac{15}{12}$
(B) $\frac{17}{12}$
(C) $\frac{13}{12}$
(D) $\frac{19}{12}$

137. Let $g(x) = \begin{vmatrix} \cos x & x & 1 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$ then the equation of normal drawn to $g(x)$ at $x = 0$, is
(A) x = 0
(B) y = 0
(C) x = 1
(D) y = 1

138. Consider the three linear equations with respect to X, Y, Z as
$(x)X + Y + 2Z = 0$
$(f(x))\ X + 3Y + x^2Z = 0$
$(5x)X + 6Y + Z = 0$
having non-trivial solution for X, Y, Z. The curve f(x)
(A) is always increasing
(B) is always decreasing
(C) has exactly one critical point
(D) has three critical points

139. The number of critical points on the graph of the function

$f(x) = \begin{cases} x^{3/5}, & \text{if} \quad x \le 2 \\ -(x-3)^3 & \text{if} \quad x > 2 \end{cases}$ is
(A) 1
(B) 2
(C) 3
(D) 4

140. The normal to the curve $x = a(\cos\theta + \theta\sin\theta)$, $y = a(\sin\theta - \theta\cos\theta)$ at any θ is such that
(A) it makes constant angle with x-axis
(B) it is a constant distance from the origin

(C) it touches circle $(x-\sin\theta)^2+(y-\cos\theta)^2=a^2$
(D) it passes through origin

141. If $f(x)=\int_{x}^{x^2}(t-1)\,dt$, $x\in[1,2]$, then global maximum value of $f(x)$ is

142. If the minimum value of the definite integral $\int_0^{\pi/2}|x\sin t-\cos t|\,dt$ $(x>0)$ can be expressed as $\sqrt{a}-\sqrt{b}$ then find the value of $(a+b)$.

143. If $f(x) = a|\cos x| + b|\sin x|$ $(a, b\in R)$ has a local minimum at $x=\frac{-\pi}{3}$ and satisfies $\int_{-\pi/2}^{\pi/2}(f(x))^2\,dx=2$. Find the value of b^2/a^2.

144. Let $f:[a,b]\to R$ be a continuous monotonic function and let $F:[a,b]\to R$, Such that

$F(x)=(x-a)\int_x^b f(t)\,dt+(x-b)\int_a^x f(t)\,dt$ then which of the following is/are correct?

(A) $F(x)=0$ must have at least one real root in (a, b)
(B) $F(x)>0\ \forall\ x\in[a,b]$ if $f(x)$ be monotonically increasing.
(C) $F(x)<0\ \forall\ x\in[a,b]$ if $f(x)$ be monotonically decreasing
(D) $F(x)$ must have the same sign throughout in $[a, b]$

Paragraph for question nos. 145 & 146

Consider the polynomial
$p(x) = 9x^8 - 18x^5 - 20x^3 + 15$

145. $p(x) = 0$ has
(A) Exactly one real root in $[0, 3^{1/3}]$
(B) Exactly two real roots in $[0, 5^{1/5}]$
(C) At least two real roots in $[0, 3^{1/3}]$
(D) No real roots in $[3^{1/5}, 5^{1/5}]$

146. If the graph of $y=p(x)$ represents the rate of change of slope of tangent at any 'x' to the curve $y=f(x)$ then number of horizontal tangents to the curve $y=f(x)$ will be [Given that $f'(1)=8$]
(A) 0
(B) 1
(C) 2
(D) 3

147. Let $f(x)$ be a differentiable function on R defined by $f(x) = 5-(x+1)^2$, and A be the point of intersection where the tangent line drawn to the graph of $y=f(x)$ at the point $P(x, f(x))$ intersects with x-axis and B be the intersection point where the tangent line intersects with y-axis. If S(x) denotes the area of ΔOAB where O is the origin, then the minimum value of $S(x)$ in the interval $(0, 1)$ is equal to $\frac{p}{q}$ (where p and q are in their lowest form). Find the value of $(p+q)$.

148. If $f(x)$ and $g(x)$ are two differentiable function and graph of $y = g(x)$ is reflection of graph of $y = f(x)$ with respect to line $y = x$. If point (a, b) lies on curve $y = f(x)$ and $f'(a) = 3$, then find the value of $\frac{670}{\gamma}$ (where g is ordinate of the point on the curve $y = g'(x)$ whose abscissa is b.)

149. If the angle between the tangent at any point 'A' of the curve $ln(x^2 + y^2) = C\tan^{-1}\frac{y}{x}$ and the line joining A to the origin is $\frac{\pi}{4}$ then find the value of $|C|$.

150. The length of a longest interval in which the function $f(x) = 3\sin x - 4\sin^3 x$ is increasing, is
(A) $\pi/3$ (B) $\pi/2$ (C) $3\pi/2$ (D) π

151. If the range of all real values of b for which the function $f(x) = (b^2 - 3b + 2)(\cos^2 x - \sin^2 x) + (b - 1)x + \sin 2$ does not possess any critical points on R is (b_1, b_2), then find the value of $(b_1 + b_2)$.

152. If the largest positive value of the function $f(x) = \sqrt{8x - x^2} - \sqrt{14x - x^2 - 48}$ is $\sqrt{k}$ where $k \in N$, then find the value of k.

153. Let $h(x) = f(x) - (f(x))^2 + (f(x))^3$ for every real number x. Then
(A) h is increasing whenever f is increasing
(B) h is increasing whenever f is decreasing
(C) h is decreasing whenever f is decreasing
(D) nothing can be said in general.

Paragraph for question nos. 154 to 156

Let $f(x) = \frac{(x^3 - 3x + 1)(x - \alpha)}{(x - \alpha)}$

If range of $f(x)$ is a proper subset of real numbers, then

154. range of α can be
(A) $(-\infty, -1) \cup (1, \infty)$
(B) $(-1, +1)$
(C) $(-\infty, -2) \cup (2, \infty)$
(D) $(-2, +2)$

155. number of integral values that α cannot take
(A) 1 (B) 2 (C) 3 (D) 5

156. $\lim_{x \to \alpha} f(x) \in A$ then set A can be
(A) $(-\infty, -1) \cup (3, \infty)$
(B) $[-1, 3]$
(C) $(-\infty, -1] \cup [3, \infty)$
(D) $(-1, 3)$

Paragraph for question nos. 157 & 158

Consider the polynomial $p(x) = 9x^8 - 18x^5 - 20x^3 + 15$

157. $p(x) = 0$ has

(A) Exactly one real root in $[0, 3^{1/3}]$ (B) Exactly two real roots in $[0, 5^{1/5}]$

(C) At least two real roots in $[0, 3^{1/3}]$ (D) No real roots in $[3^{1/5}, 5^{1/5}]$

158. If the graph of $y = p(x)$ represents the rate of change of slope of tangent at any 'x' to the curve $y = f(x)$ then number of horizontal tangents to the curve $y = f(x)$ will be [Given that $f'(1) = 8$]

(A) 0 (B) 1 (C) 2 (D) 3

159. Let $f(x) = \dfrac{1}{\sin x\sqrt{1-\cos x}}$ $(0 < x < \pi)$. If the local minimum value of $f(x)$ is $\sqrt{\dfrac{m}{n}}$ (where m and n are relatively prime). Find minimum value of $(m + n)$.

160. If the maximum value of the expression $y = \dfrac{x^4 - x^2}{x^6 + 2x^3 - 1}$ for $x > 1$, is $\dfrac{p}{q}$ and it occurs at $x = \dfrac{a+\sqrt{b}}{c}$ where p and q are in their lowest term and a, b, c are pairwise relatively prime positive numbers, find the value of $(a + b + c + p + q)$.

Paragraph for question nos. 161 to 163

Let $f(x) = \dfrac{\sin x}{x}$ and $g(x) = \int_0^x f(t)\,dt$

161. The number of local maxima of $g(x)$ in $(0, 4\pi)$ are

(A) 1 (B) 2 (C) 3 (D) 4

162. The number of local minima of $g(x)$ in $(4\pi, 10\pi)$ are

(A) 1 (B) 2 (C) 3 (D) 4

163. The number of points of inflection of $g(x)$ in $(10\pi, 20\pi)$ are

(A) 9 (B) 10 (C) 11 (D) 12

164. If $f(x) = |x - e|^{\pi} + |x - \pi|^{e}$, then find the number of values of x at which $f'''(x)$ does not exist.

165. If $a^2 + b^2 = 1$ and u is the minimum value of $\dfrac{b+1}{a+b-2}$ then find the value of u^2.

166. Let $f'(x^2) = \dfrac{1}{x}$ for $x > 0$, $f(1) = 1$ and $g'(\sin^2 x - 1) = \cos^2 x + p\ \forall\ x \in R$, $g(-1) = 0$.

If $h(x) = \begin{cases} f(x), & x > 0 \\ g(x), & -1 \le x \le 0 \end{cases}$

is a continuous function then find the absolute value of $2p$.

167. Let $f(x)$ and $g(x)$ be differentiable for $0 \le x \le 1$, such that $f(0) = 2$, $g(0) = 0$, $f(1) = 6$. If there exist a real number c in $[0, 1]$ such that $f'(c) = 2g'(c)$, then the value of $g(1)$ must be
(A) 1 (B) 2 (C) -2 (D) -1

168. If $f(x) = \left(\alpha - \frac{1}{\alpha} - x\right)(4 - 3x^2)$, $(\alpha > 0)$ has a maximum and minimum and $g(\alpha)$ is their absolute difference then the minimum value of $g(\alpha)$, for different values of α, is
(A) $\frac{32}{9}$ (B) $\frac{25}{9}$ (C) $\frac{42}{9}$ (D) $\frac{17}{9}$

Paragraph for question nos. 169 to 171

Let $f(x)$ be a real valued function defined on positive real numbers. The tangent lines drawn to the graph of $y = f(x)$ always intersect the y-axis 1 unit lower than where they meet the function. Also $f(1) = 0$.

169. Range of the function is
(A) $(-\infty, \infty)$ (B) $(0, \infty)$ (C) $[1, \infty)$ (D) $[-1, 1]$

170. If $x f(|x|) = k$ has exactly one distinct solution then true set of values of k is equal to
(A) $\left(0, \frac{1}{2}\right)$ (B) $\left(-\infty, \frac{-1}{e}\right) \cup \left(\frac{1}{e}, \infty\right)$
(C) $\left(\frac{-1}{e}, \frac{1}{e}\right)$ (D) $\left(\frac{-1}{e}, 0\right)$

171. The longest interval where the function $g(x) = \frac{x}{f(x)}$ is decreasing is denoted by J. Number of integral values in the interval J is
(A) 0 (B) 1 (C) 2 (D) 3

172. Let $f : [1, \infty) \to R$, $f(x)$ is a strictly increasing function. If $f(x)$ is a differentiable function and $f(1) = 1$, then the sum of all the solutions of the equation $f(f(f(x))) = \frac{1}{x^2 - 2x + 2}$ is equal to
(A) 1 (B) 0 (C) 2 (D) 4

173. Let $f : R \to R$ be a function defined by $f(x) = 2x^3 - 21x^2 + 78x + 24$. Number of integers in the solutions set of x satisfying the inequality $f(f(f(x) - 2x^3)) \ge f(f(2x^3 - f(x)))$ is
(A) 3 (B) 4 (C) 5 (D) 6

Paragraph for Question no. 174 to 176

Consider $f : R \to R$ be a polynomial function whose degree is greater than 1 and atmost 4. Also f is a one-one and onto function whose graph is symmetrical about A (4, 0) and f has horizontal tangent at $x = 4$.

174. The value of definite integral $\int\limits_{-2010}^{2018} f(x)\,dx$ is equal to

(A) -4 (B) 0 (C) 4 (D) 8

175. The value of $f^{-1}(2) + f^{-1}(-2)$ is equal to

(A) 0 (B) 2 (C) 4 (D) 8

176. If $f'(10) = 20$, then $f'(-2)$ is equal to

(A) 4 (B) 8 (C) 20 (D) 32

177. Let $f(x) = \begin{vmatrix} 1 & -1 \\ -x & x^2-2 \end{vmatrix} + \begin{vmatrix} -1 & 1 \\ -2x & 2x^2-4 \end{vmatrix} + \begin{vmatrix} 1 & -1 \\ -3x & 3x^2-6 \end{vmatrix} + \begin{vmatrix} -1 & 1 \\ -4x & 4x^2-8 \end{vmatrix} + \ldots\ldots$

$\ldots\ldots\ldots\ldots + \begin{vmatrix} 1 & -1 \\ -31x & 31x^2-62 \end{vmatrix}$

If P: represent minimum value of g(x) where $g(x) = \int\limits_0^x f(x-t)\,dt$, $x \in [0, 3]$

and Q : represent the value of $(a + b)$, if the set of values of k for which the equation $|f(x^2)| = k$ has six different solutions is (a, b), then find the value of $(Q - 3P)$.

178. Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and let m(b) be the minimum value of f(x). As b varies, the range of m (b) is

(A) $[0, 1]$ (B) $\left(0, \frac{1}{2}\right]$ (C) $\left[\frac{1}{2}, 1\right]$ (D) $(0, 1]$

179. Let $f(x) = \int\limits_x^{2x} (t\,\ell n\,t - t)\,dt$ for all $x > 0$. Then which of the following is/are **correct**?

(A) f(x) is strictly increasing in (0, e) and strictly decreasing in (e, ¥).

(B) f(x) is strictly decreasing in $\left(0, \frac{e}{2^{\frac{4}{3}}}\right)$ and strictly increasing in $\left(\frac{e}{2^{\frac{4}{3}}}, \infty\right)$.

(C) f(x) has a local minima but no local maxima.

(D) $\lim\limits_{x\to\infty} \frac{f'(x)}{x^2}$ is equal to zero.

180. Let f(x) be the curve passing through $(-2, 1)$ such that slope of the normal line at the point (x, y) on the curve is equal to x^2 y. Find the area bounded by the curve $y = x\,f^2(x)$ and coordinates axes.

181. Suppose f(x) is real valued polynomial function of degree 6 satisfying the following conditions ;

(a) f has minimum value at x = 0 and 2 (b) f has maximum value at x = 1

(c) for all x, $\underset{x \to 0}{\text{Limit}} \frac{1}{x} \ln \begin{vmatrix} \frac{f(x)}{x} & 1 & 0 \\ 0 & \frac{1}{x} & 1 \\ 1 & 0 & \frac{1}{x} \end{vmatrix} = 2$. Determine f(x).

Paragraph for Question nos. 182 to 184

Let P(x) be a polynomial of degree 4 with P(2) = – 1, P'(2) = P"(2) = 0, P'''(2) = – 12 and P''''(2) = 24. where dash denotes the derivative of P(x) w.r.t x.

182. P"(1) is equal to

(A) 24 (B) 26 (C) 28 (D) 30

183. $\int_0^4 P(x)\,dx$ equals

(A) $\frac{32}{5}$ (B) $\frac{42}{5}$ (C) $\frac{34}{5}$ (D) $\frac{44}{5}$

184. Least positive integral value of λ for which the equation $|P(x)| = \lambda$ has exactly two distinct solutions is

(A) 2 (B) 3 (C) 4 (D) 5

185. Let $x \in \left[-\frac{5\pi}{12}, -\frac{\pi}{3}\right]$, then find the maximum value of $y = \tan\left(x + \frac{2\pi}{3}\right) - \tan\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right)$.

186. The maximum value of $(\cos \alpha_1) \cdot (\cos \alpha_2)$.......... $(\cos \alpha_n)$, under the restrictions

$O \le \alpha_1, \alpha_2, ..., \alpha_n \le \frac{\pi}{2}$ and $\cot \alpha_1 \cdot \cot \alpha_2 .... \cot \alpha_n = 1$ is

(A) $\frac{1}{2^{n/2}}$ (B) $\frac{1}{2^n}$ (C) $\frac{1}{2n}$ (D) 1

187. Find the cosine of the angle at the vertex of an isosceles triangle having the greatest area for the given constant length *l* of the median drawn to its lateral side.

188. A trapezium ABCD is inscribed into a semicircle of radius *l* so that the base AD of the trapezium isdiameter and the vertices B and C lie on the circumference. Find the base angle θ (in degree) of the trapezium ABCD which has the greatest perimeter.

189. The circle $x^2 + y^2 = 1$ cuts the x-axis at P & Q. Another circle with centre at Q and varable radius intersects the first circle at R above the x-axis & the line segment PQ at S. Find the maximum area of the triangle QSR.

190. A curve passes through (2, 0) and the slope of tangent at any point (x, y) is $x^2 - 2x \ \forall x \in R$. The point of minimum ordinate on the curve where $x > 0$ is (a, b), then find the value of (a + 6b).

191. Let f be a differentiable function on R and satisfying $f(x) = -(x^2 - x + 1)e^x + \int_0^x e^{x-y} \cdot f'(y)\,dy$.

If $f(1) + f'(1) + f''(1) = ke$, where $k \in N$, then find k.

192. Consider an equation with x as a variable $7\sin 3x - 2\sin 9x = \sec^2\theta + 4\operatorname{cosec}^2\theta$, then find the value of $\frac{15}{\pi}\left((\text{minimum positive root}) - (\text{maximum negative root})\right)$.

Paragraph for question nos. 193 to 195

Consider the cubic $f(x) = 8x^3 + 4ax^2 + 2bx + a$ where $a, b \in R$.

193. For a = 1 if y = f (x) is strictly increasing $\forall x \in R$ then maximum range of values of b is

(A) $\left(-\infty, \frac{1}{3}\right]$ (B) $\left(\frac{1}{3}, \infty\right)$ (C) $\left[\frac{1}{3}, \infty\right)$ (D) $(-\infty, \infty)$

194. For b = 1, if y = f (x) is non monotonic then the sum of all the integral values of $a \in [1, 100]$, is

(A) 4950 (B) 5049 (C) 5050 (D) 5047

195. If the sum of the base 2 logarithms of the roots of the cubic f (x) = 0 is 5 then the value of 'a' is

(A) – 64 (B) – 8 (C) –128 (D) – 256

196. Let $f : R \to R$ be defined by $f(x) = \begin{cases} 2x^2 \ln|x| - 5x^2, & x \neq 0 \\ 0, & x = 0 \end{cases}$

Which of the following statement(s) is/are correct?

(A) f(x) has exactly one local maximum and two local minimum points.

(B) f(x) is strictly increasing in $(10, \infty)$.

(C) Absolute minimum value of f(x) exist but absolute maximum value of f(x) does not exist.

(D) In the interval $x \in (0, \infty)$, g(x) = k has two distinct roots.

197. Let $f: [0, \infty) \to [0, \infty)$ such that $f'(x) > 0$ and g is the inverse of f(x). Then the number of real roots of the equation $f^2(x) \cdot g(x) + 3x^2 = 20$, is

(A) 0 (B) 1 (C) 2 (D) more than 2

198. Number of solutions of the equation $e^{2x} = 9x^2$, is equal to

(A) 0 (B) 2 (C) 3 (D) 4

199. Let f(x) be a continuous and differentiable function satisfying the following conditions

(i) $\prod_{r=1}^{6} f(r) < 0$, $\prod_{r=1}^{3} f(2r) < 0$, $\prod_{r=1}^{2} f(3r) < 0$

(ii) $f(6) > 0$ and

(iii) f(x) is monotonic in $(n, n+1)$, $n \in I$

Let A denotes the set consisting of number of distinct possible roots of $f(x) = 0$ in $x \in (1, 6)$. Find the sum of all the elements of set A.

200. If $f(x) = \int_{-2}^{x} \left(t^4 - bt^3 + (b+1)\, t^2 - bt + b\right) dt$ strictly increases $\forall\, x \in R$ then find the number of integer in the range of b is

201. The angle between the tangent lines to the grpah of the function $f(x) = \int_{1/2}^{x} (2t + a)dt$ at the points where the graph cuts the x-axis is $\frac{\pi}{2}$. Find the sum of the squares of all values of a.

202. Which of the following statement is(are) correct?

(A) Let $f : [0, 1] \to R$ be a differentiable function with $f(0) = 0$, and $0 \le f'(x) \le 2f(x)$, then $f(x)$ is identically zero $\forall\, x \in [0, 1]$.

(B) If $f(x) = x^3 + ax^2 + bx + 5\sin^2 x$ is an increasing function $\forall\, x \in R$, then $a^2 - 3b + 15 \le 0$.

(C) If $x^a \le a^x$ $\forall\, x \in (1, \infty)$ where $a \in (1, \infty)$ then number of values of a satisfying the inequality is exactly one.

(D) Let $f(x)$ be a differentiable function in $[0, \infty)$ with $f(0) = 0$ and $f'(x)$ is an increasing function $\forall\, x \ge 0$, then $g(x) = \begin{cases} \frac{f(x)}{x}, & x > 0 \\ f'(0), & x = 0 \end{cases}$ is a decreasing function.

203. The graph of $f(x) = \frac{x^5}{20} - \frac{x^4}{12} + 5$ has

(A) no relative extrema, one point of inflection.

(B) two relative maxima, one relative minimum, two points of inflection.

(C) one relative maximum, one relative minimum, one point of inflection.

(D) one relative maximum, one relative minimum, two point of inflection.

204. Which of the following statement(s) is(are) correct?

(A) $\lim_{n\to\infty} \left(\frac{1}{\sqrt{4n-1^2}} + \frac{1}{\sqrt{8n-2^2}} + \frac{1}{\sqrt{12n-3^2}} + ...+ \frac{1}{2n} \right)$ is equal to $\frac{\pi}{2}$.

(B) If for a twice differentiable function $f(x)$, $f''(x) < 0$ $\forall\, x \in (a, b)$ then $f'(x) = 0$ has atmost one solution in (a, b).

(C) If $f(x) = ax^7 + bx^5 + cx^3 + dx + 2$, where a, b, c, d are real constants and $f(-3) = 3$ then range of $f(3) + 3\cos^2 x + 4\sin^2 x$, is equal to [4, 5].

(D) $\ln(1+x) < \dfrac{\tan^{-1} x}{1+x}$ for $x > 0$.

205. Let $x_1 = \sqrt[7]{3} + \sqrt[7]{9}$ and $x_2 = \sqrt[7]{5} + \sqrt[7]{7}$, then

(A) $x_1 = x_2$ (B) $x_1 > x_2$ (C) $x_1 < x_2$ (D) none

206. If $f(x) = \int_0^1 e^{|t-x|}\,dt$ where $(0 \le x \le 1)$, then maximum value of $f(x)$ is

(A) $e - 2$ (B) $e - 3$ (C) $e - 1$ (D) $2(\sqrt{e} - 1)$

207. Let $f(x) = \begin{cases} 14 - 10x + x^2 - x^3, & x < 1 \\ 3x + \log_{10}(p^2 - 4), & x \ge 1 \end{cases}$

Then $f(x)$ attains the absolute minimum value at $x = 1$. True set of values of p, is

(A) $\left(-\infty, -\sqrt{14}\right] \cup \left[\sqrt{14}, \infty\right)$ (B) $\left[-\sqrt{14}, -2\right] \cup \left[2, \sqrt{14}\right]$

(C) $\left[-\sqrt{14}, \sqrt{14}\right]$ (D) $\left[-\sqrt{14}, -2\right) \cup \left(2, \sqrt{14}\right]$

208. Let A be the set of real values of k for which the function $f(x) = 2x^3 - 3(k+4)x^2 + 18(k+1)x$ has exactly one local maximum and exactly one local minimum, then subsets of A can be

(A) $(-2, \infty)$ (B) $(-4, 4)$ (C) $(-\infty, 1)$ (D) $(4, \infty)$

209. If $f(x) = x^3 - 3x + \sin^{-1}(a^2 - 3a + 2)$. Then the smallest positive integer 'a' for which $f(x) = 0$ has three distinct real solution.

210. If curves $C_1 : y^2 = 2ax\ (a > 0)$ and $C_2 : xy = 4\sqrt{2}$ intersect orthogonally, then a equals

(1) $\frac{1}{2}$ (2) $\frac{2}{3}$ (3) 2 (4) $\frac{3}{2}$

Paragraph for question nos. 211 to 213

Consider f, g and h be three real valued differentiable functions defined on R. Let $g(x) = x^3 + g''(1)x^2 + \left(3g'(1) - g''(1) - 1\right)x + 3g'(1)$, $f(x) = x\,g(x) - 12x + 1$ and $f(x) = (h(x))^2$ where $h(0) = 1$.

211. The function $y = f(x)$ has

(A) Exactly one local minima and no local maxima

(B) Exactly one local maxima and no local minima

(C) Exactly one local maxima and two local minima

(D) Exactly two local maxima and one local minima

212. Which of the following is/are true for the function $y = g(x)$?

(A) g(x) monotonically decreases in $\left(-\infty, 2-\frac{1}{\sqrt{3}}\right) \cup \left(2+\frac{1}{\sqrt{3}}, \infty\right)$

(B) g(x) monotonically increases in $\left(2-\frac{1}{\sqrt{3}}, 2+\frac{1}{\sqrt{3}}\right)$

(C) There exists exactly one tangent to $y = g(x)$ which is parallel to the chord joining the points $(1, g(1))$ and $(3, g(3))$

(D) There exists exactly two distinct Lagrange's mean value in (0, 4) for the function $y = g(x)$.

213. Which one of the following does not hold good for $y = h(x)$?

(A) Exactly one critical point
(B) No point of inflection
(C) Exactly one real zero in (0, 3)
(D) Exactly one tangent parallel to x-axis

Paragraph for question nos. 214 to 216

Let $f(x)$ be function defined on the set of real numbers to real numbers, such it is continuous on R.

214. If $f(x)$ is also differentiable on R such that $f'(x) > f(x)\ \forall x \in R$ and $f(x_0) = 0$ then

(A) $f(x) < 0\ \forall x > x_0$
(B) $f(x) \geq 0\ \forall x > x_0$
(C) $f(x) > 0\ \forall x > x_0$
(D) $f(x) \leq 0\ \forall x > x_0$

215. The equation $k \cdot e^x = 5 + x + \frac{x^2}{2}$ where k is a positive constant has

(A) Exactly two roots
(B) Exactly one root
(C) No real root
(D) Many roots

216. If $f(x) = x^3 - (9-a)x^2 + 3(9-a^2)x + 7$ has points of extrema which are of opposite sign, then the values of parameter a.

(A) $(0, \infty) \cup \left(-\infty, -\frac{4}{9}\right)$
(B) $(-3, 3) \cup (4, \infty)$
(C) $\left(-3, -\frac{4}{9}\right) \cup (0, 3)$
(D) $(-\infty, -3) \cup (3, \infty)$

Paragraph for question nos. 217 & 218

Consider the function $f(x) = x^{30} \cdot (\ln x)^{20}$ for $x > 0$

217. If f is continuous at $x = 0$ then $f(0)$

(A) is equal to 0
(B) is equal to 2/3
(C) is equal to 1
(D) can not be defined to make $f(x)$ continuous at x=0

218. Maximum value of $f(x)$ occurs at

(A) $x = -\frac{2}{3}$ (B) $x = 1$ (C) $x = e^{-2/3}$ (D) $x = e^{2/3}$

219. A curve is defined parametrically by $x = e^{\sqrt{t}}$, $y = 3t - ln(t^2)$, where t is a parameter, then the equation of the tangent line drawn to the curve at $t = 1$, is

(A) $y = \frac{2}{e}x + 1$ (B) $y = \frac{2}{e}x - 1$ (C) $y = \frac{e}{2}x + 1$ (D) $y = \frac{e}{2}x - 1$

220. Let A be the set of real values of k for which the function $f(x) = 2x^3 - 3(k+4)x^2 + 18(k+1)x$ has exactly one local maximum and exactly one local minimum, then subsets of A can be

(A) $(-2, \infty)$ (B) $(-4, 4)$ (C) $(-\infty, 1)$ (D) $(4, \infty)$

221. If minimum value of $f(x) = x^2\, ln\, x$ is p, then the value of $ln\,|2p|$ is equal to

(A) 1 (B) -1 (C) $\frac{-1}{2e}$ (D) $\frac{1}{2}$

222. Consider $f(x) = \left| ln\,|x| \right| - kx^2$, $x \neq 0$. Match the column-I with the value of k in column-II.

Column-I	Column-II
(A) $f(x) = 0$ has two distinct solutions	(P) $k = 0$
(B) $f(x) = 0$ has four distinct solutions	(Q) $k = \frac{1}{2e}$
(C) $f(x) = 0$ has six distinct solutions	(R) $k \in \left(\frac{1}{2e}, \infty\right)$
(D) $f(x) =$ has no solution	(S) $k \in (-\infty, 0)$
	(T) $k \in \left(0, \frac{1}{2e}\right)$

Paragraph for question nos. 223 to 225

Let $f(x) = a(x-2)(x-b)$, where $a, b \in R$ and $a \neq 0$.

Also $f(f(x)) = a^3\left(x^2 - (2+b)x + 2b - \frac{2}{a}\right)\left(x^2 - (2+b)x + 2b - \frac{b}{a}\right)$, $a \neq 0$

has exactly one real zeroes 5.

223. Which of the following is **INCORRECT** ?

(A) The minimum value of $f(x)$ is 5 and attained at $x = -2$.
(B) The maximum value of $f(x)$ is 2 and attained at $x = 5$.
(C) The minimum value of $f(x)$ is 10 and attained at $x = 0$.
(D) The maximum value of $f(x)$ is 24 and attained at $x = 6$

.

224. The slope of the straight line passing through $O(0, 0)$ and tangent to $y = f(x)$, can be

(A) 4 (B) 2 (C) $\frac{4}{9}$ (D) $\frac{2}{9}$

225. The value of definite integral $\int_2^5 f(x)\,dx$ is equal to

(A) 4 (B) 8 (C) 10 (D) 14

226. **Column-I** **Column-II**

(A) If $\prod b$ denotes the product of all possible values of b and $\sum b$ denotes the sum of all possible values of b, for which a line tangent to the graph of $f(x) = x^3 - \frac{11x}{3}$ at the point $M\,(b, f(b))$ passes through the point $N(3, 0)$, then $\sum b - \prod b$ has the value equal to — (P) 1

(B) The real value of a for which the integral $\int_{a-1}^{a+1} e^{-(x-1)^2}\,dx$ attains its maximum value is equal to — (Q) 4

(C) Line l is tangent to the curve $y = e^x$ and is parallel to the line $x - 4y = 1$. If x-intercept of the line is $-\ln(ke)$, $k \in N$ then k is equal to — (R) 6

(S) 10

Paragraph for question nos. 227 to 229

In a triangle ABC, $\tan A + \tan B + \tan C = k$ where $k \in R$.
Assume that a triangle to be isosceles, if atleast two of its angles are equal.

227. Range of values of k for which there exists no isosceles triangle, is

(A) $[0, 3\sqrt{3})$ (B) $(3\sqrt{3}, \infty)$

(C) $(1, 3\sqrt{3})$ (D) $(-\infty, 0) \cup \{3\sqrt{3}\}$

228. Range of values of k for which there exists exactly one isosceles triangle is

(A) $[0, 3\sqrt{3}]$ (B) $(3\sqrt{3}, \infty)$

(C) $[1, 3\sqrt{3}]$ (D) $(-\infty, 0) \cup \{3\sqrt{3}\}$

229. Range of values of k for which there exists two isosceles triangles

(A) $[0, 3\sqrt{3}]$ (B) $(3\sqrt{3}, \infty)$

(C) $[1, 3\sqrt{3}]$ (D) $(-\infty, 0) \cup \{3\sqrt{3}\}$

230. Which of the following function(s) cannot exist?

(A) $f''(x) > 0$ for all $x \in R$, $f'(0) = 1$ and $f'(1) = 1$.
(B) $f''(x) > 0$ for all $x \in R$, $f'(0) = 1$ and $f'(1) = 2$.
(C) $f''(x) \geq 0$ for all $x \in R$, $f'(0) = 1$ and $f(x) \leq 100$ for all $x > 0$.
(D) $f''(x) > 0$ for all $x \in R$, $f'(0) = 1$ and $f(x) \leq 1$ for all $x < 0$.

ANSWER KEY

1. 15 **2.** 1 3. **D** 4. **C** 5. **A** 6. **149** 8. **B**

9. D **10.** g increases for x < 0 ; g decreases for all x > 0 **11.** $\left(\frac{1}{2},\frac{3}{4}\right] \cup (1,\infty)$

12.	28	**13.**	64	**14.**	3	**16.**	3	**17.**	A	**18.**	4	**19.**	2
20.	D	**21.**	12	**22.**	A	**23.**	B	**24.**	B	**25.**	A	**26.**	D

27. (A) Q ; (B) R ; (C) S **28.** C **29.** D **30.** C **31.** C **32.** 36

33. B **34.** B **35.** B **36.** A **37.** $\frac{-82\cdot 7^3}{3}$ **41.** **B**

42. C **45.** D *46.* *C* *47.* *D* *48.* *B*

49. x = 1 when t = 1, m $\to \infty$; 5x – 4y = 1 if t $\neq$ 1 then m = 1/3

50.	7	**51.**	400	**52.**	25	**53.**	C	**54.**	A	**55.**	D	**56.**	D

57. B **58.** D **59.** A **60.** $(-\infty, -3] \cup [1, \infty)$

61. **(a)** $(-\infty, 0]$;

(b) $\uparrow$ in $\left(1,\frac{5}{3}\right)$ and $\downarrow$ in $(-\infty,1)\cup\left(\frac{5}{3},\infty\right)-\{-3\}$;

(c) $x = \frac{5}{3}$

(d) removable discont. at x = –3 (missing point) and non removable discont. at x = 1 (infinite type)

(e) – 2.

62.	27	**63.**	0	**64.**	A	**65.**	B	**66.**	D	**67.**	C	**68.**	C
69.	A	**70.**	D	71.	A	**72.**	C	**73.**	A	74.	2	**75.**	D
77.	24	**78.**	0	**79.**	C	**80.**	C	**81.**	A	**82.**	B	**83.**	C
84.	125	**85.**	B	**86.**	ABCD	**87.**	12	**88.**	2010	**89.**	A	**90.**	A
91.	C	**92.**	BCD	**93.**	ABCD	**94.**	B	**95.**	B	**96.**	1	**97.**	5
98.	B	**99.**	ABCD	**100.**	B	**101.**	C	**102.**	7	**103.**	B	**104.**	B
105.	C	**106.**	C	**107.**	4	**108.**	C	**109.**	B	**110.**	C	**111.**	D
112.	C	**113.**	D	**114.**	BB	**115.**	AD	**116.**	AC				

117. 1 where a = – 1, b = 0, c = 2 **118.** 4 when a = $\sqrt{2}$ **119.** 3 **120.** 31

121.	A	**122.**	C	**123.**	B	**124.**	C	**125.**	A	**126.**	ACD	**127.**	C
128.	B	**129.**	C	**130.**	AB	**131.**	A	**132.**	$a \in (-\infty, -3) \cup \left(3, \frac{29}{7}\right)$; a = 4				
133.	D	**134.**	D	**135.**	C	**136.**	B	**137.**	A	**138.**	A	**139.**	C
140.	BC	**141.**	4	**142.**	4	**143.**	3	**144.**	BCD	**145.**	C	**146.**	D
147.	107	**148.**	2010	**149.**	2	**150.**	A	**151.**	4	**152.**	12	**153.**	AC
154.	C	**155.**	D	**156.**	A	**157.**	C	**158.**	D	**159.**	59	**160.**	15
161.	B	162	B	**163.**	B	**164.**	1, at x = π	**165.**	9	**166.**	3	**167.**	B
168.	A	**169.**	A	**170.**	B	**171.**	B	**172.**	A	**173.**	C	**174.**	B
175.	D	**176.**	C	**177.**	$P = \frac{-160}{3}$; Q = 68 ; Hence Q – 3P = 228							**178.**	D
179.	BCD	**180.**	1	**181.**	$f(x) = \frac{2}{3}x^6 - \frac{12}{5}x^5 + 2x^4$					**182.**	A	**183.**	D
184.	B	**185.**	$\frac{11\sqrt{3}}{6}$	**186.**	A	**187.**	cos A = 0.8	**188.**	θ = 60				
189.	$\frac{4}{3\sqrt{3}}$	**190.**	2	**191.**	9	**192.**	10	**193.**	C	**194.**	B	**195.**	D
196.	ABC	**197.**	A	**198.**	C	**199.**	10	**200.**	5	**201.**	4		
202.	ABC	**203.**	C	**204.**	ABC	**205.**	C	**206.**	C	**207.**	D	**208.**	CD
209.	1	**210.**	C	**211.**	C	**212.**	D	**213.**	C	**214.**	C	**215.**	B
216.	D	**217.**	A	**218.**	C	**219.**	A	**220.**	CD	**221.**	B		
222.	(A) R, P; (B) Q; (C) T; (D) S					**223.**	ACD	**224.**	AC	**225.**	A		
226.	(A) S; (B) P; (C) Q			**227.**	A	**228.**	D	**229.**	B	**230.**	AC		

SOLUTIONS

1. For domain of f(x) : $16 - x^2 \geq 0$ and $-1 \leq \dfrac{x^2}{16} \leq 1$

$\Rightarrow$ $0 \leq x^2 \leq 16$ and $-16 \leq x^2 \leq 16$

$\therefore$ common interval, $0 \leq x^2 \leq 16$

$\Rightarrow$ $-4 \leq x \leq 4$

$\therefore$ m = number of integers in $[-4, 4] = 9$

Now, as f(x) decreases when x^2 increases from 0 to 4

$$\therefore \quad f(x)_{min} = f(x^2_{max}) = \sqrt{16-16} + \cos^{-1}\left(\frac{16}{16}\right) = 0$$

$$\text{and} \quad f(x)_{max} = f(x^2_{min}) = \sqrt{16-0} + \cos^{-1}\left(\frac{0}{16}\right) = 4 + \frac{\pi}{2}$$

$$\therefore \quad \text{Range is } \left[0,\ 4+\frac{\pi}{2}\right]$$

$$\therefore \quad n = \text{number of integers in } \left[0,\ 4+\frac{\pi}{2}\right] = 6.$$

$\therefore$ $m + n = 15$

2. Given $P(x) + 2 = Q_1(x)(x-2)^2$

and $P(x) - 2 = Q_2(x)\,(x+2)^2$

Hence $P'(2) = 0$ and $P'(-2) = 0$

$\therefore$ Let $P'(x) = k(x-2)(x+2)$ (As $P'(x)$ must be a quadratic polynomial)

$P'(2) = 0$ and $P'(-2) = 0$

$\therefore$ On integrating, we get

$$P(x) = k\left(\frac{x^3}{3} - 4x\right) + C \qquad ....(1)$$

Aliter: Let $P'(x) = k(x-2)(x+2) = k(x^2 - 4)$

$$\therefore \quad P(x) = k\left(\frac{x^3}{3} - 4x\right) + c \qquad \text{........... equation (1)}$$

$\because$ P(2) = – 2 and P(–2) = 2

$\therefore$ $k\left(\frac{8}{3}-8\right)+c=-2$ .... equation (2)

and $k\left(\frac{-8}{3}+8\right)+c=2$ .... equation (3)

$\therefore$ from equation (2) + equation (3) we have 2c = 0

and $\frac{-16}{3}k=-2$ $\therefore$ $k=\frac{3}{8}$

$\therefore$ from equation (1)

$$P(x)=\frac{3}{8}\left(\frac{x^3}{3}-4x\right)+0$$

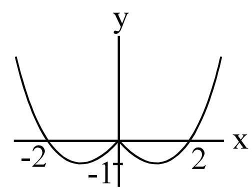
$\therefore$ $P(-3)=\frac{9}{8}$

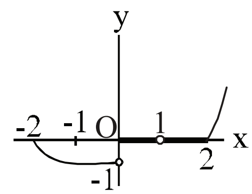
$\therefore$ $(a-b) \Rightarrow = 9-8=1$.

3 to 5.

Graph of y = f(x) is Graph of y = g(x) is

$$g(x)=\begin{cases} f(x) & \text{if } -2\le x\le -1 \\ -1 & \text{if } -1\le x<0 \\ 0 & \text{if } 0\le x<1 \\ f(x) & 1\le x\le 3 \end{cases}$$





6. $f'(x)=2c_1e^{2x}+5c_2e^{5x}$

$f''(x)=4c_1e^{2x}+25c_2e^{2x}$

$\therefore$ $(4c_1e^{2x}+25c_2e^{5x})+a(2c_1e^{2x}+5c_2e^{5x})+b(c_1e^{2x}+c_2e^{5x})=0$

$c_1(4e^{2x}+2ae^{2x}+be^{2x})+c_2(25e^{5x}+5ae^{5x}+be^{5x})=0$

$\therefore$ $c_1e^{2x}(2a+b+4)+c_2e^{5x}(25+5a+b)=0$ is true $\forall$ c_1, c_2

$\Rightarrow$ $2a+b+4=0$ and $5a+b+25=0$

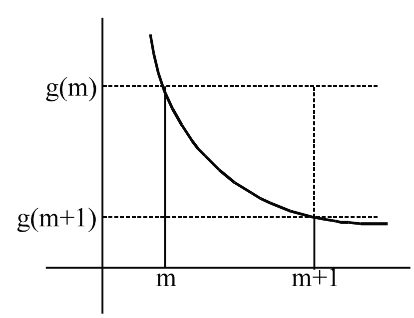
hence $3a+21=0$ $\Rightarrow$ $a=-7$ and $b=10$

$a^2+b^2=49+100=149$.]

7. $f(x)=ln(lnx)$

$$g(x)=f'(x)=\frac{1}{xlnx}$$

$$g'(x)=\frac{1}{-(xlnx)^2}[lnx+1] = -\left[\frac{1}{x^2ln^2x}+\frac{1}{x^2lnx}\right]$$



$$= -\frac{1}{x^2}\left[\frac{1}{ln^2 x} + \frac{1}{lnx}\right] \text{ which is} < 0 \text{ for } x > e$$

Hence g(x) is a decreasing function for $x > e$

hence $g(m) > g(m + 1)$ for $m > e$

From the graph

$$g(m + 1)\ (m + 1 - m) < \int_m^{m+1} g(x)\ dx < g(m)\ [m + 1 - m]$$

$$g(m + 1) \le \int_m^{m+1} f'(x)\ dx \le g(m)$$

$$\frac{1}{(m+1)ln(m+1)} \le f(m + 1) - f(m) \le \frac{1}{m \cdot ln(m)} \quad]$$

8 $f(x) = x^2\, ln\, x$

$$f'(x) = x^2 \cdot \frac{1}{x} + ln\, x \cdot 2x$$

$$= x\,(1 + 2\, ln\, x) \quad x > 0$$

f(x) is increasing in $\left(e^{-\frac{1}{2}}, \infty\right)$ and decreasing in $\left(0, e^{-\frac{1}{2}}\right)$

$$\therefore \quad f(x)\Big|_{min} = e^{-1}\, ln\, e^{-\frac{1}{2}} = -\frac{1}{2e} = p$$

$$\therefore \quad ln\ |2p| = ln\left|2\left(\frac{-1}{2e}\right)\right| = -1.$$

9. Let $f(x) = 2\tan x + (2a + 1)\, ln\, |\sec x| + (a - 2)x$

$\Rightarrow$ $f'(x) = 2\sec^2 x + (2a + 1)\tan x + (a - 2) = 2\tan^2 x + (2a + 1)\tan x + a$

For $f(x)$ to be increasing, $f'(x) \ge 0\ \ \forall\ x \in \left(0, \frac{\pi}{2}\right)$

$$\Rightarrow \quad 2\tan^2 x + (2a + 1)\tan x + a \ge 0\ \ \forall\ x \in \left(0, \frac{\pi}{2}\right)$$

$$\Rightarrow \quad (1 + 2\tan x)\ (\tan x + a) \ge 0\ \forall\ x \in \left(0, \frac{\pi}{2}\right)$$

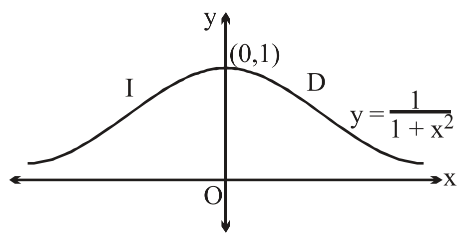
Hence $a \ge 0$. .

Note: For $a < 0$, $f(x)$ becomes non-monotonic in $\left(0, \frac{\pi}{2}\right)$ as interval of increasing and decreasing are obtained.

10. $f(x) > 0$ for all x
f is increasing for $x < 0$ and decreasing for $x > 0$
$\therefore$ $f'(x) > 0$ for $x < 0$ ....(1)
and $f'(x) < 0$ for $x > 0$ ....(2)

Now $g(x) = \dfrac{1}{f\left(\frac{1}{x}\right)}$

$$g'(x) = \frac{1}{f^2\left(\frac{1}{x}\right)} \cdot f'\left(\frac{1}{x}\right) \cdot \left(\frac{1}{x^2}\right)$$

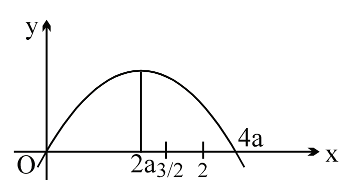


for $x < 0$, $g'(x) = $ (positive) (positive) (positive) > 0
$\Rightarrow$ g(x) is increasing for $x < 0$.
for $x > 0$ $g'(x) = $ (positive) (negative) (positive) < 0
$\therefore$ g(x) is decreasing for $x > 0$
Alternatively : consider the graph of

$f(x) = \dfrac{1}{1+x^2}$ and interpret. .

11. Case-I:
If $0 < a < 1$ (obviously 'a' can not be < 0)
then for $f(x)$ to be increasing

$4ax - x^2$ should be decreasing in $\left(\frac{3}{2}, 2\right)$



$\Rightarrow$ $\frac{3}{2} \ge 2a$ and $2 < 4a$

$\therefore$ $a \le \frac{3}{4}$ and $a > \frac{1}{2}$ $\Rightarrow$ $a \in \left(\frac{1}{2}, \frac{3}{4}\right]$

Caser-II: If $a > 1$ then for $f(x)$ to be increasing

$4ax - x^2$ increasing in $\left(\frac{3}{2}, 2\right)$

$\therefore \quad 2a \geq 2 \Rightarrow \quad a \geq 1$ but $a \neq 1$; $\quad \therefore \quad a > 1$

$\therefore$ final wer is $\left(\frac{1}{2}, \frac{3}{4}\right] \cup (1, \infty)$.

12. Given $f(3) = 5$ and $f'(3) = \frac{5}{4}$ ....(1)

$$L = \lim_{x \to 3} \frac{\left(3^{4f(x)} - 2.\left(\frac{3^{20}-1}{2}\right) - 1\right)^2}{\frac{1-\cos(ln(4-x))}{ln^2(4-x)}\, ln^2(4-x)} = \lim_{x \to 3} \frac{\left(3^{4f(x)} - 3^{20}\right)^2}{\frac{1}{2}.\frac{ln^2(1+3-x)}{(3-x)^2}(3-x)^2}$$

$$L = \lim_{x \to 3} 2\left(\frac{3^{4f(x)} - 3^{20}}{x-3}\right)^2 \qquad \text{............(1)}$$

Let $l = \lim_{x \to 3}\left(\frac{3^{4f(x)} - 3^{20}}{x-3}\right)\left(\frac{0}{0} \text{ form}\right)$ (Using L'Hospital's rule)

$$\therefore \quad l = \lim_{x \to 3} \frac{(3^{4f(x)} ln3)(4f'(x))}{1}$$

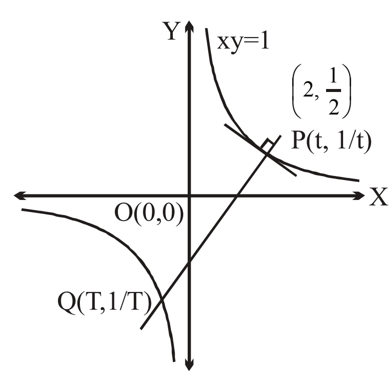
$$= \frac{(3^{4f(3)} ln3)4f'(3)}{1} = (3^{20} ln3) \times 4\left(\frac{5}{4}\right) = 5(3^{20} ln3)$$

Hence $L = 2(5 \cdot 3^{20} \cdot ln\, 3)^2 \equiv 2(a\, 3^b\, ln\, c)^2$ (Given)

So, $a = 5,\ b = 20,\ c = 3.$

Hence $(a + b + c) = 5 + 20 + 3 = 28$ **.**

13. We have $y = \frac{1}{x}$



$\frac{dy}{dx} = \frac{-1}{x^2}$ = slope of the tangent

slope of normal at $P = x^2 = t^2$

$$\therefore\ t^2 = \frac{\frac{1}{t}-\frac{1}{T}}{t-T} = \frac{T-t}{tT}\cdot\frac{1}{t-T} = \frac{-1}{tT}$$

$T = \dfrac{-1}{t^3}$. Hence slope of the curve at Q $= \dfrac{-1}{x^2} = \dfrac{-1}{T^2} = -\left(\dfrac{1}{\left(\dfrac{-1}{t^3}\right)^2}\right) = -\,t^6$

Let $t = 2$

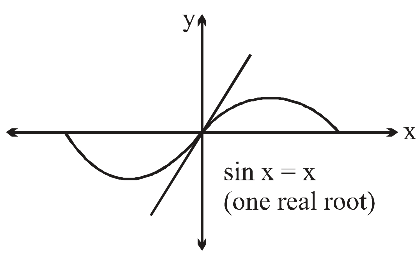
Slope of the curve at Q $= -\,2^6 = -\,64\ =\ m$ (Given)

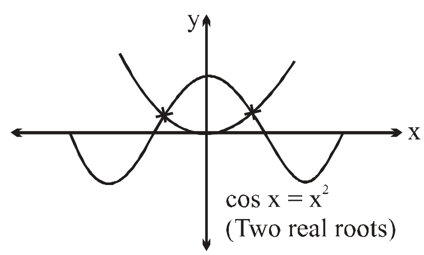
Hence, $|m|\ =\ |-64|\ =\ 64.$ **.**

14.

We have, $\begin{vmatrix} x^2+\sin x & \cos x & x(1+\sin x) \\ x+\cos x & & x+1 \end{vmatrix} = 0$

$\Rightarrow\quad \begin{vmatrix} \sin x & x \\ 1 & 1 \end{vmatrix}\begin{vmatrix} \cos x & x \\ x & 1 \end{vmatrix} = 0$





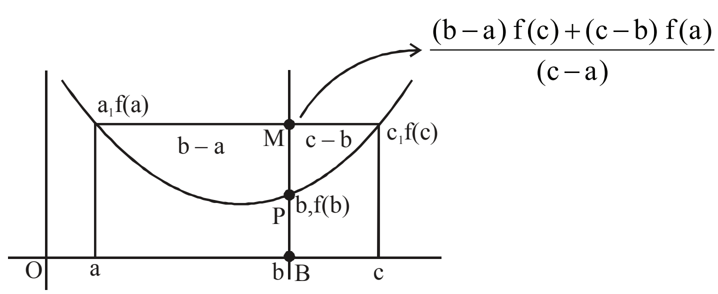
$\Rightarrow\qquad (\sin x - x)(\cos x - x^2) = 0$

$\therefore\qquad$ Either $\sin x = x$ or $\cos x = x^2$

Hence, number of real roots are 3. **.**]

15. To prove that : $(c-b)\,.\,f(a) + (b-a)\,f(c) > (c-a)\,f(b)$

Or to prove that $\qquad \dfrac{(b-a)\,f(c)+(c-b)\,f(a)}{(c-a)} > f(b)$



BM > BP $\Rightarrow \dfrac{(b-a)\,f(c)+(c-b)\,f(a)}{(c-a)} > f(b)$

Given $f''(x) > 0 \Rightarrow$ concave up in (a, c) and $f'(a) \uparrow \forall x \in [a, c]$

16. We have, $f(x) = e^{2x} - (c+1)e^x + 2x + \cos 2 + \sin 1$

$\therefore \quad f'(x) = 2e^{2x} - (c+1)e^x + 2$

Now, $\quad f'(x) \geq 0 \ \forall x \in R$

i.e., $\quad 2\left(e^x + \dfrac{1}{e^x}\right) - (c+1) \geq 0 \ \forall x \in R \Rightarrow (c+1) \leq \underbrace{2\left(e^x + \dfrac{1}{e^x}\right)}_{\text{minimum value} = 4} \ \forall x \in R$

$\therefore \quad c + 1 \leq 4 \Rightarrow c \leq 3$

$\therefore \quad c \in (-\infty, 3] \equiv (-\infty, \lambda]$

Hence $\lambda = 3$. .

17. $f(x) = e^x + \int_0^1 e^x f(t)\,dt = e^x + ke^x$ where $k = \int_0^1 f(t)\,dt$

$\therefore \quad k = \int_0^1 \left(e^t + ke^t\right)dt = e + ke - 1 - k$

$\therefore \quad k = \dfrac{e-1}{2-e}$, thus $f(x) = e^x\left(1 + \dfrac{e-1}{2-e}\right) = \dfrac{e^x}{2-e}$

obviously, $f(0) = \dfrac{1}{2-e} < 0$

also $\quad f'(x) = \dfrac{e^x}{2-e} < 0$ for $\forall x \in R$

Hence, $f(x)$ is a decreasing function.

also $\int_0^1 f(x)\,dx = \int_0^1 \dfrac{e^x}{2-e}\,dx = \left[\dfrac{e^x}{2-e}\right]_0^1 = \dfrac{e-1}{2-e} < 0.$]

18. $y = \dfrac{(x-a)^2}{4}$ and $y = e^x$

$\dfrac{dy}{dx} = \dfrac{x-a}{2}$ and $\dfrac{dy}{dx} = e^x$

$\therefore \quad \dfrac{x-a}{2} = e^x \Rightarrow (x - a) = 2e^x \quad$ ....(1)

Also $\quad \dfrac{(x-a)^2}{4} = e^x \quad$ ....(2)

$\therefore \quad \dfrac{4e^{2x}}{4} = e^x \qquad$ [Using (1)]

$\Rightarrow \quad e^x(e^x - 1) = 0.$ Hence $e^x = 1 \Rightarrow x = 0$

if $\quad x = 0$ then $y = 1$

$\therefore \quad 1 = \dfrac{a^2}{4} \Rightarrow a^2 = 4 \Rightarrow a = -2$ or $a = 2$ (rejected)

As, $\quad$ for $a = 2$ curves do not intersect.

Hence, sum of the squares of all possible values of a is 4.

19. Let $\quad f_n(x) = \int_x^{2x} e^{-t^n}\,dt$

$$f_n'(x) = 2\cdot e^{-(2x)^n} - e^{-x^n}$$

For maxima and minima, $f_n'(x) = 0 \Rightarrow 2e^{-(2x)^n} = e^{-x^n}$

$\Rightarrow \quad 2\cdot e^{-(2^n x^n)} = e^{-x^n}$

Taking log on both sides, we get

$\ln 2 - 2^n x^n = -x^n \Rightarrow \ln 2 = x^n(2^n - 1)$

$$\Rightarrow x^n = \frac{\ln 2}{2^n - 1} \Rightarrow x = \left(\frac{\ln 2}{2^n - 1}\right)^{\frac{1}{n}} = a_n$$

Also, $\quad f_n''\left(x = \left(\dfrac{\ln 2}{2^n - 1}\right)^{\frac{1}{n}}\right) < 0$

$\Rightarrow \quad f_n(x)$ is maximum at $x = \left(\dfrac{\ln 2}{2^n - 1}\right)^{\frac{1}{n}}$.

Now, $\quad \ln a_n = \dfrac{\ln\left(\dfrac{\ln 2}{2^n - 1}\right)}{n} = \dfrac{\ln(\ln 2) - \ln(2^n - 1)}{n}$

Hence $\quad L = \lim\limits_{n\to\infty} \ln(a_n) = \lim\limits_{n\to\infty} \dfrac{\ln(\ln 2) - \ln(2^n - 1)}{n} = \lim\limits_{n\to\infty}\left(\dfrac{\ln(\ln 2)}{n} - \dfrac{\ln\left(2^n\left(1 - \dfrac{1}{2^n}\right)\right)}{n}\right)$

$$L = \lim_{n\to\infty}\left(0 - \frac{n \cdot \ln 2 + \ln\left(1 - \frac{1}{2^n}\right)}{n}\right) = -\ln 2$$

Hence, $e^{-L} = 2$.

20. $g(x) = f\left((\tan x - 1)^2 + 3\right)$

$g'(x) = f'\left((\tan x - 1)^2 + 3\right)\left[2(\tan x - 1)\sec^2 x\right]$

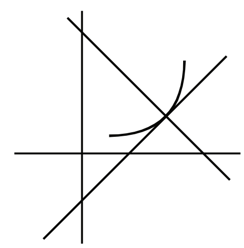
$f''(x) > 0$ $\therefore$ f' is $\uparrow$

$f'\left((\tan x - 1)^2 + 3\right) > f'(3)\ \forall x \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$(\tan x - 1) > 0\ \forall x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$\therefore$ interval of $\uparrow x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

21. Given $9y^2 = x^3$



Let the point on the curve be $x = t^2$ and $y = \frac{t^3}{3}$

$$\frac{dx}{dt} = 2t \quad ; \quad \frac{dy}{dt} = t^2$$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{t^2}{2t} = \frac{t}{2} \Rightarrow \text{slope of the normal} = -\frac{2}{t}$$

$\therefore$ normal makes equal intercept

hence $-\frac{2}{t} = -1$

$\Rightarrow$ $t = 2$

Hence $P = (4, \frac{8}{3})$

$\Rightarrow$ $a + 3b = 4 + 3 \cdot \frac{8}{3} = 4 + 8 = 12$.

22.

Let $G(x) = \frac{1}{2}\left\{\int_0^x f(t)\,dt\right\}^2$

then $G'(x) = g(x) = \frac{1}{2}\cdot 2\ f(x)\int_0^x f(t)\,dt$

$\because$ g(x) is a non-increasing function and g(0) = 0

$\therefore$ $g(x) \le g(0) \Rightarrow g(x) \le 0$ ......inequation (i)

$\therefore$ $G'(x) \le 0$ $\therefore$ G(x) is non-increasing

function $\therefore$ $G(x) \le G(0)$

$\Rightarrow$ $G(x) \le 0$ ......inequation (ii)

But $G(x) = \frac{1}{2}\left\{\int_0^x f(t)\,dt\right\}^2 \ge 0$ ......inequation (iii)

$\therefore$ from inequation (ii) and inequation (iii)

G(x) = 0 identically

$\therefore$ $\int_0^x f(t)\,dt = 0$

$\therefore$ f(x) = 0 identically

$\therefore$ from inequation (i) g(x) = 0

$\therefore$ f{g(ln 5)} + g{f(ln 5)} = 0]

23. Given $F(x) = \left(\int_a^x f(t)\,dt - \int_x^b f(t)\,dt\right)\left(2x - (a+b)\right)$ ....(1)

as f is continuous hence F (x) is also continuous. Also

put x = a

$$F(a) = \left(-\int_a^b f(t)\,dt\right)(a-b) = (b-a)\int_a^b f(t)\,dt$$

and put x = b

$$F(b) = \left(\int_a^b f(t)\,dt\right)(b-a)$$

hence F (a) = F (b)

hence Rolle's Theorem is applicable to F (x)

$\therefore$ $\exists$ some $c \in (a, b)$ such that $F'(c) = 0$

now $F'(x) = 2\left(\int_a^x f(t)\,dt - \int_x^b f(t)\,dt\right) + \left(2x - (a+b)\right)[f(x) + f(x)] = 0$

$\therefore$ $F'(c) = \left(\int_a^c f(t)\,dt - \int_c^b f(t)\,dt\right) = f(c)[(a+b) - 2c]$.

Note: Option (A) is not always true for any continuous function $f(x)$

24 to 26. Consider $f(x)$ in [0, 1]

$f'(t) = 3t^2 - 2t + 1 > 0 \ \forall\ t \in (0, 1) \quad \Rightarrow \quad f$ is $\uparrow$ in (0, 1)

maximum occurs at $t = x$.

and hence $f(x)$ is $\begin{cases} x^3 - x^2 + x + 1 & 0 \le x \le 1 \\ 3 - x & 1 < x \le 2 \end{cases}$

again consider $g(x) =$ in [0, 1)

$g(t) = \frac{3}{8}t^4 + \frac{1}{2}t^3 - \frac{3}{2}t^2 + 1$

$g'(t) = \frac{3}{2}t^3 + \frac{3}{2}t^2 - 3t = \frac{3}{2}t\,(t^2 + t - 2) = \frac{3}{2}t\,(t - 1))\,(t + 2)$

$g(t)$ decreases in [0, 1) $\Rightarrow$ maximum occurs when $t = 0$ and $g(0) = 1$

again consider $g(x)$ function in [1, 2]

$g(t) = \frac{3}{8}t + \frac{1}{32}\sin^2\pi t + \frac{5}{8}$

$g'(t) = \frac{3}{8} + \frac{\pi}{32}\sin(2\pi t) > 0 \ \forall\ t \in R$

$\therefore$ g is an increasing function in [1, 2]

$\Rightarrow$ minimum occurs when $t = 1$ and $g(1) = 1$

hence $g(x) = \begin{cases} 1 & 0 \le x < 1 \\ 1 & 1 < x \le 2 \end{cases} = 1 \quad \forall\ x \in [0, 2]$

24. $f(x)$ is continuous but not differentiable at $x = 1$. (B)

25. $\lim\limits_{x \to 1^-} fog(x) = f(1)$ and $\lim\limits_{x \to 1^+} gof(x) = 1$ also $f(1) > 1$. (A)

26. $Z(x) = \frac{d}{dx} f(x)^{g(x)} = \frac{d}{dx} f(x)^1 \ \forall\ x \in [0, 1) \cup (1, 2]$

$= f'(x) \ \forall\ x \in [0, 1) \cup (1, 2]$

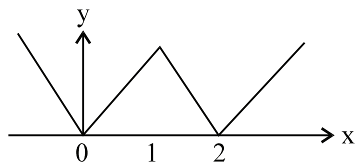
& $Y(x) = \frac{d}{dx} g(x)^{f(x)} = \frac{d}{dx} 1^{f(x)} = 0 \quad \forall\ x \in [0, 1) \cup (1, 2]$

Hence the functions $Y(x)$ and $Z(x)$ can vanish simultaneously at $f'(x) = 0$ which is not possible for any real x.

27.

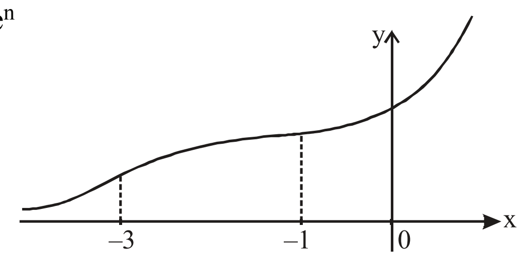
(A) If $f(0) = 0$ or $f(1) = 1$, the root is 0 or 1
If $f(0) > 1$ and $f(1) < 1$, then $g(x) = f(x) - x^3$ satifies, $g(0) > 0$, $g(1) < 1$
Hence by intermediate value theorem
$g(x) = 0$ has at least one root.

(B) If $f'(a) = f'(b) = 0$ and $f(\alpha) = f(\beta) = 0$ where $a < \alpha < \beta < b$, then by Rolle's theorem $f'(x)$ vanishes between α and β contradicting the fact that a and b are two consecutive roots of $f'(x) = 0$.

(C) 

f(x) is not differentiable at $x = 0, 1, 2$

28. Graph of $f(x) = (1 + x^2)\, e^x$
$f'(x) = (1 + x^2)\, e^n + e^n \cdot 2x$
$= e^x (x + 1)^2 \geq 0 \uparrow$
f(x) has above the x-axis and has an inflection point at $x = -1$
$f''(x) = e^x (x + 1)(x + 3)$
$\therefore$ in $(-\infty, -3)$ f(x) is concave up
$(-3, -1)$ f(x) as concave down
$(-1, \infty)$ concave up.
graph of $y = (1 + x^2)\, e^n$



hence $k > 0$, $f(x) = k$ has one solution, for $k < 0$ no solution
Note that number of inflection points on $y = f(x) = 2$.

29. Let $g(x) = f(x) - x^3$
Clearly $g(1) = g(2) = g(3) = g(4) = 0$
According to Rolle's theorem
$g'(c_1) = g'(c_2) = g'(c_3) = 0$ for some $c_1 \in (1, 2)$, $c_2 \in (2, 3)$ and $c_3 \in (3, 4)$
Again applying Rolle's theorem on $g'(x)$
$g''(k_1) = g''(k_2) = 0$ for some $k_1 \in (c_1, c_2)$ and $k_2 \in (c_2, c_3)$
Again using Rolle's theorem on $g''(x)$ on $[k_1, k_2]$
$g'''(x) = 0$ for some $x \in (k_1, k_2) \Rightarrow f'''(x) = 6$ for some $x \in (1, 4)$.

30. Let v be the most economical speed and c is the cost of fuel

$\Rightarrow c = kv^2$, where k is constant

when $v = 16, c = 48$

$\therefore\ k = \dfrac{3}{16}$

Hence the cost of fuel $= \dfrac{3v^2}{16}$

Total running cost per hour $= \left(300 + \dfrac{3}{16}v^2\right)\dfrac{d}{v} = \left(300\dfrac{d}{v} + \dfrac{3}{16}dv\right) = f(v)$ (where d is the total distance)

Now $f'(v) = 0 \Rightarrow \left(-300\dfrac{d}{v^2} + \dfrac{3}{16}d\right) = 0 \Rightarrow v^2 = (100)(16)$

$\Rightarrow v = (10)(4) = 40$ km/hr. .

31. $f(x) = ax^2 + bx + c$

$f(0) = c;\ f'(0) = b$ and $f''(0) = 2a \Rightarrow |a| + |b| + |c| = \left|\dfrac{f''(0)}{2}\right| + |f'(0)| + |f(0)|$

which is maximum when $f(0) = 1,\ f(1) = 1$ and vertex is about the line is $x = \dfrac{1}{2}$.

$A = f\left(\dfrac{1}{2}\right) = \dfrac{a}{4} + \dfrac{b}{2} + c$

$B = f(1) = a + b + c$

$C = f(0) = c$

$\Rightarrow\ |A| \le 1,\ |B| \le 1, |C| \le 1$

$a = -4A + 2B + 2C$

$b = 4A - B - 3C$

$c = C$

$$\begin{aligned}\Rightarrow\ |a| + |b| + |c| &= |-4A + 2B + 2C| + |4A - B - 3C| + |C| \\ &\le 4|A| + 2|B| + 2|C| + 4|A| + |B| + 3|C| + |C| \\ &\le 17\end{aligned}$$

Equality holds for $f(x) = 8x^2 - 8x + 1$.

32. Let $x^2 + x + 1 = t$

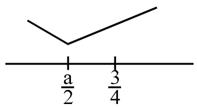
$\therefore\ t \in \left[\dfrac{3}{4}, \infty\right)$

$\Rightarrow\ t^2 - at + 2a \ge 0\ \ \forall\ t \in \left[\dfrac{3}{4}, \infty\right)$

Let $f(t) = t^2 - at + 2a$

$f'(t) = 2t - a$

Case-I: If $\frac{a}{2} \le \frac{3}{4} \quad \Rightarrow \quad a \le \frac{3}{2}$



In this case for $t \in \left[\frac{3}{4}, \infty\right)$ minimum value occur at $t = \frac{3}{4}$

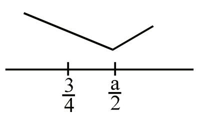
$\therefore \quad f\left(\frac{3}{4}\right) \ge 0 \;\Rightarrow\; \frac{9}{16} - \frac{3a}{4} + 2a \ge 0$

$\Rightarrow \frac{9}{16} + \frac{5a}{4} \ge 0 \Rightarrow 9 + 20a \ge 0$

$\Rightarrow a \ge \frac{-9}{20}$

$\therefore \quad a \in \left[\frac{-9}{20}, \frac{3}{2}\right]$ ....(1)

Case-II: If $\frac{a}{2} \ge \frac{3}{4} \quad \Rightarrow \quad a \ge \frac{3}{2}$



then for $t \in \left[\frac{3}{4}, \infty\right)$, minimum value occur at $t = \frac{a}{2}$

$\therefore \quad f\left(\frac{a}{2}\right) \ge 0 \;\Rightarrow\; \frac{a^2}{4} - \frac{a^2}{2} + 2a \ge 0$

$\Rightarrow \frac{a^2}{4} + 2a \ge 0 \Rightarrow a\left(2 - \frac{a}{4}\right) \ge 0 \;\Rightarrow a \in (0, 8)$

$\therefore \quad a \in \left(\frac{3}{2}, 8\right]$ ....(2)

$(1) \cup (2) \quad \Rightarrow \quad a \in \left[\frac{-9}{20}, 8\right]$

$\therefore$ Sum of all integral values

$S = 1 + 2 + ... + 8 = \frac{8 \times 9}{2} = 36$.

33. (ii) is true since $|-7| = 7$ will be maximum value of $|f(x)|$. To see why (i) and (iii) do not have to be true, consider the following:

$$f(x) = \begin{cases} 5 & \text{if } x \le -5 \\ -x & \text{if } -5 < x < 7 \\ -7 & \text{if } x \ge 7 \end{cases}$$

For $f(|x|)$, the maximum is 0 and the minimum is -7.

34 to 36.

34. $$f(x) = \int_0^1 ((12x + 20y)f(y) + 8)xy\, dy$$

$$= \int_0^1 12x^2 y f(y)\, dy + \int_0^1 20xy^2 f(y)\, dy + \int_0^1 8xy\, dy$$

$$= 12x^2 \int_0^1 y f(y)\, dy + 20x \int_0^1 y^2 f(y)\, dy + 8x\left(\frac{y^2}{2}\right)_0^1$$

$$f(x) = 12x^2 \int_0^1 y f(y)\, dy + 20x \int_0^1 y^2 f(y)\, dy + 4x$$

Let $f(x) = ax^2 + bx$

then $f(y) = ay^2 + by$

$\therefore$ $f(x) = 12x^2 \int_0^1 y(ay^2 + by)\, dy + 20x \int_0^1 y^2(ay^2 + by) dy + 4x$

$$= 12x^2\left(\frac{a}{4} + \frac{b}{3}\right) + 20x\left(\frac{a}{5} + \frac{b}{4}\right) + 4x$$

$\Rightarrow$ $ax^2 + bx = x^2(3a + 4b) + x(4a + 5b + 4)$

On comparing $a = 3a + 4b$

$\Rightarrow$ $a + 2b = 0$

$b = 4a + 5b + 4$

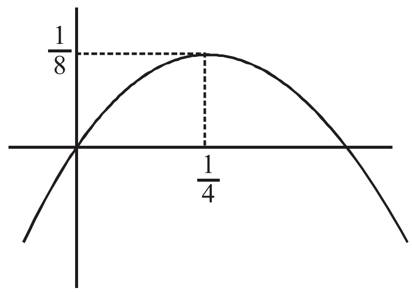
$\Rightarrow$ $a + b + 1 = 0$

On solving $a = -2, \quad b = 1$

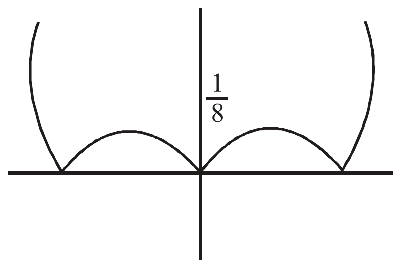
$\therefore$ $f(x) = -2x^2 + x$

$$= \frac{1}{8} - 2\left(x - \frac{1}{4}\right)^2$$

35. Graph of f(x) is



$\therefore$ Graph of | f (| x |) | is



Clearly y = | f (| x |) | will cut y = 2 at two points.

36. $f(-2^x) = -2 \cdot 2^{2x} - 2^x$

Clearly $f(-2^x)$ is always –ve

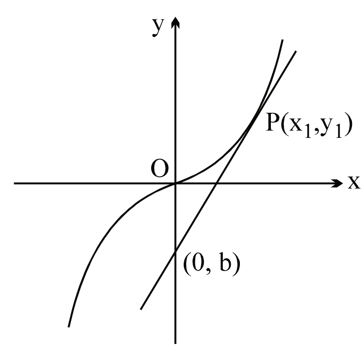
and at $x \to -\infty$ $\quad f(-2^x) \to 0$

$\therefore$ Range is $(-\infty, 0)$.

37. $f(x) = \dfrac{41x^3}{3}$

$f'(x) = 41x^2$

$f'(x)\big|_{x_1, y_1} = 41\,x_1^2$



$\therefore \qquad 41\,x_1^2 = 2009 = 7^2 \cdot 41$

$x_1^2 = 49$

$\Rightarrow \qquad x_1 = 7;\ y_1 = \dfrac{41\cdot 7^3}{3} \quad (x_1 \neq -7,\ \text{think !})$

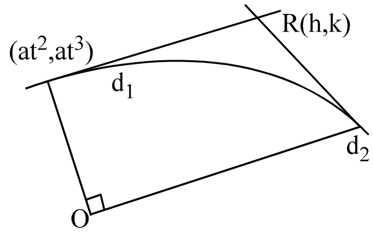
now $\dfrac{y_1 - b}{x_1 - 0} = 2009 \quad \Rightarrow \quad y_1 - b = 7 \cdot 2009 = 7^3 \cdot 41$

$b = \dfrac{41\cdot 7^3}{3} - 7^3\cdot 41 = \dfrac{41\cdot 7^3}{3}(-2) = -\dfrac{82\cdot 7^3}{3}$.

38

$x = at^2$; $y = at^3$

$\frac{dx}{dt} = 2at$; $\frac{dy}{dt} = 3at^2$



$\frac{dy}{dx} = \frac{3t}{2}$

$y - at^3 = \frac{3t}{2}(x - at^2)$ ....(1)

$2k - 2at^3 = 3th - 3at^3$

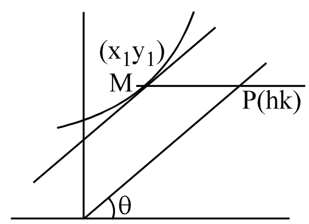
$at^3 - 3th + 2k = 0$

$t_1t_2t_3 = -\frac{2k}{a}$ (put $t_1t_2 = -1$); hence $t_3 = \frac{2k}{a}$

now t_3 must satisfy the equation (1) which gives the required locus.

39.$_8$

$|y_1| = |k|$ and $\left.\frac{dy}{dx}\right|_{x_1, y_1} = \frac{k}{h}$



now putting $y = a \sin\theta$ in the given equation

$$\frac{x + a\cos\theta}{a} = ln\left(\frac{1 + \cos\theta}{\sin\theta}\right)$$

$$\frac{x}{a} + \cos\theta = ln\cot\frac{\theta}{2}$$

differentiate w.r.t. θ

$$\frac{1}{a}\frac{dx}{d\theta} - \sin\theta = \frac{-1}{\sin\theta}$$

$$\frac{1}{a}\frac{dx}{d\theta} = \sin\theta - \frac{1}{\sin\theta} = -\frac{\cos^2\theta}{\sin\theta}$$

$$\frac{dx}{d\theta} = -\frac{a\cos^2\theta}{\sin\theta} \qquad \frac{dy}{d\theta} = a\cos\theta$$

$$\frac{dy}{dx} = -\tan\theta$$

hence $\frac{k}{h} = -\tan\theta$ ....(1)

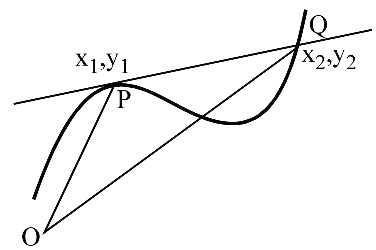
also $y = a\sin\theta$

i.e. $k = a\sin\theta$

$$k^2 = a^2 \frac{1}{1+\cot^2\theta} = \frac{a^2}{1+\frac{h^2}{k^2}}$$

$h^2 + k^2 = a^2$ Hence proved]

40. $x^2 + 3y^2\frac{dy}{dx} = 0$



$$\Rightarrow \quad \left.\frac{dy}{dx}\right]_P = -\frac{x_1^2}{y_1^2}$$

now $\quad -\frac{x_1^2}{y_1^2} = \frac{y_2 - y_1}{x_2 - x_1} = -\left(\frac{x_2^2 + x_1^2 + x_1x_2}{y_2^2 + y_1^2 + y_1y_2}\right)$

$$\boxed{\begin{array}{l} y_2^3 = a^3 - x_2^3 \\ y_1^3 = a^3 - x_1^3 \\ \hline y_2^3 - y_1^3 = -\left[x_2^3 - x_1^3\right] \\ \dfrac{y_2 - y_1}{x_2 - x_1} = -\left(\dfrac{x_2^2 + x_1x_2 + x_1^2}{y_2^2 + y_1y_2 + y_1^2}\right) \end{array}}$$

$$\frac{x_1^2}{y_1^2} = \frac{x_2^2 + x_1x_2}{y_2^2 + y_1y_2} = \frac{x_1^2\left(\frac{x_2^2}{x_1^2} + \frac{x_2}{x_1}\right)}{y_1^2\left(\frac{y_2^2}{y_1^2} + \frac{y_2}{y_1}\right)}$$

put $\frac{x_2}{x_1} = p$ and $\frac{y_2}{y_1} = q$

$p^2 + p = q^2 + q$

$\Rightarrow \quad (p - q)(p + q) + (p - q) = 0$

$\Rightarrow \quad (p - q)(p + q + 1) = 0$

$\Rightarrow \quad \frac{x_2}{x_1} + \frac{y_2}{y_1} = -1$

Note that: $a \neq n$

as $a = b$

$\Rightarrow \quad \frac{x_2}{x_1} = \frac{y_2}{y_1} \qquad$ i.e. $\frac{y_2}{x_2} = \frac{y_1}{x_1}$

$m_{OP} = m_{OQ}$

$\Rightarrow$ curve passes through origin ***i.e.*** $a = 0$ which is not possible.

41.

(A) $x \in (0, 1)$

$e^x < 1 + x$

$f(x) = e^x - x - 1$

$f'(x)\ e^x - 1$ which is $> 0\ \forall\ x \in (0, 1)$

so $f(x)\uparrow$ in $(0, 1)$

i.e. $f(x) > f(0)$

$f(x) > 0$

(B) $ln(1 + x) < x$

$f(x) = ln(1 + x) - x$

$f'(x) = \dfrac{1}{1+x} - 1$

$= \dfrac{-x}{1+x} < 0$ in $x \in (0, 1)$

i.e. $f(x)$ is $\downarrow$ is $(0, 1)$

$f(x) < f(0)$ $f(x) < 0$ is

(C) $f(x) = \sin x - x$

$f'(x) = \cos x - 1 < 0$ in $\alpha \in (0, 1)$

$f(x)$ is decreasing in $x \in (0, 1)$

$f(x) < f(0)$

$f(x) < 0$ i.e. $\sin x - x < 0;\ \sin x < x$

(D) $\ln x > x$

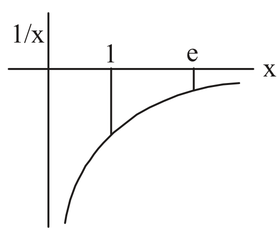
$f(x) = \ln x - x$

$f'(x) = \dfrac{1}{x} - 1 = \dfrac{1-x}{x} > 0$ in $(0, 1)$

$f(x)$ is increasing in $(0, 1)$

$f(x) < f(1)$

42 $f(x) = \int e^x (x - 1)(x - 2)dx$



43 to 45.

43. Let $\int_0^2 f(x)\,dx = \lambda$ (where λ = constant)

$\therefore$ $f'(x) = f(x) + \lambda \Rightarrow \int \dfrac{f'(x)\,dx}{f(x)+\lambda} = \int 1\,dx$

$\Rightarrow$ $ln\ (f(x) + \lambda) = x + c$ (where c is integration constant)

$\Rightarrow$ $f(x) + \lambda = e^c \cdot e^x$

$\because$ $f(0) = 1 \Rightarrow 1 + \lambda = e^c$

$\therefore$ $f(x) + \lambda = (1 + \lambda)e^x \Rightarrow f(x) = -\lambda + (1 + \lambda)e^x$

$\because$ $\lambda = \int_0^2 f(x)\,dx = \int_0^2 \left(-\lambda + (1+\lambda)e^x\right)dx$

$= -2\lambda + (1 + \lambda)(e^2 - 1) \Rightarrow 3\lambda = \lambda(e^2 - 1) + e^2 - 1$

$\Rightarrow \quad \lambda\,(4 - e^2) = e^2 - 1 \Rightarrow \lambda = \dfrac{e^2 - 1}{4 - e^2}$

$\therefore \quad f(x) = \dfrac{1 - e^2}{4 - e^2} + \left(1 + \dfrac{e^2 - 1}{4 - e^2}\right)e^x$

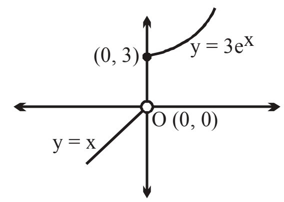
$= \dfrac{1 - e^2}{4 - e^2} + \left(\dfrac{3}{4 - e^2}\right)e^x = \left(\dfrac{1 - e^2 + 3e^x}{4 - e^2}\right)$

$\therefore$ For the number of points of intersection with x-axis

$1 - e^2 + 3e^x = 0 \Rightarrow 3e^x = e^2 - 1$

Clearly, $y = 3e^x$ is increasing function and it will cut $y = 3e^2 - 1$ at only one point.

44. $g(x) = \begin{cases} 3e^x, & x \ge 0 \\ x, & x < 0 \end{cases}$



graph of $g(x)$ will be as like as shown.

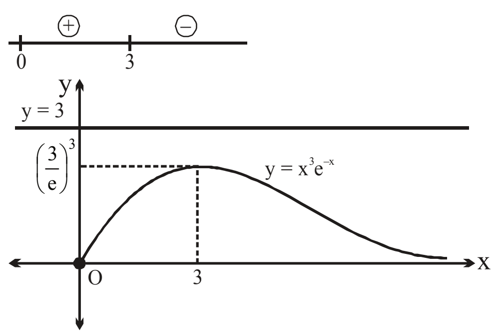
So, $g(x)$ is one one into function.

45. Point of intersection of $y = x$ and $y = x^3$ for $x < 0$ is only $x = -1$.

For $x \ge 0$, for point of intersection of $y = x^3$ and $y = 3e^x \Rightarrow x^3 = 3e^x \Rightarrow x^3 e^{-x} = 3$

So, number of solutions is same as point of intersection of $y = x^3 e^{-x}$ and $y = 3$.

Now, $y = x^3 e^{-x} \Rightarrow \dfrac{dy}{dx} = 3x^2 e^{-x} - x^3 e^{-x} = (3 - x)\,x^2 e^{-x}$



$\therefore$ no point of intersection for $x \ge 0$.

$\therefore$ Total number of points of intersection = 1.

46. $g'(x) = 2\left[f'\left(\dfrac{x^2}{2}\right) - f'(6 - x^2)\right]$

$g'(x) > 0$ if $x > 0$ and $f'\left(\dfrac{x^2}{2}\right) > f'(6 - x^2)$

$\Rightarrow \quad \dfrac{x^2}{2} > 6 - x^2 \quad \Rightarrow \quad x > 2$

($\because$ $f''(x) > 0$, $\therefore$ $f'(x)$ is an increasing function)

$g'(x) > 0$ if $x < 0$ and $f'\left(\frac{x^2}{2}\right) < f'(6 - x^2)$

$$\Rightarrow \quad \frac{x^2}{2} < 6 - x^2 \quad \Rightarrow \quad -2 < x < 0$$

$$\therefore \quad g'(x) = \begin{cases} < 0 \text{ in } (-\infty, -2) \\ > 0 \text{ in } (-2, 0) \\ < 0 \text{ in } (0, 2) \\ > 0 \text{ in } (2, \infty) \end{cases}$$

Hence g (x) attains minimum for $x = \pm 2$ and maximum for $x = 0$.

47. $f'(x) = \begin{cases} 3x^2 - 2x + 10, & x \le 1 \\ -2, & x > 1 \end{cases}$

since for the quadratic $3x^2 - 2x + 10$, $D < 0$ hence it is always positive

$\Rightarrow$ function f(x) increases in $(-\infty, 1)$ and decreasing in $(1, \infty)$

$\therefore$ for max at $x = 1$

$$f(1) \ge f(1^+)$$

$$\Rightarrow \quad 1 - 1 + 10 - 5 \ge -2 + \log_2 (a^2 - 2)$$

$$\log_2 (a^2 - 2) \le 7$$

$$a^2 - 2 \le 2^7$$

$$\Rightarrow \quad a^2 \le 130$$

Hence, $\quad 2 < a^2 \le 130$

$$\Rightarrow \quad a \in \{\pm 2, \pm 3, \pm 4, \pm 5, \pm 6, \pm 7, \pm 8, \pm 9, \pm 10, \pm 11\}$$

$\therefore$ No. of integral values of a = 20

48. $f^3(x) \Rightarrow f^3(x)\{4f'(x) - x^2\} = 0$

$f^3(x) = 0 \;\Rightarrow$ Not possible

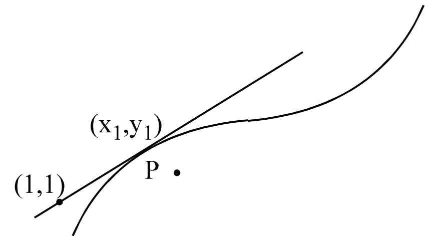
$$f'(x) = \frac{x^2}{4} \quad \Rightarrow \quad f(x) = \frac{x^3}{12} + C$$

$$\text{Also, } f(0) = \frac{-1}{4} = C \;\Rightarrow\; f(x) = \frac{x^3}{12} - \frac{3}{12}$$

$$\therefore \quad \Rightarrow f(x) = \frac{x^3 - 3}{12}$$

$$\therefore \; f(3) = \frac{27 - 3}{12} = \frac{24}{12} = 2.$$

49.



$y = 2$

$\frac{dx}{dt} = 2 - 2t;\quad \frac{dy}{dt} = 2t + 1$

$\left.\frac{dy}{dx}\right]_t = \frac{2t+1}{2(1-t)} = \frac{y_1 - 1}{x_1 - 1} = \frac{(t+t^2-1)}{(2t-t^2-1)}$

$\frac{2t+1}{2(t-1)} = \frac{t^2+t-1}{(t-1)^2}$

$\Rightarrow\ t = \frac{1}{3}$ (**note that** at t = 1, dy/dx $\to \infty$, now tanget is $\perp$ to x-axis; $\therefore$ eq^n is x=1)

when $t = \frac{1}{3}$, $\left.\frac{dy}{dx}\right]_{t=1/3} = \frac{2t+1}{2(1-t)} = \frac{(2/3)+1}{2(2/3)} = \frac{5}{3}\cdot\frac{3}{4} = \frac{5}{4}$

equation of line $y - 1 = \frac{5}{4}(x-1) \Rightarrow 4y - 4 = 5x - 5 \Rightarrow 5x - 4y = 1$. if $t = 1$ then $x = 1$.

50. $\cos^3 t = \frac{1}{8} \Rightarrow \cos t = \frac{1}{2} \Rightarrow t = \frac{\pi}{3}$

now $\frac{dy}{dx} = \frac{y'(t)}{x'(t)} = -\frac{2\sin t\cos t}{3\cos^2 t\sin t} = -\frac{2}{3}\sec t$

$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{3}} = -\frac{4}{3}$.]

51. Let $x = a\sin^4\theta$ and $y = a\cos^4\theta$, where $a = 400$

so that $\sqrt{x} + \sqrt{y} = 20\ \forall\ \theta \in R$

Now $\frac{dy}{dx} = -\frac{\cos^2\theta}{\sin^2\theta}$

So, equation of tangent is $(y - a\cos^4\theta) = -\frac{\cos^2\theta}{\sin^2\theta}(x - a\sin^4\theta)$

$\therefore$ x-intercept = $a\sin^2\theta$
y-intercept = $a\cos^2\theta$

Hence sum of intercepts = $(a\sin^2\theta) + (a\cos^2\theta) = a = 400$.

52.

We have $f(x) = ax + \cos 2x + \sin x + \cos x \Rightarrow f'(x) = a - 2\sin 2x + \cos x - \sin x$

As $f'(x) \geq 0$ for any real number $x \Rightarrow a \geq 2\sin 2x + \sin x - \cos x$ ......[]

Let $t = \sin x - \cos x = \sqrt{2}\sin\left(t - \frac{\pi}{4}\right) \Rightarrow -\sqrt{2} \leq t \leq \sqrt{2}$,

so the inequality can be written as $a \geq -2t^2 + t + 2$

Let $g(t) = -2t^2 + t + 2 = -2\left(t - \frac{1}{4}\right)^2 + \frac{17}{8}$

then range of $g(t)$ for $-\sqrt{2} \leq t \leq \sqrt{2}$ is $g\left(-\sqrt{2}\right) \leq g(t) \leq g\left(\frac{1}{4}\right)$

$\Rightarrow \quad -2-\sqrt{2} \leq g(t) \leq \frac{17}{8}$

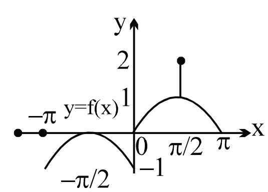
So, the range of a can be found $a \geq \max_{|t| \leq \sqrt{2}}$

$\Rightarrow a \geq \frac{17}{8} \Rightarrow a \in \left[\frac{17}{8}, \infty\right)$

Hence, $(m+n)_{\text{least}} = 17 + 8 = 25$.

53 to 56.

53. $f(x) = \begin{cases} 0, & x = \pm\pi, 0 \\ -\sin x - 1, & -\pi < x < 0 \\ \sin x, & 0 < x < \frac{\pi}{2}, \frac{\pi}{2} < x < \pi \\ 2, & x = \frac{\pi}{2} \end{cases}$



$f(x)$ has discontinuity at $x = \pm\pi, 0, \frac{\pi}{2}$

54. $(x) \neq 1, -1$ for any x.

55. $f(x)$ has maxima at $x = \pm\frac{\pi}{2}$ and $-\pi$

56. $\int_{-\pi/2}^{\pi/2} f(x)\,dx = \int_{-\pi/2}^{0} f(x)\,dx + \int_{0}^{\pi/2} f(x)\,dx = \int_{-\pi/2}^{0} (-\sin x - 1)\,dx + \int_{0}^{\pi/2} \sin x\,dx$

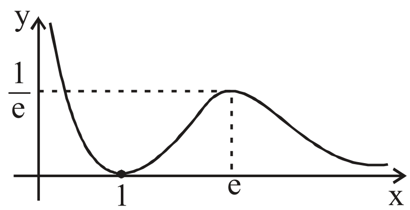
$= 1 - \frac{\pi}{2} + 1 = 2 - \frac{\pi}{2}$.

57to59.

$px = |\ln x| \Rightarrow p = \frac{|\ln x|}{x}$

$$f(x) = \frac{|\ln x|}{x} = \begin{cases} \dfrac{\ln x}{x}, & x \ge 1 \\ \dfrac{-\ln x}{x}, & 0 < x < 1 \end{cases}$$

$$f'(x) = \begin{cases} \dfrac{1-\ln x}{x^2}, & x > 1 \\ \dfrac{-(1-\ln x)}{x^2}, & 0 < x < 1 \end{cases}$$



$\therefore$ $f(x) \downarrow$ in $(0, 1)$ and (e, ∞)

and $f(x) \uparrow (1, e)$

graph of $f(x)$ will be as in diagram

Clearly line $y = p$ will cut graph at 3 points if $p \in \left(0, \frac{1}{e}\right)$

Clearly line $y = p$ will cut graph at 1 point if $p \in \left(\frac{1}{e}, \infty\right) \cup \{0\}$

Clearly line $y = p$ will cut graph at 2 points if $p = \frac{1}{e}$

60. $f(x) = \left(\frac{a^2-1}{3}\right) x^3 + (a - 1) x^2 + 2x + 1$ is $\uparrow \forall x \in R$

$(a^2 - 1) x^2 + 2(a - 1) x + 2$

$8x^2 - 8x + 2$

$2(4x^2 - 4x + 1) = (2x - 1)^2$

$f'(x) = 2 > 0$ hence $a = 1$

then for $f'(x) > 0$

$D < 0$

i.e. $4 (a - 1)^2 - 8 (a^2 - 1) < 0$

i.e. $a^2 - 2a + 1 - 2a^2 + 2 < 0$

$- a^2 - 2a + 3 < 0$

i.e. $a^2 + 2a - 3 > 0$

i.e. $(a + 3) (a - 1) > 0$

$(-\infty, -3] \cup [1, \infty)$.

more than to $a = 1$

$f(x) = 2x + 1$

which 4 always $\uparrow$

61. **(a)** $(-\infty, 0]$;

(b) $\uparrow$ in $\left(1, \frac{5}{3}\right)$ and $\downarrow$ in $(-\infty, 1) \cup \left(\frac{5}{3}, \infty\right) - \{-3\}$;

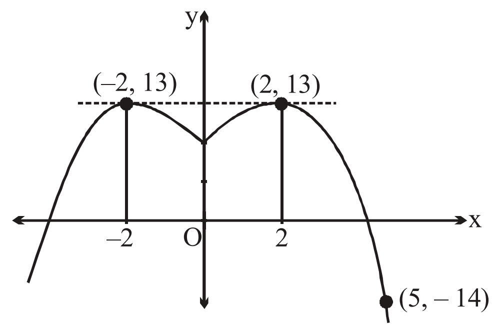
(c) $x = \frac{5}{3}$

(d) removable discont. at $x = -3$ (missing point) and non removable discont. at $x = 1$ (infinite type)

(e) -2.

62. Let $g(x) = 1 + 12x - 3x^2$

So, $g'(x) = 12 - 6x = 0 \Rightarrow x = 2$



Graph of $f(x) = 1 + 12|x| - 3x^2$

Also, $f(x) = g(|x|) = 1 + 12|x| - 3x^2$

$\therefore$ global maximum of $f(x)$ at $(x = 2 \text{ or } -2) = 13$

and global minimum of $f(x)$ occurs at $x = 5$.

Note that $f(5) = 61 - 75 = -14$.

Hence difference $= 13 - (-14) = 27$.

63. Given, $9 + f''(x) + f'(x) = x^2 + f^2(x)$ or $9 + \frac{d^2y}{dx^2} + \frac{dy}{dx} = x^2 + y^2$

As, P be the point of maxima of f(x) so $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0$ Þ $x^2 + y^2 = 9 + \frac{d^2y}{dx^2}$

So, P(x, y) will lie inside the circle $x^2 + y^2 = 9$.

Hence, no tangent is possible.

64 to 66.

Tangent to the curves are $Y - f(x) = f'(x)(X - x)$; $y - g(x) = g'(x)(X - x)$

The two tangents intersect on y-axis $f(x) - xf'(x) = g(x) - xg'(x)$

$\Rightarrow \quad f(x) - g(x) = x\{f'(x) - g'(x)\} \Rightarrow \frac{f'(x) - g'(x)}{f(x) - g(x)} = \frac{1}{x}$

$\Rightarrow \quad \log(f(x) - g(x) = Cx$ ....(i)

Normal to the two curves are $Y - f(x) = \frac{-1}{f'(x)}(X - x)$

$y - g(x) = \frac{-1}{g'(x)}(X - x)$

The two normal intersect on x - axis $x + f(x)f'(x) = x + g(x)g'(x)$

$f(x)f'(x) = g(x)g'(x)$

Integrating both sides $f(x)^2 - g(x)^2 = C_1$

$(f(x) + g(x))(f(x) - g(x) = C_1$

$f(x) + g(x) = \frac{C_1}{Cx}$ ......(ii)

Solving equation (i) and (ii)

$\Rightarrow \quad f(x) = \frac{1}{2}\left(Cx \frac{C_1}{Cx}\right) \quad$ .....(iii)

$g(x) = \frac{1}{2}\left(\frac{C_1}{Cx} - Cx\right) \quad$ .....(iv)

Satisfy (iii) by (1,1) and (iv) by (2, 3) $\Rightarrow$ $C = -2$ and $C_1 = -8$

$f(x) = -x + \frac{2}{x}$ and $g(x) = \frac{2}{x} + x$.

67 to 69.

67. $y' = 4ax^3 - 3bx^2$
$= x^2(4ax - 3b)$
$y'' = 12ax^2 - 6bx$
$= 6x(2ax - b)$
Clearly (0, 0) will be point of inflextion and (3, – 27) is minima
so $4ax - 3b = 0$ at $x = 3$ i.e. $12a - 3b = 0$ i.e. $4a - b = 0$

also at $x = 3$ $y = -27$ putting we get $a = 1$ and $b = 4$ so other point of inflextion is $x = \frac{b}{2a} = 2 = c$.

so $a = 1$, $b = 4$, $c = 2$, so $a + b + c = 7$.

68. Putting $a = 1$, $b = 4$ and $c = 2$ we get equation
as $y = 2x^3 - 9Kx^2 + 12K^2x + 1$
so $y' = 6x^2 - 18Kx + 12K^2$
whose roots are K, 2K
Since we want $x_1 = x_2^2$ so K must be
+ve so at x = K we will get local max.
and at x= 2K we will get local min.
so $x_2 = K$ and $x_1 = 2K$
Now $x_1 = x_2^2$
$2K = K^2$
i.e. $K = 2 \Rightarrow K = c = 2$ **(from question 1)**
Also K = 0 rejected as x_1, x_2 distinct. so wer is (C)

69. putting $a = 1$, $b = 4$, $c = 2$ we get

$$\int \frac{dx}{(x+1)^{8/7}(x-4)^{6/7}} = \int \frac{dx}{(x+1)^2\left(\frac{x-4}{x+1}\right)^{6/7}}$$

Put $\frac{x-4}{x+1} = t$ so $\frac{5}{(x+1)^2}dx = dt$

so $I = \frac{1}{5}\int \frac{dt}{t^{6/7}} = \frac{7}{5}t^{1/7} = \frac{7}{5}\left(\frac{x-4}{x+1}\right)^{1/7} + c$

70 to 72.

70. Since minimum occurs before maximum, so $a < 0$.

Also, a is root of $x^2 - x - 6 = 0 \Rightarrow a = -2$.

Let $g(x) = ax^3 + bx^2 + cx + d \Rightarrow g(x) = -2x^3 + bx^2 + cx + d$

So, $g'(x) = -6x^2 + 2bx + c = -6(x+2)(x-2) \Rightarrow b = 0, c = 24$.

$\Rightarrow$ $a + b + c = -2 + 0 + 24 = 22$.

71. Clearly, the equation $ax^2 + bx + c = 0$, is $-2x^2 + 0\cdot x + 24 = 0$

$\Rightarrow x = \pm 2\sqrt{3}$,

so roots of above equation are opposite in sign.

72. Since, minimum and maximum values are non-zero, so both $g(-2) > 0$ and $g(2) > 0$.

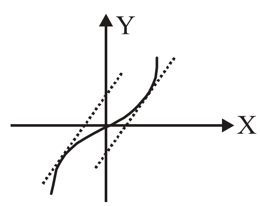
Now, $g(-2) > 0 \Rightarrow d > 32$

Also, $g(2) > 0 \Rightarrow d > -32$

Then, $a = -2, b = 0, c = 24, d > 32$

Hence, $d_{\text{smallest positive integral}} = 33$.

73. $\frac{dy}{dx} = 2$, $\frac{dy}{dx} = \frac{1}{\sqrt{(1-x^2)}}$



$2 = \frac{1}{\sqrt{1-x^2}} = 1 - x^2 = \frac{1}{4}, x^2 = \frac{3}{4}$

$x = \pm\frac{\sqrt{3}}{2}$ $\quad y = \sin^{-1}\left(\pm\frac{\sqrt{3}}{2}\right) = \pm\frac{\pi}{3}$

$\sin^{-1} x = 2x + c, c = \sin^{-1} x - 2x = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \frac{2\sqrt{3}}{2} = \frac{\pi}{3} - \sqrt{3}$

$c = \sin^{-1} x - 2x = \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) - 2\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3} + \sqrt{3}$.

74. Here $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$

$\therefore$ $f'(x) = 6x^2 - 18ax + 12a^2$

$f''(x) = 12x - 18a$

for max. / min. value, $6x^2 - 18ax + 12a^2 = 0$

$\Rightarrow$ $6(x-a)(x-2a) = 0 = 0$ Thus $x = a, x = 2a$

$f''(a) = 12a - 18a = -6a < 0$

$f''(2a) = 24a - 18a = 6a < 0$

$\therefore$ f(x) ia max at $x = a$ and minimum at $x = 2a$

$\Rightarrow$ $p = a$ and $q = 2a$

given that $p^2 = q$ so $a^2 = 2a$

$a = 2.$

75. Since C_2 is the reflection of C_1 in the origin,

we have $(x, y) \in C_1 \Leftrightarrow (-x, -y) \in C_2$

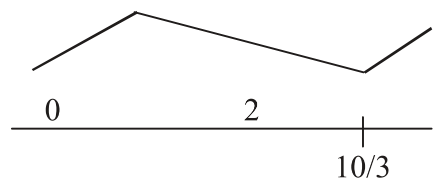
$\Leftrightarrow (-x)^2 - 13(-x) + 4(-y) = 1$

$\Leftrightarrow x^2 + 13x - 4y = 1$

76. **Proof:** $x^3 + 1 = 5x^2 + 5 \Rightarrow x^3 - 5x - 4 = 0$

Let $f(x) = x^3 - 5x - 4$

$f'(x) = 3x^2 - 10x = x(3x - 10)$

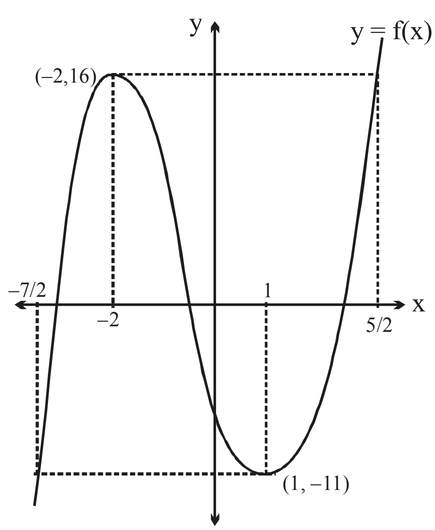


hence f is decreasing in $(0, 10/3)$

$\therefore$ $f(2) < f(0) < 0$

hence no root in $[0, 2]$ Hence proved]

77.



$m = \dfrac{-7}{2}, n = \dfrac{5}{2}$

$4(n - m) = 4\left(\dfrac{5}{2} + \dfrac{7}{2}\right) = 24.$

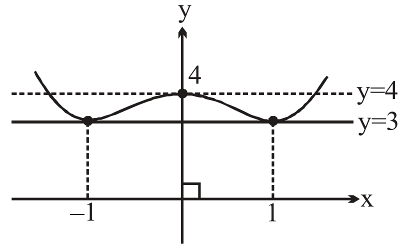
78. The degree of $f(x)$ must be 4. So degree of $f'(x)$ is 3

$\Rightarrow$ $f'(x) = k\,x\,(x - 1)(x + 1)$

By solving, $f(x) = x^4 - 2x^2 + 4$

Graph of $y = f(x)$:

For 4 distinct real solution : $k \in (3, 4)$. .]



79. $f(1) \geq f(1^+)$

$5 \geq -2 + \log_2 (b^2 - 1)$

$2^7 \geq b^2 - 1$

$\therefore \quad 1 < b^2 \leq 129$

$\therefore \quad b \in \left[-\sqrt{129}, -1\right] \cup \left(1, \sqrt{129}\right]$

$\Rightarrow \quad b_{max.} = 11.$.

80. As given $g[f(x)] = x$

$\therefore \quad g'(f(x)) \,.\, f'(x) = 1$

$\therefore \quad g'(f(x)) = \dfrac{1}{f'(x)} = \sqrt{x^2 + x + 3}$ ....(1)

If upper limit of integration is $x = 2$ then $f(2) = 0$

$\therefore$ from equation $g'f(2) = g'(0) = \sqrt{2^2 + 2 + 3} = 3$

81to83.

$\because$ f(x) be a odd cubic polynomial function,

$\therefore$ It must be of the form, $f(x) = ax^3 + bx$

Now, $\displaystyle \lim_{h \to \frac{1}{\sqrt{3}}} \frac{f'(h) - 2}{h - \frac{1}{\sqrt{3}}} = 4\sqrt{3}$

$f'\left(\dfrac{1}{\sqrt{3}}\right) = 2 \Rightarrow 3ax^2 + b \Big|_{x = \frac{1}{\sqrt{3}}} = 2$

$\Rightarrow a + b = 2$ ....(1)

$\Rightarrow \displaystyle \lim_{h \to \frac{1}{\sqrt{3}}} \frac{f''(h)}{1} = 4\sqrt{3}$

$\Rightarrow f''\left(\dfrac{1}{\sqrt{3}}\right) = 4\sqrt{3} \Rightarrow 6ax \Big|_{x = \frac{1}{\sqrt{3}}} = 4\sqrt{3}$

$\Rightarrow \quad a = 2$ ....(2)

from (1)

$b = 0$

$\therefore \quad f(x) = 2x^3$

81. $f'(x) = 6x^2$

$$f'\left(\frac{1}{\sqrt{6}}\right) = 1.$$

82. $\phi(x) = \int_1^x e^t \; f(x+1-t)\, dt$

$$\phi(x) = \int_1^x e^{x+1-t} f(t)\, dt \; = e^{x+1} \int_1^x e^{-t} f(t)\, dt$$

$$\phi'(x) = e^{x+1} \cdot e^{-x} f(x) + \int_1^x e^{-t} f(t)\, dt \cdot e^{x+1}$$

$$= e\, f(x) + e^{x+1} \cdot \int_1^x e^{-t} f(t)\, dt$$

$\phi'(1) = ef(1) + 0 \; = 2e$

83. $gof(x) = x \;\forall\; x \in R$

$\Rightarrow$ g is the inverse function of f.

$g(2) = a \;\Rightarrow\; g^{-1}(a) = 2 \;\Rightarrow\; f(a) = 2 \;\Rightarrow\; 2a^3 = 2 \Rightarrow\; a = 1$

$g(16) = b \;\Rightarrow\; g^{-1} = 16$

$\Rightarrow\; f(b) = 16 \;\Rightarrow\; 2b^3 = 16 \;\Rightarrow b = 2.$

$$\therefore \quad \int_1^2 f(x)\, dx + \int_2^{16} g(x)\, dx \; = 2 \times 16 - 1 \times 2 = 30$$

84. Let $\dfrac{x}{y} + \dfrac{y}{x} = t$ $\qquad \therefore \; |t| \ge 2$

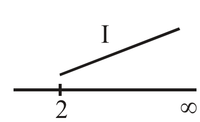
$8k^2 (t^2 - 2) + 4kt - 15 = 0$

$$k = \frac{-4t \pm \sqrt{16t^2 + 32(t^2-2)\times 15}}{2\times 8\,(t^2-2)} = \frac{-t \pm \sqrt{t^2 + 30\,(t^2-2)}}{4\;(t^2-2)}$$

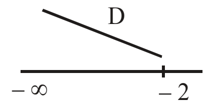
$$= \frac{-t \pm \sqrt{31t^2 - 60}}{4\,(t^2-2)} = \frac{15}{2\left(t + \sqrt{31t^2-60}\right)}$$

for $t \ge 2$ $\qquad f(t) = t + \sqrt{31t^2 - 60}$ is $\uparrow$

$f(2)\big|_{\min} = 10 \qquad \therefore\; k = \dfrac{15}{20} = \dfrac{3}{4}$



for $t \le -2$ $\quad f(t)$ is $\downarrow$ and minimum occurs at $t = -2$



$f(-2) = 6$ $\qquad \therefore\ k = \dfrac{15}{12} = \dfrac{5}{4}$

Hence $k_{max} = \dfrac{5}{4} = M$

$\therefore\ 100M = 125.$

85. Let $\cos x = t, -1 \le t \le 1$

$g(t) = \dfrac{1+t}{1+t+t^2}, -1 \le t \le 1$

$g'(t) = \dfrac{-t(t+2)}{(1+t+t^2)} = 0 \Rightarrow t = 0$

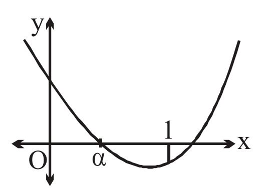
So, maximum of $g(t)$ occur at $t = 0$.
maximum value = 1

Aliternatively: $f(x) = 1 - \dfrac{\cos^2 x}{1+\cos x+\cos^2 x} \ge 1$

Hence $f(x)\,]_{max} = 1$ when $\cos x = 0$

86. 2

(A) $f(x) = x^4 + 2x^2 - 6x + 2$
$f'(x) = 4x^3 + 4x - 6 = 2(2x^3 + 2x - 3)$
$f''(x) = 2\,(6x^2 + 2) > 0 \Rightarrow f' \uparrow$
$\Rightarrow f'$ is zero at only one point i.e. between 0 and 1.]



(B) $f'(x) = 5(x^4 + 1) > 0$ range of f is $(-\infty, \infty)$.

(C) $f'(x) = nx^{n-1} + a = 0 \Rightarrow$ only one root.

(D) $f'(x) = 3(x^2 - 1) = 3(x-1)(x+1) < 0.$

87. Integrating $\dfrac{dy}{dx} = 3x^2 - 4x + A;\ \left.\dfrac{dy}{dx}\right]_{x=1} = 0$

$\Rightarrow A = 1$

Hence $\dfrac{dy}{dx} = 3x^2 - 4x + 1$; $\qquad$ Integrating again,

$y = x^3 - 2x^2 + x + B\ ; y]_{x=1}$

$\Rightarrow B = 5.$ $\qquad$ Thus $y = x^3 - 2x^2 + x + 5$

Also $\dfrac{dy}{dx} = 0$ given $x = \dfrac{1}{3}$ and $x = 1$ $\qquad f(1/3) = \dfrac{139}{27}$; $f(1) = 5$

also $\qquad f(0) = 5$; $f(2) = 7$.

88. **Method I**

$$\because\ g(x) = \left(\frac{\pi}{2}-\sin^{-1}x\right) + \left(\frac{\pi}{2}-\cos^{-1}x^2\right) + \left(\frac{\pi}{2}-\sin^{-1}x^3\right) + \ldots\ldots. + \left(\frac{\pi}{2}-\cos^{-1}x^{2n}\right)$$

$$= \frac{2n\pi}{2} - f(x) = n\pi - f(x)$$

$\therefore$ g(x) will be maximum when f(x) will be minimum

$\therefore$ $g(x)_{max} + f(x)_{min} = n\pi = 2010\pi$ (given)

$\therefore$ $n = 2010$,

Method II

Let $S(x) = \sin^{-1}x + \sin^{-1}x^3 + \sin^{-1}x^5 + \ldots\ldots.. + \sin^{-1}x^{2n-1}$

and $C(x) = \cos^{-1}x^2 + \cos^{-1}x^4 + \cos^{-1}x^6 + \ldots\ldots + \cos^{-1}x^{2n}$

$$\therefore\ f(x) = S(x) + C(x) \text{ and } g(x) = 2n\cdot\left(\frac{\pi}{2}\right) - f(x) = n\pi - f(x)$$

Clearly S(x) is monotonically increasing and c(x) is m.i. in [–1, 0) and m.d. in (0, 1]

$\therefore$ C(x) will be maximum at $x = 0$ and minimum at $x = \pm 1$

$\therefore$ $f(x) = S(x) + C(x)$ will be minimum at $x = -1$

$$\therefore\ f(x)_{min} = S(x)_{min} + C(x)_{min} = n\left(\frac{-\pi}{2}\right) + 0 = \frac{-n\pi}{2}$$

$$\therefore\ g(x)_{max} = n\pi - f(x)_{min} = n\pi - \left(\frac{-n\pi}{2}\right) = \frac{3n\pi}{2}$$

$$\therefore\ f(x)_{min} + g(x)_{max} = \frac{2n\pi}{2} = n\pi = 2010\pi \text{ (given)}$$

$\therefore$ $n = 2010$.]

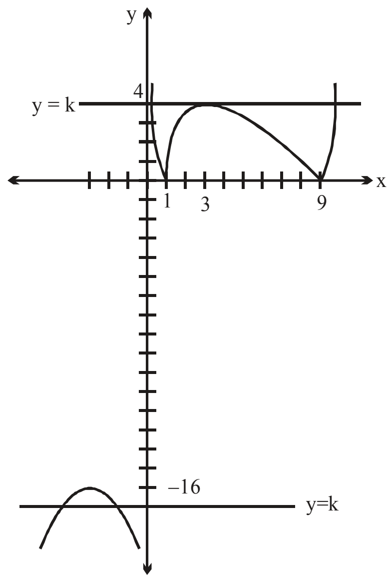
89. $$\frac{|x^2-10x+9|}{x} = k$$

$$f(x) = \frac{|x^2-10x+9|}{x} = \begin{cases} \dfrac{(x-1)(x-9)}{x} & \text{if } x \in (-\infty, 1)\cup(9,\infty) \\ \dfrac{-(x-1)(x-9)}{x} & \text{if } x \in (1,9) \end{cases}$$

$$f'(x) = \frac{x(2x-10)-(x^2-10x+9)}{x^2} = \frac{2x^2-10x-x^2+10x-9}{x^2}$$

$$= \frac{x^2-9}{x^2} = \frac{(x-3)(x+3)}{x^2}$$

$$\Rightarrow \quad f'(x) = \begin{cases} \dfrac{x^2-9}{x^2} & \text{for } x \in (-\infty, 1) \cup (9, \infty) \\ & \uparrow \text{ in } |x|>3 \text{ and } \downarrow \text{ in } |x|<3 \\ \dfrac{9-x^2}{x^2} & \text{for } x \in (1, 9) \end{cases}$$



From the graph $(-\infty, -16] \cup [4, \infty)$.

Aliter: For the given condition to be satisfied, the value of k must lie between the slopes of line OA and OB.

For slope of line OA,

$-(x^2 - 10x + 9) = kx$ must have equal roots.

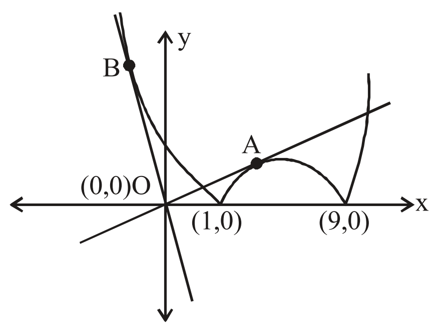
$\Rightarrow \quad (k - 10)^2 - 36 = 0 \Rightarrow k = 4, 16 \Rightarrow k = 4$

For slope of the line OB,

$x^2 - 10x + 9 = kx$ must have equal roots

$\Rightarrow \quad (k + 10)^2 - 36 = 0 \Rightarrow k = -4, -16$

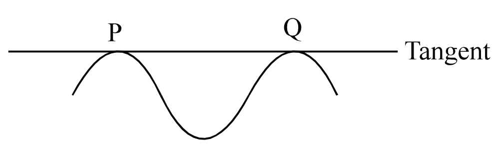
$\therefore \quad k \in [4, \infty) \cup (-\infty, -16]$.]



90. Let the 2 points on the curve be $P \equiv (a, a^4 - 2a^2 - a)$ and $Q \equiv (b, b^4 - 2b^2 - b)$

$$\left.\frac{dy}{dx}\right|_P = 4x^3 - 4x - 1 = 4a^3 - 4a - 1$$

equation of tangent line at P



$y - (a^4 - 2a^2 - a) = (4a^3 - 4a - 1)(x - a)$

$\Rightarrow$ $y = (4a^3 - 4a - 1)x + (a^4 - 2a^2 - a) - a(4a^3 - 4a - 1)$

$\therefore$ $y = (4a^3 - 4a - 1)x - 3a^4 + 2a^2$ ....(1)

|||ly tangent at Q is

$y = (4b^3 - 4b - 1)x - 3b^4 + 2b^2$ ....(2)

given (1) and (2) is same line and hence

$4a^3 - 4a - 1 = 4b^3 - 4b - 1$ (equating slope)

i.e. $a^3 - b^3 = a - b$ ....(3) and $-3a^4 + 2a^2 = -3b^4 + 2b^2$ (equating constant term)

i.e. $2(a^2 - b^2) = 3(a^4 - b^4)$ ....(4)

(3) $\Rightarrow$ $(a - b)(a^2 + ab + b^2) = (a - b)$ $(a \neq b)$

$\therefore$ $a^2 + ab + b^2 = 1$ ....(5)

(4) $\Rightarrow$ $2(a^2 - b^2) = 3(a^2 - b^2)(a^2 + b^2)$

$\therefore$ $a^2 = b^2$ or $a^2 + b^2 = \frac{2}{3}$

Case-I: If $a^2 = b^2$

$\Rightarrow$ $a = -b$ $(a \neq b)$

put in (5) $a^2 = 1 \Rightarrow a = \pm 1$

$$\left.\begin{array}{ll}\text{hence} & \text{if } a = 1,\ b = -1 \\ \text{and} & \text{if } a = -1,\ b = 1\end{array}\right\} \text{or vice versa}$$

$\therefore$ $P \equiv (1, -2)$; $Q \equiv (-1, 0)$

Case-II: If $a^2 + b^2 = \frac{2}{3}$ and $a^2 \neq b^2$

from (5) $ab = \frac{1}{3} \Rightarrow b = \frac{1}{3a}$ put in $a^2 + b^2 = \frac{2}{3}$

$\therefore$ $a^2 + \frac{1}{9a^2} = \frac{2}{3} \Rightarrow \frac{9a^4 + 1}{9a^2} = \frac{2}{3}$

$\Rightarrow$ $9a^4 - 6a^2 + 1 = 0 \Rightarrow (3a^2 - 1)^2 = 0$

$\therefore$ $a^2 = \frac{1}{3} \Rightarrow b^2 = \frac{1}{3}$

$\therefore$ $a^2 = b^2$ which is not possible

$\Rightarrow$ this case is not possible

91. Let $f(\theta) = (1 - t)\sin\theta + t\tan\theta - \theta$ in $\left(0, \frac{\pi}{2}\right)$

$f'(\theta) = (1 - t)\cos\theta + t\sec^2\theta - 1$

$$f'(\theta) = \frac{(1-t)\cos^3\theta - \cos^2\theta + t}{\cos^2\theta}$$

$$f'(\theta) = \frac{t(1-\cos^3\theta) - \cos^2\theta(1-\cos\theta)}{\cos^2\theta} = \underbrace{\frac{(1-\cos^3\theta)}{\cos^2\theta}}_{+\text{ve in }(0,\ \pi/2)}\left[t - \frac{\cos^2\theta}{\cos^2\theta+\cos\theta+1}\right]$$

$$f'(\theta) = \left(t - \frac{1}{\sec^2\theta+\sec\theta+1}\right) = \left(t - \frac{1}{\left(\sec\theta+\frac{1}{2}\right)^2+\frac{3}{4}}\right)$$

$$\therefore \quad \left.\left(\sec\theta+\frac{1}{2}\right)^2+\frac{3}{4}\right|_{\min} = \frac{12}{4} = 3 \Rightarrow \left.\frac{1}{\left(\sec\theta+\frac{1}{2}\right)^2+\frac{3}{4}}\right|_{\max} = \frac{1}{3}$$

$\therefore$ If $t \ge \frac{1}{3}$ then $f'(\theta) > 0$

$\Rightarrow$ $f \uparrow \ \forall\ \theta \in \left(0, \frac{\pi}{2}\right)$ $\Rightarrow$ $f(\theta) > f(0)$ but $f(0) = 0$

$\therefore$ $f(\theta) > 0\ \forall\, \theta \in \left(0, \frac{\pi}{2}\right)$

92. Slope of tangent at infinity ot $y = f(x)$ is 1

$$\therefore \lim_{x\to\infty} f'(x) = 1 \qquad \Rightarrow \lim_{x\to\infty}\left(\sqrt[3]{8x^3+mx^2} - nx\right) = 1$$

Let $x = \frac{1}{y}$, As $x \to \infty$, $y \to 0^+$

$$\Rightarrow \lim_{y\to 0^+}\left(\sqrt[3]{\frac{8}{y^3}+\frac{m}{y^2}} - \frac{n}{y}\right) = 1 \Rightarrow \lim_{y\to 0^+}\frac{\sqrt[3]{8+my} - n}{y} = 1$$

$$\Rightarrow \lim_{y\to 0^+}\frac{8^{\frac{1}{3}}\left[\left(1+\frac{my}{8}\right)^{\frac{1}{3}} - \frac{n}{2}\right]}{y} = 1$$

By expion

$$\Rightarrow \lim_{y\to 0^+} \frac{2\left[1+\frac{m}{8}\cdot\frac{1}{3}y+\frac{\left(\frac{1}{3}\right)\cdot\left(-\frac{2}{3}\right)}{2!}\cdot\left(\frac{my}{8}\right)^2+\ldots\infty-\frac{n}{2}\right]}{y}=1$$

$$\Rightarrow \lim_{y\to 0^+} \frac{2\left[\left(1-\frac{n}{2}\right)+\frac{m}{24}\cdot y-\frac{1}{9}\left(\frac{my}{8}\right)^2+\ldots\infty\right]}{y}=1$$

The above relation will be true if $1-\frac{n}{2}=0$ and $\frac{m}{12}=1$

$\therefore$ n = 2 and m = 12 $\quad m=\frac{1}{6}n$, $m^2+n^2=148$, $m+n=14$ and $|m-n|=10$

93. Given curve is $ax^3-y+b=0$ $\quad\Rightarrow y=ax^3+b$

Let point P_1 be (t_1, at_1^3+b)

Slope of tangent $=\left.\frac{dy}{dx}\right|_{P_1}=3at_1^2$

$\therefore$ Equation of tangent is

$$y-(at_1^3+b)=3at_1^2(x-t_1)$$

$\therefore$ Tangent meets curve at $P_2(t_2, at_2^3+b)$

$\therefore$ $(at_2^3+b)-(at_1^3+b)=3at_1^2(t_2-t_1)$

$\Rightarrow$ $a(t_2^3-t_1^3)=3at_1^2(t_2-t_1)$

$\Rightarrow$ $t_2^2+t_1^2+t_2t_1=3t_1^2$ $\quad(\therefore t_1\neq t_2)$

$\Rightarrow$ $t_2^2-t_2t_1-2t_1^2=0$

$\Rightarrow$ $(t_2+2t_1)(t_2-t_1)=0$

$\Rightarrow$ $t_2=-2t_1$

Similarly $\quad t_3=-2t_2$

$\therefore$ abscissae are in G.P. for all values of a and b.

94. If $f(x)=|x-\alpha|^\beta$, then $f'(\alpha)$ exists if $\beta>1$, $f''(\alpha)$ exist if $\beta>2$ and $f'''(\alpha)$ exists if $\beta>3$

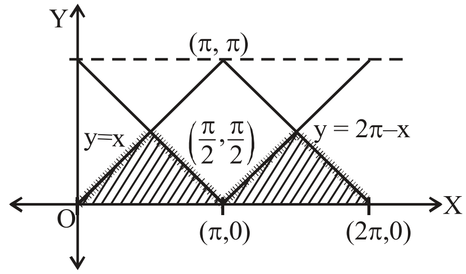
$\therefore$ $f'''(e)$ exists since $\pi>3$

$f'''(\pi)$ does not exist since $e>3$.

95. Clearly, $\int_0^{2\pi} \text{Min.}\left(|x-\pi|, \cos^{-1}(\cos x)\right)dx$

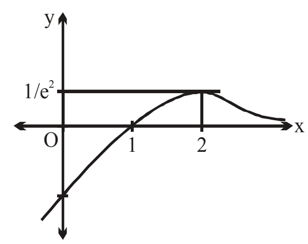
$= 2\left(\frac{1}{2}\times\pi\times\frac{\pi}{2}\right) = \frac{\pi^2}{2} = k\pi^2$ (Given)

So, $k = \frac{1}{2}$ . .]



96. $\int_0^x t\, f(x-t)\, dt = \int_x^0 -(x-t)\, f(t)\, dt = x\int_0^x f(t)\, dt - \int_0^x t\, f(t)\, dt$

$\Rightarrow \quad (x+1)\int_0^x f(t)\, dt - \int_0^x t\, f(t)\, dt = e^{-x} - 1$



$\Rightarrow \quad \int_0^x f(t)\, dt + f(x)\,(x+1) - x\, f(x) = -e^{-x}$

$\Rightarrow \quad \int_0^x f(t)\, dt + f(x) = -e^{-x}$

$\Rightarrow \quad f(0) = -1,\ f(x) + f'(x) = e^{-x}.$

$\Rightarrow \quad e^x\left[f(x) + f'(x)\right] = 1$

$\Rightarrow \quad \frac{d\left[e^x f(x)\right]}{dx} = 1$

$\Rightarrow \quad e^x f(x) = x + c,\ f(x) = e^{-x}(x + c)$

$f(0) = -1 \Rightarrow c = -1,\ f(x) = e^{-x}(x-1)$

$f(x) = e^{-x}(x-1)$ satisfies the first equation and solution is done

97. $\because \quad \int_a^x e^{f(t)}dt = \int_0^x g(x-t)dt + 2x + 3$

$\Rightarrow \quad \int_a^x e^{f(t)}dt = \int_0^x g(t)dt + 2x + 3$ (Using King)

differentiate both sides, we get

$e^{f(x)} = g(x) + 2$

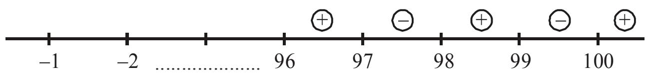
$\Rightarrow \quad g(x) = e^{f(x)} - 2$

$\Rightarrow \quad g'(x) = e^{f(x)} \cdot f'(x)$

$\because \quad e^{f(x)}$ is always greater than zero

$\therefore$ sing of $g'(x)$ is same as sign of $f'(x)$.

$\therefore$ sign of $g'(x)$



Clearly local extermum (maximum or minimum) will occur at

$x = 99, 97, 95, \ldots\ldots., 3, 1.$

$\therefore$ sum of all the values $= 1 + 3 + 5 + \ldots\ldots.. + 99$

$= \dfrac{50}{2}\,[2 \times 1 + (50 - 1) \times 2] = 2500 = 5 \times 500$

$\therefore \lambda = 5.$.

98. Consider $\phi(x) = f(x) - 2g(x)$ defined on $[0, 1]$ since $f(x)$ and $g(x)$ are differentiable for $0 \le x \le 1$, therefore $\phi(x)$ is differentiable on $(0, 1)$ and continuous on $[0, 1]$

$\therefore\ \phi(0) = f(0) - 2g(0) = 2 - 0 = 2$

$\phi(1) = f(1) - 2g(1) = 6 - 2g(1)$

Now $\phi'(x) = f'(x) - 2g'(x)$

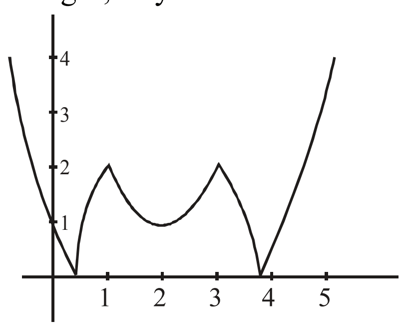
$\Rightarrow\ \phi'(c) = f'(c) - 2g'(c) = 0$ (given)

$\Rightarrow\ \phi(x)$ satisfies Rolle's theorem on $[0, 1]$ $\therefore\ \phi(0) = \phi(1)$

$\Rightarrow\ 2 = 6 - 2g(1)$

$\Rightarrow\ g(1) = 2$]

99. See graph $y = f(x) = ||x^2 - 4x + 3| - 2|$ $y = m$ is a horizontal line with intersection points, from which the x-values have different signs, only if $m > 2$.



100. $f(x) = \left(\sqrt{4 - x^2} - 3\right)^2 + \left(\sqrt{4 - x^2} + 1\right)^3$

Let $\sqrt{4 - x^2} = a \ \Rightarrow\ a \in [0, 2]$ for $x \in [-2, 2]$

$f(a) = (a - 3)^2 + (a + 1)^3$

$f'(a) = 3a^2 + 8a - 3$

$\Rightarrow\ f'(a) = 0 \ \Rightarrow\ a = 1/3,\ -3$ (to be rejected)

$\Rightarrow$ $f(a)$ will be minimum at $a = 1/3$ and correspond $x = \pm\sqrt{\dfrac{35}{9}}$

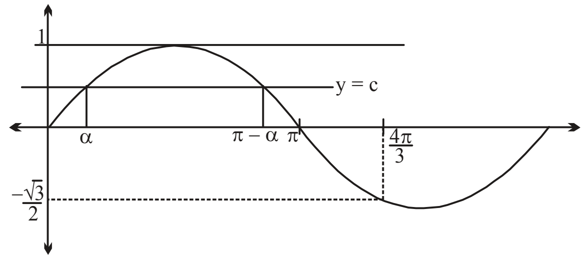
and $f(a)$ will have maxima either at $a = 0$ or at $a = 2$

now $f(0) = 10$ and $f(2) = 28$

$\Rightarrow\ f(a)|_{max} = 28$ corresponding to $x = 0$

101. $I = \int_0^{\alpha}(c - \sin x)\,dx + \int_0^{\pi-\alpha}(\sin x - c)\,dx + \int_{\pi-\alpha}^{\frac{4\pi}{3}}(c - \sin x)\,dx$

$I = c\left(4\alpha - \frac{2\pi}{3}\right) + \left(4\cos\alpha - \frac{3}{2}\right)$



$I(a) = \sin\alpha\left(4\alpha - \frac{2\pi}{3}\right) + \left(4\cos\alpha - \frac{3}{2}\right)$

$I'(\alpha) = 0 \qquad \Rightarrow \qquad a = \frac{\pi}{6}$

$\therefore \qquad c = \frac{1}{2}$.

102. Given, $f(x) = x\int_0^x 3^t(3^t - 4)\,dt - \int_0^x 3^t \cdot t(3^t - 4)\,dt$

For maxima / minima, we have

$f'(x) = 0$

$\Rightarrow \qquad \int_0^x 3^t(3^t - 4)\,dt + x \cdot 3^x(3^x - 4) - 3^x \cdot x(3^x - 4)$

$\therefore \qquad f'(x) = \int_0^x 3^t(3^t - 4)\,dt$

$f'(x) = \frac{1}{2\ln 3}(3^{2x} - 8 \cdot 3^x + 7)$

$f'(x) = \frac{1}{2\ln 3}(3^x - 1)(3^x - 7)$

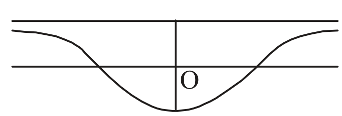
sign scheme of $f'(x)$

$f'(x) = 0 \qquad \Rightarrow \qquad x = 0,\ \log_3 7$

$\therefore \qquad x = \log_3 7$ is the point of minima.

Hence, $3^a = 3^{\log_3 7} = 7$.

103. $f(x) = \frac{x^2 - 1}{x^2 + 1}$ $\qquad f(x) = 1 - \frac{2}{1 + x^2}$; $f'(x) = \frac{4x}{(1 + x^2)^2}$



even function

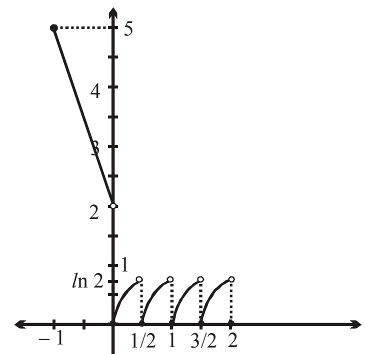
$f(0) = -1$

it $f(\infty) \Rightarrow 1$

so it is a bounded function

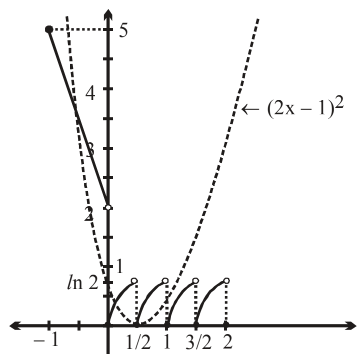
104 to 106. $_{137\text{-}38\text{-}39/aod/QZ}$

$$f(x) = \begin{cases} 2-3x, & -1 \le x < 0 \\ \ln(2x+1), & 0 \le x < \frac{1}{2} \\ \ln(2x), & \frac{1}{2} \le x < 1 \\ \ln(2x-1), & 1 \le x < \frac{3}{2} \\ \ln(2x-2), & \frac{3}{2} \le x < 2 \\ 0, & x = 2 \end{cases}$$



Graph of f(x)

104. Range of f(x) is $[0, \ln 2) \cup (2, 5]$
Integers in the range of f(x) are 0, 3, 4, 5.
$\therefore$ Number of integers in the range of f(x) is 4.



Graph of f(x)

105. $4x^2 - 4x + 1 - f(x) = 0$
$\Rightarrow f(x) = (2x-1)^2$
Clearly,
$y = f(x)$ and $y = (2x-1)^2$ meet
each other at 4 distinct points.

106. $g(x) = 3 - 2f(3-2x)$
Range of f(x) is $[0, \ln 2) \cup (2, 5]$
Range of $f(3-2x)$ is same as f(x)
Range of $2f(3-2x)$ is $[0, 2\ln 2) \cup (4, 10]$
Range of $-2f(3-2x)$ is $[-10, -4) \cup (-2\ln 2, 0]$.
Range of $3 - 2f(3-2x)$ is $[-7, -1) \cup (3 - 2\ln 2, 3]$.

107. We have $f(x) = (b^2 - 3b + 2)(\cos^2 x - \sin^2 x) + (b-1)x + \sin 2$

$\therefore \quad f'(x) = (b-1)(b-2)(-2\sin 2x) + (b-1)$

Now, $f'(x) \ne 0$ for every $x \in R$,

so $\quad (b-1)\left(1 - 2(b-2)\sin 2x\right) \ne 0 \ \forall\ x \in R$

$\therefore \quad b \ne 1$

Also, $\left|\dfrac{1}{2(b-2)}\right| > 1 \Rightarrow b \in \left(\dfrac{3}{2}, 2\right) \cup \left(2, \dfrac{5}{2}\right)$

Now, when $b = 2$, $f(x) = x + \sin 2 \Rightarrow f'(x) = 1\ (\ne 0)$.

Hence, $b \in \left(\dfrac{3}{2}, \dfrac{5}{2}\right) \quad \Rightarrow \quad b_1 = \dfrac{3}{2}$ and $b_2 = \dfrac{5}{2}$

$\Rightarrow \quad (b_1 + b_2) = \dfrac{3}{2} + \dfrac{5}{2} = \dfrac{8}{2} = 4$

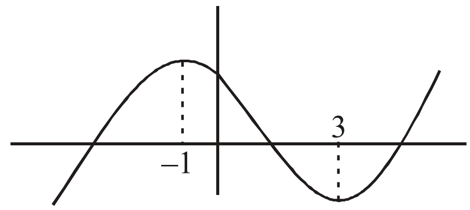
108 to 110.

(i) $f(x) = 2x^3 - 6x^2 - 18x + a$

$f'(x) = 6x^2 - 12x - 18 = 0$

$x = -1, 3$

So, $f(x)$ is injective in $x \in (-1, 3)$



in, $x \in (0, 4)$, $f(x)$ has minimum value at $x = 3$, then

$f(3) \leq 40$

$2 \cdot 3^3 - 6 \cdot 3^2 - 18 \cdot 3 + a \leq -40$

$-54 + a \leq -40$

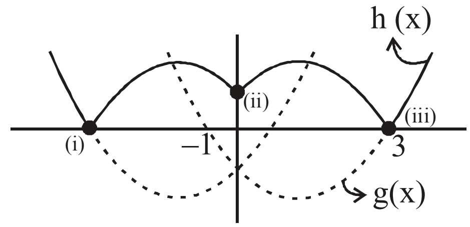
$a \leq 14$

(iii) $f'(x) = 6x^2 - 12x - 18$

$g(x) = 6(x+1)(x-3)$

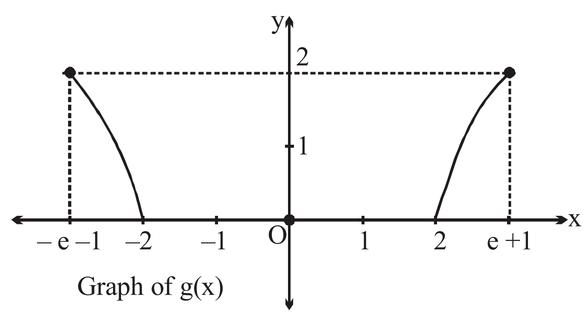
$h(x) = \left|g\left(|x|\right)\right| = \left|6|x|^2 - 12|x| - 18\right|$

number of points where h'(x) does not exist are = 3



111 to 113.

Clearly, $g(x) = \begin{cases} 2\ln(-x-1), & -e-1 \leq x < -2 \\ 0, & -2 \leq x \leq 2 \\ 2\ln(x-1), & 2 < x \leq e+1 \end{cases}$



Graph of g(x)

111. From above graph of $g(x)$, range of $g(x)$ is $[0, 2]$ and $g(x) = 0$ in interval $[-2, 2]$. Also $g(x)$ is an even continuous function.

112. Clearly, $g(x)$ is non-differentiable at $x = -2, 2$.

113. Clealry, $g(x) = k$ has exactly two distinct solutions, then integral value of k is either 1 or 2. So, sum of all possible integral values of $k = 1 + 2 = 3$.

114. $f(x) = \int_{-1}^{x} t(e^t - 1)(t-1)(t-2)^3(t-3)^5$

$f(x) = x(e^x - 1)(x-1)(x-2)^3(x-3)^5 = 0$

$n = 0, 1, 2, 3$

for $x < 0$, $(-)\,(-)\,(-)\,(-)\,(-) < 0$

$x > 0$ $(+)\,(+)\,(-)\,(-)\,(-) < 0$

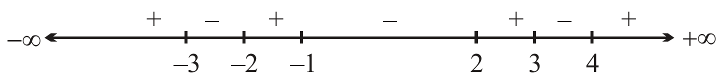
$x < 1$, $+ + - - - < 0$

$x > 1$, $+ + + - - > 0$ $\qquad$ $n = 1$ min.

|||y change]

115. We have $f'(x) = (x^2 - x + 2)(x^2 - x - 2)(x^2 - x - 6)(x^2 - x - 12)$

$$= \underbrace{(x^2 - x + 2)}_{\text{Always positive}} (x-2)(x+1)(x-3)(x+2)(x-4)(x+3)$$



$\therefore$ f(x) $\uparrow$ on $(-\infty, -3) \cup (-2, -1) \cup (2, 3) \cup (4, \infty)$

and f(x) $\downarrow$ on $(-3, -2) \cup (-1, 2) \cup (3, 4)$

Also, $f(x = -2) = 0$ and $f'(-2) = 0$

So, equation of normal is line $x + 2 = 0$

Note that f(x) has local maximum at $x = -3, -1, 3$.

So, sum of values of x at which f(x) has local maximum

$= (-3) + (-1) + 3 = -1$

116 $f(x) = \cot x - \tan x - 2\tan 2x - 4\tan 4x - 8\tan 8x = 0$

($\because \cot x - \tan x = 2\cot 2x$)

$h(x) = f(x) + g(x)$

$h(x) = x^3 + 6x - 1,\ x \in R - \left\{\frac{n\pi}{8}, n \in I\right\}$

(A) $h'(x) = 3x^2 + 6;$ $h''(x) = 6x$

$\therefore$ $h''\left(\frac{\pi}{24}\right) = \frac{\pi}{4}$

(B) $h(x) = x^3 + 6x - 1$ which is not an odd function.

(C) $h'(x) = 3x^2 + 6 > 0 \Rightarrow$ h(x) is increasing

(D) $h(x) = \lambda$, values of λ in (0, 3) for which equation has a solution if $\lambda \in (-1, 8)$]

117. $f(x) = x^3 + ax^2 + bx + c$

$f(-2) = -8 + 4a - 2b + c = -10$

$4a - 2b + c + 2 = 0$ .....(1)

$f'(x) = 3x^2 + 2ax + b$

$f'(2/3) = 0$

$\therefore 3 \cdot \frac{4}{9} + 2a\left(\frac{2}{3}\right) + b = 0 \Rightarrow \frac{4}{3} + \frac{4a}{3} + b = 0$

$\Rightarrow 4 + 4a + 3b = 0$ ....(2)

Also $f(2/3) = \frac{50}{27}$

This gives $\frac{8}{27}+\frac{4a}{9}+\frac{2b}{3}+c=\frac{50}{27}$

$8 + 12a + 18b + 27c = 50$

$12a + 18b + 27c = 42$

$4a + 6b + 9c = 14$ .....(3)

solving (1), (2) and (3) we get

$\boxed{a=-1}$, $\boxed{b=0}$ and $\boxed{c=2}$.

118. Let $\int_0^{1/a}(a^3+4x-a^5x^2)e^{ax}\,dx = \left[e^{ax}[Ax^2+Bx+C]\right]_0^{1/a}$

differentiate both sides

$e^{ax}[-a^5x^2+4x+a^3] = e^{ax}(2Ax+B)+a(Ax^2+Bx+C)\,e^{ax}$

equating coefficient

$aA = -a^5 \Rightarrow \boxed{A=-a^4}$

$aB + 2A = 4 \Rightarrow \boxed{B=\frac{2a^4+4}{a}}$

$aB - 2a^4 = 4 \Rightarrow aB = 4 + 2a^4$

$aC + B = a^3 \Rightarrow aC + \frac{2a^4+4}{a} = a^3 - \frac{2a^4+4}{a} = \frac{-a^4-4}{a}$

$\Rightarrow \boxed{C=-\frac{4+a^4}{a^2}}$

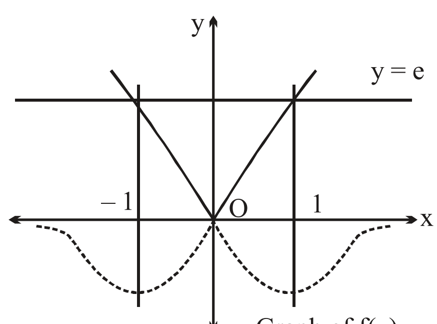
$e\left[\frac{A}{a^2}+\frac{B}{a}+C\right]-C$

$C(e-1)+\frac{A}{a^2}+\frac{B}{a}$

$\Rightarrow a^2+\frac{4}{a^2}$]

119. $f(x)=\begin{cases} xe^x, & x\ge 0 \\ -xe^{-x}, & x\le 0 \end{cases}$

For maximam area a = e



Graph of f(x)

$$A_{max} = 2\int_0^1 \left(e - xe^x\right)dx$$

$$= 2\left[ex - (xe^x - e^x)\right]_0^1$$

$$= 2(e-1)$$

Hence, [A] = 3. .

120.

$f(x) = x^3 - 3x^2 + 3x$

$f'(x) = 3x_1^2 - 6x_1 + 3$

$$\therefore \quad \frac{y_1 - 1}{x_1 - 0} = 3x_1^2 - 6x_1 + 3$$

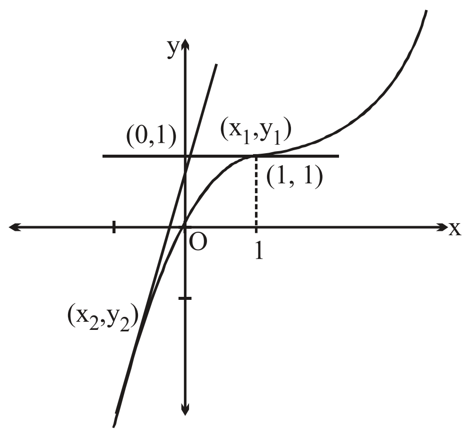
$$y_1 = 3x_1^3 - 6x_1^2 + 3x_1 + 1$$

$$\therefore \quad x_1^3 - 3x_1^2 + 3x_1 = 3x_1^3 - 6x_1^2 + 3x_1 + 1$$

$$2x_1^3 - 3x_1^2 + 1 = 0$$

$$(x_1 - 1)^2 + (2x_1 + 1) = 0$$

$$x_1 = 1 \quad \text{or} \quad x_1 = \frac{-1}{2}$$



Hence $m = 0$ or $m = \dfrac{27}{4}$

$$\text{Angle} = \tan^{-1}\left(\frac{27}{4}\right) \quad \Rightarrow \quad m + n = 27 + 4 = 31\ .]$$

121 to 123.

121. $f(x) = e^{(p+1)x} - e^x = e^x(e^{px} - 1)$

$f'(x) = (p+1)\cdot e^{(p+1)x} - e^x = 0$

$e^x[(p+1)e^{px} - 1] = 0 \qquad (e^x \neq 0)$

$$\Rightarrow \quad e^{px} = \frac{1}{p+1}$$

$$\left[\begin{array}{l} f''(x) = (p+1)^2 e^{(p+1)x} - e^x \\ f''(x) = e^x[(p+1)^2 e^{px} - 1] \\ f''\left(e^{px} = \dfrac{1}{p+1}\right) = e^x[(p+1) - 1] > 0 \\ \Rightarrow \quad \textit{minima} \end{array}\right.$$

$$px = -\ln(p+1)$$

$$x = \frac{-\ln(p+1)}{p} \qquad \text{[verify that } f''(x) > 0]$$

hence $x = s_p = \dfrac{-\ln(p+1)}{p}$ **.(i)**

122. $g(t) = \int_t^{t+1} \left(e^{px} \cdot e^x - e^x\right) e^t e^{-x}\, dx = e^t \int_t^{t+1} (e^{px} - 1)\, dx$

on integrating

$$g(t) = e^t \left[\frac{e^{px}}{p} - x\right]_t^{t+1} = e^t \left[\left\{\frac{e^{p(t+1)}}{p} - t - 1\right\} - \left\{\frac{e^{pt}}{p} - t\right\}\right] = e^t \left[\frac{e^{pt}(e^p - 1)}{p} - 1\right]$$

$$g(t) = \frac{(e^p - 1)}{p} e^{(p+1)t} - e^t$$

$$g'(t) = (p+1)\frac{(e^p - 1)}{p} e^{(p+1)t} - e^t = 0$$

$$\Rightarrow \quad \frac{(p+1)(e^p - 1)}{p} e^{pt} = 1$$

$$\Rightarrow \quad e^{pt} = \frac{p}{(p+1)(e^p - 1)}$$

$$\Rightarrow \quad pt = \ln \frac{p}{(p+1)(e^p - 1)}$$

$$\therefore \quad t = -\frac{1}{p} \ln\left(\frac{(p+1)(e^p - 1)}{p}\right) \quad \Rightarrow \quad t_p = -\frac{1}{p} \ln\left(\frac{(p+1)(e^p - 1)}{p}\right)$$

123. $s_p - t_p = \dfrac{1}{p} \ln\left(\dfrac{(p+1)(e^p - 1)}{p}\right) - \dfrac{\ln(p+1)}{p}$

$$= \frac{1}{p} \ln\left(\frac{(p+1)(e^p - 1)}{p} \cdot \frac{1}{(p+1)}\right) = \frac{1}{p} \ln\left(\frac{e^p - 1}{p}\right)$$

hence $s_p - t_p = \lim\limits_{p \to 0} \left(\dfrac{1}{p} \ln \dfrac{(e^p - 1)}{p}\right) = \lim\limits_{p \to 0} \ln\left(\dfrac{e^p - 1}{p}\right)^{1/p} \to 1^\infty$ form

$$\lambda = \lim_{p\to 0} \frac{1}{p}\left(\frac{e^p - 1}{p} - 1\right) = \lim_{p\to 0}\left(\frac{e^p - 1 - p}{p^2}\right) = \frac{1}{2}.$$

Alternatively for 123:

$$s_p - t_p = \frac{1}{p} ln\left(\frac{(p+1)(e^p-1)}{p}\right) - \frac{ln(p+1)}{p}$$

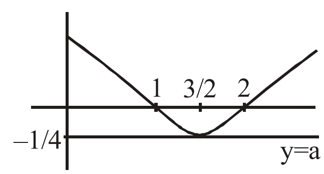
$$= \frac{1}{p} ln\left(\frac{(p+1)(e^p-1)}{p}\cdot\frac{1}{(p+1)}\right) = \frac{1}{p} ln\left(\frac{e^p-1}{p}\right)$$

$$\therefore \lim_{p\to 0^+} \frac{\ln\left(1+\frac{e^p-p-1}{p}\right)}{\left(\frac{e^p-p-1}{p}\right)\cdot p}\cdot\left(\frac{e^p-p-1}{p}\right) = \lim_{p\to 0^+} 1\times\left(\frac{e^p-p-1}{p^2}\right) = \frac{1}{2} \text{ (special limit)]}$$

124. $ax = x^3 - 3x^2 + 2x$

$a = x^2 - 3x + 2$

$\therefore \quad a \ge \frac{-1}{4}$.



125. $f'(x) = \frac{\sin|x|}{(1+\cos^2 x)} 2x = 0$, $x = 0, n\pi$

$x = -\pi, 0, \pi, 2\pi$

$\lim_{x\to 2\pi^+} f'(x) =$ positive quantity.

$\lim_{x\to 2\pi^-} f'(x) =$ Negative, so maxima at $x = 2\pi$.

$\lim_{x\to \pi^-} f'(x) = +$ve, $\lim_{x\to \pi^+} f'(x) = -$ve, so maxima.

$\lim_{x\to 0^-} f'(x) = +$ve, $\lim_{x\to 0^+} f'(x) = -$ve, $\Rightarrow$ minima

$\lim_{x\to -\pi^-} f'(x) = +$ve, $\lim_{x\to -\pi^+} f'(x) = -$ve, maxima.

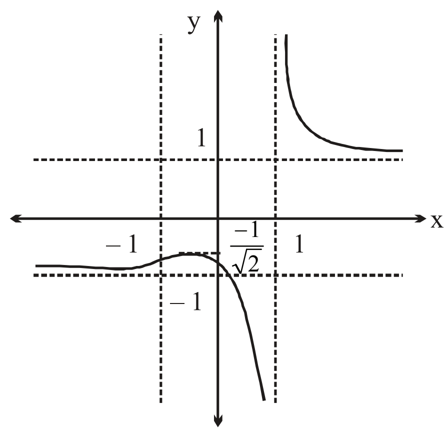
126. $f(x) = \frac{\sqrt{x^2+1}}{x-1}$; $f'(x) = \frac{(x-1)\,x}{\sqrt{x^2+1}} - \sqrt{x^2+1} = 0$

$(x - 1)\,x = x^2 + 1 \Rightarrow x = -1$

$\therefore f(-1) = -1$

$\lim_{x\to 1^+} f(x) \to \infty$; $\lim_{x\to 1^-} f(x) \to -\infty$

$$\lim_{x\to\infty} \frac{|x|\sqrt{1+\frac{1}{x^2}}}{x\left(1-\frac{1}{x}\right)} \to 1 \ ; \ \lim_{x\to -\infty} \frac{-x\sqrt{1+\frac{1}{x^2}}}{x\left(1-\frac{1}{x}\right)} \to -1$$



The graph is

Range $\left(-\infty, \frac{1}{\sqrt{2}}\right] \cup (1, \infty)$

127. At $x = 4$; $y = 7$

$y^2 = x^2 + 33$

$2yy' = 2x$

$y' = \frac{4}{7}$

$\therefore \quad m_{normal} = \frac{-7}{4}$

equation is $y - 7 = \frac{-7}{4}(x - 4)$

y-intercept $\Rightarrow x = 0$; $y = 7 + 7 = 14$.

128. $P(x) = k(x-2)^2 (x-3)^2 + 3$

$P(1) = 4k + 3 = 7$

$\therefore \quad k = 1 \Rightarrow P(x) = (x-2)^2 (x-3)^2 + 3$

$\therefore \quad P(5) = 9 \times 4 + 3 = 39$.

129. Given, $f(x) = \cos^3 x (\cos x - \lambda)$, $0 < x < \pi$

$\therefore \quad f'(x) = \cos^2 x (-\sin x) + (\cos x - \lambda)\, 2 \cos x (-\sin x)$

$\Rightarrow \quad f'(x) = \cos x \sin x (2\lambda - 3\cos x) \quad 0 < x < \pi$.

so, $f'(x) = 0$ gives

$$\cos x = 0 \Rightarrow x = \frac{\pi}{2} \quad \text{or} \quad \cos x = \frac{2\lambda}{3}$$

As, $0 < x < \pi \Rightarrow -1 < \cos x < 1$

$$\Rightarrow -1 < \frac{2\lambda}{3} < 1 \Rightarrow \frac{-3}{2} < \lambda < \frac{3}{2}$$

But $\lambda = 0$ (Rejected)

So, $\lambda \in \left(\frac{-3}{2}, 0\right) \cup \left(0, \frac{3}{2}\right)$

Hence, 2 integral values of λ exists i.e. $\lambda = -1, 1$.

130. Let $h(x) = \dfrac{g^2(x)}{2} + 3x^3 - 5$

Now, $h'(x) > 0 \Rightarrow g(x)\, g'(x) > -9x^2$

Now, integrate it.

131. Given, $f(x) = \int_{\frac{\pi}{2}}^{x} (\cos t - 2\sin t)\, dt \Rightarrow f'(x) = (\cos x - 2\sin x) < 0$, in $x \in \left[\frac{\pi}{2}, \pi\right]$.

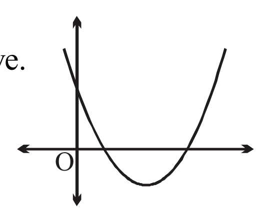
$\therefore$ $f(x)$ is decreasing ($\downarrow$) function in $\left[\frac{\pi}{2}, \pi\right]$.

Hence, $f\left(x = \frac{\pi}{2}\right) = \int_{\frac{\pi}{2}}^{\frac{\pi}{2}} (\cos t - 2\sin t)\, dt = 0$.

132. $f'(x) = 3x^2 + 6(a - 7)x + 3(a^2 - 9)$

$f'(x) = 3\,[x^2 + 2(a - 7)x + (a^2 - 9)]$ ....(1)

for positive point of maxima, both roots of $f'(x) = 0$ be must be positive.



$\therefore$ sum $> 0 \Rightarrow -2(a - 7) > 0$

i.e., $a - 7 < 0$

$a < 7$ and

Product, $P > 0 \Rightarrow a^2 - 9 > 0 \Rightarrow a > 3$ or $a < -3$ and $D > 0$

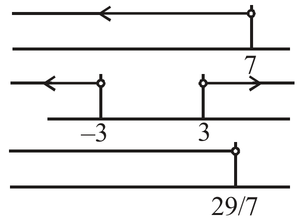
$4(a - 7)^2 - 4(a^2 - 9) > 0$

$a^2 - 14a + 49 - a^2 + 9 > 0$

$58 - 14a > 0$

$29 - 7a > 0$

$a < \frac{29}{7}$.



Hence $a \in (-\infty, -3) \cup \left(3, \frac{29}{7}\right)$.

$\therefore$ Largest natural value of 'a' $= 4$.

133. $f(x)\,f(y) + 2 = f(x) + f(y) + f(xy)\ \forall\ x, y \in R - \{0\}$ ......(1)

Put $y = \frac{1}{x}$

Hence, $f(x) \cdot f\left(\frac{1}{x}\right) + 2 = f(x) + f\left(\frac{1}{x}\right) + f(1)$

Now, put $x = y = 1$ in equation (1)
Hence, $f(1) = 2$

$f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) \Rightarrow f(x) = 1 + x^n$

Given $f'(1) = 3 \Rightarrow n$

134 to 136.

134. $F(x) = \int_0^x f(t)\,dt \quad \Rightarrow \quad F'(x) = f(x)$

for $0 \le x \le 1$

$$F(x) = \int_0^x (1-t)\,dt = \left[t - \frac{t^2}{2}\right]_0^x$$

$$F(x) = x - \frac{x^2}{2} \text{ and } \quad F(1) = \frac{1}{2}$$

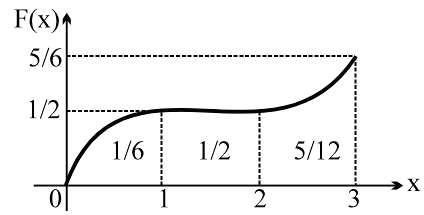
for $1 < x \le 2$

$$F(x) = \int_0^1 (1-t)\,dt + \int_1^x 0\,dt = \frac{1}{2} + 0 \quad \Rightarrow \quad F(x) \text{ is constant}$$

for $2 < x \le 3$

$$F(x) = \int_0^1 f(t)\,dt + \int_1^2 f(t)\,dt + \int_2^x f(t)\,dt = \frac{1}{2} + 0 + \int_0^x (t-2)^2\,dt = \frac{1}{2} + \frac{(x-2)^3}{3}$$

$$\therefore \quad F(x) = \begin{cases} x - \dfrac{x^2}{2} & \text{if } 0 \le x \le 1 \\ \dfrac{1}{2} & \text{if } 1 < x \le 2 \\ \dfrac{1}{2} + \dfrac{(x-2)^3}{3} & \text{if } 2 < x \le 3 \end{cases};$$



135. range of F (x) from the graph $\left[0, \dfrac{5}{6}\right]$

136. $$\text{Area} = \int_0^1 F(x)\,dx + \frac{1}{2} + \int_2^3 F(x)\,dx = \int_0^1 \left(x - \frac{x^2}{2}\right)dx + \frac{1}{2} + \int_2^3 \left(\frac{1}{2} + \left(\frac{(x-2)^3}{3}\right)\right)dx$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{6}\right]_0^1 + \frac{1}{2} + \left[\frac{x}{2} + \frac{(x-2)^4}{12}\right]_2^3$$

$$= \frac{1}{2} + \left(\frac{1}{2} - \frac{1}{6}\right) + \left[\left(\frac{3}{2} + \frac{1}{12}\right) - (1)\right]$$

$$= \frac{1}{2} + \frac{1}{3} + \frac{7}{12} = \frac{6+4+7}{12} = \frac{17}{12}$$

137. Given,

$$g(x) = \begin{vmatrix} \cos x & x & 1 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$$

$= \cos x\,(x^2 - 2x^2) - x\,(2\sin x - 2x\tan x) + 1\,(2x\sin x - x^2 \cdot \tan x)$

$= -x^2 \cdot \cos x + 2x^2 \cdot \tan x - x^2 \cdot \tan x$

$= -x^2 \cdot \cos x + x^2 \tan x$

$\therefore \quad g(x) = x^2(\tan x - \cos x)$

Also, $g(0) = 0$ and $g'(x) = x^2(\sec^2 x + \sin x) + 2x \cdot (\tan x - \cos x)$

$\therefore \quad g'(0) = 0$

So, equation of normal drawn to g(x) at x = 0, is x = 0.

138. We must have

$$\begin{vmatrix} x & 1 & 2 \\ f(x) & 3 & x^2 \\ 5x & 6 & 1 \end{vmatrix} = 0$$

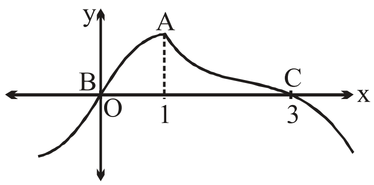
$\therefore \quad x(3 - 6x^2) - 1\,(f(x) - 5x^3) + 2\,(6f(x) - 15x) = 0$

$\Rightarrow \quad 11\,f(x) - x^3 - 27x = 0$

$\therefore \quad f(x) = \dfrac{x^3 + 27x}{11}$

$\Rightarrow \quad f'(x) = \dfrac{3x^2 + 27}{11} > 0 \ \forall \ x \in R$

139. Three critical points A, B, C



140. $\dfrac{dx}{d\theta} = a\,\theta \cos\theta\,,\ \dfrac{dy}{d\theta} = a\,\theta\sin\theta$

$\dfrac{dy}{dx} = \tan\theta$

Equation of normal $\quad y - a(\sin\theta - \theta\cos\theta) = -\cot\theta\,(x - a(\cos\theta + \theta\sin\theta))$

$\Rightarrow \quad \cos\theta\ x - \sin\theta\ y = a$

distance from origin = a

distance from $(\sin\theta, \cos\theta) = a$]

141. $f'(x) = (x^2 - 1)\cdot 2x - 1(x-1)$

$f'(x) = 2x^3 - 3x + 1 \ > 0 \quad \forall \ x \in [1, 2]$

$f(x)\big|_{max} = f(2) \ \Rightarrow \ \int_2^4 (t-1)\,dt = \dfrac{(t-1)^2}{2}\Bigg|_2^4 = \dfrac{9-1}{2} = 4$]

142. $x\sin t - \cos t \quad = \sqrt{x^2+1}\left[\dfrac{x\sin t}{\sqrt{x^2+1}} - \dfrac{\cos t}{\sqrt{x^2+1}}\right]$

$= \sqrt{x^2+1}\ [\sin t\cos\theta - \cos t\sin\theta]$

When $\theta = \tan^{-1}\dfrac{1}{x}$

$= \sqrt{x^2+1}\ [\sin(t-\theta)$

$\therefore \ I = \sqrt{x^2+1}\ \int_0^{\pi/2} |\sin(t-\theta)|\,dt$

$$I = \sqrt{x^2+1}\left[\int_0^{\pi/2}\left|\sin\left(t-\tan^{-1}\frac{1}{x}\right)\right|dt\right]$$

$$= \sqrt{x^2+1}\left[\int_0^{\tan^{-1}\frac{1}{x}}\sin\left(\tan^{-1}\frac{1}{x}-t\right)dt+\int_{\tan^{-1}\frac{1}{x}}^{\pi/2}\sin\left(t-\tan^{-1}\frac{1}{x}\right)dt\right]$$

$$= \sqrt{x^2+1}\left[\left[\cos\left(\tan^{-1}\frac{1}{x}-t\right)\right]_0^{\tan^{-1}\frac{1}{x}}-\cos\left(t-\tan^{-1}\frac{1}{x}\right)\Bigg]_{\tan^{-1}\frac{1}{x}}^{\pi/2}\right]$$

$$= \sqrt{x^2+1}\left[\left\{1-\cos\left(\tan^{-1}\frac{1}{x}\right)\right\}-\left\{\sin\left(\tan^{-1}\frac{1}{x}\right)-1\right\}\right]$$

$$= \sqrt{x^2+1}\left[2-\left\{\frac{x+1}{\sqrt{x^2+1}}\right\}\right]$$

$I = 2\sqrt{x^2+1} - (x+1) = f(x)$

$\therefore\ f'(x) = \dfrac{2x}{\sqrt{1+x^2}} - 1 = 0 \Rightarrow x = \dfrac{1}{\sqrt{3}}$ which gives minima

$\therefore\ f\left(\dfrac{1}{3}\right) = 2\sqrt{\dfrac{4}{3}} - \dfrac{1}{\sqrt{3}} - 1 = \sqrt{3} - 1 \Rightarrow (a+b) = 3+1 = 4.$]

143. $f(x) = \begin{cases} a\cos x + b\sin x & \text{if } 0 \le x \le \dfrac{\pi}{2} \\ a\cos x - b\sin x & \text{if } -\dfrac{\pi}{2} \le x \le 0 \end{cases}$ [$a = -\dfrac{1}{\sqrt{\pi+\sqrt{3}}}$; $b = -\dfrac{\sqrt{3}}{\sqrt{\pi+\sqrt{3}}}$]

for $-\pi/2 < x < 0$

$f'(x) = -a\sin x - b\cos x$ ....(1)

and $f''(x) = -a\cos x + b\sin x$ ....(2)

since $f(x)$ has a minima at $x = -\pi/3$

hence $f'(-\pi/3) = 0$ and $f''(-\pi/3) > 0$

now $f'(-\pi/3) = +a\cdot\dfrac{\sqrt{3}}{2} - \dfrac{b}{2} = 0 = \sqrt{3}a - b = 0$

and $f''(-\pi/3) = -\dfrac{a}{2} - b\cdot\dfrac{\sqrt{3}}{2} = -\dfrac{1}{2}[a + b\sqrt{3}] = -2a > 0$

hence $a < 0$ and $b < 0$

now $I = \int_{-\pi/2}^{\pi/2} (f(x))^2\, dx = \int_{-\pi/2}^{0} f^2(x)\, dx + \int_{0}^{\pi/2} f^2(x)\, dx$

$$= \int_{-\pi/2}^{0} (a^2 \cos^2 x - 2ab \sin x \cos x + b^2 \sin^2 x)\, dx + \int_{0}^{\pi/2} (a^2 \cos^2 x + 2ab \sin x \cos x + b^2 \sin^2 x)\, dx$$

hence $I = \frac{\pi a^2}{2} + \frac{\pi b^2}{2} + 2ab = 2$

$$2(\sqrt{3} + \pi)a^2 = 2 \quad \Rightarrow \quad \boxed{a = -\frac{1}{\sqrt{\pi + \sqrt{3}}}} \text{ and } \boxed{b = -\frac{\sqrt{3}}{\sqrt{\pi + \sqrt{3}}}}.$$

144. Let $\int_{x}^{b} f(t)\, dt = g(x)$

then in [x, b] where $a < x \le b$ by LMVT

We have $g'(c_1) = \frac{g(b) - g(x)}{b - x} = \frac{0 - g(x)}{b - x}$

$$\therefore \quad g(x) = \int_{x}^{b} f(t)\, dt = (b - x)\, g'(c_1) = (b - x)\, f(c_1) \qquad ....(1)$$

Similarly, for $\int_{a}^{x} f(t)\, dt = h(x)$,

in [a, x] where $a \le x < b$ by LMVT we have

$$\int_{a}^{x} f(t)\, dt = (x - a)\, f(c_2)$$

Where $c_2 < c_1 \ \forall\, x \in [a, b]$ ....(2)

$\therefore$ From equation (1) and equation (2) in given equation

$f(x) = (x - a)(b - x) f(c_1) + (x - b)(x - a) f(c_2)$

$= (x - a)(b - x)\{f(c_1) - f(c_2)\}$

$\because$ $f(x)$ is monotonic

$\therefore$ $f(c_1) - f(c_2)$ has a constant sign in [a, b]

Also $(x - a)(b - x) \ge 0 \ \forall\, x \in [a, b]$

$\therefore$ $f(x)$ must have the same sign $\forall\, x \in [a, b]$

Also if $f(x)$ be monotonically increasing

$f(c_1) - f(c_2) > 0$

$\therefore$ $f(x) > 0 \ \forall\, x \in [a, b]$

and if $f(x)$ be monotonically decreasing $\forall\, x \in [a, b]$

then $f(c_1) - f(c_2) < 0$

$\therefore \quad f(x) < 0 \;\forall\, x \in [a, b]$

145 to 146.

145. Let $g(x) = \int p(x)dx$

$= x(x^8 - 3x^5 - 5x^3 + 15) + c$

$= x(x^3 - 3)(x^5 - 5) + c$

Clearly $f(0) = f(3^{1/3}) = f(5^{1/5}) = c$

$\therefore$ Rolle's theorem is applicable on g(x) in intervals $[0, 3^{1/3}]$, $[3^{1/3}, 5^{1/5}]$ and in $[0, 5^{1/5}]$

$\because$ $y = x^{1/x}$ has a local maxima at x = e

$\therefore$ $0 < 5^{1/5} < 3^{1/3}$

$\therefore$ There exists at least one 'c', each in $[0, 5^{1/5}]$ and in $[5^{1/5}, 3^{1/3}]$ for which $g'(x) = 0$

i.e. $p(x) = 0$

$\therefore$ $p(x) = 0$ has at least two real roots in $[0, 3^{1/3}]$

146. Given $f''(x) = p(x)$

$\therefore$ $f'(x) = \int p(x)dx$

$f'(x) = x(x^3 - 3)(x^5 - 5) + c$

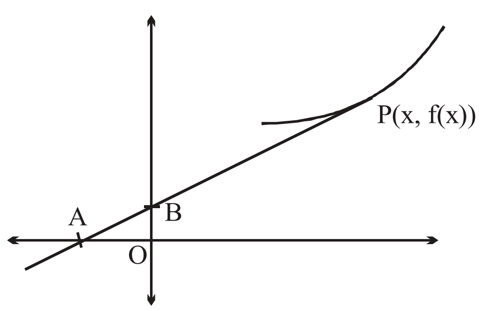
$\because$ $f'(1) = 8 \quad \therefore \; 1(1-3)(1-5) + c = 8 \quad \therefore \; c = 0$

$\therefore$ $f'(x) = x(x^3 - 3)(x^5 - 5)$

$\therefore$ $f'(x) = 0$ has exactly three real roots.

147. Equation of tangent

$Y - f(x) = f'(x)(X - x)$



$A \equiv \left(x - \dfrac{f(x)}{f'(x)}, 0\right)$

$B \equiv (0, f(x) - xf'(x))$

Area of ΔOAB, $S(x) = \dfrac{1}{2} \times OA \times OB$

$\Rightarrow S(x) = \dfrac{1}{2}\left(x - \dfrac{f(x)}{f'(x)}\right)(f(x) - x(f'(x)) = -\dfrac{1}{2}\dfrac{(f(x) - xf'(x))^2}{f'(x)}$

$$= \frac{\{(5-(x+1)^2+2x(x+1)\}^2}{4(x+1)} = \frac{\{5-x^2-2x-1+2x^2+2x\}^2}{4(x+1)} = \frac{(x^2+4)^2}{4(x+1)}$$

$$S'(x) = \frac{2(x+1)(x^2+4)\cdot 2x-(x^2+4)^2}{4(x+1)^2} = 0$$

$\Rightarrow\ 4x^2 + 4x - x^2 - 4 = 0 \ \Rightarrow\ 3x^2 + 4x - 4 = 0 \ \Rightarrow\ 3x^2 + 6x - 2x - 4 = 0$

$\Rightarrow\ (x + 2)(3x - 2) = 0 \ \Rightarrow\ x = \frac{2}{3} \quad (\because\ 0 < x < 1)$

$\therefore$ minimum value of $S(x) = \frac{80}{27} \ \Rightarrow (p+q) = 107$.

148. $\because$ Graph of $y = g(x)$ is reflection ofgraph of $y = f(x)$ with respect to line $y = x$.

$g(x)$ is inverse function of $f(x)$

$\because$ (a, b) lies on $y = f(x)$

$f(a) = b$

$g(f(x)) = x \quad \Rightarrow \quad g'(f(x))\ f'(x) = 1$

$\Rightarrow \quad g'(f(x)) = \frac{1}{f'(x)} \quad \Rightarrow \quad g'(f(\alpha)) = \frac{1}{f'(\alpha)}$

$\Rightarrow \quad g'(b) = \frac{1}{3}$

$g = \frac{1}{3} \quad \Rightarrow \quad \frac{670}{\gamma} = 2010$.

149. $ln(x^2 + y^2) = C\tan^{-1}\frac{y}{x}$

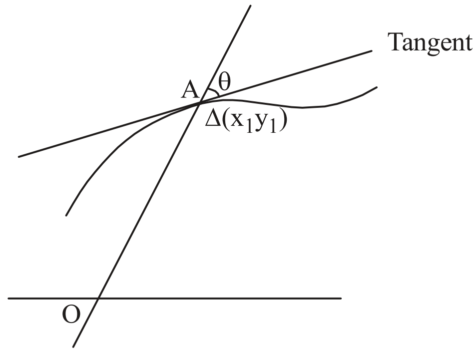
$$\Rightarrow \quad \frac{1}{x^2+y^2}\left(2x+2y\frac{dy}{dx}\right) = C\,\frac{1}{1+\frac{y^2}{x^2}}\left(\frac{x\frac{dy}{dx}-y}{x^2}\right)$$

$$\Rightarrow \quad \frac{1}{x^2+y^2}\,2\left(x+y\frac{dy}{dx}\right) = \frac{Cx^2}{(x^2+y^2)}\cdot\frac{\left(x\frac{dy}{dx}-y\right)}{x^2}$$

$$\Rightarrow \quad 2\left(x+y\frac{dy}{dx}\right) = C\left(x\frac{dy}{dx}-y\right)$$

$$\Rightarrow \quad (Cx-2y)\frac{dy}{dx} = 2x + Cy$$

$$\Rightarrow \quad \left.\frac{dy}{dx}\right]_A = \frac{2x_1 + Cy_1}{Cx_1 - 2y_1};$$



(but $\frac{y_1}{x_1} = m$)

$$\Rightarrow \quad M_1 = \frac{2+5m}{C-2m}$$

$M_2 = m$

$$\Rightarrow \quad \tan\theta = \frac{\frac{2+Cm}{C-2m} - m}{1 + \left(\frac{2m+Cm^2}{C-2m}\right)} \quad \Rightarrow \quad \tan\theta = \left|\left(\frac{2+2m^2}{C(1+m^2)}\right)\right| = \left|\frac{2}{C}\right|$$

Which is made independent of position of A.]

150. $f(x) = 3\sin x - 4\sin^3 x$

$f(x) = \sin 3x$

$f(x)\uparrow$ when $\quad 3x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$$x \in \left[-\frac{\pi}{6}, \frac{\pi}{6}\right]$$

$\therefore \quad$ Length of interval $= \frac{2\pi}{6} = \frac{\pi}{3}$

151. We have $f(x) = (b^2 - 3b + 2)(\cos^2 x - \sin^2 x) + (b-1)x + \sin 2$

$\therefore \quad f'(x) = (b-1)(b-2)(-2\sin 2x) + (b-1)$

Now, $\quad f'(x) \neq 0$ for every $x \in R$,

so $\quad (b-1)\,(1 - 2(b-2)\ \sin 2x) \neq 0 \ \forall\ x \in R$

$\therefore \quad b \neq 1$

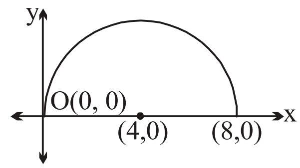
Also, $\quad \left|\frac{1}{2(b-2)}\right| > 1 \Rightarrow b \in \left(\frac{3}{2}, 2\right) \cup \left(2, \frac{5}{2}\right)$

Now, when $b = 2$, $f(x) = x + \sin 2 \Rightarrow f'(x) = 1\ (\neq 0)$.

Hence, $b \in \left(\frac{3}{2}, \frac{5}{2}\right) \quad \Rightarrow \quad b_1 = \frac{3}{2}$ and $b_2 = \frac{5}{2}$

$$\Rightarrow \quad (b_1 + b_2) = \frac{3}{2} + \frac{5}{2} = \frac{8}{2} = 4$$

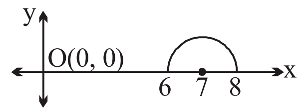
152. We have $f(x) = \sqrt{16-(x-4)^2} - \sqrt{1-(x-7)^2}$



Now consider $y = \sqrt{16-(x-4)^2}$

$\Rightarrow \quad (x-4)^2 + y^2 = 16, \; y > 0$

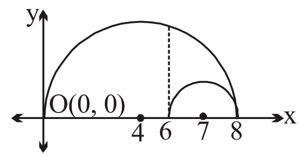
is a semi circle with cenre (4, 0) and radius = 4.



|||ly $\quad y = \sqrt{1-(x-7)^2}$

$\Rightarrow \quad (x-7)^2 + y^2 = 1, y > 0$

is a semi circle with centre (7, 0) and radius = 1



Now on combining the 2 figures, we have

$f(x)_{max.}$ = maximum vertical distance between the 2 curves which occurs when x = 6.

Hence $f(x)_{max.} = \sqrt{16-(6-4)^2} - 0 = \sqrt{12}$.

__Aliter__: We have $f(x) = \sqrt{8x - x^2} - \sqrt{14x - x^2 - 48}$

$= \sqrt{x(8-x)} - \sqrt{(8-x)(x-6)} = \sqrt{8-x}\left(\sqrt{x} - \sqrt{x-6}\right)$

$\therefore$ Domain of f(x) is [6, 8].

Now $f(x) = \sqrt{8-x}\left(\sqrt{x} - \sqrt{x-6}\right)$

$$\Rightarrow f'(x) = \sqrt{8-x}\left(\frac{1}{2\sqrt{x}} - \frac{1}{2\sqrt{x-6}}\right) + \left(\sqrt{x} - \sqrt{x-6}\right)\left(\frac{1}{2\sqrt{8-x}}\right)$$

$$= \frac{\left(\sqrt{x-6} - \sqrt{x}\right)}{2\sqrt{x}\sqrt{x-6}\sqrt{8-x}}\left(8 - x + \sqrt{x}\;\sqrt{x-6}\right) < 0 \;\forall\; x \in [6, 8]$$

Clearly f(x) is strictly decreasing in [6, 8].

Hence $f(x)_{max} = f(6) = \sqrt{2}\left(\sqrt{6} - 0\right) = \sqrt{12} = 2\sqrt{3} = m\sqrt{n}$

So m = 2 and n = 3

Hence (m + n) = 2 + 3 = 5

153. $h(x) = f(x) - f^2(x) + f^3(x)$

$h'(x) = f'(x) - 2f(x)\,f'(x) + 3f^2(x) \cdot f'(x)$

$= f'(x)\,[3f^2(x) - 2f(x) + 1\,]$ always +ve $4 - 12 < 0$

i.e. $\quad h'(x) > 0 \quad$ or $\quad < 0$

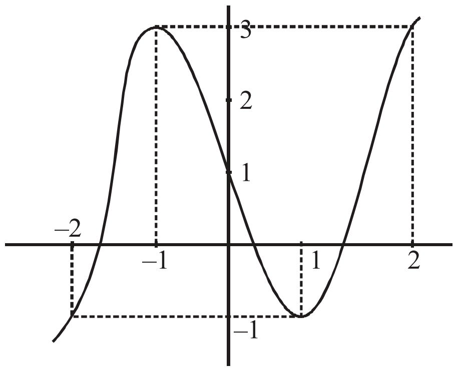
depends on wheter $f'(x) > 0$ or $f'(x) < 0$

154 to 156 :

$D_f = x \in R - \{x\}$

if $\quad x \neq \alpha \; f(x) = (x^3 - 3x + 1)$

graph of $y = x^3 - 3x + 1$



as Range of $x^3 - 3x + 1$ will be R untill we eliminate the value at α. But it can be done only if value of $x^3 - 3x + 1$ at α is not given by any other value i.e. $\alpha \notin [-2, +2]$

$\Rightarrow \quad \alpha \in (-\infty, -2) \cup (2, \infty)$

$\Rightarrow \quad \alpha \neq -2, -1, 0, 1, 2$

$\Rightarrow \quad \lim\limits_{x \to \alpha} f(x) \notin [-1, 3] \Rightarrow \lim\limits_{x \to \alpha} f(x) \in (-\infty, -1) \cup (3, \infty)$.

157 to 158.

157. Let $\quad g(x) = \int p(x)dx$

$= x(x^8 - 3x^5 - 5x^3 + 15) + c$

$= x(x^3 - 3)(x^5 - 5) + c$

Clearly $f(0) = f(3^{1/3}) = f(5^{1/5}) = c$

$\therefore \quad$ Rolle's theorem is applicable on g(x) in intervals $[0, 3^{1/3}]$, $[3^{1/3}, 5^{1/5}]$ and in $[0, 5^{1/5}]$

$\because \quad y = x^{1/x}$ has a local maxima at x = e

$\therefore \quad 0 < 5^{1/5} < 3^{1/3}$

$\therefore \quad$ There exists at least one 'c', each in $[0, 5^{1/5}]$ and in $[5^{1/5}, 3^{1/3}]$ for which

$g'(x) = 0$ i.e. $p(x) = 0$

$\therefore \quad p(x) = 0$ has at least two real roots in $[0, 3^{1/3}]$

158. Given $\quad f''(x) = p(x)$

$\therefore \quad f'(x) = \int p(x)dx$

$f'(x) = x(x^3 - 3)(x^5 - 5) + c$

$\because \quad f'(1) = 8 \qquad \therefore \quad 1(1 - 3)(1 - 5) + c = 8 \qquad \therefore \quad c = 0$

$\therefore \quad f'(x) = x(x^3 - 3)(x^5 - 5)$

$\therefore \quad f'(x) = 0$ has exactly three real roots.

159. Domain of f(x) is $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$

Let $f(x) = \frac{1}{g(x)}$ where $g(x) = \sin x\sqrt{1 - \cos x}$

$$\Rightarrow \quad g'(x)=\frac{d}{dx}g(x) = \sin x\cdot\frac{\sin x}{2\sqrt{1-\cos x}}+\sqrt{1-\cos x}\cdot\cos x = \frac{\sin^2 x+2(1-\cos x)\cdot\cos x}{2\sqrt{1-\cos x}}$$

$$= \frac{2\cos x+\sin^2 x-2\cos^2 x}{2\sqrt{1-\cos x}} = \frac{1+2\cos x-3\cos^2 x}{2\sqrt{1-\cos x}} = \frac{(1-\cos x)(1+3\cos x)}{2\sqrt{1-\cos x}}$$

$g'(x) > 0 \Rightarrow \cos x > \frac{-1}{3}$ $\quad (\because \ 1-\cos x > 0)$

and $\quad g'(x) < 0 \quad \Rightarrow \cos x < \frac{-1}{3}$

$\therefore$ $\quad$ g(x) increases on $\left(\frac{\pi}{2},\ \pi-\cos^{-1}\frac{1}{3}\right)$ and decreases on $\left(\pi-\cos^{-1}\frac{1}{3}, \pi\right)$

$\therefore$ $\quad$ g(x) has a local maxima at $x = \pi-\cos^{-1}\left(\frac{1}{3}\right)$

$\therefore$ $\quad$ f(x) has a local maxima at $x = \pi-\cos^{-1}\left(\frac{1}{3}\right)$

$$\therefore \quad f(x)_{\text{local min}} = \frac{1}{\sqrt{1-\left(\frac{1}{3}\right)^2}\cdot\sqrt{1-\left(-\frac{1}{3}\right)}} = \frac{1}{\frac{2\sqrt{2}}{3}\cdot\frac{2}{\sqrt{3}}} = \frac{3\sqrt{3}}{4\sqrt{2}}$$

160. $y = \frac{x^4-x^2}{x^6+2x^3-1} = \frac{x-\frac{1}{x}}{x^3+2-\frac{1}{x^3}}$ $\quad$ (taking x^3 common)

$$y = \frac{x-\frac{1}{x}}{\left(x-\frac{1}{x}\right)^3+3\left(x-\frac{1}{x}\right)+2} \quad ....(1)$$

Let $\left(x-\frac{1}{x}\right) = t$

$y = \frac{t}{t^3+3t+2}$; $\quad \frac{dy}{dx} = (t^3+3t+2)-t(3t^2+3) = -2t^2+2 = 0 \Rightarrow \quad t = 1$

$$\left.\frac{d^2y}{dx^2}\right]_{t=1} = -4t = -4 < 0$$

$\Rightarrow$ maxima at t = 1

maximum value = $\frac{1}{6} = \frac{p}{q}$ $\Rightarrow$ $p = 1$ and $q = 6$

It occurs when $t = 1$ $\Rightarrow$ $x - \frac{1}{x} = 1$ $\Rightarrow$ $x^2 - x - 1 = 0$

$$x = \frac{1 \pm \sqrt{5}}{2} = \frac{1+\sqrt{5}}{2} \quad \text{as } x > 1$$

$\therefore$ $p = 1;\ q = 6;\ a = 1;\ b = 5;\ c = 2$

$\therefore$ $p + q + a + b + c = 15$.

Alternatively, after step -1

using AM $\geq$ GM between 3 positive quantities $\left(x - \frac{1}{x}\right)^3$, 1, 1 $(x > 1)$

we have $$\frac{\left(x-\frac{1}{x}\right)^3 + 1 + 1}{3} \geq \left[\left(x - \frac{1}{x}\right)^3\right]^{1/3}$$

$$\therefore \quad \left(x-\frac{1}{x}\right)^3 + 2 \geq 3\left(x - \frac{1}{x}\right)$$

now $$\frac{x - \frac{1}{x}}{\left(x-\frac{1}{x}\right)^3 + 2 + 3\left(x-\frac{1}{x}\right)} \leq \frac{\left(x-\frac{1}{x}\right)}{6\left(x-\frac{1}{x}\right)} = \frac{1}{6}$$

$\therefore$ maximum value = $\frac{1}{6}$ this occurs when $x - \frac{1}{x} = 1 \Rightarrow$ same result

161 to 163.

161 $g'(x) = f(x) = \frac{\sin x}{x}$

sign of g'(x) is $(0, 4\pi)$ will be

$$\begin{array}{ccccccccc} & & & - & & + & & - & \\ 0 & & \pi & & 2\pi & & 3\pi & & 4\pi \end{array}$$

Clearly g'(x) changes its sign from positive to negative at π and 3π. Therefore two points of local max.

162. Sign of g'(x) in $(4\pi, 10\pi)$ will be

$$\begin{array}{ccccccccccccc} & + & & - & & + & & - & & + & & - & \\ 4\pi & & 5\pi & & 6\pi & & 7\pi & & 8\pi & & 9\pi & & 10 \end{array}$$

Clearly g'(x) changes sign from negative to positive at 6π and 8π. Therefore two points of min.

163 $g''(x) = \dfrac{x\cos x - \sin x}{x^2} = f'(x)$

g(x) will have point of inflection at points of local extremum of $f(x)$

$\therefore \quad f'(x) = 0 \Rightarrow \tan x = x$

Clearly there will be 10 points between $(10\pi, 20\pi)$.

164. If $f(x) = |x - \alpha|^{\beta}$, then $f'(\alpha)$ exists if $\beta > 1$, $f''(\alpha)$ exist if $\beta > 2$ and $f'''(\alpha)$ exists if $\beta > 3$

$\therefore \quad f'''(e)$ exists since $\pi > 3$

$f'''(\pi)$ does not exist since $e > 3$.

165. Put $a = \cos\theta$

and $b = \sin\theta$

Hence $E = \dfrac{b+1}{a+b-2} = \dfrac{\sin\theta + 1}{\sin\theta + \cos\theta - 2} = f(\theta)$

For maxima / minima, $f'(\theta) = 0$

$\Rightarrow \quad (\sin\theta + \cos\theta - 2)(\cos\theta) - (\sin\theta + 1)(\cos\theta - \sin\theta) = 0$

$\Rightarrow \quad \sin\theta\cos\theta + \cos^2\theta - 2\cos\theta - \sin\theta\cos\theta + \sin^2\theta - \cos\theta + \sin\theta = 0$

$\Rightarrow \quad \cos^2\theta - 2\cos\theta + \sin^2\theta - \cos\theta + \sin\theta = 0$

$\Rightarrow \quad 1 - 3\cos\theta + \sin\theta = 0$

$\Rightarrow \quad 1 + \sin\theta = 3\cos\theta$

$\Rightarrow \quad (1 + \sin\theta)^2 = 9\cos^2\theta = 9 - 9\sin^2\theta$

$\Rightarrow \quad 10\sin^2\theta + 2\sin\theta - 8 = 0$

$\Rightarrow \quad 5\sin^2\theta + \sin\theta - 4 = 0$

$\Rightarrow \quad 5\sin^2\theta + 5\sin\theta - 4\sin\theta - 4 = 0$

$\Rightarrow \quad (\sin\theta + 1)(5\sin\theta - 4) = 0$

$\Rightarrow \quad \sin\theta = -1$ or $\sin\theta = \dfrac{4}{5}$

Now value of E, when $\sin\theta = -1$ is 0.

and value of E when $\sin\theta = \dfrac{4}{5}$ is $\dfrac{\sin\theta + 1}{\sin\theta + \cos\theta - 2} = \dfrac{\frac{4}{5} + 1}{\frac{4}{5} + \frac{3}{5} - 2}$ or $\dfrac{\frac{4}{5} + 1}{\frac{4}{5} - \frac{3}{5} - 2}$

i.e $E = \dfrac{-9}{3}$ or $\dfrac{9}{-9}$

Hence $E]_{\text{minimum}} = -3$ when $\sin\theta = \dfrac{4}{5}$ and $\cos\theta = \dfrac{3}{5} \Rightarrow u = -3$

Hence, $u^2 = 9$.

166. $\because$ $f'(x^2) = \dfrac{1}{x}$ $\Rightarrow$ $f'(x) = \dfrac{1}{\sqrt{x}}$, $x > 0$

$\Rightarrow$ $f(x) = 2\sqrt{x} + c$ (c = integration constant)

$\because$ $f(1) = 1$ $\Rightarrow$ $c = -1$

$\therefore$ $f(x) = 2\sqrt{x} - 1, x > 0$

and $g'(\sin^2 x - 1) = \cos^2 x + p \ \forall\ x \in R$

$\Rightarrow$ $g'(-\cos^2 x) = \cos^2 x + p$

$\Rightarrow$ $g'(x) = p - x, \forall\ x \in [-1, 0]$

$\Rightarrow$ $g(x) = px - \dfrac{x^2}{2} + k$ (where k = integration constant)

$\because$ $g(-1) = 0 \Rightarrow 0 = -p - \dfrac{1}{2} + k \Rightarrow k = \dfrac{1}{2} + p$

$\therefore$ $g(x) = px - \dfrac{x^2}{2} + \dfrac{1}{2} + p$

$\therefore$ $h(x) = \begin{cases} 2\sqrt{x} - 1 & , x > 0 \\ px - \dfrac{x^2}{2} + \dfrac{1}{2} + p & , -1 \le x \le 0 \end{cases}$

$\therefore$ At $x = 0$

L.H.L = R.H.L = f(0)

$\Rightarrow -1 = \dfrac{1}{2} + p \Rightarrow p = \dfrac{-3}{2}$

Hence $2p = -3$

$\therefore$ Absolute value of $2p$ is 3

167. Consider $\phi(x) = f(x) - 2g(x)$ defined on [0, 1] since f(x) and g(x) are differentiable for $0 \le x \le 1$, therefore $\phi(x)$ is differentiable on (0, 1) and continuous on [0, 1]

$\therefore$ $\phi(0) = f(0) - 2g(0) = 2 - 0 = 2$ $\phi(1) = f(1) - 2g(1) = 6 - 2g(1)$

Now $\phi'(x) = f'(x) - 2g'(x)$

$\Rightarrow$ $\phi'(c) = f'(c) - 2g'(c) = 0$ (given)

$\Rightarrow$ $\phi(x)$ satisfies Rolle's theorem on [0, 1]

$\therefore$ $\phi(0) = \phi(1)$

$\Rightarrow$ $2 = 6 - 2g(1)$

$\Rightarrow$ $g(1) = 2$

168. $f'(x) = 0 \Rightarrow x = \frac{2\alpha}{3}$ or $x = -\frac{2}{3\alpha}$

now $f\left(\frac{2\alpha}{3}\right) = -\frac{4}{9\alpha}\left(\alpha^2 - 3\right)^2$ and $f\left(-\frac{2}{3\alpha}\right) = \frac{4(3\alpha^2-1)^2}{9\alpha^3}$

now $g(\alpha) = \frac{4(3\alpha^2-1)^2}{9\alpha^3} + \frac{4(\alpha^2-3)^2}{9\alpha}$

$$= \frac{4}{9\alpha^3}\left[(3\alpha^2-1)^2 + \alpha^2(\alpha^2-3)^2\right] = \frac{4}{9}\frac{(\alpha^2+1)^3}{\alpha^3} = \frac{4}{9}\left(\alpha + \frac{1}{\alpha}\right)^3 \Rightarrow \frac{32}{9}\ ]$$

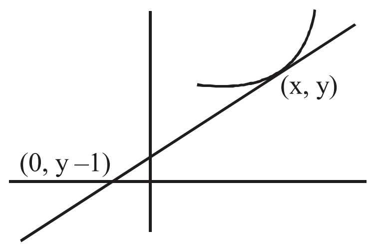
169 to 171.

$Y - y = m(X - x)$

$X = 0, Y = y - mx$

given $y - mx = y - 1$

$\therefore \quad m = \frac{1}{x}$



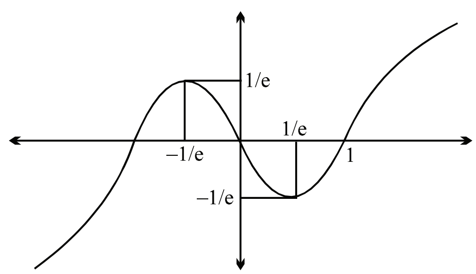
$\frac{dy}{dx} = \frac{1}{x}$

$\therefore \quad y = \ln x + c$

$x = 1\ ;\ y = 0\ ;\ c = 0$

$\therefore \quad y = f(x) = \ln x$

171. $g(x) = \frac{x}{\ln x}$



$g'(x) = \frac{\ln x - 1}{\ln^2 x}$

$\downarrow$ in $(0, 1) \cup (1, e)$

$\therefore$ Only one integral value.]

172. Let $g(x) = f(f(f(x)))$

$\Rightarrow \quad g'(x) = f'(f(f(x)))\, f'(f(x))\, f'(x) > 0$

$\therefore \quad g(x)$ is strictly increasing function and $g(1) = f(f(f(1))) = f(f(1))$

$= f(1) = 1$

$\therefore \quad g(x) \ge 1 \quad \forall\, x \in [1, \infty) \Rightarrow \quad (x^2 - 2x - 2) \le 1 \Rightarrow (x-1)^2 \le 0$

$\because \quad \frac{1}{x^2 - 2x + 2} = \frac{1}{(x-1)^2 + 1} \le 1$

$\therefore \quad f(f(f(x))) = \frac{1}{x^2 - 2x + 2} \quad \Rightarrow x = 1$

$\therefore$ sum of all the solutions = 1. Ans.

173. $f(x) = 2x^3 - 21x^2 + 78x + 24.$
$f'(x) = 6(x^2 - 7x + 13)$
$\Rightarrow f'(x) > 0 \ \forall \ x \in R$
$\Rightarrow f(x)$ is increasing function
Now, $f(f(f(x) - 2x^3)) \geq f(f(2x^3 - f(x)))$
$\Rightarrow \ f(f(x) - 2x^3) \geq f(2x^3 - f(x)) \ \Rightarrow \ f(x) - 2x^3 \geq 2x^3 - f(x) \ \Rightarrow \ f(x) \geq 2x^3 \ \Rightarrow 7x^2 - 26x - 8 \leq 0$

$\Rightarrow \ x \in \left[-\frac{2}{7}, 4\right]$. **Ans.**]

174 to 176.
From above information, we must have
$f(x) = a\,(x-4)^3 + b\,(x-4)$; a, b $\in$ R and $a \neq 0$.
and $f'(4) = 0 \Rightarrow \ b = 0$, hence $f(x) = a\,(x-4)^3$
$\therefore \ f(x) = a(x-4)^3$

174. Let $I = \int_{-2010}^{2018} f(x)\,dx = \int_{-2010}^{2018} a\,(x-4)^3\,dx$
Put $(x-4) = t \ \Rightarrow dx = dt$

So, $I = \int_{-2014}^{2014} at^3\,dt = 0 \ \Rightarrow$ **(B) is correct**.

175. We have, $f(x) = a(x-4)^3,\ a \neq 0$

$y = f(x) = a\,(x-4)^3 \ \Rightarrow \ \left(\frac{y}{a}\right)^{\frac{1}{3}} + 4 = x$

$\therefore \ f^{-1}(x) = \left(\frac{x}{a}\right)^{\frac{1}{3}} + 4$

Now, $f^{-1}(x) + f^{-1}(-x) = \left(\left(\frac{x}{a}\right)^{\frac{1}{3}} + 4\right) + \left(\left(\frac{-x}{a}\right)^{\frac{1}{3}} + 4\right) = 4 + 4 = 8, \ \forall \ x \in R.$

So, $f^{-1}(2) + f^{-1}(-2) = 8 \ \Rightarrow$ **(D) is correct**.

176. As, $f(x) = a(x-4)^3$
So, $f'(x) = 3a\,(x-4)^2$

Given $f'(10) = 20 \ \Rightarrow \ 36 \times 3a = 20 \ \Rightarrow \ a = \frac{5}{27}$

Now $f'(-2) = 3 \times \frac{5}{27} \times 36 = 20 \Rightarrow$ **(C) is correct**.]

177. $f(x) = \begin{vmatrix} 1 & -1 \\ -x & x^2-2 \end{vmatrix} - 2\begin{vmatrix} 1 & -1 \\ -x & x^2-2 \end{vmatrix} + 3\begin{vmatrix} 1 & -1 \\ -x & x^2-2 \end{vmatrix} - 4\begin{vmatrix} 1 & -1 \\ -x & x^2-2 \end{vmatrix} + ...+ 31\begin{vmatrix} 1 & -1 \\ -x & x^2-2 \end{vmatrix}$

$= (1-2+3-4+.........-30+31)\begin{vmatrix} 1 & -1 \\ -x & x^2-2 \end{vmatrix}$

$= (-15+31)\begin{vmatrix} 1 & -1 \\ -x & x^2-2 \end{vmatrix} = 16\begin{vmatrix} 1 & -1 \\ -x & x^2-2 \end{vmatrix} = 16(x^2-x-2)$

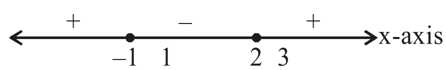
(i) $g(x) = \int\limits_0^x f(x-t)\,dt = \int\limits_0^x f(t)\,dt$

$\Rightarrow g'(x) = f(x)$

$\Rightarrow \left[\begin{array}{l} g'(x) > 0 \Rightarrow f(x) > 0 \Rightarrow x^2 - x - 2 > 0 \Rightarrow x \in (-\infty, -1) \cup (2, \infty) \\ g'(x) < 0 \Rightarrow f(x) < 0 \Rightarrow x \in (-1, 2) \end{array}\right.$

$\therefore$ g (x) is increasing in $x \in (-\infty, -1) \cup (2, \infty)$ and g (x) is decreasing in $x \in (-1, 2)$

$\therefore$ for $x \in [1, 3]$ g (x) is minimum at x = 2

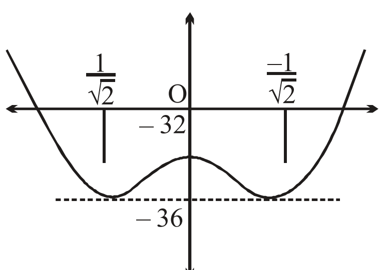


$g(2) = 16\int\limits_0^2 (x^2-x-2)\,dx = 16\left(\frac{x^3}{3} - \frac{x^2}{2} - 2x\right)_0^2$

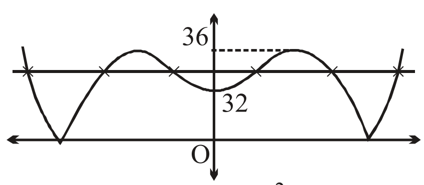
$= 16\left(\frac{8}{3} - \frac{4}{2} - 4\right) = 16\left(\frac{8}{3} - 6\right) = \frac{-160}{3}$ **Ans.**

(ii) $\left|f(x^2)\right| = k$

$\Rightarrow \quad 16\left|x^4 - x^2 - 2\right| = k$



Graph of $f(x^2)$ is



Graph of $|f(x^2)|$

$f(x^2) = 16(x^4 - x^2 - 2)$

$f'(x^2) = 16(4x^3 - 2x) = 0$; $x = 0, \pm\frac{1}{\sqrt{2}}$

$|f(x^2)| = k$ has six different solution

So, $k \in (32, 36)$

$\Rightarrow (a + b) = 68$ **Ans.**

178.

(a) $f(x) = (1 + b^2)\, x^2 + 2bx + 1$

Minimum value of $f(x)$ whould occur when $x = \dfrac{-2b}{2(1+b^2)} = \dfrac{-b}{1+b^2}$

Minimum value of $f(x) = m(b)$

$\therefore \quad m(b) = (1 + b^2)\, \dfrac{(b)^2}{(1+b^2)^2} - \dfrac{2b^2}{1+b^2} + 1$

$m(b) = \dfrac{-b^2}{1+b^2} + 1 = \dfrac{1}{1+b^2}$ as and varies m(b) would vary from (0, 1] as and varies from $(-\infty, \infty)$.]

179. $f'(x) = 2\,(2x\, ln\,(2x) - 2x) - (x\, ln\, x - x)$

$= 4x\, ln\,(2x) - x\, ln\, x - 3x$

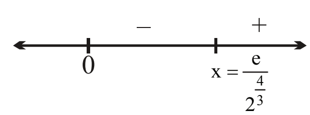
$= x\,[ln\,(16x^4) - ln\, x - 3]$

$\therefore \quad f'(x) = x\,[ln\,(16x^3) - 3]$

Now, $f'(x) > 0 \Rightarrow ln\,(16x^3) > 3$

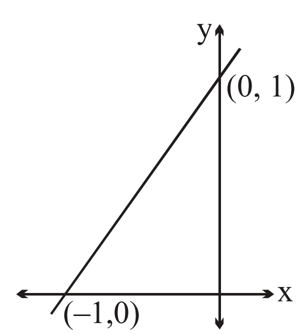
$\Rightarrow \quad 16x^3 > e^3 \Rightarrow 2^{\frac{4}{3}}\, x > e$

$\therefore \quad x > \dfrac{e}{2^{\frac{4}{3}}}$



Clearly, f(x) has a local minima at $x = \dfrac{e}{2^{\frac{4}{3}}}$. **Ans.**

180. Slope of the normal at P(x, y) $\dfrac{-1}{dy/dx} = x^2y$



$\Rightarrow \quad \dfrac{dy}{dx} = \dfrac{-1}{x^2y} \quad \Rightarrow \quad y\,dy = \dfrac{-dx}{x^2} \quad \Rightarrow \quad \dfrac{y^2}{2} = \dfrac{1}{x} + C$

at $x = -2, y = 1 \quad \Rightarrow \quad c = 1$

$\Rightarrow \quad \dfrac{y^2}{2} = \dfrac{1}{x} + 1 \quad \Rightarrow \quad y^2 = \dfrac{2}{x} + 2 = f^2(x)$

$y = x\, f^2(x) = x\left(\dfrac{2}{x} + 2\right) = 2 + 2x$, Area bounded by $y = 2x + 2$ and coordinate axes

$A = \dfrac{1}{2} \times 1 \times 2 = 1$ **Ans.**

181. $D = 1 + \dfrac{f(x)}{x^3}$ $\qquad \underset{x \to 0}{\text{Limit}}\ \ln\left(1 + \dfrac{f(x)}{x^3}\right)^{1/x} = 2$

$\Rightarrow$ f(x) have co-efficient of x^3, x^2, x and constant term zero in order that the limit may exist.

$$= \ln\ e^{\underset{x \to 0}{\text{Limit}} \frac{1}{x} \cdot \frac{f(x)}{x^3}} = \underset{x \to 0}{\text{Limit}} \frac{f(x)}{x^4} = 2$$

$$= \underset{x \to 0}{\text{Limit}} \frac{ax^6 + bx^5 + cx^4}{x^4} = 2 \quad \Rightarrow \quad c = 2.$$

Hence $\quad f(x) = ax^6 + bx^5 + 2x^4$

$\qquad f'(x) = x^3 (6ax^2 + 5bx + 8)$

$f'(1) = 0$ and $f'(2) = 0$ gives $\quad 6a + 5b + 8 = 0$ and $24a + 10b + 8 = 0$

$$\Rightarrow a = \frac{2}{3} \ ; \ b = -\frac{12}{5}$$

$$\Rightarrow \quad f(x) = \frac{2}{3}x^6 - \frac{12}{5}x^5 + 2x^4$$

Sol.182 to 184

Let $P(x) = (ax + b)(x - 2)^3 - 1$

$P'(x) = (ax + b)\,3(x - 2)^2 + (x - 2)^3.a$

$P''(x) = 3[(x - 2)^2.a + (ax + b)2(x - 2)] + 3a(x - 2)^2$

$P'''(x) = 3[2a(x - 2) + 2(ax + b).1 + 2(x - 2).a] + 6a(x - 2)$

$P''''(x) = 3[2a + 2a + 2a] + 6a$

$P''''(2) = 24 \Rightarrow 24a = 24 \Rightarrow a = 1$

$P'''(2) = 3.2\,(2a + b)$

$\Rightarrow -12 \Rightarrow 2a + b = -2$

$\Rightarrow \ b = -4$

$P(x) = (x - 4)(x - 2)^3 - 1$

182. $P''(1) = 3(1 + 6) + 3 = 24$

183. $\displaystyle\int_0^4 \left((x-4)(x-2)^3 - 1\right)dx \qquad$ Put $\ x - 2 = t \ \Rightarrow \ dx = dt$

$$\int_{-2}^{2} \left(t^3(t-2) - 1\right)dt$$

$$\int_{-2}^{2} \left(t^4 - 2t^3 - 1\right)dt = 2\int_0^2 \left(t^4 - 1\right)dt$$

$$\Rightarrow 2\left(\frac{t^5}{5} - t\right)_0^2 \Rightarrow 2\left(\frac{32}{5} - 2\right) = \frac{44}{5}$$

184. $P(x) = (x - 4)(x - 2)^3 - 1$

$P'(x) = (x - 2)^3 \cdot 1 + (x - 4)\,3(x - 2)^2$

$\Rightarrow (x-2)^2\,(x-2+3x-12)$

$\Rightarrow (x-2)^2\,(4x-14)$

$P'(x) = 0 \;\Rightarrow\; x = 2, \dfrac{7}{2}$

For $\quad x > \dfrac{7}{2}\quad P(x)$ is $\uparrow$

For $\quad x < \dfrac{7}{2}\quad P(x)$ is $\downarrow$

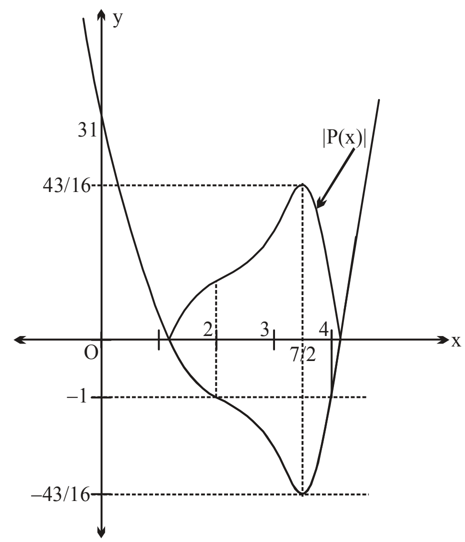
$$P\left(\frac{7}{2}\right) = \left(\frac{7}{2}-4\right)\left(\frac{7}{2}-2\right)^3 - 1$$

$$= \frac{-1}{2}\,\frac{27}{8} - 1 = \frac{-27-16}{16} = \frac{-43}{16}$$

$|P(x)| = \lambda$

$\lambda = \dfrac{43}{16}$

$\therefore \quad \lambda = 3.$ **Ans**



185 $\quad y = \tan\left(\dfrac{\pi}{2}+x+\dfrac{\pi}{6}\right) - \tan\left(x+\dfrac{\pi}{6}\right) + \cos\left(x+\dfrac{\pi}{6}\right)$

$$= -\cot\left(x+\frac{\pi}{6}\right) - \tan\left(x+\frac{\pi}{6}\right) + \cos\left(x+\frac{\pi}{6}\right)$$

$$y = \cot\left(-\left(x+\frac{\pi}{6}\right)\right) - \tan\left(-\left(x+\frac{\pi}{6}\right)\right) + \cos\left(-\left(x+\frac{\pi}{6}\right)\right)$$

let $\quad -x-\dfrac{\pi}{6} = z$

as $\; -\dfrac{5\pi}{12} \le x \le -\dfrac{\pi}{3};\;\; \dfrac{\pi}{3} \le -x \le \dfrac{5\pi}{12}\;;$

$\dfrac{\pi}{6} \le -x-\dfrac{\pi}{6} \le \dfrac{\pi}{4} \qquad \Rightarrow \qquad z \in \left[\dfrac{\pi}{6}, \dfrac{\pi}{4}\right]$

and $\;y = \cot z + \tan z + \cos z = \dfrac{1}{\sin z \cdot \cos z} + \cos z = 2\,\text{cosec}\,2z + \cos z$

as both the functions are decreasing in $\left[\dfrac{\pi}{6}, \dfrac{\pi}{4}\right]$

hence $\quad y_{max}$ occurs when $z = \dfrac{\pi}{6}$

$$y\left(\frac{\pi}{6}\right) = \cot\frac{\pi}{6} + \tan\frac{\pi}{6} + \cos\frac{\pi}{6} = \sqrt{3} + \frac{1}{\sqrt{3}} + \frac{\sqrt{3}}{2} = \frac{11\sqrt{3}}{6}$$ **Ans.**

186 Given $\cos\alpha_1 \cos\alpha_2 - \cos\alpha_3 \,.....\, \cos\alpha_n = \sin\alpha_1 \sin\alpha_2 \,.....\, \sin\alpha_n$

$y = (\cos\alpha_1)(\cos\alpha_2) \,......\, (\cos\alpha_n)(\sin\alpha_1)(\sin\alpha_2)(\sin\alpha_3) \,....\, (\sin\alpha_n)$

[M-1] $y^2 = (\cos\alpha_1)(\cos\alpha_2) \,....\, (\cos\alpha_n)(\sin\alpha_1)(\sin\alpha_2) \,...\, (\sin\alpha_n)$

$$y^2 = \frac{1}{2^n}(\sin 2\alpha_1)(\sin 2\alpha_1) \,...\, (\sin + \alpha_n)$$

$\therefore \quad y^2_{max} = \frac{1}{2^n} \quad \Rightarrow y_{max} = \frac{1}{2^{4/2}}$

[M-2] $\frac{1+\tan^2\alpha_1}{2} \geq (1+\tan^2\alpha)^{1/2} = \tan\alpha_1$

$\therefore \quad 1 + \tan^2\alpha_1 \geq 2\tan\alpha_1 \sec\alpha_1 \sin\alpha_2 \,....\, \sin\alpha_n \geq 2^{4/2}$

$\sec^2\alpha_1 \geq 2\tan\alpha_1$

$\sec^2\alpha^2 \geq 2\tan\alpha_2$

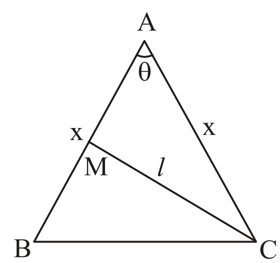
$\sec^2\alpha_n \geq 2\tan\alpha_n \qquad \therefore \qquad \cos\alpha_1 \cos\alpha_2 \,.....\, \cos\alpha_n \leq \frac{1}{2^{n/2}}$

187 $A = \frac{1}{2}x^2\sin\theta \qquad ...(1)$

$$l^2 = \frac{x^2}{4} + x^2 - 2\cdot\frac{x}{2}\cdot x\cos\theta$$

$$= \frac{5x^2}{4} - x^2\cos\theta$$

$$4l^2 = x^2(5 - 4\cos\theta)$$



$$2A = \frac{4l^2\cdot\sin\theta}{5-4\cos\theta}$$

$$\therefore\ f(\theta) = \frac{4l^2\cdot\sin\theta}{5-4\cos\theta}.$$

188 $AD = 2l\cos\theta$ from ΔABD

$\therefore \quad x = AD\cos\theta = 2l\cos^2\theta$

Perimeter $= 2l + 2l - 2x + 2AB$

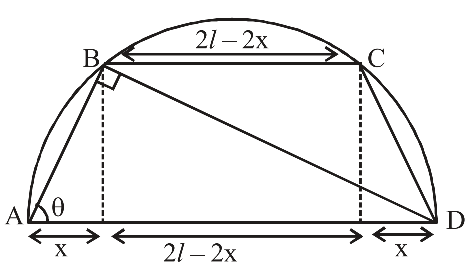
$= 4l - 2x + 4l\cos\theta$

$P(\theta) = 4l(1 - \cos^2\theta + \cos\theta)$

$\therefore \quad P'(\theta) = 4l(2\cos\theta\sin\theta - \sin\theta) = 0$

$\therefore \quad 2\cos\theta\sin\theta = \sin\theta \quad \theta \neq 0 \text{ or } \pi$

$\therefore \quad \cos\theta = \frac{1}{2}$

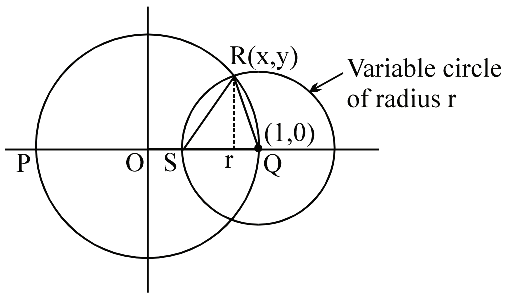


$\therefore \quad \theta = \pi/3 = 60°$

$\therefore$ Perimeter is maximum when $\theta = 60°$ **Ans.**

189 Equation of the fixed circle $x^2 + y^2 = 1$ ....(1)
equation of the variable circle is $(x-1)^2 + y^2 = r^2$ .....(2)
solving (1) and (2) $x^2 + y^2 - 2x + 1 = r^2$

$2 - 2x = r^2 \Rightarrow 2x = 2 - r^2 \Rightarrow x = \dfrac{2-r^2}{2}$



substituting in (1) $\dfrac{(2-r^2)^2}{4} + y^2 = 1$

or $y^2 = 1 - \dfrac{4+r^4-4r^2}{4} = \dfrac{4-4-r^4+4r^2}{4}$

$y^2 = \dfrac{r^2(4-r^2)}{4} \Rightarrow y = \dfrac{r\sqrt{4-r^2}}{2}$

Hence area of $\Delta QSR = \dfrac{ry}{2} = \dfrac{r^2\sqrt{4-r^2}}{4}$

$f(r) = A^2 = \dfrac{r^4(4-r^2)}{16}$

$f'(r) = \dfrac{1}{16}[16r^3 - 6r^5]\ \dfrac{r^3}{16}(16 - 6r^2)$

$r^2 = \dfrac{16}{6} = \dfrac{8}{3} \Rightarrow r = 2\sqrt{\dfrac{2}{3}}$

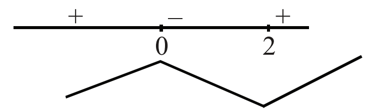
$f''(r) = \dfrac{1}{16}[48r^2 - 30r^4]$

$f''(r)\big|_{r^2=8/3} = \dfrac{1}{16}\left[48\times\dfrac{8}{3} - 30\times\dfrac{64}{9}\right] < 0 \Rightarrow f(r)$ is maximum

$A^2\big|_{max} = \dfrac{64}{9}\dfrac{(4-(8/3))}{16} = \dfrac{4}{9}\times\dfrac{4}{3} = \dfrac{16}{27}$

$A_{max} = \dfrac{4}{3\sqrt{3}}$ Ans.

190. $f'(x) = x^2 - 2x$



$f(x) = \dfrac{x^3}{3} - x^2 + c$

$f(2) = 0$

$\therefore\ \dfrac{8}{3} - 4 + c = 0$

$\Rightarrow c = \dfrac{4}{3}$

$f(x) = \dfrac{x^3}{3} - x^2 + \dfrac{4}{3}$

Minimum ordinate will be at x= 2

$\therefore\ f(2) = \dfrac{8}{3} - 4 + \dfrac{4}{3} = 0$

$\therefore$ point (a, b) = (2, 0)

a + 6b = 2 **Ans.**

191. Given, $f(x) = e^x \cdot \int_0^x e^{-y} \cdot f'(y)\, dy - (x^2 - x + 1)e^x$ ....(1)

differentiating both the sides

$f'(x) = e^x \cdot e^{-x} \cdot f'(x) + e^x \int_0^x e^{-y} \cdot f'(y)\, dy - \left[(x^2 - x + 1)e^x + e^x(2x - 1)\right]$

$f'(x) = f'(x) + \int_0^x e^{x-y} \cdot f'(y)\, dy - e^x(x^2 + x)$

$\Rightarrow 0 = f(x) + (x^2 - x + 1)\, e^x - e^x\,(x^2 + x)$ [Substituting $\int_0^x e^{x-y} f'(y)\, dy = f(x) + (x^2 - x + 1)\, e^x$ from (1)]

$f(x) = e^x\,(x^2 + x - x^2 + x - 1)$

$f(x) = e^x \cdot (2x - 1) \Rightarrow f(1) = e.$

Also, $f'(x) = e^x \cdot 2 + (2x - 1)\, e^x \Rightarrow f'(1) = 2e + e = 3e$

and $f''(x) = 2e^x + (2x - 1)e^x + 2e^x$

$\Rightarrow\ f''(1) = 2e + e + 2e = 5e.$

Hence $f(1) + f'(1) + f''(1) = 9e \Rightarrow k = 9.$ **Ans.**

192. Given, $7 \sin 3x - 2\,(3 \sin 3x - 4 \sin^3 3x) = (1 + \tan^2 \theta) + 4\,(1 + \cot^2\theta)$

Put, $\sin 3x = y$, we get

$y\,(8y^2 + 1) = 9 + (\tan\theta - 2\cot\theta)^2$

L.H.S. min occurs

minimum value = 9

when $y = -1$ and minimum value $= -9$.

maximum occurs when $y = 1$ and maximum value = 9 when $\tan^2\theta = 2$

Hence maximum of L.H.S = minimum value of R.H.S.

$\therefore \quad \sin 3x = 1 = \sin\frac{\pi}{2} \quad \Rightarrow \quad x = \frac{\pi}{6}$ (minimum positive root).

or $\quad \sin 3x = \sin\left(\frac{-3\pi}{2}\right) \quad \Rightarrow \quad x = \frac{-\pi}{2}$ (maximum negative root).

$$\text{Sum} = \frac{\pi}{6} + \frac{\pi}{2} = \frac{\pi + 3\pi}{6} = \frac{2\pi}{3}.$$

Hence, $\frac{15}{\pi}\Big(\text{(minimum positive root)} - \text{(maximum negative root)}\Big)$

$$= \left(\frac{15}{\pi}\right)\left(\frac{2\pi}{3}\right) = 10 \textbf{ Ans.}$$

Sol 193 to 195

193. $a = 1$

$f(x) = 8x^3 + 4x^2 + 2bx + 1$

$f'(x) = 24x^2 + 8x + 2b = 2(12x^2 + 4x + b)$

for increasing function, $f'(x) \geq 0 \quad \forall \ x \in R$

$\therefore \quad D \leq 0 \quad \Rightarrow \quad 16 - 48b \leq 0 \quad \Rightarrow \quad b \geq \frac{1}{3}$

194. if $b = 1$

$f(x) = 8x^3 + 4ax^2 + 2x + a$

$f'(x) = 24x^2 + 8ax + 2 \quad$ or $\quad 2(12x^2 + 4ax + 1)$

for non monotonic $f'(x) = 0$ must have distinct roots

hence $\quad D > 0 \quad$ i.e. $\quad 16a^2 - 48 > 0 \quad \Rightarrow \quad a^2 > 3;$

$\therefore \quad a > \sqrt{3}$ or $a < -\sqrt{3}$

$\therefore \quad a \in 2, 3, 4, \ldots\ldots$

sum = 5050 – 1 = 5049 Ans.

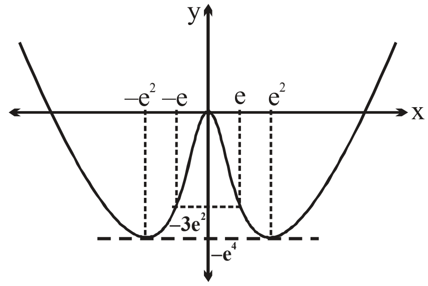
195. If x_1, x_2 and x_3 are the roots then $\log_2 x_1 + \log_2 x_2 + \log_2 x_3 = 5$

$\log_2(x_1x_2x_3) = 5$

$x_1x_2x_3 = 32$

$-\frac{a}{8} = 32 \quad \Rightarrow \quad a = -256$ **Ans.**

196.



Graph of f (x)

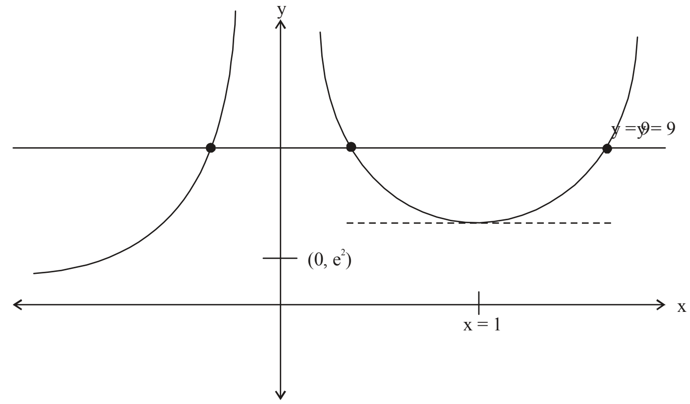
Now, verify alternatives.

197. $f^2(x) \cdot g(x) = 20 - 3x^2$

L.H.S. is increasing and R.H.S. is decreasing in $[0, \infty)$

$\therefore$ exactly onereal roots. **Ans.**

198. Let $f(x) = \dfrac{e^{2x}}{x^2}$



$\Rightarrow f'(x) = \dfrac{2e^{2x}(x-1)}{x^3}$

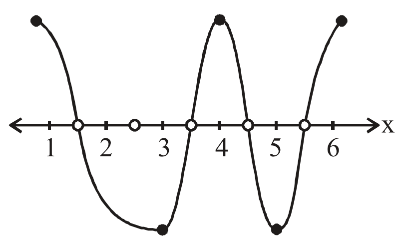
So, we get

$\dfrac{e^{2x}}{x^2} = 9$

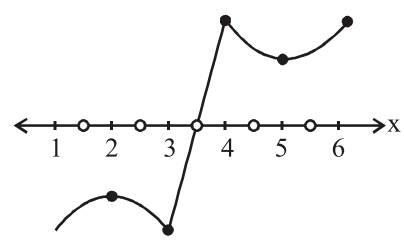
Clearly $f(x) = 9$ has three solutions.

199. $f(6) > 0$, $f(3) < 0$, $f(2)$, $f(4)$ will be of opposite sign. $f(1)$, $f(5)$ will be of opposite sign. Following 4 possible are as

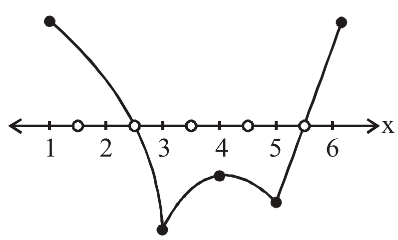
(1)



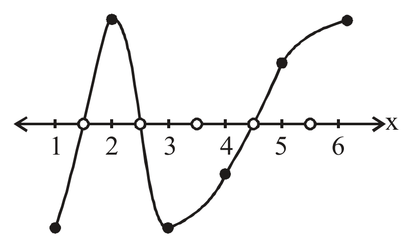
(2)



(3)



(4)



Number of possible roots 1, 2, 3, 4.

A = {1, 2, 3, 4}

Hence, $\sum n(A) = 10$. **Ans.**

200. $f'(x) = x^4 - bx^3 + (b+1)x^2 - bx + b$

$= (x^4 - bx^3 + bx^2) + (x^2 - bx + b)$

$= x^2(x^2 - bx + b) + (x^2 - bx + b)$

$= (x^2 + 1) + (x^2 - bx + b)$

$f'(x) > 0 \ \forall\ x \in R$

$\Rightarrow x^2 - bx + b > 0 \ \ \forall\ x \in R$

$\Rightarrow \Delta < 0 \Rightarrow \ b^2 - 4b < 0$

$\Rightarrow b(b-4) < 0 \Rightarrow 0 \le b \le 4$ Ans.

201. $f(x) = \int_{1/2}^{x} (2t + a)\,dt$

$f'(x) = 2x + a$ .... (1)

Now, $f(x) = x^2 + ax - \frac{1}{4} - \frac{a}{2}$

$f(x) = 0 \Rightarrow \left(x^2 - \frac{1}{4}\right) + a\left(x - \frac{1}{2}\right) = 0$

$\Rightarrow \left(x + \frac{1}{2} + a\right)\left(x - \frac{1}{2}\right) = 0 \Rightarrow x = \frac{1}{2}, -a - \frac{1}{2}$

So, $f'\left(\frac{1}{2}\right) \cdot f'\left(-a - \frac{1}{2}\right) = 1$

$\Rightarrow (1 + a)(-1 - a) = -1$

$\Rightarrow (a + 1)^2 = 1 \Rightarrow a = 0, -2.$

$\therefore a_1^2 + a_2^2 = 4.$ **Ans.**

202.

(A) Given $f'(x) - 2f(x) \le 0$ ....(1) and $f'(x) \ge 0$ ....(2)

from equation (2), $f(x)$ is an increasing function $\forall\ x \ge 0 \Rightarrow f(x) \ge f(0) \Rightarrow f(x) \ge 0$ ....(3)

from equation (1), $\frac{d}{dx}\left(f(x)e^{-2x}\right) \le 0$

$\Rightarrow \left(f(x)e^{-2x}\right)$ is decreasing function $\forall\ x \ge 0$

Hence $f(x)e^{-2x} \le f(0) \le 0$

Hence $f(x) \le 0$ ....(4)

From (3) and (4), $f(x) = 0$

(B) $f(x) = x^3 + ax^2 + bx + 5\sin^2 x$

$f'(x) = 3x^2 + 2ax + b + 5\sin 2x$

hence $3x^2 + 2ax + b - 5 \le 3x^2 + 2ax + b + 5\sin 2x \le 3x^2 + 2ax + b + 5$

For $f(x)$ to be strictly increasing $f'(x) > 0$

$\Rightarrow 3x^2 + 2ax + b - 5 \ge 0\ \forall\ x$

$\Rightarrow D \le 0 \Rightarrow a^2 - 3b + 15 \le 0$

(C) Let $f(x) = x^{1/x}$.

$f(x)$ is increasing in $(1, e)$ and decreasing (e, ∞) and hence maximum value of $f(x)$ occurs at $x = e$

$\Rightarrow f(x) \le f(e)\ \forall\ x \ge 1$

$\Rightarrow x^{1/e} < e^{1/e}$

Hence $x^a \le a^x\ \forall\ x \in (1, \infty)$

$\Rightarrow a = e$

(D) $g'(x) = \dfrac{x f'(x) - f(x)}{x^2}$ for $x > 0$

Now given $f'(x)$ is an $\uparrow$ function for $x > 0$.

$$f(x) = \int_0^x f'(t)dt < \int_0^x f'(x)dt$$

hence $f(x) < x f'(x)$

$\Rightarrow \quad g'(x) > 0 \ \forall\ x > 0$

hence $g(x)$ is an $\uparrow$ function.

203. $f'(x) = \dfrac{x^4}{4} - \dfrac{x^3}{3} = \dfrac{x^3}{12}(3x - 4)$

$f''(x) = x^3 - x^2 = x^2(x - 1)$

for $\quad x < 0,\ f'(x) > 0$ and

$\quad x > 0,\ f'(x) < 0$ hence at $x = 0$, we have maxima

Also $\quad f''(0) = 0$ when $x = 1$ and $x = 0$ but at $x = 1$ only we have an inflection point

204.

(A) $T_r = \dfrac{1}{\sqrt{4nr - r^2}}$

$$L = \lim_{n\to\infty} \sum_{r=1}^{2n} \frac{1}{\sqrt{4nr - r^2}} = \lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{2n} \frac{1}{\sqrt{\frac{4r}{n} - \left(\frac{r}{n}\right)^2}}$$

$$= \int_0^2 \frac{1}{\sqrt{4x - x^2}} = \int_0^2 \frac{1}{\sqrt{4 - (x-2)^2}}\,dx$$

$$= \sin^{-1}\left(\frac{x-2}{2}\right)\Bigg|_0^2 = \frac{\pi}{2}$$

(B) On the contrary assume $\exists$ two points x_1, x_2 in (a, b) such that $f'(x_1) = f'(x_2) = 0$. By Rolle's theorem on f' on $[x_1, x_2]$

$\Rightarrow \exists\ c \in (x_1, x_2)$ such that $f''(c) = 0$. This is contradiction to given $\Rightarrow$ $f'(x)$ is zero at atmost one point.

(C) $\because\ f(-x) = -ax^7 - bx^5 - cx^3 - dx + 2$

$\therefore\ f(x) + f(-x) = 4$

putting $x = 3$, we get

$\quad f(3) + f(-3) = 4$

$\Rightarrow \quad f(3) = 1$

$\therefore \quad f(3) + 3\cos^2 x + 4\sin^2 x = 1 + 3 + \sin^2 x$

$\quad = 4 + \sin^2 x$

$\because\ 0 \le \sin^2 x \le 1 \quad \Rightarrow 4 \le 4 + \sin^2 x \le 5$

$\therefore$ Range of f(x) will be [4, 5]

(D) $\quad f(x) = \ln(1+x) - \dfrac{\tan^{-1}x}{1+x}$

$$f'(x) = \frac{1}{1+x} - \frac{1}{(1+x)(1+x^2)} + \frac{\tan^{-1}x}{(1+x)^2}$$

$$= \frac{x^2(1+x)+(1+x^2)\tan^{-1}x}{(1+x)^2(1+x^2)} > 0 \quad \forall\, x > 0$$

$$\Rightarrow \quad f'(x) > f(0) \;\Rightarrow\; ln(1+x) > \frac{\tan^{-1}x}{1+x}$$

205. Let $f(x) = x^{1/7}$

Apply LMVT in [3, 5], $f'(a_1) = \dfrac{f(5)-f(3)}{5-3}$

$$\Rightarrow \frac{1}{7}a_1^{-6/7} = \frac{f(5)-f(3)}{2},\; a_1 \in (3, 5)$$

Again LMVT in [7, 9]

$$f'(a_2) = \frac{f(9)-f(7)}{7-9} \;\Rightarrow\; \frac{1}{7}a_2^{-6/7} = \frac{f(9)-f(7)}{2},\; a_2 \in (7, 9)$$

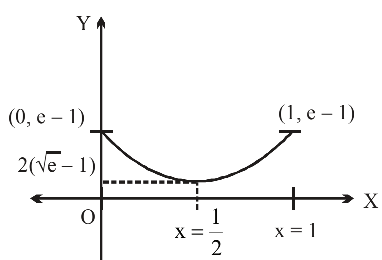
$$\frac{f'(a_1)}{f'(a_2)} = \left(\frac{a_2}{a_1}\right)^{6/7} > 1$$

$\Rightarrow f'(a_1) > f'(a_2) \;\Rightarrow\; f(5) - f(3) > f(9) - f(7)$

$\Rightarrow f(5) + f(7) > f(3) + f(9)$. **Ans.**

Aliter: $f(x) = x^{1/7} - (x-2)^{1/7}$ and $f'(x) < 0$ in [5, 9].

206. $f(x) = \displaystyle\int_0^x e^{x-t}\,dt + \int_x^1 e^{t-x}\,dt$



Graph of y = f(x) in [0, 1]

$$f(x) = e^x \int_0^x e^{-t}\,dt + e^{-x}\int_x^1 e^t\,dt$$

$f(x) = e^x(1 - e^{-x}) + e^{-x}(e - e^x)$

$\therefore \quad f(x) = e^x + e^{1-x} - 2$

$\quad\quad f'(x) = e^x - e^{1-x} = 0$

$\therefore \quad x = \dfrac{1}{2}$

$f''(x) = e^x + e^{1-x} > 0 \;\Rightarrow\;$ minima

$$\left.\begin{matrix} f(0) = e-1 \\ f(1) = e-1 \end{matrix}\right\} \text{Maximum value}$$

$f\left(\frac{1}{2}\right) = 2\left(\sqrt{e}-1\right) \rightarrow$ Minimum value.

208. $f(x) = 2x^3 - 3(k+4)x^2 + 18(k+1)x$
$f'(x) = 6x^2 - 6(k+4)x + 18(k+1)$
$f'(x) = 6(x^2 - (k+4)x + 3(k+1))$
$f'(x) = 6(x-3)(x-(k+1))$
$x = 3, k+1$
then $k+1 \neq 3 \Rightarrow k \neq 2$

209. $f'(x) = 3x^2 - 3 = 3(x^2-1)$.
$f(-1) \cdot f(1) = (1 - 3 + \sin^{-1}(a^2 - 3a + 2))(-1 + 3 + \sin^{-1}(a^2 - 3a + 2) < 0$

$$\Rightarrow \quad \underbrace{(\sin^{-1}(a^2 - 3a + 2) - 2)}_{\text{always negative}} \underbrace{(2 + \sin^{-1}(a^2 - 3a + 2)}_{\text{always positive}} < 0$$

but $\quad -1 \le a^2 - 3a + 2 \le 1$

$$\left(a - \frac{3}{2}\right)^2 - \left(\frac{\sqrt{5}}{2}\right)^2 \le 0$$

$$a \in \left[a - \left(\frac{3+\sqrt{5}}{2}\right)\right]\left[a - \left(\frac{3-\sqrt{5}}{2}\right)\right] \le 0$$

$$\Rightarrow a \in \left[\frac{3-\sqrt{5}}{2}, \frac{3+\sqrt{5}}{2}\right]$$

$$\Rightarrow \quad a \in \left[\frac{3-\sqrt{5}}{2}, \frac{3+\sqrt{5}}{2}\right], \ a = 1 \textbf{ Ans.}$$

210. Let the point of intersection of curves $y^2 = 2ax\ (a > 0)$ and $xy = 4\sqrt{2}$ be $P(x_1, y_1)$.

Now, $y^2 = 2ax \Rightarrow \left.\frac{dy}{dx}\right]_{P(x_1,y_1)} = \frac{a}{y_1} = m_1 \qquad \dots (1)$

Also, $xy = 4\sqrt{2}$

$\Rightarrow \left.\frac{dy}{dx}\right]_{P(x_1,y_1)} = \frac{-y_1}{x_1} = m_2 \qquad \dots (2)$

As, $m_1 \times m_2 = -1 \Rightarrow x_1 = a \qquad \dots (3)$

As, $x_1y_1 = 4\sqrt{2} \Rightarrow y_1 = \frac{4\sqrt{2}}{a}$

So, $P\left(x_1 = a,\ y_1 = \frac{4\sqrt{2}}{a}\right)$

Also, Point P must satisfy

$y_1^2 = 2ax_1$

$\frac{32}{a^2} = 2a^2 \Rightarrow a^4 = 16 \Rightarrow a = 2$ Ans.

Sol. 211 to 213

We have $g(x) = x^3 + g''(1)\,x^2 + \{3g'(1) - g''(1) - 1\}x + 3g'(1)$

Let $g'(1) = a$ and $g''(1) = b$ then $g(x) = x^3 + bx^2 + (3a - b - 1)\,x + 3a$

Differentiating both sides w.r.t. x

$\therefore\ g'(x) = 3x^2 + 2bx + (3a - b - 1)$

Put $x = 1 \Rightarrow g'(1) = 3 + 2b + 3a - b - 1$

$\Rightarrow a = b + 3a + 2 \Rightarrow 2a + b = -2$ .... (1)

$g''(x) = 6x + 2b$

Put $x = 1$

$g''(1) = 6 + 2b \Rightarrow b = 6 + 2b \Rightarrow b = -6$ .... (2)

$\therefore$ From equation (1), we get $a = 2$

$\therefore\ g(x) = x^3 - 6x^2 + 11x + 6$

Given $f(x) = xg(x) - 12x + 1 = x^4 - 6x^3 + 11x^2 - 6x + 1$

$= (x^2 + 1)^2 - 2x^2 + 11x^2 - 6x^3 - 6x$

$= (x^2 + 1)^2 - 6x\,(x^2 + 1) + (3x)^2$

$f(x) = (x^2 - 3x + 1)^2 = \{h(x)\}^2$ (Given)

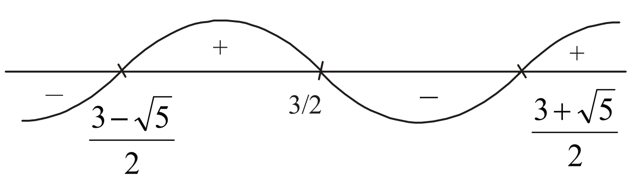
$\because\ h(0) = 1$

$\therefore\ h(x) = x^2 - 3x + 1$

211. $f(x) = (x^2 - 3x + 1)^2$

$\Rightarrow f'(x) = 2(x^2 - 3x + 1)(2x - 3) = 0 \Rightarrow x = \frac{3-\sqrt{5}}{2}, \frac{3}{2}$ and $\frac{3+\sqrt{5}}{2}$

sign scheme for f'(x) will be



Clearly f(x) has local maxima at $x = \frac{3}{2}$ and local minima at $x = \frac{3\pm\sqrt{5}}{2}$

$\therefore$ f(x) has exactly one local maxima and two local minima.

212. We have $g(x) = x^3 - 6x^2 + 11x + 6$

$$g'(x) = 3x^2 - 12x + 11 = 3(x-2)^2 - 1 = 3\left[(x-2)^2 - \frac{1}{3}\right]$$

$\therefore$ $g'(x) > 0 \Rightarrow x \in \left(-\infty, 2 - \frac{1}{\sqrt{3}}\right) \cup \left(2 + \frac{1}{\sqrt{3}}, \infty\right)$ and $g'(x) < 0$

$$\Rightarrow x \in \left(2 - \frac{1}{\sqrt{3}}, 2 + \frac{1}{\sqrt{3}}\right)$$

$\therefore$ g(x) monotonically increases for $x \in \left(-\infty, 2 - \frac{1}{\sqrt{3}}\right) \cup \left(2 + \frac{1}{\sqrt{3}}, \infty\right)$

and monotonically decreases for $x \in \left(2 - \frac{1}{\sqrt{3}}, 2 + \frac{1}{\sqrt{3}}\right)$

For $x \in [1, 3]$

$g(x) = (x-1)(x-2)(x-3) + 12$

$\Rightarrow g(1) = 12$ and $g(3) = 12$

$\therefore$ By Rolle's theorem in [1, 3] we have, $g'(c) = 0$

$\Rightarrow c = 2 \pm \frac{1}{\sqrt{3}}$ (both $\in (1, 3)$)

$\therefore$ There exists two distinct tangents to the curve $y = g(x)$ which are parallel to the chord joining $(1, g(1))$ and $(3, g(3))$

For $x \in [0, 4]$

$g(0) = 6$ and $g(4) = 18$

$\therefore$ By LMVT

$$g'(c) = \frac{18-6}{4-0} \Rightarrow 3c^2 - 12c + 11 = 3$$

$\Rightarrow 3c^2 - 12c + 8 = 0 \Rightarrow c = 2 \pm \frac{2}{\sqrt{3}}$ (both $\in (0, 4)$)

$\therefore$ There exists exactly two distinct Lagrange's mean value in (0, 4) for $y = g(x)$.

213. We have $h(x) = x^2 - 3x + 1 = \left(x - \frac{3}{2}\right)^2 - \frac{5}{4}$

The curve $y = h(x)$ is an upward parabola, intersecting x axis at two distinct points.

$\therefore$ h(x) has exactly one critical point (i.e. the vertex) and no any point of inflection.

Also $h(x) = 0 \Rightarrow x = \frac{3}{2} \pm \frac{\sqrt{5}}{2}$ (both $\in (0, 3)$)

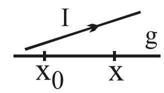
$\therefore$ $h(x) = 0$ has exactly two distinct real zeroes in (0, 3).

Every quadratic function has exactly one tangent on x-axis or parallel to x-axis.

Sol 214 to 216

214. Let $g(x) = e^{-x} f(x)$

$g'(x) = e^{-x}(f'(x) - f(x)) > 0 \quad \forall\ x \in R$



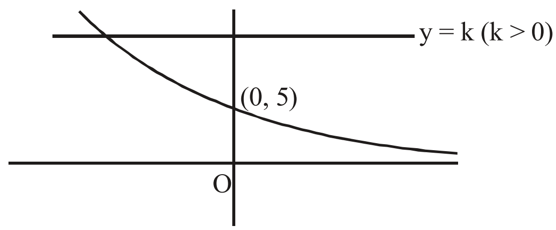
$\Rightarrow$ g(x) is increasing or R

$\therefore \quad g(x) > g(x_0)\ \forall\ x > x_0 \Rightarrow e^{-x} f(x) > \underbrace{e^{-x_0} f(x_0)}_{0}$

$\Rightarrow f(x) > 0\ \forall\ x > x_0$ [given $f(x_0) = 0$]

215. $k = \dfrac{5 + x + \dfrac{x^2}{2}}{e^x}$

Let $f(x) = \dfrac{5 + x + \dfrac{x^2}{2}}{e^x}$



$f'(x) = \dfrac{-(x^2 + 8)}{2e^x} < 0$

$\therefore$ f(x) is a decreasing function $\forall$ x

$\lim\limits_{x \to \infty} f(x) = 0$ and $\lim\limits_{x \to -\infty} f(x) = \infty$ also f(0) = 5

Hence graph of f(x) is as shown

and f(x) = k has exactly one root.

216. $g'(x) = 3x^2 - 2(9 - a)x + 3(9 - a^2) \left\langle \begin{matrix} x_1 \\ x_2 \end{matrix} \right.$

Since, x_1 and x_2 are opposite sign.

$\therefore$ Product < 0

Product of roots $9 - a^2 < 0$

$a^2 > 9 \Rightarrow a \in (-\infty, -3) \cup (3, \infty)$

Sol 217 , 218

217. $f(0) = \lim\limits_{x \to 0} f(x) = \lim\limits_{x \to 0} (\ln x)^{20} \cdot x^{30} = \lim\limits_{x \to 0} \dfrac{(\ln x)^{20}}{x^{-30}} \quad \left(\dfrac{\infty}{\infty}\right)$

$= \lim\limits_{x \to 0} \dfrac{20(\ln x)^{19}}{-30 \cdot x^{-31}} \dfrac{1}{x} = -\dfrac{20}{30} \dfrac{(\ln x)^{19}}{x^{-30}}$ .... so on

hence $\lim\limits_{x \to 0} f(x) = 0 = f(0)$

$\Rightarrow f$ is continuous at x = 0 with f(0) = 0 Ans.

now $f'(x) = x^{29}\, 20\, (\ln x)^{19} + (\ln x)^{20} \cdot 30 \cdot x^{29}$

$= (\ln x)^{19} \cdot x^{29} \cdot 10\,[2 + 3 \ln x] = 0$

this gives $x = 0$ which is rejected

or $\ln x = 0 \Rightarrow x = 1$

or $\ln x = -\frac{2}{3} \Rightarrow x = e^{-\frac{2}{3}}$

218. if $x < e^{-2/3}$ then $f'(x) > 0$

$x > e^{-2/3}$ then $f'(x) < 0$

hence $x = e^{-2/3}$ gives local maxima and $x = 1$ gives global minima i.e. 0.

219. $\frac{dx}{dt} = \frac{e^{\sqrt{t}}}{2\sqrt{t}}$; $\frac{dy}{dt} = 3 - \frac{2}{t}$. At $t = 1$, then $x = e$, $y = 3$, $\frac{dx}{dt} = \frac{e}{2}$, $\frac{dy}{dt} = 1$.

Hence, the slope of the tangent line is $\frac{2}{e}$, hence equation of tangent line is $y = \frac{2}{e}x + 1$.

220. $f(x) = 2x^3 - 3(k+4)x^2 + 18(k+1)x$

$f'(x) = 6x^2 - 6(k+4)x + 18(k+1)$

$f'(x) = 6(x^2 - (k+4)x + 3(k+1))$

$f'(x) = 6(x-3)(x-(k+1))$

$x = 3, k+1$

then $k + 1 \neq 3 \Rightarrow k \neq 2$

221. $f(x) = x^2 \ln x$

$f'(x) = x^2 \cdot \frac{1}{x} + \ln x \cdot 2x$

$= x(1 + 2\ln x) \quad x > 0$

f(x) is increasing in $\left(e^{-\frac{1}{2}}, \infty\right)$ and decreasing in $\left(0, e^{-\frac{1}{2}}\right)$

$\therefore \quad f(x)\Big|_{min} = e^{-1} \ln e^{-\frac{1}{2}} = -\frac{1}{2e} = p$

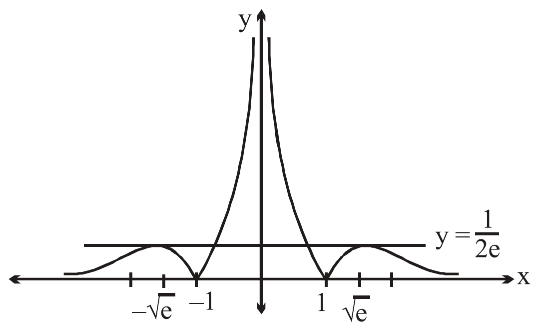
$\therefore \quad \ln|2p| = \ln\left|2\left(\frac{-1}{2e}\right)\right| = -1.$ **Ans.**

222. $f(x) = \left|\ln|x|\right| - kx^2 = 0$

$\Rightarrow \quad k = \dfrac{\left|\ln|x|\right|}{x^2}$

Consider $g(x) = \dfrac{\ln x}{x^2},\ x > 0$

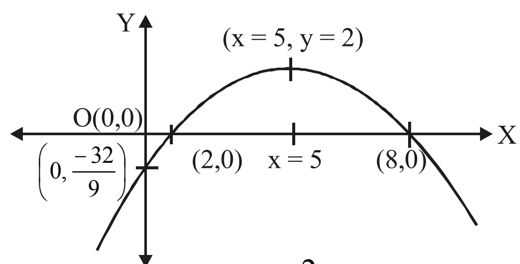
g (x) is increasing in $\left(0, \sqrt{e}\right)$ and decreasing $\left(\sqrt{e}, \infty\right)$



Sol : 223 - 225

$$f\left(f(x)\right) = a^3\left(x^2 - (2+b)x + 2b - \frac{2}{a}\right)\left(x^2 - (2+b)x + 2b - \frac{b}{a}\right),\ a \neq 0$$

Now, we shall consider 3 cases:



Graph of $f(x) = \dfrac{-2}{9}(x-2)(x-8)$

Case I: When $\underbrace{x^2 - (2+b)x}_{5\quad 5} + \left(2b - \dfrac{2}{a}\right) = 0$

$\therefore \quad 2 + b = 10 \Rightarrow b = 8$

Also, $\quad 2b - \dfrac{2}{a} = 25 \Rightarrow 16 - 25 = \dfrac{2}{a} \Rightarrow a = \dfrac{-2}{9}$

Let, $\quad D = \text{discriminant of}\left[x^2 - (2+b)x + 2b - \dfrac{b}{a}\right]$

$$= (2+b)^2 - 4\left(2b - \frac{b}{a}\right) = (2+8)^2 - 8 \times 8 + \frac{4\times 8}{-2/9} = 100 - 64 - 144 < 0$$

So, $\quad f(x) = \dfrac{-2}{9}(x-2)(x-8) = \dfrac{-2}{9}(x-5)^2 + 2$

Case II: When $\underbrace{x^2 - (2+b)x}_{5\quad 5} + \left(2b - \dfrac{b}{a}\right) = 0$

$\therefore \quad 2 + b = 10 \Rightarrow b = 8$

$\Rightarrow \quad 16 - 25 = \dfrac{8}{a} \quad \Rightarrow \quad a = \dfrac{-8}{9}$

Let $\quad D' = \text{discriminant of} \left(x^2 - (2+b)x + 2b - \dfrac{2}{a} \right)$

$= (2+b)^2 - 4\left(2b - \dfrac{2}{a}\right)$

$= (2+8)^2 - 8 \times 8 + \dfrac{8}{-8/9} = 100 - 64 - 9 > 0.$ So, above case is rejected.

Case III : When, $\underset{5\ \ \ 5}{x^2} - (2+b)x + \left(2b - \dfrac{2}{a}\right) = 0$ and $\underset{5\ \ \ 5}{x^2} - (2+b)x + \left(2b - \dfrac{b}{a}\right) = 0$

But, it is not possible for any real value of a and b. So, this possibility is also rejected.

$\therefore$ We conclude that $f(x) = \dfrac{-2}{9}(x-2)(x-8)$

223. From above graph of f(x), we can say that the maximum value of f(x) is 2 and attained at $x = 5$.

224. Let the line passing through O(0, 0) be $y = mx$.

$\therefore$ Solving $y = mx$ and $y = \dfrac{-2}{9}(x-5)^2 + 2$; we get

$mx = \dfrac{-2}{9}(x-5)^2 + 2 \Rightarrow 9mx = -2(x^2 - 10x + 25) + 18$

$\Rightarrow 2x^2 + (9m - 20)x + 32 = 0$

As, $y = mx$ is tangent line, so put $D = 0$

$\Rightarrow (9m-20)^2 = 4 \times 2 \times 16 \Rightarrow (9m - 20) = \pm 16 \Rightarrow 9m = 20 \pm 16 \Rightarrow m = \dfrac{36}{9}, \dfrac{4}{9}$

$\therefore \quad m = 4, \dfrac{4}{9}$ **Ans.**

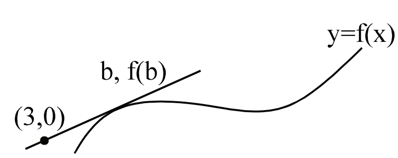
225. $\displaystyle\int_2^5 f(x)\,dx = \int_2^5 \left(\frac{-2}{9}(x-5)^2 + 2\right)dx = \frac{-2}{9}\left[\frac{(x-5)^3}{3}\right]_2^5 + 6$

$= \dfrac{-2}{27}(0 + 27) + 6 = -2 + 6 = 4$ **Ans.**

226.

(A) Given $f(x) = x^3 - \dfrac{11x}{3}$

we have $\dfrac{f(b)}{b-3} = f'(b) = 3b^2 - \dfrac{11}{3}$



$\Rightarrow \quad b^3 - \frac{11b}{3} = (b-3)\left(3b^2 - \frac{11}{3}\right) \qquad \left(f(b) = b^3 - \frac{11b}{3}\right)$

or $\quad b^3 - \frac{11b}{3} = 3b^3 - \frac{11b}{3} - 9b^2 + 11$

or $\quad 2b^3 - 9b^2 + 11 = 0$

$\therefore \quad \sum b = \frac{9}{2}; \qquad \prod b = -\frac{11}{2}$

$\Rightarrow \quad \sum b - \prod b = 10$ **Ans.**

(B) Let $\quad I(a) = \int_{a-1}^{a+1} e^{-(x-1)^2} dx \Rightarrow \qquad I'(a) = e^{-a^2} - e^{-(a-2)^2}$

Now, for maximum or minimum value of I(a), we must have

$I'(a) = 0$

$\Rightarrow \quad e^{-a^2} = e^{-(a-2)^2} \Rightarrow e^{(a-2)^2 - a^2} = 1 = e^0$

$\Rightarrow \quad (a-2)^2 - a^2 = 0 \Rightarrow -4a + 4 = 0 \Rightarrow a = 1$

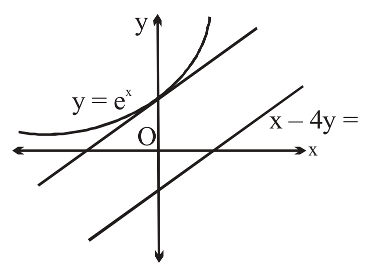
Also, $\quad I''(a) = -2a\ e^{-a^2} + 2\ e^{-(a-2)^2}(a-2)$

$= 2\left[(a-2)\,e^{-(a-2)^2} - a\ e^{-a^2}\right]$

Clearly $I''(a = 1) = 2\left[-e^{-1} - e^{-1}\right] = 2\left(\frac{-2}{e}\right) = \frac{-4}{e} < 0$

So, $\quad$ I (a) will be maximum for $a = 1$ **Ans.**

(C) We have $\frac{dy}{dx} = e^{x_1} = \frac{1}{4}$



$\Rightarrow x_1 = -\ln 4$ and $y_1 = e^{-\ln 4} = \frac{1}{e^{\ln 4}} = \frac{1}{4}$

Equation of 'l' is $\left(y - \frac{1}{4}\right) = \frac{1}{4}(x + \ln 4)$

$\Rightarrow \quad 4y - 1 = x + \ln 4 \quad .....(1)$

Put $y = 0$ in equation (1), we get

$x = -(1 + \ln 4) = -\ln(4e)$

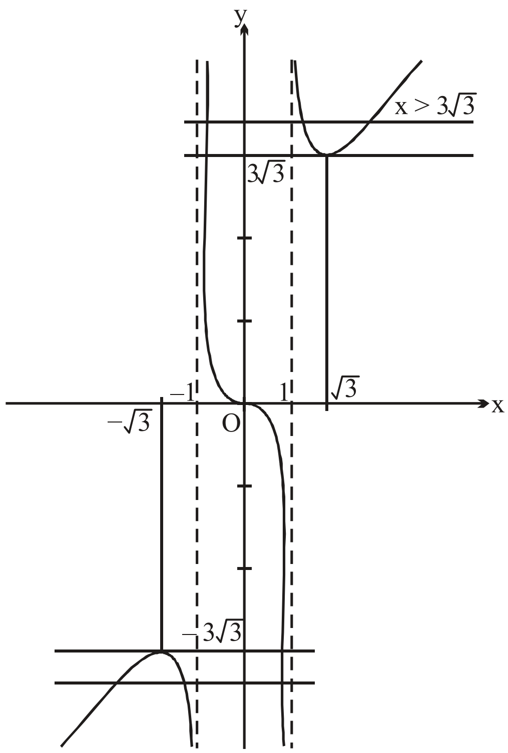
$= \ln\left(\frac{1}{4e}\right) = -\ln(4e)$

$\Rightarrow \qquad k = 4.$

Sol : 227 - 229

227. Let $A = B = \alpha$ and $C = \pi - 2\alpha$

$\therefore \quad \tan A = \tan B = \tan\alpha$ and $\tan C = -\tan 2\alpha$



$\therefore \quad 2\tan\alpha - \tan 2\alpha = k$

$$2\tan\alpha - \frac{2\tan\alpha}{1-\tan^2\alpha} = k$$

as $\quad k = 2\tan\alpha\left[1 - \frac{1}{1-\tan^2\alpha}\right]$

$$k = \frac{2\tan^3\alpha}{\tan^2\alpha - 1}, \text{ if } \tan\alpha = x \in R$$

then $\quad y = \dfrac{2x^3}{x^2-1}$ and $y = k$

$$\frac{dy}{dx} = \frac{2\left[(x^2-1)3x^2 - x^3 \cdot 2x\right]}{(x^2-1)^2} = \frac{2x^2(x^2-3)}{(x^2-1)^2}$$

Hence $\dfrac{dy}{dx} = 0 \Rightarrow x = \sqrt{3}$ or $-\sqrt{3}$ or 0

At $x = 0$, we have inflection point.

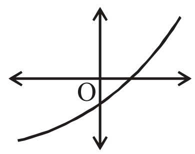
Graph of $y = \dfrac{2x^3}{x^2 - 1}$

For the graph at is clear that for $k > 3\sqrt{3}$
there exist 2 isosceles triangles. ($k \neq$ negative)
for $k = 3\sqrt{3} \cup k < 0$
There exist one isosceles triangle the negative k
there is two isoscles triangle.

230.

(A) $f''(x) > 0 \; \forall \; x \in R \Rightarrow f'(x)$ is a strictly increasing function hence $f'(1) > f'(0)$,
but given $f'(0) = f'(1) = 1$
Which is not possible.

(B) $f''(x) > 0, \; \forall \; x \in R \Rightarrow f'(x)$ is increasing
$f'(1) > f'(0)$ which is true.
$\Rightarrow$ such a function is possible.



(C) $f''(x) \geq 0 \Rightarrow f'(x)$ is increasing and $y = f(x)$ must be concave up.
Since $f(x) \leq 100$ for all $x > 0$ which means that a function always concave up and increasing should also be asymptotic to a line $y = 100$ which is not possible. (think !)
$f'(0) = 1$ is not possible

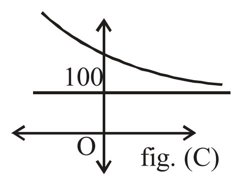
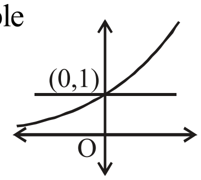


fig. (C)

(D) Exists
consider $f(x) = e^x$



01 The value of $\frac{d^2y}{dx^2}$ if $x^2 - y^4 = 6$, is

(A) $-\frac{3}{16y^7}$ (B) $\frac{y}{3x}$

(C) $\frac{y-3x}{2y^4}$ (D) $\frac{2y^4-3x^2}{4y^7}$

02 $\lim\limits_{x\to 2} \frac{\sqrt{x+2}+\sqrt{x^2+5}-5}{\sqrt{3x-2}+\sqrt{4x^2+5x+23}-9}$ equals to $\frac{p}{q}$ (p and q in lowest form). Find (p + q).

03 For the curve $\sin x + \sin y = 1$ lying on the first quadrant, if $\lim\limits_{x\to 0} x^{\alpha}\cdot\frac{d^2y}{dx^2}$ exists and has the non-zero value equa to L, find the value of $\left(\frac{\alpha}{L}\right)^2$.

04 Let $y = f(x)$ be an infinitely differentiable function on R such that

$$f(0)\neq 0 \ \ and \ \ \frac{d^n y}{dx^n}\neq 0 .$$

at $x = 0$ *for* $n = 1,2,3,4$

If

$$\lim_{x\to 0}\frac{f(4x)+af(3x)+bf(2x)+cf(x)+df(0)}{x^4}$$

exists, then find the value of (25a + 50b + 100c + 500d).

05 If $(f(x))^3 = 3x^2 - x^3$ and

$f''(x) + \frac{nx^2}{(f(x))^5} = 0$, then the value of 'n' is

(A) 1 (B) 2
(C) 3 (D) 4

06 $\lim\limits_{x\to 2} \frac{\sqrt{x+2}+\sqrt{x^2+5}-5}{\sqrt{3x-2}+\sqrt{4x^2+5x+23}-9}$ equals to $\frac{p}{q}$ (p and q in lowest form). Find (p + q).

07 Let f(x)

$$=\left(x^{\left(x^x-1\right)}-(\cot x)^{\sin x}+(\sec x)^{\operatorname{cosec} x}-\frac{\ell n(\sec x)}{x^2}\right).$$

Find the value of $\lim\limits_{x\to 0^+} f(x)$.

08 Let $f(x) = (\log_2 x)^3 + x^3 \ \ \forall\, x > 0$, then the derivative of $f^{-1}(x)$ with respect to x at x = 9 is

(A) $\frac{27}{2}$ (B*) $\frac{2}{27}$

(C) 27 (D) $\frac{1}{27}$

09(mcq) If $f(x) = \int \frac{x}{2} d\left(\frac{x^2-1}{x^2}\right)$ and $f(2) = \frac{1}{2}$ and

$g_r(x) = \underbrace{f(f(f(......f(x)....)))}_{r-\text{times}}$ i.e. $g_1(x) = f(x)$, $g_2(x) = f(f(x))$ and so on then identify the correct statement(s).

(A) $\frac{d}{dx}\left(g_{(3n-2)}(x)\right) = 1$
whenever exists, $n \in N$

(B) $\frac{d}{dx}\left(g_{3n}(x)\right) = 1$
whenever exists, $n \in N$

(C) $\lim\limits_{x\to 1}\frac{\sum\limits_{r=1}^{100}\left(g_{3r}(x)\right)^r + x - 101}{x-1} = 5050$

(D) Slope of the tangent to the graph of the function $y = g_{80}(x)$ at $x = \frac{1}{2}$ is 4.

10. If $y = y(x)$ and it follows the relation

$x^{2x} - 2x^x \cot y - 1 = 0$. Then $\left(\frac{dy}{dx}\right)_{\left(1,\frac{\pi}{2}\right)}$

equals

(A) – 1 (B) 1
(C) $ln\,2$ (D) $-ln\,2$

11 **Statement-1:**

Let $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$ and $g(x) = f^{-1}(x)$. The value of $g''(0)$ is equal to 1.

Statement-2:

If the inverse functions f and g are defined $y = f(x)$ and $x = g(y)$ and if $f'(x)$ exists and $f'(x) \neq 0$ then $g''(y) = \frac{-f''(x)}{(f'(x))^3}$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

12. For a function $f(x) =$

$ln\left(1+\sqrt{1-x^2}\right) - \left(\sqrt{1-x^2} + ln\, x\right)$

where $0 < x < 1$, the value of $f'(x)$ is

(A) $\frac{1+x\sqrt{1-x^2}}{x\sqrt{1-x^2}}$ (B) $\frac{-\sqrt{1-x^2}}{x}$

(C) $\frac{x}{1+\sqrt{1-x^2}}$ (D) None

13. Let $f: R \to R$ be defined as $f(x) = x^3 + 2x^2 + 4x + \sin\left(\frac{\pi x}{2}\right)$ and g be the inverse function of f, then $g'(8)$ equals

(A) $\frac{1}{9}$ (B) 9

(C) $\frac{1}{11}$ (D) 11

14. If $f(x)$, $g(x)$, $h(x)$ are three polynomial functions of degree two and

$$\phi(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix}$$, then the value of $\lim\limits_{x \to 2} \frac{\phi(x) - \phi(4-x)}{\sin(x-2)}$ is equal to

(A) 0 (B) 1

(C) 2 (D) 4

15. Let $f: R \to R$ defined by $f(x) = x^3 + 3x + 1$ and g is the inverse of f then the value of $g''(5)$ is equal to

(A) $\frac{-1}{6}$ (B) $\frac{-1}{36}$

(C) $\frac{-1}{216}$ (D) None of these

16 Two continuous and differentiable functions f(x) and g(x) are related as

$$f(x) = \begin{cases} \frac{g(x)-4}{x-2}, & x \neq 2 \\ k, & x = 2 \end{cases}$$

If equation of tangent to the curve $y = f(x)$ at $x = 2$ be $y = x + 3$ then find the value of $f(2) + f'(2) + g(2) + g'(2) + g''(2)$

17 Find the value of

$$\lim_{x \to 0} \frac{\left(\left(\sin^{-1} x\right)^{2010} - x^{2010}\right)\cos^{2010} x}{ln\left(1+x^{2010}\right)\left(1+\sin^{2010} x\right) \tan^2 x}.$$

18 Let $y = \cos x \cdot ln\left(\frac{1+\cos x}{1-\cos x}\right) - (a\cos x + 2)$ where a is a constant then for non-zero 'y'.

Find the value of $\dfrac{\frac{d^2y}{dx^2} + \cot x \frac{dy}{dx}}{y}$.

Paragraph for question nos 19 to 21

Consider a function $y = f(x)$. Let the functional rule for $y = f(x)$ is same as the functional rule for the height h (dependent variable) of a triangle ABC from the vertex A to the base BC (where angle A is independent variable). The triangle ABC is inscribed in a circle of radius 6 and the area of the triangle ABC is 12.

19 The least value of $f(x)$ is equal to

(A) $\frac{1}{2}$ (B) 2

(C) 4 (D) 6

20. If $g(x) = f(\sin^{-1}x)$ then $g'\left(\frac{4}{5}\right)$ is equal to

(A) $\frac{-25}{4}$ (B) $\frac{-25}{8}$

(C) $\frac{25}{8}$ (D) $\frac{25}{4}$

21 If $h(x) = \sec^{-1}\left(\frac{1}{2}f(x)\right)$ then

$$\lim_{x\to\left(\frac{\pi}{2}\right)^-} \frac{e^{2h(x)} - 2e^{\left(\frac{\pi}{2}-x\right)} + \sin x}{h(x)\cos x}$$ is equal to

(A) $\frac{1}{2}$ (B) 2

(C) $\frac{3}{2}$ (D) 0

22. Let $f(x)$ be a differentiable function in $[-1, \infty)$ and $f(0) = 1$ such that

$$\lim_{t\to x+1} \frac{t^2 f(x+1) - (x+1)^2 f(t)}{f(t) - f(x+1)} = 1.$$

Find the value of $\lim_{x\to 1} \frac{\ln(f(x)) - \ln 2}{x-1}$.

23. Find the absolute value of the limit,

$$\lim_{x\to 0} \frac{(x+1)\sqrt{2x+1} - (x^2+2009)\sqrt[3]{3x+1} + 2008}{x}.$$

24. $\lim_{x\to\infty} \left(e^{11x} - 7x\right)^{\frac{1}{3x}}$ is equal to

(A) $\frac{11}{3}$ (B) $\frac{3}{11}$

(C) $e^{\frac{3}{11}}$ (D) $e^{\frac{11}{3}}$

25. Let $f(x) = \left(2x - x^2\right)^{\frac{1}{\log_{10}\left(2x-x^2\right)}}$. The value of x for which $f'(x)$ vanishes may be

(A) 1 (B) $\frac{1}{2}$

(C) $\frac{3}{2}$ (D) 3

26. Let $f(x) = ax^3 + bx^2 + cx + 5$. If $|f(x)| \le |e^x - e^2|$ for all $x \ge 0$ and if the maximum value of $|12a + 4b + c|$ is l, then $[l]$ is equal to

(Note: $[y]$ denotes greatest integer less than or equal to y)

(A) 4 (B) 5

(C) 6 (D) 7

27. If the function $f(x) = 2 - e^{-x}$ and $g(x) = f^{-1}(x)$, then the value of $g''(1)$ is equal to

(A) – 1 (B) 0

28. $\lim_{x\to\infty} \left(\frac{x^{100}}{e^x} + \left(\cos\frac{2}{x}\right)^{x^2}\right)$ equals

(A) e^{-1} (B) e^{-4}

(C) $(1 + e^{-2})$ (D) e^{-2}

29 If $y = \frac{\cos 6x + 6\cos 4x + 15\cos 2x + 10}{\cos 5x + 5\cos 3x + 10\cos x}$,

then $\frac{dy}{dx} =$

(A) 2 sinx + cosx (B) –2sinx

(C) cos2x (D) sin2x

30. Let $x_1, x_1, x_1, x_2, x_3, x_4, \ldots\ldots\ldots\ldots, x_8$ be 10 real zeroes of the polynomial $P(x) = x^{10} + ax^2 + bx + c$ where $a, b, c \in R$. If the value of $Q(x_1) = \frac{p}{q}$ where p and q are coprime to each other, $Q(x) = (x - x_2)(x - x_3) \ldots\ldots (x - x_8)$ and $x_1 = \frac{1}{2}$, then find the value of $p + q$.

31. Let $f: (-3, 3) \to R$ be a differentiable function with $f(0) = -2$ and $f'(0) = -1$. and $g(x) = (f(3f(x)+6))^3$. Then $g'(0)$ is equal to

(A) 0 (B) 9
(C) 36 (D) – 36

32. If $x^2 - y^2 - 2x = 1$, then

(A) $yy'' + (y')^2 = 1$ (B) $yy'' - (y')^2 = 1$
(C) $yy'' + (y')^2 + 1 = 0$ (D) $yy'' - (y')^2 + 1 = 0$

33. If the value of the limit $\lim_{x\to\infty} x\left(e - \left(\frac{x+2}{x+1}\right)^x\right)$ equals $\frac{ae}{b}$ where $\frac{a}{b}$ is a rational in its lowest form, find the value of $(a^4 + b^5)$.

Paragraph for question nos. 34 to 36

Let $f: R \to R$ be a differentiable function such that $f(1) = 1$, $f(2) = 20$, $f(-4) = -4$, $f'(0) = 0$ and $f(x + y) = f(x) + f(y) + 3xy(x + y) + bxy + c(x + y) + 4 \ \forall\ x, y \in R$, where a, b, c are constants.

34 The value of $(b + c)$ is equal to

(A) 8 (B) 9
(C) 10 (D) 11

35 Number of solutions of the equation $f(x) = x^3 + 4e^x$ is equal to

(A) 0 (B) 1
(C) 2 (D) 3

36. If $f(x) \geq mx^2 + (5m + 1)x + 4m \ \forall\ x \geq 0$, then the maximum value of m is equal to

(A) 1 (B) – 1
(C) 0 (D) 2

37. If $f(x) = \ln(1 + x^2) + \tan^{-1}x$, $x > 0$ and $g(x) = f^{-1}(x)$ then find the value of $\left|27g''\left(\ln 2e^{\frac{\pi}{4}}\right)\right|$ where $g''(x)$ denotes second derivative of $g(x)$.

38. Let f(x) be a polynomial of degree three with $f(2) = 1$, $f'(2) = 1$, $f''(2) = 2$ and $f'''(2) = 6$ then

(A) $f(0) = -5$ (B) $f'(0) = 9$
(C) $f''(0) = -10$ (D*) All A, B & C are correct

39. If $y = x^5(\cos(\ln x) + \sin(\ln x))$, then find the value of $(a + b)$ in the relation $x^2y_2 + axy_1 + by = 0$.

40. If $f(x) = \sin^{-1}\{[3x+2] - \{3x + (x - \{2x\})\}\}$, $x \in \left(0, \frac{\pi}{12}\right)$ and $gof(x) = x \ \forall\ x \in \left(0, \frac{\pi}{12}\right)$ then find the absolute value of $100\, g'\left(\frac{\pi}{3}\right)$.

Note : {y} and [y] denote fractional part function and greatest integer function respectively.

41. Let $f(x) = \tan\left[\sin\left(\cos^{-1}\left(\sin(\cos^{-1} x)\right)\right) + \cos\left(\sin^{-1}\left(\cos(\sin^{-1} x)\right)\right)\right]$, $x \in (0, 1)$ and $g(x) = \tan x$, $x \in \left(0, \frac{\pi}{2}\right)$, then find the value of $\frac{d(f(x))}{d(g(x))}$ at $x = \frac{\pi}{6}$.

42. If h (x) is inverse of g (x) and $g'(x) = \frac{1}{1+x^3}$ then $h''(x)$ is equal to

(A) $\frac{-3(h(x))^2}{(1+(h(x))^3)^3}$ (B) $\frac{-3(h(x))^2}{(1+(h(x))^3)^2}$

(C) $\dfrac{-3(h(x))^2}{\left(1+(h(x))^3\right)}$

(D) $3h^2(x)\left(1+h^3(x)\right)$

43. If $g(x)=f^{-1}(x)$ and $f(x)=x^3+2x+2$, then the positive x-coordinate c of g(x) for which $g'(c)=\dfrac{1}{5}$, is

44. Two continuous and differentiable functions f(x) and g(x) are related as

$$f(x)=\begin{cases}\dfrac{g(x)-4}{x-2}, & x\neq 2\\ k, & x=2\end{cases}$$

If equation of tangent to the curve y = f(x) at x = 2 be y = x + 3 then find the value of $f(2)+f'(2)+g(2)+g'(2)+g''(2)$

45. Let $f(x)=2\tan^{-1}x$ and g(x) be a differentiable function satisfying

$$g\left(\frac{x+2y}{3}\right)=\frac{g(x)+2g(y)}{3}\quad \forall\ x,y\in R$$

and $g'(0)=1$, $g(0)=2$.
Find the number of integers satisfying

$$f^2(g(x))-5f(g(x))+4>0$$

where $x\in(-10, 10)$.

46. Let A, B, P be the points the curve $y = \ln x$ with their x coordinates as 1, 2 and t respectively $\lim\limits_{t\to\infty}\cos\angle BAP$ is

(A) $\sqrt{1+\ln^2 2}$ (B) $\ln 2$

(C*) $\dfrac{1}{\sqrt{1+\ln^2 2}}$ (D) $\dfrac{1}{1+\ln 2}$

47 If $y=1+\dfrac{x_1}{x-x_1}+\dfrac{x_2\cdot x}{(x-x_1)(x-x_2)}+\dfrac{x_3\cdot x^2}{(x-x_1)(x-x_2)(x-x_3)}+\ldots$ upto $(n+1)$ terms then prove that

$$\frac{dy}{dx}=\frac{y}{x}\left[\frac{x_1}{x_1-x}+\frac{x_2}{x_2-x}+\frac{x_3}{x_3-x}+\ldots+\frac{x_n}{x_n-x}\right]$$

48(mcq) If f(x) is a cubic polynomial such that $f(0)+2=0$ and $f'''(0)=6$ and $f(x)=0$ posses three positive integral roots then

(A*) $f'(1)=0$

(B) $2f'(0)+f''(0)=0$

(C*) $2f'(0)+f''(0)=2$

(D) $f'(2)=0$

49. Let $g(x)=f\left(\dfrac{x}{f(x)}\right)$ where f(x) is a differentiable positive function on $(0,\infty)$ such that $f(1)=f'(1)$ Determine $g'(1)$

50. If a differentiable function f(x) satisfies a functional rule $f(x)+f(x+2)+f(x+4)=0$ $\forall\ x\in R$ and $f'(12)=4$ then find the value of

$$\lim_{x\to 0}\frac{f^2(x+12)-f(x)\,f(0)-f(x+6)\,f(18)+f^2(18)}{x\left(\dfrac{\pi}{4}-\tan^{-1}(1-x)\right)}$$

51. Let $f(x)=\begin{cases}x^3, & x<1\\ px^2+qx+r, & x\geq 1\end{cases}$

If $f''(x)$ exists $\forall\ x\in R$ then the value of $|p|+|q|+|r|$, is

(A) 5 (B) 6

(C) 7 (D) 8

52. $\lim\limits_{x\to\infty}\left(\dfrac{x^2+\sin x}{x+\sin x}\right)^{\frac{1}{x}}$ equals

(A) 0 (B) 1

(C) e (D) $\dfrac{1}{e}$

53. Let f be a real valued function satisfying a relation $3f(\cos x)+2f(\sin x)=5x\ \forall\ x\in\left[0,\dfrac{\pi}{2}\right]$ and g be the inverse of f then the absolute value of $\left(\dfrac{2g'\left(\frac{2\pi}{3}\right)}{\sqrt{3}}\right)^{-1}$, is

(A) 5 (B) 7
(C) 9 (D) 11

Paragraph for question nos. 54 to 56

Let $f, g, f_1, f_2 : R \to R$ be twice differentiable function, and $f(x) \geq 0, f'(x) \geq 0, g'(x) > 0$ $\forall x \in R$

Also $\lim\limits_{x\to\infty} f_1(x) = 5$, $\lim\limits_{x\to\infty} f_2(x) = 12$,

$\lim\limits_{x\to\infty} f(x) = \infty$, $\lim\limits_{x\to\infty} g(x) = \infty$

and $\dfrac{f'(x)}{g'(x)} + f_1(x)\dfrac{f(x)}{g(x)} = f_2(x) \ \forall x > 0.$

(Also $f'(x)$ and $g'(x)$ are continuous).

54 $\lim\limits_{x\to\infty} \dfrac{g'(x)}{f'(x)}$ equals

(A) 0 (B) 2

(C) 1 (D) $\dfrac{1}{2}$

55. $\lim\limits_{x\to\infty} \dfrac{f(x)}{g(x)}$ equals

(A) 1 (B) 0

(C) $\dfrac{1}{2}$ (D) 2

56. $\lim\limits_{x\to\infty} \dfrac{\lambda f'(x) + f''(x)}{\lambda g'(x) + g''(x)}$ equals to $(\lambda > 0)$

(A) ∞ (B) 0
(C) 2 (D) 1

57. Let $g(x)$ is the only invertible function from $R \to R$ which satisfy the equation
$g^3(x) - (x^3 + 2)\, g^2(x) + (2x^3 + 1)\, g(x) - x^3 = 0.$
The value of $g'(8) \cdot (g^{-1})'(8)$ is

Answer Key

1.	D	**2.**	38	**3.**	18
4.	300	**5.**	B	**6.**	38
7.	$\frac{1}{2}$	**8.**	B	**9.**	BD
10.	A	**11.**	D	**12.**	B
13.	C	**14.**	A	**15.**	B
16.	7	**17.**	335	**18.**	– 2
19.	B	**20.**	B	**21.**	A
22.	1	**23.**	2007	**24**	D
25.	C	**26.**	D	**27.**	C
28.	D	**29.**	B	**30.**	6
31.	C	**32.**	A	**33.**	113
34.	A	**35.**	B	**36.**	B
37.	4	**38.**	D	**39.**	17
40.	25	**41.**	6	**42.**	D
44.	17	**45.**	8	**46.**	C
48.	AC	**49.**	0	**50.**	32
51.	C	**52.**	B	**53.**	A
54.	D	**55.**	D	**56.**	C
57.	16				

01 $y^4 = x^2 - 6$

$$4y^3\frac{dy}{dx} = 2x \quad \Rightarrow y^3\frac{dy}{dx} = \frac{x}{2}$$

$$\Rightarrow y^3\frac{d^2y}{dx^2} + 3y^2\left(\frac{dy}{dx}\right)^2 = \frac{1}{2}$$

$$y^3\frac{d^2y}{dx^2} + 3y^2\left(\frac{x}{2y^3}\right)^2 = \frac{1}{2}$$

$$\Rightarrow y^3\frac{d^2y}{dx^2} + 3y^2\frac{x^2}{4y^6} = \frac{1}{2}$$

$$\Rightarrow y^3\frac{d^2y}{dx^2} + \frac{3x^2}{4y^4} = \frac{1}{2}$$

$$\Rightarrow y^3\frac{d^2y}{dx^2} = \frac{1}{2} - \frac{3x^2}{4y^4}$$

$$\Rightarrow y^3\frac{d^2y}{dx^2} = \frac{2y^4 - 3x^2}{4y^4}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{2y^4 - 3x^2}{4y^7} \text{ Ans.}$$

02 Using L'Hospital's Rule

$$\lim_{x\to 2}\frac{\dfrac{1}{2\sqrt{x+2}} + \dfrac{x}{\sqrt{x^2+5}}}{\dfrac{3}{2\sqrt{3x-2}} + \dfrac{8x+5}{2\sqrt{4x^2+5x+23}}}$$

$$= \frac{\frac{1}{4}+\frac{2}{3}}{\frac{3}{4}+\frac{21}{14}} = \frac{\frac{1}{4}+\frac{2}{3}}{\frac{3}{4}+\frac{3}{2}} = \frac{\frac{3+8}{12}}{\frac{9}{4}} = \frac{11}{27} = \frac{p}{q}$$

$\Rightarrow$ p + q = 38 **Ans.**

03 Given $\sin x + \sin y = 1$

$$\cos x + \cos y\frac{dy}{dx} = 0 \qquad \text{.......(1)}$$

again

$$-\sin x + \cos y\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^2 \sin y = 0$$

$$\text{or } \cos y\frac{d^2y}{dx^2} = \sin x + \sin y\left(\frac{dy}{dx}\right)^2 = \sin x$$

$$+ \sin y\left(\frac{\cos x}{\cos y}\right)^2$$

$$\cos^3 y\frac{d^2y}{dx^2} = \sin x\cos^2 y + \sin y\cos^2 x$$

$$\frac{d^2y}{dx^2} = \frac{\sin x(1-\sin^2 y) + \sin y(1-\sin^2 x)}{(1-\sin^2 y)\sqrt{1-\sin^2 y}}$$

Using $\sin x + \sin y = 1$

$$\frac{d^2y}{dx^2} =$$

$$\lim_{x\to 0}\frac{\sin x\left(1-(1-\sin x)^2\right) + (1-\sin x)(1-\sin^2 x)}{\left(1-(1-\sin x)^2\right)\sqrt{1-(1-\sin x)^2}}$$

$$= \frac{\sin x(2\sin x - \sin^2 x) + (1-\sin x)(1-\sin^2 x)}{(2\sin x - \sin^2 x)\sqrt{2\sin x - \sin^2 x}}$$

$$= \frac{\sin^2 x(2-\sin x) + 1 - \sin^2 x - \sin x + \sin^3 x}{(\sin x)^{3/2}(2-\sin x)^{3/2}}$$

$$\frac{d^2y}{dx^2} = \frac{\sin^2 x - \sin x + 1}{(\sin x)^{3/2}(2-\cos x)^{3/2}}$$

$$\lim_{x\to 0} x^\alpha\frac{d^2y}{dx^2} = \frac{x^\alpha}{(\sin x)^{3/2}} \cdot$$

$$\frac{\sin^2 x - \sin x + 1}{(2-\sin x)^{3/2}}$$

for non-zero existence of limit $\alpha = \frac{3}{2}$

(if $a > \frac{3}{2}$ then limit will be zero)

and $\quad L = \frac{1}{2\sqrt{2}}$

$$\therefore \frac{\alpha}{L} = \frac{3}{2}\cdot\frac{2\sqrt{2}}{1} = 3\sqrt{2}$$

$\Rightarrow \left(\frac{\alpha}{L}\right)^2 = 18$ **Ans.**

04 The fact that the limit exists implies that

$$\lim_{x\to 0}\left(f(4x)+af(3x)+bf(2x)+cf(x)+df(0)\right)$$

$= (1 + a + b + c + d)\ f(0) = 0$

$\therefore a + b + c + d = -1$ ....(1)

Apply L'Hospital Rule once, then we have

$$\lim_{x\to 0}\frac{f(4x)+af(3x)+bf(2x)+cf(x)+df(0)}{x^4}$$

$$=\lim_{x\to 0}\frac{4f'(4x)+3af'(3x)+2bf'(2x)+cf'(x)}{4x^3}$$

and for the following limit to exist, we also need

$$\lim_{x\to 0}\left(4f'(4x)+3af'(3x)+2bf'(2x)+cf'(x)\right)$$

$= (4 + 3a + 2b + c)\ f'(0) = 0,$

$\therefore\quad 3a + 2b + c = -4$ ....(2)

Repeat this process twice and get another two equations as

$9a + 4b + c = -16$ ....(3)

and $27a + 8b + c = -64$ ....(4)

Now, $(4) - (3) \Rightarrow 18a + 4b + 48 = 0$

$\Rightarrow 9a + 2b + 24 = 0$ ....(5)

$(3) - (2) \Rightarrow 6a + 2b + 12 = 0$

$\Rightarrow 6a + 2b + 12 = 0$ ....(6)

$(5) - 6\ (6) \Rightarrow 3a + 12 = 0$

$\Rightarrow a = -4,\ b = 6$

From equation (2), we get $c = -4$

and from equation (1), $d = 1$.

Hence $(25a + 50b + 100c + 500d)$

$= -100 + 300 - 400 + 500 = 300$ **Ans.**

05 $y^3 = 3kx^2 - x^3$ (Differentiating twice)

$3y^2 y_1 = 6kx - 3x^2$

$y^2 y_1 = 2kx - x^2$ ....(1)

$2y y_1^2 + y^2 y_2 = 2(k - x)$ ....(2)

from (1) $\quad y_1 = \dfrac{2kx - x^2}{y^2}$

substitute in (2)

$$y^2 y_2 + 2y\left[\frac{2xk - x^2}{y^2}\right]^2 = 2(k - x)$$

$$y_2 + \frac{2(2kx - x^2)^2}{y^5} = \frac{2(k-x)}{y^2}$$

$$y_2 + \frac{2(2kx - x^2)^2}{y^5} = \frac{2(k-x)}{y^5}.y^3$$

$$y_2 + \frac{2(2kx - x^2)^2}{y^5} = \frac{2(k-x)}{y^5}.(3kx^2 - x^3)$$

$$y_2 + \frac{2}{y^5}\ [4k^2x^2 + x^4 - 4kx^3 - 3k^2x^2 + kx^3 + 3kx^3 - x^4] = 0$$

$$y_2 + \frac{2k^2x^2}{y^5} = 0$$

$\Rightarrow\ n = 2k^2$

06 Using L'Hospital's Rule

$$\lim_{x\to 2}\frac{\dfrac{1}{2\sqrt{x+2}}+\dfrac{x}{\sqrt{x^2+5}}}{\dfrac{3}{2\sqrt{3x-2}}+\dfrac{8x+5}{2\sqrt{4x^2+5x+23}}}$$

$$=\frac{\frac{1}{4}+\frac{2}{3}}{\frac{3}{4}+\frac{21}{14}}=\frac{\frac{1}{4}+\frac{2}{3}}{\frac{3}{4}+\frac{3}{2}}=\frac{\frac{3+8}{12}}{\frac{9}{4}}=\frac{11}{27}=\frac{p}{q}$$

$\Rightarrow\ p + q = 38$ **Ans.**

07 [Sol.$_{17/mod/QZ}$ $\quad l=\lim_{x\to 0^+} x^{(x^x-1)}$ (0^0 form)

$$\ln l=\lim_{x\to 0}(x^x-1).\ln x$$

$$=\underset{x\to 0}{\text{Limit}}\frac{\left(e^{x\ln x}-1\right)}{x\ln x}\ \underset{x\to 0}{\text{Limit}}\ x\ln x\ .\ln x$$

$$=\underset{x\to 0}{\text{Limit}}\ x(\ln x)^2\ \ (\text{ as } x\to 0\ x\ln x\to 0)$$

$$=\underset{x\to 0}{\text{Limit}}\frac{(\ln x)^2}{1/x}=\underset{x\to 0}{\text{Limit}}\ -\frac{2\ln x}{x}.x^2$$

(use Lopital's rule)

$$=\underset{x\to 0}{\text{Limit}}\ -2\ln x.x = 0$$

$\Rightarrow l = e^0 = 1$

$$m=\underset{x\to 0}{\text{Limit}}\ (\cot x)^{\sin x}\ \ (\infty^0 \text{ form})$$

$$\ln m=\underset{x\to 0}{\text{Limit}}\ (\sin x).\ln(\cot x)$$

$$=\underset{x\to 0}{\text{Limit}}\frac{\ln(\cot x)}{\operatorname{cosec} x}$$

$$=\underset{x\to 0}{\text{Limit}}\ \frac{\operatorname{cosec}^2 x}{\cot x}\ \frac{1}{\operatorname{cosec} x.\cot x}$$

$$=\underset{x\to 0}{\text{Limit}}\ \frac{\operatorname{cosec} x}{\cot^2 x}\ =\underset{x\to 0}{\text{Limit}}\ \frac{\sin x}{\cos^2 x}\ =0$$

Hence $m=e^0=1$

$n=\underset{x\to 0}{\text{Limit}}\ (\sec x)^{(\operatorname{cosec} x)}$ $(1^\infty$ form)

$$=e^{\underset{x\to 0}{\text{Limit}}\ \operatorname{cosec} x(\sec x-1)}\ =\ e^{\underset{x\to 0}{\text{Limit}}\left(\frac{1-\cos x}{\sin 2x}\right)^2}\ =e^0\ =1$$

$$k=\underset{x\to 0}{\text{Limit}}\frac{\ln(\sec x)}{x^2}=\underset{x\to 0}{\text{Limit}}\ \ln(\sec x)^{\frac{1}{x^2}}$$

$$=\underset{x\to 0}{\text{Limit}}\ \frac{1}{x^2}\ (\sec x-1)$$

$$=\ \underset{x\to 0}{\text{Limit}}\ \frac{1-\cos x}{x^2}=\frac{1}{2}$$

Hence $l-m+n-k\ =\frac{1}{2}$ Ans

08 $f'(x)=3\ (\log_2 x)^2\ \frac{1}{x}+3x^2$

$f(x)=9\ \Rightarrow\ x=2.$

$$\frac{d}{dx}\ f^{-1}(x)\Big|_{x=9}=\frac{1}{f'(x)}\Big|_{x=2}=\frac{1}{\frac{3}{2}+12}=\frac{2}{27}.$$

09 $f(x)=\int\frac{x}{2}d\left(1-\frac{1}{x^2}\right)=\int\frac{x}{2}\frac{2}{x^3}dx$

$$=\int\frac{1}{x^2}dx=-\frac{1}{x}+C$$

$$f(2)=-\frac{1}{2}+C\ =\frac{1}{2}$$

$\Rightarrow C=1$

$$\therefore\ f(x)=\frac{x-1}{x}$$

$$g_1(x)=f(x)=\frac{x-1}{x}\ ;$$

$$g_2(x)\ =\ f(f(x))\ =\ \frac{\frac{x-1}{x}-1}{\frac{x-1}{x}}=\frac{-1}{x-1}=\frac{1}{1-x}\ ;$$

$$g_3(x)=\ f(f(f(x)))=\frac{1}{1-\left(\frac{x-1}{x}\right)}\ =x$$

$$g_4(x)=g_1(x)=\frac{x-1}{x}\ ;$$

$$g_5(x)=g_2(x)=\frac{1}{1-x}\ ;$$

$g_6(x)=g_3(x)=x$ and so on

$$\therefore\ g_{(3n-2)}\ (x)=\frac{x-1}{x},\ g_{(3n-1)}\ (x)=\frac{1}{1-x},$$

$g_{3n}(x)=x$

$$\frac{d}{dx}\left(g_{(3n-2)}(x)\right)=\frac{d}{dx}\left(1-\frac{1}{x}\right)=\frac{1}{x^2}$$

$\therefore$ **option (A) is wrong.**

$$\frac{d}{dx}\left(g_{(3n)}(x)\right)=\frac{d}{dx}(x)=1$$

$\therefore$ **option (B) is correct.**

$$\underset{x\to 1}{\text{Lim}}\ \frac{x+x^2+.....+x^{100}+x-101}{x-1}$$

applying L'Hospital's rule

$$\underset{x\to 1}{\text{Lim}}\ \frac{1+2x+.....+100x^{99}+1}{1}$$

$=1+2+......+100+1=5051$

$\therefore$ **option (C) is wrong.**

$$y=g_{80}\ (x)=g_{(3n-1)}(x)=\frac{1}{1-x}$$

$$\therefore\ g'_{(3n-1)}(x)=\frac{1}{(1-x)^2}$$

$$g'_{(3n-1)}\ (x)\Big|_{x=\frac{1}{2}}=4$$

$\therefore$ **option (D) is also correct.**

10. $x^{2x}-2x^x\cot y-1=0$ ....(1)

Now, differentiating equation (1) w.r.t. x, we get

$2x^{2x}\ (1+ln\ x)-2$

$$\left[x^x\left(-\operatorname{cosec}^2 y\frac{dy}{dx}+(\cot y)\cdot x^x\cdot(1+ln\ x)\right)\right.$$

$=0$

Now, at $\left(1,\frac{\pi}{2}\right)$

$$2(1 + \ln 1) - 2\left(1(-1)\frac{dy}{dx}_{\left(1,\frac{\pi}{2}\right)} + 0\right) = 0$$

$$\left(\frac{dy}{dx}\right)_{\left(1,\frac{\pi}{2}\right)} = -1\text{. } \textbf{Ans.}]$$

11 $g(f(x)) = x$

$$g'(y) = \frac{1}{f'(x)}$$

$$g''(f(x)) \cdot f'(x) = \frac{-1}{(f'(x))^2} f''(x)$$

$$\therefore g''(f(x)) = \frac{-f''(x)}{(f'(x))^3}$$

Put $f(x) = 0$

$\Rightarrow x = 0$

$$g''(0) = \frac{-f''(0)}{(f'(0))^3} = -1.$$

$$\boxed{\begin{matrix} f'(0) = 1 \\ f''(0) = 1 \end{matrix}}.$$ **Ans.**

12. $$f'(x) = \frac{\frac{1(-2x)}{2\sqrt{1-x^2}}}{1+\sqrt{1-x^2}} + \frac{2x}{2\sqrt{1-x^2}} - \frac{1}{x}$$

$$= \frac{-x}{\sqrt{1-x^2}\left(1+\sqrt{1-x^2}\right)} + \frac{x}{\sqrt{1-x^2}} - \frac{1}{x}$$

$$= \frac{x}{\sqrt{1-x^2}}\left(1 - \frac{1}{1+\sqrt{1-x^2}}\right) - \frac{1}{x}$$

$$= \frac{x\sqrt{1-x^2}}{\sqrt{1-x^2}\left(1+\sqrt{1-x^2}\right)} - \frac{1}{x}$$

$$= \frac{x}{1+\sqrt{1-x^2}} - \frac{1}{x}$$

$$= \frac{x^2 - 1 - \sqrt{1-x^2}}{x\left(1+\sqrt{1-x^2}\right)} = \frac{\sqrt{1-x^2} + (1-x^2)}{-x\left(1+\sqrt{1-x^2}\right)}$$

$$= \frac{-\sqrt{1-x^2}}{x}$$ **Ans.**

13. We have, $f(x) = x^3 + 2x^2 + 4x + \sin\left(\frac{\pi x}{2}\right)$

$$\therefore f'(x) = 3x^2 + 4x + 4 + \frac{\pi}{2}\cos\left(\frac{\pi x}{2}\right)$$

$\Rightarrow f'(1) = 11$

Also, $f(1) = 8$

So, $g'(8) = \frac{1}{f'(1)} = \frac{1}{11}$. **Ans.**

14. $$\because \phi(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix}$$

$$\Rightarrow \varphi'(x) = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix} +$$

$$\begin{vmatrix} f(x) & g(x) & h(x) \\ f''(x) & g''(x) & h''(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix} +$$

$$\begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix}$$

$$= 0 + 0 + \begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ 0 & 0 & 0 \end{vmatrix}$$

($\because$ f, g, h are quadratic polynomial.

$\therefore f''' = g''' = h''' = 0$) $= 0$

$\therefore$ $\phi(x)$ must be a constant function.

$\therefore$ $\phi(x) = \phi(4 - x)$

$$\therefore \lim_{x \to 2} \frac{\phi(x) - \phi(4-x)}{\sin(x-2)} = 0.$$ **Ans.**

15. $g(5) = 1$ and $f(1) = 5$

Now, $$g''(5) = -\frac{f''(g(x))}{[f'(g(x))]^3}$$

$$\Rightarrow g''(5) = -\frac{f''(g(5))}{[f'(g(5))]^3} = -\frac{f''(1)}{[f'(1)]^3}$$

$f'(x) = 3x^2 + 3,\ f'(1) = 6$

$f''(x) = 6x,\ f''(1) = 6.$

$$g''(5) = \frac{-6}{216} = -\frac{1}{36}.$$ **Ans.**

16 Equation of tangent at $x = 2$ to $y = f(x)$ is $y = x + 3$

$\therefore$ the point $(2, f(2))$

i.e. $(2, k)$ must satisfy $y = x + 3$

$\therefore \quad k = 5$

$$\therefore \quad f(2) = 5 = \lim_{x\to 2} f(x) = \lim_{x\to 2} \frac{g(x)-4}{x-2}$$

For finite value of above limit we must have

$$\lim_{x\to 2}(g(x)-4) = 0$$

$\therefore \lim\limits_{x\to 2} g(x) = 4 = g(2)$ {As $g(x)$ is a continuous function}

$$\text{Also} \quad \lim_{x\to 2}\frac{g(x)-4}{x-2} = 5$$

$$\Rightarrow \lim_{x\to 2} g'(x) = 5$$

$\therefore \quad g(2) = 5$

(As $g(x)$ is differentiable)

$\because g(x)$ is differentiable function.

$$\therefore f'(2) = f'(2^+) = \lim_{h\to 0}\frac{f(2+h)-f(2)}{h}$$

$$= \lim_{h\to 0}\frac{\frac{g(2+h)-4}{h}-5}{h}$$

$$= \lim_{h\to 0}\frac{g(2+h)-4-5h}{h^2}$$

$$= \frac{g''(2)}{2}$$

{Applying L Hospital's rule twice}

Given, slope of tangent to $y = f(x)$ at $x = 2$ is 1.

$$\therefore \quad f'(2) = 1 \qquad \therefore \quad \frac{g''(2)}{2} = 1$$

$\therefore \quad g''(2) = 2$

$\therefore \quad f(2) + f'(2) + g(2) + g'(2) + g''(2)$

$= 5 + 1 + 4 + 5 + 2 = 17.$ **Ans.**

17 Given limit =

$$\lim_{x\to 0}\frac{\left(\left(\sin^{-1}x\right)^{2010} - x^{2010}\right)\cos^{2010}x}{x^{2010}\left(\frac{\ln\left(1+x^{2010}\right)}{x^{2010}}\right)\left(1+\sin^{2010}x\right)\left(\frac{\tan^2 x}{x^2}\right)x^2}$$

$$= \lim_{x\to 0}\frac{\left(\sin^{-1}x\right)^{2010} - x^{2010}}{x^{2012}} \quad (\text{Let } x = \sin\theta)$$

$$\therefore \text{given limit} = \lim_{\theta\to 0}\frac{\theta^{2010} - \sin^{2010}\theta}{\theta^{2012}}$$

$$= \lim_{\theta\to 0}\frac{\theta^{2010} - \left(\theta - \frac{\theta^3}{3!} + \dots\right)^{2010}}{\theta^{2012}}$$

$$= \lim_{\theta\to 0}\frac{1 - \left(1 - \frac{\theta^2}{3!} + \dots\right)^{2010}}{\theta^2}$$

$$= \lim_{\theta\to 0}\frac{1 - \left(1 - \frac{2010}{6}\theta^2 + \dots\right)}{\theta^2} = \frac{2010}{6}$$

$= 335$ **Ans.**

18 $$\frac{y}{\cos x} = \ln(1+\cos x) - \ln(1-\cos x) - a - \frac{2}{\cos x}$$

Differentiating both sides w.r.t. x

$$y\sec x\tan x + \sec x\cdot\frac{dy}{dx} = \frac{-\sin x}{1+\cos x} - \frac{\sin x}{1-\cos x} - 2\sec x\tan x$$

$$\Rightarrow \sec x\tan x\left(y + \frac{1}{\tan x}\frac{dy}{dx}\right) = -\sin x\left(\frac{1-\cos x+1+\cos x}{(1+\cos x)(1-\cos x)}\right) - 2\sec x\tan x$$

$$\Rightarrow y + \cot x\frac{dy}{dx} =$$

$$\left(\frac{-2\sin x}{\sin^2 x}-2\sec x\ \tan x\right)\frac{1}{\sec x\ \tan x}$$

$$\Rightarrow y+\cot x\ \frac{dy}{dx}=-2\cot^2 x-2=-2(1+\cot^2 x)$$

$$\Rightarrow y+\cot x\ \frac{dy}{dx}=-2\text{cosec}^2 x \qquad .....(i)$$

Differentiating both sides again w.r.t. 'x', we get

$$\frac{dy}{dx}+\cot x\cdot\left(\frac{d^2y}{dx^2}\right)+\frac{dy}{dx}(-\text{cosec}^2 x)=4$$

$$\text{cosec}x\cdot\text{cosec}x\cot x$$

$$\cot x\ \frac{d^2y}{dx}-(\text{cosec}^2 x-1)\ \frac{dy}{dx}=4\cot x$$

$$\text{cosec}^2 x$$

$$\cot x\ \frac{d^2y}{dx^2}-\cot^2 x\ \frac{dy}{dx}=4\cot x\ \text{cosec}^2 x$$

$$\Rightarrow\frac{d^2y}{dx^2}-\cot x\ \frac{dy}{dx}=4\ \text{cosec}^2 x \qquad ......(ii)$$

From equation (ii) + 2 equation (i) we have

$$2y+2\cot x\ \frac{dy}{dx}+\frac{d^2y}{dx^2}-\cot x\ \frac{dy}{dx}=0$$

$$\therefore\frac{d^2y}{dx^2}+\cot x\ \frac{dy}{dx}=-2y$$

$$\Rightarrow\frac{\frac{d^2y}{dx^2}+\cot x\frac{dy}{dx}}{y}=-2 \quad \text{Ans.}$$

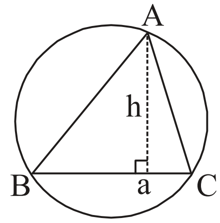
Paragraph for question nos 19 to 21

Sol. Area of ΔABC

We have, $\Delta=\frac{1}{2}\ ah=12$

$$\Rightarrow\ ah=24\ \Rightarrow h=\frac{24}{a}=\frac{24}{2R\sin A}$$

$$\Rightarrow\ h=\frac{24}{2\times 6\times\sin A}\Rightarrow\ h=2\ \text{cosec}\ A$$



So, $y=f(x)=2\ \text{cosec}\ x$

19. Clearly, least value of $f(x)$ is 2.

20. We have, $g(x)=f(\sin^{-1}x)$

$$=2\ \text{cosec}\ (\sin^{-1}x)=\frac{2}{x},$$

$$g'(x)=\frac{-2}{x^2}\ \Rightarrow g'\left(\frac{4}{5}\right)=\frac{-2.25}{16}=\frac{-25}{8}.$$

21. We have, $h(x)=\sec^{-1}\left(\frac{1}{2}2\,\text{cosec}\ x\right)$

$$=\sec^{-1}(\text{cosec}\ x)=\frac{\pi}{2}-x$$

Now, $$\lim_{x\to\left(\frac{\pi}{2}\right)^-}\frac{e^{2h(x)}-2e^{\left(\frac{\pi}{2}-x\right)}+\sin x}{h(x)\cos x}$$

$$=\lim_{x\to\left(\frac{\pi}{2}\right)^-}\frac{e^{2\left(\frac{\pi}{2}-x\right)}-2e^{\left(\frac{\pi}{2}-x\right)}+\sin x}{\left(\frac{\pi}{2}-x\right)\cos x}$$

Put, $x=\frac{\pi}{2}-h$, we get

$$\lim_{h\to 0}\frac{e^{2h}-2e^{h}+1-(1-\cos h)}{h\cdot\sin h}$$

$$=\lim_{h\to 0}\frac{\frac{(e^h-1)^2}{h^2}-\frac{(1-\cos h)}{h^2}}{\frac{\sin h}{h}}=1-\frac{1}{2}=\frac{1}{2}.$$

22. $\displaystyle \lim_{t\to x+1} \frac{t^2 f(x+1)-(x+1)^2 f(t)}{f(t)-f(x+1)} = 1 \left(\frac{0}{0}\right)$

$\displaystyle \Rightarrow \lim_{t\to x+1} \frac{2t\, f(x+1)-(x+1)^2 f'(t)}{f'(t)} = 1$

$\Rightarrow 2(x+1)\, f(x+1) - (x+1)^2\ f'(x+1)$
$= f'(x+1)$
$\Rightarrow 2(x+1)\cdot f(x+1) = [1+(x+1)^2]\, f'(x+1)$

$\displaystyle \Rightarrow \frac{f'(x+1)}{f(x+1)} = \frac{2(x+1)}{1+(x+1)^2}$

Integrating, we get

$ln\left(f(x+1)\right) = \ln\left(1+(x+1)^2\right) + ln\, c$

Put $\quad x = -1$

$\quad 0 = 0 + ln\, c \Rightarrow c = 1$

$\Rightarrow \quad f(x+1) = 1 + (x+1)^2$

So, $\quad f(x) = 1 + x^2$

Hence, $\displaystyle \lim_{x\to 1} \frac{ln\left(f(x)\right) - ln\, 2}{x-1}$

$\displaystyle = \lim_{x\to 1} \frac{ln\,(1+x^2) - ln\, 2}{(x-1)} = 1.$ **Ans.**

23. $\displaystyle \lim_{x\to 0} \frac{(x+1)\sqrt{2x+1}-(x^2+2009)\sqrt[3]{3x+1}+2008}{x}$

Using L'Hospital's rule, we have

$\displaystyle = \lim_{x\to 0} \frac{x+1}{\sqrt{2x+1}} + \sqrt{2x+1} -$

$\displaystyle \frac{-x^2+(2009)^{2/3}}{3x+1} - 2x(3x+1)^{1/3}$

$= 1 + 1 - 2009 = -2007$

Hence absolute value $= 2007$ **Ans.**

24. Let $\displaystyle L = \lim_{x\to\infty} \left(e^{11x} - 7x\right)^{1/3x}$ $\quad (\infty^\circ$ form)

$\displaystyle As \lim_{x\to\infty}\left(e^{11x}-7x\right) = \lim_{x\to\infty}\left(1-\frac{7x}{e^{11x}}\right)e^{11x}$

$\displaystyle = (1-0)\times\infty \to \infty \ and \ \lim_{x\to\infty}\frac{1}{3x} \to 0$

$\displaystyle ln\, L = \lim_{x\to\infty} \frac{ln\left(e^{11x}-7x\right)}{3x}$

(Using L-Hospital rule)

$\displaystyle ln\, L = \lim_{x\to\infty} \frac{\left(11e^{11x}-7\right)}{3\left(e^{11x}-7x\right)}$

$\displaystyle = \lim_{x\to\infty} \frac{\left(11-7e^{-11x}\right)}{3\left(1-7xe^{-11x}\right)}$

Hence $L = e^{\frac{11}{3}}$

25. $f(x) = \left(2x-x^2\right)^{\log_{\left(2x-x^2\right)}10} = 10$

$\therefore\ f(x) = 10$

$\Rightarrow\ f'(x) = 0 \ \ \forall\ x \in D_g$

$2x - x^2 > 0 \quad$ and $\quad 2x - x^2 \neq 1$

$x^2 - 2x < 0$

$(x-1)^2 \neq 0 \Rightarrow\ x \neq 1 \qquad$ ....(1)

$x\,(x-2) < 0$

$x \in (0, 2) \qquad$ ....(2)

(1) and (2) $\Rightarrow\ x \in (0, 2) - \{1\}$. **Ans.**

26. $\left|\, f(x)\,\right| \le \left|\, e^x - e^2\,\right| \ \ \forall\ x \ge 0$

$\left|\, f(2)\,\right| \le 0 \qquad \Rightarrow \qquad f(2) = 0$

$\because \left|\, f(x)\,\right| \le \left|\, e^x - e^2\,\right|$

$\displaystyle \Rightarrow \left|\frac{f(x)-f(2)}{x-2}\right| \le \lim_{x\to 2}\left|\frac{e^2(e^{x-2}-1)}{x-2}\right|$

$\Rightarrow \left|\, f'(2)\,\right| \le e^2 \Rightarrow \left|\, 12a+4b+c\,\right| \le e^2$

$\therefore l = e^2 \Rightarrow [l] = 7.$ **Ans.**

27. $y = 2 - e^{-x}$

$e^{-x} = 2 - y$

$-x = ln(2-y)$

$g\,(x) = -ln(2-x)$

$\displaystyle g'(x) = \frac{1}{2-x}\ ;\ g''(x) = \frac{1}{(2-x)^2}$

$\displaystyle g''(1) = \frac{1}{(1)^2} = 1$ **Ans.**

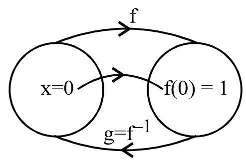
Aliter: $gof(x) = x$

$$g'(f(x))f'(x) = 1$$

$$\Rightarrow g'(f(x)) = \frac{1}{f'(x)}$$

$$g''(f(x))f'(x) = \frac{-1}{(f'(x))^2} f''(x)$$

Hence $g''(f(x)) = \dfrac{-1}{(f'(x))^3} f''(x)$



Put $f(x) = 1 \quad \Rightarrow \quad x = 0$

Hence $g''(1) = \dfrac{-f''(0)}{(f'(0))^3} = \dfrac{-(-1)}{(1)^3} = 1$ **Ans.**

Aliter: We know that $\dfrac{d^2x}{dy^2} = \dfrac{-\dfrac{d^2y}{dx^2}}{\left(\dfrac{dy}{dx}\right)^3}$

Hence $g''(1) = \dfrac{-f''(0)}{(f'(0))^3} = \dfrac{-(-1)}{(1)^3} = 1$

28. $\lim\limits_{x\to\infty} \dfrac{x^{100}}{e^x} = 0$ (using L'Hospital's Rule)

$\lim\limits_{x\to\infty} \left(\cos\dfrac{2}{x}\right)^{x^2}$ (1^∞)

$= e^{\lim\limits_{x\to\infty} x^2\left(\cos\frac{2}{x}-1\right)} = e^{-\lim\limits_{x\to 0}\left(\frac{1-\cos t}{t^2}\right)\cdot 4}$;

put $\dfrac{2}{x} = t \quad \Rightarrow \quad x = \dfrac{2}{t} = e^{-2}$ **Ans.**

29 $N^r = \cos 6x + (1+5)\cos 4x + (5+10)\cos 2x + 10$
$= \cos 6x + \cos 4x + 5(\cos 4x + \cos 2x) + 10(1 + \cos 2x)$
$= 2\cos 5x\ \cos x + 10\cos 3x\ \cos x + 20\cos^2 x$
$= 2\cos x\,[\cos 5x + 5\cos 3x + 10\cos x]$
------Denominator------

$\therefore\ y = \dfrac{N^r}{D^r} = 2\cos x \quad \therefore \dfrac{dy}{dx} = -2\sin x$

30. $P(x) = (x-x_1)^3 (x-x_2)(x-x_3) \ldots\ldots. (x-x_8)$
$\Rightarrow P(x) = (x-x_1)^3, Q(x)$

$\therefore Q(x) = \dfrac{P(x)}{(x-x_1)^3}$

As Q(x) is a polynomial
$\therefore$ Q(x) must be continuous at $x = x_1$.

$\therefore \quad Q(x_1) = \lim\limits_{x\to x_1} Q(x)$

$= \lim\limits_{x\to x_1} \dfrac{x^{10} + ax^2 + bx + c}{(x-x_1)^3}$

$\therefore$ by L Hospital's rule , we have

$$Q(x_1) = \frac{10.9.8}{3.2.1} x_1^7$$

$\therefore Q\left(\dfrac{1}{2}\right) = 120.\left(\dfrac{1}{2}\right)^7 = \dfrac{15}{16} = \dfrac{p}{q}$

$\therefore p + q = 15 + 16 = 31$**Ans.**

31. $\because\ g(x) = (f(3f(x)+6))^3$

$g'(x) = 3(f(3f(x)+6))^2\ f'(3f(x)+6)\cdot 3f'(x)$

$g'(0) = 3(f(3f(0)+6))^2 \cdot f'(3f(0)+6)\cdot 3f'(0)$

$= 9(f(-6+6))^2\ f'(-6+6)\ f'(0)$

$= 9(f(0))^2 (f'(0))^2 = 9 \times 4 \times 1 = 36$

32. $\because\ x^2 - y^2 - 2x = 1$
differentiating w.r.t. x, we get
$2x - 2y\,y' - 2 = 0$
again differentiating w.r.t.x,
$2 - 2(yy'' + (y')^2) = 0$
$\Rightarrow yy'' + (y')^2 = 1.$

33. $l = -\lim\limits_{x\to\infty} x\left(e^{x\,ln\left(\frac{x+2}{x+1}\right)} - e\right)$

$= -\,e\lim\limits_{x\to\infty} x\left(e^{x\,ln\left(\frac{x+2}{x+1}\right)-1} - 1\right)$;

Let $x\,ln\left(\dfrac{x+2}{x+1}\right) - 1 = M$,

as $x \to \infty,\ M \to 0$

$$l = - e \lim_{x\to\infty} x\left(\frac{e^M - 1}{M}\right)\cdot M$$

$$- e \lim_{x\to\infty} x\left(x\,ln\left(\frac{x+2}{x+1}\right) - 1\right)$$

put $x = 1/t$

$$- e \lim_{t\to 0} \frac{ln\left(\frac{1+2t}{1+t}\right) - t}{t^2} = - e$$

$$\lim_{t\to 0}\frac{ln(1+2t) - ln(1+t) - t}{t^2} \quad ; \quad \left(\frac{0}{0}\right) \text{ form}$$

Using L'Hospitals Rule

$$- e \lim_{t\to 0} \frac{\frac{2}{1+2t} - \frac{1}{1+t} - 1}{2t}$$

$$= - e \lim_{t\to 0} \frac{\frac{-4}{(1+2t)^2} + \frac{1}{(1+t)^2}}{2} = \frac{3e}{2} = \frac{ae}{b}$$

$\Rightarrow$ a = 3 and b = 2

$\therefore \quad a^4 + b^5 = 81 + 32 = 113$ **Ans.**

Paragraph for question nos. 34 to 36

Sol. Given $f(1) = 1, f(2) = 20, f(-4) = -4$

and $f(x + y) = f(x) + f(y) + axy(x + y) + bxy + c(x + y) + 4 \;\; \forall\, x, y \in R$,

where a, b, c are constants.

Put $x = y = 0$

$\Rightarrow f(0) = 2f(0) + 4 \Rightarrow f(0) = -4$

Now, put x = 1 and y = 0, we get c = 0

put x = 1 and y = 1, we get

$f(2) = 2f(1) + 6 + b + 4$

$\Rightarrow 20 = 2(1) + 6 + b + 4$

$\Rightarrow 14 = 6 + b$ .....(1)

$\Rightarrow b = 8$

34. Hence b = 8 and c = 0 $\Rightarrow$ b + c = 8

35. $f(x + y) = f(x) + f(y) + 3xy(x + y) + 8xy + 4$

Now $f'(x) = \lim_{h\to 0}\frac{f(x+h) - f(x)}{h}$

$$= \lim_{h\to 0}\frac{f(h) + 3xh(x+h) + 8xh + 4}{h}$$

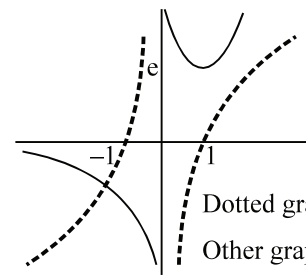
$$= \lim_{h\to 0}\frac{f(h) + 3xh(x+h) + 8xh - f(0)}{h}$$

$= f'(0) + 3x^2 + 8x = 3x^2 + 8x$

Integrating both sides w.r.t. x, we get

$f(x) = x^3 + 4x^2 + C$

$f(0) = -4 \Rightarrow C = -4$



Dotted graph is of $y = x - \frac{1}{x}$

Other graph is of $y = \frac{e^x}{x}$

Hence $f(x) = x^3 + 4x^2 - 4$

$\therefore f(x) = x^3 + 4e^x$

$x^3 + 4x^2 - 4 = x^3 + 4e^x$

$x^2 - 1 = e^x$

Dividing by x, we get $x - \frac{1}{x} = \frac{e^x}{x}$

Hence the above equation will have two solutions.

36. $f(x) \geq mx^2 + (5m + 1)x + 4m \;\; \forall\, x \geq 0.$

$(x + 4)(x + 1)(x - 1) \geq m(x + 1)(x + 4)$

$\Rightarrow (x - 1) \geq m$

$\Rightarrow m + 1 \leq x \;\; \forall\, x \geq 0$

$\Rightarrow m + 1 \leq 0 \Rightarrow m \leq -1$

Hence maximum value of m = – 1 **Ans.**

37. $f(x) = ln(1 + x^2) + \tan^{-1}x$

$g(f(x)) = x$

$g'(f(x))\,f'(x) = 1$

$$\Rightarrow g'(f(x)) = \frac{1}{f'(x)}$$

$$\Rightarrow g''(f(x))f'(x) = -\frac{f''(x)}{(f'(x))^2}$$

$$\Rightarrow g''(f(x)) = -\frac{f''(x)}{(f'(x))^3}$$

$$f(x) = ln\left(2e^{\frac{\pi}{4}}\right) = ln(1 + x^2) + \tan^{-1}x$$

$\Rightarrow\ ln\,2 + \dfrac{\pi}{4} = \ln(1 + x^2) + \tan^{-1}x$

$\Rightarrow\ x = 1$

$$g''\left(ln\left(2e^{\frac{\pi}{4}}\right)\right) = \frac{-f''(1)}{(f'(1))^3}$$

$$f'(x) = \frac{1}{1+x^2} + \frac{2x}{1+x^2} = \frac{2x+1}{1+x^2}$$

$$\Rightarrow\ f'(1) = \frac{3}{2}$$

$$f''(x) = \frac{(1+x^2)2 - (2x+1).2x}{(1+x^2)^2} =$$

$$\frac{2(1-x-x^2)}{(1+x^2)^2} \Rightarrow f''(1) = \frac{-1}{2}$$

$$\therefore\ g''\left(ln\left(2e^{\frac{\pi}{4}}\right)\right) = \frac{\frac{-1}{2}}{\frac{27}{8}} = \frac{-4}{27}$$

$$\left|27.g''\left(ln\left(2e^{\frac{\pi}{4}}\right)\right)\right| = 4$$

Alternative: $\dfrac{d^2x}{dy^2} = -\dfrac{\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx}\right)^3}$

Hence $g''(y) = -\dfrac{f''(x)}{(f(x))^3}$

38. $f(x) = ax^3 + bx^2 + cx + d$

$\Rightarrow\ 8a + 4b + 2c + d = 1$

$\Rightarrow\ d = -5$

$f'(x) = 3ax^2 + 2bx + c$

$\Rightarrow\ f'(2) = 12a + 4b + c = 1\ \Rightarrow\ c = 9$

$f''(x) = 6ax + 2b$

$\Rightarrow\ f''(2) = 12a + 2b \Rightarrow\ 12a + 2b = 2$

$\Rightarrow\ b = -5$

$f'''(x) = 6a \Rightarrow\ 6a = 6 \Rightarrow\ a = 1$

Hence $f(x) = x^3 - 5x^2 + 9x - 5$

$\Rightarrow\ f(0) = -5\ ;\ f'(0) = 9\ ;\ f''(0) = -10$

39. We have $\dfrac{dy}{dx} = 5x^4\left(\cos(ln\,x) + \sin(ln\,x)\right) +$

$x^5\left(\dfrac{-\sin(ln\,x)}{x} + \dfrac{\cos(ln\,x)}{x}\right),$

$\Rightarrow xy_1 = 5y + x^5\left(\cos(ln\,x) - \sin(ln\,x)\right)$

$\Rightarrow xy_2 + y_1 = 5y_1 +$

$\left(\cos(\ln x) - \sin(\ln x)\right) +$

$x^5\left(\dfrac{-\sin(\ln x)}{x} - \dfrac{\cos(\ln x)}{x}\right)$

$\Rightarrow x^2y_2 + xy_1 = 5xy_1$

$+5x^5\left(\cos(ln\,x) - \sin(ln\,x)\right) -$

$x^5\left(\sin(ln\,x) + \cos(ln\,x)\right)$

$\Rightarrow\quad x^2y_2 - 4xy_1 = 5(xy_1 - 5y) - y$

$\Rightarrow\quad x^2y_2 - 4xy_1 = 5xy_1 - 26y$

$\Rightarrow\quad x^2y_2 - 9xy_1 + 26y = 0$

$\equiv x^2y_2 + axy_1 + by = 0$

$\therefore\quad a = -9$ and $b = 26$

Hence $(a + b) = 17$ **Ans.**

40. $f(x) = \sin^{-1}\{[3x + 2] - \{3x + (x - \{2x\})\}\}$

$\because\ x \in \left(0, \dfrac{\pi}{12}\right) \Rightarrow\quad 2x \in \left(0, \dfrac{\pi}{6}\right)$

$\therefore\ \{2x\} = 2x$ and $3x \in \left(0, \dfrac{\pi}{4}\right)$

$f(x) = \sin^{-1}\{2 - \{3x - x\}\}$

$= \sin^{-1}\{2 - 2x\}\ \left(\text{As } \{2 - 2x\} = 1 - 2x\right)$

$f(x) = \sin^{-1}(1 - 2x) = y$

$\Rightarrow\ 1 - 2x = \sin y$

$\Rightarrow\ x = \dfrac{1 - \sin y}{2} = f^{-1}(y)$

$\therefore\ f^{-1}(x) = \dfrac{1 - \sin x}{2} = g(x)$

$g'(x) = \dfrac{-1}{2}\cos x$

$g'\left(\dfrac{\pi}{3}\right) = \dfrac{-1}{2}\left(\dfrac{1}{2}\right) = \dfrac{-1}{4}$

$\Rightarrow\ \left|100 \times \dfrac{-1}{4}\right| = 25$ **Ans.**

41. $\cos^{-1}x = \dfrac{\pi}{2} - \sin^{-1} x,$

$\sin(\cos^{-1}x) = \cos(\sin^{-1}x)$

$\sin\left(\cos^{-1}\left(\sin\left(\cos^{-1} x\right)\right)\right)$

$= \cos\left(\sin^{-1}\left(\cos\left(\sin^{-1} x\right)\right)\right) = x$

$f(x) = \tan 2x$

$$\frac{d(\tan 2x)}{d(\tan x)} = \frac{2\sec^2 2x}{\sec^2 x} = 6.\ \textbf{Ans.}$$

42. $\because g(h(x)) = x$

$\Rightarrow \quad g'(h(x))\, h'(x) = 1$

$\Rightarrow \quad \dfrac{1}{1+(h(x))^3} \cdot h'(x) = 1$

$\Rightarrow \quad h'(x) = 1 + (h(x))^3$

$\Rightarrow \quad h''(x) = 3(h(x))^2 h'(x)$

$\Rightarrow \quad h''(x) = 3h^2(x)\left(1 + h^3(x)\right)$ **Ans.**

43. $g(f(x)) = x$

$\therefore\ g'(f(x))\, f'(x) = 1$

$\Rightarrow g'(c) = \dfrac{1}{f'(c)}$

$\Rightarrow \dfrac{1}{5} = \dfrac{1}{3c^2+2} \quad \Rightarrow c = 1$

$\therefore\ y = 5.$ **Ans.**

44. Equation of tangent at $x = 2$ to $y = f(x)$ is $y = x + 3$

$\therefore$ the point $(2, f(2))$ i.e.

$(2, k)$ must satisfy $y = x + 3$

$\therefore \quad k = 5$

$\therefore \quad f(2) = 5 = \lim\limits_{x\to 2} f(x) = \lim\limits_{x\to 2} \dfrac{g(x)-4}{x-2}$

For finite value of above limit we must have

$\lim\limits_{x\to 2} (g(x) - 4) = 0$

$\therefore \lim\limits_{x\to 2} g(x) = 4 = g(2)$

{As $g(x)$ is a continuous function}

Also $\quad \lim\limits_{x\to 2} \dfrac{g(x)-4}{x-2} = 5$

$\Rightarrow \lim\limits_{x\to 2} g'(x) = 5$

$\therefore g(2) = 5$

(As $g(x)$ is differentiable)

$\because \quad g(x)$ is differentiable function.

$\therefore f'(2) = f'(2^+) = \lim\limits_{h\to 0} \dfrac{f(2+h) - f(2)}{h}$

$$= \lim_{h\to 0} \frac{\dfrac{g(2+h)-4}{h} - 5}{h}$$

$$= \lim_{h\to 0} \frac{g(2+h) - 4 - 5h}{h^2}$$

$$= \frac{g''(2)}{2}$$

{Applying L Hospital's rule twice}

Given, slope of tangent to $y = f(x)$ at $x = 2$ is 1.

$\therefore\ f'(2) = 1$

$\therefore \quad \dfrac{g''(2)}{2} = 1$

$\therefore \quad g''(2) = 2$

$\therefore \quad f(2) + f'(2) + g(2) + g'(2) + g''(2)$

$= 5 + 1 + 4 + 5 + 2 = 17.$

45. $f(x) = 2\tan^{-1}x$ & $g(x) = x + 2$

$\Rightarrow f(g(x)) = 2\tan^{-1}(x + 2)$ solution of inequality

$f^2(g(x)) - 5 f(g(x)) + 4 > 0$ is

$f(g(x)) < 1$ or $f(g(x)) > 4$

$\Rightarrow \tan^{-1}(x+2) < \dfrac{1}{2}$ or $\tan^{-1}(x+2) > 2$

$\Rightarrow \tan^{-1}(x+2) < \dfrac{1}{2} \quad \left[\text{As } \tan^{-1}(x+2) < \dfrac{\pi}{2}\right]$

or $x + 2 < \tan\left(\dfrac{1}{2}\right)$

$\Rightarrow x \in \left(-10, \tan\left(\dfrac{1}{2}\right) - 2\right)$

As $\quad \dfrac{1}{2} < \dfrac{\pi}{6} \quad \Rightarrow \tan\dfrac{1}{2} < \dfrac{1}{\sqrt{3}}$

$\Rightarrow \tan\dfrac{1}{2} - 2 < \dfrac{1}{\sqrt{3}} - 2$

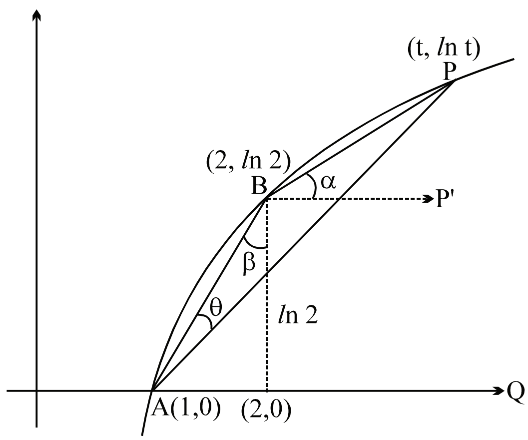
Hence total integer in the range are $\{-9, -8, -7, -6, -5, -4, -3, -2\} \Rightarrow 8$ integer

46. $\tan\alpha_{t\to\infty} = \lim_{t\to\infty} \dfrac{\ln t - \ln 2}{t-2}$

using Lopital rule

$= \lim_{t\to\infty} \dfrac{1/t}{1} \to 0$

hence when the situation is limiting



$BP \to BP'$ (along horizontal)
and $AP \to AQ$

$\Rightarrow \theta \to \dfrac{\pi}{2} - \beta = \cos\theta$

$\cos\theta \to \cos\left(\dfrac{\pi}{2} - \beta\right) = \sin\beta$

but $\quad \tan\beta = \dfrac{1}{\ln 2}$

$$\lim_{t\to\infty} \cos\theta = \lim_{t\to\infty} \cos\left(\frac{\pi}{2} - \beta\right) = \sin\beta = \frac{1}{\sqrt{1+\ln^2 2}}$$

47 adding term by term

$$y = \frac{x^n}{(x-x_1)(x-x_2)\,(x-x_3)\,....(x-x_n)}$$

$$y = \frac{x}{(x-x_1)} \cdot \frac{x}{(x-x_2)} \cdot \frac{x}{(x-x_3)} \;.......\; \frac{x}{(x-x_n)}$$

$$\ln y = \ln\frac{x}{(x-x_1)} + \ln\frac{x}{(x-x_2)} + \ln\frac{x}{(x-x_3)} + ...+ \ln\frac{x}{(x-x_n)}$$

now $D\left(\dfrac{x}{x-x_n}\right) = \dfrac{x-x_n}{x}\left(\dfrac{(x-x_n)-x}{(x-x_n)^2}\right)$

$= \dfrac{1}{x}\left(\dfrac{x_n}{x_n - x}\right)$

Hence $\dfrac{1}{y}\dfrac{dy}{dx} =$

$$\frac{1}{x}\left[\frac{x_1}{x_1-x} + \frac{x_2}{x_2-x} + ...+ \frac{x_n}{x_n-x}\right]$$

$$\frac{dy}{dx} = \frac{y}{x}\left[\frac{x_1}{x_1-x} + \frac{x_2}{x_2-x} + ...+ \frac{x_n}{x_n-x}\right].$$

48 Consider $f(x) = x^3 + ax^2 + bx - 2$
$\alpha\beta\gamma = 2$
$\alpha, \beta, \gamma \in N$
$\Rightarrow$ two of them are 1 and other is 2

$\Rightarrow a = -(\alpha+\beta+\gamma) = -4 = \dfrac{f''(0)}{2}$

$b = (\alpha\beta + \beta\gamma + \alpha\gamma) = 5 = f'(0)$
Hence, $f(x) = x^3 - 4x^2 + 5x - 2$.
Again, $\because$ 1 is a repeated root of $f(x) = 0$
$\Rightarrow f'(1) = 0$.

49. We have $g(x) = f\left(\dfrac{x}{f(x)}\right)$

on differentiating w.r.t. x, we get

$g'(x) = f'\left(\dfrac{x}{f(x)}\right) \times \left(\dfrac{f(x) - xf'(x)}{f^2(x)}\right)$

$\therefore f'(1) = f'\left(\dfrac{1}{f(1)}\right) \times \left(\dfrac{f(1) - f'(1)}{f^2(1)}\right)$

As $f(1) = f'(1)$
$\Rightarrow \quad g'(1) = 0$

50. $f(x) + f(x+2) + f(x+4) = 0$ ....(1)
Replacing x by $x+2$
$f(x+2) + f(x+4) + f(x+6) = 0$ ....(2)
(1) – (2)
$f(x) - f(x+6) = 0$
$f(x) = f(x+6)\ \forall\ x \in R$
$\therefore$ f(x) is periodic with period 6.
Differentiating above equation
$f'(x) = f'(x+6)\ \forall\ x \in R$
f'(x) is also periodic with same period 6.

$\therefore\ f(x) = f(x+6) = f(x+12) = f(x+18)$

$f(0) = f(6) = f(12) = f(18) = ...$

and $f'(x) = f'(x+6) = f'(x+12)$

$= f'(x+18) = ...$

$f'(0) = f'(6) = f'(12) = f'(18) = ...$

$$\lim_{x\to 0}\frac{f^2(x+12) - f(x)\,f(0) - f(x+6)\,f(18) + f^2(18)}{x\left(\frac{\pi}{4} - \tan^{-1}(1-x)\right)}$$

Now, $= \lim_{x\to 0}\dfrac{f^2(x) - 2f(x)\,f(0) + f^2(0)}{x\left(\tan^{-1}1 - \tan^{-1}(1-x)\right)}$

$$= \lim_{x\to 0}\frac{\left(f(x) - f(0)\right)^2}{x\cdot\tan^{-1}\left(\frac{1-(1-x)}{1+1-x}\right)}$$

$$= \lim_{x\to 0}\left(\frac{f(x)-f(0)}{x}\right)^2\cdot\frac{x}{\tan^{-1}\left(\frac{x}{2-x}\right)}\cdot\frac{2-x}{2-x}$$

$= \left(f'(0)\right)^2\cdot 2 = 32$. **Ans.**

51. $f(x) = \begin{cases} x^3, & x < 1 \\ px^2 + qx + r, & x \ge 1 \end{cases}$

$\Rightarrow\ f'(x) = \begin{cases} 3x^2, & x < 1 \\ 2px + q, & x > 1 \end{cases}$

$\Rightarrow\ f''(x) = \begin{cases} 6x, & x < 1 \\ 2p, & x > 1 \end{cases}$

If $f''(x)$ exist $\in x \in R$ then $(p+q+r) = 1$

$\Rightarrow\ 2p + q = 3\ \Rightarrow\ 2p = 6$

$\Rightarrow\ p = 3,\ q = -3,\ r = 1$

$\Rightarrow\ |p| + |q| + |r| = 7$. **Ans.**

52. $\lim_{x\to\infty}\left(\dfrac{x^2 + \sin x}{x + \sin x}\right)^{\frac{1}{x}}$ $\left(\infty^0 \text{ form}\right)$

$$= \lim_{x\to\infty}\left(\frac{x^2\left(1 + \frac{\sin x}{x^2}\right)}{x\left(1 + \frac{\sin x}{x}\right)}\right)^{\frac{1}{x}}$$

$= \lim_{x\to\infty} x^{\frac{1}{x}}$ $\left(\infty^0 \text{ form}\right) = l$ (say)

Now, $\ln l = \lim_{x\to\infty}\dfrac{\ln x}{x}$ $\left(\dfrac{\infty}{\infty}\text{ form}\right)$

$= \lim_{x\to\infty}\dfrac{1}{x} = 0$

Hence $l = 1$.

Alternatively : Let $\lim_{x\to\infty}\left(\dfrac{x^2 + \sin x}{x + \sin x}\right)^{\frac{1}{x}} = L$

$\because\ \lim_{x\to\infty}\left(\dfrac{x^2 + \sin x}{x + \sin x}\right)$

$$= \lim_{x\to\infty}\left(\frac{x + \sin x}{1 + \frac{\sin x}{x}}\right) \to \infty$$

$\therefore$ Given limit possesses indeterminate form of the type ∞^0

$\therefore\ \ln L = \lim_{x\to\infty}\dfrac{1}{x}\ln\left(\dfrac{x^2 + \sin x}{x + \sin x}\right)$

$$= \lim_{x\to\infty}\frac{\ln\left(x^2 + \sin x\right) - \ln\left(x + \sin x\right)}{x}$$

Applying L Hospital's rule once

$$\ln L = \lim_{x\to\infty}\frac{\frac{2x + \cos x}{x^2 + \sin x} - \frac{1 + \cos x}{x + \sin x}}{1}$$

$$= \lim_{x\to\infty}\left\{\frac{1}{x}\left(\frac{2 + \frac{\cos x}{x}}{1 + \frac{\sin x}{x^2}}\right) - \frac{1}{x}\left(\frac{1 + \cos x}{1 + \frac{\sin x}{x}}\right)\right\} =$$

$0 - 0 = 0$

$\therefore\ L = e^0 = 1$

53. $3f(\cos x) + 2f(\sin x) = 5x$ ... (1)

$x \to \dfrac{\pi}{2} - x$

$3f(\sin x) + 2f(\cos x) = 5\left(\dfrac{\pi}{2} - x\right)$ ...(2)

From (1) & (2)

$$f(\sin x) = \frac{3\pi}{2} - 5x$$

$$f(x) = \frac{3\pi}{2} - 5\sin^{-1} x = y$$

$$\Rightarrow \frac{3\pi}{2} - y = 5\sin^{-1} x$$

$$\Rightarrow \sin\left[\frac{1}{5}\left(\frac{3\pi}{2} - y\right)\right] = x$$

$$\therefore f^{-1}(x) = g(x) = \sin\left[\frac{1}{5}\left(\frac{3\pi}{2} - y\right)\right]$$

$$g(y) = \sin\left(\frac{3\pi}{2} - y\right)$$

$$g'(y) = \frac{-1}{5}\sin\left(\frac{\frac{3\pi}{2} - y}{5}\right)$$

$$g'\left(\frac{2\pi}{3}\right) = \frac{-1}{5}\cdot\frac{\sqrt{3}}{2}$$

$$\text{Hence } \left(\frac{2\,g'\left(\frac{2\pi}{3}\right)}{\sqrt{3}}\right)^{-1} = \left(\frac{2\left(\frac{-1}{5}\cdot\frac{\sqrt{3}}{2}\right)}{\sqrt{3}}\right)^{-1}$$

$$= \left(\frac{-1}{5}\right)^{-1} = (-5)$$

Hence absolute value is 5 **Ans.**

Paragraph for question nos. 54 to 56

Sol. $$\lim_{x\to\infty}\left(\frac{f'(x)}{g'(x)} + f_1(x)\frac{f(x)}{g(x)}\right) = \lim_{x\to\infty} f_2(x)$$

$$\Rightarrow \lim_{x\to\infty}\frac{f'(x)}{g'(x)} + 5\lim_{x\to\infty}\frac{f(x)}{g(x)} = 12 \qquad \text{........(1)}$$

Now, Let $\lim_{x\to\infty}\frac{f(x)}{g(x)} = L \left(\frac{\infty}{\infty}\right)$

$$\Rightarrow \lim_{x\to\infty}\frac{f'(x)}{g'(x)} = L \text{ (From L' Hospital)}$$

From (1),

$$L + 5L = 12$$
$$L = 2$$

Now, $\lim_{x\to\infty}\frac{f'(x)}{g'(x)} = 2$

$$\Rightarrow \lim_{x\to\infty}\frac{e^{\lambda x} f'(x)}{e^{\lambda x} g'(x)} = 2 \left(\frac{\infty}{\infty}\right), \lambda > 0$$

$$\Rightarrow \lim_{x\to\infty}\frac{e^{\lambda x}\left[f''(x) + \lambda f'(x)\right]}{e^{\lambda x}\left[g''(x) + \lambda g'(x)\right]} = 2$$

$$\therefore \lim_{x\to\infty}\frac{\lambda f'(x) + f''(x)}{\lambda g'(x) + g''(x)} = 2. \textbf{ Ans.}$$

57. $g^3(x) - (x^3 + 2)\,g^2(x) + (2x^3 + 1)\,g(x) - x^3 = 0$

$$\Rightarrow (g(x) - 1)^2\left(g(x) - x^3\right) = 0$$

So, invertible function is

$$g(x) = x^3$$
$$g^{-1}(x) = x^{1/3}$$

$g'(8)\cdot(g^{-1})'(8) = 16.$ **Ans.**

Questions

01 If the value of $\int_0^{\frac{\pi}{2}} (\cos^{2006} x \cdot \sin 2008x)\,dx = \frac{p}{q}$ (where p and q are in their lowest form), then find the value of $(3p+q)$.

02 Number of distinct real roots satisfying the equation

$$\left(\int_0^1 \frac{1}{t^2+2t\cos a+1}\,dt\right)x^2 - \left(\int_{-3}^{3} \frac{t^2\sin 2t}{t^2+1}\,dt\right)x+2 = 0 \text{ is}$$

(A) 0 (B) 1
(C) 2 (D) more than 2

03 If $\int_0^{\pi}\left(\frac{x}{1+\sin x}\right)^2 dx = A$, then the value for $\int_0^{\pi}\frac{2x^2\cdot\cos^2(x/2)}{(1+\sin x)^2}\,dx$, is equal to

(A) $A + 2\pi - \pi^2$ (B) $A - 2\pi + \pi^2$
(C) $2\pi - A - \pi^2$ (D) None of these

04. Evaluate: $\int_0^1 \frac{x}{1+x}\sqrt{1-x^2}\,dx$

05. Evaluate $\int_0^{\frac{\pi}{4}} \frac{\sin\theta(\sin\theta\cos\theta+2)}{\cos^4\theta}\,d\theta$

06 The value of definite integral $I = \int_0^{\frac{\pi}{2}} x\left|\sin^2 x - \frac{1}{2}\right|dx$, is

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$
(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{8}$

07. Let f(x) be a continuous function defined from $[0, 2] \to R$ and satisfying the equation $\int_0^2 f(x)(x-f(x))\,dx = \frac{2}{3}$. Find the value of $f(1)$.

08

Column-I

(A) $\int_{\pi/4}^{\pi/3}\left(\frac{\sin^2 3x}{\sin^2 x}-\frac{\cos^2 3x}{\cos^2 x}\right)dx$

(B) $\int_{\pi/2}^{\pi/4}\frac{1}{1+\sin 2x}\sqrt[2009]{\frac{\tan x-1}{\tan x+1}}\,dx$

(C) $\int_0^{\pi/2}\frac{x^2}{(x\sin x+\cos x)^2}\,dx$

(D) $\int_1^e \frac{\ln x}{x\left(\sqrt{1-\ln x}+\sqrt{\ln x+1}\right)}\,dx$

Column-II

(P) $\frac{-1}{2}$

(Q) $\frac{2\sqrt{2}-2}{3}$

(R) $2\sqrt{3}-4$

(S) $\frac{2}{\pi}$

(T) None

09 Evaluate: $\int_0^{2\pi}\frac{x^2\sin x}{8+\sin^2 x}\,dx$

10. Evaluate: $\int_1^{16}\tan^{-1}\sqrt{\sqrt{x}-1}\,dx$

11. Evaluate: $\int_0^{2\pi}\frac{dx}{2+\sin 2x}$

12. If a_1, a_2 and a_3 are the three values of a which satisfy the equation

$$\int_0^{\pi/2}(\sin x + a\cos x)^3 dx - \frac{4a}{\pi-2}\int_0^{\pi/2} x\cos x\, dx = 2$$

then find the value of $1000(a_1^2 + a_2^2 + a_3^2)$.

13. If the value of definite integral $\int_{\frac{\pi}{4}}^{\frac{65\pi}{4}} \frac{dx}{(1+2^{\cos x})(1+2^{\sin x})}$ equals $k\pi$, where $k \in N$ then find k.

14. Find the number of values of x satisfying

$$\int_0^x t^2 \sin(x-t)\, dt = x^2 \text{ in } [0, 100].$$

15. If the function $\int_0^x f(t)dt \to 5$ as $|x| \to 1$, where f is continuous then find the number of integers in the range of p so that the equation $2x + \int_0^x f(t)dt = p$ has two roots of opposite sign in $(-1, 1)$.

16. If $\lim_{x\to\infty} x^{-\left(\frac{3}{2}+n\right)}\left(\int_0^x t^n \sqrt{t+1}\, dt\right)$ has the value equal $\frac{2}{11}$ then find the value of $n \in N$.

17. If the value of the definite integral $\int_{-1}^{1} \frac{1}{1+x+x^2+\sqrt{x^4+3x^2+1}} dx$ is equal to $\frac{\pi}{k}$ $(k \in N)$, find the value of k.

18. If $\int_0^1 \frac{dx}{\sqrt{1+x}+\sqrt{1-x}+2}$ can be expressed in the form $a\sqrt{b} - \frac{\pi}{c} - 1$ where a, b, c are prime numbers. Find the value of $(a+b+c)$.

19. Given that $I_n = \int_0^1 \left(1+x^2\right)^n dx$ then find the value of $13I_6 - 12I_5$.

20. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2^{\sin x} dx$ + $\int_{\frac{5}{2}}^{4} \sin^{-1}\left(\log_2(x-2)\right) dx$ is equal to

(A) $\frac{5\pi}{4}$ (B) π

(C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{4}$

21. Let $f(x) = \int_x^{\frac{\pi}{4}-x} \log_4(1+\tan t)\, dt$ $\left(0 < x \le \frac{\pi}{8}\right)$. Find the value of $\frac{\pi - 16f(x)}{x}$.

22. If $\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} k\, ln\left(\frac{n^2+(k-1)^2}{n^2+k^2}\right)$ exists and is equal to L. Find the absolute value of [L].
where [y] denotes greatest integer less than or equals to y.

23. **Column-I**

(A) $\int_0^1 \frac{x^3-x^2}{\sqrt[3]{3x^4-4x^3-1}} dx$

(B) $\int_{0}^{\pi/2} \frac{\sin 2x}{3+4\sin x-\cos 2x}dx$

(C) $\int_{0}^{2} \frac{x^4-x+1}{x^2+4}dx$

Column-II

(P) $ln(2)-\frac{1}{2}$

(Q) $\frac{1}{4}ln(2)-\frac{1}{8}$

(R) $\frac{17\pi}{8}-\frac{1}{2}ln 2-\frac{16}{3}$

(S) None

24. The value of definite integral $\int_{0}^{1}\left(\sqrt{1-x^2}\cdot ln x\right)dx$ is equal to

(A) $\frac{-\pi}{4}ln 2$ (B) $\frac{-\pi}{8}ln 2$

(C*) $\frac{-\pi}{4}ln 2-\frac{\pi}{8}$ (D) $\frac{-\pi}{8}ln 2-\frac{\pi}{4}$

25. Let a be a real number. If the value of definite integral $\int_{-\pi+a}^{3\pi+a} |x-a-\pi|\sin\left(\frac{x}{2}\right)dx$ is equal to -16 then sum of all the values of a in $[0, 314]$ is $k\pi$. Find the value of k.

26. Let $f(t)$ be a cubic polynomial such that $\cos 3x = f(\cos x)$ holds $\forall x \in R$ and $J = \int_{0}^{1} f^2(t)\sqrt{1-t^2}\,dt$. Find the value of $\frac{(2016)J}{\pi}$.

27. **Column-I**

(A) $\int_{0}^{1}\left(\frac{x-1}{x^2+1}\right)^2 e^x\,dx$

(B) $\int_{-1}^{0} x\left(e^{2x}+\sqrt[3]{x+1}\right)dx$

(C) $\int_{0}^{\pi/4}\left(\tan x+e^{\sin x}\cos x\right)dx$

Column-II

(P) $\frac{1}{2}ln(2)+\sqrt{e^{\sqrt{2}}}-1$

(Q) $\frac{e}{2}-1$

(R) $\frac{e}{2}-2$

(S) $\frac{21-16e^2}{28e^2}$

28. If $f:[0, 1]\to R$ is a continuous function satisfying $\int_{0}^{1} f(x)\,dx=\frac{1}{3}+\int_{0}^{1}\left(f(x^2)\right)^2 dx$, then find the reciprocal of $f\left(\frac{1}{4}\right)$.

29. Let $I_1=\int_{0}^{1}\frac{3^x dx}{3+x}$ and $I_2=\int_{0}^{1}\frac{x^3dx}{3^{x^4}(4-x^4)}$ then find the value of $\frac{I_1}{I_2}$.

30. Let $f(x)=\begin{cases} x-[x]-\frac{1}{2}; & \text{if } x\notin I \\ 0; & \text{if } x\in I\end{cases}$

and $P(x)=\int_{0}^{x} f(t)\,dt\ \forall x\in R$. Find the absolute value of $\int_{0}^{24} P(x)\,dx$.

31. The value of definite integral $\int_0^{\pi/2}\left(\sqrt{3+6\cos^2 x+\sin^4 x}+\sqrt{\cos^4 x+4\sin^2 x}\right)$

dx is equal to

(A) $\frac{\pi}{2}$ (B) π
(C) 2π (D) 4π

32. Let $f(x)$ be a continuous function on R. If

$\int_0^1 (f(x)- f(2x))dx = 5$ and

$\int_0^2 (f(x)- f(4x))dx = 10$

then the value of $\int_0^1 (f(x)- f(8x))dx$ is equal to

(A) 0 (B) 5
(C) 10 (D) 15

33. $\int_0^1 \frac{dx}{\left(1+x^{1/4}\right)\sqrt{\sqrt{x}-x}} =$

(A) $2\pi - 8$ (B) $2\pi - 4$
(C) $8 - 2\pi$ (D) $4 + 2\pi$

34. The graph of $f(x) = x^2 + ax + b$ intersects the x-axis at 2 distinct points A, B and y-axis at C. The centroid of ΔABC lie on the line $y = x$. If $I = \int_0^6 f(x)\, dx$, then the value of I cannot be

(A) – 50 (B) 50
(C) 100 (D) – 100

35(mcq) Let $f: R \to R^+$ be a positive function such that $1 \le \frac{f(x)}{f(y)} \le e^{(x-y)^2}$ $\forall\ x \ge y$ and $x, y \in R$. If $f(1) = 1$ then

(A) $\int_{-1}^{1} f(x)\, dx = 2$ (B) $\int_0^2 f(x)\, dx = 2$

(C) $\int_1^3 (f(x))^2\, dx = 2$ (D) $\int_{-1}^{1} \ln f(x)\, dx = 2$

36. The value of the definite integral $\int_0^1 \frac{x^{\cos\alpha}-1}{\ln x}dx$, where $\alpha \ne (2n + 1)\pi$ is

(A) $\ln(1 - \sin\alpha)$ (B) $\ln(1 + \sin\alpha)$
(C) $\ln(1 - \cos\alpha)$ (D) $\ln(1 + \cos\alpha)$

37. If $f(x) =$

$$\lim_{n\to\infty} x\left(\frac{1}{1\cdot 3}+\frac{2}{1\cdot 3\cdot 5}+....+\frac{n}{1\cdot 3\cdot 5.....(2n+1)}\right),$$

then $\int_0^4 [f(x)]dx$ (where [ ] represent greatest integer function) is equal to

(A) 1 (B) 2
(C) 3 (D) none of these

38. Let a_n is a positive number such that $\int_0^{a_n} \frac{e^x-1}{e^x+1}dx = \ln n$.

If $\lim_{n\to\infty}(a_n - \ln n)$ exists and has the value equal to $\ln k$, find k.

39. Compute $\int_1^{\sqrt{3}} \frac{x^2-1}{x^4+x^3+3x^2+x+1}dx$

40. **Statement-1:** $f(x) = \int_1^x \frac{\ln t}{1+t+t^2}\, dt\,(x>0)$

then $f(x) = -f\left(\frac{1}{x}\right)$

Statement-2: If $f(x) = \int_1^x \frac{\ln t\, dt}{t+1}$ then $f(x) + f\left(\frac{1}{x}\right) = \frac{1}{2}(\ln x)^2$

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

41. If f(x) is differentiable function such that f(x) $e^{f(x)} = x$ then find $\int_0^e f(x)\, dx$

42. $\left(\int_0^a x\left(e^x - x - 1\right) dx\right) t^2 + \left(\int_0^a (\tan x - x)\, dx\right) t + \int_0^a (\sin x - x)\, dx = 0$ has roots $\alpha(a)$ and $\beta(a)$. Then $\lim_{a\to 0^+} \left|\frac{1}{\alpha(a)} - \frac{1}{\beta(a)}\right|$

43. Let f be a real valued twice differentiable function defined on the interval (– 1, 1) such that $f(x) + e^x \int_0^x f(x)\, dx = 2e^x + e^x \ln\left(x + \sqrt{x^2+1}\right) + e^x \int_x^0 t\, f(t)\, dt$.
Then $f(0) + f'(0) + f''(0) =$ ___

44. Find the value of the definite integral

$$\int_0^{\pi/4} \left\{ \begin{array}{l} (x\sqrt{\sin x} + 2\sqrt{\cos x})\sqrt{\tan x} + \\ (x\sqrt{\cos x} - 2\sqrt{\sin x})\sqrt{\cot x} \end{array} \right\} dx$$

45. If

$$\int_{1/5}^{5} \frac{1}{x}\left(\left\{\frac{x}{4x^2 - 2x + 9}\right\} + \left\{-\frac{x}{9x^2 - 2x + 4}\right\}\right) dx$$

$= p \ln q + 3 \tan^{-1} r$
where {.} denotes fractional part function, p & q are relatively prime numbers and r is an integer, then find the value of $p + q + r$.

46 If $\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} k \ln\left(\frac{n^2 + (k-1)^2}{n^2 + k^2}\right)$ exists and is equal to L. Find the absolute value of [L]. where [y] denotes greatest integer less than or equals to y.

47. The value of the definite integral

$$\int_{-\pi}^{\pi} (\cos 2x \cdot \cos 2^2 x \cdot \cos 2^3 x \cdot \cos 2^4 x \cdot \cos 2^5 x)\, dx,$$

is equal to
(A) π (B) 2π
(C) 0 (D) none

48. $\int_{-1}^{1} \frac{e^{2x} + 1 - (x+1)(e^x + e^{-x})}{x(e^x - 1)} dx$ is equal to

(A) $e + \frac{1}{e}$ (B) $e - \frac{1}{e}$
(C) 1 (D) $e - \frac{2}{e}$

49(mcq) Let $L = \lim_{n\to\infty} \int_a^{\infty} \frac{n}{1 + n^2 x^2} dx$ $(a \in R)$, then L can be

(A) 0 (B) $\frac{\pi}{2}$

(C) π (D) 2π

50 Let $f(k) = \int_{-1}^{1} \frac{\sqrt{1-x^2}}{\sqrt{k+1} - x} dx$.

If $\sum_{k=0}^{99} f(k) = p\pi$, then find the value of p.

51. Let $f: \left[\frac{1}{2}, 2\right] \to R$ be a function such that $f(x) + f\left(\frac{1}{x}\right) = 10$. If the value of the definite integral $\int_{1/2}^{2} \frac{f(x)\cdot(1+x)}{x\sqrt{x^2+1}} dx$ is $k\, ln\left(\frac{3+\sqrt{5}}{3-\sqrt{5}}\right)$, find the value of k.

52$_{(mcq)}$ Given f is an odd function defined everywhere, periodic with period 2 and integrable on every interval. Let $g(x) = \int_0^x f(t)\, dt$. Then, which of the following is(are) correct?

(A) $g(2n) = 0$ for every integer n

(B) g(x) and f(x) have the same period

(C) g(x) is an even function

(D) g (x) is odd.

53 Consider a parabola $y = \frac{x^2}{4}$ and the point F (0, 1). Let $A_1(x_1, y_1)$, $A_2(x_2, y_2)$, $A_3(x_3, y_3)$,........, $A_n(x_n, y_n)$ are 'n' points on the parabola such $x_k > 0$ and $\angle OFA_k = \frac{k\pi}{2n}$ (k = 1, 2, 3, ..., n). Then find the value of $\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} FA_k$.

Paragraph for question nos. 54 to 56

Suppose f is an odd differentiable function on R satisfying

$$\int_x^{x+1} f(t)\,dt = k\left(x^3 + \frac{3}{2}x^2 - \frac{1}{4}\right)$$

where k is a fixed non-zero real number and f(2) = 6.

54 The function f(x) will be

(A) linear (B) quadratic

(C) cubic (D) biquadratic

55. The value of $\sum_{r=1}^{10} f(r)$, is

(A) 2550 (B) 4950

(C) 5050 (D) 2970

56. The value of definite integral $\int_{-1}^{0} \{f(t)\}\, dt$ is equal to

(where {y} denotes fractional part of y respectively)

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{2}{3}$ (D) $\frac{1}{3}$

57. Consider a non-constant continuous function f(x) such that $f^2(x) = \int_0^x \frac{4t\,f(t)}{(1+t^2)} dt$.

Let L_1 and L_2 are two perpendicular tangent lines drawn from the point $A(0, \alpha)$ to the curve y = f(x) and P, Q are points of contact respectively. Let R is the point of intersection of normals at P and Q to the curve y = f(x).

ColumnI

(A) The value of $f'(1) + f'(-1)$ lies in the interval

(B) The value of α lies in the interval

(C) The y-coordinate of R lies in the interval

(D) The area of quadrilateral APRQ lies in the interval

Column II

(P) $\left(-1, \frac{-1}{2}\right)$

(Q) $\left(\frac{-1}{2}, \frac{1}{2}\right)$

(R) (0, 1)

(S) $\left(1, \frac{3}{2}\right)$

(T) $\left(\frac{3}{2}, \frac{5}{2}\right)$

58. If $f(y) = y^2 - 2y - \frac{k^2}{\pi^2}$ and $k \in (a, b)$ for which $f\left(\sqrt{5}+1\right) > 0$, then find the value of

$$35 \int_{\frac{a}{4}}^{\frac{b}{4}} \sin^3 x \cos^3 x \left(\sin x + \cos x\right) dx$$

59. Let g be a differentiable function satisfying

$\int_0^x (x - t + 1)\, g(t)\, dt = x^4 + x^2$ for all $x \ge 0$.

The value of $\int_0^1 \frac{12}{g'(x) + g(x) + 10} dx$ is equal to

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

60. Given $\int_{f(y)}^{f(x)} e^t\, dt = \int_y^x \left(\frac{1}{t}\right) dt$,

$\forall\, x, y \in \left(\frac{1}{e^2}, \infty\right)$

Where $f(x)$ is continuous and differentiable function and $f\left(\frac{1}{e}\right) = 0$.

If $g(x) = \begin{cases} e^x; & x \ge k \\ e^{x^2}; & 0 < x < k \end{cases}$

then find the value of k for which $f(g(x))$ is continuous $\forall\, x \in R^+$.

61. Let $f(x) = \frac{1}{x^2} \int_4^x \left(4t^2 - 2f'(t)\right) dt$ then the value of $f'(4)$ is equal to

(A) $\frac{32}{9}$ (B) $\frac{64}{3}$

(C) $\frac{64}{9}$ (D) $\frac{81}{4}$

62. If the value of definite integral

$\int_0^{\frac{\pi}{6}} ln\left(1 + \tan x \tan\frac{\pi}{6}\right) dx$ is $\frac{\pi}{a} ln\left(\frac{b}{c}\right)$

(where a, b, c $\in$ N and b & c are relatively prime), then find the value of $(a + b + c)$.

63. Let $y = k$ be a line that intersects the curve $y = f(x)$ atleast at two points.

If $\int_2^x f(t)\, dt = \frac{x^2}{2} + \int_x^2 t^2 f(t)\, dt$, then

(A) $k \in (-1, 1) - \{0\}$

(B) $k \in R$

(C) $k \in \left(\frac{-1}{2}, \frac{1}{2}\right) - \{0\}$

(D) $k \in R^+$

64. Let $f(x) =$

$\sqrt{1 + x\sqrt{1 + (x+1)\sqrt{1 + (x+2)(x+4)}}}$, then

$\int_0^{100} f(x)dx$ is

(A) 4950 (B) 5050
(C) 5100 (D) 5150

65. $\lim_{n \to \infty} \int_0^1 |\sin nx|^3 dx$ is equal to

(A) $\frac{4}{3\pi}$ (B) $\frac{3}{4\pi}$

(C) $\frac{4}{3}$ (D) $\frac{3}{4}$

66. If

$$\int_0^{\pi/4} \frac{\ln(\cot x)}{\left((\sin x)^{2009} + (\cos x)^{2009}\right)^2} \cdot (\sin 2x)^{2008}\, dx =$$

$\frac{a^b \ln a}{c^2}$ (where a, b, c are in their lowest form) then find the value of $(a+b+c)$.

67. Let $I_n =$

$$\int_{-1}^{1} |x| \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3} + ... + \frac{x^{2n}}{2n}\right) dx.$$ If

$\lim_{n\to\infty} I_n$ can be expressed as rational $\frac{p}{q}$ in the lowest form, then find the value of $pq(p^3 + q^2)$.

68. Let a and b be two positive real numbers. Find the value of the definite integral $\int_a^b \frac{e^{x/a} - e^{b/x}}{x} dx$.

69. Evaluate $I =$

$$\int_0^{\pi/4} (\sin^6 2x + \cos^6 2x) \cdot \ln(1 + \tan x)\, dx.$$

70. Let $f(x)$ be a function defined in

$0 < x < \frac{\pi}{2}$ satisfying

(i) $f\left(\frac{\pi}{6}\right) = 0$

(ii) $f'(x) \tan x = \int_{\pi/6}^{x} \frac{2\cos t}{\sin t} dt$

Find $f(x)$.

71$_{(mcq)}$ Consider : $f(x) = 1 - 2p \cos x + p^2 = g(p)$;

$h(x) = g(-p) \cdot g(p)$; $K(p) = \int_0^{\pi} \ell n\, g(p)\, dx$

for $p, x \in R$. Then which of the following is(are) correct?

(A) $g(p^2) = h\left(\frac{x}{2}\right)$

(B) K(p) is an even function

(C) $K(x) = \frac{1}{2} K(x^2)$

(D) $K(x) = \frac{1}{4} K(x^2)$

72. Let $J = \int_0^1 \frac{2x - (1+x^2)^2 \cot^{-1} x}{(1+x^2)\left(1 - (1+x^2)\cot^{-1} x\right)} dx$.

Find the value of $100(J - \ln 2)$.

73. If

$$\lim_{x\to 1} \left(\frac{\sin(n\cos^{-1} x)}{\sqrt{1-x^2}} + \frac{1 - \cos(n\cos^{-1} x)}{1-x^2} \right)$$

(where $n \in N$) exist and is equal to $\frac{3}{2}$, then find the sum of all possible values of n.

74. The value of

$$\lim_{n\to\infty} \frac{1^6+2^6+3^6.......n^6}{(1^2+2^2+3^2.......n^2)\left(1^3+2^3+3^3+.......n^3\right)}$$

is equal to

(A) $\frac{14}{7}$ (B) $\frac{21}{8}$

(C) $\frac{132}{17}$ (D) $\frac{12}{7}$

75. If the value of definite integral $\int_e^{e^2} \frac{4\,ln^2 x+1}{(ln\,x)^{3/2}}\,dx$ is expressed in the form $\sqrt{a}\,e^2-\sqrt{b}\,e$, then find the minimum value of $(a+b)$.

76. If $\dfrac{\int_0^1 \left(1-\left(1-x^2\right)^{100}\right)^{201}\cdot x\,dx}{\int_0^1 \left(1-(1-x^2)^{100}\right)^{202}\cdot x\,dx}=\dfrac{p}{q}$ where p, $q \in N$, then find the least value of $(p-q)$.

77. Let $f(x)=\sin x\,\cos(\cos x)\,(1-\sin x)$ and $g(x)=\cos x\sin(\cos x)\,(1-\sin x)$.

If $l=\lim_{x\to 1^+} \sin\left(2^{-2^{\frac{1}{1-x}}}\right)$ and

$$m=\lim_{x\to 1^-}\frac{\sin\left(x-[1+x]\right)\sin\left(1+[x]\right)}{x-1}$$

where [y] denotes greatest integer function of y, then find the value of

$$\int_l^{100\pi+m} f(x)\,dx+\int_m^{100\pi+l} g(x)\,dx.$$

78. If {x} denotes the fractional part of x, then find the value of $\int_0^{100} 3\{\sqrt{x}\}\,dx$.

79. If the value of definite integral $\int_0^2 \frac{e^{-\sin^{-1}(\sqrt{x/2})}}{e^{\sin^{-1}(\sqrt{x/2})}+e^{\cos^{-1}(\sqrt{x/2})}}\,dx$ is $ke^{l\pi}$, then find the value $(6k-4l)$.

80. If x_1 and x_2 $(x_1<x_2)$ are two values of x satisfying the equation

$$\left|2\left(x^2+\frac{1}{x^2}\right)+\left|1-x^2\right|\right| = 4\left(\frac{3}{2}-2^{x^2-3}-\frac{1}{2^{x^2+1}}\right),$$

then find the value of

$$\int_{x_1+x_2}^{3x_2-x_1}\left\{\frac{x}{4}\right\}\left(1+\left[\tan\left(\frac{\{x\}}{1+\{x\}}\right)\right]\right)dx.$$

[**Note:** [y] and {y} denote greatest integer and fractional part functions respectively.]

81. Let f be a differentiable function defined such that $f:[0,27]\to\left[\frac{1}{3},6\right]$ and $f'(x)<0\ \forall\ x\in D_f$.

If $\int_0^{27} x\,f'(x)\,dx=\lambda-3\int_0^3 x^2\,f(x^3)\,dx$ then find the value of λ.

[**Note :** D_f denotes the domain of the function.]

82. Let g(x) be a real valued function defined on the interval $\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$ such that $g(x)=e^{2x}+$

$$\int_0^{\sin x}\frac{e^t}{\cos^2 x+2t\sin x-t^2}\,dt\ \forall\ x\in\left(\frac{-\pi}{2},\frac{\pi}{2}\right).$$

Also f(x) be the inverse function of g(x), where $0\le x\le\frac{\pi}{2}$. Find the value of

$$\frac{1}{\left(f'(1)\right)^2}+g(0)+g'(0)+g''(0).$$

83. Let $P(x)$ be a polynomial with real coefficients such that
$(x^2+x+1)\,P(x-1)=(x^2-x+1)\,P(x)\ \forall\ x \in R$ and $P(1)=3$.
If

$$\int_0^1 \tan^{-1}\left(\frac{2x}{1+P\left(x^2\right)}\right)dx+\int_0^1 \tan^{-1}(x+1)\,dx$$

$=\dfrac{k}{16}(\pi-ln\,4)$ then find the value of k.

84. If $\displaystyle\int_{-\infty}^{\infty} e^{-x^2}\,dx=\sqrt{\pi}$, then find the value of

$$\int_{-\infty}^{\infty} e^{-(x^2+4x+1)}\,dx.$$

85. If $\displaystyle\lim_{t\to\infty}\left(\int_1^t \frac{dx}{1+\sqrt[3]{x^2}}-at^b\right)$ exists and equals to non-zero finite number L, where a and b are positive real numbers, then find the value of $(ab-4L-3\pi)$.

86 The value of

$$\int_{-\pi}^{\pi}\left(\begin{gathered}1+\cos x+\cos 2x+\\ \ldots..+\cos(2013x)\end{gathered}\right)\left(\begin{gathered}1+\sin x+\sin 2x+\\ \ldots..+\sin(2013x)\end{gathered}\right)dx,$$

is
(A) 2π (B) $2013\,\pi$
(C) 0 (D) π

87. Let $J=\displaystyle\int_0^1 \cot^{-1}(1-x+x^2)\,dx$ and

$K=\displaystyle\int_0^1 \tan^{-1}x\,dx$. If $J=\lambda K\ (\lambda\in N)$, then λ equals

88. A sequence $a_1, a_2, \ldots\ldots.$ is explicitly defined as $a_n=\displaystyle\int_0^{\pi}\cot\left(\frac{x}{2}\right)\sin(nx)\,dx$.

If $S=\displaystyle\sum_{n=1}^{2013} a_n$, then the value of $\cos S$, is

89 The value of $\displaystyle\lim_{n\to\infty}\sum_{k=1}^{n}\sqrt[3]{\frac{k^4}{n^7}}$ is equal to

(A) $\dfrac{1}{8}$ (B) $\dfrac{3}{7}$
(C) $\dfrac{4}{9}$ (D) $\dfrac{3}{4}$

90. Let $f(x)=\displaystyle\int_0^{\infty} e^t\,e^{-xt}\,dt$. Given that the domain of $f(x)$ is $(1,\infty)$, then the range of $f(x)$ is
(A) $(-\infty,\infty)$ (B) $(0, 1)$
(C) $(-\infty, 0)$ (D) $(0,\infty)$

91. If $g(x)=\displaystyle\int_1^x e^{t^2}\,dt$ then the value of $\displaystyle\int_3^{x^3} e^{t^2}\,dt$ equals
(A) $g(x^3)-g(3)$ (B) $g(x^3)+g(3)$
(C) $g(x^3)-3$ (D) $g(x^3)-3g(x)$

92(mcq) Let $I_n=\displaystyle\int_0^{\sqrt{3}}\frac{dx}{1+x^n}$ $(n = 1, 2, 3 \ldots\ldots)$ and $\displaystyle\lim_{n\to\infty} I_n=I_0$ (say), then which of the following tatement(s)is/are correct?
(Given : $e=2.71828$)
(A) $I_1>I_0$ (B) $I_2<I_0$
(C) $I_0+I_1+I_2>3$ (D) $I_0+I_1>2$

93 The value of the definite integral $\displaystyle\int_0^{2\pi}(\sin^2 2x-\sin^4 2x)\,dx$, is

94. Let $f(x)$ be a positive differentiable function defined on [0, 1] such that $f(1) = 0$ and for any $a \in (0, 1)$, $\int_0^a f(x)\,dx - \int_a^1 f(x)\,dx = 2f(a) + 3a + b$, where b is a constant. Find $f(x)$.

95. The value of the definite integral $\int_{-1}^{1} \frac{dx}{(1+e^x)(1+x^2)}$ is

(A) $\pi/2$ (B) $\pi/4$ (C) $\pi/8$ (D) $\pi/16$

96. If $L = \lim_{n\to\infty} \prod_{k=1}^{n} \left(\frac{k+n}{k}\right)^{\left(e^{1/n}-1\right)}$, find (225)L

97 If f(x) is differentiable function such that $f(x)\,e^{f(x)} = x$ then find $\int_0^e f(x)\,dx$

98. The value of $\int_{e^{e^e}}^{\infty} \frac{dx}{x\,\ln x\,\ln\ln x\,(\ln\ln\ln x)^{\frac{-11}{10}}}$ is equal to

(A) 7 (B) 8
(C) 9 (D) 10

99. The value of the integral $\int_1^{16} \frac{dx}{\sqrt{x}+\sqrt[4]{x}}$ is

(A) $4\ln\frac{3}{2}$ (B) $4\ln\frac{3}{2}+1$

(C) $4\ln\frac{3}{2}+2$ (D) none

100. Value of the definite integral

$$\int_{-1/2}^{1/2} \left(\sin^{-1}(3x-4x^3) - \cos^{-1}(4x^3-3x)\right)dx$$

(A) 0 (B) $-\frac{\pi}{2}$ (C) $\frac{7\pi}{2}$ (D) $\frac{\pi}{2}$

101 The value of $\lim_{n\to\infty} \int_1^n \ln\left(1+\frac{1}{\sqrt{x}}\right)dx$ is equal to

(A) 1 (B) $\frac{1}{2}$
(C) 0 (D) 2

102 Let $I_1 = \int_0^{\pi/2} \cos\theta \cdot f\left(\sin\theta + \cos^2\theta\right)d\theta$ and $I_2 = \int_0^{\pi/2} \sin 2\theta \cdot f\left(\sin\theta + \cos^2\theta\right)d\theta$ then $\frac{I_1}{I_2}$ is equal to

(A) 1/2 (B) 2
(C) 1 (D) 1/4

103 The value of definite integral $\int_0^1 \frac{1}{\sqrt{x}\sqrt{1+\sqrt{x}}\sqrt{1+\sqrt{1+\sqrt{x}}}}dx$ is equal to

(A) $\sqrt{1+\sqrt{2}}-\sqrt{2}$

(B) $2\left(\sqrt{1+\sqrt{2}}-\sqrt{2}\right)$

(C) $4\left(\sqrt{1+\sqrt{2}}-\sqrt{2}\right)$

(D) $8\left(\sqrt{1+\sqrt{2}}-\sqrt{2}\right)$

104. If the value of the definite integral $\int_{5/4}^{10} \frac{\sqrt{x+2\sqrt{x-1}}+\sqrt{x-2\sqrt{x+1}+2}}{\sqrt{x^2-1}}dx$ can be expressed in the form $a\left(\sqrt{b}+\sqrt{c}\right)$ where a, b, c $\in$ N and pair wise coprime. Find (a + b + c).

105 Let $F(t) = \frac{1}{t}\int_0^{\frac{\pi t}{2}} |\cos 2x|\,dx$ for $0 < t \le 1$.

The value of $\lim_{t\to 0} F(t)$ equals

(A) π (B) $\frac{\pi}{2}$

(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

106 If $\int_0^{\pi/6} x\sec x \cdot \sec\left(\frac{\pi}{6} - x\right)dx = \alpha\, ln \sec\beta$. where β is the smallest positive such angle in radians and $\frac{\alpha}{\beta}$ be a rational number, then find the value of $\frac{335\pi}{|\alpha - \beta|}$.

107. The value of the definite integral $\int_0^1 \left(\frac{2^{\log_{2^{1/4}} x} - 3^{\log_{27}(x^2+1)^3} - 2x}{7^{4\log_{49} x} - x - 1}\right)dx$ equals

(A) 3 (B) 2

(C) 11/6 (D) 5/6

108$_{(mcq)}$ Let f(x) be a non zero continuous function for all x, such that

$$(f(x))^2 = \int_0^x f(t)\cdot\frac{2\sec^2 t}{4+\tan t}dt \text{ and}$$

$f(0) = 0$, then

(A) $f\left(\frac{\pi}{4}\right) = ln\frac{5}{4}$ (B) $f\left(\frac{\pi}{4}\right) = \frac{3}{4}$

(C) $f\left(\frac{\pi}{3}\right) = 2$ (D) $f\left(\frac{\pi}{3}\right) = ln\frac{4+\sqrt{3}}{4}$

109 $\int_0^1 \left(\sqrt[4]{1-x^7} - \sqrt[7]{1-x^4}\right)dx$ is equal to

(A) $\frac{1}{2}$ (B) 1

(C) 0 (D) None

Paragraph for question nos. 110 to 113

Let $n \in N$. The A.M., G.M., H.M. and R.M.S. (root mean square) of the *n* numbers $(n+1), (n+2), (n+3), ..., (n+n)$ are A_n, G_n, H_n, R_n respectively. Then

110. $\lim_{n\to\infty}\frac{A_n}{n} =$

(A) 1 (B) 3/2

(C) 2 (D) 1/2

111. $\lim_{n\to\infty}\frac{G_n}{n} =$

(A) 1/e (B) 2/e

(C) 3/e (D) 4/e

112. $\lim_{n\to\infty}\frac{H_n}{n} =$

(A) $ln\,2$ (B) $\frac{1}{ln\,2}$

(C) 1 (D) e

113. $\lim_{n\to\infty}\frac{R_n}{n} =$

(A) $\sqrt{3}$ (B) $\sqrt{5/3}$

(C) $\sqrt{7/3}$ (D) 3

114. Let $I = \int_1^3 \left|(x-1)(x-2)(x-3)\right|dx$. The value of I^{-1}, is equal to

(A) 1 (B) 2 (C) 3 (D) 4

115. Let $0 < t < \frac{\pi}{2}$. Find the absolute value of

$$\lim_{t\to\frac{\pi}{2}}\int_0^t \tan\theta\sqrt{\cos\theta}\; ln(\cos\theta)\,d\theta.$$

116 If $\int_x^{xy} f(t)\,dt$ is independent of x and f(2) = 2, then $\int_1^x \{f(t)\}\,dt$ is equal to

(A) 2 *l*n x (B) 3 *l*n x
(C) 4 *l*n x (D) None

117 The value of definite integral $\int_1^3 \frac{ln\,(x+1)}{x^2}\,dx$ is equal to

(A) $\frac{1}{3}(ln27 - ln4)$ (B) $\frac{1}{3}(ln9 - ln4)$
(C) $(ln9 - ln4)$ (D) $(ln27 - ln4)$

Paragraph for question nos. 118 to 120

Let, $f(x) = \begin{cases} \frac{8}{\pi}\tan^{-1}(-|x|+3), & |x| > 2 \\ \left[\frac{3x^2 - |x| + 3}{x^2+1}\right], & |x| \le 2 \end{cases}$

where [x] denotes largest integer less than or equal to x.

118 Number of points where the function f(x) is discontinuous is equal to

(A) 0 (B) 1
(C) 2 (D) 4

119 Range of f(x) is equal to

(A) (– 4, 2] ∪ {3} ∪ [4, ∞)
(B) (– 4 , 4)
(C) (– 4, 3]
(D) (– 4, 2] ∪ {3}

120 The value of the definite integral $\int_{-2}^{1} f(x)\,dx$ is equal to

(A) 3 (B) 6
(C) 9 (D) 12

121 Find the value of the definite integral

$$2^{2010}\frac{\int_0^1 x^{1004}(1-x)^{1004}dx}{\int_0^1 x^{1004}(1-x^{2010})^{1004}dx}.$$

122 The value of the definite integral $\int_0^1 \frac{4x^3\left(1+x^{4(2009)}\right)}{(1+x^4)^{2011}}dx$, is equal to

(A) $\frac{1}{2009}$ (B) $\frac{1}{2010}$
(C) $\frac{1}{2011}$ (D) $\frac{1}{2012}$

123 Let g(x) be a polynomial function satisfying g(x) g(y) = g(x) + g(y) + g(xy) – 2 ∀ x ∈ R. Also g(1) ≠ 1 and g(2) = 5. If the value of the integral $\int_1^{\infty} \frac{dx}{x\,g(x)}$ is $\frac{p}{q} ln\,2$ where p and q are co-prime, then find the value of $(p^2 + q^2)$.

124 $\sum_{k=1}^{\infty} \frac{1}{2k\,(2k-1)}$ is equal to

(A) 2 *l*n 2 (B) 3 *l*n 2
(C) *l*n 2 (D) $ln\,\frac{2}{e}$

125. Let $f(x) = \frac{1}{x+1}$ and $f(g(x)) = \frac{x-1}{x}$ then the value of $\int_2^1 g(f(x))\,dx$ is equal to

(A) *l*n 2 (B) 1 + *l*n 2
(C) 2 + *l*n 2 (D) 1 – *l*n 2

Paragraph for Question no.126 to 128

Let $f(x)$ is a derivable function satisfying $f(x) - \int_0^x f(t)\,dt = x + ln\left(\sqrt{x^2+1} - x\right)$ with $f(0) = ln\,2$. Let $g(x) = x f'(x)$ then

126. Range of g (x) is
(A) $[0, \infty)$ (B) $[0, 1)$
(C) $[1, \infty)$ (D) $(-\infty, \infty)$

127. For the function f which one of the following is correct?
(A) f is neither odd nor even
(B) f is transcendental
(C) f is injective
(D) f is symmetric w.r.t. origin

128. $\int_0^1 f(x)\,dx$ equals
(A) $ln\left(3+2\sqrt{2}\right)-1$ (B) $2\,ln\left(1+\sqrt{2}\right)$
(C) $ln\left(1+\sqrt{2}\right)-1$ (D) 1

129$_{(mcq)}$ Let $f(x) = \int_0^x e^t \cdot \sin(x-t)\,dt$ and $g(x) = f(x) + f''(x)$. Which of the following statements are correct?
(A) g (x) is positive $\forall\ x \in R$.
(B) g (x) is a constant function
(C) $g(1) = e$
(D) $g'(x) = g(x)$

130. If the value of the definite integral $\int_0^{2\pi} \frac{dx}{\sin^2 x + (100)^2 \cdot \cos^2 x} = \frac{a\pi}{b}$ where a and b are coprime find $(a + b)$.

131. If f(x) is a non-zero differentiable function such that $\int_0^x f(t)\,dt = (f(x))^2\ \forall\ x \in R$, then f(4) equals

132. For $\alpha \in R$, the range of the function

$$f(\alpha) = \int_{\tan^{-1}\alpha}^{\cot^{-1}\alpha} \left(\frac{\tan x}{\tan x + \cot x}\right) dx$$ is equal to

(A) $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ (B) $(0, \pi)$
(C) $\left(\frac{-\pi}{4}, \frac{3\pi}{4}\right)$ (D) $\left(0, \frac{\pi}{2}\right)$

Paragraph for question nos. 133 to 135

Let $f(x) = f'(x) + 2x \int_0^{a(x)} \left(\frac{f(t)}{t^2}\right) dt - x$,

where $a(x) = \int_0^x \frac{f(t)}{t^2}\,dt$,

Also f(x) is an even differentiable function on R and $g(x) = \int_0^{a(x)} \frac{f(t)}{t^2}\,dt$.

133. The value of f(3), is
(A) $\frac{3}{2}$ (B) $\frac{5}{2}$
(C) $\frac{7}{2}$ (D) $\frac{9}{2}$

134. The value of g'(2), is
(A) $\frac{1}{4}$ (B) $\frac{1}{2}$
(C) $\frac{3}{2}$ (D) $\frac{11}{4}$

135. The area bounded by f(x) and g(x), is
(A) $\frac{1}{28}$ (B) $\frac{1}{96}$
(C) $\frac{1}{112}$ (D) $\frac{1}{132}$

136. Find the value of definite integral

$$\int_1^4 \frac{(x-2)}{(x^2+4)\sqrt{x}}\,dx.$$

137. If

$$\int_0^{\pi} \frac{x}{\sqrt{1+\sin^3 x}}\left((3\pi\cos x+4\sin x)\sin^2 x+4\right)dx =$$

$k\pi^2$, then find the value of k.

138. Consider a function of the form $f(x) = \alpha e^{2x} + \beta e^x - \gamma x$, where α, β, γ are independent of x and f(x) satisfies the following conditions $f(0) = -1$,

$f'(\ln 2) = 30$ and $\int_0^{\ln 4} (f(x)+\gamma x)\,dx = 24.$

The value of $(\alpha + \beta + \gamma)$ is equal to

139 **(mcq)** Let r be a positive integer. If $L = \lim_{n\to\infty} \frac{\sum_{k=1}^{n} k^r}{n^a}$ has a non zero finite value, then which of the following relation is/are correct?

(A) $a - r - 1 = 0$ (B) $a + r - 1 = 0$

(C) $r + 1 - \frac{1}{L} = 0$ (D) $r - 1 - \frac{1}{L} = 0$

140. If $f(x) = \begin{vmatrix} \cos x & 1 & 0 \\ 1 & 2\cos x & 1 \\ 0 & 1 & 2\cos x \end{vmatrix}$ then the value of definite integral $\int_0^{\pi/2} f(x)\,dx$ is equal to

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) none

141. Let $I_1 = \int_0^x e^{tx}\cdot e^{-t^2}dt$ and $I_2 = \int_0^x e^{-t^2/4}dt$

where $x > 0$ then the value of $\frac{I_1}{I_2}$ is

(A) $e^{-x^2/2}$ (B) $e^{x^2/4}$

(C) $e^{-x^2/4}$ (D) $e^{x^2/2}$

142. A function f(x) satisfies $f(x) = f\left(\frac{c}{x}\right)$ for some real number c $(c > 1)$ and $\forall\ x > 0$. If $\int_1^{\sqrt{c}} \frac{f(x)}{x}dx = 3$, then the value of $\int_1^{c} \frac{f(x)}{x}dx$ is

Paragraph for Que Nos. 143 to 145

If $f(x) = x^2 - 2|x|$

$g(x) = \begin{cases} \text{minimum } \{f(t) : -2 \le t \le x, -2 \le x < 0\} \\ \text{maximum } \{f(t) : 0 \le t \le x, 0 \le x \le 3\} \end{cases}$

then

143. The function $y = |f(x)|$ is differentiable for

(A) $x \in R$ (B) $x \in R - \{0\}$

(C) $x \in R - \{0,2\}$ (D) None of these

144. Number of values of x in [–2, 3] where g(x) is non-derivable is

(A) 0 (B) 1

(C) 2 (D) 3

145. $\int_0^1 fog(x)\,dx$ is equal to

(A) 0 (B) $\frac{33}{2}$

(C) 21 (D) None of these

146. Let

$$\int (\cos x)^n\,dx = \frac{(\cos x)^{n-1}\cdot \sin x}{n} + \frac{n-1}{n}\int (\cos x)^{n-2}\,dx.$$

If $J_1 = \int_0^{\pi/2} (\cos x)^{1003}\, dx$ and

$J_2 = \int_0^{\pi/2} (\cos x)^{1004}\, dx$ then $(J_1 \cdot J_2)$ has the value equal to

(A) $\frac{\pi}{2008}$ (B) $\frac{\pi}{2006}$

(C) $\frac{\pi}{1004}$ (D) $\frac{\pi}{1003}$

147. Let p(x) be a differentiable function on R such that $p'(x) = p'(5 - x)\ \forall\ x \in [0, 5]$, $p(0) = 1$ and $p(5) = 7$ then $\int_1^4 p(x)\, dx$ equals

(A) 8 (B) 12

(C) 6 (D) 4

148$_{(mcq)}$ If $f(x) = \text{Min}(\tan x, \cot x)$ then

(A) f(x) is discontinuous at $x = 0, \frac{\pi}{4}$ and $\frac{5\pi}{4}$

(B) f(x) is continuous at $x = \frac{\pi}{2}$ and $\frac{3\pi}{2}$

(C) $\int_0^{\pi/2} f(x)\, dx = \ln 2$

(D) f(x) is periodic with period π.

149. Let $f(x) = e^x + 2x + 1$ then find the value of $\int_2^{e+3} f^{-1}(x)\, dx$.

150. The value of

$$\underset{n\to\infty}{\text{Limit}} \frac{1}{n}\sum_{k=1}^{n}\left(e^{\left(1+\frac{k}{n}\right)^2} - \frac{3e^{\left(1+\frac{3k}{n}\right)}}{2\sqrt{1+\frac{3k}{n}}}\right) \text{ equals}$$

Paragraph for Question no. 151 & 152

Let $\int \frac{g(x)\left[(1+x)\ln x - 5\right]}{x(\ln x)^6} dx$

$= \frac{g(x)}{(\ln x)^5} + C$, $g(1) = e$ and

$g : (0, \infty) \to (0, \infty)$.

151. If $f(x) = \ln\left(g(x)+1\right) - \ln\left(g(x)\right)\ \forall\ x \in R^+$ then the value of $(2013)^2 \sum_{x=1}^{2012} f''(x)$ is

(A) (2011) (2013) (B) (2012) (2014)

(C) (2013) (2014) (D) (2011) (2012)

152. Let $I_n = \int_0^{\infty} \frac{x^{n+1}}{g(x)} dx$, then the value of I_{2012} is

(A) (2012) ! (B) 2012

(C) $\frac{1}{2012}$ (D) $(2012)^2$

153. Find the value of $\int_0^{\pi} \frac{x^2\cos^2 x - x\sin x - \cos x - 1}{(1 + x\sin x)^2} dx$.

154. $\int_0^{\pi} \frac{x\sin^3 x}{4 - \cos^2 x} dx$

155. The value of $\int_{-1/2}^{1/2} \sin\left(\tan^{-1}\left(\frac{x^2}{\sqrt{1-x^4}}\right)\right) dx$ equals

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{1}{6}$ (D) $\frac{1}{12}$

156. The value of definite integral

$$\int_{-3}^{-1} (x+1)^4 (x+2)^3 (x+3)^4 \, dx$$, is equal to

(A) – 2 (B*) 0
(C) 2 (D) 4

Paragraph for Question no. 157 to 159

Consider $I_1 = \int_0^{\frac{\pi}{2}} \frac{x^3 \cos x \, dx}{(3\sin x - \sin 3x)}$,

$I_2 = \int_0^{\frac{\pi}{2}} x^2 \operatorname{cosec}^2 x \, dx$ and $I_3 = \int_0^{\frac{\pi}{2}} x \cot x \, dx$

157. If $I_1 = \frac{\pi^3}{k_1} + k_2 \pi \ln 2$, where $k_1, k_2 \in Q$ then $(k_1 + k_2)$ equals

(A) $\frac{-509}{8}$ (B) $\frac{509}{8}$

(C) $\frac{511}{8}$ (D) $\frac{-511}{8}$

158. If $I_2 = k_1 + k_2 \pi \ln 2$, where $k_1, k_2 \in Q$, then $(k_1^2 + k_2^2)$ equals

(A) $\frac{1}{4}$ (B) $\frac{5}{32}$

(C) 1 (D) 9

159. If $I_3 = k_1 + k_2 \pi \ln 2$, where $k_1 k_2 \in Q$, then $(k_1^3 + k_2^3)$ equals

(A) $\frac{1}{8}$ (B) $\frac{1}{27}$

(C) $\frac{1}{64}$ (D) $\frac{3}{32}$

[**Note :** Q denotes the set of rational numbers]

160. If $\frac{626 \int_0^{\infty} e^{-x} \cdot \sin^{25} x \, dx}{\int_0^{\infty} e^{-x} \sin^{23} x \, dx} = \lambda$, then find the sum of digits of λ.

161. Let θ be a constant number such that $0 \le \theta \le \pi$. If $f(\theta) = \int_0^{2\pi} \sin 8x \, |\sin(x-\theta)| \, dx$, then find the maximum value of $f(\theta)$.

162. The value of the definite integral

$$\int_0^{\frac{\pi^2}{4}} \frac{dx}{1 + \sin\sqrt{x} + \cos\sqrt{x}}$$ is

(A) $\pi \ln 2$ (B) $\frac{\pi \ln 2}{2}$

(C) $\frac{\pi \ln 2}{4}$ (D) $2\pi \ln 2$

163. Let $f:(0,\infty) \to R$ be a continuous function, satisfying $f(x) + f(y) = f(xy)$.
Then the value of

$$\int_{\sqrt{f(e)}}^{\sqrt{f(e^2)}} \frac{x \tan x^2}{\tan x^2 + \tan\left(f(e^3) - x^2\right)} dx$$ is equal to

(A) $\frac{1}{2} f(e^2)$ (B) $\frac{1}{4} f(e^2)$

(C) $\frac{1}{2} f(e)$ (D) $\frac{f(e)}{4}$

164 Let $f(x) = \int_0^4 e^{|t-x|} dt$, $x \in [0, 4]$. If the range of $f(x)$ is $[a, b]$, then $\sqrt{b-a}$ is

(A) e^2 (B) $e^2 + 1$

(C) $e^2 - 1$ (D) $\frac{e^2}{2}$

165. Let p(x) be a polynomial with real coefficients such that $(x + 10)\, p(2x) = (8x - 32)\, p(x + 6)$ for all real x and $p(1) = 210$. Find the value of

$$\frac{3}{11}\int_{-1}^{1} p(x)\,dx.$$

166. If

$$\int_{0}^{\pi} (\cos x \cdot \cos 2x \cdot \cos 3x \ \ldots\ldots\ldots\ldots \cos mx)\,dx$$

vanishes, then the number of integral values of
$m \in [1, 10]$, is

(A) 4 (B) 6
(C) 8 (D) 10

167. Let $f(x) = \displaystyle\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln(1 + x\cos\theta)}{\cos\theta}\,d\theta$, then the value of $f'(1)$ is

(A) 0 (B) 1
(C) 2 (D) 3

168. Let $I_n = \displaystyle\int_{0}^{\frac{\pi}{4}} \tan^n x\ dx$ $(n = 0, 1, 2, 3, 4, \ldots\ldots\ldots)$

and $S_n =$

$$\sum_{n=0}^{n} \left(I_n I_{n+1} + I_n I_{n+3} + I_{n+1} I_{n+2} + I_{n+2} I_{n+3}\right).$$

Find the value of $\lim\limits_{n\to\infty} 100\,(S_n)$.

169. Let the functions f(x) and g(x) be differentiable and strictly increasing $\forall\, x \in R$. Also $f(0) = 2$, $f(1) = 3$ and $f'(x)$ is continuous.
If $f^2(x) = 1 + g^2(x)$ and J =

$$\int_{0}^{1} \left(\frac{g(x)}{f(x)}\right)' \cdot \frac{1}{g(x)}\,dx,$$ then the value of

$(6J + 1)$ is equal to $\ln\dfrac{a}{b}$,
where $a, b \in N$. Find the minimum value of $(a + b)$.

170. Find the value of definite integral

$$\int_{0}^{\frac{\pi}{2}} \frac{\sin^2 10\theta}{\sin^2\theta}\,d\theta.$$

171 Let $U = \displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \min.\left(\sqrt{3}\sin x,\ \cos x\right) dx$ and

$V = \displaystyle\int_{-3}^{5} x^2\,\text{sgn}\,(x - 1)\ dx$. If $V = \lambda U$,

then find the value of λ.
[**Note:** sgn k denotes the signum function of k.]

172. If $\lim\limits_{x\to\infty} x^{-\left(\frac{3}{2}+n\right)} \left(\displaystyle\int_{0}^{x} t^n \sqrt{t+1}\ dt\right)$ has the value equal $\dfrac{2}{11}$ then find the value of $n \in N$.

173. If the value of the definite integral

$$\int_{-1}^{1} \cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)\cdot\left(\cot^{-1}\frac{x}{\sqrt{1-(x^2)^{|x|}}}\right)dx = \frac{\pi^2\left(\sqrt{a}-\sqrt{b}\right)}{\sqrt{c}}$$

where $a, b, c \in N$ in their lowest form, then find the value of $(a + b + c)$.

174. Evaluate: $\displaystyle\int_{1}^{e}\left(\frac{x^2 \ln x - 1}{x}\right) e^x\,dx$

Paragraph for question nos. 175 to 176

Let, $f(x)=\begin{cases}\dfrac{8}{\pi}\tan^{-1}(-|x|+3), & |x|>2\\ \left[\dfrac{3x^2-|x|+3}{x^2+1}\right], & |x|\le 2\end{cases}$

where [x] denotes largest integer less than or equal to x.

175. Number of points where the function f(x) is discontinuous is equal to
(A) 0 (B) 1
(C) 2 (D) 4

176. Range of f(x) is equal to
(A) $(-4, 2]\cup\{3\}\cup[4,\infty)$
(B) $(-4, 4)$
(C) $(-4, 3]$
(D) $(-4, 2]\cup\{3\}$

177. The value of the definite integral $\int_{-2}^{1} f(x)\,dx$ is equal to
(A) 3 (B) 6
(C) 9 (D) 12

178. The value of definite integral

$$\int_0^1 \frac{dx}{\sqrt{(x+1)^3\cdot(3x+1)}} \text{ equals}$$

(A) $\tan\dfrac{\pi}{8}$ (B) $\tan\dfrac{\pi}{12}$
(C) $\tan\dfrac{3\pi}{8}$ (D) $\tan\dfrac{5\pi}{12}$

179. $\int_0^1 \dfrac{1-x^2}{1+x^2+x^4}dx$ is equal to

(A) $\dfrac{1}{2}\ln 3$ (B) $\dfrac{1}{3}\ln 2$
(C) $\dfrac{1}{3}\ln 3$ (D) $\dfrac{1}{2}\ln 2$

Paragraph for question nos.180 to 182

Let f and g be two real-valued differentiable functions on R satisfying

$$\int_0^x g(t)\,dt = 3x+\int_x^0 \cos^2 t\; g(t)\,dt \quad\text{and } f(x)=$$

$$\lim_{\alpha\to 0}\frac{1}{\alpha^4}\int_0^{\alpha}\frac{\left(e^{x+t}-e^x\right)\ln^2(1+t)}{2t^3+3}dt$$

180$_{(mcq)}$ $f(\ln 2)$ is greater than

(A) 0 (B) $\dfrac{1}{6}$
(C) $\dfrac{3}{20}$ (D) $\dfrac{4}{25}$

181. Range of g(x) is equal to

(A) $\left[\dfrac{3}{2}, 3\right]$ (B) $\left[\dfrac{1}{2}, 3\right]$
(C) $\left[\dfrac{3}{2}, \infty\right)$ (D) $[-3, 3]$

182$_{(mcq)}$ The value of definite integral $\int_0^{\frac{\pi}{2}} g(x)\,dx$ lies in the interval

(A) $\left(\dfrac{2\pi}{3}, \dfrac{4\pi}{5}\right)$ (B) $\left(\dfrac{\pi}{2}, \dfrac{5\pi}{6}\right)$
(C) $\left(\dfrac{3\pi}{4}, \dfrac{6\pi}{5}\right)$ (D) $\left(\pi, \dfrac{3\pi}{2}\right)$

183. The value of the definite integral

$$\int_{-1}^{1} e^{-x^4}\left(1+\ln\left(x+\sqrt{x^2+1}\right)+5x^3-4x^4\right)dx$$

is equal to

(A) 4e (B) $\dfrac{4}{e}$
(C) 2e (D) $\dfrac{2}{e}$

184. Let **L** denotes the value of $\lim_{x\to0^+}\frac{1}{x}\cos^{-1}\left(\frac{\sin x}{x}\right)$ and **U** denotes the value of $f''(0)$ where $f(x)=\int_0^x \sin(t^2-t+x)\,dt$, Compute the value of $\frac{120U}{L^2}$.

185. Let $I=\int_1^2\left(\cot^{-1}\sqrt{x-1}\right)^2 dx$. Find the value of $\frac{\pi^2+8I-8\ln 2}{\pi}$.

186. **Column I**

(A) The function $f(x)=\frac{e^{x\cos x}-1-x}{\sin x^2}$ is not defined at $x=0$. The value of $f(0)$ so that f is continuous at $x=0$ is

(B) The value of the definite integral $\int_0^1\frac{dx}{\sqrt{x}+\sqrt[3]{x}}$ equals $a+b\ln 2$, where a and b are integers then $(a+b)$ equals

(C) Given $e^n\int_0^n\frac{\sec^2\theta-\tan\theta}{e^\theta}d\theta=1$ then the value of $\tan(n)$ is equal to

(D) Let $a_n=\int_{\frac{1}{n+1}}^{\frac{1}{n}}\tan^{-1}(nx)\,dx$ and

$b_n=\int_{\frac{1}{n+1}}^{\frac{1}{n}}\sin^{-1}(nx)\,dx$ then $\lim_{n\to\infty}\frac{a_n}{b_n}$ has the value equal to

Column II

(P) -1

(Q) 0

(R) $1/2$

(S) 1

187. Let f and g be two positive real valued functions defined on $[-1, 1]$ such that

$f(-x)=\frac{1}{f(x)}$ and

$g(x)$ is an even function such that

$\int_{-1}^{1}g(x)\,dx=1$, then $I=\int_{-1}^{1}f(x)g(x)\,dx$

satisfies

(A) $0<I<\frac{1}{3}$ (B) $\frac{1}{3}<I<\frac{1}{2}$

(C) $\frac{1}{2}<I<1$ (D) $I\ge 1$

188. Let f(x) be a differentiable injective function such that $f'(x)\ne0$ and $F(x)=\int_1^x f(t)\,dt$.

Also $f(1)=f'(1)=1$ and g(x) is the inverse of f(x). Let $G(x)=x^2g(x)-xF(g(x))\ \forall x\in R$. The value of $G''(1)$ is

(A) 0 (B) 1

(C) 2 (D) 3

189. The value of definite integral $\int_{1/3}^{1}\left\{2x\left[\frac{1}{x}\right]\right\}dx$ is

[Note: $\{\cdot\}$ and $[x]$ denote fractional part and greatest integer functions respectively.]

(A) $\frac{13}{36}$ (B) $\frac{8}{36}$

(C) $\frac{15}{36}$ (D) $\frac{9}{36}$

Paragraph for question nos. 190. to 191

Let $f: R \to R$ be a bijective function. Let $g(x)$ be a continuous function such that

$$\int_0^{f(x)} g(t)dt = g(f(x)) - 1 \quad \forall \; x \in R$$

190. Value of $\int_0^1 g(\{x\}) \cdot \left(\dfrac{1-x}{(1+x)^3}\right) dx$ is

Note : $\{y\}$ denotes fractional part of y.

(A) $1 - \dfrac{e}{4}$ (B) 1

(C) $1 + \dfrac{e}{4}$ (D) $\dfrac{e}{4} - 1$

191. The value of $\lim\limits_{x \to \infty} e^{-x} \int_{\frac{1}{x}}^{x+\frac{1}{x}} g(t)\, dt$ equal to

(A) 0 (B) 1

(C) 2 (D) DNE

ANSWER KEY

1.	2010	2.	A	3.	A	4.	$1-\frac{\pi}{4}$
5.	$=\frac{2}{3}\left(2\sqrt{2}-1\right)+\frac{1}{\sqrt{2}}-\frac{1}{2}\ln\left(\sqrt{2}+1\right)$.	6.	D	7.	1/2		
8.	(A) R; (B) T ; (C) S ; (D) Q	9.	$-\frac{2\pi^2}{3}\,ln\,2$	10.	$\frac{16\pi}{3}-2\sqrt{3}$		
11.	$\frac{2\pi}{\sqrt{3}}$	12.	5250	13.	4	14.	16
15.	2	16.	4	17.	4	18.	6 (where a = b = c = 2)
19.	64	20.	A	21.	8	22.	1
23.	(A) Q; (B) P; (C) R			24.	C	25.	1200
26.	252			27.	(A) Q ; (B) S ; (C) P		
28.	2	29.	12	30.	2	31.	D
32.	C	33.	C	34.	B	35.	ABC
36.	D	37.	B	38.	4	39.	$I=\frac{2\sqrt{3}}{3}\text{ arctan }\frac{13\sqrt{3}-21}{44}$
40.	D	41.	e – 1	42.	4	43.	2
44.	0	45.	7	46.	1	47.	C
48.	B	49.	ABC	50.	10	51.	5
52.	ABC	53	$\frac{4}{\pi}$	54.	C	55.	D
56.	B	57.	(A) Q ; (B) Q ; (C) T (D) T			58.	4
59.	C	60.	1	61.	A	62.	19
64.	C	65.	A	66.	4019	67.	186
68.	0	69.	$\frac{5}{64}\pi\,ln\,2$	70.	$ln^2(2\sin x)$		
71.	ABC	72.	100	73.	1	74.	D
75.	22	76.	1	77.	0	78.	155
79.	8	80.	2	81.	9	82.	18
83.	12	84.	$e^3\sqrt{\pi}$	85.	$a=3\,;\,b=\frac{1}{3}\,;\,L=\frac{-3\pi}{2}+\frac{3\pi}{4}-3$		
86.	A	87.	2	88.	– 1	89.	B
90.	D	91.	A	92.	ACD	93.	$\frac{\pi}{4}$
94.	$\frac{3}{2}\left(1-e^{x-1}\right)$	95.	B	96.	900	97.	e – 1
98.	D	99.	C	100.	B	101.	D
102.	C	103.	D	104.	14	105.	B
106.	2010	107.	C	108.	AD	109.	C

110.	B	111.	D	112.	B	113.	C	114.	B	115.	4	116.	C
117.	A	118.	B	119.	D	120.	B	121.	4020	122.	B	123.	5
124.	C	125.	D	126.	B	127.	B	128.	A	129.	ACD	130.	51
131.	2	132.	C	133.	D	134.	A	135.	B	136.	0	137.	2
138.	3	139.	AC	140.	D	141.	B	142.	6	143.	D	144.	C
145.	A	146.	A	147.	B	148.	CD	149.	2	150.	0	151.	B
152.	A	153.	0	154.	$1-\frac{3\ln 3}{4}$			155.	D	156.	B	157.	A
158.	C	159.	A	160.	6	161.	$\frac{-4}{63}\sin 8\theta$			162.	B	163.	D
164.	C	165.	32	166.	B	167.	2			168.	100	169.	35
170.	$\frac{n\pi}{2}$	171.	64	172.	4	173.	7	174.	$(e-2)\,e^e + e$			175.	B
176.	D	177.	B	178.	A	179.	A	180.	ACD	181.	A	182.	CD
183.	2 /e	184.	360	185.	4								
186.	(A) R; (B) P; (C) S; (D) R					187.	D	188.	D	189.	A	190.	A
191.	B												

SOLUTIONS

01 Let $I = \int \cos^{2006} x \cdot \sin(2008x)\, dx$

$$= \int \cos^{2006} x\, [\sin x \cdot \cos(2007\, x) + \cos x \cdot \sin(2007\, x)]\, dx$$

$$= \int \cos(2007x)\, [\cos^{2006} x \cdot \sin x] + (\cos^{2007} x)\, [\sin(2007\, x)]\, dx$$

$$= -\frac{1}{2007}\int \cos(2007x)\, d(\cos^{2007} x) + (\cos^{2007} x)\, d\left(\cos(2007\, x)\right)$$

Since $d(uv) = u\, dv + v\, du$

$$I = -\frac{1}{2007}\int d\left(\cos(2007\, x)\cdot(\cos^{2007} x)\right)$$

and so, the answer to the original question is the current year, 2007.

02 Let $\int_0^1 \frac{1}{t^2 + 2t\cos a + 1}\, dt = p$ and $\int_{-3}^{3} \frac{t^2 \sin 2t}{t^2+1}\, dt = q$

$t^2 + 2t \cos a + 1 = t^2 + 2t \cos a + \cos^2 a + \sin^2 a = (t + \cos a)^2 + \sin^2 a > 0$
$\forall\, t \in (0, 1)$

$$\Rightarrow \quad \frac{1}{t^2 + 2t\cos a + 1} > 0 \quad \forall\, t \in (0, 1)$$

$$\Rightarrow \quad \int_0^1 \frac{1}{t^2 + 2t\cos a + 1}\, dt > 0 \Rightarrow p > 0$$

and $\frac{t^2 \sin 2t}{t^2+1}$ is an odd function $\Rightarrow \int_{-3}^{3} \frac{t^2 \sin 2t}{t^2+1}\, dt = 0 \Rightarrow q = 0$

Now given quadratic become $px^2 + 2 = 0 \Rightarrow$ no real solution for x.

03 $A = \int_0^{\pi}\left(\frac{x}{1+\sin x}\right)^2 dx$; $I = \int_0^{\pi} \frac{x^2(1+\cos x)}{(1+\sin x)^2}$; $I = A + \int_0^{\pi} \frac{\underset{I}{x^2}\ \underset{II}{(\cos x)}}{(1+\sin x)^2}$;

$$A + \left(\frac{-x^2}{1+\sin x}\right)_0^{\pi} + 2\underbrace{\int_0^{\pi} \frac{x}{1+\sin x} dx}_{I'} \ ; \qquad I' = \int_0^{\pi} \frac{x}{1+\sin x} dx$$

using king $\quad I' = \int_0^{\pi} \frac{\pi - x}{1+\sin x} dx$; $2I' = 2\pi \int_0^{\pi/2} \frac{dx}{1+\sin x}$

$$I' = \pi \int_0^{\pi/2} \frac{1-\sin x}{\cos^2 x} dx \ ; \ I' = \pi\, (\tan x - \sec x)_0^{\pi/2} \ ; \ I' = \pi \left[\lim_{x\to\pi/2}(\tan x - \sec x) - (0-1)\right] = \pi$$

$\therefore \quad I = A + 2\pi - \pi^2$

04. $I = \int_0^1 \sqrt{\frac{1-x}{1+x}}\,dx$

Put $x = \cos 2y$ to get

$$I = \int_0^{\frac{\pi}{4}} 2\cos 2y \frac{\sin y}{\cos y} \sin 2y\,dy = \int_0^{\frac{\pi}{4}} 4\sin^2 y\,dy - \frac{1}{4}\int_0^{\frac{\pi}{4}} 32\sin^4 y\,dy$$

$$= \left(2y - \sin 2y\Big|_0^{\pi/4}\right) - \frac{1}{4}\left(12y - 8\sin 2y + \sin 4y\Big|_0^{\pi/4}\right) = 1 - \frac{\pi}{4}$$

05. $I = \int_0^{\frac{\pi}{4}} \frac{\sin\theta(\sin\theta\cos\theta + 2)}{\cos^4\theta}\,d\theta = \int_0^{\frac{\pi}{4}} \left(2\sec^2\theta + \sqrt{\sec^2\theta - 1}\right)\sec\theta\tan\theta\,d\theta$

for $\sec\theta = t$

$$I = \int_1^{\sqrt{2}} 2t^2 + \sqrt{t^2-1}\,dt = \left|\frac{2t^3}{3} + \frac{t}{2}\sqrt{t^2-1} - \frac{1}{2}\ln\left(t + \sqrt{t^2-1}\right)\right|_1^{\sqrt{2}}$$

$$= \frac{2}{3}\left(2\sqrt{2} - 1\right) + \frac{1}{\sqrt{2}} - \frac{1}{2}\ln\left(\sqrt{2} + 1\right).$$

06 $I = \frac{1}{2}\int_0^{\frac{\pi}{2}} x\,|\cos 2x|\,dx \quad$ put $2x = t$

$$I = \frac{1}{8}\int_0^{\pi} t\,|\cos t|\,dt \qquad ....(1)$$

$$I = \frac{1}{8}\int_0^{\pi} (\pi - t)|\cos t|\,dt \quad ....(2)$$

$$2I = \frac{\pi}{8}\int_0^{\pi} |\cos t|\,dt = \frac{\pi}{8}\times 2 \qquad \Rightarrow \qquad I = \frac{\pi}{8} \text{ Ans.}$$

07. [Sol.$_{85/def/QZ}$ $\int_0^2 \left(x\,f(x) - f^2(x)\right)dx = \frac{2}{3}$

$$= \int_0^2 \left(\frac{x^2}{4} - \left(f(x) - \frac{x}{2} \right)^2 \right) dx = \frac{2}{3} \quad = \int_0^2 \frac{x^2}{4} dx - \int_0^2 \left(f(x) - \frac{x}{2} \right)^2 dx = \frac{2}{3}$$

$$= \frac{2}{3} - \int_0^2 \left(f(x) - \frac{x}{2} \right)^2 dx = \frac{2}{3}$$

$$\therefore \quad \int_0^2 \left(f(x) - \frac{x}{2} \right)^2 dx = 0 \Rightarrow \quad f(x) = \frac{x}{2}; \quad \therefore \quad f(1) = \frac{1}{2}$$ **Ans.**]

08

(A) $$\left(\frac{\sin^2 3x}{\sin^2 x} - \frac{\cos^2 3x}{\cos^2 x} \right) = \left(\frac{\sin^2 3x \cos^2 x - \sin^2 x \cos^2 3x}{\sin^2 x \cos^2 x} \right)$$

$$= \left(\frac{(\sin 3x \cos x + \sin x \cos 3x)(\sin 3x \cos x - \sin x \cos 3x)}{\sin^2 x \cos^2 x} \right) = \frac{4 \sin 4x \sin 2x}{\sin^2 2x} = 8 \cos 2x$$

$$\int_{\pi/4}^{\pi/3} \left(\frac{\sin^2 3x}{\sin^2 x} - \frac{\cos^2 3x}{\cos^2 x} \right) dx = \int_{\pi/4}^{\pi/3} 8 \cos 2x \, dx = 4 \sin 2x \Big|_{\pi/4}^{\pi/3} = 2\sqrt{3} - 4$$ **Ans.**

(B) $$\frac{1}{1 + \sin 2x} = \frac{1}{1 + \frac{2 \tan x}{1 + \tan^2 x}} = \frac{\sec^2 x}{(1 + \tan x)^2}$$

Let $u = \frac{\tan x - 1}{\tan x + 1} \Rightarrow du = \frac{(1 + \tan x)\sec^2 x - (1 - \tan x)\sec^2 x}{(1 + \tan x)^2} = \frac{2 \sec^2 x}{(1 + \tan x)^2}$

$$\therefore \quad \int_{\pi/2}^{\pi/4} \frac{1}{1 + \sin 2x} \sqrt[2009]{\frac{\tan x - 1}{\tan x + 1}} \, dx = -\frac{1}{2} \int_0^1 u^{\frac{1}{2009}} du = \frac{-1}{2} \times \frac{2009}{2010}$$ **Ans.**

(C) put $x = \tan \theta$

$$\Rightarrow \quad I = \int \frac{x^2}{(x \sin x + \cos x)^2} dx = \int \frac{\tan^2 \theta}{\cos^2(\tan \theta - \theta)} d\theta; \quad \text{put } \tan \theta - \theta = y$$

$$\Rightarrow \quad \int \sec^2 y \, dy = \tan y = \tan(\tan \theta - \theta) = \tan\left(x - \tan^{-1} x\right) = \frac{\tan x - x}{1 + x \tan x}$$

$$\frac{\tan x - x}{1 + x \tan x} \Big|_0^{\pi/2} = \frac{1}{\pi/2} - 0 = \frac{2}{\pi}$$ **Ans.**

(D) $$\int_1^e \frac{\ln x}{x\left(\sqrt{1 - \ln x} + \sqrt{\ln x + 1}\right)} dx$$

rationalising the denominator, we get

$$\int_1^e \frac{\ln x\left(\sqrt{1-\ln x}-\sqrt{\ln x+1}\right)}{-2x\ln x}dx = \int_1^e \frac{\ln x\left(\sqrt{1-\ln x}-\sqrt{\ln x+1}\right)}{-2x}dx$$

now put $t = \ln x \quad \Rightarrow \quad dt = \frac{1}{x}dx$

At $x = 1,\ t = 0$
at $x = e,\ t = 1$

$$= \frac{1}{2}\int_0^1 \left(\sqrt{1+t}-\sqrt{1-t}\right)dt = \frac{1}{3}\left((1+t)^{3/2}\Big|_0^1 + (1-t)^{3/2}\Big|_0^1\right) = \frac{2\sqrt{2}-2}{3} \quad \textbf{Ans.}$$

09 $I = \int_0^{2\pi} \underbrace{\frac{\sin x}{9-\cos^2 x}}_{II}\cdot \underbrace{x^2}_{I}\,dx$ (using integration by parts)

$$I = -\frac{x^2}{6}\ln\frac{3+\cos x}{3-\cos x}\Bigg|_0^{2\pi} + \underbrace{\frac{2}{6}\int_0^{2\pi} x\ln\frac{3+\cos x}{3-\cos x}dx}_{I_1} = -\frac{2\pi^2}{3}\ln 2 + \frac{1}{3}I_1$$

now $\quad I_1 = \int_0^{2\pi} x\ln\frac{3+\cos x}{3-\cos x}dx$

$$I_1 = \int_0^{2\pi} (2\pi - x)\ln\frac{3+\cos x}{3-\cos x}dx$$

$$2I_1 = 2\pi\int_0^{2\pi} \ln\frac{3+\cos x}{3-\cos x}dx = 4\pi\int_0^{\pi} \ln\frac{3+\cos x}{3-\cos x}dx$$

$$I_1 = 2\pi\int_0^{\pi} \ln\frac{3+\cos x}{3-\cos x}dx$$

$$I_1 = 2\pi\int_0^{\pi} \ln\frac{3-\cos x}{3+\cos x}dx$$

$$2I_1 = 2\pi\int_0^{\pi} 0\,dx = 0 \quad \Rightarrow \quad I = -\frac{2\pi^2}{3}\ln 2 \ \text{Ans.}$$

Method 2: $\quad I = \int_0^{2\pi} \frac{x^2\sin x}{8+\sin^2 x}dx = \int_0^{2\pi} -\frac{(2\pi - x)^2\sin x}{9-\cos^2 x}dx$

$$2I = \int_0^{2\pi} \frac{\sin x(x^2-(2\pi-x)^2)}{8+\sin^2 x}dx = 2\pi\int_0^{2\pi}\frac{2(x-\pi)\sin x}{8+\sin^2 x}dx$$

$$I = 2\pi\int_0^{2\pi}\frac{(x-\pi)\sin x}{8+\sin^2 x}dx = 2\pi\underbrace{\int_0^{2\pi}\frac{x\sin x}{8+\sin^2 x}dx}_{I_1} - 2\pi^2\underbrace{\int_0^{2\pi}\frac{\sin x}{8+\sin^2 x}dx}_{I_2=\text{zero}}$$

$$I_1 = \int_0^{2\pi}\underbrace{x}_{I}\,\underbrace{\frac{\sin x}{9-\cos^2 x}}_{II}dx = -\frac{x}{6}\,ln\frac{3+\cos x}{3-\cos x}\Bigg|_0^{2\pi} + \underbrace{\frac{1}{6}\int_0^{2\pi} ln\frac{3+\cos x}{3-\cos x}dx}_{\text{zero}}$$

$$I_1 = -\frac{\pi}{3}ln\left(\frac{4}{2}\right) + \frac{2}{6}\underbrace{\int_0^{\pi} ln\frac{3+\cos x}{3-\cos x}dx}_{\text{zero}};$$

$$I = 2\pi\cdot I_1 = 2\pi\cdot\left(-\frac{\pi}{3}ln\left(\frac{4}{2}\right)\right) = -\frac{2\pi^2}{3}\,ln\,2\ \text{Ans.}$$

10. $$\tan^{-1}\sqrt{\sqrt{x}-1}\cdot x\Big]_1^{16} - \int_1^{16}\frac{dx}{1+\left(\sqrt{x}-1\right)}\cdot\frac{x}{2\sqrt{\sqrt{x}-1}}\cdot\frac{1}{2\sqrt{x}} - \int_1^{16}\frac{dx}{4x\sqrt{\sqrt{x}-1}}$$

$\sqrt{x}-1 = t^2;\quad \frac{1}{2\sqrt{x}}dx = 2t\,dt$

$$-\frac{1}{4}\int_0^{\sqrt{3}}\frac{2t\,dt}{(1+t^2)\,t}.$$

11. $$I = \int_0^{2\pi}\frac{dx}{2+\sin 2x} = \int_0^{\pi}\frac{dx}{2+\sin 2x} + \int_0^{\pi}\frac{dx}{2-\sin 2x} = \int_0^{\pi}\frac{4}{4-\sin^2 2x}$$

(using Queen and adding)

$$= 8\int_0^{\pi/2}\frac{dx}{4-\sin^2 2x}$$

put $2x = t \quad\Rightarrow\quad dx = \frac{dt}{2}$

$$I = 4\int_0^{\pi}\frac{dt}{4-\sin^2 t} = 8\int_0^{\pi/2}\frac{dt}{4-\sin^2 t}\ ]$$

12. $$\int_0^{\pi/2} (\sin x + a\cos x)^3 dx - \frac{4a}{\pi-2}\int_0^{\pi/2} x\cos x\, dx = 2$$

let $$I_1 = \int_0^{\pi/2} (\sin x + a\cos x)^3 dx = \int_0^{\pi/2} (\sin^3 x + a^3\cos^3 x + 3a\sin^2 x\cos x + 3a^2 \sin x\cos^2 x)dx$$

$$= \int_0^{\pi/2} \sin^3 x\, dx + a^3\int_0^{\pi/2}\cos^3 x\, dx + 3a\int_0^{\pi/2}\sin^2 x\cos x\, dx + 3a^2\int_0^{\pi/2}\sin x\cos^2 x\, dx$$

$$= \frac{2}{3} + a^3\left(\frac{2}{3}\right) + 3a\int_0^1 t^2\, dt + 3a^2\int_0^1 t^2\, dt \qquad \left(\sin x = t \Rightarrow \cos x\, dx = dt;\ \int_0^1 t^2\, dt\right)$$

$$= \frac{2}{3}(1+a^3) + a + a^2 = \frac{2}{3} + \frac{2a^3}{3} + a + a^2$$

$$I_1 = \frac{2a^3}{3} + a^2 + a + \frac{2}{3}$$

$$I_2 = \int_0^{\pi/2} \underbrace{x}_{\mathrm{I}}\cdot\underbrace{\cos x}_{\mathrm{II}}\, dx = x\sin x\Big|_0^{\pi/2} - \int_0^{\pi/2}\sin x\, dx$$

$$I_2 = x\sin x + \cos x\Big|_0^{\pi/2}$$

$$I_2 = \frac{\pi}{2} - 1 = \frac{\pi-2}{2}$$

$$I = \frac{2a^3}{3} + a^2 + a + \frac{2}{3} - \frac{4a}{\pi-2}\cdot\frac{\pi-2}{2} \qquad \Rightarrow \qquad \frac{2a^3}{3} + a^2 - a + \frac{2}{3} = 2$$

$\Rightarrow$ $2a^3 + 3a^2 - 3a + 2 = 6$

$\Rightarrow$ $2a^3 + 3a^2 - 3a - 4 = 0$

$$a_1 + a_2 + a_3 = -\frac{3}{2}$$

$\Rightarrow$ $$\sum a_1a_2 = -\frac{3}{2}$$

$\therefore$ $$\sum a_1^2 = \frac{9}{4} + \frac{6}{2} = \frac{21}{4}$$

$\therefore$ $$1000\sum a_1^2 = 1000 \times \frac{21}{4} = 250 \times 21 = 5250$$ **Ans.**

13. Let $I = \int_{\frac{\pi}{4}}^{\frac{65\pi}{4}} \frac{dx}{(1+2^{\cos x})(1+2^{\sin x})} = \int_{\frac{\pi}{4}}^{16\pi+\frac{\pi}{4}} \frac{dx}{(1+2^{\cos x})(1+2^{\sin x})}$

$$= \int_0^{16\pi} \underbrace{\frac{dx}{(1+2^{\cos x})(1+2^{\sin x})}}_{\text{periodic with period } =2\pi} = 8\int_0^{2\pi} \frac{dx}{(1+2^{\cos x})(1+2^{\sin x})} = 8I_1$$

where $I_1 = \int_0^{2\pi} \frac{dx}{(1+2^{\cos x})(1+2^{\sin x})}$

Using King and add, $2I_1 = \int_0^{2\pi} \frac{dx}{(1+2^{\cos x})}$

Again using King and add, $4I_1 = \int_0^{2\pi} dx = 2\pi$

Hence $I_1 = \frac{\pi}{2}$

$\therefore$ $I = 8I_1 = 8 \times \frac{\pi}{2} = 4\pi \equiv k\pi.$ Hence $k = 4$. **Ans.**

14. Given $\int_0^x t^2 \underset{\substack{\downarrow \\ \text{King}}}{\sin(x-t)}\, dt = x^2$

$\Rightarrow$ $\int_0^x (x-t)^2 \sin t\, dt = x^2$

$\Rightarrow$ $x^2 \int_0^x \sin t\, dt - 2x \int_0^x t \sin t + \int_0^x t^2 \sin t = x^2$

$\Rightarrow$ $x^2 (1 - \cos x) - 2x(-x\cos x + \sin x) + (-x^2 \cos x + 2x \sin + \cos x - 1) = x^2$

$\Rightarrow$ $\cos x = 1$

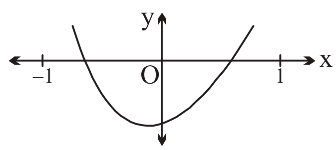
So, $x = 2n\pi$; $n \in I$

Hence, number of values of $0, 2\pi, 4\pi, \ldots\ldots, 30\pi$ equals 16. **Ans.**

15. Let $F(x) = 2x + \int_0^x f(t)dt - p$ in $(-1, 1)$

Now $F(0) = -p$,

$F(1^-) = 2 - p + \int_0^1 f(t)\, dt = 2 - p + 5 = 7 - p$



Graph of F(x)

$$F(-1^+) = -2 - p + \int_0^{-1} f(t)\,dt = -2 - p + 5 = 3 - p$$

$F(-1)\,F(0) < 0$ and $F(0)\,F(1) < 0$

i.e. $(3-p)(-p) < 0$ and $(-p)(7-p) < 0$

$p(p-3) < 0$ and $p(p-7) < 0$

$\Rightarrow$ $p \in (0, 3)$ and $p \in (0, 7)$

$\Rightarrow$ $p \in (0, 3)$

$\therefore$ Number of integral values of p is 2. **Ans.**

16. $$l = \lim_{x\to\infty} \frac{\int_0^x t^n \sqrt{t+1}\,dt}{x^{\left(\frac{3}{2}+n\right)}}$$ Using lopital's rule

$$l = \lim_{x\to\infty} \frac{x^n\sqrt{x+1}}{\left(\frac{3}{2}+n\right)x^{\left(n+\frac{1}{2}\right)}} = \lim_{x\to\infty} \frac{2\sqrt{x+1}}{(2x+3)\sqrt{x}} = \lim_{x\to\infty} \frac{2}{2n+3}\sqrt{1+\frac{1}{x}}$$

$$l = \frac{2}{2n+3} = \frac{2}{11} \qquad \therefore\ 2n+3 = 11 \Rightarrow n = 4.$$ **Ans.**

17. $$I = \int_{-1}^{1} \frac{1}{1+x+x^2+\sqrt{x^4+3x^2+1}}\,dx = \int_{-1}^{1} \frac{\sqrt{x^4+3x^2+1}-(x^2+x+1)}{x^4+3x^2+1-(x^2+x+1)^2}\,dx$$

$$I = \int_{-1}^{1} \frac{\sqrt{x^4+3x^2+1}-(x^2+x+1)}{-2x(x^2+1)}\,dx$$

$$I_1 = \int_{-1}^{1} \frac{\sqrt{x^4+3x^2+1}}{-2x(x^2+1)}\,dx = \int_{-1}^{1} \frac{x\sqrt{(x^2+1)(x^2+2)}}{-2x^2(x^2+1)}\,dx = \int_{-1}^{1} \frac{-1}{2x^2}\sqrt{\frac{x^2+2}{x^2+1}}\;x\,dx = 0$$

After the substitution $\sqrt{x^2+1} = t$

$$I_2 = \int_{-1}^{1} \frac{x^2+x+1}{2x(x^2+1)}\,dx = \frac{1}{2}\Big[\ln|x| + \tan^{-1}x\Big]_{-1}^{1} = \frac{\pi}{4}.$$ **Ans.**

`

18. Put $x = \sin 2t \Rightarrow dx = 2\cos 2t$

now $I = \int_0^{\pi/4} \frac{2(2\cos^2 t - 1)\,dt}{\cos t + \sin t + \cos t - \sin t + 2} = \int_0^{\pi/4} \frac{2\cos^2 t - 1}{\cos t + 1}dt = \int_0^{\pi/4} \frac{1 - 2(1-\cos^2 t)}{(1+\cos t)}dt$

$= 2\int_0^{\pi/4} (\cos t - 1)\,dt + \int_0^{\pi/4} \frac{1}{2\cos^2(t/2)}dt$

$= 2\,[\sin t - t]_0^{\pi/4} + \left.\frac{1}{2}\cdot 2\cdot \tan\frac{t}{2}\right]_0^{\pi/4} = 2\left[\frac{1}{\sqrt{2}} - \frac{\pi}{4}\right] + \tan\frac{\pi}{8} = \sqrt{2} - \frac{\pi}{2} + \sqrt{2} - 1$

$= 2\sqrt{2} - \frac{\pi}{2} - 1 \Rightarrow a = b = c = 2 \Rightarrow a + b + c = 6$ **Ans.**

19 $I_n = \int_0^1 \underset{II}{1} \cdot \underset{I}{\left(1+x^2\right)^n} dx = \left.\left(1+x^2\right)^n \cdot x\right|_0^1 - 2n\int_0^1 \left(1+x^2\right)^{n-1} x^2\,dx$

$= 2^n - 2n\int_0^1 \left(1+x^2\right)^{n-1} (1 + x^2 - 1)\,dx$

$I_n = 2^n - 2n\left[\int_0^1 \left(1+x^2\right)^n dx - \int_0^1 \left(1+x^2\right)^{n-1} dx\right]$

$I_n = 2^n - 2n\,I_n + 2n\,I_{n-1}$

$(2n+1)\,I_n = 2^n + 2n \cdot I_{n-1}$

or $(2n+1)\,I_n - 2n\,I_{n-1} = 2^n$

Put $n = 6$,

$\therefore$ $13\,I_6 - 12\,I_5 = 2^6 = 64$. **Ans.**

20. As we know that $\int_a^b f(x)\,dx + \int_c^d f^{-1}(x)\,dx = bd - ac$, where f(x) is increasing and invertible function.

Also $f(b) = d$, $f(a) = c$.

Now, $g(x) = 2 + 2^{\sin x}$, then $g^{-1}(x) = \sin^{-1}\left(\log_2(x-2)\right)$

Hence, $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (2 + 2^{\sin x})\,dx + \int_{\frac{5}{2}}^{4} \sin^{-1}\left(\log_2(x-2)\right)dx = 2\pi + \frac{5\pi}{4}$

$\Rightarrow$ $2\pi + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2^{\sin x}\,dx + \int_{\frac{5}{2}}^{4} \sin^{-1}\left(\log_2(x-2)\right)dx = 2\pi + \frac{5\pi}{4}$

$$\Rightarrow \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2^{\sin x}\, dx + \int_{\frac{5}{2}}^{4} \sin^{-1}\left(\log_2(x-2)\right) dx = \frac{5\pi}{4}.$$ **Ans.**

21. Let $f(x) = I = \int_{x}^{\frac{\pi}{4}-x} \frac{\ln(1+\tan t)\,dt}{\ln 4}\,dt$ ....(1) **[**

Using King, $I = \frac{1}{\ln 4}\int_{x}^{\frac{\pi}{4}-x} \ln\left(1+\tan\left(\frac{\pi}{4}-t\right)\right) dt$

$\therefore \quad I = \frac{1}{\ln 4}\int_{x}^{\frac{\pi}{4}-x} \ln\left(\frac{2}{1+\tan t}\right) dt$ ....(2)

(1) and (2) $\Rightarrow \quad 2I = \frac{1}{\ln 4}\int_{x}^{\frac{\pi}{4}-x} \ln 2\, dt$

$\therefore \quad 2I = \frac{1}{2}\int_{x}^{\frac{\pi}{4}-x} dt$

$\therefore \quad I = \frac{1}{4}\left(\frac{\pi}{4}-2x\right)$

$\Rightarrow f(x) = \frac{\pi}{16} - \frac{x}{2} = \frac{\pi - 8x}{16} \Rightarrow \frac{\pi - 16 f(x)}{x} = 8.$ **Ans.**

22. $\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} k \ln \frac{\left(n^2+(k-1)^2\right)}{n^2+k^2}$

$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} k \left[\ln (n^2+(k-1)^2 - \ln (n^2+k^2)\right]$

$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} k \ln \left(n^2+(k-1)^2\right) - k \ln (n^2+k^2)$

$\lim_{n\to\infty} \frac{1}{n}\left[\sum_{k=0}^{n-1} (k+1)\ln (n^2+k^2) - \sum_{k=0}^{n} k \ln (n^2+k^2)\right]$

$\lim_{n\to\infty} \frac{1}{n}\left[\sum_{k=0}^{n-1} \ln(n^2+k^2) + \sum_{k=0}^{n-1} k \ln (n^2+k^2) - \sum_{k=0}^{n-1} k \ln (n^2+k^2) - n \ln (n^2+n^2)\right]$

$$\lim_{n\to\infty}\frac{1}{n}\left[\sum_{k=0}^{n-1} ln(n^2+k^2) - n\, ln\,(n^2+n^2)\right]$$

$$\lim_{n\to\infty}\frac{1}{n}\left[\sum_{k=0}^{n-1} ln(n^2+k^2) - \sum_{k=0}^{n-1} ln\,(n^2+n^2)\right]$$

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=0}^{n-1} ln\left(\frac{n^2+k^2}{n^2+n^2}\right) = \lim_{n\to\infty}\frac{1}{n}\sum_{k=0}^{n-1} ln\left(\frac{1+\frac{k^2}{n^2}}{2}\right)$$

$$\therefore \qquad S = \int_0^1 \underbrace{1}_{II}.\, ln\underbrace{\left(\frac{1+x^2}{2}\right)}_{I} dx = x\, ln\left(\frac{1+x^2}{2}\right)\Bigg]_0^1 - \int_0^1 \frac{2x^2}{1+x^2}dx$$

$$= 0 - 2\left[\int_0^1 \frac{x^2+1}{1+x^2}dx - \int_0^1 \frac{dx}{1+x^2}\right] = -2\left[x - \tan^{-1}x\right]_0^1 = -2\left[1-\frac{\pi}{4}\right] = \frac{\pi}{2} - 2$$

23.

(A) $$\int_0^1 \frac{x^3 - x^2}{\sqrt[3]{3x^4 - 4x^3 - 1}}dx$$

put $u^3 = 3x^4 - 4x^3$

$$\frac{1}{4}\int_0^{-1}\frac{u^2}{u-1}du = -\frac{1}{4}\int_{-1}^{0}\left[u+1+\frac{1}{u-1}\right]du = -\frac{1}{4}\left[\frac{1}{2}u^2 + u + \ln|u-1|\right]_{-1}^{0} = \frac{1}{4}\left[\frac{1}{2}u - \ln 2\right]$$

$$= \frac{1}{4}ln(2) - \frac{1}{8}$$ **Ans.**

(B) $$\int_0^{\pi/2}\frac{\sin 2x}{3+4\sin x - \cos 2x}dx = \int_0^{\pi/2}\frac{\sin 2x}{2\sin^2 x + 4\sin x + 2}dx = \int_0^{\pi/2}\frac{\sin x\cos x}{\sin^2 x + 2\sin x + 1}dx$$

$$= \int_0^{\pi/2}\frac{\sin x\cos x}{(\sin x + 1)^2}dx = \int_1^2 \frac{u-1}{u^2}du \quad (\text{where } u = \sin x + 1)$$

$$= \left[\ln|u| + \frac{1}{u}\right]_1^2 = ln\,(2) - \frac{1}{2}$$ **Ans.**

(C) $\displaystyle\int_0^2 \frac{x^4 - x + 1 + 16 - 16}{x^2+4}dx = \int_0^2 \frac{(x^4-16)+17-x}{x^2+4}dx = \int_0^2 (x^2-4)\,dx - \int_0^2 \frac{x}{x^2+4}dx + \int_0^2 \frac{17}{x^2+4}dx$

$$\left(\frac{x^3}{3} - 4x\right)_0^2 - \frac{1}{2}\left(\ln(x^2+4)\right)_0^2 + \left(\frac{17}{2}\tan^{-1}\left(\frac{x}{2}\right)\right)_0^1 = \frac{17\pi}{8} - \frac{1}{2}\ln 2 - \frac{16}{3} \textbf{ Ans.}$$

24. Put $x = \sin\theta \Rightarrow dx = \cos\theta\, d\theta$

$$I = \frac{1}{2}\int_0^{\pi/2} \ln(\sin\theta)\cdot 2\cos^2\theta\, d\theta = \frac{1}{2}\int_0^{\pi/2} \ln(\sin\theta)\cdot(1+\cos 2\theta)\, d\theta$$

$$= \frac{1}{2}\int_0^{\pi/2} \ln(\sin\theta)\, d\theta + \frac{1}{2}\int_0^{\pi/2} \ln(\sin\theta)\cos 2\theta\, d\theta$$

$$I = -\frac{\pi}{4}\ln 2 + \frac{1}{2} I_1 \qquad \left(\text{As } \int_0^{\pi/2} \ln(\sin\theta)\, d\theta = \frac{-\pi}{2}\ln 2\right)$$

$$I_1 = \int_0^{\pi/2} \underbrace{\ln(\sin\theta)}_{\text{I}}\underbrace{\cos 2\theta}_{\text{II}}\, d\theta = \underbrace{\frac{1}{2}\ln(\sin\theta)\sin 2\theta\ \Big]_0^{\pi/2}}_{\text{vanishes}} - \frac{1}{2}\int_0^{\pi/2} \frac{\cos\theta}{\sin\theta}2\sin\theta\cos\theta\, d\theta$$

$$= 0 - \int_0^{\pi/2} \cos^2\theta\, d\theta = -\frac{\pi}{4}$$

$$\left(\text{As } \lim_{\theta\to 0^+} \frac{\ln(\sin\theta)}{\operatorname{cosec} 2\theta}\left(\frac{\infty}{\infty}\text{ form}\right) = \lim_{\theta\to 0^+} \frac{\cot\theta}{-2\operatorname{cosec} 2\theta\cot 2\theta} = \lim_{\theta\to 0^+} \frac{\tan 2\theta\sin 2\theta}{-2\tan\theta} = 0\right)$$

$$I = -\frac{\pi}{4}\ln 2 - \frac{\pi}{8} = \frac{-\pi}{4}\left(\ln 2 + \frac{1}{2}\right) \qquad \text{Ans.}$$

25. [Sol.$_{\text{51055/def/OMR}}$ $\displaystyle I = \int_{-\pi+a}^{3\pi+a} |x - a - \pi| \sin\left(\frac{x}{2}\right)dx$

Put $(x - a - \pi) = t$

$$I = \int_{-2\pi}^{2\pi} |t| \sin\left(\frac{\pi}{2} + \frac{a+t}{2}\right)dt = \int_{-2\pi}^{2\pi} |t| \cos\left(\frac{a+t}{2}\right)dt$$

$$= \underbrace{\cos\frac{a}{2}\int_{-2\pi}^{2\pi} |t|\cos\left(\frac{t}{2}\right)dt}_{\text{even}} - \underbrace{\sin\frac{a}{2}\int_{-2\pi}^{2\pi} |t|\sin\left(\frac{t}{2}\right)dt}_{\text{zero being odd}}$$

$$I = 2\cos\frac{a}{2}\int_0^{2\pi} t\cos\frac{t}{2}\,dt$$

Put $\frac{t}{2} = y$, we get

$$I = 8\cos\frac{a}{2}\int_0^{\pi} \underset{\mathrm{I}}{y}\,\underbrace{\cos y}_{\mathrm{II}}\,dy = 8\cos\frac{a}{2}\left[\underbrace{y\sin y\Big|_0^{\pi}}_{\text{zero}} - \int_0^{\pi}\sin y\,dy\right]$$

$\therefore \quad I = -16\cos\frac{a}{2} \Rightarrow -16\cos\frac{a}{2} = -16$

$\therefore \quad \cos\frac{a}{2} = 1 \Rightarrow \frac{a}{2} = 2n\pi \Rightarrow a = 4n\pi,\ n \in I$

Sum $= \pi[4 + 8 + 12 + \ldots\ldots.. + 96] = \pi\left[\frac{24}{2}\times 100\right] = \pi[24 \times 50] \Rightarrow k = 1200$ **Ans.**

26. $f(\cos x) = \cos 3x = 4\cos^3 x - 3\cos x$

$\cos x = t;\quad t \in [-1, 1]$

$f(t) = 4t^3 - 3t;\ t \in [-1, 1]$

$\therefore\ J = \int_0^1 (4t^3 - 3t)^2\sqrt{1-t^2}\,dt \qquad$ put $t = \cos\theta$

$$J = \int_0^{\frac{\pi}{2}} \cos^2 3\theta\cdot\sin^2\theta\,d\theta \qquad \ldots(1)$$

$$J = \frac{1}{4}\int_0^{\frac{\pi}{2}} (2\sin\theta\cos 3\theta)^2\,d\theta = \frac{1}{4}\int_0^{\frac{\pi}{2}} (\sin 4\theta - \sin 2\theta)^2\,d\theta$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} (\sin^2 4\theta + \sin^2 2\theta - 2\sin 2\theta\sin 4\theta)\,d\theta$$

$$J = \frac{1}{4}\int_0^{\frac{\pi}{2}}\sin^2 4\theta\,d\theta + \frac{1}{4}\int_0^{\frac{\pi}{2}}\sin^2 2\theta\,d\theta - \underbrace{\frac{2}{4}\int_0^{\frac{\pi}{2}}\sin 2\theta\cdot\sin 4\theta\,d\theta}_{\text{using King and add} \Rightarrow \text{zero}}$$

$$J = \frac{1}{4}\cdot\frac{1}{4}\int_0^{2\pi}\sin^2 t\,dt + \frac{1}{4}\cdot\frac{1}{2}\int_0^{\pi}\sin^2 t\,dt$$

$$J = \frac{1}{4}\int_0^{\frac{\pi}{2}} \sin^2 t\,dt + \frac{1}{4}\int_0^{\frac{\pi}{2}} \sin^2 t\,dt = \frac{1}{4}\cdot\frac{\pi}{4} + \frac{1}{4}\cdot\frac{\pi}{4} = \frac{\pi}{8}$$

$$\therefore \frac{(2016)\,J}{\pi} = \frac{2016}{\pi}\cdot\frac{\pi}{8} = 252 \textbf{ Ans.}$$

27.

(A) $$\int_0^1 \left(\frac{x-1}{x^2+1}\right)^2 e^x\,dx = \int_0^1 \left(\frac{x^2-2x+1}{(x^2+1)^2}\right) e^x\,dx = \int_0^1 \left(\frac{1}{x^2+1} - \frac{2x}{(x^2+1)^2}\right) e^x\,dx = e^x\left(\frac{1}{x^2+1}\right)_0^1 = \frac{e}{2} - 1$$

(B) $$\int_{-1}^{0} x\left(e^{2x} + \sqrt[3]{x+1}\right)dx = \int_{-1}^{0} xe^{2x}\,dx + \int_{-1}^{0} x\,\sqrt[3]{x+1}\,dx$$

Let $I_1 = \int_{-1}^{0} xe^{2x}\,dx$ and $I_2 = \int_{-1}^{0} x\,\sqrt[3]{x+1}\,dx$

Integrating I_1 by parts, we get

$$\left.\frac{xe^{2x}}{2}\right|_{-1}^{0} - \left.\frac{e^{2x}}{4}\right|_{-1}^{0} = \frac{3-e^2}{4e^2}$$

compute I_2 and add $I_1 + I_2$, we get $\dfrac{21-16e^2}{28e^2}$ Ans.

(D) $$\int_0^{\pi/4} \left(\tan x + e^{\sin x}\cos x\right)dx = \int_0^{\pi/4} \frac{\sin x\,dx}{doxx} + \int_0^{\pi/4} e^{\sin x}\cos x\,dx = \int_1^{1/\sqrt{2}} -\frac{du}{u} + \int_0^{1/\sqrt{2}} e^v\,dv$$

$$= \left[ln 1 - ln\left(\frac{1}{\sqrt{2}}\right)\right] + \left[e^{\frac{1}{\sqrt{2}}} - e^0\right] = \frac{1}{2} ln(2) + \sqrt{e^{\sqrt{2}}} - 1 \textbf{ Ans.}$$

28. Given,

$$\int_0^1 f(x)\,dx = \frac{1}{3} + \int_0^1 \left(f(x^2)\right)^2 dx \qquad \ldots (1)$$

Consider,

$$I_1 = \int_0^1 \left(f(x^2)\right)^2 dx$$

Put $\quad x^2 = t \Rightarrow 2x\,dx = dt$

$$= \frac{1}{2}\int_0^1 \frac{(f(t))^2}{\sqrt{t}}\,dt$$

$\therefore$ Equation (1) becomes

$$\int_0^1 \frac{t}{2\sqrt{t}}\,dt + \int_0^1 \frac{(f(t))^2 - 2\sqrt{t}\,f(t)}{2\sqrt{t}}\,dt \Rightarrow \int_0^1 \left(f(t) - \sqrt{t}\right)^2 dt = 0$$

$\therefore$ $f(t) = \sqrt{t}$

So, $f\left(\frac{1}{4}\right) = \frac{1}{2}$

So, reciprocal of $f\left(\frac{1}{4}\right) = 2$

29. $I_1 = \int_0^1 \frac{3^x\,dx}{3+x}$

Now, $I_2 = \int_0^1 \frac{x^3 dx}{3^{x^4} \times (4 - x^4)}$

Let $x^4 = 7 \Rightarrow 4x^3 dx = dt$

$$I^2 = \frac{1}{4}\int_0^1 \frac{dt}{3^t \times (4-t)} = \frac{1}{4}\int_0^1 \frac{dt}{3^{1-t}(4-(1-t))}$$

$$= \frac{1}{12}\int_0^1 \frac{3^t\,dt}{3+t} = \frac{1}{12}\int_0^1 \frac{3^x\,dx}{3+x} = \frac{1}{12} I_1$$

$\frac{I_1}{I_2} = 12$ Ans.

30. We have $f(x) = \begin{bmatrix} \{x\} - \frac{1}{2}\,; & x \in I \\ 0\,; & x \in I \end{bmatrix}$

Clearly, f(x) is periodic with period = 1.

Now, $\int_0^{24} P(x)\,dx = \int_0^{24} \underbrace{1}_{II} \cdot \underbrace{P(x)}_{I}dx = P(x)\cdot x\Big|_0^{24} - \int_0^{24} x\,P'(x)dx = 24\,P(24) - \int_0^{24} x\,P'(x)dx$
(I.B.P)

As, $P(24) = \int_0^{24} f(t)dt = 24\int_0^1 f(t)\,dt = 24\int_0^1 \left(t - \frac{1}{2}\right)dt = 0$

and $\int_0^{24} x\,P'(x)\,dx = \int_0^{24} x\,f(x)\,dx = \int_0^{1} x\left(x-\frac{1}{2}\right)dx + \int_1^{2} x\left(x-\frac{3}{2}\right)dx + ... + \int_{23}^{24} x\left(x-\frac{47}{2}\right)dx$

$\int_0^{24} x\,P'(x) = 24 \times \frac{1}{12} = 2$ $\quad\left(\text{As value of every definite integral is } \frac{1}{12}\right)$

So, $\left|\int_0^{24} P(x)\,dx\right| = |0-2| = 2.$ Ans.]

31. $\sqrt{3+6(1-\sin^2 x)+\sin^4 x} + \sqrt{\cos^4 x+4(1-\cos^2 x)}$

$= \sqrt{9-6\sin^2 x+\sin^4 x} + \sqrt{\cos^4 x-4\cos^2 x+4} = \sqrt{(3-\sin^2 x)^2} + \sqrt{(2-\cos^2 x)^2}$

$= 3 - \sin^2 x + 2 - \cos^2 x = 4.$ **Ans.**

32. $\int_0^2 (f(x)-f(4x))dx = 10.$ $\quad$ Let $x = 2t \Rightarrow dx = 2dt$

So, $2\int_0^1 [f(2t)-f(8t)]dt = 10 \Rightarrow \int_0^1 [f(2x)-f(8x)]dx = 5$ $\quad$ ... (1)

Given that $\int_0^1 (f(x)-f(2x))dx = 5$ $\quad$ ... (2)

From (1) + (2), we get

$\int_0^1 (f(x)-f(8x))dx = 10$ **Ans.**

33. $x = t^4 \Rightarrow$ $\quad I = \int_0^1 \frac{4t^2\,dt}{(1+t)\sqrt{1-t^2}}$, $t = \sin\theta$

$= 4\int_0^{\pi/2} \frac{\sin^2\theta\,d\theta}{1+\sin\theta} = 4\int_0^{\pi/2}\left(\sin\theta - 1 + \frac{1}{1+\sin\theta}\right)d\theta = 4 - 2\pi + 4\int_0^{\pi/2}\frac{d\theta}{1+\sin\theta}$, $u = \tan\frac{\theta}{2}$

$= 4 - 2\pi + 8\int_0^1 \frac{dt}{(t+1)^2} = 4 - 2\pi - \left.\frac{8}{t+1}\right|_0^1 = 8 - 2\pi$ **Ans.**

34. [Sol.$_{232/def/QZ}$ Let A (α, 0), B(β, 0) and (0, b) centroid $= \left[\frac{\alpha+\beta}{3}, \frac{b}{3}\right]$

$\Rightarrow$ $\quad \alpha + \beta = b \Rightarrow -a = b$

Now, $\quad f(x) = x^2 - bx + b$

Since, $\quad y = f(x)$ has distinct roots.

$\Rightarrow \quad b^2 - 4b > 0 \Rightarrow b \in (-\infty, 0) \cup (4, \infty)$

Now, $\quad I = \int_0^6 \left(x^2 - bx + b\right)dx = 72 - 12b$

But value of b cannot lie in [0, 4]
So, value of I cannot lie in [24, 72]

35. $x \ge y,\ 0 \le \ln f(x) - \ln f(y) \le (x-y)^2$

Let $\quad g(x) = \ln f(x)$

$0 \le g(x) - g(y) \le (x-y)^2$

$$\Rightarrow \quad \frac{|g(x) - g(y)|}{|x-y|} \le |x-y| \ \forall\ x \ge y$$

$$\left|g'(x^+)\right| = \lim_{h \to 0} \frac{|g(x+h) - g(x)|}{|h|} \le \lim_{h \to 0} |h|$$

$g'(x^+) = 0$

$$|g'(x^-)| = \lim_{h\to 0} \frac{|g(x-h) - g(x)|}{|h|}$$

$$= \lim_{h\to 0} \frac{|g(x) - g(x-h)|}{|h|} \le \lim_{h \to 0} |h|$$

$g'(x^-) = 0 \Rightarrow g'(x) = 0$

$\Rightarrow \quad g(x) = c \Rightarrow \ln f(x) = c \Rightarrow f(x) = e^c \Rightarrow f(x) = 1.$ **Ans.**

36. $$f(\alpha) = \int_0^1 \frac{x^{\cos\alpha} - 1}{\ln x} dx,\ f\left(\frac{\pi}{2}\right) = 0$$

$$\frac{df}{d\alpha} = \int_0^1 -\sin\alpha \cdot x^{\cos\alpha}\, dx = -\sin\alpha \left.\frac{x^{\cos\alpha + 1}}{\cos\alpha + 1}\right|_0^1 = \frac{-\sin\alpha}{1 + \cos\alpha}$$

Integrating, $\quad f(\alpha) = \ln(1 + \cos\alpha) + C$

$$f\left(\frac{\pi}{2}\right) = 0 \quad \Rightarrow \quad C = 0$$

$\therefore \quad f(\alpha) = \ln(1 + \cos\alpha)$ **Ans.]**

37. $f(x) = \frac{x}{2} \lim_{n\to\infty} \sum_{r=1}^{n} \left(\frac{1}{1\cdot3\cdot5....(2r-1)} - \frac{1}{1\cdot3\cdot5.....(2r+1)} \right)$

$= \frac{x}{2} \lim_{n\to\infty} \left(1 - \frac{1}{1.3.5....(2n+1)} \right) = \frac{x}{2}$

Hence $\int_0^4 [f(x)] d(x-[x]) = \int_0^4 \left[\frac{x}{2} \right] dx = \int_0^2 0 dx \quad \int_2^4 1 dx = 2$ **Ans.**

38. $I = \int_0^{a_n} \frac{e^x + 1 - 2}{e^x + 1} dx$ where $I = \ln n$. Also since $\lim_{n\to\infty} (a_n - \ln x)$ exists hence $\lim_{n\to\infty} a_n \to \infty$

$\therefore \quad I = \int_0^{a_n} \left(1 - \frac{2e^{-x}}{1+e^{-x}} \right) dx = \left[x + 2\ln\left(1+e^{-x}\right) \right]_0^{a_n} = a_n + 2\ln\left(1+e^{-a_n}\right) - [0 + 2\ln 2]$

Hence $\ln n = a_n + 2\ln\left(1+e^{-a_n}\right) - \ln 4$

$\therefore \quad a_n - \ln n = \ln 4 - 2\ln\left(1+e^{-a_n}\right)$

$\lim_{n\to\infty} (a_n - \ln n) = \ln 4 \Rightarrow \quad k = 4$ **Ans.**

39. $I = \int_1^{\sqrt{3}} \frac{x^2-1}{x^4+x^3+3x^2+x+1} dx = \int_1^{\sqrt{3}} \frac{1}{x^2+x+3+\frac{1}{x}+\frac{1}{x^2}} \frac{x^2-1}{x^2} dx$

Substitute $y = x + \frac{1}{x}$, $y^2 = x^2 + 2 + \frac{1}{x^2}$, $dy = \left(1 - \frac{1}{x^2}\right) dx = \frac{x^2-1}{x^2} dx$

$$I = \int_2^{\frac{4\sqrt{3}}{3}} \frac{dy}{y^2+y+1}$$

Substitute $z = y + \frac{1}{2}$, $z^2 = y^2 + y + \frac{1}{4}$, $dz = dy$

$$I = \int_{\frac{5}{2}}^{\frac{3+8+\sqrt{3}}{6}} \frac{dz}{z^2 + \frac{3}{4}}$$

Substitute $v = \frac{2}{\sqrt{3}} z$, $v^2 = \frac{4}{3} z^2$, $dv = \frac{2}{\sqrt{3}} dz$

$$I = \frac{2\sqrt{3}}{3}\int_{\frac{5\sqrt{3}}{3}}^{\frac{8+\sqrt{3}}{3}} \frac{dv}{v^2+1} = \left[\frac{2\sqrt{3}}{3}\arctan v\right]_{\frac{5\sqrt{3}}{3}}^{\frac{8+\sqrt{3}}{3}} = \frac{2\sqrt{3}}{3}\left(\arctan\frac{8+\sqrt{3}}{3} - \arctan\frac{5\sqrt{3}}{3}\right)$$

Let $\alpha = \arctan\frac{8+\sqrt{3}}{3}$, $\beta = \arctan\frac{5\sqrt{3}}{3}$

$$\tan(\alpha-\beta) = \frac{\tan\alpha - \tan\beta}{1+\tan\alpha\tan\beta} = \frac{3(8+\sqrt{3}-5\sqrt{3})}{9+5\sqrt{3}(8+\sqrt{3})} = \frac{3}{2}\,\frac{2-\sqrt{3}}{3+5\sqrt{3}} = \frac{13\sqrt{3}-21}{44}$$

$$I = \frac{2\sqrt{3}}{3}\arctan\frac{13\sqrt{3}-21}{44}$$

40. $f\left(\frac{1}{x}\right) = \int_1^{1/x}\frac{\ln t}{1+t+t^2}dt$; putting $t = \frac{1}{Z}$; $f\left(\frac{1}{x}\right) = f(x)$.]

41. Integrate by parts

$$I = [xf(x)]_0^e - \int_0^e x\ f'(x)\ dx$$

$$f'(x)e^{f(x)} + f(x)\cdot e^{f(x)}\cdot f'(x) = 1$$

$$x\ f'(x) = 1 - f'(x)\cdot e^{f(x)}$$

$$I = e\ f(e) - \int_0^e 1\ dx + \int_0^e e^{f(x)}\cdot f'(x)\ dx$$

$$I = e\ f(e) - (e-0) + \left[e^{f(x)}\right]_0^e$$

$$I = e\ f(e) - e + e^{f(e)} - e^{f(0)}$$

$$I = e - e + e - 1$$

$$I = e - 1$$

Put $x = 0$ in original equation

$f(0) = 0$

Let $f(x) = 1$ in original equation

$e = x$

$\Rightarrow f(e) = 1.$]

42.
$$\lim_{a\to 0^+}\frac{\int_0^a x\left(e^x - x - 1\right)dx}{a^4} = \lim_{a\to 0^+}\frac{e^a - a - 1}{4a^2} = \frac{1}{8}$$

$$\lim_{a\to 0^+}\frac{\int_0^a (\tan x - x)\,dx}{a^4} = \lim_{a\to 0^+}\frac{\tan a - a}{4a^3} = \frac{1}{12}$$

$$\lim_{a \to 0^+} \frac{\int_0^a (\sin x - x)\,dx}{a^4} = \lim_{a \to 0^+} \frac{\sin a - a}{4a^3} = \frac{-1}{24}$$

$$\frac{1}{8}t^2 + \frac{1}{12}\cdot t - \frac{1}{24} = 0$$

$3t^2 + 2t - 1 = 0$

$(3t - 1)(t + 1) = 0$

$$t = \frac{1}{3}, -1$$

$$\lim_{a \to 0^+} \left| \frac{1}{\alpha(a)} - \frac{1}{\beta(a)} \right| = |3+1| = 4. \textbf{ Ans.}$$

43. $e^{-x} f(x) + \int_0^x f(x)\,dx = 2 + ln\left(x + \sqrt{x^2+1}\right) + \int_x^0 t\,f(t)\,dt$

Put $x = 0$

$f(0) = 2$

$$e^{-x} f'(x) - e^{-x} f(x) + f(x) = \frac{1}{\sqrt{x^2+1}} - x\,f(x)$$

Put $x = 0$

$f'(0) = 1$

$$e^{-x} f''(x) - 2e^{-x} f(x) + e^{-x} f(x) + f'(x) = \frac{-x}{\sqrt{x^2+1}} - \left(x\,f'(x) + f(x)\right)$$

Put $x = 0,\quad f''(0) - 2f(0) + f(0) + f'(0) = -f(0)$

$f''(0) = -f'(0) = -1$

$\therefore\quad f(0) + f'(0) + f''(0) = 2.$ **Ans.**

44. $$I = \int_0^{\pi/4} \left[\left(x\frac{\sin x}{\sqrt{\cos x}} + 2\sqrt{\sin x}\right)\left(\frac{x\cos x}{\sqrt{\sin x}} - 2\sqrt{\cos x}\right)\right] dx$$

$$= \int_0^{\pi/4} \left[\left(\underbrace{2\sqrt{\sin x}}_{f(x)} + \underbrace{x\frac{\cos x}{\sqrt{\sin x}}}_{x f'(x)}\right)\left(\underbrace{-2\sqrt{\cos x}}_{f(x)} + \underbrace{x\frac{\sin x}{\sqrt{\cos x}}}_{x f'(x)}\right)\right] dx$$

$$\therefore\quad I = x\left(2\sqrt{\sin x}\right) + x\left(-2\sqrt{\cos x}\right)\Big]_0^{\pi/4} = 2x\left(\sqrt{\sin x} - \sqrt{\cos x}\right)\Big]_0^{\pi/4} = 0 \text{ Ans.}$$

45. $\dfrac{x}{4x^2-2x+9} = \dfrac{1}{4x+\dfrac{9}{x}-2} \le \dfrac{1}{10}$ $\left\{\because 4x+\dfrac{9}{x} \ge 12 \quad \text{here } x \in \left[\dfrac{1}{5}, 5\right]\right\}$

$$I = \underbrace{\int_{1/5}^{5} \frac{1}{x}\left\{\frac{x}{4x^2-2x+9}\right\}dx}_{I_1} + \underbrace{\int_{1/5}^{5} \frac{1}{x}\left\{-\frac{x}{9x^2-2x+4}\right\}dx}_{I_2}$$

$$I_2 = \int_{1/5}^{5} \frac{1}{x}\left\{-\frac{x}{9x^2-2x+4}\right\}dx$$

putting $x = \dfrac{1}{t} \Rightarrow dx = -\dfrac{1}{t^2}dt$

$$I_2 = -\int_{5}^{1/5} t\left\{-\frac{\frac{1}{t}}{\frac{9}{t^2}-\frac{2}{t}+4}\right\}\frac{1}{t^2}dt = \int_{1/5}^{5} \frac{1}{t}\left\{-\frac{t}{4t^2-2t+9}\right\}dt$$

Now, $I = \displaystyle\int_{1/5}^{5} \frac{1}{x}\left(\left\{\frac{x}{4x^2-2x+9}\right\} + \left\{-\frac{x}{4x^2-2x+9}\right\}\right)dx$

$= I = \displaystyle\int_{1/5}^{5} \frac{1}{x}dx$ $\{\because \{x\} + \{-x\} = 1 \text{ when } x \notin I\}$

$\therefore I = (\ln x)_{1/5}^{5} = 2\ln 5 = p \ln q + 3\tan^{-1} r$

$\Rightarrow p = 2, q = 5 \ \& \ r = 0$

$\therefore p + q + r = 7$

46 $\displaystyle\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} \ln\left(\frac{n^2+(k-1)^2}{n^2+k^2}\right)^k$

$$= \lim_{n\to\infty} \frac{1}{n}\left(\ln\left(\left(\frac{n^2+0^2}{n^2+1^2}\right)^1 \cdot \left(\frac{n^2+1^2}{n^2+2^2}\right)^2 \cdot \left(\frac{n^2+2^2}{n^2+3^2}\right)^3 \cdot \ldots\ldots\ldots \cdot \left(\frac{n^2+(n-1)^2}{n^2+n^2}\right)^n\right)\right)$$

$$= \lim_{n\to\infty} \frac{1}{n}\left[\ln\left(\frac{n^2\cdot(n^2+1^2)\cdot(n^2+2^2)\cdot\ldots\ldots\ldots\cdot\left(n^2+(n-1)^2\right)}{(2n^2)^n}\right)\right]$$

$$= \lim_{n\to\infty} \frac{1}{n}\left[\ln\left(\frac{1}{2}\cdot\left(\frac{1}{2}+\frac{1^2}{2n^2}\right)\cdot\left(\frac{1}{2}+\frac{2^2}{2n^2}\right)\cdot\ldots\ldots\ldots\cdot\left(\frac{1}{2}+\frac{(n-1)^2}{2n^2}\right)\right)\right]$$

$$= \lim_{n\to\infty} \frac{1}{n}\sum_{r=0}^{n-1} \ln\left(\frac{1}{2}+\frac{r^2}{2n^2}\right) = \int_0^1 \underbrace{1}_{I} \cdot \underbrace{\ln\left(\frac{1+x^2}{2}\right)}_{I} dx$$

$$\therefore S = \int_0^1 \underbrace{1}_{II} . \underbrace{\ln\left(\frac{1+x^2}{2}\right)}_{I} dx \;=\; x\ln\left(\frac{1+x^2}{2}\right)\Bigg]_0^1 - \int_0^1 \frac{2x^2}{1+x^2}dx$$

$$= 0 - 2\left[\int_0^1 \frac{x^2+1}{1+x^2}dx - \int_0^1 \frac{dx}{1+x^2}\right] = -2\left[x - \tan^{-1}x\right]_0^1 = -2\left[1-\frac{\pi}{4}\right] = \frac{\pi}{2} - 2$$

$\therefore$ 1 **Ans.**

47. $I = \int_{-\pi}^{\pi} \underbrace{(\cos 2x\cdot\cos 2^2x\cdot\cos 2^3x\cdot\cos 2^4x\cdot\cos 2^5x)}_{f(x)}dx$

$I = 2\int_0^{\pi} f(x)\,dx$ [f (x) is even]

$I = 2\cdot 2\int_0^{\frac{\pi}{2}} f(x)dx$ (using Queen)

$I = 4\int_0^{\frac{\pi}{2}} f(x)\,dx = 4I_1$

Now $I_1 = \int_0^{\frac{\pi}{2}} f(x)\,dx$

using King

$I_1 = -I_1$

$\therefore$ $I_1 = 0$

$\Rightarrow$ I = 0 **Ans.**

48. $I = \int_{-1}^{1} \frac{e^x(e^x+e^{-x})-(x+1)(e^x+e^{-x})}{x(e^x-1)}dx$

$$I = \underbrace{\int_{-1}^{1} \frac{(e^x+e^{-x})(e^x-x-1)}{x(e^x-1)}dx}_{\downarrow \text{king}} \qquad \ldots (1)$$

$$I = \int_{-1}^{1} \frac{(e^x + e^{-x})(e^{-x} + x - 1)e^x}{x(e^x - 1)} dx$$

$$I = \int_{-1}^{1} \frac{(e^x + e^{-x})(1 + xe^x - e^x)}{x(e^x - 1)} dx \qquad \ldots (2)$$

(1) & (2)

$$2I = \int_{-1}^{1} \frac{(e^x + e^{-x})\,(xe^x - x)}{x(e^x - 1)} dx$$

$$2I = 2\int_{0}^{1} (e^x + e^{-x})\, dx$$

$$I = e^x - e^{-x}\Big|_0^1 = e - \frac{1}{e} \ \textbf{Ans.}$$

49 $\quad L = \quad \displaystyle\lim_{n\to\infty} \frac{1}{n}\int_a^{\infty} \frac{dx}{x^2 + \frac{1}{n^2}} = \lim_{n\to\infty} \frac{1}{n}\, n\tan^{-1} nx\Big]_a^{\infty} = \lim_{n\to\infty}\left(\frac{\pi}{2} - \tan^{-1} na\right)$

$$L = \begin{bmatrix} 0 & \text{if } R > 0 \\ \pi/2 & \text{if } R = 0 \\ \pi & \text{if } R < 0 \end{bmatrix}]$$

50 $\quad \displaystyle I = f(k) = \int_{-1}^{1} \frac{\sqrt{1-x^2}\left(\sqrt{k+1} + x\right)}{(k+1) - x^2} dx$

$$f(x) = \int_{-1}^{1} \frac{\left(\sqrt{k+1}\right)\left(\sqrt{1-x^2}\right)}{k+1-x^2} dx + \underbrace{\int_{-1}^{1} \frac{x\sqrt{1-x^2}}{k+1-x^2} dx}_{\text{zero (odd function)}}$$

$$f(k) = 2\sqrt{k+1}\int_0^1 \frac{\sqrt{1-x^2}}{k + (1-x^2)} dx$$

Put $x = \sin\theta$

$$f(k) = 2\sqrt{k+1}\int_0^{\pi/2} \frac{\cos^2\theta}{k + \cos^2\theta} d\theta = 2\sqrt{k+1}\left[\int_0^{\pi/2} d\theta - k\int_0^{\pi/2} \frac{d\theta}{k + \cos^2\theta}\right]$$

$$= 2\sqrt{k+1}\left[\frac{\pi}{2} - k\int_0^{\pi/2} \frac{\sec^2\theta\, d\theta}{1 + k(1 + \tan^2\theta)}\right] = 2\sqrt{k+1}\left[\frac{\pi}{2} - \frac{\pi}{2}\frac{\sqrt{k}}{\sqrt{k+1}}\right]$$

$\therefore \quad f(k) = \pi\left(\sqrt{k+1}-\sqrt{k}\right)$

Now $\displaystyle\sum_{k=0}^{99} f(k) = \pi \sum_{k=0}^{99}\left(\sqrt{k+1}-\sqrt{k}\right)$

$$= \pi\left[(1-0)+\left(\sqrt{2}-\sqrt{1}\right)+\left(\sqrt{3}-\sqrt{2}\right)+........+\left(\sqrt{100}-\sqrt{99}\right)\right]$$

$$= \pi\left(\sqrt{100}-0\right) = 10\pi \Rightarrow \quad p = 10 \textbf{ Ans.}$$

51. $I = \displaystyle\int_{1/2}^{2} \frac{f(x)\cdot(1+x)}{x\sqrt{x^2+1}}dx \qquad ....(1)$

put $x = \dfrac{1}{t}$

$I = \displaystyle\int_{1/2}^{2} \frac{f\left(\frac{1}{x}\right)(1+x)}{x\sqrt{x^2+1}}dx \qquad ....(2)$

1 + 2

$$2I = \int_{1/2}^{2} \frac{(1+x)}{x\sqrt{1+x^2}}\left(f(x)+f\left(\frac{1}{x}\right)\right)dx$$

$$2I = 10\int_{1/2}^{2} \frac{x+1}{x\sqrt{1+x^2}}dx$$

Consider $I_1 = \displaystyle\int_{1/2}^{2} \frac{x+1}{x\sqrt{1+x^2}}dx \qquad x = \tan\theta$

$$I_1 = \int_{\tan^{-1}\frac{1}{2}}^{\tan^{-1}2} \frac{(1+\tan\theta)}{\tan\theta}\sec\theta\, d\theta = \int_{\tan^{-1}\frac{1}{2}}^{\tan^{-1}2} \frac{(\sin\theta+\cos\theta)}{\sin\theta\cos\theta}d\theta = \int_{\tan^{-1}\frac{1}{2}}^{\tan^{-1}2} (\sec\theta+\operatorname{cosec}\theta)\, d\theta$$

$$I_1 = \ln(\sec\theta+\tan\theta)+\ln(\operatorname{cosec}\theta-\cot\theta)\Big]_{\tan^{-1}\frac{1}{2}}^{\tan^{-1}2}$$

$$I_1 = \left(\ln\left(\sqrt{5}+2\right)+\ln\left(\frac{\sqrt{5}}{2}-\frac{1}{2}\right)\right) - \left(\ln\left(\frac{\sqrt{5}}{2}+\frac{1}{2}\right)+\ln\left(\sqrt{5}-2\right)\right)$$

$$I_1 = \left(\ln\left(\sqrt{5}+2\right)+\ln\left(\sqrt{5}-1\right)-\ln 2\right) - \left(\ln\left(\sqrt{5}+1\right)+\ln\left(\sqrt{5}-2\right)-\ln 2\right) = \ln\left(\frac{\left(\sqrt{5}+2\right)\left(\sqrt{5}-1\right)}{\left(\sqrt{5}+1\right)\left(\sqrt{5}-2\right)}\right)$$

$$I_1 = \ln\left(\frac{3+\sqrt{5}}{3-\sqrt{5}}\right)$$

$$\therefore\ I = 5I_1 = 5\ln\left(\frac{3+\sqrt{5}}{3-\sqrt{5}}\right) \Rightarrow k = 5 \textbf{ Ans.}$$

52. [Sol.$_{41431/def}$ $g(x) = \int_0^x f(t)\,dt$; $g'(x) = f(x) \Rightarrow g'(x)$ is odd $\Rightarrow g(x)$ is even

$$g(2n) = \int_0^{2n} f(t)\,dt = n\int_0^2 f(t)\,dt \quad = n\left[\int_0^1 f(t)\,dt + \int_0^1 f(-t)\,dt\right] = 0 \quad]$$

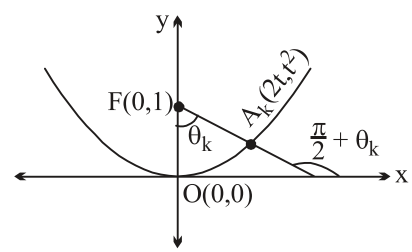
Again $g(x+2) = \int_0^{x+2} f(t) = \int_0^2 f(t)\,dt + \int_2^{x+2} f(t)\,dt = g(2) + \int_0^x f(y+2)\,dy = 0$

$$= \int_0^x f(y)\,dy \quad \Rightarrow \quad g(x+2) = g(x) \quad (\text{as } g(2) = 0)$$

53 Let $A_k = (2t, t^2)$

$\therefore$ Slope of $FA_k = \dfrac{t^2-1}{2t-0} = \tan\left(\dfrac{\pi}{2}+\theta_k\right)$

$\therefore$ $\tan\theta_k = \dfrac{2t}{1-t^2} = \tan(2\phi)$ (Say)



$\Rightarrow$ $\phi = \dfrac{\theta_k}{2} = \dfrac{k\pi}{4n}$ where $\tan\phi = t$

Also $FA_k = \sqrt{(t^2-1)^2+(2t)^2} = t^2 + 1 = 1 + \tan^2\phi = \sec^2\left(\dfrac{k\pi}{4n}\right)$

$$\therefore \lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n} FA_k = \lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\sec^2\left(\frac{\pi}{4}\left(\frac{k}{n}\right)\right) = \int_0^1 \sec^2\left(\frac{\pi x}{4}\right)dx = \frac{4}{\pi}\left[\tan\frac{\pi x}{4}\right]_0^1 = \frac{4}{\pi} \textbf{ Ans.}$$

Paragraph for question nos. 54 to 56

Sol

.54. We have $\int_{x}^{x+1} f(t)\,dt = k\left(x^3+\frac{3}{2}x^2-\frac{1}{4}\right)$ ......(1)

Now, on differentiating both the sides with respect to x, we get

$f(x+1) - f(x) = 3kx^2 + 3kx$

$\therefore$ $f(x+1) = f(x) + 3kx(x+1)$ ....(2)

Put x = 0 in equation (2), we get

$f(1) = f(0) = 0$ (As f is an odd function on R, so $f(0) = 0$.)

Also, $f(-1) = 0$

Hence $f(x) = kx(x-1)(x+1)$

As $f(2) = 6$ (Given), so $k = 1$

$\therefore$ $f(x) = (x^3 - x)$

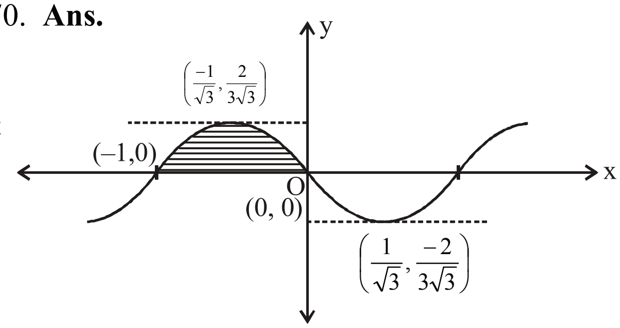
Clearly f(x) is a cubic function.

55 $\sum_{r=1}^{10}\left(r^3-r\right) = \left(\sum_{r=1}^{10} r^3\right) - \left(\sum_{r=1}^{10} r\right)$

$= \frac{(10)^2(11)^2}{4} - \frac{(10)(11)}{2} = 3025 - 55 = 2970.$ **Ans.**

56 Also, $\int_{-1}^{0}\{f(t)\}\,dt = \int_{-1}^{0}\{t^3-t\}\,dt = \int_{-1}^{0}\left(t^3-t\right)dt$

$= \left(\frac{t^4}{4}-\frac{t^2}{2}\right)_{-1}^{0} = 0 - \left(\frac{1}{4}-\frac{1}{2}\right) = \frac{1}{4}$ Ans.



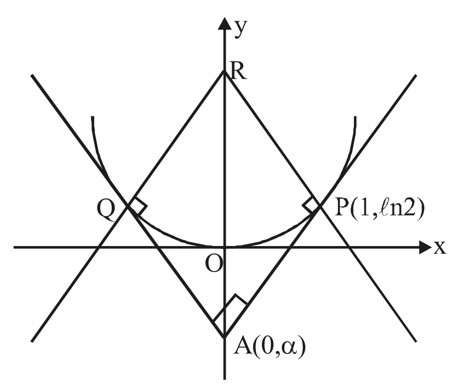
57. $f^2(x) = \int_0^x \frac{4t\,f(t)}{(1+t^2)}\,dt$

$\Rightarrow 2f(x).f'(x) = \frac{4x\,f(x)}{(1+x^2)}$

$\Rightarrow f'(x) = \frac{2x}{1+x^2}$ (f(x) is not constant)

$\Rightarrow$ $f(x) = ln(1+x^2)$

$\Rightarrow f'(1) + f'(-1) = 0$



Since f(x) is symmetrical about y-axis, hence $\angle OAP = 45°$

$\therefore$ slope of AP = 1

$$\frac{2x}{1+x^2}=1 \Rightarrow x=1$$

Equation of AP is , $y - ln\,2 = (x-1)$

$\Rightarrow \alpha = ln\,2 - 1$

Quadrilateral APRQ is square.

AP = $\sqrt{2}$, R = [0, 1 + ln 2]. **Ans.**

58. $f(y) = y^2 - 2y - \frac{k^2}{\pi^2}$

$= (y-1)^2 - \frac{k^2}{\pi^2} - 1.$

Now,

$$\Rightarrow \quad \left(\sqrt{5}+1-1\right)^2 - \frac{k^2}{\pi^2} - 1 > 0$$

$$\Rightarrow \quad \frac{k^2}{\pi^2} < 4 \Rightarrow k \in (-2\pi, 2\pi)$$

$\therefore \quad a = -2\pi$ and $b = 2\pi$

$$I = \int_{\frac{a}{4}}^{\frac{b}{4}} \sin^3 x \cos^3 x \left(\sin x + \cos x\right) dx = \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \sin^3 x \cos^3 x \left(\sin x + \cos x\right) dx$$

$$= 2\int_0^{\frac{\pi}{2}} \sin^4 x \cos^3 x \, dx = 2 \frac{3\cdot 1\cdot 2}{7\cdot 5\cdot 3\cdot 1} = \frac{4}{35}$$

$35I = 4$. **Ans.**

59. $x\int_0^x g(t)\,dt + \int_0^x (1-t)\,g(t)\,dt = x^4 + x^2$

differentiate w.r.t. x

$$x\, g(x) + \int_0^x g(t)\,dt + (1-x)\,g(x) = 4x^3 + 2x$$

Again differentiate w.r.t x

$g'(x) + g(x) = 12x^2 + 2$

Now, $\int_0^1 \frac{12}{g'+g+10}dx = \int_0^1 \frac{dx}{x^2+1} = \tan^{-1} x\Big]_0^1 = \frac{\pi}{4}$. **Ans.**

60. From $\int\limits_{f(y)}^{f(x)} e^t\,dt = \int\limits_{y}^{x}\left(\frac{1}{t}\right)dt$, we get

$e^{f(x)} - e^{f(y)} = ln\,x - ln\,y$

$\Rightarrow \quad e^{f(x)} - ln\,x = c \Rightarrow \quad f(x) = ln(ln\,x + c)$

As, $\quad f\left(\frac{1}{e}\right) = 0 \quad \Rightarrow \quad c = 2$

$\therefore \quad f(x) = ln(ln\,x + 2)$

Now, $\quad f(g(x)) = \begin{cases} ln(x+2);\ x \ge k \\ ln(2+x^2);\ 0 < x < k \end{cases}$

$\therefore$ for containing at $x = k$, $ln(k+2) = ln(2+k^2)$

$\Rightarrow \quad k^2 = k \Rightarrow \quad k = 0,\ 1.$ But $k > 0$

so, $\quad k = 1$ **Ans.**

61. $f(x) = \frac{1}{x^2}\int\limits_4^x \left(4t^2 - 2f'(t)\right)dt$

$x^2 f(x) = \int\limits_4^x \left(4t^2 - 2f'(t)\right)dt$

Different both sides

$x^2 f'(x) + 2x\,f(x) = 4x^2 - 2\,f'(x)$

put $\quad x = 4$

$16\,f'(4) + 0 = 64 - 2\,f'(4)$

$f'(4) = \frac{64}{18} = \frac{32}{9}$. **Ans.**

62. [Sol.$_{206/def/QZ}$ $\quad I = \int\limits_0^{\frac{\pi}{6}} \underbrace{ln\,(\sqrt{3} + \tan x)\,dx}_{I_1} - \frac{\pi}{12}\,ln\,3$

Using King in I_1

$$I_1 = \int\limits_0^{\frac{\pi}{6}} ln\,\left(\sqrt{3} + \tan\left(\frac{\pi}{6} - x\right)\right)dx$$

$$I_1 = \int\limits_0^{\frac{\pi}{6}} ln\,\left(\sqrt{3} + \frac{1-\sqrt{3}\,\tan x}{\sqrt{3} + \tan x}\right)dx$$

$$I_1 = \int_0^{\frac{\pi}{6}} ln\left(\frac{4}{13+\tan x}\right)dx$$

$$\therefore \qquad 2I_1 = (ln\,4)\,\frac{\pi}{6}$$

$$I_1 = \frac{\pi}{12}\,ln\,4$$

$$\therefore \qquad I = \frac{\pi}{12}\,ln\left(\frac{4}{3}\right).$$ **Ans.**

63. $\int_2^x f(t)\,dt = \frac{x^2}{2} + \int_x^2 t^2 f(t)\,dt \Rightarrow f(x) = x - x^2 f(x) \Rightarrow (1+x^2)\,f(x) = x \Rightarrow f(x) = \frac{x}{1+x^2}$. **Ans.**

64. $f(x) = \sqrt{1+x\sqrt{1+(x+1)\sqrt{1+(x+2)(x+4)}}} = \sqrt{1+x\sqrt{1+(x+1)\sqrt{x^2+6x+9}}}$

$= \sqrt{1+x\sqrt{1+(x+1)(x+3)}} = \sqrt{1+x\sqrt{x^2+4x+4}} = \sqrt{1+x(x+2)}$

$= \sqrt{x^2+2x+1} = (x+1)$

$$\therefore \qquad I = \int_0^{100}(x+1)\,dx = \left.\frac{(x+1)^2}{2}\right|_0^{100} = \frac{(101)^2-1^2}{2} = \frac{100\times 102}{2} = 5100.$$ **Ans.**

65. $nx = t$

$$\lim_{n\to\infty}\frac{1}{n}\int_0^n |\sin t|^3\,dt$$

Put $\quad n = a$

$$\lim_{a\to\infty}\frac{\int_0^a |\sin t|^3\,dt}{a}$$

$a = n\pi, \qquad n \in I$

$$\lim_{n\to\infty}\frac{\int_0^{n\pi} |\sin t|^3\,dt}{n\pi}$$

$$\lim_{n\to\infty}\frac{n\int_0^{\pi}|\sin t|^3\,dt}{n\pi}$$

$$2\cdot\frac{2}{3}\cdot\frac{1}{\pi}=\frac{4}{3\pi}$$

Note : if $n \neq I$

then $$\lim_{n\to\infty}\frac{\int_0^{n\pi}|\sin t|^3\,dt}{n\pi}+\underbrace{\frac{\int_{n\pi}^{n\pi+h}|\sin t|^3\,dt}{n\pi}}_{\text{zero}}=\frac{4}{3\pi}.$$ **Ans.**

66. $$I=\int_0^{\pi/4}\frac{\ln(\cot x)}{\left((\sin x)^{2009}+(\cos x)^{2009}\right)^2}\cdot(\sin 2x)^{2008}\,dx$$

$$=\int_0^{\pi/4}\frac{\ln(\cot x)}{\left(1+(\cot x)^{2009}\right)^2}\cdot\frac{2^{2008}(\sin x)^{2008}(\cos x)^{2008}}{(\sin x)^{2009}(\sin x)^{2009}}dx$$

$$=-\frac{2^{2008}}{(2009)^2}\int_0^{\pi/4}\frac{\ln(\cot x)^{2009}}{\left(1+(\cot x)^{2009}\right)^2}\cdot\frac{2009(\cot x)^{2008}}{(-\sin^2 x)}dx$$

Let $(\cot x)^{2009}=u$, $\dfrac{-2009(\cot x)^{2008}}{\sin^2 x}dx=du$

$$=\frac{2^{2008}}{(2009)^2}\int_{\infty}^{1}-\frac{\ln(u)}{(1+u)^2}du=\frac{2^{2008}}{(2009)^2}\left[\frac{\ln u}{1+u}\right]_{\infty}^{1}-\frac{2^{2008}}{(2009)^2}\int_{\infty}^{1}\frac{du}{u(1+u)}$$

$$=-\frac{2^{2008}}{(2009)^2}\int_{\infty}^{1}\left[\frac{1}{u}-\frac{1}{u+1}\right]du=-\frac{2^{2008}}{(2009)^2}\left[\ln\frac{u}{u+1}\right]_{\infty}^{1}=\frac{2^{2008}\ln 2}{(2009)^2}=\frac{a^b\ln a}{c^2}$$

$\Rightarrow$ $a=2$, $b=2008$ and $c=2009$ $\Rightarrow$ $(a+b+c)=4019$ **Ans.**

67. We have $$I_n=2\int_0^1 x\left(1+\frac{x^2}{2}+\frac{x^4}{4}+\frac{x^6}{6}+\ldots\ldots+\frac{x^{2n}}{2n}\right)dx \qquad \left(\int_{-1}^{1}(\text{odd})\,dx=0\right)$$

$$=2\left[\frac{x^2}{1\cdot 2}+\frac{x^4}{2\cdot 4}+\frac{x^6}{4\cdot 6}+\ldots\ldots+\frac{x^{2n+2}}{2n(2n+2)}\right]_0^1=2\left[\frac{1}{1\cdot 2}+\frac{1}{2\cdot 4}+\frac{1}{4\cdot 6}+\ldots\ldots+\frac{1}{2n(2n+2)}\right]$$

$$=1+\frac{1}{2}\left[\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)++\left(\frac{1}{3}-\frac{1}{4}\right)+\ldots..+\left(\frac{1}{n}-\frac{1}{n+1}\right)\right]$$

Hence $\lim\limits_{n\to\infty}\left[1+\frac{1}{2}\left(1-\frac{1}{n+1}\right)\right] = \frac{3}{2}$

$\therefore \quad p = 3 \;;\; q = 2 \;\Rightarrow\; pq(p^3 + q^2) = (3)(2)[27 + 4] = 186$ **Ans.**

68. $I = \displaystyle\int_a^b \frac{e^{x/a} - e^{b/x}}{x}\,dx$

let $\quad x = at \;\Rightarrow\; dx = a\,dt$

$$= a\int_1^{b/a} \frac{e^t - e^{b/at}}{at}\,dt$$

$$I = \int_1^{\alpha} \frac{e^t - e^{\alpha/t}}{t}\,dt \quad ....(1) \qquad (\text{where } b/a = \alpha)$$

put $\quad t = \frac{\alpha}{y} \;\Rightarrow\; dt = -\frac{\alpha}{y^2}\,dy$

$$I = -\int_{\alpha}^{1} \frac{(e^{\alpha/y} - e^y)y}{\alpha}\cdot\frac{\alpha}{y^2}\,dy$$

$$I = \int_{\alpha}^{1} \frac{(e^y - e^{\alpha/y})dy}{y} \quad \text{or} \quad I = -\int_1^{\alpha} \frac{(e^t - e^{\alpha/t})dt}{t} \quad ....(2)$$

from (1) and (2) $\quad 2I = 0 \;\Rightarrow\; I = 0$ Ans.

69. $x = \frac{\pi}{4} - \theta \;\Rightarrow\; dx = -\theta,\ 2x = \frac{\pi}{2} - 2\theta.$

$$I = \int_{\pi/4}^{0} (\cos^6 2\theta + \sin^6 2\theta)\, ln\left[1+\tan\left(\frac{\pi}{4}-\theta\right)\right] d(-\theta) = \int_0^{\pi/4} (\cos^6 2\theta + \sin^6 2\theta)\, ln\,\frac{2}{1+\tan\theta}\,d\theta$$

$$= (ln\,2)\int_0^{\pi/4}(\cos^6 2\theta + \sin^6 \theta)\,d\theta - I$$

$$2I = (ln\,2)\int_0^{\pi/4}(\cos^6 2\theta + \sin^6 \theta)\,d\theta$$

Here, we have

$$\sin^6\theta + \cos^6\theta = (\sin^2\theta + \cos^2\theta)^3 - 3\sin^2\theta\cos^2\theta\,(\sin^2\theta + \cos^2\theta)$$

$$= 1 - 3\sin^2\theta\cos^2\theta = 1 - \frac{3}{4}\sin^2 2\theta = 1 - \frac{3}{4}\left(\frac{1-\cos 4\theta}{2}\right)$$

$= \frac{1}{8}(5 + 3\cos 4\theta)$

$\therefore\ 2I = \frac{\ln 2}{8}\int_0^{\pi/4}(5+3\cos 8\theta)\,d\theta = \frac{5\pi \ln 2}{32} \Rightarrow I = \frac{5\pi \ln 2}{64}$. **Ans.**

70. $f'(x) = \int_{\pi/6}^{x}\frac{2\cos t}{\sin t}\,dt \Leftrightarrow f'(x)\tan x = 2\left[\ln(\sin t)\right]_{\pi/6}^{x} \Leftrightarrow f'(x)\tan x = 2\ln(\sin x) + 2\ln 2$

$\Leftrightarrow f'(x) = 2\cot x \ln(\sin x) + 2\cot x \ln 2 \Leftrightarrow f'(x) = [\ln^2(\sin x) + 2\ln 2\,\ln(\sin x)]$

$\Leftrightarrow f(x) = \ln^2(\sin x) + 2\ln 2\,\ln(\sin x) + C$

For $x = \frac{\pi}{6} \Rightarrow 0 = \ln^2 2 - 2\ln^2 2 + C \Leftrightarrow C = \ln^2 2$

So, $f(x) = \ln^2(\sin x) + 2\ln 2\,\ln(\sin x) + \ln^2 2 \Leftrightarrow f(x) = [\ln 2 + \ln(\sin x)]^2 = \ln^2(2\sin x)$. **Ans.**

71. $g(p) = 1 - 2p\cos x + p^2$

$g(-p) = 1 + 2p\cos x + p^2$

$h(x) = g(p)\cdot g(-p) = (1+p^2)^2 - 4p^2\cos^2 x = 1 + 2p^2 + p^4 - 4p^2\cos^2 x$

$= 1 - 2p^2(2\cos^2 x - 1) + p^4$

$h(x) = 1 - 2p^2\cos 2x + p^4$ ....(1)

also $g(p^2) = 1 - 2p^2\cos x + p^4$ ....(2)

from (1) $h\left(\frac{x}{2}\right) = 1 - 2p^2\cos x + p^4 = g(p^2)$ ....(i)

again $K(p) = \int_0^{\pi}\ell n(1 - 2p\cos x + p^2)\,dx$ ....(3)

$= \int_0^{\pi}\ell n(1 + 2p\cos x + p^2)\,dx \Rightarrow 2K(p) = \int_0^{\pi}\ell n\left((p^2+1)^2 - 4p^2\cos^2 x\right)dx$

or $2K(p) = \int_0^{\pi}\ell n\left(p^4 + 2p^2 + 1 - 4p^2\cos^2 x\right)dx$

$= \int_0^{\pi}\ell n\,(1 - 2p^2\cos 2x + p^4)\,dx$ put $2x = t$

$K(p) = \frac{1}{2}\int_0^{2\pi}\ell n\left(1 - 2p^2\cos t + p^4\right)dt = \frac{1}{4}\int_0^{2\pi}\left(1 - 2p^2\cos t + p^4\right)dt$

$K(-p) = \frac{1}{4}\int_0^{2\pi}\ell n\left(1 - 2p^2\cos t + p^4\right)dt \Rightarrow K(p) = K(-p) \Rightarrow K$ is even

$K(p) + K(-p) = \frac{1}{2}\int_0^{2\pi}\ell n\left(1 - 2p^2\cos t + p^4\right)dt$

or $\quad 2K(p) = \int_0^{\pi} \ell n\left(1-2p^2\cos t + p^4\right) dt = \int_0^{\pi} \ell n\, g\left(p^2\right) dx = K(p^2)$

$\therefore \quad K(p) = \frac{1}{2}K(p^2)$

$K(x) = \frac{1}{2}K(x^2)$

72. put $\quad x = \cot\theta \quad \Rightarrow \quad dx = -\operatorname{cosec}^2\theta = d\theta$

$$J = \int_{\pi/4}^{\pi/2} \frac{2\cot\theta - (\operatorname{cosec}^2\theta)^2\cdot\theta}{1-(\operatorname{cosec}^2\theta)\theta} d\theta = \int_{\pi/4}^{\pi/2} \frac{2\sin\theta\cdot\cos\theta - (\operatorname{cosec}^2\theta)\cdot\theta}{\sin^2\theta - \theta} d\theta$$

$$= \int_{\pi/4}^{\pi/2} \frac{(2\sin\theta\cdot\cos\theta - 1) + 1 - \theta\cdot\operatorname{cosec}^2\theta}{\sin^2\theta - \theta} d\theta = ln(\sin^2\theta - \theta)\Big|_{\pi/4}^{\pi/2} + \int_{\pi/4}^{\pi/2} \frac{\sin^2\theta - \theta}{\sin^2\theta(\sin^2\theta - \theta)} d\theta$$

$$= ln\left(\frac{1-\frac{\pi}{2}}{\frac{1}{2}-\frac{\pi}{4}}\right) + \int_{\pi/4}^{\pi/2} \operatorname{cosec}^2\theta\, d\theta = (ln\, 2) + 1 \quad \Rightarrow \quad 100(1 + ln\, 2 - ln\, 2) = 100 \text{ Ans.}$$

Aliter: $J = \int_0^1 \frac{2x - (1+x^2)^2\cot^{-1}x}{(1+x^2)\left(1-(1+x^2)\cot^{-1}x\right)} dx$

Divide by $(1 + x^2)^2$

$$J = \int_0^1 \frac{\frac{2x}{(1+x^2)^2} - \cot^{-1}x}{\frac{1}{1+x^2} - \cot^{-1}x} dx = \int_0^1 \frac{\left(\frac{2x}{(1+x^2)^2} - \frac{1}{1+x^2}\right) + \left(\frac{1}{1+x^2} - \cot^{-1}x\right)}{\frac{1}{1+x^2} - \cot^{-1}x} dx$$

$$= \int_0^1 \frac{\left(\frac{2x}{(1+x^2)^2} - \frac{1}{1+x^2}\right)}{\left(\frac{1}{1+x^2} - \cot^{-1}x\right)} dx + \int_0^1 \frac{\left(\frac{1}{1+x^2} - \cot^{-1}x\right)}{\left(\frac{1}{1+x^2} - \cot^{-1}x\right)} dx \text{ . Now proceed.]}$$

73. $L_1 = \lim_{x\to 1} \frac{\sin(n\cos^{-1}x)}{\sqrt{1-x}\,\sqrt{1+x}} = \lim_{x\to 1} \frac{n}{\sqrt{1+x}}\left(\frac{\cos^{-1}x}{\sqrt{1-x}}\right) = n$

$L_2 = \lim_{x\to 1} \frac{1-\cos(n\cos^{-1}x)}{(1-x)(1+x)} = \lim_{x\to 1} \frac{n^2}{2}\frac{(\cos^{-1}x)^2}{(1-x)}\left(\frac{1}{2}\right)$

$\therefore\ L_2 = \frac{n^2}{2}$

Given, $\frac{n^2}{2} + n = \frac{3}{2}$

$\therefore\ n^2 + 2n = 3 \Rightarrow n^2 + 2n - 3 = 0 \Rightarrow (n+3)(n-1) = 0$

$\therefore\ n = 1$ or $n = -3$

Since, $n \in N$

$\therefore\ n = 1 \Rightarrow$ Sum = 1. **Ans.**

74. Required limit $\lim\limits_{n\to\infty} \dfrac{\sum\limits_{r=1}^{n} \frac{r^6}{n^7}}{\sum\limits_{r=1}^{n} \frac{r^2}{n^3} \sum\limits_{r=1}^{n} \frac{r^3}{n^4}} = \dfrac{\int_0^1 x^6\,dx}{\int_0^1 x^2\,dx \int_0^1 x^3\,dx}$]

75. $\ln x = t \quad \Rightarrow \quad x = e^t$

$$I = \int_1^2 \frac{(4t^2+1)}{t\cdot\sqrt{t}} e^t\,dt$$

$$I = \int_1^2 e^t\left(4t^{\frac{1}{2}} + t^{\frac{-3}{2}}\right)dt = \int_1^2 e^t\left(4t^{\frac{1}{2}} + 2t^{\frac{-1}{2}}\right)dt - \int_1^2 e^t\left(2t^{\frac{-1}{2}} - t^{\frac{-3}{2}}\right)dt$$

$$= e^t\cdot 4t^{\frac{1}{2}}\Big]_1^2 - e^t\left(2t^{\frac{-1}{2}}\right)_1^2 = \left(e^2\cdot 4\sqrt{2} - 4e\right) - \left(\sqrt{2}e^2 - 2e\right) = 3\sqrt{2}e^2 - 2e = \sqrt{18}\,e^2 - \sqrt{4}\,e$$

Hence minimum value of $(a+b) = 22$. **Ans.**

76. Let $\quad I_1 = \int_0^1 \left(1-(1-x^2)^{100}\right)^{200} x\,dx$

Put $\quad 1 - x^2 = t \quad$ then $\quad x\,dx = -\frac{1}{2}dt$

$$\therefore\quad I_1 = \int_1^0 (1-t^{100})^{200}\left(-\frac{1}{2}\right)dt = \int_0^1 (1-t^{100})^{200}\frac{1}{2}dt \qquad ...(1)$$

and similarly $I_2 = \int_0^1 (1-t^{100})^{202}\frac{1}{2}dt = \left[(1-t^{100})^{202}\cdot t\right]_0^1 - \int_0^1 202\cdot 100(1-t^{100})^{201}(-t^{99})\,t\,dt$

(Integrating by parts)

$$= 0-20200\int_0^1 (1-t^{100})^{201}(1-t^{100}-1)\,dt = -20200\int_0^1 (1-t^{100})^{202} + 20200\int_0^1 (1-t^{100})^{201}dt$$

$\therefore \quad I_2 = -20200 I_2 + 20200 I_1$

$\therefore \quad \frac{I_1}{I_2} = \frac{20200}{20201} = \frac{p}{q}$

$\therefore \quad (q-p)$ least $= 1$ **Ans.**

77. $l = \lim_{x \to 1^+} \sin\left(2^{-2^{\frac{1}{1-x}}}\right) = \sin 1$

$$m = \lim_{x \to 1^-} \frac{\sin(x-[1+x])\sin(1+[x])}{x+1} = \sin 1$$

$$\int_{\sin 1}^{100\pi+\sin 1} \sin x \cos(\cos x)\,(1-\sin x)\,dx + \int_{\sin 1}^{100\pi+\sin 1} \cos x \sin(\cos x)\,(1-\sin x)\,dx$$

$$= \int_{\sin 1}^{100\pi+\sin 1} (\sin x \cos(\cos x) + \cos x \sin(\cos x))(1-\sin x)\,dx$$

$$= \int_{\sin 1}^{100\pi+\sin 1} \underbrace{\sin(x+\cos x)(1-\sin x)}_{\text{Periodic with period } 2\pi}dx$$

$$= 50\int_0^{2\pi} \sin(x+\cos x)(1-\sin x)\,dx \qquad x+\cos x = t$$

$\Rightarrow \quad (1-\sin x)\,dx = dt$

$$= 50\int_1^{1+2\pi} \sin t\,dt = 50\int_0^{2\pi} \sin t\,dt = 0.$$ **Ans.**

78. $I = \int_0^{100}\{\sqrt{x}\}dx = \int_0^{100}(\sqrt{x}-[\sqrt{x}])dx = \int_0^{100}\sqrt{x}\,dx - \int_0^{100}[\sqrt{x}]dx = \frac{2000}{3} - \int_0^{100}[\sqrt{x}]dx$

$$= \int_0^1 0\,dx + \int_1^4 1\,dx + \int_4^9 2\,dx + \ldots + \int_{81}^{100} 9\,dx = 615$$

$\therefore \quad I = \frac{2003}{3} - 615 = \frac{155}{3}$, so $3I = 155$]

79. $I_1 = \int_0^2 \frac{e^{-\sin^{-1}\sqrt{\frac{x}{2}}}}{e^{\sin^{-1}\sqrt{\frac{x}{2}}}+e^{\cos^{-1}\sqrt{\frac{x}{2}}}}dx = e^{\frac{-\pi}{2}}\int_0^2 \frac{e^{\cos^{-1}\sqrt{\frac{x}{2}}}}{e^{\sin^{-1}\sqrt{\frac{x}{2}}}+e^{\cos^{-1}\sqrt{\frac{x}{2}}}}dx = e^{\frac{-\pi}{2}} I$

$$I = \int_0^2 \frac{e^{\cos^{-1}\sqrt{\frac{x}{2}}}}{e^{\sin^{-1}\sqrt{\frac{x}{2}}}+e^{\cos^{-1}\sqrt{\frac{x}{2}}}}dx = \frac{1}{2}\int_0^2\left\{\frac{e^{\cos^{-1}\sqrt{\frac{x}{2}}}}{e^{\sin^{-1}\sqrt{\frac{x}{2}}}+e^{\cos^{-1}\sqrt{\frac{x}{2}}}}+\frac{e^{\cos^{-1}\sqrt{1-\frac{x}{2}}}}{e^{\sin^{-1}\sqrt{1-\frac{x}{2}}}+e^{\cos^{-1}\sqrt{1-\frac{x}{2}}}}\right\}dx$$

$$\Rightarrow \quad \frac{1}{2}\int_0^2\left\{\frac{e^{\cos^{-1}\sqrt{\frac{x}{2}}}}{e^{\sin^{-1}\sqrt{\frac{x}{2}}}+e^{\cos^{-1}\sqrt{\frac{x}{2}}}}+\frac{e^{\sin^{-1}\sqrt{\frac{x}{2}}}}{e^{\cos^{-1}\sqrt{\frac{x}{2}}}+e^{\sin^{-1}\sqrt{\frac{x}{2}}}}\right\}dx = \frac{1}{2}\int_0^2 1\,dx = 1$$

$k = 1, l = \frac{-1}{2}$. **Ans.**

80. $\left|2\left(x^2+\frac{1}{x^2}\right)+\left|1-x^2\right|\right| = 6 - 2^{x^2-1} - \frac{1}{2^{x^2-1}}$

$= \left|2\left(x^2+\frac{1}{x^2}\right)+\left|1-x^2\right|\right| = 6 - \left(2^{x^2-1} + \frac{1}{2^{x^2-1}}\right)$

L.H.S. ≥ 4 and R.H.S. ≤ 4

equation will satisfy at $x = \pm 1$

$\therefore \quad x_1 = -1$ and $x_2 = 1$

Now, $\int_0^4 \left\{\frac{x}{4}\right\}\left(1+\left[\tan\left(\frac{\{x\}}{1+\{x\}}\right)\right]\right)dx$

$\therefore \quad \frac{\{x\}}{1+\{x\}} = 1 - \frac{1}{1+\{x\}} \Rightarrow \frac{\{x\}}{1+\{x\}} \in \left[0, \frac{1}{2}\right)$

$\therefore \quad \tan\left(\frac{\{x\}}{1+\{x\}}\right) < 1$ and $\left[\tan\left(\frac{\{x\}}{1+\{x\}}\right)\right] = 0$

$\therefore \quad \int_0^4 \left\{\frac{x}{4}\right\}dx = \int_0^4 \frac{x}{4}dx \qquad 0 \leq x \leq 4;\ 0 \leq \frac{x}{4} \leq 1;\ \therefore \left\{\frac{x}{4}\right\} = \frac{x}{4}$

$= \frac{1}{4}\left(\frac{x^2}{2}\right)_0^4 = \frac{1}{8}\cdot 16 = 2$ **Ans.**

81. $\int_0^{27} x\, f'(x)\, dx = \lambda - 3\int_0^{3} x^2 f(x^3)\, dx$

$x^3 = t \Rightarrow 3x^2\, dx = dt$

$\int_0^{27} x\, f'(x)\, dx = \lambda - \int_0^{27} f(t)\, dt$

$\int_0^{27} x\, f'(x)\, dx + \int_0^{27} f(x)\, dx = \lambda$

$\lambda = \int_0^{27} \left(f(x) + x\, f'(x)\right) dx = \left(x\, f(x)\right)_0^{27}$

$\lambda = 27\, f(27) - 0$

$= 27 \cdot \frac{1}{3} = 9$ $\qquad \{\because f \text{ is decreasing } \therefore f(27) = \frac{1}{3}\}$

82. $g(x) = e^{2x} + \int_0^{\sin x} \frac{e^t dt}{\cos^2 x + 2\underset{\substack{\downarrow \\ \text{king}}}{t} \sin x - t^2}$

$\Rightarrow \qquad g(x) = e^{2x} + e^{\sin x} \int_0^{\sin x} \frac{e^{-t} dt}{1 - t^2}$

$g(0) = 1 \Rightarrow f(1) = 0$

$g'(x) = 2e^{2x} + \sec x + e^{\sin x} \cdot \cos x \int_0^{\sin x} \frac{e^{-t} dt}{1 - t^2}$

$g'(0) = 3$

$g(f(x)) = x$

$g\left(f(x)\right) \cdot f'(x) = 1$

$g'\left(f(1)\right) \cdot f'(1) = 1$

$g'(0) \cdot f'(1) = 1 \Rightarrow f'(1) = \frac{1}{3}.$

$g''(x) = 4e^{2x} + \sec x \cdot \tan x + \sec x + e^{\sin x} \cdot \cos x \int_0^{\sin x} \frac{e^{-t} dt}{1 - t^2}$

$g(0) = 1$

$g'(0) = 3$

$g''(0) = 5.$ **Ans.**

83. $(x^2 + x + 1)\,P(x-1) = (x^2 - x + 1)\,P(x)$ ....(1)

$$\frac{P(x)}{x^2+x+1} = \frac{P(x-1)}{x^2-x+1} = \frac{P(x-1)}{(x-1)^2+(x-1)+1}$$

Hence, $\dfrac{P(x)}{x^2+x+1} = \dfrac{P(x-1)}{(x-1)^2+(x-1)+1} = \text{constant} = k$

$\Rightarrow \quad P(x) = k\,(x^2 + x + 1),\ P(1) = 3$

$\Rightarrow \quad P(x) = x^2 + x + 1$

Now, $\displaystyle\int_0^1 \tan^{-1}\left(\frac{2x}{1+P(x^2)}\right)dx = \int_0^1 \tan^{-1}\left(\frac{2x}{1+x^4+x^2+1}\right)dx$

$$= \int_0^1 \tan^{-1}\left(\frac{2x}{1+(x^2+x+1)(x^2-x+2)}\right)dx$$

$$= \int_0^1 \left(\tan^{-1}(x^2+x+1) - \tan^{-1}(x^2-x+1)\right)dx$$

$$= \int_0^1 \left(\frac{\pi}{2} - \cot^{-1}(x^2+x+1) - \frac{\pi}{2} + \cot^{-1}(x^2-x+1)\right)dx$$

$$= \int_0^1 \left(\cot^{-1}(x^2-x+1) - \cot^{-1}(x^2+x+1)\right)dx$$

$$= \int_0^1 \left(\tan^{-1}\left(\frac{1}{1+x^2-x}\right) - \tan^{-1}\left(\frac{1}{1+x^2+x}\right)\right)dx$$

$$= \int_0^1 \left(\tan^{-1}x - \underset{\substack{\downarrow \\ \text{King}}}{\tan^{-1}(x-1)} - \tan^{-1}(1+x) + \tan^{-1}x\right)dx$$

$$= \int_0^1 \tan^{-1}x\,dx + \int_0^1 \tan^{-1}x\,dx - \int_0^1 \tan^{-1}(x+1) + \int_0^1 \tan^{-1}x$$

$$\int_0^1 \tan^{-1}\left(\frac{2x}{1+P(x^2)}\right)dx + \int_0^1 \tan^{-1}(1+x)\,dx = 3\int_0^1 \tan^{-1}x\,dx = 3\left[\frac{\pi}{4} - \frac{1}{2}\ln 2\right]$$

$$= 3\left[\tan^{-1}x \cdot x\Big|_0^1 - \int_0^1 \frac{x}{1+x^2}\right] = 3\left[\frac{\pi}{4} - \frac{1}{2}\ln 2\right]$$

$$= \frac{3}{4}[\pi - 2\ln 2] = \frac{3}{4}[\pi - \ln 4]$$

$$= \frac{3}{4}[\pi - \ln 4] \Rightarrow k = 12 \textbf{ Ans.}$$

84. $I = \int_{-\infty}^{\infty} e^{-(x^2+4x+1)}\,dx = \int_{-\infty}^{\infty} e^{-\left((x+2)^2-3\right)}\,dx = e^3 \int_{-\infty}^{\infty} e^{-(x+2)^2}\,dx$

Put $x + 2 = t$

$I = e^3 \int_{-\infty}^{\infty} e^{-t^2}\,dt = e^3\sqrt{\pi}$. **Ans.**

85. Consider $I = \int_1^t \frac{dx}{1+x^{2/3}}$ (put, $x = y^3$)

So, $I = \int_1^{t^{1/3}} \frac{3y^2}{1+y^2}\,dy = \int_1^{t^{1/3}} \frac{3(y^2+1)-3}{1+y^2}\,dy$

$\Rightarrow \quad I = (3t^{1/3} - 3) - 3\tan^{-1}\left(t^{1/3}\right) + \frac{3\pi}{4}$

Now, $L = \lim_{t\to\infty}\left(3t^{1/3} - 3 - 3\tan^{-1}\left(t^{1/3}\right) + \frac{3\pi}{4} - at^b\right)$

For L to exist, $a = 3$ and $b = \frac{1}{3}$

and hence $L = \frac{3\pi}{4} - \frac{3\pi}{2} - 3 = \frac{-3\pi}{4} - 3$

So, $4L = -3\pi - 12 \Rightarrow 12 = -4L - 3\pi$

Hence, $ab - 4L - 3\pi = 1 + 12 = 13$. **Ans.**

86 $I =$

$$\int_{-\pi}^{\pi}\left(1+\cos x+\cos 2x+\ldots..+\cos(2013x)\right)\left(1+\sin x+\sin 2x+\ldots..+\sin(2013x)\right)dx$$

Using King and add

$$2I = \int_{-\pi}^{\pi} 2\left(1 + \cos x + \cos 2x + \ldots\ldots + \cos(2013x)\right) dx$$

$$\Rightarrow \quad I = \int_{-\pi}^{\pi} 1\,dx + \int_{-\pi}^{\pi} \left(\cos x + \cos 2x + \ldots\ldots + \cos(2013x)\right) dx$$

$$= 2\int_{0}^{\pi} 1\,dx + 2\underbrace{\int_{0}^{\pi} \left(\cos x + \cos 2x + \ldots\ldots + \cos(2013x)\right) dx}_{\text{zero}} = 2\pi$$

$\Rightarrow \quad I = 2\pi$. **Ans.**

87. $\cot^{-1}(1 - x + x^2) = \tan^{-1}\dfrac{1}{1-x+x^2} = \tan^{-1}\dfrac{x-(x-1)}{1+x(x-1)^2} = \tan^{-1} x - \tan^{-1}(x-1)$.

Also, $\displaystyle\int_0^1 \tan^{-1}(x-1)dx = -\int_0^1 \tan^{-1} x\; dx$

Therefore, $\displaystyle\int_0^1 \cot^{-1}(1-x+x^2)\; dx = \int_0^1 \left(\tan^{-1} x - \tan^{-1}(x-1)\right) dx = 2\int_0^1 \tan^{-1} x\; dx$,

Hence $\dfrac{\displaystyle\int_0^1 \cot^{-1}(1-x+x^2)\,dx}{\displaystyle\int_0^1 \tan^{-1} x\,dx} = 2$ **Ans.**

88. $1 + 2\cos x + 2\cos(2x) + \ldots\ldots\ldots + 2\cos(nx)$

$$= \frac{\sin\left(\left(n+\frac{1}{2}\right)x\right)}{\sin\left(\frac{x}{2}\right)} = \sin(nx)\cot\left(\frac{x}{2}\right) + \cos(nx),$$

we have $\displaystyle a_n = \int_0^{\pi} \cot\left(\frac{x}{2}\right)\sin(nx)\; dx = \int_0^{\pi} \left((1 + 2\cos x + 2\cos(2x) + \ldots\ldots + \cos(nx)\right)dx$

$$= \int_0^{\pi} 1\,dx + 0 = \pi\ .$$

Thus, $S = 2013\,\pi$ and $\cos S = \cos(2013\pi) = -1$

89 $\lim_{n\to\infty}\sum_{k=1}^{n}\sqrt[3]{\frac{k^4}{n^7}} = \lim_{n\to\infty}\sum_{k=1}^{n}\left(\frac{1}{n}\right)\left(\frac{k}{n}\right)^{\frac{4}{3}} = \int_0^1 x^{\frac{4}{3}}\,dx = \frac{3}{7}$. **Ans.**

90. $\int_0^\infty e^t e^{-xt}\,dt = \int_0^\infty e^{(1-x)t}\,dt \Rightarrow f(x) = \left[\frac{e^{(1-x)t}}{1-x}\right]_{t=0}^{\infty} = \frac{1}{x-1}$

(Since, $x > 1$, $e^{(1-x)t} \to 0$) which has range $(0, \infty)$]

91. [Sol.$_{36/def/SC}$ $\int_3^{x^3} e^{t^2}\,dt = \int_3^1 e^{t^2}\,dt + \int_1^{x^3} e^{t^2}\,dt = \int_1^{x^3} e^{t^2}\,dt - \int_1^3 e^{t^2}\,dt = g(x^3) - g(3)$ **Ans.**

92. We have $I_1 = \ln(1+\sqrt{3})$; $I_2 = \int_0^{\sqrt{3}} \frac{dx}{1+x^2}$

$I_2 = \frac{\pi}{3}$

$I_0 = \lim_{n\to\infty} I_n = \lim_{n\to\infty}\left(\int_0^1 \frac{dx}{1+x^n} + \underbrace{\int_1^{\sqrt{3}} \frac{dx}{1+x^n}}_{\text{zero}}\right) = \int_0^1 dx = 1$

Hence $I_0 = 1$. Now verify all alternatives.

93 $I = 4\int_0^{\pi/2} (\sin^2 2x - \sin^4 2x)\,dx$; $2x = t \Rightarrow dx = \frac{dt}{2}$

hence $2\int_0^{\pi} (\sin^2 t - \sin^4 t)\,dt = 4\int_0^{\pi/2} (\cos^2 t - \sin^4 t)\,dt$; $4\left[\frac{\pi}{4} - \frac{3\pi}{16}\right] = \pi - \frac{3\pi}{4} = \frac{\pi}{4}$ Ans.

94. $\int_0^a f(x)\,dx - \int_a^1 f(x)\,dx = 2f(a) + 3a + b$

Differentiating w.r.t. by a Leibnitz rule,

$$2f(a) = 2f'(a) + 3 \quad \text{or} \quad f'(a) - f(a) = \frac{-3}{2}$$

a linear differential equation with integrating factor e^{-a}

$$\therefore \quad f(a)e^{-a} = \int \frac{-3}{2}e^{-a}\,da = \frac{3}{2}e^{-a} + A$$

$$f(a) = \frac{3}{2} + Ae^{a}$$

$$f(1) = 0 \quad \Rightarrow \quad \frac{3}{2} + Ae = 0 \quad \Rightarrow \quad A = \frac{-3}{2}e^{-1}$$

$$\therefore \quad f(a) = \frac{3}{2}\left(1 - e^{a-1}\right)$$

$$f(x) = \frac{3}{2}\left(1 - e^{x-1}\right) \textbf{ Ans.}$$

95. $$I = \int_{-1}^{1} \frac{dx}{(1+e^{x})(1+x^{2})} \quad ....(1)$$

$$= \int_{-1}^{1} \frac{dx}{1+e^{-x}} \cdot \frac{1}{1+x^{2}} \quad \text{(using King)}$$

$$I = \int_{-1}^{1} \frac{e^{x}\,dx}{(1+e^{x})(1+x^{2})} \quad ....(2)$$

adding (1) and (2)

$$2I = \int_{-1}^{1} \frac{(1+e^{x})\,dx}{(1+e^{x})(1+x^{2})} = \int_{-1}^{1} \frac{dx}{(1+x^{2})} = 2\int_{0}^{1} \frac{dx}{(1+x^{2})}$$

$$I = \int_{0}^{1} \frac{dx}{(1+x^{2})} = \tan^{-1}(1) = \pi/4$$

[convert it into value of definite integral 'I' is same as]

96. Let $$L = \lim_{n\to\infty} \prod_{k=1}^{n} \left(\frac{k+n}{k}\right)^{\left(e^{1/n}-1\right)}$$

$$\ln L = \lim_{n\to\infty} \left(e^{1/n} - 1\right) \sum_{k=1}^{n} \ln\left(\frac{k+n}{k}\right) = \lim_{n\to\infty} \frac{e^{1/n}-1}{1/n} \cdot \frac{1}{n} \sum_{k=1}^{n} \ln\left(\frac{1+(k/n)}{k/n}\right)$$

$$= \lim_{n\to\infty} \frac{e^{x}-1}{x} \cdot \int_{0}^{1} \ln\left(\frac{x+1}{x}\right) dx$$

$$\ln L = \int_{0}^{1} \ln(x+1)\,dx - \int_{0}^{1} \ln x\,dx$$

$$\ln L = \left[(x+1)\ln(x+1) - (x+1)\right]_0^1 - \left(x\ln x - x\right)\Big]_0^1$$

$$= [\,(2\ln 2 - 2) - (-1)] - [(-1) - 0] \qquad \lim_{x\to 0} x\ln x = 0$$

$$= 2\ln 2 - 1 + 1 = \ln 4$$

$\Rightarrow L = 4$

$\Rightarrow \quad 225 \times 4 = 900$ Ans.

$= 2\ln 2 - 1 + 1 = \ln 4 \quad \Rightarrow \quad L = 4$ Ans.

97 $f(x)\cdot e^{f(x)} = x$ (Given)

$\therefore \ \ln(f(x)) + f(x) = \ln x$

$\therefore \ f(x) = \ln x - \ln(f(x))$

$$\int_0^e f(x)\,dx = \underbrace{\int_0^e \ln x\,dx}_{\text{zero}} - \int_0^e \ln f(x)\,dx \ \Rightarrow \ \int_0^e f(x)\,dx = -\int_0^e \underbrace{1}_{\text{II}}\cdot\underbrace{\ln f(x)}_{\text{I}}\,dx$$

$$\Rightarrow \int_0^e f(x)\,dx = -\ln(f(x))\cdot x\Big|_0^e + \int_0^e \frac{1}{f(x)}f'(x)\cdot x\,dx = (-0+0) + \int_0^e f'(x)\cdot e^{f(x)}\,dx$$

$= e^{f(x)}\Big]_0^e = e - 1.$ **Ans**

Note: $f(e) = 1$ and $f(0) = 0$.

Put $x = 0$ in original eq.

$f(0) = 0$

Let $f(x) = 1$ in original equation

$e = x$

$\Rightarrow f(e) = 1.$

98. $t = \ln\ln\ln x$

$$\text{Given integral} = \int_1^{\infty} t^{-11/10}\,dt = \left(\frac{t^{-1/10}}{-1/10}\right)_1^{\infty} = 10.$$

99. Put $x = t^4 \Rightarrow dx = 4t^3dt$

$$I = \int_1^2 \frac{4t^3}{t^2+t}dt = \int_1^2 \frac{4t^2}{1+t}dt = 4\left[\int_1^2 \frac{(t^2-1)+1}{1+t}dt\right] = 4\left[\int_1^2 (t-1)dt + \int_1^2 \frac{dt}{1+t}\right]$$

$$= 4\left[\left(\frac{t^2}{2} - t\right)_1^2 + \ln(1+t)\Big|_1^2\right] = 4\left[(0) - \left(-\frac{1}{2}\right) + \ln 3 - \ln 2\right] = \left[\frac{1}{2} + \ln\frac{3}{2}\right] = 4\ln\frac{3}{2} + 2 \text{ Ans.}$$

100. Note that in $\left(-\frac{1}{2}, \frac{1}{2}\right)$, $\sin^{-1}(3x - 4x^3) = 3\sin^{-1}x$ and $\cos^{-1}(4x^3 - 3x) = 2\pi - 3\cos^{-1}x$

hence $f(x) = 3\sin^{-1}x - 2\pi + 3\cos^{-1}x = -\frac{\pi}{2}$

$$\therefore I = -\frac{\pi}{2}\int_{-1/2}^{1/2} dx = -\frac{\pi}{2} \;]$$

[Alternate:

$f(x) = \sin^{-1}(3x - 4x^3) - [\pi - \cos^{-1}(3x - 4x^3)]$

$= -\pi + (\sin^{-1}(3x - 4x^3) + \cos^{-1}(3x - 4x^3)) = -\frac{\pi}{2}$

101 $\lim\limits_{n\to\infty} \dfrac{\int_0^n ln\left(1+\frac{1}{\sqrt{x}}\right)dx}{\sqrt{n}}$ $\left(\frac{\infty}{\infty}\right)$

$\lim\limits_{n\to\infty} \dfrac{ln\left(1+\frac{1}{\sqrt{n}}\right)}{\frac{1}{2\sqrt{n}}} = 2$. **Ans.**

102 $I_1 = \int_0^{\pi/2} \cos\theta \cdot f\left(\sin\theta + 1 - \sin^2\theta\right)d\theta$

put $\sin\theta = t \Rightarrow \cos\theta\, d\theta = dt$

$I_1 = \int_0^1 f\left(t + 1 - t^2\right)dt$ ....(1)

Now, $I_2 = \int_0^{\pi/2} 2\sin\theta\cos\theta \cdot f\left(\sin\theta + 1 - \sin^2\theta\right)d\theta$

put $\sin\theta = t \Rightarrow \cos\theta\, d\theta = dt$

$I_2 = \int_0^1 2t \cdot f\left(t + 1 - t^2\right)dt$ .....(2)

$= \int_0^1 (1-t)\cdot f\left(1 - t + 1 - (1-t)^2\right)dt$ (Using King)

$I_2 = 2\int_0^1 (1-t)\cdot f\left(1 + t - t^2\right)dt$, $I_2 = 2\left[\int_0^1 f\left(1 + t - t^2\right)dt - \int_0^1 t\cdot f\left(1 + t - t^2\right)dt\right]$ ....(3)

(2) + (3), $2I_2 = 2\int_0^1 f\left(1 + t - t^2\right)dt$

$\Rightarrow I_2 = \int_0^1 f\left(1 + t - t^2\right)dt \Rightarrow I_1 = I_2 \Rightarrow \frac{I_1}{I_2} = 1$ **Ans.**

103. $$\int_0^1 \frac{1}{\sqrt{x}\sqrt{1+\sqrt{x}}\sqrt{1+\sqrt{1+\sqrt{x}}}\,dx} = 2\int_0^1 \frac{\left(\sqrt{x}\right)}{\sqrt{1+\sqrt{x}}\sqrt{1+\sqrt{1+\sqrt{x}}}\,dx}\,dx$$

$$= 4\int_0^1 \frac{\left(\sqrt{1+\sqrt{x}}\right)'}{\sqrt{1+\sqrt{1+\sqrt{x}}}\,dx} = 8\left[\sqrt{1+\sqrt{1+\sqrt{x}}}\right]_0^1 = 8\left(\sqrt{1+\sqrt{2}}-\sqrt{2}\right).$$

104. $$I = \int_{5/4}^{10} \frac{\sqrt{\left(\sqrt{x-1}\right)^2+2\sqrt{x-1}+1}+\sqrt{\left(\sqrt{x+1}\right)^2-2\sqrt{x+1}+1}}{\sqrt{x^2-1}}\,dx$$

$$= \int_{5/4}^{10} \frac{\left|\sqrt{x-1}+1\right|+\left|\sqrt{x+1}-1\right|}{\left(\sqrt{x-1}\right)\left(\sqrt{x+1}\right)}\,dx$$

$$= \int_{5/4}^{10} \frac{\sqrt{x-1}+\sqrt{x+1}}{\sqrt{x-1}\cdot\sqrt{x+1}}\,dx = \int_{5/4}^{10} \frac{dx}{\sqrt{x+1}} + \int_{5/4}^{10} \frac{dx}{\sqrt{x-1}}$$

$$= 2(x+1)^{1/2} + 2(x-1)^{1/2}\Big]_{5/4}^{10}$$

$$= 2\left[\left(\sqrt{11}+3\right)-\left(\frac{3}{2}+\frac{1}{2}\right)\right]_{5/4}^{10} = 2\left[\sqrt{11}+1\right]$$

$\Rightarrow$ $a = 2;\ b = 11$ and $c = 1$

$\Rightarrow$ $a + b + c = 14$ **Ans.**

105. $$F(t) = \frac{\int_0^{\pi t/2} |\cos 2x|\,dx}{t} \qquad \left(\text{put } 2x = V \Rightarrow dx = \frac{dV}{2}\right)$$

$$= \frac{\int_0^{\pi t} |\cos V|\,dV}{2t} = \frac{\pi}{2t}\int_0^t |\cos V|\,dV = \frac{\pi}{2t}\int_0^t \cos V\,dV \quad \text{as } t \in (0, 1]$$

$$= \frac{\pi}{2t}\left(\sin V\right)_0^t = \frac{\pi \sin t}{2t}$$

hence $\displaystyle\lim_{t\to 0} F(t) = \lim_{t\to 0} \frac{\pi}{2}\frac{\sin t}{t} = \frac{\pi}{2}$ **Ans.**

106. $$I = \int_0^{\pi/6} x \sec x \cdot \sec(\pi/6 - x)\,dx = \int_0^{\pi/6} \frac{x}{\cos x \cos(\pi/6 - x)}\,dx$$

$$\Rightarrow \qquad I = \int_0^{\pi/6} \frac{x}{\sin \pi/6} \cdot \frac{\sin\{x + (\pi/6 - x)\}}{\cos x \cdot \cos(\pi/6 - x)} dx$$

$$\Rightarrow \qquad I = 2\int_0^{\pi/6} x \cdot \frac{\sin x \cdot \cos\left(\frac{\pi}{6} - x\right) + \cos x \cdot \sin\left(\frac{\pi}{6} - x\right)}{\cos x \cos\left(\frac{\pi}{6} - x\right)} dx$$

$$\Rightarrow \qquad I = 2\int_0^{\pi/6} x \cdot \left\{\tan x + \tan\left(\frac{\pi}{6} - x\right)\right\} dx$$

Applying king properly and adding, we have

$$2I = 2\int_0^{\pi/6} \left(x + \frac{\pi}{6} - x\right) \cdot \left\{\tan x + \tan\left(\frac{\pi}{6} - x\right)\right\} dx$$

$$\Rightarrow \qquad I = \frac{\pi}{6}\int_0^{\pi/6} 2\tan x \; dx \qquad \left\{\because \int_0^{\pi/6} \tan\left(\frac{\pi}{6} - x\right) dx = \int_0^{\pi/6} \tan x \, dx\right\}$$

$$= \frac{\pi}{3}[\ln \sec x]_0^{\pi/6} = \frac{\pi}{3}\left(\ln \sec \frac{\pi}{6}\right) = \alpha \ln \sec \beta$$

Clearly $\alpha = \frac{\pi}{3}$ and $\beta = \frac{\pi}{6}$

$$\therefore \qquad \frac{335\pi}{|\alpha - \beta|} = \frac{335\pi}{\left|\frac{\pi}{3} - \frac{\pi}{6}\right|} = 2010 \textbf{ Ans.}$$

107. $f(x)$ simplifies to $x^2 + x + 1$

$$\therefore \qquad I = \int_0^1 (x^2 + x + 1) dx = \frac{1}{3} + \frac{1}{2} + 1 = \frac{11}{6} \text{ Ans.}$$

108 On differentiation $(f(x))^2 = \int_0^x f(t) \cdot \frac{2\sec^2 t}{4 + \tan t} dt$, we get $f'(x) = \frac{\sec^2 x}{4 + \tan x}$

$\Rightarrow \qquad f(x) = ln(4 + \tan x) + C;$

$f(0) = 0 \qquad \Rightarrow \qquad C = -ln\,4$

$$\therefore \qquad f(x) = ln\frac{(4 + \tan x)}{4}$$

109. $\int_0^1 (1 - x^7)^{1/4} dx - \int_0^1 (1 - x^4)^{1/7} dx$

$y = f(x) = (1 - x^7)^{1/4}$

$\Rightarrow \quad y^4 = (1 - x^7) \quad \Rightarrow \quad x = (1 - y^4)^{1/7}$

hence functions are inverse of each other

$$I = \int_0^1 f(x)\,dx - \int_0^1 g(y)\,dy = \int_0^1 f(x)\,dx - \int_1^0 x\,f'(x)\,dx$$

$$\text{Hence } I = \int_0^1 \left(f(x) + x\,f'(x)\right)dx = x\,f(x)\Big]_0^1 = f(1) = 0 \;]$$

Paragraph for question nos. 110 to 113

110 $A_n = \dfrac{(n+1)+(n+2)+.....+(n+n)}{n} = \dfrac{3n+1}{2}$;

$$\lim_{n\to\infty} \frac{A_n}{n} = \lim_{n\to\infty} \frac{3n+1}{2n} = \frac{3}{2} \textbf{ Ans.}$$

111. $G_n = \left((n+1)(n+2)....(n+n)\right)^{1/n}$;

$$\frac{G_n}{n} = \left(\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)....\left(1+\frac{n}{n}\right)^{1/n}\right)$$

$$\lim_{n\to\infty} ln\left(\frac{G_n}{n}\right) = \lim_{n\to\infty} \frac{1}{n}\sum_{r=1}^{n} \ln\left(1+\frac{r}{n}\right) = \int_0^1 ln(1+x)\,dx = 2ln\,2 - 1 = ln\left(\frac{4}{e}\right)$$

$$\therefore \quad \lim_{n\to\infty} \frac{G_n}{n} = \frac{4}{e} \textbf{ Ans.}$$

Here use is made of the formula: $\lim\limits_{n\to\infty} \dfrac{1}{n}\sum\limits_{r=1}^{n} f\left(\dfrac{r}{n}\right) = \displaystyle\int_0^1 f(x)\,dx$

112 $H_n = \dfrac{n}{\dfrac{1}{n+1}+\dfrac{1}{n+2}+.....+\dfrac{1}{n+n}}$;

$$\lim_{n\to\infty} \frac{n}{H_n} = \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{1+\frac{r}{n}} = \int_0^1 \frac{dn}{1+x} = ln\,2$$

$$\therefore \quad \lim_{n\to\infty} \frac{H_n}{n} = (ln\,2)^{-1} \textbf{ Ans.}$$

113 $R_n = \left(\dfrac{(n+1)^2+(n+2)^2+.....+(n+n)^2}{n}\right)^{1/2}$;

$$\lim_{n\to\infty}\left(\frac{R_n}{n}\right)^2 = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\left(1+\frac{r}{n}\right)^2 = \int_0^1 (1+x)^2\,dx = \frac{7}{3};$$

$$\therefore\quad \lim_{n\to\infty}\frac{R_n}{n} = \sqrt{\frac{7}{3}}\ \textbf{Ans.}$$

114. $I = \int_1^3 \left|(x-1)(3-x)(x-2)\right| dx$

let $x = \cos^2\theta + 3\sin^2\theta$

$dx = 2\sin 2\theta\, d\theta$

$x-1 = 2\sin^2\theta$; $3-x = 2\cos^2\theta$ and

$x-2 = \cos^2\theta + 3\sin^2\theta - 2 = 2\sin^2\theta - 1 = -\cos 2\theta$

$$I = \int_0^{\pi/2}\left|2\sin\theta\cdot 2\cos^2\theta\cdot\cos 2\theta\right| 2\sin 2\theta\, d\theta$$

$$= \int_0^{\pi/2} 4\sin^2\theta\cdot\cos^2\theta\cdot 2\sin 2\theta\left|\cos 2\theta\right| d\theta$$

$$= \int_0^{\pi/2} 2\sin^3 2\theta\left|\cos 2\theta\right| d\theta$$

put $2\theta = t$

$$I = \int_0^{\pi}\cancel{2}\sin^3 t\,|\cos t|\frac{dt}{\cancel{2}} = 2\int_0^{\pi/2}(\sin^3 t\cdot\cos t)\,dt$$

put $\sin t = y$

$$I = 2\int_0^1 y^3\,dy = 2\cdot\left.\frac{y^4}{4}\right|_0^1 = \frac{1}{2}\ \text{Ans.}$$

115. Consider $I = \int_0^t \tan\theta\sqrt{\cos\theta}\ ln(\cos\theta)\,d\theta = \int_0^t \underbrace{\frac{\sin\theta}{\sqrt{\cos\theta}}}_{\text{II}}\ \underbrace{ln(\cos\theta)}_{\text{II}}\,d\theta$

$$I = ln(\cos\theta)\left(-2\sqrt{\cos\theta}\right)\Big]_0^t - \int_0^t\left(\frac{-\sin\theta}{\sqrt{\cos\theta}}\right)(-2)\,d\theta$$

$$= \left(-2\sqrt{\cos t}\ ln(\cos t) - 0\right) + 2\int_0^t\frac{\sin\theta}{\sqrt{\cos\theta}}\,d\theta$$

$$= -2\sqrt{\cos t}\ ln(\cos t) + 4\left((\cos t)^{\frac{1}{2}}\right)_0^t$$

$$= -2\sqrt{\cos t}\ ln(\cos t) + 4\left((\cos t)^{\frac{1}{2}} - 1\right)$$

$$= -2\sqrt{\cos t}\ ln(\cos t) + 4(\cos t)^{\frac{1}{2}} - 4$$

Now $\lim\limits_{t \to \frac{\pi}{2}} I = \underbrace{\lim\limits_{t \to \frac{\pi}{2}} 4\sqrt{\cos t}}_{0} - 2\overbrace{\lim\limits_{t \to \frac{\pi}{2}} \sqrt{\cos t}\ ln(\cos t)}^{\text{zero}} - 4$

$\therefore$ $\quad -4$ Ans.

116 Given $F(y) = \int\limits_{x}^{xy} f(t)\, dt$ with $f(2) = 2$.

Find $\int\limits_{1}^{x} f(t)\, dt$

$F(y) = \int\limits_{x}^{xy} f(t)\, dt \Rightarrow F'(y) = f(xy)\, x \Rightarrow \dfrac{F'(y)}{x} = f(xy).$

$f(2) = 2 \Rightarrow F'(y) = \dfrac{4}{y} \Rightarrow F(y) = 4 \log y$

Hence, $F\left(\dfrac{1}{x}\right) = -4 \log x \Rightarrow \int\limits_{1}^{x} f(t)\, dt = 4\, ln\, x$ Ans.

Aliter: Given $\int\limits_{x}^{xy} f(t)\, dt$ is independent of x

hence its derivative w.r.t. x is zero

$\Rightarrow \quad y\, f(xy) - f(x) = 0$

$\Rightarrow \quad f(xy) = \dfrac{f(x)}{y}$

put $xy = 2 \quad \Rightarrow \quad y = \dfrac{2}{x}$. Hence $f(x) = \dfrac{4}{x}$

$\Rightarrow \int\limits_{1}^{x} \{f(t)\}\, dt = \int\limits_{1}^{x} \dfrac{4}{t}\, dt = 4\, ln\, x$ **Ans.**

117. By parts, $\int\limits_{1}^{3} \dfrac{ln\,(x+1)}{x^2}\, dx = ln(x+1)\left(-\dfrac{1}{x}\right)\Bigg|_{1}^{3} + \int\limits_{1}^{3} \dfrac{1}{x(x+1)}\, dx$

$$= \left(\frac{-1}{3} ln 4 + ln 2\right) + \left[ln x - ln(x+1)\right]_1^3$$

$$= \frac{1}{3} ln 2 + (ln 3 - ln 4) - (ln 2) = ln 3 - \frac{2}{3} ln 2$$

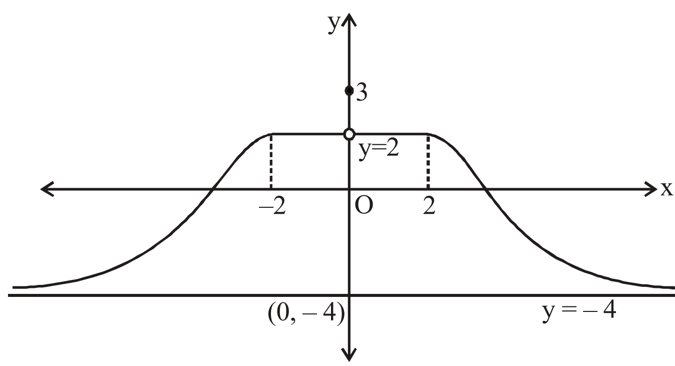
$$= \frac{1}{3} (ln 27 - ln 4) \text{ Ans.}$$

Paragraph for question nos. 118 to 120

Sol. We have $H(x) = \left[\frac{3(x^2+1) - |x|}{x^2+1}\right] = \left[3 - \frac{|x|}{x^2+1}\right]$ $\left(\text{As } 0 < \frac{|x|}{x^2+1} = \frac{1}{|x| + \frac{1}{|x|}} \le \frac{1}{2}\right)$

So, $H(x) = \begin{cases} \frac{8}{\pi} \tan^{-1}(-|x|+3), & x \in (-\infty, -2) \cup (2, \infty) \\ 3, & x = 0 \\ 2, & x \in [-2, 2] - \{0\} \end{cases}$

Also, H(x) is an even function on R, so graph of H(x) is symmetrical about y-axis.



Graph of H(x)

Note : $H(x) = \frac{8}{\pi} \tan^{-1}(3 - |x|), |x| > 2$

Also for $|x| > 2$, $3 - |x| \in (-\infty, 1)$, so

$$H(x) \in \frac{8}{\pi}\left(\frac{-\pi}{2}, \frac{\pi}{4}\right) = (-4, 2)$$

118 From the above graph, it is clear that H(x) is discontinuous at only one point i.e. x = 0.

119 Range of H(x) is $(-4, 2] \cup \{3\}$.

120 $\int_{-2}^{1} H(x)\, dx = \int_{-2}^{1} 2\, dx = 2(3) = 6.$

121 Consider $I_2 = \int_0^1 x^{1004}(1-x^{2010})^{1004}dx$; Put $x^{1005} = t$

$\Rightarrow 1005\, x^{1004}\, dx = dt$

So, $I_2 = \frac{1}{1005}\int_0^1 (1-t^2)^{1004}dt$ .... (1)

Also $I_2 = \frac{1}{1005}\int_0^1 \left[1-(1-t)^2\right]^{1004} dt$ .... (2) (Using King)

$\Rightarrow I_2 = \frac{1}{1005}\int_0^1 \left(t(2-t)\right)^{1004}dt = \frac{1}{1005}\int_0^1 t^{1004}(2-t)^{2004}dt$;

Put $t = 2y \Rightarrow dt = 2dy$

So, $I_2 = \frac{1}{1005}\int_0^{1/2} (2y)^{1004}(2-2y)^{1004}dt$

$= \frac{1}{1005} 2\cdot 2^{1004}\cdot 2^{1004}\int_0^{1/2} y^{1004}(1-y)^{1004}dy$

$I_2 = \frac{1}{1005} 2^{2009}\int_0^{1/2} y^{1004}(1-y)^{1004}dy$ .... (3)

Now $I_1 = \int_0^1 x^{1004}(1-x)^{1004}dx = 2\int_0^{1/2} x^{1004}(1-x)^{1004}dx$ .... (4) (using Queen)

$\therefore$ From (3) and (4), we get

$I_2 = \frac{1}{1005} 2^{2010}\frac{I_1}{4} \Rightarrow 2^{2010}\frac{I_1}{I_2} = 4020$ **Ans.**

122 $I = \int_1^2 \frac{1+(u-1)^{2009}}{u^{2011}}du$ (where $1 + x^4 = u$)

$= \int_1^2 u^{-2011}du + \int_1^2 \left(1-\frac{1}{u}\right)^{2009}\cdot\frac{1}{u^2}du$

$= \left.\frac{u^{-2010}}{2010}\right|_1^2 + \int_0^{\frac{1}{2}} t^{2009}dt$ (where $1 - \frac{1}{u} = t$)

$$= \left.\frac{u^{-2010}}{-2010}\right|_1^2 + \left.\frac{t^{2010}}{2010}\right|_0^{1/2}$$

$$= \frac{-1}{2010}\left[\frac{1}{2^{2010}} - 1\right] + \frac{1}{2010}\left[\frac{1}{2^{2010}} - 0\right] = \frac{1}{2010}. \text{ Ans.}$$

123 Put $x = 1$ & $y = 1 \rightarrow g(1) = 1$ or $g(1) = 2$

but $g(1) \neq 1 \Rightarrow g(1) = 2$

Put $y = \frac{1}{x}$ to get

$$g(x)\, g\left(\frac{1}{x}\right) = g(x) + g\left(\frac{1}{x}\right) + g(1) - 2 \Rightarrow g(x) = x^4 + 1$$

$g(2) = 5 \Rightarrow x = 2$

$\therefore \quad g(x) = 1 + x^2$

$$\therefore \quad I = \int_1^{\infty} \frac{dx}{x(1+x^2)} \text{ ; put } x = \tan\theta$$

$$\int_{\pi/4}^{\pi/2} \cot\theta\, dt = \ln \sin\theta \Big|_{\pi/4}^{\pi/2}$$

$$= (0) - \ln \frac{1}{1^2} = \frac{1}{2} \ln 2 \equiv \frac{p}{q} \ln 2$$

$\Rightarrow p^2 + q^2 = 5$. Ans.

124 $$\sum_{k=1}^{\infty}\left(\frac{1}{2k-1} - \frac{1}{2k}\right)$$

$$\lim_{n \to \infty}\left(\left(1 - \frac{1}{2}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \text{........} + \left(\frac{1}{2n-1} - \frac{1}{2n}\right)\right)$$

$$\lim_{n \to \infty}\left(\left(1 + \frac{1}{3} + \frac{1}{5} + \text{........} + \frac{1}{2n-1}\right) - \left(\frac{1}{2} + \frac{1}{4} + \text{.......} + \frac{1}{2n}\right)\right)$$

$$\lim_{n \to \infty}\left(\left(1 + \frac{1}{2} + \frac{1}{3} + \text{........} + \frac{1}{2n}\right) - 2\left(\frac{1}{2} + \frac{1}{4} + \text{......} + \frac{1}{2n}\right)\right)$$

$$\lim_{n \to \infty}\left(\left(1 + \frac{1}{2} + \text{.......} + \frac{1}{2n}\right) - \left(1 + \frac{1}{2} + \text{......} + \frac{1}{n}\right)\right)$$

$$\lim_{n \to \infty}\left(\frac{1}{n+1} + \frac{1}{n+2} + \text{......} + \frac{1}{n+n}\right)$$

$$= \lim_{n \to \infty} \sum_{r=1}^{n} \frac{1}{n+r} = \int_0^1 \frac{1}{1+x} dx = ln\, 2$$ **Ans.**

125. $f(x) = \frac{1}{x+1}$

$\therefore \quad f(g(x)) = \frac{1}{g(x)+1} \Rightarrow \frac{1}{g(x)+1} = \frac{x-1}{x}$

$\therefore \quad g(x) + 1 = \frac{x}{x-1} \Rightarrow g(x) = \frac{x}{x-1} - 1 = \frac{1}{x-1}$

$\Rightarrow \quad g(f(x)) = \frac{1}{f(x)-1} = \frac{1}{\frac{1}{x+1}-1} = -\frac{x+1}{x}$

$\therefore \quad I = \int_2^1 -\left(\frac{x+1}{x}\right) dx = \int_1^2 \left(1+\frac{1}{x}\right) dx = x + ln\, x\Big]_1^2$

$= (2 + ln\, 2) - (1) = 1 + ln\, 2.$ **Ans.**

Paragraph for Question no.126 to 128

126 $x f(x) - \int_0^x f(t)\,dt = x + ln\left(\sqrt{x^2+1} - x\right)$

differentiating

$$x f'(x) + f(x) = 1 + \frac{\frac{x}{\sqrt{x^2+1}} - 1}{\left(\sqrt{x^2+1} - x\right)}$$

$\Rightarrow \quad x f'(x) = 1 - \frac{1}{\sqrt{x^2+1}}$ ....(1)

$\therefore \quad$ range of $g(x) = x f'(x)$ is $[0, 1)$ **Ans.**

$f'(x) = \frac{\sqrt{x^2+1} - 1}{x\sqrt{x^2+1}}$

$\Rightarrow \quad$ antiderivative of an odd function.

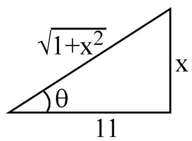
Integrating (1), i.e. $\quad f'(x) = \frac{1}{x} - \frac{1}{x\sqrt{x^2+1}}$

$$f(x) = \int \frac{dx}{x} - \int \frac{dx}{x\sqrt{x^2+1}}$$

$$f(x) = \ln(x) - I,$$

where $I = \int \frac{dx}{x\sqrt{x^2+1}}$; put $x = \tan\theta$

$\Rightarrow \quad dx = \sec^2 d\theta$



$$I = \int \frac{\sec^2\theta d\theta}{\tan\theta \sec\theta} = \int \operatorname{cosec}\theta\, d\theta = \ln(\operatorname{cosec}\theta - \cot\theta) = \ln\left(\frac{\sqrt{1+x^2}-1}{x}\right) + C$$

$$\therefore \quad f(x) = \ln x - \ln\left(\frac{\sqrt{1+x^2}-1}{x}\right) + C$$

$$\therefore \quad f(x) = \ln x - \ln\left(\frac{x}{\sqrt{1+x^2}+1}\right) + C$$

$$f(x) = \ln\left(\sqrt{1+x^2}+1\right) + C; \quad \text{put } x = 0,\ f(0) = \ln 2$$

$$f(0) = \ln 2 \quad \Rightarrow \quad C = 0$$

$$f(x) = \ln\left(\sqrt{1+x^2}+1\right) \Rightarrow f \text{ is even}$$

now $\displaystyle\int_0^1 f(x)\,dx = \int_0^1 \underbrace{1}_{II} \cdot \underbrace{\ln\left(\sqrt{1+x^2}+1\right)}_{I} dx$

integrating by parts

$$\left. f(x)\cdot x\right]_0^1 - \int_0^1 x f'(x)\,dx = f(1) - \int_0^1 \left(1 - \frac{1}{\sqrt{x^2+1}}\right) dx$$

$$= \ln\left(1+\sqrt{2}\right) - \left[x - \ln\left(x + \sqrt{x^2+1}\right)\right]_0^1$$

$$= \ln\left(1+\sqrt{2}\right) - \left[x - \ln\left(x + \sqrt{x^2+1}\right)\right]_0^1 = \ln\left(1+\sqrt{2}\right) - \left[\{1 - \ln(1+\sqrt{2})\} - \{0\}\right]$$

$$= 2\ln\left(1+\sqrt{2}\right) - 1 = \ln\left(3+2\sqrt{2}\right) - 1 \ \textbf{Ans.}$$

129. $f(x) = \int_0^x e^t (\sin x \cos t - \cos x \sin t)\, dt$

$$f(x) = \sin x \int_0^x e^t \cos t\, dt - \cos x \int_0^x e^t \sin t\, dt \quad \text{....(1)}$$

$$f'(x) = \sin x \cdot e^x \cos x + \left(\int_0^x e^t \cos t\, dt\right)\cos x - \left[\cos x \cdot e^x \sin x - \left(\int_0^x e^t \sin t\, dt\right)\sin x\right]$$

$$f'(x) = \cos x \int_0^x e^t \cos t\, dt + \sin x \int_0^x e^t \sin t\, dt$$

$$f''(x) = \cos x \cdot e^x \cdot \cos x - \sin x \int_0^x e^t \cos t\, dt + \sin x \cdot e^x \cdot \sin x + \left(\int_0^x e^t \sin t\, dt\right)\cos x$$

$$f''(x) = e^x + \cos x \int_0^x e^t \sin t\, dt - \sin x \int_0^x e^t \cos t\, dt \quad \text{....(2)}$$

(1) + (2)

$$f(x) + f''(x) = e^x = g(x)$$

130. $I = 4\int_0^{\pi/2} \frac{dx}{\sin^2 x + k^2 \cos^2 x}$ where $k = 100$

$$= 4\int_0^{\pi/2} \frac{\sin^2 x\, dx}{\tan^2 x + k^2} \quad (\text{put } \tan x = t)$$

$$= 4\int_0^{\infty} \frac{dt}{t^2 + k^2} = \frac{4}{k}\tan^{-1}\frac{t}{k}\Big]_0^{\infty} = \frac{4}{k}\cdot\frac{\pi}{2} = \frac{2\pi}{k}$$

$\therefore \quad I = \frac{2\pi}{100} = \frac{\pi}{50}$

$\therefore \quad a + b = 51$. Ans.

131. Given $\int_0^x f(t)\, dt = (f(x))^2$ ....(1)

Put $x = 0$, we get $f(0) = 0$

Now, differentiable both sides of equation (1) w.r.t x, we get

$f(x) = 2 f(x)\; f'(x)$

As, $f(x) \neq 0\ \forall\ x \in R$, so

$$f'(x) = \frac{1}{2} \Rightarrow f(x) = \frac{x}{2} + C$$

As, $\quad f(0) = 0 \Rightarrow C = 0$

$\therefore \quad f(x) = \frac{x}{2}$

Hence, $f(4) = 2$. **Ans.**

132. We have

$$f(\alpha) = \int_{\tan^{-1}\alpha}^{\cot^{-1}\alpha} \left(\frac{\tan x}{\tan x + \cot x} \right) dx \qquad \ldots (1)$$

$$\Rightarrow \quad f(\alpha) = \int_{\tan^{-1}\alpha}^{\cot^{-1}\alpha} \frac{\cot x}{\cot x + \tan x} dx \qquad \ldots (2)$$

(Using king property)

$\therefore$ Adding (1) and (2), we get

$$f(\alpha) = \frac{1}{2}\ (\cot^{-1}\alpha - \tan^{-1}\alpha) = \frac{1}{2}\left(\frac{\pi}{2} - 2\tan^{-1}\alpha\right)$$

Clearly, range of $f(\alpha)$ is $\left(-\frac{\pi}{4}, \frac{3\pi}{4}\right)$ Ans.

Paragraph for question nos. 133 to 135

Sol. Given, $a(x) = \int_0^x \frac{f(t)}{t^2} dt$

$$\therefore \ a(-x) = \int_0^{-x} \frac{f(t)}{t^2} dt$$

Put $t = -y \Rightarrow dt = -dy$

$$= -\int_0^x \frac{f(-y)}{y^2} dy = -\int_0^x \frac{f(y)}{y^2} dy \quad \text{(As } f \text{ is even)}$$

$\Rightarrow a(-x) = -a(x) \Rightarrow$ odd function

$$\text{Now, } \underbrace{f(x)}_{\text{even}} = \underbrace{f'(x)}_{\text{odd}} + \underbrace{2x\int_0^{a(x)} \left(\frac{f(t)}{t^2}\right) dt}_{\text{even}} - \underbrace{x}_{\text{odd}}$$

So, $f'(x) = x \qquad ....(1)$

$\Rightarrow f(x) = \frac{x^2}{2} + C$ ....(2)

Also, $f(0) = f'(0) = 0$ [From (1)]

$\therefore C = 0$

Hence, $f(x) = \frac{x^2}{2}$

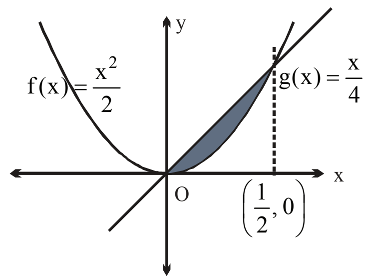
Now, $a(x) = \int_0^x \left(\frac{\frac{t^2}{2}}{t^2} \right) dt$. $\therefore a(x) = \frac{x}{2}$

Also, $g(x) = \int_0^{x/2} \left(\frac{t^2/2}{t^2} \right) dt$. $\therefore g(x) = \frac{x}{4}$

133 The value of $f(3) = \frac{9}{2} \Rightarrow$ Option (D) is correct.

134 As, $g'(x) = \frac{1}{4} \Rightarrow g'(2) = \frac{1}{4} \Rightarrow$ Option (A) is correct.

135 So, required shaded area $= \int_0^{1/2} \left(\frac{x}{4} - \frac{x^2}{2} \right) dx$



$= \left(\frac{x^2}{8} - \frac{x^3}{6} \right)_0^{1/2} = \frac{1}{32} - \frac{1}{48}$

$= \frac{3-2}{96} = \frac{1}{96}$. **Ans.**

136. $I = \int_1^4 \frac{(x-2)}{(x^2+4)\sqrt{x}} dx$ $\quad x = 2\tan\theta$

$I = \int_{\cot^{-1}2}^{\tan^{-1}2} \frac{(\tan\theta - 1)}{\sqrt{2}\sqrt{\tan\theta}} d\theta$

$I = \frac{1}{\sqrt{2}} \int_\alpha^\beta \left(\sqrt{\tan\theta} - \sqrt{\cot\theta} \right) d\theta$, $\quad \left(\alpha + \beta = \frac{\pi}{2} \right)$

Using King

$$I = \frac{1}{\sqrt{2}} \int_{\alpha}^{\beta} \left(\sqrt{\cot\theta} - \sqrt{\tan\theta}\right) d\theta$$

$I = -I \quad \Rightarrow \qquad I = 0.$ **Ans.**

137. Let $I = \int_{0}^{\pi} \frac{x}{\sqrt{1+\sin^3 x}} \left((3\pi\cos x + 4\sin x)\sin^2 x + 4\right) dx$

$$\Rightarrow \qquad I = \int_{0}^{\pi/2} \frac{x\left((3\pi\cos x + 4\sin x)\sin^2 x + 4\right)}{\sqrt{1+\sin^3 x}} + \frac{(\pi - x)\left((-3\pi\cos x + 4\sin x)\sin^2 x + 4\right)}{\sqrt{1+\sin^3 x}} dx$$

$$\Rightarrow \qquad I = \int_{0}^{\pi/2} \frac{(2x-\pi)3\pi\cos x\sin^2 x}{\sqrt{1+\sin^3 x}} + 4\pi\sqrt{1+\sin^3 x}\, dx$$

$$\Rightarrow \qquad I = \int_{0}^{\pi/2} \left(\underbrace{\frac{-6x\pi\sin x\cos^2 x}{\sqrt{1+\cos^3 x}}}_{x f'(x)} + \underbrace{4\pi\sqrt{1+\cos^3 x}}_{f(x)} \right) dx$$

$$\Rightarrow \qquad I = \left(4\pi x\sqrt{1+\cos^3 x}\right)_0^{\pi/2} = 2\pi^2$$

$\Rightarrow \qquad k = 2$

138. We have $f(x) = \alpha e^{2x} + \beta e^x - \gamma x$

Now $f(0) = -1 \Rightarrow \alpha + \beta = -1$ ....(1)

$f'(\ln 2) = 30 \Rightarrow 8\alpha + 2\beta - \gamma = 30$ ....(2)

and $\int_{0}^{\ln 4} (f(x) + \gamma x)\, dx = 24 \Rightarrow \int_{0}^{\ln 4} \left(\alpha e^{2x} + \beta e^x\right) dx = 24$

$\Rightarrow \left(\frac{\alpha e^{2x}}{2} + \beta e^x\right)_0^{\ln 4} = 24 \Rightarrow 15\alpha + 6\beta = 48$ ....(3)

$\therefore$ On solving (1), (2), (3), we get $\alpha = 6, \beta = -7, \gamma = 4$

Hence $(\alpha + \beta + \gamma) = 6 - 7 + 4 = 3$ **Ans.**

139. Let $\lim\limits_{n\to\infty} \frac{\sum\limits_{k=1}^{n} k^r}{n^a} = L$ = a finite non-zero quantity then

$$L = \lim_{n\to\infty} \left(\frac{1}{n}\right)^{a-(r+1)} \cdot \frac{1}{n} \cdot \sum_{k=1}^{n} \left(\frac{k}{n}\right)^r$$

$\because \quad \lim\limits_{n\to\infty} \frac{1}{n}\sum\limits_{k=1}^{n}\left(\frac{k}{n}\right)^r = \int\limits_0^1 x^r dx = \frac{1}{r+1}$ = a finite non-zero quantity

$\therefore$ L can be evaluated only if

$a - (r + 1) = 0$

$\therefore \quad a = r + 1 \;\; \& \;\; L = \frac{1}{r+1}$

140. $f(x) = \cos 3x$

$$I = \int\limits_0^{\pi/2} \cos 3x \, dx = \frac{1}{3}(\sin 3x)\Big|_0^{\pi/2} = \frac{-1}{3}$$

141. $I_1 = \int\limits_0^x e^{t(x-t)} dt$

$t(x - t) = -[t^2 - tx]$

$$= -\left[\left(t - \frac{x}{2}\right)^2 - \frac{x^2}{4}\right] = \frac{x^2}{4} - \left(t - \frac{x}{2}\right)^2$$

$$I_1 = \int\limits_0^x e^{\frac{x^2}{4} - \left(t - \frac{x}{2}\right)^2} dt = e^{\frac{x^2}{4}} \int\limits_0^x e^{-\frac{(2t-x)^2}{4}} dt \quad ;2t - x = y$$

$$\Rightarrow dt = \frac{dy}{2}$$

$$= \frac{e^{x^2/4}}{2} \int\limits_{-x}^{x} e^{-\frac{y^2}{4}} dy$$

142. Let $u = \frac{c}{x}$, so $du = \frac{-c}{x^2}dx$, so $\int\limits_1^{\sqrt{c}} \frac{f(x)}{x} dx$

$$= \int\limits_c^{\sqrt{c}} \frac{u f(u)}{c}\left(\frac{-x^2}{c}\right) du = \int\limits_{\sqrt{c}}^{c} \frac{f(u)}{u} du$$

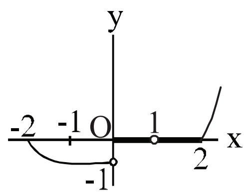
Therefore, $\int\limits_1^c \frac{f(x)}{x} dx = \int\limits_1^{\sqrt{c}} \frac{f(x)}{x} dx + \int\limits_{\sqrt{c}}^{c} \frac{f(u)}{u} du = 3 + 3 = 6$ **Ans.**

Paragraph for Question Nos. 143 to 145

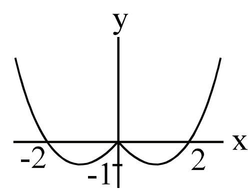
Sol

$$g(x) = \begin{cases} f(x) & \text{if } -2 \le x \le -1 \\ -1 & \text{if } -1 \le x < 0 \\ 0 & \text{if } 0 \le x < 1 \\ f(x) & 1 \le x \le 3 \end{cases}$$

Graph of y = f(x) is



Graph of y = g(x) is



146. $J_1 = \frac{1002}{1003}\int_0^{\pi/2}(\cos x)^{1001}\,dx = \frac{(1002)(1000)}{(1003)(1001)}.........\frac{2}{3}\underbrace{\int_0^{\pi/2}\cos x\,dx}_{1}$

$J_1 = \frac{2\cdot 4\cdot 6..........1002}{3\cdot 5\cdot 7..........1003}$ ....(1)

|||ly $J_2 = \frac{1003}{1004}\int_0^{\pi/2}(\cos x)^{2002}\,dx$

$= \frac{1003}{1004}\cdot\frac{1001}{1002}\cdot\frac{999}{1000}.....\frac{3}{6}\cdot\frac{2}{4}\cdot\frac{1}{2}-\frac{\pi}{2}$ ....(2)

$\therefore$ $J_1\cdot J_2 = \frac{\pi}{2008}$ **Ans.**

<u>Alternatively</u>: Use Wali's formula

147 $I = \int_1^4 p(x)\,dx$

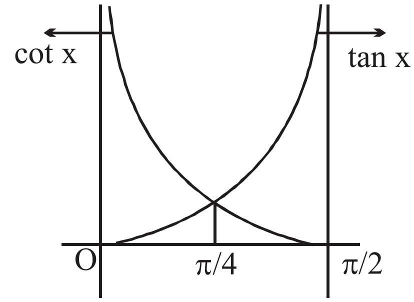
$I = \int_1^4 p(5-x)\,dx$

$2I = \int_1^4 p(x)+p(5-x)\,dx$

Now, $p'(x) = p'(5-x) \Rightarrow p(x)+p(5-x) = C$

$\Rightarrow 2I = \int_1^4 8\,dx = 24 \Rightarrow I = 12.$ **Ans.**

148. Clearly from graph



$$I = \int_0^{\pi/4} \tan x\, dx + \int_{\pi/4}^{\pi/2} \cot x\, dx$$

$$= \left[ln\,|\sec x|\right]_0^{\pi/4} + \left[ln\,|\sin x|\right]_{\pi/4}^{\pi/2} = 2ln\sqrt{2}\,.$$

(D) obviously the period of $\tan x$ and $\cot x$ is π.

149. $$\int_2^{e+3} f^{-1}(x)\, dx = \int_0^1 tf'(t)\, dt$$

$$= \left| tf(t) \right|_0^1 - \int_0^1 f(t)\, dt = e + 3 - \left| \left(e^{-1} + t^2 + t\right) \right|_0^1$$

$= e + 3 - (e + 2 - 1) = 2$. **Ans.**

150 $$\underset{n\to\infty}{\text{Limit}} \frac{1}{n}\sum_{k=1}^{n}\left(e^{\left(1+\frac{k}{n}\right)^2} - \frac{3e^{\left(1+\frac{3k}{n}\right)}}{2\sqrt{1+\frac{3k}{n}}} \right)$$

$$= \int_0^1 e^{(1+x)^2} dx - \frac{3}{2}\int_0^1 \frac{e^{(1+3x)}}{\sqrt{1+3x}} dx$$

put $t = x + 1$ and $\sqrt{1+3x} = u$

$= \int_1^2 e^{t^2} dt - \int_1^2 e^{u^2} du = 0$ **Ans.**

Paragraph for Question no. 151 & 152

Sol. $$\frac{g(x)\left[(1+x)\,ln\,x - 5\right]}{x\,(ln\,x)^6} = \left[\frac{g(x)}{(ln\,x)^5}\right]'$$

$$\Rightarrow \frac{g(x)\left[(1+x)\,ln\,x - 5\right]}{x\,(ln\,x)^6} = \frac{x \cdot ln\,x \cdot g'(x) - 5g(x)}{x\,(ln\,x)^6}$$

$\Rightarrow g(x)\,(1 + x) = x\, g'(x)$

$$\Rightarrow \frac{g'(x)}{g(x)} = 1 + \frac{1}{x}$$

$\Rightarrow ln\, g(x) = x + ln\, x \Rightarrow g(x) = k \cdot xe^x$

$\Rightarrow g(1) = k \cdot e = e \Rightarrow k = 1$

$g(x) = xe^x$.

151. $g^n(x) = n \cdot e^x + x\, e^x$

$g^n(1) = (n+1)e = (n+1)\, g(1)$

$f(x) = ln\,(x+1) - ln\,(x) + 1$

$$f''(x) = \frac{1}{x^2} - \frac{1}{(x+1)^2}.$$

152. $I_n = \int_0^\infty x^n e^{-x} dx = n \cdot I_{n-1} = n\,!$

$I_{2012} = (2012)\,!$. **Ans.**

153. $$I = \int_0^\pi \frac{(x^2\cos^2 x - x\sin x - \cos x - 1)}{x^2\left(\frac{1}{x} + \sin x\right)^2} dx$$

$$I = \int_0^\pi \frac{\left(\cos^2 x - \frac{1}{x}\sin x - \frac{1}{x^2}\cos x - \frac{1}{x^2}\right)dx}{\left(\frac{1}{x} + \sin x\right)^2}$$

$$I = \int_0^\pi \underbrace{\cos x}_{\mathrm{I}} \underbrace{\frac{\left(\cos x - \frac{1}{x^2}\right)dx}{\left(\frac{1}{x} + \sin x\right)^2}}_{\mathrm{II}} - \int_0^\pi \frac{\frac{1}{x}\left(\frac{1}{x} + \sin x\right)dx}{\left(\frac{1}{x} + \sin x\right)^2}$$

$$I = \left[\cos x\left(\frac{-1}{\frac{1}{x} + \sin x}\right)\right]_0^\pi - \int_0^\pi \frac{x\sin x\, dx}{1 + x\sin x} - \int \frac{dx}{1 + x\sin x}$$

$$I = \left[\frac{-x\cos x}{1 + x\sin x}\right]_0^\pi - \int_0^\pi \frac{x\sin x + 1}{x\sin x + 1} dx$$

$I = (\pi - 0) - [x]_0^{\pi}$

$I = \pi - \pi = 0$. **Ans.**

154. $$I = \int_0^{\pi} \frac{x\sin^3 x}{4-\cos^2 x}\,dx$$

Let $y = \pi - x$, and the integral becomes

$$I = \int_0^{\pi} \frac{(\pi - y)\sin^3 y}{4-\cos^2 y}\,dy$$

Hence, $2I = \int_0^{\pi} \frac{\sin^2 x}{4-\cos^2 x}\,d(-\cos x) = \int_{-1}^{1} \frac{1-t^2}{4-t^2}\,dt = 2 - \frac{3\ln 3}{2}$

$I = 1 - \frac{3\ln 3}{4}$. **Ans.**]

155. $$\int_{-1/2}^{1/2} \sin\left(\tan^{-1}\left(\frac{x^2}{\sqrt{1-x^4}}\right)\right)dx$$

$$\int_{-1/2}^{1/2} \sin\left(\sin^{-1} x^2\right)dx = \int_{-1/2}^{1/2} x^2\,dx$$

$$= 2\cdot \int_0^{1/2} x^2\,dx$$

$$= 2\cdot\left(\frac{x^3}{3}\right)_0^{1/2} = \frac{2}{3}\cdot\frac{1}{8} = \frac{1}{12}.$$ **Ans.**

156. Let $I = \int_{-3}^{-1} (x+1)^4(x+2)^3(x+3)^4\,dx$ Put $x + 2 = t \Rightarrow dx = dt$

$$I = \int_{-1}^{1} (t-1)^4\cdot t^3\cdot(t+1)^4\,dt = \int_{-1}^{1} \underbrace{t^3\cdot\left(t^2-1\right)^4}_{\text{odd function}}\,dt = 0.$$ **Ans.**

Paragraph for Question no.157 to 159

Sol. $I_3 = \int_0^{\frac{\pi}{2}} \underbrace{x}_{(I)} \underbrace{\cot x}_{(II)} dx = \left(x \cdot \ln(\sin x)\right)_0^{\pi/2} - \int_0^{\frac{\pi}{2}} \ln(\sin x)\, dx$

I.B.P.

$\therefore \quad I_3 = 0 - \left(-\frac{\pi}{2}\ln 2\right) = \frac{\pi}{2}\ln 2$

Now, $I_2 = \int_0^{\frac{\pi}{2}} \underbrace{x^2}_{(I)} \cdot \underbrace{\left(\operatorname{cosec}^2 x\right)}_{\substack{(II)\\(I.B.P)}} dx = \left(x^2 \cdot (-\cot x)\right)_0^{\pi/2} + \int_0^{\frac{\pi}{2}} 2x \cdot \cot x\, dx$

$I_2 = 0 + 2\left(\frac{\pi}{2}\ln 2\right) = \pi \ln 2$

Also, $I_1 = \int_0^{\frac{\pi}{2}} \frac{x^3 \cdot \cos x\, dx}{4\sin^3 x} = \frac{1}{4}\int_0^{\frac{\pi}{2}} \underbrace{x^3}_{(I)} \underbrace{\left(\frac{\cos x}{\sin^3 x}\right)}_{\substack{(II)\\(I.B.P)}} dx$

$= \frac{1}{4}\left[\left(x^3 \cdot \left(\frac{-1}{2\sin^2 x}\right)\right)_0^{\frac{\pi}{2}} + \frac{3}{2}\int_0^{\frac{\pi}{2}} x^2 \operatorname{cosec}^2 x\, dx\right]$

$= \frac{1}{4}\left[\left(\frac{\frac{\pi^3}{8}}{-2} - 0\right) + \frac{3}{2}(\pi \ln 2)\right]$

$I_3 = \frac{-\pi^3}{64} + \frac{3}{8}\pi \cdot \ln 2.$ **Ans.**

Now, Verify alternatives.

160. $\int_0^{\infty} \underbrace{e^{-x}}_{(II)} \underbrace{\sin^n x}_{(I)} dx$

$= \left(-\sin^n x \cdot e^{-x}\right)_0^{\infty} + \int_0^{\infty} n \cdot \sin^{n-1} x \cdot \cos x \cdot e^{-x}\, dx$

$$= n\int_0^\infty \underbrace{(\sin^{n-1} x \cdot \cos x)}_{(I)} \underbrace{e^{-x}}_{(II)} dx$$

$$= n\left[\left((\sin^{n-1} x \cdot \cos x)\,(-e^{-x})\right)_0^\infty + \int_0^\infty e^{-x}\left\{(n-1)\cdot \sin^{n-2} \cdot \cos^2 x - \sin^{n-1} \cdot \sin x\right\}\right]$$

$$= n\left[\int_0^\infty e^{-x}\left\{(n-1)\cdot \sin^{n-2} x \cdot (1-\sin^2 x) - \sin^4 x\right\}dx\right]$$

$$= n(n-1)\int_0^\infty e^{-x}\cdot \sin^{n-2} x\, dx \; - n(n-1)\int_0^\infty e^{-x}\cdot \sin^n x\, dx \; - n\int_0^\infty e^{-x}\cdot \sin^4 x\, dx$$

$$(1 + n\,(n-1) + n)\int_0^\infty e^{-x}\cdot \sin^n x\, dx = n(n-1)\int_0^\infty e^{-x}\cdot \sin^{n-2} x\, dx$$

$$\Rightarrow (1+n^2)\int_0^\infty e^{-x} \sin^n x\, dx = n(n-1)\int_0^\infty e^{-x} \sin^{n-2} x\, dx$$

$$\therefore \frac{(1+n^2)\int_0^\infty e^{-x} \sin^n x\, dx}{\int_0^\infty e^{-x} \sin^{n-2} x\, dx} = n(n-1)$$

Put $n = 25$, we get

$$\frac{626\int_0^\infty e^{-x} \sin^{25} x\, dx}{\int_0^\infty e^{-x} \sin^{23} x\, dx} = 25 \times 24 = 600 = \lambda,$$

So sum of digits of λ equals 6.

161. $I = \int_0^{2\pi} \sin 8x\,|\sin(x-\theta)|\,dx$ Put $x - \theta = t$

$$I = \int_{-\theta}^{2\pi-\theta} \sin 8\,(t+\theta)\,|\sin t|\,dt = \int_{-\pi}^{\pi} \sin 8\,(t+\theta)\,|\sin t|\,dt$$

($\because$ the integrand is a periodic function with period 2π.)

$$I = \cos 8\theta \int_{-\pi}^{\pi} \sin 8t\,|\sin t|\,dt + \sin 8\theta \int_{-\pi}^{\pi} \cos 8t\,|\sin t|\,dt = 2\sin 8\theta \int_{0}^{\pi} \cos 8t\,|\sin t|\,dt$$

($\because$ $\sin 8t\,|\sin t|$ is odd function)

$$I = 2\sin 8\theta \int_{0}^{\pi} \cos 8t \sin t\,dt \qquad (\because\ 0 \le t \le \pi)$$

$$I = 2\sin 8\theta \int_{0}^{\pi} \frac{1}{2}(\sin 9t - \sin 7t)\,dt = \sin 8\theta \left[\frac{-\cos 9t}{9} + \frac{\cos 7t}{7}\right]_0^{\pi} = \frac{-4}{63}\sin 8\theta.$$

162. Now, $I = \int_{0}^{\pi/2} \frac{2t}{\underset{\underset{\text{King \& add}}{\downarrow}}{1+\sin t+\cos t}}\,dt$

$$2I = \int_{0}^{\pi/2} \frac{\pi}{1+\sin t+\cos t}\,dt$$

$$= \int_{0}^{\pi/2} \frac{\pi\left(1+\tan^2\frac{t}{2}\right)}{\left(1+\tan^2\frac{t}{2}\right)+2\tan\frac{t}{2}+\left(1-\tan^2\frac{t}{2}\right)}\,dt$$

$$\Rightarrow 2I = \pi\int_{0}^{\pi/2} \frac{\sec^2\frac{t}{2}}{2\left(1+\tan\frac{t}{2}\right)}\,dt\,;$$

Put $\tan\frac{t}{2} = y \;\Rightarrow\; \frac{1}{2}\sec^2\frac{t}{2}\,dt = dy$

So, $I = \frac{\pi}{4}\int_{0}^{1} \frac{2}{1+y}\,dy = \frac{\pi}{2} ln\,2$. **Ans.**

163. $I = \frac{1}{2}\int_{f(e)}^{f(e^2)} \frac{\tan t\,dt}{\tan t+\tan\left(f(e^3)-t\right)} = \frac{1}{2}\int_{f(e)}^{f(e^2)} \frac{\tan\left(f(e^3)-t\right)dt}{\tan t+\tan\left(f(e^3)-t\right)}$ (By King)

$$2I = \frac{1}{2}\int_{f(e)}^{f(e^2)} 1\,dt = \frac{f(e^2)-f(e)}{2}$$

$I = \frac{1}{4}\left(f(e^2)-f(e)\right) = \frac{f(e)}{4}$. **Ans.**

164. $f(x) = \int_0^4 e^{|t-x|}\,dt = \int_0^x e^{(x-t)}\,dt + \int_x^4 e^{(t-x)}\,dt \Rightarrow f(x) = e^x + e^{4-x} - 2$

$\therefore\ f'(x) = e^x - e^{4-x} = 0 \Rightarrow x = 2$

So, $f(0) = f(4) = e^4 - 1$ and $f(2) = 2(e^2 - 1)$

$\Rightarrow \sqrt{b-a} = \sqrt{(e^4-1) - 2e^2 + 2} = e^2 - 1$. **Ans.**

165. Let n - degree p and m be leading coefficient p(x). Thus the leading coefficient of $(x + 10)\,p(2x)$ is $m \cdot 2^n$ and the leading coefficient of $(8x - 32)\,p(x + 6)$ is 8m. Thus, n = 3.

Substituting $x = -10$ gives $p(-4) = 0$. Substituting $x = -2$ gives

$-48\,p(4) = 8p(-4) = 0,$

so $p(4) = 0$. Substituting $x = 4$ gives $14\,p(8) = 0 \cdot p(10) = 0$

so $p(8) = 0$

$\therefore\ p(x) = a(x-4)(x+4)(x-8)$ for some a. Substituting $p(1) = 210$ gives $a = 2$.

So, $p(x) = 2(x-4)(x+4)(x-8)$. **Ans.**

166. If m is even and not a multiple of 4 then integral vanishes (using king & add)

$\therefore\ m = 2, 6, 10$

and if m is odd but not $(4n - 1)$, then $I = 0$

$\therefore\ m = 1, 5, 9$

$\Rightarrow$ Total vlaues 6 i.e. 1, 2, 5, 6, 9, 10.

167. $f(x) = \int_{-\pi/2}^{\pi/2} \frac{\ln(1 + x\cos\theta)}{\cos\theta}\,d\theta \Rightarrow f'(x)$

$$= \int_{-\pi/2}^{\pi/2} \frac{\cos\theta\,d\theta}{(1 + x\cos\theta)\cos\theta} = \int_{-\pi/2}^{\pi/2} \frac{d\theta}{1 + x\cos\theta}$$

$$f'(1) = \int_{-\pi/2}^{\pi/2} \frac{d\theta}{1 + x\cos\theta} = 2\int_0^{\pi/2} \frac{d\theta}{2\cos^2\frac{\theta}{2}}$$

$$= \int_0^{\pi/2} \sec^2\frac{\theta}{2}\,d\theta = 2\left[\tan\frac{\theta}{2}\right]_0^{\pi/2} = 2\,[1 - 0] = 2.\ \textbf{Ans.}$$

168. We have

$$I_n + I_{n+2} = \int_0^{\pi/4} \tan^n x \; dx + \int_0^{\pi/4} \tan^{n+2} x \; dx$$

$$= \int_0^{\pi/4} \tan^n x\,(1+\tan^2 x)\,dx = \int_0^{\pi/4} \tan^n x \sec^2 x\,dx$$

$$= \int_0^1 u^n\,du = \frac{1}{n+1} \quad \text{where } u = \tan x$$

So, $I_n + I_{n+2} = \dfrac{1}{n+1}$ which implies that $I_{n+1} + I_{n+3} = \dfrac{1}{n+2}$

Hence $\displaystyle\sum_{n=0}^{\infty}\left(I_n I_{n+1} + I_n I_{n+3} + I_{n+1} I_{n+2} + I_{n+2} I_{n+3}\right) = \sum_{n=0}^{\infty}\left(I_n + I_{n+2}\right)\left(I_{n+1} + I_{n+3}\right)$

$$= \sum_{n=0}^{\infty}\left(\frac{1}{n+1}\right)\left(\frac{1}{n+2}\right) = \sum_{n=0}^{\infty}\left(\frac{1}{n+1} - \frac{1}{n+2}\right)$$

$$\begin{aligned} = 1 &- \frac{1}{2} \\ \frac{1}{2} &- \frac{1}{3} \\ \frac{1}{3} &- \frac{1}{4} \\ \vdots & \quad \vdots \\ \frac{1}{n+1} &- \frac{1}{n+2} \end{aligned}$$

$$S_n = 1 - \frac{1}{n+2} \;\Rightarrow\; \lim_{n\to\infty}\left(1 - \frac{1}{n+2}\right) = 1 \;\Rightarrow\; 100 \lim_{n\to\infty}\left(1 - \frac{1}{n+2}\right) = 100 \textbf{ Ans.}]$$

169. Let $J = \displaystyle\int_0^1 \underbrace{\left(\frac{g(x)}{f(x)}\right)'}_{\text{II}} \cdot \underbrace{\frac{1}{g(x)}}_{\text{I}}\; dx$

(I.B.P.)

Now, Integrating by parts, $\displaystyle \left.\frac{1}{g(x)} \cdot \frac{g(x)}{f(x)}\right|_0^1 + \int_0^1 \frac{g'(x)}{g^2(x)} \cdot \frac{g(x)}{f(x)}\,dx$

$$J = \left(\frac{1}{f(1)} - \frac{1}{f(0)}\right) + \int_0^1 \frac{g'(x)}{g(x)\cdot f(x)}\,dx \qquad \text{...........(1)}$$

Now given, $f^2(x) = 1 + g^2(x)$

$$\Rightarrow f(x)\cdot f'(x) = g(x)\cdot g'(x) \qquad \Rightarrow \qquad g'(x) = \frac{f(x)\cdot f'(x)}{g(x)}$$

$$\therefore \qquad J = \left(\frac{1}{3} - \frac{1}{2}\right) + \int_0^1 \frac{f(x)\cdot f'(x)}{g^2(x)\cdot f(x)}\,dx$$

$$J = \frac{-1}{6} + \int_0^1 \frac{f'(x)}{f^2(x)-1}\,dx$$

Now put, $f(x) = t$, we get

$$J = \frac{-1}{6} + \int_2^3 \frac{dt}{t^2-1} = \frac{-1}{6} + \frac{1}{2}\,ln\left.\frac{t-1}{t+1}\right]_2^3 = \frac{-1}{6} + \frac{1}{2}\left[ln\frac{1}{2} - ln\frac{1}{3}\right]$$

$$J = \frac{-1}{6} + \frac{1}{2}\,ln\frac{3}{2} \Rightarrow J = \frac{1}{2}\,ln\frac{3}{2} - \frac{1}{6}.$$

$$\Rightarrow \qquad 6J + 1 = ln\,\frac{27}{8} \qquad \therefore$$

$\Rightarrow \qquad a + b = 27 + 8 = 35$ **Ans.**

170. $$I_n = \int_0^{\frac{\pi}{2}} \frac{\sin^2 n\theta}{\sin^2\theta}\,d\theta$$

$$\Rightarrow I_{n+1} - I_n = \int_0^{\frac{\pi}{2}} \frac{\sin^2(n+1)\,\theta - \sin^2 n\theta}{\sin^2\theta}\,d\theta$$

Now, note that $\sin^2(n+1)\,\theta - \sin^2 n\theta = [\sin(n+1)\,\theta - \sin n\theta]\,[\sin(n+1)\,\theta + \sin n\theta]$

$$= 2\cos\left(\frac{2n+1}{2}\right)\theta\,\sin\left(\frac{\theta}{2}\right)\cdot 2\sin\left(\frac{2n+1}{2}\right)\theta\,\cos\left(\frac{\theta}{2}\right) = \sin\theta\,\sin(2n+1)\,\theta$$

$$\therefore\; I_{n+1} - I_n = \int_0^{\frac{\pi}{2}} \frac{\sin(2n+1)\,\theta}{\sin\theta}\,d\theta = B_n$$

$$\text{Again, } B_{n+1} - B_n = \int_0^{\frac{\pi}{2}} \frac{\sin(2n+3)\theta - \sin(2n+1)\,\theta}{\sin\theta}\,d\theta$$

$$\Rightarrow B_{n+1} - B_n = \int_0^{\frac{\pi}{2}} \frac{2\cos(2n+2)\,\theta \sin\theta}{\sin\theta}\, d\theta = \frac{1}{n+1} \sin 2(n+1)\,\theta \Big|_0^{\frac{\pi}{2}}$$

$\therefore \quad B_{n+1} = B_n \ \forall\, n \geq 1$

$$\text{Also, } B_1 = \int_0^{\frac{\pi}{2}} \frac{\sin 3\theta}{\sin\theta}\, d\theta = \int_0^{\frac{\pi}{2}} (3 - 4\sin^2\theta)\, d\theta$$

$$\Rightarrow B_1 = \int_0^{\frac{\pi}{2}} 3\, d\theta - 2\int_0^{\frac{\pi}{2}} (1-\cos 2\theta)\, d\theta = \int_0^{\frac{\pi}{2}} d\theta + \sin 2\theta \Big|_0^{\frac{\pi}{2}} = \frac{\pi}{2}$$

Thus, $B_n = \dfrac{\pi}{2} \ \forall\, n \geq 1$

So, we finally have $I_{n+1} - I_n = \dfrac{\pi}{2} \ \forall\, n \geq 1$

A simple induction on n shows that $I_{n+1} = \dfrac{n\pi}{2} + I_1$.

$$\text{But } I_1 = \int_0^{\frac{\pi}{2}} d\theta = \frac{\pi}{2} \Rightarrow I_{n+1} = (n+1)\frac{\pi}{2} \Rightarrow I_n = \frac{n\pi}{2}.$$

171. We have $U = \displaystyle\int_{\pi/6}^{\pi/2} \cos x\, dx = \sin x \Big]_{\pi/6}^{\pi/2} = 1 - \frac{1}{2} = \frac{1}{2}$

$$V = \int_{-3}^{5} x^2 \operatorname{sgn}(x-1)\, dx$$

$$= -\int_{-3}^{1} x^2\, dx + \int_{1}^{5} x^2 dx = \frac{x^3}{3}\Bigg]_1^{-3} + \frac{x^3}{3}\Bigg]_1^{5}$$

$$= \frac{-27}{3} - \frac{1}{3} + \frac{125}{3} - \frac{1}{3}$$

$$= \frac{-29+125}{3} = \frac{96}{3} = 32.$$

So, $V = 32$; $U = \dfrac{1}{2}$

$\Rightarrow \quad 32 = \dfrac{\lambda}{2}$

$\Rightarrow \quad \lambda = 64$

172. $l = \lim_{x\to\infty} \dfrac{\int_0^x t^n \sqrt{t+1}\, dt}{x^{\left(\frac{3}{2}+n\right)}}$ Using lopital's rule

$$l = \lim_{x\to\infty} \frac{x^n \sqrt{x+1}}{\left(\frac{3}{2}+n\right) x^{\left(n+\frac{1}{2}\right)}} = \lim_{x\to\infty} \frac{2\sqrt{x+1}}{(2x+3)\sqrt{x}}$$

$$= \lim_{x\to\infty} \frac{2}{2n+3}\sqrt{1+\frac{1}{x}}$$

$$l = \frac{2}{2n+3} = \frac{2}{11}$$

$\therefore \quad 2n + 3 = 11 \Rightarrow n = 4.$ **Ans.**

173. Let $I = \int_{-1}^{1} \cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)\left(\cot^{-1}\frac{x}{\sqrt{1-(x^2)^{|x|}}}\right)dx$ ......(1)

Using King

$$I = \int_{-1}^{1}\left(\cot^{-1}\frac{1}{\sqrt{1-x^2}}\right)\left(\cot^{-1}\frac{-x}{\sqrt{1-(x^2)^{|x|}}}\right)dx \quad ....(2)$$

On adding

$$2I = \int_{-1}^{1}\cot^{-1}\frac{1}{\sqrt{1-x^2}}\left\{\cot^{-1}\frac{x}{\sqrt{1-(x^2)^{|x|}}} + \pi - \cot^{-1}\frac{x}{\sqrt{1-(x^2)^{|x|}}}\right\}dx$$

$$2I = \int_{-1}^{1}\pi\cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)dx = \int_{-1}^{1}\pi\tan^{-1}\sqrt{1-x^2}\,dx = 2\pi\int_0^1 \tan^{-1}\sqrt{1-x^2}\,dx \quad ....(3)$$

$\left(\text{As } \tan^{-1}\sqrt{1-x^2} \text{ is even function}\right)$

$$\therefore \quad I = \pi\int_0^1 \underbrace{1}_{\text{II}}\ \underbrace{\tan^{-1}\left(\sqrt{1-x^2}\right)}_{\text{I}}dx \quad ....(4)$$

Integrating by parts

$$I = \underbrace{\pi\tan^{-1}\left(\sqrt{1-x^2}\right)\cdot x\Big]_0^1}_{\text{zero}} - \int_0^1 \frac{x}{(1+1-x^2)}\,\frac{(-x)}{\sqrt{1-x^2}}dx = \pi\int_0^1\frac{x^2}{(2-x^2)\sqrt{1-x^2}}dx$$

Let $x = \sin\theta \Rightarrow dx = \cos\theta\, d\theta$

$$I = \pi \int_0^{\pi/2} \frac{\sin^2\theta}{(2-\sin^2\theta)} d\theta$$

$$= -\pi \int_0^{\pi/2} \frac{2-\sin^2\theta-2}{2-\sin^2\theta} d\theta$$

$$\therefore \qquad I = 2\pi \int_0^{\pi/2} \frac{d\theta}{2-\sin^2\theta} - \frac{\pi^2}{2}$$

$$= 2\pi \int_0^{\pi/2} \frac{\sec^2\theta d\theta}{2+2\tan^2\theta-\tan^2\theta} - \frac{\pi^2}{2}$$

$$= 2\pi \int_0^{\pi/2} \frac{\sec^2\theta\, d\theta}{2+\tan^2\theta} - \frac{\pi^2}{2}$$

put $\tan\theta = t$

$$I = 2\pi \int_0^{\infty} \frac{dt}{2+t^2} - \frac{\pi^2}{2} = 2\pi \frac{1}{\sqrt{2}} \tan^{-1} \frac{t}{\sqrt{2}} \Big]_0^{\infty} - \frac{\pi^2}{2}$$

$$= \frac{\pi^2}{\sqrt{2}} - \frac{\pi^2}{2} = \frac{\pi^2\left(\sqrt{2}-1\right)}{2} = \frac{\pi^2\left(\sqrt{a}-\sqrt{b}\right)}{\sqrt{c}}$$

$\Rightarrow \qquad a = 2, b = 1$ and $c = 4 \Rightarrow \qquad a + b + c = 2 + 1 + 4 = 7$ **Ans.**

174. $$\int_1^e \left(\frac{x^2 ln\, x - 1}{x}\right) e^x\, dx = \int_1^e e^x \left(x\, ln\, x - \frac{1}{x}\right) dx$$

$$= \int_1^e e^x x\, ln\, x\, dx - \int_1^e \frac{e^x}{x} dx$$

$$= (x\, ln\, x)\, e^x \Big]_1^e - \int_1^e (1+ln\, x)\, e^x\, dx - e^x\, ln\, x \Big]_1^e + \int_1^e e^x\, ln\, x\, dx$$

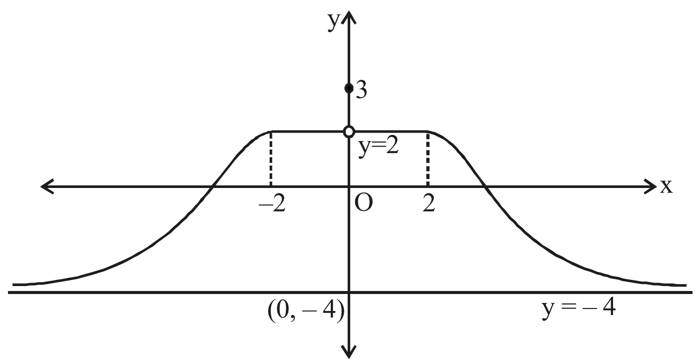
$= (e\, e^e - e^e + e - e^e)$
$= e\, e^e + e - 2e^e$
$= (e - 2)\, e^e + e.$ **Ans.**

175 to 177 Paragraph Based.

We have $H(x) = \left[\dfrac{3(x^2+1)-|x|}{x^2+1}\right] = \left[3 - \dfrac{|x|}{x^2+1}\right]$ $\left(\text{As } 0 < \dfrac{|x|}{x^2+1} = \dfrac{1}{|x|+\dfrac{1}{|x|}} \le \dfrac{1}{2}\right)$

So, $H(x) = \begin{cases} \dfrac{8}{\pi}\tan^{-1}(-|x|+3), & x \in (-\infty, -2) \cup (2, \infty) \\ 3, & x = 0 \\ 2, & x \in [-2, 2] - \{0\} \end{cases}$

Also, H(x) is an even function on R, so graph of H(x) is symmetrical about y-axis.



Graph of H(x)

Note : $H(x) = \dfrac{8}{\pi}\tan^{-1}(3 - |x|), |x| > 2$

Also for $|x| > 2$, $3 - |x| \in (-\infty, 1)$, so

$H(x) \in \dfrac{8}{\pi}\left(\dfrac{-\pi}{2}, \dfrac{\pi}{4}\right) = (-4, 2)$

175 From the above graph, it is clear that H(x) is discontinuous at only one point i.e. $x = 0$.

176. Range of H(x) is $(-4, 2] \cup \{3\}$.

177 $\int_{-2}^{1} H(x)\,dx = \int_{-2}^{1} 2\,dx = 2(3) = 6.$

178. Let $I = \int_0^1 \dfrac{dx}{\sqrt{(x+1)^3 \cdot (3x+1)}}$

Put, $x + 1 = u \Rightarrow dx = du \Rightarrow 3x + 1 = 3(u - 1) + 1 = 3u - 2$

So, $I = \int_1^2 \frac{du}{\sqrt{u^3 \cdot (3u-2)}} = \int_1^2 \frac{du}{u^2\sqrt{\left(3-\frac{2}{u}\right)}}$

Put, $3 - \frac{2}{u} = t^2 \Rightarrow \frac{2}{u^2}\,du = 2t\,dt$

Hence, $I = \int_1^{\sqrt{2}} \frac{t\,dt}{t} = t\Big]_1^{\sqrt{2}} = \sqrt{2} - 1$. **Ans.**

Aliter: Let $I = \int_0^1 \frac{dx}{(x+1)\sqrt{(x+1)(3x+1)}}$

Put $(x+1) = \frac{1}{t}$, so $I = \int_{1/2}^{1} \frac{dt}{\sqrt{3-2t}} = \left(\sqrt{3-2t}\right)_1^{1/2} = \sqrt{2} - 1$.

179. $I = -\int_0^1 \frac{(x^2-1)}{x^2\left(x^2+\frac{1}{x^2}+1\right)}\,dx$

$I = -\int_0^1 \frac{\left(1-\frac{1}{x^2}\right)}{\left(x+\frac{1}{x}\right)^2-1}\,dx$; put $x + \frac{1}{x} = t$

$I = \int_2^{\infty} \frac{dt}{t^2-1} = \frac{1}{2} ln\left(\frac{t-1}{t+1}\right)\Bigg|_2^{\infty} = \frac{1}{2}\left[0 - \ln\frac{1}{3}\right) = \frac{1}{2}\,ln\,3$ **Ans.**

Paragraph for question nos. 180 to 182

We have $f(x) = \lim_{\alpha \to 0} \frac{1}{\alpha^4}\int_0^{\alpha} \frac{\left(e^{x+t}-e^x\right)ln^2(1+t)}{2t^3+3}\,dt$

$f(x) = \lim_{\alpha\to 0} \frac{e^x \int_0^{\alpha} \frac{(e^t-1)ln^2(1+t)}{(2t^3+3)}\,dt}{\alpha^4} = \lim_{\alpha\to 0} e^x \frac{(e^{\alpha}-1)ln^2(1+\alpha)}{(2\alpha^3+3)\,4\alpha^3}$

$= e^x \lim_{\alpha\to 0} \frac{(e^{\alpha}-1)}{\alpha}\cdot\frac{ln^2(1+\alpha)}{\alpha^2}\cdot\frac{1}{4(2\alpha^3+3)} = \frac{e^x}{12}$

180. Clearly $f(\ln 2) = \dfrac{e^{\ln 2}}{12} = \dfrac{2}{12} = \dfrac{1}{6}$ **Ans.**

181 We have $\int_0^x g(t)\,dt = 3x + \int_x^0 \cos^2 t\, g(t)\,dt$

Now, on differentiating both the sides with respect to x, we get

$$g(x) = 3 - \cos^2 x\, g(x) \Rightarrow \quad g(x) = \frac{3}{1+\cos^2 x}$$

Clearly $\dfrac{3}{2} \le \dfrac{3}{1+\cos^2 x} \le 3.$

Hence range of $g(x) = \left[\dfrac{3}{2}, 3\right]$ **Ans.**

182 Let $I = \int_0^{\frac{\pi}{2}} g(x)\,dx = \int_0^{\frac{\pi}{2}} \frac{3}{1+\cos^2 x}dx = \int_0^{\frac{\pi}{2}} \frac{3\sec^2 x}{\sec^2 x + 1}dx = \int_0^{\frac{\pi}{2}} \frac{3\sec^2 x}{\tan^2 x + 2}dx$

Put $\tan x = t \Rightarrow \sec^2 x\,dx = dt$,

so $I = \int_0^{\infty} \frac{3dt}{t^2+2} = \frac{3}{\sqrt{2}}\left(\tan^{-1}\frac{t}{\sqrt{2}}\right)_0^{\infty} = \frac{3}{\sqrt{2}}\left(\frac{\pi}{2} - 0\right) = \frac{3\pi}{2\sqrt{2}}$ **Ans.**

183. As, $f(x) = 1 + \ln\left(x + \sqrt{x^2+1}\right) + 5x^3 - 4x^4$

Clearly, $\ln\left(\sqrt{x^2+1} + x\right)$, $5x^3$ are odd functions, so

$$I = \int_{-1}^{1} \left(1 - 4x^4\right)e^{-x^4}\,dx$$

$$= 2\int_0^1 \left(1 - 4x^4\right)e^{-x^4}\,dx$$

$$= 2\int_0^1 \frac{d}{dx}\left(x.e^{-x^4}\right)dx = 2\left(\frac{x}{e^{x^4}}\right)_0^1 = 2\left(\frac{1}{e} - 0\right) = \frac{2}{e}$$

184. $\displaystyle\lim_{x\to 0} \frac{\cos^{-1}\left(\frac{\sin x}{x}\right)}{x}$

$$= \lim_{x\to 0} \frac{\sin^{-1}\left(\sqrt{\frac{x^2-\sin^2 x}{x^2}}\right)}{x}$$

$$= \lim_{x\to 0} \frac{\sin^{-1} M}{M} \cdot \lim_{x\to 0} \frac{M}{x}$$

(using $\cos^{-1}t = \sin^{-1}\sqrt{1-t^2}$)

$$= 1 \cdot \lim_{x\to 0} \sqrt{\frac{x^2-\sin^2 x}{x^2 \cdot x^2}} = 1 \cdot \lim_{x\to 0} \sqrt{\frac{(x-\sin x)}{x^3} \cdot \frac{(x+\sin x)}{x}}$$

$$= \lim_{x\to 0} \sqrt{\frac{1}{6}\cdot 2} = \frac{1}{\sqrt{3}} \Rightarrow L = \frac{1}{\sqrt{3}}$$

U : $f(x) = \int_0^x \left(\sin(t^2-t)\cos x + \cos(t^2-t)\sin x\right)dt$

or $f(x) = \cos x \int_0^x \sin(t^2-t)\,dt + \sin x \cdot \int_0^x \cos(t^2-t)\,dt$

differentiating using Leibintiz rule

$$f'(x) = \cos x \cdot \sin(x^2-x) - \left(\int_0^x \sin(t^2-t)\,dt\right)\sin x + \sin x \cdot \cos(x^2-x) + \left(\int_0^x \cos(t^2-t)\,dt\right)\cos x$$

$$f''(x) = [-\sin(x^2-x)\sin x + \cos x \cdot \cos(x^2-x)(2x-1)] -$$

$$\left[\left(\int_0^x \sin(t^2-t)\,dt\right)\cos x + \sin x \cdot \sin(x^2-x)\right]$$

$$+ [\cos(x^2-x)\cos x - \sin x \cdot \sin(x^2-x)\cdot(2x-1)] + \left[\cos x \cdot \cos(x^2-x) - \left(\int_0^x \cos(t^2-t)\,dt\right)\sin x\right]$$

hence, $f''(0) = (-1) + (1) + (1) = 1$

$\Rightarrow \quad U = 1$

hence $\dfrac{120U}{L^2} = 360$ Ans.

185. $I = \int_1^2 \left(\cot^{-1}\sqrt{x-1}\right)^2 dx$; Put, $x - 1 = t^2$

$\Rightarrow dx = 2t\,dt$

$$= 2\int_0^1 \underbrace{t}_{\mathrm{I}}\ \underbrace{(\cot^{-1} t)^2}_{\substack{\mathrm{II} \\ (\mathrm{I.B.P.})}} dt$$

$$= 2\left[\left((\cot^{-1} t)^2 \frac{t^2}{2}\right)_0^1 + \int_0^1 \left(\frac{2\cot^{-1} t}{1+t^2}\right)\frac{t^2}{2} dt\right]$$

$$= 2\left[\left(\frac{\pi^2}{32} - 0\right) + \int_0^1 \left((1+t^2)-1\right)\frac{\cot^{-1} t}{1+t^2} dt\right]$$

$$= 2\left[\frac{\pi^2}{32} + \int_0^1 \underset{(\mathrm{II})}{1}.\underbrace{\underset{(\mathrm{I})}{\cot^{-1} t}}_{(\mathrm{I.B.P.})}\, dt - \int_0^1 \frac{\cot^{-1} t}{1+t^2} dt\right]$$

$$= 2\left[\frac{\pi^2}{32} + \left(t\cot^{-1} t\right)_0^1 + \int_0^1 \frac{t}{1+t^2} dt + \left[\frac{1}{2}(\cot^{-1} t)^2\right]_0^1\right]$$

$$= 2\left[\frac{\pi^2}{32} + \frac{\pi}{4} + \frac{1}{2}ln 2 + \frac{1}{2}\left(\frac{\pi^2}{16} - \frac{\pi^2}{4}\right)\right]$$

$$= 2\left[-\frac{\pi^2}{16} + \frac{\pi}{4} + \frac{1}{2}ln\, 2\right] = \frac{-\pi^2}{8} + \frac{\pi}{2} + ln 2 .$$

Hence, $\dfrac{\pi^2 + 8I - 8\, ln\, 2}{\pi} = 4.$ **Ans.**

186.

(A) Apply L'Hospital's rule twice or use expansion of $e^{x\cos x}$

(B) $x = u^6$

$\Rightarrow dx = 6u^5\, du$

$$I = \int_0^1 \frac{6u^5 du}{u^3 + u^2} = 6\int_0^1 \frac{u^3 + 1 - 1}{u+1} du = 5 - 6\, ln\, 2$$

$\Rightarrow$ $a + b = 5 - 6 = -1$ Ans.

(C) $e^n \displaystyle\int_0^n e^{-\theta}(\sec^2\theta - \tan\theta)\, d\theta = 1$

put $-\theta = t$; $d\theta = -dt$

$-e^n \displaystyle\int_0^{-n} e^t(\sec^2 t + \tan t)\, dt = 1$ [use $\int e^x\left(f(x) + f'(x)\right) = e^x f(x)$]

$$-e^{n}\left[e^{t}\tan t\right]_{0}^{-n}=1$$

$\Rightarrow \quad -e^{n}[-e^{-n}\tan n]=1$

$\Rightarrow \quad \tan n=+1$ Ans.

(D) $$\lim_{n\to\infty}\frac{\displaystyle\int_{\frac{1}{n+1}}^{\frac{1}{n}}\tan^{-1}(nx)\,dx}{\displaystyle\int_{\frac{1}{n+1}}^{\frac{1}{n}}\sin^{-1}(nx)\,dx}\qquad \text{(put } nx=t\text{);}$$

$$\therefore \lim_{n\to\infty}\frac{\displaystyle\frac{1}{n}\int_{\frac{n}{n+1}}^{1}\tan^{-1}(t)\,dt}{\displaystyle\frac{1}{n}\int_{\frac{n}{n+1}}^{1}\sin^{-1}(t)\,dt}\left(\frac{0}{0}\right)\quad \text{use L'Hospital's rule}$$

$$\lim_{n\to\infty}\frac{\tan^{-1}\left(\frac{n}{n+1}\right)}{\sin^{-1}\left(\frac{n}{n+1}\right)}=\frac{\frac{\pi}{4}}{\frac{\pi}{2}}=\frac{1}{2}\ \text{Ans.]}$$

187. $I=\displaystyle\int_{-1}^{1} f(x)\,g(x)\,dx$ .... (1)

$I=\displaystyle\int_{-1}^{1} f(-x)\,g(-x)\,dx=\int_{-1}^{1}\frac{1}{f(x)}g(x)\,dx$ .... (2)

adding (1) & (2)

$$2I=\int_{-1}^{1}\left(f(x)+\frac{1}{f(x)}\right)g(x)dx\geq 2\int_{-1}^{1}g(x)\,dx=2$$

$\Rightarrow$ **I ≥ 1 Ans.**

188. $F'(x)=f(x)$

$F''(x)=f'(x)$

$F(1)=0,\ f(1)=f'(1)=F'(1)=F''(1)=1=g(1)$

$f(g(x))=g(f(x))=x$

$\Rightarrow f'(g(x)).g'(x)=1$

$\Rightarrow f'(g(1)).g'(1) = 1 \Rightarrow g'(1) = 1$

$G(x) = x^2 g(x) - xF(g(x))$

$G'(x) = 2x\,g(x) + x^2 g'(x) - F(g(x)) - xF'(g(x)).g'(x)$

$\quad = 2xg(x) + x^2 g'(x) - F(g(x)) - x^2 g'(x))$

$\quad = 2xg(x) - F(g(x))$

$G''(x) = 2g(x) + 2xg'(x) - F'(g(x))\cdot g'(x)$

$\quad = 2g(x) + xg'(x)$

$\Rightarrow\ G''(1) = 2g(1) + g'(1) = 2 \times 1 + 1 = 3$. **Ans.**

189. $\frac{1}{3} < x < 1$

$1 < \frac{1}{3} < 3$

$[x] = 1$ or 2

Case I : $\frac{1}{3} < x < \frac{1}{2}$

$\Rightarrow 2 < \frac{1}{x} < 3$

$\Rightarrow \left[\frac{1}{x}\right] = 2$

Case II : $\frac{1}{2} < x < 1$

$\Rightarrow 1 < \frac{1}{x} < 2 \Rightarrow \left[\frac{1}{x}\right] = 1$

Now, $n = \int_{1/3}^{1/2} \left\{2x\left[\frac{1}{x}\right]\right\}dx + \int_{1/2}^{1} \left\{2x\left[\frac{1}{x}\right]\right\}dx$

$= \int_{1/3}^{1/2} \{4x\}dx + \int_{1/2}^{1} \{2x\}dx$

$\frac{1}{3} < x < \frac{1}{2} \qquad \frac{1}{2} < x < 1$

$\frac{4}{3} < 4x < 2 \qquad 1 < 2x < 2$

$\therefore \quad m = \int_{1/3}^{1/2} (4x-1)dx + \int_{1/2}^{1} (2x-1)\,dx$

$= (2x^2 - x)_{1/3}^{1/2} + (x^2 - x)_{1/2} = \frac{13}{36}$

Paragraph for question nos. 190 to 191

Sol. $\int_0^{f(x)} g(t)dt = g(f(x)) - 1$

$\Rightarrow g(f(x)) \cdot f'(x) = g'(f(x)) \cdot f'(x)$

$\Rightarrow \dfrac{g'(f(x)) \cdot f'(x)}{g(f(x))} = f'(x)$

$\Rightarrow \ln g(f(x)) = f(x) + c$

$\Rightarrow g(x) = e^x$ [If $f(x) = 0 \Rightarrow g(0) = 1 \;\therefore\; c = 0$]

190. $I = \int_0^1 e^x \left[\dfrac{1-x}{(1+x)^3}\right] dx = \int_0^1 e^x \left[-\dfrac{1}{(1+x)^2} + \dfrac{2}{(1-x)^3}\right] dx$

$= \left[\dfrac{-e^x}{(1+x)^2}\right]_0^1 = \dfrac{-e}{4} + 1$

191. $\lim\limits_{x\to\infty} e^{-x} \int_{\frac{1}{x}}^{x+\frac{1}{x}} g(t)dt = \lim\limits_{x\to\infty} \dfrac{\int_{\frac{1}{x}}^{x+\frac{1}{x}} e^t dt}{e^x}$

$= \lim\limits_{x\to\infty} \dfrac{e^{x+\frac{1}{x}} - e^{\frac{1}{x}}}{e^x} = \lim\limits_{x\to\infty} \dfrac{e^{\frac{1}{x}}(e^x - 1)}{e^x}$

$= \lim\limits_{x\to\infty} \dfrac{1-e^{-x}}{1} = 1.$ **Ans.**

Questions

01. $\int \sin 51x\,(\sin x)^{49}\,dx$ equals

(A) $\dfrac{\sin 50x\,(\sin x)^{50}}{50} + C$ (B) $\dfrac{\cos 50x\,(\sin x)^{50}}{50} + C$

(C) $\dfrac{\cos 50x\,(\cos x)^{50}}{50} + C$ (D) $\dfrac{\sin 50x\,(\sin x)^{51}}{51} + C$

02. $\int P(x) \cdot e^{kx}\,dx = Q(x)\,e^{4x} + C$. Where $P(x)$ is polynomial of degree n and $Q(x)$ is polynomial of degree 7. Then the value of $n + 7 + k + \lim\limits_{x \to \infty} \dfrac{P(x)}{Q(x)}$ is

(A) 18 (B) 19 (C) 20 (D) 22

Paragraph for question nos. 03 to 05

Let a differentiable function 'f' satisfies the functional rule

$$f(xy) = f(x) + f(y) + xy - x - y \;\forall\; x, y > 0 \text{ and } f'(1) = 4.$$

03. If $f(x_0) = 0$ then x_0 lies in the interval

(A) $(0, 1)$ (B) $(1, e)$ (C) (e, e^2) (D) (e^2, e^3)

04 $\int \dfrac{f(x)}{x}\,dx$ is equal to

(A) $3\,(\ln x)^2 + x + c$ (B) $3\ln x + x + c$ (C) $\dfrac{3}{2}\ln x + x + c$ (D) $\dfrac{3}{2}(\ln x)^2 + x + c$

05. If $\int e^{f(x)}dx = e^x(ax^3 + bx^2 + cx + d) + \lambda$ then the value of $(a + b + c + d)$ is equal to

(A) -1 (B) -2 (C) 3 (D) 6

06 Let $f(x)$ be a cubic polynomial with leading coefficient unity such that $f(0) = 1$ and all the roots of $f'(x) = 0$ are also roots of $f(x) = 0$. If $\int f(x)\,dx = g(x) + C$, where $g(0) = \dfrac{1}{4}$ and C is constant of integration, then $g(3) - g(1)$ is equal to

(A) 27 (B) 48 (C) 60 (D) 81

07. If primitive of $\sqrt{\sin^4 x + 4(1 - \sin^2 x)} + \sqrt{\cos^4 x + 4(1 - \cos^2 x)}$ with respect to x is $f(x) + C$, where $f(0) = 0$, then the value of $f(3)$ is

(A) 3 (B) 6 (C) 9 (D) 12

`

08. Let $f(x)$ be a twice differentiable function such that $f''(x) = 14 + \cos x$; $f'(1) = 14 + \sin 1$ and $f(1) = 9 - \cos 1$, then the number of solutions of the equation $f(x) = 1$ is

(A) 0 (B) 1 (C) 2 (D) 3

09. Let $\int \sec^{-1}[-\sin^2 x]\,dx = f(x)+C$ where $[y]$ denotes largest integer $\le y$ and $f(0)=0$, then the value of $\left(f\left(\frac{8}{\pi x}\right)\right)''$ at $x=2$ is

(A) 2 (B) 4 (C) 8 (D) 16

10. If $\int \frac{dx}{x\sqrt{4-\ln^2 x}} = F(x)+C$, then the value of $F(e)-F(1)$ equals

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$ (C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{4}$

11. Let $I = \int \frac{e^x}{e^{8x}+2e^{4x}+1}dx$ & $J = \int \frac{3e^x}{\left(e^{4x}+2+e^{-4x}\right)}dx$

Then, for any arbitary constant C, J – I equals

(A) $C - \frac{e^x}{\left(e^{3x}+1\right)}$ (B) $C - \frac{e^{2x}}{\left(e^{4x}-1\right)}$ (C) $C - \frac{e^x}{\left(e^{4x}+1\right)}$ (D) $C - \left(\frac{e^{3x}}{e^{4x}-1}\right)$

12. If $\int \left(x^{2010}+x^{804}+x^{402}\right)\left(2x^{1608}+5x^{402}+10\right)^{\frac{1}{402}}dx = \frac{1}{10a}\left(2x^{2010}+5x^{804}+10x^{402}\right)^{\frac{a}{402}}$, then find the value of a.

13. If $I = \int \frac{ax^2+2bx+c}{\left(Ax^2+2Bx+C\right)^2}dx$ (where $B^2 \ne AC$) is a rational function then which one of the following condition must be necessary?

(A) $2Bb = Ac + aC$ (B) $Aa + Bb = Cc$ (C) $Bb = aC + cA$ (D) $\frac{A}{c}+\frac{C}{a}=\frac{2B}{b}$

14. Let a, b be constant numbers such that $a^2 \le b$ and $I = \int \frac{dx}{x^2+2ax+b}$, then I can be

(A) $\frac{1}{\sqrt{b-a^2}}\tan^{-1}\left(\frac{x+a}{\sqrt{b-a^2}}\right)+C$ (B) $\frac{1}{2\sqrt{a^2-b}}\ln\left(\frac{x+a-\sqrt{a^2-b}}{x+a+\sqrt{a^2-b}}\right)+C$

(C) $C-\frac{1}{x+a}$ (D) $\tan^{-1}\left(\frac{x+a}{\sqrt{b-a}}\right)+C$

15. Let $f(x) = \tan x + 2\tan 2x + 4\tan 4x + 8\cot 8x$, then primitive of $f(x)$ with respect to x is

(A) $8\ln(\sin 8x)+C$ (B) $\ln(\sin 8x)+C$

(C) $\ln(\sin x)+C$ (D) $\ln(\sec x)+C$

where C is constant of integration.

16. $\int \frac{x^3}{\sqrt{1+x^2}} dx$ is equal to

(A) $\frac{(1+x^2)^{\frac{3}{2}}}{3} - \sqrt{1+x^2} + C$ (B) $\left(1+x^2\right)^{1/2} \frac{(x^2+2)}{3} + C$

(C) $x^2\sqrt{1+x^2} - \frac{2}{3}(1+x^2)^{\frac{3}{2}} + C$ (D) $\sqrt{1+x^2}\left(\frac{x^2-2}{3}\right) + C$

(Where C is constant of integration.)

17. Let $\int \frac{(1+x^4)\,dx}{\left(1-x^4\right)^{\frac{3}{2}}} = f(x) + C_1$ where $f(0) = 0$ and $\int f(x)\,dx = g(x) + C_2$ with $g(0) = 0$.

If $g\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{k}$. Find k.

18. Let $y = f(x)$ be such that $\frac{d}{dx}(x^2y) = x - 1$ where $x \neq 0$ and $y = 0$ when $x = 1$.

Find the smallest natural number x for which $\frac{dy}{dx}$ is positive.

19. $\int \frac{\sqrt{\sin^4 x + \cos^4 x}}{\sin^3 x \cos x} dx$, $x \in \left(0, \frac{\pi}{2}\right)$

20. Let $f(x)$ be a polynomial function of degree 2 satisfying

$$\int \frac{f(x)}{x^3-1} dx = \ln\left|\frac{x^2+x+1}{x-1}\right| + \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + C,$$

where C is indefinite integration constant.

(a) Find the value of $f(1)$.

(b) Let $\int \frac{1-6\operatorname{cosec} x}{6+f(\sin x)} d(\sin x) = g(x) + K$, where $g(x)$ contains no constant term.

Then find the value of $\lim_{t \to \frac{\pi}{2}} g(t)$. (where K is indefinite integration constant.)

(c) Let $\int \frac{5+f(\sin x)+f(\cos x)}{\sin x + \cos x} dx = h(x) + \lambda$, where $h(1) = -1$.

Find the value of $\tan^{-1}\left(h(2)\right) + \tan^{-1}\left(h(3)\right)$. (where λ is indefinite integration constant.)

21. Let $I_1 = \int \tan x \ \tan(ax+b)\,dx$ and $I_2 = \int \cot x \ \cot(ax+b)\,dx$

Column-I		Column-II	
(A)	value of I_1 for $a = 1$ is	(P)	$x - \cot b \ ln \dfrac{\cos(x-b)}{\cos x} + C$
(B)	value of I_2 for $a = 1$ is	(Q)	$\cot b \ ln \dfrac{\sin x}{\sin(x+b)} - x + C$
(C)	value of I_1 for $a = -1$ is	(R)	$\cot b \ ln \left(\dfrac{\cos x}{\cos(x+b)}\right) - x + C$
(D)	value of I_2 for $a = -1$ is	(S)	$x + \cot b \ ln \left(\dfrac{\sin x}{\sin(b-x)}\right) + C$

22. Evaluate: $\displaystyle\int \frac{dx}{\cos(x-1)\cos(x-2)\cos(x-3)}$

23. $\displaystyle\int \frac{(x^3-1)}{(x^4+1)(x+1)}\,dx$, is

(A) $\dfrac{1}{4} ln(1+x^4) + \dfrac{1}{3} ln(1+x^3) + c$
(B) $\dfrac{1}{4} ln(1+x^4) - \dfrac{1}{3} ln(1+x^3) + c$
(C*) $\dfrac{1}{4} ln(1+x^4) - ln(1+x) + c$
(D) $\dfrac{1}{4} ln(1+x^4) + ln(1+x) + c$

24. Let $f'(x^2) = \dfrac{1}{x}$ for $x > 0$, $f(1) = 1$ and $g'(\sin^2 x - 1) = \cos^2 x + p \ \forall \ x \in R$, $g(-1) = 0$.

If $h(x) = \begin{cases} f(x), & x > 0 \\ g(x), & -1 \le x \le 0 \end{cases}$

is a continuous function then find the absolute value of $2p$.

25. If $\displaystyle\int \frac{dx}{x + x^{2011}} = f(x) + C_1$ and $\displaystyle\int \frac{x^{2009}}{1 + x^{2010}}\,dx = g(x) + C_2$ (where C_1 and C_2 are constants of integration).
Let $h(x) = f(x) + g(x)$. If $h(1) = 0$ then $h(e)$ is equal to
(A) 0 (B) 1 (C) e (D) 2

26. Let $A = 2^{\left(\frac{\log_{10}\left(\frac{100}{x}-1\right)}{-\log_{10} 2}\right)}$ then $\displaystyle\int ln\,10 \cdot \log_{10}\left(\frac{1}{A}\right) dx$ is equal to
(A) $(x-100)\,ln\,(100-x) - x\,ln\,x + C$
(B) $(x-100)\,ln\,(100-x) + x - x\,ln\,x + C$
(C) $(100-x)\,ln\,(100-x) - x\,ln\,x + C$
(D) $(100-x)\,ln\,(100-x) + x\,ln\,x + C$

27. If $f(x)=\int \frac{x}{2}\,d\left(\frac{x^2-1}{x^2}\right)$ and $f(2)=\frac{1}{2}$ and $g_r(x)=\underbrace{f(f(f(.....f(x)....)))}_{r-\text{times}}$ i.e. $g_1(x)=f(x)$, $g_2(x)=f(f(x))$ and so on then identify the correct statement(s).

(A) $\frac{d}{dx}\left(g_{(3n-2)}(x)\right)=1$ whenever exists, $n \in N$

(B) $\frac{d}{dx}\left(g_{3n}(x)\right)=1$ whenever exists, $n \in N$

(C) $\lim\limits_{x\to 1}\frac{\sum\limits_{r=1}^{100}(g_{3r}(x))^r+x-101}{x-1}=5050$

(D) Slope of the tangent to the graph of the function $y=g_{80}(x)$ at $x=\frac{1}{2}$ is 4.

28. $\int \frac{(\cos^4 x-\cos^6 x)^{1/4}}{(1-\sin^3 x)^{1/2}}dx$ is equal to

(A) $\frac{1}{3}\sin^{-1}(\sin^{3/2}x)+c$

(B) $\frac{2}{3}\sin^{-1}(\sin^{3/2}x)+c$

(C) $\frac{4}{9}\sin^{-1}(\sin^{3/2}x)+c$

(D) $x+c$

29. Let $\int \frac{g(x)\left[(1+x)\ln x-5\right]}{x\,(\ln x)^6}dx=\frac{g(x)}{(\ln x)^5}+C$, $g(1)=e$ and $g:(0,\infty)\to(0,\infty)$.

If $f(x)=\ln\left(g(x+1)\right)-\ln\left(g(x)\right)\ \forall\, x\in R^+$ then find the value of $\frac{(2013)^2}{(2014)}\sum\limits_{x=1}^{2012}f''(x)$.

30. Evaluate $\int \frac{dx}{\sqrt{\cos^3(x+\alpha)\sin(x+\beta)}}dx$

31. If $\int \frac{(x-1)\,dx}{(x+x\sqrt{x}+\sqrt{x})\sqrt{\sqrt{x}(x+1)}}=4\tan^{-1}\left(g(x)\right)+C$, where C is an arbitrary constant of integration. Find $g^2(1)$.

32. If $\int\left(\cot 2x\cot 3x-\tan 2x\tan 7x\right)\tan 5x=a\ln(\tan 2x)+b\ln(\sin 3x)+c\ln(\sec 5x)+d\ln(\cos 7x)+C$ and $a, b, c, d\in Q$ and C is the contant of integration. If $(a+b+c+d)$ can be expressed as m/n in the lowest form, find $(m+n)$.

Paragraph for Question no. 33 & 34

A polynomial $f(x)$ when divided by $(x-1)$ leaves remainder 2 and when divided by $(x+2)$ leaves remainder 3 and the same polynomial when divided by $(x-1)(x+2)(x-3)$ leaves remainder $g(x)$ where $g(x) > 0 \ \forall\ x$. Also graph of $g(x)$ is symmetric about the line $x = \frac{-1}{3}$.

33 $\int (g(x))^{-1}\,dx$ is equal to

(A) $\frac{3}{\sqrt{2}} \tan^{-1}\left(\frac{3x+1}{\sqrt{2}}\right) + C$ (B) $\frac{\sqrt{2}}{3} \tan^{-1}\left(\frac{3x+1}{\sqrt{2}}\right) + C$

(C) $\tan^{-1}\left(\frac{3x+1}{\sqrt{2}}\right) + C$ (D) $\frac{\sqrt{3}}{\sqrt{2}} \tan^{-1}\left(\frac{3x+1}{\sqrt{2}}\right) + C$

34. Remainder when $g(x)$ is divided by $(x-3)$ is

(A) $\frac{34}{3}$ (B) 11 (C) 12 (D) $\frac{35}{3}$

35. The value of $\int \frac{(9\tan^5\theta + 7\tan^3\theta)(1+\tan^2\theta)}{(\tan^3\theta + \tan\theta)^{\frac{1}{2}}}\,d\theta$ is equal to

(A) $2(\tan\theta)^{5/2} + C$ (B) $2(\tan\theta)^{7/2}\sec\theta + C$
(C) $2(\tan\theta)^{7/2} + C$ (D) none

ANSWER KEY

1.	A	**2.**	D	**3.**	A	**4.**	D	**5.**	B	**6.**	C
7.	C	**8.**	B	**9.**	A	**10.**	B	**11.**	C	**12.**	403
13.	A	**14.**	AC	**15.**	C	**16.**	ACD	**17.**	12	**18.**	2

19. $C - \frac{\sqrt{1+t^2}}{2} - \frac{1}{4} ln \frac{\sqrt{t^2+1}-1}{\sqrt{t^2+1}+1}$, where $t = \cot^2 x$

20. **(a)** – 3; **(b)** $ln\,3$; **(c)** $\frac{-3\pi}{4}$

21. (A) R ; (B) Q ; (C) P ; (D) S

22. $\sec^2 1 \left(\frac{\sec 1}{2} ln \left| \frac{\cos 1 + \sin(x-2)}{\cos 1 - \sin(x-2)} \right| - \frac{1}{\sin(x-2)} \right) + C.$

23.	C	**24.**	3	**25.**	B	**26.**	A	**27.**	BD	**28.**	B	**29.**	2012

30. $\frac{2\sqrt{\tan(x+\alpha)+\tan(\beta-\alpha)}}{\sqrt{\cos(\beta-\alpha)}} + C$

31.	2	**32.**	499	**33.**	A
34.	A	**35.**	B		

01. $\sin 51x\,(\sin x)^{49} = [\sin x \cdot \cos 50x\,(\sin x)^{49} + \cos x \sin 50x\,(\sin x)^{49}]$

$$= \frac{\frac{d}{dx}\left((\sin x)^{50} \sin 50x\right)}{50}.\ \textbf{Ans.}$$

02. $n = 7,\ k = 4$

$\int P(x) \cdot e^{kx}\,dx = Q(x) \cdot e^{4x} + C$

$\therefore \quad k = 4$

and $\quad P(x)\,e^{4x} = e^{4x}\,[Q'(x) + 4Q(x)]$

$$\frac{P(x)}{Q(x)} = \frac{Q'(x)}{Q(x)} + 4$$

$$\therefore \quad \lim_{x \to \infty} \frac{P(x)}{Q(x)} = 0 + 4 = 4.$$

Alternative:

As $Q(x)$ will be a polynomial of same degree as that of $P(x)$ and the leading coefficient of $Q(x)$ is equal to leading coefficient of $\frac{P(x)}{k}$

$$\therefore \quad \lim_{x \to \infty} \frac{P(x)}{Q(x)} = \lim_{x \to \infty} k = 4$$

required $= 7 + 7 + 4 + 4 = 22$. **Ans.**

Paragraph for question nos. 03 to 05

Sol. $f(xy) = f(x) + f(y) + xy - x - y$

Partially differentiating w.r.t y (taking x as a constant)

$f'(xy) \cdot x = f'(y) + x - 1$

putting $y = 1$

$f'(x) \cdot x = f'(1) + x - 1$

$f'(x) \cdot x = 3 + x$

$f'(x) = \frac{3}{x} + 1$

Integrating both sides

$f(x) = 3\,\ln x + x + c$

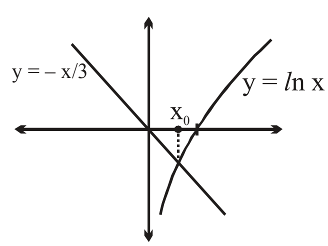
from the given functional rule

$x = y = 1,\ f(1) = 1.$

$f(x) = 3\,\ln x + x + c$

$x = 1,\ c = 0$

$\therefore \quad f(x) = 3\,\ln x + x$



3. $f(x) = 0 \Rightarrow 3\,\ln x + x = 0 \Rightarrow \ln x = \frac{-x}{3}$

Clearly, x_0 lies in the interval $(0, 1)$.

4. $\int \frac{f(x)}{x}\,dx = \int \frac{3\ln x + x}{x}\,dx = \int \left(\frac{3\ln x}{x} + 1\right) dx = \frac{3}{2}(\ln x)^2 + x + c.$

5. $\int e^{f(x)}\,dx = e^x(ax^3 + bx^2 + cx + d) + \lambda$

$\Rightarrow \int e^{3\ln x + x}\,dx = e^x(ax^3 + bx^2 + cx + d) + \lambda$

$\Rightarrow \int x^3 e^x\,dx = e^x(ax^3 + bx^2 - cx + d) + \lambda$

Differentiating both sides, w.r.t x

$x^3e^x = e^x(ax^3 + bx^2 + cx + d) + e^x(3ax^2 + 2bx + c)$

Comparing coefficients on both sides

$a = 1,\ 3a + b = 0 \Rightarrow b = -3$

$2b + c = 0 \Rightarrow c = 6$

$c + d = 0 \Rightarrow d = -6$

$\therefore\quad a + b + c + d = 1 - 3 + 6 - 6 = -2.$ **Ans.**

06 We have $f(x) = (x + 1)^3$

Now, $\int f(x)\,dx = \int (x+1)^3\,dx = \dfrac{(x+1)^4}{4} + C \Rightarrow g(x) = \dfrac{(x+1)^4}{4}$

Hence $g(3) - g(1) = \dfrac{4^4}{4} - \dfrac{2^4}{4} = 64 - 4 = 60$ **Ans.**

07. Consider $\sqrt{\sin^4 x - 4\sin^2 x + 4} + \sqrt{\cos^4 x - 4\cos^2 x + 4}$

$= (2 - \sin^2 x) + (2 - \cos^2 x) = 3$

$\therefore\quad \int 3\,dx = 3x + C \Rightarrow f(x) = 3x$

$\therefore\quad f(3) = 9.$ **Ans.**

`

08. $f''(x) = 14 + \cos x$

$f'(x) = 14x + \sin x + C$

Now $f'(1) = 14 + \sin 1 \Rightarrow C = 0$

$\therefore\quad f'(x) = 14x + \sin x$

$f(x) = \dfrac{14x^2}{2} - \cos x + C$

$f(1) = 9 - \cos 1$

$\Rightarrow\quad 9 - \cos 1 = 7 - \cos 1 + C \quad\Rightarrow\quad C = 2$

$\therefore\quad f(x) = 7x^2 - \cos x + 2$

$\therefore\quad f(x) = 1 \Rightarrow 7x^2 - \cos x + 2 = 1 \Rightarrow \underbrace{7x^2 + 1}_{\ge 1} = \underbrace{\cos x}_{\le 1}$

$\Rightarrow$ possible only if $x = 0$

09. $-1 \le \sin^2 x \le 0$

$\therefore\quad [-\sin^2 x] = -1$ or 0

$\Rightarrow\quad \sec^{-1}[-\sin^2 x] = \pi$ as $\sec^{-1}(0)$ is not defined.

$\int \sec^{-1}[-\sin^2 x]\,dx = \pi x + C \qquad \therefore\ f(x) = \pi x$

$f\left(\dfrac{8}{\pi x}\right) = \pi\,\dfrac{8}{\pi x} = \dfrac{8}{x}$

$$\left(f\left(\frac{8}{\pi x}\right)\right)'' = \frac{16}{x^3} \Rightarrow \left(f\left(\frac{8}{\pi x}\right)\right)''\Bigg|_{x=2} = \frac{16}{8} = 2$$

10. Put $\ln x = t$

$$\int_0^1 \frac{dt}{\sqrt{4-t^2}} = \sin^{-1}\frac{t}{2}\Big]_0^1 = \frac{\pi}{6} \textbf{ Ans.}]$$

11. $J = \int \frac{3e^{5x}}{e^{8x}+2e^{4x}+1}dx$

$\therefore \quad J - I = \int \frac{3e^{5x}-e^{x}}{e^{8x}+2e^{4x}+1}dx$

$= \int \frac{\left(3e^{3x}-e^{-x}\right)}{e^{6x}+2e^{2x}+e^{-2x}}dx \qquad$ (dividing by e^{2x})

$= \int \frac{\left(3e^{3x}-e^{-x}\right)}{\left(e^{3x}+e^{-x}\right)^2}dx$

Let $\quad e^{3x} + e^{-x} = t \quad \Rightarrow \left(3e^{3x}-e^{-x}\right)dx = dt$

$\therefore \quad J - I = \int \frac{dt}{t^2} = \frac{-1}{t} + C = C - \frac{1}{3e^{3x}-e^{-x}} = C - \frac{e^x}{\left(e^{4x}+1\right)}$

12. $\int x\left(x^{2009}+x^{803}+x^{401}\right)\left(2x^{1608}+5x^{402}+10\right)^{\frac{1}{402}}dx$

$\int \left(x^{2009}+x^{803}+x^{401}\right)\left(2x^{2010}+5x^{804}+10x^{402}\right)^{\frac{1}{402}}dx$

put $2x^{2010} + 5x^{804} + 10x^{402} = t$; $\quad 4020\left(x^{2009}+x^{803}+x^{401}\right)dx = dt$

$\left(x^{2009}+x^{803}+x^{401}\right)dx = \frac{dt}{4020}$;

$= \frac{1}{4020}\int t^{\frac{1}{402}}dt \qquad = \frac{1}{4020}\left(\frac{t^{\frac{1}{402}+1}}{\frac{1}{402}+1}\right)$

$= \frac{1}{4020}\left(\frac{t^{\frac{403}{402}}}{\frac{403}{402}}\right) = \frac{1}{4020}\times\frac{402}{403}t^{\frac{403}{402}} = \frac{1}{4030}\left(2x^{2010}+5x^{804}+10x^{402}\right)^{\frac{403}{402}} \Rightarrow a = 403$

13. Let $\int \frac{ax^2+2bx+c}{\left(Ax^2+2Bx+C\right)^2}dx = \frac{f(x)}{g(x)}$ where $f(x)$ and $g(x)$ are polynomials ;

($B^2 \neq AC \Rightarrow Ax^2 + 2Bx + C$ is not a perfect square)

differentiate w.r.t. x

$$\frac{ax^2+2bx+c}{(Ax^2+2Bx+C)^2} = \frac{g(x)f'(x)-f(x)g'(x)}{g^2(x)} \qquad ...(1)$$

Hence $g(x) \cong Ax^2 + 2Bx + C$

If N^r of RHS in equation (1), has to be a quadratic function then f(x) must be linear function (think !)

i.e. $\quad f(x) = px + q$

$\therefore\ p(Ax^2 + 2Bx + C) - (px + q)(2Ax + 2B) \equiv ax^2 + 2bx + c$

Comparing coefficient of x^2

$Ap - 2Ap = a \Rightarrow -Ap = a \Rightarrow p = \frac{-a}{A}$

coefficient of x

$2Bp - (2Bp + 2Aq) = 2b \Rightarrow -Aq = b \Rightarrow q = \frac{-b}{A}$

constant term

$\quad pC - 2Bq = c$

substituting the value of p and q

$$\frac{-aC}{A} + \frac{2Bb}{A} = c$$

$\Rightarrow\ 2Bb - aC = Ac \Rightarrow\ 2Bb = Ac + aC$

Hence $2Bb = Ac + aC$ **Ans.**

14. If $b > a^2$

$$I = \int \frac{dx}{(x+a)^2 + b - a^2}$$

$$I = \frac{1}{\sqrt{b-a^2}} \tan^{-1}\left(\frac{x+a}{\sqrt{b-a^2}}\right) + C$$

If $b = a^2$ then $I = \int \frac{dx}{(x+a)^2}$

$C - \frac{1}{x+a}$. **Ans.**

15. $\int f(x)\,dx = ln(\sec x) + ln(\sec 2x) + ln(\sec 4x) + ln(\sin 8x) + C$

$$= ln\left(\frac{\sin 8x}{\cos x \cos 2x \cos 4x}\right) + C = ln\left(\frac{8\sin x \cdot \sin 8x}{8\sin x \cdot \cos x \cos 2x \cos 4x}\right)$$

$$= ln\left(\frac{8\sin x\sin 8x}{\sin 8x}\right) = ln(8\sin x) = ln\,8 + ln(\sin x) + C = ln(\sin x) + C.$$ **Ans.**

16. Let $I = \int \frac{x^3}{\sqrt{1+x^2}}\,dx$

Put, $1 + x^2 = t$

$$\Rightarrow \quad I = \frac{1}{2}\int\left(\frac{t-1}{\sqrt{t}}\right)dt = \frac{1}{2}\int\left(\sqrt{t}-\frac{1}{\sqrt{t}}\right)dt = \frac{1}{2}\left[\frac{2}{3}t^{\frac{3}{2}} - 2\sqrt{t}\right] + C = \frac{1}{3}(1+x^2)^{\frac{3}{2}} - \sqrt{1+x^2} + C$$

$$= \sqrt{1+x^2}\left[\frac{(1+x^2)}{3} - 1\right] + C = \frac{\sqrt{1+x^2}}{3}(x^2-2) + C$$ **Ans.**

(C) $$x^2\sqrt{1+x^2} - \frac{2}{3}(1+x^2)^{\frac{3}{2}} + C = (1+x^2)\sqrt{1+x^2} - \sqrt{1+x^2} - \frac{2}{3}(1+x^2)^{3/2} + C$$

$$= \frac{(1+x^2)^{\frac{3}{2}}}{3} - \sqrt{1+x^2} + C.$$ **Ans.**

Aliter: Let $I = \int \frac{x^3}{\sqrt{1+x^2}}\,dx$

Put $x = \tan\theta \quad \Rightarrow \quad dx = \sec^2\theta\, d\theta$

Now, $I = \int \frac{\tan^3\theta}{\sec\theta}\sec^2\theta\, d\theta = \int(\sec^2\theta - 1)\sec\theta\tan\theta\, d\theta$

Put $\sec\theta = y$

$$= \int(y^2-1)\,dy = \frac{y^3}{3} - y + C = \frac{\sec^3\theta}{3} - \sec\theta + C$$

$$= \frac{(1+x^2)\sqrt{1+x^2}}{3} - \sqrt{1+x^2} + C$$ **Ans.**

17. $\int \frac{1+x^4}{(1-x^4)^{3/2}}\,dx$

Take out x^2 from the D^r, it will come out as x^3

$$\int \frac{x+\frac{1}{x^3}}{\left(\frac{1}{x^2}-x^2\right)^{3/2}}\,dx \qquad \text{Put } \frac{1}{x^2} - x^2 = t^2 \quad -2\left(x+\frac{1}{x^3}\right)dx = 2\,t\,dt$$

$$-\int\frac{t\,dt}{t^3}dt = \frac{1}{t} + C = \frac{x}{\sqrt{1-x^4}} + C_1; \quad \text{as } f(0) = 0 \Rightarrow C_1 = 0.$$

Now $\int \frac{x}{\sqrt{1-x^4}}\,dx = \frac{1}{2}\sin^{-1}x^2 + C_2$

but $g(0) = 0 \Rightarrow \quad C_2 = 0$

$$g(x) = \frac{1}{2}\sin^{-1} x^2$$

Hence $g\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{12} \equiv \frac{\pi}{k} \quad \Rightarrow \quad k = 12$ **Ans**

Alternatively : $\quad I = \int \frac{1-x^4+2x^4}{(1-x^4)^{3/2}}dx = \int\left(\frac{1}{(1-x^4)^{1/2}} + \frac{2x^4}{(1-x^4)^{3/2}}\right)dx$

This is of the form $f(x) + x f'(x)$

hence $I = x f(x) = \frac{x}{\sqrt{1-x^4}} + C$

18. $\frac{d}{dx}(x^2y) = x - 1$

Integrating both sides

$$x^2 y = \int (x-1)\,dx = \frac{x^2}{2} - x + C$$

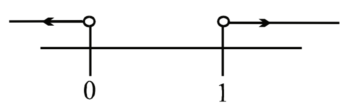
If $x = 1, y = 0$

$\therefore \quad 0 = \frac{1}{2} - 1 + C \Rightarrow C = \frac{1}{2}$

$$x^2 y = \frac{x^2}{2} - x + \frac{1}{2}$$

$$y = \frac{1}{2} - \frac{1}{x} + \frac{1}{2x^2}$$

$$\frac{dy}{dx} = \frac{1}{x^2} - \frac{1}{x^3} = \frac{x-1}{x^3}$$



$\therefore \quad \frac{x-1}{x^3} > 0$

$\therefore \quad x \in (-\infty, 0) \cup (1, \infty)$

19. $I = \int \frac{\sqrt{\sin^4 x + \cos^4 x}}{\sin^3 x \cos x}dx, \quad x \in \left(0, \frac{\pi}{2}\right)$

$$= \int \frac{\cos^2 x\sqrt{1+\tan^4 x}}{\sin^3 x \cos x}dx = \int \frac{\cos x\sqrt{1+\tan^4 x}}{\sin^3 x}dx$$

$$= \int \frac{\sqrt{1+\cot^4 x}}{\cot^2 x}\cdot \cot x \cdot \operatorname{cosec}^2 x\, dx$$

put $\cot^2 x = t \Rightarrow \quad 2\cot x \cdot \operatorname{cosec}^2 x\, dx = -dt$

$$I = -\frac{1}{2}\int \frac{\sqrt{1+t^2}}{t}dt$$

put $\quad 1 + t^2 = y^2$

$\Rightarrow \quad t\,dt = y\,dy$

$$I = -\frac{1}{2}\int \frac{y\cdot y}{t^2}dy = -\frac{1}{2}\int \frac{y^2-1+1}{y^2-1}dy$$

$$= -\frac{1}{2}\left(\int dy + \int \frac{dy}{y^2-1}\right) = C - \frac{y}{2} - \frac{1}{4}\ln\frac{y-1}{y+1}$$

$$= C - \frac{\sqrt{1+t^2}}{2} - \frac{1}{4}\ln\frac{\sqrt{t^2+1}-1}{\sqrt{t^2+1}+1} \quad \text{where } t = \cot^2 x \quad \text{Ans.}$$

20.

(i) $$\int \frac{f(x)}{x^3-1}dx = \ln\left|\frac{x^2+x+1}{x-1}\right| + \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + C$$

Differentiating both sides, we get

$$\frac{f(x)}{x^3-1} = \frac{x-1}{x^2+x+1}\,\frac{(x-1)(2x+1)-(x^2+x+1).1}{(x-1)^2} + \frac{2}{\sqrt{3}}\cdot\frac{1}{1+\left(\frac{2x+1}{\sqrt{3}}\right)^2}\cdot\frac{2}{\sqrt{3}}$$

$$= \frac{2x^2-x-1-x^2-x-1}{x^3-1} + \frac{4}{3}\cdot\frac{3}{3+(4x^2+4x+1)}$$

$$= \frac{x^2-2x-2}{x^3-1} + \frac{1}{x^2+x+1}$$

$f(x) = (x^2-2x-2)+(x-1) = x^2-x-3 \Rightarrow f(1) = -3$ **Ans.**

(ii) $$I = \int \frac{1-6\operatorname{cosec} x}{6+f(\sin x)}d(\sin x) = \int \frac{1-\frac{6}{\sin x}}{6+\sin^2 x-\sin x-3}d(\sin x)$$

$$\int \frac{1-\frac{6}{\sin x}}{\sin^2 x-\sin x+3}d(\sin x) \quad (\text{Put } \sin x = t)$$

$$= \int \frac{1-\frac{6}{t}}{t^2-t+3}dt = \int \frac{\frac{1}{t^2}-\frac{6}{t^3}}{1-\frac{1}{t}+\frac{3}{t^2}}dt = \ln\left(1-\frac{1}{t}+\frac{3}{t^2}\right)+K$$

$$= \ln\left(1-\frac{1}{\sin x}+\frac{3}{\sin^2 x}\right)+K$$

$$\Rightarrow \quad g(x) = \ln\left(1-\frac{1}{\sin x}+\frac{3}{\sin^2 x}\right)$$

$$\therefore \quad g(t) = \ln\left(1-\frac{1}{\sin t}+\frac{3}{\sin^2 t}\right)$$

Now, $\lim_{t\to\frac{\pi}{2}} g(t) = \ln 3$ **Ans.**

(iii) $I = \int \frac{5+f(\sin x)+f(\cos x)}{\sin x+\cos x}dx = \int \frac{5+\sin^2 x-\sin x-3+\cos^2 x-\cos x-3}{\sin x+\cos x}dx$

$= \int -dx = -x+\lambda$

$\therefore\ h(x) = -x \quad (\text{since } h(1) = -1)$

Hence $\tan^{-1}(h(2)) + \tan^{-1}(h(3)) = \tan^{-1}(-2) + \tan^{-1}(-3) = \frac{-3\pi}{4}$ **Ans.**

21. $a = 1 \quad I_1 = \int \tan(x)\tan(x+b)dx$

Now $\tan(x+b-x) = \frac{\tan(x+b)-\tan x}{1+\tan(x+b)\tan x}$

i.e. $1+\tan(x+b)\tan x = \frac{\tan(x+b)-\tan x}{\tan b}$

so $I_1 = \int \frac{1}{\tan b}(\tan(x+b)-\tan x)-1\ dx$

i.e. $\frac{1}{\tan b}(\ln\sec(x+b)-\ln\sec x)-x+C$

i.e. $\cot b(\ln\cos x-\ln\cos(x+b))-x+C$

(B) $a = 1 \quad I_2 = \int \cot x\ \cot(x+b)dx$

Now $\cot(x+b-x) = \frac{\cot(x+b)\cot x+1}{\cot x-\cot(x+b)}$

i.e. $\cot(x+b)\cot x+1 = \cot b(\cot x-\cot(x+b))$

$I_2 = \int \frac{1}{\tan b}(\cot x-\cot(x+b))-1\ dx$

$= \cot b(\ln\sin x-\ln\sin(x+b))-x+C$

(C) I_1 = for $a = -1$ is

i.e. $I_1 = \int \tan x\ \tan(b-x)dx$

Now $\tan(x+b-x) = \frac{\tan x+\tan(b-x)}{1-\tan x\ \tan(b-x)}$

i.e. $1-\tan x\tan(b-x) = \frac{\tan x+\tan(b-x)}{\tan b}$

$I_1 = \int 1-\frac{1}{\tan b}(\tan x+\tan(b-x))\ dx$

$= x-\frac{1}{\tan b}(\ln\sec x-\ln\sec(b-x))+C$

$$= x - \frac{1}{\tan b}\ (ln \cos(b-x) - ln \cos x) + C$$

$$= x - \cot b \left(ln \frac{\cos(x-b)}{\cos x} \right) + C$$

(D) I_2 for $a = -1$

$$I_1 = \int \cot x \ \cot(b-x)\, dx$$

Now $\cot(x + b - x) = \dfrac{\cot x \ \cot(b-x) - 1}{\cot x + \cot(b-x)}$

i.e. $\cot x \cot(b-x) = 1 + \cot b\,(\cot x + \cot(b-x))$

$$I_2 = \int 1 + \cot b\,(\cot x + \cot(b-x))\ dx$$

$= x + \cot b\,(ln \sin x - ln \sin(b-x)) + C$]

22. $\displaystyle\int \frac{dx}{\cos(x-1)\cos(x-2)\cos(x-3)}$ Put $x - 2 = y$

$$\int \frac{dy}{\cos(y-1)\cos y \cos(y+1)} = \int \frac{dy}{(\cos^2 1 - \sin^2 y)\cos y}$$ Put $\sin y = t$

$$\int \frac{dt}{\left(\cos^2 1 - t^2\right)t^2} = \sec^2 1 \int \left(\frac{1}{\cos^2 1 - t^2} + \frac{1}{t^2} \right) dt$$

$$= \sec^2 1 \left(\frac{\sec 1}{2} ln \left| \frac{\cos 1 + t}{\cos 1 - t} \right| - \frac{1}{t} \right)$$

$$= \sec^2 1 \left(\frac{\sec 1}{2} ln \left| \frac{\cos 1 + \sin(x-2)}{\cos 1 - \sin(x-2)} \right| - \frac{1}{\sin(x-2)} \right) + C.$$

23. $\displaystyle\int \frac{x^3 - 1}{(x^4+1)(x+1)}\, dx$

$$= \int \frac{(x^4 + x^3) - (x^4+1)}{(x^4+1)(x+1)}\, dx = \int \frac{x^3}{x^4+1}\, dx - \int \frac{1}{x+1}\, dx$$

$= \dfrac{1}{4} ln(x^4 + 1) - ln(x+1) + c$ **Ans.**

24. $\because$ $f'(x^2) = \dfrac{1}{x}$ $\Rightarrow$ $f'(x) = \dfrac{1}{\sqrt{x}}$, $x > 0$

$\Rightarrow$ $f(x) = 2\sqrt{x} + c$ (c = integration constant)

$\because$ $f(1) = 1$ $\Rightarrow$ $c = -1$

$\therefore$ $f(x) = 2\sqrt{x} - 1,\ x > 0$

and $g'(\sin^2 x - 1) = \cos^2 x + p\ \forall\ x \in R$

$\Rightarrow$ $g'(-\cos^2 x) = \cos^2 x + p$

$\Rightarrow$ $g'(x) = p - x,\ \forall\ x \in [-1, 0]$

$\Rightarrow \qquad g(x) = px - \dfrac{x^2}{2} + k \qquad$ (where k = integration constant)

$\because \qquad g(-1) = 0 \Rightarrow 0 = -p - \dfrac{1}{2} + k \Rightarrow k = \dfrac{1}{2} + p$

$\therefore \qquad g(x) = px - \dfrac{x^2}{2} + \dfrac{1}{2} + p$

$$\therefore \qquad h(x) = \begin{cases} 2\sqrt{x} - 1 & , \ x > 0 \\ px - \dfrac{x^2}{2} + \dfrac{1}{2} + p & , -1 \le x \le 0 \end{cases}$$

$\therefore \qquad$ At $x = 0$

L.H.L = R.H.L = f(0)

$\Rightarrow -1 = \dfrac{1}{2} + p \Rightarrow \quad p = \dfrac{-3}{2}$

Hence $2p = -3$

$\therefore \qquad$ Absolute value of $2p$ is 3

25. $\because \displaystyle\int \frac{x^{2009}}{1+x^{2010}}dx = \int \frac{x^{2010}}{x+x^{2011}}dx = \int \frac{1+x^{2010}}{x+x^{2011}}dx - \int \frac{1}{x+x^{2011}}dx$

$\Rightarrow \displaystyle\int \frac{x^{2009}}{1+x^{2010}}dx + \int \frac{1}{x+x^{2011}}dx = \int \frac{1+x^{2010}}{x\left(1+x^{2010}\right)}dx$

$\Rightarrow h(x) = \displaystyle\int \frac{dx}{x} = \ln x + C$

$\because \ h(1) = 0 \ \therefore \ C = 0$

$\therefore \ h(x) = \ln x \ \Rightarrow \ h(e) = \ln e = 1.$ **Ans.**]

26. $A = 2^{\left(\frac{-\log_{10}\left(\frac{100-x}{x}\right)}{\log_{10}2}\right)} \Rightarrow A = 2^{\left(\log_2\left(\frac{x}{100-x}\right)\right)}$

$\Rightarrow A = \dfrac{x}{100-x} \quad \Rightarrow \quad \dfrac{1}{A} = \dfrac{100-x}{x}$

$I = \displaystyle\int \log_e 10 \cdot \log_{10}\left(\frac{100-x}{x}\right)dx = \int \left(\ln(100-x) - \ln x\right)dx$

$= (x-100)\ln(100-x) + (100-x) - x\ln x + x + C$

$I = (x-100)\ln(100-x) - x\ln x + C.$ **Ans.**

27. $f(x) = \displaystyle\int \frac{x}{2} d\left(1 - \frac{1}{x^2}\right) = \int \frac{x}{2}\frac{2}{x^3}dx = \int \frac{1}{x^2}dx = -\frac{1}{x} + C$

$f(2) = -\dfrac{1}{2} + C = \dfrac{1}{2} \qquad \Rightarrow C = 1$

$\therefore\ f(x) = \dfrac{x-1}{x}$

$g_1(x) = f(x) = \dfrac{x-1}{x}$; $g_2(x) = f(f(x)) = \dfrac{\dfrac{x-1}{x}-1}{\dfrac{x-1}{x}} = \dfrac{-1}{x-1} = \dfrac{1}{1-x}$;

$g_3(x) = f\left(f\left(f(x)\right)\right) = \dfrac{1}{1-\left(\dfrac{x-1}{x}\right)} = x$

$g_4(x) = g_1(x) = \dfrac{x-1}{x}$; $g_5(x) = g_2(x) = \dfrac{1}{1-x}$;

$g_6(x) = g_3(x) = x$ and so on

$\therefore\ g_{(3n-2)}(x) = \dfrac{x-1}{x}$, $g_{(3n-1)}(x) = \dfrac{1}{1-x}$, $g_{3n}(x) = x$

$\dfrac{d}{dx}\left(g_{(3n-2)}(x)\right) = \dfrac{d}{dx}\left(1-\dfrac{1}{x}\right) = \dfrac{1}{x^2}$ $\qquad \therefore$ option (A) is worng.

$\dfrac{d}{dx}\left(g_{(3n)}(x)\right) = \dfrac{d}{dx}(x) = 1$ $\qquad \therefore$ option (B) is correct.

$\displaystyle \lim_{x\to 1} \frac{x + x^2 + \ldots\ldots + x^{100} + x - 101}{x-1}$

applying L' Hospital's rule

$\displaystyle \lim_{x\to 1} \frac{1 + 2x + \ldots\ldots + 100x^{99} + 1}{1} = 1 + 2 + \ldots\ldots + 100 + 1 = 5051$ $\qquad \therefore$ option (C) is wrong.

$y = g_{80}(x) = g_{(3n-1)}(x) = \dfrac{1}{1-x}$

$\therefore\ g'_{(3n-1)}(x) = \dfrac{1}{(1-x)^2}$

$g'_{(3n-1)}(x)\Big|_{x=\frac{1}{2}} = 4$ $\qquad \therefore$ option (D) is also correct.

28. $\displaystyle \int \frac{(\cos^4 x - \cos^6 x)^{1/4}}{(1-\sin^3 x)^{1/2}}\,dx = \int \frac{\left\{\cos^4 x(1-\cos^2 x)\right\}^{\frac{1}{4}}}{(1-\sin^3 x)^{1/2}}\,dx = \int \frac{\cos x\ \sin^{1/2} x}{(1-\sin^3 x)^{1/2}}\,dx$

Let $\sin^{3/2}x = t \Rightarrow \dfrac{3}{2}\sin^{\frac{1}{2}} x \cos x\ dx = dt$

$\therefore$ Given integral $= \dfrac{2}{3}\displaystyle\int \frac{dt}{(1-t^2)^{1/2}} = \frac{2}{3}\sin^{-1} t + c = \frac{2}{3}\sin^{-1}(\sin^{3/2} x) + c$

29. $$\frac{g(x)\left[(1+x)\,ln\,x-5\right]}{x\,(ln\,x)^6} = \left[\frac{g(x)}{(ln\,x)^5}\right]'$$

$$\Rightarrow \quad \frac{g(x)\left[(1+x)\,ln\,x-5\right]}{x\,(ln\,x)^6} = \frac{x\cdot ln\,x\cdot g'(x)-5g(x)}{x\,(ln\,x)^6}$$

$$\Rightarrow g(x)\,(1+x) = x\,g'(x) \Rightarrow \frac{g'(x)}{g(x)} = 1+\frac{1}{x}$$

$\Rightarrow \quad ln\,g(x) = x + ln\,x \Rightarrow g(x) = k\cdot xe^x$

$\Rightarrow g(1) = k\cdot e = e \Rightarrow k = 1$

$g(x) = xe^x$.

$\therefore f(x) = ln\,(x+1) - ln\,(x) + 1$

$$f''(x) = \frac{1}{x^2} - \frac{1}{(x+1)^2}$$

Now, $$\sum_{x=1}^{2012} f''(x) = \sum_{x=1}^{2012}\left(\frac{1}{x^2}-\frac{1}{(x+1)^2}\right)$$

$$= \left(\frac{1}{1^2}-\frac{1}{2^2}\right)+\left(\frac{1}{2^2}-\frac{1}{3^2}\right)+\left(\frac{1}{3^2}-\frac{1}{4^2}\right)+\,........\,+\left(\frac{1}{2012^2}-\frac{1}{2013^2}\right)$$

Hence $$\sum_{x=1}^{2012} f''(x) = 1-\frac{1}{(2013)^2} = \frac{2012\cdot 2014}{(2013)^2}$$

Hence, $$\frac{(2013)^2}{(2014)}\sum_{x=1}^{2012} f''(x) = 2012.$$ **Ans.**

30. $$I = \int \frac{dx}{\sqrt{\cos^3(x+\alpha)\sin(x+\alpha+\beta-\alpha)}}$$

$$\int \frac{dx}{\sqrt{\cos^3(x+\alpha)\left(\sin(x+\alpha)\cos(\beta-\alpha)+\cos(x+\alpha)\sin(\beta-\alpha)\right)}}$$

$$= \int \frac{\sec^2(x+\alpha)}{\sqrt{\cos(\beta-\alpha)\tan(x+\alpha)+\sin(\beta-\alpha)}}$$

$$\frac{1}{\sqrt{\cos(\beta-\alpha)}}\int \frac{\sec^2(x+\alpha)}{\sqrt{\tan(x+\alpha)+\tan(\beta-\alpha)}}\,dx$$

Put $\tan(x+\alpha)+\tan(\beta-\alpha)=t^2$
$\sec^2(x+\alpha)\,dx = 2t\,dt$

$$\int \frac{2t\ dt}{t}$$

$$\frac{2\sqrt{\tan(x+\alpha)+\tan(\beta-\alpha)}}{\sqrt{\cos(\beta-\alpha)}} + C$$

31. $I = 2\int \frac{(t^2-1)\,t\,dt}{t(t+t^2+1)\sqrt{t(t^2+1)}} = 2\int \frac{(t^2-1)\,dt}{t(t+\frac{1}{t}+1)\,t\sqrt{t+\frac{1}{t}}}$

$$= 2\int \frac{1-\frac{1}{t^2}}{(t+\frac{1}{t}+1)\sqrt{t+\frac{1}{t}}}\,dt$$

put $\quad t+\frac{1}{t} = y^2;\quad \left(1-\frac{1}{t^2}\right)dt = 2y\,dy$

$$I = 4\int \frac{y\,dy}{(y^2+1)\,y} = 4\tan^{-1}\sqrt{t+\frac{1}{t}} + C = 4\tan^{-1}\sqrt{\sqrt{x}+\frac{1}{\sqrt{x}}} + C$$

$g(x) = \sqrt{\sqrt{x}+\frac{1}{\sqrt{x}}}\,;\quad g(1) = \sqrt{2}$ **Ans.**

32. $I = \underbrace{\int \frac{\cot 2x \cot 3x}{\cot 5x}\,dx}_{I_1} - \underbrace{\int \tan 2x \tan 5x \tan 7x\,dx}_{I_2}$

consider $I_2 = \int (\tan 7x - \tan 5x - \tan 2x)\,dx \quad ....(1)$

again $\quad I_1 = \int \frac{\cot 2x \cot 3x}{\cot 5x}\,dx$

$5x = 2x + 3x;$

$$\cot 5x = \cot(A+B) = \frac{\cot A \cot B - 1}{\cot B + \cot A}$$

$$\cot 5x = \frac{\cot 2x \cot 3x - 1}{\cot 2x + \cot 3x}$$

$$\cot 5x \cot 2x + \cot 5x \cot 3x + 1 = \cot 2x \cot 3x$$

$\therefore \qquad I_1 = \int \frac{\cot 5x \cot 2x + \cot 5x \cot 3x + 1}{\cot 5x}\,dx$

$$I_1 = \int (\cot 2x + \cot 3x + \tan 5x)\,dx$$

Now $I = I_1 - I_2$

$$I = \int (\cot 2x + \cot 3x + \tan 5x)\,dx - \int (\tan 7x - \tan 5x - \tan 2x)\,dx$$

$$= \int (\cot 2x + \tan 2x)\,dx + \int (\cot 3x)\,dx + 2\int (\tan 5x)\,dx - \int (\tan 7x)\,dx$$

$$= 2\int (\operatorname{cosec} 4x)\,dx + \frac{ln(\sin 3x)}{3} + \frac{2}{5}ln(\sec 5x) + \frac{1}{7}ln(\cos 7x)$$

$$= \frac{1}{2}ln(\operatorname{cosec} 4x - \cot 4x) + \frac{ln(\sin 3x)}{3} + \frac{2}{5}ln(\sec 5x) + \frac{1}{7}ln(\cos 7x) + C$$

$$= \frac{1}{2}ln(\tan 2x) + \frac{1}{3}ln(\sin 3x) + \frac{2}{5}ln(\sec 5x) + \frac{1}{7}ln(\cos 7x) + C$$

Hence $a + b + c + d = \frac{1}{2} + \frac{1}{3} + \frac{2}{5} + \frac{1}{7} = \frac{105 + 70 + 84 + 30}{210}$

$$= \frac{289}{210} = \frac{m}{n}$$

$\Rightarrow \quad m + n = 499$ **Ans.**

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Paragraph for Question no. 33 & 34

Sol. Given $f(1) = 2$ and $f(-2) = 3$

$f(x) = P(x) \cdot (x^3 - 2x^2 - 5x + 6) + R(x)$

$\Rightarrow \quad f(x) = P(x) \cdot (x - 1)(x + 2)(x - 3) + R(x)$

Let $\quad R(x) = Ax^2 + Bx + C$

$f(x) = P(x)(x - 1)(x + 2)(x - 3) + Ax^2 + Bx + C$

$f(1) = \quad A + B + C = 2$

$f(-2) = 4A - 2B + C = 3$

On solving $A - B = 1/3$

and $\quad C = \left(\frac{7}{3} - 2A\right)$

Also $\quad g(x) = Ax^2 + \left(A - \frac{1}{3}\right)x + \left(\frac{7}{3} - 2A\right)$

Given x-coordinate of vertex is

Hence, $A = 1$

so, $g(x) = x^2 + \frac{2}{3}x + \frac{1}{3}.2.$

$f(3) = R(3) = \frac{34}{3}$. **Ans**

35. Put $\tan\theta = t \Rightarrow \sec^2\theta \, d\theta = dt \Rightarrow (1+\tan^2\theta)\, d\theta = dt$

$$\int \frac{(9t^5+7t^3)}{(t^3+t)^{1/2}}\,dt = \int \frac{t^3\,(9t^2+7)}{\left(t\,(t^2+t)\right)^{1/2}}\,dt$$

Multiplying by t^3

$$\int \frac{(9t^8+7t^6)}{(t^9+t^7)^{1/2}}\,dt \qquad \text{Put } t^9+t^7=y^2$$

$$\int \frac{2y}{y}\,dy = 2y + C = 2\,t^{\frac{7}{2}}\sqrt{t^2+1} + C$$

$= 2\,(\tan\theta)^{7/2}\sec\theta + C$. **Ans.**

01 If the area bounded by the curves $f(x) = x^2$ and $g(x) = a$ is $\frac{256}{3}$ then the value of a is

(A) 4 (B) 8 (C) 16 (D) 32

02 If the area bounded by the curve $y = x^2$, $y = -x^2$ and $y^2 = 4x - 3$ is A, then the value of 12A, is

(A) 2 (B) 4 (C) 6 (D) 8

03 If the area enclosed by $y = x^3$, its normal at P(1, 1) and x-axis is $\frac{A}{8}$, then A equals

(A) 7 (B) 14 (C) 2 (D) 3

04 The area enclosed between the curve $f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$ and x-axis where $x \in [0, 2\pi]$, is

(A) π^2 (B) $\frac{9}{4}\pi^2$ (C) $\frac{5}{4}\pi^2$ (D) $\frac{1}{4}\pi^2$

Paragraph for Question no. 5 to 6

Let $f(x) = \sin 3x + \cos x$;
$g(x) = \cos 3x + \sin x$

05 If $\int_0^{2\pi}\left(f^2(x)+g^2(x)\right)dx = k\pi$, then the value of k is

(A) $\frac{1}{2}$ (B) 2 (C) 4 (D) $\frac{1}{4}$

06 If the area of the region bounded by the two curves $y = f(x)$ and $y = g(x)$ (where x varies from 0 to π) is $k_1 \cos\frac{\pi}{8} + k_2$ then $(k_1 + k_2)$ is equal to

(A) 1 (B) 2 (C) 4 (D) none

07 If the area bounded by the curves

$f(x) = \left(\cos^{-1}|\cos x|\right)^2$,

$g(x) = \left(\cos^{-1}|\cos x|\right)$ and

$|x| = \frac{\pi}{2}$ is $a\pi^3 + b\pi^2 + c$, then find the minimum value of $(|a| + |b| + |c|)$.

Paragraph for question nos. 8 to 11

Let f(x) be a differentiable function such that $f(x + y) = e^x f(y) + e^y f(x)$ all x, y and $f'(0) = 1$.

08. f(x) has

(A) maximum
(B) minimum
(C) both maximum and minimum
(D) neither maximum nor minimum

09. The range of f(x) is :

(A) R (B) $[0, \infty)$

(C) $\left(-\frac{1}{e}, 1\right)$ (D) $\left[-\frac{1}{e}, \infty\right)$

10. $\lim\limits_{x\to-\infty} f(x)$ is

(A) 0 (B) 1
(C) –1 (D) non existent

11 The area bounded by the curve $y = f(x)$ and the x-axis is :

(A) 1 (B) $\frac{1}{2}$ (C) 2 (D) e

12. Let f(x) be a polynomial of degree 3. If the curve $y = f(x)$ has relative extrema at $x = \frac{\pm 2}{\sqrt{3}}$ and passes through (0, 0) and (1, – 2) dividing the circle $x^2 + y^2 = 4$ in two parts, then the area bounded by $x^2 + y^2 = 4$ and $y \ge f(x)$ is $\frac{k\pi}{2}$. Find the value of k.

13 If $\int_0^1 \left(4x^3 - f(x)\right)f(x)dx = \frac{4}{7}$ then the area of region bounded by $y = f(x)$, x-axis and ordinate $x = 1$ and $x = 2$, is

(A) $\frac{11}{2}$ (B) $\frac{13}{2}$

(C) $\frac{15}{2}$ (D) $\frac{17}{2}$

14 Let $R = \{(x, y) : |x - y| < 2\}$ and $S = \{(x, y) : |x + y| < 2\}$. Then the value of the area of the region formed by $(R \cap S)$ is

(A) 2 (B) $2\sqrt{2}$
(C) 4 (D) 8

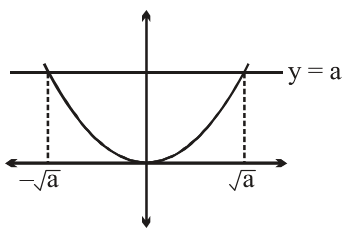
Answer Key

1.	C	2.	B	3.	B	4.	A	5.	C	6.	C
7.	1	8.	B	9.	D	10.	A	11.	A	12.	4
13.	C	14.	D								

01 $A = 2\int_0^{\sqrt{a}}\left(a - x^2\right)dx = \frac{256}{3}$

$\Rightarrow 2\left(ax - \frac{x^3}{3}\right)\Big]_0^{\sqrt{a}} = \frac{256}{3}$

$\Rightarrow 2\left(a\sqrt{a} - \frac{a\sqrt{a}}{3}\right) = \frac{256}{3}$



$4a\sqrt{a} = 256$

$a^{\frac{3}{2}} = 64$

$\therefore a = 16.$ **Ans.**

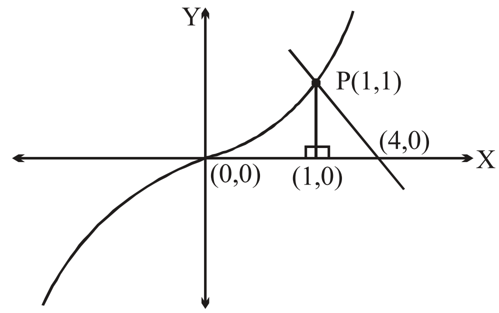
02 Required area

$= 2\left[\int_0^1 x^2dx - \int_{\frac{3}{4}}^1 \sqrt{4x-3}\right] = \frac{1}{3} = A(\text{Given})$

$\therefore\ 12A = 12 \times \frac{1}{3} = 4$

03 $y = x^3,\ \frac{dy}{dx} = 3x^2$

Now, $\left.\frac{dy}{dx}\right]_{(1,\,1)} = 3$



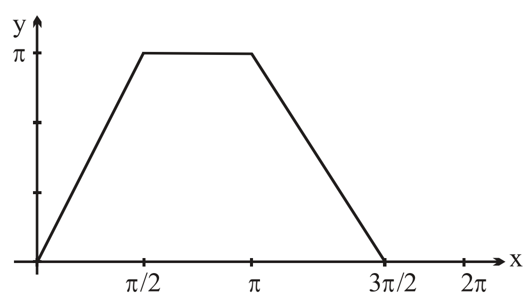
$\therefore$ Normal at $P(1, 1)$ is $(y-1) = \frac{-1}{3}(x-1)$

$\Rightarrow x + 3y = 4$, intersects x-axis at (4, 0)

So, required area $= \int_0^1 x^3dx + \frac{1}{2}(3\times 1)$

$= \frac{7}{4} = \frac{A}{8}$ (Given) $\Rightarrow A = 14.$ **Ans.**

04 Enclosed Area $= 2\pi \times \frac{\pi}{2} = \pi^2$



$f(x) = x + x = 2x \qquad ; \qquad 0 < x < \frac{\pi}{2}$

$= (\pi - x) + x = \pi \qquad ; \qquad \frac{\pi}{2} < x < \pi$

$= \pi - x + 2\pi - x = 3\pi - 2x; \quad \pi < x < \frac{3\pi}{2}$

$$= x - 2\pi + 2\pi - x = 0 \ ; \frac{3\pi}{2} < x < 2\pi$$

Paragraph for question nos. 5 to 6

(i) $\int_0^{2\pi}\left(f^2(x)+g^2(x)\right)dx = 2\int_0^{2\pi}(\sin 4x + 1)\,dx$

$$= \left(\frac{-\cos 4x}{2} + 2x\right)\Bigg|_0^{2\pi} = 4\pi$$

(ii) $f(x) = g(x) \Leftrightarrow \sin\left(\frac{\pi}{4} - x\right)\sin\left(\frac{\pi}{4} + 2x\right)$

$$= \sin\left(\frac{\pi}{4} + x\right)\sin\left(\frac{\pi}{4} + 2x\right)$$

$$\Leftrightarrow \sin\left(\frac{\pi}{4} + 2x\right) = 0$$

$$\Leftrightarrow x = \frac{3\pi}{8} \ ; x = \frac{7\pi}{8}$$

$$\sin\left(\frac{\pi}{4} - x\right) = \sin\left(\frac{\pi}{4} + x\right)$$

$\Rightarrow x = 0, \pi$ where $x \in [0, \pi]$

$$\Rightarrow A = \left|\int_0^{3\pi/8}(\sin 3x + \cos x - \cos 3x - \sin x)\,dx\right|$$

$$+ \left|\int_{3\pi/8}^{7\pi/8}(\sin 3x + \cos x - \cos 3x - \sin x)\,dx\right|$$

$$+ \left|\int_{7\pi/8}^{\pi}(\sin 3x + \cos x - \cos 3x - \sin x)\,dx\right|$$

Hence $A = A_1 + A_2 + A_3$

where $A_1 = \frac{-2}{3} + \frac{4}{3}\left(\sin\frac{\pi}{8} + \cos\frac{\pi}{8}\right)$;

$$A_2 = \frac{8}{3}\cos\frac{\pi}{8} \ ; A_3 = \frac{-2}{3} + \frac{4}{3}\left(-\sin\frac{\pi}{8} + \cos\frac{\pi}{8}\right)$$

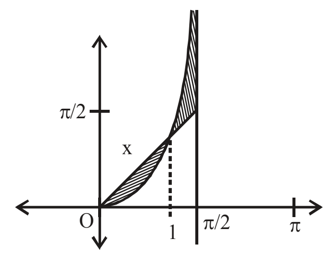
$$\Rightarrow A = \frac{-4}{3} + \frac{16}{3}\cos\frac{\pi}{8}.$$

$k_1 = \frac{16}{3}$ and $k_2 = \frac{-4}{3}$.

Hence $k_1 + k_2 = 4$. **Ans.**

07 We have $f(x) = \begin{cases} x^2 ; & x \in \left[0, \frac{\pi}{2}\right] \\ (\pi - x)^2 ; & x \in \left(\frac{\pi}{2}, \pi\right) \end{cases}$;

$$g(x) = \begin{cases} x ; & x \in \left[0, \frac{\pi}{2}\right] \\ (\pi - x) ; & x \in \left(\frac{\pi}{2}, \pi\right) \end{cases}$$



both f & g are periodic
with period = π & symmetric
about x = π/2

$\therefore$ Required area = 2

$$\left(\int_0^1 (x - x^2)\,dx + \int_1^{\pi/2}(x^2 - x)\,dx\right)$$

$$= 2\left(\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{\pi^3}{24} - \frac{1}{3}\right) - \left(\frac{\pi^2}{8} - \frac{1}{2}\right)\right)$$

$$= 2\left(\frac{\pi^3}{24} - \frac{\pi^2}{8} + \frac{1}{3}\right) = \frac{\pi^3}{12} - \frac{\pi^2}{4} + \frac{2}{3}$$

$\equiv a\pi^3 + b\pi^2 + c$ (Given)

So, $a = \frac{1}{12}$, $b = \frac{-1}{4}$, $c = \frac{2}{3}$

Hence, $(|a| + |b| + |c|) =$

$$\left|\frac{1}{12}\right| + \left|\frac{-1}{4}\right| + \left|\frac{2}{3}\right|$$

$$= \frac{1}{12} + \frac{1}{4} + \frac{2}{3} = \frac{1+3+8}{12} = 1 \textbf{ Ans}$$

Paragraph for question nos. 8 to 11

8.

(1) $f(x + y) = e^x f(y) + e^y f(x)$ .... (i)

$x = y = 0$

$\Rightarrow f(0) = 2f(0) \rightarrow f(0) = 0$

Differentiating (1) w.r.t. x partially

$f'(x+y) = e^x f(y) + e^y f'(x)$ .... (ii)

Differentiating (1) w.r.t. y partially

$f'(x+y) = e^x f'(y) + e^y f(x)$ .... (iii)

(ii) – (iii) $\Rightarrow$

$$\frac{f'(x)-f(x)}{e^x} = \frac{f'(y)-f(y)}{e^y} \rightarrow f'(x) - f(x)$$

$= Ae^x$, where A is a constant. It is linear d.e. with integrating factor e^{-x}.

$\therefore \quad f(x)e^{-x} = \int Ae^x \cdot e^{-x} dx = Ax + B$

$f(0) = 0 \rightarrow B = 0$

$\therefore \quad f(x) = Axe^x$

$f'(x) = A(x+1)e^x$

$f'(0) = 1 \rightarrow A = 1$

$\therefore \quad f(x) = xe^x,\ f'(x) = (x+1)e^x$

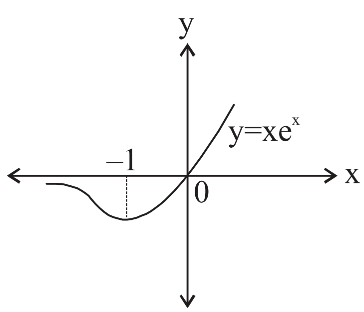
$f'(x) = 0 \rightarrow x = -1$

$f''(x) = (x+2)\,e^x,\ f''(-1) = e^{-1} > 0$

f(x) has minimum at $x = -1$.

(2) $f(x) \rightarrow \infty$ as $x \rightarrow \infty$

The range is $[f(-1), f(\infty)) = \left[-\frac{1}{e}, \infty\right)$



(3) $\lim\limits_{x\to-\infty} \frac{x}{e^{-x}} = \lim\limits_{x\to-\infty} \frac{1}{-e^{-x}} = 0$ by L.H. Rule.

(4) The area $= \int_{-\infty}^{0} -f(x)dx$

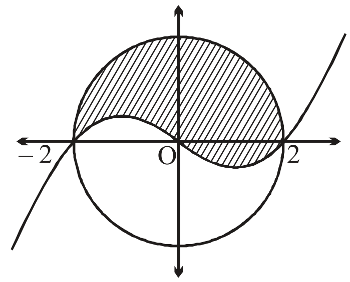
$$= \int_{-\infty}^{0} -xe^x dx = -(x-1)e^x\Big|_{-\infty}^{0} = 1$$

12. $f'(x) = a\left(x^2 - \frac{4}{3}\right)$

$\Rightarrow f(x) = a\left(\frac{x^3}{3} - \frac{4x}{3}\right) + b$ passes through

(0, 0) and (1, – 2)

$\therefore \quad b = 0,\ a = 2$



$f(x) = \frac{2x}{3}(x^2 - 4)$

required area $= \frac{\pi(2)^2}{2} = 2\pi = 4\frac{\pi}{2}$

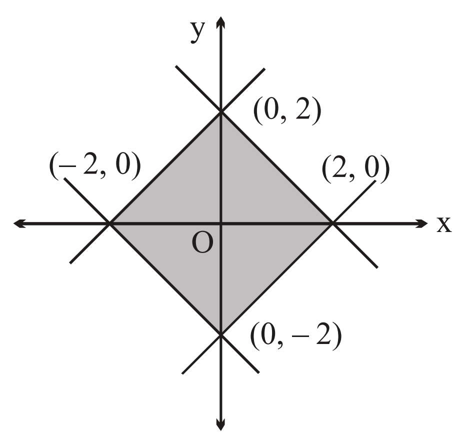
$\Rightarrow$ k = 4. **Ans.**

13 $\int_0^1 \left(4x^6 + \left(4x^3 - f(x)\right)f(x) - 4x^6\right)dx = \frac{4}{7}$

$\int_0^1 \left(f(x) - 2x^3\right)^2 dx = 0$

$\therefore \quad f(x) = 2x^3$

14



Area $= 4\left(\frac{1}{2}\right) \times 2 \times 2 = 8$. **Ans.**

01 Let f(x) $(f(x) > 0)$ be a differentiable function satisfying $f^2(x) =$

$$\int_0^x \left(f^2(t) - f^4(t) + (f'(t))^2\right)dt + 100$$

where $f^2(0) = 100$, then $\lim_{x \to \infty} f(x)$ can be

(A) 0 (B*) 1

(C) $\frac{10}{9}$ (D) 10

02 Let y = f(x) defined in [0, 2] satisfies the differential equation $y^3 y'' + 1 = 0$ where f(x) $\ge 0\ \forall\ x \in D_f$ and $f'(1) = 0$, $f(1) = 1$ then find the maximum value of f(x).
[Note : D_f denotes the domain of the function and y'' denotes the 2nd derivative of y w.r.t x.]

03 If $(x^2 + y^2)dy = xy\,dx$ and $y(1) = 1$, $y(\alpha) = e$, then $\alpha^4 = ke^4$, where k equals

(A) 9 (B) $\frac{9}{4}$

(C) 3 (D) $\frac{3}{2}$

04 If y(x) satisfies the differential equation

$$\frac{dy}{dx} + 2y\tan x = \sin x \text{ and } y\left(\frac{\pi}{3}\right) = 0,$$

then which of the following statement(s) is(are) correct ?

(A) $y'\left(\frac{\pi}{6}\right) = \frac{2\sqrt{3}-1}{2}$

(B) $y(\pi) = 3$

(C) Number of values of x for which $y(x) = -1$ in $(0, 2\pi)$ are two.

(D) $\int_0^{\frac{\pi}{2}} y(x)\,dx = 1 - \frac{\pi}{4}$.

05. A function $y = f(x)$ satisfy $f'(x) = 2f(x) - x$ with $f(0) = b$. If $f(x) > 0$ for all $x \ge 0$, then find the least integral value of b.

06 The general solution y(x) of the differential equation $\frac{d\left(\int_x^y dt\right)}{dy} = x$, is

(A) $y = ln|1 - x| + C$
(B) $y = -ln|1 - x| + C$
(C) $y = -ln|1 + x| + C$
(D) $y = ln|1 + x| + C$

07. Let f(x) b differentiable an the interval $(0, \infty)$ such that $f(1) = 1$ and

$$\lim_{t \to x} \frac{t^3 f(x) - x^3 f(t)}{t^2 - x^2} = 1 \text{ for each } x > 0,$$

then find the value of $3f(3)$.

08 Family of curves satisfying the differential equation,
$x(1 - x^2)\,dy + (2x^2y - y - 5x^3)dx = 0$, is

(A) $y + 5x = cx\sqrt{1-x^2}$

(B) $y + 5x = cx\sqrt{1+x^2}$

(C) $y - 5x = cx\sqrt{1-x^2}$

(D) $y - 5x = cx\sqrt{1+x^2}$

09. A curve y = f(x) passing through (1, 2) is such that the intercept a tangent cuts off on the ordinate axis is half the sum of the co-ordinates of the tangency point. Find the value of $f(9)$.

10 Let y = y(x) satisfy the equation

$y(x) + \int_1^x y(t)\,dt = x^2$. The value of y(e) is equal to

(A) $2e - 2 + e^{1-e}$ (B) $2e - 1 + e^{1-e}$
(C) $2e + 2 + e^{1-e}$ (D) $2e + 1 + e^{1-e}$

11(mcq) A particle moves in a straight line. Its position at time t is f(t). When $0 \le t \le 1$, the position is given by the integral

$$f(t) = \int_0^t \frac{1 + 2\sin\pi x\cos\pi x}{1 + x^2}dx. \text{ For } t \ge 1,$$

the particle moves with constant acceleration. Which of the following is(are) correct ?

(A) it's acceleration at time t = 2 is $\pi - \frac{1}{2}$

(B) it's velocity when t = 1, is $\frac{1}{2}$

(C) it's velocity when t > 1 is

$\frac{1}{2} + (\pi - \frac{1}{2})\,(t-1)$

(D) the difference $f(t) - f(1)$ when $t > 1$, is

$\frac{1}{2}(t-1) + (\pi - \frac{1}{2})(t-1)^2 . \frac{1}{2}$

12. Let $f(x)$ be a positive differentiable function defined on $[0, 1]$ such that $f(1) = 0$ and for any $a \in (0, 1)$,

$$\int_0^a f(x)\,dx - \int_a^1 f(x)\,dx = 2f(a) + 3a + b,$$

where b is a constant. Find $f(x)$.

Paragraph for question nos. 13 to 15

Let C be a curve $f(x, y) = 0$ passing through (1, 1) such that it is orthogonal to family of circles $x^2 + y^2 - ax = 0$, where a being parameter.

13. If the curve C intersects the circle $x^2 + y^2 + 2x + 4y + k = 0$ orthogonally then 'k' is equal to

(A) 4 (B) 2

(C) 0 (D) – 4

14 The area enclosed by the curve C and its tangent at (1, 1) and x-axis is equal to

(A) $1 - \cot^{-1}\sqrt{3}$ (B) $\tan^{-1}\sqrt{3} - 1$

(C) $1 - \sec^{-1}\sqrt{2}$ (D) $\sin^{-1}(1) - 1$

15 The shortest distance between the curve C and the parabola $x^2 = 2y + 1$ is equal to

(A) $\sqrt{2} - 1$ (B) $\sqrt{2} + 1$

(C) $\sqrt{6} - 1$ (D) $\sqrt{5} - 1$

ANSWER KEY

1.	B	**2.**	1	**3.**	A
4.	AC	**5.**	1	**6.**	B
7.	41	**8.**	C	**9.**	0
10.	A	**11.**	ABCD	**12.**	$\frac{3}{2}\left(1-e^{x-1}\right)$
13.	D	**14.**	C	**15.**	A

SOLUTIONS

01 Differentiate

$$2f(x)\,f'(x) = f^2(x) - f^4(x) + \left(f'(x)\right)^2$$

$$\Rightarrow f^4(x) = f^2(x) + \left(f'(x)\right)^2 - 2f(x)\,f'(x)$$

$$\therefore f^4(x) = \left(f(x) - f'(x)\right)^2$$

$$\Rightarrow f(x) - f'(x) = \pm f^2(x)$$

$$\therefore \frac{dy}{dx} = y \pm y^2$$

$$\oplus \quad \frac{dy}{y(1+y)} = x + c$$

$$ln\left(\frac{y}{y+1}\right) = x + c$$

$$\frac{y}{y+1} = Ke^x$$

at $x = 0$; $y = 10$

$$\therefore \quad K = \frac{10}{11}$$

$$\Rightarrow \frac{y}{y+1} = Ke^x$$

$$\Rightarrow \frac{y+1}{y} = \frac{1}{K}e^{-x}$$

$$1 - \frac{e^x}{K} = \frac{1}{y}$$

$$\therefore \quad y = \frac{K}{K - e^{-x}}$$

$$\therefore \quad \lim_{x \to \infty} y = 1$$

$$\ominus \quad \frac{dy}{y(1-y)} = x + c$$

$$ln\left(\frac{y}{1-y}\right) = x + c$$

$$\frac{y}{1-y} = Ke^x$$

at $x = 0$; $y = 10$

$$\therefore \quad k = \frac{-10}{9}$$

$$\frac{1-y}{y} = \frac{1}{K}e^{-x}$$

$$\frac{1}{y} = \frac{K + e^{-x}}{K}$$

$$y = \frac{K}{K + e^{-x}}$$

$\lim\limits_{x \to \infty} f(x) = 1$. **Ans.**

02 $y^3 y'' = -1$

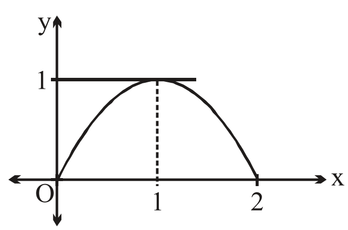
$$y' y'' = \frac{-y'}{y^3}$$

Integrating both sides

$$\frac{(y')^2}{2} = \frac{1}{2y^2} + c$$

putting x = 1, y = 1 and y'(1) = 0

$$0 = \frac{1}{2} + c \Rightarrow c = \frac{-1}{2}$$



$$\frac{(y')^2}{2} = \frac{1}{2y^2} - \frac{1}{2}$$

$$(y')^2 = \frac{1}{y^2} - 1 \Rightarrow (y')^2 = \frac{1-y^2}{y^2}$$

$$\Rightarrow y' = \pm \frac{\sqrt{1-y^2}}{y} \Rightarrow \frac{y}{\sqrt{1-y^2}} y' = \pm 1$$

Taking as +ve sign

$$\int \frac{y}{1-y^2} y'dx = \int 1.dx$$

Putting $y = t$

$\Rightarrow y'dx = dt$

$$\int \frac{t}{\sqrt{1-t^2}} dt = x + \lambda \Rightarrow 1 - t^2 = z^2$$

$\Rightarrow -2tdt = 2zdz$

$$-\int \frac{z\,dz}{z} = x + \lambda \Rightarrow -z = x + \lambda$$

$$\Rightarrow -\sqrt{1-t^2} = x + \lambda$$

$$\Rightarrow -\sqrt{1-y^2} = x + \lambda$$

putting x = 1, y(1) = 1

$\therefore \lambda = -1 \Rightarrow -\sqrt{1-y^2} = x - 1$

$\Rightarrow 1 - y^2 = x^2 - 2x + 1$

$y^2 = 2x - x^2$

$y = \sqrt{2x - x^2} = f(x) \quad \{\because y \geq 0 \ \forall\ x \in D_f\}$

Taking as negative sign we will obtain the same function.

$$f'(x) = \frac{2-2x}{2\sqrt{2x-x^2}} = 0 \Rightarrow x = 1$$

which is point of maxima

$\therefore$ maximum value of f(x) is

$= f(1) = \sqrt{2-1} = 1$. **Ans.**

03 We have, $\dfrac{dy}{dx} = \left(\dfrac{xy}{x^2+y^2}\right) = \dfrac{\frac{y}{x}}{1+\left(\frac{y}{x}\right)^2}$

Put y = vx

$$\Rightarrow \frac{dy}{dx} = v + \frac{xdv}{dx}$$

$$\Rightarrow v + \frac{xdv}{dx} = \frac{v}{1+v^2}$$

$$\Rightarrow x \cdot \frac{dv}{dx} = \left(\frac{v}{1+v^2} - v\right)$$

$$\Rightarrow x \cdot \frac{dv}{dx} = \frac{v - v - v^3}{1+v^2}$$

$$\Rightarrow \int \left(\frac{1+v^2}{v^3}\right) dv = -\int \frac{dx}{x}$$

$$\Rightarrow \int \left(v^{-3} + \frac{1}{v}\right) dv = -\int \frac{dx}{x}$$

$$\Rightarrow \frac{v^{-2}}{-2} + \ln v = -\ln x + C$$

$$\Rightarrow \frac{-1 \cdot x^2}{2y^2} + \ln\left(\frac{y}{x}\right) = -\ln x + C$$

Now, y(1) = 1

$\Rightarrow \dfrac{-1}{2} + 0 = -0 + C$

$\Rightarrow C = \dfrac{-1}{2}.$

$\therefore \dfrac{-x^2}{2y^2} + ln\left(\dfrac{y}{x}\right) + ln x = \dfrac{-1}{2}$

$\Rightarrow \dfrac{-x^2}{2y^2} + ln\ y = \dfrac{-1}{2}$

Also, $y(\alpha) = e$

$\Rightarrow \dfrac{-\alpha^2}{2e^2} + 1 = \dfrac{-1}{2}$

$\Rightarrow \dfrac{\alpha^2}{2e^2} = \dfrac{3}{2}$

$\Rightarrow \alpha^2 = 3e^2$

$\therefore \quad \alpha^4 = 9e^4$

So, $k = 9$. **Ans.**

04 We have, $\dfrac{dy}{dx} + (2\tan x)y = \sin x$

(Linear differetial equation)

$\therefore$ I.F. $= e^{2\int \tan x\, dx} = e^{2\, ln\,(\sec x)} = \sec^2 x.$

So, general solution is

$y \cdot \sec^2 x = \displaystyle\int \frac{\sin x}{\cos^2 x}\, dx + C$

$\Rightarrow y \cdot \sec^2 x = \sec x + C$

Now, $y\left(\dfrac{\pi}{3}\right) = 0 \Rightarrow C = -2.$

$\Rightarrow y(x) = \cos x - 2\cos^2 x$

(A) $y'(x) = -\sin x + 4\cos x \cdot \sin x$

$\therefore y'\left(x = \dfrac{\pi}{6}\right) = \dfrac{-1}{2} + 4\,\dfrac{\sqrt{3}}{2} \times \dfrac{1}{2}$

$= \dfrac{-1}{2} + \sqrt{3} = \dfrac{2\sqrt{3}-1}{2}.$

(B) $y(\pi) = \cos\pi - 2\cos^2\pi$

$= (-1) - 2(-1)^2 = (-1) - 2(1)$

$= -1 - 2 = -3.$

(C) Consider the equation,

$y(x) = -1$

$\Rightarrow \quad \cos x - 2\cos^2 x$

$= -1 \Rightarrow 2\cos^2 x - \cos x - 1 = 0$

$\Rightarrow \quad (2\cos x + 1)(\cos x - 1) = 0$

Either $\cos x = \dfrac{-1}{2}$ or $\cos x = 1$

But $x \in (0, 2\pi)$, so $x = \dfrac{2\pi}{3}, \dfrac{4\pi}{3}.$

(D) $\displaystyle\int_0^{\frac{\pi}{2}} y(x)\, dx = \int_0^{\frac{\pi}{2}} (\cos x - 2\cos^2 x)\, dx$

$= 1 - 2\left(\dfrac{\pi}{4}\right) = 1 - \dfrac{\pi}{2}.$ **Ans.**

05. $\dfrac{dy}{dx} - 2y = -x$ (Linear differential equation)

I.F. $= e^{-2x}$

$\therefore \quad y \cdot e^{-2x} =$

$-\displaystyle\int \underbrace{x}_{\mathrm{I}}\, \underbrace{e^{-2x}}_{\mathrm{II}}\, dx = -\left[-\frac{x}{2}e^{-2x} + \frac{1}{2}\int e^{-2x}\, dx\right]$

$= \dfrac{x}{2}e^{-2x} + \dfrac{1}{4}e^{-2x} + C$

$f(x) = \dfrac{x}{2} + \dfrac{1}{4} + Ce^{2x}$

$y(0) = b \Rightarrow C = b - \dfrac{1}{4}$

Hence $f(x) = \dfrac{x}{2} + \dfrac{1}{4} + \left(b - \dfrac{1}{4}\right)e^{2x}.$

Now $f(x) > 0 \quad \forall\ x \geq 0$

$\Rightarrow \quad b \geq \dfrac{1}{4}$

$\Rightarrow$ least integer value of $b = 1$ **Ans.**

06 $\dfrac{d\left(\displaystyle\int_x^y dt\right)}{dy} = x$

$\Rightarrow \dfrac{d(y-x)}{dy} = x$

$\Rightarrow 1 - \dfrac{dx}{dy} = x$

$\Rightarrow \dfrac{1}{1-x} = \dfrac{dy}{dx} \Rightarrow \displaystyle\int \frac{dx}{1-x} = \int dy$

$\Rightarrow y = -ln(1-x) + C$

07. $\lim\limits_{t\to x}\dfrac{t^3f(x)-x^3f(t)}{t^2-x^2}=1$

$$\lim_{t\to x}\frac{3t^2f(x)-x^3f'(t)}{2t}=1$$

$3x^2 f(x) - x^3 f'(x) = 2x$

$x^3 f'(x) - 3x^2 f(x) = -2x$

$$f'(x)-\frac{3}{x}f(x)=-\frac{2}{x^2}$$

$$\text{I.F.} = e^{-\int\frac{3}{x}dx} = e^{-3\ln x} = x^{-3}$$

$$yx^{-3} = -\int\frac{2}{x^2}\cdot\frac{1}{x^3}dx = -2\int x^{-5}dx$$

$$= -2\frac{(x^{-4})}{(-4)}+c$$

$$yx^{-3} = \frac{1}{2}x^{-4}+c$$

$$1 = \frac{1}{2}+c$$

$$y = \frac{1}{2x}+cx^3$$

$$c = \frac{1}{2}$$

$$f(x) = y = \frac{1}{2x}+\frac{x^3}{2} = \frac{(1+x^4)}{2x}$$

$$f(3) = \frac{(1+3^4)}{(2)(3)} = \frac{(1+81)}{(2)(3)} = \frac{82}{(2)(3)} = \frac{41}{3}$$

$3f(3) = 41$

08 $x(1-x^2)\dfrac{dy}{dx}+(2x^2-1)\,y-5x^3=0$

$$\Rightarrow \frac{dy}{dx}+\left(\frac{2x^2-1}{x(1-x^2)}\right)y$$

$$= \frac{5x^3}{x(1-x^2)} = \frac{5x^2}{1-x^2}$$

$$\text{I.F.} = e^{\int\frac{2x^2-1}{x(1-x^2)}dx} = e^{\int\frac{x^2+x^2-1}{x(1-x^2)}dx} = e^{-\ln(x\sqrt{1-x^2})}$$

$$= \frac{1}{x\sqrt{1-x^2}}$$

$$\text{Hence } y\cdot\frac{1}{x\sqrt{1-x^2}} = 5\int\frac{x\,dx}{x\sqrt{1-x^2}}$$

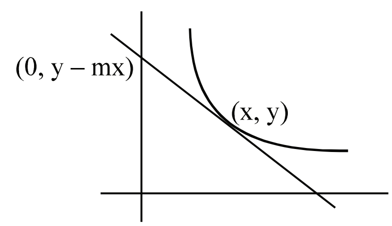
$$= \int\frac{x\,dx}{(1-x^2)^{3/2}} = -\frac{5}{\sqrt{1-x^2}}+c$$

or $y = cx\sqrt{1-x^2}-5x$

$\Rightarrow y+5x = cx\sqrt{1-x^2}$

09. $y - mx = \dfrac{x+y}{2}$

$$\Rightarrow \quad \frac{dy}{dx}-\frac{1}{2x}y = -\frac{1}{2}$$



$y = c\sqrt{x}-x$

$\Rightarrow y = 3\sqrt{x}-x$

10 Differentiate w.r.t. x

$y'(x)+y(x) = 2x$

$$\frac{dy}{dx}+y = 2x$$

I.F. $= e^x$

$$ye^x = 2\int xe^x\,dx$$

$$ye^x = 2\left[xe^x-e^x\right]+C$$

$y = 2(x-1)+ce^{-x}$

at $x = 1$; $y(1) = 1 \Rightarrow c = e$

$y = 2(x-1)+e^{1-x}$. **Ans.**

11. $V(t) = f'(t) = \dfrac{1+\sin 2\pi t}{1+t^2} \Rightarrow f'(1) = \dfrac{1}{2}$

Also

$$\frac{dV}{dt} = \frac{(1+t^2)2\pi\cos 2\pi t-(1+\sin 2\pi t)2t}{(1+t^2)^2}$$

$$\therefore \quad \left.\frac{dV}{dt}\right]_{t=1} = \frac{4\pi-2}{4} = \pi-\frac{1}{2}$$

(which is independent of t)

This remains constant

$\Rightarrow$ hence when t = 2 then acceleration

$= \pi - \frac{1}{2}$

now velocity at t = 1 is $\frac{1}{2} = V_0$ say

(using $v = u + at$)

Velocity when t > 1 is given by

$V(t > 1) = V_0 + a(t - 1)$

$= \frac{1}{2} + (\pi - \frac{1}{2})(t - 1)$

for t > 1, f(t) – f(1)

$$= \int_0^1 f(x)\,dx + \int_1^t f(x)\,dx - \int_0^1 f(x)\,dx$$

$$= \int_1^t f(x)\,dx = V_0(t-1) + \frac{1}{2}a(t-1)^2$$

(using $s = ut + \frac{1}{2}at^2$)

$$f(t) - f(1) = \frac{1}{2}(t-1) + \frac{1}{2}(\pi - \frac{1}{2})(t-1)^2$$

where t > 1

12. $\int_0^a f(x)\,dx - \int_a^1 f(x)\,dx = 2f(a) + 3a + b$

Differentiating w.r.t. by a Leibnitz rule,

$2f(a) = 2f'(a) + 3$ or $f'(a) - f(a) = \frac{-3}{2}$

a linear differential equation with integrating factor e^{-a}

$$\therefore \quad f(a)e^{-a} = \int \frac{-3}{2}e^{-a}\,da = \frac{3}{2}e^{-a} + A$$

$$f(a) = \frac{3}{2} + Ae^{a}$$

$$f(1) = 0$$

$$\Rightarrow \frac{3}{2} + Ae = 0 \Rightarrow A = \frac{-3}{2}e^{-1}$$

$$\therefore \quad f(a) = \frac{3}{2}\left(1 - e^{a-1}\right)$$

$$f(x) = \frac{3}{2}\left(1 - e^{x-1}\right)$$ **Ans.**

Paragraph for question nos. 13 to 15

Sol :

$x^2 + y^2 - ax = 0$

$a = \frac{x^2 + y^2}{x}$

Differentiating

$$0 = \frac{x(2x + 2yy') - (x^2 + y^2)}{x^2}$$

$\Rightarrow x^2 + 2xy\,y' - y^2 = 0$

orthogonal curve

$$x^2 - \frac{2xy}{y'} - y^2 = 0$$

$$\Rightarrow y' = \frac{2xy}{x^2 - y^2} \qquad \ldots (1)$$

Put y = vx

$$v + x\frac{dv}{dx} = \frac{2v}{1 - v^2}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{2v}{1 - v^2} - v = \frac{v + v^3}{1 - v^2}$$

$$\Rightarrow \frac{1 - v^2}{v + v^3}\,dv = \frac{dx}{x}$$

$$\Rightarrow \frac{(1 + v^2) - 2v^2}{v(1 + v^2)}\,dv = \frac{dx}{x}$$

$$\left(\frac{1}{v} - \frac{2v}{1 + v^2}\right)dv = \frac{dx}{x}$$

$\Rightarrow \ln v - \ln(1 + v^2) = \ln x + \ln c$

$$\Rightarrow \ln\left(\frac{v}{1 + v^2}\right) = \ln(cx)$$

$$\Rightarrow \frac{v}{1 + v^2} = cx \Rightarrow \frac{xy}{x^2 + y^2} = cx$$

$$\Rightarrow x^2 + y^2 - \frac{y}{c} = 0.$$

This passes through (1, 1)

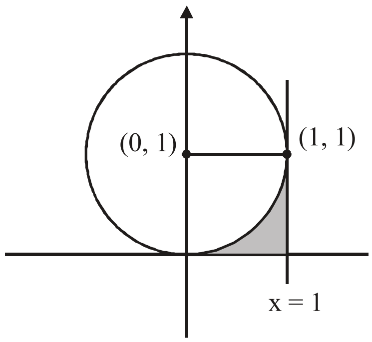
$\therefore \; c = \frac{1}{2}$

$\therefore \; C : x^2 + y^2 - 2y = 0$

(13) Since $x^2 + y^2 + 2x + 4y + k = 0$
and $x^2 + y^2 - 2y = 0$ are orthogonal
$\therefore\ 2g_1g_2 + 2f_1f_2 = c_1 + c_2$
$\therefore\ k = -4$

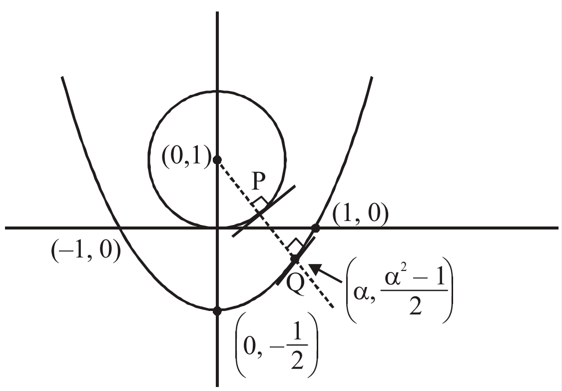
(14) Required area

$= 1 - \frac{\pi}{4}$



(15) Let $Q\left(\alpha, \frac{\alpha^2 - 1}{2}\right)$

Slope of tangent at P and Q are same
m_{PQ} = slope of normal at Q



$$\frac{-1}{\alpha} = \frac{\frac{\alpha^2 - 1}{2} - 1}{\alpha - 0}$$

$\Rightarrow \alpha^3 - \alpha = 0 \ \Rightarrow \alpha = 0, \pm 1$
$\therefore Q(1, 0)$
$PQ = \sqrt{2} - 1.$

TREND ANALYSIS OF JEE Main 2010-2019

Mathematics
Gavesh Bhardwaj (GB Sir)
Etoos Faculty, Ex. Motion,
Ex.Career Point, Ex. Allen Faculty
B.Tech (Hons.), Exp: 15 Years

MATHEMATICS

S.N.	Chapters	Years										
		19	18	17	16	15	14	13	12	11		10
1.	Sets, Relations and Functions	2	1	2	1	1	1	-	1	2	3	1
2.	Complex Numbers and Quadratic Equations	2	2	2	2	2	3	3	2	2	1	2
3.	Permutations and Combinations	1	1	1	1	1	-	3	2	1	1	2
4.	Mathematical Induction	-	-	-	-	-	-	1	-	-	1	-
5.	Binomial Theorem and Its Applications	1	1	1	1	1	1	-	1	1	-	-
6.	Sequences and Series	1	2	1	2	2	2	2	2	1	1	1
7.	Exponential and Logarithmic Series	-	-	-	-	-	-	-	-	-	-	-
8.	Matrices and Determinants	2	2	2	2	2	2	2	2	2	3	3
9.	Statistics and Probability	2	2	3	2	2	2	1	2	3	2	3
10.	Trigonometry	2	2	2	2	2	2	2	1	1	1	2
11.	Mathematical Reasoning	1	1	1	1	1	1	-	1	1	1	1
12.	Coordinate Geometry	5	5	4	5	5	5	7	5	4	5	3
13.	Limits, Continuity and Differentiability	2	2	2	2	2	4	1	5	4	4	1
14.	Integral Calculus	2	2	2	3	3	3	3	3	2	2	2
15.	Differential Equations	1	1	1	1	1	1	-	-	2	1	1
16.	Three Dimensional Geometry	1	2	2	2	2	2	2	2	2	2	2
17.	Vector Algebra	2	1	1	1	1	1	1	2	2	2	2
18.	Area Bounded by Curve	1	1	1	-	-	-	1	-	-	-	-
19.	Application of Derivative	2	2	2	2	2	-	-	-	-	-	-